

APPENDIX G

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/439,492		Filing Date 09/15/2021		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	09/15/2021	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 25	Minus	** 20	= 5			x \$100 =		500	
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0			x \$480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		500	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/SHERRY A DAVIS/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".											
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.											

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Box No. VIII (ii) DECLARATION: ENTITLEMENT TO APPLY FOR AND BE GRANTED A PATENT

The declaration must conform to the standardized wording provided for in Section 212; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (ii). If this Box is not used, this sheet should not be included in the request.

Declaration as to the applicant's entitlement, as at the international filing date, to apply for and be granted a patent (Rules 4.17(ii) and 51bis.1(a)(ii)), in a case where the declaration under Rule 4.17(iv) is not appropriate:

in relation to this international application,

trinamiX GmbH is/are entitled to apply for and be granted a patent by virtue of the following:

Assignment from Diesselberg, Lars to Hengen Ingenieurbüro GbR., dated 19 August 2019

Assignment from Hengen, Heiko to Hengen Ingenieurbüro GbR., dated 19 August 2019

Assignment from Hengen Ingenieurbüro GbR. to *trinamiX GmbH*, dated 26 August 2019

Assignment from Schick, Friedrich to *trinamiX GmbH*, dated 09 April 2019

Assignment from Schillen, Peter to *trinamiX GmbH*, dated 09 April 2019

Assignment from Schindler, Patrick to *trinamiX GmbH*, dated 09 April 2019

Assignment from Schmidt, Andre to *trinamiX GmbH*, dated 11 April 2019

Assignment from Eberspach, Michael to *trinamiX GmbH*, dated 11 April 2019

Assignment from Lennartz, Christian to *trinamiX GmbH*, dated 11 April 2019

Assignment from Send, Robert to *trinamiX GmbH*, dated 12 October 2019

Assignment from Bruder, Ingmar to *trinamiX GmbH*, dated 12 April 2019

Assignment from Unger, Jakob to *trinamiX GmbH*, dated 13 August 2019

Assignment from Bonsignore, Christian to *trinamiX GmbH*, dated 14 August 2019

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

PCT REQUEST

(Original in Electronic Form)

0	For receiving Office use only	
0-1	International Application No.	PCT/EP2020/056759
0-2	International Filing Date	13 MAR 2020 (13.03.2020)
0-3	Name of receiving Office and "PCT International Application"	RO/EP
0-4	Form PCT/RO/101 PCT Request	
0-4-1	Prepared Using	PCT Online Filing Version 3.51.000.265e MT/FOP 20141031/0.20.5.20
0-5	Petition The undersigned requests that the present international application be processed according to the Patent Cooperation Treaty	
0-6	Receiving Office (specified by the applicant)	European Patent Office (EPO) (RO/EP)
0-7	Applicant's or agent's file reference	B182037PC
I	Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY
II	Applicant	
II-1	This person is	Applicant only
II-2	Applicant for	All designated States
II-4	Name	trinamiX GmbH
II-5	Address	Industriestrasse 35 67063 Ludwigshafen am Rhein Germany
II-6	State of nationality	DE
II-7	State of residence	DE
III-1	Applicant and/or inventor	
III-1-1	This person is	Inventor only
III-1-3	Inventor for	All designated States
III-1-4	Name (LAST, First)	SCHICK, Friedrich
III-1-5	Address	Industriestraße 35 Ludwigshafen Ludwigshafen Germany
III-2	Applicant and/or inventor	
III-2-1	This person is	Inventor only
III-2-3	Inventor for	All designated States
III-2-4	Name (LAST, First)	SCHILLEN, Peter
III-2-5	Address	Industriestraße 35 Ludwigshafen Ludwigshafen Germany

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(Original in Electronic Form)

III-3	Applicant and/or inventor	
III-3-1	This person is	Inventor only
III-3-3	Inventor for	All designated States
III-3-4	Name (LAST, First)	SCHINDLER, Patrick
III-3-5	Address	Industriestraße 35 67063 Ludwigshafen Germany
III-4	Applicant and/or inventor	
III-4-1	This person is	Inventor only
III-4-3	Inventor for	All designated States
III-4-4	Name (LAST, First)	SCHMIDT, Andre
III-4-5	Address	Industriestraße 35 67063 Ludwigshafen Germany
III-5	Applicant and/or inventor	
III-5-1	This person is	Inventor only
III-5-3	Inventor for	All designated States
III-5-4	Name (LAST, First)	EBERSPACH, Michael
III-5-5	Address	Industriestraße 35 67063 Ludwigshafen Germany
III-6	Applicant and/or inventor	
III-6-1	This person is	Inventor only
III-6-3	Inventor for	All designated States
III-6-4	Name (LAST, First)	LENNARTZ, Christian
III-6-5	Address	Industriestraße 35 67063 Ludwigshafen Germany
III-7	Applicant and/or inventor	
III-7-1	This person is	Inventor only
III-7-3	Inventor for	All designated States
III-7-4	Name (LAST, First)	SEND, Robert
III-7-5	Address	Luisenstraße 25 76137 Karlsruhe Germany
III-8	Applicant and/or inventor	
III-8-1	This person is	Inventor only
III-8-3	Inventor for	All designated States
III-8-4	Name (LAST, First)	DIESSELBERG, Lars
III-8-5	Address	c/o Hengen Ingenieurbuero GbR Barthelsmühlring 26 76870 Kandel Germany

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(Original in Electronic Form)

III-9	Applicant and/or inventor	
III-9-1	This person is	Inventor only
III-9-3	Inventor for	All designated States
III-9-4	Name (LAST, First)	HENGEN, Heiko
III-9-5	Address	c/o Hengen Ingenieurbuero GbR Barthelsmühlring 26 76870 Kandel Germany
III-10	Applicant and/or inventor	
III-10-1	This person is	Inventor only
III-10-3	Inventor for	All designated States
III-10-4	Name (LAST, First)	BRUDER, Ingmar
III-10-5	Address	Industriestraße 35 67063 Ludwigshafen Germany
III-11	Applicant and/or inventor	
III-11-1	This person is	Inventor only
III-11-3	Inventor for	All designated States
III-11-4	Name (LAST, First)	UNGER, Jakob
III-11-5	Address	Industriestraße 35 67063 Ludwigshafen Germany
III-12	Applicant and/or inventor	
III-12-1	This person is	Inventor only
III-12-3	Inventor for	All designated States
III-12-4	Name (LAST, First)	BONSIGNORE, Christian
III-12-5	Address	Industriestraße 35 67063 Ludwigshafen am Rhein Germany

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(Original in Electronic Form)

IV-1	Agent or common representative; or address for correspondence The person identified below is hereby/ has been appointed to act on behalf of the applicant(s) before the competent International Authorities as:	Agent
IV-1-1	Name	Altmann Stösel Dick Patentanwälte PartG mbB
IV-1-2	Address	Dr. Matthias Stösel Dudenstraße 46 68167 Mannheim Germany
IV-1-3	Telephone No.	+49 (0) 621-391813400
IV-1-4	Facsimile No.	+49 (0) 621-391813444
IV-1-5	e-mail	BASF-Post@asd-patent.com
IV-1-5(a))	E-mail authorization The receiving Office, the International Searching Authority, the International Bureau and the International Preliminary Examining Authority are authorized to use this e-mail address, if the Office or Authority so wishes, to send notifications issued in respect of this international application:	exclusively in electronic form (no paper notifications will be sent)
IV-1-6	Agent's registration No.	446
V	DESIGNATIONS	
V-1	The filing of this request constitutes under Rule 4.9(a), the designation of all Contracting States bound by the PCT on the international filing date, for the grant of every kind of protection available and, where applicable, for the grant of both regional and national patents.	
VI-1	Priority claim of earlier regional application	
VI-1-1	Filing date	15 March 2019 (15.03.2019)
VI-1-2	Number	19163250.4
VI-1-3	Regional Office	EP
VI-2	Priority document request The International Bureau is requested to obtain from a digital library a certified copy of the earlier application(s) identified above as item(s), using, where applicable, the access code(s) indicated:	VI-1 Access code: CCEF
VI-3	Incorporation by reference : where an element of the international application referred to in Article 11(1)(iii)(d) or (e) or a part of the description, claims or drawings referred to in Rule 20.5(a) is not otherwise contained in this international application but is completely contained in an earlier application whose priority is claimed on the date on which one or more elements referred to in Article 11(1)(iii) were first received by the receiving Office, that element or part is, subject to confirmation under Rule 20.6, incorporated by reference in this international application for the purposes of Rule 20.6.	
VII-1	International Searching Authority Chosen	European Patent Office (EPO) (ISA/EP)

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VII-2	Request to use results of earlier search; reference to that search		
VII-2-1	Filing date	15 March 2019 (15.03.2019)	
VII-2-2	Application Number	19 163 250.4	
VII-2-3	Country (or regional Office)	EP	
VII-2-5	Documents are available to the ISA in a form and a manner acceptable to it, and therefore do not need to be submitted by the applicant to the receiving Office, or to the ISA (Rules 12bis1.(c) and (d) and 12bis.2(b)):	A copy of the results of the earlier search A copy of the earlier application A copy of any document cited in the earlier search results	
VIII	Declarations	Number of declarations	
VIII-1	Declaration as to the identity of the inventor	—	
VIII-2	Declaration as to the applicant's entitlement, as at the international filing date, to apply for and be granted a patent	—	
VIII-3	Declaration as to the applicant's entitlement, as at the international filing date, to claim the priority of the earlier application	—	
VIII-4	Declaration of inventorship (only for the purposes of the designation of the United States of America)	—	
VIII-5	Declaration as to non-prejudicial disclosures or exceptions to lack of novelty	—	
IX	Check list	Number of sheets	Electronic file(s) attached
IX-1	Request (including declaration sheets)	7	✓
IX-2	Description	130	✓
IX-3	Claims	5	✓
IX-4	Abstract	1	✓
IX-5	Drawings	2	✓
IX-6a	Sequence listing part of the description (also to be used for the purposes of international search)	—	—
IX-7	TOTAL	145	

PCT REQUEST

(Original in Electronic Form)

	Accompanying Items	Paper document(s) attached	Electronic file(s) attached
IX-8	Fee calculation sheet	—	✓
IX-19	Other	Pre-conversion archive Statement (Section 706): This is a complete and accurate copy of the international application prior to its conversion into the electronic document format in which it is filed.	✓
IX-19	Other	Rule 4.17 Applicant Declaration	✓
IX-19	Other	Rule 4.17_Inventor_Declaration	✓
IX-20	Figure of the drawings which should accompany the abstract	1	
IX-21	Language of filing of the international application	English	
X-1	Signature of applicant, agent or common representative	/Dr. Matthias Stösel/	
X-1-1	Name	Altmann Stösel Dick Patentanwälte PartG mbB	
X-1-2	Name of signatory	Dr. Matthias Stösel	
X-1-3	Capacity (if such capacity is not obvious from reading the request)	(Representative)	

FOR RECEIVING OFFICE USE ONLY

10-1	Date of actual receipt of the purported international application	13 MAR 2020 (13.03.2020)
10-2	Drawings:	
10-2-1	Received	Received
10-2-2	Not received	
10-3	Corrected date of actual receipt due to later but timely received papers or drawings completing the purported international application	
10-4	Date of timely receipt of the required corrections under PCT Article 11(2)	
10-5	International Searching Authority	ISA/EP
10-6	Transmittal of search copy delayed until search fee is paid	

PCT REQUEST

(Original in Electronic Form)

FOR INTERNATIONAL BUREAU USE ONLY

11-1	Date of receipt of the record copy by the International Bureau	
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Box No. VIII (iv) DECLARATION: INVENTORSHIP (only for the purposes of the designation of the United States of America)
The declaration must conform to the following standardized wording provided for in Section 214; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (iv). If this Box is not used, this sheet should not be included in the request.

**Declaration of inventorship (Rules 4.17(iv) and 51bis.1(a)(iv))
for the purposes of the designation of the United States of America:**

I hereby declare that I believe I am the original inventor or an original joint inventor of a claimed invention in the application.

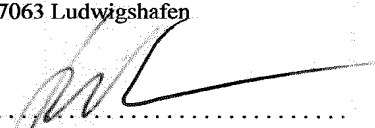
This declaration is directed to this international application of which it forms a part (if filing declaration with application).

This declaration is directed to international application No. PCT/..... (if furnishing declaration pursuant to Rule 26ter).

I hereby declare that the above-identified international application was made or authorized to be made by me.

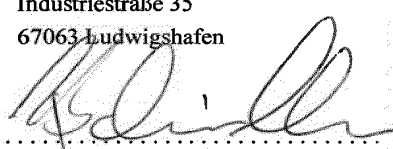
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Name: SCHICK, Friedrich
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

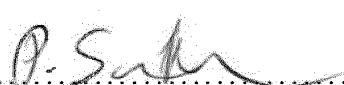
Date: 09.04.19

Name: SCHILLEN, Peter
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 09.04.19

Name: SCHINDLER, Patrick
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

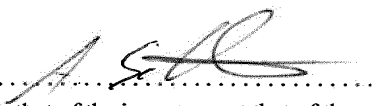
Date: 09.04.19

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: SCHMIDT, Andre
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

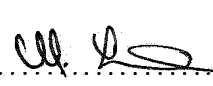
Date: 11.04.2019

Name: EBERSPACH, Michael
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 11.04.2019

Name: LENNARTZ, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

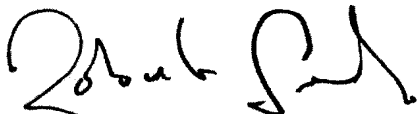
Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 11.4.2019

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: SEND, Robert
Residence: Germany
Mailing Address: Luisenstr. 25



Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date: ... 2019/09/11

Name: DIESSELBERG, Lars
Residence: Germany
Mailing Address: Geibelstr. 22
76185 Karlsruhe

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: HENGGEN, Heiko
Residence: Germany
Mailing Address: Siedlerweg 8
76872 Steinweiler

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Continuation of Box No. VIII (i) to (v) DECLARATION

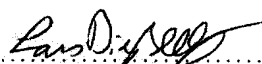
If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request

Name: SEND, Robert
Residence: Germany
Mailing Address: Luisenstr. 25
 76137 Karlsruhe

Inventor's Signature:
 (The signature must be that of the inventor, not that of the agent)

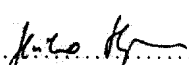
Date:

Name: DIESELBERG, Lars
Residence: Germany
Mailing Address: Geibelstr. 22
 76185 Karlsruhe

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

Date: 2019-09-05

Name: HENGEL, Heiko
Residence: Germany
Mailing Address: STEINERWEG 8
 76872 STEINWEILER

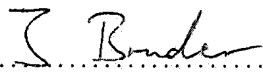
Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

Date: 2019-09-05

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: BRUDER, Ingmar
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 12.09.2019

Name: UNGER, Jakob
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: BONSIGNORE, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Continuation of Box No. VIII (i) to (v) DECLARATION

If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.

Name: BRUDER, Ingmar
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: UNGER, Jakob
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date: 13. 8. 19.....

Name: BONSIGNORE, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date: 14.08. 19.....

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

CERTIFICATE OF AVAILABILITY OF A CERTIFIED PATENT DOCUMENT IN A DIGITAL LIBRARY

The International Bureau certifies that a copy of the patent application indicated below has been available to the WIPO Digital Access Service since the date of availability indicated, and that the patent application has been available to the indicated Office(s) as of the date specified following the relevant Office code:

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Detector for identifying at least one material property

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Detector for identifying at least one material property

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining at least one material property of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

20 A large number of methods for material classification and identification are known from the prior art. For example, material identification can be used in machine vision applications, medical applications and security applications, for image classification purposes or in gesture recognition algorithms.

25 For example, US 2016/0206216 A1 describes a device, system and method for skin detection. The device comprises a thermal sensor input for obtaining thermal sensor data of a scene, a light sensor input for obtaining light sensor data of the scene, and an evaluation unit for analyzing the obtained thermal sensor data and the obtained light sensor data and for detecting skin areas within the scene based on said analysis. US 2016/155006 A1 describes a device and a
30 corresponding method for skin detection. The device comprises an illumination unit configured to project a predetermined illumination pattern onto a scene, an imaging unit configured to acquire an image of the scene, and an evaluation unit configured to evaluate the acquired image by analyzing the imaged illumination pattern as reproduced in the image and to detect skin areas within the image and distinguish them from non-skin areas within the image based on said
35 analysis.

Material classification and identification is usually done in through beam mode, namely shining light through a sample and analyzing extinction. Further, material classification and identification can be done in reflective mode in controlled environments. However, in reflective mode usually
40 wavelengths larger than 1000 nm are necessary in order to yield reliable results. Thus, despite the advantages of known methods for material identification, new concepts for reliable identification of materials are desired. Specifically, identifying a material from its reflection beam profile would be highly desirable. However, most image filters used for beam profile analysis (BPA) yield features that depend on both, distance and material such that reliable identification and

classification of the material is not possible. A typical example is the width of a beam profile reflected from a translucent material.

Problem addressed by the invention

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It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may identify at least one material property of an object, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

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Summary of the invention

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This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

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As used in the following, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

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Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

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Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way.

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The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without

any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for identifying at least one material property m is disclosed.

As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “identifying at least one material property” refers to one or more of determining and assigning the material property to the object. The detector may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by the user of the detector. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The detector may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

Specifically, the detector may be configured for detection of biological tissue, in particular human skin. As used herein, the term “biological tissue” generally refers to biological material comprising living cells. The detector may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The term “detection of biological tissue” refers to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector may be adapted to distinguish animal tissue or parts thereof from

one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector may be adapted to distinguish plant tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector may be adapted to distinguish different types of food such as a fruit, meat, and fish. The detector may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from human skin. The detector may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector may be adapted to distinguish human skin from cloth. The detector may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector may be adapted to distinguish organic material from inorganic material. The detector may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector may be used, in particular, for non-therapeutic and non-diagnostic applications.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter, or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be

an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

As used herein, an "optical sensor" generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a "light-sensitive area" generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The detector may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term "the optical sensors each having at least one light sensitive area" refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term "optical sensor" furthermore refers to a light-sensitive device configured for generating at least one output signal. In case the detector comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise

at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

- 5 The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

- Preferably, the light sensitive area may be oriented essentially perpendicular to an optical axis of the detector. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

- The light-sensitive areas specifically may be oriented towards the object. As used herein, the term "is oriented towards the object" generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term "essentially perpendicular" refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term "essentially parallel" refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The sensor element is configured for recording at least one reflection image of a light beam originating from at least one object. As used herein, the term “reflection image” refers to an image determined by the sensor element comprising at least one reflection feature. As furthermore used herein, without limitation, the term “reflection image” specifically may relate to data recorded by using the sensor element, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element. The reflection image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to “pixels”, reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. As used herein, the term “reflection feature” refers to a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. As used herein, the term “determining at least one reflection feature” refers to imaging and/or recording at least one light beam generated by the object in response to illumination with a light beam, in particular with the at least one illumination feature. In particular, the sensor element may be configured for determining and/or imaging and/or recording at least reflection image. The reflection image may comprise at least one reflection pattern comprising the at least one reflection feature.

The detector comprises at least one illumination source. The detector may comprise at least one illumination source configured for illuminating the object with at least one illumination beam. The illumination source may be configured for projecting at least one illumination pattern comprising the at least one illumination feature on at least one surface of the object. As used herein, the term “at least one illumination source” refers to at least one arbitrary device configured for providing at least one illumination light beam, specifically the at least one illumination pattern, for illumination of the object. The illumination source may be adapted to directly or indirectly illuminating the object, wherein the illumination light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam.

The illumination source may comprise at least one light source. The illumination source may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in

other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

- 5 Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transis-
- 10 tor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern.
- 15 For example, the illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of
- 20 handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source may be integrated into a housing of the detector.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alterna-

25 tively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light

30 beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation fre-

35 quencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

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The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The detector may be configured such that the illumination light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the illumination light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illumination light beam may be on the optical axis or off the optical axis. As an example, the illumination light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

Specifically, the illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.

The illumination source may be configured for generating at least one illumination pattern for illumination of the object. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the object. As used herein, the term "illumination feature" refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. For example, the illumination pattern may comprise at least one line pattern. For example, the illumination pattern may comprise at least one stripe pattern. For example, the illumination pattern may comprise at least one checkerboard pattern. For example, the illumination pattern may comprise at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing

lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. For example, the illumination source may be adapted to generate and/or to project a cloud of points. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. The illumination source may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, for generating and projecting the illumination pattern, the illumination source may comprise at least one laser source and at least one diffractive optical element (DOE). The detector may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one point pattern. As further used herein, the term "projecting at least one illumination pattern" refers to providing the at least one illumination pattern for illuminating the at least one object. The projected illumination pattern may be scarce, as only a single illumination feature, such as a single point, may be present. For increasing reliability, the illumination pattern may comprise several illumination features such as several points. If the pattern is scarce, a single image can be used for detection of biological tissue and face identification simultaneously.

For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be integrated in a mobile device or attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

As used herein, the term "ray" generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term "beam" generally refers to a collection of rays. In the following, the terms "ray" and "beam" will be used as synonyms. As further used herein, the term "light beam" generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

- As used herein, the term “object” refers to a point or region emitting at least one light beam, in particular the at least one reflection pattern. For example, the object may be at least one object selected from the group consisting of: a scene, a human such as a human, wood, carpet, foam, an animal such as a cow, a plant, a piece of tissue, a metal, a toy, a metallic object, a beverage, a food such as a fruit, meat, fish, a dish, a cosmetics product, an applied cosmetics product, cloth, fur, hair, a maintenance product, a cream, an oil, a powder, a carpet, a juice, a suspension, a paint, a plant, a body, a part of a body, organic material, inorganic material, a reflective material, a screen, a display, a wall, a sheet of paper, such as photograph. The object may comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These material properties may be used to distinguish skin from other materials, specifically by analyzing the back scattering profile.
- The least one light beam may propagate from the object towards the detector. The light beam may originate from the object or may originate from an illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. Additionally or alternatively to the at least one illumination source, the detector may use radiation already present in the scene such as from at least one ambient light source.
- The sensor element may be configured for recording a beam profile of at least one reflection feature of the reflection image. The evaluation device may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element. The evaluation device may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature.
- The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals

by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

For example, the illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device may be adapted to remove influences from background light from the reflection beam profile, for example, by an imaging without further illumination.

The evaluation device may be configured for determining the material property m by evaluating the beam profile of the reflection image. As used herein, the term "beam profile of the reflection image" refers to at least one intensity distribution of at least one of the reflection features of the

reflection image, such as of a light spot on the sensor element, as a function of the pixel. The beam profile of the reflection image, also denoted reflection beam profile, may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. As used herein, the term “evaluating the beam profile” refers to applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. As used herein, the term “image” refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. As used herein, the term “distance” refers to the distance of the object, specifically between the object and the detector. As used herein, the term “distance dependent” image filter refers to an image having a distance dependent output. The output of the distance dependent image filter is denoted herein “distance feature φ_{1z} ” or “distance dependent feature φ_{1z} ”. The distance feature may be or may comprise at least one information about the distance of the object such as at least one measure for the distance of the object, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter Φ_{1other} may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{cov\left((\Phi_i(f(x,y)) - \mu_i)(\Phi_j(f(x,y)) - \mu_j)\right)}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values

between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. As used herein, a "sensor signal" generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. As generally used herein, the term "combine" generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least one comparison signal or comparison result. As used herein, the term "combined signal Q " refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. In particular, the combined signal may be a quotient signal. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy} ,$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance L(z_o) and beam shape S(x,y;z_o). Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature ϕ_{1z} may be a function of the combined signal Q, $\phi_{1z} = \phi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q. Further, as an example, the object distance z_o may be a function of the distance dependent feature ϕ_{1z} , $z_o = z_o(\phi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in ϕ_{1z} . Thus, the object distance z_o may be a function of the combined signal Q, $z_o = z_o(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q.

The light-sensitive areas of at least two optical sensors may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q.

The evaluation device may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e.

proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor

signals or by dividing linear combinations of the first and second sensor signals. As an example, Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1 / b \cdot s_2$ or $Q = b \cdot s_2 / a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a , b , c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

Typically, the combined signal Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the setup described above, both the first signal s_1 and the second signal s_2 decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the first signal s_1 decreases more rapidly than the second signal s_2 , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the first and second sensor signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the incident light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

With respect to further details and embodiments of evaluation of the combined signal Q reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device may be configured for determining from the image the distance feature ϕ_{1z} of the object by optimizing at least one blurring

function f_a . The determined distance feature φ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature φ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. As used herein, the term “blurring function f_a ”, also referred to as blur kernel or point spread function, refers to a response function of the detector to the illumination from the object. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature φ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min\|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. As used herein, the term “sharp” or “sharp image” refers to a blurred image having a maximum contrast. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature φ_{1z} .

The evaluation device may be configured for determining at least one combined distance information z considering the distance feature φ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature φ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature φ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature φ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature φ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature φ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield advantageous distance measurement results with enhanced precision.

The evaluation device may be configured for determining at least one combined distance information using at least one recursive filter. The recursive filter may be at least one Kalman filter or at least one Extended Kalman filter (EKF). The combined distance information z may be obtained using a real function $z=f(z_{DPR}, z_{DFD})$, wherein z_{DPR} is the distance feature φ_{1z} determined by applying the depth-from-photon-ratio filter and z_{DFD} is the distance feature φ_{1z} determined by applying the depth-from-defocus filter, such as the arithmetic or geometric mean, a polynomial,

preferably a polynomial up to the eighth order in z_{DPR} and z_{DFD} . The function f may be or may be based on a look-up table of prerecorded values. For example, the evaluation device may comprise at least one data storage device configured for storing the prerecorded values and/or one or more look-up tables. The function f may be based on a look-up table in combination with an interpolation scheme for interpolation between the values in the look-up table. The interpolation scheme may be a linear interpolation, a splines interpolation, or the like. The z_{DFD} and z_{DPR} may be used as input variables within the recursive filter, in combination with a model or model function such as the function f , concerning the relationship between z , z_{DFD} , and z_{DPR} . The model or model function may comprise statistical assumptions and/or statistical models concerning the distances z , z_{DFD} , and z_{DPR} such as a distribution, such as a Gaussian distribution of a measured distance z , z_{DFD} , and/or z_{DPR} around a real distance z_{real} .

The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters. The further parameters may comprise further information from the sensor element, for example the CMOS sensor, such as information about quality and/or noise of recorded data and/or information about overexposure and/or information about underexposure or the like. The detector may comprise at least one further sensor configured for determining the further sensor data. The recursive filter may be configured for determining the combined distance information considering the further sensor data. The further sensor may be at least one sensor selected from the group consisting of: a temperature sensor, an illumination sensor such as a control sensor for determining illumination information, an inertial measurement unit; a gyroscope. The model and/or the Kalman filter may comprise further input parameters such as the further sensor data, e.g. temperature and/or detector motion from a gyroscope and/or information from an inertial measurement unit, and/or information from an illumination sensor) and/or further parameters such as the quality/noise of recorded data, overexposure, underexposure, or the like. The further sensor data may be provided by the sensor element, in particular by at least one CMOS sensor, and/or by further image analysis. The model and/or the Kalman filter may comprise the at least one material feature ϕ_{2m} as input variable.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and for selecting at least one image determined by the other one of the first sensor element or the second sensor element as reference image. As used herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference

image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature ϕ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm \varepsilon$.

The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term “essentially identical” refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. As used herein, the term “displacement region” refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to

the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term "reference feature" refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device may determine the displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\phi_{12} \pm \epsilon$.

The evaluation device may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. As used herein, the term "matching" refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displace-

ment region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device may be configured for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. As used herein, the term “displacement” refers to a difference between a position in the reference image to a position in the reflection image. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device may be adapted to determine the predetermined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device.

Additionally or alternatively, the evaluation device may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ε -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or

reference feature in shorter distance and/or ε -weighted distance to the reference feature and/or reflection feature.

The evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the object.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term “beam profile image” refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k :

$(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as ‘Material

A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

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The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

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Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m': \mu_m \neq \mu_{m'}$. As used herein, the term "not distinguish between material classifiers" refers to that the expectation values of the material classifiers are identical. As used herein, the term "distinguishes material classifiers" refers to that at least two expectation values of the material classifiers differ. As used herein "distinguishes at least two material classifiers" is used synonymous to "suitable material classifier". The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materi-

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als, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

- 10 For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

- 15 wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

- 20 For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

- 25 For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

- 30 For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse

scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,p}(g_1 g_2) = [p_{g_1, g_2}]$, whereas p_{g_1, g_2} is the occurrence rate of the grey combination $(g_1, g_2) = [f(x_1, y_1), f(x_2, y_2)]$, and the relation p defines the distance between (x_1, y_1) and (x_2, y_2) , which is $p(x, y) = (x+a, y+b)$ with a and b selected from $0, 1$.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij} (i - j)^2.$$

For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

- 5 For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i - j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

- 10 The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

- 15 and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

- 20 For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ϵ times

the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The material property m and/or the longitudinal coordinate z may be determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m . The evaluation device may be configured for determining the material property m and/or the longitudinal coordinate z by evaluating the features φ_{2m} , φ_{1z} . For example, the evaluation device may be configured for using at least one predetermined relationship between the distance feature φ_{1z} and the longitudinal coordinate of the object for determining the longitudinal coordinate z of the object. The evaluation device may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the object for determining the material property of the object. The predetermined relationship may be one or more of an empirical relationship, a semi-empirical relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. For example, the material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} . Specifically, the material property m and/or the longitudinal coordinate z may be determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or by a function $m(\varphi_{1z}, \varphi_{2m})$. The function may be predefined and/or predetermined. For example, the function may be a linear function.

In the ideal case, an image filter would yield features that are only dependent on distance or material properties. However, image filters used in beam profile analysis may yield features that depend on distance and material properties, such as translucency. At least one of the material dependent image filter or the distance dependent image filter may be a function of the feature φ_{1z} or φ_{2m} of the at least one other image filter. At least one of the material dependent image filter or the distance dependent image filter may be a function of the at least one other image filter. The evaluation device may be configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter and/or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter. Specifically, the evaluation device may be configured for determining a correlation coefficient of the material dependent image filter and the distance dependent image filter. In case the correlation coefficient of the material dependent image filter with the distance dependent image filter is close to 1 or -1, the distance may be projected out, by projecting the material feature on the principal axis with the lowest variance. As an example, the material feature may be projected onto an axis orthogonal to the correlating main component. In other words, the material feature may be projected onto the second main component. This may be done using a principal component analysis as known to the person skilled in the art. The evaluation device may be configured for applying the distance dependent image

filter and the material dependent image filter simultaneous to the reflection image. In particular, the evaluation device may be configured for simultaneously determining features that are strongly correlated with distance and weakly correlated with material properties and features that are weakly correlated with distance and strongly correlated with material properties. Alternatively, the evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image. The evaluation device may be configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

As further used herein, the term “evaluation device” generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. The above-mentioned operations, including analysis of the beam profile, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured for performing the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

The detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated

rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordi-

nate I. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As outlined above, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. For example, the detector may be configured for determining the longitudinal coordinate of the object by using at least one distance dependent filter using the depth-from-photon-ratio technique and/or the depth-from-defocus technique as outlined above. The detector may be configured for determining a position of the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. At least one of the optical sensors may be determined from which a center signal arises. This may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device,

or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured for determining at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, a center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object. The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
 - at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
- wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter;

a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,

wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

- 5 With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

10 In a further aspect of the present invention, a detector system is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

35 In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

40 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or

more of the embodiments disclosed in further detail below. The entertainment device is configured for enabling at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured for varying the entertainment function in accordance with the information.

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In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

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In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

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In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

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Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised

by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may

be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3D model by connecting adjacent points with planar surfaces. The 3D model may be stored and processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may

be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

- 10 As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points as described above and in more detail below.

- As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.

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As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

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The transfer device can be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. This feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

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For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

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The light emerging from the one or more optional beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of ex-

ample, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

$|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

With respect to further uses of the detector and devices of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

Specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device configured for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements configured for generating electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device configured for performing 3D photography. The camera generally may be configured for acquiring a single image, such as a single 3D image, or may be configured for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera configured for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be configured for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each

other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device configured for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be configured for performing at least one computer program, such as at least one computer program configured for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an
5 example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the
10 optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or
15 communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or
20 the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.
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The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue
30 interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or
35 more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors,
40 VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication
5 device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of
10 the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display
15 device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

20 The human-machine interface may comprise a plurality of beacon devices which are configured for being at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry
25 the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

30 The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device configured for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon
35 device, such as by providing one or more reflective elements configured for reflecting a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device
40 by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall

be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device.

5 Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of
10 the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

15 Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine
20 and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be configured in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of
25 the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

30 The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source configured for generating at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam proper-
35 ties. Additionally or alternatively, the beacon device may comprise at least one reflector configured for reflecting light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape.
40 Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club,

a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command configured for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is configured for gathering information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be configured for providing information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of

the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

5 The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is configured for tracking a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time in-
10 formation.

The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the ob-
15 ject itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly at-
tachable to or integratable into the object to be tracked.

The tracking system may be configured for initiating one or more actions of the tracking system
20 itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be configured for initiating at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group
25 consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

As an example of application of a tracking system, the tracking system may be used for contin-
30 uously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Addi-
tionally or alternatively, the tracking system might be used for illumination purposes, such as for
35 continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

40 Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 2: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

Embodiment 3: The detector according to the preceding embodiment, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

5 Embodiment 4: The detector according to any one of the preceding embodiments, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

10 Embodiment 5: The detector according to any one of the preceding embodiments, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.

15 Embodiment 6: The detector according to any one of the preceding embodiments, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

20 Embodiment 7: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

Embodiment 8: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

25 Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

30 Embodiment 10: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

35 Embodiment 11: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and/or φ_{2m} to the reflection image.

40 Embodiment 12: The detector according to any one of the preceding embodiments, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sen-

sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .

- 5 Embodiment 13: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 10 wherein x and y are transversal coordinates, A_1 and A_2 are different areas of the beam profile and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

- 15 Embodiment 14: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- 20 Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises
25 information relating to a number of photons in the second area of the beam profile.

- 30 Embodiment 16: The detector according to any one of the preceding embodiments, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

- 35 Embodiment 17: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 18: The detector according to any one of the two preceding embodiments, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

Embodiment 19: The detector according to any one of the preceding embodiments, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 21: The detector according to the preceding embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 23: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm \varepsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched fea-

tures using a predetermined relationship between the longitudinal coordinate and the displacement.

5 Embodiment 24: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \varepsilon$.

Embodiment 25: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 26: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.

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Embodiment 27: The detector according to any one of embodiments 25 and 26, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 28: The detector according to any one of embodiments 25 to 27, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

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Embodiment 29: The detector according to any one of embodiments 25 to 28, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.

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Embodiment 30: The detector according to any one of embodiments 25 to 29, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

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Embodiment 31: The detector according to any one of embodiments 25 to 30, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by

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$|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

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Embodiment 32: The detector according to any one of embodiments 25 to 31, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

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Embodiment 33: The detector according to any one of embodiments 25 to 32, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

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Embodiment 34: The detector according to any one of embodiments 25 to 33, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

Embodiment 35: The detector according to any one of embodiments 25 to 34, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

Embodiment 36: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

Embodiment 37: The detector according to any one of embodiments 25 to 36, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the distance feature φ_{1z} .

Embodiment 38: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are different areas of the beam profile and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

Embodiment 39: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

Embodiment 40: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a

number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

- Embodiment 41: The detector according to any one of embodiments 25 to 40, wherein the
 5 depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.
- Embodiment 42: The detector according to the preceding embodiment, wherein the reflection
 10 image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,
 15
$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

 by varying the parameters σ of the blurring function.

- Embodiment 43: The detector according to any one of embodiments 25 to 42, wherein the at
 least one blurring function f_a is a function or composite function composed from at least one
 20 function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

- Embodiment 44: The detector according to any one of embodiments 25 to 43, wherein the sen-
 25 sor element comprises at least one CMOS sensor.

- Embodiment 45: The detector according to any one of embodiments 25 to 44, wherein the de-
 tector comprises at least one illumination source, wherein the illumination source is configured
 for generating at least one illumination pattern for illumination of the object, wherein the illumina-
 30 tion pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at
 35 least one line pattern comprising at least two lines such as parallel or crossing lines.

- Embodiment 46: The detector according to the preceding embodiment, wherein the illumination
 source comprises at least one laser source and at least one diffractive optical element.

- Embodiment 47: The detector according to any one of the two preceding embodiments, wherein
 40 the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are posi-

tioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

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Embodiment 48: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 49: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 50: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,

wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

- 5 Embodiment 51: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a
10 pre-defined level of significance.

Embodiment 52: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.

- 15 Embodiment 53: The detector according to any one of embodiments 50 and 52, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 54: The detector according to any one of embodiments 50 to 53, wherein the at
20 least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

- Embodiment 55: The detector according to any one of embodiments 50 to 54, wherein the further distance dependent image filter Φ_{1other} correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$.

Embodiment 56: The detector according to any one of embodiments 50 to 55, wherein the further material dependent image filter Φ_{2other} correlates to one or more of the material dependent
30 image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

- Embodiment 57: The detector according to any one of embodiments 50 to 56, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness
35 filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof, or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by
40 $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared

norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

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Embodiment 58: The detector according to any one of embodiments 50 to 57, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

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Embodiment 59: The detector according to any one of embodiments 50 to 58, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

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Embodiment 60: The detector according to any one of embodiments 50 to 59, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

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Embodiment 61: The detector according to any one of embodiments 50 to 60, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

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Embodiment 62: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

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Embodiment 63: The detector according to any one of embodiments 50 to 62, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the distance feature φ_{1z} .

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Embodiment 64: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are different areas of the beam profile and $E(x,y,z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

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Embodiment 65: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

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Embodiment 66: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

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Embodiment 67: The detector according to any one of embodiments 50 to 66, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

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Embodiment 68: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

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$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 69: The detector according to any one of embodiments 50 to 68, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

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Embodiment 70: The detector according to any one of embodiments 50 to 69, wherein the sensor element comprises at least one CMOS sensor.

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Embodiment 71: The detector according to any one of embodiments 50 to 70, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 72: The detector according to the embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 73: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 74: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 75: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 76: A detector system, the detector system comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

Embodiment 77: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 78: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 79: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 80: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of embodiments 1 to 24, 25 to 49, or 50 to 75, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 81: A camera for imaging at least one object, the camera comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75.

Embodiment 82: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data

from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

- 5 Embodiment 83: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 1 to 24, the method comprising the following steps:
- 10 a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
- 15 b1) determining at least one distance feature ϕ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by
- 20 $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
- b2) determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as
- 25 a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-
- 30 occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-
- 35 level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,
- b3) determining a longitudinal coordinate z and the material property m by evaluating
- 40 the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

Embodiment 84: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 25 to 49, the method comprising the following steps:

- 5 a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - 10 b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by
 - 15 $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,
 - 20 b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 85: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 50 to 75, the method comprising the following steps:

- 30 a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - 35 b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by
 - 40 $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

- b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
- b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

5 Embodiment 86: A use of the detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

Brief description of the figures

15 Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary
20 embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

- 25 Figure 1 shows an embodiment of a detector according to the present invention;
- Figure 2 shows the statistics of the resulting material features of the spot shape material dependent image filter applied to the data set; and
- 30 Figure 3 shows a beam profile image.

Detailed description of the embodiments:

- 35 Fig. 1 shows in a highly schematic fashion an embodiment of a detector 110 for identifying at least one material property m of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.
- 40 The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the ele-

ments are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

5 The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other
10 embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors 120 may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

15 The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical
20 sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photo-
25 diodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

30 The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As
35 an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

40 The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more prefer-

ably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

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The detector 110 may further comprise an illumination source 124. As an example, the illumination source 124 may be configured for generating an illumination light beam 126 for illuminating the object 112. The detector 110 may be configured such that the illumination light beam 126 propagates from the detector 110 towards the object 112 along an optical axis 128 of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis 128.

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The illumination source 124 may comprise at least one light source. The illumination source 124 may comprise a plurality of light sources. The illumination source 124 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 124 may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source 124 may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

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Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 124 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points, for example the illumination source 124 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least

one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source 124 may be integrated into a housing of the detector 110.

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The illumination light beam 126 may propagate from the object 112 towards the detector 110. The detector 110 may be used in active and/or passive illumination scenarios. For example, the at least one illumination source 124 may be adapted to illuminate the object 112, for example, by directing a light beam towards the object 112, which reflects the light beam. Additionally or

10 alternatively to the at least one illumination source 124, the detector 110 may use radiation already present in the scene such as from at least one ambient light source.

The illumination source 124 may illuminate the at least one object 112 with at least one illumination pattern. The illumination pattern may comprise a plurality of points as image features.

15 These points are illustrated as light beam 126 emerging from the illumination source 124.

The detector 110 may comprise at least one transfer device 129 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel

20 lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 129 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

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Each optical sensor 120 may be designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam 130 propagating from the object 112 to the detector 110. The sensor element 116 may be configured for determining at least one reflection image of at least one reflection feature generated by the reflection light beam 130 on the optical sensors 120. The reflection image may be data recorded by

30 using the sensor element 116, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element 116. The reflection image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix 118 of the sensor element 116. The matrix 118 may comprise the reflection image 126. For example, in case of illumination with a point pattern, the reflection image may comprise points as reflection features. These points may

35 result from reflection light beams 130 originating from the at least one object 112.

The sensor element 116 may be configured for recording a beam profile of at least one reflection feature of the reflection image. The detector 110 comprises at least one evaluation device

40 132. The evaluation device 132 may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element 116. The evaluation device 132 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image anal-

ysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element 116. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

The evaluation device 132 may be configured for determining the material property m by evaluating the beam profile of the reflection image. The beam profile of the reflection image may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device 132 may be configured for applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device 132 is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance feature may be or may comprise at least one information about the distance of the object 112 such as at least one measure for the distance of the object 112, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter Φ_{1other} may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{cov\left((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j)\right)}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device 132, as an example, may comprise at least one divider 134, wherein the divider 134 is configured for deriving the quotient signal. The divider 134 may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear com-

binations of the sensor signals. The evaluation device 132 may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by

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$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy} ,$$

wherein x and y are transversal coordinates, A_1 and A_2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A_1 and area A_2 may differ. In particular, A_1 and A_2 are not congruent. Thus, A_1 and A_2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object 112 and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature φ_{1z} may be a function of the combined signal Q , $\varphi_{1z} = \varphi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q . Further, as an example, the object distance z_o may be a function of the distance dependent feature φ_{1z} , $z_o = z_o(\varphi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in φ_{1z} . Thus, the object distance z_o may be a function of the combined signal Q , $z_o = z_o(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q .

The light-sensitive areas 122 of at least two optical sensors 120 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

The evaluation device 132 may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the

center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 132 may be adapted for determining an area integral of the beam profile. The evaluation device 132 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 132 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device 132 specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example,

Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1/b \cdot s_2$ or $Q = b \cdot s_2/a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a, b, c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

With respect to further details and embodiments of evaluation of the combined signal Q and determination of the longitudinal coordinate z reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device 132 may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device 132 may be configured for determining from the image the distance feature ϕ_{1z} of the object by optimizing at least one blurring function f_a . The determined distance feature ϕ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature ϕ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature ϕ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature ϕ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature ϕ_{1z} .

The evaluation device 132 may be configured for determining at least one combined distance information z considering the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ

z_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield to advantageous distance measurement results with enhanced precision.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. The detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device 132 may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device 132 may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature ϕ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm \epsilon$.

The evaluation device 132 may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device 132 may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device 132 may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The evaluation device 132 may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device 132 may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the

reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device 132 may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object 112, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device 132 may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \epsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device 132 may determine the displacement region around the image position of the reflection feature. The evaluation device 132 may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\varphi_{1z} \pm \epsilon$.

The evaluation device 132 may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The evaluation device 132 may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device 132 may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device 132 may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to

determine an epipolar line before determining a corresponding reference feature. The evaluation device 132 may determine a displacement region around the image position of each reflection feature. The evaluation device 132 may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device 132 may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device 132 may be configured for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 132 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device 132 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device 132 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device 132.

Additionally or alternatively, the evaluation device 132 may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material feature may be or may comprise at least one information about the at least one material property of the object 112.

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The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$. With respect to the description of exemplary material dependent image filter, reference is made to the description of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter such as a Gaussian filter or median filter, the grey-level-occurrence-based contrast filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, and the threshold area filter given above.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. The beam profile image may be given by a sum of N_B Gaussian radial basis functions,

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$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all beam profile images. The center-positions, x_{lk}, y_{lk} , are identical for all images $f_k: (x_0, x_1, \dots, x_{N_B}), (y_0, y_1, \dots, y_{N_B})$. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

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Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed. Figure 3 shows an embodiment of a beam profile image defined by an exemplary data set.

An exemplary data set is presented in the following. The following parameter tables list the center positions, $x_{lk} = x_l, y_{lk} = y_l$, of all Gaussian functions g_{lk} for all images f_k :

l	0	1	2	3	4	5	6	7
x_l	14	11	10	19	15	20	12	17
y_l	17	11	17	19	20	20	12	20
l	8	9	10	11	12	13	14	15
x_l	13	18	10	14	19	15	16	12
y_l	17	19	8	18	14	11	22	17
l	16	17	18	19	20	21	22	23
x_l	25	13	18	8	9	10	14	19
y_l	15	20	20	15	14	13	24	24
l	24	25	26	27	28	29	30	31
x_l	10	9	14	12	19	20	17	15
y_l	24	16	15	11	13	19	13	14
l	32	33	34	35	36	37	38	39
x_l	16	13	12	20	18	17	21	13
y_l	19	10	22	14	10	18	15	23
l	40	41	42	43	44	45	46	47
x_l	8	10	8	14	19	11	15	14
y_l	12	14	18	21	23	15	16	8
l	48	49	50	51	52	53	54	55
x_l	16	20	11	17	16	15	21	13
y_l	13	16	16	8	16	13	21	13

l	56	57	58	59	60	61	62	63
x_l	22	18	23	14	19	11	15	16
y_l	23	15	12	22	18	10	23	10

The following parameter tables list the material classifiers, in particular for a white Teflon target (denoted control), a fabric, a dark skin (denoted as dark_skin), a pale skin (denoted pale_skin), and a highly translucent skin (denoted trans), and the distance z for each image f_k , referenced

5 by the image index k :

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
0	control	0.4	40	dark_skin	0.4	80	control	0.6
1	control	0.4	41	dark_skin	0.4	81	control	0.6
2	control	0.4	42	dark_skin	0.4	82	control	0.6
3	control	0.4	43	dark_skin	0.4	83	control	0.6
4	control	0.4	44	dark_skin	0.4	84	control	0.6
5	control	0.4	45	dark_skin	0.4	85	control	0.6
6	control	0.4	46	dark_skin	0.4	86	control	0.6
7	control	0.4	47	dark_skin	0.4	87	control	0.6
8	control	0.4	48	pale_skin	0.4	88	control	0.6
9	control	0.4	49	pale_skin	0.4	89	control	0.6
10	control	0.4	50	pale_skin	0.4	90	control	0.6
11	control	0.4	51	pale_skin	0.4	91	control	0.6
12	control	0.4	52	pale_skin	0.4	92	control	0.6
13	control	0.4	53	pale_skin	0.4	93	control	0.6
14	control	0.4	54	pale_skin	0.4	94	control	0.6
15	control	0.4	55	pale_skin	0.4	95	control	0.6
16	fabric	0.4	56	pale_skin	0.4	96	fabric	0.6
17	fabric	0.4	57	pale_skin	0.4	97	fabric	0.6
18	fabric	0.4	58	pale_skin	0.4	98	fabric	0.6
19	fabric	0.4	59	pale_skin	0.4	99	fabric	0.6
20	fabric	0.4	60	pale_skin	0.4	100	fabric	0.6
21	fabric	0.4	61	pale_skin	0.4	101	fabric	0.6
22	fabric	0.4	62	pale_skin	0.4	102	fabric	0.6
23	fabric	0.4	63	pale_skin	0.4	103	fabric	0.6
24	fabric	0.4	64	trans	0.4	104	fabric	0.6
25	fabric	0.4	65	trans	0.4	105	fabric	0.6
26	fabric	0.4	66	trans	0.4	106	fabric	0.6
27	fabric	0.4	67	trans	0.4	107	fabric	0.6
28	fabric	0.4	68	trans	0.4	108	fabric	0.6
29	fabric	0.4	69	trans	0.4	109	fabric	0.6
30	fabric	0.4	70	trans	0.4	110	fabric	0.6
31	fabric	0.4	71	trans	0.4	111	fabric	0.6
32	dark_skin	0.4	72	trans	0.4	112	dark_skin	0.6
33	dark_skin	0.4	73	trans	0.4	113	dark_skin	0.6
34	dark_skin	0.4	74	trans	0.4	114	dark_skin	0.6
35	dark_skin	0.4	75	trans	0.4	115	dark_skin	0.6
36	dark_skin	0.4	76	trans	0.4	116	dark_skin	0.6
37	dark_skin	0.4	77	trans	0.4	117	dark_skin	0.6
38	dark_skin	0.4	78	trans	0.4	118	dark_skin	0.6
39	dark_skin	0.4	79	trans	0.4	119	dark_skin	0.6

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
120	dark_skin	0.6	160	control	0.8	200	dark_skin	0.8
121	dark_skin	0.6	161	control	0.8	201	dark_skin	0.8
122	dark_skin	0.6	162	control	0.8	202	dark_skin	0.8
123	dark_skin	0.6	163	control	0.8	203	dark_skin	0.8
124	dark_skin	0.6	164	control	0.8	204	dark_skin	0.8
125	dark_skin	0.6	165	control	0.8	205	dark_skin	0.8
126	dark_skin	0.6	166	control	0.8	206	dark_skin	0.8
127	dark_skin	0.6	167	control	0.8	207	dark_skin	0.8
128	pale_skin	0.6	168	control	0.8	208	pale_skin	0.8
129	pale_skin	0.6	169	control	0.8	209	pale_skin	0.8
130	pale_skin	0.6	170	control	0.8	210	pale_skin	0.8
131	pale_skin	0.6	171	control	0.8	211	pale_skin	0.8
132	pale_skin	0.6	172	control	0.8	212	pale_skin	0.8
133	pale_skin	0.6	173	control	0.8	213	pale_skin	0.8
134	pale_skin	0.6	174	control	0.8	214	pale_skin	0.8
135	pale_skin	0.6	175	control	0.8	215	pale_skin	0.8
136	pale_skin	0.6	176	fabric	0.8	216	pale_skin	0.8
137	pale_skin	0.6	177	fabric	0.8	217	pale_skin	0.8
138	pale_skin	0.6	178	fabric	0.8	218	pale_skin	0.8
139	pale_skin	0.6	179	fabric	0.8	219	pale_skin	0.8
140	pale_skin	0.6	180	fabric	0.8	220	pale_skin	0.8
141	pale_skin	0.6	181	fabric	0.8	221	pale_skin	0.8
142	pale_skin	0.6	182	fabric	0.8	222	pale_skin	0.8
143	pale_skin	0.6	183	fabric	0.8	223	pale_skin	0.8
144	trans	0.6	184	fabric	0.8	224	trans	0.8
145	trans	0.6	185	fabric	0.8	225	trans	0.8
146	trans	0.6	186	fabric	0.8	226	trans	0.8
147	trans	0.6	187	fabric	0.8	227	trans	0.8
148	trans	0.6	188	fabric	0.8	228	trans	0.8
149	trans	0.6	189	fabric	0.8	229	trans	0.8
150	trans	0.6	190	fabric	0.8	230	trans	0.8
151	trans	0.6	191	fabric	0.8	231	trans	0.8
152	trans	0.6	192	dark_skin	0.8	232	trans	0.8
153	trans	0.6	193	dark_skin	0.8	233	trans	0.8
154	trans	0.6	194	dark_skin	0.8	234	trans	0.8
155	trans	0.6	195	dark_skin	0.8	235	trans	0.8
156	trans	0.6	196	dark_skin	0.8	236	trans	0.8
157	trans	0.6	197	dark_skin	0.8	237	trans	0.8
158	trans	0.6	198	dark_skin	0.8	238	trans	0.8
159	trans	0.6	199	dark_skin	0.8	239	trans	0.8

The following parameter tables list the prefactors a_{lk} for the Gaussian functions g_{lk} in each image f_k referenced by the image index k and the Gaussian index l :

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
0	291.94	-7.20	-14.72	-22.75	32.57	16.92	-45.50	-23.31
1	300.91	-7.58	-15.22	-14.41	32.18	15.96	-44.41	-23.92
2	297.04	-6.42	-13.94	-10.30	31.19	16.44	-43.68	-22.27
3	289.98	-7.67	-14.20	-13.60	31.68	17.16	-44.04	-21.41
4	295.94	-7.72	-12.93	-16.81	32.77	15.92	-45.15	-22.30
5	295.37	-6.87	-13.30	-15.51	33.10	16.70	-45.18	-22.29
6	298.03	-6.79	-14.89	-12.68	32.43	14.46	-44.47	-22.71

7	296.95	-7.09	-14.98	-13.54	31.56	15.69	-45.09	-20.03
8	306.69	-6.91	-14.94	-14.27	33.28	16.02	-43.93	-20.21
9	291.40	-7.73	-13.91	-14.95	30.51	16.18	-45.86	-21.61
10	301.21	-7.21	-14.05	-17.46	32.02	17.26	-45.02	-22.89
11	309.78	-6.93	-13.47	-19.23	33.01	14.94	-43.61	-20.86
12	298.62	-7.09	-14.75	-18.43	32.59	18.97	-44.11	-22.73
13	296.09	-7.18	-14.25	-15.83	31.94	14.62	-43.72	-19.77
14	301.13	-7.24	-14.44	-19.11	33.54	17.01	-44.08	-21.39
15	304.28	-7.08	-14.10	-17.26	32.54	16.81	-44.40	-21.14
16	246.71	-3.77	-13.62	1.43	19.38	-2.03	-52.50	-19.62
17	258.85	-3.11	-12.88	-3.91	19.02	-1.05	-49.64	-21.50
18	263.06	-2.91	-13.72	1.65	19.03	-1.26	-51.02	-21.58
19	251.12	-4.15	-14.47	-6.19	19.00	-1.26	-50.65	-22.40
20	250.05	-2.91	-13.84	0.30	19.48	-1.79	-49.89	-20.92
21	249.35	-2.80	-13.64	-0.25	18.90	0.42	-51.29	-19.21
22	260.34	-2.57	-14.30	1.83	17.72	-1.18	-51.27	-24.07
23	246.64	-3.68	-13.92	-2.29	19.79	-0.20	-50.33	-23.09
24	245.58	-2.91	-15.25	-6.59	18.34	-1.75	-50.89	-24.38
25	249.09	-3.09	-14.13	3.85	19.58	-1.58	-50.05	-22.87
26	255.98	-2.91	-13.67	6.77	18.09	-2.15	-48.40	-21.74
27	249.87	-3.43	-13.89	1.80	17.14	-2.32	-50.95	-25.65
28	247.42	-3.34	-13.32	2.32	17.69	-1.29	-50.40	-23.82
29	253.00	-3.32	-12.60	4.98	17.09	-1.16	-49.54	-25.89
30	254.40	-3.44	-14.69	1.64	20.18	-1.15	-49.66	-22.73
31	256.40	-3.75	-13.62	-8.47	19.57	-1.08	-52.04	-21.62
32	24.07	-2.06	2.98	-22.09	14.88	8.52	3.28	-4.27
33	24.00	-1.30	3.15	-22.28	14.65	9.22	3.94	-4.75
34	20.76	-1.69	3.24	-22.34	14.81	8.95	3.76	-4.69
35	22.23	-2.51	3.10	-22.85	14.01	9.71	3.29	-4.61
36	20.33	-2.23	3.04	-21.40	15.61	8.99	3.00	-4.12
37	21.12	-2.09	3.25	-26.46	14.68	8.75	4.67	-5.05
38	21.89	-1.92	3.05	-22.32	14.65	8.25	3.77	-4.15
39	14.81	-2.07	2.82	-20.92	14.96	9.05	3.73	-5.40

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
0	-95.90	-116.79	0.49	-92.61	-76.01	36.70	0.46	78.40
1	-99.07	-117.11	0.57	-89.09	-76.31	37.12	-0.60	86.28
2	-92.44	-115.78	0.59	-93.89	-76.95	35.15	0.42	83.74
3	-96.50	-115.93	0.49	-92.28	-75.29	36.48	-0.17	70.45
4	-100.83	-120.58	0.57	-95.39	-73.94	37.36	-0.26	79.80
5	-94.89	-117.93	0.59	-93.45	-75.67	36.88	0.22	74.29
6	-94.27	-119.97	0.51	-91.14	-74.61	35.76	-0.72	73.46
7	-94.78	-114.15	0.55	-91.40	-71.49	36.55	-0.61	73.29
8	-97.53	-112.61	0.54	-91.30	-70.94	36.62	0.39	75.30
9	-97.85	-108.91	0.57	-94.07	-72.52	36.34	-0.32	76.34
10	-97.23	-115.13	0.64	-95.38	-74.62	36.22	-0.33	74.11
11	-96.18	-115.91	0.45	-93.13	-77.72	37.20	-0.07	81.82
12	-100.79	-113.16	0.52	-90.33	-72.38	36.46	-1.02	74.17
13	-98.70	-124.58	0.53	-93.48	-74.52	36.90	-0.78	74.38
14	-93.07	-116.00	0.52	-92.98	-75.28	36.84	-0.77	76.49
15	-106.83	-117.44	0.52	-94.22	-75.28	36.42	-0.51	79.82
16	-88.58	-87.41	0.63	-70.05	-37.06	27.50	3.58	84.26
17	-87.71	-82.32	0.54	-68.18	-40.98	28.10	3.20	88.34
18	-76.19	-80.81	0.53	-66.95	-35.36	27.23	3.61	87.52
19	-80.42	-84.95	0.55	-67.91	-38.39	27.70	3.39	90.71

20	-87.87	-76.07	0.74	-66.04	-39.03	27.55	2.84	83.83
21	-93.60	-80.04	0.45	-69.48	-38.83	27.01	3.71	83.32
22	-88.04	-78.17	0.62	-72.00	-39.30	27.24	3.08	89.23
23	-88.92	-76.16	0.62	-66.79	-39.81	27.64	3.01	84.46
24	-90.47	-77.18	0.58	-66.41	-35.84	27.85	3.13	94.66
25	-89.84	-83.96	0.70	-64.36	-36.65	26.40	3.38	86.08
26	-81.82	-72.08	0.59	-66.39	-38.13	27.45	3.47	83.27
27	-80.57	-82.11	0.65	-66.61	-35.78	28.23	3.99	86.32
28	-93.40	-81.19	0.66	-68.45	-39.71	26.68	3.19	88.66
29	-92.92	-79.14	0.68	-64.90	-38.61	28.11	3.79	83.05
30	-91.82	-79.09	0.57	-66.39	-37.74	27.24	4.20	82.26
31	-90.60	-82.20	0.51	-67.61	-35.58	28.10	3.02	89.40
32	-0.54	-16.54	-0.31	-3.67	-33.80	4.91	1.87	11.78
33	-0.17	-16.06	-0.31	-2.39	-32.09	4.41	1.86	11.66
34	-3.04	-15.03	-0.34	-5.36	-31.98	4.17	1.76	11.91
35	-5.65	-15.79	-0.35	-2.71	-32.21	4.77	1.90	10.71
36	-6.54	-15.97	-0.29	-3.29	-33.09	4.26	1.69	14.99
37	-1.22	-16.20	-0.29	-2.79	-32.90	4.42	1.93	9.92
38	-3.85	-13.17	-0.22	-2.98	-33.98	4.68	1.84	11.96
39	-5.99	-12.87	-0.34	-2.53	-34.59	3.97	1.80	11.77

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
0	3.86	18.38	66.68	70.89	46.30	51.83	63.61	6.33
1	-1.41	18.18	66.65	-116.27	47.93	53.00	-19.71	6.14
2	5.49	17.64	67.40	86.57	47.99	51.08	-20.20	6.06
3	1.41	18.29	68.83	-73.86	49.40	52.06	-9.49	6.14
4	0.58	17.86	70.11	73.41	47.45	52.04	19.20	6.06
5	-0.96	18.19	71.29	12.08	48.40	52.10	36.63	6.18
6	-0.98	18.97	69.33	107.17	48.12	51.68	-30.34	6.21
7	1.44	17.76	67.10	-119.11	46.64	51.64	-19.85	6.11
8	2.83	18.10	66.62	42.19	47.28	52.29	-24.88	6.17
9	-0.87	17.50	68.71	-129.91	45.89	51.63	-43.44	6.23
10	1.24	18.26	66.88	-12.32	47.49	52.45	45.87	6.14
11	0.39	18.50	68.03	-103.43	47.42	52.40	-25.17	6.01
12	1.60	18.13	67.42	-159.30	48.41	52.43	-8.06	6.08
13	2.92	17.75	68.36	139.40	45.89	52.74	-16.56	6.19
14	1.69	17.80	68.48	-16.76	46.43	52.86	-15.34	6.20
15	0.66	17.51	68.30	-16.84	46.61	52.23	-12.07	6.00
16	1.53	10.12	45.53	-105.48	51.47	53.43	29.32	3.91
17	-0.83	10.67	47.22	-65.54	49.74	55.32	-3.14	3.90
18	-0.05	11.56	46.07	-38.72	50.65	54.17	-71.99	3.87
19	1.11	10.54	47.56	-22.99	50.63	54.47	19.05	3.81
20	-2.77	10.45	46.29	102.12	50.39	54.05	-21.67	3.88
21	1.51	11.03	46.20	-61.41	52.28	55.89	51.76	4.01
22	-2.08	11.82	45.96	121.36	49.18	54.28	-44.05	3.89
23	-0.13	10.89	46.21	34.94	52.82	54.93	26.48	3.82
24	-3.52	10.89	49.08	97.01	50.10	55.70	4.07	3.65
25	2.07	9.39	46.08	-53.59	51.65	53.65	4.94	3.97
26	0.44	11.04	47.23	-188.70	49.01	55.21	30.06	4.08
27	-4.62	9.33	47.59	141.42	51.09	54.20	-1.25	3.95
28	-1.78	10.70	45.70	-47.29	49.84	54.81	40.79	3.71
29	-2.45	10.65	47.01	-32.51	51.18	53.56	11.77	3.93
30	0.03	10.49	43.45	-149.80	50.93	53.95	14.36	3.83
31	3.37	10.57	46.51	20.87	49.67	54.60	22.79	3.96
32	0.86	7.55	19.63	-6.16	7.11	9.03	-20.02	1.08

33	-1.54	7.76	20.22	38.99	7.26	8.99	-10.41	1.12
34	-0.19	7.43	18.78	21.86	7.75	8.86	-16.09	1.15
35	-2.22	7.79	19.15	4.23	7.44	8.75	19.94	1.06
36	-1.11	7.83	20.15	20.92	6.25	8.80	-31.48	1.06
37	-0.31	7.90	19.64	8.33	7.22	9.07	18.83	1.12
38	-1.59	7.27	19.37	-18.56	6.82	8.51	21.93	1.10
39	-1.18	7.41	19.11	85.48	7.72	8.69	-17.48	1.01

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
0	-2.45	11.03	80.28	37.44	26.08	-32.86	-12.63	179.34
1	4.01	11.52	78.78	37.44	24.32	-34.91	-12.16	180.51
2	-3.35	10.90	81.50	37.26	24.80	-33.69	-10.77	180.89
3	1.73	10.84	80.38	37.51	25.47	-36.57	-15.88	183.66
4	1.27	9.42	81.14	38.14	25.13	-35.66	-13.87	180.51
5	-1.05	10.89	78.62	35.99	24.44	-34.31	-10.65	181.52
6	-0.46	10.30	81.92	35.89	24.74	-34.63	-15.73	180.07
7	2.40	10.57	78.99	37.01	23.94	-34.87	-12.35	183.67
8	1.32	10.56	80.87	37.45	24.70	-35.58	-16.41	180.05
9	1.05	9.87	79.26	37.63	24.52	-33.58	-12.25	178.55
10	-0.78	9.63	81.05	38.38	25.20	-33.14	-13.41	176.95
11	3.17	10.63	81.55	37.98	24.25	-32.39	-8.97	178.12
12	-0.12	10.57	81.11	38.18	25.06	-32.51	-14.47	180.98
13	-3.92	11.23	81.59	37.07	24.28	-35.06	-13.10	179.20
14	-1.14	11.04	81.63	36.75	25.25	-34.70	-11.65	180.54
15	1.46	11.05	80.55	36.57	23.89	-33.64	-12.62	181.91
16	-5.70	11.70	54.00	34.44	12.44	-14.39	-12.01	171.54
17	-0.70	12.09	55.21	34.71	9.92	-17.13	-11.70	168.02
18	-3.57	11.69	53.52	34.81	11.95	-13.55	-12.37	172.64
19	-3.32	12.04	54.84	35.03	11.31	-16.56	-8.97	169.62
20	2.59	12.21	55.48	35.95	11.23	-16.53	-9.56	169.59
21	2.54	12.19	54.25	35.20	10.76	-16.90	-12.28	169.58
22	-3.44	11.69	55.10	35.89	12.10	-18.59	-8.83	171.87
23	0.56	12.26	53.50	35.39	10.43	-17.85	-6.90	169.62
24	1.18	11.85	54.90	34.61	11.57	-14.31	-8.59	170.29
25	4.59	13.16	55.42	35.65	11.43	-20.08	-11.70	170.01
26	-0.12	11.06	54.09	34.94	11.37	-15.16	-13.76	172.27
27	1.10	11.81	55.45	34.85	10.99	-17.00	-9.50	170.91
28	2.78	11.51	56.06	34.44	10.57	-13.57	-8.79	172.45
29	-1.28	11.48	55.08	35.02	10.89	-15.74	-9.63	169.75
30	2.04	12.16	55.11	35.67	10.86	-17.71	-6.55	170.86
31	0.32	12.82	53.12	34.92	10.54	-18.42	-8.44	169.18
32	0.20	0.74	50.97	-0.31	16.61	-0.13	-0.95	4.08
33	-2.15	1.02	49.63	-0.32	16.22	0.23	0.22	1.95
34	1.20	0.57	50.53	-0.69	16.60	0.05	-0.14	2.81
35	2.07	0.84	50.16	-0.49	16.35	0.75	-0.83	2.72
36	0.80	0.74	49.51	0.39	16.42	-0.53	-1.24	3.37
37	-1.45	1.05	52.96	-0.54	16.22	0.77	-0.23	3.04
38	2.48	0.88	49.48	0.67	16.29	-0.46	-1.51	3.83
39	-0.81	0.65	50.20	-0.45	16.38	-0.34	-0.14	3.97

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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0	-19.43	-15.41	-0.54	35.39	4.01	146.63	-31.68	-3.15
1	-18.88	-15.15	-0.37	34.72	4.18	148.07	-31.76	-3.56
2	-16.02	-15.31	-0.38	33.72	4.18	145.90	-32.03	-3.18
3	-17.80	-15.75	-0.39	36.39	4.08	145.63	-31.54	-3.47
4	-15.90	-15.69	-0.30	34.77	4.13	149.16	-31.20	-3.47
5	-18.98	-15.68	-0.58	33.97	4.12	147.36	-31.71	-3.33
6	-15.90	-15.37	-0.49	34.31	4.29	146.96	-30.78	-3.23
7	-14.81	-15.73	-0.31	36.06	4.14	150.82	-32.18	-3.67
8	-15.48	-15.58	-0.58	36.41	4.13	146.32	-31.10	-3.39
9	-16.55	-15.89	-0.37	32.95	3.99	141.73	-30.59	-3.45
10	-18.58	-15.48	-0.48	33.49	4.40	148.23	-30.74	-3.45
11	-15.63	-15.39	-0.38	35.93	4.06	144.15	-30.84	-3.40
12	-17.34	-15.78	-0.58	35.76	4.10	140.54	-32.14	-3.58
13	-16.01	-15.44	-0.51	35.03	4.19	142.21	-31.70	-3.44
14	-19.00	-15.39	-0.63	35.03	4.06	146.35	-31.59	-3.67
15	-16.05	-15.61	-0.51	35.06	4.04	144.24	-31.00	-3.59
16	-1.73	-11.69	0.83	17.68	4.06	115.00	-27.19	-5.03
17	-4.29	-12.41	1.18	16.91	3.80	117.37	-26.45	-5.33
18	-2.73	-12.25	1.21	17.51	3.83	119.84	-28.87	-5.24
19	-1.43	-12.78	1.09	16.03	3.87	118.57	-27.48	-5.28
20	-1.08	-12.82	1.18	16.30	3.94	118.99	-26.64	-4.93
21	-3.12	-12.39	0.98	17.09	3.77	116.73	-27.35	-5.06
22	-2.46	-12.29	1.12	15.90	3.83	119.51	-27.39	-4.75
23	-1.88	-12.16	1.06	16.70	3.81	120.08	-27.74	-5.30
24	-3.35	-12.24	1.09	14.93	3.90	117.65	-27.21	-5.13
25	-1.85	-11.74	1.20	18.98	3.86	118.21	-27.50	-5.19
26	-2.53	-12.60	1.34	18.04	3.83	118.75	-26.64	-5.50
27	-4.14	-11.95	1.01	18.05	3.80	120.99	-27.10	-4.89
28	-2.49	-12.69	1.15	18.06	3.69	117.75	-26.73	-5.11
29	-2.52	-12.69	1.30	15.46	4.07	116.11	-27.32	-5.01
30	-1.51	-12.10	1.15	17.59	3.63	123.35	-27.39	-5.32
31	-4.63	-12.59	1.03	17.39	3.88	119.73	-27.34	-5.17
32	-5.22	1.64	2.40	15.00	2.34	28.20	-5.04	-5.62
33	-5.96	1.34	2.52	14.98	2.36	28.53	-4.99	-5.55
34	-5.47	1.64	2.42	14.89	2.39	29.51	-4.77	-5.28
35	-4.58	1.75	2.52	14.33	2.30	31.32	-5.18	-5.29
36	-6.27	1.70	2.46	14.36	2.42	31.69	-5.05	-5.67
37	-5.85	1.50	2.37	14.83	2.33	29.47	-5.19	-5.62
38	-4.86	1.37	2.45	14.20	2.32	30.31	-5.13	-5.42
39	-6.24	1.66	2.42	14.65	2.30	31.65	-5.26	-5.40

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
0	-10.74	-128.40	-12.06	-27.68	-47.57	129.47	-144.26	5.82
1	-10.79	-126.25	-6.60	-27.37	30.05	129.30	-151.57	0.18
2	-10.74	-126.91	22.40	-24.92	8.00	127.04	-148.75	4.11
3	-10.77	-125.83	-0.47	-27.97	-22.77	127.09	-144.81	-5.16
4	-10.78	-123.99	31.65	-28.65	24.12	128.49	-154.50	13.78
5	-10.58	-125.15	-1.09	-27.15	-24.40	127.57	-149.54	5.15
6	-10.85	-124.37	-9.75	-27.32	-4.55	126.41	-145.43	-4.92
7	-10.65	-127.00	0.46	-27.02	-13.23	129.75	-144.09	-13.95
8	-11.02	-126.29	-13.90	-26.51	-7.75	127.78	-152.92	8.36
9	-10.68	-124.07	-7.82	-27.67	-23.63	127.87	-150.02	8.26
10	-10.83	-124.06	47.68	-28.79	7.39	124.57	-149.34	-3.62

11	-10.80	-128.19	-2.30	-25.63	-6.57	129.60	-150.58	-9.25
12	-10.87	-124.73	-54.90	-24.31	41.58	134.82	-151.79	-4.73
13	-10.82	-127.07	-0.98	-28.09	7.94	130.29	-148.24	-10.15
14	-11.01	-125.00	-2.61	-26.15	-9.59	129.00	-148.98	-19.14
15	-10.69	-125.55	6.16	-27.16	5.81	130.83	-152.90	-9.64
16	-11.43	-133.31	27.27	-18.30	-26.33	148.91	-121.02	7.67
17	-11.74	-134.46	-52.34	-16.89	-23.99	151.23	-120.04	-19.73
18	-11.45	-130.80	18.52	-16.85	-48.77	150.98	-122.92	10.85
19	-11.27	-133.08	-29.76	-17.58	3.69	149.84	-121.37	-4.93
20	-11.27	-134.65	8.68	-16.64	34.24	149.00	-118.06	-9.55
21	-11.71	-137.87	30.52	-14.52	0.44	149.28	-119.23	7.09
22	-11.44	-133.43	27.53	-14.99	8.31	149.32	-122.39	-19.43
23	-11.34	-135.85	-13.46	-14.85	0.51	149.46	-118.80	12.65
24	-11.46	-134.89	-32.30	-18.64	3.14	145.98	-119.97	5.91
25	-10.92	-135.14	-16.97	-16.50	9.98	148.08	-126.66	0.91
26	-11.38	-132.17	38.54	-18.23	23.72	150.56	-114.45	1.06
27	-11.44	-134.82	-43.20	-14.82	-21.37	150.11	-123.74	-7.24
28	-11.64	-136.08	-8.20	-17.05	0.87	153.36	-125.50	18.53
29	-11.25	-132.85	-28.20	-17.77	-16.83	146.70	-118.61	-2.81
30	-11.21	-133.55	-22.12	-16.42	-9.17	150.89	-123.39	7.29
31	-11.50	-135.59	16.35	-17.71	-39.44	149.92	-117.21	-17.87
32	-1.75	-17.34	-4.56	-15.38	-4.53	27.52	-4.76	-15.17
33	-1.66	-18.40	-1.44	-15.69	-11.54	27.41	-2.83	-8.66
34	-1.72	-17.15	3.10	-14.93	-13.85	27.41	0.29	18.03
35	-1.60	-18.29	10.53	-16.01	10.69	27.36	-0.67	-6.10
36	-1.70	-17.23	-4.85	-14.95	7.00	25.49	-2.44	-2.75
37	-1.65	-17.09	-18.55	-15.00	3.78	27.40	1.75	3.17
38	-1.68	-17.34	10.25	-15.93	-2.93	27.08	-9.32	12.10
39	-1.66	-16.95	6.80	-16.81	-3.07	27.33	-0.14	-2.79

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
0	37.14	79.32	-51.17	-1.76	107.54	-136.45	-0.70	49.45
1	39.73	78.51	-53.02	-3.67	107.24	-137.40	-0.92	48.53
2	38.32	79.55	-53.33	6.10	109.08	-137.60	-0.44	47.48
3	43.04	78.76	-49.78	-5.20	109.63	-137.79	-0.41	49.72
4	41.97	78.51	-51.25	-3.82	108.30	-134.67	-0.72	47.28
5	43.91	79.20	-53.25	5.02	104.46	-134.65	-0.72	50.25
6	41.71	79.71	-51.24	9.31	106.65	-135.02	-0.97	48.74
7	44.52	80.29	-48.51	-2.75	107.04	-140.33	0.13	49.50
8	39.20	80.04	-52.02	-1.21	109.14	-135.03	-0.87	48.70
9	42.96	79.10	-48.95	0.10	107.27	-135.84	-1.15	51.11
10	42.98	80.11	-51.37	2.86	107.63	-136.54	-1.02	49.77
11	41.67	78.52	-54.87	9.25	107.78	-137.56	-0.36	49.93
12	41.54	78.66	-49.72	12.58	108.48	-138.05	-1.51	50.82
13	45.53	79.40	-53.00	-0.90	106.29	-138.06	-1.22	48.46
14	41.05	78.05	-50.05	5.44	108.17	-134.77	-1.00	49.65
15	40.71	80.07	-56.73	-3.43	106.52	-136.43	-0.06	50.09
16	37.90	83.28	-64.08	-0.55	90.49	-120.94	4.79	58.73
17	39.35	84.15	-61.34	14.09	91.22	-115.20	4.42	58.38
18	39.25	83.87	-64.28	-2.73	90.15	-119.81	5.17	59.14
19	44.51	83.85	-64.77	-5.83	91.64	-120.52	4.96	59.80
20	39.23	84.11	-64.08	-0.47	90.61	-119.79	4.41	58.04
21	33.64	84.97	-66.24	2.00	90.84	-119.05	4.36	55.65

22	40.09	84.70	-68.59	8.97	89.54	-122.30	4.21	58.89
23	39.94	84.58	-64.95	9.90	90.45	-118.55	4.20	58.47
24	41.44	83.21	-61.49	21.96	86.40	-121.01	5.88	59.02
25	43.13	84.06	-71.09	5.92	91.68	-121.74	5.43	58.39
26	40.08	84.02	-62.57	-16.45	90.01	-124.94	5.37	59.22
27	42.77	83.17	-60.93	-3.80	90.21	-119.34	4.16	59.00
28	38.19	83.14	-68.75	0.93	89.45	-118.83	4.65	58.62
29	40.28	84.08	-66.06	2.04	90.85	-121.14	4.95	58.58
30	38.60	83.57	-63.85	4.02	90.34	-121.73	5.02	59.52
31	38.72	83.98	-62.23	-8.14	90.87	-118.15	4.47	57.21
32	11.57	21.55	-10.03	-6.33	68.62	-1.53	-1.26	5.29
33	12.52	21.60	-9.90	5.14	67.28	-2.03	-1.17	6.06
34	10.52	21.27	-9.92	7.48	62.26	-1.26	-1.10	6.49
35	9.72	21.00	-9.55	4.65	67.09	-2.90	-1.42	5.32
36	10.18	21.30	-10.46	0.53	68.50	-1.87	-1.28	5.86
37	11.41	21.45	-10.21	-5.58	69.19	-2.94	-0.93	6.38
38	10.20	21.55	-10.48	-0.02	72.00	-2.62	-1.45	5.82
39	10.76	21.90	-9.25	2.27	70.02	-3.28	-1.69	5.49

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
0	-12.86	129.34	-0.04	14.92	138.37	-1.55	89.62	30.73
1	-5.85	128.69	0.01	15.19	137.24	-1.35	-3.02	-80.51
2	-4.35	128.38	0.04	15.14	134.51	-1.28	41.45	-64.49
3	-12.07	129.28	0.00	15.99	138.44	-1.79	43.17	-32.41
4	-29.61	130.64	0.05	15.92	138.34	-1.07	-61.32	-48.06
5	-8.28	128.31	0.04	16.59	134.68	-1.42	22.30	2.66
6	19.35	130.30	0.02	13.87	136.49	-1.67	-84.92	0.79
7	24.36	127.83	0.04	15.94	138.86	-1.90	0.65	-3.65
8	-14.01	128.40	0.03	15.30	137.88	-1.44	-4.38	-31.18
9	0.91	129.12	0.02	14.12	135.10	-1.55	69.96	4.18
10	23.68	129.32	0.07	16.18	140.35	-1.81	-95.92	-40.96
11	-17.89	127.87	-0.02	14.63	137.22	-1.64	-53.97	-21.93
12	-16.16	127.54	0.03	15.68	139.52	-1.33	-2.36	-18.42
13	-9.14	128.00	0.04	15.45	137.55	-1.47	-154.65	-37.01
14	6.32	128.51	0.01	15.55	135.27	-1.61	-75.40	17.78
15	-56.04	128.95	-0.01	15.81	140.72	-1.50	15.30	-0.58
16	11.07	103.86	0.32	12.40	106.49	-3.15	-17.66	-6.34
17	1.20	103.57	0.30	11.80	105.25	-3.35	158.30	-11.96
18	15.81	104.02	0.31	12.73	105.76	-3.14	27.25	38.46
19	-38.30	105.03	0.26	13.24	105.45	-3.08	55.65	-53.71
20	-11.59	103.10	0.28	12.71	104.03	-3.53	-163.96	49.02
21	-12.52	102.37	0.36	13.19	102.12	-2.92	-173.39	-5.65
22	-9.62	103.25	0.33	12.39	108.93	-2.99	52.22	-161.59
23	25.92	103.48	0.29	11.74	106.99	-2.55	105.11	-85.85
24	-5.50	103.73	0.35	12.51	104.40	-3.33	69.39	-5.04
25	-10.31	105.43	0.32	12.82	105.99	-3.71	-26.84	-118.93
26	18.71	106.00	0.30	10.96	106.89	-3.08	-32.36	-84.52
27	20.64	104.18	0.31	10.98	105.86	-2.84	-60.72	-27.19
28	-15.68	105.50	0.30	12.70	106.94	-2.97	88.69	23.52
29	-13.79	104.57	0.35	11.66	106.15	-3.25	26.48	-34.70
30	0.58	103.49	0.33	12.68	105.58	-2.79	-41.33	6.57
31	-41.18	104.89	0.34	12.31	105.97	-3.75	29.53	-6.35
32	1.27	44.57	0.12	13.53	42.01	2.37	17.83	3.10

33	6.12	45.81	0.10	13.35	42.85	2.35	-40.89	-6.28
34	-2.92	44.37	0.10	13.49	42.40	2.50	118.60	-8.15
35	-5.72	45.65	0.09	13.59	42.95	2.35	-24.50	42.41
36	8.16	46.19	0.13	13.25	42.19	2.58	-23.66	-14.21
37	-2.52	45.21	0.10	14.30	42.49	2.21	10.46	8.05
38	1.03	44.35	0.08	13.11	42.30	2.30	-19.72	-34.53
39	7.84	44.14	0.07	12.77	41.49	2.14	-87.32	-9.49

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
40	19.90	-2.13	3.40	-23.58	14.75	9.48	3.86	-5.28
41	18.83	-2.16	2.85	-22.88	14.32	8.82	4.18	-4.88
42	20.70	-1.65	3.27	-20.35	15.25	9.24	3.58	-4.35
43	20.25	-1.77	3.20	-22.22	15.27	9.16	3.48	-3.80
44	24.14	-2.06	3.32	-21.09	15.40	8.40	4.18	-4.42
45	18.27	-1.69	3.27	-23.09	14.59	8.75	3.83	-4.73
46	25.79	-2.16	2.67	-23.50	13.93	9.50	3.72	-4.28
47	20.38	-2.29	3.08	-22.68	15.06	8.65	2.52	-5.19
48	50.37	-1.77	3.59	-24.60	18.28	10.50	5.85	-7.96
49	49.85	-2.31	3.47	-28.05	18.13	10.17	7.16	-7.75
50	41.67	-2.30	3.79	-26.10	18.67	10.55	6.25	-8.25
51	42.09	-2.00	3.53	-25.53	18.63	10.87	5.39	-5.86
52	50.69	-2.14	3.14	-26.33	19.35	9.35	6.52	-7.00
53	42.49	-1.43	3.63	-27.22	19.41	9.91	6.84	-7.48
54	53.71	-2.44	3.50	-31.61	18.14	11.45	6.06	-9.48
55	46.99	-2.19	3.48	-29.49	18.31	10.81	6.44	-6.38
56	51.56	-2.37	2.97	-27.93	19.01	11.39	5.33	-6.22
57	46.94	-2.41	3.00	-30.06	17.38	9.09	6.34	-7.10
58	41.31	-2.75	3.65	-25.85	19.08	9.83	6.07	-6.96
59	44.49	-2.22	3.61	-32.55	18.31	10.01	6.66	-7.11
60	45.40	-2.42	3.22	-27.27	18.00	11.33	6.80	-7.85
61	48.69	-2.05	3.04	-27.02	19.11	10.47	6.21	-8.31
62	51.91	-2.10	3.23	-24.02	18.61	10.14	6.24	-8.18
63	49.12	-2.03	3.79	-26.05	19.92	9.39	6.41	-7.48
64	14.82	-1.26	2.54	-14.92	10.38	5.70	4.36	-2.13
65	18.85	-1.68	2.35	-15.32	10.60	5.67	4.25	-2.41
66	12.30	-1.35	2.98	-13.25	10.11	6.36	5.11	-2.88
67	10.43	-0.99	2.47	-13.28	10.38	6.20	5.13	-2.15
68	10.66	-1.07	2.81	-16.65	10.22	5.83	3.98	-2.43
69	14.27	-1.20	2.24	-13.79	9.68	4.91	3.57	-2.61
70	14.63	-0.95	2.44	-13.66	10.38	4.72	4.73	-2.55
71	13.43	-1.33	2.57	-16.19	9.82	5.72	4.41	-2.84
72	12.29	-1.25	2.24	-13.42	9.93	5.64	4.34	-2.66
73	13.65	-1.66	2.41	-16.32	10.52	5.40	4.92	-2.50
74	13.37	-1.27	2.47	-15.49	9.98	5.94	3.65	-1.62
75	14.10	-1.64	2.34	-14.76	10.05	5.40	4.82	-2.77
76	16.53	-1.24	1.83	-18.07	9.51	6.44	4.40	-2.85
77	15.73	-1.15	2.18	-13.81	9.36	5.62	4.61	-2.36
78	17.20	-1.32	2.68	-13.10	9.52	5.70	4.78	-2.81
79	18.10	-1.03	2.61	-14.46	10.11	5.73	4.83	-2.49

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
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40	-0.48	-14.13	-0.26	-4.85	-32.91	4.45	1.62	14.53
41	-2.55	-14.69	-0.26	-2.59	-32.83	4.48	1.64	10.87
42	-2.32	-13.82	-0.25	-3.21	-32.11	4.71	1.92	12.41
43	0.25	-13.86	-0.32	-0.74	-32.76	4.46	2.18	10.08
44	-2.98	-14.88	-0.29	-0.15	-33.19	4.33	1.78	12.83
45	0.26	-14.82	-0.34	-3.73	-35.01	4.39	1.93	11.13
46	-0.55	-13.52	-0.27	-2.21	-32.26	4.23	1.58	13.57
47	-2.58	-14.57	-0.27	-1.27	-32.66	4.42	1.55	11.73
48	-9.49	-17.26	-0.40	-7.16	-36.29	5.87	2.34	25.11
49	-15.26	-17.26	-0.39	-9.71	-37.64	6.04	2.21	24.19
50	-17.74	-16.05	-0.38	-8.14	-37.86	5.44	2.03	27.35
51	-7.51	-17.69	-0.44	-7.78	-36.26	6.10	2.27	26.16
52	-16.09	-16.80	-0.45	-7.39	-35.79	6.14	2.29	22.75
53	-21.20	-18.29	-0.46	-6.47	-37.56	6.01	2.73	22.73
54	-14.84	-18.42	-0.34	-6.94	-35.71	6.25	2.47	23.41
55	-16.75	-17.24	-0.39	-7.01	-38.93	6.20	2.67	26.48
56	-8.39	-22.20	-0.43	-6.67	-40.21	5.83	2.38	22.94
57	-14.66	-19.71	-0.36	-8.63	-40.12	6.05	2.57	23.51
58	-16.77	-15.89	-0.40	-8.32	-38.01	6.03	2.53	23.62
59	-16.13	-20.60	-0.48	-7.39	-36.88	6.09	2.39	23.28
60	-14.21	-16.48	-0.41	-8.28	-39.00	5.52	2.26	23.17
61	-14.30	-22.10	-0.43	-8.26	-37.30	6.59	2.03	22.66
62	-17.72	-18.12	-0.41	-7.89	-37.81	6.13	2.04	25.89
63	-17.68	-18.26	-0.39	-6.30	-36.11	5.40	2.04	21.67
64	2.73	-5.49	-0.33	1.00	-13.27	1.08	2.61	11.70
65	-4.20	-5.79	-0.38	-1.97	-13.62	1.34	2.28	11.01
66	-5.39	-6.55	-0.36	0.72	-12.79	0.98	2.48	13.80
67	-4.50	-5.76	-0.33	-1.87	-13.93	0.96	2.58	11.78
68	-4.01	-5.60	-0.37	-1.22	-14.53	1.28	2.37	12.48
69	-6.29	-5.07	-0.33	0.39	-14.47	1.50	2.31	13.61
70	-6.11	-5.86	-0.40	-0.83	-14.26	1.23	2.63	10.62
71	-8.03	-5.00	-0.32	-0.78	-12.45	1.14	2.62	12.27
72	-6.81	-6.73	-0.38	0.34	-14.42	1.52	2.49	12.46
73	-7.12	-6.95	-0.33	0.12	-13.53	1.04	2.45	10.32
74	-6.57	-3.28	-0.31	0.22	-14.17	1.09	2.41	13.68
75	-6.00	-3.61	-0.33	-0.67	-13.40	1.23	2.67	12.09
76	-5.23	-7.88	-0.38	-0.19	-14.11	1.19	2.52	13.16
77	-7.27	-3.96	-0.32	1.76	-14.15	1.23	2.79	11.77
78	-3.39	-7.77	-0.34	-0.85	-13.00	1.32	2.61	10.48
79	-4.11	-6.10	-0.36	-1.01	-13.69	1.30	2.55	11.87

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
40	-1.34	7.31	20.08	-43.84	7.06	8.83	3.34	1.08
41	-1.10	7.79	19.71	8.41	7.32	8.92	15.86	1.04
42	1.31	8.00	19.19	-20.28	6.83	8.82	24.92	1.08
43	1.45	7.98	19.87	46.25	6.78	8.45	-2.96	1.00
44	0.33	7.34	19.54	-38.90	6.92	8.67	-12.94	1.11
45	1.75	7.10	20.14	-21.37	6.82	8.73	-27.46	1.13
46	0.61	8.00	20.03	-28.34	7.09	8.93	31.83	1.12
47	-0.64	7.87	19.53	-46.57	7.31	8.89	-30.39	1.07
48	-0.15	11.06	25.54	-2.80	12.05	11.55	-13.03	1.27
49	-0.12	11.11	25.52	8.12	12.20	12.65	-6.52	1.32
50	-1.32	10.83	24.11	37.99	12.63	11.38	32.37	1.35

51	0.13	10.49	25.74	8.31	12.13	11.86	-5.37	1.28
52	-1.83	11.17	26.01	33.77	12.42	11.00	34.57	1.29
53	-1.54	11.10	25.30	42.41	11.53	11.16	50.78	1.36
54	0.86	10.54	23.84	-10.95	12.27	11.29	7.27	1.27
55	0.11	11.02	25.47	-32.01	11.40	11.13	0.00	1.32
56	1.25	10.62	26.27	11.39	11.63	11.39	0.65	1.32
57	-1.16	10.55	25.32	76.10	12.37	11.06	17.89	1.41
58	-0.85	11.14	24.88	25.65	11.81	11.47	-34.03	1.31
59	-1.57	10.72	25.21	-19.84	11.67	12.12	-12.65	1.17
60	0.05	10.51	24.90	-78.84	12.63	12.01	-6.74	1.37
61	-1.47	10.82	25.26	46.37	12.04	11.15	-18.96	1.39
62	1.03	10.49	25.79	18.87	12.11	11.18	-7.43	1.35
63	0.25	10.96	26.25	59.58	11.55	11.69	-2.09	1.42
64	-0.42	5.86	11.16	66.99	8.70	6.34	31.58	1.06
65	1.35	5.34	11.18	14.04	8.27	6.43	59.34	1.01
66	0.05	5.54	10.70	-52.21	8.29	6.36	6.84	1.05
67	0.45	5.43	11.31	18.29	7.77	6.55	8.15	1.01
68	-0.61	6.04	9.77	28.80	7.99	6.69	3.61	1.10
69	-0.84	5.57	10.58	42.30	8.11	6.44	-41.16	1.11
70	0.34	5.48	10.22	22.47	8.03	6.57	-21.22	1.11
71	-0.40	6.13	10.13	-19.36	8.25	6.63	-9.41	1.03
72	-0.74	5.96	10.71	15.47	7.99	6.60	-11.79	1.10
73	0.30	5.77	10.78	-35.45	8.02	7.23	-7.79	1.12
74	0.31	6.04	9.85	-2.67	7.43	6.23	-16.70	0.99
75	0.07	5.67	11.17	-30.70	7.82	6.03	-41.32	1.03
76	0.91	6.31	10.37	65.63	7.65	6.38	-19.29	1.11
77	-0.55	5.90	10.02	41.19	7.65	6.85	26.33	1.08
78	0.38	5.93	10.95	-33.73	8.60	6.71	15.83	1.07
79	0.08	5.82	10.18	55.66	8.03	6.07	15.49	1.06

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
40	-1.53	1.02	52.10	-0.48	16.66	0.51	-0.48	3.17
41	1.12	0.85	50.58	-0.66	16.94	-0.20	-1.60	1.47
42	-0.14	0.54	49.36	-0.28	16.62	-0.54	-0.84	2.34
43	1.89	-0.17	50.32	-0.77	16.71	0.11	-0.84	2.40
44	2.72	0.93	51.66	0.03	16.64	-0.52	-0.93	3.31
45	0.90	0.89	50.79	-0.49	16.85	0.29	-0.30	2.02
46	0.92	1.06	49.05	-0.55	15.95	-0.36	-0.75	2.36
47	-1.10	0.50	50.20	-0.34	16.34	-0.29	-0.96	2.86
48	1.43	2.74	66.38	-2.26	21.67	0.95	-5.11	7.11
49	0.16	3.22	63.74	-1.00	21.20	1.23	-4.04	5.34
50	1.56	2.91	65.05	-1.22	20.65	1.57	-3.43	4.36
51	-0.08	2.84	63.90	-2.64	21.54	-0.16	-4.06	5.01
52	-1.22	3.14	64.73	-2.15	20.73	1.26	-3.17	6.76
53	2.12	3.35	63.77	-2.31	20.64	1.90	-2.79	4.15
54	-0.09	3.01	65.03	-2.19	21.60	0.00	-3.75	3.41
55	2.71	3.52	62.85	-2.18	21.41	1.02	-3.95	4.22
56	-1.83	3.25	64.90	-2.23	20.87	1.17	-5.24	6.48
57	-1.32	3.01	65.44	-1.67	20.99	0.86	-4.19	4.77
58	-1.32	3.42	67.01	-1.82	20.89	0.54	-2.96	6.01
59	-1.20	3.13	67.59	-2.06	21.62	0.29	-4.53	7.54
60	3.64	2.66	65.10	-1.51	20.53	2.64	-2.72	4.43
61	-2.11	2.97	65.31	-2.06	21.06	1.06	-5.37	5.61

62	1.09	2.90	65.18	-2.28	21.55	1.93	-3.25	4.31
63	3.76	3.61	62.63	-3.22	19.95	0.38	-4.80	6.77
64	2.01	3.49	35.08	-1.73	7.78	-0.78	0.64	-7.62
65	0.65	3.75	37.57	-1.59	8.16	0.25	1.16	-6.40
66	-0.59	3.12	35.67	-1.59	8.70	-0.65	0.44	-7.70
67	0.84	3.74	36.42	-1.44	8.46	0.59	0.32	-7.58
68	1.82	3.81	36.20	-0.96	8.04	0.50	0.78	-6.98
69	-3.57	3.82	34.70	-1.26	8.63	0.05	0.45	-6.48
70	-0.26	3.34	35.18	-1.92	8.23	-0.19	0.54	-6.45
71	0.42	3.34	36.22	-1.78	7.93	0.71	1.11	-6.50
72	-2.08	3.40	36.98	-1.86	8.30	-0.10	-0.07	-6.00
73	0.88	3.39	36.43	-1.80	8.82	1.05	2.09	-6.28
74	0.13	3.49	37.23	-1.68	7.89	0.03	0.84	-6.37
75	0.72	3.63	34.17	-1.43	8.40	-0.15	1.45	-6.18
76	1.16	3.70	37.28	-1.47	8.24	0.62	0.36	-6.41
77	-1.66	3.54	36.23	-1.32	8.74	-0.46	1.47	-6.34
78	0.57	3.54	37.16	-1.52	8.39	0.23	0.68	-5.44
79	1.19	3.07	36.90	-2.02	8.87	0.19	0.45	-6.90

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
40	-5.34	1.56	2.37	14.21	2.38	29.25	-5.24	-4.95
41	-5.33	1.96	2.51	14.89	2.30	29.91	-4.95	-5.64
42	-4.23	1.47	2.50	14.89	2.39	30.13	-5.49	-5.67
43	-5.61	1.39	2.40	14.13	2.39	30.04	-5.16	-5.43
44	-5.26	1.42	2.47	14.70	2.47	29.51	-5.46	-5.65
45	-4.85	1.38	2.49	14.53	2.35	31.15	-5.49	-5.35
46	-5.17	1.42	2.66	14.21	2.40	31.82	-5.15	-5.79
47	-5.68	1.57	2.53	15.07	2.40	29.24	-5.09	-5.72
48	-6.00	2.58	2.50	14.63	3.20	38.48	-3.62	-5.61
49	-6.67	2.35	2.72	14.89	3.23	37.63	-3.69	-6.09
50	-5.81	2.30	2.52	15.66	3.28	38.95	-4.26	-6.18
51	-5.13	2.44	2.67	15.54	3.18	36.41	-4.09	-5.84
52	-4.42	1.99	2.54	15.71	3.13	39.60	-4.01	-6.18
53	-5.04	2.06	2.66	17.16	3.12	36.76	-4.33	-6.06
54	-6.10	2.53	2.50	16.41	3.27	37.65	-4.36	-6.06
55	-5.57	2.26	2.60	16.96	3.16	40.02	-3.85	-6.00
56	-4.10	1.92	2.64	14.69	3.17	39.48	-4.48	-6.07
57	-4.38	2.29	2.61	15.62	3.22	38.06	-3.97	-6.03
58	-5.50	2.09	2.58	16.36	3.27	40.47	-3.28	-5.79
59	-4.45	2.46	2.78	16.60	3.15	39.11	-3.78	-6.16
60	-3.34	2.51	2.52	15.15	3.23	38.56	-4.02	-5.90
61	-5.04	2.20	2.58	16.64	3.16	38.05	-4.02	-5.69
62	-4.07	2.51	2.61	16.88	3.27	38.29	-3.92	-6.25
63	-5.47	2.46	2.56	17.19	3.20	36.88	-3.81	-5.98
64	-3.26	1.93	2.58	6.64	1.79	15.55	-0.68	-6.62
65	-1.96	1.80	2.75	6.53	1.80	14.47	-1.32	-6.52
66	-2.93	1.95	2.58	6.37	1.80	13.31	-1.55	-6.16
67	-2.63	1.77	2.78	6.77	1.79	14.86	-0.93	-6.19
68	-3.09	2.07	2.62	6.63	1.85	13.31	-0.92	-6.44
69	-2.86	1.95	2.66	6.79	1.74	13.74	-1.00	-6.61
70	-1.71	1.92	2.68	5.20	1.78	15.31	-1.48	-6.46
71	-3.02	1.93	2.72	6.27	1.83	13.72	-1.27	-6.62
72	-2.50	2.08	2.62	6.55	1.75	12.96	-1.31	-6.45

73	-2.41	2.11	2.70	6.32	1.82	15.33	-1.06	-6.63
74	-2.92	2.10	2.73	6.78	1.77	15.18	-1.17	-6.35
75	-2.50	1.93	2.62	6.95	1.85	15.54	-1.22	-6.55
76	-2.98	2.03	2.66	6.13	1.77	14.15	-1.40	-6.31
77	-2.17	1.67	2.73	6.52	1.86	12.99	-0.97	-6.39
78	-3.26	1.85	2.79	5.93	1.78	15.29	-1.24	-6.51
79	-2.63	2.13	2.65	6.26	1.82	14.10	-1.09	-6.49

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
40	-1.63	-17.33	-15.14	-16.17	-29.42	28.83	-1.30	-12.03
41	-1.62	-16.73	15.47	-15.91	-10.45	27.24	-4.18	-15.77
42	-1.76	-17.69	-4.99	-16.03	12.60	28.26	-1.18	19.16
43	-1.73	-16.97	12.09	-15.62	-11.85	26.94	-0.37	0.46
44	-1.61	-17.12	-5.19	-15.90	2.02	26.26	-0.46	-5.74
45	-1.70	-17.11	2.87	-15.47	-1.41	27.27	2.06	2.50
46	-1.71	-18.19	-5.68	-15.33	-1.19	27.62	1.22	2.53
47	-1.69	-17.28	-11.69	-15.44	-22.75	27.70	-1.62	1.81
48	-2.35	-23.98	-5.23	-19.65	22.91	35.28	-12.08	-13.87
49	-2.43	-27.18	5.93	-18.98	-27.08	38.11	-20.01	-4.01
50	-2.31	-25.96	-1.21	-17.47	-3.80	36.94	-16.15	0.74
51	-2.43	-25.54	-5.39	-18.91	2.69	37.51	-17.32	-10.40
52	-2.34	-25.63	-0.54	-16.74	-13.52	38.11	-12.87	1.63
53	-2.28	-26.61	18.39	-19.47	38.54	36.67	-15.14	-8.79
54	-2.24	-25.23	0.96	-18.66	-4.01	37.97	-12.54	3.16
55	-2.46	-26.20	3.85	-20.41	-20.02	38.86	-9.05	-6.67
56	-2.20	-26.94	-5.01	-19.81	20.93	36.49	-12.64	1.44
57	-2.24	-26.94	-1.68	-18.68	-1.71	36.78	-9.21	13.52
58	-2.27	-24.52	0.95	-18.90	-12.42	39.65	-21.43	-15.86
59	-2.33	-25.67	-10.70	-19.74	1.12	39.45	-18.24	7.89
60	-2.30	-26.36	11.17	-18.81	-2.69	39.28	-17.71	-4.19
61	-2.29	-24.66	3.78	-19.38	-19.97	38.21	-17.79	-5.47
62	-2.42	-24.80	11.78	-19.14	3.75	38.29	-17.86	-0.73
63	-2.44	-24.51	-21.78	-18.50	23.88	36.49	-15.04	-0.89
64	-1.83	-13.40	-11.14	-13.47	6.68	20.28	-11.60	8.74
65	-1.62	-15.70	1.68	-12.90	6.44	19.59	-9.48	-7.91
66	-1.80	-13.18	-15.05	-13.18	1.99	18.97	-9.93	-3.74
67	-1.63	-13.75	-7.33	-13.37	-23.16	18.82	-6.33	-1.89
68	-1.76	-13.83	8.46	-12.67	5.10	20.29	-9.76	5.80
69	-1.74	-13.90	1.52	-12.82	-22.21	18.64	-11.50	-0.64
70	-1.78	-14.92	1.24	-12.89	4.97	19.29	-7.71	-4.83
71	-1.90	-13.09	-9.94	-13.40	-18.68	19.39	-5.61	6.31
72	-1.81	-14.30	-9.16	-13.84	13.20	18.53	-6.02	-1.96
73	-1.76	-14.66	-2.27	-12.31	-4.24	18.90	-6.41	14.15
74	-1.79	-13.88	-6.26	-12.21	-0.23	19.15	-10.59	-0.15
75	-1.64	-13.15	2.92	-13.45	-9.89	19.10	-10.12	-5.30
76	-1.77	-13.07	4.78	-13.52	-1.98	19.49	-8.17	-0.11
77	-1.67	-13.32	0.73	-13.84	5.22	20.52	-11.56	-5.77
78	-1.77	-13.99	-4.65	-12.55	-8.03	18.97	-7.84	0.31
79	-1.70	-13.62	3.32	-13.01	5.30	20.11	-7.49	7.78

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
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40	10.47	21.27	-10.21	0.10	68.25	-3.70	-1.40	6.15
41	10.05	21.29	-10.73	-5.80	66.81	-3.54	-1.12	5.55
42	9.65	21.44	-10.41	3.16	64.95	-2.16	-1.26	5.51
43	10.09	20.98	-9.71	-1.86	66.84	-1.27	-1.51	5.68
44	10.76	21.46	-10.07	-3.57	67.32	-1.94	-1.25	6.11
45	12.36	21.09	-11.19	-6.00	66.15	-1.00	-1.45	5.81
46	9.57	21.06	-10.42	-7.61	64.58	-2.32	-1.27	4.69
47	10.48	21.27	-8.46	-5.23	67.39	-2.78	-1.50	5.91
48	19.85	26.07	-17.78	5.77	76.70	-5.97	-1.33	7.34
49	17.76	26.10	-18.03	-11.44	73.40	-5.48	-1.67	5.96
50	19.37	25.92	-17.34	-2.21	75.97	-4.95	-1.61	6.77
51	19.60	26.19	-15.93	1.24	77.29	-6.90	-0.94	6.74
52	20.65	25.47	-17.51	-0.40	74.81	-5.15	-1.53	6.92
53	20.51	24.90	-17.03	6.78	78.03	-4.57	-1.38	7.08
54	19.00	25.89	-16.67	-8.22	78.44	-7.80	-0.90	5.93
55	19.61	26.07	-16.10	-4.31	75.29	-5.28	-0.84	7.33
56	17.19	25.63	-17.16	-4.83	76.12	-6.28	-1.30	7.15
57	16.12	26.62	-18.85	-10.03	76.11	-7.80	-1.48	6.26
58	22.83	25.80	-18.76	0.78	79.84	-6.48	-1.62	6.51
59	18.77	25.66	-19.39	-3.45	75.39	-5.64	-1.89	5.89
60	18.65	26.05	-17.83	9.61	75.07	-4.95	-1.50	7.25
61	20.10	25.50	-17.72	5.46	79.24	-5.64	-0.75	6.04
62	18.06	26.18	-16.77	-0.60	75.60	-5.70	-1.29	5.48
63	20.15	26.14	-18.30	0.52	77.40	-5.65	-1.63	6.15
64	4.17	11.78	-11.14	-6.19	47.76	4.84	-0.10	1.22
65	4.95	11.91	-11.29	-7.40	49.57	5.36	-0.13	0.55
66	5.23	11.69	-11.24	2.25	47.65	3.93	-0.44	1.66
67	4.34	11.68	-12.65	-2.83	44.20	6.00	-0.18	1.83
68	4.77	11.73	-10.81	-7.30	48.16	4.80	0.07	1.50
69	5.63	12.02	-11.26	-5.67	46.34	4.37	-0.32	1.27
70	6.80	11.44	-12.79	4.52	46.72	4.09	-0.01	1.03
71	4.40	11.61	-10.90	3.28	46.91	5.31	-0.15	1.20
72	5.63	11.86	-9.84	-2.76	47.55	4.03	-0.09	1.07
73	5.57	11.65	-11.73	4.00	48.09	5.29	-0.14	1.55
74	6.56	12.02	-11.78	5.11	47.54	4.43	-0.34	1.50
75	4.76	11.61	-12.34	-1.54	46.82	4.11	-0.51	1.09
76	3.83	11.41	-11.62	5.85	45.35	5.55	-0.03	1.42
77	5.23	11.93	-10.66	-2.98	46.92	4.34	-0.39	1.13
78	6.95	12.23	-11.66	2.36	44.43	5.72	-0.37	1.19
79	5.55	11.92	-10.93	2.02	45.36	5.01	-0.28	1.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
40	-13.21	46.06	0.12	12.76	40.97	2.21	-5.81	30.19
41	5.60	43.87	0.12	12.55	40.42	2.20	-5.66	10.59
42	1.30	43.78	0.08	13.33	42.36	2.34	104.95	11.44
43	-7.10	45.74	0.12	12.70	41.80	2.17	53.45	1.42
44	-15.01	43.41	0.08	13.65	40.16	2.34	7.38	7.04
45	-4.50	44.35	0.10	13.38	42.10	2.24	-60.08	29.55
46	-6.08	43.93	0.11	12.88	41.53	2.30	43.67	-37.28
47	-1.42	43.77	0.10	12.83	42.02	2.03	-33.22	-1.93
48	7.40	55.67	0.34	15.31	49.80	3.58	-32.69	-7.42
49	0.03	55.98	0.35	15.00	48.94	3.07	-12.52	-12.64
50	13.40	56.33	0.38	16.14	50.79	2.89	79.93	-57.05

51	-14.93	53.37	0.35	15.22	51.09	2.85	70.33	27.06
52	2.81	54.09	0.35	15.14	49.42	3.15	3.52	-55.33
53	-6.01	56.32	0.35	14.90	50.48	2.85	-83.24	-53.13
54	-22.63	56.03	0.32	15.27	49.90	2.96	131.84	-18.47
55	-12.38	57.33	0.34	14.90	48.67	2.74	61.03	8.29
56	15.92	55.65	0.35	16.30	50.40	2.95	-128.30	-29.08
57	5.86	54.69	0.34	14.83	50.68	3.04	60.23	14.95
58	6.32	56.35	0.37	16.09	50.08	3.27	-61.65	-31.02
59	-4.18	54.81	0.39	15.12	49.89	2.74	48.95	-25.32
60	-1.82	55.12	0.32	16.16	49.81	2.53	23.54	38.64
61	-6.60	56.15	0.31	14.82	50.61	3.48	-15.81	-62.06
62	13.52	55.60	0.36	14.65	50.81	2.75	-55.36	2.84
63	6.26	55.11	0.35	16.22	50.09	2.78	-38.96	-9.46
64	-6.41	21.29	0.06	13.13	25.53	2.19	3.75	28.78
65	5.40	20.54	0.03	12.43	26.21	2.26	-53.85	-56.62
66	-1.36	21.26	0.08	12.95	24.86	1.82	19.76	-12.86
67	2.78	19.53	0.06	13.77	25.62	1.69	-28.92	2.03
68	-2.41	19.89	0.07	12.87	24.40	2.26	4.00	8.27
69	-0.42	20.76	0.08	13.75	25.35	2.40	-7.41	-16.89
70	2.68	21.11	0.06	12.80	26.02	1.63	-34.09	-1.22
71	-2.07	20.29	0.06	13.23	24.64	2.14	-0.73	27.99
72	13.77	19.43	0.03	12.95	25.40	2.05	-41.90	41.49
73	6.38	21.39	0.07	12.67	25.45	2.27	64.46	10.51
74	14.28	20.99	0.05	12.33	26.63	1.89	72.91	18.93
75	11.50	20.47	0.07	13.27	25.94	1.95	95.35	4.58
76	-2.82	20.91	0.06	12.71	25.71	2.04	58.62	12.36
77	-11.93	21.90	0.02	13.81	25.86	2.21	-10.84	12.70
78	-11.93	20.49	0.07	13.10	25.82	2.10	20.80	15.62
79	-4.90	21.84	0.05	13.09	25.78	1.86	-24.02	21.73

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
80	237.62	-19.42	-15.00	22.16	76.44	7.36	55.13	20.11
81	245.81	-19.54	-15.13	18.05	77.32	7.05	55.20	19.51
82	239.03	-19.32	-14.10	14.18	76.56	7.65	54.42	21.17
83	239.85	-19.51	-14.78	16.79	75.94	7.74	53.67	19.67
84	236.54	-19.33	-14.34	17.57	76.25	5.88	54.22	20.28
85	233.79	-19.10	-14.73	18.82	77.02	7.74	55.42	19.18
86	244.29	-19.38	-14.24	21.06	76.84	7.05	53.39	21.57
87	245.46	-19.23	-14.63	19.52	76.89	6.20	55.06	19.29
88	237.27	-19.52	-15.00	11.51	75.62	6.84	54.76	18.98
89	246.30	-19.38	-14.77	19.82	76.40	7.87	53.51	19.31
90	239.99	-19.66	-14.51	23.01	78.38	7.62	54.41	20.58
91	246.68	-19.32	-14.48	19.30	76.38	6.57	54.02	19.59
92	244.90	-18.74	-14.76	19.33	75.50	8.02	53.86	19.84
93	240.25	-19.46	-14.08	15.51	77.25	6.68	54.41	20.96
94	236.25	-19.15	-14.29	15.40	77.92	8.29	54.72	20.92
95	234.53	-19.51	-14.79	15.95	76.96	7.71	53.95	19.60
96	225.80	-14.96	-12.69	32.62	79.10	-8.15	47.15	29.30
97	224.72	-15.06	-12.31	38.11	78.67	-8.34	47.15	28.12
98	219.05	-15.08	-12.67	33.06	78.99	-8.10	46.83	29.35
99	225.28	-15.07	-12.13	34.86	78.98	-7.64	46.64	29.02
100	219.18	-15.13	-12.51	32.24	79.59	-8.61	46.55	28.06
101	223.01	-14.98	-12.59	32.56	78.79	-7.85	47.48	29.74

102	226.92	-15.06	-12.82	35.50	78.13	-8.46	47.53	29.39
103	218.47	-15.15	-12.70	31.58	77.22	-8.28	47.09	28.69
104	217.40	-15.08	-12.47	31.83	77.80	-8.14	47.10	29.56
105	211.59	-15.13	-12.39	34.55	78.57	-8.92	46.36	27.44
106	218.37	-14.92	-13.15	31.41	78.23	-8.59	47.33	28.58
107	213.20	-15.28	-12.69	32.20	78.63	-8.15	47.25	28.56
108	211.17	-15.16	-12.35	33.61	78.13	-8.64	46.45	29.12
109	225.61	-15.22	-12.14	30.36	76.95	-7.72	47.83	29.32
110	225.64	-15.32	-12.32	30.59	79.54	-8.44	47.85	29.54
111	224.91	-15.05	-12.61	33.73	78.88	-8.69	47.23	29.07
112	-89.51	-0.54	3.33	-33.39	1.42	11.15	2.84	-3.79
113	-86.30	-0.64	3.48	-32.33	1.63	11.26	2.88	-3.91
114	-80.17	-0.41	3.36	-33.65	1.40	11.45	2.87	-2.39
115	-86.94	-0.54	3.66	-33.07	1.89	10.91	3.66	-2.75
116	-80.07	-0.67	3.73	-32.76	1.05	11.32	2.95	-2.93
117	-87.45	-0.91	3.55	-31.94	1.72	11.40	3.03	-3.38
118	-80.27	-0.56	3.49	-31.59	1.38	11.19	3.53	-3.49
119	-90.36	-0.70	3.58	-32.58	1.11	10.70	3.54	-3.50

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
80	-186.45	-104.53	-2.49	-80.09	-134.45	10.24	-14.72	95.88
81	-192.97	-107.51	-2.51	-80.49	-133.61	10.69	-14.16	94.57
82	-187.71	-108.95	-2.46	-79.35	-133.97	10.30	-14.36	97.20
83	-196.02	-105.38	-2.44	-78.86	-135.96	9.68	-14.04	92.95
84	-194.61	-105.27	-2.45	-79.94	-133.74	10.20	-14.61	95.05
85	-198.68	-106.99	-2.42	-81.39	-132.72	10.63	-14.47	94.83
86	-194.46	-107.19	-2.42	-82.50	-134.72	10.43	-14.56	95.38
87	-200.13	-106.43	-2.49	-78.22	-134.53	10.86	-14.80	93.82
88	-193.98	-106.70	-2.47	-81.14	-133.25	10.47	-14.52	93.65
89	-196.03	-107.53	-2.45	-80.79	-133.78	9.95	-14.23	95.79
90	-197.87	-106.99	-2.47	-79.13	-135.12	10.49	-14.67	92.70
91	-196.90	-110.92	-2.43	-82.52	-133.55	10.59	-14.29	95.36
92	-197.47	-106.85	-2.47	-82.24	-133.42	10.28	-14.49	95.94
93	-195.18	-104.82	-2.46	-82.04	-134.55	10.47	-14.71	94.09
94	-195.83	-106.77	-2.42	-79.90	-131.30	10.06	-14.41	92.75
95	-195.81	-104.92	-2.43	-82.97	-133.36	10.49	-14.20	92.97
96	-181.47	-115.06	-2.02	-70.47	-131.47	11.38	-17.41	86.99
97	-188.01	-114.36	-2.00	-72.63	-131.68	10.99	-17.24	89.31
98	-180.37	-115.61	-1.98	-70.70	-131.42	10.72	-17.68	87.15
99	-180.37	-114.48	-2.02	-67.98	-131.22	10.86	-17.79	89.21
100	-182.21	-111.90	-2.00	-70.02	-132.86	10.41	-17.62	92.41
101	-178.91	-117.06	-2.03	-71.92	-129.11	11.31	-17.42	90.89
102	-181.48	-113.66	-2.01	-69.24	-130.05	11.09	-17.57	94.75
103	-180.28	-117.23	-2.03	-71.54	-128.29	10.24	-17.24	92.37
104	-185.23	-115.73	-2.01	-69.82	-129.67	11.10	-17.39	87.03
105	-182.95	-111.77	-2.03	-71.93	-130.92	11.05	-17.50	90.66
106	-180.42	-115.91	-2.02	-71.93	-132.11	10.58	-17.60	89.10
107	-184.49	-115.82	-2.04	-70.93	-133.27	10.63	-17.39	92.69
108	-182.04	-114.89	-2.02	-73.68	-130.74	10.27	-17.30	88.48
109	-187.39	-116.94	-2.00	-70.45	-133.71	10.58	-17.37	88.44
110	-183.45	-115.03	-2.00	-72.48	-131.80	10.55	-17.51	87.44
111	-178.97	-116.76	-2.03	-70.39	-131.76	11.18	-17.83	90.02
112	53.55	15.02	-0.01	26.67	-5.88	-9.21	0.67	-27.37

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113	55.15	13.52	-0.06	23.88	-7.50	-9.22	0.32	-26.84
114	55.34	13.76	-0.05	25.76	-6.52	-9.00	0.45	-23.41
115	56.13	12.27	-0.04	26.72	-7.43	-8.65	0.23	-25.40
116	58.20	12.54	0.00	27.33	-8.21	-9.11	0.39	-25.47
117	53.40	11.55	-0.04	24.89	-6.52	-9.32	0.47	-26.03
118	57.18	15.39	-0.04	26.82	-6.54	-9.12	0.17	-26.77
119	53.75	13.14	-0.04	25.75	-5.81	-9.26	0.44	-26.64

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
80	1.25	31.48	36.45	5.03	-4.19	-13.70	34.65	3.63
81	1.58	31.20	35.32	3.71	-4.06	-13.75	-25.86	3.60
82	0.39	32.65	36.13	-66.31	-4.02	-13.44	2.89	3.60
83	1.13	32.53	35.72	9.35	-4.30	-13.69	-12.59	3.59
84	1.32	32.05	35.43	77.97	-4.23	-13.75	-11.80	3.64
85	0.40	32.67	35.83	10.64	-4.27	-13.54	-13.35	3.59
86	1.24	31.25	36.79	-0.20	-4.24	-13.69	-7.93	3.61
87	0.63	32.68	36.03	-9.71	-3.66	-13.65	28.19	3.65
88	1.29	31.34	35.88	45.80	-4.03	-13.27	-1.63	3.60
89	-1.15	32.21	35.54	38.92	-4.00	-13.62	-23.70	3.68
90	-0.43	31.99	35.51	-10.68	-4.34	-13.83	-29.95	3.70
91	0.06	31.39	37.08	16.14	-3.99	-13.78	-12.77	3.64
92	-2.56	32.15	36.26	-18.55	-3.82	-13.57	-2.25	3.67
93	-1.69	31.85	35.62	-46.41	-3.96	-13.64	-21.34	3.64
94	0.62	32.69	35.98	-64.99	-4.18	-13.56	-7.76	3.68
95	2.32	32.37	36.68	2.72	-3.94	-13.79	-10.48	3.62
96	1.91	27.17	35.46	31.42	-6.83	-14.35	-4.68	3.04
97	-0.84	28.04	36.09	66.63	-6.64	-14.40	-26.19	3.11
98	-0.22	28.28	35.61	-27.73	-6.88	-14.78	-1.46	3.08
99	0.40	28.54	36.22	-57.05	-6.45	-14.68	10.02	3.12
100	-1.97	28.51	35.25	-1.24	-6.95	-14.54	-11.66	3.11
101	-0.27	28.64	35.64	-32.40	-6.88	-14.72	-4.98	3.09
102	-0.47	27.32	35.97	27.27	-6.83	-14.57	10.59	3.08
103	0.44	28.14	35.71	-30.29	-7.15	-14.77	-9.52	3.05
104	1.54	28.04	35.70	17.22	-7.02	-14.61	-8.51	3.08
105	1.01	27.15	36.02	15.69	-6.67	-14.79	-14.83	3.10
106	-0.91	27.98	35.25	12.21	-6.68	-14.71	-3.81	3.04
107	0.73	26.90	35.44	-45.54	-7.21	-14.30	-8.87	3.12
108	0.21	28.51	35.12	-4.38	-6.82	-14.75	-32.65	3.15
109	1.95	28.09	35.19	-12.43	-7.11	-14.39	17.18	3.04
110	-0.93	28.33	36.14	-55.64	-6.91	-14.73	9.19	3.13
111	1.20	28.73	35.63	-23.48	-6.74	-14.51	0.11	3.05
112	-0.02	-4.26	4.39	-21.84	-1.95	-1.90	4.25	0.50
113	-0.19	-5.10	4.80	-4.81	-1.96	-1.52	0.62	0.54
114	-0.55	-5.07	4.62	20.67	-2.08	-1.80	1.84	0.56
115	-0.12	-4.51	4.51	41.62	-1.85	-1.93	2.57	0.56
116	-0.88	-3.89	4.43	-8.71	-1.84	-1.74	-0.13	0.55
117	-0.21	-4.95	4.50	-8.92	-1.94	-1.54	-3.25	0.54
118	-0.11	-4.62	4.46	15.90	-1.87	-1.91	1.65	0.54
119	-0.29	-4.89	4.58	-6.43	-1.68	-1.77	-5.02	0.52

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
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80	-0.72	13.41	156.23	-22.05	54.69	-21.12	-36.70	-16.35
81	1.23	12.76	154.12	-21.44	53.34	-21.60	-37.59	-15.05
82	1.64	13.48	158.87	-21.61	53.88	-20.64	-36.38	-19.94
83	-0.66	13.19	153.77	-22.15	54.43	-20.25	-34.46	-17.11
84	0.56	13.61	156.37	-21.57	54.48	-19.82	-35.80	-16.25
85	-0.14	13.52	158.07	-21.30	54.41	-20.91	-34.52	-12.68
86	0.41	13.54	156.75	-21.79	53.12	-20.37	-36.33	-13.44
87	-1.00	12.92	156.99	-21.66	53.95	-19.99	-35.01	-12.89
88	-0.16	14.20	155.55	-21.40	54.41	-20.89	-34.07	-15.57
89	-1.85	12.45	156.90	-21.26	53.80	-21.84	-36.75	-17.41
90	0.51	13.56	157.68	-21.81	54.28	-20.43	-34.55	-16.46
91	2.46	13.10	155.86	-22.04	54.75	-20.86	-34.88	-13.51
92	-1.96	13.68	155.14	-22.09	53.98	-20.80	-37.23	-20.04
93	4.18	13.40	158.36	-21.82	54.19	-20.46	-36.36	-16.43
94	3.18	13.43	156.65	-22.12	53.60	-20.05	-35.90	-16.26
95	-1.71	13.55	155.07	-22.08	53.90	-20.33	-36.81	-16.48
96	-1.87	11.55	132.79	-19.54	51.83	-3.17	-27.20	2.49
97	1.90	11.74	132.36	-19.78	51.36	-2.55	-26.79	-1.40
98	2.83	11.98	133.42	-19.47	52.29	-2.64	-26.69	2.37
99	-2.49	11.82	130.84	-19.85	52.62	-2.99	-27.51	0.39
100	-2.51	12.23	132.00	-19.92	52.38	-2.79	-27.31	1.30
101	-0.33	11.25	135.64	-19.17	52.55	-3.21	-28.44	0.79
102	-0.66	11.59	132.50	-19.63	51.89	-2.65	-28.62	0.35
103	4.76	11.46	133.48	-19.67	51.71	-3.21	-25.00	0.87
104	3.28	11.84	132.09	-19.71	52.27	-3.36	-27.45	2.13
105	0.21	11.47	129.46	-19.72	51.54	-2.97	-27.22	-2.11
106	1.65	12.36	134.76	-19.93	52.29	-2.91	-28.74	3.66
107	5.53	11.96	132.76	-19.63	52.03	-3.21	-29.41	-1.15
108	-3.77	11.16	131.28	-19.55	51.37	-2.87	-27.01	1.87
109	-3.12	11.83	134.62	-19.65	52.18	-2.48	-26.46	0.83
110	2.85	12.08	133.21	-19.44	52.47	-3.09	-27.62	-1.18
111	0.07	11.77	131.28	-19.90	52.21	-2.69	-27.81	2.83
112	2.49	-2.66	30.15	-1.56	4.27	-0.80	-2.30	-49.05
113	-0.21	-3.10	31.04	-1.60	4.31	-0.43	-2.86	-49.25
114	1.94	-2.82	31.03	-1.86	4.12	-1.00	-2.97	-50.39
115	1.08	-2.63	31.21	-1.55	3.77	-0.67	-3.06	-50.02
116	-0.40	-2.46	31.05	-1.70	4.43	-0.82	-2.46	-50.67
117	-0.06	-2.74	31.50	-1.52	4.41	-0.77	-2.77	-50.13
118	-0.80	-2.66	31.91	-1.62	3.94	-0.64	-2.97	-49.29
119	-0.25	-2.75	32.83	-1.54	3.88	-0.49	-2.99	-49.55

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
80	-95.69	0.82	-1.10	46.86	-0.54	130.43	-0.31	-6.35
81	-95.93	0.92	-0.98	46.06	-0.58	127.11	0.13	-6.43
82	-96.16	0.86	-1.01	45.27	-0.54	127.80	-0.38	-6.37
83	-95.92	0.73	-0.94	45.00	-0.54	128.49	0.19	-6.34
84	-95.27	0.81	-1.04	44.91	-0.64	126.65	-0.16	-6.39
85	-92.81	0.66	-0.86	47.10	-0.53	130.24	0.06	-6.42
86	-96.29	0.72	-0.91	45.81	-0.46	128.86	-0.40	-6.39
87	-96.00	1.01	-0.93	46.00	-0.58	128.57	0.25	-6.31
88	-94.99	0.62	-0.94	45.13	-0.55	128.39	0.16	-6.34
89	-96.60	0.78	-0.89	45.47	-0.52	126.84	-0.11	-6.40
90	-94.71	1.23	-1.19	45.58	-0.62	128.54	-0.77	-6.34

91	-94.84	0.65	-1.13	45.76	-0.67	128.77	-0.04	-6.34
92	-98.94	0.90	-1.10	46.93	-0.58	126.04	0.20	-6.35
93	-95.55	0.79	-1.08	45.13	-0.51	126.68	0.11	-6.38
94	-95.94	0.89	-0.91	46.23	-0.61	127.94	-0.21	-6.32
95	-95.34	1.31	-1.05	44.16	-0.61	129.03	-0.25	-6.34
96	-107.90	0.76	0.00	43.98	-0.72	132.70	4.65	-5.94
97	-107.59	0.99	-0.20	44.15	-0.85	132.48	4.73	-5.90
98	-107.87	0.85	-0.02	44.18	-0.77	134.56	4.47	-5.93
99	-107.13	0.84	-0.11	44.50	-0.88	132.73	5.06	-5.89
100	-107.14	0.81	-0.29	43.22	-0.80	132.40	4.81	-5.89
101	-107.45	1.18	-0.09	44.08	-0.83	129.23	4.52	-5.89
102	-107.72	0.94	-0.04	43.68	-0.76	128.11	4.76	-5.93
103	-108.87	0.85	-0.20	44.01	-0.84	131.58	4.83	-5.90
104	-106.70	0.96	-0.25	42.66	-0.82	129.83	5.01	-5.99
105	-106.40	0.98	-0.08	43.16	-0.85	130.13	4.95	-5.86
106	-106.63	1.09	-0.06	44.91	-0.82	129.33	4.92	-5.93
107	-107.55	1.00	-0.13	43.78	-0.79	131.18	4.41	-6.02
108	-105.69	1.12	-0.27	44.38	-0.87	131.80	5.34	-5.86
109	-105.87	1.07	0.08	43.11	-0.79	133.55	4.85	-5.86
110	-107.71	0.90	-0.20	43.99	-0.87	133.70	4.92	-6.02
111	-106.95	0.78	-0.10	44.61	-0.76	130.77	4.70	-5.93
112	-4.44	1.70	1.70	0.99	-0.98	-0.86	2.56	-2.05
113	-4.14	1.83	1.66	1.66	-0.97	-1.69	2.43	-2.09
114	-4.10	1.81	1.72	0.63	-0.94	-2.98	2.44	-1.99
115	-4.14	1.83	1.69	0.96	-0.99	-3.20	2.68	-2.10
116	-4.28	1.80	1.62	0.80	-0.98	-3.96	2.41	-2.02
117	-3.84	1.93	1.78	0.41	-1.00	-4.36	2.44	-2.03
118	-4.50	1.70	1.76	0.90	-1.00	-2.13	2.54	-2.05
119	-4.48	1.88	1.66	0.14	-0.95	-2.51	2.49	-2.04

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
80	3.40	19.84	3.52	-78.11	-37.18	-14.13	-71.01	6.01
81	3.35	20.61	-0.60	-78.24	-9.91	-13.29	-75.61	-1.14
82	3.41	20.64	0.82	-76.73	19.98	-13.42	-70.88	-6.79
83	3.46	20.20	-0.67	-78.09	-14.48	-13.16	-78.82	0.37
84	3.44	21.50	0.15	-78.53	-27.77	-12.11	-74.33	-4.33
85	3.42	20.07	-1.55	-77.34	-15.72	-13.50	-71.87	-5.76
86	3.47	19.94	-3.51	-80.03	9.79	-13.72	-76.54	-2.25
87	3.49	20.55	6.58	-77.18	14.38	-13.78	-73.69	-8.28
88	3.47	20.61	10.20	-77.82	-7.27	-12.83	-76.26	-5.58
89	3.41	20.55	2.07	-76.65	5.56	-12.96	-75.26	-0.05
90	3.43	20.12	-1.73	-76.33	-3.90	-13.35	-72.26	5.49
91	3.44	20.60	0.77	-77.71	-11.89	-13.52	-68.06	-2.05
92	3.46	20.20	3.82	-77.63	-0.28	-12.97	-79.25	-10.64
93	3.43	20.64	1.29	-77.80	-18.28	-12.80	-72.32	0.38
94	3.44	20.07	0.60	-76.68	15.97	-13.62	-76.04	6.19
95	3.41	20.74	-1.12	-77.81	2.76	-13.11	-72.34	-2.04
96	3.65	25.16	2.76	-72.70	-2.57	-15.00	-62.60	-2.69
97	3.65	25.10	3.51	-72.06	-1.31	-16.34	-66.97	5.31
98	3.66	25.35	0.27	-73.24	8.88	-15.21	-61.74	2.10
99	3.62	25.43	-5.10	-73.18	-11.54	-15.52	-62.32	1.04
100	3.61	25.28	0.40	-72.53	-1.10	-15.47	-70.69	-4.99
101	3.62	24.49	-3.88	-73.96	-0.97	-15.72	-68.40	2.03

102	3.58	25.52	2.92	-73.28	4.49	-15.85	-61.30	-5.42
103	3.58	25.33	1.26	-73.93	-13.86	-15.93	-64.63	-4.88
104	3.65	25.41	3.75	-73.70	4.01	-15.94	-63.18	5.70
105	3.65	25.65	-0.10	-71.82	-4.99	-15.83	-71.39	-2.33
106	3.63	24.95	1.65	-72.28	-7.68	-15.71	-61.19	0.91
107	3.64	25.07	0.78	-73.83	-7.90	-15.93	-61.00	-4.62
108	3.65	24.83	-0.31	-74.01	5.44	-15.62	-62.16	0.71
109	3.62	24.79	1.58	-72.87	-16.26	-15.41	-54.83	-3.32
110	3.57	24.99	4.36	-73.64	3.01	-16.03	-57.92	2.51
111	3.63	24.88	2.53	-72.63	-2.20	-15.87	-69.05	2.72
112	0.31	4.79	-0.41	-1.00	4.16	-7.16	62.77	-1.29
113	0.36	4.94	-1.85	-0.77	-0.50	-6.81	58.70	2.31
114	0.36	5.23	1.31	-1.09	-3.95	-6.47	60.91	-2.13
115	0.37	5.08	-0.23	-1.37	-5.48	-6.74	63.68	2.69
116	0.37	4.81	-0.94	-1.59	-1.18	-7.42	64.90	2.51
117	0.32	5.26	-0.40	-1.36	7.83	-6.88	69.96	1.83
118	0.34	4.88	-1.21	-1.55	-8.41	-6.50	60.24	2.64
119	0.30	5.04	-0.69	-1.16	-8.28	-6.90	62.01	-0.35

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
80	35.18	-18.92	-15.82	-4.04	123.00	-19.42	-1.04	-51.71
81	34.09	-18.91	-16.07	6.61	118.59	-17.87	-1.15	-52.53
82	36.45	-18.80	-17.25	2.41	120.53	-15.35	-0.15	-51.71
83	35.43	-19.22	-15.75	-0.26	121.11	-14.13	-1.33	-51.60
84	35.61	-19.31	-15.80	3.78	122.22	-13.50	-0.46	-51.12
85	35.84	-19.29	-15.60	-3.07	113.52	-19.69	-1.38	-51.49
86	34.84	-19.11	-15.75	6.88	119.39	-19.36	-1.24	-51.76
87	31.00	-18.93	-16.37	2.25	118.69	-14.80	-0.99	-52.10
88	34.90	-19.56	-16.97	-5.46	121.17	-18.96	-1.06	-52.00
89	34.10	-19.04	-15.91	1.89	118.31	-15.26	-0.99	-51.50
90	34.94	-18.97	-16.30	5.87	119.97	-19.67	-1.19	-52.26
91	29.87	-19.33	-16.96	4.69	119.99	-15.82	-1.76	-51.64
92	34.26	-18.90	-16.35	-0.85	119.09	-13.64	-1.06	-51.17
93	35.63	-18.87	-16.09	-2.07	123.06	-16.23	-0.80	-51.82
94	37.07	-19.29	-16.14	11.96	121.62	-15.06	-1.21	-51.56
95	36.78	-19.43	-16.06	-2.11	119.46	-17.76	-1.23	-51.70
96	21.23	-22.94	-15.56	-2.98	85.63	-20.85	2.54	-46.27
97	24.91	-22.30	-14.76	2.60	81.48	-21.58	2.96	-44.93
98	21.70	-22.33	-14.50	-3.51	88.57	-21.05	1.94	-46.62
99	22.56	-21.99	-14.26	-10.16	85.86	-20.91	2.38	-45.87
100	22.17	-22.94	-14.66	8.34	86.53	-21.30	2.37	-46.47
101	21.08	-23.03	-14.09	-0.27	87.85	-22.21	2.51	-46.73
102	19.67	-22.16	-14.80	3.10	85.87	-21.61	2.82	-47.10
103	22.49	-22.60	-14.67	3.54	87.94	-21.15	2.48	-46.57
104	22.32	-22.83	-15.55	-3.03	89.65	-19.85	2.31	-46.25
105	22.19	-22.25	-14.09	5.15	84.82	-18.96	2.62	-47.25
106	21.33	-22.63	-15.21	-6.48	86.04	-20.04	2.40	-47.84
107	21.61	-22.56	-13.73	2.23	87.92	-19.20	2.10	-46.74
108	19.32	-22.17	-15.69	-1.33	86.07	-18.90	2.44	-46.14
109	21.27	-22.66	-13.80	0.11	86.80	-20.63	2.40	-46.11
110	21.06	-22.97	-14.68	-2.90	86.30	-22.66	2.28	-46.75
111	22.19	-22.73	-15.73	-5.40	87.87	-19.19	2.68	-46.05
112	2.40	-6.68	6.28	-2.72	70.14	32.73	-4.02	-3.00

113	1.66	-6.52	6.68	-1.88	71.24	32.05	-4.00	-1.76
114	2.74	-6.41	6.80	-2.46	72.66	32.14	-4.04	-3.13
115	2.97	-6.78	6.95	-1.73	71.59	31.97	-3.94	-3.26
116	1.87	-6.57	6.62	-2.56	71.55	32.38	-3.96	-3.04
117	2.38	-6.59	7.29	0.80	73.77	32.78	-4.12	-2.99
118	2.23	-6.52	7.40	4.06	73.25	32.26	-4.00	-2.40
119	2.04	-6.41	6.41	-1.61	69.82	31.86	-3.80	-3.61

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
80	7.22	128.52	-1.88	43.51	39.31	15.41	-14.23	23.79
81	16.34	128.59	-1.91	43.36	39.03	15.48	-28.20	5.78
82	-6.02	129.83	-1.88	43.69	37.71	15.66	-12.79	-13.12
83	0.26	129.91	-1.89	43.46	38.12	15.62	39.21	11.40
84	-13.98	128.38	-1.87	43.41	37.06	15.59	13.00	-4.79
85	-20.46	130.06	-1.87	43.02	38.18	15.67	-23.98	-72.21
86	-2.59	127.98	-1.90	43.07	37.21	15.58	-30.51	7.52
87	11.87	128.01	-1.88	43.97	37.46	15.56	-27.53	39.33
88	-8.42	128.67	-1.91	43.50	40.10	15.60	-39.33	-13.06
89	-19.59	130.39	-1.86	43.48	38.13	15.66	-15.58	38.94
90	4.00	129.94	-1.85	43.36	37.51	15.62	-16.66	20.91
91	-15.39	128.60	-1.89	43.80	38.56	15.60	-8.93	13.06
92	13.67	129.15	-1.87	42.93	37.77	15.50	-15.23	5.92
93	3.75	129.33	-1.92	43.30	37.36	15.57	3.32	-17.65
94	-15.76	130.12	-1.87	43.29	37.20	15.39	-11.95	-15.64
95	-20.12	129.29	-1.85	43.60	40.08	15.90	-60.06	-10.83
96	-17.10	114.33	-2.02	39.62	9.75	12.73	52.83	-29.46
97	0.00	113.68	-2.03	39.64	9.45	12.67	-31.55	15.89
98	-5.74	115.79	-2.01	39.91	10.43	13.02	45.48	-21.84
99	1.37	115.20	-2.04	40.55	7.81	12.62	7.80	-103.58
100	4.74	115.96	-2.03	40.13	11.11	12.59	40.56	-60.33
101	9.00	115.12	-2.03	40.05	9.51	12.81	8.19	18.79
102	15.33	113.84	-2.02	39.69	9.57	12.73	29.45	79.46
103	9.99	114.81	-2.04	40.46	9.24	12.55	-24.76	-12.46
104	2.29	114.70	-2.04	40.01	10.31	12.96	8.96	24.34
105	-3.84	114.69	-2.03	39.86	9.44	12.77	-20.44	-33.68
106	-16.27	115.65	-2.04	40.05	9.97	12.69	31.92	-4.97
107	-8.85	114.18	-2.02	40.49	9.01	12.48	6.07	-18.59
108	13.66	113.88	-2.07	39.23	11.45	12.60	38.66	-56.12
109	-8.74	114.98	-2.03	39.73	10.32	12.72	-4.35	-32.95
110	7.85	116.66	-2.01	40.22	9.71	12.66	-0.79	-36.94
111	3.91	115.17	-1.99	39.78	9.93	12.55	9.97	15.23
112	1.14	13.18	-0.14	2.79	24.72	0.07	-16.74	8.49
113	4.25	13.58	-0.14	2.10	24.18	0.17	17.27	13.28
114	-3.59	13.49	-0.16	2.96	24.03	0.26	6.52	41.70
115	5.11	14.28	-0.14	2.68	24.04	0.36	10.37	38.29
116	6.99	13.92	-0.14	2.68	24.65	0.11	10.23	-3.54
117	5.35	14.04	-0.16	2.83	24.69	0.32	-1.11	7.77
118	-14.75	13.75	-0.12	2.85	24.55	0.22	-16.30	0.78
119	7.24	13.73	-0.13	2.73	24.81	0.13	-1.81	10.69

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
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120	-86.29	-0.63	3.88	-32.02	1.84	11.04	2.67	-3.79
121	-85.57	-0.73	3.64	-32.07	1.65	11.44	2.57	-2.83
122	-80.08	-0.26	3.98	-32.67	1.94	10.85	2.31	-3.57
123	-89.15	-0.56	3.69	-32.93	1.34	11.29	2.70	-3.13
124	-93.09	-0.87	3.69	-33.69	1.66	11.05	2.62	-2.62
125	-79.10	-0.35	3.46	-34.24	1.35	11.14	3.71	-2.73
126	-79.96	-0.48	3.30	-34.24	1.28	11.20	2.45	-3.40
127	-88.22	-0.65	3.53	-36.03	1.47	11.34	3.62	-3.03
128	-64.37	-4.04	1.18	-12.23	6.38	4.37	5.12	-0.76
129	-69.64	-4.14	1.10	-13.64	6.02	4.69	5.44	-0.43
130	-63.61	-4.12	0.95	-10.49	6.13	4.01	4.92	-0.32
131	-66.53	-4.06	0.94	-13.49	5.92	4.79	5.35	-0.99
132	-70.96	-4.00	1.10	-13.01	6.39	3.96	5.16	-0.30
133	-67.75	-4.00	1.23	-15.14	6.04	3.95	5.44	-0.43
134	-64.48	-4.20	0.82	-13.91	5.82	4.06	5.23	-0.18
135	-61.97	-4.07	0.68	-13.86	5.92	4.81	5.69	-0.76
136	-68.31	-4.09	0.82	-13.67	5.52	4.21	5.11	-0.62
137	-57.85	-4.15	1.00	-14.37	5.78	4.21	5.63	-1.43
138	-68.14	-4.06	0.87	-13.57	5.80	4.44	5.36	-0.51
139	-71.48	-4.31	1.21	-16.35	6.10	4.03	5.38	-0.61
140	-65.76	-4.29	0.82	-14.07	6.50	4.40	5.54	-0.17
141	-61.81	-4.12	1.10	-12.99	5.91	4.76	4.79	-0.75
142	-66.55	-4.48	1.08	-12.19	5.67	4.64	5.21	-0.94
143	-63.27	-4.03	1.02	-14.10	5.86	4.08	6.23	-1.19
144	-38.06	-4.80	-0.40	-7.99	5.14	1.07	4.09	-2.19
145	-34.73	-5.10	-0.37	-9.37	4.70	0.55	3.89	-1.54
146	-36.22	-4.93	-0.66	-8.77	4.53	0.87	3.70	-1.89
147	-38.97	-4.91	-0.26	-8.19	4.97	0.85	4.47	-1.42
148	-39.77	-4.98	-0.42	-9.31	5.15	0.76	4.33	-2.09
149	-40.83	-4.93	-0.29	-7.86	4.61	0.53	4.00	-1.78
150	-42.55	-4.97	-0.37	-8.84	4.88	1.17	4.23	-1.88
151	-40.43	-4.83	-0.39	-10.02	4.55	0.64	3.83	-2.11
152	-35.57	-4.98	-0.28	-9.08	4.79	1.04	4.22	-1.39
153	-36.15	-4.86	-0.47	-10.73	4.58	1.24	3.96	-1.78
154	-40.29	-4.90	-0.50	-9.89	4.86	1.22	3.86	-2.27
155	-36.78	-5.15	-0.39	-9.03	5.19	0.90	4.12	-1.66
156	-39.53	-5.04	-0.58	-9.80	5.07	0.85	4.06	-1.30
157	-42.15	-4.89	-0.60	-9.48	4.90	0.85	3.62	-2.15
158	-38.22	-5.10	-0.56	-7.54	4.48	0.71	4.17	-1.92
159	-40.67	-5.02	-0.56	-9.52	4.98	0.99	3.72	-1.58

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
120	53.48	10.31	-0.06	26.05	-6.81	-9.45	0.35	-27.38
121	58.44	14.49	-0.03	26.18	-6.15	-9.19	0.45	-24.02
122	55.08	11.57	-0.02	23.35	-9.94	-9.11	0.39	-25.86
123	56.58	12.31	-0.04	26.82	-7.44	-9.01	0.26	-25.55
124	54.35	10.39	-0.05	24.98	-8.46	-9.02	0.44	-25.96
125	53.81	15.02	-0.04	27.99	-7.80	-9.58	0.44	-25.95
126	53.27	14.24	-0.04	27.91	-7.71	-9.26	0.47	-25.63
127	58.03	12.68	-0.01	25.01	-6.40	-8.74	0.40	-28.05
128	39.98	-1.05	-0.33	21.96	-22.04	-4.80	-0.41	-15.66
129	41.22	-2.03	-0.36	22.19	-21.99	-5.06	-0.56	-15.90
130	40.95	-2.24	-0.34	21.97	-22.07	-5.14	-0.57	-19.52

131	40.26	-0.35	-0.34	20.92	-20.71	-5.18	-0.39	-18.20
132	41.86	-2.16	-0.37	22.34	-21.90	-4.85	-0.56	-16.02
133	43.88	-2.36	-0.34	21.20	-21.58	-5.34	-0.51	-16.43
134	36.87	-2.70	-0.33	20.90	-21.32	-5.10	-0.61	-19.40
135	39.83	-2.87	-0.36	18.90	-21.75	-4.79	-0.38	-17.56
136	45.23	-0.60	-0.35	22.43	-21.35	-5.04	-0.36	-17.73
137	41.16	-0.89	-0.36	23.49	-20.47	-5.32	-0.67	-14.10
138	38.19	-1.59	-0.35	21.84	-23.38	-5.16	-0.48	-17.71
139	41.58	-2.39	-0.39	21.34	-24.13	-4.83	-0.53	-17.91
140	42.51	-2.42	-0.38	23.72	-21.10	-5.44	-0.54	-15.10
141	40.30	0.10	-0.36	22.77	-22.23	-5.06	-0.67	-17.92
142	41.11	0.05	-0.35	22.22	-20.58	-4.96	-0.59	-18.54
143	42.21	-2.62	-0.33	21.58	-22.83	-5.20	-0.53	-18.43
144	23.17	3.07	-0.39	11.72	-4.83	-2.20	0.03	-8.73
145	20.93	1.49	-0.37	13.64	-5.71	-2.24	-0.03	-7.18
146	23.18	3.13	-0.37	12.07	-6.08	-2.29	-0.04	-9.87
147	24.10	2.17	-0.37	12.71	-6.07	-1.98	0.18	-8.85
148	21.38	3.16	-0.34	12.79	-5.25	-2.21	-0.02	-7.81
149	23.10	3.56	-0.37	13.95	-6.65	-2.46	-0.06	-9.00
150	19.33	3.00	-0.38	12.67	-5.73	-2.00	-0.07	-7.75
151	25.90	2.23	-0.36	11.71	-6.53	-2.38	0.09	-8.28
152	24.52	2.63	-0.37	14.16	-6.56	-2.29	0.02	-8.56
153	24.20	3.68	-0.34	12.92	-5.60	-2.00	-0.12	-8.80
154	20.10	2.21	-0.36	13.40	-5.76	-2.46	0.13	-8.69
155	23.10	1.82	-0.36	12.73	-5.94	-2.06	0.12	-8.08
156	24.71	2.42	-0.37	12.42	-5.34	-2.50	-0.12	-9.17
157	23.11	3.73	-0.39	12.30	-6.50	-2.19	-0.03	-9.57
158	24.10	3.93	-0.35	14.40	-6.18	-2.17	0.00	-8.30
159	22.82	4.16	-0.34	13.30	-6.05	-2.27	0.10	-7.50

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
120	0.31	-5.03	4.62	-32.56	-1.82	-1.70	4.33	0.53
121	0.72	-4.01	4.85	1.24	-1.88	-1.74	-1.33	0.54
122	0.45	-4.10	4.84	-2.36	-1.67	-1.69	-2.20	0.55
123	0.11	-4.61	4.79	14.98	-1.97	-1.76	-1.86	0.53
124	-0.04	-4.58	4.63	0.19	-1.97	-1.92	1.82	0.55
125	0.56	-4.69	4.83	-20.06	-1.93	-1.77	3.49	0.53
126	-0.20	-4.71	4.49	4.85	-1.85	-1.65	-2.92	0.54
127	-0.06	-4.54	4.62	-14.70	-1.93	-1.83	-4.25	0.55
128	-0.59	-1.46	4.32	-19.70	1.52	0.78	1.79	0.51
129	-0.28	-1.64	4.09	-34.57	1.15	0.54	5.08	0.51
130	0.18	-1.25	4.46	22.73	1.33	0.70	7.53	0.48
131	1.03	-1.55	4.43	8.79	1.40	0.59	5.14	0.47
132	-0.07	-1.65	4.37	-34.22	1.36	0.55	-0.52	0.48
133	0.09	-1.60	4.25	6.16	1.45	0.59	7.45	0.50
134	0.59	-1.67	4.41	-25.72	1.17	0.73	6.02	0.48
135	-1.51	-1.42	4.83	-0.65	1.22	0.56	7.84	0.49
136	-0.64	-1.43	4.64	-25.29	1.16	0.77	2.70	0.47
137	0.38	-1.91	4.26	29.50	1.25	0.50	3.59	0.47
138	-0.51	-1.60	4.68	-16.14	1.52	0.61	-4.79	0.49
139	-0.07	-1.17	4.53	42.81	1.29	0.66	6.75	0.50
140	-0.70	-1.19	4.06	6.12	1.09	0.81	-3.05	0.55
141	-0.06	-1.58	4.40	6.81	1.20	0.47	3.59	0.52

142	-0.30	-1.72	3.98	-8.17	1.33	0.76	4.02	0.50
143	-0.08	-1.32	4.27	-7.54	1.36	0.36	-7.33	0.50
144	0.45	0.82	3.10	11.79	3.12	2.48	-2.30	0.22
145	-0.01	0.44	2.79	-4.24	3.08	2.36	-0.35	0.25
146	0.37	0.43	2.61	-14.16	3.16	2.41	0.10	0.20
147	0.50	0.65	2.30	16.23	3.18	2.47	13.38	0.22
148	-0.08	0.27	2.63	15.55	3.20	2.50	6.03	0.22
149	0.77	0.39	2.89	11.11	3.10	2.45	1.68	0.23
150	-0.07	0.70	2.73	-7.13	3.16	2.45	6.97	0.23
151	-0.14	0.72	2.70	16.50	3.19	2.47	7.78	0.23
152	-0.05	0.58	3.11	5.90	3.31	2.60	5.82	0.21
153	-0.41	0.77	2.48	-3.59	3.23	2.64	1.59	0.21
154	-0.34	0.61	2.63	13.77	3.13	2.70	5.05	0.23
155	-0.58	0.21	2.43	-4.03	3.19	2.42	0.11	0.21
156	-0.25	0.71	2.85	-12.11	3.07	2.63	6.59	0.24
157	0.54	0.53	2.69	2.39	3.20	2.50	-3.10	0.21
158	0.28	0.84	2.99	-14.40	3.08	2.52	-6.42	0.19
159	0.26	0.25	2.33	8.40	3.20	2.36	5.99	0.25

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
120	-3.15	-2.74	32.68	-1.28	4.17	-0.54	-2.97	-49.03
121	0.37	-2.71	32.30	-1.60	4.71	-0.70	-2.92	-49.08
122	-1.57	-2.50	33.88	-1.53	4.70	-0.59	-2.22	-49.68
123	-2.35	-2.72	30.56	-1.41	4.83	-0.77	-2.27	-49.43
124	-1.34	-2.41	29.44	-1.53	4.20	-0.64	-3.11	-49.42
125	-0.38	-2.90	32.38	-1.44	3.98	-0.92	-3.34	-49.54
126	-0.34	-2.87	30.15	-1.49	4.13	-0.79	-3.50	-50.50
127	-1.26	-2.59	31.97	-1.33	3.99	-0.36	-2.67	-50.02
128	1.14	-0.88	32.09	-0.37	9.19	-2.91	-3.97	-36.02
129	0.01	-0.90	29.34	-1.01	8.94	-2.71	-3.49	-35.29
130	-0.35	-0.83	32.11	-0.90	9.10	-3.05	-4.29	-34.94
131	-1.21	-1.12	32.67	-0.71	9.25	-3.00	-3.36	-36.80
132	-0.17	-1.05	30.63	-0.91	8.77	-2.89	-4.80	-36.50
133	1.87	-0.79	31.14	-0.52	9.94	-2.94	-3.83	-36.11
134	-0.47	-1.01	30.51	-0.75	9.13	-3.33	-3.84	-35.71
135	0.30	-0.80	30.54	-1.22	9.54	-2.83	-3.54	-35.94
136	-1.37	-0.96	32.28	-0.65	9.08	-2.91	-3.37	-36.61
137	2.06	-1.05	29.71	-0.50	9.28	-2.99	-3.87	-35.73
138	-0.62	-0.65	28.44	-0.93	9.22	-2.73	-3.47	-36.62
139	-1.69	-0.79	32.28	-0.62	9.13	-2.77	-3.37	-36.46
140	-1.42	-0.48	31.05	-0.86	9.25	-3.21	-3.49	-35.39
141	2.17	-0.73	31.65	-0.97	9.05	-3.07	-4.55	-35.42
142	-1.91	-0.93	29.79	-0.57	8.76	-2.92	-3.90	-35.86
143	0.66	-0.81	33.30	-0.84	9.12	-3.15	-4.24	-37.51
144	1.11	0.42	18.05	0.99	2.98	-0.40	1.34	-22.51
145	0.18	0.19	18.46	0.91	2.43	-0.95	1.18	-21.18
146	-0.04	0.32	17.97	0.82	2.36	-0.36	0.97	-22.37
147	-0.54	0.58	18.34	1.07	2.36	-0.56	0.77	-22.28
148	0.51	0.19	18.43	0.91	2.45	-0.37	0.93	-22.38
149	0.27	0.47	18.05	0.97	2.15	-0.67	1.19	-22.81
150	0.84	0.36	17.25	1.26	2.25	-0.60	1.62	-22.12
151	1.21	0.45	18.18	1.26	2.77	-0.41	1.26	-22.47
152	-1.01	0.39	18.00	1.08	2.27	-0.42	1.60	-23.23

153	-0.96	0.54	17.76	1.28	2.68	-0.65	1.19	-22.35
154	-1.06	0.43	18.31	0.95	1.97	-0.58	1.72	-22.32
155	-1.09	0.56	16.71	0.46	2.04	-0.37	1.65	-21.98
156	-1.60	0.60	18.48	0.96	2.26	-0.62	1.03	-22.74
157	1.04	0.44	16.36	0.86	2.27	-0.58	1.05	-23.32
158	-0.46	0.44	16.89	0.97	2.06	-0.66	1.13	-21.35
159	0.16	0.47	18.03	1.13	2.51	-0.59	1.34	-22.34

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
120	-4.42	1.84	1.76	1.01	-0.98	-2.13	2.45	-2.01
121	-4.37	2.00	1.74	0.71	-0.97	-1.82	2.59	-2.07
122	-3.96	1.87	1.66	0.88	-0.98	-1.29	2.49	-2.04
123	-3.97	1.99	1.80	0.12	-0.92	-1.94	2.43	-2.10
124	-4.29	1.75	1.75	0.79	-0.97	-3.66	2.39	-2.04
125	-4.25	1.90	1.82	1.10	-0.98	-2.15	2.39	-2.09
126	-4.33	1.92	1.73	1.23	-0.96	-3.61	2.43	-2.04
127	-4.28	1.86	1.88	1.40	-1.02	-3.55	2.51	-1.98
128	-7.95	0.11	1.68	8.51	-0.87	10.98	-2.80	-2.96
129	-8.25	0.12	1.63	9.09	-0.88	8.94	-2.75	-3.10
130	-8.36	0.16	1.71	8.72	-0.86	9.22	-2.60	-3.06
131	-8.59	0.00	1.59	7.62	-0.86	11.42	-2.57	-2.99
132	-8.02	0.21	1.68	8.66	-0.87	9.90	-2.69	-3.05
133	-8.27	0.18	1.74	8.45	-0.92	9.04	-2.93	-3.06
134	-8.28	-0.01	1.69	8.82	-0.93	9.85	-2.73	-3.04
135	-8.91	0.14	1.71	8.75	-0.83	9.35	-2.74	-3.04
136	-8.32	0.14	1.68	8.33	-0.91	9.73	-2.51	-3.02
137	-8.23	0.27	1.76	8.53	-0.86	10.43	-2.72	-2.95
138	-8.15	0.11	1.60	8.97	-0.93	10.95	-2.78	-2.98
139	-8.33	0.06	1.70	8.68	-0.89	10.59	-3.01	-3.00
140	-7.64	0.16	1.68	8.39	-0.82	9.51	-2.84	-3.02
141	-9.02	0.15	1.58	8.08	-0.92	10.09	-2.75	-3.09
142	-8.73	0.14	1.68	8.98	-0.89	10.21	-2.67	-3.04
143	-7.95	0.22	1.63	9.15	-0.98	9.25	-3.00	-3.00
144	-2.45	-1.17	0.94	4.61	-0.40	1.78	-3.48	-2.51
145	-2.39	-1.16	1.04	4.32	-0.39	2.64	-3.46	-2.56
146	-1.88	-1.16	0.98	4.39	-0.41	0.79	-3.46	-2.50
147	-2.14	-1.10	1.02	4.42	-0.42	2.73	-3.50	-2.53
148	-2.58	-1.01	0.97	4.80	-0.36	2.52	-3.48	-2.52
149	-2.46	-1.06	0.97	4.32	-0.38	2.31	-3.48	-2.54
150	-2.92	-1.09	1.00	4.66	-0.42	2.15	-3.46	-2.52
151	-2.89	-1.10	0.90	4.61	-0.37	2.57	-3.54	-2.54
152	-2.62	-1.10	0.98	4.88	-0.37	2.24	-3.62	-2.56
153	-3.08	-1.12	1.01	4.37	-0.35	3.16	-3.55	-2.56
154	-2.80	-1.10	0.94	4.54	-0.41	2.04	-3.56	-2.52
155	-2.13	-1.11	0.95	4.54	-0.41	2.28	-3.54	-2.48
156	-2.77	-1.04	0.95	4.39	-0.36	1.86	-3.63	-2.55
157	-3.33	-1.13	1.01	3.85	-0.38	3.13	-3.50	-2.49
158	-2.72	-0.98	0.94	4.57	-0.36	2.95	-3.54	-2.53
159	-2.54	-1.18	1.01	4.96	-0.35	1.73	-3.52	-2.48

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
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120	0.33	4.92	-0.31	-1.59	0.68	-6.93	56.35	1.36
121	0.30	4.77	0.00	-1.50	4.89	-6.96	59.88	-3.65
122	0.34	4.82	4.31	-1.16	-7.78	-6.96	67.78	-1.90
123	0.31	4.96	-0.12	-2.05	-5.38	-7.09	66.01	0.16
124	0.36	4.79	3.53	-0.49	6.42	-6.72	66.01	2.43
125	0.35	5.13	-0.44	-1.41	-1.40	-6.90	63.79	3.73
126	0.33	4.75	-1.99	-1.23	-9.02	-6.86	58.33	3.46
127	0.36	4.73	0.36	-1.53	4.56	-7.29	61.10	3.11
128	-0.07	-1.38	2.12	-8.54	-9.27	-1.78	54.45	-0.97
129	-0.13	-1.34	2.14	-8.46	0.16	-2.53	47.17	-2.51
130	-0.06	-1.74	-2.76	-8.60	2.08	-2.47	57.05	-0.39
131	-0.15	-1.22	0.09	-8.08	5.34	-2.74	64.38	-0.47
132	-0.10	-2.01	-3.16	-8.12	-3.11	-2.73	54.72	3.38
133	-0.13	-1.45	2.02	-8.04	2.46	-2.24	68.54	1.40
134	-0.11	-1.36	-2.75	-7.68	0.87	-2.20	62.77	0.46
135	-0.10	-1.66	0.80	-7.99	-6.50	-2.85	58.11	-3.03
136	-0.10	-1.39	-0.66	-8.05	-0.64	-2.58	60.67	-1.20
137	-0.08	-2.04	-0.70	-8.62	5.92	-2.03	63.08	2.07
138	-0.11	-1.44	-2.66	-8.23	-3.00	-2.64	52.05	-12.58
139	-0.09	-1.34	1.73	-7.92	1.25	-2.09	54.21	-4.23
140	-0.13	-1.72	-1.14	-8.65	3.68	-2.46	60.90	-3.56
141	-0.14	-1.29	-1.05	-8.42	-4.40	-1.80	55.03	-1.73
142	-0.10	-1.33	-4.62	-9.38	-5.19	-2.68	56.49	-0.74
143	-0.09	-1.35	-1.42	-9.07	3.70	-2.50	61.63	0.05
144	-0.44	-6.62	-0.08	-8.43	3.03	2.65	34.50	2.06
145	-0.47	-6.28	1.57	-9.29	-6.12	2.33	29.93	-1.97
146	-0.44	-6.15	-0.35	-8.90	-6.12	2.76	31.77	-0.39
147	-0.47	-6.03	1.02	-8.50	-3.98	2.55	27.37	6.41
148	-0.47	-6.47	0.58	-9.16	-4.25	2.53	29.28	-2.16
149	-0.47	-6.41	0.58	-8.59	-7.69	2.66	32.78	-4.26
150	-0.44	-6.63	0.33	-8.96	5.06	2.56	29.31	0.25
151	-0.45	-6.32	4.26	-8.83	-3.55	2.68	35.36	-1.39
152	-0.44	-6.65	0.05	-8.71	-11.51	2.71	30.30	3.43
153	-0.44	-6.47	-0.37	-8.79	7.18	2.80	30.82	-6.53
154	-0.44	-6.12	0.06	-8.76	3.86	2.95	28.19	-2.98
155	-0.44	-6.12	1.80	-9.11	-9.21	2.68	30.34	6.30
156	-0.46	-6.68	-0.77	-8.69	1.26	2.63	30.94	-0.36
157	-0.47	-6.39	-0.08	-8.15	7.13	2.80	28.23	1.12
158	-0.45	-6.39	-0.75	-8.70	8.69	2.90	31.24	1.50
159	-0.46	-6.57	-1.82	-8.71	-3.29	2.75	28.59	-2.87

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
120	2.78	-6.70	6.71	-4.07	75.91	32.50	-4.02	-2.35
121	1.75	-6.51	6.99	1.30	76.01	32.64	-3.86	-1.98
122	2.33	-6.45	7.21	-2.74	73.44	32.65	-3.94	-3.11
123	2.80	-6.68	7.26	-2.89	72.83	33.33	-3.87	-2.74
124	3.47	-6.47	6.87	1.60	66.53	32.72	-3.73	-1.95
125	1.90	-6.45	7.49	0.13	71.55	33.59	-3.90	-2.40
126	2.54	-6.66	7.75	-4.69	70.82	32.24	-3.83	-3.09
127	1.80	-6.47	6.30	-0.37	75.11	32.34	-3.76	-2.51
128	5.08	-2.11	4.54	-3.75	56.90	24.33	-1.11	-3.41
129	6.93	-2.21	4.69	-1.04	53.20	22.82	-0.86	-3.26
130	5.71	-2.02	4.77	-2.04	59.19	22.72	-1.09	-3.87

131	6.18	-2.16	5.19	-1.55	56.71	22.38	-0.87	-3.38
132	4.92	-2.36	4.70	-3.26	53.43	21.56	-0.71	-3.77
133	5.37	-2.04	5.37	4.33	55.12	23.02	-1.00	-3.41
134	4.82	-2.12	4.95	2.78	57.33	21.93	-1.02	-4.04
135	5.17	-2.10	4.70	2.48	52.38	21.10	-0.78	-3.61
136	5.59	-2.43	5.28	-2.07	55.91	21.57	-1.07	-3.74
137	6.13	-2.32	4.42	-3.71	55.69	22.77	-0.81	-3.72
138	4.85	-2.26	4.81	6.46	55.44	22.67	-0.71	-3.47
139	4.82	-2.22	5.09	2.24	54.98	22.90	-0.79	-3.88
140	6.04	-2.27	4.38	-0.44	59.54	22.51	-0.85	-3.18
141	6.36	-2.39	5.49	-0.20	54.86	21.78	-0.90	-3.71
142	5.02	-2.24	5.11	2.02	57.42	21.87	-0.84	-3.51
143	5.52	-2.22	4.91	1.56	58.08	23.02	-1.00	-3.42
144	-0.21	1.06	1.30	-0.44	38.34	16.26	0.17	-2.55
145	-0.70	1.07	0.82	0.28	39.73	16.77	0.10	-2.79
146	-0.67	0.88	0.94	0.56	41.10	17.62	0.23	-2.75
147	-0.68	1.10	1.41	0.32	40.17	16.63	0.19	-2.42
148	-0.41	1.17	1.07	0.73	37.75	15.98	0.12	-2.83
149	-0.55	1.15	1.39	3.93	39.29	16.75	0.30	-3.12
150	-1.68	1.29	1.09	2.30	41.93	16.56	0.33	-2.48
151	-0.46	1.00	1.37	-0.76	41.86	16.89	0.23	-3.06
152	-1.16	0.71	0.72	-3.59	36.96	15.92	0.09	-2.85
153	-0.73	1.16	1.43	-2.06	37.10	16.71	0.20	-2.38
154	-0.77	1.19	1.48	0.25	39.82	16.00	0.06	-1.96
155	-0.23	1.11	0.91	0.10	39.59	16.68	0.02	-2.58
156	-0.49	1.09	1.59	1.28	37.71	16.57	0.07	-2.28
157	-0.44	1.01	0.76	0.78	41.09	16.19	0.21	-2.43
158	-0.30	1.03	1.11	-2.55	38.76	15.61	0.29	-2.24
159	-1.59	1.10	1.12	-0.03	39.03	16.28	0.16	-2.11

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
120	-3.49	13.40	-0.17	2.77	24.03	0.22	-1.92	8.36
121	-1.56	13.53	-0.15	2.91	24.00	0.19	-0.13	9.30
122	-0.33	13.08	-0.14	2.69	23.29	-0.11	-10.57	21.22
123	4.41	13.72	-0.15	2.98	24.76	0.23	-14.55	8.52
124	-5.43	13.28	-0.15	2.53	23.69	0.36	3.17	30.92
125	-2.68	13.25	-0.13	2.61	24.53	0.23	43.45	1.59
126	7.50	15.48	-0.16	2.78	24.04	0.28	-3.80	25.95
127	4.32	14.32	-0.14	2.77	23.76	0.04	2.42	-25.97
128	-13.42	27.45	-0.24	7.34	17.51	2.47	0.94	9.49
129	10.58	26.06	-0.24	7.23	17.53	2.31	14.43	46.44
130	4.21	24.93	-0.25	7.19	17.44	2.41	-21.84	-14.67
131	1.14	27.25	-0.25	7.89	17.62	2.36	-6.28	-1.76
132	-1.71	25.57	-0.28	7.46	17.35	2.48	0.91	32.36
133	12.79	24.80	-0.28	7.83	17.59	2.47	-0.89	-14.20
134	3.23	24.22	-0.24	7.25	17.33	2.29	7.74	36.11
135	5.29	24.94	-0.25	7.41	17.48	2.41	0.58	13.47
136	-3.92	26.67	-0.25	7.53	17.72	2.42	-8.16	22.44
137	-3.58	26.44	-0.26	7.61	16.62	2.49	-1.47	-16.57
138	4.11	25.58	-0.24	7.62	17.52	2.36	-16.44	-25.70
139	-1.61	25.23	-0.24	7.50	17.71	2.41	-13.21	25.78
140	-2.54	27.55	-0.25	7.87	18.15	2.40	1.36	18.40
141	-12.06	26.71	-0.24	7.31	17.41	2.28	-12.71	9.79

142	1.45	25.83	-0.25	7.40	17.14	2.38	-6.48	-4.14
143	3.09	26.96	-0.25	7.34	17.09	2.50	-5.15	17.91
144	-4.85	8.77	-0.03	7.05	9.88	2.53	14.66	9.71
145	-3.68	8.85	-0.04	7.36	9.83	2.65	13.40	23.05
146	-2.12	8.74	-0.03	7.62	10.26	2.57	-4.56	-13.00
147	0.77	9.60	-0.04	7.20	10.29	2.55	-1.43	5.03
148	1.14	9.42	-0.03	7.40	10.21	2.61	-10.34	1.62
149	1.93	8.85	-0.04	7.68	9.93	2.31	-10.78	11.24
150	-1.34	10.06	-0.04	7.44	10.26	2.50	2.13	6.41
151	3.92	9.28	-0.03	7.41	10.52	2.47	-23.89	21.08
152	1.39	9.20	-0.02	7.35	10.53	2.59	16.81	5.07
153	-8.77	8.14	-0.02	7.68	10.60	2.67	2.72	-9.55
154	-0.43	9.26	-0.04	7.60	10.54	2.54	-2.97	14.19
155	0.07	9.38	-0.03	7.27	10.46	2.65	2.90	14.79
156	-0.81	9.37	-0.05	7.54	10.21	2.54	10.72	0.67
157	-1.46	9.47	-0.05	7.26	10.23	2.60	-7.04	-39.70
158	-5.91	8.46	-0.04	7.27	9.78	2.38	0.29	9.16
159	-3.15	8.99	-0.05	7.41	10.58	2.70	-10.17	2.93

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
160	-183.25	2.14	5.62	-25.74	3.13	10.32	3.56	11.78
161	-200.77	1.99	5.47	-27.03	3.73	10.43	3.14	12.40
162	-189.77	1.88	5.81	-25.08	3.29	10.28	2.50	12.23
163	-180.47	1.92	5.11	-24.66	3.56	9.96	3.94	11.75
164	-190.80	2.23	5.54	-24.51	1.97	10.54	4.33	11.66
165	-202.82	2.32	5.41	-25.31	2.83	10.24	3.98	11.72
166	-191.96	2.42	5.77	-26.22	4.19	10.76	2.92	12.35
167	-202.76	2.17	6.14	-25.94	2.82	10.63	3.41	11.44
168	-200.75	1.98	4.80	-24.85	2.05	10.13	4.60	11.72
169	-203.33	2.07	4.43	-25.07	3.21	10.43	4.22	11.70
170	-210.89	2.59	5.25	-26.95	2.38	10.55	4.53	11.48
171	-182.98	1.78	5.53	-26.40	2.18	10.46	3.14	12.03
172	-198.69	1.99	5.10	-24.36	2.57	10.33	3.05	11.84
173	-188.40	1.90	5.59	-24.79	3.64	10.55	3.01	11.43
174	-214.04	2.14	5.54	-24.92	2.11	10.22	2.96	11.82
175	-184.63	2.21	5.32	-25.17	2.94	10.15	4.37	12.25
176	-200.57	2.15	5.04	-22.81	1.93	9.60	4.05	11.62
177	-192.89	1.82	5.99	-23.73	1.29	9.49	3.28	11.57
178	-203.63	1.64	5.83	-20.94	-0.29	9.36	3.67	12.18
179	-188.18	2.06	5.25	-23.68	1.11	9.95	5.50	11.80
180	-209.99	1.94	5.75	-22.03	1.03	9.56	4.07	11.49
181	-186.37	1.67	5.88	-21.16	0.82	9.68	4.08	11.39
182	-179.41	2.00	4.79	-22.47	1.65	9.46	3.93	11.91
183	-205.64	1.85	6.10	-21.08	0.69	9.89	2.51	12.14
184	-191.97	1.98	5.27	-22.05	0.35	9.36	3.29	11.41
185	-189.61	1.88	5.68	-22.09	1.18	9.34	3.93	11.81
186	-194.95	2.16	5.47	-23.46	2.22	9.99	4.03	11.41
187	-199.69	1.98	5.83	-21.48	0.05	9.70	3.21	11.41
188	-186.98	1.60	5.00	-23.36	1.17	9.80	2.65	11.23
189	-194.99	1.63	5.53	-23.27	1.00	9.60	4.56	12.00
190	-195.44	1.23	5.91	-21.99	1.57	9.26	3.87	12.39
191	-197.60	1.64	5.88	-21.36	0.88	9.76	2.72	10.91
192	-80.99	1.13	5.78	-41.96	-7.61	14.12	4.56	-2.34

193	-89.68	0.98	5.35	-40.67	-7.33	14.32	4.86	-3.09
194	-87.61	1.04	5.68	-40.81	-7.47	14.54	5.72	-3.11
195	-75.78	0.82	5.89	-41.56	-7.69	14.38	4.84	-3.28
196	-81.87	1.14	5.62	-39.90	-7.61	14.36	5.82	-3.01
197	-86.38	1.03	5.52	-40.84	-7.57	14.22	4.74	-2.69
198	-90.67	1.07	5.62	-40.01	-7.85	14.09	4.38	-2.72
199	-85.90	0.90	5.82	-39.91	-7.87	14.58	4.61	-3.11

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
160	93.69	50.35	-0.06	43.76	-7.96	-11.45	-1.65	-39.83
161	94.19	51.22	0.05	41.97	-8.07	-11.77	-1.83	-35.89
162	96.08	50.24	-0.01	44.07	-8.94	-11.94	-1.74	-33.20
163	91.01	50.82	-0.02	41.62	-9.97	-11.61	-1.79	-35.37
164	94.91	51.64	-0.03	44.52	-10.47	-11.14	-1.66	-38.97
165	99.80	49.04	-0.02	49.32	-10.67	-11.51	-1.78	-35.32
166	100.32	49.55	0.01	48.96	-8.32	-11.16	-1.87	-35.89
167	91.74	50.26	-0.03	47.65	-8.62	-11.54	-1.66	-34.51
168	99.23	48.64	-0.05	42.04	-8.21	-11.68	-1.67	-34.09
169	91.16	51.29	-0.01	48.34	-11.59	-11.22	-1.69	-38.04
170	84.86	50.02	0.02	47.47	-8.47	-12.16	-1.65	-39.32
171	104.41	49.99	-0.04	46.22	-7.03	-11.70	-1.61	-36.51
172	95.70	51.92	0.01	43.17	-11.26	-11.33	-1.77	-42.20
173	99.92	49.58	0.00	46.70	-8.98	-12.03	-1.72	-36.09
174	92.96	50.20	-0.05	47.19	-10.20	-11.95	-1.65	-39.79
175	87.99	50.84	-0.06	47.82	-10.64	-11.26	-1.78	-40.49
176	92.57	45.83	-0.03	46.46	-3.54	-10.29	-1.66	-40.48
177	103.07	45.08	-0.08	46.56	-6.48	-10.76	-1.59	-34.80
178	89.56	45.20	-0.04	45.63	-3.64	-10.46	-1.56	-43.17
179	100.68	43.71	-0.03	49.09	-2.93	-10.14	-1.61	-40.29
180	105.61	44.15	-0.06	48.19	-3.47	-9.72	-1.61	-40.95
181	91.34	45.93	-0.02	50.35	-3.56	-10.55	-1.62	-40.77
182	87.76	44.55	0.00	46.54	-7.49	-10.17	-1.51	-43.01
183	97.67	44.55	-0.04	49.44	-4.48	-10.41	-1.59	-36.79
184	93.30	44.01	-0.02	47.30	-2.87	-10.32	-1.78	-37.03
185	101.12	47.15	0.00	45.58	-6.35	-9.75	-1.68	-42.95
186	92.30	43.85	-0.03	49.68	-5.53	-10.36	-1.73	-39.98
187	90.84	43.94	-0.02	44.30	-1.86	-10.72	-1.55	-37.11
188	101.37	42.91	-0.04	48.77	-4.67	-9.64	-1.62	-45.48
189	95.02	44.73	-0.04	45.22	-2.35	-10.64	-1.65	-39.73
190	93.66	43.89	-0.01	45.99	-6.29	-9.88	-1.58	-39.34
191	103.16	44.83	-0.01	47.51	-5.19	-10.19	-1.53	-36.73
192	59.69	57.64	-0.04	16.47	24.54	-9.22	2.64	-37.59
193	69.07	58.35	-0.09	18.14	26.55	-9.27	2.42	-38.58
194	64.30	59.42	-0.09	19.73	26.26	-9.01	2.47	-37.31
195	64.31	57.75	-0.08	18.64	26.38	-9.27	2.49	-39.18
196	66.35	57.50	-0.09	17.61	26.14	-9.63	2.47	-34.66
197	70.49	57.92	-0.09	19.08	25.79	-9.25	2.47	-36.50
198	67.45	59.20	-0.07	19.74	26.43	-9.27	2.38	-38.96
199	69.13	57.68	-0.11	19.25	26.73	-9.21	2.41	-38.46

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
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160	0.05	-6.82	-24.67	-24.74	-2.61	-2.86	-2.71	-1.60
161	0.00	-6.42	-24.61	-59.26	-2.55	-2.53	6.00	-1.60
162	-0.30	-6.70	-24.98	8.36	-2.47	-2.78	20.37	-1.59
163	-0.11	-5.77	-24.59	50.59	-2.62	-2.67	6.97	-1.61
164	0.19	-6.04	-24.78	-12.92	-2.65	-2.69	-11.54	-1.60
165	-0.02	-5.73	-24.78	-25.13	-2.64	-2.84	8.52	-1.63
166	-0.06	-7.15	-24.72	10.36	-2.90	-2.60	11.33	-1.60
167	0.05	-5.02	-24.62	-18.12	-2.65	-2.70	-9.13	-1.60
168	-0.18	-5.74	-24.65	-43.31	-2.61	-2.74	2.60	-1.63
169	0.05	-3.92	-24.69	-0.11	-2.62	-2.81	-0.57	-1.59
170	-0.22	-6.42	-24.74	31.59	-2.54	-2.42	4.93	-1.60
171	0.38	-6.29	-24.79	17.32	-2.47	-2.80	3.16	-1.58
172	-0.20	-4.47	-24.71	-76.32	-2.72	-2.81	14.78	-1.58
173	-0.02	-3.41	-24.55	12.62	-2.59	-2.77	13.40	-1.61
174	-0.19	-6.36	-24.80	-30.09	-2.58	-2.56	11.87	-1.60
175	0.01	-5.58	-24.80	-0.33	-2.57	-2.98	8.06	-1.59
176	-0.21	-7.35	-22.87	-2.52	-2.79	-3.04	-3.04	-1.43
177	0.42	-6.96	-22.77	-14.25	-2.87	-2.93	-3.93	-1.45
178	-0.02	-6.32	-22.67	6.85	-2.92	-3.19	-7.99	-1.45
179	-0.09	-6.74	-22.78	-7.19	-2.85	-2.83	8.51	-1.45
180	0.17	-6.39	-22.66	56.84	-2.89	-2.75	1.92	-1.46
181	0.19	-5.99	-22.80	8.81	-2.85	-2.82	-0.75	-1.45
182	0.14	-5.99	-22.76	24.31	-2.86	-2.90	-4.31	-1.46
183	-0.08	-7.56	-22.53	13.39	-2.79	-2.74	16.60	-1.45
184	-0.11	-6.77	-22.95	19.90	-3.09	-2.74	1.85	-1.47
185	0.08	-6.63	-22.79	28.47	-2.86	-3.04	-8.57	-1.44
186	-0.04	-6.29	-22.94	53.97	-2.89	-3.03	-0.93	-1.44
187	-0.14	-6.33	-22.59	99.70	-3.14	-2.75	-4.33	-1.43
188	0.00	-7.06	-22.71	-0.85	-2.92	-2.89	0.54	-1.45
189	-0.07	-8.13	-22.59	-18.23	-2.91	-3.01	-8.11	-1.44
190	-0.17	-7.08	-22.39	23.48	-2.85	-2.87	-3.17	-1.46
191	-0.06	-7.27	-22.67	41.89	-2.92	-2.76	-3.93	-1.46
192	-0.08	-4.09	-16.51	-9.54	-3.59	-2.93	-0.63	-1.16
193	-0.14	-3.11	-16.34	18.48	-3.60	-3.00	1.15	-1.14
194	0.04	-3.51	-16.64	17.91	-3.51	-2.91	1.10	-1.13
195	-0.17	-3.94	-16.52	15.91	-3.51	-2.89	-1.44	-1.13
196	0.08	-3.88	-16.58	-20.06	-3.47	-2.90	5.56	-1.13
197	0.09	-3.67	-16.49	-16.57	-3.57	-2.88	3.05	-1.13
198	0.05	-3.94	-16.42	-16.93	-3.42	-2.97	1.59	-1.11
199	0.02	-4.02	-16.51	-25.71	-3.60	-2.90	2.24	-1.14

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
160	-1.72	-4.57	-3.31	-5.64	1.91	1.82	6.79	-74.96
161	2.72	-4.07	0.55	-5.64	1.76	1.89	6.87	-73.97
162	-4.32	-4.19	-2.77	-5.46	2.01	1.92	7.05	-73.97
163	-1.39	-4.01	-0.95	-5.48	2.46	1.69	6.74	-74.71
164	-0.61	-4.33	-1.97	-5.78	2.93	1.84	6.85	-75.67
165	2.85	-4.36	-3.46	-5.41	2.81	1.98	6.35	-75.37
166	-1.58	-3.98	-1.06	-5.35	1.94	1.90	6.59	-74.94
167	-0.38	-5.08	-3.13	-5.36	2.77	1.81	6.39	-75.00
168	2.47	-3.87	-0.88	-5.81	2.49	1.71	7.18	-75.15
169	-2.51	-4.45	-6.66	-4.91	1.83	2.03	6.90	-76.07
170	6.57	-3.95	-4.60	-5.75	1.82	2.01	6.96	-73.68

171	2.24	-3.92	-1.15	-5.47	2.71	1.61	6.63	-73.51
172	3.40	-4.54	-2.52	-5.54	2.40	2.00	6.80	-73.48
173	-0.75	-4.31	2.15	-5.78	2.35	1.56	6.88	-74.65
174	3.78	-4.76	-3.69	-5.35	1.90	1.99	7.05	-75.07
175	2.70	-4.13	-3.26	-5.61	1.83	1.74	6.62	-75.22
176	-3.40	-4.70	-1.81	-4.85	1.49	1.10	5.28	-69.91
177	3.89	-4.52	-2.20	-5.42	1.47	1.05	5.44	-68.29
178	-0.89	-4.57	-0.54	-5.06	1.72	1.23	5.32	-68.69
179	0.34	-4.88	-0.94	-5.10	1.60	1.22	5.19	-67.29
180	2.91	-4.43	-2.20	-5.38	1.18	1.28	5.03	-68.04
181	0.20	-4.51	0.35	-5.67	1.11	1.13	5.20	-69.65
182	-5.24	-4.31	-1.03	-5.08	1.14	1.21	4.73	-67.72
183	2.46	-4.05	0.58	-5.67	0.91	1.03	5.01	-68.14
184	-1.50	-4.78	-1.48	-5.34	1.04	1.23	5.58	-68.25
185	0.24	-5.14	0.73	-4.87	1.21	1.05	5.17	-68.14
186	-0.78	-4.30	-4.96	-5.25	1.03	1.33	5.06	-68.46
187	1.72	-5.38	0.77	-4.80	1.35	0.99	5.47	-67.65
188	-0.43	-5.19	-0.03	-5.43	1.03	1.19	5.13	-71.14
189	4.51	-4.64	-3.26	-4.93	0.78	0.95	5.52	-68.34
190	0.38	-4.65	0.48	-4.86	1.19	1.08	5.19	-69.56
191	-3.32	-4.37	4.83	-5.59	1.22	1.00	5.24	-71.35
192	-1.10	-6.19	10.44	-5.27	-10.64	1.78	8.50	-46.49
193	-0.22	-5.99	10.70	-5.10	-10.32	1.84	8.95	-46.44
194	1.19	-6.00	10.41	-5.23	-9.90	1.72	8.65	-45.75
195	-0.99	-6.04	11.34	-5.36	-10.16	1.78	8.59	-44.99
196	-1.78	-5.87	9.88	-5.37	-10.11	1.82	8.86	-46.13
197	0.12	-6.17	11.67	-5.32	-10.56	1.82	8.33	-46.32
198	0.29	-6.03	10.23	-5.05	-10.16	1.82	8.42	-44.92
199	-2.02	-5.95	9.82	-5.22	-9.75	1.57	8.69	-45.94

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
160	-7.10	3.84	2.29	7.25	-1.41	-42.89	-2.47	-2.31
161	-7.16	4.00	2.40	6.79	-1.29	-44.10	-2.43	-2.37
162	-6.88	3.82	2.18	6.67	-1.35	-44.13	-2.49	-2.35
163	-7.18	3.48	2.31	6.99	-1.38	-42.27	-2.53	-2.32
164	-6.67	3.67	2.34	6.50	-1.34	-42.33	-2.46	-2.34
165	-8.06	3.80	2.15	6.43	-1.41	-42.56	-2.48	-2.36
166	-6.97	3.84	2.07	6.30	-1.37	-41.59	-2.35	-2.30
167	-7.50	3.63	2.27	8.27	-1.29	-43.63	-2.40	-2.33
168	-7.23	3.84	2.14	7.07	-1.36	-42.42	-2.44	-2.32
169	-6.82	3.72	2.20	6.12	-1.31	-43.38	-2.53	-2.33
170	-6.81	3.71	2.21	7.09	-1.38	-41.57	-2.45	-2.37
171	-6.97	3.74	2.16	7.13	-1.35	-42.14	-2.49	-2.32
172	-6.01	3.77	2.13	7.48	-1.41	-42.50	-2.42	-2.30
173	-6.58	3.62	2.20	6.64	-1.30	-41.40	-2.46	-2.35
174	-6.09	3.80	2.19	6.27	-1.46	-42.84	-2.53	-2.33
175	-7.31	3.70	2.41	6.68	-1.30	-41.21	-2.56	-2.35
176	-6.83	3.47	2.53	4.38	-1.11	-39.26	-2.01	-2.17
177	-7.14	3.45	2.16	4.46	-1.08	-38.15	-2.01	-2.17
178	-6.84	3.45	2.44	4.82	-1.20	-37.87	-1.96	-2.14
179	-7.17	3.28	2.32	4.76	-1.15	-38.48	-2.04	-2.17
180	-6.73	3.38	2.39	3.85	-1.07	-36.51	-1.99	-2.13
181	-6.45	3.47	2.29	5.05	-0.93	-37.26	-1.92	-2.14

182	-6.41	3.52	2.16	4.22	-1.01	-40.58	-2.04	-2.16
183	-6.37	3.34	2.26	4.47	-1.11	-38.40	-1.98	-2.14
184	-6.07	3.42	2.22	5.10	-1.07	-38.98	-1.95	-2.18
185	-6.69	3.57	2.23	4.02	-1.03	-37.01	-2.01	-2.19
186	-6.10	3.26	2.21	4.04	-1.15	-37.83	-1.92	-2.15
187	-6.41	3.36	2.25	4.21	-1.12	-41.20	-2.00	-2.14
188	-6.48	3.30	2.49	4.27	-1.03	-39.02	-1.98	-2.14
189	-6.73	3.32	2.34	4.37	-1.12	-37.91	-1.96	-2.18
190	-6.61	3.66	2.16	4.43	-1.14	-37.43	-1.86	-2.15
191	-7.03	3.45	2.56	4.75	-1.13	-38.65	-1.94	-2.17
192	14.34	3.23	0.52	-6.89	-0.44	-54.93	-0.40	-0.38
193	15.28	3.27	0.54	-6.69	-0.47	-54.20	-0.34	-0.37
194	15.25	3.25	0.65	-6.55	-0.50	-55.28	-0.39	-0.36
195	15.22	3.25	0.61	-6.99	-0.47	-54.89	-0.36	-0.38
196	15.23	3.32	0.62	-6.76	-0.44	-55.82	-0.36	-0.31
197	14.98	3.30	0.73	-7.01	-0.45	-54.51	-0.36	-0.36
198	14.34	3.25	0.68	-7.27	-0.47	-54.67	-0.36	-0.37
199	14.44	3.17	0.51	-6.82	-0.48	-54.34	-0.37	-0.36

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
160	0.37	6.16	-4.89	-3.95	-0.18	-5.80	216.68	-4.24
161	0.44	6.30	-5.02	-1.35	9.89	-6.94	215.70	0.94
162	0.47	5.80	-2.58	-2.61	8.27	-5.37	202.79	-3.53
163	0.41	6.05	2.06	-4.03	5.60	-6.26	205.63	-4.57
164	0.40	5.86	4.56	-3.05	-1.66	-6.13	201.34	-2.44
165	0.43	6.14	-4.27	-1.20	-3.83	-6.40	219.57	-2.51
166	0.35	6.13	-2.92	-2.09	8.70	-6.09	207.33	1.18
167	0.39	5.87	-2.87	-4.76	8.99	-7.35	200.76	5.54
168	0.38	6.11	2.58	-1.78	-3.50	-5.84	212.19	-5.32
169	0.41	5.65	-0.03	-1.87	9.47	-6.14	212.81	2.41
170	0.44	5.74	-2.44	-1.68	6.05	-6.48	202.50	3.86
171	0.45	6.48	-1.42	-1.59	4.23	-6.51	218.85	-1.59
172	0.38	6.75	2.44	-0.51	1.69	-6.58	204.05	-0.83
173	0.46	5.72	-0.38	-0.25	5.15	-5.50	191.42	0.98
174	0.45	6.33	3.98	-2.61	-5.38	-5.59	205.05	-3.88
175	0.40	5.97	6.85	-2.28	7.55	-6.99	200.65	-0.70
176	0.45	7.06	-0.46	-1.72	1.09	-6.70	193.26	3.23
177	0.44	6.59	0.91	-1.08	-0.45	-7.42	197.52	2.04
178	0.46	7.05	-2.54	-0.60	1.25	-6.31	187.34	0.15
179	0.47	6.80	-3.35	-0.78	2.85	-7.77	196.66	2.32
180	0.45	6.54	-0.56	-2.66	5.28	-7.34	204.22	-2.50
181	0.49	6.72	2.50	-0.06	5.80	-6.91	198.68	-4.05
182	0.51	7.05	-4.15	-2.54	-1.36	-7.48	201.35	0.50
183	0.45	7.01	-1.19	-0.48	3.29	-8.19	205.06	-2.27
184	0.50	6.51	1.42	-1.52	2.04	-7.31	184.98	5.86
185	0.46	6.88	0.90	-0.33	1.45	-7.19	200.08	-1.33
186	0.41	6.38	-0.92	0.12	-2.64	-8.02	193.03	4.59
187	0.48	7.11	-1.69	-1.28	-2.29	-7.25	195.79	-4.40
188	0.44	6.54	3.58	-0.63	5.67	-6.94	198.26	-5.30
189	0.42	6.39	-4.58	-1.29	7.96	-7.29	198.65	-3.19
190	0.46	6.64	-4.68	-1.11	-4.70	-7.46	196.34	3.04
191	0.45	6.81	2.51	-2.38	0.31	-7.21	195.56	1.44
192	0.51	7.88	-0.86	5.86	-1.23	-9.89	31.77	-1.15

193	0.49	7.70	0.24	5.63	-4.70	-9.82	32.51	-0.43
194	0.54	7.71	-0.61	5.51	-5.12	-9.75	32.83	1.99
195	0.49	7.83	0.62	5.83	2.58	-9.67	33.03	0.79
196	0.52	7.78	-1.62	5.24	-1.38	-9.49	37.38	1.85
197	0.54	7.94	0.26	4.88	0.93	-9.66	28.25	-0.89
198	0.48	7.63	-0.16	6.08	1.97	-9.63	26.78	-1.78
199	0.50	7.86	-1.26	5.43	1.97	-9.74	29.21	-1.97

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
160	-15.09	1.43	8.38	2.09	67.10	50.59	-4.05	-2.61
161	-15.99	1.74	8.50	-0.09	74.46	49.89	-3.80	-2.33
162	-15.23	1.34	8.48	-1.80	66.68	50.94	-3.90	-2.70
163	-14.71	1.41	6.35	-3.76	65.56	52.11	-3.89	-3.17
164	-15.64	1.35	8.94	-11.87	69.07	50.02	-3.94	-2.36
165	-15.55	1.51	9.37	4.37	68.67	53.08	-3.92	-2.80
166	-15.40	1.37	8.38	2.83	69.57	52.07	-3.76	-1.98
167	-15.25	1.65	7.20	-1.24	69.80	49.58	-4.03	-3.15
168	-15.21	1.29	7.64	-5.76	65.67	51.80	-3.99	-4.23
169	-15.65	1.44	7.37	0.61	67.24	51.04	-3.94	-3.61
170	-15.53	1.54	7.44	-3.07	72.23	52.70	-4.07	-2.45
171	-14.94	1.54	9.01	-0.48	66.04	51.64	-4.11	-2.67
172	-15.34	1.52	8.43	2.06	71.91	50.70	-4.02	-2.48
173	-14.78	1.57	7.45	-2.85	62.87	50.65	-3.87	-1.81
174	-15.51	1.51	9.59	0.14	65.71	49.51	-4.02	-1.82
175	-15.48	1.26	6.97	7.15	69.96	53.52	-3.95	-3.26
176	-12.35	1.49	9.88	-9.25	57.89	46.03	-3.59	-1.89
177	-12.31	1.61	9.42	-3.73	67.80	45.21	-3.64	-2.45
178	-12.14	1.49	10.23	-0.63	58.49	45.33	-3.73	-2.72
179	-12.13	1.61	10.38	0.24	55.05	46.72	-3.63	-2.00
180	-11.85	1.51	9.89	-9.35	58.31	44.55	-3.63	-3.61
181	-11.39	1.68	10.30	-3.55	55.82	44.55	-3.69	-0.74
182	-11.71	1.58	9.81	-3.65	60.66	45.97	-3.65	-2.52
183	-11.84	1.62	7.87	-3.91	62.48	45.42	-3.52	-2.08
184	-12.30	1.57	9.70	8.09	55.78	46.36	-3.52	-2.01
185	-11.78	1.63	8.33	-0.93	55.84	45.39	-3.58	-2.29
186	-11.46	1.71	9.94	-4.37	60.19	44.13	-3.66	-3.12
187	-12.17	1.61	8.29	-5.53	58.65	44.83	-3.78	-2.80
188	-11.84	1.68	7.48	-2.86	65.02	45.54	-3.61	-2.11
189	-11.77	1.64	9.14	2.22	55.82	44.24	-3.61	-2.37
190	-12.38	1.71	7.76	-4.37	63.61	45.04	-3.67	-2.30
191	-11.86	1.57	8.30	0.81	62.45	45.01	-3.51	-2.05
192	-14.53	1.99	13.18	1.33	106.76	38.07	-5.34	-5.15
193	-13.82	2.08	13.41	1.91	104.39	37.64	-5.16	-4.44
194	-14.37	2.15	13.46	-0.20	110.70	38.01	-5.34	-5.04
195	-13.91	2.12	13.46	0.42	102.96	38.66	-5.33	-4.93
196	-14.24	2.07	14.12	2.08	97.99	38.20	-5.46	-5.31
197	-14.21	2.13	13.13	-1.64	100.89	37.80	-5.44	-4.59
198	-14.39	2.09	13.96	-2.45	103.11	38.34	-5.33	-5.41
199	-13.74	2.08	13.50	-2.52	105.29	38.24	-5.41	-4.78

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
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160	4.80	-3.26	-0.21	3.57	2.42	-0.68	-12.67	-24.28
161	-3.84	-4.72	-0.18	2.94	2.29	-0.37	1.69	35.79
162	2.38	-4.79	-0.18	3.21	2.09	-0.33	4.88	-13.73
163	-0.25	-2.02	-0.17	2.96	2.59	-0.43	27.32	49.32
164	3.73	-5.32	-0.15	3.77	2.77	-0.50	-9.12	57.33
165	0.28	-2.82	-0.17	3.31	2.63	-0.50	22.97	31.74
166	-4.59	-3.20	-0.18	3.25	3.45	-0.72	-15.61	33.03
167	-1.01	-4.60	-0.17	4.16	2.50	-0.33	-27.43	-11.64
168	-4.66	-1.98	-0.19	4.12	2.54	-0.22	-1.92	-10.90
169	3.96	-7.01	-0.20	2.86	2.56	-0.62	17.28	-12.73
170	2.95	-4.19	-0.21	3.43	2.63	-0.19	-14.50	-1.16
171	-7.27	-3.21	-0.20	3.49	2.62	-0.49	1.07	-12.08
172	7.69	-6.61	-0.16	2.74	2.46	-0.33	12.29	-21.91
173	0.38	-3.77	-0.20	3.10	2.67	-0.35	-8.43	25.41
174	1.04	-2.66	-0.18	3.00	2.98	-0.48	-6.63	-62.55
175	1.60	-1.83	-0.18	3.19	2.77	-0.62	-10.72	-93.17
176	-0.52	-7.31	-0.07	2.13	2.59	-0.40	-6.00	14.91
177	-0.77	-6.06	-0.12	1.95	3.14	-0.16	-9.91	-11.31
178	-1.38	-6.04	-0.11	1.95	2.90	-0.10	-14.84	-5.31
179	3.62	-5.16	-0.12	2.88	2.42	-0.29	-2.65	24.60
180	-0.82	-5.22	-0.08	2.50	2.59	-0.01	-9.33	68.11
181	1.07	-4.69	-0.09	1.89	2.66	-0.23	5.36	72.62
182	1.54	-7.98	-0.10	2.35	2.91	-0.19	-19.95	-12.65
183	0.38	-6.18	-0.10	2.44	2.23	-0.51	30.27	7.42
184	-1.62	-7.10	-0.09	1.94	2.99	-0.47	-3.11	-34.92
185	2.42	-5.25	-0.08	2.10	2.45	-0.43	3.54	3.01
186	-5.80	-6.76	-0.10	1.44	2.25	-0.49	-21.05	17.21
187	7.44	-5.48	-0.12	2.46	2.44	-0.27	23.77	30.52
188	-5.44	-6.90	-0.11	2.41	2.25	-0.51	10.61	-0.47
189	0.88	-6.94	-0.09	2.24	2.12	-0.02	20.18	-19.77
190	8.28	-6.66	-0.10	2.54	2.34	-0.33	21.36	57.54
191	-3.19	-5.28	-0.09	2.05	2.63	-0.25	7.47	31.17
192	5.63	-31.18	0.32	-1.71	12.41	0.20	-0.23	7.04
193	2.70	-30.14	0.28	-2.06	12.10	0.16	-29.14	20.81
194	-1.45	-29.85	0.28	-1.81	12.41	0.08	17.53	16.41
195	-2.76	-29.39	0.31	-1.97	12.28	0.17	-0.96	15.02
196	-3.77	-29.74	0.29	-2.19	12.65	0.09	-9.79	31.38
197	-3.05	-29.86	0.30	-1.86	12.48	0.13	-18.07	9.58
198	-2.17	-30.42	0.30	-1.83	12.21	0.18	-5.19	14.38
199	-2.93	-30.44	0.29	-1.97	12.62	0.04	16.43	-0.26

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
200	-84.84	1.13	5.88	-40.53	-7.81	13.76	5.16	-2.61
201	-86.46	1.17	5.92	-41.73	-7.49	14.37	4.96	-2.65
202	-85.65	0.92	5.61	-41.84	-7.73	14.39	4.88	-3.04
203	-83.58	0.92	5.57	-41.07	-7.79	14.00	5.78	-2.87
204	-84.03	0.90	5.59	-40.04	-8.07	15.05	5.36	-2.80
205	-88.01	1.00	5.82	-40.27	-7.37	14.65	5.64	-2.95
206	-91.14	1.09	5.67	-38.94	-7.49	13.90	5.00	-3.05
207	-87.84	1.13	5.79	-39.92	-7.62	14.06	4.61	-3.12
208	-139.32	2.88	8.38	-29.54	-7.87	11.87	0.06	1.37
209	-136.35	2.65	8.20	-29.63	-7.90	11.88	0.14	1.88
210	-136.39	2.88	8.19	-29.62	-7.91	11.54	0.09	1.74

211	-140.86	2.90	8.40	-28.91	-8.02	11.83	1.40	1.81
212	-141.28	2.64	8.02	-30.90	-8.11	11.87	0.91	1.38
213	-147.44	3.03	8.55	-29.96	-8.18	12.28	0.27	0.89
214	-137.06	2.83	8.21	-28.00	-7.95	11.69	-0.13	1.43
215	-136.69	2.81	8.31	-31.18	-8.32	12.24	-0.51	1.76
216	-144.43	3.02	8.72	-30.56	-7.94	11.59	0.02	1.81
217	-137.63	2.89	8.13	-30.66	-7.76	11.71	0.40	1.39
218	-132.91	2.90	8.74	-30.41	-8.04	11.61	0.24	2.17
219	-146.30	2.70	8.31	-32.57	-8.00	12.00	0.33	1.31
220	-141.57	2.69	8.20	-30.05	-8.02	11.79	0.65	1.73
221	-143.00	2.79	8.25	-29.94	-7.79	11.74	-0.08	1.70
222	-138.77	2.83	8.47	-28.20	-7.64	11.82	0.14	1.48
223	-140.99	2.63	8.27	-30.91	-7.93	11.74	0.18	1.55
224	-54.29	0.60	4.08	-17.30	-4.02	6.24	2.53	-0.47
225	-49.96	0.52	4.11	-17.63	-3.57	6.25	2.57	-0.65
226	-53.86	0.63	4.07	-16.88	-3.88	6.15	2.39	-0.17
227	-53.38	0.55	4.12	-16.78	-3.97	6.27	2.64	-0.44
228	-52.04	0.58	4.25	-17.54	-3.97	6.26	2.50	-0.44
229	-49.03	0.59	4.08	-16.65	-3.89	6.26	2.41	-0.68
230	-51.57	0.47	3.92	-17.80	-4.05	6.13	2.83	-0.45
231	-51.91	0.73	4.07	-16.88	-3.72	6.31	2.68	-0.76
232	-53.06	0.54	4.08	-17.42	-3.88	6.13	2.80	-0.66
233	-55.52	0.57	4.00	-17.43	-3.79	6.01	2.67	-0.74
234	-52.68	0.62	3.76	-16.64	-3.85	6.32	2.48	-0.51
235	-57.68	0.48	4.06	-16.80	-3.86	5.94	2.56	-0.56
236	-53.28	0.60	4.17	-17.07	-3.56	6.08	2.93	-0.65
237	-50.64	0.50	4.15	-16.55	-3.98	5.93	2.86	-0.63
238	-54.90	0.57	3.85	-17.16	-3.98	6.12	2.89	-0.43
239	-54.82	0.69	3.85	-16.27	-3.91	6.14	2.75	-0.44

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
200	66.06	57.96	-0.06	20.15	25.87	-9.12	2.49	-40.56
201	68.69	59.17	-0.05	18.98	26.20	-9.42	2.57	-38.06
202	65.93	58.25	-0.06	17.31	26.06	-9.06	2.45	-37.70
203	66.02	57.57	-0.06	17.54	24.57	-9.56	2.48	-37.48
204	62.62	60.56	-0.05	18.59	26.34	-9.47	2.50	-37.12
205	69.42	59.25	-0.07	18.44	25.76	-9.14	2.51	-38.09
206	69.43	58.91	-0.08	17.90	25.60	-9.19	2.56	-38.37
207	66.76	59.12	-0.06	17.58	23.78	-9.08	2.57	-36.53
208	105.25	45.16	0.14	30.65	17.19	-11.84	1.41	-56.05
209	100.97	49.11	0.17	33.24	19.20	-11.96	1.36	-54.82
210	110.23	47.17	0.14	32.19	18.96	-11.95	1.33	-57.81
211	99.46	47.41	0.17	34.11	19.23	-11.51	1.40	-58.91
212	100.89	47.64	0.20	32.97	18.91	-11.64	1.29	-55.22
213	105.89	47.15	0.17	34.74	19.38	-11.71	1.35	-54.68
214	100.66	48.39	0.12	33.05	18.56	-11.70	1.47	-55.39
215	105.40	48.05	0.14	35.47	19.87	-11.73	1.47	-56.98
216	107.76	46.39	0.15	32.78	18.55	-11.55	1.40	-56.60
217	102.67	46.58	0.16	34.43	19.08	-11.72	1.35	-56.71
218	106.31	45.50	0.13	35.30	18.10	-11.87	1.42	-56.77
219	103.64	46.85	0.14	34.19	18.24	-11.49	1.25	-56.12
220	105.64	46.90	0.17	34.57	17.92	-11.69	1.39	-57.90
221	104.76	45.29	0.14	35.97	21.36	-11.91	1.36	-56.05

222	103.79	46.27	0.14	33.43	19.41	-11.53	1.41	-56.73
223	101.88	45.23	0.17	34.42	19.64	-12.02	1.45	-53.76
224	44.53	25.82	-0.06	11.68	10.35	-5.06	0.95	-25.85
225	45.43	25.95	-0.04	11.59	10.33	-4.95	1.01	-27.25
226	42.76	25.29	-0.04	11.55	10.68	-4.99	1.00	-26.64
227	44.83	25.97	-0.05	12.62	10.83	-5.07	0.98	-26.88
228	43.61	25.36	-0.03	12.76	10.30	-4.92	1.10	-27.08
229	46.60	25.62	-0.05	11.49	10.21	-5.09	1.00	-26.17
230	43.98	25.50	-0.06	11.29	10.13	-5.14	0.85	-26.86
231	43.86	25.36	-0.04	11.30	10.41	-4.94	1.01	-25.41
232	46.44	25.68	-0.05	10.66	10.53	-5.00	0.95	-27.32
233	47.20	25.26	-0.02	10.75	11.18	-5.01	0.97	-26.35
234	45.19	26.43	-0.02	11.44	10.41	-5.05	0.94	-25.67
235	46.18	25.03	-0.03	11.83	10.34	-4.91	1.00	-26.54
236	45.28	25.60	-0.04	11.79	10.74	-5.00	0.99	-27.22
237	45.71	26.50	-0.06	11.87	10.86	-5.00	0.99	-28.24
238	44.83	25.30	-0.04	11.05	10.70	-5.05	0.95	-26.50
239	45.99	25.08	-0.06	11.38	10.68	-4.93	0.88	-26.89

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
200	-0.14	-3.90	-16.42	18.94	-3.52	-2.90	2.97	-1.12
201	-0.01	-3.17	-16.60	20.24	-3.57	-2.91	4.30	-1.14
202	0.14	-3.90	-16.67	6.48	-3.59	-2.84	-0.10	-1.11
203	-0.24	-3.89	-16.69	4.00	-3.46	-3.00	-2.50	-1.12
204	-0.10	-3.70	-16.54	0.10	-3.60	-2.94	-1.46	-1.13
205	-0.10	-3.49	-16.60	1.65	-3.53	-2.91	1.37	-1.12
206	0.11	-3.34	-16.64	5.16	-3.55	-2.70	-1.01	-1.15
207	0.02	-4.27	-16.57	-0.97	-3.50	-2.94	9.24	-1.12
208	0.27	-7.69	-16.18	9.78	-3.44	-1.72	2.71	-1.02
209	-0.04	-7.39	-15.94	-9.32	-3.51	-1.63	1.56	-1.04
210	-0.07	-7.15	-16.15	14.88	-3.47	-1.77	-0.45	-1.06
211	0.08	-7.65	-16.49	5.04	-3.52	-1.64	-0.63	-1.05
212	-0.03	-7.56	-16.13	-45.52	-3.43	-1.85	-3.97	-1.04
213	0.07	-6.72	-16.27	-34.23	-3.53	-1.73	-1.11	-1.03
214	0.17	-7.68	-16.15	2.33	-3.41	-1.95	2.65	-1.04
215	0.23	-7.81	-16.12	8.35	-3.41	-1.63	-1.63	-1.05
216	-0.06	-7.14	-16.07	-6.69	-3.47	-1.52	1.36	-1.04
217	-0.18	-7.44	-16.05	-10.69	-3.52	-1.66	4.16	-1.04
218	-0.16	-7.64	-16.02	4.38	-3.55	-1.58	-0.43	-1.03
219	-0.09	-7.83	-16.27	51.83	-3.50	-1.72	2.93	-1.05
220	-0.08	-7.89	-16.36	-26.86	-3.50	-1.95	1.99	-1.04
221	-0.24	-7.62	-16.20	-13.42	-3.47	-1.61	-0.73	-1.07
222	0.22	-7.37	-16.13	-6.98	-3.48	-1.67	5.77	-1.05
223	-0.05	-8.48	-16.17	7.62	-3.47	-1.88	1.22	-1.02
224	0.08	-2.72	-7.93	-0.84	-2.21	-1.51	1.68	-0.53
225	0.06	-2.53	-7.92	0.54	-2.21	-1.54	1.26	-0.55
226	-0.03	-2.53	-8.02	3.16	-2.19	-1.67	1.70	-0.53
227	-0.03	-2.64	-7.78	3.03	-2.22	-1.61	-2.87	-0.53
228	-0.07	-2.56	-7.82	-6.44	-2.22	-1.52	-0.46	-0.53
229	0.06	-2.67	-7.95	7.79	-2.17	-1.65	-0.30	-0.54
230	0.03	-2.67	-7.73	-7.24	-2.21	-1.63	-0.68	-0.53
231	0.08	-2.32	-7.83	-8.27	-2.18	-1.51	0.09	-0.52
232	-0.01	-2.60	-7.85	-3.83	-2.19	-1.56	-0.25	-0.54

233	0.14	-2.88	-7.84	2.24	-2.19	-1.51	0.55	-0.54
234	-0.05	-2.42	-7.90	-4.40	-2.19	-1.60	0.13	-0.54
235	-0.02	-2.55	-7.79	-11.70	-2.22	-1.55	-0.60	-0.53
236	0.19	-2.48	-7.84	12.76	-2.20	-1.54	3.63	-0.53
237	-0.02	-2.83	-7.95	2.25	-2.23	-1.51	-0.31	-0.53
238	-0.12	-2.61	-7.96	13.30	-2.23	-1.60	-0.31	-0.51
239	0.10	-2.52	-7.69	18.92	-2.22	-1.58	0.44	-0.54

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
200	3.99	-5.95	11.60	-5.28	-9.88	1.58	8.77	-45.86
201	0.55	-5.96	10.24	-5.13	-9.75	1.70	8.73	-46.07
202	-1.77	-6.09	11.31	-4.91	-9.69	1.63	8.59	-45.79
203	1.56	-5.98	10.38	-5.02	-10.51	1.79	8.69	-45.10
204	1.56	-6.08	10.34	-5.40	-10.12	1.67	8.71	-45.35
205	2.13	-6.31	10.69	-5.05	-10.29	1.70	8.93	-45.65
206	2.79	-6.28	10.42	-5.44	-10.01	1.75	8.49	-46.18
207	-0.59	-6.19	11.15	-5.23	-10.73	1.85	8.51	-46.18
208	-0.86	-7.96	-3.95	-3.60	-9.52	0.68	11.12	-49.79
209	0.81	-8.49	-5.59	-3.29	-8.62	0.65	11.07	-50.16
210	-1.82	-8.57	-4.86	-3.74	-8.55	0.75	11.01	-50.88
211	-0.18	-8.60	-5.31	-3.43	-8.14	0.70	11.15	-50.58
212	2.61	-8.70	-6.29	-3.63	-8.86	0.61	10.85	-50.54
213	-3.08	-8.69	-3.95	-3.63	-9.03	0.68	11.10	-50.64
214	-1.21	-8.67	-3.89	-3.44	-8.30	0.57	10.92	-50.48
215	-0.63	-8.21	-5.17	-3.56	-8.01	0.58	10.75	-51.30
216	-2.33	-8.45	-6.57	-3.49	-8.58	0.62	10.57	-49.36
217	-1.75	-8.63	-4.28	-3.72	-9.07	0.52	11.17	-51.38
218	-1.16	-8.33	-3.96	-3.48	-8.78	0.65	11.05	-52.15
219	-1.30	-8.25	-5.48	-3.26	-8.99	0.67	10.97	-50.97
220	-2.13	-8.82	-5.83	-3.54	-8.76	0.61	10.94	-50.33
221	0.26	-8.28	-5.24	-4.00	-8.47	0.67	10.96	-50.23
222	1.12	-8.32	-6.96	-3.18	-9.03	0.48	11.17	-52.19
223	2.79	-8.82	-5.29	-3.80	-8.83	0.67	10.90	-50.60
224	0.34	-4.36	2.59	-2.85	-4.74	0.74	5.72	-22.17
225	-0.31	-4.18	2.99	-3.02	-4.17	0.66	5.46	-22.03
226	-0.12	-4.41	2.85	-2.84	-4.73	0.61	5.85	-21.93
227	-0.33	-4.36	3.11	-2.91	-4.70	0.71	5.44	-21.77
228	0.44	-4.44	3.35	-2.94	-4.84	0.57	5.52	-22.25
229	0.33	-4.39	3.32	-2.77	-4.87	0.60	5.59	-22.24
230	-0.06	-4.13	2.39	-2.85	-4.78	0.58	5.64	-21.99
231	0.24	-4.30	4.62	-2.89	-4.63	0.62	5.66	-22.01
232	0.96	-4.38	4.07	-3.03	-4.86	0.60	5.48	-22.08
233	-0.65	-4.41	3.56	-2.99	-4.53	0.68	5.63	-21.70
234	0.36	-4.36	3.33	-2.77	-4.45	0.63	5.65	-21.98
235	-0.71	-4.32	3.30	-2.85	-4.91	0.59	5.67	-22.12
236	-0.29	-4.26	3.75	-2.72	-4.75	0.66	5.69	-22.65
237	0.22	-4.19	2.49	-2.91	-4.69	0.71	5.68	-22.50
238	-1.44	-4.36	3.56	-2.73	-5.08	0.70	5.66	-22.30
239	-0.57	-4.42	2.22	-2.74	-4.73	0.58	5.58	-22.22

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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200	14.32	3.22	0.58	-6.43	-0.50	-53.19	-0.36	-0.36
201	14.99	3.34	0.54	-6.74	-0.49	-54.35	-0.39	-0.36
202	14.82	3.28	0.68	-7.03	-0.52	-54.80	-0.41	-0.36
203	14.71	3.23	0.63	-6.98	-0.46	-54.42	-0.37	-0.36
204	14.96	3.30	0.60	-7.13	-0.47	-55.09	-0.35	-0.36
205	14.23	3.28	0.66	-6.96	-0.49	-55.95	-0.36	-0.36
206	15.18	3.28	0.69	-7.34	-0.50	-55.62	-0.37	-0.35
207	14.69	3.23	0.61	-6.88	-0.52	-53.88	-0.36	-0.35
208	9.47	3.48	1.57	-4.65	-0.96	-45.31	-1.00	-0.74
209	9.10	3.48	1.40	-3.70	-0.97	-43.24	-1.04	-0.68
210	9.37	3.51	1.54	-3.31	-0.90	-42.54	-0.99	-0.72
211	9.83	3.51	1.58	-3.74	-0.92	-42.03	-0.94	-0.72
212	9.76	3.52	1.42	-3.73	-0.93	-43.64	-0.96	-0.71
213	9.51	3.50	1.54	-4.16	-0.90	-44.19	-1.03	-0.72
214	9.23	3.38	1.59	-3.62	-0.89	-43.99	-1.02	-0.71
215	9.61	3.50	1.54	-3.80	-0.93	-44.32	-1.01	-0.74
216	8.98	3.46	1.50	-3.79	-0.98	-44.82	-0.98	-0.72
217	9.21	3.52	1.53	-3.92	-0.92	-44.28	-0.94	-0.73
218	9.94	3.55	1.52	-4.54	-0.91	-41.51	-1.02	-0.72
219	9.63	3.53	1.40	-3.79	-0.93	-43.59	-0.99	-0.73
220	9.23	3.46	1.49	-3.60	-0.89	-45.16	-0.99	-0.73
221	8.94	3.48	1.46	-3.51	-0.91	-44.60	-0.97	-0.72
222	9.57	3.47	1.57	-3.09	-0.87	-43.51	-1.02	-0.73
223	9.36	3.49	1.43	-3.84	-0.88	-43.84	-0.98	-0.72
224	6.36	1.72	0.50	-2.51	-0.38	-24.29	-0.56	-0.24
225	6.20	1.80	0.47	-1.97	-0.35	-25.26	-0.57	-0.23
226	6.18	1.80	0.46	-2.30	-0.37	-24.28	-0.51	-0.24
227	6.37	1.78	0.42	-2.20	-0.34	-23.88	-0.53	-0.23
228	6.45	1.78	0.51	-1.83	-0.37	-24.04	-0.54	-0.23
229	7.00	1.71	0.46	-1.80	-0.35	-24.35	-0.53	-0.23
230	6.17	1.71	0.53	-2.11	-0.33	-25.11	-0.55	-0.21
231	6.09	1.76	0.48	-2.36	-0.36	-23.07	-0.53	-0.23
232	5.82	1.77	0.43	-2.38	-0.37	-24.22	-0.56	-0.22
233	5.85	1.74	0.46	-2.27	-0.32	-24.46	-0.55	-0.25
234	6.48	1.79	0.52	-2.15	-0.37	-23.83	-0.56	-0.25
235	6.59	1.79	0.45	-2.00	-0.36	-24.26	-0.56	-0.23
236	6.53	1.80	0.54	-2.13	-0.35	-24.30	-0.53	-0.25
237	6.32	1.77	0.46	-2.42	-0.32	-24.11	-0.54	-0.23
238	6.42	1.76	0.48	-2.25	-0.34	-24.32	-0.51	-0.22
239	6.80	1.81	0.45	-2.12	-0.34	-25.08	-0.53	-0.24

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
200	0.46	7.85	-0.13	6.00	-0.66	-9.63	27.73	-0.71
201	0.49	7.88	1.56	5.01	5.66	-9.86	37.21	-0.38
202	0.50	7.82	-0.21	5.62	3.39	-9.78	36.01	-3.02
203	0.47	7.87	1.17	5.57	6.66	-9.76	28.22	-1.35
204	0.47	7.73	-1.49	4.73	-0.17	-9.85	34.81	-0.69
205	0.51	7.78	0.66	4.81	0.90	-9.79	27.69	-1.68
206	0.48	7.68	0.46	5.55	0.83	-9.68	33.16	0.42
207	0.49	7.68	1.72	5.43	-3.05	-9.64	30.82	2.49
208	0.15	6.36	-2.14	8.22	-2.43	-9.98	99.11	-0.05
209	0.19	6.61	0.32	8.42	3.47	-9.67	83.51	-0.45
210	0.14	6.61	-1.32	7.86	6.02	-10.02	86.68	1.50

211	0.16	6.66	2.90	7.31	0.41	-9.81	87.51	1.78
212	0.22	6.31	-1.30	7.70	-0.04	-9.87	87.42	-0.67
213	0.15	6.49	-1.79	7.85	5.42	-9.54	85.96	-0.59
214	0.22	6.46	0.30	8.60	2.92	-9.67	90.41	0.31
215	0.20	6.38	-0.86	8.71	4.58	-9.76	83.46	-0.82
216	0.19	6.15	0.25	7.76	-2.22	-9.68	91.93	-0.68
217	0.20	6.57	-1.20	8.05	-1.02	-9.91	86.11	1.29
218	0.17	6.27	-1.14	7.68	2.32	-10.04	91.11	-1.23
219	0.15	6.39	-0.80	7.44	5.12	-9.89	88.36	0.45
220	0.15	6.49	-3.00	8.07	4.47	-9.78	95.29	-0.28
221	0.19	6.40	0.75	7.85	6.16	-9.63	90.57	-0.67
222	0.18	6.41	1.30	8.33	0.11	-9.71	81.50	3.06
223	0.22	6.24	-1.01	8.04	5.86	-9.95	86.36	-0.02
224	0.27	4.44	-0.04	3.08	2.50	-6.38	20.25	-1.03
225	0.24	4.52	-0.34	3.01	-1.29	-6.25	16.20	-0.21
226	0.25	4.55	-0.39	3.51	-0.60	-6.37	18.03	-1.33
227	0.26	4.52	0.03	3.32	-1.50	-6.28	24.04	-0.27
228	0.26	4.70	-0.29	3.18	-0.44	-6.27	17.92	-1.90
229	0.25	4.57	0.92	3.34	1.93	-6.40	22.77	-0.80
230	0.28	4.73	-0.66	2.98	4.13	-6.28	15.11	-0.47
231	0.24	4.58	0.43	3.54	-0.58	-6.26	25.20	0.44
232	0.27	4.64	0.84	3.11	0.14	-6.38	22.13	-0.19
233	0.25	4.51	0.22	3.69	-2.72	-6.35	21.97	0.80
234	0.27	4.51	-0.11	3.55	1.86	-6.42	25.28	-0.06
235	0.26	4.67	1.15	3.35	0.05	-6.42	23.04	-1.52
236	0.27	4.61	-0.75	3.40	-1.85	-6.07	16.17	1.21
237	0.26	4.65	-2.07	3.25	1.95	-6.27	18.98	-0.86
238	0.22	4.73	-0.73	3.27	-0.72	-6.28	20.99	0.14
239	0.24	4.67	1.08	3.59	-0.16	-6.19	18.79	-0.29

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
200	-14.58	2.15	13.52	1.39	98.94	38.13	-5.24	-4.83
201	-14.85	2.12	13.25	-1.46	105.91	38.10	-5.40	-4.96
202	-14.77	2.04	13.42	-0.83	108.07	37.94	-5.43	-4.34
203	-14.29	2.04	13.65	1.22	104.62	38.41	-5.42	-4.76
204	-14.52	2.02	13.38	-2.48	103.69	38.64	-5.38	-5.42
205	-13.83	2.00	13.13	-1.90	100.76	38.11	-5.22	-5.16
206	-14.72	2.09	13.54	1.63	99.44	36.76	-5.42	-5.11
207	-14.24	2.13	13.38	-2.68	106.04	38.62	-5.44	-5.01
208	-18.04	2.04	17.33	-4.13	80.49	45.11	-4.33	-0.42
209	-18.35	1.98	18.32	0.33	75.08	44.55	-4.26	-0.29
210	-18.34	1.91	17.45	-5.83	78.06	45.51	-4.19	0.29
211	-17.86	2.06	18.77	-1.57	76.22	44.80	-4.37	-0.99
212	-18.15	2.07	17.42	3.49	68.79	44.77	-4.29	-0.51
213	-17.90	1.96	18.27	2.48	78.99	45.19	-4.26	-1.08
214	-18.00	2.17	16.74	1.35	81.10	45.50	-4.36	0.34
215	-18.07	2.03	18.09	-0.23	80.85	45.97	-4.42	-0.63
216	-17.82	2.01	17.98	-1.08	76.05	44.86	-4.26	0.54
217	-18.26	1.97	17.85	2.39	76.52	45.77	-4.35	-0.24
218	-18.22	1.93	17.46	-2.42	78.00	44.64	-4.34	-0.75
219	-18.12	2.02	17.20	-2.55	79.55	45.31	-4.45	-0.57
220	-18.32	2.05	18.01	-0.15	73.51	44.83	-4.23	-1.39
221	-18.65	1.85	17.38	1.55	75.38	45.01	-4.24	0.04

222	-18.24	2.01	17.33	3.21	79.54	44.83	-4.33	-0.20
223	-17.86	2.09	17.70	-2.31	78.34	45.15	-4.28	0.51
224	-9.08	1.04	9.54	-0.38	45.23	20.55	-2.26	-2.70
225	-8.99	1.11	9.50	-2.62	47.46	20.68	-2.31	-2.31
226	-8.93	1.09	9.73	-1.20	48.83	20.40	-2.28	-2.68
227	-8.58	1.14	9.31	-0.97	45.76	20.14	-2.25	-2.35
228	-8.73	1.14	9.72	1.59	49.19	20.34	-2.30	-2.72
229	-9.14	1.21	9.50	0.67	47.88	20.41	-2.25	-2.50
230	-8.91	1.09	9.32	0.39	49.39	20.28	-2.25	-2.31
231	-9.06	1.07	9.52	1.36	48.38	20.82	-2.31	-2.84
232	-8.97	1.08	9.76	0.66	47.42	20.49	-2.30	-2.82
233	-8.95	1.16	9.71	-0.24	48.65	20.33	-2.32	-2.82
234	-8.79	1.05	9.83	0.41	50.36	20.50	-2.29	-2.90
235	-8.84	1.09	9.42	0.66	45.98	20.80	-2.30	-2.95
236	-9.18	1.08	9.80	-2.83	45.60	19.75	-2.31	-2.90
237	-9.04	1.19	9.56	1.14	49.97	20.36	-2.42	-2.38
238	-9.17	1.08	9.84	2.18	49.18	19.80	-2.27	-2.61
239	-8.69	1.11	9.76	2.64	47.25	20.56	-2.31	-2.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
200	-1.36	-29.64	0.29	-1.93	12.05	0.24	6.43	22.33
201	3.10	-29.13	0.30	-1.91	12.36	0.07	-7.30	-0.91
202	-5.52	-29.28	0.29	-2.07	12.24	0.09	-9.28	8.15
203	-4.47	-29.11	0.30	-1.96	12.32	0.14	-6.20	8.77
204	-2.98	-29.56	0.32	-1.95	12.15	0.15	7.99	31.91
205	-1.35	-29.64	0.30	-1.68	12.64	0.24	-18.47	-13.38
206	1.26	-30.90	0.30	-1.91	12.30	0.25	6.12	4.06
207	2.52	-29.07	0.29	-1.96	12.47	0.22	3.27	-13.81
208	-0.08	-22.82	0.23	-3.30	8.90	-1.48	9.45	-22.79
209	-0.46	-23.73	0.20	-3.31	9.41	-1.13	3.74	-10.82
210	3.20	-22.42	0.22	-3.21	9.23	-1.46	-3.21	9.83
211	-1.51	-22.56	0.20	-3.15	8.81	-1.43	-20.84	-1.75
212	3.07	-23.26	0.20	-3.21	9.20	-1.39	0.80	-30.30
213	-2.33	-23.08	0.23	-2.69	8.98	-1.30	-19.53	8.55
214	-6.54	-23.51	0.22	-3.41	8.76	-1.14	27.04	6.01
215	-3.79	-21.32	0.21	-3.00	8.93	-1.33	-17.66	-2.79
216	-0.24	-23.24	0.22	-3.03	9.09	-1.24	6.07	24.80
217	3.44	-24.06	0.20	-3.67	8.63	-1.29	-8.15	19.18
218	-1.92	-22.05	0.20	-3.29	9.00	-1.18	6.54	10.41
219	0.30	-22.68	0.21	-3.36	8.67	-1.31	-10.55	1.54
220	-2.80	-23.71	0.20	-3.40	8.54	-1.20	16.87	-20.57
221	-4.02	-23.59	0.23	-3.23	8.89	-1.33	-29.66	-8.53
222	3.71	-22.32	0.22	-3.65	9.42	-1.27	-7.45	20.01
223	4.01	-22.74	0.20	-3.23	9.05	-1.11	-0.92	8.73
224	-0.07	-13.13	0.10	-1.31	4.94	0.10	-2.81	9.80
225	3.00	-13.34	0.13	-1.17	5.09	0.13	9.49	1.20
226	1.83	-13.14	0.12	-1.12	4.86	0.09	-6.43	7.09
227	2.44	-12.89	0.11	-1.27	4.92	0.14	11.11	3.43
228	-1.82	-12.71	0.11	-1.30	5.04	0.23	5.82	2.18
229	2.60	-13.68	0.12	-1.18	4.89	0.13	-13.54	-2.96
230	-0.58	-12.59	0.13	-1.32	5.16	0.14	5.68	2.50
231	-1.66	-13.08	0.12	-1.37	5.20	0.05	0.12	4.41
232	4.51	-13.15	0.12	-1.50	4.62	0.10	-3.69	-7.06

233	1.30	-12.99	0.10	-1.29	5.15	0.15	-8.75	-0.68
234	3.05	-13.52	0.12	-1.26	5.30	0.06	-5.26	-12.29
235	1.63	-13.65	0.11	-1.20	5.12	0.00	-9.46	4.53
236	4.38	-13.38	0.12	-1.23	5.11	0.11	-0.87	-0.28
237	-2.09	-14.06	0.12	-1.23	4.84	0.01	-7.24	-4.51
238	2.60	-13.88	0.11	-1.47	4.69	0.15	-10.41	-5.46
239	1.90	-12.57	0.11	-1.22	5.22	0.13	12.39	8.11

For all Gaussian functions g_{lk} for all images f_k , the exponential factor $\alpha = 1/\epsilon$ is given by $\epsilon = 2.0615528128088303$.

- 5 Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not describe the material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . The hypothesis testing may use as alternative hypothesis that the filter does describe the material classifier. The alternative hypothesis may be given by $H_1: \exists m, m': \mu_m \neq \mu_{m'}$. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each material, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for material for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:
- 15 $mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J)$. The hypothesis testing may comprise determining a Mean Sum of Squares between, $mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1)$. The hypothesis testing may comprise performing an F-Test:
- 20

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

- 25 Herein, I is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For exam-

ple, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.05$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

5

Figure 2 shows experimental results of using the spot shape filter to calculate the material feature for different material groups. Specifically, the material feature φ_{2m} is shown for different materials, namely from left to right for a wooden board denoted board, a fabric, a dark skin, a pale skin, and a highly translucent skin denoted trans, and illustrates the separation of skin materials (right) from other materials (left). The spot shape filter passes the hypothesis testing. The p-value computed from the F-test is 0.0076. Thus, the Null-hypothesis H_0 can be rejected with respect to a significance level of 0.01. Further, the alternative hypothesis H_1 can be accepted with respect to a significance level of 0.01. Thus, the image filter is distinguishes at least two material classifiers. In addition, Figure 2 shows the mean value of the feature per material, $\bar{\varphi}_m$, as the line within the respective box. Vertical bars mark the Q1-1.5 interquartile range (IQR) and Q3+1.5 IQR, wherein $IQR = Q3 - Q1$, with Q1 and Q3 being the first and third quartiles. The box marks the Q1 and Q3 quartiles. Outliers are plotted as points. Figure 2 shows, that this material feature separates all skin samples from the non-skin samples. In other words, the expectation values of the Skin Features for skin and non-skin materials differ.

20

Figure 1 moreover shows, in a highly schematic illustration, an exemplary embodiment of the detector 110, wherein the detector 110 specifically may be embodied as a camera 136 and/or may be part of a camera 136. The camera 136 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 1 further shows an embodiment of a detector system 138, which, besides the at least one detector 110, comprises one or more beacon devices 140, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 1 further shows an exemplary embodiment of a human-machine interface 142, which comprises the at least one detector system 138 and, further, an entertainment device 144, which comprises the human-machine interface 142. The figure further shows an embodiment of a tracking system 146 for tracking a position of the object 112, which comprises the detector system 138. The components of the devices and systems shall be explained in further detail below.

35

Fig. 1 further shows an exemplary embodiment of a scanning system 148 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 148 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well

as, optionally, at least one further illumination source, not depicted here. The illumination source 124, generally, is configured for emitting at least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 140 and/or on a surface of the object 112. The scanning system 148 may be designed to
5 generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 148, specifically the detector 110, by using the at least one detector 110.

10 The detector 110, besides the optical sensors 120, comprises the at least one evaluation device 132, having e.g. at least one image analysis device 150 and/or the at least one position evaluation device 152, as symbolically depicted in Fig. 1. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially
15 combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 154 and/or by one or more interfaces, as symbolically depicted in Fig. 1. Further, the one or more connectors 154 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connect-
20 or 154, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 156 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be
25 designed as an article of sports equipment and/or may form a control element or a control device 158, the position of which may be manipulated by a user 160. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 160 himself or herself may be considered as the object 112, the position of which shall be detected.

30 As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 156. Further, the detector 110 may comprise the at least one transfer device 129 such as one or more optical systems, preferably comprising one or more lenses. An opening 162 inside the housing 156, which, preferably, is located concentrically with
35 regard to the optical axis 128 of the detector 110, preferably defines a direction of view 164 of the detector 110. A coordinate system 166 may be defined, in which a direction parallel or anti-parallel to the optical axis 128 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 128 may be defined as transversal directions. In the coordinate system 166, symbolically depicted in Fig. 1, a longitudinal direction is denoted by z, and trans-
40 versal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 142, in order to provide at least one item of information to a machine 168. In the embodiments schematically depicted in Fig. 1, the machine 168 may be a computer and/or may comprise a computer. Other
5 embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 168, such as into the computer.

As outlined above, Fig. 1 also depicts an example of a tracking system 146, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system
10 146 comprises the detector 110 and at least one track controller 170. The track controller 170 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 170 may be an independent device and/or may be fully or partially integrated into the machine 168, specifically the computer, as indicated in Fig. 1 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 142 may form part of an entertainment device 144. The machine 168, specifically the computer, may also form part of the entertainment device 144. Thus, by means of the user 160 functioning as the object 112 and/or by means of the user 160 handling a control device functioning as the object 112, the user 160
20 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Fig. 1 also depicts an example of an inertial measurement unit 172 for use in an electronic device. The inertial measurement unit 172 is adapted to receive data determined by the detector
25 110. The inertial measurement unit 172 further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is
30 adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

List of reference numbers

110	detector
112	object
116	sensor element
118	matrix
120	optical sensor
122	light-sensitive area
124	illumination source
126	illumination light beam
128	optical axis
129	transfer device
130	reflection light beam
132	evaluation device
134	divider
136	camera
138	detector system
140	beacon device
142	human-machine-interface
144	entertainment device
146	tracking system
148	scanning system
150	image analysis device
152	position evaluation device
154	connector
156	housing
158	control device
160	user
162	opening
164	direction of view
166	coordinate system
168	machine
170	track controller
172	inertial measurement unit

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Claims

1. A detector (110) for identifying at least one material property m comprising
- 5 - at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein the sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112);
- 10 - at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device (132) is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a
- 15 linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device (132) is configured for determining at least one material
- 20 feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter;
- 25 a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based
- 30 dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,
- 35 wherein the evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .
- 40
2. The detector (110) according to the preceding claim, , wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifi-

ers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

- 5 3. The detector (110) according to the preceding claim, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.
- 10 4. The detector (110) according to any one of the preceding claims, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.
- 15 5. The detector (110) according to any one of the preceding claims, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.
- 20 6. The detector (110) according to any one of the preceding claims, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.
7. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between ϕ_{1z} , ϕ_{2m} and z , m .
- 25 8. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\phi_{1z}, \phi_{2m})$ and/or $m(\phi_{1z}, \phi_{2m})$.
- 30 9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.
- 35 10. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature ϕ_{1z} or ϕ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.
- 40 11. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of ϕ_{1z} and ϕ_{2m} to the reflection image.

12. The detector (110) according to any one of the preceding claims, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element (116), wherein the evaluation device (132) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the distance feature φ_{1z} .
- 10 13. The detector (110) according to any one of the preceding claims, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.
- 15 14. The detector (110) according to any one of the preceding claims, wherein the sensor element (116) comprises at least one CMOS sensor.
- 20 15. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.
- 30 16. The detector (110) according to the preceding claim, wherein the illumination source (124) comprises at least one laser source and at least one diffractive optical element.
- 35 17. A detector system (138), the detector system (138) comprising at least one detector (110) according to any one of the preceding claims, the detector system (138) further comprising at least one beacon device (140) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (140) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
- 40 18. A human-machine interface(142) for exchanging at least one item of information between a user (160) and a machine (168), wherein the human-machine interface (142) comprises at least one detector system (138) according to the preceding claim, wherein the at least one beacon device (140) is adapted to be at least one of directly or indirectly attached to

the user (160) and held by the user (160), wherein the human-machine interface (142) is designed to determine at least one position of the user (160) by means of the detector system (138), wherein the human-machine interface (142) is designed to assign to the position at least one item of information.

- 5
19. An entertainment device (144) for carrying out at least one entertainment function, wherein the entertainment device (144) comprises at least one human-machine interface (142) according to the preceding claim, wherein the entertainment device (144) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (142), wherein the entertainment device (144) is designed to vary the entertainment function in accordance with the information.
- 10
20. A tracking system (146) for tracking a position of at least one movable object, the tracking system (146) comprising at least one detector system (138) according to any one of the preceding claims referring to a detector system, the tracking system (146) further comprising at least one track controller (170), wherein the track controller (170) is adapted to track a series of positions of the object (112) at specific points in time.
- 15
21. A scanning system (148) for determining a depth profile of a scenery, the scanning system (148) comprising at least one detector (110) according to any of the preceding claims referring to a detector, the scanning system (148) further comprising at least one illumination source adapted to scan the scenery with at least one light beam.
- 20
22. A camera (136) for imaging at least one object (112), the camera (136) comprising at least one detector (110) according to any one of the preceding claims referring to a detector.
- 25
23. An inertial measurement unit (172) for use in an electronic device, wherein the inertial measurement unit (172) is adapted to receive data determined by at least one detector (110) according to any one of the preceding claims referring to a detector, wherein the inertial measurement unit (172) further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.
- 30
- 35
- 40 24. A method for determining at least one material property of at least one object (112) by using at least one detector (110) according to any one of the preceding claims referring to a detector, the method comprising the following steps:

- a) determining at least one reflection image of the object (112) by using at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122);
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device (132), wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

25. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

Detector for identifying at least one material property**Abstract**

5 A detector (110) for identifying at least one material property m is proposed. The detector (110) comprises at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122). The sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112). The detector (110) comprises at least one evaluation device (132) configured

10 for determining the material property by evaluation of at least one beam profile of the reflection image. The evaluation device (132) is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further

15 distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The evaluation device (132) is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image.

20 The material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area

25 filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, by

30 $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The evaluation device (132) is configured for determining a longitudinal coordinate

35 z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

40 (Figure 1)

FIG. 1

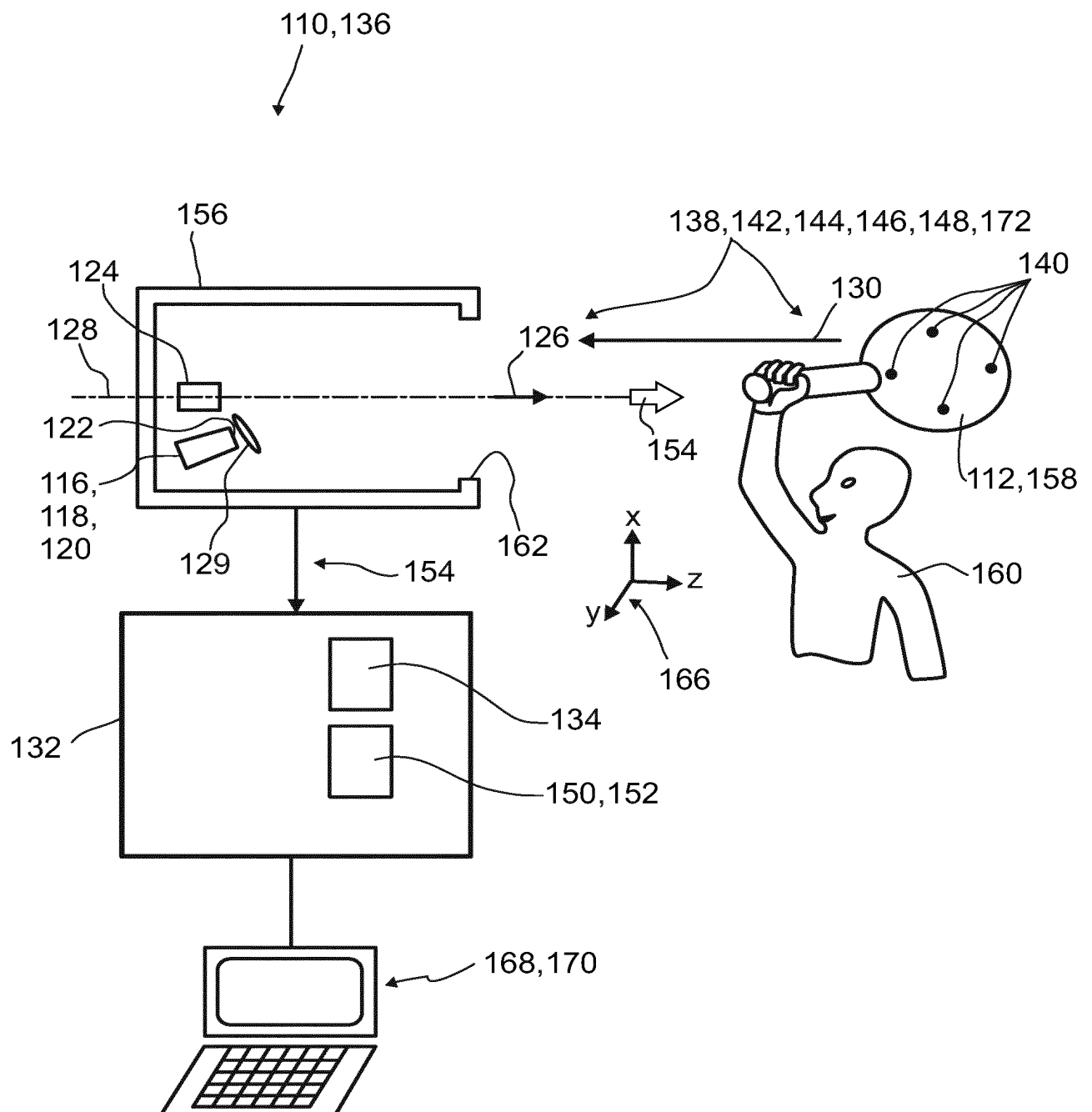


FIG.2

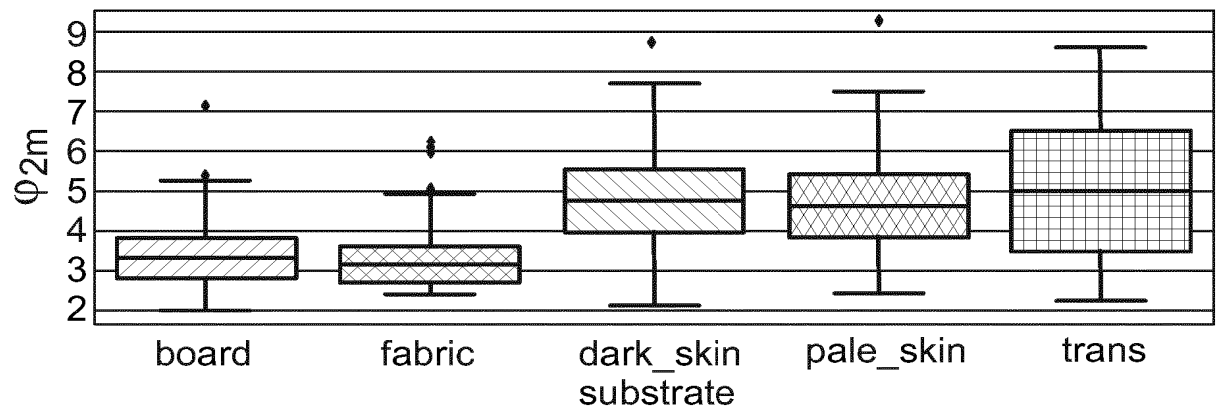
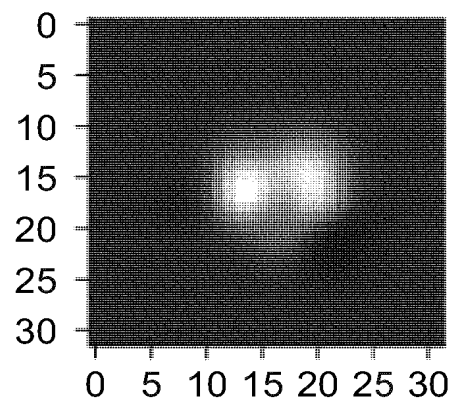


FIG.3



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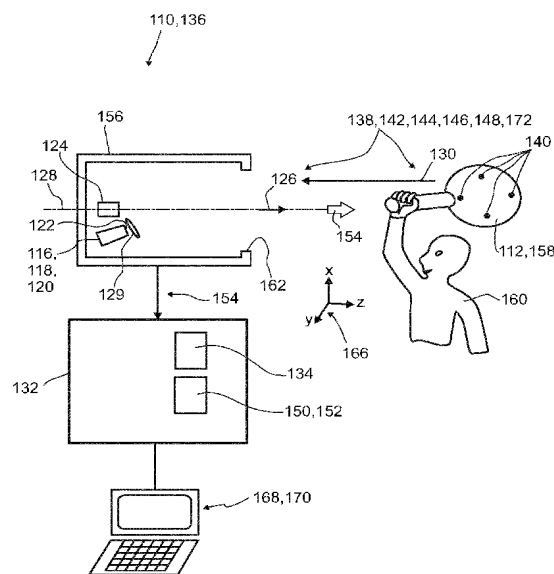
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(54) Title: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

FIG. 1



(57) **Abstract:** A detector (110) for identifying at least one material property m is proposed. The detector (110) comprises at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122). The sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112). The detector (110) comprises at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image. The evaluation device (132) is configured for determining at least one distance feature ϕ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The evaluation device (132) is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a

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grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

Claims

1. A detector (110) for identifying at least one material property m comprising
 - at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein the sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112);
 - at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device (132) is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device (132) is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .
2. The detector (110) according to the preceding claim, , wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifi-

ers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

- 5 3. The detector (110) according to the preceding claim, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.
4. The detector (110) according to any one of the preceding claims, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.
- 10 5. The detector (110) according to any one of the preceding claims, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.
- 15 6. The detector (110) according to any one of the preceding claims, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.
- 20 7. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .
- 25 8. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.
9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.
- 30 10. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.
- 35 11. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.
- 40

12. The detector (110) according to any one of the preceding claims, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element (116), wherein the evaluation device (132) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .
- 10 13. The detector (110) according to any one of the preceding claims, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.
- 15 14. The detector (110) according to any one of the preceding claims, wherein the sensor element (116) comprises at least one CMOS sensor.
- 20 15. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.
- 30 16. The detector (110) according to the preceding claim, wherein the illumination source (124) comprises at least one laser source and at least one diffractive optical element.
- 35 17. A detector system (138), the detector system (138) comprising at least one detector (110) according to any one of the preceding claims, the detector system (138) further comprising at least one beacon device (140) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (140) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
- 40 18. A human-machine interface (142) for exchanging at least one item of information between a user (160) and a machine (168), wherein the human-machine interface (142) comprises at least one detector system (138) according to the preceding claim, wherein the at least one beacon device (140) is adapted to be at least one of directly or indirectly attached to

the user (160) and held by the user (160), wherein the human-machine interface (142) is designed to determine at least one position of the user (160) by means of the detector system (138), wherein the human-machine interface (142) is designed to assign to the position at least one item of information.

- 5
19. An entertainment device (144) for carrying out at least one entertainment function, wherein the entertainment device (144) comprises at least one human-machine interface (142) according to the preceding claim, wherein the entertainment device (144) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (142), wherein the entertainment device (144) is designed to vary the entertainment function in accordance with the information.
- 10
20. A tracking system (146) for tracking a position of at least one movable object, the tracking system (146) comprising at least one detector system (138) according to any one of the preceding claims referring to a detector system, the tracking system (146) further comprising at least one track controller (170), wherein the track controller (170) is adapted to track a series of positions of the object (112) at specific points in time.
- 15
21. A scanning system (148) for determining a depth profile of a scenery, the scanning system (148) comprising at least one detector (110) according to any of the preceding claims referring to a detector, the scanning system (148) further comprising at least one illumination source adapted to scan the scenery with at least one light beam.
- 20
22. A camera (136) for imaging at least one object (112), the camera (136) comprising at least one detector (110) according to any one of the preceding claims referring to a detector.
- 25
23. An inertial measurement unit (172) for use in an electronic device, wherein the inertial measurement unit (172) is adapted to receive data determined by at least one detector (110) according to any one of the preceding claims referring to a detector, wherein the inertial measurement unit (172) further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.
- 30
- 35
- 40 24. A method for determining at least one material property of at least one object (112) by using at least one detector (110) according to any one of the preceding claims referring to a detector, the method comprising the following steps:

a) determining at least one reflection image of the object (112) by using at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122);

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device (132), wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

25. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

Detector for identifying at least one material property

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining at least one material property of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

- 20 A large number of methods for material classification and identification are known from the prior art. For example, material identification can be used in machine vision applications, medical applications and security applications, for image classification purposes or in gesture recognition algorithms.
- 25 For example, US 2016/0206216 A1 describes a device, system and method for skin detection. The device comprises a thermal sensor input for obtaining thermal sensor data of a scene, a light sensor input for obtaining light sensor data of the scene, and an evaluation unit for analyzing the obtained thermal sensor data and the obtained light sensor data and for detecting skin areas within the scene based on said analysis. US 2016/155006 A1 describes a device and a
- 30 corresponding method for skin detection. The device comprises an illumination unit configured to project a predetermined illumination pattern onto a scene, an imaging unit configured to acquire an image of the scene, and an evaluation unit configured to evaluate the acquired image by analyzing the imaged illumination pattern as reproduced in the image and to detect skin areas within the image and distinguish them from non-skin areas within the image based on said
- 35 analysis.

Material classification and identification is usually done in through beam mode, namely shining light through a sample and analyzing extinction. Further, material classification and identification can be done in reflective mode in controlled environments. However, in reflective mode usually wavelengths larger than 1000 nm are necessary in order to yield reliable results. Thus, despite the advantages of known methods for material identification, new concepts for reliable identification of materials are desired. Specifically, identifying a material from its reflection beam profile would be highly desirable. However, most image filters used for beam profile analysis (BPA) yield features that depend on both, distance and material such that reliable identification and

classification of the material is not possible. A typical example is the width of a beam profile reflected from a translucent material.

DE 198 46 619 A1 describes structured surface appearance quality determining equipment, which evaluates electrical measurement signal from photosensor array to derive structure code characterizing structure-dependent characteristic of measurement surface.

CN 108 363 482 A describes a method for controlling smart televisions via three-dimensional gestures on basis of binocular structured light. The method comprises the following steps of: synchronously acquiring images through a binocular camera; reconstructing a three-dimensional image according to an obtained left-right view; carrying out preprocessing; recognizing a gesture action after three-dimensional gestures are segmented; and converting the recognized gesture action into an operation instruction for a smart television, and executing the operation instruction.

US 2018/033146 A1 describes systems and methods for determining a depth map and a reflectivity map from a structured light image. The depth map can be determined by capturing the structured light image and then using a triangulation method to determine a depth map based on the dots in the captured structured light image. The reflectivity map can be determined based on the depth map and based on performing additional analysis of the dots in the captured structured light image.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may identify at least one material property of an object, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are

present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for identifying at least one material property m is disclosed.

As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “identifying at least one material property” refers to one or more of determining and assigning the material property to the object. The detector may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by

the user of the detector. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The detector may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

Specifically, the detector may be configured for detection of biological tissue, in particular human skin. As used herein, the term "biological tissue" generally refers to biological material comprising living cells. The detector may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The term "detection of biological tissue" refers to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector may be adapted to distinguish animal tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector may be adapted to distinguish plant tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector may be adapted to distinguish different types of food such as a fruit, meat, and fish. The detector may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from human skin. The detector may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector may be adapted to distinguish human skin from cloth. The detector may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector may be adapted to distinguish organic material from inorganic material. The detector may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector may be used, in particular, for non-therapeutic and non-diagnostic applications.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter, or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented

essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The detector may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured for generating at least one output signal. In case the detector comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly,

as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

Preferably, the light sensitive area may be oriented essentially perpendicular to an optical axis of the detector. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The sensor element is configured for recording at least one reflection image of a light beam originating from at least one object. As used herein, the term “reflection image” refers to an image determined by the sensor element comprising at least one reflection feature. As furthermore used herein, without limitation, the term “reflection image” specifically may relate to data recorded by using the sensor element, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element. The reflection image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to “pixels”, reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. As used herein, the term “reflection feature” refers to a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. As used herein, the term “determining at least one reflection feature” refers to imaging and/or recording at least one light beam generated by the object in response to illumination with a light beam, in particular with the at least one illumination feature. In particular, the sensor element may be configured for determining and/or imaging and/or recording at least reflection image. The reflection image may comprise at least one reflection pattern comprising the at least one reflection feature.

The detector comprises at least one illumination source. The detector may comprise at least one illumination source configured for illuminating the object with at least one illumination beam.

The illumination source may be configured for projecting at least one illumination pattern comprising the at least one illumination feature on at least one surface of the object. As used herein, the term “at least one illumination source” refers to at least one arbitrary device configured for providing at least one illumination light beam, specifically the at least one illumination pattern, for illumination of the object. The illumination source may be adapted to directly or indirectly illuminating the object, wherein the illumination light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam.

The illumination source may comprise at least one light source. The illumination source may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of

handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source may be integrated into a housing of the detector.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The detector may be configured such that the illumination light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the illumination light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illumination light beam may be on the optical axis or off the optical axis. As an example, the illumination light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

Specifically, the illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.

The illumination source may be configured for generating at least one illumination pattern for illumination of the object. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one
5 quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. As used herein, the term "pattern" refers to an arbitrary known or pre-
10 determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the object.
15 As used herein, the term "illumination feature" refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. For example, the illumination pattern may comprise at least one line pattern. For example, the illumination pattern may comprise at least one stripe pattern. For example, the illumination pattern may comprise at least
20 one checkerboard pattern. For example, the illumination pattern may comprise at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing
25 lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. For example, the illumination source may be adapted to generate and/or to project a cloud of points. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. The illumination source may comprise the at least one light
30 source configured for generating the at least one illumination pattern. Specifically, for generating and projecting the illumination pattern, the illumination source may comprise at least one laser source and at least one diffractive optical element (DOE). The detector may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at
35 least one point pattern. As further used herein, the term "projecting at least one illumination pattern" refers to providing the at least one illumination pattern for illuminating the at least one object. The projected illumination pattern may be scarce, as only a single illumination feature, such as a single point, may be present. For increasing reliability, the illumination pattern may comprise several illumination features such as several points. If the pattern is scarce, a single image
40 can be used for detection of biological tissue and face identification simultaneously.

For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The

illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern
5 may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination
10 source may be integrated in a mobile device or attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers
15 to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may
20 be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining
25 and determining the beam profile, in particular a shape of the beam profile.

As used herein, the term “object” refers to a point or region emitting at least one light beam, in particular the at least one reflection pattern. For example, the object may be at least one object selected from the group consisting of: a scene, a human such as a human, wood, carpet, foam,
30 an animal such as a cow, a plant, a piece of tissue, a metal, a toy, a metallic object, a beverage, a food such as a fruit, meat, fish, a dish, a cosmetics product, an applied cosmetics product, cloth, fur, hair, a maintenance product, a cream, an oil, a powder, a carpet, a juice, a suspension, a paint, a plant, a body, a part of a body, organic material, inorganic material, a reflective material, a screen, a display, a wall, a sheet of paper, such as photograph. The object may
35 comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrat-
40 ing the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase

with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These material properties may be used to distinguish skin from other materials, specifically by analyzing the back scattering profile.

The least one light beam may propagate from the object towards the detector. The light beam may originate from the object or may originate from an illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. Additionally or alternatively to the at least one illumination source, the detector may use radiation already present in the scene such as from at least one ambient light source.

The sensor element may be configured for recording a beam profile of at least one reflection feature of the reflection image. The evaluation device may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element. The evaluation device may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-

transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

For example, the illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device may be adapted to remove influences from background light from the reflection beam profile, for example, by an imaging without further illumination.

The evaluation device may be configured for determining the material property m by evaluating the beam profile of the reflection image. As used herein, the term "beam profile of the reflection image" refers to at least one intensity distribution of at least one of the reflection features of the reflection image, such as of a light spot on the sensor element, as a function of the pixel. The beam profile of the reflection image, also denoted reflection beam profile, may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. As used herein, the term "evaluating the beam profile" refers to applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. As used herein, the term "image" refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term "image filter" refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. As used herein, the term “distance” refers to the distance of the object, specifically between the object and the detector.

As used herein, the term “distance dependent” image filter refers to an image having a distance dependent output. The output of the distance dependent image filter is denoted herein “distance feature φ_{1z} ” or “distance dependent feature φ_{1z} ”. The distance feature may be or may comprise at least one information about the distance of the object such as at least one measure for the distance of the object, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter $\Phi_{1\text{other}}$ may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{\text{cov}((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j))}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one cur-

rent signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. As generally used herein, the term “combine” generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least one comparison signal or comparison result. As used herein, the term “combined signal Q” refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. In particular, the combined signal may be a quotient signal. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature ϕ_{1z} for determining the distance feature ϕ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy} ,$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing

precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature ϕ_{1z} may be a function of the combined signal Q , $\phi_{1z} = \phi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q . Further, as an example, the object distance z_0 may be a function of the distance dependent feature ϕ_{1z} , $z_0 = z_0(\phi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in ϕ_{1z} . Thus, the object distance z_0 may be a function of the combined signal Q , $z_0 = z_0(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q .

The light-sensitive areas of at least two optical sensors may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q .

The evaluation device may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for de-

termining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example, Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1 / b \cdot s_2$ or $Q = b \cdot s_2 / a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a , b , c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

Typically, the combined signal Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the setup described above, both the first signal s_1 and the second signal s_2 decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the first signal s_1 decreases more rapidly than the second signal s_2 , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus,

is spread over a larger area. The quotient of the first and second sensor signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the incident light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

With respect to further details and embodiments of evaluation of the combined signal Q reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device may be configured for determining from the image the distance feature φ_{1z} of the object by optimizing at least one blurring function f_a . The determined distance feature φ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature φ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. As used herein, the term “blurring function f_a ”, also referred to as blur kernel or point spread function, refers to a response function of the detector to the illumination from the object. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature φ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min\|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. As used herein, the term “sharp” or “sharp image” refers to a blurred image having a maximum contrast. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature ϕ_{1z} .

The evaluation device may be configured for determining at least one combined distance information z considering the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield advantageous distance measurement results with enhanced precision.

The evaluation device may be configured for determining at least one combined distance information using at least one recursive filter. The recursive filter may be at least one Kalman filter or at least one Extended Kalman filter (EKF). The combined distance information z may be obtained using a real function $z=f(z_{DPR}, z_{DFD})$, wherein z_{DPR} is the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and z_{DFD} is the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter, such as the arithmetic or geometric mean, a polynomial, preferably a polynomial up to the eighth order in z_{DPR} and z_{DFD} . The function f may be or may be based on a look-up table of prerecorded values. For example, the evaluation device may comprise at least one data storage device configured for storing the prerecorded values and/or one or more look-up tables. The function f may be based on a look-up table in combination with an interpolation scheme for interpolation between the values in the look-up table. The interpolation scheme may be a linear interpolation, a splines interpolation, or the like. The z_{DFD} and z_{DPR} may be used as input variables within the recursive filter, in combination with a model or model function such as the function f , concerning the relationship between z , z_{DFD} , and z_{DPR} . The model or model function may comprise statistical assumptions and/or statistical models concerning the distances z , z_{DFD} , and z_{DPR} such as a distribution, such as a Gaussian distribution of a measured distance z , z_{DFD} , and/or z_{DPR} around a real distance z_{real} .

The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters. The further parameters may comprise further information from the sensor element, for example the CMOS sensor, such as information about quality and/or noise of recorded data and/or information about overexposure and/or information about underexposure or the like. The detector may comprise at least one further sensor configured for determining the further sensor data. The recursive filter may be configured for

determining the combined distance information considering the further sensor data. The further sensor may be at least one sensor selected from the group consisting of: a temperature sensor, an illumination sensor such as a control sensor for determining illumination information, an inertial measurement unit; a gyroscope. The model and/or the Kalman filter may comprise further input parameters such as the further sensor data, e.g. temperature and/or detector motion from a gyroscope and/or information from an inertial measurement unit, and/or information from an illumination sensor) and/or further parameters such as the quality/noise of recorded data, overexposure, underexposure, or the like. The further sensor data may be provided by the sensor element, in particular by at least one CMOS sensor, and/or by further image analysis. The model and/or the Kalman filter may comprise the at least one material feature ϕ_{2m} as input variable.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and for selecting at least one image determined by the other one of the first sensor element or the second sensor element as reference image. As used herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature ϕ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm\epsilon$.

The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Hei-

delberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. As used herein, the term “displacement region” refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term “reference feature” refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \epsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evalua-

tion device may determine the displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\varphi_{1z} \pm \varepsilon$.

5 The evaluation device may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at
10 least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the
15 extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the
20 epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region
25 along the assigned epipolar line.

The evaluation device may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device may be configured
30 for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. As used herein, the term “displacement” refers to a difference between a position in the reference image to a position in the reflection image. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a
35 distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device may be adapted to determine the predetermined relationship by using triangulation methods. In case position of the selected
40 reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent

and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device.

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Additionally or alternatively, the evaluation device may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ε -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or reference feature in shorter distance and/or ε -weighted distance to the reference feature and/or reflection feature.

The evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. As used herein, the term "material dependent" image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein "material feature φ_{2m} " or "material dependent feature φ_{2m} ". The material feature may be or may comprise at least one information about the at least one material property of the object.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof

by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter $\Phi_{2\text{other}}$ may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term "passes a hypothesis testing" refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term "beam profile image" refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k :

$(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J, $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values

from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k .

The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term “not distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical. As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq$

0.075, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

- 5 For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1g_2) = [p_{g_1,g_2}]$, whereas p_{g_1,g_2} is the occurrence rate of the grey combination $(g_1,g_2)=[f(x_1,y_1),f(x_2,y_2)]$, and the relation ρ defines the distance between (x_1,y_1) and (x_2,y_2) , which is $\rho(x,y)=(x+a,y+b)$ with a and b selected from 0,1.

10

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

- For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

15

The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

- For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

20

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

- 25 The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

5 and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ε times the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The material property m and/or the longitudinal coordinate z may be determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m . The evaluation device may be configured for determining the material property m and/or the longitudinal coordinate z by evaluating the features φ_{2m} , φ_{1z} . For example, the evaluation device may be configured for using at least one predetermined relationship between the distance feature φ_{1z} and the longitudinal coordinate of the object for determining the longitudinal coordinate z of the object. The evaluation device may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the object for determining the material property of the object. The predetermined relationship may be one or more of an empirical relationship, a semi-empirical relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. For example, the material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z such that the information about the

longitudinal coordinate z can be considered for evaluating of ϕ_{2m} . Specifically, the material property m and/or the longitudinal coordinate z may be determined by a function $z(\phi_{1z}, \phi_{2m})$ and/or by a function $m(\phi_{1z}, \phi_{2m})$. The function may be predefined and/or predetermined. For example, the function may be a linear function.

5

In the ideal case, an image filter would yield features that are only dependent on distance or material properties. However, image filters used in beam profile analysis may yield features that depend on distance and material properties, such as translucency. At least one of the material dependent image filter or the distance dependent image filter may be a function of the feature ϕ_{1z} or ϕ_{2m} of the at least one other image filter. At least of the material dependent image filter or the distance dependent image filter may be a function of the at least one other image filter. The evaluation device may be configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature ϕ_{1z} or ϕ_{1m} of the other image filter and/or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter. Specifically, the evaluation device may be configured for determining a correlation coefficient of the material dependent image filter and the distance dependent image filter. In case the correlation coefficient of the material dependent image filter with the distance dependent image filter is close to 1 or -1, the distance may be projected out, by projecting the material feature on the principal axis with the lowest variance. As an example, the material feature may be projected onto an axis orthogonal to the correlating main component. In other words, the material feature may be projected onto the second main component. This may be done using a principal component analysis as known to the person skilled in the art. The evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image. In particular, the evaluation device may be configured for simultaneously determining features that are strongly correlated with distance and weakly correlated with material properties and features that are weakly correlated with distance and strongly correlated with material properties. Alternatively, the evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image. The evaluation device may be configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of ϕ_{1z} and ϕ_{2m} to the reflection image.

As further used herein, the term “evaluation device” generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. The above-mentioned operations, including analysis of the beam profile, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more

programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured for performing the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

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The detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

10 The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by
15 modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element,
20 preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more
25 imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging
30 thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on
35 the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

40 The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having

at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As outlined above, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. For example, the detector may be configured for determining the longitudinal coordinate of the object by using at least one distance dependent filter using the depth-from-photon-ratio technique and/or the depth-from-defocus technique as outlined above. The detector may be configured for determining a position of the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. At least one of the optical sensors may
5 determined from which a center signal arises. This may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal
10 sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also
15 be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one
20 transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created.
25 Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured for determining at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

30 The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, a center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. The use of the matrix of optical
35 sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object. The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at
40 least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p, is smaller or equal than a pre-defined level of significance, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

In a further aspect of the present invention, a detector system is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect

one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured for enabling at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured for varying the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As

used herein, the term “scenery” generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term “scan” generally refers to a consecutive measurement in different regions.

Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term “dot” refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a

case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using

the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

- 5 As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mir-
10 rors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

- Thus, the scanning system may provide a static arrangement of the one or more dots placed on
15 the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors
20 comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or
25 more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices
30 as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety
35 curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenu-
40 ate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated

out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

5 As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system
10 and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3D model by connecting adjacent points with planar surfaces. The 3D model may be stored and
15 processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser
20 sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters
25 or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

30 As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection
35 in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system
40 may be adapted to determine at least one longitudinal coordinate for each of the light points as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. This feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor.

5 The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least
10 one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination
15 source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively
20 or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic
25 radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence
30 and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be
35 regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the
40 visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of

3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the

grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
 - b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking

application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

With respect to further uses of the detector and devices of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

Specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device configured for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements configured for generating electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device configured for performing 3D photography. The camera generally may be configured for acquiring a single image, such as a single 3D image, or may be configured for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera configured for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be configured for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression “position” generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device configured for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be configured for performing at least one computer program, such as at least one computer program configured for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

The human-machine interface may comprise a plurality of beacon devices which are configured for being at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or

by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device configured for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements configured for reflecting a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be configured in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the

user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

- 5 The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source configured for generating at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam proper-
- 10 ties. Additionally or alternatively, the beacon device may comprise at least one reflector configured for reflecting light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

- The object, which may form part of the detector system, may generally have an arbitrary shape.
- 15 Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be
- 20 part of the human-machine interface and/or of the entertainment device.

- As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably
- 25 computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

- The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one
- 35 item of information preferably may comprise at least one command configured for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements
- 40 may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is configured for gathering information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be configured for providing information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is configured for tracking a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.

The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.

The tracking system may be configured for initiating one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be configured for initiating at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards

the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

In a further aspect of the present invention, an inertial measurement unit for use in an electronic device disclosed. The inertial measurement unit is adapted to receive data determined by at least one detector according to the present invention. The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

The inertial measurement unit may comprise the detector according to the present invention and/or may be connected to the detector, via at least one data connection. The evaluation device and/or at least one processing device of the inertial measurement unit may be configured for determining the at least one combined distance information, in particular using the at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters such as further sensor data from the further sensor of the inertial measurement unit. With respect to definitions and embodiments of the inertial measurement unit reference is made to the description of the detector.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

- Embodiment 1: A detector for identifying at least one material property m comprising
- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;

- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 2: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

Embodiment 3: The detector according to the preceding embodiment, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

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Embodiment 5: The detector according to any one of the preceding embodiments, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.80$.

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Embodiment 6: The detector according to any one of the preceding embodiments, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}},\Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}},\Phi_m}| \geq 0.80$.

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Embodiment 7: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

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Embodiment 8: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

25

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

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Embodiment 10: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

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Embodiment 11: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and/or φ_{2m} to the reflection image.

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Embodiment 12: The detector according to any one of the preceding embodiments, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .

Embodiment 13: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

5 wherein x and y are transversal coordinates, A1 and A2 are different areas of the beam profile and E(x,y,z_o) denotes the beam profile given at the object distance z_o, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

10 Embodiment 14: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

15 Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

25 Embodiment 16: The detector according to any one of the preceding embodiments, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

30 Embodiment 17: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

35
$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 18: The detector according to any one of the two preceding embodiments, wherein the at least one blurring function f_a is a function or composite function composed from at least

one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

- 5 Embodiment 19: The detector according to any one of the preceding embodiments, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the
10 illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at
15 least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 21: The detector according to the preceding embodiment, wherein the illumination
20 source comprises at least one laser source and at least one diffractive optical element.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at
25 least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 23: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement
30 region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.
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Embodiment 24: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \epsilon$.

Embodiment 25: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
 - at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
- wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 26: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.

Embodiment 27: The detector according to any one of embodiments 25 and 26, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 28: The detector according to any one of embodiments 25 to 27, wherein the at least one material property is a property selected from the group consisting of: a scattering coef-

ficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

Embodiment 29: The detector according to any one of embodiments 25 to 28, wherein the further distance dependent image filter $\Phi_{10\text{ther}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{10\text{ther}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{10\text{ther}}, \Phi_z}| \geq 0.80$.

Embodiment 30: The detector according to any one of embodiments 25 to 29, wherein the further material dependent image filter $\Phi_{20\text{ther}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.80$.

Embodiment 31: The detector according to any one of embodiments 25 to 30, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{20\text{ther}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

Embodiment 32: The detector according to any one of embodiments 25 to 31, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

Embodiment 33: The detector according to any one of embodiments 25 to 32, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

Embodiment 34: The detector according to any one of embodiments 25 to 33, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

Embodiment 35: The detector according to any one of embodiments 25 to 34, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the

at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

- 5 Embodiment 36: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

- 10 Embodiment 37: The detector according to any one of embodiments 25 to 36, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .
- 15

Embodiment 38: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 20 wherein x and y are transversal coordinates, A_1 and A_2 are different areas of the beam profile and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

- 25 Embodiment 39: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- 30 Embodiment 40: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
- 35

Embodiment 41: The detector according to any one of embodiments 25 to 40, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

Embodiment 42: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 43: The detector according to any one of embodiments 25 to 42, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

Embodiment 44: The detector according to any one of embodiments 25 to 43, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 45: The detector according to any one of embodiments 25 to 44, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 46: The detector according to the preceding embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 47: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflec-

tion image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 48: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 49: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 50: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,

wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

5 Embodiment 51: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a
10 pre-defined level of significance.

Embodiment 52: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.
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Embodiment 53: The detector according to any one of embodiments 50 and 52, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 54: The detector according to any one of embodiments 50 to 53, wherein the at
20 least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

Embodiment 55: The detector according to any one of embodiments 50 to 54, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.
25

Embodiment 56: The detector according to any one of embodiments 50 to 55, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.
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Embodiment 57: The detector according to any one of embodiments 50 to 56, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness
35 filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof, or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by
40 $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared

norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

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Embodiment 58: The detector according to any one of embodiments 50 to 57, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

10 Embodiment 59: The detector according to any one of embodiments 50 to 58, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

15 Embodiment 60: The detector according to any one of embodiments 50 to 59, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

20 Embodiment 61: The detector according to any one of embodiments 50 to 60, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

25 Embodiment 62: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

30 Embodiment 63: The detector according to any one of embodiments 50 to 62, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q
35 and the distance feature φ_{1z} for determining the distance feature φ_{1z} .

Embodiment 64: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are different areas of the beam profile and $E(x,y,z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

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Embodiment 65: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

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Embodiment 66: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

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Embodiment 67: The detector according to any one of embodiments 50 to 66, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

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Embodiment 68: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

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$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 69: The detector according to any one of embodiments 50 to 68, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

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Embodiment 70: The detector according to any one of embodiments 50 to 69, wherein the sensor element comprises at least one CMOS sensor.

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Embodiment 71: The detector according to any one of embodiments 50 to 70, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 72: The detector according to the embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 73: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 74: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 75: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 76: A detector system, the detector system comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

Embodiment 77: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 78: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 79: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 80: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of embodiments 1 to 24, 25 to 49, or 50 to 75, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 81: A camera for imaging at least one object, the camera comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75.

Embodiment 82: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data

from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

5 Embodiment 83: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 1 to 24, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- 10 b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

$$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$$
 with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - 15 b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by

$$|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$$
 with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,
 - 20 b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 84: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 25 to 49, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,
 - b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 85: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 50 to 75, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

- b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
- b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

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Embodiment 86: A use of the detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

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Brief description of the figures

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Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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Figure 1 shows an embodiment of a detector according to the present invention;

Figure 2 shows the statistics of the resulting material features of the spot shape material dependent image filter applied to the data set; and

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Figure 3 shows a beam profile image.

Detailed description of the embodiments:

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Fig. 1 shows in a highly schematic fashion an embodiment of a detector 110 for identifying at least one material property m of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.

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The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the ele-

ments are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

5 The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other
10 embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors 120 may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

15 The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical
20 sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photo-
25 diodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

30 The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As
35 an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

40 The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more prefer-

ably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

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The detector 110 may further comprise an illumination source 124. As an example, the illumination source 124 may be configured for generating an illumination light beam 126 for illuminating the object 112. The detector 110 may be configured such that the illumination light beam 126 propagates from the detector 110 towards the object 112 along an optical axis 128 of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis 128.

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The illumination source 124 may comprise at least one light source. The illumination source 124 may comprise a plurality of light sources. The illumination source 124 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 124 may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source 124 may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

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Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 124 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points, for example the illumination source 124 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least

one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source 124 may be integrated into a housing of the detector 110.

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The illumination light beam 126 may propagate from the object 112 towards the detector 110. The detector 110 may be used in active and/or passive illumination scenarios. For example, the at least one illumination source 124 may be adapted to illuminate the object 112, for example, by directing a light beam towards the object 112, which reflects the light beam. Additionally or
10 alternatively to the at least one illumination source 124, the detector 110 may use radiation already present in the scene such as from at least one ambient light source.

The illumination source 124 may illuminate the at least one object 112 with at least one illumination pattern. The illumination pattern may comprise a plurality of points as image features.

15 These points are illustrated as light beam 126 emerging from the illumination source 124.

The detector 110 may comprise at least one transfer device 129 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel
20 lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 129 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

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Each optical sensor 120 may be designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam 130 propagating from the object 112 to the detector 110. The sensor element 116 may be configured for determining at least one reflection image of at least one reflection feature generated by the reflection light beam 130 on the optical sensors 120. The reflection image may be data recorded by
30 using the sensor element 116, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element 116. The reflection image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix 118 of the sensor element 116. The matrix 118 may comprise the reflection image 126. For example, in case of illumination with a point pattern, the reflection image may comprise points as reflection features. These points may
35 result from reflection light beams 130 originating from the at least one object 112.

The sensor element 116 may be configured for recording a beam profile of at least one reflection feature of the reflection image. The detector 110 comprises at least one evaluation device
40 132. The evaluation device 132 may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element 116. The evaluation device 132 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image anal-

ysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element 116. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

The evaluation device 132 may be configured for determining the material property m by evaluating the beam profile of the reflection image. The beam profile of the reflection image may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device 132 may be configured for applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device 132 is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance feature may be or may comprise at least one information about the distance of the object 112 such as at least one measure for the distance of the object 112, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter Φ_{1other} may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{cov\left((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j)\right)}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device 132, as an example, may comprise at least one divider 134, wherein the divider 134 is configured for deriving the quotient signal. The divider 134 may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear com-

binations of the sensor signals. The evaluation device 132 may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance L(z_o) and beam shape S(x,y;z_o). Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object 112 and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature φ_{1z} may be a function of the combined signal Q, $\varphi_{1z} = \varphi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q. Further, as an example, the object distance z_o may be a function of the distance dependent feature φ_{1z} , $z_o = z_o(\varphi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in φ_{1z} . Thus, the object distance z_o may be a function of the combined signal Q, $z_o = z_o(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q.

The light-sensitive areas 122 of at least two optical sensors 120 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

The evaluation device 132 may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the

center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 132 may be adapted for determining an area integral of the beam profile. The evaluation device 132 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 132 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device 132 specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example,

Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1/b \cdot s_2$ or $Q = b \cdot s_2/a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a, b, c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

With respect to further details and embodiments of evaluation of the combined signal Q and determination of the longitudinal coordinate z reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device 132 may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device 132 may be configured for determining from the image the distance feature ϕ_{1z} of the object by optimizing at least one blurring function f_a . The determined distance feature ϕ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature ϕ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature ϕ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature ϕ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature ϕ_{1z} .

The evaluation device 132 may be configured for determining at least one combined distance information z considering the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ

z_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature φ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature φ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield to advantageous distance measurement results with enhanced precision.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. The detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device 132 may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device 132 may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm \epsilon$.

The evaluation device 132 may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device 132 may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device 132 may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The evaluation device 132 may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device 132 may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the

reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device 132 may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object 112, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device 132 may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \varepsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device 132 may determine the displacement region around the image position of the reflection feature. The evaluation device 132 may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\varphi_{1z} \pm \varepsilon$.

The evaluation device 132 may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The evaluation device 132 may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device 132 may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device 132 may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to

determine an epipolar line before determining a corresponding reference feature. The evaluation device 132 may determine a displacement region around the image position of each reflection feature. The evaluation device 132 may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device 132 may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device 132 may be configured for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 132 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device 132 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device 132 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device 132.

Additionally or alternatively, the evaluation device 132 may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material feature may be or may comprise at least one information about the at least one material property of the object 112.

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The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$. With respect to the description of exemplary material dependent image filter, reference is made to the description of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter such as a Gaussian filter or median filter, the grey-level-occurrence-based contrast filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, and the threshold area filter given above.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. The beam profile image may be given by a sum of N_B Gaussian radial basis functions,

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$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all beam profile images. The center-positions, x_{lk} , y_{lk} , are identical for all images $f_k: (x_0, x_1, \dots, x_{N_B}), (y_0, y_1, \dots, y_{N_B})$. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

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Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed. Figure 3 shows an embodiment of a beam profile image defined by an exemplary data set.

An exemplary data set is presented in the following. The following parameter tables list the center positions, $x_{lk} = x_l, y_{lk} = y_l$, of all Gaussian functions g_{lk} for all images f_k :

l	0	1	2	3	4	5	6	7
x_l	14	11	10	19	15	20	12	17
y_l	17	11	17	19	20	20	12	20
l	8	9	10	11	12	13	14	15
x_l	13	18	10	14	19	15	16	12
y_l	17	19	8	18	14	11	22	17
l	16	17	18	19	20	21	22	23
x_l	25	13	18	8	9	10	14	19
y_l	15	20	20	15	14	13	24	24
l	24	25	26	27	28	29	30	31
x_l	10	9	14	12	19	20	17	15
y_l	24	16	15	11	13	19	13	14
l	32	33	34	35	36	37	38	39
x_l	16	13	12	20	18	17	21	13
y_l	19	10	22	14	10	18	15	23
l	40	41	42	43	44	45	46	47
x_l	8	10	8	14	19	11	15	14
y_l	12	14	18	21	23	15	16	8
l	48	49	50	51	52	53	54	55
x_l	16	20	11	17	16	15	21	13
y_l	13	16	16	8	16	13	21	13

l	56	57	58	59	60	61	62	63
x_l	22	18	23	14	19	11	15	16
y_l	23	15	12	22	18	10	23	10

The following parameter tables list the material classifiers, in particular for a white Teflon target (denoted control), a fabric, a dark skin (denoted as dark_skin), a pale skin (denoted pale_skin), and a highly translucent skin (denoted trans), and the distance z for each image f_k , referenced

5 by the image index k :

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
0	control	0.4	40	dark_skin	0.4	80	control	0.6
1	control	0.4	41	dark_skin	0.4	81	control	0.6
2	control	0.4	42	dark_skin	0.4	82	control	0.6
3	control	0.4	43	dark_skin	0.4	83	control	0.6
4	control	0.4	44	dark_skin	0.4	84	control	0.6
5	control	0.4	45	dark_skin	0.4	85	control	0.6
6	control	0.4	46	dark_skin	0.4	86	control	0.6
7	control	0.4	47	dark_skin	0.4	87	control	0.6
8	control	0.4	48	pale_skin	0.4	88	control	0.6
9	control	0.4	49	pale_skin	0.4	89	control	0.6
10	control	0.4	50	pale_skin	0.4	90	control	0.6
11	control	0.4	51	pale_skin	0.4	91	control	0.6
12	control	0.4	52	pale_skin	0.4	92	control	0.6
13	control	0.4	53	pale_skin	0.4	93	control	0.6
14	control	0.4	54	pale_skin	0.4	94	control	0.6
15	control	0.4	55	pale_skin	0.4	95	control	0.6
16	fabric	0.4	56	pale_skin	0.4	96	fabric	0.6
17	fabric	0.4	57	pale_skin	0.4	97	fabric	0.6
18	fabric	0.4	58	pale_skin	0.4	98	fabric	0.6
19	fabric	0.4	59	pale_skin	0.4	99	fabric	0.6
20	fabric	0.4	60	pale_skin	0.4	100	fabric	0.6
21	fabric	0.4	61	pale_skin	0.4	101	fabric	0.6
22	fabric	0.4	62	pale_skin	0.4	102	fabric	0.6
23	fabric	0.4	63	pale_skin	0.4	103	fabric	0.6
24	fabric	0.4	64	trans	0.4	104	fabric	0.6
25	fabric	0.4	65	trans	0.4	105	fabric	0.6
26	fabric	0.4	66	trans	0.4	106	fabric	0.6
27	fabric	0.4	67	trans	0.4	107	fabric	0.6
28	fabric	0.4	68	trans	0.4	108	fabric	0.6
29	fabric	0.4	69	trans	0.4	109	fabric	0.6
30	fabric	0.4	70	trans	0.4	110	fabric	0.6
31	fabric	0.4	71	trans	0.4	111	fabric	0.6
32	dark_skin	0.4	72	trans	0.4	112	dark_skin	0.6
33	dark_skin	0.4	73	trans	0.4	113	dark_skin	0.6
34	dark_skin	0.4	74	trans	0.4	114	dark_skin	0.6
35	dark_skin	0.4	75	trans	0.4	115	dark_skin	0.6
36	dark_skin	0.4	76	trans	0.4	116	dark_skin	0.6
37	dark_skin	0.4	77	trans	0.4	117	dark_skin	0.6
38	dark_skin	0.4	78	trans	0.4	118	dark_skin	0.6
39	dark_skin	0.4	79	trans	0.4	119	dark_skin	0.6

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
120	dark_skin	0.6	160	control	0.8	200	dark_skin	0.8
121	dark_skin	0.6	161	control	0.8	201	dark_skin	0.8
122	dark_skin	0.6	162	control	0.8	202	dark_skin	0.8
123	dark_skin	0.6	163	control	0.8	203	dark_skin	0.8
124	dark_skin	0.6	164	control	0.8	204	dark_skin	0.8
125	dark_skin	0.6	165	control	0.8	205	dark_skin	0.8
126	dark_skin	0.6	166	control	0.8	206	dark_skin	0.8
127	dark_skin	0.6	167	control	0.8	207	dark_skin	0.8
128	pale_skin	0.6	168	control	0.8	208	pale_skin	0.8
129	pale_skin	0.6	169	control	0.8	209	pale_skin	0.8
130	pale_skin	0.6	170	control	0.8	210	pale_skin	0.8
131	pale_skin	0.6	171	control	0.8	211	pale_skin	0.8
132	pale_skin	0.6	172	control	0.8	212	pale_skin	0.8
133	pale_skin	0.6	173	control	0.8	213	pale_skin	0.8
134	pale_skin	0.6	174	control	0.8	214	pale_skin	0.8
135	pale_skin	0.6	175	control	0.8	215	pale_skin	0.8
136	pale_skin	0.6	176	fabric	0.8	216	pale_skin	0.8
137	pale_skin	0.6	177	fabric	0.8	217	pale_skin	0.8
138	pale_skin	0.6	178	fabric	0.8	218	pale_skin	0.8
139	pale_skin	0.6	179	fabric	0.8	219	pale_skin	0.8
140	pale_skin	0.6	180	fabric	0.8	220	pale_skin	0.8
141	pale_skin	0.6	181	fabric	0.8	221	pale_skin	0.8
142	pale_skin	0.6	182	fabric	0.8	222	pale_skin	0.8
143	pale_skin	0.6	183	fabric	0.8	223	pale_skin	0.8
144	trans	0.6	184	fabric	0.8	224	trans	0.8
145	trans	0.6	185	fabric	0.8	225	trans	0.8
146	trans	0.6	186	fabric	0.8	226	trans	0.8
147	trans	0.6	187	fabric	0.8	227	trans	0.8
148	trans	0.6	188	fabric	0.8	228	trans	0.8
149	trans	0.6	189	fabric	0.8	229	trans	0.8
150	trans	0.6	190	fabric	0.8	230	trans	0.8
151	trans	0.6	191	fabric	0.8	231	trans	0.8
152	trans	0.6	192	dark_skin	0.8	232	trans	0.8
153	trans	0.6	193	dark_skin	0.8	233	trans	0.8
154	trans	0.6	194	dark_skin	0.8	234	trans	0.8
155	trans	0.6	195	dark_skin	0.8	235	trans	0.8
156	trans	0.6	196	dark_skin	0.8	236	trans	0.8
157	trans	0.6	197	dark_skin	0.8	237	trans	0.8
158	trans	0.6	198	dark_skin	0.8	238	trans	0.8
159	trans	0.6	199	dark_skin	0.8	239	trans	0.8

The following parameter tables list the prefactors a_{lk} for the Gaussian functions g_{lk} in each image f_k referenced by the image index k and the Gaussian index l :

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
0	291.94	-7.20	-14.72	-22.75	32.57	16.92	-45.50	-23.31
1	300.91	-7.58	-15.22	-14.41	32.18	15.96	-44.41	-23.92
2	297.04	-6.42	-13.94	-10.30	31.19	16.44	-43.68	-22.27
3	289.98	-7.67	-14.20	-13.60	31.68	17.16	-44.04	-21.41
4	295.94	-7.72	-12.93	-16.81	32.77	15.92	-45.15	-22.30
5	295.37	-6.87	-13.30	-15.51	33.10	16.70	-45.18	-22.29
6	298.03	-6.79	-14.89	-12.68	32.43	14.46	-44.47	-22.71

7	296.95	-7.09	-14.98	-13.54	31.56	15.69	-45.09	-20.03
8	306.69	-6.91	-14.94	-14.27	33.28	16.02	-43.93	-20.21
9	291.40	-7.73	-13.91	-14.95	30.51	16.18	-45.86	-21.61
10	301.21	-7.21	-14.05	-17.46	32.02	17.26	-45.02	-22.89
11	309.78	-6.93	-13.47	-19.23	33.01	14.94	-43.61	-20.86
12	298.62	-7.09	-14.75	-18.43	32.59	18.97	-44.11	-22.73
13	296.09	-7.18	-14.25	-15.83	31.94	14.62	-43.72	-19.77
14	301.13	-7.24	-14.44	-19.11	33.54	17.01	-44.08	-21.39
15	304.28	-7.08	-14.10	-17.26	32.54	16.81	-44.40	-21.14
16	246.71	-3.77	-13.62	1.43	19.38	-2.03	-52.50	-19.62
17	258.85	-3.11	-12.88	-3.91	19.02	-1.05	-49.64	-21.50
18	263.06	-2.91	-13.72	1.65	19.03	-1.26	-51.02	-21.58
19	251.12	-4.15	-14.47	-6.19	19.00	-1.26	-50.65	-22.40
20	250.05	-2.91	-13.84	0.30	19.48	-1.79	-49.89	-20.92
21	249.35	-2.80	-13.64	-0.25	18.90	0.42	-51.29	-19.21
22	260.34	-2.57	-14.30	1.83	17.72	-1.18	-51.27	-24.07
23	246.64	-3.68	-13.92	-2.29	19.79	-0.20	-50.33	-23.09
24	245.58	-2.91	-15.25	-6.59	18.34	-1.75	-50.89	-24.38
25	249.09	-3.09	-14.13	3.85	19.58	-1.58	-50.05	-22.87
26	255.98	-2.91	-13.67	6.77	18.09	-2.15	-48.40	-21.74
27	249.87	-3.43	-13.89	1.80	17.14	-2.32	-50.95	-25.65
28	247.42	-3.34	-13.32	2.32	17.69	-1.29	-50.40	-23.82
29	253.00	-3.32	-12.60	4.98	17.09	-1.16	-49.54	-25.89
30	254.40	-3.44	-14.69	1.64	20.18	-1.15	-49.66	-22.73
31	256.40	-3.75	-13.62	-8.47	19.57	-1.08	-52.04	-21.62
32	24.07	-2.06	2.98	-22.09	14.88	8.52	3.28	-4.27
33	24.00	-1.30	3.15	-22.28	14.65	9.22	3.94	-4.75
34	20.76	-1.69	3.24	-22.34	14.81	8.95	3.76	-4.69
35	22.23	-2.51	3.10	-22.85	14.01	9.71	3.29	-4.61
36	20.33	-2.23	3.04	-21.40	15.61	8.99	3.00	-4.12
37	21.12	-2.09	3.25	-26.46	14.68	8.75	4.67	-5.05
38	21.89	-1.92	3.05	-22.32	14.65	8.25	3.77	-4.15
39	14.81	-2.07	2.82	-20.92	14.96	9.05	3.73	-5.40

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
0	-95.90	-116.79	0.49	-92.61	-76.01	36.70	0.46	78.40
1	-99.07	-117.11	0.57	-89.09	-76.31	37.12	-0.60	86.28
2	-92.44	-115.78	0.59	-93.89	-76.95	35.15	0.42	83.74
3	-96.50	-115.93	0.49	-92.28	-75.29	36.48	-0.17	70.45
4	-100.83	-120.58	0.57	-95.39	-73.94	37.36	-0.26	79.80
5	-94.89	-117.93	0.59	-93.45	-75.67	36.88	0.22	74.29
6	-94.27	-119.97	0.51	-91.14	-74.61	35.76	-0.72	73.46
7	-94.78	-114.15	0.55	-91.40	-71.49	36.55	-0.61	73.29
8	-97.53	-112.61	0.54	-91.30	-70.94	36.62	0.39	75.30
9	-97.85	-108.91	0.57	-94.07	-72.52	36.34	-0.32	76.34
10	-97.23	-115.13	0.64	-95.38	-74.62	36.22	-0.33	74.11
11	-96.18	-115.91	0.45	-93.13	-77.72	37.20	-0.07	81.82
12	-100.79	-113.16	0.52	-90.33	-72.38	36.46	-1.02	74.17
13	-98.70	-124.58	0.53	-93.48	-74.52	36.90	-0.78	74.38
14	-93.07	-116.00	0.52	-92.98	-75.28	36.84	-0.77	76.49
15	-106.83	-117.44	0.52	-94.22	-75.28	36.42	-0.51	79.82
16	-88.58	-87.41	0.63	-70.05	-37.06	27.50	3.58	84.26
17	-87.71	-82.32	0.54	-68.18	-40.98	28.10	3.20	88.34
18	-76.19	-80.81	0.53	-66.95	-35.36	27.23	3.61	87.52
19	-80.42	-84.95	0.55	-67.91	-38.39	27.70	3.39	90.71

20	-87.87	-76.07	0.74	-66.04	-39.03	27.55	2.84	83.83
21	-93.60	-80.04	0.45	-69.48	-38.83	27.01	3.71	83.32
22	-88.04	-78.17	0.62	-72.00	-39.30	27.24	3.08	89.23
23	-88.92	-76.16	0.62	-66.79	-39.81	27.64	3.01	84.46
24	-90.47	-77.18	0.58	-66.41	-35.84	27.85	3.13	94.66
25	-89.84	-83.96	0.70	-64.36	-36.65	26.40	3.38	86.08
26	-81.82	-72.08	0.59	-66.39	-38.13	27.45	3.47	83.27
27	-80.57	-82.11	0.65	-66.61	-35.78	28.23	3.99	86.32
28	-93.40	-81.19	0.66	-68.45	-39.71	26.68	3.19	88.66
29	-92.92	-79.14	0.68	-64.90	-38.61	28.11	3.79	83.05
30	-91.82	-79.09	0.57	-66.39	-37.74	27.24	4.20	82.26
31	-90.60	-82.20	0.51	-67.61	-35.58	28.10	3.02	89.40
32	-0.54	-16.54	-0.31	-3.67	-33.80	4.91	1.87	11.78
33	-0.17	-16.06	-0.31	-2.39	-32.09	4.41	1.86	11.66
34	-3.04	-15.03	-0.34	-5.36	-31.98	4.17	1.76	11.91
35	-5.65	-15.79	-0.35	-2.71	-32.21	4.77	1.90	10.71
36	-6.54	-15.97	-0.29	-3.29	-33.09	4.26	1.69	14.99
37	-1.22	-16.20	-0.29	-2.79	-32.90	4.42	1.93	9.92
38	-3.85	-13.17	-0.22	-2.98	-33.98	4.68	1.84	11.96
39	-5.99	-12.87	-0.34	-2.53	-34.59	3.97	1.80	11.77

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
0	3.86	18.38	66.68	70.89	46.30	51.83	63.61	6.33
1	-1.41	18.18	66.65	-116.27	47.93	53.00	-19.71	6.14
2	5.49	17.64	67.40	86.57	47.99	51.08	-20.20	6.06
3	1.41	18.29	68.83	-73.86	49.40	52.06	-9.49	6.14
4	0.58	17.86	70.11	73.41	47.45	52.04	19.20	6.06
5	-0.96	18.19	71.29	12.08	48.40	52.10	36.63	6.18
6	-0.98	18.97	69.33	107.17	48.12	51.68	-30.34	6.21
7	1.44	17.76	67.10	-119.11	46.64	51.64	-19.85	6.11
8	2.83	18.10	66.62	42.19	47.28	52.29	-24.88	6.17
9	-0.87	17.50	68.71	-129.91	45.89	51.63	-43.44	6.23
10	1.24	18.26	66.88	-12.32	47.49	52.45	45.87	6.14
11	0.39	18.50	68.03	-103.43	47.42	52.40	-25.17	6.01
12	1.60	18.13	67.42	-159.30	48.41	52.43	-8.06	6.08
13	2.92	17.75	68.36	139.40	45.89	52.74	-16.56	6.19
14	1.69	17.80	68.48	-16.76	46.43	52.86	-15.34	6.20
15	0.66	17.51	68.30	-16.84	46.61	52.23	-12.07	6.00
16	1.53	10.12	45.53	-105.48	51.47	53.43	29.32	3.91
17	-0.83	10.67	47.22	-65.54	49.74	55.32	-3.14	3.90
18	-0.05	11.56	46.07	-38.72	50.65	54.17	-71.99	3.87
19	1.11	10.54	47.56	-22.99	50.63	54.47	19.05	3.81
20	-2.77	10.45	46.29	102.12	50.39	54.05	-21.67	3.88
21	1.51	11.03	46.20	-61.41	52.28	55.89	51.76	4.01
22	-2.08	11.82	45.96	121.36	49.18	54.28	-44.05	3.89
23	-0.13	10.89	46.21	34.94	52.82	54.93	26.48	3.82
24	-3.52	10.89	49.08	97.01	50.10	55.70	4.07	3.65
25	2.07	9.39	46.08	-53.59	51.65	53.65	4.94	3.97
26	0.44	11.04	47.23	-188.70	49.01	55.21	30.06	4.08
27	-4.62	9.33	47.59	141.42	51.09	54.20	-1.25	3.95
28	-1.78	10.70	45.70	-47.29	49.84	54.81	40.79	3.71
29	-2.45	10.65	47.01	-32.51	51.18	53.56	11.77	3.93
30	0.03	10.49	43.45	-149.80	50.93	53.95	14.36	3.83
31	3.37	10.57	46.51	20.87	49.67	54.60	22.79	3.96
32	0.86	7.55	19.63	-6.16	7.11	9.03	-20.02	1.08

33	-1.54	7.76	20.22	38.99	7.26	8.99	-10.41	1.12
34	-0.19	7.43	18.78	21.86	7.75	8.86	-16.09	1.15
35	-2.22	7.79	19.15	4.23	7.44	8.75	19.94	1.06
36	-1.11	7.83	20.15	20.92	6.25	8.80	-31.48	1.06
37	-0.31	7.90	19.64	8.33	7.22	9.07	18.83	1.12
38	-1.59	7.27	19.37	-18.56	6.82	8.51	21.93	1.10
39	-1.18	7.41	19.11	85.48	7.72	8.69	-17.48	1.01

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
0	-2.45	11.03	80.28	37.44	26.08	-32.86	-12.63	179.34
1	4.01	11.52	78.78	37.44	24.32	-34.91	-12.16	180.51
2	-3.35	10.90	81.50	37.26	24.80	-33.69	-10.77	180.89
3	1.73	10.84	80.38	37.51	25.47	-36.57	-15.88	183.66
4	1.27	9.42	81.14	38.14	25.13	-35.66	-13.87	180.51
5	-1.05	10.89	78.62	35.99	24.44	-34.31	-10.65	181.52
6	-0.46	10.30	81.92	35.89	24.74	-34.63	-15.73	180.07
7	2.40	10.57	78.99	37.01	23.94	-34.87	-12.35	183.67
8	1.32	10.56	80.87	37.45	24.70	-35.58	-16.41	180.05
9	1.05	9.87	79.26	37.63	24.52	-33.58	-12.25	178.55
10	-0.78	9.63	81.05	38.38	25.20	-33.14	-13.41	176.95
11	3.17	10.63	81.55	37.98	24.25	-32.39	-8.97	178.12
12	-0.12	10.57	81.11	38.18	25.06	-32.51	-14.47	180.98
13	-3.92	11.23	81.59	37.07	24.28	-35.06	-13.10	179.20
14	-1.14	11.04	81.63	36.75	25.25	-34.70	-11.65	180.54
15	1.46	11.05	80.55	36.57	23.89	-33.64	-12.62	181.91
16	-5.70	11.70	54.00	34.44	12.44	-14.39	-12.01	171.54
17	-0.70	12.09	55.21	34.71	9.92	-17.13	-11.70	168.02
18	-3.57	11.69	53.52	34.81	11.95	-13.55	-12.37	172.64
19	-3.32	12.04	54.84	35.03	11.31	-16.56	-8.97	169.62
20	2.59	12.21	55.48	35.95	11.23	-16.53	-9.56	169.59
21	2.54	12.19	54.25	35.20	10.76	-16.90	-12.28	169.58
22	-3.44	11.69	55.10	35.89	12.10	-18.59	-8.83	171.87
23	0.56	12.26	53.50	35.39	10.43	-17.85	-6.90	169.62
24	1.18	11.85	54.90	34.61	11.57	-14.31	-8.59	170.29
25	4.59	13.16	55.42	35.65	11.43	-20.08	-11.70	170.01
26	-0.12	11.06	54.09	34.94	11.37	-15.16	-13.76	172.27
27	1.10	11.81	55.45	34.85	10.99	-17.00	-9.50	170.91
28	2.78	11.51	56.06	34.44	10.57	-13.57	-8.79	172.45
29	-1.28	11.48	55.08	35.02	10.89	-15.74	-9.63	169.75
30	2.04	12.16	55.11	35.67	10.86	-17.71	-6.55	170.86
31	0.32	12.82	53.12	34.92	10.54	-18.42	-8.44	169.18
32	0.20	0.74	50.97	-0.31	16.61	-0.13	-0.95	4.08
33	-2.15	1.02	49.63	-0.32	16.22	0.23	0.22	1.95
34	1.20	0.57	50.53	-0.69	16.60	0.05	-0.14	2.81
35	2.07	0.84	50.16	-0.49	16.35	0.75	-0.83	2.72
36	0.80	0.74	49.51	0.39	16.42	-0.53	-1.24	3.37
37	-1.45	1.05	52.96	-0.54	16.22	0.77	-0.23	3.04
38	2.48	0.88	49.48	0.67	16.29	-0.46	-1.51	3.83
39	-0.81	0.65	50.20	-0.45	16.38	-0.34	-0.14	3.97

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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0	-19.43	-15.41	-0.54	35.39	4.01	146.63	-31.68	-3.15
1	-18.88	-15.15	-0.37	34.72	4.18	148.07	-31.76	-3.56
2	-16.02	-15.31	-0.38	33.72	4.18	145.90	-32.03	-3.18
3	-17.80	-15.75	-0.39	36.39	4.08	145.63	-31.54	-3.47
4	-15.90	-15.69	-0.30	34.77	4.13	149.16	-31.20	-3.47
5	-18.98	-15.68	-0.58	33.97	4.12	147.36	-31.71	-3.33
6	-15.90	-15.37	-0.49	34.31	4.29	146.96	-30.78	-3.23
7	-14.81	-15.73	-0.31	36.06	4.14	150.82	-32.18	-3.67
8	-15.48	-15.58	-0.58	36.41	4.13	146.32	-31.10	-3.39
9	-16.55	-15.89	-0.37	32.95	3.99	141.73	-30.59	-3.45
10	-18.58	-15.48	-0.48	33.49	4.40	148.23	-30.74	-3.45
11	-15.63	-15.39	-0.38	35.93	4.06	144.15	-30.84	-3.40
12	-17.34	-15.78	-0.58	35.76	4.10	140.54	-32.14	-3.58
13	-16.01	-15.44	-0.51	35.03	4.19	142.21	-31.70	-3.44
14	-19.00	-15.39	-0.63	35.03	4.06	146.35	-31.59	-3.67
15	-16.05	-15.61	-0.51	35.06	4.04	144.24	-31.00	-3.59
16	-1.73	-11.69	0.83	17.68	4.06	115.00	-27.19	-5.03
17	-4.29	-12.41	1.18	16.91	3.80	117.37	-26.45	-5.33
18	-2.73	-12.25	1.21	17.51	3.83	119.84	-28.87	-5.24
19	-1.43	-12.78	1.09	16.03	3.87	118.57	-27.48	-5.28
20	-1.08	-12.82	1.18	16.30	3.94	118.99	-26.64	-4.93
21	-3.12	-12.39	0.98	17.09	3.77	116.73	-27.35	-5.06
22	-2.46	-12.29	1.12	15.90	3.83	119.51	-27.39	-4.75
23	-1.88	-12.16	1.06	16.70	3.81	120.08	-27.74	-5.30
24	-3.35	-12.24	1.09	14.93	3.90	117.65	-27.21	-5.13
25	-1.85	-11.74	1.20	18.98	3.86	118.21	-27.50	-5.19
26	-2.53	-12.60	1.34	18.04	3.83	118.75	-26.64	-5.50
27	-4.14	-11.95	1.01	18.05	3.80	120.99	-27.10	-4.89
28	-2.49	-12.69	1.15	18.06	3.69	117.75	-26.73	-5.11
29	-2.52	-12.69	1.30	15.46	4.07	116.11	-27.32	-5.01
30	-1.51	-12.10	1.15	17.59	3.63	123.35	-27.39	-5.32
31	-4.63	-12.59	1.03	17.39	3.88	119.73	-27.34	-5.17
32	-5.22	1.64	2.40	15.00	2.34	28.20	-5.04	-5.62
33	-5.96	1.34	2.52	14.98	2.36	28.53	-4.99	-5.55
34	-5.47	1.64	2.42	14.89	2.39	29.51	-4.77	-5.28
35	-4.58	1.75	2.52	14.33	2.30	31.32	-5.18	-5.29
36	-6.27	1.70	2.46	14.36	2.42	31.69	-5.05	-5.67
37	-5.85	1.50	2.37	14.83	2.33	29.47	-5.19	-5.62
38	-4.86	1.37	2.45	14.20	2.32	30.31	-5.13	-5.42
39	-6.24	1.66	2.42	14.65	2.30	31.65	-5.26	-5.40

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
0	-10.74	-128.40	-12.06	-27.68	-47.57	129.47	-144.26	5.82
1	-10.79	-126.25	-6.60	-27.37	30.05	129.30	-151.57	0.18
2	-10.74	-126.91	22.40	-24.92	8.00	127.04	-148.75	4.11
3	-10.77	-125.83	-0.47	-27.97	-22.77	127.09	-144.81	-5.16
4	-10.78	-123.99	31.65	-28.65	24.12	128.49	-154.50	13.78
5	-10.58	-125.15	-1.09	-27.15	-24.40	127.57	-149.54	5.15
6	-10.85	-124.37	-9.75	-27.32	-4.55	126.41	-145.43	-4.92
7	-10.65	-127.00	0.46	-27.02	-13.23	129.75	-144.09	-13.95
8	-11.02	-126.29	-13.90	-26.51	-7.75	127.78	-152.92	8.36
9	-10.68	-124.07	-7.82	-27.67	-23.63	127.87	-150.02	8.26
10	-10.83	-124.06	47.68	-28.79	7.39	124.57	-149.34	-3.62

11	-10.80	-128.19	-2.30	-25.63	-6.57	129.60	-150.58	-9.25
12	-10.87	-124.73	-54.90	-24.31	41.58	134.82	-151.79	-4.73
13	-10.82	-127.07	-0.98	-28.09	7.94	130.29	-148.24	-10.15
14	-11.01	-125.00	-2.61	-26.15	-9.59	129.00	-148.98	-19.14
15	-10.69	-125.55	6.16	-27.16	5.81	130.83	-152.90	-9.64
16	-11.43	-133.31	27.27	-18.30	-26.33	148.91	-121.02	7.67
17	-11.74	-134.46	-52.34	-16.89	-23.99	151.23	-120.04	-19.73
18	-11.45	-130.80	18.52	-16.85	-48.77	150.98	-122.92	10.85
19	-11.27	-133.08	-29.76	-17.58	3.69	149.84	-121.37	-4.93
20	-11.27	-134.65	8.68	-16.64	34.24	149.00	-118.06	-9.55
21	-11.71	-137.87	30.52	-14.52	0.44	149.28	-119.23	7.09
22	-11.44	-133.43	27.53	-14.99	8.31	149.32	-122.39	-19.43
23	-11.34	-135.85	-13.46	-14.85	0.51	149.46	-118.80	12.65
24	-11.46	-134.89	-32.30	-18.64	3.14	145.98	-119.97	5.91
25	-10.92	-135.14	-16.97	-16.50	9.98	148.08	-126.66	0.91
26	-11.38	-132.17	38.54	-18.23	23.72	150.56	-114.45	1.06
27	-11.44	-134.82	-43.20	-14.82	-21.37	150.11	-123.74	-7.24
28	-11.64	-136.08	-8.20	-17.05	0.87	153.36	-125.50	18.53
29	-11.25	-132.85	-28.20	-17.77	-16.83	146.70	-118.61	-2.81
30	-11.21	-133.55	-22.12	-16.42	-9.17	150.89	-123.39	7.29
31	-11.50	-135.59	16.35	-17.71	-39.44	149.92	-117.21	-17.87
32	-1.75	-17.34	-4.56	-15.38	-4.53	27.52	-4.76	-15.17
33	-1.66	-18.40	-1.44	-15.69	-11.54	27.41	-2.83	-8.66
34	-1.72	-17.15	3.10	-14.93	-13.85	27.41	0.29	18.03
35	-1.60	-18.29	10.53	-16.01	10.69	27.36	-0.67	-6.10
36	-1.70	-17.23	-4.85	-14.95	7.00	25.49	-2.44	-2.75
37	-1.65	-17.09	-18.55	-15.00	3.78	27.40	1.75	3.17
38	-1.68	-17.34	10.25	-15.93	-2.93	27.08	-9.32	12.10
39	-1.66	-16.95	6.80	-16.81	-3.07	27.33	-0.14	-2.79

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
0	37.14	79.32	-51.17	-1.76	107.54	-136.45	-0.70	49.45
1	39.73	78.51	-53.02	-3.67	107.24	-137.40	-0.92	48.53
2	38.32	79.55	-53.33	6.10	109.08	-137.60	-0.44	47.48
3	43.04	78.76	-49.78	-5.20	109.63	-137.79	-0.41	49.72
4	41.97	78.51	-51.25	-3.82	108.30	-134.67	-0.72	47.28
5	43.91	79.20	-53.25	5.02	104.46	-134.65	-0.72	50.25
6	41.71	79.71	-51.24	9.31	106.65	-135.02	-0.97	48.74
7	44.52	80.29	-48.51	-2.75	107.04	-140.33	0.13	49.50
8	39.20	80.04	-52.02	-1.21	109.14	-135.03	-0.87	48.70
9	42.96	79.10	-48.95	0.10	107.27	-135.84	-1.15	51.11
10	42.98	80.11	-51.37	2.86	107.63	-136.54	-1.02	49.77
11	41.67	78.52	-54.87	9.25	107.78	-137.56	-0.36	49.93
12	41.54	78.66	-49.72	12.58	108.48	-138.05	-1.51	50.82
13	45.53	79.40	-53.00	-0.90	106.29	-138.06	-1.22	48.46
14	41.05	78.05	-50.05	5.44	108.17	-134.77	-1.00	49.65
15	40.71	80.07	-56.73	-3.43	106.52	-136.43	-0.06	50.09
16	37.90	83.28	-64.08	-0.55	90.49	-120.94	4.79	58.73
17	39.35	84.15	-61.34	14.09	91.22	-115.20	4.42	58.38
18	39.25	83.87	-64.28	-2.73	90.15	-119.81	5.17	59.14
19	44.51	83.85	-64.77	-5.83	91.64	-120.52	4.96	59.80
20	39.23	84.11	-64.08	-0.47	90.61	-119.79	4.41	58.04
21	33.64	84.97	-66.24	2.00	90.84	-119.05	4.36	55.65

22	40.09	84.70	-68.59	8.97	89.54	-122.30	4.21	58.89
23	39.94	84.58	-64.95	9.90	90.45	-118.55	4.20	58.47
24	41.44	83.21	-61.49	21.96	86.40	-121.01	5.88	59.02
25	43.13	84.06	-71.09	5.92	91.68	-121.74	5.43	58.39
26	40.08	84.02	-62.57	-16.45	90.01	-124.94	5.37	59.22
27	42.77	83.17	-60.93	-3.80	90.21	-119.34	4.16	59.00
28	38.19	83.14	-68.75	0.93	89.45	-118.83	4.65	58.62
29	40.28	84.08	-66.06	2.04	90.85	-121.14	4.95	58.58
30	38.60	83.57	-63.85	4.02	90.34	-121.73	5.02	59.52
31	38.72	83.98	-62.23	-8.14	90.87	-118.15	4.47	57.21
32	11.57	21.55	-10.03	-6.33	68.62	-1.53	-1.26	5.29
33	12.52	21.60	-9.90	5.14	67.28	-2.03	-1.17	6.06
34	10.52	21.27	-9.92	7.48	62.26	-1.26	-1.10	6.49
35	9.72	21.00	-9.55	4.65	67.09	-2.90	-1.42	5.32
36	10.18	21.30	-10.46	0.53	68.50	-1.87	-1.28	5.86
37	11.41	21.45	-10.21	-5.58	69.19	-2.94	-0.93	6.38
38	10.20	21.55	-10.48	-0.02	72.00	-2.62	-1.45	5.82
39	10.76	21.90	-9.25	2.27	70.02	-3.28	-1.69	5.49

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
0	-12.86	129.34	-0.04	14.92	138.37	-1.55	89.62	30.73
1	-5.85	128.69	0.01	15.19	137.24	-1.35	-3.02	-80.51
2	-4.35	128.38	0.04	15.14	134.51	-1.28	41.45	-64.49
3	-12.07	129.28	0.00	15.99	138.44	-1.79	43.17	-32.41
4	-29.61	130.64	0.05	15.92	138.34	-1.07	-61.32	-48.06
5	-8.28	128.31	0.04	16.59	134.68	-1.42	22.30	2.66
6	19.35	130.30	0.02	13.87	136.49	-1.67	-84.92	0.79
7	24.36	127.83	0.04	15.94	138.86	-1.90	0.65	-3.65
8	-14.01	128.40	0.03	15.30	137.88	-1.44	-4.38	-31.18
9	0.91	129.12	0.02	14.12	135.10	-1.55	69.96	4.18
10	23.68	129.32	0.07	16.18	140.35	-1.81	-95.92	-40.96
11	-17.89	127.87	-0.02	14.63	137.22	-1.64	-53.97	-21.93
12	-16.16	127.54	0.03	15.68	139.52	-1.33	-2.36	-18.42
13	-9.14	128.00	0.04	15.45	137.55	-1.47	-154.65	-37.01
14	6.32	128.51	0.01	15.55	135.27	-1.61	-75.40	17.78
15	-56.04	128.95	-0.01	15.81	140.72	-1.50	15.30	-0.58
16	11.07	103.86	0.32	12.40	106.49	-3.15	-17.66	-6.34
17	1.20	103.57	0.30	11.80	105.25	-3.35	158.30	-11.96
18	15.81	104.02	0.31	12.73	105.76	-3.14	27.25	38.46
19	-38.30	105.03	0.26	13.24	105.45	-3.08	55.65	-53.71
20	-11.59	103.10	0.28	12.71	104.03	-3.53	-163.96	49.02
21	-12.52	102.37	0.36	13.19	102.12	-2.92	-173.39	-5.65
22	-9.62	103.25	0.33	12.39	108.93	-2.99	52.22	-161.59
23	25.92	103.48	0.29	11.74	106.99	-2.55	105.11	-85.85
24	-5.50	103.73	0.35	12.51	104.40	-3.33	69.39	-5.04
25	-10.31	105.43	0.32	12.82	105.99	-3.71	-26.84	-118.93
26	18.71	106.00	0.30	10.96	106.89	-3.08	-32.36	-84.52
27	20.64	104.18	0.31	10.98	105.86	-2.84	-60.72	-27.19
28	-15.68	105.50	0.30	12.70	106.94	-2.97	88.69	23.52
29	-13.79	104.57	0.35	11.66	106.15	-3.25	26.48	-34.70
30	0.58	103.49	0.33	12.68	105.58	-2.79	-41.33	6.57
31	-41.18	104.89	0.34	12.31	105.97	-3.75	29.53	-6.35
32	1.27	44.57	0.12	13.53	42.01	2.37	17.83	3.10

33	6.12	45.81	0.10	13.35	42.85	2.35	-40.89	-6.28
34	-2.92	44.37	0.10	13.49	42.40	2.50	118.60	-8.15
35	-5.72	45.65	0.09	13.59	42.95	2.35	-24.50	42.41
36	8.16	46.19	0.13	13.25	42.19	2.58	-23.66	-14.21
37	-2.52	45.21	0.10	14.30	42.49	2.21	10.46	8.05
38	1.03	44.35	0.08	13.11	42.30	2.30	-19.72	-34.53
39	7.84	44.14	0.07	12.77	41.49	2.14	-87.32	-9.49

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
40	19.90	-2.13	3.40	-23.58	14.75	9.48	3.86	-5.28
41	18.83	-2.16	2.85	-22.88	14.32	8.82	4.18	-4.88
42	20.70	-1.65	3.27	-20.35	15.25	9.24	3.58	-4.35
43	20.25	-1.77	3.20	-22.22	15.27	9.16	3.48	-3.80
44	24.14	-2.06	3.32	-21.09	15.40	8.40	4.18	-4.42
45	18.27	-1.69	3.27	-23.09	14.59	8.75	3.83	-4.73
46	25.79	-2.16	2.67	-23.50	13.93	9.50	3.72	-4.28
47	20.38	-2.29	3.08	-22.68	15.06	8.65	2.52	-5.19
48	50.37	-1.77	3.59	-24.60	18.28	10.50	5.85	-7.96
49	49.85	-2.31	3.47	-28.05	18.13	10.17	7.16	-7.75
50	41.67	-2.30	3.79	-26.10	18.67	10.55	6.25	-8.25
51	42.09	-2.00	3.53	-25.53	18.63	10.87	5.39	-5.86
52	50.69	-2.14	3.14	-26.33	19.35	9.35	6.52	-7.00
53	42.49	-1.43	3.63	-27.22	19.41	9.91	6.84	-7.48
54	53.71	-2.44	3.50	-31.61	18.14	11.45	6.06	-9.48
55	46.99	-2.19	3.48	-29.49	18.31	10.81	6.44	-6.38
56	51.56	-2.37	2.97	-27.93	19.01	11.39	5.33	-6.22
57	46.94	-2.41	3.00	-30.06	17.38	9.09	6.34	-7.10
58	41.31	-2.75	3.65	-25.85	19.08	9.83	6.07	-6.96
59	44.49	-2.22	3.61	-32.55	18.31	10.01	6.66	-7.11
60	45.40	-2.42	3.22	-27.27	18.00	11.33	6.80	-7.85
61	48.69	-2.05	3.04	-27.02	19.11	10.47	6.21	-8.31
62	51.91	-2.10	3.23	-24.02	18.61	10.14	6.24	-8.18
63	49.12	-2.03	3.79	-26.05	19.92	9.39	6.41	-7.48
64	14.82	-1.26	2.54	-14.92	10.38	5.70	4.36	-2.13
65	18.85	-1.68	2.35	-15.32	10.60	5.67	4.25	-2.41
66	12.30	-1.35	2.98	-13.25	10.11	6.36	5.11	-2.88
67	10.43	-0.99	2.47	-13.28	10.38	6.20	5.13	-2.15
68	10.66	-1.07	2.81	-16.65	10.22	5.83	3.98	-2.43
69	14.27	-1.20	2.24	-13.79	9.68	4.91	3.57	-2.61
70	14.63	-0.95	2.44	-13.66	10.38	4.72	4.73	-2.55
71	13.43	-1.33	2.57	-16.19	9.82	5.72	4.41	-2.84
72	12.29	-1.25	2.24	-13.42	9.93	5.64	4.34	-2.66
73	13.65	-1.66	2.41	-16.32	10.52	5.40	4.92	-2.50
74	13.37	-1.27	2.47	-15.49	9.98	5.94	3.65	-1.62
75	14.10	-1.64	2.34	-14.76	10.05	5.40	4.82	-2.77
76	16.53	-1.24	1.83	-18.07	9.51	6.44	4.40	-2.85
77	15.73	-1.15	2.18	-13.81	9.36	5.62	4.61	-2.36
78	17.20	-1.32	2.68	-13.10	9.52	5.70	4.78	-2.81
79	18.10	-1.03	2.61	-14.46	10.11	5.73	4.83	-2.49

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
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40	-0.48	-14.13	-0.26	-4.85	-32.91	4.45	1.62	14.53
41	-2.55	-14.69	-0.26	-2.59	-32.83	4.48	1.64	10.87
42	-2.32	-13.82	-0.25	-3.21	-32.11	4.71	1.92	12.41
43	0.25	-13.86	-0.32	-0.74	-32.76	4.46	2.18	10.08
44	-2.98	-14.88	-0.29	-0.15	-33.19	4.33	1.78	12.83
45	0.26	-14.82	-0.34	-3.73	-35.01	4.39	1.93	11.13
46	-0.55	-13.52	-0.27	-2.21	-32.26	4.23	1.58	13.57
47	-2.58	-14.57	-0.27	-1.27	-32.66	4.42	1.55	11.73
48	-9.49	-17.26	-0.40	-7.16	-36.29	5.87	2.34	25.11
49	-15.26	-17.26	-0.39	-9.71	-37.64	6.04	2.21	24.19
50	-17.74	-16.05	-0.38	-8.14	-37.86	5.44	2.03	27.35
51	-7.51	-17.69	-0.44	-7.78	-36.26	6.10	2.27	26.16
52	-16.09	-16.80	-0.45	-7.39	-35.79	6.14	2.29	22.75
53	-21.20	-18.29	-0.46	-6.47	-37.56	6.01	2.73	22.73
54	-14.84	-18.42	-0.34	-6.94	-35.71	6.25	2.47	23.41
55	-16.75	-17.24	-0.39	-7.01	-38.93	6.20	2.67	26.48
56	-8.39	-22.20	-0.43	-6.67	-40.21	5.83	2.38	22.94
57	-14.66	-19.71	-0.36	-8.63	-40.12	6.05	2.57	23.51
58	-16.77	-15.89	-0.40	-8.32	-38.01	6.03	2.53	23.62
59	-16.13	-20.60	-0.48	-7.39	-36.88	6.09	2.39	23.28
60	-14.21	-16.48	-0.41	-8.28	-39.00	5.52	2.26	23.17
61	-14.30	-22.10	-0.43	-8.26	-37.30	6.59	2.03	22.66
62	-17.72	-18.12	-0.41	-7.89	-37.81	6.13	2.04	25.89
63	-17.68	-18.26	-0.39	-6.30	-36.11	5.40	2.04	21.67
64	2.73	-5.49	-0.33	1.00	-13.27	1.08	2.61	11.70
65	-4.20	-5.79	-0.38	-1.97	-13.62	1.34	2.28	11.01
66	-5.39	-6.55	-0.36	0.72	-12.79	0.98	2.48	13.80
67	-4.50	-5.76	-0.33	-1.87	-13.93	0.96	2.58	11.78
68	-4.01	-5.60	-0.37	-1.22	-14.53	1.28	2.37	12.48
69	-6.29	-5.07	-0.33	0.39	-14.47	1.50	2.31	13.61
70	-6.11	-5.86	-0.40	-0.83	-14.26	1.23	2.63	10.62
71	-8.03	-5.00	-0.32	-0.78	-12.45	1.14	2.62	12.27
72	-6.81	-6.73	-0.38	0.34	-14.42	1.52	2.49	12.46
73	-7.12	-6.95	-0.33	0.12	-13.53	1.04	2.45	10.32
74	-6.57	-3.28	-0.31	0.22	-14.17	1.09	2.41	13.68
75	-6.00	-3.61	-0.33	-0.67	-13.40	1.23	2.67	12.09
76	-5.23	-7.88	-0.38	-0.19	-14.11	1.19	2.52	13.16
77	-7.27	-3.96	-0.32	1.76	-14.15	1.23	2.79	11.77
78	-3.39	-7.77	-0.34	-0.85	-13.00	1.32	2.61	10.48
79	-4.11	-6.10	-0.36	-1.01	-13.69	1.30	2.55	11.87

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
40	-1.34	7.31	20.08	-43.84	7.06	8.83	3.34	1.08
41	-1.10	7.79	19.71	8.41	7.32	8.92	15.86	1.04
42	1.31	8.00	19.19	-20.28	6.83	8.82	24.92	1.08
43	1.45	7.98	19.87	46.25	6.78	8.45	-2.96	1.00
44	0.33	7.34	19.54	-38.90	6.92	8.67	-12.94	1.11
45	1.75	7.10	20.14	-21.37	6.82	8.73	-27.46	1.13
46	0.61	8.00	20.03	-28.34	7.09	8.93	31.83	1.12
47	-0.64	7.87	19.53	-46.57	7.31	8.89	-30.39	1.07
48	-0.15	11.06	25.54	-2.80	12.05	11.55	-13.03	1.27
49	-0.12	11.11	25.52	8.12	12.20	12.65	-6.52	1.32
50	-1.32	10.83	24.11	37.99	12.63	11.38	32.37	1.35

51	0.13	10.49	25.74	8.31	12.13	11.86	-5.37	1.28
52	-1.83	11.17	26.01	33.77	12.42	11.00	34.57	1.29
53	-1.54	11.10	25.30	42.41	11.53	11.16	50.78	1.36
54	0.86	10.54	23.84	-10.95	12.27	11.29	7.27	1.27
55	0.11	11.02	25.47	-32.01	11.40	11.13	0.00	1.32
56	1.25	10.62	26.27	11.39	11.63	11.39	0.65	1.32
57	-1.16	10.55	25.32	76.10	12.37	11.06	17.89	1.41
58	-0.85	11.14	24.88	25.65	11.81	11.47	-34.03	1.31
59	-1.57	10.72	25.21	-19.84	11.67	12.12	-12.65	1.17
60	0.05	10.51	24.90	-78.84	12.63	12.01	-6.74	1.37
61	-1.47	10.82	25.26	46.37	12.04	11.15	-18.96	1.39
62	1.03	10.49	25.79	18.87	12.11	11.18	-7.43	1.35
63	0.25	10.96	26.25	59.58	11.55	11.69	-2.09	1.42
64	-0.42	5.86	11.16	66.99	8.70	6.34	31.58	1.06
65	1.35	5.34	11.18	14.04	8.27	6.43	59.34	1.01
66	0.05	5.54	10.70	-52.21	8.29	6.36	6.84	1.05
67	0.45	5.43	11.31	18.29	7.77	6.55	8.15	1.01
68	-0.61	6.04	9.77	28.80	7.99	6.69	3.61	1.10
69	-0.84	5.57	10.58	42.30	8.11	6.44	-41.16	1.11
70	0.34	5.48	10.22	22.47	8.03	6.57	-21.22	1.11
71	-0.40	6.13	10.13	-19.36	8.25	6.63	-9.41	1.03
72	-0.74	5.96	10.71	15.47	7.99	6.60	-11.79	1.10
73	0.30	5.77	10.78	-35.45	8.02	7.23	-7.79	1.12
74	0.31	6.04	9.85	-2.67	7.43	6.23	-16.70	0.99
75	0.07	5.67	11.17	-30.70	7.82	6.03	-41.32	1.03
76	0.91	6.31	10.37	65.63	7.65	6.38	-19.29	1.11
77	-0.55	5.90	10.02	41.19	7.65	6.85	26.33	1.08
78	0.38	5.93	10.95	-33.73	8.60	6.71	15.83	1.07
79	0.08	5.82	10.18	55.66	8.03	6.07	15.49	1.06

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
40	-1.53	1.02	52.10	-0.48	16.66	0.51	-0.48	3.17
41	1.12	0.85	50.58	-0.66	16.94	-0.20	-1.60	1.47
42	-0.14	0.54	49.36	-0.28	16.62	-0.54	-0.84	2.34
43	1.89	-0.17	50.32	-0.77	16.71	0.11	-0.84	2.40
44	2.72	0.93	51.66	0.03	16.64	-0.52	-0.93	3.31
45	0.90	0.89	50.79	-0.49	16.85	0.29	-0.30	2.02
46	0.92	1.06	49.05	-0.55	15.95	-0.36	-0.75	2.36
47	-1.10	0.50	50.20	-0.34	16.34	-0.29	-0.96	2.86
48	1.43	2.74	66.38	-2.26	21.67	0.95	-5.11	7.11
49	0.16	3.22	63.74	-1.00	21.20	1.23	-4.04	5.34
50	1.56	2.91	65.05	-1.22	20.65	1.57	-3.43	4.36
51	-0.08	2.84	63.90	-2.64	21.54	-0.16	-4.06	5.01
52	-1.22	3.14	64.73	-2.15	20.73	1.26	-3.17	6.76
53	2.12	3.35	63.77	-2.31	20.64	1.90	-2.79	4.15
54	-0.09	3.01	65.03	-2.19	21.60	0.00	-3.75	3.41
55	2.71	3.52	62.85	-2.18	21.41	1.02	-3.95	4.22
56	-1.83	3.25	64.90	-2.23	20.87	1.17	-5.24	6.48
57	-1.32	3.01	65.44	-1.67	20.99	0.86	-4.19	4.77
58	-1.32	3.42	67.01	-1.82	20.89	0.54	-2.96	6.01
59	-1.20	3.13	67.59	-2.06	21.62	0.29	-4.53	7.54
60	3.64	2.66	65.10	-1.51	20.53	2.64	-2.72	4.43
61	-2.11	2.97	65.31	-2.06	21.06	1.06	-5.37	5.61

62	1.09	2.90	65.18	-2.28	21.55	1.93	-3.25	4.31
63	3.76	3.61	62.63	-3.22	19.95	0.38	-4.80	6.77
64	2.01	3.49	35.08	-1.73	7.78	-0.78	0.64	-7.62
65	0.65	3.75	37.57	-1.59	8.16	0.25	1.16	-6.40
66	-0.59	3.12	35.67	-1.59	8.70	-0.65	0.44	-7.70
67	0.84	3.74	36.42	-1.44	8.46	0.59	0.32	-7.58
68	1.82	3.81	36.20	-0.96	8.04	0.50	0.78	-6.98
69	-3.57	3.82	34.70	-1.26	8.63	0.05	0.45	-6.48
70	-0.26	3.34	35.18	-1.92	8.23	-0.19	0.54	-6.45
71	0.42	3.34	36.22	-1.78	7.93	0.71	1.11	-6.50
72	-2.08	3.40	36.98	-1.86	8.30	-0.10	-0.07	-6.00
73	0.88	3.39	36.43	-1.80	8.82	1.05	2.09	-6.28
74	0.13	3.49	37.23	-1.68	7.89	0.03	0.84	-6.37
75	0.72	3.63	34.17	-1.43	8.40	-0.15	1.45	-6.18
76	1.16	3.70	37.28	-1.47	8.24	0.62	0.36	-6.41
77	-1.66	3.54	36.23	-1.32	8.74	-0.46	1.47	-6.34
78	0.57	3.54	37.16	-1.52	8.39	0.23	0.68	-5.44
79	1.19	3.07	36.90	-2.02	8.87	0.19	0.45	-6.90

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
40	-5.34	1.56	2.37	14.21	2.38	29.25	-5.24	-4.95
41	-5.33	1.96	2.51	14.89	2.30	29.91	-4.95	-5.64
42	-4.23	1.47	2.50	14.89	2.39	30.13	-5.49	-5.67
43	-5.61	1.39	2.40	14.13	2.39	30.04	-5.16	-5.43
44	-5.26	1.42	2.47	14.70	2.47	29.51	-5.46	-5.65
45	-4.85	1.38	2.49	14.53	2.35	31.15	-5.49	-5.35
46	-5.17	1.42	2.66	14.21	2.40	31.82	-5.15	-5.79
47	-5.68	1.57	2.53	15.07	2.40	29.24	-5.09	-5.72
48	-6.00	2.58	2.50	14.63	3.20	38.48	-3.62	-5.61
49	-6.67	2.35	2.72	14.89	3.23	37.63	-3.69	-6.09
50	-5.81	2.30	2.52	15.66	3.28	38.95	-4.26	-6.18
51	-5.13	2.44	2.67	15.54	3.18	36.41	-4.09	-5.84
52	-4.42	1.99	2.54	15.71	3.13	39.60	-4.01	-6.18
53	-5.04	2.06	2.66	17.16	3.12	36.76	-4.33	-6.06
54	-6.10	2.53	2.50	16.41	3.27	37.65	-4.36	-6.06
55	-5.57	2.26	2.60	16.96	3.16	40.02	-3.85	-6.00
56	-4.10	1.92	2.64	14.69	3.17	39.48	-4.48	-6.07
57	-4.38	2.29	2.61	15.62	3.22	38.06	-3.97	-6.03
58	-5.50	2.09	2.58	16.36	3.27	40.47	-3.28	-5.79
59	-4.45	2.46	2.78	16.60	3.15	39.11	-3.78	-6.16
60	-3.34	2.51	2.52	15.15	3.23	38.56	-4.02	-5.90
61	-5.04	2.20	2.58	16.64	3.16	38.05	-4.02	-5.69
62	-4.07	2.51	2.61	16.88	3.27	38.29	-3.92	-6.25
63	-5.47	2.46	2.56	17.19	3.20	36.88	-3.81	-5.98
64	-3.26	1.93	2.58	6.64	1.79	15.55	-0.68	-6.62
65	-1.96	1.80	2.75	6.53	1.80	14.47	-1.32	-6.52
66	-2.93	1.95	2.58	6.37	1.80	13.31	-1.55	-6.16
67	-2.63	1.77	2.78	6.77	1.79	14.86	-0.93	-6.19
68	-3.09	2.07	2.62	6.63	1.85	13.31	-0.92	-6.44
69	-2.86	1.95	2.66	6.79	1.74	13.74	-1.00	-6.61
70	-1.71	1.92	2.68	5.20	1.78	15.31	-1.48	-6.46
71	-3.02	1.93	2.72	6.27	1.83	13.72	-1.27	-6.62
72	-2.50	2.08	2.62	6.55	1.75	12.96	-1.31	-6.45

73	-2.41	2.11	2.70	6.32	1.82	15.33	-1.06	-6.63
74	-2.92	2.10	2.73	6.78	1.77	15.18	-1.17	-6.35
75	-2.50	1.93	2.62	6.95	1.85	15.54	-1.22	-6.55
76	-2.98	2.03	2.66	6.13	1.77	14.15	-1.40	-6.31
77	-2.17	1.67	2.73	6.52	1.86	12.99	-0.97	-6.39
78	-3.26	1.85	2.79	5.93	1.78	15.29	-1.24	-6.51
79	-2.63	2.13	2.65	6.26	1.82	14.10	-1.09	-6.49

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
40	-1.63	-17.33	-15.14	-16.17	-29.42	28.83	-1.30	-12.03
41	-1.62	-16.73	15.47	-15.91	-10.45	27.24	-4.18	-15.77
42	-1.76	-17.69	-4.99	-16.03	12.60	28.26	-1.18	19.16
43	-1.73	-16.97	12.09	-15.62	-11.85	26.94	-0.37	0.46
44	-1.61	-17.12	-5.19	-15.90	2.02	26.26	-0.46	-5.74
45	-1.70	-17.11	2.87	-15.47	-1.41	27.27	2.06	2.50
46	-1.71	-18.19	-5.68	-15.33	-1.19	27.62	1.22	2.53
47	-1.69	-17.28	-11.69	-15.44	-22.75	27.70	-1.62	1.81
48	-2.35	-23.98	-5.23	-19.65	22.91	35.28	-12.08	-13.87
49	-2.43	-27.18	5.93	-18.98	-27.08	38.11	-20.01	-4.01
50	-2.31	-25.96	-1.21	-17.47	-3.80	36.94	-16.15	0.74
51	-2.43	-25.54	-5.39	-18.91	2.69	37.51	-17.32	-10.40
52	-2.34	-25.63	-0.54	-16.74	-13.52	38.11	-12.87	1.63
53	-2.28	-26.61	18.39	-19.47	38.54	36.67	-15.14	-8.79
54	-2.24	-25.23	0.96	-18.66	-4.01	37.97	-12.54	3.16
55	-2.46	-26.20	3.85	-20.41	-20.02	38.86	-9.05	-6.67
56	-2.20	-26.94	-5.01	-19.81	20.93	36.49	-12.64	1.44
57	-2.24	-26.94	-1.68	-18.68	-1.71	36.78	-9.21	13.52
58	-2.27	-24.52	0.95	-18.90	-12.42	39.65	-21.43	-15.86
59	-2.33	-25.67	-10.70	-19.74	1.12	39.45	-18.24	7.89
60	-2.30	-26.36	11.17	-18.81	-2.69	39.28	-17.71	-4.19
61	-2.29	-24.66	3.78	-19.38	-19.97	38.21	-17.79	-5.47
62	-2.42	-24.80	11.78	-19.14	3.75	38.29	-17.86	-0.73
63	-2.44	-24.51	-21.78	-18.50	23.88	36.49	-15.04	-0.89
64	-1.83	-13.40	-11.14	-13.47	6.68	20.28	-11.60	8.74
65	-1.62	-15.70	1.68	-12.90	6.44	19.59	-9.48	-7.91
66	-1.80	-13.18	-15.05	-13.18	1.99	18.97	-9.93	-3.74
67	-1.63	-13.75	-7.33	-13.37	-23.16	18.82	-6.33	-1.89
68	-1.76	-13.83	8.46	-12.67	5.10	20.29	-9.76	5.80
69	-1.74	-13.90	1.52	-12.82	-22.21	18.64	-11.50	-0.64
70	-1.78	-14.92	1.24	-12.89	4.97	19.29	-7.71	-4.83
71	-1.90	-13.09	-9.94	-13.40	-18.68	19.39	-5.61	6.31
72	-1.81	-14.30	-9.16	-13.84	13.20	18.53	-6.02	-1.96
73	-1.76	-14.66	-2.27	-12.31	-4.24	18.90	-6.41	14.15
74	-1.79	-13.88	-6.26	-12.21	-0.23	19.15	-10.59	-0.15
75	-1.64	-13.15	2.92	-13.45	-9.89	19.10	-10.12	-5.30
76	-1.77	-13.07	4.78	-13.52	-1.98	19.49	-8.17	-0.11
77	-1.67	-13.32	0.73	-13.84	5.22	20.52	-11.56	-5.77
78	-1.77	-13.99	-4.65	-12.55	-8.03	18.97	-7.84	0.31
79	-1.70	-13.62	3.32	-13.01	5.30	20.11	-7.49	7.78

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
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40	10.47	21.27	-10.21	0.10	68.25	-3.70	-1.40	6.15
41	10.05	21.29	-10.73	-5.80	66.81	-3.54	-1.12	5.55
42	9.65	21.44	-10.41	3.16	64.95	-2.16	-1.26	5.51
43	10.09	20.98	-9.71	-1.86	66.84	-1.27	-1.51	5.68
44	10.76	21.46	-10.07	-3.57	67.32	-1.94	-1.25	6.11
45	12.36	21.09	-11.19	-6.00	66.15	-1.00	-1.45	5.81
46	9.57	21.06	-10.42	-7.61	64.58	-2.32	-1.27	4.69
47	10.48	21.27	-8.46	-5.23	67.39	-2.78	-1.50	5.91
48	19.85	26.07	-17.78	5.77	76.70	-5.97	-1.33	7.34
49	17.76	26.10	-18.03	-11.44	73.40	-5.48	-1.67	5.96
50	19.37	25.92	-17.34	-2.21	75.97	-4.95	-1.61	6.77
51	19.60	26.19	-15.93	1.24	77.29	-6.90	-0.94	6.74
52	20.65	25.47	-17.51	-0.40	74.81	-5.15	-1.53	6.92
53	20.51	24.90	-17.03	6.78	78.03	-4.57	-1.38	7.08
54	19.00	25.89	-16.67	-8.22	78.44	-7.80	-0.90	5.93
55	19.61	26.07	-16.10	-4.31	75.29	-5.28	-0.84	7.33
56	17.19	25.63	-17.16	-4.83	76.12	-6.28	-1.30	7.15
57	16.12	26.62	-18.85	-10.03	76.11	-7.80	-1.48	6.26
58	22.83	25.80	-18.76	0.78	79.84	-6.48	-1.62	6.51
59	18.77	25.66	-19.39	-3.45	75.39	-5.64	-1.89	5.89
60	18.65	26.05	-17.83	9.61	75.07	-4.95	-1.50	7.25
61	20.10	25.50	-17.72	5.46	79.24	-5.64	-0.75	6.04
62	18.06	26.18	-16.77	-0.60	75.60	-5.70	-1.29	5.48
63	20.15	26.14	-18.30	0.52	77.40	-5.65	-1.63	6.15
64	4.17	11.78	-11.14	-6.19	47.76	4.84	-0.10	1.22
65	4.95	11.91	-11.29	-7.40	49.57	5.36	-0.13	0.55
66	5.23	11.69	-11.24	2.25	47.65	3.93	-0.44	1.66
67	4.34	11.68	-12.65	-2.83	44.20	6.00	-0.18	1.83
68	4.77	11.73	-10.81	-7.30	48.16	4.80	0.07	1.50
69	5.63	12.02	-11.26	-5.67	46.34	4.37	-0.32	1.27
70	6.80	11.44	-12.79	4.52	46.72	4.09	-0.01	1.03
71	4.40	11.61	-10.90	3.28	46.91	5.31	-0.15	1.20
72	5.63	11.86	-9.84	-2.76	47.55	4.03	-0.09	1.07
73	5.57	11.65	-11.73	4.00	48.09	5.29	-0.14	1.55
74	6.56	12.02	-11.78	5.11	47.54	4.43	-0.34	1.50
75	4.76	11.61	-12.34	-1.54	46.82	4.11	-0.51	1.09
76	3.83	11.41	-11.62	5.85	45.35	5.55	-0.03	1.42
77	5.23	11.93	-10.66	-2.98	46.92	4.34	-0.39	1.13
78	6.95	12.23	-11.66	2.36	44.43	5.72	-0.37	1.19
79	5.55	11.92	-10.93	2.02	45.36	5.01	-0.28	1.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
40	-13.21	46.06	0.12	12.76	40.97	2.21	-5.81	30.19
41	5.60	43.87	0.12	12.55	40.42	2.20	-5.66	10.59
42	1.30	43.78	0.08	13.33	42.36	2.34	104.95	11.44
43	-7.10	45.74	0.12	12.70	41.80	2.17	53.45	1.42
44	-15.01	43.41	0.08	13.65	40.16	2.34	7.38	7.04
45	-4.50	44.35	0.10	13.38	42.10	2.24	-60.08	29.55
46	-6.08	43.93	0.11	12.88	41.53	2.30	43.67	-37.28
47	-1.42	43.77	0.10	12.83	42.02	2.03	-33.22	-1.93
48	7.40	55.67	0.34	15.31	49.80	3.58	-32.69	-7.42
49	0.03	55.98	0.35	15.00	48.94	3.07	-12.52	-12.64
50	13.40	56.33	0.38	16.14	50.79	2.89	79.93	-57.05

51	-14.93	53.37	0.35	15.22	51.09	2.85	70.33	27.06
52	2.81	54.09	0.35	15.14	49.42	3.15	3.52	-55.33
53	-6.01	56.32	0.35	14.90	50.48	2.85	-83.24	-53.13
54	-22.63	56.03	0.32	15.27	49.90	2.96	131.84	-18.47
55	-12.38	57.33	0.34	14.90	48.67	2.74	61.03	8.29
56	15.92	55.65	0.35	16.30	50.40	2.95	-128.30	-29.08
57	5.86	54.69	0.34	14.83	50.68	3.04	60.23	14.95
58	6.32	56.35	0.37	16.09	50.08	3.27	-61.65	-31.02
59	-4.18	54.81	0.39	15.12	49.89	2.74	48.95	-25.32
60	-1.82	55.12	0.32	16.16	49.81	2.53	23.54	38.64
61	-6.60	56.15	0.31	14.82	50.61	3.48	-15.81	-62.06
62	13.52	55.60	0.36	14.65	50.81	2.75	-55.36	2.84
63	6.26	55.11	0.35	16.22	50.09	2.78	-38.96	-9.46
64	-6.41	21.29	0.06	13.13	25.53	2.19	3.75	28.78
65	5.40	20.54	0.03	12.43	26.21	2.26	-53.85	-56.62
66	-1.36	21.26	0.08	12.95	24.86	1.82	19.76	-12.86
67	2.78	19.53	0.06	13.77	25.62	1.69	-28.92	2.03
68	-2.41	19.89	0.07	12.87	24.40	2.26	4.00	8.27
69	-0.42	20.76	0.08	13.75	25.35	2.40	-7.41	-16.89
70	2.68	21.11	0.06	12.80	26.02	1.63	-34.09	-1.22
71	-2.07	20.29	0.06	13.23	24.64	2.14	-0.73	27.99
72	13.77	19.43	0.03	12.95	25.40	2.05	-41.90	41.49
73	6.38	21.39	0.07	12.67	25.45	2.27	64.46	10.51
74	14.28	20.99	0.05	12.33	26.63	1.89	72.91	18.93
75	11.50	20.47	0.07	13.27	25.94	1.95	95.35	4.58
76	-2.82	20.91	0.06	12.71	25.71	2.04	58.62	12.36
77	-11.93	21.90	0.02	13.81	25.86	2.21	-10.84	12.70
78	-11.93	20.49	0.07	13.10	25.82	2.10	20.80	15.62
79	-4.90	21.84	0.05	13.09	25.78	1.86	-24.02	21.73

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
80	237.62	-19.42	-15.00	22.16	76.44	7.36	55.13	20.11
81	245.81	-19.54	-15.13	18.05	77.32	7.05	55.20	19.51
82	239.03	-19.32	-14.10	14.18	76.56	7.65	54.42	21.17
83	239.85	-19.51	-14.78	16.79	75.94	7.74	53.67	19.67
84	236.54	-19.33	-14.34	17.57	76.25	5.88	54.22	20.28
85	233.79	-19.10	-14.73	18.82	77.02	7.74	55.42	19.18
86	244.29	-19.38	-14.24	21.06	76.84	7.05	53.39	21.57
87	245.46	-19.23	-14.63	19.52	76.89	6.20	55.06	19.29
88	237.27	-19.52	-15.00	11.51	75.62	6.84	54.76	18.98
89	246.30	-19.38	-14.77	19.82	76.40	7.87	53.51	19.31
90	239.99	-19.66	-14.51	23.01	78.38	7.62	54.41	20.58
91	246.68	-19.32	-14.48	19.30	76.38	6.57	54.02	19.59
92	244.90	-18.74	-14.76	19.33	75.50	8.02	53.86	19.84
93	240.25	-19.46	-14.08	15.51	77.25	6.68	54.41	20.96
94	236.25	-19.15	-14.29	15.40	77.92	8.29	54.72	20.92
95	234.53	-19.51	-14.79	15.95	76.96	7.71	53.95	19.60
96	225.80	-14.96	-12.69	32.62	79.10	-8.15	47.15	29.30
97	224.72	-15.06	-12.31	38.11	78.67	-8.34	47.15	28.12
98	219.05	-15.08	-12.67	33.06	78.99	-8.10	46.83	29.35
99	225.28	-15.07	-12.13	34.86	78.98	-7.64	46.64	29.02
100	219.18	-15.13	-12.51	32.24	79.59	-8.61	46.55	28.06
101	223.01	-14.98	-12.59	32.56	78.79	-7.85	47.48	29.74

102	226.92	-15.06	-12.82	35.50	78.13	-8.46	47.53	29.39
103	218.47	-15.15	-12.70	31.58	77.22	-8.28	47.09	28.69
104	217.40	-15.08	-12.47	31.83	77.80	-8.14	47.10	29.56
105	211.59	-15.13	-12.39	34.55	78.57	-8.92	46.36	27.44
106	218.37	-14.92	-13.15	31.41	78.23	-8.59	47.33	28.58
107	213.20	-15.28	-12.69	32.20	78.63	-8.15	47.25	28.56
108	211.17	-15.16	-12.35	33.61	78.13	-8.64	46.45	29.12
109	225.61	-15.22	-12.14	30.36	76.95	-7.72	47.83	29.32
110	225.64	-15.32	-12.32	30.59	79.54	-8.44	47.85	29.54
111	224.91	-15.05	-12.61	33.73	78.88	-8.69	47.23	29.07
112	-89.51	-0.54	3.33	-33.39	1.42	11.15	2.84	-3.79
113	-86.30	-0.64	3.48	-32.33	1.63	11.26	2.88	-3.91
114	-80.17	-0.41	3.36	-33.65	1.40	11.45	2.87	-2.39
115	-86.94	-0.54	3.66	-33.07	1.89	10.91	3.66	-2.75
116	-80.07	-0.67	3.73	-32.76	1.05	11.32	2.95	-2.93
117	-87.45	-0.91	3.55	-31.94	1.72	11.40	3.03	-3.38
118	-80.27	-0.56	3.49	-31.59	1.38	11.19	3.53	-3.49
119	-90.36	-0.70	3.58	-32.58	1.11	10.70	3.54	-3.50

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
80	-186.45	-104.53	-2.49	-80.09	-134.45	10.24	-14.72	95.88
81	-192.97	-107.51	-2.51	-80.49	-133.61	10.69	-14.16	94.57
82	-187.71	-108.95	-2.46	-79.35	-133.97	10.30	-14.36	97.20
83	-196.02	-105.38	-2.44	-78.86	-135.96	9.68	-14.04	92.95
84	-194.61	-105.27	-2.45	-79.94	-133.74	10.20	-14.61	95.05
85	-198.68	-106.99	-2.42	-81.39	-132.72	10.63	-14.47	94.83
86	-194.46	-107.19	-2.42	-82.50	-134.72	10.43	-14.56	95.38
87	-200.13	-106.43	-2.49	-78.22	-134.53	10.86	-14.80	93.82
88	-193.98	-106.70	-2.47	-81.14	-133.25	10.47	-14.52	93.65
89	-196.03	-107.53	-2.45	-80.79	-133.78	9.95	-14.23	95.79
90	-197.87	-106.99	-2.47	-79.13	-135.12	10.49	-14.67	92.70
91	-196.90	-110.92	-2.43	-82.52	-133.55	10.59	-14.29	95.36
92	-197.47	-106.85	-2.47	-82.24	-133.42	10.28	-14.49	95.94
93	-195.18	-104.82	-2.46	-82.04	-134.55	10.47	-14.71	94.09
94	-195.83	-106.77	-2.42	-79.90	-131.30	10.06	-14.41	92.75
95	-195.81	-104.92	-2.43	-82.97	-133.36	10.49	-14.20	92.97
96	-181.47	-115.06	-2.02	-70.47	-131.47	11.38	-17.41	86.99
97	-188.01	-114.36	-2.00	-72.63	-131.68	10.99	-17.24	89.31
98	-180.37	-115.61	-1.98	-70.70	-131.42	10.72	-17.68	87.15
99	-180.37	-114.48	-2.02	-67.98	-131.22	10.86	-17.79	89.21
100	-182.21	-111.90	-2.00	-70.02	-132.86	10.41	-17.62	92.41
101	-178.91	-117.06	-2.03	-71.92	-129.11	11.31	-17.42	90.89
102	-181.48	-113.66	-2.01	-69.24	-130.05	11.09	-17.57	94.75
103	-180.28	-117.23	-2.03	-71.54	-128.29	10.24	-17.24	92.37
104	-185.23	-115.73	-2.01	-69.82	-129.67	11.10	-17.39	87.03
105	-182.95	-111.77	-2.03	-71.93	-130.92	11.05	-17.50	90.66
106	-180.42	-115.91	-2.02	-71.93	-132.11	10.58	-17.60	89.10
107	-184.49	-115.82	-2.04	-70.93	-133.27	10.63	-17.39	92.69
108	-182.04	-114.89	-2.02	-73.68	-130.74	10.27	-17.30	88.48
109	-187.39	-116.94	-2.00	-70.45	-133.71	10.58	-17.37	88.44
110	-183.45	-115.03	-2.00	-72.48	-131.80	10.55	-17.51	87.44
111	-178.97	-116.76	-2.03	-70.39	-131.76	11.18	-17.83	90.02
112	53.55	15.02	-0.01	26.67	-5.88	-9.21	0.67	-27.37

113	55.15	13.52	-0.06	23.88	-7.50	-9.22	0.32	-26.84
114	55.34	13.76	-0.05	25.76	-6.52	-9.00	0.45	-23.41
115	56.13	12.27	-0.04	26.72	-7.43	-8.65	0.23	-25.40
116	58.20	12.54	0.00	27.33	-8.21	-9.11	0.39	-25.47
117	53.40	11.55	-0.04	24.89	-6.52	-9.32	0.47	-26.03
118	57.18	15.39	-0.04	26.82	-6.54	-9.12	0.17	-26.77
119	53.75	13.14	-0.04	25.75	-5.81	-9.26	0.44	-26.64

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
80	1.25	31.48	36.45	5.03	-4.19	-13.70	34.65	3.63
81	1.58	31.20	35.32	3.71	-4.06	-13.75	-25.86	3.60
82	0.39	32.65	36.13	-66.31	-4.02	-13.44	2.89	3.60
83	1.13	32.53	35.72	9.35	-4.30	-13.69	-12.59	3.59
84	1.32	32.05	35.43	77.97	-4.23	-13.75	-11.80	3.64
85	0.40	32.67	35.83	10.64	-4.27	-13.54	-13.35	3.59
86	1.24	31.25	36.79	-0.20	-4.24	-13.69	-7.93	3.61
87	0.63	32.68	36.03	-9.71	-3.66	-13.65	28.19	3.65
88	1.29	31.34	35.88	45.80	-4.03	-13.27	-1.63	3.60
89	-1.15	32.21	35.54	38.92	-4.00	-13.62	-23.70	3.68
90	-0.43	31.99	35.51	-10.68	-4.34	-13.83	-29.95	3.70
91	0.06	31.39	37.08	16.14	-3.99	-13.78	-12.77	3.64
92	-2.56	32.15	36.26	-18.55	-3.82	-13.57	-2.25	3.67
93	-1.69	31.85	35.62	-46.41	-3.96	-13.64	-21.34	3.64
94	0.62	32.69	35.98	-64.99	-4.18	-13.56	-7.76	3.68
95	2.32	32.37	36.68	2.72	-3.94	-13.79	-10.48	3.62
96	1.91	27.17	35.46	31.42	-6.83	-14.35	-4.68	3.04
97	-0.84	28.04	36.09	66.63	-6.64	-14.40	-26.19	3.11
98	-0.22	28.28	35.61	-27.73	-6.88	-14.78	-1.46	3.08
99	0.40	28.54	36.22	-57.05	-6.45	-14.68	10.02	3.12
100	-1.97	28.51	35.25	-1.24	-6.95	-14.54	-11.66	3.11
101	-0.27	28.64	35.64	-32.40	-6.88	-14.72	-4.98	3.09
102	-0.47	27.32	35.97	27.27	-6.83	-14.57	10.59	3.08
103	0.44	28.14	35.71	-30.29	-7.15	-14.77	-9.52	3.05
104	1.54	28.04	35.70	17.22	-7.02	-14.61	-8.51	3.08
105	1.01	27.15	36.02	15.69	-6.67	-14.79	-14.83	3.10
106	-0.91	27.98	35.25	12.21	-6.68	-14.71	-3.81	3.04
107	0.73	26.90	35.44	-45.54	-7.21	-14.30	-8.87	3.12
108	0.21	28.51	35.12	-4.38	-6.82	-14.75	-32.65	3.15
109	1.95	28.09	35.19	-12.43	-7.11	-14.39	17.18	3.04
110	-0.93	28.33	36.14	-55.64	-6.91	-14.73	9.19	3.13
111	1.20	28.73	35.63	-23.48	-6.74	-14.51	0.11	3.05
112	-0.02	-4.26	4.39	-21.84	-1.95	-1.90	4.25	0.50
113	-0.19	-5.10	4.80	-4.81	-1.96	-1.52	0.62	0.54
114	-0.55	-5.07	4.62	20.67	-2.08	-1.80	1.84	0.56
115	-0.12	-4.51	4.51	41.62	-1.85	-1.93	2.57	0.56
116	-0.88	-3.89	4.43	-8.71	-1.84	-1.74	-0.13	0.55
117	-0.21	-4.95	4.50	-8.92	-1.94	-1.54	-3.25	0.54
118	-0.11	-4.62	4.46	15.90	-1.87	-1.91	1.65	0.54
119	-0.29	-4.89	4.58	-6.43	-1.68	-1.77	-5.02	0.52

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
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80	-0.72	13.41	156.23	-22.05	54.69	-21.12	-36.70	-16.35
81	1.23	12.76	154.12	-21.44	53.34	-21.60	-37.59	-15.05
82	1.64	13.48	158.87	-21.61	53.88	-20.64	-36.38	-19.94
83	-0.66	13.19	153.77	-22.15	54.43	-20.25	-34.46	-17.11
84	0.56	13.61	156.37	-21.57	54.48	-19.82	-35.80	-16.25
85	-0.14	13.52	158.07	-21.30	54.41	-20.91	-34.52	-12.68
86	0.41	13.54	156.75	-21.79	53.12	-20.37	-36.33	-13.44
87	-1.00	12.92	156.99	-21.66	53.95	-19.99	-35.01	-12.89
88	-0.16	14.20	155.55	-21.40	54.41	-20.89	-34.07	-15.57
89	-1.85	12.45	156.90	-21.26	53.80	-21.84	-36.75	-17.41
90	0.51	13.56	157.68	-21.81	54.28	-20.43	-34.55	-16.46
91	2.46	13.10	155.86	-22.04	54.75	-20.86	-34.88	-13.51
92	-1.96	13.68	155.14	-22.09	53.98	-20.80	-37.23	-20.04
93	4.18	13.40	158.36	-21.82	54.19	-20.46	-36.36	-16.43
94	3.18	13.43	156.65	-22.12	53.60	-20.05	-35.90	-16.26
95	-1.71	13.55	155.07	-22.08	53.90	-20.33	-36.81	-16.48
96	-1.87	11.55	132.79	-19.54	51.83	-3.17	-27.20	2.49
97	1.90	11.74	132.36	-19.78	51.36	-2.55	-26.79	-1.40
98	2.83	11.98	133.42	-19.47	52.29	-2.64	-26.69	2.37
99	-2.49	11.82	130.84	-19.85	52.62	-2.99	-27.51	0.39
100	-2.51	12.23	132.00	-19.92	52.38	-2.79	-27.31	1.30
101	-0.33	11.25	135.64	-19.17	52.55	-3.21	-28.44	0.79
102	-0.66	11.59	132.50	-19.63	51.89	-2.65	-28.62	0.35
103	4.76	11.46	133.48	-19.67	51.71	-3.21	-25.00	0.87
104	3.28	11.84	132.09	-19.71	52.27	-3.36	-27.45	2.13
105	0.21	11.47	129.46	-19.72	51.54	-2.97	-27.22	-2.11
106	1.65	12.36	134.76	-19.93	52.29	-2.91	-28.74	3.66
107	5.53	11.96	132.76	-19.63	52.03	-3.21	-29.41	-1.15
108	-3.77	11.16	131.28	-19.55	51.37	-2.87	-27.01	1.87
109	-3.12	11.83	134.62	-19.65	52.18	-2.48	-26.46	0.83
110	2.85	12.08	133.21	-19.44	52.47	-3.09	-27.62	-1.18
111	0.07	11.77	131.28	-19.90	52.21	-2.69	-27.81	2.83
112	2.49	-2.66	30.15	-1.56	4.27	-0.80	-2.30	-49.05
113	-0.21	-3.10	31.04	-1.60	4.31	-0.43	-2.86	-49.25
114	1.94	-2.82	31.03	-1.86	4.12	-1.00	-2.97	-50.39
115	1.08	-2.63	31.21	-1.55	3.77	-0.67	-3.06	-50.02
116	-0.40	-2.46	31.05	-1.70	4.43	-0.82	-2.46	-50.67
117	-0.06	-2.74	31.50	-1.52	4.41	-0.77	-2.77	-50.13
118	-0.80	-2.66	31.91	-1.62	3.94	-0.64	-2.97	-49.29
119	-0.25	-2.75	32.83	-1.54	3.88	-0.49	-2.99	-49.55

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
80	-95.69	0.82	-1.10	46.86	-0.54	130.43	-0.31	-6.35
81	-95.93	0.92	-0.98	46.06	-0.58	127.11	0.13	-6.43
82	-96.16	0.86	-1.01	45.27	-0.54	127.80	-0.38	-6.37
83	-95.92	0.73	-0.94	45.00	-0.54	128.49	0.19	-6.34
84	-95.27	0.81	-1.04	44.91	-0.64	126.65	-0.16	-6.39
85	-92.81	0.66	-0.86	47.10	-0.53	130.24	0.06	-6.42
86	-96.29	0.72	-0.91	45.81	-0.46	128.86	-0.40	-6.39
87	-96.00	1.01	-0.93	46.00	-0.58	128.57	0.25	-6.31
88	-94.99	0.62	-0.94	45.13	-0.55	128.39	0.16	-6.34
89	-96.60	0.78	-0.89	45.47	-0.52	126.84	-0.11	-6.40
90	-94.71	1.23	-1.19	45.58	-0.62	128.54	-0.77	-6.34

91	-94.84	0.65	-1.13	45.76	-0.67	128.77	-0.04	-6.34
92	-98.94	0.90	-1.10	46.93	-0.58	126.04	0.20	-6.35
93	-95.55	0.79	-1.08	45.13	-0.51	126.68	0.11	-6.38
94	-95.94	0.89	-0.91	46.23	-0.61	127.94	-0.21	-6.32
95	-95.34	1.31	-1.05	44.16	-0.61	129.03	-0.25	-6.34
96	-107.90	0.76	0.00	43.98	-0.72	132.70	4.65	-5.94
97	-107.59	0.99	-0.20	44.15	-0.85	132.48	4.73	-5.90
98	-107.87	0.85	-0.02	44.18	-0.77	134.56	4.47	-5.93
99	-107.13	0.84	-0.11	44.50	-0.88	132.73	5.06	-5.89
100	-107.14	0.81	-0.29	43.22	-0.80	132.40	4.81	-5.89
101	-107.45	1.18	-0.09	44.08	-0.83	129.23	4.52	-5.89
102	-107.72	0.94	-0.04	43.68	-0.76	128.11	4.76	-5.93
103	-108.87	0.85	-0.20	44.01	-0.84	131.58	4.83	-5.90
104	-106.70	0.96	-0.25	42.66	-0.82	129.83	5.01	-5.99
105	-106.40	0.98	-0.08	43.16	-0.85	130.13	4.95	-5.86
106	-106.63	1.09	-0.06	44.91	-0.82	129.33	4.92	-5.93
107	-107.55	1.00	-0.13	43.78	-0.79	131.18	4.41	-6.02
108	-105.69	1.12	-0.27	44.38	-0.87	131.80	5.34	-5.86
109	-105.87	1.07	0.08	43.11	-0.79	133.55	4.85	-5.86
110	-107.71	0.90	-0.20	43.99	-0.87	133.70	4.92	-6.02
111	-106.95	0.78	-0.10	44.61	-0.76	130.77	4.70	-5.93
112	-4.44	1.70	1.70	0.99	-0.98	-0.86	2.56	-2.05
113	-4.14	1.83	1.66	1.66	-0.97	-1.69	2.43	-2.09
114	-4.10	1.81	1.72	0.63	-0.94	-2.98	2.44	-1.99
115	-4.14	1.83	1.69	0.96	-0.99	-3.20	2.68	-2.10
116	-4.28	1.80	1.62	0.80	-0.98	-3.96	2.41	-2.02
117	-3.84	1.93	1.78	0.41	-1.00	-4.36	2.44	-2.03
118	-4.50	1.70	1.76	0.90	-1.00	-2.13	2.54	-2.05
119	-4.48	1.88	1.66	0.14	-0.95	-2.51	2.49	-2.04

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
80	3.40	19.84	3.52	-78.11	-37.18	-14.13	-71.01	6.01
81	3.35	20.61	-0.60	-78.24	-9.91	-13.29	-75.61	-1.14
82	3.41	20.64	0.82	-76.73	19.98	-13.42	-70.88	-6.79
83	3.46	20.20	-0.67	-78.09	-14.48	-13.16	-78.82	0.37
84	3.44	21.50	0.15	-78.53	-27.77	-12.11	-74.33	-4.33
85	3.42	20.07	-1.55	-77.34	-15.72	-13.50	-71.87	-5.76
86	3.47	19.94	-3.51	-80.03	9.79	-13.72	-76.54	-2.25
87	3.49	20.55	6.58	-77.18	14.38	-13.78	-73.69	-8.28
88	3.47	20.61	10.20	-77.82	-7.27	-12.83	-76.26	-5.58
89	3.41	20.55	2.07	-76.65	5.56	-12.96	-75.26	-0.05
90	3.43	20.12	-1.73	-76.33	-3.90	-13.35	-72.26	5.49
91	3.44	20.60	0.77	-77.71	-11.89	-13.52	-68.06	-2.05
92	3.46	20.20	3.82	-77.63	-0.28	-12.97	-79.25	-10.64
93	3.43	20.64	1.29	-77.80	-18.28	-12.80	-72.32	0.38
94	3.44	20.07	0.60	-76.68	15.97	-13.62	-76.04	6.19
95	3.41	20.74	-1.12	-77.81	2.76	-13.11	-72.34	-2.04
96	3.65	25.16	2.76	-72.70	-2.57	-15.00	-62.60	-2.69
97	3.65	25.10	3.51	-72.06	-1.31	-16.34	-66.97	5.31
98	3.66	25.35	0.27	-73.24	8.88	-15.21	-61.74	2.10
99	3.62	25.43	-5.10	-73.18	-11.54	-15.52	-62.32	1.04
100	3.61	25.28	0.40	-72.53	-1.10	-15.47	-70.69	-4.99
101	3.62	24.49	-3.88	-73.96	-0.97	-15.72	-68.40	2.03

102	3.58	25.52	2.92	-73.28	4.49	-15.85	-61.30	-5.42
103	3.58	25.33	1.26	-73.93	-13.86	-15.93	-64.63	-4.88
104	3.65	25.41	3.75	-73.70	4.01	-15.94	-63.18	5.70
105	3.65	25.65	-0.10	-71.82	-4.99	-15.83	-71.39	-2.33
106	3.63	24.95	1.65	-72.28	-7.68	-15.71	-61.19	0.91
107	3.64	25.07	0.78	-73.83	-7.90	-15.93	-61.00	-4.62
108	3.65	24.83	-0.31	-74.01	5.44	-15.62	-62.16	0.71
109	3.62	24.79	1.58	-72.87	-16.26	-15.41	-54.83	-3.32
110	3.57	24.99	4.36	-73.64	3.01	-16.03	-57.92	2.51
111	3.63	24.88	2.53	-72.63	-2.20	-15.87	-69.05	2.72
112	0.31	4.79	-0.41	-1.00	4.16	-7.16	62.77	-1.29
113	0.36	4.94	-1.85	-0.77	-0.50	-6.81	58.70	2.31
114	0.36	5.23	1.31	-1.09	-3.95	-6.47	60.91	-2.13
115	0.37	5.08	-0.23	-1.37	-5.48	-6.74	63.68	2.69
116	0.37	4.81	-0.94	-1.59	-1.18	-7.42	64.90	2.51
117	0.32	5.26	-0.40	-1.36	7.83	-6.88	69.96	1.83
118	0.34	4.88	-1.21	-1.55	-8.41	-6.50	60.24	2.64
119	0.30	5.04	-0.69	-1.16	-8.28	-6.90	62.01	-0.35

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
80	35.18	-18.92	-15.82	-4.04	123.00	-19.42	-1.04	-51.71
81	34.09	-18.91	-16.07	6.61	118.59	-17.87	-1.15	-52.53
82	36.45	-18.80	-17.25	2.41	120.53	-15.35	-0.15	-51.71
83	35.43	-19.22	-15.75	-0.26	121.11	-14.13	-1.33	-51.60
84	35.61	-19.31	-15.80	3.78	122.22	-13.50	-0.46	-51.12
85	35.84	-19.29	-15.60	-3.07	113.52	-19.69	-1.38	-51.49
86	34.84	-19.11	-15.75	6.88	119.39	-19.36	-1.24	-51.76
87	31.00	-18.93	-16.37	2.25	118.69	-14.80	-0.99	-52.10
88	34.90	-19.56	-16.97	-5.46	121.17	-18.96	-1.06	-52.00
89	34.10	-19.04	-15.91	1.89	118.31	-15.26	-0.99	-51.50
90	34.94	-18.97	-16.30	5.87	119.97	-19.67	-1.19	-52.26
91	29.87	-19.33	-16.96	4.69	119.99	-15.82	-1.76	-51.64
92	34.26	-18.90	-16.35	-0.85	119.09	-13.64	-1.06	-51.17
93	35.63	-18.87	-16.09	-2.07	123.06	-16.23	-0.80	-51.82
94	37.07	-19.29	-16.14	11.96	121.62	-15.06	-1.21	-51.56
95	36.78	-19.43	-16.06	-2.11	119.46	-17.76	-1.23	-51.70
96	21.23	-22.94	-15.56	-2.98	85.63	-20.85	2.54	-46.27
97	24.91	-22.30	-14.76	2.60	81.48	-21.58	2.96	-44.93
98	21.70	-22.33	-14.50	-3.51	88.57	-21.05	1.94	-46.62
99	22.56	-21.99	-14.26	-10.16	85.86	-20.91	2.38	-45.87
100	22.17	-22.94	-14.66	8.34	86.53	-21.30	2.37	-46.47
101	21.08	-23.03	-14.09	-0.27	87.85	-22.21	2.51	-46.73
102	19.67	-22.16	-14.80	3.10	85.87	-21.61	2.82	-47.10
103	22.49	-22.60	-14.67	3.54	87.94	-21.15	2.48	-46.57
104	22.32	-22.83	-15.55	-3.03	89.65	-19.85	2.31	-46.25
105	22.19	-22.25	-14.09	5.15	84.82	-18.96	2.62	-47.25
106	21.33	-22.63	-15.21	-6.48	86.04	-20.04	2.40	-47.84
107	21.61	-22.56	-13.73	2.23	87.92	-19.20	2.10	-46.74
108	19.32	-22.17	-15.69	-1.33	86.07	-18.90	2.44	-46.14
109	21.27	-22.66	-13.80	0.11	86.80	-20.63	2.40	-46.11
110	21.06	-22.97	-14.68	-2.90	86.30	-22.66	2.28	-46.75
111	22.19	-22.73	-15.73	-5.40	87.87	-19.19	2.68	-46.05
112	2.40	-6.68	6.28	-2.72	70.14	32.73	-4.02	-3.00

113	1.66	-6.52	6.68	-1.88	71.24	32.05	-4.00	-1.76
114	2.74	-6.41	6.80	-2.46	72.66	32.14	-4.04	-3.13
115	2.97	-6.78	6.95	-1.73	71.59	31.97	-3.94	-3.26
116	1.87	-6.57	6.62	-2.56	71.55	32.38	-3.96	-3.04
117	2.38	-6.59	7.29	0.80	73.77	32.78	-4.12	-2.99
118	2.23	-6.52	7.40	4.06	73.25	32.26	-4.00	-2.40
119	2.04	-6.41	6.41	-1.61	69.82	31.86	-3.80	-3.61

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
80	7.22	128.52	-1.88	43.51	39.31	15.41	-14.23	23.79
81	16.34	128.59	-1.91	43.36	39.03	15.48	-28.20	5.78
82	-6.02	129.83	-1.88	43.69	37.71	15.66	-12.79	-13.12
83	0.26	129.91	-1.89	43.46	38.12	15.62	39.21	11.40
84	-13.98	128.38	-1.87	43.41	37.06	15.59	13.00	-4.79
85	-20.46	130.06	-1.87	43.02	38.18	15.67	-23.98	-72.21
86	-2.59	127.98	-1.90	43.07	37.21	15.58	-30.51	7.52
87	11.87	128.01	-1.88	43.97	37.46	15.56	-27.53	39.33
88	-8.42	128.67	-1.91	43.50	40.10	15.60	-39.33	-13.06
89	-19.59	130.39	-1.86	43.48	38.13	15.66	-15.58	38.94
90	4.00	129.94	-1.85	43.36	37.51	15.62	-16.66	20.91
91	-15.39	128.60	-1.89	43.80	38.56	15.60	-8.93	13.06
92	13.67	129.15	-1.87	42.93	37.77	15.50	-15.23	5.92
93	3.75	129.33	-1.92	43.30	37.36	15.57	3.32	-17.65
94	-15.76	130.12	-1.87	43.29	37.20	15.39	-11.95	-15.64
95	-20.12	129.29	-1.85	43.60	40.08	15.90	-60.06	-10.83
96	-17.10	114.33	-2.02	39.62	9.75	12.73	52.83	-29.46
97	0.00	113.68	-2.03	39.64	9.45	12.67	-31.55	15.89
98	-5.74	115.79	-2.01	39.91	10.43	13.02	45.48	-21.84
99	1.37	115.20	-2.04	40.55	7.81	12.62	7.80	-103.58
100	4.74	115.96	-2.03	40.13	11.11	12.59	40.56	-60.33
101	9.00	115.12	-2.03	40.05	9.51	12.81	8.19	18.79
102	15.33	113.84	-2.02	39.69	9.57	12.73	29.45	79.46
103	9.99	114.81	-2.04	40.46	9.24	12.55	-24.76	-12.46
104	2.29	114.70	-2.04	40.01	10.31	12.96	8.96	24.34
105	-3.84	114.69	-2.03	39.86	9.44	12.77	-20.44	-33.68
106	-16.27	115.65	-2.04	40.05	9.97	12.69	31.92	-4.97
107	-8.85	114.18	-2.02	40.49	9.01	12.48	6.07	-18.59
108	13.66	113.88	-2.07	39.23	11.45	12.60	38.66	-56.12
109	-8.74	114.98	-2.03	39.73	10.32	12.72	-4.35	-32.95
110	7.85	116.66	-2.01	40.22	9.71	12.66	-0.79	-36.94
111	3.91	115.17	-1.99	39.78	9.93	12.55	9.97	15.23
112	1.14	13.18	-0.14	2.79	24.72	0.07	-16.74	8.49
113	4.25	13.58	-0.14	2.10	24.18	0.17	17.27	13.28
114	-3.59	13.49	-0.16	2.96	24.03	0.26	6.52	41.70
115	5.11	14.28	-0.14	2.68	24.04	0.36	10.37	38.29
116	6.99	13.92	-0.14	2.68	24.65	0.11	10.23	-3.54
117	5.35	14.04	-0.16	2.83	24.69	0.32	-1.11	7.77
118	-14.75	13.75	-0.12	2.85	24.55	0.22	-16.30	0.78
119	7.24	13.73	-0.13	2.73	24.81	0.13	-1.81	10.69

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
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120	-86.29	-0.63	3.88	-32.02	1.84	11.04	2.67	-3.79
121	-85.57	-0.73	3.64	-32.07	1.65	11.44	2.57	-2.83
122	-80.08	-0.26	3.98	-32.67	1.94	10.85	2.31	-3.57
123	-89.15	-0.56	3.69	-32.93	1.34	11.29	2.70	-3.13
124	-93.09	-0.87	3.69	-33.69	1.66	11.05	2.62	-2.62
125	-79.10	-0.35	3.46	-34.24	1.35	11.14	3.71	-2.73
126	-79.96	-0.48	3.30	-34.24	1.28	11.20	2.45	-3.40
127	-88.22	-0.65	3.53	-36.03	1.47	11.34	3.62	-3.03
128	-64.37	-4.04	1.18	-12.23	6.38	4.37	5.12	-0.76
129	-69.64	-4.14	1.10	-13.64	6.02	4.69	5.44	-0.43
130	-63.61	-4.12	0.95	-10.49	6.13	4.01	4.92	-0.32
131	-66.53	-4.06	0.94	-13.49	5.92	4.79	5.35	-0.99
132	-70.96	-4.00	1.10	-13.01	6.39	3.96	5.16	-0.30
133	-67.75	-4.00	1.23	-15.14	6.04	3.95	5.44	-0.43
134	-64.48	-4.20	0.82	-13.91	5.82	4.06	5.23	-0.18
135	-61.97	-4.07	0.68	-13.86	5.92	4.81	5.69	-0.76
136	-68.31	-4.09	0.82	-13.67	5.52	4.21	5.11	-0.62
137	-57.85	-4.15	1.00	-14.37	5.78	4.21	5.63	-1.43
138	-68.14	-4.06	0.87	-13.57	5.80	4.44	5.36	-0.51
139	-71.48	-4.31	1.21	-16.35	6.10	4.03	5.38	-0.61
140	-65.76	-4.29	0.82	-14.07	6.50	4.40	5.54	-0.17
141	-61.81	-4.12	1.10	-12.99	5.91	4.76	4.79	-0.75
142	-66.55	-4.48	1.08	-12.19	5.67	4.64	5.21	-0.94
143	-63.27	-4.03	1.02	-14.10	5.86	4.08	6.23	-1.19
144	-38.06	-4.80	-0.40	-7.99	5.14	1.07	4.09	-2.19
145	-34.73	-5.10	-0.37	-9.37	4.70	0.55	3.89	-1.54
146	-36.22	-4.93	-0.66	-8.77	4.53	0.87	3.70	-1.89
147	-38.97	-4.91	-0.26	-8.19	4.97	0.85	4.47	-1.42
148	-39.77	-4.98	-0.42	-9.31	5.15	0.76	4.33	-2.09
149	-40.83	-4.93	-0.29	-7.86	4.61	0.53	4.00	-1.78
150	-42.55	-4.97	-0.37	-8.84	4.88	1.17	4.23	-1.88
151	-40.43	-4.83	-0.39	-10.02	4.55	0.64	3.83	-2.11
152	-35.57	-4.98	-0.28	-9.08	4.79	1.04	4.22	-1.39
153	-36.15	-4.86	-0.47	-10.73	4.58	1.24	3.96	-1.78
154	-40.29	-4.90	-0.50	-9.89	4.86	1.22	3.86	-2.27
155	-36.78	-5.15	-0.39	-9.03	5.19	0.90	4.12	-1.66
156	-39.53	-5.04	-0.58	-9.80	5.07	0.85	4.06	-1.30
157	-42.15	-4.89	-0.60	-9.48	4.90	0.85	3.62	-2.15
158	-38.22	-5.10	-0.56	-7.54	4.48	0.71	4.17	-1.92
159	-40.67	-5.02	-0.56	-9.52	4.98	0.99	3.72	-1.58

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
120	53.48	10.31	-0.06	26.05	-6.81	-9.45	0.35	-27.38
121	58.44	14.49	-0.03	26.18	-6.15	-9.19	0.45	-24.02
122	55.08	11.57	-0.02	23.35	-9.94	-9.11	0.39	-25.86
123	56.58	12.31	-0.04	26.82	-7.44	-9.01	0.26	-25.55
124	54.35	10.39	-0.05	24.98	-8.46	-9.02	0.44	-25.96
125	53.81	15.02	-0.04	27.99	-7.80	-9.58	0.44	-25.95
126	53.27	14.24	-0.04	27.91	-7.71	-9.26	0.47	-25.63
127	58.03	12.68	-0.01	25.01	-6.40	-8.74	0.40	-28.05
128	39.98	-1.05	-0.33	21.96	-22.04	-4.80	-0.41	-15.66
129	41.22	-2.03	-0.36	22.19	-21.99	-5.06	-0.56	-15.90
130	40.95	-2.24	-0.34	21.97	-22.07	-5.14	-0.57	-19.52

131	40.26	-0.35	-0.34	20.92	-20.71	-5.18	-0.39	-18.20
132	41.86	-2.16	-0.37	22.34	-21.90	-4.85	-0.56	-16.02
133	43.88	-2.36	-0.34	21.20	-21.58	-5.34	-0.51	-16.43
134	36.87	-2.70	-0.33	20.90	-21.32	-5.10	-0.61	-19.40
135	39.83	-2.87	-0.36	18.90	-21.75	-4.79	-0.38	-17.56
136	45.23	-0.60	-0.35	22.43	-21.35	-5.04	-0.36	-17.73
137	41.16	-0.89	-0.36	23.49	-20.47	-5.32	-0.67	-14.10
138	38.19	-1.59	-0.35	21.84	-23.38	-5.16	-0.48	-17.71
139	41.58	-2.39	-0.39	21.34	-24.13	-4.83	-0.53	-17.91
140	42.51	-2.42	-0.38	23.72	-21.10	-5.44	-0.54	-15.10
141	40.30	0.10	-0.36	22.77	-22.23	-5.06	-0.67	-17.92
142	41.11	0.05	-0.35	22.22	-20.58	-4.96	-0.59	-18.54
143	42.21	-2.62	-0.33	21.58	-22.83	-5.20	-0.53	-18.43
144	23.17	3.07	-0.39	11.72	-4.83	-2.20	0.03	-8.73
145	20.93	1.49	-0.37	13.64	-5.71	-2.24	-0.03	-7.18
146	23.18	3.13	-0.37	12.07	-6.08	-2.29	-0.04	-9.87
147	24.10	2.17	-0.37	12.71	-6.07	-1.98	0.18	-8.85
148	21.38	3.16	-0.34	12.79	-5.25	-2.21	-0.02	-7.81
149	23.10	3.56	-0.37	13.95	-6.65	-2.46	-0.06	-9.00
150	19.33	3.00	-0.38	12.67	-5.73	-2.00	-0.07	-7.75
151	25.90	2.23	-0.36	11.71	-6.53	-2.38	0.09	-8.28
152	24.52	2.63	-0.37	14.16	-6.56	-2.29	0.02	-8.56
153	24.20	3.68	-0.34	12.92	-5.60	-2.00	-0.12	-8.80
154	20.10	2.21	-0.36	13.40	-5.76	-2.46	0.13	-8.69
155	23.10	1.82	-0.36	12.73	-5.94	-2.06	0.12	-8.08
156	24.71	2.42	-0.37	12.42	-5.34	-2.50	-0.12	-9.17
157	23.11	3.73	-0.39	12.30	-6.50	-2.19	-0.03	-9.57
158	24.10	3.93	-0.35	14.40	-6.18	-2.17	0.00	-8.30
159	22.82	4.16	-0.34	13.30	-6.05	-2.27	0.10	-7.50

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
120	0.31	-5.03	4.62	-32.56	-1.82	-1.70	4.33	0.53
121	0.72	-4.01	4.85	1.24	-1.88	-1.74	-1.33	0.54
122	0.45	-4.10	4.84	-2.36	-1.67	-1.69	-2.20	0.55
123	0.11	-4.61	4.79	14.98	-1.97	-1.76	-1.86	0.53
124	-0.04	-4.58	4.63	0.19	-1.97	-1.92	1.82	0.55
125	0.56	-4.69	4.83	-20.06	-1.93	-1.77	3.49	0.53
126	-0.20	-4.71	4.49	4.85	-1.85	-1.65	-2.92	0.54
127	-0.06	-4.54	4.62	-14.70	-1.93	-1.83	-4.25	0.55
128	-0.59	-1.46	4.32	-19.70	1.52	0.78	1.79	0.51
129	-0.28	-1.64	4.09	-34.57	1.15	0.54	5.08	0.51
130	0.18	-1.25	4.46	22.73	1.33	0.70	7.53	0.48
131	1.03	-1.55	4.43	8.79	1.40	0.59	5.14	0.47
132	-0.07	-1.65	4.37	-34.22	1.36	0.55	-0.52	0.48
133	0.09	-1.60	4.25	6.16	1.45	0.59	7.45	0.50
134	0.59	-1.67	4.41	-25.72	1.17	0.73	6.02	0.48
135	-1.51	-1.42	4.83	-0.65	1.22	0.56	7.84	0.49
136	-0.64	-1.43	4.64	-25.29	1.16	0.77	2.70	0.47
137	0.38	-1.91	4.26	29.50	1.25	0.50	3.59	0.47
138	-0.51	-1.60	4.68	-16.14	1.52	0.61	-4.79	0.49
139	-0.07	-1.17	4.53	42.81	1.29	0.66	6.75	0.50
140	-0.70	-1.19	4.06	6.12	1.09	0.81	-3.05	0.55
141	-0.06	-1.58	4.40	6.81	1.20	0.47	3.59	0.52

142	-0.30	-1.72	3.98	-8.17	1.33	0.76	4.02	0.50
143	-0.08	-1.32	4.27	-7.54	1.36	0.36	-7.33	0.50
144	0.45	0.82	3.10	11.79	3.12	2.48	-2.30	0.22
145	-0.01	0.44	2.79	-4.24	3.08	2.36	-0.35	0.25
146	0.37	0.43	2.61	-14.16	3.16	2.41	0.10	0.20
147	0.50	0.65	2.30	16.23	3.18	2.47	13.38	0.22
148	-0.08	0.27	2.63	15.55	3.20	2.50	6.03	0.22
149	0.77	0.39	2.89	11.11	3.10	2.45	1.68	0.23
150	-0.07	0.70	2.73	-7.13	3.16	2.45	6.97	0.23
151	-0.14	0.72	2.70	16.50	3.19	2.47	7.78	0.23
152	-0.05	0.58	3.11	5.90	3.31	2.60	5.82	0.21
153	-0.41	0.77	2.48	-3.59	3.23	2.64	1.59	0.21
154	-0.34	0.61	2.63	13.77	3.13	2.70	5.05	0.23
155	-0.58	0.21	2.43	-4.03	3.19	2.42	0.11	0.21
156	-0.25	0.71	2.85	-12.11	3.07	2.63	6.59	0.24
157	0.54	0.53	2.69	2.39	3.20	2.50	-3.10	0.21
158	0.28	0.84	2.99	-14.40	3.08	2.52	-6.42	0.19
159	0.26	0.25	2.33	8.40	3.20	2.36	5.99	0.25

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
120	-3.15	-2.74	32.68	-1.28	4.17	-0.54	-2.97	-49.03
121	0.37	-2.71	32.30	-1.60	4.71	-0.70	-2.92	-49.08
122	-1.57	-2.50	33.88	-1.53	4.70	-0.59	-2.22	-49.68
123	-2.35	-2.72	30.56	-1.41	4.83	-0.77	-2.27	-49.43
124	-1.34	-2.41	29.44	-1.53	4.20	-0.64	-3.11	-49.42
125	-0.38	-2.90	32.38	-1.44	3.98	-0.92	-3.34	-49.54
126	-0.34	-2.87	30.15	-1.49	4.13	-0.79	-3.50	-50.50
127	-1.26	-2.59	31.97	-1.33	3.99	-0.36	-2.67	-50.02
128	1.14	-0.88	32.09	-0.37	9.19	-2.91	-3.97	-36.02
129	0.01	-0.90	29.34	-1.01	8.94	-2.71	-3.49	-35.29
130	-0.35	-0.83	32.11	-0.90	9.10	-3.05	-4.29	-34.94
131	-1.21	-1.12	32.67	-0.71	9.25	-3.00	-3.36	-36.80
132	-0.17	-1.05	30.63	-0.91	8.77	-2.89	-4.80	-36.50
133	1.87	-0.79	31.14	-0.52	9.94	-2.94	-3.83	-36.11
134	-0.47	-1.01	30.51	-0.75	9.13	-3.33	-3.84	-35.71
135	0.30	-0.80	30.54	-1.22	9.54	-2.83	-3.54	-35.94
136	-1.37	-0.96	32.28	-0.65	9.08	-2.91	-3.37	-36.61
137	2.06	-1.05	29.71	-0.50	9.28	-2.99	-3.87	-35.73
138	-0.62	-0.65	28.44	-0.93	9.22	-2.73	-3.47	-36.62
139	-1.69	-0.79	32.28	-0.62	9.13	-2.77	-3.37	-36.46
140	-1.42	-0.48	31.05	-0.86	9.25	-3.21	-3.49	-35.39
141	2.17	-0.73	31.65	-0.97	9.05	-3.07	-4.55	-35.42
142	-1.91	-0.93	29.79	-0.57	8.76	-2.92	-3.90	-35.86
143	0.66	-0.81	33.30	-0.84	9.12	-3.15	-4.24	-37.51
144	1.11	0.42	18.05	0.99	2.98	-0.40	1.34	-22.51
145	0.18	0.19	18.46	0.91	2.43	-0.95	1.18	-21.18
146	-0.04	0.32	17.97	0.82	2.36	-0.36	0.97	-22.37
147	-0.54	0.58	18.34	1.07	2.36	-0.56	0.77	-22.28
148	0.51	0.19	18.43	0.91	2.45	-0.37	0.93	-22.38
149	0.27	0.47	18.05	0.97	2.15	-0.67	1.19	-22.81
150	0.84	0.36	17.25	1.26	2.25	-0.60	1.62	-22.12
151	1.21	0.45	18.18	1.26	2.77	-0.41	1.26	-22.47
152	-1.01	0.39	18.00	1.08	2.27	-0.42	1.60	-23.23

153	-0.96	0.54	17.76	1.28	2.68	-0.65	1.19	-22.35
154	-1.06	0.43	18.31	0.95	1.97	-0.58	1.72	-22.32
155	-1.09	0.56	16.71	0.46	2.04	-0.37	1.65	-21.98
156	-1.60	0.60	18.48	0.96	2.26	-0.62	1.03	-22.74
157	1.04	0.44	16.36	0.86	2.27	-0.58	1.05	-23.32
158	-0.46	0.44	16.89	0.97	2.06	-0.66	1.13	-21.35
159	0.16	0.47	18.03	1.13	2.51	-0.59	1.34	-22.34

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
120	-4.42	1.84	1.76	1.01	-0.98	-2.13	2.45	-2.01
121	-4.37	2.00	1.74	0.71	-0.97	-1.82	2.59	-2.07
122	-3.96	1.87	1.66	0.88	-0.98	-1.29	2.49	-2.04
123	-3.97	1.99	1.80	0.12	-0.92	-1.94	2.43	-2.10
124	-4.29	1.75	1.75	0.79	-0.97	-3.66	2.39	-2.04
125	-4.25	1.90	1.82	1.10	-0.98	-2.15	2.39	-2.09
126	-4.33	1.92	1.73	1.23	-0.96	-3.61	2.43	-2.04
127	-4.28	1.86	1.88	1.40	-1.02	-3.55	2.51	-1.98
128	-7.95	0.11	1.68	8.51	-0.87	10.98	-2.80	-2.96
129	-8.25	0.12	1.63	9.09	-0.88	8.94	-2.75	-3.10
130	-8.36	0.16	1.71	8.72	-0.86	9.22	-2.60	-3.06
131	-8.59	0.00	1.59	7.62	-0.86	11.42	-2.57	-2.99
132	-8.02	0.21	1.68	8.66	-0.87	9.90	-2.69	-3.05
133	-8.27	0.18	1.74	8.45	-0.92	9.04	-2.93	-3.06
134	-8.28	-0.01	1.69	8.82	-0.93	9.85	-2.73	-3.04
135	-8.91	0.14	1.71	8.75	-0.83	9.35	-2.74	-3.04
136	-8.32	0.14	1.68	8.33	-0.91	9.73	-2.51	-3.02
137	-8.23	0.27	1.76	8.53	-0.86	10.43	-2.72	-2.95
138	-8.15	0.11	1.60	8.97	-0.93	10.95	-2.78	-2.98
139	-8.33	0.06	1.70	8.68	-0.89	10.59	-3.01	-3.00
140	-7.64	0.16	1.68	8.39	-0.82	9.51	-2.84	-3.02
141	-9.02	0.15	1.58	8.08	-0.92	10.09	-2.75	-3.09
142	-8.73	0.14	1.68	8.98	-0.89	10.21	-2.67	-3.04
143	-7.95	0.22	1.63	9.15	-0.98	9.25	-3.00	-3.00
144	-2.45	-1.17	0.94	4.61	-0.40	1.78	-3.48	-2.51
145	-2.39	-1.16	1.04	4.32	-0.39	2.64	-3.46	-2.56
146	-1.88	-1.16	0.98	4.39	-0.41	0.79	-3.46	-2.50
147	-2.14	-1.10	1.02	4.42	-0.42	2.73	-3.50	-2.53
148	-2.58	-1.01	0.97	4.80	-0.36	2.52	-3.48	-2.52
149	-2.46	-1.06	0.97	4.32	-0.38	2.31	-3.48	-2.54
150	-2.92	-1.09	1.00	4.66	-0.42	2.15	-3.46	-2.52
151	-2.89	-1.10	0.90	4.61	-0.37	2.57	-3.54	-2.54
152	-2.62	-1.10	0.98	4.88	-0.37	2.24	-3.62	-2.56
153	-3.08	-1.12	1.01	4.37	-0.35	3.16	-3.55	-2.56
154	-2.80	-1.10	0.94	4.54	-0.41	2.04	-3.56	-2.52
155	-2.13	-1.11	0.95	4.54	-0.41	2.28	-3.54	-2.48
156	-2.77	-1.04	0.95	4.39	-0.36	1.86	-3.63	-2.55
157	-3.33	-1.13	1.01	3.85	-0.38	3.13	-3.50	-2.49
158	-2.72	-0.98	0.94	4.57	-0.36	2.95	-3.54	-2.53
159	-2.54	-1.18	1.01	4.96	-0.35	1.73	-3.52	-2.48

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
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120	0.33	4.92	-0.31	-1.59	0.68	-6.93	56.35	1.36
121	0.30	4.77	0.00	-1.50	4.89	-6.96	59.88	-3.65
122	0.34	4.82	4.31	-1.16	-7.78	-6.96	67.78	-1.90
123	0.31	4.96	-0.12	-2.05	-5.38	-7.09	66.01	0.16
124	0.36	4.79	3.53	-0.49	6.42	-6.72	66.01	2.43
125	0.35	5.13	-0.44	-1.41	-1.40	-6.90	63.79	3.73
126	0.33	4.75	-1.99	-1.23	-9.02	-6.86	58.33	3.46
127	0.36	4.73	0.36	-1.53	4.56	-7.29	61.10	3.11
128	-0.07	-1.38	2.12	-8.54	-9.27	-1.78	54.45	-0.97
129	-0.13	-1.34	2.14	-8.46	0.16	-2.53	47.17	-2.51
130	-0.06	-1.74	-2.76	-8.60	2.08	-2.47	57.05	-0.39
131	-0.15	-1.22	0.09	-8.08	5.34	-2.74	64.38	-0.47
132	-0.10	-2.01	-3.16	-8.12	-3.11	-2.73	54.72	3.38
133	-0.13	-1.45	2.02	-8.04	2.46	-2.24	68.54	1.40
134	-0.11	-1.36	-2.75	-7.68	0.87	-2.20	62.77	0.46
135	-0.10	-1.66	0.80	-7.99	-6.50	-2.85	58.11	-3.03
136	-0.10	-1.39	-0.66	-8.05	-0.64	-2.58	60.67	-1.20
137	-0.08	-2.04	-0.70	-8.62	5.92	-2.03	63.08	2.07
138	-0.11	-1.44	-2.66	-8.23	-3.00	-2.64	52.05	-12.58
139	-0.09	-1.34	1.73	-7.92	1.25	-2.09	54.21	-4.23
140	-0.13	-1.72	-1.14	-8.65	3.68	-2.46	60.90	-3.56
141	-0.14	-1.29	-1.05	-8.42	-4.40	-1.80	55.03	-1.73
142	-0.10	-1.33	-4.62	-9.38	-5.19	-2.68	56.49	-0.74
143	-0.09	-1.35	-1.42	-9.07	3.70	-2.50	61.63	0.05
144	-0.44	-6.62	-0.08	-8.43	3.03	2.65	34.50	2.06
145	-0.47	-6.28	1.57	-9.29	-6.12	2.33	29.93	-1.97
146	-0.44	-6.15	-0.35	-8.90	-6.12	2.76	31.77	-0.39
147	-0.47	-6.03	1.02	-8.50	-3.98	2.55	27.37	6.41
148	-0.47	-6.47	0.58	-9.16	-4.25	2.53	29.28	-2.16
149	-0.47	-6.41	0.58	-8.59	-7.69	2.66	32.78	-4.26
150	-0.44	-6.63	0.33	-8.96	5.06	2.56	29.31	0.25
151	-0.45	-6.32	4.26	-8.83	-3.55	2.68	35.36	-1.39
152	-0.44	-6.65	0.05	-8.71	-11.51	2.71	30.30	3.43
153	-0.44	-6.47	-0.37	-8.79	7.18	2.80	30.82	-6.53
154	-0.44	-6.12	0.06	-8.76	3.86	2.95	28.19	-2.98
155	-0.44	-6.12	1.80	-9.11	-9.21	2.68	30.34	6.30
156	-0.46	-6.68	-0.77	-8.69	1.26	2.63	30.94	-0.36
157	-0.47	-6.39	-0.08	-8.15	7.13	2.80	28.23	1.12
158	-0.45	-6.39	-0.75	-8.70	8.69	2.90	31.24	1.50
159	-0.46	-6.57	-1.82	-8.71	-3.29	2.75	28.59	-2.87

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
120	2.78	-6.70	6.71	-4.07	75.91	32.50	-4.02	-2.35
121	1.75	-6.51	6.99	1.30	76.01	32.64	-3.86	-1.98
122	2.33	-6.45	7.21	-2.74	73.44	32.65	-3.94	-3.11
123	2.80	-6.68	7.26	-2.89	72.83	33.33	-3.87	-2.74
124	3.47	-6.47	6.87	1.60	66.53	32.72	-3.73	-1.95
125	1.90	-6.45	7.49	0.13	71.55	33.59	-3.90	-2.40
126	2.54	-6.66	7.75	-4.69	70.82	32.24	-3.83	-3.09
127	1.80	-6.47	6.30	-0.37	75.11	32.34	-3.76	-2.51
128	5.08	-2.11	4.54	-3.75	56.90	24.33	-1.11	-3.41
129	6.93	-2.21	4.69	-1.04	53.20	22.82	-0.86	-3.26
130	5.71	-2.02	4.77	-2.04	59.19	22.72	-1.09	-3.87

131	6.18	-2.16	5.19	-1.55	56.71	22.38	-0.87	-3.38
132	4.92	-2.36	4.70	-3.26	53.43	21.56	-0.71	-3.77
133	5.37	-2.04	5.37	4.33	55.12	23.02	-1.00	-3.41
134	4.82	-2.12	4.95	2.78	57.33	21.93	-1.02	-4.04
135	5.17	-2.10	4.70	2.48	52.38	21.10	-0.78	-3.61
136	5.59	-2.43	5.28	-2.07	55.91	21.57	-1.07	-3.74
137	6.13	-2.32	4.42	-3.71	55.69	22.77	-0.81	-3.72
138	4.85	-2.26	4.81	6.46	55.44	22.67	-0.71	-3.47
139	4.82	-2.22	5.09	2.24	54.98	22.90	-0.79	-3.88
140	6.04	-2.27	4.38	-0.44	59.54	22.51	-0.85	-3.18
141	6.36	-2.39	5.49	-0.20	54.86	21.78	-0.90	-3.71
142	5.02	-2.24	5.11	2.02	57.42	21.87	-0.84	-3.51
143	5.52	-2.22	4.91	1.56	58.08	23.02	-1.00	-3.42
144	-0.21	1.06	1.30	-0.44	38.34	16.26	0.17	-2.55
145	-0.70	1.07	0.82	0.28	39.73	16.77	0.10	-2.79
146	-0.67	0.88	0.94	0.56	41.10	17.62	0.23	-2.75
147	-0.68	1.10	1.41	0.32	40.17	16.63	0.19	-2.42
148	-0.41	1.17	1.07	0.73	37.75	15.98	0.12	-2.83
149	-0.55	1.15	1.39	3.93	39.29	16.75	0.30	-3.12
150	-1.68	1.29	1.09	2.30	41.93	16.56	0.33	-2.48
151	-0.46	1.00	1.37	-0.76	41.86	16.89	0.23	-3.06
152	-1.16	0.71	0.72	-3.59	36.96	15.92	0.09	-2.85
153	-0.73	1.16	1.43	-2.06	37.10	16.71	0.20	-2.38
154	-0.77	1.19	1.48	0.25	39.82	16.00	0.06	-1.96
155	-0.23	1.11	0.91	0.10	39.59	16.68	0.02	-2.58
156	-0.49	1.09	1.59	1.28	37.71	16.57	0.07	-2.28
157	-0.44	1.01	0.76	0.78	41.09	16.19	0.21	-2.43
158	-0.30	1.03	1.11	-2.55	38.76	15.61	0.29	-2.24
159	-1.59	1.10	1.12	-0.03	39.03	16.28	0.16	-2.11

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
120	-3.49	13.40	-0.17	2.77	24.03	0.22	-1.92	8.36
121	-1.56	13.53	-0.15	2.91	24.00	0.19	-0.13	9.30
122	-0.33	13.08	-0.14	2.69	23.29	-0.11	-10.57	21.22
123	4.41	13.72	-0.15	2.98	24.76	0.23	-14.55	8.52
124	-5.43	13.28	-0.15	2.53	23.69	0.36	3.17	30.92
125	-2.68	13.25	-0.13	2.61	24.53	0.23	43.45	1.59
126	7.50	15.48	-0.16	2.78	24.04	0.28	-3.80	25.95
127	4.32	14.32	-0.14	2.77	23.76	0.04	2.42	-25.97
128	-13.42	27.45	-0.24	7.34	17.51	2.47	0.94	9.49
129	10.58	26.06	-0.24	7.23	17.53	2.31	14.43	46.44
130	4.21	24.93	-0.25	7.19	17.44	2.41	-21.84	-14.67
131	1.14	27.25	-0.25	7.89	17.62	2.36	-6.28	-1.76
132	-1.71	25.57	-0.28	7.46	17.35	2.48	0.91	32.36
133	12.79	24.80	-0.28	7.83	17.59	2.47	-0.89	-14.20
134	3.23	24.22	-0.24	7.25	17.33	2.29	7.74	36.11
135	5.29	24.94	-0.25	7.41	17.48	2.41	0.58	13.47
136	-3.92	26.67	-0.25	7.53	17.72	2.42	-8.16	22.44
137	-3.58	26.44	-0.26	7.61	16.62	2.49	-1.47	-16.57
138	4.11	25.58	-0.24	7.62	17.52	2.36	-16.44	-25.70
139	-1.61	25.23	-0.24	7.50	17.71	2.41	-13.21	25.78
140	-2.54	27.55	-0.25	7.87	18.15	2.40	1.36	18.40
141	-12.06	26.71	-0.24	7.31	17.41	2.28	-12.71	9.79

142	1.45	25.83	-0.25	7.40	17.14	2.38	-6.48	-4.14
143	3.09	26.96	-0.25	7.34	17.09	2.50	-5.15	17.91
144	-4.85	8.77	-0.03	7.05	9.88	2.53	14.66	9.71
145	-3.68	8.85	-0.04	7.36	9.83	2.65	13.40	23.05
146	-2.12	8.74	-0.03	7.62	10.26	2.57	-4.56	-13.00
147	0.77	9.60	-0.04	7.20	10.29	2.55	-1.43	5.03
148	1.14	9.42	-0.03	7.40	10.21	2.61	-10.34	1.62
149	1.93	8.85	-0.04	7.68	9.93	2.31	-10.78	11.24
150	-1.34	10.06	-0.04	7.44	10.26	2.50	2.13	6.41
151	3.92	9.28	-0.03	7.41	10.52	2.47	-23.89	21.08
152	1.39	9.20	-0.02	7.35	10.53	2.59	16.81	5.07
153	-8.77	8.14	-0.02	7.68	10.60	2.67	2.72	-9.55
154	-0.43	9.26	-0.04	7.60	10.54	2.54	-2.97	14.19
155	0.07	9.38	-0.03	7.27	10.46	2.65	2.90	14.79
156	-0.81	9.37	-0.05	7.54	10.21	2.54	10.72	0.67
157	-1.46	9.47	-0.05	7.26	10.23	2.60	-7.04	-39.70
158	-5.91	8.46	-0.04	7.27	9.78	2.38	0.29	9.16
159	-3.15	8.99	-0.05	7.41	10.58	2.70	-10.17	2.93

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
160	-183.25	2.14	5.62	-25.74	3.13	10.32	3.56	11.78
161	-200.77	1.99	5.47	-27.03	3.73	10.43	3.14	12.40
162	-189.77	1.88	5.81	-25.08	3.29	10.28	2.50	12.23
163	-180.47	1.92	5.11	-24.66	3.56	9.96	3.94	11.75
164	-190.80	2.23	5.54	-24.51	1.97	10.54	4.33	11.66
165	-202.82	2.32	5.41	-25.31	2.83	10.24	3.98	11.72
166	-191.96	2.42	5.77	-26.22	4.19	10.76	2.92	12.35
167	-202.76	2.17	6.14	-25.94	2.82	10.63	3.41	11.44
168	-200.75	1.98	4.80	-24.85	2.05	10.13	4.60	11.72
169	-203.33	2.07	4.43	-25.07	3.21	10.43	4.22	11.70
170	-210.89	2.59	5.25	-26.95	2.38	10.55	4.53	11.48
171	-182.98	1.78	5.53	-26.40	2.18	10.46	3.14	12.03
172	-198.69	1.99	5.10	-24.36	2.57	10.33	3.05	11.84
173	-188.40	1.90	5.59	-24.79	3.64	10.55	3.01	11.43
174	-214.04	2.14	5.54	-24.92	2.11	10.22	2.96	11.82
175	-184.63	2.21	5.32	-25.17	2.94	10.15	4.37	12.25
176	-200.57	2.15	5.04	-22.81	1.93	9.60	4.05	11.62
177	-192.89	1.82	5.99	-23.73	1.29	9.49	3.28	11.57
178	-203.63	1.64	5.83	-20.94	-0.29	9.36	3.67	12.18
179	-188.18	2.06	5.25	-23.68	1.11	9.95	5.50	11.80
180	-209.99	1.94	5.75	-22.03	1.03	9.56	4.07	11.49
181	-186.37	1.67	5.88	-21.16	0.82	9.68	4.08	11.39
182	-179.41	2.00	4.79	-22.47	1.65	9.46	3.93	11.91
183	-205.64	1.85	6.10	-21.08	0.69	9.89	2.51	12.14
184	-191.97	1.98	5.27	-22.05	0.35	9.36	3.29	11.41
185	-189.61	1.88	5.68	-22.09	1.18	9.34	3.93	11.81
186	-194.95	2.16	5.47	-23.46	2.22	9.99	4.03	11.41
187	-199.69	1.98	5.83	-21.48	0.05	9.70	3.21	11.41
188	-186.98	1.60	5.00	-23.36	1.17	9.80	2.65	11.23
189	-194.99	1.63	5.53	-23.27	1.00	9.60	4.56	12.00
190	-195.44	1.23	5.91	-21.99	1.57	9.26	3.87	12.39
191	-197.60	1.64	5.88	-21.36	0.88	9.76	2.72	10.91
192	-80.99	1.13	5.78	-41.96	-7.61	14.12	4.56	-2.34

193	-89.68	0.98	5.35	-40.67	-7.33	14.32	4.86	-3.09
194	-87.61	1.04	5.68	-40.81	-7.47	14.54	5.72	-3.11
195	-75.78	0.82	5.89	-41.56	-7.69	14.38	4.84	-3.28
196	-81.87	1.14	5.62	-39.90	-7.61	14.36	5.82	-3.01
197	-86.38	1.03	5.52	-40.84	-7.57	14.22	4.74	-2.69
198	-90.67	1.07	5.62	-40.01	-7.85	14.09	4.38	-2.72
199	-85.90	0.90	5.82	-39.91	-7.87	14.58	4.61	-3.11

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
160	93.69	50.35	-0.06	43.76	-7.96	-11.45	-1.65	-39.83
161	94.19	51.22	0.05	41.97	-8.07	-11.77	-1.83	-35.89
162	96.08	50.24	-0.01	44.07	-8.94	-11.94	-1.74	-33.20
163	91.01	50.82	-0.02	41.62	-9.97	-11.61	-1.79	-35.37
164	94.91	51.64	-0.03	44.52	-10.47	-11.14	-1.66	-38.97
165	99.80	49.04	-0.02	49.32	-10.67	-11.51	-1.78	-35.32
166	100.32	49.55	0.01	48.96	-8.32	-11.16	-1.87	-35.89
167	91.74	50.26	-0.03	47.65	-8.62	-11.54	-1.66	-34.51
168	99.23	48.64	-0.05	42.04	-8.21	-11.68	-1.67	-34.09
169	91.16	51.29	-0.01	48.34	-11.59	-11.22	-1.69	-38.04
170	84.86	50.02	0.02	47.47	-8.47	-12.16	-1.65	-39.32
171	104.41	49.99	-0.04	46.22	-7.03	-11.70	-1.61	-36.51
172	95.70	51.92	0.01	43.17	-11.26	-11.33	-1.77	-42.20
173	99.92	49.58	0.00	46.70	-8.98	-12.03	-1.72	-36.09
174	92.96	50.20	-0.05	47.19	-10.20	-11.95	-1.65	-39.79
175	87.99	50.84	-0.06	47.82	-10.64	-11.26	-1.78	-40.49
176	92.57	45.83	-0.03	46.46	-3.54	-10.29	-1.66	-40.48
177	103.07	45.08	-0.08	46.56	-6.48	-10.76	-1.59	-34.80
178	89.56	45.20	-0.04	45.63	-3.64	-10.46	-1.56	-43.17
179	100.68	43.71	-0.03	49.09	-2.93	-10.14	-1.61	-40.29
180	105.61	44.15	-0.06	48.19	-3.47	-9.72	-1.61	-40.95
181	91.34	45.93	-0.02	50.35	-3.56	-10.55	-1.62	-40.77
182	87.76	44.55	0.00	46.54	-7.49	-10.17	-1.51	-43.01
183	97.67	44.55	-0.04	49.44	-4.48	-10.41	-1.59	-36.79
184	93.30	44.01	-0.02	47.30	-2.87	-10.32	-1.78	-37.03
185	101.12	47.15	0.00	45.58	-6.35	-9.75	-1.68	-42.95
186	92.30	43.85	-0.03	49.68	-5.53	-10.36	-1.73	-39.98
187	90.84	43.94	-0.02	44.30	-1.86	-10.72	-1.55	-37.11
188	101.37	42.91	-0.04	48.77	-4.67	-9.64	-1.62	-45.48
189	95.02	44.73	-0.04	45.22	-2.35	-10.64	-1.65	-39.73
190	93.66	43.89	-0.01	45.99	-6.29	-9.88	-1.58	-39.34
191	103.16	44.83	-0.01	47.51	-5.19	-10.19	-1.53	-36.73
192	59.69	57.64	-0.04	16.47	24.54	-9.22	2.64	-37.59
193	69.07	58.35	-0.09	18.14	26.55	-9.27	2.42	-38.58
194	64.30	59.42	-0.09	19.73	26.26	-9.01	2.47	-37.31
195	64.31	57.75	-0.08	18.64	26.38	-9.27	2.49	-39.18
196	66.35	57.50	-0.09	17.61	26.14	-9.63	2.47	-34.66
197	70.49	57.92	-0.09	19.08	25.79	-9.25	2.47	-36.50
198	67.45	59.20	-0.07	19.74	26.43	-9.27	2.38	-38.96
199	69.13	57.68	-0.11	19.25	26.73	-9.21	2.41	-38.46

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
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160	0.05	-6.82	-24.67	-24.74	-2.61	-2.86	-2.71	-1.60
161	0.00	-6.42	-24.61	-59.26	-2.55	-2.53	6.00	-1.60
162	-0.30	-6.70	-24.98	8.36	-2.47	-2.78	20.37	-1.59
163	-0.11	-5.77	-24.59	50.59	-2.62	-2.67	6.97	-1.61
164	0.19	-6.04	-24.78	-12.92	-2.65	-2.69	-11.54	-1.60
165	-0.02	-5.73	-24.78	-25.13	-2.64	-2.84	8.52	-1.63
166	-0.06	-7.15	-24.72	10.36	-2.90	-2.60	11.33	-1.60
167	0.05	-5.02	-24.62	-18.12	-2.65	-2.70	-9.13	-1.60
168	-0.18	-5.74	-24.65	-43.31	-2.61	-2.74	2.60	-1.63
169	0.05	-3.92	-24.69	-0.11	-2.62	-2.81	-0.57	-1.59
170	-0.22	-6.42	-24.74	31.59	-2.54	-2.42	4.93	-1.60
171	0.38	-6.29	-24.79	17.32	-2.47	-2.80	3.16	-1.58
172	-0.20	-4.47	-24.71	-76.32	-2.72	-2.81	14.78	-1.58
173	-0.02	-3.41	-24.55	12.62	-2.59	-2.77	13.40	-1.61
174	-0.19	-6.36	-24.80	-30.09	-2.58	-2.56	11.87	-1.60
175	0.01	-5.58	-24.80	-0.33	-2.57	-2.98	8.06	-1.59
176	-0.21	-7.35	-22.87	-2.52	-2.79	-3.04	-3.04	-1.43
177	0.42	-6.96	-22.77	-14.25	-2.87	-2.93	-3.93	-1.45
178	-0.02	-6.32	-22.67	6.85	-2.92	-3.19	-7.99	-1.45
179	-0.09	-6.74	-22.78	-7.19	-2.85	-2.83	8.51	-1.45
180	0.17	-6.39	-22.66	56.84	-2.89	-2.75	1.92	-1.46
181	0.19	-5.99	-22.80	8.81	-2.85	-2.82	-0.75	-1.45
182	0.14	-5.99	-22.76	24.31	-2.86	-2.90	-4.31	-1.46
183	-0.08	-7.56	-22.53	13.39	-2.79	-2.74	16.60	-1.45
184	-0.11	-6.77	-22.95	19.90	-3.09	-2.74	1.85	-1.47
185	0.08	-6.63	-22.79	28.47	-2.86	-3.04	-8.57	-1.44
186	-0.04	-6.29	-22.94	53.97	-2.89	-3.03	-0.93	-1.44
187	-0.14	-6.33	-22.59	99.70	-3.14	-2.75	-4.33	-1.43
188	0.00	-7.06	-22.71	-0.85	-2.92	-2.89	0.54	-1.45
189	-0.07	-8.13	-22.59	-18.23	-2.91	-3.01	-8.11	-1.44
190	-0.17	-7.08	-22.39	23.48	-2.85	-2.87	-3.17	-1.46
191	-0.06	-7.27	-22.67	41.89	-2.92	-2.76	-3.93	-1.46
192	-0.08	-4.09	-16.51	-9.54	-3.59	-2.93	-0.63	-1.16
193	-0.14	-3.11	-16.34	18.48	-3.60	-3.00	1.15	-1.14
194	0.04	-3.51	-16.64	17.91	-3.51	-2.91	1.10	-1.13
195	-0.17	-3.94	-16.52	15.91	-3.51	-2.89	-1.44	-1.13
196	0.08	-3.88	-16.58	-20.06	-3.47	-2.90	5.56	-1.13
197	0.09	-3.67	-16.49	-16.57	-3.57	-2.88	3.05	-1.13
198	0.05	-3.94	-16.42	-16.93	-3.42	-2.97	1.59	-1.11
199	0.02	-4.02	-16.51	-25.71	-3.60	-2.90	2.24	-1.14

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
160	-1.72	-4.57	-3.31	-5.64	1.91	1.82	6.79	-74.96
161	2.72	-4.07	0.55	-5.64	1.76	1.89	6.87	-73.97
162	-4.32	-4.19	-2.77	-5.46	2.01	1.92	7.05	-73.97
163	-1.39	-4.01	-0.95	-5.48	2.46	1.69	6.74	-74.71
164	-0.61	-4.33	-1.97	-5.78	2.93	1.84	6.85	-75.67
165	2.85	-4.36	-3.46	-5.41	2.81	1.98	6.35	-75.37
166	-1.58	-3.98	-1.06	-5.35	1.94	1.90	6.59	-74.94
167	-0.38	-5.08	-3.13	-5.36	2.77	1.81	6.39	-75.00
168	2.47	-3.87	-0.88	-5.81	2.49	1.71	7.18	-75.15
169	-2.51	-4.45	-6.66	-4.91	1.83	2.03	6.90	-76.07
170	6.57	-3.95	-4.60	-5.75	1.82	2.01	6.96	-73.68

171	2.24	-3.92	-1.15	-5.47	2.71	1.61	6.63	-73.51
172	3.40	-4.54	-2.52	-5.54	2.40	2.00	6.80	-73.48
173	-0.75	-4.31	2.15	-5.78	2.35	1.56	6.88	-74.65
174	3.78	-4.76	-3.69	-5.35	1.90	1.99	7.05	-75.07
175	2.70	-4.13	-3.26	-5.61	1.83	1.74	6.62	-75.22
176	-3.40	-4.70	-1.81	-4.85	1.49	1.10	5.28	-69.91
177	3.89	-4.52	-2.20	-5.42	1.47	1.05	5.44	-68.29
178	-0.89	-4.57	-0.54	-5.06	1.72	1.23	5.32	-68.69
179	0.34	-4.88	-0.94	-5.10	1.60	1.22	5.19	-67.29
180	2.91	-4.43	-2.20	-5.38	1.18	1.28	5.03	-68.04
181	0.20	-4.51	0.35	-5.67	1.11	1.13	5.20	-69.65
182	-5.24	-4.31	-1.03	-5.08	1.14	1.21	4.73	-67.72
183	2.46	-4.05	0.58	-5.67	0.91	1.03	5.01	-68.14
184	-1.50	-4.78	-1.48	-5.34	1.04	1.23	5.58	-68.25
185	0.24	-5.14	0.73	-4.87	1.21	1.05	5.17	-68.14
186	-0.78	-4.30	-4.96	-5.25	1.03	1.33	5.06	-68.46
187	1.72	-5.38	0.77	-4.80	1.35	0.99	5.47	-67.65
188	-0.43	-5.19	-0.03	-5.43	1.03	1.19	5.13	-71.14
189	4.51	-4.64	-3.26	-4.93	0.78	0.95	5.52	-68.34
190	0.38	-4.65	0.48	-4.86	1.19	1.08	5.19	-69.56
191	-3.32	-4.37	4.83	-5.59	1.22	1.00	5.24	-71.35
192	-1.10	-6.19	10.44	-5.27	-10.64	1.78	8.50	-46.49
193	-0.22	-5.99	10.70	-5.10	-10.32	1.84	8.95	-46.44
194	1.19	-6.00	10.41	-5.23	-9.90	1.72	8.65	-45.75
195	-0.99	-6.04	11.34	-5.36	-10.16	1.78	8.59	-44.99
196	-1.78	-5.87	9.88	-5.37	-10.11	1.82	8.86	-46.13
197	0.12	-6.17	11.67	-5.32	-10.56	1.82	8.33	-46.32
198	0.29	-6.03	10.23	-5.05	-10.16	1.82	8.42	-44.92
199	-2.02	-5.95	9.82	-5.22	-9.75	1.57	8.69	-45.94

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
160	-7.10	3.84	2.29	7.25	-1.41	-42.89	-2.47	-2.31
161	-7.16	4.00	2.40	6.79	-1.29	-44.10	-2.43	-2.37
162	-6.88	3.82	2.18	6.67	-1.35	-44.13	-2.49	-2.35
163	-7.18	3.48	2.31	6.99	-1.38	-42.27	-2.53	-2.32
164	-6.67	3.67	2.34	6.50	-1.34	-42.33	-2.46	-2.34
165	-8.06	3.80	2.15	6.43	-1.41	-42.56	-2.48	-2.36
166	-6.97	3.84	2.07	6.30	-1.37	-41.59	-2.35	-2.30
167	-7.50	3.63	2.27	8.27	-1.29	-43.63	-2.40	-2.33
168	-7.23	3.84	2.14	7.07	-1.36	-42.42	-2.44	-2.32
169	-6.82	3.72	2.20	6.12	-1.31	-43.38	-2.53	-2.33
170	-6.81	3.71	2.21	7.09	-1.38	-41.57	-2.45	-2.37
171	-6.97	3.74	2.16	7.13	-1.35	-42.14	-2.49	-2.32
172	-6.01	3.77	2.13	7.48	-1.41	-42.50	-2.42	-2.30
173	-6.58	3.62	2.20	6.64	-1.30	-41.40	-2.46	-2.35
174	-6.09	3.80	2.19	6.27	-1.46	-42.84	-2.53	-2.33
175	-7.31	3.70	2.41	6.68	-1.30	-41.21	-2.56	-2.35
176	-6.83	3.47	2.53	4.38	-1.11	-39.26	-2.01	-2.17
177	-7.14	3.45	2.16	4.46	-1.08	-38.15	-2.01	-2.17
178	-6.84	3.45	2.44	4.82	-1.20	-37.87	-1.96	-2.14
179	-7.17	3.28	2.32	4.76	-1.15	-38.48	-2.04	-2.17
180	-6.73	3.38	2.39	3.85	-1.07	-36.51	-1.99	-2.13
181	-6.45	3.47	2.29	5.05	-0.93	-37.26	-1.92	-2.14

182	-6.41	3.52	2.16	4.22	-1.01	-40.58	-2.04	-2.16
183	-6.37	3.34	2.26	4.47	-1.11	-38.40	-1.98	-2.14
184	-6.07	3.42	2.22	5.10	-1.07	-38.98	-1.95	-2.18
185	-6.69	3.57	2.23	4.02	-1.03	-37.01	-2.01	-2.19
186	-6.10	3.26	2.21	4.04	-1.15	-37.83	-1.92	-2.15
187	-6.41	3.36	2.25	4.21	-1.12	-41.20	-2.00	-2.14
188	-6.48	3.30	2.49	4.27	-1.03	-39.02	-1.98	-2.14
189	-6.73	3.32	2.34	4.37	-1.12	-37.91	-1.96	-2.18
190	-6.61	3.66	2.16	4.43	-1.14	-37.43	-1.86	-2.15
191	-7.03	3.45	2.56	4.75	-1.13	-38.65	-1.94	-2.17
192	14.34	3.23	0.52	-6.89	-0.44	-54.93	-0.40	-0.38
193	15.28	3.27	0.54	-6.69	-0.47	-54.20	-0.34	-0.37
194	15.25	3.25	0.65	-6.55	-0.50	-55.28	-0.39	-0.36
195	15.22	3.25	0.61	-6.99	-0.47	-54.89	-0.36	-0.38
196	15.23	3.32	0.62	-6.76	-0.44	-55.82	-0.36	-0.31
197	14.98	3.30	0.73	-7.01	-0.45	-54.51	-0.36	-0.36
198	14.34	3.25	0.68	-7.27	-0.47	-54.67	-0.36	-0.37
199	14.44	3.17	0.51	-6.82	-0.48	-54.34	-0.37	-0.36

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
160	0.37	6.16	-4.89	-3.95	-0.18	-5.80	216.68	-4.24
161	0.44	6.30	-5.02	-1.35	9.89	-6.94	215.70	0.94
162	0.47	5.80	-2.58	-2.61	8.27	-5.37	202.79	-3.53
163	0.41	6.05	2.06	-4.03	5.60	-6.26	205.63	-4.57
164	0.40	5.86	4.56	-3.05	-1.66	-6.13	201.34	-2.44
165	0.43	6.14	-4.27	-1.20	-3.83	-6.40	219.57	-2.51
166	0.35	6.13	-2.92	-2.09	8.70	-6.09	207.33	1.18
167	0.39	5.87	-2.87	-4.76	8.99	-7.35	200.76	5.54
168	0.38	6.11	2.58	-1.78	-3.50	-5.84	212.19	-5.32
169	0.41	5.65	-0.03	-1.87	9.47	-6.14	212.81	2.41
170	0.44	5.74	-2.44	-1.68	6.05	-6.48	202.50	3.86
171	0.45	6.48	-1.42	-1.59	4.23	-6.51	218.85	-1.59
172	0.38	6.75	2.44	-0.51	1.69	-6.58	204.05	-0.83
173	0.46	5.72	-0.38	-0.25	5.15	-5.50	191.42	0.98
174	0.45	6.33	3.98	-2.61	-5.38	-5.59	205.05	-3.88
175	0.40	5.97	6.85	-2.28	7.55	-6.99	200.65	-0.70
176	0.45	7.06	-0.46	-1.72	1.09	-6.70	193.26	3.23
177	0.44	6.59	0.91	-1.08	-0.45	-7.42	197.52	2.04
178	0.46	7.05	-2.54	-0.60	1.25	-6.31	187.34	0.15
179	0.47	6.80	-3.35	-0.78	2.85	-7.77	196.66	2.32
180	0.45	6.54	-0.56	-2.66	5.28	-7.34	204.22	-2.50
181	0.49	6.72	2.50	-0.06	5.80	-6.91	198.68	-4.05
182	0.51	7.05	-4.15	-2.54	-1.36	-7.48	201.35	0.50
183	0.45	7.01	-1.19	-0.48	3.29	-8.19	205.06	-2.27
184	0.50	6.51	1.42	-1.52	2.04	-7.31	184.98	5.86
185	0.46	6.88	0.90	-0.33	1.45	-7.19	200.08	-1.33
186	0.41	6.38	-0.92	0.12	-2.64	-8.02	193.03	4.59
187	0.48	7.11	-1.69	-1.28	-2.29	-7.25	195.79	-4.40
188	0.44	6.54	3.58	-0.63	5.67	-6.94	198.26	-5.30
189	0.42	6.39	-4.58	-1.29	7.96	-7.29	198.65	-3.19
190	0.46	6.64	-4.68	-1.11	-4.70	-7.46	196.34	3.04
191	0.45	6.81	2.51	-2.38	0.31	-7.21	195.56	1.44
192	0.51	7.88	-0.86	5.86	-1.23	-9.89	31.77	-1.15

193	0.49	7.70	0.24	5.63	-4.70	-9.82	32.51	-0.43
194	0.54	7.71	-0.61	5.51	-5.12	-9.75	32.83	1.99
195	0.49	7.83	0.62	5.83	2.58	-9.67	33.03	0.79
196	0.52	7.78	-1.62	5.24	-1.38	-9.49	37.38	1.85
197	0.54	7.94	0.26	4.88	0.93	-9.66	28.25	-0.89
198	0.48	7.63	-0.16	6.08	1.97	-9.63	26.78	-1.78
199	0.50	7.86	-1.26	5.43	1.97	-9.74	29.21	-1.97

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
160	-15.09	1.43	8.38	2.09	67.10	50.59	-4.05	-2.61
161	-15.99	1.74	8.50	-0.09	74.46	49.89	-3.80	-2.33
162	-15.23	1.34	8.48	-1.80	66.68	50.94	-3.90	-2.70
163	-14.71	1.41	6.35	-3.76	65.56	52.11	-3.89	-3.17
164	-15.64	1.35	8.94	-11.87	69.07	50.02	-3.94	-2.36
165	-15.55	1.51	9.37	4.37	68.67	53.08	-3.92	-2.80
166	-15.40	1.37	8.38	2.83	69.57	52.07	-3.76	-1.98
167	-15.25	1.65	7.20	-1.24	69.80	49.58	-4.03	-3.15
168	-15.21	1.29	7.64	-5.76	65.67	51.80	-3.99	-4.23
169	-15.65	1.44	7.37	0.61	67.24	51.04	-3.94	-3.61
170	-15.53	1.54	7.44	-3.07	72.23	52.70	-4.07	-2.45
171	-14.94	1.54	9.01	-0.48	66.04	51.64	-4.11	-2.67
172	-15.34	1.52	8.43	2.06	71.91	50.70	-4.02	-2.48
173	-14.78	1.57	7.45	-2.85	62.87	50.65	-3.87	-1.81
174	-15.51	1.51	9.59	0.14	65.71	49.51	-4.02	-1.82
175	-15.48	1.26	6.97	7.15	69.96	53.52	-3.95	-3.26
176	-12.35	1.49	9.88	-9.25	57.89	46.03	-3.59	-1.89
177	-12.31	1.61	9.42	-3.73	67.80	45.21	-3.64	-2.45
178	-12.14	1.49	10.23	-0.63	58.49	45.33	-3.73	-2.72
179	-12.13	1.61	10.38	0.24	55.05	46.72	-3.63	-2.00
180	-11.85	1.51	9.89	-9.35	58.31	44.55	-3.63	-3.61
181	-11.39	1.68	10.30	-3.55	55.82	44.55	-3.69	-0.74
182	-11.71	1.58	9.81	-3.65	60.66	45.97	-3.65	-2.52
183	-11.84	1.62	7.87	-3.91	62.48	45.42	-3.52	-2.08
184	-12.30	1.57	9.70	8.09	55.78	46.36	-3.52	-2.01
185	-11.78	1.63	8.33	-0.93	55.84	45.39	-3.58	-2.29
186	-11.46	1.71	9.94	-4.37	60.19	44.13	-3.66	-3.12
187	-12.17	1.61	8.29	-5.53	58.65	44.83	-3.78	-2.80
188	-11.84	1.68	7.48	-2.86	65.02	45.54	-3.61	-2.11
189	-11.77	1.64	9.14	2.22	55.82	44.24	-3.61	-2.37
190	-12.38	1.71	7.76	-4.37	63.61	45.04	-3.67	-2.30
191	-11.86	1.57	8.30	0.81	62.45	45.01	-3.51	-2.05
192	-14.53	1.99	13.18	1.33	106.76	38.07	-5.34	-5.15
193	-13.82	2.08	13.41	1.91	104.39	37.64	-5.16	-4.44
194	-14.37	2.15	13.46	-0.20	110.70	38.01	-5.34	-5.04
195	-13.91	2.12	13.46	0.42	102.96	38.66	-5.33	-4.93
196	-14.24	2.07	14.12	2.08	97.99	38.20	-5.46	-5.31
197	-14.21	2.13	13.13	-1.64	100.89	37.80	-5.44	-4.59
198	-14.39	2.09	13.96	-2.45	103.11	38.34	-5.33	-5.41
199	-13.74	2.08	13.50	-2.52	105.29	38.24	-5.41	-4.78

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
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160	4.80	-3.26	-0.21	3.57	2.42	-0.68	-12.67	-24.28
161	-3.84	-4.72	-0.18	2.94	2.29	-0.37	1.69	35.79
162	2.38	-4.79	-0.18	3.21	2.09	-0.33	4.88	-13.73
163	-0.25	-2.02	-0.17	2.96	2.59	-0.43	27.32	49.32
164	3.73	-5.32	-0.15	3.77	2.77	-0.50	-9.12	57.33
165	0.28	-2.82	-0.17	3.31	2.63	-0.50	22.97	31.74
166	-4.59	-3.20	-0.18	3.25	3.45	-0.72	-15.61	33.03
167	-1.01	-4.60	-0.17	4.16	2.50	-0.33	-27.43	-11.64
168	-4.66	-1.98	-0.19	4.12	2.54	-0.22	-1.92	-10.90
169	3.96	-7.01	-0.20	2.86	2.56	-0.62	17.28	-12.73
170	2.95	-4.19	-0.21	3.43	2.63	-0.19	-14.50	-1.16
171	-7.27	-3.21	-0.20	3.49	2.62	-0.49	1.07	-12.08
172	7.69	-6.61	-0.16	2.74	2.46	-0.33	12.29	-21.91
173	0.38	-3.77	-0.20	3.10	2.67	-0.35	-8.43	25.41
174	1.04	-2.66	-0.18	3.00	2.98	-0.48	-6.63	-62.55
175	1.60	-1.83	-0.18	3.19	2.77	-0.62	-10.72	-93.17
176	-0.52	-7.31	-0.07	2.13	2.59	-0.40	-6.00	14.91
177	-0.77	-6.06	-0.12	1.95	3.14	-0.16	-9.91	-11.31
178	-1.38	-6.04	-0.11	1.95	2.90	-0.10	-14.84	-5.31
179	3.62	-5.16	-0.12	2.88	2.42	-0.29	-2.65	24.60
180	-0.82	-5.22	-0.08	2.50	2.59	-0.01	-9.33	68.11
181	1.07	-4.69	-0.09	1.89	2.66	-0.23	5.36	72.62
182	1.54	-7.98	-0.10	2.35	2.91	-0.19	-19.95	-12.65
183	0.38	-6.18	-0.10	2.44	2.23	-0.51	30.27	7.42
184	-1.62	-7.10	-0.09	1.94	2.99	-0.47	-3.11	-34.92
185	2.42	-5.25	-0.08	2.10	2.45	-0.43	3.54	3.01
186	-5.80	-6.76	-0.10	1.44	2.25	-0.49	-21.05	17.21
187	7.44	-5.48	-0.12	2.46	2.44	-0.27	23.77	30.52
188	-5.44	-6.90	-0.11	2.41	2.25	-0.51	10.61	-0.47
189	0.88	-6.94	-0.09	2.24	2.12	-0.02	20.18	-19.77
190	8.28	-6.66	-0.10	2.54	2.34	-0.33	21.36	57.54
191	-3.19	-5.28	-0.09	2.05	2.63	-0.25	7.47	31.17
192	5.63	-31.18	0.32	-1.71	12.41	0.20	-0.23	7.04
193	2.70	-30.14	0.28	-2.06	12.10	0.16	-29.14	20.81
194	-1.45	-29.85	0.28	-1.81	12.41	0.08	17.53	16.41
195	-2.76	-29.39	0.31	-1.97	12.28	0.17	-0.96	15.02
196	-3.77	-29.74	0.29	-2.19	12.65	0.09	-9.79	31.38
197	-3.05	-29.86	0.30	-1.86	12.48	0.13	-18.07	9.58
198	-2.17	-30.42	0.30	-1.83	12.21	0.18	-5.19	14.38
199	-2.93	-30.44	0.29	-1.97	12.62	0.04	16.43	-0.26

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
200	-84.84	1.13	5.88	-40.53	-7.81	13.76	5.16	-2.61
201	-86.46	1.17	5.92	-41.73	-7.49	14.37	4.96	-2.65
202	-85.65	0.92	5.61	-41.84	-7.73	14.39	4.88	-3.04
203	-83.58	0.92	5.57	-41.07	-7.79	14.00	5.78	-2.87
204	-84.03	0.90	5.59	-40.04	-8.07	15.05	5.36	-2.80
205	-88.01	1.00	5.82	-40.27	-7.37	14.65	5.64	-2.95
206	-91.14	1.09	5.67	-38.94	-7.49	13.90	5.00	-3.05
207	-87.84	1.13	5.79	-39.92	-7.62	14.06	4.61	-3.12
208	-139.32	2.88	8.38	-29.54	-7.87	11.87	0.06	1.37
209	-136.35	2.65	8.20	-29.63	-7.90	11.88	0.14	1.88
210	-136.39	2.88	8.19	-29.62	-7.91	11.54	0.09	1.74

211	-140.86	2.90	8.40	-28.91	-8.02	11.83	1.40	1.81
212	-141.28	2.64	8.02	-30.90	-8.11	11.87	0.91	1.38
213	-147.44	3.03	8.55	-29.96	-8.18	12.28	0.27	0.89
214	-137.06	2.83	8.21	-28.00	-7.95	11.69	-0.13	1.43
215	-136.69	2.81	8.31	-31.18	-8.32	12.24	-0.51	1.76
216	-144.43	3.02	8.72	-30.56	-7.94	11.59	0.02	1.81
217	-137.63	2.89	8.13	-30.66	-7.76	11.71	0.40	1.39
218	-132.91	2.90	8.74	-30.41	-8.04	11.61	0.24	2.17
219	-146.30	2.70	8.31	-32.57	-8.00	12.00	0.33	1.31
220	-141.57	2.69	8.20	-30.05	-8.02	11.79	0.65	1.73
221	-143.00	2.79	8.25	-29.94	-7.79	11.74	-0.08	1.70
222	-138.77	2.83	8.47	-28.20	-7.64	11.82	0.14	1.48
223	-140.99	2.63	8.27	-30.91	-7.93	11.74	0.18	1.55
224	-54.29	0.60	4.08	-17.30	-4.02	6.24	2.53	-0.47
225	-49.96	0.52	4.11	-17.63	-3.57	6.25	2.57	-0.65
226	-53.86	0.63	4.07	-16.88	-3.88	6.15	2.39	-0.17
227	-53.38	0.55	4.12	-16.78	-3.97	6.27	2.64	-0.44
228	-52.04	0.58	4.25	-17.54	-3.97	6.26	2.50	-0.44
229	-49.03	0.59	4.08	-16.65	-3.89	6.26	2.41	-0.68
230	-51.57	0.47	3.92	-17.80	-4.05	6.13	2.83	-0.45
231	-51.91	0.73	4.07	-16.88	-3.72	6.31	2.68	-0.76
232	-53.06	0.54	4.08	-17.42	-3.88	6.13	2.80	-0.66
233	-55.52	0.57	4.00	-17.43	-3.79	6.01	2.67	-0.74
234	-52.68	0.62	3.76	-16.64	-3.85	6.32	2.48	-0.51
235	-57.68	0.48	4.06	-16.80	-3.86	5.94	2.56	-0.56
236	-53.28	0.60	4.17	-17.07	-3.56	6.08	2.93	-0.65
237	-50.64	0.50	4.15	-16.55	-3.98	5.93	2.86	-0.63
238	-54.90	0.57	3.85	-17.16	-3.98	6.12	2.89	-0.43
239	-54.82	0.69	3.85	-16.27	-3.91	6.14	2.75	-0.44

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
200	66.06	57.96	-0.06	20.15	25.87	-9.12	2.49	-40.56
201	68.69	59.17	-0.05	18.98	26.20	-9.42	2.57	-38.06
202	65.93	58.25	-0.06	17.31	26.06	-9.06	2.45	-37.70
203	66.02	57.57	-0.06	17.54	24.57	-9.56	2.48	-37.48
204	62.62	60.56	-0.05	18.59	26.34	-9.47	2.50	-37.12
205	69.42	59.25	-0.07	18.44	25.76	-9.14	2.51	-38.09
206	69.43	58.91	-0.08	17.90	25.60	-9.19	2.56	-38.37
207	66.76	59.12	-0.06	17.58	23.78	-9.08	2.57	-36.53
208	105.25	45.16	0.14	30.65	17.19	-11.84	1.41	-56.05
209	100.97	49.11	0.17	33.24	19.20	-11.96	1.36	-54.82
210	110.23	47.17	0.14	32.19	18.96	-11.95	1.33	-57.81
211	99.46	47.41	0.17	34.11	19.23	-11.51	1.40	-58.91
212	100.89	47.64	0.20	32.97	18.91	-11.64	1.29	-55.22
213	105.89	47.15	0.17	34.74	19.38	-11.71	1.35	-54.68
214	100.66	48.39	0.12	33.05	18.56	-11.70	1.47	-55.39
215	105.40	48.05	0.14	35.47	19.87	-11.73	1.47	-56.98
216	107.76	46.39	0.15	32.78	18.55	-11.55	1.40	-56.60
217	102.67	46.58	0.16	34.43	19.08	-11.72	1.35	-56.71
218	106.31	45.50	0.13	35.30	18.10	-11.87	1.42	-56.77
219	103.64	46.85	0.14	34.19	18.24	-11.49	1.25	-56.12
220	105.64	46.90	0.17	34.57	17.92	-11.69	1.39	-57.90
221	104.76	45.29	0.14	35.97	21.36	-11.91	1.36	-56.05

222	103.79	46.27	0.14	33.43	19.41	-11.53	1.41	-56.73
223	101.88	45.23	0.17	34.42	19.64	-12.02	1.45	-53.76
224	44.53	25.82	-0.06	11.68	10.35	-5.06	0.95	-25.85
225	45.43	25.95	-0.04	11.59	10.33	-4.95	1.01	-27.25
226	42.76	25.29	-0.04	11.55	10.68	-4.99	1.00	-26.64
227	44.83	25.97	-0.05	12.62	10.83	-5.07	0.98	-26.88
228	43.61	25.36	-0.03	12.76	10.30	-4.92	1.10	-27.08
229	46.60	25.62	-0.05	11.49	10.21	-5.09	1.00	-26.17
230	43.98	25.50	-0.06	11.29	10.13	-5.14	0.85	-26.86
231	43.86	25.36	-0.04	11.30	10.41	-4.94	1.01	-25.41
232	46.44	25.68	-0.05	10.66	10.53	-5.00	0.95	-27.32
233	47.20	25.26	-0.02	10.75	11.18	-5.01	0.97	-26.35
234	45.19	26.43	-0.02	11.44	10.41	-5.05	0.94	-25.67
235	46.18	25.03	-0.03	11.83	10.34	-4.91	1.00	-26.54
236	45.28	25.60	-0.04	11.79	10.74	-5.00	0.99	-27.22
237	45.71	26.50	-0.06	11.87	10.86	-5.00	0.99	-28.24
238	44.83	25.30	-0.04	11.05	10.70	-5.05	0.95	-26.50
239	45.99	25.08	-0.06	11.38	10.68	-4.93	0.88	-26.89

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
200	-0.14	-3.90	-16.42	18.94	-3.52	-2.90	2.97	-1.12
201	-0.01	-3.17	-16.60	20.24	-3.57	-2.91	4.30	-1.14
202	0.14	-3.90	-16.67	6.48	-3.59	-2.84	-0.10	-1.11
203	-0.24	-3.89	-16.69	4.00	-3.46	-3.00	-2.50	-1.12
204	-0.10	-3.70	-16.54	0.10	-3.60	-2.94	-1.46	-1.13
205	-0.10	-3.49	-16.60	1.65	-3.53	-2.91	1.37	-1.12
206	0.11	-3.34	-16.64	5.16	-3.55	-2.70	-1.01	-1.15
207	0.02	-4.27	-16.57	-0.97	-3.50	-2.94	9.24	-1.12
208	0.27	-7.69	-16.18	9.78	-3.44	-1.72	2.71	-1.02
209	-0.04	-7.39	-15.94	-9.32	-3.51	-1.63	1.56	-1.04
210	-0.07	-7.15	-16.15	14.88	-3.47	-1.77	-0.45	-1.06
211	0.08	-7.65	-16.49	5.04	-3.52	-1.64	-0.63	-1.05
212	-0.03	-7.56	-16.13	-45.52	-3.43	-1.85	-3.97	-1.04
213	0.07	-6.72	-16.27	-34.23	-3.53	-1.73	-1.11	-1.03
214	0.17	-7.68	-16.15	2.33	-3.41	-1.95	2.65	-1.04
215	0.23	-7.81	-16.12	8.35	-3.41	-1.63	-1.63	-1.05
216	-0.06	-7.14	-16.07	-6.69	-3.47	-1.52	1.36	-1.04
217	-0.18	-7.44	-16.05	-10.69	-3.52	-1.66	4.16	-1.04
218	-0.16	-7.64	-16.02	4.38	-3.55	-1.58	-0.43	-1.03
219	-0.09	-7.83	-16.27	51.83	-3.50	-1.72	2.93	-1.05
220	-0.08	-7.89	-16.36	-26.86	-3.50	-1.95	1.99	-1.04
221	-0.24	-7.62	-16.20	-13.42	-3.47	-1.61	-0.73	-1.07
222	0.22	-7.37	-16.13	-6.98	-3.48	-1.67	5.77	-1.05
223	-0.05	-8.48	-16.17	7.62	-3.47	-1.88	1.22	-1.02
224	0.08	-2.72	-7.93	-0.84	-2.21	-1.51	1.68	-0.53
225	0.06	-2.53	-7.92	0.54	-2.21	-1.54	1.26	-0.55
226	-0.03	-2.53	-8.02	3.16	-2.19	-1.67	1.70	-0.53
227	-0.03	-2.64	-7.78	3.03	-2.22	-1.61	-2.87	-0.53
228	-0.07	-2.56	-7.82	-6.44	-2.22	-1.52	-0.46	-0.53
229	0.06	-2.67	-7.95	7.79	-2.17	-1.65	-0.30	-0.54
230	0.03	-2.67	-7.73	-7.24	-2.21	-1.63	-0.68	-0.53
231	0.08	-2.32	-7.83	-8.27	-2.18	-1.51	0.09	-0.52
232	-0.01	-2.60	-7.85	-3.83	-2.19	-1.56	-0.25	-0.54

233	0.14	-2.88	-7.84	2.24	-2.19	-1.51	0.55	-0.54
234	-0.05	-2.42	-7.90	-4.40	-2.19	-1.60	0.13	-0.54
235	-0.02	-2.55	-7.79	-11.70	-2.22	-1.55	-0.60	-0.53
236	0.19	-2.48	-7.84	12.76	-2.20	-1.54	3.63	-0.53
237	-0.02	-2.83	-7.95	2.25	-2.23	-1.51	-0.31	-0.53
238	-0.12	-2.61	-7.96	13.30	-2.23	-1.60	-0.31	-0.51
239	0.10	-2.52	-7.69	18.92	-2.22	-1.58	0.44	-0.54

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
200	3.99	-5.95	11.60	-5.28	-9.88	1.58	8.77	-45.86
201	0.55	-5.96	10.24	-5.13	-9.75	1.70	8.73	-46.07
202	-1.77	-6.09	11.31	-4.91	-9.69	1.63	8.59	-45.79
203	1.56	-5.98	10.38	-5.02	-10.51	1.79	8.69	-45.10
204	1.56	-6.08	10.34	-5.40	-10.12	1.67	8.71	-45.35
205	2.13	-6.31	10.69	-5.05	-10.29	1.70	8.93	-45.65
206	2.79	-6.28	10.42	-5.44	-10.01	1.75	8.49	-46.18
207	-0.59	-6.19	11.15	-5.23	-10.73	1.85	8.51	-46.18
208	-0.86	-7.96	-3.95	-3.60	-9.52	0.68	11.12	-49.79
209	0.81	-8.49	-5.59	-3.29	-8.62	0.65	11.07	-50.16
210	-1.82	-8.57	-4.86	-3.74	-8.55	0.75	11.01	-50.88
211	-0.18	-8.60	-5.31	-3.43	-8.14	0.70	11.15	-50.58
212	2.61	-8.70	-6.29	-3.63	-8.86	0.61	10.85	-50.54
213	-3.08	-8.69	-3.95	-3.63	-9.03	0.68	11.10	-50.64
214	-1.21	-8.67	-3.89	-3.44	-8.30	0.57	10.92	-50.48
215	-0.63	-8.21	-5.17	-3.56	-8.01	0.58	10.75	-51.30
216	-2.33	-8.45	-6.57	-3.49	-8.58	0.62	10.57	-49.36
217	-1.75	-8.63	-4.28	-3.72	-9.07	0.52	11.17	-51.38
218	-1.16	-8.33	-3.96	-3.48	-8.78	0.65	11.05	-52.15
219	-1.30	-8.25	-5.48	-3.26	-8.99	0.67	10.97	-50.97
220	-2.13	-8.82	-5.83	-3.54	-8.76	0.61	10.94	-50.33
221	0.26	-8.28	-5.24	-4.00	-8.47	0.67	10.96	-50.23
222	1.12	-8.32	-6.96	-3.18	-9.03	0.48	11.17	-52.19
223	2.79	-8.82	-5.29	-3.80	-8.83	0.67	10.90	-50.60
224	0.34	-4.36	2.59	-2.85	-4.74	0.74	5.72	-22.17
225	-0.31	-4.18	2.99	-3.02	-4.17	0.66	5.46	-22.03
226	-0.12	-4.41	2.85	-2.84	-4.73	0.61	5.85	-21.93
227	-0.33	-4.36	3.11	-2.91	-4.70	0.71	5.44	-21.77
228	0.44	-4.44	3.35	-2.94	-4.84	0.57	5.52	-22.25
229	0.33	-4.39	3.32	-2.77	-4.87	0.60	5.59	-22.24
230	-0.06	-4.13	2.39	-2.85	-4.78	0.58	5.64	-21.99
231	0.24	-4.30	4.62	-2.89	-4.63	0.62	5.66	-22.01
232	0.96	-4.38	4.07	-3.03	-4.86	0.60	5.48	-22.08
233	-0.65	-4.41	3.56	-2.99	-4.53	0.68	5.63	-21.70
234	0.36	-4.36	3.33	-2.77	-4.45	0.63	5.65	-21.98
235	-0.71	-4.32	3.30	-2.85	-4.91	0.59	5.67	-22.12
236	-0.29	-4.26	3.75	-2.72	-4.75	0.66	5.69	-22.65
237	0.22	-4.19	2.49	-2.91	-4.69	0.71	5.68	-22.50
238	-1.44	-4.36	3.56	-2.73	-5.08	0.70	5.66	-22.30
239	-0.57	-4.42	2.22	-2.74	-4.73	0.58	5.58	-22.22

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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200	14.32	3.22	0.58	-6.43	-0.50	-53.19	-0.36	-0.36
201	14.99	3.34	0.54	-6.74	-0.49	-54.35	-0.39	-0.36
202	14.82	3.28	0.68	-7.03	-0.52	-54.80	-0.41	-0.36
203	14.71	3.23	0.63	-6.98	-0.46	-54.42	-0.37	-0.36
204	14.96	3.30	0.60	-7.13	-0.47	-55.09	-0.35	-0.36
205	14.23	3.28	0.66	-6.96	-0.49	-55.95	-0.36	-0.36
206	15.18	3.28	0.69	-7.34	-0.50	-55.62	-0.37	-0.35
207	14.69	3.23	0.61	-6.88	-0.52	-53.88	-0.36	-0.35
208	9.47	3.48	1.57	-4.65	-0.96	-45.31	-1.00	-0.74
209	9.10	3.48	1.40	-3.70	-0.97	-43.24	-1.04	-0.68
210	9.37	3.51	1.54	-3.31	-0.90	-42.54	-0.99	-0.72
211	9.83	3.51	1.58	-3.74	-0.92	-42.03	-0.94	-0.72
212	9.76	3.52	1.42	-3.73	-0.93	-43.64	-0.96	-0.71
213	9.51	3.50	1.54	-4.16	-0.90	-44.19	-1.03	-0.72
214	9.23	3.38	1.59	-3.62	-0.89	-43.99	-1.02	-0.71
215	9.61	3.50	1.54	-3.80	-0.93	-44.32	-1.01	-0.74
216	8.98	3.46	1.50	-3.79	-0.98	-44.82	-0.98	-0.72
217	9.21	3.52	1.53	-3.92	-0.92	-44.28	-0.94	-0.73
218	9.94	3.55	1.52	-4.54	-0.91	-41.51	-1.02	-0.72
219	9.63	3.53	1.40	-3.79	-0.93	-43.59	-0.99	-0.73
220	9.23	3.46	1.49	-3.60	-0.89	-45.16	-0.99	-0.73
221	8.94	3.48	1.46	-3.51	-0.91	-44.60	-0.97	-0.72
222	9.57	3.47	1.57	-3.09	-0.87	-43.51	-1.02	-0.73
223	9.36	3.49	1.43	-3.84	-0.88	-43.84	-0.98	-0.72
224	6.36	1.72	0.50	-2.51	-0.38	-24.29	-0.56	-0.24
225	6.20	1.80	0.47	-1.97	-0.35	-25.26	-0.57	-0.23
226	6.18	1.80	0.46	-2.30	-0.37	-24.28	-0.51	-0.24
227	6.37	1.78	0.42	-2.20	-0.34	-23.88	-0.53	-0.23
228	6.45	1.78	0.51	-1.83	-0.37	-24.04	-0.54	-0.23
229	7.00	1.71	0.46	-1.80	-0.35	-24.35	-0.53	-0.23
230	6.17	1.71	0.53	-2.11	-0.33	-25.11	-0.55	-0.21
231	6.09	1.76	0.48	-2.36	-0.36	-23.07	-0.53	-0.23
232	5.82	1.77	0.43	-2.38	-0.37	-24.22	-0.56	-0.22
233	5.85	1.74	0.46	-2.27	-0.32	-24.46	-0.55	-0.25
234	6.48	1.79	0.52	-2.15	-0.37	-23.83	-0.56	-0.25
235	6.59	1.79	0.45	-2.00	-0.36	-24.26	-0.56	-0.23
236	6.53	1.80	0.54	-2.13	-0.35	-24.30	-0.53	-0.25
237	6.32	1.77	0.46	-2.42	-0.32	-24.11	-0.54	-0.23
238	6.42	1.76	0.48	-2.25	-0.34	-24.32	-0.51	-0.22
239	6.80	1.81	0.45	-2.12	-0.34	-25.08	-0.53	-0.24

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
200	0.46	7.85	-0.13	6.00	-0.66	-9.63	27.73	-0.71
201	0.49	7.88	1.56	5.01	5.66	-9.86	37.21	-0.38
202	0.50	7.82	-0.21	5.62	3.39	-9.78	36.01	-3.02
203	0.47	7.87	1.17	5.57	6.66	-9.76	28.22	-1.35
204	0.47	7.73	-1.49	4.73	-0.17	-9.85	34.81	-0.69
205	0.51	7.78	0.66	4.81	0.90	-9.79	27.69	-1.68
206	0.48	7.68	0.46	5.55	0.83	-9.68	33.16	0.42
207	0.49	7.68	1.72	5.43	-3.05	-9.64	30.82	2.49
208	0.15	6.36	-2.14	8.22	-2.43	-9.98	99.11	-0.05
209	0.19	6.61	0.32	8.42	3.47	-9.67	83.51	-0.45
210	0.14	6.61	-1.32	7.86	6.02	-10.02	86.68	1.50

211	0.16	6.66	2.90	7.31	0.41	-9.81	87.51	1.78
212	0.22	6.31	-1.30	7.70	-0.04	-9.87	87.42	-0.67
213	0.15	6.49	-1.79	7.85	5.42	-9.54	85.96	-0.59
214	0.22	6.46	0.30	8.60	2.92	-9.67	90.41	0.31
215	0.20	6.38	-0.86	8.71	4.58	-9.76	83.46	-0.82
216	0.19	6.15	0.25	7.76	-2.22	-9.68	91.93	-0.68
217	0.20	6.57	-1.20	8.05	-1.02	-9.91	86.11	1.29
218	0.17	6.27	-1.14	7.68	2.32	-10.04	91.11	-1.23
219	0.15	6.39	-0.80	7.44	5.12	-9.89	88.36	0.45
220	0.15	6.49	-3.00	8.07	4.47	-9.78	95.29	-0.28
221	0.19	6.40	0.75	7.85	6.16	-9.63	90.57	-0.67
222	0.18	6.41	1.30	8.33	0.11	-9.71	81.50	3.06
223	0.22	6.24	-1.01	8.04	5.86	-9.95	86.36	-0.02
224	0.27	4.44	-0.04	3.08	2.50	-6.38	20.25	-1.03
225	0.24	4.52	-0.34	3.01	-1.29	-6.25	16.20	-0.21
226	0.25	4.55	-0.39	3.51	-0.60	-6.37	18.03	-1.33
227	0.26	4.52	0.03	3.32	-1.50	-6.28	24.04	-0.27
228	0.26	4.70	-0.29	3.18	-0.44	-6.27	17.92	-1.90
229	0.25	4.57	0.92	3.34	1.93	-6.40	22.77	-0.80
230	0.28	4.73	-0.66	2.98	4.13	-6.28	15.11	-0.47
231	0.24	4.58	0.43	3.54	-0.58	-6.26	25.20	0.44
232	0.27	4.64	0.84	3.11	0.14	-6.38	22.13	-0.19
233	0.25	4.51	0.22	3.69	-2.72	-6.35	21.97	0.80
234	0.27	4.51	-0.11	3.55	1.86	-6.42	25.28	-0.06
235	0.26	4.67	1.15	3.35	0.05	-6.42	23.04	-1.52
236	0.27	4.61	-0.75	3.40	-1.85	-6.07	16.17	1.21
237	0.26	4.65	-2.07	3.25	1.95	-6.27	18.98	-0.86
238	0.22	4.73	-0.73	3.27	-0.72	-6.28	20.99	0.14
239	0.24	4.67	1.08	3.59	-0.16	-6.19	18.79	-0.29

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
200	-14.58	2.15	13.52	1.39	98.94	38.13	-5.24	-4.83
201	-14.85	2.12	13.25	-1.46	105.91	38.10	-5.40	-4.96
202	-14.77	2.04	13.42	-0.83	108.07	37.94	-5.43	-4.34
203	-14.29	2.04	13.65	1.22	104.62	38.41	-5.42	-4.76
204	-14.52	2.02	13.38	-2.48	103.69	38.64	-5.38	-5.42
205	-13.83	2.00	13.13	-1.90	100.76	38.11	-5.22	-5.16
206	-14.72	2.09	13.54	1.63	99.44	36.76	-5.42	-5.11
207	-14.24	2.13	13.38	-2.68	106.04	38.62	-5.44	-5.01
208	-18.04	2.04	17.33	-4.13	80.49	45.11	-4.33	-0.42
209	-18.35	1.98	18.32	0.33	75.08	44.55	-4.26	-0.29
210	-18.34	1.91	17.45	-5.83	78.06	45.51	-4.19	0.29
211	-17.86	2.06	18.77	-1.57	76.22	44.80	-4.37	-0.99
212	-18.15	2.07	17.42	3.49	68.79	44.77	-4.29	-0.51
213	-17.90	1.96	18.27	2.48	78.99	45.19	-4.26	-1.08
214	-18.00	2.17	16.74	1.35	81.10	45.50	-4.36	0.34
215	-18.07	2.03	18.09	-0.23	80.85	45.97	-4.42	-0.63
216	-17.82	2.01	17.98	-1.08	76.05	44.86	-4.26	0.54
217	-18.26	1.97	17.85	2.39	76.52	45.77	-4.35	-0.24
218	-18.22	1.93	17.46	-2.42	78.00	44.64	-4.34	-0.75
219	-18.12	2.02	17.20	-2.55	79.55	45.31	-4.45	-0.57
220	-18.32	2.05	18.01	-0.15	73.51	44.83	-4.23	-1.39
221	-18.65	1.85	17.38	1.55	75.38	45.01	-4.24	0.04

222	-18.24	2.01	17.33	3.21	79.54	44.83	-4.33	-0.20
223	-17.86	2.09	17.70	-2.31	78.34	45.15	-4.28	0.51
224	-9.08	1.04	9.54	-0.38	45.23	20.55	-2.26	-2.70
225	-8.99	1.11	9.50	-2.62	47.46	20.68	-2.31	-2.31
226	-8.93	1.09	9.73	-1.20	48.83	20.40	-2.28	-2.68
227	-8.58	1.14	9.31	-0.97	45.76	20.14	-2.25	-2.35
228	-8.73	1.14	9.72	1.59	49.19	20.34	-2.30	-2.72
229	-9.14	1.21	9.50	0.67	47.88	20.41	-2.25	-2.50
230	-8.91	1.09	9.32	0.39	49.39	20.28	-2.25	-2.31
231	-9.06	1.07	9.52	1.36	48.38	20.82	-2.31	-2.84
232	-8.97	1.08	9.76	0.66	47.42	20.49	-2.30	-2.82
233	-8.95	1.16	9.71	-0.24	48.65	20.33	-2.32	-2.82
234	-8.79	1.05	9.83	0.41	50.36	20.50	-2.29	-2.90
235	-8.84	1.09	9.42	0.66	45.98	20.80	-2.30	-2.95
236	-9.18	1.08	9.80	-2.83	45.60	19.75	-2.31	-2.90
237	-9.04	1.19	9.56	1.14	49.97	20.36	-2.42	-2.38
238	-9.17	1.08	9.84	2.18	49.18	19.80	-2.27	-2.61
239	-8.69	1.11	9.76	2.64	47.25	20.56	-2.31	-2.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
200	-1.36	-29.64	0.29	-1.93	12.05	0.24	6.43	22.33
201	3.10	-29.13	0.30	-1.91	12.36	0.07	-7.30	-0.91
202	-5.52	-29.28	0.29	-2.07	12.24	0.09	-9.28	8.15
203	-4.47	-29.11	0.30	-1.96	12.32	0.14	-6.20	8.77
204	-2.98	-29.56	0.32	-1.95	12.15	0.15	7.99	31.91
205	-1.35	-29.64	0.30	-1.68	12.64	0.24	-18.47	-13.38
206	1.26	-30.90	0.30	-1.91	12.30	0.25	6.12	4.06
207	2.52	-29.07	0.29	-1.96	12.47	0.22	3.27	-13.81
208	-0.08	-22.82	0.23	-3.30	8.90	-1.48	9.45	-22.79
209	-0.46	-23.73	0.20	-3.31	9.41	-1.13	3.74	-10.82
210	3.20	-22.42	0.22	-3.21	9.23	-1.46	-3.21	9.83
211	-1.51	-22.56	0.20	-3.15	8.81	-1.43	-20.84	-1.75
212	3.07	-23.26	0.20	-3.21	9.20	-1.39	0.80	-30.30
213	-2.33	-23.08	0.23	-2.69	8.98	-1.30	-19.53	8.55
214	-6.54	-23.51	0.22	-3.41	8.76	-1.14	27.04	6.01
215	-3.79	-21.32	0.21	-3.00	8.93	-1.33	-17.66	-2.79
216	-0.24	-23.24	0.22	-3.03	9.09	-1.24	6.07	24.80
217	3.44	-24.06	0.20	-3.67	8.63	-1.29	-8.15	19.18
218	-1.92	-22.05	0.20	-3.29	9.00	-1.18	6.54	10.41
219	0.30	-22.68	0.21	-3.36	8.67	-1.31	-10.55	1.54
220	-2.80	-23.71	0.20	-3.40	8.54	-1.20	16.87	-20.57
221	-4.02	-23.59	0.23	-3.23	8.89	-1.33	-29.66	-8.53
222	3.71	-22.32	0.22	-3.65	9.42	-1.27	-7.45	20.01
223	4.01	-22.74	0.20	-3.23	9.05	-1.11	-0.92	8.73
224	-0.07	-13.13	0.10	-1.31	4.94	0.10	-2.81	9.80
225	3.00	-13.34	0.13	-1.17	5.09	0.13	9.49	1.20
226	1.83	-13.14	0.12	-1.12	4.86	0.09	-6.43	7.09
227	2.44	-12.89	0.11	-1.27	4.92	0.14	11.11	3.43
228	-1.82	-12.71	0.11	-1.30	5.04	0.23	5.82	2.18
229	2.60	-13.68	0.12	-1.18	4.89	0.13	-13.54	-2.96
230	-0.58	-12.59	0.13	-1.32	5.16	0.14	5.68	2.50
231	-1.66	-13.08	0.12	-1.37	5.20	0.05	0.12	4.41
232	4.51	-13.15	0.12	-1.50	4.62	0.10	-3.69	-7.06

233	1.30	-12.99	0.10	-1.29	5.15	0.15	-8.75	-0.68
234	3.05	-13.52	0.12	-1.26	5.30	0.06	-5.26	-12.29
235	1.63	-13.65	0.11	-1.20	5.12	0.00	-9.46	4.53
236	4.38	-13.38	0.12	-1.23	5.11	0.11	-0.87	-0.28
237	-2.09	-14.06	0.12	-1.23	4.84	0.01	-7.24	-4.51
238	2.60	-13.88	0.11	-1.47	4.69	0.15	-10.41	-5.46
239	1.90	-12.57	0.11	-1.22	5.22	0.13	12.39	8.11

For all Gaussian functions g_{lk} for all images f_k , the exponential factor $\alpha = 1/\epsilon$ is given by $\epsilon = 2.0615528128088303$.

- 5 Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not describe the material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . The hypothesis testing may use as alternative hypothesis that the filter does describe the material classifier. The alternative hypothesis may be given by $H_1: \exists m, m': \mu_m \neq \mu_{m'}$. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each material, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for material for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:
- 15 $mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J)$. The hypothesis testing may comprise determining a Mean Sum of Squares between, $mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1)$. The hypothesis testing may comprise performing an F-Test:
- 20

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

- 25 Herein, I is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For exam-

ple, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.05$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

5

Figure 2 shows experimental results of using the spot shape filter to calculate the material feature for different material groups. Specifically, the material feature φ_{2m} is shown for different materials, namely from left to right for a wooden board denoted board, a fabric, a dark skin, a pale skin, and a highly translucent skin denoted trans, and illustrates the separation of skin materials (right) from other materials (left). The spot shape filter passes the hypothesis testing. The p-value computed from the F-test is 0.0076. Thus, the Null-hypothesis H_0 can be rejected with respect to a significance level of 0.01. Further, the alternative hypothesis H_1 can be accepted with respect to a significance level of 0.01. Thus, the image filter is distinguishes at least two material classifiers. In addition, Figure 2 shows the mean value of the feature per material, $\bar{\varphi}_m$, as the line within the respective box. Vertical bars mark the Q1-1.5 interquartile range (IQR) and Q3+1.5 IQR, wherein $IQR = Q3 - Q1$, with Q1 and Q3 being the first and third quartiles. The box marks the Q1 and Q3 quartiles. Outliers are plotted as points. Figure 2 shows, that this material feature separates all skin samples from the non-skin samples. In other words, the expectation values of the Skin Features for skin and non-skin materials differ.

20

Figure 1 moreover shows, in a highly schematic illustration, an exemplary embodiment of the detector 110, wherein the detector 110 specifically may be embodied as a camera 136 and/or may be part of a camera 136. The camera 136 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 1 further shows an embodiment of a detector system 138, which, besides the at least one detector 110, comprises one or more beacon devices 140, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 1 further shows an exemplary embodiment of a human-machine interface 142, which comprises the at least one detector system 138 and, further, an entertainment device 144, which comprises the human-machine interface 142. The figure further shows an embodiment of a tracking system 146 for tracking a position of the object 112, which comprises the detector system 138. The components of the devices and systems shall be explained in further detail below.

35

Fig. 1 further shows an exemplary embodiment of a scanning system 148 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 148 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well

as, optionally, at least one further illumination source, not depicted here. The illumination source 124, generally, is configured for emitting at least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 140 and/or on a surface of the object 112. The scanning system 148 may be designed to
5 generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 148, specifically the detector 110, by using the at least one detector 110.

10 The detector 110, besides the optical sensors 120, comprises the at least one evaluation device 132, having e.g. at least one image analysis device 150 and/or the at least one position evaluation device 152, as symbolically depicted in Fig. 1. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially
15 combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 154 and/or by one or more interfaces, as symbolically depicted in Fig. 1. Further, the one or more connectors 154 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connect-
20 or 154, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 156 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

25 In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 158, the position of which may be manipulated by a user 160. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 160 himself or herself may be considered as the object 112, the position of which shall be detected.

30 As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 156. Further, the detector 110 may comprise the at least one transfer device 129 such as one or more optical systems, preferably comprising one or more lenses. An opening 162 inside the housing 156, which, preferably, is located concentrically with
35 regard to the optical axis 128 of the detector 110, preferably defines a direction of view 164 of the detector 110. A coordinate system 166 may be defined, in which a direction parallel or anti-parallel to the optical axis 128 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 128 may be defined as transversal directions. In the coordinate system 166, symbolically depicted in Fig. 1, a longitudinal direction is denoted by z, and trans-
40 versal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 142, in order to provide at least one item of information to a machine 168. In the embodiments schematically depicted in Fig. 1, the machine 168 may be a computer and/or may comprise a computer. Other
5 embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 168, such as into the computer.

As outlined above, Fig. 1 also depicts an example of a tracking system 146, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system
10 146 comprises the detector 110 and at least one track controller 170. The track controller 170 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 170 may be an independent device and/or may be fully or partially integrated into the machine 168, specifically the computer, as indicated in Fig. 1 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 142 may form part of an entertainment device 144. The machine 168, specifically the computer, may also form part of the entertainment device 144. Thus, by means of the user 160 functioning as the object 112 and/or by means of the user 160 handling a control device functioning as the object 112, the user 160
20 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Fig. 1 also depicts an example of an inertial measurement unit 172 for use in an electronic device. The inertial measurement unit 172 is adapted to receive data determined by the detector
25 110. The inertial measurement unit 172 further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is
30 adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

The inertial measurement unit 172 may comprise the detector 110 and/or may be connected to the detector 110, via at least one data connection. The evaluation device 132 and/or at least one processing device of the inertial measurement unit 172 may be configured for determining at least one combined distance information, in particular using at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters such as further sensor data from the further
40 sensor of the inertial measurement unit 172.

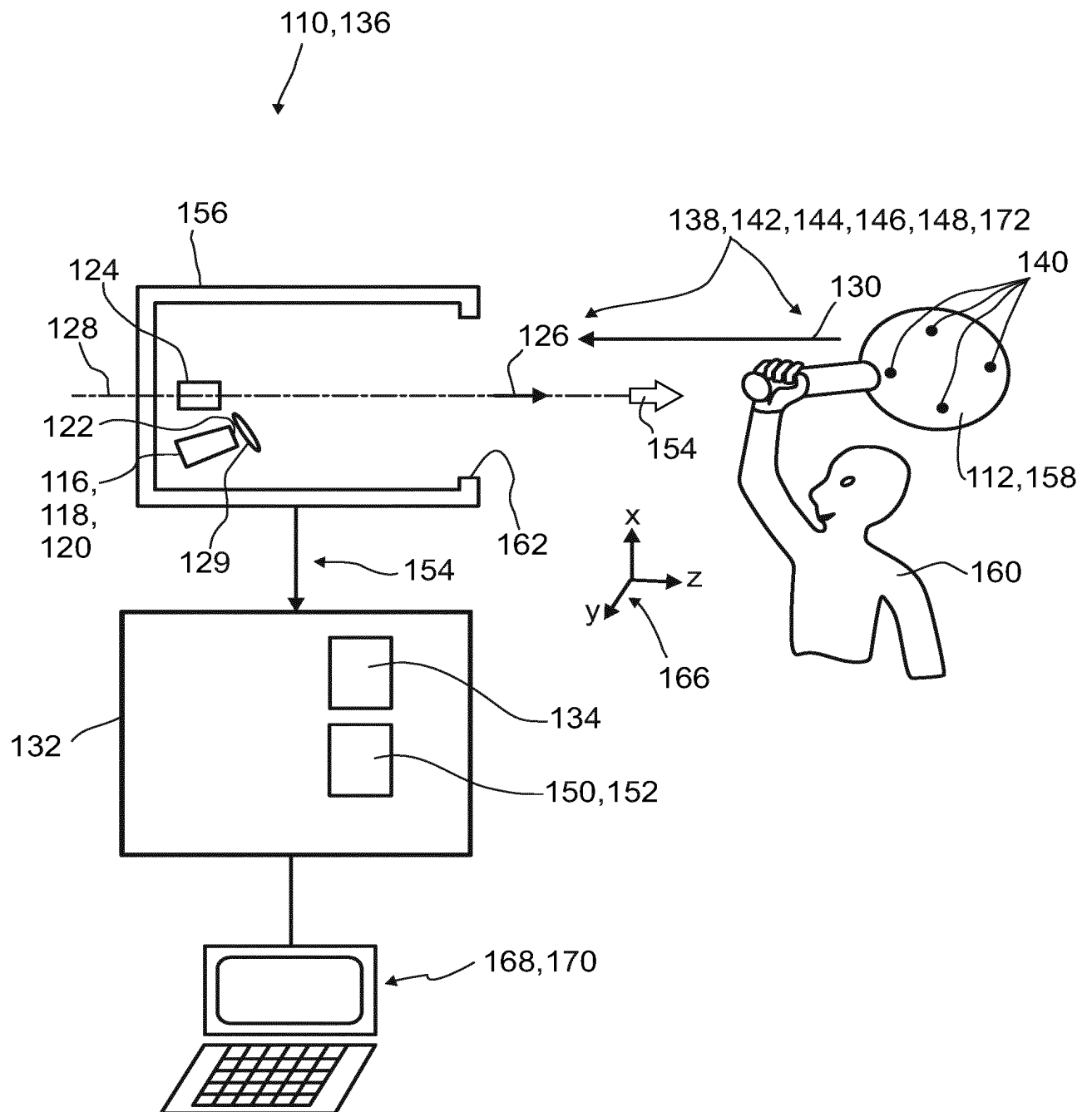
List of reference numbers

110	detector
112	object
116	sensor element
118	matrix
120	optical sensor
122	light-sensitive area
124	illumination source
126	illumination light beam
128	optical axis
129	transfer device
130	reflection light beam
132	evaluation device
134	divider
136	camera
138	detector system
140	beacon device
142	human-machine-interface
144	entertainment device
146	tracking system
148	scanning system
150	image analysis device
152	position evaluation device
154	connector
156	housing
158	control device
160	user
162	opening
164	direction of view
166	coordinate system
168	machine
170	track controller
172	inertial measurement unit

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FIG.1



2/2

FIG.2

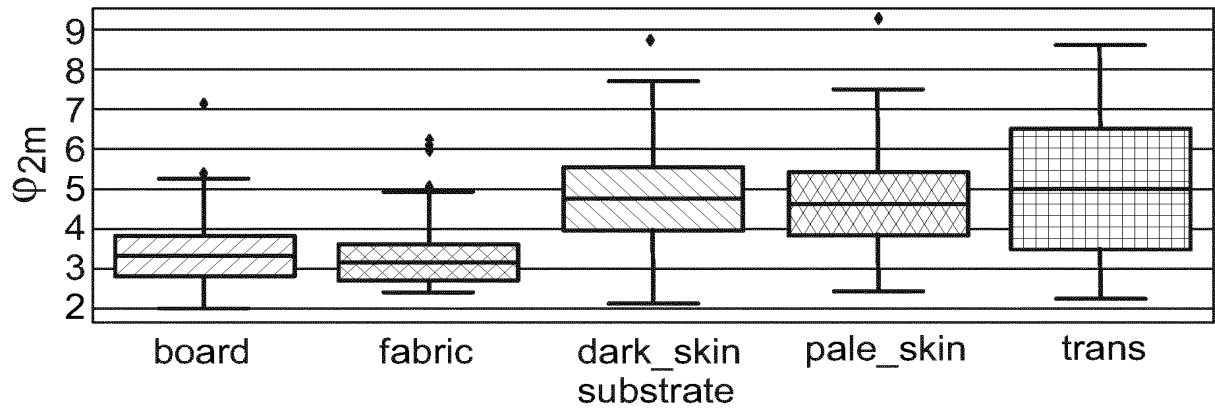
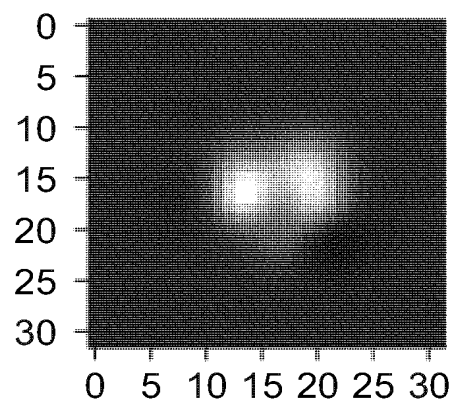


FIG.3



SCORE Placeholder Sheet for IFW Content

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TRANSMITTAL LETTER TO THE UNITED STATES DESIGNATED/ELECTED OFFICE (DO/EO/US) CONCERNING A SUBMISSION UNDER 35 U.S.C. 371		Attorney Docket No. 27843-1809 182037US01
		U.S. Application No. (if known, see 37 CFR 1.5)
International Application No. PCT/EP2020/056759	International Filing Date 03/13/2020	Priority Date Claimed 03/15/2019
Title of Invention DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		
First Named Inventor Schick, Friedrich		
Applicant herewith submits to the United States Designated/Elected Office (DO/EO/US) the following items and other information.		
1. ☒ This is an express request to begin national examination procedures (35 U.S.C. 371(f)). NOTE: The express request under 35 U.S.C. 371(f) will not be effective unless the requirements under 35 U.S.C. 371(c)(1), (2), and (4) for payment of the basic national fee, copy of the International Application and English translation thereof (if required), and the oath or declaration of the inventor(s) have been received.		
2. ☒ A copy of the International Application (35 U.S.C. 371(c)(2)) is attached hereto (not required if the International Application was previously communicated by the International Bureau or was filed in the United States Receiving Office (RO/US)).		
3. An English language translation of the International Application (35 U.S.C. 371(c)(2))		
a. ☐ is attached hereto.		
b. ☐ has been previously submitted under 35 U.S.C. 154(d)(4).		
4. An oath or declaration of the inventor(s) (35 U.S.C. 371(c)(4))		
a. ☒ is attached.		
b. ☐ was previously filed in the international phase under PCT Rule 4.17(iv).		
Items 5 to 8 below concern amendments made in the international phase.		
<u>PCT Article 19 and 34 amendments</u>		
5. ☐ Amendments to the claims under PCT Article 19 are attached (not required if communicated by the International Bureau) (35 U.S.C. 371(c)(3)).		
6. ☐ English translation of the PCT Article 19 amendment is attached (35 U.S.C. 371(c)(3)).		
7. ☐ English translation of annexes (Article 19 and/or 34 amendments only) of the International Preliminary Examination Report is attached (35 U.S.C. 371(c)(5)).		
<u>Cancellation of amendments made in the international phase</u>		
8a. ☐ Do not enter the amendment made in the international phase under PCT Article 19.		
8b. ☐ Do not enter the amendment made in the international phase under PCT Article 34.		
NOTE: A proper amendment made in English under Article 19 or 34 will be entered in the U.S. national phase application absent a clear instruction from applicant not to enter the amendment(s).		
The following items 9 to 17 concern a document(s) or information included.		
9. ☒ An Information Disclosure Statement under 37 CFR 1.97 and 1.98.		
10. ☒ A preliminary amendment.		
11. ☒ An Application Data Sheet under 37 CFR 1.76.		
12. ☐ A substitute specification. NOTE: A substitute specification cannot include claims. See 37 CFR 1.125(b).		
13. ☒ A power of attorney and/or change of address letter.		
14. ☐ A computer-readable form of the sequence listing in accordance with PCT Rule 13ter.3 and 37 CFR 1.821-1.825 (not required if sequence listing in text format was indicated on the PCT Request as part of the International Application and the sequence listing was published as part of the international application).		
15. ☐ Assignment papers (cover sheet and document(s)). Name of Assignee: _____		
16. ☐ 37 CFR 3.73(c) Statement (when there is an Assignee). _____		

This collection of information is required by 37 CFR 1.414 and 1.491-1.492. The information is required to obtain or retain a benefit by the public, which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 15 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Mail Stop PCT, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

U.S. APPLN. No. (if known – see 37 CFR 1.5)	INTERNATIONAL APPLICATION No. PCT/EP2020/056759	ATTORNEY DOCKET No. 27843-1809 182037US01
17. ☐ Other items or information: International Search Report and Written Opinion for corresponding PCT/EP2020/056759 mailed May 29, 2021, 11 pages.		
The following fees have been submitted.		CALCULATIONS
18. ☐ Basic national fee (37 CFR 1.492(a)) \$320		\$ 320.00
19. ☐ Examination fee (37 CFR 1.492(c)) • If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)..... \$0 • All other situations \$800		\$ 800.00
20. ☐ Search fee (37 CFR 1.492(b)) • If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)..... \$0 • Search fee (37 CFR 1.445(a)(2)) has been paid on the international application to the USPTO as an International Searching Authority \$140 • International Search Report prepared by an ISA other than the US and provided to the Office or previously communicated to the US by the IB..... \$540 • All other situations \$700		\$ 540.00
TOTAL OF 18, 19, and 20 =		\$ 1660.00
☐ Additional fee for specification and drawings filed in paper over 100 sheets (excluding sequence listing in compliance with 37 CFR 1.821(c) or (e) in an electronic medium or computer program listing in an electronic medium) (37 CFR 1.492(j)). Fee for each additional 50 sheets of paper or fraction thereof \$420		\$ 420.00
Total Sheets	Extra Sheets	Number of each additional 50 or fraction thereof (round up to a whole number)
139 - 100 =	39 / 50 =	1
		x \$420
Surcharge for furnishing any of the search fee, examination fee, or the oath or declaration after the date of commencement of the national stage (37 CFR 1.492(h))..... \$160		\$ 0
CLAIMS	NUMBER FILED	NUMBER EXTRA
Total claims	25 - 20 =	5
Independent claims	1 - 3 =	0
MULTIPLE DEPENDENT CLAIM(S) (if applicable)		+ \$860
Fee for submission of Sequence Listing text file of 300 MB to 800 MB (37 CFR 1.21(o)(1))..... \$1,060		\$
Fee for submission of Sequence Listing text file of more than 800 MB (37 CFR 1.21(o)(2))..... \$10,500		\$
Processing fee for furnishing the English translation later than 30 months from the earliest claimed priority date (37 CFR 1.492(i))..... \$140 +		\$
TOTAL OF ABOVE CALCULATIONS =		\$ 920.00
☐ Applicant asserts small entity status. See 37 CFR 1.27. Fees above are reduced by 1/2.		
☐ Applicant certifies micro entity status. See 37 CFR 1.29. Fees above are reduced by 3/4. Applicant must attach form PTO/SB/15A or B or equivalent.		
TOTAL NATIONAL FEE =		\$ 2580.00
Fee for recording the enclosed assignment (37 CFR 1.21(h)). The assignment must be accompanied by an appropriate cover sheet (37 CFR 3.28, 3.31)..... \$50.00 per property +		\$
TOTAL FEES ENCLOSED =		\$ 2580.00

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

- a. ☐ A check in the amount of \$ _____ to cover the above fees is enclosed.
- b. ☒ Please charge my Deposit Account No. 012384 in the amount of \$ 2580.00 to cover the above fees.
- c. ☒ The Director is hereby authorized to charge additional fees which may be required, or credit any overpayment, to Deposit Account No. 012384 as follows:
- i. ☐ any required fee.
- ii. ☒ any required fee except for excess claims fees required under 37 CFR 1.492(d) and (e) and multiple dependent claim fee required under 37 CFR 1.492(f).
- d. ☐ Fees are to be charged to a credit card. **WARNING:** Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. The PTO-2038 should only be mailed or faxed to the USPTO. However, when paying the basic national fee, the PTO-2038 may NOT be faxed to the USPTO.

ADVISORY: If filing by EFS-Web, do NOT attach the PTO-2038 form as a PDF along with your EFS-Web submission. Please be advised that this is not recommended and by doing so your credit card information may be displayed via PAIR. To protect your information, it is recommended to pay fees online by using the electronic payment method.

NOTE: Where an appropriate time limit under 37 CFR 1.495 has not been met, a petition to revive (37 CFR 1.137(a) or (b)) must be filed and granted to restore the International Application to pending status.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

- ☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013, and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE 1: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

NOTE 2: A U.S. national stage application may not claim priority to the international application of which it is the national phase. The filing date of a U.S. national stage application is the international filing date. See 35 U.S.C. 363.

Correspondence Address

- ☒ The address associated with Customer Number: 76656 OR ☐ Correspondence address below

Name					
Address					
City		State		Zip Code	
Country				Telephone	
Email					

Signature	/Christopher M. Goff/	Date	2021-09-15
Name (Print/Type)	Christopher M. Goff	Registration No. (Attorney/Agent)	41785

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Box No. VIII (iv) DECLARATION: INVENTORSHIP (only for the purposes of the designation of the United States of America)
The declaration must conform to the following standardized wording provided for in Section 214; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (iv). If this Box is not used, this sheet should not be included in the request.

**Declaration of inventorship (Rules 4.17(iv) and 51bis.1(a)(iv))
 for the purposes of the designation of the United States of America:**

I hereby declare that I believe I am the original inventor or an original joint inventor of a claimed invention in the application.

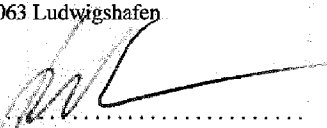
This declaration is directed to this international application of which it forms a part (if filing declaration with application).

This declaration is directed to international application No. PCT/..... (if furnishing declaration pursuant to Rule 26ter).

I hereby declare that the above-identified international application was made or authorized to be made by me.

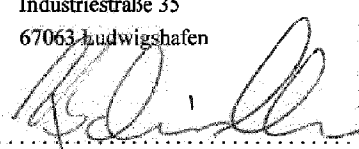
I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Name: SCHICK, Friedrich
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

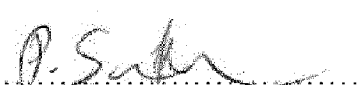
Date: 09.05.19

Name: SCHILLEN, Peter
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

Date: 09.04.19

Name: SCHINDLER, Patrick
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

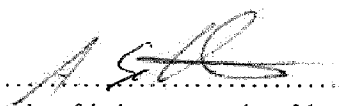
Date: 09.04.19

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: SCHMIDT, Andre
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

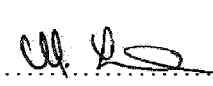
Date: 11.04.2019

Name: EBERSPACH, Michael
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 11.04.2019

Name: LENNARTZ, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

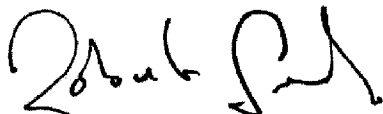
Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 11.4.2019

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

Name: SEND, Robert
Residence: Germany
Mailing Address: Luisenstr. 25



Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date: ... 2019/09/11

Name: DIESSELBERG, Lars
Residence: Germany
Mailing Address: Geibelstr. 22
76185 Karlsruhe

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: HENGEN, Heiko
Residence: Germany
Mailing Address: Siedlerweg 8
76872 Steinweiler

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Continuation of Box No. VIII (i) to (v) DECLARATION

If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.

Name: SEND, Robert
Residence: Germany
Mailing Address: Luisenstr. 25
 76137 Karlsruhe

Inventor's Signature:
 (The signature must be that of the inventor, not that of the agent)

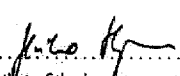
Date:

Name: DIESELBERG, Lars
Residence: Germany
Mailing Address: Geibelstr. 22
 76185 Karlsruhe

Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

Date: 2019-09-05

Name: HENGEN, Heiko
Residence: Germany
Mailing Address: SIEDLERWEG 8
 76872 STEINWEILER

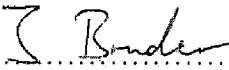
Inventor's Signature: 
 (The signature must be that of the inventor, not that of the agent)

Date: 2019-09-05

Continuation of Box No. VIII (i) to (v) DECLARATION

If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.

Name: BRUDER, Ingmar
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature: 

Date: 12.09.2019

(The signature must be that of the inventor, not that of the agent)

Name: UNGER, Jakob
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:

Date:

(The signature must be that of the inventor, not that of the agent)

Name: BONSIGNORE, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:

Date:

(The signature must be that of the inventor, not that of the agent)

Continuation of Box No. VIII (i) to (v) DECLARATION

If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.

Name: BRUDER, Ingmar
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: UNGER, Jakob
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date: 13.8.19

Name: BONSIGNORE, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date: 14.08.19

Continuation of Box No. VIII (i) to (v) DECLARATION

*If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where **more than three inventors are to be named** in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.*

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Friedrich Schick

Serial No.: TBD

Filed: September 15, 2021

Confirmation No.: TBD

For: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

September 15, 2021

PRELIMINARY AMENDMENT

TO THE COMMISSIONER FOR PATENTS:

Please enter the following amendments prior to calculation of fees due in the present application.

Amendments to the Specification are on page 2.

Amendments to the Abstract are on page 3.

Amendments to the Claims begin on page 5.

Remarks are on page 13.

AMENDMENTS TO THE SPECIFICATION

Please enter the following amendments to the specification.

After the TITLE, please add the following heading and paragraph:

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Phase Application of International Patent Application No. PCT/EP2020/056759, filed March 13, 2020, which claims priority to European Patent Application No. 19163250.4, filed March 15, 2019, the entire contents of which are hereby incorporated by reference herein.

AMENDMENTS TO THE ABSTRACT

Please amend the Abstract as follows:

[[A]] Described herein is a detector (110) for identifying at least one material property m is proposed. The detector (110) comprises includes at least one sensor element (116) comprising including a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122). The sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112). The detector (110) comprises includes at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image. ~~The evaluation device (132) is configured for determining at least one distance feature ϕ_{1x} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_x}| \geq 0.40$ with Φ_x being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The evaluation device (132) is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter,~~

~~the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .~~

(Figure 1)

AMENDMENTS TO THE CLAIMS

1. (Currently Amended) A detector (110) for identifying at least one material property m comprising
 - at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein the sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112); **and**
 - at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the evaluation device (132) is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; **[[or]] and** a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

wherein the evaluation device (132) is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter; **such as** a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; **[[or]] and** a further material dependent image filter Φ_{2other} which correlates to one or more of

the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, **and**

wherein the evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

2. (Currently Amended) The detector (110) according to ~~the preceding claim,~~ **claim 1**, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.
3. (Currently Amended) The detector (110) according to ~~the preceding claim~~ **claim 1**, wherein $p \leq 0.075$, ~~preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.~~
4. (Currently Amended) The detector (110) according to ~~any one of the preceding claims~~ **claim 1**, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, **and** a speckle, ~~and the like.~~

5. (Currently Amended) The detector (110) according to ~~any one of the preceding claims claim 1~~, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, ~~preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.~~
6. (Currently Amended) The detector (110) according to ~~any one of the preceding claims claim 1~~, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, ~~preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.~~
7. (Currently Amended) The detector (110) according to ~~any one of the preceding claims claim 1~~, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .
8. (Currently Amended) The detector (110) according to ~~any one of the preceding claims claim 1~~, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.
9. (Currently Amended) The detector (110) according to ~~any one of the preceding claims claim 1~~, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.
10. (Currently Amended) The detector (110) according to ~~any one of the preceding claims claim 1~~, wherein the evaluation device (132) is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.
11. (Currently Amended) The detector (110) according to ~~the preceding claim claim 10~~, wherein the evaluation device (132) is configured for determining at least one

of z and/or m by applying at least one further filter depending on at least one of ϕ_{1z} and ϕ_{2m} to the reflection image.

12. (Currently Amended) The detector (110) according to ~~any one of the preceding claims~~ **claim 1**, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element (116), wherein the evaluation device (132) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the combined signal Q and the distance feature ϕ_{1z} for determining the distance feature ϕ_{1z} .
13. (Currently Amended) The detector (110) according to ~~any one of the preceding claims~~ **claim 1**, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature ϕ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.
14. (Currently Amended) The detector (110) according to ~~any one of the preceding claims~~ **claim 1**, wherein the sensor element (116) comprises at least one CMOS sensor.
15. (Currently Amended) The detector (110) according to ~~any one of the preceding claims~~ **claim 1**, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one

hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; **and at least one line pattern comprising such as** parallel or crossing lines.

16. (Currently Amended) The detector (110) according to ~~the preceding claim~~ **claim 15**, wherein the illumination source (124) comprises at least one laser source and at least one diffractive optical element.
17. (Currently Amended) A detector system (138), the detector system (138) comprising at least one detector (110) according to ~~any one of the preceding claims~~ **claim 1**, the detector system (138) further comprising at least one beacon device (140) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (140) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
18. (Currently Amended) A human-machine interface(142) for exchanging at least one item of information between a user (160) and a machine (168), wherein the human-machine interface (142) comprises at least one detector system (138) according to ~~the preceding claim~~ **claim 17**, wherein the at least one beacon device (140) is adapted to be at least one of directly or indirectly attached to the user (160) and held by the user (160), wherein the human-machine interface (142) is designed to determine at least one position of the user (160) by means of the detector system (138), wherein the human-machine interface (142) is designed to assign to the position at least one item of information.
19. (Currently Amended) An entertainment device (144) for carrying out at least one entertainment function, wherein the entertainment device (144) comprises at least one human-machine interface (142) according to ~~the preceding claim~~ **claim 18**, wherein the entertainment device (144) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (142), wherein the entertainment device (144) is designed to vary the entertainment function in accordance with the information.

20. (Currently Amended) A tracking system (146) for tracking a position of at least one movable object, the tracking system (146) comprising at least one detector system (138) according to ~~any one of the preceding claims~~ claim 17 referring to a detector system, the tracking system (146) further comprising at least one track controller (170), wherein the track controller (170) is adapted to track a series of positions of the object (112) at specific points in time.
21. (Currently Amended) A scanning system (148) for determining a depth profile of a scenery, the scanning system (148) comprising at least one detector (110) according to ~~any of the preceding claims referring to a detector~~ claim 1, the scanning system (148) further comprising at least one illumination source adapted to scan the scenery with at least one light beam.
22. (Currently Amended) A camera (136) for imaging at least one object (112), the camera (136) comprising at least one detector (110) according to ~~any one of the preceding claims referring to a detector~~ claim 1.
23. (Currently Amended) An inertial measurement unit (172) for use in an electronic device, wherein the inertial measurement unit (172) is adapted to receive data determined by at least one detector (110) according to ~~any one of the preceding claims referring to a detector~~ claim 1, wherein the inertial measurement unit (172) further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; and a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, and speed.

24. (Currently Amended) A method for determining at least one material property of at least one object (112) by using at least one detector (110) according to ~~any one of the preceding claims referring to a detector~~ claim 1, the method comprising the following steps:

a) determining at least one reflection image of the object (112) by using at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122); and

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device (132), wherein the evaluating comprises:

b1) determining at least one distance feature ϕ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof [[or]] and a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter; ~~such as~~ a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; [[or]] and a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the

spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, **and**

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

25. (Currently Amended) A **use of method of using** the detector (110) according to **any one of the preceding claims relating to a detector, claim 1, the method comprising using the detector (110)** for a purpose of use[[,]] selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; **and** a manufacturing application.

REMARKS

The Specification has been amended to include the priority paragraph.

Claims 1-25 will be pending after entry of this Preliminary Amendment A. Claims 1-25 are amended herein. No new matter has been added in this amendment.

The Commissioner is hereby authorized to charge any fees under 37 CFR 1.16 and 1.17 which may be required during the entire pendency of this application to Account No. 012384.

Respectfully submitted,

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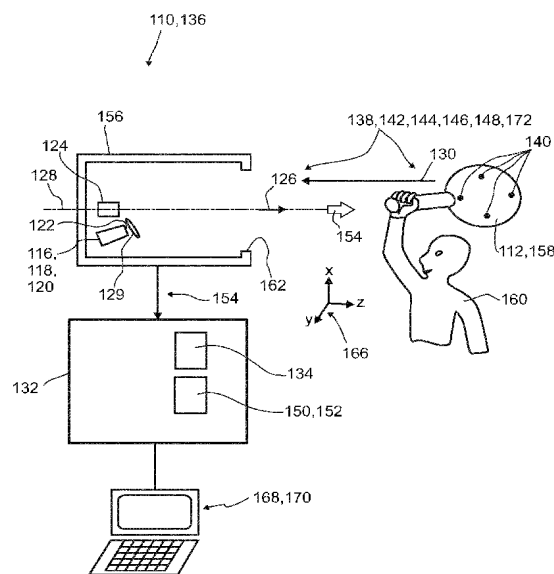
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(54) Title: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

FIG. 1



(57) **Abstract:** A detector (110) for identifying at least one material property m is proposed. The detector (110) comprises at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122). The sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112). The detector (110) comprises at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image. The evaluation device (132) is configured for determining at least one distance feature ϕ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The evaluation device (132) is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a



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grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

Detector for identifying at least one material property

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining at least one material property of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

- 20 A large number of methods for material classification and identification are known from the prior art. For example, material identification can be used in machine vision applications, medical applications and security applications, for image classification purposes or in gesture recognition algorithms.
- 25 For example, US 2016/0206216 A1 describes a device, system and method for skin detection. The device comprises a thermal sensor input for obtaining thermal sensor data of a scene, a light sensor input for obtaining light sensor data of the scene, and an evaluation unit for analyzing the obtained thermal sensor data and the obtained light sensor data and for detecting skin areas within the scene based on said analysis. US 2016/155006 A1 describes a device and a
- 30 corresponding method for skin detection. The device comprises an illumination unit configured to project a predetermined illumination pattern onto a scene, an imaging unit configured to acquire an image of the scene, and an evaluation unit configured to evaluate the acquired image by analyzing the imaged illumination pattern as reproduced in the image and to detect skin areas within the image and distinguish them from non-skin areas within the image based on said
- 35 analysis.

Material classification and identification is usually done in through beam mode, namely shining light through a sample and analyzing extinction. Further, material classification and identification can be done in reflective mode in controlled environments. However, in reflective mode usually wavelengths larger than 1000 nm are necessary in order to yield reliable results. Thus, despite the advantages of known methods for material identification, new concepts for reliable identification of materials are desired. Specifically, identifying a material from its reflection beam profile would be highly desirable. However, most image filters used for beam profile analysis (BPA) yield features that depend on both, distance and material such that reliable identification and

classification of the material is not possible. A typical example is the width of a beam profile reflected from a translucent material.

DE 198 46 619 A1 describes structured surface appearance quality determining equipment, which evaluates electrical measurement signal from photosensor array to derive structure code characterizing structure-dependent characteristic of measurement surface.

CN 108 363 482 A describes a method for controlling smart televisions via three-dimensional gestures on basis of binocular structured light. The method comprises the following steps of: synchronously acquiring images through a binocular camera; reconstructing a three-dimensional image according to an obtained left-right view; carrying out preprocessing; recognizing a gesture action after three-dimensional gestures are segmented; and converting the recognized gesture action into an operation instruction for a smart television, and executing the operation instruction.

US 2018/033146 A1 describes systems and methods for determining a depth map and a reflectivity map from a structured light image. The depth map can be determined by capturing the structured light image and then using a triangulation method to determine a depth map based on the dots in the captured structured light image. The reflectivity map can be determined based on the depth map and based on performing additional analysis of the dots in the captured structured light image.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may identify at least one material property of an object, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are

present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for identifying at least one material property m is disclosed.

As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “identifying at least one material property” refers to one or more of determining and assigning the material property to the object. The detector may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement using the detector according to the present invention, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by

the user of the detector. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The detector may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

Specifically, the detector may be configured for detection of biological tissue, in particular human skin. As used herein, the term "biological tissue" generally refers to biological material comprising living cells. The detector may be a device for detection, in particular optical detection, of biological tissue, in particular of human skin. The term "detection of biological tissue" refers to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces, and/or distinguishing different types of biological tissue such as distinguishing different types of human tissue e.g. muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise human tissue or parts thereof such as skin, hair, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise animal tissue or a part thereof such as skin, fur, muscle, fat, organs, or the like. For example, the biological tissue may be or may comprise plant tissue or a part thereof. The detector may be adapted to distinguish animal tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines or milking machines. The detector may be adapted to distinguish plant tissue or parts thereof from one or more of inorganic tissue, metal surfaces, plastics surfaces, for example of farming machines. The detector may be adapted to distinguish food and/or beverage from dish and/or glasses. The detector may be adapted to distinguish different types of food such as a fruit, meat, and fish. The detector may be adapted to distinguish a cosmetics product and/or, an applied cosmetics product from human skin. The detector may be adapted to distinguish human skin from foam, paper, wood, a display, a screen. The detector may be adapted to distinguish human skin from cloth. The detector may be adapted to distinguish a maintenance product from material of machine components such metal components etc. The detector may be adapted to distinguish organic material from inorganic material. The detector may be adapted to distinguish human biological tissue from surfaces of artificial or non-living objects. The detector may be used, in particular, for non-therapeutic and non-diagnostic applications.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter, or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented

essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The detector may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured for generating at least one output signal. In case the detector comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one inorganic photodiode which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly,

as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

Preferably, the light sensitive area may be oriented essentially perpendicular to an optical axis of the detector. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The sensor element is configured for recording at least one reflection image of a light beam originating from at least one object. As used herein, the term “reflection image” refers to an image determined by the sensor element comprising at least one reflection feature. As furthermore used herein, without limitation, the term “reflection image” specifically may relate to data recorded by using the sensor element, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element. The reflection image itself, thus, may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to “pixels”, reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. As used herein, the term “reflection feature” refers to a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. As used herein, the term “determining at least one reflection feature” refers to imaging and/or recording at least one light beam generated by the object in response to illumination with a light beam, in particular with the at least one illumination feature. In particular, the sensor element may be configured for determining and/or imaging and/or recording at least reflection image. The reflection image may comprise at least one reflection pattern comprising the at least one reflection feature.

The detector comprises at least one illumination source. The detector may comprise at least one illumination source configured for illuminating the object with at least one illumination beam.

The illumination source may be configured for projecting at least one illumination pattern comprising the at least one illumination feature on at least one surface of the object. As used herein, the term “at least one illumination source” refers to at least one arbitrary device configured for providing at least one illumination light beam, specifically the at least one illumination pattern, for illumination of the object. The illumination source may be adapted to directly or indirectly illuminating the object, wherein the illumination light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam.

The illumination source may comprise at least one light source. The illumination source may comprise a plurality of light sources. The illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of

handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source may be integrated into a housing of the detector.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The detector may be configured such that the illumination light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the illumination light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illumination light beam may be on the optical axis or off the optical axis. As an example, the illumination light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

Specifically, the illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.

The illumination source may be configured for generating at least one illumination pattern for illumination of the object. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the object. As used herein, the term "illumination feature" refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. For example, the illumination pattern may comprise at least one line pattern. For example, the illumination pattern may comprise at least one stripe pattern. For example, the illumination pattern may comprise at least one checkerboard pattern. For example, the illumination pattern may comprise at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. For example, the illumination source may be adapted to generate and/or to project a cloud of points. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. The illumination source may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, for generating and projecting the illumination pattern, the illumination source may comprise at least one laser source and at least one diffractive optical element (DOE). The detector may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one point pattern. As further used herein, the term "projecting at least one illumination pattern" refers to providing the at least one illumination pattern for illuminating the at least one object. The projected illumination pattern may be scarce, as only a single illumination feature, such as a single point, may be present. For increasing reliability, the illumination pattern may comprise several illumination features such as several points. If the pattern is scarce, a single image can be used for detection of biological tissue and face identification simultaneously.

For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The

illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern
5 may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination
10 source may be integrated in a mobile device or attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers
15 to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may
20 be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining
25 and determining the beam profile, in particular a shape of the beam profile.

As used herein, the term “object” refers to a point or region emitting at least one light beam, in particular the at least one reflection pattern. For example, the object may be at least one object selected from the group consisting of: a scene, a human such as a human, wood, carpet, foam,
30 an animal such as a cow, a plant, a piece of tissue, a metal, a toy, a metallic object, a beverage, a food such as a fruit, meat, fish, a dish, a cosmetics product, an applied cosmetics product, cloth, fur, hair, a maintenance product, a cream, an oil, a powder, a carpet, a juice, a suspension, a paint, a plant, a body, a part of a body, organic material, inorganic material, a reflective material, a screen, a display, a wall, a sheet of paper, such as photograph. The object may
35 comprise at least one surface on which the illumination pattern is projected. The surface may be adapted to at least partially reflect the illumination pattern back towards the detector. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrat-
40 ing the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to “Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen”, “Wirkung von Laserstrahlung auf Gewebe”, 1991, pages 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase

with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These material properties may be used to distinguish skin from other materials, specifically by analyzing the back scattering profile.

The least one light beam may propagate from the object towards the detector. The light beam may originate from the object or may originate from an illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. Additionally or alternatively to the at least one illumination source, the detector may use radiation already present in the scene such as from at least one ambient light source.

The sensor element may be configured for recording a beam profile of at least one reflection feature of the reflection image. The evaluation device may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element. The evaluation device may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-

transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

For example, the illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The evaluation device may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The evaluation device may be adapted to remove influences from background light from the reflection beam profile, for example, by an imaging without further illumination.

The evaluation device may be configured for determining the material property m by evaluating the beam profile of the reflection image. As used herein, the term "beam profile of the reflection image" refers to at least one intensity distribution of at least one of the reflection features of the reflection image, such as of a light spot on the sensor element, as a function of the pixel. The beam profile of the reflection image, also denoted reflection beam profile, may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. As used herein, the term "evaluating the beam profile" refers to applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. As used herein, the term "image" refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term "image filter" refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. As used herein, the term “distance” refers to the distance of the object, specifically between the object and the detector.

As used herein, the term “distance dependent” image filter refers to an image having a distance dependent output. The output of the distance dependent image filter is denoted herein “distance feature φ_{1z} ” or “distance dependent feature φ_{1z} ”. The distance feature may be or may comprise at least one information about the distance of the object such as at least one measure for the distance of the object, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter $\Phi_{1\text{other}}$ may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{\text{cov}((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j))}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one cur-

rent signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by
 5 filtering or the like. As generally used herein, the term “combine” generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least one comparison signal or comparison result. As used herein, the term “combined signal Q” refers to a signal which is generated by combining the sensor signals, in particular by
 10 one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. In particular, the combined signal may be a quotient signal. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the eval-
 15 uation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device may be configured for deriving the combined signal Q by one or more of
 20 dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature ϕ_{1z} for determining the distance feature ϕ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by
 25

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy} ,$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and
 30 $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally
 35 the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering proper-
 40 ties of the object and independent from alterations of the light source such as by manufacturing

precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature ϕ_{1z} may be a function of the combined signal Q , $\phi_{1z} = \phi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q . Further, as an example, the object distance z_0 may be a function of the distance dependent feature ϕ_{1z} , $z_0 = z_0(\phi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in ϕ_{1z} . Thus, the object distance z_0 may be a function of the combined signal Q , $z_0 = z_0(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q .

The light-sensitive areas of at least two optical sensors may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q .

The evaluation device may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for de-

termining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example, Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1 / b \cdot s_2$ or $Q = b \cdot s_2 / a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a , b , c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

Typically, the combined signal Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the setup described above, both the first signal s_1 and the second signal s_2 decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the first signal s_1 decreases more rapidly than the second signal s_2 , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus,

is spread over a larger area. The quotient of the first and second sensor signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the incident light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

With respect to further details and embodiments of evaluation of the combined signal Q reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device may be configured for determining from the image the distance feature φ_{1z} of the object by optimizing at least one blurring function f_a . The determined distance feature φ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature φ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. As used herein, the term “blurring function f_a ”, also referred to as blur kernel or point spread function, refers to a response function of the detector to the illumination from the object. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature φ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min\|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. As used herein, the term “sharp” or “sharp image” refers to a blurred image having a maximum contrast. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature ϕ_{1z} .

The evaluation device may be configured for determining at least one combined distance information z considering the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield advantageous distance measurement results with enhanced precision.

The evaluation device may be configured for determining at least one combined distance information using at least one recursive filter. The recursive filter may be at least one Kalman filter or at least one Extended Kalman filter (EKF). The combined distance information z may be obtained using a real function $z=f(z_{DPR}, z_{DFD})$, wherein z_{DPR} is the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and z_{DFD} is the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter, such as the arithmetic or geometric mean, a polynomial, preferably a polynomial up to the eighth order in z_{DPR} and z_{DFD} . The function f may be or may be based on a look-up table of prerecorded values. For example, the evaluation device may comprise at least one data storage device configured for storing the prerecorded values and/or one or more look-up tables. The function f may be based on a look-up table in combination with an interpolation scheme for interpolation between the values in the look-up table. The interpolation scheme may be a linear interpolation, a splines interpolation, or the like. The z_{DFD} and z_{DPR} may be used as input variables within the recursive filter, in combination with a model or model function such as the function f , concerning the relationship between z , z_{DFD} , and z_{DPR} . The model or model function may comprise statistical assumptions and/or statistical models concerning the distances z , z_{DFD} , and z_{DPR} such as a distribution, such as a Gaussian distribution of a measured distance z , z_{DFD} , and/or z_{DPR} around a real distance z_{real} .

The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters. The further parameters may comprise further information from the sensor element, for example the CMOS sensor, such as information about quality and/or noise of recorded data and/or information about overexposure and/or information about underexposure or the like. The detector may comprise at least one further sensor configured for determining the further sensor data. The recursive filter may be configured for

determining the combined distance information considering the further sensor data. The further sensor may be at least one sensor selected from the group consisting of: a temperature sensor, an illumination sensor such as a control sensor for determining illumination information, an inertial measurement unit; a gyroscope. The model and/or the Kalman filter may comprise further
5 input parameters such as the further sensor data, e.g. temperature and/or detector motion from a gyroscope and/or information from an inertial measurement unit, and/or information from an illumination sensor) and/or further parameters such as the quality/noise of recorded data, overexposure, underexposure, or the like. The further sensor data may be provided by the sensor element, in particular by at least one CMOS sensor, and/or by further image analysis. The model
10 el and/or the Kalman filter may comprise the at least one material feature φ_{2m} as input variable.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At
15 least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and for selecting at least one image determined by the other one of the first sensor element or the second sensor element as reference image. As used
20 herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation
25 device may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm \varepsilon$.

35 The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection
40 feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Hei-

delberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. As used herein, the term “displacement region” refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term “reference feature” refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \epsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evalua-

tion device may determine the displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\varphi_{1z} \pm \varepsilon$.

5 The evaluation device may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at
10 least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the
15 extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the
20 epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region
25 along the assigned epipolar line.

The evaluation device may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device may be configured
30 for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. As used herein, the term “displacement” refers to a difference between a position in the reference image to a position in the reflection image. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a
35 distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device may be adapted to determine the predetermined relationship by using triangulation methods. In case position of the selected
40 reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent

and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device.

5

Additionally or alternatively, the evaluation device may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ε -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or reference feature in shorter distance and/or ε -weighted distance to the reference feature and/or reflection feature.

The evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the object.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof

by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter $\Phi_{2\text{other}}$ may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term "passes a hypothesis testing" refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term "beam profile image" refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k :

$(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J, $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values

from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k .

The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term “not distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical. As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq$

0.075, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

- 5 For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1g_2) = [p_{g_1,g_2}]$, whereas p_{g_1,g_2} is the occurrence rate of the grey combination $(g_1,g_2)=[f(x_1,y_1),f(x_2,y_2)]$, and the relation ρ defines the distance between (x_1,y_1) and (x_2,y_2) , which is $\rho(x,y)=(x+a,y+b)$ with a and b selected from $0,1$.

10

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

- For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

15

The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

- For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

20

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

- 25 The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

5 and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ε times the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The material property m and/or the longitudinal coordinate z may be determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m . The evaluation device may be configured for determining the material property m and/or the longitudinal coordinate z by evaluating the features φ_{2m} , φ_{1z} . For example, the evaluation device may be configured for using at least one predetermined relationship between the distance feature φ_{1z} and the longitudinal coordinate of the object for determining the longitudinal coordinate z of the object. The evaluation device may be configured for using at least one predetermined relationship between the material feature φ_{2m} and the material property of the object for determining the material property of the object. The predetermined relationship may be one or more of an empirical relationship, a semi-empirical relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. For example, the material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z such that the information about the

longitudinal coordinate z can be considered for evaluating of ϕ_{2m} . Specifically, the material property m and/or the longitudinal coordinate z may be determined by a function $z(\phi_{1z}, \phi_{2m})$ and/or by a function $m(\phi_{1z}, \phi_{2m})$. The function may be predefined and/or predetermined. For example, the function may be a linear function.

5

In the ideal case, an image filter would yield features that are only dependent on distance or material properties. However, image filters used in beam profile analysis may yield features that depend on distance and material properties, such as translucency. At least one of the material dependent image filter or the distance dependent image filter may be a function of the feature ϕ_{1z} or ϕ_{2m} of the at least one other image filter. At least of the material dependent image filter or the distance dependent image filter may be a function of the at least one other image filter. The evaluation device may be configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature ϕ_{1z} or ϕ_{1m} of the other image filter and/or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter. Specifically, the evaluation device may be configured for determining a correlation coefficient of the material dependent image filter and the distance dependent image filter. In case the correlation coefficient of the material dependent image filter with the distance dependent image filter is close to 1 or -1, the distance may be projected out, by projecting the material feature on the principal axis with the lowest variance. As an example, the material feature may be projected onto an axis orthogonal to the correlating main component. In other words, the material feature may be projected onto the second main component. This may be done using a principal component analysis as known to the person skilled in the art. The evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image. In particular, the evaluation device may be configured for simultaneously determining features that are strongly correlated with distance and weakly correlated with material properties and features that are weakly correlated with distance and strongly correlated with material properties. Alternatively, the evaluation device may be configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image. The evaluation device may be configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of ϕ_{1z} and ϕ_{2m} to the reflection image.

As further used herein, the term “evaluation device” generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. The above-mentioned operations, including analysis of the beam profile, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more

programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured for performing the above-mentioned evaluation. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

5

The detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

10 The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a
15 direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element,
20 preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more
25 imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging
30 thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on
35 the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

40 The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having

at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As outlined above, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. For example, the detector may be configured for determining the longitudinal coordinate of the object by using at least one distance dependent filter using the depth-from-photon-ratio technique and/or the depth-from-defocus technique as outlined above. The detector may be configured for determining a position of the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. The distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. At least one of the optical sensors may
5 determined from which a center signal arises. This may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal
10 sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also
15 be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one
20 transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created.
25 Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured for determining at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

30 The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, a center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. The use of the matrix of optical
35 sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object. The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, the detector according to the pre-
40 sent invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

The present invention discloses in a further aspect, a detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p, is smaller or equal than a pre-defined level of significance, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

With respect to definitions and embodiments reference is made to the description of the detector described in a first aspect of the present invention.

In a further aspect of the present invention, a detector system is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect

one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured for enabling at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured for varying the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As

used herein, the term “scenery” generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term “scan” generally refers to a consecutive measurement in different regions.

Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term “dot” refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a

case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using

the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

- 5 As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mir-
10 rors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

- 15 Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors
20 comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or
25 more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices
30 as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety
35 curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenu-
40 ate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated

out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

5 As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system
10 and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3D model by connecting adjacent points with planar surfaces. The 3D model may be stored and
15 processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser
20 sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters
25 or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

30 As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection
35 in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system
40 may be adapted to determine at least one longitudinal coordinate for each of the light points as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. This feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor.

5 The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least
10 one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination
15 source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively
20 or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic
25 radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence
30 and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be
35 regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the
40 visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of

3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the

grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect, the present invention discloses a method for determining at least one material property of at least one object by using at least one detector according to the present invention. The method comprises the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
 - b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly. For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking

application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

With respect to further uses of the detector and devices of the present invention reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the content of which is included by reference.

Specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device configured for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements configured for generating electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device configured for performing 3D photography. The camera generally may be configured for acquiring a single image, such as a single 3D image, or may be configured for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera configured for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be configured for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression “position” generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device configured for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be configured for performing at least one computer program, such as at least one computer program configured for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

The human-machine interface may comprise a plurality of beacon devices which are configured for being at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or

by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device configured for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements configured for reflecting a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be configured in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the

user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

- 5 The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source configured for generating at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam proper-
- 10 ties. Additionally or alternatively, the beacon device may comprise at least one reflector configured for reflecting light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

- The object, which may form part of the detector system, may generally have an arbitrary shape.
- 15 Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be
- 20 part of the human-machine interface and/or of the entertainment device.

- As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably
- 25 computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

- The entertainment device comprises at least one human-machine interface according to the
- 30 present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one
- 35 item of information preferably may comprise at least one command configured for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements
- 40 may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is configured for gathering information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be configured for providing information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is configured for tracking a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.

The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.

The tracking system may be configured for initiating one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be configured for initiating at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards

the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

In a further aspect of the present invention, an inertial measurement unit for use in an electronic device disclosed. The inertial measurement unit is adapted to receive data determined by at least one detector according to the present invention. The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

The inertial measurement unit may comprise the detector according to the present invention and/or may be connected to the detector, via at least one data connection. The evaluation device and/or at least one processing device of the inertial measurement unit may be configured for determining the at least one combined distance information, in particular using the at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters such as further sensor data from the further sensor of the inertial measurement unit. With respect to definitions and embodiments of the inertial measurement unit reference is made to the description of the detector.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;

- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 2: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

Embodiment 3: The detector according to the preceding embodiment, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

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Embodiment 5: The detector according to any one of the preceding embodiments, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.80$.

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Embodiment 6: The detector according to any one of the preceding embodiments, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}},\Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}},\Phi_m}| \geq 0.80$.

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Embodiment 7: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

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Embodiment 8: The detector according to any one of the preceding embodiments, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

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Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

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Embodiment 10: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

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Embodiment 11: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and/or φ_{2m} to the reflection image.

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Embodiment 12: The detector according to any one of the preceding embodiments, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .

Embodiment 13: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

5 wherein x and y are transversal coordinates, A1 and A2 are different areas of the beam profile and E(x,y,z_o) denotes the beam profile given at the object distance z_o, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

10 Embodiment 14: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

15 Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

25 Embodiment 16: The detector according to any one of the preceding embodiments, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

30 Embodiment 17: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

35
$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 18: The detector according to any one of the two preceding embodiments, wherein the at least one blurring function f_a is a function or composite function composed from at least

one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

- 5 Embodiment 19: The detector according to any one of the preceding embodiments, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 21: The detector according to the preceding embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 23: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 24: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein
 5 in the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 25: A detector for identifying at least one material property m comprising

- 10 - at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,
- 15 wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which
- 20 correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection
- 25 image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,
- 30 wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 26: The detector according to the preceding embodiment, wherein the hypothesis
 35 testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.

Embodiment 27: The detector according to any one of embodiments 25 and 26, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 28: The detector according to any one of embodiments 25 to 27, wherein the at
 40 least one material property is a property selected from the group consisting of: a scattering coef-

ficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

Embodiment 29: The detector according to any one of embodiments 25 to 28, wherein the further distance dependent image filter $\Phi_{10\text{ther}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{10\text{ther}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{10\text{ther}}, \Phi_z}| \geq 0.80$.

Embodiment 30: The detector according to any one of embodiments 25 to 29, wherein the further material dependent image filter $\Phi_{20\text{ther}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.80$.

Embodiment 31: The detector according to any one of embodiments 25 to 30, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{20\text{ther}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{20\text{ther}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

Embodiment 32: The detector according to any one of embodiments 25 to 31, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

Embodiment 33: The detector according to any one of embodiments 25 to 32, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

Embodiment 34: The detector according to any one of embodiments 25 to 33, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

Embodiment 35: The detector according to any one of embodiments 25 to 34, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the

at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

- 5 Embodiment 36: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

- 10 Embodiment 37: The detector according to any one of embodiments 25 to 36, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .
- 15

Embodiment 38: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 20 wherein x and y are transversal coordinates, A_1 and A_2 are different areas of the beam profile and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

- 25 Embodiment 39: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- 30 Embodiment 40: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
- 35

Embodiment 41: The detector according to any one of embodiments 25 to 40, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

Embodiment 42: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 43: The detector according to any one of embodiments 25 to 42, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

Embodiment 44: The detector according to any one of embodiments 25 to 43, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 45: The detector according to any one of embodiments 25 to 44, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 46: The detector according to the preceding embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 47: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflec-

tion image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 48: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 49: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 50: A detector for identifying at least one material property m comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object;
- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the evaluation device is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,

wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

5 Embodiment 51: The detector according to the preceding embodiment, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a
10 pre-defined level of significance.

Embodiment 52: The detector according to the preceding embodiment, wherein the hypothesis testing is based on a dataset of beam profile images, wherein each beam profile image corresponds to a material classifier and a distance.
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Embodiment 53: The detector according to any one of embodiments 50 and 52, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.

Embodiment 54: The detector according to any one of embodiments 50 to 53, wherein the at
20 least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.

Embodiment 55: The detector according to any one of embodiments 50 to 54, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.
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Embodiment 56: The detector according to any one of embodiments 50 to 55, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.
30

Embodiment 57: The detector according to any one of embodiments 50 to 56, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness
35 filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof, or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by
40 $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared

norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

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Embodiment 58: The detector according to any one of embodiments 50 to 57, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .

10 Embodiment 59: The detector according to any one of embodiments 50 to 58, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.

15 Embodiment 60: The detector according to any one of embodiments 50 to 59, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

20 Embodiment 61: The detector according to any one of embodiments 50 to 60, wherein the evaluation device is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

25 Embodiment 62: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.

30 Embodiment 63: The detector according to any one of embodiments 50 to 62, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .

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Embodiment 64: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are different areas of the beam profile and $E(x,y,z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the light beam propagating from the object to the detector.

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Embodiment 65: The detector according to the preceding embodiment, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

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Embodiment 66: The detector according to the preceding embodiment, wherein the evaluation device is configured for determining the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

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Embodiment 67: The detector according to any one of embodiments 50 to 66, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

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Embodiment 68: The detector according to the preceding embodiment, wherein the reflection image is a blurred image i_b , wherein the evaluation device is configured for reconstructing the distance feature φ_{1z} from the blurred image i_b and the blurring function f_a , wherein the distance feature φ_{1z} is determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

30

$$\min \|(i'_b * f_a(\sigma(\varphi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function.

Embodiment 69: The detector according to any one of embodiments 50 to 68, wherein the at least one blurring function f_a is a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

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Embodiment 70: The detector according to any one of embodiments 50 to 69, wherein the sensor element comprises at least one CMOS sensor.

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Embodiment 71: The detector according to any one of embodiments 50 to 70, wherein the detector comprises at least one illumination source, wherein the illumination source is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 72: The detector according to the embodiment, wherein the illumination source comprises at least one laser source and at least one diffractive optical element.

Embodiment 73: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors, wherein at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein the evaluation device is configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

Embodiment 74: The detector according to the preceding embodiment, wherein the evaluation device is configured for selecting at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature φ_{1z} and an error interval $\pm\epsilon$, wherein the evaluation device is configured for determining at least one displacement region in the reference image corresponding to the distance estimate, wherein the evaluation device is configured for matching the selected reflection feature with at least one reference feature within the displacement region, wherein the evaluation device is configured for determining a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is configured for determining a longitudinal coordinate of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement.

Embodiment 75: The detector according the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the distance feature φ_{1z} and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 76: A detector system, the detector system comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

Embodiment 77: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 78: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 79: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 80: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of embodiments 1 to 24, 25 to 49, or 50 to 75, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 81: A camera for imaging at least one object, the camera comprising at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75.

Embodiment 82: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data

from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

5 Embodiment 83: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 1 to 24, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- 10 b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

$$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$$
 with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - 15 b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by

$$|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$$
 with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,
 - 20 b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 84: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 25 to 49, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,
 - b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance,
 - b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

Embodiment 85: A method for determining at least one material property of at least one object by using at least one detector according to any one of embodiments 50 to 75, the method comprising the following steps:

- a) determining at least one reflection image of the object by using at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area;
- b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device, wherein the evaluating comprises:
 - b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

- b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image,
- b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

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Embodiment 86: A use of the detector according to any one of embodiments 1 to 24, 25 to 49, or 50 to 75, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

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Brief description of the figures

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Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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Figure 1 shows an embodiment of a detector according to the present invention;

Figure 2 shows the statistics of the resulting material features of the spot shape material dependent image filter applied to the data set; and

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Figure 3 shows a beam profile image.

Detailed description of the embodiments:

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Fig. 1 shows in a highly schematic fashion an embodiment of a detector 110 for identifying at least one material property m of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.

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The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the ele-

ments are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

5 The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other
10 embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors 120 may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

15 The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical
20 sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photo-
25 diodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

30 The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As
35 an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

40 The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more prefer-

ably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

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The detector 110 may further comprise an illumination source 124. As an example, the illumination source 124 may be configured for generating an illumination light beam 126 for illuminating the object 112. The detector 110 may be configured such that the illumination light beam 126 propagates from the detector 110 towards the object 112 along an optical axis 128 of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis 128.

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The illumination source 124 may comprise at least one light source. The illumination source 124 may comprise a plurality of light sources. The illumination source 124 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 124 may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used. Using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors. The illumination source 124 may be adapted to emit light at a single wavelength. In other embodiments, the illumination may be adapted to emit light with a plurality of wavelengths allowing additional measurements in other wavelengths channels. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

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Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 124 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points, for example the illumination source 124 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least

one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. The illumination source 124 may be integrated into a housing of the detector 110.

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The illumination light beam 126 may propagate from the object 112 towards the detector 110. The detector 110 may be used in active and/or passive illumination scenarios. For example, the at least one illumination source 124 may be adapted to illuminate the object 112, for example, by directing a light beam towards the object 112, which reflects the light beam. Additionally or
10 alternatively to the at least one illumination source 124, the detector 110 may use radiation already present in the scene such as from at least one ambient light source.

The illumination source 124 may illuminate the at least one object 112 with at least one illumination pattern. The illumination pattern may comprise a plurality of points as image features.

15 These points are illustrated as light beam 126 emerging from the illumination source 124.

The detector 110 may comprise at least one transfer device 129 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel
20 lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 129 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

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Each optical sensor 120 may be designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam 130 propagating from the object 112 to the detector 110. The sensor element 116 may be configured for determining at least one reflection image of at least one reflection feature generated by the reflection light beam 130 on the optical sensors 120. The reflection image may be data recorded by
30 using the sensor element 116, such as a plurality of electronic readings from an imaging device, such as the pixels of the sensor element 116. The reflection image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix 118 of the sensor element 116. The matrix 118 may comprise the reflection image 126. For example, in case of illumination with a point pattern, the reflection image may comprise points as reflection features. These points may
35 result from reflection light beams 130 originating from the at least one object 112.

The sensor element 116 may be configured for recording a beam profile of at least one reflection feature of the reflection image. The detector 110 comprises at least one evaluation device
40 132. The evaluation device 132 may be configured for identifying and/or selecting at least one reflection feature in the reflection image, specifically at least one light spot, provided by the sensor element 116. The evaluation device 132 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection feature. The image anal-

ysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. Specifically, the evaluation of the reflection image comprises selecting the region of interest in the reflection image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the sensor element 116. For example, in case of a spot-like reflection feature the region of interest may be selected as a region around the spot profile.

The evaluation device 132 may be configured for determining the material property m by evaluating the beam profile of the reflection image. The beam profile of the reflection image may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. The evaluation device 132 may be configured for applying at least one distance dependent image filter and at least one material dependent image filter to the beam profile and/or to at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a distance feature in case of distance dependent image filters and a material feature in case of material dependent image filters. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The evaluation device 132 is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image. The distance feature may be or may comprise at least one information about the distance of the object 112 such as at least one measure for the distance of the object 112, a distance value, a longitudinal coordinate of the object or the like. The distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof. The further distance dependent image filter Φ_{1other} may correlate to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.80$. The similarity of two image filters Φ_i and Φ_j may be assessed through the correlation of their features, specifically, by calculating Pearson's correlation coefficients,

$$\rho_{\Phi_i \Phi_j} = \frac{cov\left((\Phi_i(f(x, y)) - \mu_i)(\Phi_j(f(x, y)) - \mu_j)\right)}{\sigma_i \sigma_j},$$

wherein μ and σ are mean value and standard deviation of the obtained features. The testing of correlation of filters may be performed using a set of random test images, specifically matrices filled with random numbers. The number of random test images may be chosen as such that the results of the correlation test are statistically significant. The correlation coefficient takes values between -1 and 1, whereas 0 means there is no linear correlation. The correlation coefficients are well suited to determine whether two filters are similar or even equivalent. To measure whether the features of a filter correlate with a given property, for example the distance, the test images may be chosen as such, that a correlated filter actually yields the property. As an example, in order to measure whether the features of a filter correlate with distance, beam profiles recorded in different distances may be used as test images. To obtain a comparable, transferable, and transparent assessment, a fixed test-set of test images may be defined.

For example, the distance dependent image filter may be a depth-from-photon-ratio filter. The depth-from-photon-ratio filter may comprise evaluating a combined signal Q from at least two sensor signals of the sensor element. The evaluation device may be configured for determining the distance feature φ_{1z} by evaluation of the combined signal Q . The distance feature determined by evaluation of the combined signal Q may correspond directly to the longitudinal coordinate of the object. The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device 132, as an example, may comprise at least one divider 134, wherein the divider 134 is configured for deriving the quotient signal. The divider 134 may fully or partially be embodied as one or both of a software divider or a hardware divider.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear com-

binations of the sensor signals. The evaluation device 132 may be configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} . For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the light beam propagating from the object to the detector at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance L(z_o) and beam shape S(x,y;z_o). Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object 112 and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. As an example, the distance dependent feature φ_{1z} may be a function of the combined signal Q, $\varphi_{1z} = \varphi_{1z}(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q. Further, as an example, the object distance z_o may be a function of the distance dependent feature φ_{1z} , $z_o = z_o(\varphi_{1z})$, whereas the function may be a linear, quadratic, or higher order polynomial in φ_{1z} . Thus, the object distance z_o may be a function of the combined signal Q, $z_o = z_o(Q)$, whereas the function may be a linear, quadratic, or higher order polynomial in Q.

The light-sensitive areas 122 of at least two optical sensors 120 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

The evaluation device 132 may be configured for determining and/or for selecting the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the

center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 132 may be adapted for determining an area integral of the beam profile. The evaluation device 132 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 132 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device 132 may be configured for deriving the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The evaluation device 132 specifically may be configured for deriving the combined signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example,

Q may simply be determined as $Q = s_1/s_2$ or $Q = s_2/s_1$, with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as $Q = a \cdot s_1/b \cdot s_2$ or $Q = b \cdot s_2/a \cdot s_1$, with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as $Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2)$, with a, b, c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as $Q = s_1 / (s_1 + s_2)$. Other combined or quotient signals are feasible.

With respect to further details and embodiments of evaluation of the combined signal Q and determination of the longitudinal coordinate z reference e.g. may be made to WO 2018/091640, WO 2018/091649 A1 and WO 2018/091638 A2 the full disclosure of which is incorporated herein by reference.

For example, the distance dependent image filter may be a depth-from-defocus filter. As outlined above, the evaluation device 132 may be configured for determining at least one image of the region of interest from the sensor signals. The evaluation device 132 may be configured for determining from the image the distance feature ϕ_{1z} of the object by optimizing at least one blurring function f_a . The determined distance feature ϕ_{1z} may correspond directly to the longitudinal coordinate of the object. The distance feature ϕ_{1z} may be determined by using at least one convolution-based algorithm such as a depth-from-defocus algorithm. To obtain the distance from the image, the depth-from-defocus algorithm estimates the defocus of the object. For this estimation, the blurring function is assumed. Specifically, the blurring function models the blur of a defocused object. The at least one blurring function f_a may be a function or composite function composed from at least one function from the group consisting of: a Gaussian, a sinc function, a pillbox function, a square function, a Lorentzian function, a radial function, a polynomial, a Hermite polynomial, a Zernike polynomial, a Legendre polynomial.

The blurring function may be optimized by varying the parameters of the at least one blurring function. The reflection image may be a blurred image i_b . The evaluation device may be configured for reconstructing the distance feature ϕ_{1z} from the blurred image i_b and the blurring function f_a . The distance feature ϕ_{1z} may be determined by minimizing a difference between the blurred image i_b and the convolution of the blurring function f_a with at least one further image i'_b ,

$$\min \|(i'_b * f_a(\sigma(\phi_{1z})) - i_b)\|,$$

by varying the parameters σ of the blurring function. $\sigma(z)$ is a set of distance dependent blurring parameters. The further image may be blurred or sharp. The at least one further image may be generated from the blurred image i_b by a convolution with a known blurring function. Thus, the depth-from-defocus algorithm may be used to obtain the distance feature ϕ_{1z} .

The evaluation device 132 may be configured for determining at least one combined distance information z considering the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a real function depending on the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ

z_{1z} determined by applying the depth-from-defocus filter. The combined distance information z may be a rational or irrational polynomial of the distance feature ϕ_{1z} determined by applying the depth-from-photon-ratio filter and the distance feature ϕ_{1z} determined by applying the depth-from-defocus filter. Depth-from-defocus is a complementary method to depth-from-photon-ratio but uses a similar hardware set up. Further, depth-from-defocus distance measurements may have a similar accuracy. Combining both technologies may yield to advantageous distance measurement results with enhanced precision.

For example, the distance dependent image filter may be a structured-light-filter in combination with a depth-from-photon-ratio filter and/or a depth-from-defocus image filter. The detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device 132 may be configured for selecting at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device 132 may be adapted to select the at least one reflection feature in the reflection image and to determine at least one distance estimate of the selected reflection feature of the reflection image given by the distance feature ϕ_{1z} determined by applying the depth-from-photon ratio image filter and/or the depth-from-defocus image filter and an error interval $\pm \epsilon$.

The evaluation device 132 may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. The evaluation device 132 may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device 132 may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The evaluation device 132 may be adapted to determine an epipolar line in the reference image. The relative position of the reference image and reflection image may be known. For example, the relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device 132 may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the

reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

The evaluation device 132 may be configured for determining at least one displacement region in the reference image corresponding to the distance estimate. Specifically, the displacement region may be a region in the reference image in which the reference feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object 112, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device 132 may be adapted to determine the reference feature along the epipolar line corresponding to the distance feature and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \varepsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance estimate may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device 132 may determine the displacement region around the image position of the reflection feature. The evaluation device 132 may be adapted to determine the distance estimate and to determine the displacement region along the epipolar line corresponding to $\varphi_{12} \pm \varepsilon$.

The evaluation device 132 may be configured for matching the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The evaluation device 132 may be configured for matching the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined distance estimate. The evaluation algorithm may be a linear scaling algorithm. The evaluation device 132 may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device 132 may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to

determine an epipolar line before determining a corresponding reference feature. The evaluation device 132 may determine a displacement region around the image position of each reflection feature. The evaluation device 132 may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device 132 may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 may be configured for determining a displacement of the matched reference feature and the selected reflection feature. The evaluation device 132 may be configured for determining a longitudinal information of the matched features using a predetermined relationship between the longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 132 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table. The evaluation device 132 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device 132 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device 132.

Additionally or alternatively, the evaluation device 132 may be configured for performing the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to the displacement region and/or within the displacement region and/or closest to the displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

The evaluation device 132 is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image. The material feature may be or may comprise at least one information about the at least one material property of the object 112.

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The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$. With respect to the description of exemplary material dependent image filter, reference is made to the description of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter such as a Gaussian filter or median filter, the grey-level-occurrence-based contrast filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, and the threshold area filter given above.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. The beam profile image may be given by a sum of N_B Gaussian radial basis functions,

35

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all beam profile images. The center-positions, x_{lk} , y_{lk} , are identical for all images $f_k: (x_0, x_1, \dots, x_{N_B}), (y_0, y_1, \dots, y_{N_B})$. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

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Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed. Figure 3 shows an embodiment of a beam profile image defined by an exemplary data set.

An exemplary data set is presented in the following. The following parameter tables list the center positions, $x_{lk} = x_l, y_{lk} = y_l$, of all Gaussian functions g_{lk} for all images f_k :

l	0	1	2	3	4	5	6	7
x_l	14	11	10	19	15	20	12	17
y_l	17	11	17	19	20	20	12	20
l	8	9	10	11	12	13	14	15
x_l	13	18	10	14	19	15	16	12
y_l	17	19	8	18	14	11	22	17
l	16	17	18	19	20	21	22	23
x_l	25	13	18	8	9	10	14	19
y_l	15	20	20	15	14	13	24	24
l	24	25	26	27	28	29	30	31
x_l	10	9	14	12	19	20	17	15
y_l	24	16	15	11	13	19	13	14
l	32	33	34	35	36	37	38	39
x_l	16	13	12	20	18	17	21	13
y_l	19	10	22	14	10	18	15	23
l	40	41	42	43	44	45	46	47
x_l	8	10	8	14	19	11	15	14
y_l	12	14	18	21	23	15	16	8
l	48	49	50	51	52	53	54	55
x_l	16	20	11	17	16	15	21	13
y_l	13	16	16	8	16	13	21	13

l	56	57	58	59	60	61	62	63
x_l	22	18	23	14	19	11	15	16
y_l	23	15	12	22	18	10	23	10

The following parameter tables list the material classifiers, in particular for a white Teflon target (denoted control), a fabric, a dark skin (denoted as dark_skin), a pale skin (denoted pale_skin), and a highly translucent skin (denoted trans), and the distance z for each image f_k , referenced

5 by the image index k :

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
0	control	0.4	40	dark_skin	0.4	80	control	0.6
1	control	0.4	41	dark_skin	0.4	81	control	0.6
2	control	0.4	42	dark_skin	0.4	82	control	0.6
3	control	0.4	43	dark_skin	0.4	83	control	0.6
4	control	0.4	44	dark_skin	0.4	84	control	0.6
5	control	0.4	45	dark_skin	0.4	85	control	0.6
6	control	0.4	46	dark_skin	0.4	86	control	0.6
7	control	0.4	47	dark_skin	0.4	87	control	0.6
8	control	0.4	48	pale_skin	0.4	88	control	0.6
9	control	0.4	49	pale_skin	0.4	89	control	0.6
10	control	0.4	50	pale_skin	0.4	90	control	0.6
11	control	0.4	51	pale_skin	0.4	91	control	0.6
12	control	0.4	52	pale_skin	0.4	92	control	0.6
13	control	0.4	53	pale_skin	0.4	93	control	0.6
14	control	0.4	54	pale_skin	0.4	94	control	0.6
15	control	0.4	55	pale_skin	0.4	95	control	0.6
16	fabric	0.4	56	pale_skin	0.4	96	fabric	0.6
17	fabric	0.4	57	pale_skin	0.4	97	fabric	0.6
18	fabric	0.4	58	pale_skin	0.4	98	fabric	0.6
19	fabric	0.4	59	pale_skin	0.4	99	fabric	0.6
20	fabric	0.4	60	pale_skin	0.4	100	fabric	0.6
21	fabric	0.4	61	pale_skin	0.4	101	fabric	0.6
22	fabric	0.4	62	pale_skin	0.4	102	fabric	0.6
23	fabric	0.4	63	pale_skin	0.4	103	fabric	0.6
24	fabric	0.4	64	trans	0.4	104	fabric	0.6
25	fabric	0.4	65	trans	0.4	105	fabric	0.6
26	fabric	0.4	66	trans	0.4	106	fabric	0.6
27	fabric	0.4	67	trans	0.4	107	fabric	0.6
28	fabric	0.4	68	trans	0.4	108	fabric	0.6
29	fabric	0.4	69	trans	0.4	109	fabric	0.6
30	fabric	0.4	70	trans	0.4	110	fabric	0.6
31	fabric	0.4	71	trans	0.4	111	fabric	0.6
32	dark_skin	0.4	72	trans	0.4	112	dark_skin	0.6
33	dark_skin	0.4	73	trans	0.4	113	dark_skin	0.6
34	dark_skin	0.4	74	trans	0.4	114	dark_skin	0.6
35	dark_skin	0.4	75	trans	0.4	115	dark_skin	0.6
36	dark_skin	0.4	76	trans	0.4	116	dark_skin	0.6
37	dark_skin	0.4	77	trans	0.4	117	dark_skin	0.6
38	dark_skin	0.4	78	trans	0.4	118	dark_skin	0.6
39	dark_skin	0.4	79	trans	0.4	119	dark_skin	0.6

k	material classifier	Distance z	k	material classifier	Distance z	k	material classifier	Distance z
120	dark_skin	0.6	160	control	0.8	200	dark_skin	0.8
121	dark_skin	0.6	161	control	0.8	201	dark_skin	0.8
122	dark_skin	0.6	162	control	0.8	202	dark_skin	0.8
123	dark_skin	0.6	163	control	0.8	203	dark_skin	0.8
124	dark_skin	0.6	164	control	0.8	204	dark_skin	0.8
125	dark_skin	0.6	165	control	0.8	205	dark_skin	0.8
126	dark_skin	0.6	166	control	0.8	206	dark_skin	0.8
127	dark_skin	0.6	167	control	0.8	207	dark_skin	0.8
128	pale_skin	0.6	168	control	0.8	208	pale_skin	0.8
129	pale_skin	0.6	169	control	0.8	209	pale_skin	0.8
130	pale_skin	0.6	170	control	0.8	210	pale_skin	0.8
131	pale_skin	0.6	171	control	0.8	211	pale_skin	0.8
132	pale_skin	0.6	172	control	0.8	212	pale_skin	0.8
133	pale_skin	0.6	173	control	0.8	213	pale_skin	0.8
134	pale_skin	0.6	174	control	0.8	214	pale_skin	0.8
135	pale_skin	0.6	175	control	0.8	215	pale_skin	0.8
136	pale_skin	0.6	176	fabric	0.8	216	pale_skin	0.8
137	pale_skin	0.6	177	fabric	0.8	217	pale_skin	0.8
138	pale_skin	0.6	178	fabric	0.8	218	pale_skin	0.8
139	pale_skin	0.6	179	fabric	0.8	219	pale_skin	0.8
140	pale_skin	0.6	180	fabric	0.8	220	pale_skin	0.8
141	pale_skin	0.6	181	fabric	0.8	221	pale_skin	0.8
142	pale_skin	0.6	182	fabric	0.8	222	pale_skin	0.8
143	pale_skin	0.6	183	fabric	0.8	223	pale_skin	0.8
144	trans	0.6	184	fabric	0.8	224	trans	0.8
145	trans	0.6	185	fabric	0.8	225	trans	0.8
146	trans	0.6	186	fabric	0.8	226	trans	0.8
147	trans	0.6	187	fabric	0.8	227	trans	0.8
148	trans	0.6	188	fabric	0.8	228	trans	0.8
149	trans	0.6	189	fabric	0.8	229	trans	0.8
150	trans	0.6	190	fabric	0.8	230	trans	0.8
151	trans	0.6	191	fabric	0.8	231	trans	0.8
152	trans	0.6	192	dark_skin	0.8	232	trans	0.8
153	trans	0.6	193	dark_skin	0.8	233	trans	0.8
154	trans	0.6	194	dark_skin	0.8	234	trans	0.8
155	trans	0.6	195	dark_skin	0.8	235	trans	0.8
156	trans	0.6	196	dark_skin	0.8	236	trans	0.8
157	trans	0.6	197	dark_skin	0.8	237	trans	0.8
158	trans	0.6	198	dark_skin	0.8	238	trans	0.8
159	trans	0.6	199	dark_skin	0.8	239	trans	0.8

The following parameter tables list the prefactors a_{lk} for the Gaussian functions g_{lk} in each image f_k referenced by the image index k and the Gaussian index l :

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
0	291.94	-7.20	-14.72	-22.75	32.57	16.92	-45.50	-23.31
1	300.91	-7.58	-15.22	-14.41	32.18	15.96	-44.41	-23.92
2	297.04	-6.42	-13.94	-10.30	31.19	16.44	-43.68	-22.27
3	289.98	-7.67	-14.20	-13.60	31.68	17.16	-44.04	-21.41
4	295.94	-7.72	-12.93	-16.81	32.77	15.92	-45.15	-22.30
5	295.37	-6.87	-13.30	-15.51	33.10	16.70	-45.18	-22.29
6	298.03	-6.79	-14.89	-12.68	32.43	14.46	-44.47	-22.71

7	296.95	-7.09	-14.98	-13.54	31.56	15.69	-45.09	-20.03
8	306.69	-6.91	-14.94	-14.27	33.28	16.02	-43.93	-20.21
9	291.40	-7.73	-13.91	-14.95	30.51	16.18	-45.86	-21.61
10	301.21	-7.21	-14.05	-17.46	32.02	17.26	-45.02	-22.89
11	309.78	-6.93	-13.47	-19.23	33.01	14.94	-43.61	-20.86
12	298.62	-7.09	-14.75	-18.43	32.59	18.97	-44.11	-22.73
13	296.09	-7.18	-14.25	-15.83	31.94	14.62	-43.72	-19.77
14	301.13	-7.24	-14.44	-19.11	33.54	17.01	-44.08	-21.39
15	304.28	-7.08	-14.10	-17.26	32.54	16.81	-44.40	-21.14
16	246.71	-3.77	-13.62	1.43	19.38	-2.03	-52.50	-19.62
17	258.85	-3.11	-12.88	-3.91	19.02	-1.05	-49.64	-21.50
18	263.06	-2.91	-13.72	1.65	19.03	-1.26	-51.02	-21.58
19	251.12	-4.15	-14.47	-6.19	19.00	-1.26	-50.65	-22.40
20	250.05	-2.91	-13.84	0.30	19.48	-1.79	-49.89	-20.92
21	249.35	-2.80	-13.64	-0.25	18.90	0.42	-51.29	-19.21
22	260.34	-2.57	-14.30	1.83	17.72	-1.18	-51.27	-24.07
23	246.64	-3.68	-13.92	-2.29	19.79	-0.20	-50.33	-23.09
24	245.58	-2.91	-15.25	-6.59	18.34	-1.75	-50.89	-24.38
25	249.09	-3.09	-14.13	3.85	19.58	-1.58	-50.05	-22.87
26	255.98	-2.91	-13.67	6.77	18.09	-2.15	-48.40	-21.74
27	249.87	-3.43	-13.89	1.80	17.14	-2.32	-50.95	-25.65
28	247.42	-3.34	-13.32	2.32	17.69	-1.29	-50.40	-23.82
29	253.00	-3.32	-12.60	4.98	17.09	-1.16	-49.54	-25.89
30	254.40	-3.44	-14.69	1.64	20.18	-1.15	-49.66	-22.73
31	256.40	-3.75	-13.62	-8.47	19.57	-1.08	-52.04	-21.62
32	24.07	-2.06	2.98	-22.09	14.88	8.52	3.28	-4.27
33	24.00	-1.30	3.15	-22.28	14.65	9.22	3.94	-4.75
34	20.76	-1.69	3.24	-22.34	14.81	8.95	3.76	-4.69
35	22.23	-2.51	3.10	-22.85	14.01	9.71	3.29	-4.61
36	20.33	-2.23	3.04	-21.40	15.61	8.99	3.00	-4.12
37	21.12	-2.09	3.25	-26.46	14.68	8.75	4.67	-5.05
38	21.89	-1.92	3.05	-22.32	14.65	8.25	3.77	-4.15
39	14.81	-2.07	2.82	-20.92	14.96	9.05	3.73	-5.40

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
0	-95.90	-116.79	0.49	-92.61	-76.01	36.70	0.46	78.40
1	-99.07	-117.11	0.57	-89.09	-76.31	37.12	-0.60	86.28
2	-92.44	-115.78	0.59	-93.89	-76.95	35.15	0.42	83.74
3	-96.50	-115.93	0.49	-92.28	-75.29	36.48	-0.17	70.45
4	-100.83	-120.58	0.57	-95.39	-73.94	37.36	-0.26	79.80
5	-94.89	-117.93	0.59	-93.45	-75.67	36.88	0.22	74.29
6	-94.27	-119.97	0.51	-91.14	-74.61	35.76	-0.72	73.46
7	-94.78	-114.15	0.55	-91.40	-71.49	36.55	-0.61	73.29
8	-97.53	-112.61	0.54	-91.30	-70.94	36.62	0.39	75.30
9	-97.85	-108.91	0.57	-94.07	-72.52	36.34	-0.32	76.34
10	-97.23	-115.13	0.64	-95.38	-74.62	36.22	-0.33	74.11
11	-96.18	-115.91	0.45	-93.13	-77.72	37.20	-0.07	81.82
12	-100.79	-113.16	0.52	-90.33	-72.38	36.46	-1.02	74.17
13	-98.70	-124.58	0.53	-93.48	-74.52	36.90	-0.78	74.38
14	-93.07	-116.00	0.52	-92.98	-75.28	36.84	-0.77	76.49
15	-106.83	-117.44	0.52	-94.22	-75.28	36.42	-0.51	79.82
16	-88.58	-87.41	0.63	-70.05	-37.06	27.50	3.58	84.26
17	-87.71	-82.32	0.54	-68.18	-40.98	28.10	3.20	88.34
18	-76.19	-80.81	0.53	-66.95	-35.36	27.23	3.61	87.52
19	-80.42	-84.95	0.55	-67.91	-38.39	27.70	3.39	90.71

20	-87.87	-76.07	0.74	-66.04	-39.03	27.55	2.84	83.83
21	-93.60	-80.04	0.45	-69.48	-38.83	27.01	3.71	83.32
22	-88.04	-78.17	0.62	-72.00	-39.30	27.24	3.08	89.23
23	-88.92	-76.16	0.62	-66.79	-39.81	27.64	3.01	84.46
24	-90.47	-77.18	0.58	-66.41	-35.84	27.85	3.13	94.66
25	-89.84	-83.96	0.70	-64.36	-36.65	26.40	3.38	86.08
26	-81.82	-72.08	0.59	-66.39	-38.13	27.45	3.47	83.27
27	-80.57	-82.11	0.65	-66.61	-35.78	28.23	3.99	86.32
28	-93.40	-81.19	0.66	-68.45	-39.71	26.68	3.19	88.66
29	-92.92	-79.14	0.68	-64.90	-38.61	28.11	3.79	83.05
30	-91.82	-79.09	0.57	-66.39	-37.74	27.24	4.20	82.26
31	-90.60	-82.20	0.51	-67.61	-35.58	28.10	3.02	89.40
32	-0.54	-16.54	-0.31	-3.67	-33.80	4.91	1.87	11.78
33	-0.17	-16.06	-0.31	-2.39	-32.09	4.41	1.86	11.66
34	-3.04	-15.03	-0.34	-5.36	-31.98	4.17	1.76	11.91
35	-5.65	-15.79	-0.35	-2.71	-32.21	4.77	1.90	10.71
36	-6.54	-15.97	-0.29	-3.29	-33.09	4.26	1.69	14.99
37	-1.22	-16.20	-0.29	-2.79	-32.90	4.42	1.93	9.92
38	-3.85	-13.17	-0.22	-2.98	-33.98	4.68	1.84	11.96
39	-5.99	-12.87	-0.34	-2.53	-34.59	3.97	1.80	11.77

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
0	3.86	18.38	66.68	70.89	46.30	51.83	63.61	6.33
1	-1.41	18.18	66.65	-116.27	47.93	53.00	-19.71	6.14
2	5.49	17.64	67.40	86.57	47.99	51.08	-20.20	6.06
3	1.41	18.29	68.83	-73.86	49.40	52.06	-9.49	6.14
4	0.58	17.86	70.11	73.41	47.45	52.04	19.20	6.06
5	-0.96	18.19	71.29	12.08	48.40	52.10	36.63	6.18
6	-0.98	18.97	69.33	107.17	48.12	51.68	-30.34	6.21
7	1.44	17.76	67.10	-119.11	46.64	51.64	-19.85	6.11
8	2.83	18.10	66.62	42.19	47.28	52.29	-24.88	6.17
9	-0.87	17.50	68.71	-129.91	45.89	51.63	-43.44	6.23
10	1.24	18.26	66.88	-12.32	47.49	52.45	45.87	6.14
11	0.39	18.50	68.03	-103.43	47.42	52.40	-25.17	6.01
12	1.60	18.13	67.42	-159.30	48.41	52.43	-8.06	6.08
13	2.92	17.75	68.36	139.40	45.89	52.74	-16.56	6.19
14	1.69	17.80	68.48	-16.76	46.43	52.86	-15.34	6.20
15	0.66	17.51	68.30	-16.84	46.61	52.23	-12.07	6.00
16	1.53	10.12	45.53	-105.48	51.47	53.43	29.32	3.91
17	-0.83	10.67	47.22	-65.54	49.74	55.32	-3.14	3.90
18	-0.05	11.56	46.07	-38.72	50.65	54.17	-71.99	3.87
19	1.11	10.54	47.56	-22.99	50.63	54.47	19.05	3.81
20	-2.77	10.45	46.29	102.12	50.39	54.05	-21.67	3.88
21	1.51	11.03	46.20	-61.41	52.28	55.89	51.76	4.01
22	-2.08	11.82	45.96	121.36	49.18	54.28	-44.05	3.89
23	-0.13	10.89	46.21	34.94	52.82	54.93	26.48	3.82
24	-3.52	10.89	49.08	97.01	50.10	55.70	4.07	3.65
25	2.07	9.39	46.08	-53.59	51.65	53.65	4.94	3.97
26	0.44	11.04	47.23	-188.70	49.01	55.21	30.06	4.08
27	-4.62	9.33	47.59	141.42	51.09	54.20	-1.25	3.95
28	-1.78	10.70	45.70	-47.29	49.84	54.81	40.79	3.71
29	-2.45	10.65	47.01	-32.51	51.18	53.56	11.77	3.93
30	0.03	10.49	43.45	-149.80	50.93	53.95	14.36	3.83
31	3.37	10.57	46.51	20.87	49.67	54.60	22.79	3.96
32	0.86	7.55	19.63	-6.16	7.11	9.03	-20.02	1.08

33	-1.54	7.76	20.22	38.99	7.26	8.99	-10.41	1.12
34	-0.19	7.43	18.78	21.86	7.75	8.86	-16.09	1.15
35	-2.22	7.79	19.15	4.23	7.44	8.75	19.94	1.06
36	-1.11	7.83	20.15	20.92	6.25	8.80	-31.48	1.06
37	-0.31	7.90	19.64	8.33	7.22	9.07	18.83	1.12
38	-1.59	7.27	19.37	-18.56	6.82	8.51	21.93	1.10
39	-1.18	7.41	19.11	85.48	7.72	8.69	-17.48	1.01

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
0	-2.45	11.03	80.28	37.44	26.08	-32.86	-12.63	179.34
1	4.01	11.52	78.78	37.44	24.32	-34.91	-12.16	180.51
2	-3.35	10.90	81.50	37.26	24.80	-33.69	-10.77	180.89
3	1.73	10.84	80.38	37.51	25.47	-36.57	-15.88	183.66
4	1.27	9.42	81.14	38.14	25.13	-35.66	-13.87	180.51
5	-1.05	10.89	78.62	35.99	24.44	-34.31	-10.65	181.52
6	-0.46	10.30	81.92	35.89	24.74	-34.63	-15.73	180.07
7	2.40	10.57	78.99	37.01	23.94	-34.87	-12.35	183.67
8	1.32	10.56	80.87	37.45	24.70	-35.58	-16.41	180.05
9	1.05	9.87	79.26	37.63	24.52	-33.58	-12.25	178.55
10	-0.78	9.63	81.05	38.38	25.20	-33.14	-13.41	176.95
11	3.17	10.63	81.55	37.98	24.25	-32.39	-8.97	178.12
12	-0.12	10.57	81.11	38.18	25.06	-32.51	-14.47	180.98
13	-3.92	11.23	81.59	37.07	24.28	-35.06	-13.10	179.20
14	-1.14	11.04	81.63	36.75	25.25	-34.70	-11.65	180.54
15	1.46	11.05	80.55	36.57	23.89	-33.64	-12.62	181.91
16	-5.70	11.70	54.00	34.44	12.44	-14.39	-12.01	171.54
17	-0.70	12.09	55.21	34.71	9.92	-17.13	-11.70	168.02
18	-3.57	11.69	53.52	34.81	11.95	-13.55	-12.37	172.64
19	-3.32	12.04	54.84	35.03	11.31	-16.56	-8.97	169.62
20	2.59	12.21	55.48	35.95	11.23	-16.53	-9.56	169.59
21	2.54	12.19	54.25	35.20	10.76	-16.90	-12.28	169.58
22	-3.44	11.69	55.10	35.89	12.10	-18.59	-8.83	171.87
23	0.56	12.26	53.50	35.39	10.43	-17.85	-6.90	169.62
24	1.18	11.85	54.90	34.61	11.57	-14.31	-8.59	170.29
25	4.59	13.16	55.42	35.65	11.43	-20.08	-11.70	170.01
26	-0.12	11.06	54.09	34.94	11.37	-15.16	-13.76	172.27
27	1.10	11.81	55.45	34.85	10.99	-17.00	-9.50	170.91
28	2.78	11.51	56.06	34.44	10.57	-13.57	-8.79	172.45
29	-1.28	11.48	55.08	35.02	10.89	-15.74	-9.63	169.75
30	2.04	12.16	55.11	35.67	10.86	-17.71	-6.55	170.86
31	0.32	12.82	53.12	34.92	10.54	-18.42	-8.44	169.18
32	0.20	0.74	50.97	-0.31	16.61	-0.13	-0.95	4.08
33	-2.15	1.02	49.63	-0.32	16.22	0.23	0.22	1.95
34	1.20	0.57	50.53	-0.69	16.60	0.05	-0.14	2.81
35	2.07	0.84	50.16	-0.49	16.35	0.75	-0.83	2.72
36	0.80	0.74	49.51	0.39	16.42	-0.53	-1.24	3.37
37	-1.45	1.05	52.96	-0.54	16.22	0.77	-0.23	3.04
38	2.48	0.88	49.48	0.67	16.29	-0.46	-1.51	3.83
39	-0.81	0.65	50.20	-0.45	16.38	-0.34	-0.14	3.97

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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0	-19.43	-15.41	-0.54	35.39	4.01	146.63	-31.68	-3.15
1	-18.88	-15.15	-0.37	34.72	4.18	148.07	-31.76	-3.56
2	-16.02	-15.31	-0.38	33.72	4.18	145.90	-32.03	-3.18
3	-17.80	-15.75	-0.39	36.39	4.08	145.63	-31.54	-3.47
4	-15.90	-15.69	-0.30	34.77	4.13	149.16	-31.20	-3.47
5	-18.98	-15.68	-0.58	33.97	4.12	147.36	-31.71	-3.33
6	-15.90	-15.37	-0.49	34.31	4.29	146.96	-30.78	-3.23
7	-14.81	-15.73	-0.31	36.06	4.14	150.82	-32.18	-3.67
8	-15.48	-15.58	-0.58	36.41	4.13	146.32	-31.10	-3.39
9	-16.55	-15.89	-0.37	32.95	3.99	141.73	-30.59	-3.45
10	-18.58	-15.48	-0.48	33.49	4.40	148.23	-30.74	-3.45
11	-15.63	-15.39	-0.38	35.93	4.06	144.15	-30.84	-3.40
12	-17.34	-15.78	-0.58	35.76	4.10	140.54	-32.14	-3.58
13	-16.01	-15.44	-0.51	35.03	4.19	142.21	-31.70	-3.44
14	-19.00	-15.39	-0.63	35.03	4.06	146.35	-31.59	-3.67
15	-16.05	-15.61	-0.51	35.06	4.04	144.24	-31.00	-3.59
16	-1.73	-11.69	0.83	17.68	4.06	115.00	-27.19	-5.03
17	-4.29	-12.41	1.18	16.91	3.80	117.37	-26.45	-5.33
18	-2.73	-12.25	1.21	17.51	3.83	119.84	-28.87	-5.24
19	-1.43	-12.78	1.09	16.03	3.87	118.57	-27.48	-5.28
20	-1.08	-12.82	1.18	16.30	3.94	118.99	-26.64	-4.93
21	-3.12	-12.39	0.98	17.09	3.77	116.73	-27.35	-5.06
22	-2.46	-12.29	1.12	15.90	3.83	119.51	-27.39	-4.75
23	-1.88	-12.16	1.06	16.70	3.81	120.08	-27.74	-5.30
24	-3.35	-12.24	1.09	14.93	3.90	117.65	-27.21	-5.13
25	-1.85	-11.74	1.20	18.98	3.86	118.21	-27.50	-5.19
26	-2.53	-12.60	1.34	18.04	3.83	118.75	-26.64	-5.50
27	-4.14	-11.95	1.01	18.05	3.80	120.99	-27.10	-4.89
28	-2.49	-12.69	1.15	18.06	3.69	117.75	-26.73	-5.11
29	-2.52	-12.69	1.30	15.46	4.07	116.11	-27.32	-5.01
30	-1.51	-12.10	1.15	17.59	3.63	123.35	-27.39	-5.32
31	-4.63	-12.59	1.03	17.39	3.88	119.73	-27.34	-5.17
32	-5.22	1.64	2.40	15.00	2.34	28.20	-5.04	-5.62
33	-5.96	1.34	2.52	14.98	2.36	28.53	-4.99	-5.55
34	-5.47	1.64	2.42	14.89	2.39	29.51	-4.77	-5.28
35	-4.58	1.75	2.52	14.33	2.30	31.32	-5.18	-5.29
36	-6.27	1.70	2.46	14.36	2.42	31.69	-5.05	-5.67
37	-5.85	1.50	2.37	14.83	2.33	29.47	-5.19	-5.62
38	-4.86	1.37	2.45	14.20	2.32	30.31	-5.13	-5.42
39	-6.24	1.66	2.42	14.65	2.30	31.65	-5.26	-5.40

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
0	-10.74	-128.40	-12.06	-27.68	-47.57	129.47	-144.26	5.82
1	-10.79	-126.25	-6.60	-27.37	30.05	129.30	-151.57	0.18
2	-10.74	-126.91	22.40	-24.92	8.00	127.04	-148.75	4.11
3	-10.77	-125.83	-0.47	-27.97	-22.77	127.09	-144.81	-5.16
4	-10.78	-123.99	31.65	-28.65	24.12	128.49	-154.50	13.78
5	-10.58	-125.15	-1.09	-27.15	-24.40	127.57	-149.54	5.15
6	-10.85	-124.37	-9.75	-27.32	-4.55	126.41	-145.43	-4.92
7	-10.65	-127.00	0.46	-27.02	-13.23	129.75	-144.09	-13.95
8	-11.02	-126.29	-13.90	-26.51	-7.75	127.78	-152.92	8.36
9	-10.68	-124.07	-7.82	-27.67	-23.63	127.87	-150.02	8.26
10	-10.83	-124.06	47.68	-28.79	7.39	124.57	-149.34	-3.62

11	-10.80	-128.19	-2.30	-25.63	-6.57	129.60	-150.58	-9.25
12	-10.87	-124.73	-54.90	-24.31	41.58	134.82	-151.79	-4.73
13	-10.82	-127.07	-0.98	-28.09	7.94	130.29	-148.24	-10.15
14	-11.01	-125.00	-2.61	-26.15	-9.59	129.00	-148.98	-19.14
15	-10.69	-125.55	6.16	-27.16	5.81	130.83	-152.90	-9.64
16	-11.43	-133.31	27.27	-18.30	-26.33	148.91	-121.02	7.67
17	-11.74	-134.46	-52.34	-16.89	-23.99	151.23	-120.04	-19.73
18	-11.45	-130.80	18.52	-16.85	-48.77	150.98	-122.92	10.85
19	-11.27	-133.08	-29.76	-17.58	3.69	149.84	-121.37	-4.93
20	-11.27	-134.65	8.68	-16.64	34.24	149.00	-118.06	-9.55
21	-11.71	-137.87	30.52	-14.52	0.44	149.28	-119.23	7.09
22	-11.44	-133.43	27.53	-14.99	8.31	149.32	-122.39	-19.43
23	-11.34	-135.85	-13.46	-14.85	0.51	149.46	-118.80	12.65
24	-11.46	-134.89	-32.30	-18.64	3.14	145.98	-119.97	5.91
25	-10.92	-135.14	-16.97	-16.50	9.98	148.08	-126.66	0.91
26	-11.38	-132.17	38.54	-18.23	23.72	150.56	-114.45	1.06
27	-11.44	-134.82	-43.20	-14.82	-21.37	150.11	-123.74	-7.24
28	-11.64	-136.08	-8.20	-17.05	0.87	153.36	-125.50	18.53
29	-11.25	-132.85	-28.20	-17.77	-16.83	146.70	-118.61	-2.81
30	-11.21	-133.55	-22.12	-16.42	-9.17	150.89	-123.39	7.29
31	-11.50	-135.59	16.35	-17.71	-39.44	149.92	-117.21	-17.87
32	-1.75	-17.34	-4.56	-15.38	-4.53	27.52	-4.76	-15.17
33	-1.66	-18.40	-1.44	-15.69	-11.54	27.41	-2.83	-8.66
34	-1.72	-17.15	3.10	-14.93	-13.85	27.41	0.29	18.03
35	-1.60	-18.29	10.53	-16.01	10.69	27.36	-0.67	-6.10
36	-1.70	-17.23	-4.85	-14.95	7.00	25.49	-2.44	-2.75
37	-1.65	-17.09	-18.55	-15.00	3.78	27.40	1.75	3.17
38	-1.68	-17.34	10.25	-15.93	-2.93	27.08	-9.32	12.10
39	-1.66	-16.95	6.80	-16.81	-3.07	27.33	-0.14	-2.79

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
0	37.14	79.32	-51.17	-1.76	107.54	-136.45	-0.70	49.45
1	39.73	78.51	-53.02	-3.67	107.24	-137.40	-0.92	48.53
2	38.32	79.55	-53.33	6.10	109.08	-137.60	-0.44	47.48
3	43.04	78.76	-49.78	-5.20	109.63	-137.79	-0.41	49.72
4	41.97	78.51	-51.25	-3.82	108.30	-134.67	-0.72	47.28
5	43.91	79.20	-53.25	5.02	104.46	-134.65	-0.72	50.25
6	41.71	79.71	-51.24	9.31	106.65	-135.02	-0.97	48.74
7	44.52	80.29	-48.51	-2.75	107.04	-140.33	0.13	49.50
8	39.20	80.04	-52.02	-1.21	109.14	-135.03	-0.87	48.70
9	42.96	79.10	-48.95	0.10	107.27	-135.84	-1.15	51.11
10	42.98	80.11	-51.37	2.86	107.63	-136.54	-1.02	49.77
11	41.67	78.52	-54.87	9.25	107.78	-137.56	-0.36	49.93
12	41.54	78.66	-49.72	12.58	108.48	-138.05	-1.51	50.82
13	45.53	79.40	-53.00	-0.90	106.29	-138.06	-1.22	48.46
14	41.05	78.05	-50.05	5.44	108.17	-134.77	-1.00	49.65
15	40.71	80.07	-56.73	-3.43	106.52	-136.43	-0.06	50.09
16	37.90	83.28	-64.08	-0.55	90.49	-120.94	4.79	58.73
17	39.35	84.15	-61.34	14.09	91.22	-115.20	4.42	58.38
18	39.25	83.87	-64.28	-2.73	90.15	-119.81	5.17	59.14
19	44.51	83.85	-64.77	-5.83	91.64	-120.52	4.96	59.80
20	39.23	84.11	-64.08	-0.47	90.61	-119.79	4.41	58.04
21	33.64	84.97	-66.24	2.00	90.84	-119.05	4.36	55.65

22	40.09	84.70	-68.59	8.97	89.54	-122.30	4.21	58.89
23	39.94	84.58	-64.95	9.90	90.45	-118.55	4.20	58.47
24	41.44	83.21	-61.49	21.96	86.40	-121.01	5.88	59.02
25	43.13	84.06	-71.09	5.92	91.68	-121.74	5.43	58.39
26	40.08	84.02	-62.57	-16.45	90.01	-124.94	5.37	59.22
27	42.77	83.17	-60.93	-3.80	90.21	-119.34	4.16	59.00
28	38.19	83.14	-68.75	0.93	89.45	-118.83	4.65	58.62
29	40.28	84.08	-66.06	2.04	90.85	-121.14	4.95	58.58
30	38.60	83.57	-63.85	4.02	90.34	-121.73	5.02	59.52
31	38.72	83.98	-62.23	-8.14	90.87	-118.15	4.47	57.21
32	11.57	21.55	-10.03	-6.33	68.62	-1.53	-1.26	5.29
33	12.52	21.60	-9.90	5.14	67.28	-2.03	-1.17	6.06
34	10.52	21.27	-9.92	7.48	62.26	-1.26	-1.10	6.49
35	9.72	21.00	-9.55	4.65	67.09	-2.90	-1.42	5.32
36	10.18	21.30	-10.46	0.53	68.50	-1.87	-1.28	5.86
37	11.41	21.45	-10.21	-5.58	69.19	-2.94	-0.93	6.38
38	10.20	21.55	-10.48	-0.02	72.00	-2.62	-1.45	5.82
39	10.76	21.90	-9.25	2.27	70.02	-3.28	-1.69	5.49

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
0	-12.86	129.34	-0.04	14.92	138.37	-1.55	89.62	30.73
1	-5.85	128.69	0.01	15.19	137.24	-1.35	-3.02	-80.51
2	-4.35	128.38	0.04	15.14	134.51	-1.28	41.45	-64.49
3	-12.07	129.28	0.00	15.99	138.44	-1.79	43.17	-32.41
4	-29.61	130.64	0.05	15.92	138.34	-1.07	-61.32	-48.06
5	-8.28	128.31	0.04	16.59	134.68	-1.42	22.30	2.66
6	19.35	130.30	0.02	13.87	136.49	-1.67	-84.92	0.79
7	24.36	127.83	0.04	15.94	138.86	-1.90	0.65	-3.65
8	-14.01	128.40	0.03	15.30	137.88	-1.44	-4.38	-31.18
9	0.91	129.12	0.02	14.12	135.10	-1.55	69.96	4.18
10	23.68	129.32	0.07	16.18	140.35	-1.81	-95.92	-40.96
11	-17.89	127.87	-0.02	14.63	137.22	-1.64	-53.97	-21.93
12	-16.16	127.54	0.03	15.68	139.52	-1.33	-2.36	-18.42
13	-9.14	128.00	0.04	15.45	137.55	-1.47	-154.65	-37.01
14	6.32	128.51	0.01	15.55	135.27	-1.61	-75.40	17.78
15	-56.04	128.95	-0.01	15.81	140.72	-1.50	15.30	-0.58
16	11.07	103.86	0.32	12.40	106.49	-3.15	-17.66	-6.34
17	1.20	103.57	0.30	11.80	105.25	-3.35	158.30	-11.96
18	15.81	104.02	0.31	12.73	105.76	-3.14	27.25	38.46
19	-38.30	105.03	0.26	13.24	105.45	-3.08	55.65	-53.71
20	-11.59	103.10	0.28	12.71	104.03	-3.53	-163.96	49.02
21	-12.52	102.37	0.36	13.19	102.12	-2.92	-173.39	-5.65
22	-9.62	103.25	0.33	12.39	108.93	-2.99	52.22	-161.59
23	25.92	103.48	0.29	11.74	106.99	-2.55	105.11	-85.85
24	-5.50	103.73	0.35	12.51	104.40	-3.33	69.39	-5.04
25	-10.31	105.43	0.32	12.82	105.99	-3.71	-26.84	-118.93
26	18.71	106.00	0.30	10.96	106.89	-3.08	-32.36	-84.52
27	20.64	104.18	0.31	10.98	105.86	-2.84	-60.72	-27.19
28	-15.68	105.50	0.30	12.70	106.94	-2.97	88.69	23.52
29	-13.79	104.57	0.35	11.66	106.15	-3.25	26.48	-34.70
30	0.58	103.49	0.33	12.68	105.58	-2.79	-41.33	6.57
31	-41.18	104.89	0.34	12.31	105.97	-3.75	29.53	-6.35
32	1.27	44.57	0.12	13.53	42.01	2.37	17.83	3.10

33	6.12	45.81	0.10	13.35	42.85	2.35	-40.89	-6.28
34	-2.92	44.37	0.10	13.49	42.40	2.50	118.60	-8.15
35	-5.72	45.65	0.09	13.59	42.95	2.35	-24.50	42.41
36	8.16	46.19	0.13	13.25	42.19	2.58	-23.66	-14.21
37	-2.52	45.21	0.10	14.30	42.49	2.21	10.46	8.05
38	1.03	44.35	0.08	13.11	42.30	2.30	-19.72	-34.53
39	7.84	44.14	0.07	12.77	41.49	2.14	-87.32	-9.49

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
40	19.90	-2.13	3.40	-23.58	14.75	9.48	3.86	-5.28
41	18.83	-2.16	2.85	-22.88	14.32	8.82	4.18	-4.88
42	20.70	-1.65	3.27	-20.35	15.25	9.24	3.58	-4.35
43	20.25	-1.77	3.20	-22.22	15.27	9.16	3.48	-3.80
44	24.14	-2.06	3.32	-21.09	15.40	8.40	4.18	-4.42
45	18.27	-1.69	3.27	-23.09	14.59	8.75	3.83	-4.73
46	25.79	-2.16	2.67	-23.50	13.93	9.50	3.72	-4.28
47	20.38	-2.29	3.08	-22.68	15.06	8.65	2.52	-5.19
48	50.37	-1.77	3.59	-24.60	18.28	10.50	5.85	-7.96
49	49.85	-2.31	3.47	-28.05	18.13	10.17	7.16	-7.75
50	41.67	-2.30	3.79	-26.10	18.67	10.55	6.25	-8.25
51	42.09	-2.00	3.53	-25.53	18.63	10.87	5.39	-5.86
52	50.69	-2.14	3.14	-26.33	19.35	9.35	6.52	-7.00
53	42.49	-1.43	3.63	-27.22	19.41	9.91	6.84	-7.48
54	53.71	-2.44	3.50	-31.61	18.14	11.45	6.06	-9.48
55	46.99	-2.19	3.48	-29.49	18.31	10.81	6.44	-6.38
56	51.56	-2.37	2.97	-27.93	19.01	11.39	5.33	-6.22
57	46.94	-2.41	3.00	-30.06	17.38	9.09	6.34	-7.10
58	41.31	-2.75	3.65	-25.85	19.08	9.83	6.07	-6.96
59	44.49	-2.22	3.61	-32.55	18.31	10.01	6.66	-7.11
60	45.40	-2.42	3.22	-27.27	18.00	11.33	6.80	-7.85
61	48.69	-2.05	3.04	-27.02	19.11	10.47	6.21	-8.31
62	51.91	-2.10	3.23	-24.02	18.61	10.14	6.24	-8.18
63	49.12	-2.03	3.79	-26.05	19.92	9.39	6.41	-7.48
64	14.82	-1.26	2.54	-14.92	10.38	5.70	4.36	-2.13
65	18.85	-1.68	2.35	-15.32	10.60	5.67	4.25	-2.41
66	12.30	-1.35	2.98	-13.25	10.11	6.36	5.11	-2.88
67	10.43	-0.99	2.47	-13.28	10.38	6.20	5.13	-2.15
68	10.66	-1.07	2.81	-16.65	10.22	5.83	3.98	-2.43
69	14.27	-1.20	2.24	-13.79	9.68	4.91	3.57	-2.61
70	14.63	-0.95	2.44	-13.66	10.38	4.72	4.73	-2.55
71	13.43	-1.33	2.57	-16.19	9.82	5.72	4.41	-2.84
72	12.29	-1.25	2.24	-13.42	9.93	5.64	4.34	-2.66
73	13.65	-1.66	2.41	-16.32	10.52	5.40	4.92	-2.50
74	13.37	-1.27	2.47	-15.49	9.98	5.94	3.65	-1.62
75	14.10	-1.64	2.34	-14.76	10.05	5.40	4.82	-2.77
76	16.53	-1.24	1.83	-18.07	9.51	6.44	4.40	-2.85
77	15.73	-1.15	2.18	-13.81	9.36	5.62	4.61	-2.36
78	17.20	-1.32	2.68	-13.10	9.52	5.70	4.78	-2.81
79	18.10	-1.03	2.61	-14.46	10.11	5.73	4.83	-2.49

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
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40	-0.48	-14.13	-0.26	-4.85	-32.91	4.45	1.62	14.53
41	-2.55	-14.69	-0.26	-2.59	-32.83	4.48	1.64	10.87
42	-2.32	-13.82	-0.25	-3.21	-32.11	4.71	1.92	12.41
43	0.25	-13.86	-0.32	-0.74	-32.76	4.46	2.18	10.08
44	-2.98	-14.88	-0.29	-0.15	-33.19	4.33	1.78	12.83
45	0.26	-14.82	-0.34	-3.73	-35.01	4.39	1.93	11.13
46	-0.55	-13.52	-0.27	-2.21	-32.26	4.23	1.58	13.57
47	-2.58	-14.57	-0.27	-1.27	-32.66	4.42	1.55	11.73
48	-9.49	-17.26	-0.40	-7.16	-36.29	5.87	2.34	25.11
49	-15.26	-17.26	-0.39	-9.71	-37.64	6.04	2.21	24.19
50	-17.74	-16.05	-0.38	-8.14	-37.86	5.44	2.03	27.35
51	-7.51	-17.69	-0.44	-7.78	-36.26	6.10	2.27	26.16
52	-16.09	-16.80	-0.45	-7.39	-35.79	6.14	2.29	22.75
53	-21.20	-18.29	-0.46	-6.47	-37.56	6.01	2.73	22.73
54	-14.84	-18.42	-0.34	-6.94	-35.71	6.25	2.47	23.41
55	-16.75	-17.24	-0.39	-7.01	-38.93	6.20	2.67	26.48
56	-8.39	-22.20	-0.43	-6.67	-40.21	5.83	2.38	22.94
57	-14.66	-19.71	-0.36	-8.63	-40.12	6.05	2.57	23.51
58	-16.77	-15.89	-0.40	-8.32	-38.01	6.03	2.53	23.62
59	-16.13	-20.60	-0.48	-7.39	-36.88	6.09	2.39	23.28
60	-14.21	-16.48	-0.41	-8.28	-39.00	5.52	2.26	23.17
61	-14.30	-22.10	-0.43	-8.26	-37.30	6.59	2.03	22.66
62	-17.72	-18.12	-0.41	-7.89	-37.81	6.13	2.04	25.89
63	-17.68	-18.26	-0.39	-6.30	-36.11	5.40	2.04	21.67
64	2.73	-5.49	-0.33	1.00	-13.27	1.08	2.61	11.70
65	-4.20	-5.79	-0.38	-1.97	-13.62	1.34	2.28	11.01
66	-5.39	-6.55	-0.36	0.72	-12.79	0.98	2.48	13.80
67	-4.50	-5.76	-0.33	-1.87	-13.93	0.96	2.58	11.78
68	-4.01	-5.60	-0.37	-1.22	-14.53	1.28	2.37	12.48
69	-6.29	-5.07	-0.33	0.39	-14.47	1.50	2.31	13.61
70	-6.11	-5.86	-0.40	-0.83	-14.26	1.23	2.63	10.62
71	-8.03	-5.00	-0.32	-0.78	-12.45	1.14	2.62	12.27
72	-6.81	-6.73	-0.38	0.34	-14.42	1.52	2.49	12.46
73	-7.12	-6.95	-0.33	0.12	-13.53	1.04	2.45	10.32
74	-6.57	-3.28	-0.31	0.22	-14.17	1.09	2.41	13.68
75	-6.00	-3.61	-0.33	-0.67	-13.40	1.23	2.67	12.09
76	-5.23	-7.88	-0.38	-0.19	-14.11	1.19	2.52	13.16
77	-7.27	-3.96	-0.32	1.76	-14.15	1.23	2.79	11.77
78	-3.39	-7.77	-0.34	-0.85	-13.00	1.32	2.61	10.48
79	-4.11	-6.10	-0.36	-1.01	-13.69	1.30	2.55	11.87

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
40	-1.34	7.31	20.08	-43.84	7.06	8.83	3.34	1.08
41	-1.10	7.79	19.71	8.41	7.32	8.92	15.86	1.04
42	1.31	8.00	19.19	-20.28	6.83	8.82	24.92	1.08
43	1.45	7.98	19.87	46.25	6.78	8.45	-2.96	1.00
44	0.33	7.34	19.54	-38.90	6.92	8.67	-12.94	1.11
45	1.75	7.10	20.14	-21.37	6.82	8.73	-27.46	1.13
46	0.61	8.00	20.03	-28.34	7.09	8.93	31.83	1.12
47	-0.64	7.87	19.53	-46.57	7.31	8.89	-30.39	1.07
48	-0.15	11.06	25.54	-2.80	12.05	11.55	-13.03	1.27
49	-0.12	11.11	25.52	8.12	12.20	12.65	-6.52	1.32
50	-1.32	10.83	24.11	37.99	12.63	11.38	32.37	1.35

51	0.13	10.49	25.74	8.31	12.13	11.86	-5.37	1.28
52	-1.83	11.17	26.01	33.77	12.42	11.00	34.57	1.29
53	-1.54	11.10	25.30	42.41	11.53	11.16	50.78	1.36
54	0.86	10.54	23.84	-10.95	12.27	11.29	7.27	1.27
55	0.11	11.02	25.47	-32.01	11.40	11.13	0.00	1.32
56	1.25	10.62	26.27	11.39	11.63	11.39	0.65	1.32
57	-1.16	10.55	25.32	76.10	12.37	11.06	17.89	1.41
58	-0.85	11.14	24.88	25.65	11.81	11.47	-34.03	1.31
59	-1.57	10.72	25.21	-19.84	11.67	12.12	-12.65	1.17
60	0.05	10.51	24.90	-78.84	12.63	12.01	-6.74	1.37
61	-1.47	10.82	25.26	46.37	12.04	11.15	-18.96	1.39
62	1.03	10.49	25.79	18.87	12.11	11.18	-7.43	1.35
63	0.25	10.96	26.25	59.58	11.55	11.69	-2.09	1.42
64	-0.42	5.86	11.16	66.99	8.70	6.34	31.58	1.06
65	1.35	5.34	11.18	14.04	8.27	6.43	59.34	1.01
66	0.05	5.54	10.70	-52.21	8.29	6.36	6.84	1.05
67	0.45	5.43	11.31	18.29	7.77	6.55	8.15	1.01
68	-0.61	6.04	9.77	28.80	7.99	6.69	3.61	1.10
69	-0.84	5.57	10.58	42.30	8.11	6.44	-41.16	1.11
70	0.34	5.48	10.22	22.47	8.03	6.57	-21.22	1.11
71	-0.40	6.13	10.13	-19.36	8.25	6.63	-9.41	1.03
72	-0.74	5.96	10.71	15.47	7.99	6.60	-11.79	1.10
73	0.30	5.77	10.78	-35.45	8.02	7.23	-7.79	1.12
74	0.31	6.04	9.85	-2.67	7.43	6.23	-16.70	0.99
75	0.07	5.67	11.17	-30.70	7.82	6.03	-41.32	1.03
76	0.91	6.31	10.37	65.63	7.65	6.38	-19.29	1.11
77	-0.55	5.90	10.02	41.19	7.65	6.85	26.33	1.08
78	0.38	5.93	10.95	-33.73	8.60	6.71	15.83	1.07
79	0.08	5.82	10.18	55.66	8.03	6.07	15.49	1.06

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
40	-1.53	1.02	52.10	-0.48	16.66	0.51	-0.48	3.17
41	1.12	0.85	50.58	-0.66	16.94	-0.20	-1.60	1.47
42	-0.14	0.54	49.36	-0.28	16.62	-0.54	-0.84	2.34
43	1.89	-0.17	50.32	-0.77	16.71	0.11	-0.84	2.40
44	2.72	0.93	51.66	0.03	16.64	-0.52	-0.93	3.31
45	0.90	0.89	50.79	-0.49	16.85	0.29	-0.30	2.02
46	0.92	1.06	49.05	-0.55	15.95	-0.36	-0.75	2.36
47	-1.10	0.50	50.20	-0.34	16.34	-0.29	-0.96	2.86
48	1.43	2.74	66.38	-2.26	21.67	0.95	-5.11	7.11
49	0.16	3.22	63.74	-1.00	21.20	1.23	-4.04	5.34
50	1.56	2.91	65.05	-1.22	20.65	1.57	-3.43	4.36
51	-0.08	2.84	63.90	-2.64	21.54	-0.16	-4.06	5.01
52	-1.22	3.14	64.73	-2.15	20.73	1.26	-3.17	6.76
53	2.12	3.35	63.77	-2.31	20.64	1.90	-2.79	4.15
54	-0.09	3.01	65.03	-2.19	21.60	0.00	-3.75	3.41
55	2.71	3.52	62.85	-2.18	21.41	1.02	-3.95	4.22
56	-1.83	3.25	64.90	-2.23	20.87	1.17	-5.24	6.48
57	-1.32	3.01	65.44	-1.67	20.99	0.86	-4.19	4.77
58	-1.32	3.42	67.01	-1.82	20.89	0.54	-2.96	6.01
59	-1.20	3.13	67.59	-2.06	21.62	0.29	-4.53	7.54
60	3.64	2.66	65.10	-1.51	20.53	2.64	-2.72	4.43
61	-2.11	2.97	65.31	-2.06	21.06	1.06	-5.37	5.61

62	1.09	2.90	65.18	-2.28	21.55	1.93	-3.25	4.31
63	3.76	3.61	62.63	-3.22	19.95	0.38	-4.80	6.77
64	2.01	3.49	35.08	-1.73	7.78	-0.78	0.64	-7.62
65	0.65	3.75	37.57	-1.59	8.16	0.25	1.16	-6.40
66	-0.59	3.12	35.67	-1.59	8.70	-0.65	0.44	-7.70
67	0.84	3.74	36.42	-1.44	8.46	0.59	0.32	-7.58
68	1.82	3.81	36.20	-0.96	8.04	0.50	0.78	-6.98
69	-3.57	3.82	34.70	-1.26	8.63	0.05	0.45	-6.48
70	-0.26	3.34	35.18	-1.92	8.23	-0.19	0.54	-6.45
71	0.42	3.34	36.22	-1.78	7.93	0.71	1.11	-6.50
72	-2.08	3.40	36.98	-1.86	8.30	-0.10	-0.07	-6.00
73	0.88	3.39	36.43	-1.80	8.82	1.05	2.09	-6.28
74	0.13	3.49	37.23	-1.68	7.89	0.03	0.84	-6.37
75	0.72	3.63	34.17	-1.43	8.40	-0.15	1.45	-6.18
76	1.16	3.70	37.28	-1.47	8.24	0.62	0.36	-6.41
77	-1.66	3.54	36.23	-1.32	8.74	-0.46	1.47	-6.34
78	0.57	3.54	37.16	-1.52	8.39	0.23	0.68	-5.44
79	1.19	3.07	36.90	-2.02	8.87	0.19	0.45	-6.90

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
40	-5.34	1.56	2.37	14.21	2.38	29.25	-5.24	-4.95
41	-5.33	1.96	2.51	14.89	2.30	29.91	-4.95	-5.64
42	-4.23	1.47	2.50	14.89	2.39	30.13	-5.49	-5.67
43	-5.61	1.39	2.40	14.13	2.39	30.04	-5.16	-5.43
44	-5.26	1.42	2.47	14.70	2.47	29.51	-5.46	-5.65
45	-4.85	1.38	2.49	14.53	2.35	31.15	-5.49	-5.35
46	-5.17	1.42	2.66	14.21	2.40	31.82	-5.15	-5.79
47	-5.68	1.57	2.53	15.07	2.40	29.24	-5.09	-5.72
48	-6.00	2.58	2.50	14.63	3.20	38.48	-3.62	-5.61
49	-6.67	2.35	2.72	14.89	3.23	37.63	-3.69	-6.09
50	-5.81	2.30	2.52	15.66	3.28	38.95	-4.26	-6.18
51	-5.13	2.44	2.67	15.54	3.18	36.41	-4.09	-5.84
52	-4.42	1.99	2.54	15.71	3.13	39.60	-4.01	-6.18
53	-5.04	2.06	2.66	17.16	3.12	36.76	-4.33	-6.06
54	-6.10	2.53	2.50	16.41	3.27	37.65	-4.36	-6.06
55	-5.57	2.26	2.60	16.96	3.16	40.02	-3.85	-6.00
56	-4.10	1.92	2.64	14.69	3.17	39.48	-4.48	-6.07
57	-4.38	2.29	2.61	15.62	3.22	38.06	-3.97	-6.03
58	-5.50	2.09	2.58	16.36	3.27	40.47	-3.28	-5.79
59	-4.45	2.46	2.78	16.60	3.15	39.11	-3.78	-6.16
60	-3.34	2.51	2.52	15.15	3.23	38.56	-4.02	-5.90
61	-5.04	2.20	2.58	16.64	3.16	38.05	-4.02	-5.69
62	-4.07	2.51	2.61	16.88	3.27	38.29	-3.92	-6.25
63	-5.47	2.46	2.56	17.19	3.20	36.88	-3.81	-5.98
64	-3.26	1.93	2.58	6.64	1.79	15.55	-0.68	-6.62
65	-1.96	1.80	2.75	6.53	1.80	14.47	-1.32	-6.52
66	-2.93	1.95	2.58	6.37	1.80	13.31	-1.55	-6.16
67	-2.63	1.77	2.78	6.77	1.79	14.86	-0.93	-6.19
68	-3.09	2.07	2.62	6.63	1.85	13.31	-0.92	-6.44
69	-2.86	1.95	2.66	6.79	1.74	13.74	-1.00	-6.61
70	-1.71	1.92	2.68	5.20	1.78	15.31	-1.48	-6.46
71	-3.02	1.93	2.72	6.27	1.83	13.72	-1.27	-6.62
72	-2.50	2.08	2.62	6.55	1.75	12.96	-1.31	-6.45

73	-2.41	2.11	2.70	6.32	1.82	15.33	-1.06	-6.63
74	-2.92	2.10	2.73	6.78	1.77	15.18	-1.17	-6.35
75	-2.50	1.93	2.62	6.95	1.85	15.54	-1.22	-6.55
76	-2.98	2.03	2.66	6.13	1.77	14.15	-1.40	-6.31
77	-2.17	1.67	2.73	6.52	1.86	12.99	-0.97	-6.39
78	-3.26	1.85	2.79	5.93	1.78	15.29	-1.24	-6.51
79	-2.63	2.13	2.65	6.26	1.82	14.10	-1.09	-6.49

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
40	-1.63	-17.33	-15.14	-16.17	-29.42	28.83	-1.30	-12.03
41	-1.62	-16.73	15.47	-15.91	-10.45	27.24	-4.18	-15.77
42	-1.76	-17.69	-4.99	-16.03	12.60	28.26	-1.18	19.16
43	-1.73	-16.97	12.09	-15.62	-11.85	26.94	-0.37	0.46
44	-1.61	-17.12	-5.19	-15.90	2.02	26.26	-0.46	-5.74
45	-1.70	-17.11	2.87	-15.47	-1.41	27.27	2.06	2.50
46	-1.71	-18.19	-5.68	-15.33	-1.19	27.62	1.22	2.53
47	-1.69	-17.28	-11.69	-15.44	-22.75	27.70	-1.62	1.81
48	-2.35	-23.98	-5.23	-19.65	22.91	35.28	-12.08	-13.87
49	-2.43	-27.18	5.93	-18.98	-27.08	38.11	-20.01	-4.01
50	-2.31	-25.96	-1.21	-17.47	-3.80	36.94	-16.15	0.74
51	-2.43	-25.54	-5.39	-18.91	2.69	37.51	-17.32	-10.40
52	-2.34	-25.63	-0.54	-16.74	-13.52	38.11	-12.87	1.63
53	-2.28	-26.61	18.39	-19.47	38.54	36.67	-15.14	-8.79
54	-2.24	-25.23	0.96	-18.66	-4.01	37.97	-12.54	3.16
55	-2.46	-26.20	3.85	-20.41	-20.02	38.86	-9.05	-6.67
56	-2.20	-26.94	-5.01	-19.81	20.93	36.49	-12.64	1.44
57	-2.24	-26.94	-1.68	-18.68	-1.71	36.78	-9.21	13.52
58	-2.27	-24.52	0.95	-18.90	-12.42	39.65	-21.43	-15.86
59	-2.33	-25.67	-10.70	-19.74	1.12	39.45	-18.24	7.89
60	-2.30	-26.36	11.17	-18.81	-2.69	39.28	-17.71	-4.19
61	-2.29	-24.66	3.78	-19.38	-19.97	38.21	-17.79	-5.47
62	-2.42	-24.80	11.78	-19.14	3.75	38.29	-17.86	-0.73
63	-2.44	-24.51	-21.78	-18.50	23.88	36.49	-15.04	-0.89
64	-1.83	-13.40	-11.14	-13.47	6.68	20.28	-11.60	8.74
65	-1.62	-15.70	1.68	-12.90	6.44	19.59	-9.48	-7.91
66	-1.80	-13.18	-15.05	-13.18	1.99	18.97	-9.93	-3.74
67	-1.63	-13.75	-7.33	-13.37	-23.16	18.82	-6.33	-1.89
68	-1.76	-13.83	8.46	-12.67	5.10	20.29	-9.76	5.80
69	-1.74	-13.90	1.52	-12.82	-22.21	18.64	-11.50	-0.64
70	-1.78	-14.92	1.24	-12.89	4.97	19.29	-7.71	-4.83
71	-1.90	-13.09	-9.94	-13.40	-18.68	19.39	-5.61	6.31
72	-1.81	-14.30	-9.16	-13.84	13.20	18.53	-6.02	-1.96
73	-1.76	-14.66	-2.27	-12.31	-4.24	18.90	-6.41	14.15
74	-1.79	-13.88	-6.26	-12.21	-0.23	19.15	-10.59	-0.15
75	-1.64	-13.15	2.92	-13.45	-9.89	19.10	-10.12	-5.30
76	-1.77	-13.07	4.78	-13.52	-1.98	19.49	-8.17	-0.11
77	-1.67	-13.32	0.73	-13.84	5.22	20.52	-11.56	-5.77
78	-1.77	-13.99	-4.65	-12.55	-8.03	18.97	-7.84	0.31
79	-1.70	-13.62	3.32	-13.01	5.30	20.11	-7.49	7.78

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
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40	10.47	21.27	-10.21	0.10	68.25	-3.70	-1.40	6.15
41	10.05	21.29	-10.73	-5.80	66.81	-3.54	-1.12	5.55
42	9.65	21.44	-10.41	3.16	64.95	-2.16	-1.26	5.51
43	10.09	20.98	-9.71	-1.86	66.84	-1.27	-1.51	5.68
44	10.76	21.46	-10.07	-3.57	67.32	-1.94	-1.25	6.11
45	12.36	21.09	-11.19	-6.00	66.15	-1.00	-1.45	5.81
46	9.57	21.06	-10.42	-7.61	64.58	-2.32	-1.27	4.69
47	10.48	21.27	-8.46	-5.23	67.39	-2.78	-1.50	5.91
48	19.85	26.07	-17.78	5.77	76.70	-5.97	-1.33	7.34
49	17.76	26.10	-18.03	-11.44	73.40	-5.48	-1.67	5.96
50	19.37	25.92	-17.34	-2.21	75.97	-4.95	-1.61	6.77
51	19.60	26.19	-15.93	1.24	77.29	-6.90	-0.94	6.74
52	20.65	25.47	-17.51	-0.40	74.81	-5.15	-1.53	6.92
53	20.51	24.90	-17.03	6.78	78.03	-4.57	-1.38	7.08
54	19.00	25.89	-16.67	-8.22	78.44	-7.80	-0.90	5.93
55	19.61	26.07	-16.10	-4.31	75.29	-5.28	-0.84	7.33
56	17.19	25.63	-17.16	-4.83	76.12	-6.28	-1.30	7.15
57	16.12	26.62	-18.85	-10.03	76.11	-7.80	-1.48	6.26
58	22.83	25.80	-18.76	0.78	79.84	-6.48	-1.62	6.51
59	18.77	25.66	-19.39	-3.45	75.39	-5.64	-1.89	5.89
60	18.65	26.05	-17.83	9.61	75.07	-4.95	-1.50	7.25
61	20.10	25.50	-17.72	5.46	79.24	-5.64	-0.75	6.04
62	18.06	26.18	-16.77	-0.60	75.60	-5.70	-1.29	5.48
63	20.15	26.14	-18.30	0.52	77.40	-5.65	-1.63	6.15
64	4.17	11.78	-11.14	-6.19	47.76	4.84	-0.10	1.22
65	4.95	11.91	-11.29	-7.40	49.57	5.36	-0.13	0.55
66	5.23	11.69	-11.24	2.25	47.65	3.93	-0.44	1.66
67	4.34	11.68	-12.65	-2.83	44.20	6.00	-0.18	1.83
68	4.77	11.73	-10.81	-7.30	48.16	4.80	0.07	1.50
69	5.63	12.02	-11.26	-5.67	46.34	4.37	-0.32	1.27
70	6.80	11.44	-12.79	4.52	46.72	4.09	-0.01	1.03
71	4.40	11.61	-10.90	3.28	46.91	5.31	-0.15	1.20
72	5.63	11.86	-9.84	-2.76	47.55	4.03	-0.09	1.07
73	5.57	11.65	-11.73	4.00	48.09	5.29	-0.14	1.55
74	6.56	12.02	-11.78	5.11	47.54	4.43	-0.34	1.50
75	4.76	11.61	-12.34	-1.54	46.82	4.11	-0.51	1.09
76	3.83	11.41	-11.62	5.85	45.35	5.55	-0.03	1.42
77	5.23	11.93	-10.66	-2.98	46.92	4.34	-0.39	1.13
78	6.95	12.23	-11.66	2.36	44.43	5.72	-0.37	1.19
79	5.55	11.92	-10.93	2.02	45.36	5.01	-0.28	1.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
40	-13.21	46.06	0.12	12.76	40.97	2.21	-5.81	30.19
41	5.60	43.87	0.12	12.55	40.42	2.20	-5.66	10.59
42	1.30	43.78	0.08	13.33	42.36	2.34	104.95	11.44
43	-7.10	45.74	0.12	12.70	41.80	2.17	53.45	1.42
44	-15.01	43.41	0.08	13.65	40.16	2.34	7.38	7.04
45	-4.50	44.35	0.10	13.38	42.10	2.24	-60.08	29.55
46	-6.08	43.93	0.11	12.88	41.53	2.30	43.67	-37.28
47	-1.42	43.77	0.10	12.83	42.02	2.03	-33.22	-1.93
48	7.40	55.67	0.34	15.31	49.80	3.58	-32.69	-7.42
49	0.03	55.98	0.35	15.00	48.94	3.07	-12.52	-12.64
50	13.40	56.33	0.38	16.14	50.79	2.89	79.93	-57.05

51	-14.93	53.37	0.35	15.22	51.09	2.85	70.33	27.06
52	2.81	54.09	0.35	15.14	49.42	3.15	3.52	-55.33
53	-6.01	56.32	0.35	14.90	50.48	2.85	-83.24	-53.13
54	-22.63	56.03	0.32	15.27	49.90	2.96	131.84	-18.47
55	-12.38	57.33	0.34	14.90	48.67	2.74	61.03	8.29
56	15.92	55.65	0.35	16.30	50.40	2.95	-128.30	-29.08
57	5.86	54.69	0.34	14.83	50.68	3.04	60.23	14.95
58	6.32	56.35	0.37	16.09	50.08	3.27	-61.65	-31.02
59	-4.18	54.81	0.39	15.12	49.89	2.74	48.95	-25.32
60	-1.82	55.12	0.32	16.16	49.81	2.53	23.54	38.64
61	-6.60	56.15	0.31	14.82	50.61	3.48	-15.81	-62.06
62	13.52	55.60	0.36	14.65	50.81	2.75	-55.36	2.84
63	6.26	55.11	0.35	16.22	50.09	2.78	-38.96	-9.46
64	-6.41	21.29	0.06	13.13	25.53	2.19	3.75	28.78
65	5.40	20.54	0.03	12.43	26.21	2.26	-53.85	-56.62
66	-1.36	21.26	0.08	12.95	24.86	1.82	19.76	-12.86
67	2.78	19.53	0.06	13.77	25.62	1.69	-28.92	2.03
68	-2.41	19.89	0.07	12.87	24.40	2.26	4.00	8.27
69	-0.42	20.76	0.08	13.75	25.35	2.40	-7.41	-16.89
70	2.68	21.11	0.06	12.80	26.02	1.63	-34.09	-1.22
71	-2.07	20.29	0.06	13.23	24.64	2.14	-0.73	27.99
72	13.77	19.43	0.03	12.95	25.40	2.05	-41.90	41.49
73	6.38	21.39	0.07	12.67	25.45	2.27	64.46	10.51
74	14.28	20.99	0.05	12.33	26.63	1.89	72.91	18.93
75	11.50	20.47	0.07	13.27	25.94	1.95	95.35	4.58
76	-2.82	20.91	0.06	12.71	25.71	2.04	58.62	12.36
77	-11.93	21.90	0.02	13.81	25.86	2.21	-10.84	12.70
78	-11.93	20.49	0.07	13.10	25.82	2.10	20.80	15.62
79	-4.90	21.84	0.05	13.09	25.78	1.86	-24.02	21.73

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
80	237.62	-19.42	-15.00	22.16	76.44	7.36	55.13	20.11
81	245.81	-19.54	-15.13	18.05	77.32	7.05	55.20	19.51
82	239.03	-19.32	-14.10	14.18	76.56	7.65	54.42	21.17
83	239.85	-19.51	-14.78	16.79	75.94	7.74	53.67	19.67
84	236.54	-19.33	-14.34	17.57	76.25	5.88	54.22	20.28
85	233.79	-19.10	-14.73	18.82	77.02	7.74	55.42	19.18
86	244.29	-19.38	-14.24	21.06	76.84	7.05	53.39	21.57
87	245.46	-19.23	-14.63	19.52	76.89	6.20	55.06	19.29
88	237.27	-19.52	-15.00	11.51	75.62	6.84	54.76	18.98
89	246.30	-19.38	-14.77	19.82	76.40	7.87	53.51	19.31
90	239.99	-19.66	-14.51	23.01	78.38	7.62	54.41	20.58
91	246.68	-19.32	-14.48	19.30	76.38	6.57	54.02	19.59
92	244.90	-18.74	-14.76	19.33	75.50	8.02	53.86	19.84
93	240.25	-19.46	-14.08	15.51	77.25	6.68	54.41	20.96
94	236.25	-19.15	-14.29	15.40	77.92	8.29	54.72	20.92
95	234.53	-19.51	-14.79	15.95	76.96	7.71	53.95	19.60
96	225.80	-14.96	-12.69	32.62	79.10	-8.15	47.15	29.30
97	224.72	-15.06	-12.31	38.11	78.67	-8.34	47.15	28.12
98	219.05	-15.08	-12.67	33.06	78.99	-8.10	46.83	29.35
99	225.28	-15.07	-12.13	34.86	78.98	-7.64	46.64	29.02
100	219.18	-15.13	-12.51	32.24	79.59	-8.61	46.55	28.06
101	223.01	-14.98	-12.59	32.56	78.79	-7.85	47.48	29.74

102	226.92	-15.06	-12.82	35.50	78.13	-8.46	47.53	29.39
103	218.47	-15.15	-12.70	31.58	77.22	-8.28	47.09	28.69
104	217.40	-15.08	-12.47	31.83	77.80	-8.14	47.10	29.56
105	211.59	-15.13	-12.39	34.55	78.57	-8.92	46.36	27.44
106	218.37	-14.92	-13.15	31.41	78.23	-8.59	47.33	28.58
107	213.20	-15.28	-12.69	32.20	78.63	-8.15	47.25	28.56
108	211.17	-15.16	-12.35	33.61	78.13	-8.64	46.45	29.12
109	225.61	-15.22	-12.14	30.36	76.95	-7.72	47.83	29.32
110	225.64	-15.32	-12.32	30.59	79.54	-8.44	47.85	29.54
111	224.91	-15.05	-12.61	33.73	78.88	-8.69	47.23	29.07
112	-89.51	-0.54	3.33	-33.39	1.42	11.15	2.84	-3.79
113	-86.30	-0.64	3.48	-32.33	1.63	11.26	2.88	-3.91
114	-80.17	-0.41	3.36	-33.65	1.40	11.45	2.87	-2.39
115	-86.94	-0.54	3.66	-33.07	1.89	10.91	3.66	-2.75
116	-80.07	-0.67	3.73	-32.76	1.05	11.32	2.95	-2.93
117	-87.45	-0.91	3.55	-31.94	1.72	11.40	3.03	-3.38
118	-80.27	-0.56	3.49	-31.59	1.38	11.19	3.53	-3.49
119	-90.36	-0.70	3.58	-32.58	1.11	10.70	3.54	-3.50

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
80	-186.45	-104.53	-2.49	-80.09	-134.45	10.24	-14.72	95.88
81	-192.97	-107.51	-2.51	-80.49	-133.61	10.69	-14.16	94.57
82	-187.71	-108.95	-2.46	-79.35	-133.97	10.30	-14.36	97.20
83	-196.02	-105.38	-2.44	-78.86	-135.96	9.68	-14.04	92.95
84	-194.61	-105.27	-2.45	-79.94	-133.74	10.20	-14.61	95.05
85	-198.68	-106.99	-2.42	-81.39	-132.72	10.63	-14.47	94.83
86	-194.46	-107.19	-2.42	-82.50	-134.72	10.43	-14.56	95.38
87	-200.13	-106.43	-2.49	-78.22	-134.53	10.86	-14.80	93.82
88	-193.98	-106.70	-2.47	-81.14	-133.25	10.47	-14.52	93.65
89	-196.03	-107.53	-2.45	-80.79	-133.78	9.95	-14.23	95.79
90	-197.87	-106.99	-2.47	-79.13	-135.12	10.49	-14.67	92.70
91	-196.90	-110.92	-2.43	-82.52	-133.55	10.59	-14.29	95.36
92	-197.47	-106.85	-2.47	-82.24	-133.42	10.28	-14.49	95.94
93	-195.18	-104.82	-2.46	-82.04	-134.55	10.47	-14.71	94.09
94	-195.83	-106.77	-2.42	-79.90	-131.30	10.06	-14.41	92.75
95	-195.81	-104.92	-2.43	-82.97	-133.36	10.49	-14.20	92.97
96	-181.47	-115.06	-2.02	-70.47	-131.47	11.38	-17.41	86.99
97	-188.01	-114.36	-2.00	-72.63	-131.68	10.99	-17.24	89.31
98	-180.37	-115.61	-1.98	-70.70	-131.42	10.72	-17.68	87.15
99	-180.37	-114.48	-2.02	-67.98	-131.22	10.86	-17.79	89.21
100	-182.21	-111.90	-2.00	-70.02	-132.86	10.41	-17.62	92.41
101	-178.91	-117.06	-2.03	-71.92	-129.11	11.31	-17.42	90.89
102	-181.48	-113.66	-2.01	-69.24	-130.05	11.09	-17.57	94.75
103	-180.28	-117.23	-2.03	-71.54	-128.29	10.24	-17.24	92.37
104	-185.23	-115.73	-2.01	-69.82	-129.67	11.10	-17.39	87.03
105	-182.95	-111.77	-2.03	-71.93	-130.92	11.05	-17.50	90.66
106	-180.42	-115.91	-2.02	-71.93	-132.11	10.58	-17.60	89.10
107	-184.49	-115.82	-2.04	-70.93	-133.27	10.63	-17.39	92.69
108	-182.04	-114.89	-2.02	-73.68	-130.74	10.27	-17.30	88.48
109	-187.39	-116.94	-2.00	-70.45	-133.71	10.58	-17.37	88.44
110	-183.45	-115.03	-2.00	-72.48	-131.80	10.55	-17.51	87.44
111	-178.97	-116.76	-2.03	-70.39	-131.76	11.18	-17.83	90.02
112	53.55	15.02	-0.01	26.67	-5.88	-9.21	0.67	-27.37

113	55.15	13.52	-0.06	23.88	-7.50	-9.22	0.32	-26.84
114	55.34	13.76	-0.05	25.76	-6.52	-9.00	0.45	-23.41
115	56.13	12.27	-0.04	26.72	-7.43	-8.65	0.23	-25.40
116	58.20	12.54	0.00	27.33	-8.21	-9.11	0.39	-25.47
117	53.40	11.55	-0.04	24.89	-6.52	-9.32	0.47	-26.03
118	57.18	15.39	-0.04	26.82	-6.54	-9.12	0.17	-26.77
119	53.75	13.14	-0.04	25.75	-5.81	-9.26	0.44	-26.64

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
80	1.25	31.48	36.45	5.03	-4.19	-13.70	34.65	3.63
81	1.58	31.20	35.32	3.71	-4.06	-13.75	-25.86	3.60
82	0.39	32.65	36.13	-66.31	-4.02	-13.44	2.89	3.60
83	1.13	32.53	35.72	9.35	-4.30	-13.69	-12.59	3.59
84	1.32	32.05	35.43	77.97	-4.23	-13.75	-11.80	3.64
85	0.40	32.67	35.83	10.64	-4.27	-13.54	-13.35	3.59
86	1.24	31.25	36.79	-0.20	-4.24	-13.69	-7.93	3.61
87	0.63	32.68	36.03	-9.71	-3.66	-13.65	28.19	3.65
88	1.29	31.34	35.88	45.80	-4.03	-13.27	-1.63	3.60
89	-1.15	32.21	35.54	38.92	-4.00	-13.62	-23.70	3.68
90	-0.43	31.99	35.51	-10.68	-4.34	-13.83	-29.95	3.70
91	0.06	31.39	37.08	16.14	-3.99	-13.78	-12.77	3.64
92	-2.56	32.15	36.26	-18.55	-3.82	-13.57	-2.25	3.67
93	-1.69	31.85	35.62	-46.41	-3.96	-13.64	-21.34	3.64
94	0.62	32.69	35.98	-64.99	-4.18	-13.56	-7.76	3.68
95	2.32	32.37	36.68	2.72	-3.94	-13.79	-10.48	3.62
96	1.91	27.17	35.46	31.42	-6.83	-14.35	-4.68	3.04
97	-0.84	28.04	36.09	66.63	-6.64	-14.40	-26.19	3.11
98	-0.22	28.28	35.61	-27.73	-6.88	-14.78	-1.46	3.08
99	0.40	28.54	36.22	-57.05	-6.45	-14.68	10.02	3.12
100	-1.97	28.51	35.25	-1.24	-6.95	-14.54	-11.66	3.11
101	-0.27	28.64	35.64	-32.40	-6.88	-14.72	-4.98	3.09
102	-0.47	27.32	35.97	27.27	-6.83	-14.57	10.59	3.08
103	0.44	28.14	35.71	-30.29	-7.15	-14.77	-9.52	3.05
104	1.54	28.04	35.70	17.22	-7.02	-14.61	-8.51	3.08
105	1.01	27.15	36.02	15.69	-6.67	-14.79	-14.83	3.10
106	-0.91	27.98	35.25	12.21	-6.68	-14.71	-3.81	3.04
107	0.73	26.90	35.44	-45.54	-7.21	-14.30	-8.87	3.12
108	0.21	28.51	35.12	-4.38	-6.82	-14.75	-32.65	3.15
109	1.95	28.09	35.19	-12.43	-7.11	-14.39	17.18	3.04
110	-0.93	28.33	36.14	-55.64	-6.91	-14.73	9.19	3.13
111	1.20	28.73	35.63	-23.48	-6.74	-14.51	0.11	3.05
112	-0.02	-4.26	4.39	-21.84	-1.95	-1.90	4.25	0.50
113	-0.19	-5.10	4.80	-4.81	-1.96	-1.52	0.62	0.54
114	-0.55	-5.07	4.62	20.67	-2.08	-1.80	1.84	0.56
115	-0.12	-4.51	4.51	41.62	-1.85	-1.93	2.57	0.56
116	-0.88	-3.89	4.43	-8.71	-1.84	-1.74	-0.13	0.55
117	-0.21	-4.95	4.50	-8.92	-1.94	-1.54	-3.25	0.54
118	-0.11	-4.62	4.46	15.90	-1.87	-1.91	1.65	0.54
119	-0.29	-4.89	4.58	-6.43	-1.68	-1.77	-5.02	0.52

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
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80	-0.72	13.41	156.23	-22.05	54.69	-21.12	-36.70	-16.35
81	1.23	12.76	154.12	-21.44	53.34	-21.60	-37.59	-15.05
82	1.64	13.48	158.87	-21.61	53.88	-20.64	-36.38	-19.94
83	-0.66	13.19	153.77	-22.15	54.43	-20.25	-34.46	-17.11
84	0.56	13.61	156.37	-21.57	54.48	-19.82	-35.80	-16.25
85	-0.14	13.52	158.07	-21.30	54.41	-20.91	-34.52	-12.68
86	0.41	13.54	156.75	-21.79	53.12	-20.37	-36.33	-13.44
87	-1.00	12.92	156.99	-21.66	53.95	-19.99	-35.01	-12.89
88	-0.16	14.20	155.55	-21.40	54.41	-20.89	-34.07	-15.57
89	-1.85	12.45	156.90	-21.26	53.80	-21.84	-36.75	-17.41
90	0.51	13.56	157.68	-21.81	54.28	-20.43	-34.55	-16.46
91	2.46	13.10	155.86	-22.04	54.75	-20.86	-34.88	-13.51
92	-1.96	13.68	155.14	-22.09	53.98	-20.80	-37.23	-20.04
93	4.18	13.40	158.36	-21.82	54.19	-20.46	-36.36	-16.43
94	3.18	13.43	156.65	-22.12	53.60	-20.05	-35.90	-16.26
95	-1.71	13.55	155.07	-22.08	53.90	-20.33	-36.81	-16.48
96	-1.87	11.55	132.79	-19.54	51.83	-3.17	-27.20	2.49
97	1.90	11.74	132.36	-19.78	51.36	-2.55	-26.79	-1.40
98	2.83	11.98	133.42	-19.47	52.29	-2.64	-26.69	2.37
99	-2.49	11.82	130.84	-19.85	52.62	-2.99	-27.51	0.39
100	-2.51	12.23	132.00	-19.92	52.38	-2.79	-27.31	1.30
101	-0.33	11.25	135.64	-19.17	52.55	-3.21	-28.44	0.79
102	-0.66	11.59	132.50	-19.63	51.89	-2.65	-28.62	0.35
103	4.76	11.46	133.48	-19.67	51.71	-3.21	-25.00	0.87
104	3.28	11.84	132.09	-19.71	52.27	-3.36	-27.45	2.13
105	0.21	11.47	129.46	-19.72	51.54	-2.97	-27.22	-2.11
106	1.65	12.36	134.76	-19.93	52.29	-2.91	-28.74	3.66
107	5.53	11.96	132.76	-19.63	52.03	-3.21	-29.41	-1.15
108	-3.77	11.16	131.28	-19.55	51.37	-2.87	-27.01	1.87
109	-3.12	11.83	134.62	-19.65	52.18	-2.48	-26.46	0.83
110	2.85	12.08	133.21	-19.44	52.47	-3.09	-27.62	-1.18
111	0.07	11.77	131.28	-19.90	52.21	-2.69	-27.81	2.83
112	2.49	-2.66	30.15	-1.56	4.27	-0.80	-2.30	-49.05
113	-0.21	-3.10	31.04	-1.60	4.31	-0.43	-2.86	-49.25
114	1.94	-2.82	31.03	-1.86	4.12	-1.00	-2.97	-50.39
115	1.08	-2.63	31.21	-1.55	3.77	-0.67	-3.06	-50.02
116	-0.40	-2.46	31.05	-1.70	4.43	-0.82	-2.46	-50.67
117	-0.06	-2.74	31.50	-1.52	4.41	-0.77	-2.77	-50.13
118	-0.80	-2.66	31.91	-1.62	3.94	-0.64	-2.97	-49.29
119	-0.25	-2.75	32.83	-1.54	3.88	-0.49	-2.99	-49.55

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
80	-95.69	0.82	-1.10	46.86	-0.54	130.43	-0.31	-6.35
81	-95.93	0.92	-0.98	46.06	-0.58	127.11	0.13	-6.43
82	-96.16	0.86	-1.01	45.27	-0.54	127.80	-0.38	-6.37
83	-95.92	0.73	-0.94	45.00	-0.54	128.49	0.19	-6.34
84	-95.27	0.81	-1.04	44.91	-0.64	126.65	-0.16	-6.39
85	-92.81	0.66	-0.86	47.10	-0.53	130.24	0.06	-6.42
86	-96.29	0.72	-0.91	45.81	-0.46	128.86	-0.40	-6.39
87	-96.00	1.01	-0.93	46.00	-0.58	128.57	0.25	-6.31
88	-94.99	0.62	-0.94	45.13	-0.55	128.39	0.16	-6.34
89	-96.60	0.78	-0.89	45.47	-0.52	126.84	-0.11	-6.40
90	-94.71	1.23	-1.19	45.58	-0.62	128.54	-0.77	-6.34

91	-94.84	0.65	-1.13	45.76	-0.67	128.77	-0.04	-6.34
92	-98.94	0.90	-1.10	46.93	-0.58	126.04	0.20	-6.35
93	-95.55	0.79	-1.08	45.13	-0.51	126.68	0.11	-6.38
94	-95.94	0.89	-0.91	46.23	-0.61	127.94	-0.21	-6.32
95	-95.34	1.31	-1.05	44.16	-0.61	129.03	-0.25	-6.34
96	-107.90	0.76	0.00	43.98	-0.72	132.70	4.65	-5.94
97	-107.59	0.99	-0.20	44.15	-0.85	132.48	4.73	-5.90
98	-107.87	0.85	-0.02	44.18	-0.77	134.56	4.47	-5.93
99	-107.13	0.84	-0.11	44.50	-0.88	132.73	5.06	-5.89
100	-107.14	0.81	-0.29	43.22	-0.80	132.40	4.81	-5.89
101	-107.45	1.18	-0.09	44.08	-0.83	129.23	4.52	-5.89
102	-107.72	0.94	-0.04	43.68	-0.76	128.11	4.76	-5.93
103	-108.87	0.85	-0.20	44.01	-0.84	131.58	4.83	-5.90
104	-106.70	0.96	-0.25	42.66	-0.82	129.83	5.01	-5.99
105	-106.40	0.98	-0.08	43.16	-0.85	130.13	4.95	-5.86
106	-106.63	1.09	-0.06	44.91	-0.82	129.33	4.92	-5.93
107	-107.55	1.00	-0.13	43.78	-0.79	131.18	4.41	-6.02
108	-105.69	1.12	-0.27	44.38	-0.87	131.80	5.34	-5.86
109	-105.87	1.07	0.08	43.11	-0.79	133.55	4.85	-5.86
110	-107.71	0.90	-0.20	43.99	-0.87	133.70	4.92	-6.02
111	-106.95	0.78	-0.10	44.61	-0.76	130.77	4.70	-5.93
112	-4.44	1.70	1.70	0.99	-0.98	-0.86	2.56	-2.05
113	-4.14	1.83	1.66	1.66	-0.97	-1.69	2.43	-2.09
114	-4.10	1.81	1.72	0.63	-0.94	-2.98	2.44	-1.99
115	-4.14	1.83	1.69	0.96	-0.99	-3.20	2.68	-2.10
116	-4.28	1.80	1.62	0.80	-0.98	-3.96	2.41	-2.02
117	-3.84	1.93	1.78	0.41	-1.00	-4.36	2.44	-2.03
118	-4.50	1.70	1.76	0.90	-1.00	-2.13	2.54	-2.05
119	-4.48	1.88	1.66	0.14	-0.95	-2.51	2.49	-2.04

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
80	3.40	19.84	3.52	-78.11	-37.18	-14.13	-71.01	6.01
81	3.35	20.61	-0.60	-78.24	-9.91	-13.29	-75.61	-1.14
82	3.41	20.64	0.82	-76.73	19.98	-13.42	-70.88	-6.79
83	3.46	20.20	-0.67	-78.09	-14.48	-13.16	-78.82	0.37
84	3.44	21.50	0.15	-78.53	-27.77	-12.11	-74.33	-4.33
85	3.42	20.07	-1.55	-77.34	-15.72	-13.50	-71.87	-5.76
86	3.47	19.94	-3.51	-80.03	9.79	-13.72	-76.54	-2.25
87	3.49	20.55	6.58	-77.18	14.38	-13.78	-73.69	-8.28
88	3.47	20.61	10.20	-77.82	-7.27	-12.83	-76.26	-5.58
89	3.41	20.55	2.07	-76.65	5.56	-12.96	-75.26	-0.05
90	3.43	20.12	-1.73	-76.33	-3.90	-13.35	-72.26	5.49
91	3.44	20.60	0.77	-77.71	-11.89	-13.52	-68.06	-2.05
92	3.46	20.20	3.82	-77.63	-0.28	-12.97	-79.25	-10.64
93	3.43	20.64	1.29	-77.80	-18.28	-12.80	-72.32	0.38
94	3.44	20.07	0.60	-76.68	15.97	-13.62	-76.04	6.19
95	3.41	20.74	-1.12	-77.81	2.76	-13.11	-72.34	-2.04
96	3.65	25.16	2.76	-72.70	-2.57	-15.00	-62.60	-2.69
97	3.65	25.10	3.51	-72.06	-1.31	-16.34	-66.97	5.31
98	3.66	25.35	0.27	-73.24	8.88	-15.21	-61.74	2.10
99	3.62	25.43	-5.10	-73.18	-11.54	-15.52	-62.32	1.04
100	3.61	25.28	0.40	-72.53	-1.10	-15.47	-70.69	-4.99
101	3.62	24.49	-3.88	-73.96	-0.97	-15.72	-68.40	2.03

102	3.58	25.52	2.92	-73.28	4.49	-15.85	-61.30	-5.42
103	3.58	25.33	1.26	-73.93	-13.86	-15.93	-64.63	-4.88
104	3.65	25.41	3.75	-73.70	4.01	-15.94	-63.18	5.70
105	3.65	25.65	-0.10	-71.82	-4.99	-15.83	-71.39	-2.33
106	3.63	24.95	1.65	-72.28	-7.68	-15.71	-61.19	0.91
107	3.64	25.07	0.78	-73.83	-7.90	-15.93	-61.00	-4.62
108	3.65	24.83	-0.31	-74.01	5.44	-15.62	-62.16	0.71
109	3.62	24.79	1.58	-72.87	-16.26	-15.41	-54.83	-3.32
110	3.57	24.99	4.36	-73.64	3.01	-16.03	-57.92	2.51
111	3.63	24.88	2.53	-72.63	-2.20	-15.87	-69.05	2.72
112	0.31	4.79	-0.41	-1.00	4.16	-7.16	62.77	-1.29
113	0.36	4.94	-1.85	-0.77	-0.50	-6.81	58.70	2.31
114	0.36	5.23	1.31	-1.09	-3.95	-6.47	60.91	-2.13
115	0.37	5.08	-0.23	-1.37	-5.48	-6.74	63.68	2.69
116	0.37	4.81	-0.94	-1.59	-1.18	-7.42	64.90	2.51
117	0.32	5.26	-0.40	-1.36	7.83	-6.88	69.96	1.83
118	0.34	4.88	-1.21	-1.55	-8.41	-6.50	60.24	2.64
119	0.30	5.04	-0.69	-1.16	-8.28	-6.90	62.01	-0.35

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
80	35.18	-18.92	-15.82	-4.04	123.00	-19.42	-1.04	-51.71
81	34.09	-18.91	-16.07	6.61	118.59	-17.87	-1.15	-52.53
82	36.45	-18.80	-17.25	2.41	120.53	-15.35	-0.15	-51.71
83	35.43	-19.22	-15.75	-0.26	121.11	-14.13	-1.33	-51.60
84	35.61	-19.31	-15.80	3.78	122.22	-13.50	-0.46	-51.12
85	35.84	-19.29	-15.60	-3.07	113.52	-19.69	-1.38	-51.49
86	34.84	-19.11	-15.75	6.88	119.39	-19.36	-1.24	-51.76
87	31.00	-18.93	-16.37	2.25	118.69	-14.80	-0.99	-52.10
88	34.90	-19.56	-16.97	-5.46	121.17	-18.96	-1.06	-52.00
89	34.10	-19.04	-15.91	1.89	118.31	-15.26	-0.99	-51.50
90	34.94	-18.97	-16.30	5.87	119.97	-19.67	-1.19	-52.26
91	29.87	-19.33	-16.96	4.69	119.99	-15.82	-1.76	-51.64
92	34.26	-18.90	-16.35	-0.85	119.09	-13.64	-1.06	-51.17
93	35.63	-18.87	-16.09	-2.07	123.06	-16.23	-0.80	-51.82
94	37.07	-19.29	-16.14	11.96	121.62	-15.06	-1.21	-51.56
95	36.78	-19.43	-16.06	-2.11	119.46	-17.76	-1.23	-51.70
96	21.23	-22.94	-15.56	-2.98	85.63	-20.85	2.54	-46.27
97	24.91	-22.30	-14.76	2.60	81.48	-21.58	2.96	-44.93
98	21.70	-22.33	-14.50	-3.51	88.57	-21.05	1.94	-46.62
99	22.56	-21.99	-14.26	-10.16	85.86	-20.91	2.38	-45.87
100	22.17	-22.94	-14.66	8.34	86.53	-21.30	2.37	-46.47
101	21.08	-23.03	-14.09	-0.27	87.85	-22.21	2.51	-46.73
102	19.67	-22.16	-14.80	3.10	85.87	-21.61	2.82	-47.10
103	22.49	-22.60	-14.67	3.54	87.94	-21.15	2.48	-46.57
104	22.32	-22.83	-15.55	-3.03	89.65	-19.85	2.31	-46.25
105	22.19	-22.25	-14.09	5.15	84.82	-18.96	2.62	-47.25
106	21.33	-22.63	-15.21	-6.48	86.04	-20.04	2.40	-47.84
107	21.61	-22.56	-13.73	2.23	87.92	-19.20	2.10	-46.74
108	19.32	-22.17	-15.69	-1.33	86.07	-18.90	2.44	-46.14
109	21.27	-22.66	-13.80	0.11	86.80	-20.63	2.40	-46.11
110	21.06	-22.97	-14.68	-2.90	86.30	-22.66	2.28	-46.75
111	22.19	-22.73	-15.73	-5.40	87.87	-19.19	2.68	-46.05
112	2.40	-6.68	6.28	-2.72	70.14	32.73	-4.02	-3.00

113	1.66	-6.52	6.68	-1.88	71.24	32.05	-4.00	-1.76
114	2.74	-6.41	6.80	-2.46	72.66	32.14	-4.04	-3.13
115	2.97	-6.78	6.95	-1.73	71.59	31.97	-3.94	-3.26
116	1.87	-6.57	6.62	-2.56	71.55	32.38	-3.96	-3.04
117	2.38	-6.59	7.29	0.80	73.77	32.78	-4.12	-2.99
118	2.23	-6.52	7.40	4.06	73.25	32.26	-4.00	-2.40
119	2.04	-6.41	6.41	-1.61	69.82	31.86	-3.80	-3.61

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
80	7.22	128.52	-1.88	43.51	39.31	15.41	-14.23	23.79
81	16.34	128.59	-1.91	43.36	39.03	15.48	-28.20	5.78
82	-6.02	129.83	-1.88	43.69	37.71	15.66	-12.79	-13.12
83	0.26	129.91	-1.89	43.46	38.12	15.62	39.21	11.40
84	-13.98	128.38	-1.87	43.41	37.06	15.59	13.00	-4.79
85	-20.46	130.06	-1.87	43.02	38.18	15.67	-23.98	-72.21
86	-2.59	127.98	-1.90	43.07	37.21	15.58	-30.51	7.52
87	11.87	128.01	-1.88	43.97	37.46	15.56	-27.53	39.33
88	-8.42	128.67	-1.91	43.50	40.10	15.60	-39.33	-13.06
89	-19.59	130.39	-1.86	43.48	38.13	15.66	-15.58	38.94
90	4.00	129.94	-1.85	43.36	37.51	15.62	-16.66	20.91
91	-15.39	128.60	-1.89	43.80	38.56	15.60	-8.93	13.06
92	13.67	129.15	-1.87	42.93	37.77	15.50	-15.23	5.92
93	3.75	129.33	-1.92	43.30	37.36	15.57	3.32	-17.65
94	-15.76	130.12	-1.87	43.29	37.20	15.39	-11.95	-15.64
95	-20.12	129.29	-1.85	43.60	40.08	15.90	-60.06	-10.83
96	-17.10	114.33	-2.02	39.62	9.75	12.73	52.83	-29.46
97	0.00	113.68	-2.03	39.64	9.45	12.67	-31.55	15.89
98	-5.74	115.79	-2.01	39.91	10.43	13.02	45.48	-21.84
99	1.37	115.20	-2.04	40.55	7.81	12.62	7.80	-103.58
100	4.74	115.96	-2.03	40.13	11.11	12.59	40.56	-60.33
101	9.00	115.12	-2.03	40.05	9.51	12.81	8.19	18.79
102	15.33	113.84	-2.02	39.69	9.57	12.73	29.45	79.46
103	9.99	114.81	-2.04	40.46	9.24	12.55	-24.76	-12.46
104	2.29	114.70	-2.04	40.01	10.31	12.96	8.96	24.34
105	-3.84	114.69	-2.03	39.86	9.44	12.77	-20.44	-33.68
106	-16.27	115.65	-2.04	40.05	9.97	12.69	31.92	-4.97
107	-8.85	114.18	-2.02	40.49	9.01	12.48	6.07	-18.59
108	13.66	113.88	-2.07	39.23	11.45	12.60	38.66	-56.12
109	-8.74	114.98	-2.03	39.73	10.32	12.72	-4.35	-32.95
110	7.85	116.66	-2.01	40.22	9.71	12.66	-0.79	-36.94
111	3.91	115.17	-1.99	39.78	9.93	12.55	9.97	15.23
112	1.14	13.18	-0.14	2.79	24.72	0.07	-16.74	8.49
113	4.25	13.58	-0.14	2.10	24.18	0.17	17.27	13.28
114	-3.59	13.49	-0.16	2.96	24.03	0.26	6.52	41.70
115	5.11	14.28	-0.14	2.68	24.04	0.36	10.37	38.29
116	6.99	13.92	-0.14	2.68	24.65	0.11	10.23	-3.54
117	5.35	14.04	-0.16	2.83	24.69	0.32	-1.11	7.77
118	-14.75	13.75	-0.12	2.85	24.55	0.22	-16.30	0.78
119	7.24	13.73	-0.13	2.73	24.81	0.13	-1.81	10.69

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
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120	-86.29	-0.63	3.88	-32.02	1.84	11.04	2.67	-3.79
121	-85.57	-0.73	3.64	-32.07	1.65	11.44	2.57	-2.83
122	-80.08	-0.26	3.98	-32.67	1.94	10.85	2.31	-3.57
123	-89.15	-0.56	3.69	-32.93	1.34	11.29	2.70	-3.13
124	-93.09	-0.87	3.69	-33.69	1.66	11.05	2.62	-2.62
125	-79.10	-0.35	3.46	-34.24	1.35	11.14	3.71	-2.73
126	-79.96	-0.48	3.30	-34.24	1.28	11.20	2.45	-3.40
127	-88.22	-0.65	3.53	-36.03	1.47	11.34	3.62	-3.03
128	-64.37	-4.04	1.18	-12.23	6.38	4.37	5.12	-0.76
129	-69.64	-4.14	1.10	-13.64	6.02	4.69	5.44	-0.43
130	-63.61	-4.12	0.95	-10.49	6.13	4.01	4.92	-0.32
131	-66.53	-4.06	0.94	-13.49	5.92	4.79	5.35	-0.99
132	-70.96	-4.00	1.10	-13.01	6.39	3.96	5.16	-0.30
133	-67.75	-4.00	1.23	-15.14	6.04	3.95	5.44	-0.43
134	-64.48	-4.20	0.82	-13.91	5.82	4.06	5.23	-0.18
135	-61.97	-4.07	0.68	-13.86	5.92	4.81	5.69	-0.76
136	-68.31	-4.09	0.82	-13.67	5.52	4.21	5.11	-0.62
137	-57.85	-4.15	1.00	-14.37	5.78	4.21	5.63	-1.43
138	-68.14	-4.06	0.87	-13.57	5.80	4.44	5.36	-0.51
139	-71.48	-4.31	1.21	-16.35	6.10	4.03	5.38	-0.61
140	-65.76	-4.29	0.82	-14.07	6.50	4.40	5.54	-0.17
141	-61.81	-4.12	1.10	-12.99	5.91	4.76	4.79	-0.75
142	-66.55	-4.48	1.08	-12.19	5.67	4.64	5.21	-0.94
143	-63.27	-4.03	1.02	-14.10	5.86	4.08	6.23	-1.19
144	-38.06	-4.80	-0.40	-7.99	5.14	1.07	4.09	-2.19
145	-34.73	-5.10	-0.37	-9.37	4.70	0.55	3.89	-1.54
146	-36.22	-4.93	-0.66	-8.77	4.53	0.87	3.70	-1.89
147	-38.97	-4.91	-0.26	-8.19	4.97	0.85	4.47	-1.42
148	-39.77	-4.98	-0.42	-9.31	5.15	0.76	4.33	-2.09
149	-40.83	-4.93	-0.29	-7.86	4.61	0.53	4.00	-1.78
150	-42.55	-4.97	-0.37	-8.84	4.88	1.17	4.23	-1.88
151	-40.43	-4.83	-0.39	-10.02	4.55	0.64	3.83	-2.11
152	-35.57	-4.98	-0.28	-9.08	4.79	1.04	4.22	-1.39
153	-36.15	-4.86	-0.47	-10.73	4.58	1.24	3.96	-1.78
154	-40.29	-4.90	-0.50	-9.89	4.86	1.22	3.86	-2.27
155	-36.78	-5.15	-0.39	-9.03	5.19	0.90	4.12	-1.66
156	-39.53	-5.04	-0.58	-9.80	5.07	0.85	4.06	-1.30
157	-42.15	-4.89	-0.60	-9.48	4.90	0.85	3.62	-2.15
158	-38.22	-5.10	-0.56	-7.54	4.48	0.71	4.17	-1.92
159	-40.67	-5.02	-0.56	-9.52	4.98	0.99	3.72	-1.58

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
120	53.48	10.31	-0.06	26.05	-6.81	-9.45	0.35	-27.38
121	58.44	14.49	-0.03	26.18	-6.15	-9.19	0.45	-24.02
122	55.08	11.57	-0.02	23.35	-9.94	-9.11	0.39	-25.86
123	56.58	12.31	-0.04	26.82	-7.44	-9.01	0.26	-25.55
124	54.35	10.39	-0.05	24.98	-8.46	-9.02	0.44	-25.96
125	53.81	15.02	-0.04	27.99	-7.80	-9.58	0.44	-25.95
126	53.27	14.24	-0.04	27.91	-7.71	-9.26	0.47	-25.63
127	58.03	12.68	-0.01	25.01	-6.40	-8.74	0.40	-28.05
128	39.98	-1.05	-0.33	21.96	-22.04	-4.80	-0.41	-15.66
129	41.22	-2.03	-0.36	22.19	-21.99	-5.06	-0.56	-15.90
130	40.95	-2.24	-0.34	21.97	-22.07	-5.14	-0.57	-19.52

131	40.26	-0.35	-0.34	20.92	-20.71	-5.18	-0.39	-18.20
132	41.86	-2.16	-0.37	22.34	-21.90	-4.85	-0.56	-16.02
133	43.88	-2.36	-0.34	21.20	-21.58	-5.34	-0.51	-16.43
134	36.87	-2.70	-0.33	20.90	-21.32	-5.10	-0.61	-19.40
135	39.83	-2.87	-0.36	18.90	-21.75	-4.79	-0.38	-17.56
136	45.23	-0.60	-0.35	22.43	-21.35	-5.04	-0.36	-17.73
137	41.16	-0.89	-0.36	23.49	-20.47	-5.32	-0.67	-14.10
138	38.19	-1.59	-0.35	21.84	-23.38	-5.16	-0.48	-17.71
139	41.58	-2.39	-0.39	21.34	-24.13	-4.83	-0.53	-17.91
140	42.51	-2.42	-0.38	23.72	-21.10	-5.44	-0.54	-15.10
141	40.30	0.10	-0.36	22.77	-22.23	-5.06	-0.67	-17.92
142	41.11	0.05	-0.35	22.22	-20.58	-4.96	-0.59	-18.54
143	42.21	-2.62	-0.33	21.58	-22.83	-5.20	-0.53	-18.43
144	23.17	3.07	-0.39	11.72	-4.83	-2.20	0.03	-8.73
145	20.93	1.49	-0.37	13.64	-5.71	-2.24	-0.03	-7.18
146	23.18	3.13	-0.37	12.07	-6.08	-2.29	-0.04	-9.87
147	24.10	2.17	-0.37	12.71	-6.07	-1.98	0.18	-8.85
148	21.38	3.16	-0.34	12.79	-5.25	-2.21	-0.02	-7.81
149	23.10	3.56	-0.37	13.95	-6.65	-2.46	-0.06	-9.00
150	19.33	3.00	-0.38	12.67	-5.73	-2.00	-0.07	-7.75
151	25.90	2.23	-0.36	11.71	-6.53	-2.38	0.09	-8.28
152	24.52	2.63	-0.37	14.16	-6.56	-2.29	0.02	-8.56
153	24.20	3.68	-0.34	12.92	-5.60	-2.00	-0.12	-8.80
154	20.10	2.21	-0.36	13.40	-5.76	-2.46	0.13	-8.69
155	23.10	1.82	-0.36	12.73	-5.94	-2.06	0.12	-8.08
156	24.71	2.42	-0.37	12.42	-5.34	-2.50	-0.12	-9.17
157	23.11	3.73	-0.39	12.30	-6.50	-2.19	-0.03	-9.57
158	24.10	3.93	-0.35	14.40	-6.18	-2.17	0.00	-8.30
159	22.82	4.16	-0.34	13.30	-6.05	-2.27	0.10	-7.50

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
120	0.31	-5.03	4.62	-32.56	-1.82	-1.70	4.33	0.53
121	0.72	-4.01	4.85	1.24	-1.88	-1.74	-1.33	0.54
122	0.45	-4.10	4.84	-2.36	-1.67	-1.69	-2.20	0.55
123	0.11	-4.61	4.79	14.98	-1.97	-1.76	-1.86	0.53
124	-0.04	-4.58	4.63	0.19	-1.97	-1.92	1.82	0.55
125	0.56	-4.69	4.83	-20.06	-1.93	-1.77	3.49	0.53
126	-0.20	-4.71	4.49	4.85	-1.85	-1.65	-2.92	0.54
127	-0.06	-4.54	4.62	-14.70	-1.93	-1.83	-4.25	0.55
128	-0.59	-1.46	4.32	-19.70	1.52	0.78	1.79	0.51
129	-0.28	-1.64	4.09	-34.57	1.15	0.54	5.08	0.51
130	0.18	-1.25	4.46	22.73	1.33	0.70	7.53	0.48
131	1.03	-1.55	4.43	8.79	1.40	0.59	5.14	0.47
132	-0.07	-1.65	4.37	-34.22	1.36	0.55	-0.52	0.48
133	0.09	-1.60	4.25	6.16	1.45	0.59	7.45	0.50
134	0.59	-1.67	4.41	-25.72	1.17	0.73	6.02	0.48
135	-1.51	-1.42	4.83	-0.65	1.22	0.56	7.84	0.49
136	-0.64	-1.43	4.64	-25.29	1.16	0.77	2.70	0.47
137	0.38	-1.91	4.26	29.50	1.25	0.50	3.59	0.47
138	-0.51	-1.60	4.68	-16.14	1.52	0.61	-4.79	0.49
139	-0.07	-1.17	4.53	42.81	1.29	0.66	6.75	0.50
140	-0.70	-1.19	4.06	6.12	1.09	0.81	-3.05	0.55
141	-0.06	-1.58	4.40	6.81	1.20	0.47	3.59	0.52

142	-0.30	-1.72	3.98	-8.17	1.33	0.76	4.02	0.50
143	-0.08	-1.32	4.27	-7.54	1.36	0.36	-7.33	0.50
144	0.45	0.82	3.10	11.79	3.12	2.48	-2.30	0.22
145	-0.01	0.44	2.79	-4.24	3.08	2.36	-0.35	0.25
146	0.37	0.43	2.61	-14.16	3.16	2.41	0.10	0.20
147	0.50	0.65	2.30	16.23	3.18	2.47	13.38	0.22
148	-0.08	0.27	2.63	15.55	3.20	2.50	6.03	0.22
149	0.77	0.39	2.89	11.11	3.10	2.45	1.68	0.23
150	-0.07	0.70	2.73	-7.13	3.16	2.45	6.97	0.23
151	-0.14	0.72	2.70	16.50	3.19	2.47	7.78	0.23
152	-0.05	0.58	3.11	5.90	3.31	2.60	5.82	0.21
153	-0.41	0.77	2.48	-3.59	3.23	2.64	1.59	0.21
154	-0.34	0.61	2.63	13.77	3.13	2.70	5.05	0.23
155	-0.58	0.21	2.43	-4.03	3.19	2.42	0.11	0.21
156	-0.25	0.71	2.85	-12.11	3.07	2.63	6.59	0.24
157	0.54	0.53	2.69	2.39	3.20	2.50	-3.10	0.21
158	0.28	0.84	2.99	-14.40	3.08	2.52	-6.42	0.19
159	0.26	0.25	2.33	8.40	3.20	2.36	5.99	0.25

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
120	-3.15	-2.74	32.68	-1.28	4.17	-0.54	-2.97	-49.03
121	0.37	-2.71	32.30	-1.60	4.71	-0.70	-2.92	-49.08
122	-1.57	-2.50	33.88	-1.53	4.70	-0.59	-2.22	-49.68
123	-2.35	-2.72	30.56	-1.41	4.83	-0.77	-2.27	-49.43
124	-1.34	-2.41	29.44	-1.53	4.20	-0.64	-3.11	-49.42
125	-0.38	-2.90	32.38	-1.44	3.98	-0.92	-3.34	-49.54
126	-0.34	-2.87	30.15	-1.49	4.13	-0.79	-3.50	-50.50
127	-1.26	-2.59	31.97	-1.33	3.99	-0.36	-2.67	-50.02
128	1.14	-0.88	32.09	-0.37	9.19	-2.91	-3.97	-36.02
129	0.01	-0.90	29.34	-1.01	8.94	-2.71	-3.49	-35.29
130	-0.35	-0.83	32.11	-0.90	9.10	-3.05	-4.29	-34.94
131	-1.21	-1.12	32.67	-0.71	9.25	-3.00	-3.36	-36.80
132	-0.17	-1.05	30.63	-0.91	8.77	-2.89	-4.80	-36.50
133	1.87	-0.79	31.14	-0.52	9.94	-2.94	-3.83	-36.11
134	-0.47	-1.01	30.51	-0.75	9.13	-3.33	-3.84	-35.71
135	0.30	-0.80	30.54	-1.22	9.54	-2.83	-3.54	-35.94
136	-1.37	-0.96	32.28	-0.65	9.08	-2.91	-3.37	-36.61
137	2.06	-1.05	29.71	-0.50	9.28	-2.99	-3.87	-35.73
138	-0.62	-0.65	28.44	-0.93	9.22	-2.73	-3.47	-36.62
139	-1.69	-0.79	32.28	-0.62	9.13	-2.77	-3.37	-36.46
140	-1.42	-0.48	31.05	-0.86	9.25	-3.21	-3.49	-35.39
141	2.17	-0.73	31.65	-0.97	9.05	-3.07	-4.55	-35.42
142	-1.91	-0.93	29.79	-0.57	8.76	-2.92	-3.90	-35.86
143	0.66	-0.81	33.30	-0.84	9.12	-3.15	-4.24	-37.51
144	1.11	0.42	18.05	0.99	2.98	-0.40	1.34	-22.51
145	0.18	0.19	18.46	0.91	2.43	-0.95	1.18	-21.18
146	-0.04	0.32	17.97	0.82	2.36	-0.36	0.97	-22.37
147	-0.54	0.58	18.34	1.07	2.36	-0.56	0.77	-22.28
148	0.51	0.19	18.43	0.91	2.45	-0.37	0.93	-22.38
149	0.27	0.47	18.05	0.97	2.15	-0.67	1.19	-22.81
150	0.84	0.36	17.25	1.26	2.25	-0.60	1.62	-22.12
151	1.21	0.45	18.18	1.26	2.77	-0.41	1.26	-22.47
152	-1.01	0.39	18.00	1.08	2.27	-0.42	1.60	-23.23

153	-0.96	0.54	17.76	1.28	2.68	-0.65	1.19	-22.35
154	-1.06	0.43	18.31	0.95	1.97	-0.58	1.72	-22.32
155	-1.09	0.56	16.71	0.46	2.04	-0.37	1.65	-21.98
156	-1.60	0.60	18.48	0.96	2.26	-0.62	1.03	-22.74
157	1.04	0.44	16.36	0.86	2.27	-0.58	1.05	-23.32
158	-0.46	0.44	16.89	0.97	2.06	-0.66	1.13	-21.35
159	0.16	0.47	18.03	1.13	2.51	-0.59	1.34	-22.34

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
120	-4.42	1.84	1.76	1.01	-0.98	-2.13	2.45	-2.01
121	-4.37	2.00	1.74	0.71	-0.97	-1.82	2.59	-2.07
122	-3.96	1.87	1.66	0.88	-0.98	-1.29	2.49	-2.04
123	-3.97	1.99	1.80	0.12	-0.92	-1.94	2.43	-2.10
124	-4.29	1.75	1.75	0.79	-0.97	-3.66	2.39	-2.04
125	-4.25	1.90	1.82	1.10	-0.98	-2.15	2.39	-2.09
126	-4.33	1.92	1.73	1.23	-0.96	-3.61	2.43	-2.04
127	-4.28	1.86	1.88	1.40	-1.02	-3.55	2.51	-1.98
128	-7.95	0.11	1.68	8.51	-0.87	10.98	-2.80	-2.96
129	-8.25	0.12	1.63	9.09	-0.88	8.94	-2.75	-3.10
130	-8.36	0.16	1.71	8.72	-0.86	9.22	-2.60	-3.06
131	-8.59	0.00	1.59	7.62	-0.86	11.42	-2.57	-2.99
132	-8.02	0.21	1.68	8.66	-0.87	9.90	-2.69	-3.05
133	-8.27	0.18	1.74	8.45	-0.92	9.04	-2.93	-3.06
134	-8.28	-0.01	1.69	8.82	-0.93	9.85	-2.73	-3.04
135	-8.91	0.14	1.71	8.75	-0.83	9.35	-2.74	-3.04
136	-8.32	0.14	1.68	8.33	-0.91	9.73	-2.51	-3.02
137	-8.23	0.27	1.76	8.53	-0.86	10.43	-2.72	-2.95
138	-8.15	0.11	1.60	8.97	-0.93	10.95	-2.78	-2.98
139	-8.33	0.06	1.70	8.68	-0.89	10.59	-3.01	-3.00
140	-7.64	0.16	1.68	8.39	-0.82	9.51	-2.84	-3.02
141	-9.02	0.15	1.58	8.08	-0.92	10.09	-2.75	-3.09
142	-8.73	0.14	1.68	8.98	-0.89	10.21	-2.67	-3.04
143	-7.95	0.22	1.63	9.15	-0.98	9.25	-3.00	-3.00
144	-2.45	-1.17	0.94	4.61	-0.40	1.78	-3.48	-2.51
145	-2.39	-1.16	1.04	4.32	-0.39	2.64	-3.46	-2.56
146	-1.88	-1.16	0.98	4.39	-0.41	0.79	-3.46	-2.50
147	-2.14	-1.10	1.02	4.42	-0.42	2.73	-3.50	-2.53
148	-2.58	-1.01	0.97	4.80	-0.36	2.52	-3.48	-2.52
149	-2.46	-1.06	0.97	4.32	-0.38	2.31	-3.48	-2.54
150	-2.92	-1.09	1.00	4.66	-0.42	2.15	-3.46	-2.52
151	-2.89	-1.10	0.90	4.61	-0.37	2.57	-3.54	-2.54
152	-2.62	-1.10	0.98	4.88	-0.37	2.24	-3.62	-2.56
153	-3.08	-1.12	1.01	4.37	-0.35	3.16	-3.55	-2.56
154	-2.80	-1.10	0.94	4.54	-0.41	2.04	-3.56	-2.52
155	-2.13	-1.11	0.95	4.54	-0.41	2.28	-3.54	-2.48
156	-2.77	-1.04	0.95	4.39	-0.36	1.86	-3.63	-2.55
157	-3.33	-1.13	1.01	3.85	-0.38	3.13	-3.50	-2.49
158	-2.72	-0.98	0.94	4.57	-0.36	2.95	-3.54	-2.53
159	-2.54	-1.18	1.01	4.96	-0.35	1.73	-3.52	-2.48

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
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120	0.33	4.92	-0.31	-1.59	0.68	-6.93	56.35	1.36
121	0.30	4.77	0.00	-1.50	4.89	-6.96	59.88	-3.65
122	0.34	4.82	4.31	-1.16	-7.78	-6.96	67.78	-1.90
123	0.31	4.96	-0.12	-2.05	-5.38	-7.09	66.01	0.16
124	0.36	4.79	3.53	-0.49	6.42	-6.72	66.01	2.43
125	0.35	5.13	-0.44	-1.41	-1.40	-6.90	63.79	3.73
126	0.33	4.75	-1.99	-1.23	-9.02	-6.86	58.33	3.46
127	0.36	4.73	0.36	-1.53	4.56	-7.29	61.10	3.11
128	-0.07	-1.38	2.12	-8.54	-9.27	-1.78	54.45	-0.97
129	-0.13	-1.34	2.14	-8.46	0.16	-2.53	47.17	-2.51
130	-0.06	-1.74	-2.76	-8.60	2.08	-2.47	57.05	-0.39
131	-0.15	-1.22	0.09	-8.08	5.34	-2.74	64.38	-0.47
132	-0.10	-2.01	-3.16	-8.12	-3.11	-2.73	54.72	3.38
133	-0.13	-1.45	2.02	-8.04	2.46	-2.24	68.54	1.40
134	-0.11	-1.36	-2.75	-7.68	0.87	-2.20	62.77	0.46
135	-0.10	-1.66	0.80	-7.99	-6.50	-2.85	58.11	-3.03
136	-0.10	-1.39	-0.66	-8.05	-0.64	-2.58	60.67	-1.20
137	-0.08	-2.04	-0.70	-8.62	5.92	-2.03	63.08	2.07
138	-0.11	-1.44	-2.66	-8.23	-3.00	-2.64	52.05	-12.58
139	-0.09	-1.34	1.73	-7.92	1.25	-2.09	54.21	-4.23
140	-0.13	-1.72	-1.14	-8.65	3.68	-2.46	60.90	-3.56
141	-0.14	-1.29	-1.05	-8.42	-4.40	-1.80	55.03	-1.73
142	-0.10	-1.33	-4.62	-9.38	-5.19	-2.68	56.49	-0.74
143	-0.09	-1.35	-1.42	-9.07	3.70	-2.50	61.63	0.05
144	-0.44	-6.62	-0.08	-8.43	3.03	2.65	34.50	2.06
145	-0.47	-6.28	1.57	-9.29	-6.12	2.33	29.93	-1.97
146	-0.44	-6.15	-0.35	-8.90	-6.12	2.76	31.77	-0.39
147	-0.47	-6.03	1.02	-8.50	-3.98	2.55	27.37	6.41
148	-0.47	-6.47	0.58	-9.16	-4.25	2.53	29.28	-2.16
149	-0.47	-6.41	0.58	-8.59	-7.69	2.66	32.78	-4.26
150	-0.44	-6.63	0.33	-8.96	5.06	2.56	29.31	0.25
151	-0.45	-6.32	4.26	-8.83	-3.55	2.68	35.36	-1.39
152	-0.44	-6.65	0.05	-8.71	-11.51	2.71	30.30	3.43
153	-0.44	-6.47	-0.37	-8.79	7.18	2.80	30.82	-6.53
154	-0.44	-6.12	0.06	-8.76	3.86	2.95	28.19	-2.98
155	-0.44	-6.12	1.80	-9.11	-9.21	2.68	30.34	6.30
156	-0.46	-6.68	-0.77	-8.69	1.26	2.63	30.94	-0.36
157	-0.47	-6.39	-0.08	-8.15	7.13	2.80	28.23	1.12
158	-0.45	-6.39	-0.75	-8.70	8.69	2.90	31.24	1.50
159	-0.46	-6.57	-1.82	-8.71	-3.29	2.75	28.59	-2.87

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
120	2.78	-6.70	6.71	-4.07	75.91	32.50	-4.02	-2.35
121	1.75	-6.51	6.99	1.30	76.01	32.64	-3.86	-1.98
122	2.33	-6.45	7.21	-2.74	73.44	32.65	-3.94	-3.11
123	2.80	-6.68	7.26	-2.89	72.83	33.33	-3.87	-2.74
124	3.47	-6.47	6.87	1.60	66.53	32.72	-3.73	-1.95
125	1.90	-6.45	7.49	0.13	71.55	33.59	-3.90	-2.40
126	2.54	-6.66	7.75	-4.69	70.82	32.24	-3.83	-3.09
127	1.80	-6.47	6.30	-0.37	75.11	32.34	-3.76	-2.51
128	5.08	-2.11	4.54	-3.75	56.90	24.33	-1.11	-3.41
129	6.93	-2.21	4.69	-1.04	53.20	22.82	-0.86	-3.26
130	5.71	-2.02	4.77	-2.04	59.19	22.72	-1.09	-3.87

131	6.18	-2.16	5.19	-1.55	56.71	22.38	-0.87	-3.38
132	4.92	-2.36	4.70	-3.26	53.43	21.56	-0.71	-3.77
133	5.37	-2.04	5.37	4.33	55.12	23.02	-1.00	-3.41
134	4.82	-2.12	4.95	2.78	57.33	21.93	-1.02	-4.04
135	5.17	-2.10	4.70	2.48	52.38	21.10	-0.78	-3.61
136	5.59	-2.43	5.28	-2.07	55.91	21.57	-1.07	-3.74
137	6.13	-2.32	4.42	-3.71	55.69	22.77	-0.81	-3.72
138	4.85	-2.26	4.81	6.46	55.44	22.67	-0.71	-3.47
139	4.82	-2.22	5.09	2.24	54.98	22.90	-0.79	-3.88
140	6.04	-2.27	4.38	-0.44	59.54	22.51	-0.85	-3.18
141	6.36	-2.39	5.49	-0.20	54.86	21.78	-0.90	-3.71
142	5.02	-2.24	5.11	2.02	57.42	21.87	-0.84	-3.51
143	5.52	-2.22	4.91	1.56	58.08	23.02	-1.00	-3.42
144	-0.21	1.06	1.30	-0.44	38.34	16.26	0.17	-2.55
145	-0.70	1.07	0.82	0.28	39.73	16.77	0.10	-2.79
146	-0.67	0.88	0.94	0.56	41.10	17.62	0.23	-2.75
147	-0.68	1.10	1.41	0.32	40.17	16.63	0.19	-2.42
148	-0.41	1.17	1.07	0.73	37.75	15.98	0.12	-2.83
149	-0.55	1.15	1.39	3.93	39.29	16.75	0.30	-3.12
150	-1.68	1.29	1.09	2.30	41.93	16.56	0.33	-2.48
151	-0.46	1.00	1.37	-0.76	41.86	16.89	0.23	-3.06
152	-1.16	0.71	0.72	-3.59	36.96	15.92	0.09	-2.85
153	-0.73	1.16	1.43	-2.06	37.10	16.71	0.20	-2.38
154	-0.77	1.19	1.48	0.25	39.82	16.00	0.06	-1.96
155	-0.23	1.11	0.91	0.10	39.59	16.68	0.02	-2.58
156	-0.49	1.09	1.59	1.28	37.71	16.57	0.07	-2.28
157	-0.44	1.01	0.76	0.78	41.09	16.19	0.21	-2.43
158	-0.30	1.03	1.11	-2.55	38.76	15.61	0.29	-2.24
159	-1.59	1.10	1.12	-0.03	39.03	16.28	0.16	-2.11

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
120	-3.49	13.40	-0.17	2.77	24.03	0.22	-1.92	8.36
121	-1.56	13.53	-0.15	2.91	24.00	0.19	-0.13	9.30
122	-0.33	13.08	-0.14	2.69	23.29	-0.11	-10.57	21.22
123	4.41	13.72	-0.15	2.98	24.76	0.23	-14.55	8.52
124	-5.43	13.28	-0.15	2.53	23.69	0.36	3.17	30.92
125	-2.68	13.25	-0.13	2.61	24.53	0.23	43.45	1.59
126	7.50	15.48	-0.16	2.78	24.04	0.28	-3.80	25.95
127	4.32	14.32	-0.14	2.77	23.76	0.04	2.42	-25.97
128	-13.42	27.45	-0.24	7.34	17.51	2.47	0.94	9.49
129	10.58	26.06	-0.24	7.23	17.53	2.31	14.43	46.44
130	4.21	24.93	-0.25	7.19	17.44	2.41	-21.84	-14.67
131	1.14	27.25	-0.25	7.89	17.62	2.36	-6.28	-1.76
132	-1.71	25.57	-0.28	7.46	17.35	2.48	0.91	32.36
133	12.79	24.80	-0.28	7.83	17.59	2.47	-0.89	-14.20
134	3.23	24.22	-0.24	7.25	17.33	2.29	7.74	36.11
135	5.29	24.94	-0.25	7.41	17.48	2.41	0.58	13.47
136	-3.92	26.67	-0.25	7.53	17.72	2.42	-8.16	22.44
137	-3.58	26.44	-0.26	7.61	16.62	2.49	-1.47	-16.57
138	4.11	25.58	-0.24	7.62	17.52	2.36	-16.44	-25.70
139	-1.61	25.23	-0.24	7.50	17.71	2.41	-13.21	25.78
140	-2.54	27.55	-0.25	7.87	18.15	2.40	1.36	18.40
141	-12.06	26.71	-0.24	7.31	17.41	2.28	-12.71	9.79

142	1.45	25.83	-0.25	7.40	17.14	2.38	-6.48	-4.14
143	3.09	26.96	-0.25	7.34	17.09	2.50	-5.15	17.91
144	-4.85	8.77	-0.03	7.05	9.88	2.53	14.66	9.71
145	-3.68	8.85	-0.04	7.36	9.83	2.65	13.40	23.05
146	-2.12	8.74	-0.03	7.62	10.26	2.57	-4.56	-13.00
147	0.77	9.60	-0.04	7.20	10.29	2.55	-1.43	5.03
148	1.14	9.42	-0.03	7.40	10.21	2.61	-10.34	1.62
149	1.93	8.85	-0.04	7.68	9.93	2.31	-10.78	11.24
150	-1.34	10.06	-0.04	7.44	10.26	2.50	2.13	6.41
151	3.92	9.28	-0.03	7.41	10.52	2.47	-23.89	21.08
152	1.39	9.20	-0.02	7.35	10.53	2.59	16.81	5.07
153	-8.77	8.14	-0.02	7.68	10.60	2.67	2.72	-9.55
154	-0.43	9.26	-0.04	7.60	10.54	2.54	-2.97	14.19
155	0.07	9.38	-0.03	7.27	10.46	2.65	2.90	14.79
156	-0.81	9.37	-0.05	7.54	10.21	2.54	10.72	0.67
157	-1.46	9.47	-0.05	7.26	10.23	2.60	-7.04	-39.70
158	-5.91	8.46	-0.04	7.27	9.78	2.38	0.29	9.16
159	-3.15	8.99	-0.05	7.41	10.58	2.70	-10.17	2.93

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
160	-183.25	2.14	5.62	-25.74	3.13	10.32	3.56	11.78
161	-200.77	1.99	5.47	-27.03	3.73	10.43	3.14	12.40
162	-189.77	1.88	5.81	-25.08	3.29	10.28	2.50	12.23
163	-180.47	1.92	5.11	-24.66	3.56	9.96	3.94	11.75
164	-190.80	2.23	5.54	-24.51	1.97	10.54	4.33	11.66
165	-202.82	2.32	5.41	-25.31	2.83	10.24	3.98	11.72
166	-191.96	2.42	5.77	-26.22	4.19	10.76	2.92	12.35
167	-202.76	2.17	6.14	-25.94	2.82	10.63	3.41	11.44
168	-200.75	1.98	4.80	-24.85	2.05	10.13	4.60	11.72
169	-203.33	2.07	4.43	-25.07	3.21	10.43	4.22	11.70
170	-210.89	2.59	5.25	-26.95	2.38	10.55	4.53	11.48
171	-182.98	1.78	5.53	-26.40	2.18	10.46	3.14	12.03
172	-198.69	1.99	5.10	-24.36	2.57	10.33	3.05	11.84
173	-188.40	1.90	5.59	-24.79	3.64	10.55	3.01	11.43
174	-214.04	2.14	5.54	-24.92	2.11	10.22	2.96	11.82
175	-184.63	2.21	5.32	-25.17	2.94	10.15	4.37	12.25
176	-200.57	2.15	5.04	-22.81	1.93	9.60	4.05	11.62
177	-192.89	1.82	5.99	-23.73	1.29	9.49	3.28	11.57
178	-203.63	1.64	5.83	-20.94	-0.29	9.36	3.67	12.18
179	-188.18	2.06	5.25	-23.68	1.11	9.95	5.50	11.80
180	-209.99	1.94	5.75	-22.03	1.03	9.56	4.07	11.49
181	-186.37	1.67	5.88	-21.16	0.82	9.68	4.08	11.39
182	-179.41	2.00	4.79	-22.47	1.65	9.46	3.93	11.91
183	-205.64	1.85	6.10	-21.08	0.69	9.89	2.51	12.14
184	-191.97	1.98	5.27	-22.05	0.35	9.36	3.29	11.41
185	-189.61	1.88	5.68	-22.09	1.18	9.34	3.93	11.81
186	-194.95	2.16	5.47	-23.46	2.22	9.99	4.03	11.41
187	-199.69	1.98	5.83	-21.48	0.05	9.70	3.21	11.41
188	-186.98	1.60	5.00	-23.36	1.17	9.80	2.65	11.23
189	-194.99	1.63	5.53	-23.27	1.00	9.60	4.56	12.00
190	-195.44	1.23	5.91	-21.99	1.57	9.26	3.87	12.39
191	-197.60	1.64	5.88	-21.36	0.88	9.76	2.72	10.91
192	-80.99	1.13	5.78	-41.96	-7.61	14.12	4.56	-2.34

193	-89.68	0.98	5.35	-40.67	-7.33	14.32	4.86	-3.09
194	-87.61	1.04	5.68	-40.81	-7.47	14.54	5.72	-3.11
195	-75.78	0.82	5.89	-41.56	-7.69	14.38	4.84	-3.28
196	-81.87	1.14	5.62	-39.90	-7.61	14.36	5.82	-3.01
197	-86.38	1.03	5.52	-40.84	-7.57	14.22	4.74	-2.69
198	-90.67	1.07	5.62	-40.01	-7.85	14.09	4.38	-2.72
199	-85.90	0.90	5.82	-39.91	-7.87	14.58	4.61	-3.11

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
160	93.69	50.35	-0.06	43.76	-7.96	-11.45	-1.65	-39.83
161	94.19	51.22	0.05	41.97	-8.07	-11.77	-1.83	-35.89
162	96.08	50.24	-0.01	44.07	-8.94	-11.94	-1.74	-33.20
163	91.01	50.82	-0.02	41.62	-9.97	-11.61	-1.79	-35.37
164	94.91	51.64	-0.03	44.52	-10.47	-11.14	-1.66	-38.97
165	99.80	49.04	-0.02	49.32	-10.67	-11.51	-1.78	-35.32
166	100.32	49.55	0.01	48.96	-8.32	-11.16	-1.87	-35.89
167	91.74	50.26	-0.03	47.65	-8.62	-11.54	-1.66	-34.51
168	99.23	48.64	-0.05	42.04	-8.21	-11.68	-1.67	-34.09
169	91.16	51.29	-0.01	48.34	-11.59	-11.22	-1.69	-38.04
170	84.86	50.02	0.02	47.47	-8.47	-12.16	-1.65	-39.32
171	104.41	49.99	-0.04	46.22	-7.03	-11.70	-1.61	-36.51
172	95.70	51.92	0.01	43.17	-11.26	-11.33	-1.77	-42.20
173	99.92	49.58	0.00	46.70	-8.98	-12.03	-1.72	-36.09
174	92.96	50.20	-0.05	47.19	-10.20	-11.95	-1.65	-39.79
175	87.99	50.84	-0.06	47.82	-10.64	-11.26	-1.78	-40.49
176	92.57	45.83	-0.03	46.46	-3.54	-10.29	-1.66	-40.48
177	103.07	45.08	-0.08	46.56	-6.48	-10.76	-1.59	-34.80
178	89.56	45.20	-0.04	45.63	-3.64	-10.46	-1.56	-43.17
179	100.68	43.71	-0.03	49.09	-2.93	-10.14	-1.61	-40.29
180	105.61	44.15	-0.06	48.19	-3.47	-9.72	-1.61	-40.95
181	91.34	45.93	-0.02	50.35	-3.56	-10.55	-1.62	-40.77
182	87.76	44.55	0.00	46.54	-7.49	-10.17	-1.51	-43.01
183	97.67	44.55	-0.04	49.44	-4.48	-10.41	-1.59	-36.79
184	93.30	44.01	-0.02	47.30	-2.87	-10.32	-1.78	-37.03
185	101.12	47.15	0.00	45.58	-6.35	-9.75	-1.68	-42.95
186	92.30	43.85	-0.03	49.68	-5.53	-10.36	-1.73	-39.98
187	90.84	43.94	-0.02	44.30	-1.86	-10.72	-1.55	-37.11
188	101.37	42.91	-0.04	48.77	-4.67	-9.64	-1.62	-45.48
189	95.02	44.73	-0.04	45.22	-2.35	-10.64	-1.65	-39.73
190	93.66	43.89	-0.01	45.99	-6.29	-9.88	-1.58	-39.34
191	103.16	44.83	-0.01	47.51	-5.19	-10.19	-1.53	-36.73
192	59.69	57.64	-0.04	16.47	24.54	-9.22	2.64	-37.59
193	69.07	58.35	-0.09	18.14	26.55	-9.27	2.42	-38.58
194	64.30	59.42	-0.09	19.73	26.26	-9.01	2.47	-37.31
195	64.31	57.75	-0.08	18.64	26.38	-9.27	2.49	-39.18
196	66.35	57.50	-0.09	17.61	26.14	-9.63	2.47	-34.66
197	70.49	57.92	-0.09	19.08	25.79	-9.25	2.47	-36.50
198	67.45	59.20	-0.07	19.74	26.43	-9.27	2.38	-38.96
199	69.13	57.68	-0.11	19.25	26.73	-9.21	2.41	-38.46

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
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160	0.05	-6.82	-24.67	-24.74	-2.61	-2.86	-2.71	-1.60
161	0.00	-6.42	-24.61	-59.26	-2.55	-2.53	6.00	-1.60
162	-0.30	-6.70	-24.98	8.36	-2.47	-2.78	20.37	-1.59
163	-0.11	-5.77	-24.59	50.59	-2.62	-2.67	6.97	-1.61
164	0.19	-6.04	-24.78	-12.92	-2.65	-2.69	-11.54	-1.60
165	-0.02	-5.73	-24.78	-25.13	-2.64	-2.84	8.52	-1.63
166	-0.06	-7.15	-24.72	10.36	-2.90	-2.60	11.33	-1.60
167	0.05	-5.02	-24.62	-18.12	-2.65	-2.70	-9.13	-1.60
168	-0.18	-5.74	-24.65	-43.31	-2.61	-2.74	2.60	-1.63
169	0.05	-3.92	-24.69	-0.11	-2.62	-2.81	-0.57	-1.59
170	-0.22	-6.42	-24.74	31.59	-2.54	-2.42	4.93	-1.60
171	0.38	-6.29	-24.79	17.32	-2.47	-2.80	3.16	-1.58
172	-0.20	-4.47	-24.71	-76.32	-2.72	-2.81	14.78	-1.58
173	-0.02	-3.41	-24.55	12.62	-2.59	-2.77	13.40	-1.61
174	-0.19	-6.36	-24.80	-30.09	-2.58	-2.56	11.87	-1.60
175	0.01	-5.58	-24.80	-0.33	-2.57	-2.98	8.06	-1.59
176	-0.21	-7.35	-22.87	-2.52	-2.79	-3.04	-3.04	-1.43
177	0.42	-6.96	-22.77	-14.25	-2.87	-2.93	-3.93	-1.45
178	-0.02	-6.32	-22.67	6.85	-2.92	-3.19	-7.99	-1.45
179	-0.09	-6.74	-22.78	-7.19	-2.85	-2.83	8.51	-1.45
180	0.17	-6.39	-22.66	56.84	-2.89	-2.75	1.92	-1.46
181	0.19	-5.99	-22.80	8.81	-2.85	-2.82	-0.75	-1.45
182	0.14	-5.99	-22.76	24.31	-2.86	-2.90	-4.31	-1.46
183	-0.08	-7.56	-22.53	13.39	-2.79	-2.74	16.60	-1.45
184	-0.11	-6.77	-22.95	19.90	-3.09	-2.74	1.85	-1.47
185	0.08	-6.63	-22.79	28.47	-2.86	-3.04	-8.57	-1.44
186	-0.04	-6.29	-22.94	53.97	-2.89	-3.03	-0.93	-1.44
187	-0.14	-6.33	-22.59	99.70	-3.14	-2.75	-4.33	-1.43
188	0.00	-7.06	-22.71	-0.85	-2.92	-2.89	0.54	-1.45
189	-0.07	-8.13	-22.59	-18.23	-2.91	-3.01	-8.11	-1.44
190	-0.17	-7.08	-22.39	23.48	-2.85	-2.87	-3.17	-1.46
191	-0.06	-7.27	-22.67	41.89	-2.92	-2.76	-3.93	-1.46
192	-0.08	-4.09	-16.51	-9.54	-3.59	-2.93	-0.63	-1.16
193	-0.14	-3.11	-16.34	18.48	-3.60	-3.00	1.15	-1.14
194	0.04	-3.51	-16.64	17.91	-3.51	-2.91	1.10	-1.13
195	-0.17	-3.94	-16.52	15.91	-3.51	-2.89	-1.44	-1.13
196	0.08	-3.88	-16.58	-20.06	-3.47	-2.90	5.56	-1.13
197	0.09	-3.67	-16.49	-16.57	-3.57	-2.88	3.05	-1.13
198	0.05	-3.94	-16.42	-16.93	-3.42	-2.97	1.59	-1.11
199	0.02	-4.02	-16.51	-25.71	-3.60	-2.90	2.24	-1.14

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
160	-1.72	-4.57	-3.31	-5.64	1.91	1.82	6.79	-74.96
161	2.72	-4.07	0.55	-5.64	1.76	1.89	6.87	-73.97
162	-4.32	-4.19	-2.77	-5.46	2.01	1.92	7.05	-73.97
163	-1.39	-4.01	-0.95	-5.48	2.46	1.69	6.74	-74.71
164	-0.61	-4.33	-1.97	-5.78	2.93	1.84	6.85	-75.67
165	2.85	-4.36	-3.46	-5.41	2.81	1.98	6.35	-75.37
166	-1.58	-3.98	-1.06	-5.35	1.94	1.90	6.59	-74.94
167	-0.38	-5.08	-3.13	-5.36	2.77	1.81	6.39	-75.00
168	2.47	-3.87	-0.88	-5.81	2.49	1.71	7.18	-75.15
169	-2.51	-4.45	-6.66	-4.91	1.83	2.03	6.90	-76.07
170	6.57	-3.95	-4.60	-5.75	1.82	2.01	6.96	-73.68

171	2.24	-3.92	-1.15	-5.47	2.71	1.61	6.63	-73.51
172	3.40	-4.54	-2.52	-5.54	2.40	2.00	6.80	-73.48
173	-0.75	-4.31	2.15	-5.78	2.35	1.56	6.88	-74.65
174	3.78	-4.76	-3.69	-5.35	1.90	1.99	7.05	-75.07
175	2.70	-4.13	-3.26	-5.61	1.83	1.74	6.62	-75.22
176	-3.40	-4.70	-1.81	-4.85	1.49	1.10	5.28	-69.91
177	3.89	-4.52	-2.20	-5.42	1.47	1.05	5.44	-68.29
178	-0.89	-4.57	-0.54	-5.06	1.72	1.23	5.32	-68.69
179	0.34	-4.88	-0.94	-5.10	1.60	1.22	5.19	-67.29
180	2.91	-4.43	-2.20	-5.38	1.18	1.28	5.03	-68.04
181	0.20	-4.51	0.35	-5.67	1.11	1.13	5.20	-69.65
182	-5.24	-4.31	-1.03	-5.08	1.14	1.21	4.73	-67.72
183	2.46	-4.05	0.58	-5.67	0.91	1.03	5.01	-68.14
184	-1.50	-4.78	-1.48	-5.34	1.04	1.23	5.58	-68.25
185	0.24	-5.14	0.73	-4.87	1.21	1.05	5.17	-68.14
186	-0.78	-4.30	-4.96	-5.25	1.03	1.33	5.06	-68.46
187	1.72	-5.38	0.77	-4.80	1.35	0.99	5.47	-67.65
188	-0.43	-5.19	-0.03	-5.43	1.03	1.19	5.13	-71.14
189	4.51	-4.64	-3.26	-4.93	0.78	0.95	5.52	-68.34
190	0.38	-4.65	0.48	-4.86	1.19	1.08	5.19	-69.56
191	-3.32	-4.37	4.83	-5.59	1.22	1.00	5.24	-71.35
192	-1.10	-6.19	10.44	-5.27	-10.64	1.78	8.50	-46.49
193	-0.22	-5.99	10.70	-5.10	-10.32	1.84	8.95	-46.44
194	1.19	-6.00	10.41	-5.23	-9.90	1.72	8.65	-45.75
195	-0.99	-6.04	11.34	-5.36	-10.16	1.78	8.59	-44.99
196	-1.78	-5.87	9.88	-5.37	-10.11	1.82	8.86	-46.13
197	0.12	-6.17	11.67	-5.32	-10.56	1.82	8.33	-46.32
198	0.29	-6.03	10.23	-5.05	-10.16	1.82	8.42	-44.92
199	-2.02	-5.95	9.82	-5.22	-9.75	1.57	8.69	-45.94

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
160	-7.10	3.84	2.29	7.25	-1.41	-42.89	-2.47	-2.31
161	-7.16	4.00	2.40	6.79	-1.29	-44.10	-2.43	-2.37
162	-6.88	3.82	2.18	6.67	-1.35	-44.13	-2.49	-2.35
163	-7.18	3.48	2.31	6.99	-1.38	-42.27	-2.53	-2.32
164	-6.67	3.67	2.34	6.50	-1.34	-42.33	-2.46	-2.34
165	-8.06	3.80	2.15	6.43	-1.41	-42.56	-2.48	-2.36
166	-6.97	3.84	2.07	6.30	-1.37	-41.59	-2.35	-2.30
167	-7.50	3.63	2.27	8.27	-1.29	-43.63	-2.40	-2.33
168	-7.23	3.84	2.14	7.07	-1.36	-42.42	-2.44	-2.32
169	-6.82	3.72	2.20	6.12	-1.31	-43.38	-2.53	-2.33
170	-6.81	3.71	2.21	7.09	-1.38	-41.57	-2.45	-2.37
171	-6.97	3.74	2.16	7.13	-1.35	-42.14	-2.49	-2.32
172	-6.01	3.77	2.13	7.48	-1.41	-42.50	-2.42	-2.30
173	-6.58	3.62	2.20	6.64	-1.30	-41.40	-2.46	-2.35
174	-6.09	3.80	2.19	6.27	-1.46	-42.84	-2.53	-2.33
175	-7.31	3.70	2.41	6.68	-1.30	-41.21	-2.56	-2.35
176	-6.83	3.47	2.53	4.38	-1.11	-39.26	-2.01	-2.17
177	-7.14	3.45	2.16	4.46	-1.08	-38.15	-2.01	-2.17
178	-6.84	3.45	2.44	4.82	-1.20	-37.87	-1.96	-2.14
179	-7.17	3.28	2.32	4.76	-1.15	-38.48	-2.04	-2.17
180	-6.73	3.38	2.39	3.85	-1.07	-36.51	-1.99	-2.13
181	-6.45	3.47	2.29	5.05	-0.93	-37.26	-1.92	-2.14

182	-6.41	3.52	2.16	4.22	-1.01	-40.58	-2.04	-2.16
183	-6.37	3.34	2.26	4.47	-1.11	-38.40	-1.98	-2.14
184	-6.07	3.42	2.22	5.10	-1.07	-38.98	-1.95	-2.18
185	-6.69	3.57	2.23	4.02	-1.03	-37.01	-2.01	-2.19
186	-6.10	3.26	2.21	4.04	-1.15	-37.83	-1.92	-2.15
187	-6.41	3.36	2.25	4.21	-1.12	-41.20	-2.00	-2.14
188	-6.48	3.30	2.49	4.27	-1.03	-39.02	-1.98	-2.14
189	-6.73	3.32	2.34	4.37	-1.12	-37.91	-1.96	-2.18
190	-6.61	3.66	2.16	4.43	-1.14	-37.43	-1.86	-2.15
191	-7.03	3.45	2.56	4.75	-1.13	-38.65	-1.94	-2.17
192	14.34	3.23	0.52	-6.89	-0.44	-54.93	-0.40	-0.38
193	15.28	3.27	0.54	-6.69	-0.47	-54.20	-0.34	-0.37
194	15.25	3.25	0.65	-6.55	-0.50	-55.28	-0.39	-0.36
195	15.22	3.25	0.61	-6.99	-0.47	-54.89	-0.36	-0.38
196	15.23	3.32	0.62	-6.76	-0.44	-55.82	-0.36	-0.31
197	14.98	3.30	0.73	-7.01	-0.45	-54.51	-0.36	-0.36
198	14.34	3.25	0.68	-7.27	-0.47	-54.67	-0.36	-0.37
199	14.44	3.17	0.51	-6.82	-0.48	-54.34	-0.37	-0.36

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
160	0.37	6.16	-4.89	-3.95	-0.18	-5.80	216.68	-4.24
161	0.44	6.30	-5.02	-1.35	9.89	-6.94	215.70	0.94
162	0.47	5.80	-2.58	-2.61	8.27	-5.37	202.79	-3.53
163	0.41	6.05	2.06	-4.03	5.60	-6.26	205.63	-4.57
164	0.40	5.86	4.56	-3.05	-1.66	-6.13	201.34	-2.44
165	0.43	6.14	-4.27	-1.20	-3.83	-6.40	219.57	-2.51
166	0.35	6.13	-2.92	-2.09	8.70	-6.09	207.33	1.18
167	0.39	5.87	-2.87	-4.76	8.99	-7.35	200.76	5.54
168	0.38	6.11	2.58	-1.78	-3.50	-5.84	212.19	-5.32
169	0.41	5.65	-0.03	-1.87	9.47	-6.14	212.81	2.41
170	0.44	5.74	-2.44	-1.68	6.05	-6.48	202.50	3.86
171	0.45	6.48	-1.42	-1.59	4.23	-6.51	218.85	-1.59
172	0.38	6.75	2.44	-0.51	1.69	-6.58	204.05	-0.83
173	0.46	5.72	-0.38	-0.25	5.15	-5.50	191.42	0.98
174	0.45	6.33	3.98	-2.61	-5.38	-5.59	205.05	-3.88
175	0.40	5.97	6.85	-2.28	7.55	-6.99	200.65	-0.70
176	0.45	7.06	-0.46	-1.72	1.09	-6.70	193.26	3.23
177	0.44	6.59	0.91	-1.08	-0.45	-7.42	197.52	2.04
178	0.46	7.05	-2.54	-0.60	1.25	-6.31	187.34	0.15
179	0.47	6.80	-3.35	-0.78	2.85	-7.77	196.66	2.32
180	0.45	6.54	-0.56	-2.66	5.28	-7.34	204.22	-2.50
181	0.49	6.72	2.50	-0.06	5.80	-6.91	198.68	-4.05
182	0.51	7.05	-4.15	-2.54	-1.36	-7.48	201.35	0.50
183	0.45	7.01	-1.19	-0.48	3.29	-8.19	205.06	-2.27
184	0.50	6.51	1.42	-1.52	2.04	-7.31	184.98	5.86
185	0.46	6.88	0.90	-0.33	1.45	-7.19	200.08	-1.33
186	0.41	6.38	-0.92	0.12	-2.64	-8.02	193.03	4.59
187	0.48	7.11	-1.69	-1.28	-2.29	-7.25	195.79	-4.40
188	0.44	6.54	3.58	-0.63	5.67	-6.94	198.26	-5.30
189	0.42	6.39	-4.58	-1.29	7.96	-7.29	198.65	-3.19
190	0.46	6.64	-4.68	-1.11	-4.70	-7.46	196.34	3.04
191	0.45	6.81	2.51	-2.38	0.31	-7.21	195.56	1.44
192	0.51	7.88	-0.86	5.86	-1.23	-9.89	31.77	-1.15

193	0.49	7.70	0.24	5.63	-4.70	-9.82	32.51	-0.43
194	0.54	7.71	-0.61	5.51	-5.12	-9.75	32.83	1.99
195	0.49	7.83	0.62	5.83	2.58	-9.67	33.03	0.79
196	0.52	7.78	-1.62	5.24	-1.38	-9.49	37.38	1.85
197	0.54	7.94	0.26	4.88	0.93	-9.66	28.25	-0.89
198	0.48	7.63	-0.16	6.08	1.97	-9.63	26.78	-1.78
199	0.50	7.86	-1.26	5.43	1.97	-9.74	29.21	-1.97

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
160	-15.09	1.43	8.38	2.09	67.10	50.59	-4.05	-2.61
161	-15.99	1.74	8.50	-0.09	74.46	49.89	-3.80	-2.33
162	-15.23	1.34	8.48	-1.80	66.68	50.94	-3.90	-2.70
163	-14.71	1.41	6.35	-3.76	65.56	52.11	-3.89	-3.17
164	-15.64	1.35	8.94	-11.87	69.07	50.02	-3.94	-2.36
165	-15.55	1.51	9.37	4.37	68.67	53.08	-3.92	-2.80
166	-15.40	1.37	8.38	2.83	69.57	52.07	-3.76	-1.98
167	-15.25	1.65	7.20	-1.24	69.80	49.58	-4.03	-3.15
168	-15.21	1.29	7.64	-5.76	65.67	51.80	-3.99	-4.23
169	-15.65	1.44	7.37	0.61	67.24	51.04	-3.94	-3.61
170	-15.53	1.54	7.44	-3.07	72.23	52.70	-4.07	-2.45
171	-14.94	1.54	9.01	-0.48	66.04	51.64	-4.11	-2.67
172	-15.34	1.52	8.43	2.06	71.91	50.70	-4.02	-2.48
173	-14.78	1.57	7.45	-2.85	62.87	50.65	-3.87	-1.81
174	-15.51	1.51	9.59	0.14	65.71	49.51	-4.02	-1.82
175	-15.48	1.26	6.97	7.15	69.96	53.52	-3.95	-3.26
176	-12.35	1.49	9.88	-9.25	57.89	46.03	-3.59	-1.89
177	-12.31	1.61	9.42	-3.73	67.80	45.21	-3.64	-2.45
178	-12.14	1.49	10.23	-0.63	58.49	45.33	-3.73	-2.72
179	-12.13	1.61	10.38	0.24	55.05	46.72	-3.63	-2.00
180	-11.85	1.51	9.89	-9.35	58.31	44.55	-3.63	-3.61
181	-11.39	1.68	10.30	-3.55	55.82	44.55	-3.69	-0.74
182	-11.71	1.58	9.81	-3.65	60.66	45.97	-3.65	-2.52
183	-11.84	1.62	7.87	-3.91	62.48	45.42	-3.52	-2.08
184	-12.30	1.57	9.70	8.09	55.78	46.36	-3.52	-2.01
185	-11.78	1.63	8.33	-0.93	55.84	45.39	-3.58	-2.29
186	-11.46	1.71	9.94	-4.37	60.19	44.13	-3.66	-3.12
187	-12.17	1.61	8.29	-5.53	58.65	44.83	-3.78	-2.80
188	-11.84	1.68	7.48	-2.86	65.02	45.54	-3.61	-2.11
189	-11.77	1.64	9.14	2.22	55.82	44.24	-3.61	-2.37
190	-12.38	1.71	7.76	-4.37	63.61	45.04	-3.67	-2.30
191	-11.86	1.57	8.30	0.81	62.45	45.01	-3.51	-2.05
192	-14.53	1.99	13.18	1.33	106.76	38.07	-5.34	-5.15
193	-13.82	2.08	13.41	1.91	104.39	37.64	-5.16	-4.44
194	-14.37	2.15	13.46	-0.20	110.70	38.01	-5.34	-5.04
195	-13.91	2.12	13.46	0.42	102.96	38.66	-5.33	-4.93
196	-14.24	2.07	14.12	2.08	97.99	38.20	-5.46	-5.31
197	-14.21	2.13	13.13	-1.64	100.89	37.80	-5.44	-4.59
198	-14.39	2.09	13.96	-2.45	103.11	38.34	-5.33	-5.41
199	-13.74	2.08	13.50	-2.52	105.29	38.24	-5.41	-4.78

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
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160	4.80	-3.26	-0.21	3.57	2.42	-0.68	-12.67	-24.28
161	-3.84	-4.72	-0.18	2.94	2.29	-0.37	1.69	35.79
162	2.38	-4.79	-0.18	3.21	2.09	-0.33	4.88	-13.73
163	-0.25	-2.02	-0.17	2.96	2.59	-0.43	27.32	49.32
164	3.73	-5.32	-0.15	3.77	2.77	-0.50	-9.12	57.33
165	0.28	-2.82	-0.17	3.31	2.63	-0.50	22.97	31.74
166	-4.59	-3.20	-0.18	3.25	3.45	-0.72	-15.61	33.03
167	-1.01	-4.60	-0.17	4.16	2.50	-0.33	-27.43	-11.64
168	-4.66	-1.98	-0.19	4.12	2.54	-0.22	-1.92	-10.90
169	3.96	-7.01	-0.20	2.86	2.56	-0.62	17.28	-12.73
170	2.95	-4.19	-0.21	3.43	2.63	-0.19	-14.50	-1.16
171	-7.27	-3.21	-0.20	3.49	2.62	-0.49	1.07	-12.08
172	7.69	-6.61	-0.16	2.74	2.46	-0.33	12.29	-21.91
173	0.38	-3.77	-0.20	3.10	2.67	-0.35	-8.43	25.41
174	1.04	-2.66	-0.18	3.00	2.98	-0.48	-6.63	-62.55
175	1.60	-1.83	-0.18	3.19	2.77	-0.62	-10.72	-93.17
176	-0.52	-7.31	-0.07	2.13	2.59	-0.40	-6.00	14.91
177	-0.77	-6.06	-0.12	1.95	3.14	-0.16	-9.91	-11.31
178	-1.38	-6.04	-0.11	1.95	2.90	-0.10	-14.84	-5.31
179	3.62	-5.16	-0.12	2.88	2.42	-0.29	-2.65	24.60
180	-0.82	-5.22	-0.08	2.50	2.59	-0.01	-9.33	68.11
181	1.07	-4.69	-0.09	1.89	2.66	-0.23	5.36	72.62
182	1.54	-7.98	-0.10	2.35	2.91	-0.19	-19.95	-12.65
183	0.38	-6.18	-0.10	2.44	2.23	-0.51	30.27	7.42
184	-1.62	-7.10	-0.09	1.94	2.99	-0.47	-3.11	-34.92
185	2.42	-5.25	-0.08	2.10	2.45	-0.43	3.54	3.01
186	-5.80	-6.76	-0.10	1.44	2.25	-0.49	-21.05	17.21
187	7.44	-5.48	-0.12	2.46	2.44	-0.27	23.77	30.52
188	-5.44	-6.90	-0.11	2.41	2.25	-0.51	10.61	-0.47
189	0.88	-6.94	-0.09	2.24	2.12	-0.02	20.18	-19.77
190	8.28	-6.66	-0.10	2.54	2.34	-0.33	21.36	57.54
191	-3.19	-5.28	-0.09	2.05	2.63	-0.25	7.47	31.17
192	5.63	-31.18	0.32	-1.71	12.41	0.20	-0.23	7.04
193	2.70	-30.14	0.28	-2.06	12.10	0.16	-29.14	20.81
194	-1.45	-29.85	0.28	-1.81	12.41	0.08	17.53	16.41
195	-2.76	-29.39	0.31	-1.97	12.28	0.17	-0.96	15.02
196	-3.77	-29.74	0.29	-2.19	12.65	0.09	-9.79	31.38
197	-3.05	-29.86	0.30	-1.86	12.48	0.13	-18.07	9.58
198	-2.17	-30.42	0.30	-1.83	12.21	0.18	-5.19	14.38
199	-2.93	-30.44	0.29	-1.97	12.62	0.04	16.43	-0.26

k	a0k	a1k	a2k	a3k	a4k	a5k	a6k	a7k
200	-84.84	1.13	5.88	-40.53	-7.81	13.76	5.16	-2.61
201	-86.46	1.17	5.92	-41.73	-7.49	14.37	4.96	-2.65
202	-85.65	0.92	5.61	-41.84	-7.73	14.39	4.88	-3.04
203	-83.58	0.92	5.57	-41.07	-7.79	14.00	5.78	-2.87
204	-84.03	0.90	5.59	-40.04	-8.07	15.05	5.36	-2.80
205	-88.01	1.00	5.82	-40.27	-7.37	14.65	5.64	-2.95
206	-91.14	1.09	5.67	-38.94	-7.49	13.90	5.00	-3.05
207	-87.84	1.13	5.79	-39.92	-7.62	14.06	4.61	-3.12
208	-139.32	2.88	8.38	-29.54	-7.87	11.87	0.06	1.37
209	-136.35	2.65	8.20	-29.63	-7.90	11.88	0.14	1.88
210	-136.39	2.88	8.19	-29.62	-7.91	11.54	0.09	1.74

211	-140.86	2.90	8.40	-28.91	-8.02	11.83	1.40	1.81
212	-141.28	2.64	8.02	-30.90	-8.11	11.87	0.91	1.38
213	-147.44	3.03	8.55	-29.96	-8.18	12.28	0.27	0.89
214	-137.06	2.83	8.21	-28.00	-7.95	11.69	-0.13	1.43
215	-136.69	2.81	8.31	-31.18	-8.32	12.24	-0.51	1.76
216	-144.43	3.02	8.72	-30.56	-7.94	11.59	0.02	1.81
217	-137.63	2.89	8.13	-30.66	-7.76	11.71	0.40	1.39
218	-132.91	2.90	8.74	-30.41	-8.04	11.61	0.24	2.17
219	-146.30	2.70	8.31	-32.57	-8.00	12.00	0.33	1.31
220	-141.57	2.69	8.20	-30.05	-8.02	11.79	0.65	1.73
221	-143.00	2.79	8.25	-29.94	-7.79	11.74	-0.08	1.70
222	-138.77	2.83	8.47	-28.20	-7.64	11.82	0.14	1.48
223	-140.99	2.63	8.27	-30.91	-7.93	11.74	0.18	1.55
224	-54.29	0.60	4.08	-17.30	-4.02	6.24	2.53	-0.47
225	-49.96	0.52	4.11	-17.63	-3.57	6.25	2.57	-0.65
226	-53.86	0.63	4.07	-16.88	-3.88	6.15	2.39	-0.17
227	-53.38	0.55	4.12	-16.78	-3.97	6.27	2.64	-0.44
228	-52.04	0.58	4.25	-17.54	-3.97	6.26	2.50	-0.44
229	-49.03	0.59	4.08	-16.65	-3.89	6.26	2.41	-0.68
230	-51.57	0.47	3.92	-17.80	-4.05	6.13	2.83	-0.45
231	-51.91	0.73	4.07	-16.88	-3.72	6.31	2.68	-0.76
232	-53.06	0.54	4.08	-17.42	-3.88	6.13	2.80	-0.66
233	-55.52	0.57	4.00	-17.43	-3.79	6.01	2.67	-0.74
234	-52.68	0.62	3.76	-16.64	-3.85	6.32	2.48	-0.51
235	-57.68	0.48	4.06	-16.80	-3.86	5.94	2.56	-0.56
236	-53.28	0.60	4.17	-17.07	-3.56	6.08	2.93	-0.65
237	-50.64	0.50	4.15	-16.55	-3.98	5.93	2.86	-0.63
238	-54.90	0.57	3.85	-17.16	-3.98	6.12	2.89	-0.43
239	-54.82	0.69	3.85	-16.27	-3.91	6.14	2.75	-0.44

k	a8k	a9k	a10k	a11k	a12k	a13k	a14k	a15k
200	66.06	57.96	-0.06	20.15	25.87	-9.12	2.49	-40.56
201	68.69	59.17	-0.05	18.98	26.20	-9.42	2.57	-38.06
202	65.93	58.25	-0.06	17.31	26.06	-9.06	2.45	-37.70
203	66.02	57.57	-0.06	17.54	24.57	-9.56	2.48	-37.48
204	62.62	60.56	-0.05	18.59	26.34	-9.47	2.50	-37.12
205	69.42	59.25	-0.07	18.44	25.76	-9.14	2.51	-38.09
206	69.43	58.91	-0.08	17.90	25.60	-9.19	2.56	-38.37
207	66.76	59.12	-0.06	17.58	23.78	-9.08	2.57	-36.53
208	105.25	45.16	0.14	30.65	17.19	-11.84	1.41	-56.05
209	100.97	49.11	0.17	33.24	19.20	-11.96	1.36	-54.82
210	110.23	47.17	0.14	32.19	18.96	-11.95	1.33	-57.81
211	99.46	47.41	0.17	34.11	19.23	-11.51	1.40	-58.91
212	100.89	47.64	0.20	32.97	18.91	-11.64	1.29	-55.22
213	105.89	47.15	0.17	34.74	19.38	-11.71	1.35	-54.68
214	100.66	48.39	0.12	33.05	18.56	-11.70	1.47	-55.39
215	105.40	48.05	0.14	35.47	19.87	-11.73	1.47	-56.98
216	107.76	46.39	0.15	32.78	18.55	-11.55	1.40	-56.60
217	102.67	46.58	0.16	34.43	19.08	-11.72	1.35	-56.71
218	106.31	45.50	0.13	35.30	18.10	-11.87	1.42	-56.77
219	103.64	46.85	0.14	34.19	18.24	-11.49	1.25	-56.12
220	105.64	46.90	0.17	34.57	17.92	-11.69	1.39	-57.90
221	104.76	45.29	0.14	35.97	21.36	-11.91	1.36	-56.05

222	103.79	46.27	0.14	33.43	19.41	-11.53	1.41	-56.73
223	101.88	45.23	0.17	34.42	19.64	-12.02	1.45	-53.76
224	44.53	25.82	-0.06	11.68	10.35	-5.06	0.95	-25.85
225	45.43	25.95	-0.04	11.59	10.33	-4.95	1.01	-27.25
226	42.76	25.29	-0.04	11.55	10.68	-4.99	1.00	-26.64
227	44.83	25.97	-0.05	12.62	10.83	-5.07	0.98	-26.88
228	43.61	25.36	-0.03	12.76	10.30	-4.92	1.10	-27.08
229	46.60	25.62	-0.05	11.49	10.21	-5.09	1.00	-26.17
230	43.98	25.50	-0.06	11.29	10.13	-5.14	0.85	-26.86
231	43.86	25.36	-0.04	11.30	10.41	-4.94	1.01	-25.41
232	46.44	25.68	-0.05	10.66	10.53	-5.00	0.95	-27.32
233	47.20	25.26	-0.02	10.75	11.18	-5.01	0.97	-26.35
234	45.19	26.43	-0.02	11.44	10.41	-5.05	0.94	-25.67
235	46.18	25.03	-0.03	11.83	10.34	-4.91	1.00	-26.54
236	45.28	25.60	-0.04	11.79	10.74	-5.00	0.99	-27.22
237	45.71	26.50	-0.06	11.87	10.86	-5.00	0.99	-28.24
238	44.83	25.30	-0.04	11.05	10.70	-5.05	0.95	-26.50
239	45.99	25.08	-0.06	11.38	10.68	-4.93	0.88	-26.89

k	a16k	a17k	a18k	a19k	a20k	a21k	a22k	a23k
200	-0.14	-3.90	-16.42	18.94	-3.52	-2.90	2.97	-1.12
201	-0.01	-3.17	-16.60	20.24	-3.57	-2.91	4.30	-1.14
202	0.14	-3.90	-16.67	6.48	-3.59	-2.84	-0.10	-1.11
203	-0.24	-3.89	-16.69	4.00	-3.46	-3.00	-2.50	-1.12
204	-0.10	-3.70	-16.54	0.10	-3.60	-2.94	-1.46	-1.13
205	-0.10	-3.49	-16.60	1.65	-3.53	-2.91	1.37	-1.12
206	0.11	-3.34	-16.64	5.16	-3.55	-2.70	-1.01	-1.15
207	0.02	-4.27	-16.57	-0.97	-3.50	-2.94	9.24	-1.12
208	0.27	-7.69	-16.18	9.78	-3.44	-1.72	2.71	-1.02
209	-0.04	-7.39	-15.94	-9.32	-3.51	-1.63	1.56	-1.04
210	-0.07	-7.15	-16.15	14.88	-3.47	-1.77	-0.45	-1.06
211	0.08	-7.65	-16.49	5.04	-3.52	-1.64	-0.63	-1.05
212	-0.03	-7.56	-16.13	-45.52	-3.43	-1.85	-3.97	-1.04
213	0.07	-6.72	-16.27	-34.23	-3.53	-1.73	-1.11	-1.03
214	0.17	-7.68	-16.15	2.33	-3.41	-1.95	2.65	-1.04
215	0.23	-7.81	-16.12	8.35	-3.41	-1.63	-1.63	-1.05
216	-0.06	-7.14	-16.07	-6.69	-3.47	-1.52	1.36	-1.04
217	-0.18	-7.44	-16.05	-10.69	-3.52	-1.66	4.16	-1.04
218	-0.16	-7.64	-16.02	4.38	-3.55	-1.58	-0.43	-1.03
219	-0.09	-7.83	-16.27	51.83	-3.50	-1.72	2.93	-1.05
220	-0.08	-7.89	-16.36	-26.86	-3.50	-1.95	1.99	-1.04
221	-0.24	-7.62	-16.20	-13.42	-3.47	-1.61	-0.73	-1.07
222	0.22	-7.37	-16.13	-6.98	-3.48	-1.67	5.77	-1.05
223	-0.05	-8.48	-16.17	7.62	-3.47	-1.88	1.22	-1.02
224	0.08	-2.72	-7.93	-0.84	-2.21	-1.51	1.68	-0.53
225	0.06	-2.53	-7.92	0.54	-2.21	-1.54	1.26	-0.55
226	-0.03	-2.53	-8.02	3.16	-2.19	-1.67	1.70	-0.53
227	-0.03	-2.64	-7.78	3.03	-2.22	-1.61	-2.87	-0.53
228	-0.07	-2.56	-7.82	-6.44	-2.22	-1.52	-0.46	-0.53
229	0.06	-2.67	-7.95	7.79	-2.17	-1.65	-0.30	-0.54
230	0.03	-2.67	-7.73	-7.24	-2.21	-1.63	-0.68	-0.53
231	0.08	-2.32	-7.83	-8.27	-2.18	-1.51	0.09	-0.52
232	-0.01	-2.60	-7.85	-3.83	-2.19	-1.56	-0.25	-0.54

233	0.14	-2.88	-7.84	2.24	-2.19	-1.51	0.55	-0.54
234	-0.05	-2.42	-7.90	-4.40	-2.19	-1.60	0.13	-0.54
235	-0.02	-2.55	-7.79	-11.70	-2.22	-1.55	-0.60	-0.53
236	0.19	-2.48	-7.84	12.76	-2.20	-1.54	3.63	-0.53
237	-0.02	-2.83	-7.95	2.25	-2.23	-1.51	-0.31	-0.53
238	-0.12	-2.61	-7.96	13.30	-2.23	-1.60	-0.31	-0.51
239	0.10	-2.52	-7.69	18.92	-2.22	-1.58	0.44	-0.54

k	a24k	a25k	a26k	a27k	a28k	a29k	a30k	a31k
200	3.99	-5.95	11.60	-5.28	-9.88	1.58	8.77	-45.86
201	0.55	-5.96	10.24	-5.13	-9.75	1.70	8.73	-46.07
202	-1.77	-6.09	11.31	-4.91	-9.69	1.63	8.59	-45.79
203	1.56	-5.98	10.38	-5.02	-10.51	1.79	8.69	-45.10
204	1.56	-6.08	10.34	-5.40	-10.12	1.67	8.71	-45.35
205	2.13	-6.31	10.69	-5.05	-10.29	1.70	8.93	-45.65
206	2.79	-6.28	10.42	-5.44	-10.01	1.75	8.49	-46.18
207	-0.59	-6.19	11.15	-5.23	-10.73	1.85	8.51	-46.18
208	-0.86	-7.96	-3.95	-3.60	-9.52	0.68	11.12	-49.79
209	0.81	-8.49	-5.59	-3.29	-8.62	0.65	11.07	-50.16
210	-1.82	-8.57	-4.86	-3.74	-8.55	0.75	11.01	-50.88
211	-0.18	-8.60	-5.31	-3.43	-8.14	0.70	11.15	-50.58
212	2.61	-8.70	-6.29	-3.63	-8.86	0.61	10.85	-50.54
213	-3.08	-8.69	-3.95	-3.63	-9.03	0.68	11.10	-50.64
214	-1.21	-8.67	-3.89	-3.44	-8.30	0.57	10.92	-50.48
215	-0.63	-8.21	-5.17	-3.56	-8.01	0.58	10.75	-51.30
216	-2.33	-8.45	-6.57	-3.49	-8.58	0.62	10.57	-49.36
217	-1.75	-8.63	-4.28	-3.72	-9.07	0.52	11.17	-51.38
218	-1.16	-8.33	-3.96	-3.48	-8.78	0.65	11.05	-52.15
219	-1.30	-8.25	-5.48	-3.26	-8.99	0.67	10.97	-50.97
220	-2.13	-8.82	-5.83	-3.54	-8.76	0.61	10.94	-50.33
221	0.26	-8.28	-5.24	-4.00	-8.47	0.67	10.96	-50.23
222	1.12	-8.32	-6.96	-3.18	-9.03	0.48	11.17	-52.19
223	2.79	-8.82	-5.29	-3.80	-8.83	0.67	10.90	-50.60
224	0.34	-4.36	2.59	-2.85	-4.74	0.74	5.72	-22.17
225	-0.31	-4.18	2.99	-3.02	-4.17	0.66	5.46	-22.03
226	-0.12	-4.41	2.85	-2.84	-4.73	0.61	5.85	-21.93
227	-0.33	-4.36	3.11	-2.91	-4.70	0.71	5.44	-21.77
228	0.44	-4.44	3.35	-2.94	-4.84	0.57	5.52	-22.25
229	0.33	-4.39	3.32	-2.77	-4.87	0.60	5.59	-22.24
230	-0.06	-4.13	2.39	-2.85	-4.78	0.58	5.64	-21.99
231	0.24	-4.30	4.62	-2.89	-4.63	0.62	5.66	-22.01
232	0.96	-4.38	4.07	-3.03	-4.86	0.60	5.48	-22.08
233	-0.65	-4.41	3.56	-2.99	-4.53	0.68	5.63	-21.70
234	0.36	-4.36	3.33	-2.77	-4.45	0.63	5.65	-21.98
235	-0.71	-4.32	3.30	-2.85	-4.91	0.59	5.67	-22.12
236	-0.29	-4.26	3.75	-2.72	-4.75	0.66	5.69	-22.65
237	0.22	-4.19	2.49	-2.91	-4.69	0.71	5.68	-22.50
238	-1.44	-4.36	3.56	-2.73	-5.08	0.70	5.66	-22.30
239	-0.57	-4.42	2.22	-2.74	-4.73	0.58	5.58	-22.22

k	a32k	a33k	a34k	a35k	a36k	a37k	a38k	a39k
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200	14.32	3.22	0.58	-6.43	-0.50	-53.19	-0.36	-0.36
201	14.99	3.34	0.54	-6.74	-0.49	-54.35	-0.39	-0.36
202	14.82	3.28	0.68	-7.03	-0.52	-54.80	-0.41	-0.36
203	14.71	3.23	0.63	-6.98	-0.46	-54.42	-0.37	-0.36
204	14.96	3.30	0.60	-7.13	-0.47	-55.09	-0.35	-0.36
205	14.23	3.28	0.66	-6.96	-0.49	-55.95	-0.36	-0.36
206	15.18	3.28	0.69	-7.34	-0.50	-55.62	-0.37	-0.35
207	14.69	3.23	0.61	-6.88	-0.52	-53.88	-0.36	-0.35
208	9.47	3.48	1.57	-4.65	-0.96	-45.31	-1.00	-0.74
209	9.10	3.48	1.40	-3.70	-0.97	-43.24	-1.04	-0.68
210	9.37	3.51	1.54	-3.31	-0.90	-42.54	-0.99	-0.72
211	9.83	3.51	1.58	-3.74	-0.92	-42.03	-0.94	-0.72
212	9.76	3.52	1.42	-3.73	-0.93	-43.64	-0.96	-0.71
213	9.51	3.50	1.54	-4.16	-0.90	-44.19	-1.03	-0.72
214	9.23	3.38	1.59	-3.62	-0.89	-43.99	-1.02	-0.71
215	9.61	3.50	1.54	-3.80	-0.93	-44.32	-1.01	-0.74
216	8.98	3.46	1.50	-3.79	-0.98	-44.82	-0.98	-0.72
217	9.21	3.52	1.53	-3.92	-0.92	-44.28	-0.94	-0.73
218	9.94	3.55	1.52	-4.54	-0.91	-41.51	-1.02	-0.72
219	9.63	3.53	1.40	-3.79	-0.93	-43.59	-0.99	-0.73
220	9.23	3.46	1.49	-3.60	-0.89	-45.16	-0.99	-0.73
221	8.94	3.48	1.46	-3.51	-0.91	-44.60	-0.97	-0.72
222	9.57	3.47	1.57	-3.09	-0.87	-43.51	-1.02	-0.73
223	9.36	3.49	1.43	-3.84	-0.88	-43.84	-0.98	-0.72
224	6.36	1.72	0.50	-2.51	-0.38	-24.29	-0.56	-0.24
225	6.20	1.80	0.47	-1.97	-0.35	-25.26	-0.57	-0.23
226	6.18	1.80	0.46	-2.30	-0.37	-24.28	-0.51	-0.24
227	6.37	1.78	0.42	-2.20	-0.34	-23.88	-0.53	-0.23
228	6.45	1.78	0.51	-1.83	-0.37	-24.04	-0.54	-0.23
229	7.00	1.71	0.46	-1.80	-0.35	-24.35	-0.53	-0.23
230	6.17	1.71	0.53	-2.11	-0.33	-25.11	-0.55	-0.21
231	6.09	1.76	0.48	-2.36	-0.36	-23.07	-0.53	-0.23
232	5.82	1.77	0.43	-2.38	-0.37	-24.22	-0.56	-0.22
233	5.85	1.74	0.46	-2.27	-0.32	-24.46	-0.55	-0.25
234	6.48	1.79	0.52	-2.15	-0.37	-23.83	-0.56	-0.25
235	6.59	1.79	0.45	-2.00	-0.36	-24.26	-0.56	-0.23
236	6.53	1.80	0.54	-2.13	-0.35	-24.30	-0.53	-0.25
237	6.32	1.77	0.46	-2.42	-0.32	-24.11	-0.54	-0.23
238	6.42	1.76	0.48	-2.25	-0.34	-24.32	-0.51	-0.22
239	6.80	1.81	0.45	-2.12	-0.34	-25.08	-0.53	-0.24

k	a40k	a41k	a42k	a43k	a44k	a45k	a46k	a47k
200	0.46	7.85	-0.13	6.00	-0.66	-9.63	27.73	-0.71
201	0.49	7.88	1.56	5.01	5.66	-9.86	37.21	-0.38
202	0.50	7.82	-0.21	5.62	3.39	-9.78	36.01	-3.02
203	0.47	7.87	1.17	5.57	6.66	-9.76	28.22	-1.35
204	0.47	7.73	-1.49	4.73	-0.17	-9.85	34.81	-0.69
205	0.51	7.78	0.66	4.81	0.90	-9.79	27.69	-1.68
206	0.48	7.68	0.46	5.55	0.83	-9.68	33.16	0.42
207	0.49	7.68	1.72	5.43	-3.05	-9.64	30.82	2.49
208	0.15	6.36	-2.14	8.22	-2.43	-9.98	99.11	-0.05
209	0.19	6.61	0.32	8.42	3.47	-9.67	83.51	-0.45
210	0.14	6.61	-1.32	7.86	6.02	-10.02	86.68	1.50

211	0.16	6.66	2.90	7.31	0.41	-9.81	87.51	1.78
212	0.22	6.31	-1.30	7.70	-0.04	-9.87	87.42	-0.67
213	0.15	6.49	-1.79	7.85	5.42	-9.54	85.96	-0.59
214	0.22	6.46	0.30	8.60	2.92	-9.67	90.41	0.31
215	0.20	6.38	-0.86	8.71	4.58	-9.76	83.46	-0.82
216	0.19	6.15	0.25	7.76	-2.22	-9.68	91.93	-0.68
217	0.20	6.57	-1.20	8.05	-1.02	-9.91	86.11	1.29
218	0.17	6.27	-1.14	7.68	2.32	-10.04	91.11	-1.23
219	0.15	6.39	-0.80	7.44	5.12	-9.89	88.36	0.45
220	0.15	6.49	-3.00	8.07	4.47	-9.78	95.29	-0.28
221	0.19	6.40	0.75	7.85	6.16	-9.63	90.57	-0.67
222	0.18	6.41	1.30	8.33	0.11	-9.71	81.50	3.06
223	0.22	6.24	-1.01	8.04	5.86	-9.95	86.36	-0.02
224	0.27	4.44	-0.04	3.08	2.50	-6.38	20.25	-1.03
225	0.24	4.52	-0.34	3.01	-1.29	-6.25	16.20	-0.21
226	0.25	4.55	-0.39	3.51	-0.60	-6.37	18.03	-1.33
227	0.26	4.52	0.03	3.32	-1.50	-6.28	24.04	-0.27
228	0.26	4.70	-0.29	3.18	-0.44	-6.27	17.92	-1.90
229	0.25	4.57	0.92	3.34	1.93	-6.40	22.77	-0.80
230	0.28	4.73	-0.66	2.98	4.13	-6.28	15.11	-0.47
231	0.24	4.58	0.43	3.54	-0.58	-6.26	25.20	0.44
232	0.27	4.64	0.84	3.11	0.14	-6.38	22.13	-0.19
233	0.25	4.51	0.22	3.69	-2.72	-6.35	21.97	0.80
234	0.27	4.51	-0.11	3.55	1.86	-6.42	25.28	-0.06
235	0.26	4.67	1.15	3.35	0.05	-6.42	23.04	-1.52
236	0.27	4.61	-0.75	3.40	-1.85	-6.07	16.17	1.21
237	0.26	4.65	-2.07	3.25	1.95	-6.27	18.98	-0.86
238	0.22	4.73	-0.73	3.27	-0.72	-6.28	20.99	0.14
239	0.24	4.67	1.08	3.59	-0.16	-6.19	18.79	-0.29

k	a48k	a49k	a50k	a51k	a52k	a53k	a54k	a55k
200	-14.58	2.15	13.52	1.39	98.94	38.13	-5.24	-4.83
201	-14.85	2.12	13.25	-1.46	105.91	38.10	-5.40	-4.96
202	-14.77	2.04	13.42	-0.83	108.07	37.94	-5.43	-4.34
203	-14.29	2.04	13.65	1.22	104.62	38.41	-5.42	-4.76
204	-14.52	2.02	13.38	-2.48	103.69	38.64	-5.38	-5.42
205	-13.83	2.00	13.13	-1.90	100.76	38.11	-5.22	-5.16
206	-14.72	2.09	13.54	1.63	99.44	36.76	-5.42	-5.11
207	-14.24	2.13	13.38	-2.68	106.04	38.62	-5.44	-5.01
208	-18.04	2.04	17.33	-4.13	80.49	45.11	-4.33	-0.42
209	-18.35	1.98	18.32	0.33	75.08	44.55	-4.26	-0.29
210	-18.34	1.91	17.45	-5.83	78.06	45.51	-4.19	0.29
211	-17.86	2.06	18.77	-1.57	76.22	44.80	-4.37	-0.99
212	-18.15	2.07	17.42	3.49	68.79	44.77	-4.29	-0.51
213	-17.90	1.96	18.27	2.48	78.99	45.19	-4.26	-1.08
214	-18.00	2.17	16.74	1.35	81.10	45.50	-4.36	0.34
215	-18.07	2.03	18.09	-0.23	80.85	45.97	-4.42	-0.63
216	-17.82	2.01	17.98	-1.08	76.05	44.86	-4.26	0.54
217	-18.26	1.97	17.85	2.39	76.52	45.77	-4.35	-0.24
218	-18.22	1.93	17.46	-2.42	78.00	44.64	-4.34	-0.75
219	-18.12	2.02	17.20	-2.55	79.55	45.31	-4.45	-0.57
220	-18.32	2.05	18.01	-0.15	73.51	44.83	-4.23	-1.39
221	-18.65	1.85	17.38	1.55	75.38	45.01	-4.24	0.04

222	-18.24	2.01	17.33	3.21	79.54	44.83	-4.33	-0.20
223	-17.86	2.09	17.70	-2.31	78.34	45.15	-4.28	0.51
224	-9.08	1.04	9.54	-0.38	45.23	20.55	-2.26	-2.70
225	-8.99	1.11	9.50	-2.62	47.46	20.68	-2.31	-2.31
226	-8.93	1.09	9.73	-1.20	48.83	20.40	-2.28	-2.68
227	-8.58	1.14	9.31	-0.97	45.76	20.14	-2.25	-2.35
228	-8.73	1.14	9.72	1.59	49.19	20.34	-2.30	-2.72
229	-9.14	1.21	9.50	0.67	47.88	20.41	-2.25	-2.50
230	-8.91	1.09	9.32	0.39	49.39	20.28	-2.25	-2.31
231	-9.06	1.07	9.52	1.36	48.38	20.82	-2.31	-2.84
232	-8.97	1.08	9.76	0.66	47.42	20.49	-2.30	-2.82
233	-8.95	1.16	9.71	-0.24	48.65	20.33	-2.32	-2.82
234	-8.79	1.05	9.83	0.41	50.36	20.50	-2.29	-2.90
235	-8.84	1.09	9.42	0.66	45.98	20.80	-2.30	-2.95
236	-9.18	1.08	9.80	-2.83	45.60	19.75	-2.31	-2.90
237	-9.04	1.19	9.56	1.14	49.97	20.36	-2.42	-2.38
238	-9.17	1.08	9.84	2.18	49.18	19.80	-2.27	-2.61
239	-8.69	1.11	9.76	2.64	47.25	20.56	-2.31	-2.38

k	a56k	a57k	a58k	a59k	a60k	a61k	a62k	a63k
200	-1.36	-29.64	0.29	-1.93	12.05	0.24	6.43	22.33
201	3.10	-29.13	0.30	-1.91	12.36	0.07	-7.30	-0.91
202	-5.52	-29.28	0.29	-2.07	12.24	0.09	-9.28	8.15
203	-4.47	-29.11	0.30	-1.96	12.32	0.14	-6.20	8.77
204	-2.98	-29.56	0.32	-1.95	12.15	0.15	7.99	31.91
205	-1.35	-29.64	0.30	-1.68	12.64	0.24	-18.47	-13.38
206	1.26	-30.90	0.30	-1.91	12.30	0.25	6.12	4.06
207	2.52	-29.07	0.29	-1.96	12.47	0.22	3.27	-13.81
208	-0.08	-22.82	0.23	-3.30	8.90	-1.48	9.45	-22.79
209	-0.46	-23.73	0.20	-3.31	9.41	-1.13	3.74	-10.82
210	3.20	-22.42	0.22	-3.21	9.23	-1.46	-3.21	9.83
211	-1.51	-22.56	0.20	-3.15	8.81	-1.43	-20.84	-1.75
212	3.07	-23.26	0.20	-3.21	9.20	-1.39	0.80	-30.30
213	-2.33	-23.08	0.23	-2.69	8.98	-1.30	-19.53	8.55
214	-6.54	-23.51	0.22	-3.41	8.76	-1.14	27.04	6.01
215	-3.79	-21.32	0.21	-3.00	8.93	-1.33	-17.66	-2.79
216	-0.24	-23.24	0.22	-3.03	9.09	-1.24	6.07	24.80
217	3.44	-24.06	0.20	-3.67	8.63	-1.29	-8.15	19.18
218	-1.92	-22.05	0.20	-3.29	9.00	-1.18	6.54	10.41
219	0.30	-22.68	0.21	-3.36	8.67	-1.31	-10.55	1.54
220	-2.80	-23.71	0.20	-3.40	8.54	-1.20	16.87	-20.57
221	-4.02	-23.59	0.23	-3.23	8.89	-1.33	-29.66	-8.53
222	3.71	-22.32	0.22	-3.65	9.42	-1.27	-7.45	20.01
223	4.01	-22.74	0.20	-3.23	9.05	-1.11	-0.92	8.73
224	-0.07	-13.13	0.10	-1.31	4.94	0.10	-2.81	9.80
225	3.00	-13.34	0.13	-1.17	5.09	0.13	9.49	1.20
226	1.83	-13.14	0.12	-1.12	4.86	0.09	-6.43	7.09
227	2.44	-12.89	0.11	-1.27	4.92	0.14	11.11	3.43
228	-1.82	-12.71	0.11	-1.30	5.04	0.23	5.82	2.18
229	2.60	-13.68	0.12	-1.18	4.89	0.13	-13.54	-2.96
230	-0.58	-12.59	0.13	-1.32	5.16	0.14	5.68	2.50
231	-1.66	-13.08	0.12	-1.37	5.20	0.05	0.12	4.41
232	4.51	-13.15	0.12	-1.50	4.62	0.10	-3.69	-7.06

233	1.30	-12.99	0.10	-1.29	5.15	0.15	-8.75	-0.68
234	3.05	-13.52	0.12	-1.26	5.30	0.06	-5.26	-12.29
235	1.63	-13.65	0.11	-1.20	5.12	0.00	-9.46	4.53
236	4.38	-13.38	0.12	-1.23	5.11	0.11	-0.87	-0.28
237	-2.09	-14.06	0.12	-1.23	4.84	0.01	-7.24	-4.51
238	2.60	-13.88	0.11	-1.47	4.69	0.15	-10.41	-5.46
239	1.90	-12.57	0.11	-1.22	5.22	0.13	12.39	8.11

For all Gaussian functions g_{lk} for all images f_k , the exponential factor $\alpha = 1/\epsilon$ is given by $\epsilon = 2.0615528128088303$.

- 5 Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not describe the material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . The hypothesis testing may use as alternative hypothesis that the filter does describe the material classifier. The alternative hypothesis may be given by $H_1: \exists m, m': \mu_m \neq \mu_{m'}$. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each material, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for material for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:
- 10 $mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J)$. The hypothesis testing may comprise determining a Mean Sum of Squares between, $mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1)$. The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

- 25 Herein, I is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For exam-

ple, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.05$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

5

Figure 2 shows experimental results of using the spot shape filter to calculate the material feature for different material groups. Specifically, the material feature φ_{2m} is shown for different materials, namely from left to right for a wooden board denoted board, a fabric, a dark skin, a pale skin, and a highly translucent skin denoted trans, and illustrates the separation of skin materials (right) from other materials (left). The spot shape filter passes the hypothesis testing. The p-value computed from the F-test is 0.0076. Thus, the Null-hypothesis H_0 can be rejected with respect to a significance level of 0.01. Further, the alternative hypothesis H_1 can be accepted with respect to a significance level of 0.01. Thus, the image filter is distinguishes at least two material classifiers. In addition, Figure 2 shows the mean value of the feature per material, $\bar{\varphi}_m$, as the line within the respective box. Vertical bars mark the Q1-1.5 interquartile range (IQR) and Q3+1.5 IQR, wherein $IQR = Q3 - Q1$, with Q1 and Q3 being the first and third quartiles. The box marks the Q1 and Q3 quartiles. Outliers are plotted as points. Figure 2 shows, that this material feature separates all skin samples from the non-skin samples. In other words, the expectation values of the Skin Features for skin and non-skin materials differ.

20

Figure 1 moreover shows, in a highly schematic illustration, an exemplary embodiment of the detector 110, wherein the detector 110 specifically may be embodied as a camera 136 and/or may be part of a camera 136. The camera 136 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 1 further shows an embodiment of a detector system 138, which, besides the at least one detector 110, comprises one or more beacon devices 140, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 1 further shows an exemplary embodiment of a human-machine interface 142, which comprises the at least one detector system 138 and, further, an entertainment device 144, which comprises the human-machine interface 142. The figure further shows an embodiment of a tracking system 146 for tracking a position of the object 112, which comprises the detector system 138. The components of the devices and systems shall be explained in further detail below.

35

Fig. 1 further shows an exemplary embodiment of a scanning system 148 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 148 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well

as, optionally, at least one further illumination source, not depicted here. The illumination source 124, generally, is configured for emitting at least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 140 and/or on a surface of the object 112. The scanning system 148 may be designed to
5 generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 148, specifically the detector 110, by using the at least one detector 110.

10 The detector 110, besides the optical sensors 120, comprises the at least one evaluation device 132, having e.g. at least one image analysis device 150 and/or the at least one position evaluation device 152, as symbolically depicted in Fig. 1. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially
15 combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 154 and/or by one or more interfaces, as symbolically depicted in Fig. 1. Further, the one or more connectors 154 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connect-
20 or 154, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 156 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

25 In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 158, the position of which may be manipulated by a user 160. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 160 himself or herself may be considered as the object 112, the position of which shall be detected.

30 As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 156. Further, the detector 110 may comprise the at least one transfer device 129 such as one or more optical systems, preferably comprising one or more lenses. An opening 162 inside the housing 156, which, preferably, is located concentrically with
35 regard to the optical axis 128 of the detector 110, preferably defines a direction of view 164 of the detector 110. A coordinate system 166 may be defined, in which a direction parallel or anti-parallel to the optical axis 128 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 128 may be defined as transversal directions. In the coordinate system 166, symbolically depicted in Fig. 1, a longitudinal direction is denoted by z, and trans-
40 versal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 142, in order to provide at least one item of information to a machine 168. In the embodiments schematically depicted in Fig. 1, the machine 168 may be a computer and/or may comprise a computer. Other
5 embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 168, such as into the computer.

As outlined above, Fig. 1 also depicts an example of a tracking system 146, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system
10 146 comprises the detector 110 and at least one track controller 170. The track controller 170 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 170 may be an independent device and/or may be fully or partially integrated into the machine 168, specifically the computer, as indicated in Fig. 1 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 142 may form part of an entertainment device 144. The machine 168, specifically the computer, may also form part of the entertainment device 144. Thus, by means of the user 160 functioning as the object 112 and/or by means of the user 160 handling a control device functioning as the object 112, the user 160
20 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Fig. 1 also depicts an example of an inertial measurement unit 172 for use in an electronic device. The inertial measurement unit 172 is adapted to receive data determined by the detector
25 110. The inertial measurement unit 172 further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is
30 adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

The inertial measurement unit 172 may comprise the detector 110 and/or may be connected to the detector 110, via at least one data connection. The evaluation device 132 and/or at least one processing device of the inertial measurement unit 172 may be configured for determining at least one combined distance information, in particular using at least one recursive filter. The recursive filter may be configured for determining the combined distance information considering further sensor data and/or further parameters such as further sensor data from the further
40 sensor of the inertial measurement unit 172.

List of reference numbers

110	detector
112	object
116	sensor element
118	matrix
120	optical sensor
122	light-sensitive area
124	illumination source
126	illumination light beam
128	optical axis
129	transfer device
130	reflection light beam
132	evaluation device
134	divider
136	camera
138	detector system
140	beacon device
142	human-machine-interface
144	entertainment device
146	tracking system
148	scanning system
150	image analysis device
152	position evaluation device
154	connector
156	housing
158	control device
160	user
162	opening
164	direction of view
166	coordinate system
168	machine
170	track controller
172	inertial measurement unit

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Claims

1. A detector (110) for identifying at least one material property m comprising
 - at least one sensor element (116) comprising a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein the sensor element (116) is configured for recording at least one reflection image of a light beam originating from at least one object (112);
 - at least one evaluation device (132) configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device (132) is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter $\Phi_{1\text{other}}$ which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device (132) is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter $\Phi_{2\text{other}}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, wherein the evaluation device (132) is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .
2. The detector (110) according to the preceding claim, , wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifi-

ers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

- 5 3. The detector (110) according to the preceding claim, wherein $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, most preferably $p \leq 0.01$.
4. The detector (110) according to any one of the preceding claims, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like.
- 10 5. The detector (110) according to any one of the preceding claims, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.60$, preferably by $|\rho_{\Phi_{1\text{other}}, \Phi_z}| \geq 0.80$.
- 15 6. The detector (110) according to any one of the preceding claims, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.
- 20 7. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between φ_{1z} , φ_{2m} and z , m .
- 25 8. The detector (110) according to any one of the preceding claims, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\varphi_{1z}, \varphi_{2m})$ and/or $m(\varphi_{1z}, \varphi_{2m})$.
9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.
- 30 10. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature φ_{1z} or φ_{1m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.
- 35 11. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of φ_{1z} and φ_{2m} to the reflection image.
- 40

12. The detector (110) according to any one of the preceding claims, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element (116), wherein the evaluation device (132) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the combined signal Q and the distance feature φ_{1z} for determining the distance feature φ_{1z} .
13. The detector (110) according to any one of the preceding claims, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature φ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.
14. The detector (110) according to any one of the preceding claims, wherein the sensor element (116) comprises at least one CMOS sensor.
15. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.
16. The detector (110) according to the preceding claim, wherein the illumination source (124) comprises at least one laser source and at least one diffractive optical element.
17. A detector system (138), the detector system (138) comprising at least one detector (110) according to any one of the preceding claims, the detector system (138) further comprising at least one beacon device (140) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (140) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
18. A human-machine interface (142) for exchanging at least one item of information between a user (160) and a machine (168), wherein the human-machine interface (142) comprises at least one detector system (138) according to the preceding claim, wherein the at least one beacon device (140) is adapted to be at least one of directly or indirectly attached to

the user (160) and held by the user (160), wherein the human-machine interface (142) is designed to determine at least one position of the user (160) by means of the detector system (138), wherein the human-machine interface (142) is designed to assign to the position at least one item of information.

- 5
19. An entertainment device (144) for carrying out at least one entertainment function, wherein the entertainment device (144) comprises at least one human-machine interface (142) according to the preceding claim, wherein the entertainment device (144) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (142), wherein the entertainment device (144) is designed to vary the entertainment function in accordance with the information.
- 10
20. A tracking system (146) for tracking a position of at least one movable object, the tracking system (146) comprising at least one detector system (138) according to any one of the preceding claims referring to a detector system, the tracking system (146) further comprising at least one track controller (170), wherein the track controller (170) is adapted to track a series of positions of the object (112) at specific points in time.
- 15
21. A scanning system (148) for determining a depth profile of a scenery, the scanning system (148) comprising at least one detector (110) according to any of the preceding claims referring to a detector, the scanning system (148) further comprising at least one illumination source adapted to scan the scenery with at least one light beam.
- 20
22. A camera (136) for imaging at least one object (112), the camera (136) comprising at least one detector (110) according to any one of the preceding claims referring to a detector.
- 25
23. An inertial measurement unit (172) for use in an electronic device, wherein the inertial measurement unit (172) is adapted to receive data determined by at least one detector (110) according to any one of the preceding claims referring to a detector, wherein the inertial measurement unit (172) further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.
- 30
- 35
- 40 24. A method for determining at least one material property of at least one object (112) by using at least one detector (110) according to any one of the preceding claims referring to a detector, the method comprising the following steps:

a) determining at least one reflection image of the object (112) by using at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122);

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device (132), wherein the evaluating comprises:

b1) determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof or a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by

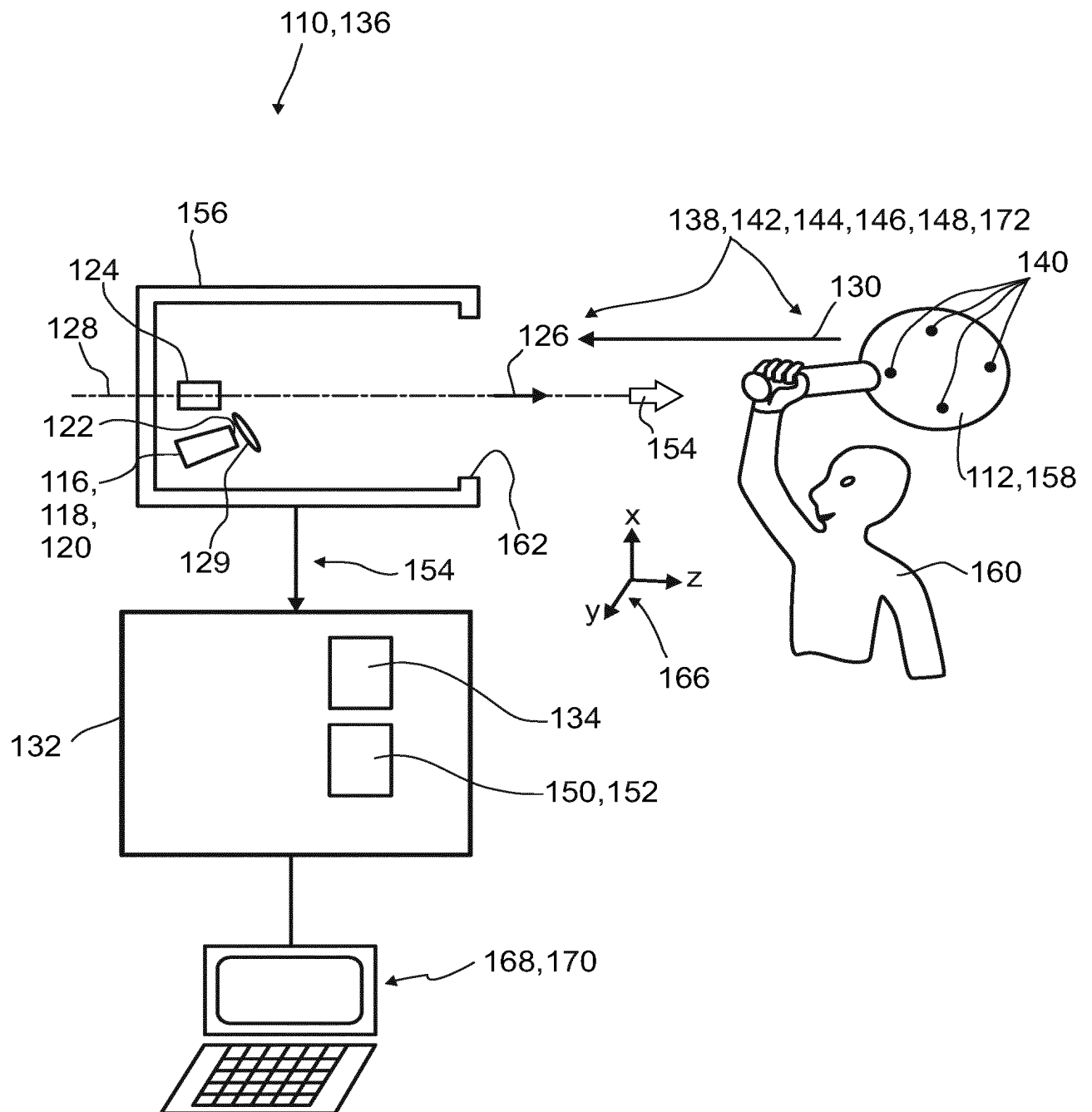
$|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof,

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

25. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

FIG. 1



2/2

FIG.2

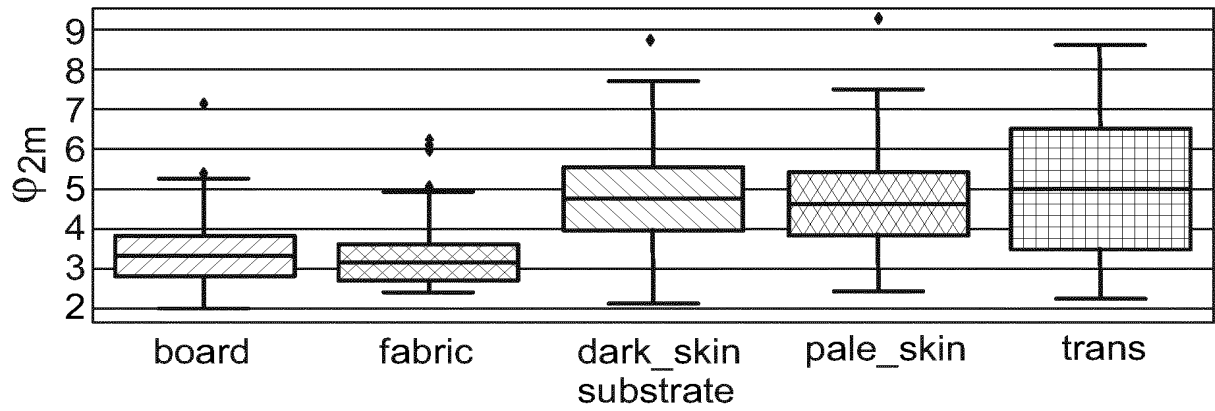
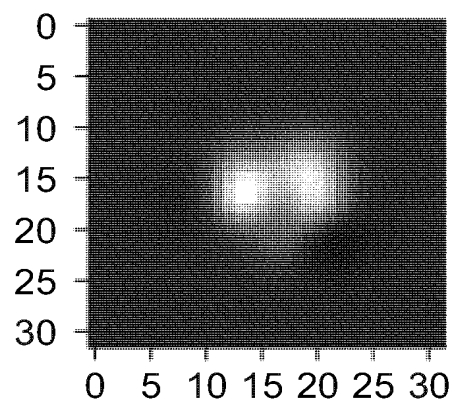


FIG.3



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Application Number	
Filing Date	2021-09-15
First Named Inventor	Schick, Friedrich
Title	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY
Art Unit	
Examiner Name	
Attorney Docket Number	27843-1809 182037US01

SIGNATURE of Applicant or Patent Practitioner

Signature	/Christopher M. Goff/	Date (Optional)	
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NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.



*Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Doc Code: PA..

Document Description: Power of Attorney

PTO/AIA/82B (07-13)

Approved for use through 03/31/2021. OMB 0651-0035

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number

Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)



I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

76656

OR



I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:



The address associated with the above-mentioned Customer Number

OR



The address associated with Customer Number:

OR



Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

trinamiX GmbH



Inventor or Joint Inventor (title not required below)



Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)



Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)



Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

2020/06/09

Name

Dr. Hans Graf

Title

Patent Attorney

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.



Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Friedrich		Schick		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen		Country of Residence ⁱ	DE	
Mailing Address of Inventor:					
Address 1		Industriestrasse 35			
Address 2					
City	Ludwigshafen		State/Province		
Postal Code		67063	Country ⁱ	DE	
Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Peter		Schillen		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen		Country of Residence ⁱ	DE	
Mailing Address of Inventor:					
Address 1		Industriestrasse 35			
Address 2					
City	Ludwigshafen		State/Province		
Postal Code		67063	Country ⁱ	DE	
Inventor	3				Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Patrick		Schindler	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Ludwigshafen	Country of Residence ⁱ	DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country ⁱ	DE	
Inventor	4	Remove		
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Andre		Schmidt	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Ludwigshafen	Country of Residence ⁱ	DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country ⁱ	DE	
Inventor	5	Remove		
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Michael		Eberspach	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Ludwigshafen	Country of Residence ⁱ	DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country ⁱ	DE	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Inventor	6				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Christian		Lennartz		
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service					
City	Ludwigshafen	Country of Residence ⁱ		DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35				
Address 2					
City	Ludwigshafen	State/Province			
Postal Code	67063	Country ⁱ	DE		

Inventor	7				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Robert		Send		
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service					
City	Karlsruhe	Country of Residence ⁱ		DE	

Mailing Address of Inventor:

Address 1	Luisenstr. 25				
Address 2					
City	Karlsruhe	State/Province			
Postal Code	76137	Country ⁱ	DE		

Inventor	8				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Lars		Diesselberg		
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service					
City	Karlsruhe	Country of Residence ⁱ		DE	

Mailing Address of Inventor:

Address 1	Geibelstr. 22				
Address 2					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

City	Karlsruhe	State/Province	
Postal Code	76185	Country ⁱ	DE
Inventor	9	Remove	
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Heiko		Hengen
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service			
City	Steinweiler	Country of Residence ⁱ	DE

Mailing Address of Inventor:

Address 1	Siedlerweg 8		
Address 2			
City	Steinweiler	State/Province	
Postal Code	76872	Country ⁱ	DE
Inventor	10	Remove	
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Ingmar		Bruder
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service			
City	Ludwigshafen	Country of Residence ⁱ	DE

Mailing Address of Inventor:

Address 1	Industriestrasse 35		
Address 2			
City	Ludwigshafen	State/Province	
Postal Code	67063	Country ⁱ	DE
Inventor	11	Remove	
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Jakob		Unger
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service			
City	Ludwigshafen	Country of Residence ⁱ	DE

Mailing Address of Inventor:

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01	
		Application Number		
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY			

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country i	DE	
Inventor	12	Remove		
Legal Name				
Prefix	Given Name	Middle Name	Family Name	Suffix
	Christian		Bonsignore	
Residence Information (Select One) US Residency ☒ Non US Residency Active US Military Service				
City	Ludwigshafen	Country of Residence i	DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country i	DE	
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add				

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	76656		
Email Address	uspatents@armstrongteasdale.com	Add Email	Remove Email

Application Information:

Title of the Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY			
Attorney Docket Number	27843-1809 182037US01	Small Entity Status Claimed	☐	
Application Type	Nonprovisional			
Subject Matter	Utility			
Total Number of Drawing Sheets (if any)	2	Suggested Figure for Publication (if any)	1	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not be** the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	76656		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	a 371 of international	PCT/EP2020/056759	2020-03-13
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
19163250.4	EP	2019-03-15	

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	trinamiX GmbH	
Mailing Address Information For Applicant:		
Address 1	Industriestrasse 35	
Address 2		
City	Ludwigshafen am Rhein	State/Province
Country	DE	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the **INITIAL** filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Christopher M. Goff/		Date (YYYY-MM-DD)	2021-09-15
First Name	Christopher	Last Name	M. Goff	Registration Number
Additional Signature may be generated within this form by selecting the Add button.				Add

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		
	Filing Date		2021-09-15
	First Named Inventor	Schick, Friedrich	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	27843-1809 182037US01	

U.S.PATENTS						Remove
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Patentee or Applicant of cited Document	Pages,Columns,Lines where Relevant Passages or Relevant Figures Appear
	1					

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**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number		
Filing Date		2021-09-15
First Named Inventor	Schick, Friedrich	
Art Unit		
Examiner Name		
Attorney Docket Number	27843-1809 182037US01	

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Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2021-09-15
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①9 **BUNDESREPUBLIK
DEUTSCHLAND**



**DEUTSCHES
PATENT- UND
MARKENAMT**

⑫ **Offenlegungsschrift**
⑩ **DE 198 46 619 A 1**

②1 Aktenzeichen: 198 46 619.6
②2 Anmeldetag: 9. 10. 1998
④3 Offenlegungstag: 27. 4. 2000

⑤1 Int. Cl. 7:
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Die folgenden Angaben sind den vom Anmelder eingereichten Unterlagen entnommen

⑤4 Vorrichtung und Verfahren zur Bestimmung der Qualität strukturierter Oberflächen

⑤7 Die Erfindung betrifft eine Vorrichtung zur quantifizierten Bestimmung der Qualität strukturierter Oberflächen, die eine erste optische Einrichtung aufweist, welche Licht unter einem vorbestimmten Winkel auf die Meßfläche ausstrahlt, und eine zweite optische Einrichtung beinhaltet, welche mit wenigstens einem Fotosensor das von der Meßfläche reflektierte Licht aufnimmt. Die optischen Einrichtungen sind derart beschaffen, daß das reflektierte Licht durch die Struktur der Meßfläche beeinflußt wird, und in einer Auswerteeinrichtung, die zur Steuerung des Meßablaufs vorgesehen ist und die eine Prozessoreinrichtung und eine Speichereinrichtung aufweist, wird das reflektierte Licht ausgewertet und daraus wenigstens eine Strukturkennzahl abgeleitet, welche wenigstens eine strukturbedingte Eigenschaft der Oberfläche charakterisiert.

DE 198 46 619 A 1

Beschreibung

Die vorliegende Erfindung betrifft eine Vorrichtung und ein Verfahren zur Bestimmung der Qualität strukturierter Oberflächen.

Unter der Qualität einer Oberfläche bzw. einer strukturierten Oberfläche sollen hier die physikalischen Eigenschaften einer Oberfläche verstanden werden, die das Aussehen einer Oberfläche für den menschlichen Betrachter bestimmen.

Insbesondere Eigenschaften wie Struktur, Farbe, Helligkeit der Farbe, Glanz, Abbildungsschärfe (engl. DOI), Glanzschleier, Oberflächentexturen und Oberflächenwelligkeiten (engl. orange peel) etc. kennzeichnen strukturierte Oberflächen.

Die Beschaffenheit der sichtbaren Oberflächen ist eine wesentliche Eigenschaft von Gegenständen des täglichen Lebens, wie beispielsweise Einrichtungsgegenständen und Gebrauchsgegenständen, wie Autos und dergleichen.

Im Bereich der Kraftfahrzeugtechnik wird viel Wert auf den visuellen Eindruck der sichtbaren Oberflächen gelegt. Im folgenden werden die technischen Probleme, die bei der Gestaltung von Oberflächen im Kraftfahrzeugbereich entstehen, näher erläutert, ohne jedoch die vorliegende Erfindung in ihrer Anwendung in irgendeiner Weise einzuschränken.

Kraftfahrzeugkarosserien werden üblicherweise mit einer Hochglanz- oder Metalllackierung versehen, deren Reflexionsvermögen oder Glanzkennwert den entsprechenden Werten von anderen Flächen, wie z. B. Möbeln, weit überlegen ist. Der hohe Glanz der verwendeten Lacke und die relativ großen, ebenen Flächen erfordern eine außerordentlich sorgfältige Vorbereitung der zu lackierenden Flächen und eine besonders sorgfältige Auftragung des Lacks.

Im Stand der Technik sind verschiedene Verfahren, Vorrichtungen und Apparate bekannt, wie z. B. in der DE 44 34 203 A1 beschrieben, mit denen die visuellen Eigenschaften und speziell das Reflexionsverhalten von Oberflächen bestimmt werden können.

Weiterhin werden in der Automobil-Industrie im Innenraum von Kraftfahrzeugen eine Vielzahl von strukturierten Oberflächen verarbeitet, die aus natürlichen Stoffen, wie z. B. Holz oder Kunststoff bestehen.

Der visuelle Eindruck strukturierter Oberflächen hängt wesentlich von der Ausgeprägtheit der Struktur der Oberfläche ab, wird jedoch auch von den anderen obengenannten optischen Kenngrößen bestimmt.

Die Automobil-Hersteller beschäftigen heute eine Vielzahl von Prüfern und verwenden eine Vielzahl von aufwendigen, großbauenden Meßapparaturen, die die Qualität der Flächen visuell oder automatisch prüfen, um Qualitätsmängel von Oberflächen schon bei der Produktion zu erkennen. Diese Methode bringt jedoch eine ganze Reihe von Nachteilen mit sich.

Die Arbeit der visuellen Prüfer ist sehr anstrengend und erfordert Werkräume, deren Lichtverhältnisse immer exakt definiert sind. Dennoch wurden große Unterschiede bei der Bewertung von gleichen Oberflächen durch verschiedene Prüfer festgestellt, da zum einen der jeweilige physiologische Eindruck von Prüfer zu Prüfer unterschiedlich ist, und da zum anderen das Sehvermögen des individuellen Prüfers auch von dessen jeweiliger physischer Verfassung abhängt.

Es bereitet somit große Schwierigkeiten, eine untere Qualitätsgrenze zu definieren, deren Unterschreiten nicht mehr hingenommen werden kann.

Weiterhin ist es auch für erfahrene Prüfer schwierig festzustellen, welche Ursachen ein ungenügender visueller Eindruck hat. Besonders bei strukturierten Oberflächen wird der visuelle Eindruck durch eine Kombination unterschiedlicher

Parameter bestimmt, so daß es schwierig ist, aufgrund der gemachten Beobachtungen und Messungen z. B. Steuerungswerte einer automatischen Lackiereinrichtung zur Verbesserung der Qualität zu verändern.

Strukturierte Kunststoffoberflächen werden häufig durch ein Prägeverfahren hergestellt, bei dem die verwendeten Formen bzw. Prägerollen etc. einem Verschleiß unterliegen. Mit zunehmender Anzahl geprägter Oberflächen wird die Struktur der Prägeform und somit auch die Struktur der Kunststoffoberfläche immer geringer. Durch einen regelmäßigen Austausch der Prägewerkzeuge kann die Produktion hinsichtlich Qualität und Quantität optimiert werden; allerdings ist die Lebensdauer der verwendeten Werkzeuge bei Schwankungen in der Materialzusammensetzung oder Schwankungen der Umgebungsbedingungen unterschiedlich, so daß die Werkzeuge im allgemeinen schon ausgetauscht werden, obwohl ihre Qualität zur Herstellung fertiger strukturierter Oberflächen noch ausreicht.

Werden zur Prüfung der Qualität strukturierter Oberflächen automatisch arbeitende Geräte verwendet, um z. B. den Glanz oder die Abbildungsschärfe der Oberfläche zu bestimmen, so können quantitative Zahlenwerte erhalten werden. Ein entscheidender Nachteil ist jedoch bei den visuellen und auch bei den automatischen Prüfungen, daß zwar Werte für die untersuchten, optischen Eigenschaften bestimmt werden, der Benutzer jedoch keinerlei Angaben über die Ursache hat.

Ein weiterer Nachteil herkömmlicher Meßapparaturen ist, daß sie einen erheblichen Raumbedarf aufweisen, und somit nicht vom Benutzer mitgenommen werden können.

Es ist deshalb die Aufgabe der vorliegenden Erfindung, ein Verfahren und eine Vorrichtung der eingangs genannten Art zur Verfügung zu stellen, so daß damit eine reproduzierbare Bewertung der Qualität von Oberflächen und insbesondere strukturierter Oberflächen erfolgen kann und daß derartige physikalische Größen der zu messenden Oberfläche bestimmt werden, daß eine charakteristische strukturbedingte Eigenschaft dieser Oberfläche bestimmt wird.

Ein weiterer Aspekt der Aufgabe ist es, eine Vorrichtung zu schaffen, die kleinbauend und leicht gestaltet und derart beschaffen ist, daß sie von einer Bedienungsperson leicht mitgenommen und ohne weitere Hilfsmittel zur Bewertung einer Oberfläche herangezogen werden kann.

Ein weiterer Aspekt der Aufgabe ist es, eine Vorrichtung zum Messen visueller Eigenschaften zu schaffen, bei welcher trotz des kompakten Aufbaus gemäß dem vorherigen Aspekt der Aufgabe der Erfindung die Möglichkeiten zur Messung gegenüber den im Stand der Technik bekannten Vorrichtungen erheblich erweitert ist.

Ein weiterer Aspekt der Aufgabe ist es, ein Verfahren anzugeben, welches eine vorteilhafte Erfassung einer charakteristischen strukturbedingten Eigenschaft von Oberflächen ermöglicht.

Diese Aufgabe wird erfindungsgemäß durch eine Vorrichtung gelöst, wie sie in Anspruch 1 definiert ist. Das erfindungsgemäße Verfahren ist Gegenstand des Anspruchs 45.

Zu bevorzugende Weiterbildungen der Erfindung sind Gegenstand der Unteransprüche.

Bei der erfindungsgemäßen Vorrichtung wird mit einer ersten optischen Einrichtung, die eine oder mehrere Lichtquellen aufweist, Licht unter einem vorbestimmten Winkel auf eine Meßfläche gerichtet, die Teil der zu messenden Oberfläche ist.

Mit einer unter einem zweiten, vorbestimmten Winkel zu dieser Meßfläche ausgerichteten, zweiten optischen Einrichtung, die einen oder mehrere lichtempfindliche Fotosensoren aufweist, wird das von der Meßfläche reflektierte Licht aufgenommen.

Dieser Fotosensor bzw. diese Fotosensoren, der bzw. die mindestens eine lichtempfindliche Fläche aufweisen, geben ein elektrisches Meßsignal aus, welches für das reflektierte Licht charakteristisch ist, wobei die erste und die zweite optische Einrichtung so beschaffen sind, daß das reflektierte Licht durch die Struktur dieser Meßfläche beeinflusst wird.

In der erfindungsgemäßen Vorrichtung ist eine Auswerteinrichtung angeordnet, die zur Steuerung des Meßablaufs vorgesehen ist und die wenigstens eine Prozessoreinrichtung und wenigstens eine Speichereinrichtung aufweist und die das reflektierte Licht auswertet und daraus wenigstens eine Strukturkennzahl ableitet, welche wenigstens eine der strukturbedingten Eigenschaften dieser Oberfläche charakterisiert.

Eine Ausgabereinrichtung, die in der erfindungsgemäßen Vorrichtung angeordnet ist, dient zur Ausgabe der Meßwerte.

Die Auswerteinrichtung wertet über ein in der Speichereinrichtung gespeichertes Programm die Meßsignale aus und speichert in einer bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung die Meßsignale und/oder die Meßwerte in dieser Speichereinrichtung.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung sind in der zweiten optischen Einrichtung eine Vielzahl von Fotosensoren angeordnet, welche in Reihen und/oder Spalten angeordnet sein können.

Im Unterschied zu konventionellen Vorrichtungen können bei der erfindungsgemäßen Vorrichtung die Fotosensoren auf einem gemeinsamen Substrat angeordnet werden. Die einzelnen Fotosensoren können auch durch Verschaltung lichtempfindlicher Flächen auf einem gemeinsamen Substrat gebildet werden, wobei die Fläche jedes lichtempfindlichen Sensors so gewählt oder wählbar ist, daß sie in der Meßvorrichtung einem vorgegebenen Winkelbereich des Reflexionswinkels entspricht.

Die erfindungsgemäße Vorrichtung hat erhebliche Vorteile:

Eine wesentliche Voraussetzung zur Steuerung des Fertigungsablaufs bei der Herstellung von strukturierten und nichtstrukturierten Oberflächen ist die Bestimmung der optischen Qualität oder des visuellen Eindrucks der Oberfläche. Durch die Analyse des an der Oberfläche reflektierten Lichts wird eine Strukturkennzahl abgeleitet, die die Qualität der Oberfläche beschreibt. Eine Kennzahl für die Struktur strukturierter Oberflächen wird von herkömmlichen visuellen oder automatischen Meßmethoden nicht bestimmt.

Dies ist besonders vorteilhaft, da das Profil der Struktur einer strukturierten Oberfläche wesentlich zum visuellen Gesamteindruck dieser Oberfläche beiträgt.

Bei der Verwendung von Präge-, Guß- oder anderen Verfahren unterliegen die verwendeten Formen einem Verschleiß, der u. a. von der Anzahl hergestellter, strukturierter Oberflächen abhängt. Neben der Anzahl hergestellter Bauteile mit strukturierten Oberflächen spielen aber auch Umweltbedingungen, wie Luftfeuchtigkeit, Material und Lufttemperatur eine entscheidende Rolle für die Qualität der herzustellenden Oberfläche.

Bei herkömmlichen Herstellungsverfahren müssen die verwendeten Formen nach einer bestimmten Anzahl geprägter oder gegossener Bauteile oder nach einer bestimmten Zeitdauer ausgetauscht werden, um eine gleichbleibend hohe Qualität der hergestellten Bauteile zu sichern.

Dieses Verfahren führt zu einem unnötig hohen Aufwand, da oftmals auch Werkzeuge mit noch ausreichender Qualität ausgewechselt werden, während es andererseits auch vorkommt, daß durch Umwelteinflüsse oder Materialinhomogenitäten bzw. -unterschiede die Werkzeuge vor dem vorgesehenen Wechselzeitpunkt verschlissen sind.

Eine Möglichkeit zur Vermeidung eines zu großen Ausschusses ist die regelmäßige Kontrolle. Dazu muß der Produktionsablauf zur Überprüfung der Werkzeuge angehalten werden, so daß sich kostspielige Stillstandszeiten ergeben.

Mit der erfindungsgemäßen Vorrichtung ist es möglich, einfach, schnell und zuverlässig während der laufenden Produktion die Qualität strukturierter Oberflächen zu bestimmen, so daß ein unzulässiger Verschleiß der Werkzeuge frühzeitig festgestellt werden kann und unnötige Stillstandszeiten der Produktionsanlage vermieden werden.

Ein weiterer Vorteil der Verwendung der erfindungsgemäßen Vorrichtung ist, daß unnötig häufige Wechsel der verwendeten Werkzeuge, Formen und Prägerollen und dergl. unterbleiben können, was zu geringeren Produktionskosten führt.

Ein weiterer wesentlicher Vorteil ist, daß die erfindungsgemäße Vorrichtung kleinbauend ist und so vom Benutzer jederzeit mitgenommen werden kann.

Deshalb eignet sich die erfindungsgemäße Vorrichtung auch für die Endkontrolle an schlecht zugänglichen Orten oder in Innenräumen von z. B. Kraftfahrzeugen, wo eine Vielzahl strukturierter Oberflächen eingesetzt wird.

Ein weiterer Vorteil einer solchen kleinbauenden Vorrichtung ist, daß auch an konkav oder konvex gekrümmten Oberflächen die Qualität einfach und zuverlässig bestimmt werden kann.

Vorteilhaft ist auch, daß durch die erfindungsgemäße Vorrichtung unterschiedlichste Stellen einer zu untersuchenden Oberfläche schnell und einfach gemessen werden können und so die Qualitätsverteilung über einer Oberfläche oder auch die Reproduzierbarkeit getestet werden kann.

In einer bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist wenigstens ein erster Teil dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts ein Lichtmuster auf. Die erste optische Einrichtung kann eine Mustereinrichtung enthalten. Das von dieser wenigstens einen Lichtquelle dieser ersten optischen Einrichtung ausgestrahlte Licht wird bei dieser Weiterbildung der erfindungsgemäßen Vorrichtung wenigstens teilweise auf diese Mustereinrichtung gestrahlt. Das durch diese Mustereinrichtung transmittierte Licht wird entsprechend dem Muster der Mustereinrichtung in der Amplitude und/oder Phase beeinflusst.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist das von dieser Mustereinrichtung ausgestrahlte Lichtmuster wenigstens eine Hell-/Dunkel-Kante, oder eine Vielzahl von Hell-/Dunkel-Kanten auf, wobei in einer weiteren Weiterbildung der erfindungsgemäßen Vorrichtung wenigstens ein Teil dieser Vielzahl von Hell-/Dunkel-Kanten wenigstens abschnittsweise parallel zueinander verläuft.

Das Muster dieser Mustereinrichtung kann symmetrisch zur optischen Achse angeordnet sein. In dem Fall, daß das ausgestrahlte Lichtmuster eine Vielzahl von Hell-/Dunkel-Kanten aufweist, können die einzelnen Hell-/Dunkel-Kanten einen geraden, kreis- oder sinusförmigen Verlauf aufweisen.

In einer bevorzugten Weiterbildung ist wenigstens ein Teil dieser Vielzahl von Hell-/Dunkel-Kanten gitter-, kreuzgitter- oder kreisförmig angeordnet.

Die Projektion eines Lichtmusters auf die zu untersuchende Oberfläche ist besonders vorteilhaft, da die Auswertung des von der zu untersuchenden Oberfläche bzw. der Meßfläche reflektierten Lichtmusters Rückschlüsse auf die Qualität und die Struktur der zu untersuchenden Meßfläche erlaubt.

Es ist möglich, daß das Lichtmuster keine scharfen Übergänge von hell nach dunkel aufweist, sondern die Intensität

des projizierten Lichts sinus- oder sägezahnförmig beeinflußt.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist ein erster Teil eines von dieser ersten optischen Einrichtung ausgestrahlten Lichts dieses Lichtmuster auf, während ein zweiter Teil ohne Lichtmuster auf die Meßfläche gerichtet ist.

Es ist aber auch möglich, daß bei der erfindungsgemäßen Vorrichtung die Mustereinrichtung zuschaltbar ist, so daß ein abwechselndes Messen mit Lichtmuster und ohne Lichtmuster möglich ist.

Dazu ist es möglich, daß diese Mustereinrichtung z. B. eine LCD-Einrichtung aufweist, bei welcher einzelne Bereiche oder einzelne Pixel je nach Bedarf zu unterschiedlichen Lichtmustern verschaltet werden können, so daß das Lichtmuster schnell und einfach an die gegebenen Erfordernisse angepaßt bzw. abgeschaltet werden kann.

Durch Auswertung des Signals dieses wenigstens einen Fotosensors, welcher wenigstens einen Teil des reflektierten Lichtmusters aufnimmt, kann eine Strukturkennzahl von dieser Auswerteeinrichtung bestimmt werden, welche wenigstens für eine strukturbedingte Eigenschaft dieser Oberfläche charakteristisch ist.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung bestimmt diese Auswerteeinrichtung wenigstens für einen Teil dieser Vielzahl von Fotosensoren eine Steigung des elektrischen und/oder digitalen Meßsignals, indem eine Differenz des elektrischen und/oder digitalen Meßsignals des Fotosensors mit diesem elektrischen und/oder digitalen Meßsignal des nächsten Fotosensors in dieser Reihe und/oder Spalte gebildet wird, oder indem Differenzen des elektrischen und/oder digitalen Meßsignals dieses Fotosensors mit diesem elektrischen und/oder digitalen Meßsignal wenigstens eines Teils der direkt oder diagonal benachbarten Fotosensoren gebildet werden.

Dies ist besonders vorteilhaft, da die Steigungen der Meßsignale und insbesondere die Steigungen der Meßsignale bei Bestrahlung mit einem Lichtmuster dazu geeignet sind, eine Strukturkennzahl abzuleiten, welche für wenigstens eine strukturbedingte Eigenschaft der zu untersuchenden Meßfläche charakteristisch ist.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung bestimmt diese Auswerteeinrichtung wenigstens einen Durchschnitt wenigstens eines Teils dieser Steigungen der Meßsignale dieser Fotosensoren und bestimmt aus diesem wenigstens einen Durchschnitt diese charakteristische Strukturkennzahl für diese wenigstens eine strukturbedingte Eigenschaft dieser Oberfläche.

Aus diesen Steigungen kann aber auch ein Histogramm bestimmt werden, in welchem für vorbestimmte und/oder wählbare Bereiche die Steigung über der Anzahl dieser Steigungen in den entsprechenden Intervallen bzw. Bereichen aufgetragen wird. Mit der Bildung des Schwerpunkts und/oder des Mittelwerts der Steigungen kann eine charakteristische Strukturkennzahl in dieser Auswerteeinrichtung abgeleitet werden, welche die strukturbedingten Eigenschaften dieser Oberfläche charakterisiert.

Diese bevorzugte Weiterbildung der erfindungsgemäßen Vorrichtung ist besonders vorteilhaft, da durch die Bestimmung wenigstens eines Teils der Steigungen und durch die Bildung der Durchschnittssteigung schnell und zuverlässig eine charakteristische Strukturkennzahl für diese Meßfläche bestimmt werden kann.

Abgerundete Oberflächenstrukturen, wie sie z. B. bei Verwendung von abgenutzten oder verschlissenen Werkzeugen bei der Produktion strukturierter Oberflächen entstehen, reflektieren auf diese Oberfläche projiziertes Licht diffuser als Oberflächen mit schärfer ausgeprägtem Profil.

Durch die Verwendung einer solchen Weiterbildung der erfindungsgemäßen Vorrichtung ist es möglich, die strukturbedingten Eigenschaften zu bestimmen.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung werden als optische Kenngrößen neben dieser charakteristischen Strukturkennzahl noch wenigstens eine, zwei, drei oder mehr charakteristische, optische Kenngrößen dieser Meßfläche bestimmt.

In bevorzugten Weiterbildungen der Erfindung wird der Glanz (engl. gloss) oder der Schleierglanz (engl. haze) oder die Abbildungsschärfe (engl. distinction of image DOI) oder ein repräsentatives Maß für die typische Wellenlänge und deren Amplitude (engl. orange peel) der Topologie der Schichtdicke dieser Meßfläche bestimmt.

Die Bestimmung des orange peel erfolgt in einem vorbestimmten Wellenlängenintervall; die Auswertung zur Bestimmung des orange peel kann aber auch in zwei oder mehr Wellenlängenbereichen erfolgen. Weiterhin ist es möglich, die Farbe der Meßfläche oder die Helligkeit der Farbe zu bestimmen.

In einer weiteren bevorzugten Weiterbildung dieser Erfindung werden neben der Strukturkennzahl noch weitere der obengenannten charakteristischen optischen Kenngrößen bestimmt.

Eine solche Weiterbildung der erfindungsgemäßen Vorrichtung ist besonders vorteilhaft, da mit dieser kleinbauenden Vorrichtung u. U. alle interessierenden, visuellen und physikalischen Eigenschaften dieser Meßfläche bestimmt werden können.

Bei einer vorteilhaften Weiterbildung aller vorstehend beschriebenen Ausführungsbeispiele ist eine dritte optische Einrichtung angeordnet, welche wenigstens eine Lichtquelle aufweist und Licht mit einer vorgegebenen, spektralen Charakteristik ausstrahlt, welches in einem vorbestimmten Winkel auf diese Meßfläche gerichtet ist.

In den bisher genannten bevorzugten Weiterbildungen der erfindungsgemäßen Vorrichtung kann der vorbestimmte Winkel, in welchem das von der ersten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist und/oder der vorbestimmte Winkel, unter welchem die zweite optische Einrichtung das von dieser Meßfläche reflektierte Licht aufnimmt, und/oder der vorbestimmte Winkel, in welchem das von der dritten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist, frei eingestellt werden und umfaßt insbesondere die Winkel 5°, 10°, 15°, 20°, 30°, 45°, 60°, 75°, 80° und 85° zwischen der Normalen der Meßfläche und dem entsprechenden Winkel. Es sei jedoch darauf hingewiesen, daß auch andere als die hier genannten Winkel möglich sind.

In einer weiteren bevorzugten Weiterbildung der Erfindung ist dieses von dieser dritten optischen Einrichtung ausgestrahlte Licht in einem derartigen Winkel auf die Oberfläche gerichtet, daß das unmittelbar von der Meßfläche gemäß der Fresnel'schen Reflexion gerichtet reflektierte Licht gegenüber dieser Meßfläche einen anderen Winkel aufweist als der Winkel zwischen dieser Meßfläche und dem von dieser Meßfläche gerichtet reflektierten Licht, welches von dieser ersten optischen Einrichtung ausgestrahlt wird, so daß die zweite optische Einrichtung bzw. die Fotosensoren im wesentlichen das von der dritten optischen Einrichtung ausgestrahlte Licht und diffus von dieser Meßfläche reflektierte Licht aufnimmt.

Diese wenigstens eine Lichtquelle dieser dritten optischen Einrichtung kann eine, zwei und besonders bevorzugt drei lichtausstrahlende Elemente aufweisen, deren Licht sich in der emittierten spektralen Charakteristik derart unterscheidet, daß die emittierten Wellenlängenbereiche unterschiedlich sind.

Die emittierten Wellenlängenbereiche dieser lichtausstrahlenden Elemente können sich im sichtbaren Bereich des Lichtspektrums zumindest teilweise überlappen, und in einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung sind die emittierten, spektralen Charakteristika dieser lichtausstrahlenden Elemente voneinander linear unabhängig.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung sind wenigstens zwei, vorzugsweise drei oder mehr Fotosensoren derart gestaltet, daß sie sich in ihrer spektralen Charakteristik unterscheiden, so daß als optische Kenngröße dieser Meßfläche die Farbe des reflektierten Lichts erfaßbar ist.

In einer anderen, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung weist wenigstens ein Fotosensor wenigstens zwei, vorzugsweise drei oder mehr fotosensitive Elemente auf, deren elektrische Ausgangssignale einzeln erfaßbar sind und die sich in ihrer spektralen Charakteristik unterscheiden, so daß die Farbe des von dieser Meßfläche reflektierten Lichts erfaßbar ist.

Eine Weiterbildung der erfindungsgemäßen Vorrichtung, die es ermöglicht, die Farbe des von dieser Meßfläche reflektierten Lichts zu erfassen, ist besonders vorteilhaft, da die Farbe und die Helligkeit der Farbe wesentliche Merkmale von Oberflächen und auch von strukturierten Oberflächen sind.

In weiteren bevorzugten Weiterbildungen der erfindungsgemäßen Vorrichtung sind in dieser ersten und/oder dieser dritten optischen Einrichtung eine oder mehrere Leuchtdioden oder Laserlichtquellen angeordnet.

Wenigstens ein Teil des von dieser ersten optischen Einrichtung und/oder dieser dritten optischen Einrichtung ausgestrahlten Lichts ist im wesentlichen parallel, divergent oder konvergent.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung strahlt diese erste und/oder dritte optische Einrichtung wenigstens einen Lichtfleck mit vorbestimmtem Durchmesser senkrecht zur Ausbreitungsrichtung des ausgestrahlten Lichts oder wenigstens einen Lichtstreifen mit vorbestimmter Länge und Breite senkrecht zur Ausbreitungsrichtung des ausgestrahlten Lichts aus.

Dieses von der ersten optischen Einrichtung projizierte Lichtmuster kann auch aus verschiedenen Lichtflecken oder Lichtstreifen bestehen.

Es sei darauf hingewiesen, daß diese wenigstens eine optische Kenngröße bzw. diese Strukturkennzahl auch aus den elektrischen und/oder digitalen Meßsignalen dieser Fotosensoren bestimmt werden kann, welche diesen von dieser Meßfläche reflektierten zweiten Teil ohne Lichtmuster dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts aufnehmen.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung ist in möglichst unmittelbarer Nähe wenigstens einer Lichtquelle dieser ersten optischen Einrichtung und/oder dieser wenigstens einen Lichtquelle bzw. dieses wenigstens einen lichtausstrahlenden Elementes dieser dritten optischen Einrichtung und/oder dieses wenigstens einen Fotosensors bzw. dieses wenigstens einen fotosensitiven Elementes wenigstens eine Temperaturmeßeinrichtung angeordnet, welcher zur Bestimmung der charakteristischen Temperatur der jeweiligen Lichtquelle, lichtausstrahlenden Elemente, Fotosensoren oder lichtsensitiven Elementen vorgesehen ist, damit eine temperaturkorrigierte Bestimmung dieses wenigstens einen Strukturkennwerts und/oder dieser wenigstens einen optischen Kenngröße erfolgen kann.

Die Bestimmung der charakteristischen Temperaturen der Lichtquellen und Fotosensoren ist besonders vorteilhaft, da

diese elektrischen Einrichtungen temperaturabhängig arbeiten und durch eine temperaturkorrigierte Bestimmung der Kenngrößen und Kennwerte die Reproduzierbarkeit von Messungen erhöht wird; denn mit zunehmender Meßdauer oder bei wechselndem Meßapparat kann sich die Temperatur und deren Verteilung innerhalb der erfindungsgemäßen Vorrichtung ändern.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird wenigstens ein Teil des Verlaufs des auf dieser Vielzahl von Fotosensoren abgebildeten Lichtmusters bestimmt. Durch eine Abweichung des gemessenen Verlaufs vom idealen Verlauf kann ein charakteristischer Profilhöhenkennwert dieser Meßfläche bestimmt werden.

In einer besonders bevorzugten Weiterbildung wird der Verlauf wenigstens einer Hell-/Dunkel-Kante abschnittsweise auf dieser Vielzahl von Fotosensoren bzw. auf dem CCD-Chip bestimmt.

Wird eine helle oder dunkle Linie auf eine zu untersuchende Oberfläche projiziert, welche ein Rechteckprofil mit einer Profilhöhe h aufweist, so findet eine räumliche Trennung des Reflexes dieser Linie auf dem CCD-Chip der zweiten optischen Einrichtung statt.

Die von den unteren Kanten des Rechteckprofils reflektierten Teile der dunklen oder hellen Linie werden räumlich parallel versetzt zu denen reflektiert, die an den oberen Kanten des Rechteckprofils reflektiert werden.

Das reflektierte Abbild einer idealen, rechteckförmig strukturierten Oberfläche sind zwei gestrichelte helle bzw. dunkle Linien auf dem CCD-Chip.

Mit dem bekannten Einstrahl- und Reflexionswinkel und dem Abstand dieser zwei gestrichelten Linien auf dem CCD-Chip kann die Höhe des Profils an jedem Ort der Meßfläche bestimmt werden.

Bei der Vermessung von nicht rechteckförmigen Oberflächenprofilen kann analog mit den entsprechenden geometrischen Beziehungen ein Profilhöhenkennwert dieser Meßfläche bestimmt werden.

Die Bestimmung eines Profilhöhenkennwertes ist besonders vorteilhaft, da beim Verschleifen der Herstellungswerkzeuge auch die Profilhöhe der hergestellten Bauteile abnehmen kann.

In einer weiteren bevorzugten Weiterbildung der Erfindung wird diese Strukturkennzahl mittels eines Triangulationsverfahrens bestimmt, wobei diese Bestimmung dieser Strukturkennzahl ausschließlich oder auch zusätzlich mittels dieses Triangulationsverfahrens erfolgen kann. Es sei darauf hingewiesen, daß eines der gängigen Triangulationsverfahren, wie sie im Stand der Technik bekannt geworden sind, eingesetzt werden kann.

Beispielhaft sei auf die aus den amerikanischen Patenten US 5,546,189 und US 5,744,793 dargestellten Triangulationsverfahren hingewiesen, die hiermit Teil der Offenbarung der vorliegenden Erfindung werden.

In einer weiteren bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung wird im Gegensatz zu dem in dem US-Patent 5,546,189 hergestellten Verfahren kein bzw. nicht ausschließlich ein Grauwert bestimmt, welcher repräsentativ für die Höhe der Meßfläche am Meßort ist, sondern es wird die Oberflächenhöhe am Meßort in größtmöglicher Präzision bestimmt.

In einer weiteren, bevorzugten Weiterbildung der Erfindung ist eine zusätzliche optische Einrichtung mit wenigstens einem Fotosensor vorgesehen, welche unter einem vorbestimmten Winkel zu dieser Meßfläche ausgerichtet ist. In einer besonders bevorzugten Weiterbildung ist diese zusätzliche optische Einrichtung in der Ebene angeordnet, die durch den Meßpunkt, die zweite optische Einrichtung und die dritte optische Einrichtung aufgespannt wird, und beson-

ders bevorzugt ist, daß der Winkel, unter dem diese zusätzliche optische Einrichtung das reflektierte Licht aufnimmt, gleich dem Winkel ist, unter dem die zweite optische Einrichtung das reflektierte Licht aufnimmt, so daß die zweite und die zusätzliche optische Einrichtung symmetrisch zu dem von der ersten optischen Einrichtung oder symmetrisch zu dem von der dritten optischen Einrichtung ausgestrahlten Licht angeordnet sind.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird die Intensität und der Ort des Auftreffpunktes der reflektierten Intensität auf dieser zweiten optischen Einrichtung und dieser zusätzlichen optischen Einrichtung bestimmt, und besonders bevorzugt wird von dieser Auswerteeinheit ein charakteristischer Ortskennwert für den Ort der Messung und die gemessene Intensität in dieser Speichereinrichtung gespeichert, und diese Auswerteeinheit bestimmt einen charakteristischen Abstandskennwert für den Abstand zum Meßpunkt auf der zu untersuchenden Oberfläche und speichert diesen in dieser Speichereinrichtung.

In einer weiteren bevorzugten Weiterbildung der Erfindung weist die erste optische Einrichtung eine Scanneinrichtung auf, welche die zu untersuchende Oberfläche derart abtastet, daß das von dieser ersten optischen Einrichtung ausgestrahlte Licht zeitlich nacheinander die zu untersuchende Oberfläche abtastet.

Besonders bevorzugt bestimmt diese Auswerteeinrichtung mit diesen Abstandskennwerten und diesen Ortskennwerten wenigstens eine dieser wenigstens einen Strukturkennzahl.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung wird wenigstens ein quantitativer Steilheitskennwert für die Steilheit der Struktur der zu untersuchenden Oberfläche bestimmt.

In einer weiteren, bevorzugten Weiterbildung der Erfindung bestimmt diese Auswerteeinrichtung wenigstens einen quantitativen Schärfenkennwert für die Schärfe der Kanten der zu untersuchenden Oberfläche.

Diese Weiterbildungen sind besonders vorteilhaft, da mit der Abnutzung der Werkzeuge die Struktur der Oberfläche der herzustellenden Bauteile abnimmt und insbesondere durch Abnutzung und Verschleiß die Steilheit der Struktur abnimmt und insbesondere auch die Schärfe der Kanten abnimmt, da mit zunehmendem Verschleiß das Profil des Werkzeugs abrundet.

In einer weiteren, bevorzugten Weiterbildung wird aus dieser wenigstens einen Strukturkennzahl und wenigstens einer der genannten charakteristischen optischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, orange peel) eine Bewertungsstrukturkennzahl abgeleitet, welche wenigstens für einen Teil der visuellen Eigenschaften dieser Oberfläche charakteristisch ist.

Bei der Bewertung von Oberflächen und insbesondere von strukturierten Oberflächen hängt der visuelle Eindruck nicht nur von der Ausprägung der Struktur, sondern auch von den anderen genannten charakteristischen, optischen Größen ab. Beispielsweise sind bei besonders glatten und hochreflektierenden Flächen die Anforderungen an den Herstellungsprozeß besonders hoch, da auf solchen Oberflächen das menschliche Auge auch kleinste Störungen, Unebenheiten und Veränderungen wahrnimmt. Bei Kraftfahrzeugen mit sogenannter "Metallic"-Lackierung ist es nahezu unmöglich, auf einer großen Fläche einen Lackschaden derart nachzubessern, daß dieser Schaden für den Betrachter unsichtbar bleibt.

Im Gegensatz dazu sind die Anforderungen bei matten und schwach reflektierenden Flächen an die Beschichtungsqualität des Lacks und der darunter liegenden Oberfläche erheblich geringer.

Während Oberflächenrauigkeit und orange peel schon im geringen Maß bei hochglänzenden Flächen das physiologische Empfinden eines Betrachters stören, wirken sich weitaus größere Rauigkeiten und Oberflächenwelligkeiten bei schwach reflektierenden und matten Flächen auf den visuellen Eindruck eines Betrachters nur wenig oder gar nicht aus. Bei strukturierten Oberflächen ist der visuelle Eindruck durch die Struktur bedingt weniger abhängig von der Oberflächenrauigkeit, dem Glanz oder den anderen optischen Kenngrößen.

Gemäß dieser bevorzugten Weiterbildung der Erfindung wird deshalb wenigstens eine Bewertungsstrukturkennzahl und wenigstens eine der genannten charakteristischen Größen gebildet, welche ein Maß für die Qualität der Meßfläche darstellt.

In einer bevorzugten Weiterbildung der Erfindung wird aus diesem Kennwert für den Glanz der Meßfläche und dieser Strukturkennzahl eine Bewertungsstrukturkennzahl derart gebildet, daß für Flächen mit geringem Glanz auch bei einer nicht idealen Strukturkennzahl eine ausreichende bzw. befriedigende Bewertungsstrukturkennzahl gebildet wird.

In einer weiteren, bevorzugten Weiterbildung ist vorgesehen, daß diese Bewertungsstrukturkennzahl derart gebildet wird, daß z. B. ein höherer Glanz eine schlechtere Strukturkennzahl ausgleicht, so daß diese Bewertungsstrukturkennzahl ein qualitativ ausreichendes Ergebnis erzielt.

In einer weiteren, bevorzugten Weiterbildung ist vorgesehen, daß für wenigstens zwei oder alle der genannten optischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, orange peel) eine Bewertungsstrukturkennzahl gebildet wird oder daß aus zwei oder mehr der genannten charakteristischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, orange peel) und dieser Strukturkennzahl eine Bewertungsstrukturkennzahl abgeleitet wird, welche für den visuellen Eindruck dieser Meßfläche charakteristisch ist.

Die Bildung wenigstens einer Bewertungsstrukturkennzahl aus dieser wenigstens einen Strukturkennzahl und wenigstens einer der charakteristischen optischen Größen, wie Glanz etc., ist besonders vorteilhaft, da eine besonders einfache und automatische bzw. automatisierte Bestimmung der Qualität der zu untersuchenden Oberfläche möglich ist.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung ist wenigstens ein unterer bzw. oberer Grenzwert für diese wenigstens eine Strukturkennzahl und/oder diese wenigstens eine charakteristische optische Kenngröße und/oder diese wenigstens eine Bewertungsstrukturkennzahl festlegbar, bei dessen Unter- bzw. Überschreitung eine Nachricht auf diese Ausgabereinrichtung oder ein Alarmsignal von akustischer oder optischer Art ausgegeben wird.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird dieser wenigstens ein Grenzwert durch Messung wenigstens einer Referenzoberfläche automatisch bestimmt. Es ist z. B. möglich, daß eine, zwei oder mehr bewertete Referenzoberflächen unterschiedlicher Qualität gemessen werden, wobei vorteilhafterweise wenigstens eine dieser Referenzoberflächen eine hohe Qualität und wenigstens eine weitere eine unzureichende Qualität aufweist.

Durch eine derartige teilautomatisierte Kalibrierung dieser Vorrichtung können schnell und zuverlässig verschiedene Meßpunkte auf einer Oberfläche oder verschiedenen zu untersuchenden Oberflächen durchgeführt werden.

In einer weiteren, bevorzugten Weiterbildung ist ein Ein- bzw. Ausschalter angeordnet.

Des weiteren kann an dieser erfindungsgemäßen Vorrichtung eine Eingabevorrichtung vorgesehen sein, wobei diese u. a. dafür vorgesehen sein kann, daß der Benutzer ein Typkennzeichen für den Typ der zu messenden Oberfläche ein-

geben kann, wobei nachfolgende Messungen mit Bezug auf dieses Typkennzeichen von dieser Auswerteeinrichtung ausgewertet und in dieser Speichereinrichtung abgelegt werden.

In einer weiteren, bevorzugten Weiterbildung der Erfindung wird der Benutzer beim Einschalten durch ein akustisches oder optisches Signal dazu aufgefordert werden, in dieser Eingabeeinrichtung ein Typkennzeichen für den zu messenden Oberflächentyp einzugeben, wobei in einer besonders bevorzugten Weiterbildung der Benutzer jederzeit durch Eingabe eines anderen Typkennzeichens den Oberflächentyp ändern kann, auf den sich die nachfolgenden Messungen beziehen.

Die Eingabe bzw. die Wahl eines Typkennzeichens vor der Messung eines bestimmten Oberflächentyps ist besonders vorteilhaft, da bei der Bewertung verschiedener Oberflächentypen die einzelnen Kennwerte eine unterschiedliche Bedeutung und bei Bildung einer Bewertungsstrukturkennzahl eine unterschiedliche Gewichtung haben.

In einer weiteren, bevorzugten Weiterbildung der Erfindung kann der Benutzer bei jeder Messung durch Eingabe der Benutzer-Beurteilungszahl in dieser Eingabeeinrichtung diese Meßfläche gemäß seinem physiologischen Eindruck bewerten. Diese Benutzer-Beurteilungszahl wird besonders bevorzugt zusammen mit dieser wenigstens einen charakteristischen, optischen Kenngröße und dieser wenigstens einen Strukturkennzahl und dieser wenigstens einen Bewertungsstrukturkennzahl in dieser Speichereinrichtung dauerhaft abgelegt, wobei der Wertebereich dieser Benutzer-Beurteilungszahl wenigstens zwei unterschiedliche Qualitätswerte, die "gut" und "schlecht" entsprechen, umfaßt.

In einer weiteren, bevorzugten Weiterbildung der Erfindung werden wenigstens ein Teil dieser in dieser Speichereinrichtung abgelegten Benutzer-Beurteilungszahlen mit zugehörigen Kenngrößen und Kennwerte und entsprechenden Messungen zu automatischen Bestimmungen wenigstens eines dieses wenigstens eines Grenzwerts für dieses Typkennzeichen der zu untersuchenden Oberfläche verwendet.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung berücksichtigt diese Auswerteeinrichtung im wesentlichen alle in dieser Speichereinrichtung für dieses Typkennzeichen der zu messenden Oberfläche abgelegten Benutzer-Beurteilungszahlen und die entsprechenden Kenngrößen und Kennwerte und Strukturkennwerte zur Bestimmung dieses wenigstens einen Grenzwerts und zur Bestimmung der Gewichtungsfaktoren, welche zur Bestimmung der Bewertungsstrukturkennzahl verwendet werden, so daß diese Gewichtungsfaktoren und dieser wenigstens eine Grenzwert mit zunehmender Anzahl von Messungen derart angepaßt werden, daß diese Bewertungsstrukturkennzahl dem visuellen Eindruck des Benutzers angepaßt wird.

Diese bevorzugte Weiterbildung der Erfindung ist besonders vorteilhaft, da diese Vorrichtung bei der Bestimmung dieser Kenngrößen und Kennwerte und dieser Bewertungsstrukturkennzahl lernfähig ist und auf der Basis eines Expertensystems die Zuverlässigkeit bei der Bestimmung dieser wenigstens einen Bewertungsstrukturkennzahl mit der Anzahl der Messungen zunimmt.

Bei dieser bevorzugten Weiterbildung der Erfindung ist es weiterhin vorgesehen, daß bei der Bestimmung dieser Grenzwerte und Gewichtungsfaktoren nicht alle gemessenen und gespeicherten Werte für dieses Typkennzeichen verwendet werden, da sich der physiologische Zustand des Benutzers auf die Bewertung dieser Meßfläche auswirkt, und da durch irrtümliche Eingabe falscher Benutzer-Beurteilungszahlen ungünstige Grenzwerte und Gewich-

tungsfaktoren bestimmt werden können.

Möglich ist es z. B., daß bei einer Vielzahl von vorliegenden Meßergebnissen eine bestimmte Prozentzahl von z. B. 5% oder 10% bei der Bildung dieser Gewichtungsfaktoren und dieser Grenzwerte unberücksichtigt bleiben, wobei die Auswahl der unberücksichtigt bleibenden Werte nach bekannten, statistischen Verfahren erfolgt.

Diese Weiterbildung der Erfindung ist besonders vorteilhaft, da auch bei irrtümlicher Eingabe falscher Benutzer-Beurteilungszahlen diese im wesentlichen unberücksichtigt bleiben, so daß durch die Lernfähigkeit dieser Auswerteeinrichtung bedingt die Zuverlässigkeit der automatisch bestimmten Bewertungsstrukturkennzahlen mit zunehmender Anzahl von Messungen steigt.

Bei einer vorteilhaften Weiterbildung aller vorstehend beschriebenen Ausführungsbeispiele ist diese Vorrichtung relativ zur Meßfläche in einer im wesentlichen abstandsgleichen Richtung verschiebbar, und es ist eine Wegstrecken-Meßeinrichtung vorgesehen, welche diese relative Verschiebung quantitativ erfaßt. In dieser Speichereinrichtung werden die entlang vorgegebenen Meßschwellen auf dieser Oberfläche gemessenen Strukturkennwerte und/oder charakteristischen optischen Kenngrößen und/oder Bewertungsstrukturkennzahlen gespeichert.

In einer weiteren, bevorzugten Weiterbildung ist wenigstens ein Meßrad vorgesehen, welches während der Messung auf der zu messenden Oberfläche aufgesetzt ist und sich während der Relativbewegung zwischen der Vorrichtung und der zu messenden Oberfläche dreht, wobei dieses wenigstens eine Meßrad mit einem Drehwinkelgeber verbunden sein kann, welcher ein elektrisches Drehwinkelsignal ausgibt, welches für den vom Meßrad zurückgelegten Drehwinkel repräsentativ ist.

Diese bevorzugte Weiterbildung der Erfindung ist besonders vorteilhaft, da zu untersuchende Oberflächen z. B. systematisch über die gesamte Oberfläche gemessen werden können.

In der bevorzugten Weiterbildung der Erfindung, in welcher eine Vielzahl von Fotosensoren angeordnet sind, wird besonders bevorzugterweise ein CCD-Chip eingesetzt, welcher in einer weiteren, bevorzugten Weiterbildung der Erfindung als Farb-CCD-Chip ausgeführt sein kann.

Bei Einsatz eines zweidimensionalen CCD-Chips wird ein zweidimensionales Abbild dieser Meßfläche in der zweiten optischen Einrichtung auf den CCD-Chip abgebildet.

In einer bevorzugten Weiterbildung der Erfindung wird vor Beginn der Messungen oder vor Beginn jeder einzelnen Messung ein Dunkelbild aufgenommen, und die Werte werden gemittelt. Bei der Messung wird das Bild mit Beleuchtung aufgenommen, ebenfalls gemittelt und davon der Mittelwert des Dunkelbildes abgezogen. Es wird ein Wert für den Kontrast dieses Bildes bestimmt. Mit den Mittelwerten des Dunkelbildes, des beleuchteten Bildes und dem Wert für den Kontrast wird die Helligkeit des gemessenen Bildes normiert.

Die weitere Auswertung erfolgt mit den normierten Werten bzw. mit dem normierten Bild.

In einer weiteren, bevorzugten Weiterbildung der Erfindung erfolgt die Auswertung über eine oder mehrere Fouriertransformationen. Besonders bevorzugt werden für jede Zeile die Ortsfrequenzkoeffizienten aus einer Fouriertransformation dieser Zeile bestimmt und die Ortsfrequenzkoeffizienten für das gesamte Bild werden aus dem Mittelwert der Frequenzen aller Zeilen bestimmt.

In einer weiteren, bevorzugten Weiterbildung der erfindungsgemäßen Vorrichtung wird dieser Strukturkennwert über ein übliches Verfahren zur Rauheitsmessung von Oberflächen bestimmt, wie z. B. in DIN 4768 (Ausgabe 1990-

05), ISO 13565-1 (Ausgabe 1996-12) oder ISO 13565-2 (Ausgabe 1996-12) beschrieben, die somit Teil der Offenbarung werden.

Es sei darauf hingewiesen, daß das Zusammenwirken der einzelnen erfindungsgemäßen Merkmale in jeder beliebigen Kombination bevorzugt ist. Insbesondere sind auch die durch die unabhängigen und abhängigen Ansprüche offenbarten Merkmalskombinationen unter Weglassung eines oder mehrerer Merkmale bevorzugt. Für den Fachmann ist ersichtlich, daß über die hier dargestellten Ausführungsbeispiele hinaus eine Vielzahl weiterer Modifikationen und Ausführungen denkbar sind, die von der Erfindung erfaßt sind. Die Erfindung beschränkt sich insbesondere nicht nur auf die hier dargestellten Ausführungsformen.

Weitere Vorteile, Merkmale und Anwendungsmöglichkeiten der vorliegenden Erfindung ergeben sich aus der nachfolgenden Beschreibung von Ausführungsbeispielen im Zusammenhang mit den Zeichnungen.

Darin zeigen:

Fig. 1 einen Schnitt eines ersten Ausführungsbeispiels gemäß der vorliegenden Erfindung;

Fig. 2 ein weiteres Ausführungsbeispiel eines Sensorelements, bei welchem die zur Lichtmessung herangezogenen Flächen variiert werden können;

Fig. 3 ein weiteres Ausführungsbeispiel eines Sensorelements, bei welchem sieben Fotosensoren mit den lichtempfindlichen Flächen gebildet werden;

Fig. 4 ein Diagramm, welches das Reflexionsverhalten bei Beleuchtung ohne Lichtmuster zeigt, wobei auf der Ordinate die gemessene Lichtintensität und auf der Abszisse die Winkelabweichung bezüglich des idealen Reflexionswinkels aufgetragen ist;

Fig. 5 ein Abbild des auf die Meßfläche projizierten Lichtmusters,

Fig. 6 den prinzipiellen, schaltungstechnischen Aufbau einer Meßeinrichtung, wie er bei den Ausführungsbeispielen gemäß **Fig. 1-5** zu verwenden ist; und

Fig. 7 ein weiteres Ausführungsbeispiel der Erfindung, welches auch zu weitergehenden Messungen geeignet ist.

Ein erstes Ausführungsbeispiel der Erfindung wird nun in bezug auf die **Fig. 1** beschrieben.

Die in der **Fig. 1** schematisch dargestellte Meßeinrichtung zur Bestimmung der Qualität strukturierter Oberflächen weist ein Gehäuse auf, in welchem ein erster Belichtungstubus 2 angeordnet ist.

In diesem Belichtungstubus ist, schematisch angedeutet, eine Lichtquelle 3 angeordnet, eine Blende 4, eine Lichtmustereinrichtung 6 und eine Linseneinrichtung 5.

Das von der punktförmigen Lichtquelle 3 ausgestrahlte Licht wird in seiner Apertur durch die Blende 4 begrenzt und trifft auf die Lichtmustereinrichtung 6. Je nach Ausführungsform kann in der Lichtmustereinrichtung 6 eine teilweise transparente Lichtmusterplatte 9 angeordnet sein, bei welcher auf einem Teil der ausgeleuchteten Fläche die Phase und/oder die Amplitude des auftreffenden Lichts derart beeinflusst wird, daß das transmittierte Licht ein charakteristisches Lichtmuster aufweist. In diesem Ausführungsbeispiel wird die Amplitude des von der Lichtquelle 3 ausgestrahlten Lichts durch Hell-/Dunkel-Kanten auf der Lichtmusterplatte 9 beeinflusst.

Weiterhin wird das ausgestrahlte Licht von der Linse 5 gebündelt und trifft durch die Öffnung 7 auf die Meßfläche 8 auf. Von der Meßfläche 8 wird das Licht reflektiert und tritt in den Meßtubus 10 ein, welcher ebenfalls eine Linse 11, eine Blende 12 und den eigentlichen Sensor 13 aufweist.

Die Reflexionsmeßvorrichtung weist weiterhin eine (nicht dargestellte) Steuereinrichtung auf, durch welche der Betrieb der Vorrichtung gesteuert wird und eine ebenfalls

nicht dargestellte Anzeigeeinrichtung, durch welche die gemessenen Strukturkennwerte und charakteristischen optischen Kenngrößen angezeigt werden.

In **Fig. 2** ist der Sensor eines weiteren Ausführungsbeispiels dargestellt. Dieses Ausführungsbeispiel ist mit Ausnahme des Sensors genauso aufgebaut wie in **Fig. 1**. **Fig. 2** zeigt die Sensoreinrichtung 20, bei welcher eine Vielzahl von fotosensitiven Elementen 21 in Reihen und Spalten angeordnet ist.

Bei diesem Beispiel wird ein CCD-Chip eingesetzt. Auf der Fläche des CCD-Chips werden einzelne Flächenelemente zu unterschiedlichen Fotosensoren 22, 23, 24 verschaltet, so daß bei der Messung nur die Signale dieser Fotosensoren 22, 23, 24 vorliegen.

Es sei darauf hingewiesen, daß sich durch die Verschaltung einzelner Flächenelemente zu Fotosensoren die unterschiedlichsten Meßgeometrien realisieren lassen. Weiterhin müssen zur Messung nicht alle Flächenelemente herangezogen werden.

In **Fig. 3** ist ein weiteres Ausführungsbeispiel der Sensoreinrichtung dargestellt. Der weitere Aufbau und der Betrieb dieser Vorrichtung erfolgt in gleicher Weise wie bei **Fig. 1**.

In **Fig. 3** ist die Sensoreinrichtung 30 dargestellt, welche wie bei dem Ausführungsbeispiel gemäß **Fig. 2** als CCD-Chip ausgeführt ist.

Auf der Fläche des CCD-Chips werden durch Verschaltung einzelner Flächenelemente unterschiedliche Fotosensoren 31, 32, 33 und 34 gebildet. Durch die Verschaltung einzelner Flächenelemente zu Fotosensoren kann schnell und einfach nach unterschiedlichen Meßnormen, wie z. B. der amerikanischen ASTM E 430 gemessen werden.

In einem weiteren Ausführungsbeispiel ist der prinzipielle Aufbau gleich dem des Ausführungsbeispiels nach **Fig. 1**, aber es wird kein Lichtmuster auf die Meßfläche projiziert. Als Sensoreinrichtung wird ein CCD-Chip eingesetzt, mit dem alle Flächenelemente einzeln gemessen werden.

Im Diagramm gemäß **Fig. 4** ist auf der Ordinate die gemessene Lichtintensität aufgetragen und auf der Abszisse die Winkelabweichung bezüglich des idealen Reflexionswinkels. Die gemessene Intensität ist im Bereich des idealen Reflexionswinkels am höchsten und nimmt dann mit zunehmenden Winkelabstand ab. Aus der so erfaßten Kurve kann auf einfache Weise das Reflexionsverhalten an der Oberfläche beurteilt werden.

Außerdem ist es möglich, aus der Kurve die Kennwerte abzuleiten, welche gemäß den verschiedenen Normen das Reflexionsverhalten von Oberflächen kennzeichnen.

In **Fig. 5** ist das von einer Meßfläche auf die Sensoreinrichtung reflektierte Bild dargestellt. Das Lichtmuster 50 weist dunkle Kanten 51 und helle Linien 52 auf. Ein zweiter Teil des Reflexes 53 weist kein Lichtmuster auf.

Mit zunehmender Rauigkeit nimmt der Kontrast im Lichtmuster ab, während ein schlechterer Glanz sich auf die Intensität auswirkt. Orange peel führt zu einer Verzerrung der einzelnen dunklen und hellen Linien, und eine strukturierte Oberfläche führt im Fall eines Rechteckprofils zum Versatz der einzelnen Linien im Bereich der Vertiefungen. Bei Sägezahn- oder Dreiecksprofilen können an den entsprechenden Abschnitten korrelierende Neigungen der Linien festgestellt werden.

Bei dem aufgenommenen Bild wird die Intensität und der Kontrast ausgewertet, und es werden die Gradienten im Bereich des Lichtmusters bestimmt. Durch Mittelwertbildung der Gradienten wird ein charakteristisches Maß für die Struktur der Meßfläche bestimmt.

In **Fig. 7** ist der prinzipielle, spaltungstechnische Aufbau einer Meßeinrichtung, wie sie bei den Ausführungsbeispielen

len gemäß **Fig. 1-6** zu verwenden ist, dargestellt.

Der prinzipielle Meßaufbau ist für alle gezeigten Ausführungsbeispiele derselbe; lediglich die Programmierung weicht je nach verwendeter Sensorart ab.

Die Meßeinrichtung weist insgesamt eine Steuereinrichtung **60** auf, welche einen handelsüblichen Mikroprozessor enthält, der durch ein Programm gesteuert wird, welches in einem Speicher **61** abgelegt ist. Der Kommunikation der Steuereinrichtung **60** mit dem Benutzer dient eine Eingabe-einrichtung **62**, die eine Anzahl von Schaltern enthält, um den Betrieb der Steuereinrichtung zu Starten und um (bei den entsprechenden Ausführungsbeispielen) zwischen einzelnen Betriebsarten umschalten zu können. Weiterhin kann der Benutzer ein Typkennzeichen für die zu messende, strukturierte Oberfläche in die Eingabeeinrichtung **61** eingeben, auf das sich nachfolgende Messungen beziehen.

Neben dem Mikroprozessor weist die Steuereinrichtung Ein-/Ausgabeeinrichtungen auf, welche der Verbindung der Steuereinrichtung mit den einzelnen Bauelementen der Vorrichtung dienen.

Die Steuereinrichtung ist mit der Lichtquelle **3** und dem Sensor **14** verbunden. Die Ergebnisse der Messung werden in einem Display **65** angezeigt, welcher vorzugsweise als LCD-Display ausgebildet ist. Für die weitere Auswertung der Messung ist eine Verbindung zu einem externen Computer **76** vorgesehen; die Meßergebnisse werden vorzugsweise auch in der Speichereinrichtung **61** abgelegt.

Die Meßvorrichtung wird durch eine (nicht dargestellte) Batterie mit Strom versorgt.

Die Meßvorrichtung wird insgesamt vorzugsweise in einem Gehäuse **1** angeordnet, welches in etwa die Abmessung eines Taschenbuchs aufweist.

Ein weiteres Ausführungsbeispiel der vorliegenden Erfindung wird nun in Bezug auf **Fig. 7** beschrieben.

Bei diesem Ausführungsbeispiel ist die Vorrichtung insgesamt in einem Gehäuse **100** angeordnet, welches eine Öffnung **101** aufweist, mit der das Gerät auf die zu messende Oberfläche aufgesetzt wird.

Im Unterschied zu den vorangegangenen Ausführungsbeispielen wird das Gerät jedoch nicht direkt auf die Oberfläche aufgesetzt, sondern mittels (schematisch angedeuteter) mindestens zwei Gummiwalzen **103**, **104** oder mindestens vier Gummirädern **103**, **104**, welche drehbeweglich (nicht gezeigt) im Gehäuse **100** gelagert sind.

Wenigstens eines der Gummiräder oder Walzen ist mit einer (nicht dargestellten) Wegstreckenmeßeinrichtung versehen, welche die Winkelbewegungen der Gummiräder **103**, **104** erfaßt und ein dafür repräsentatives elektrisches Signal ausgibt.

Die Vorrichtung weist weiterhin eine erste optische Einrichtung **110** auf, in welcher eine punktförmige Lichtquelle **111** und eine Linse **112** angeordnet sind. Diese erste optische Einrichtung **110** ist derart ausgerichtet, daß die optische Achse in einem vorbestimmten Winkel (im gezeigten Beispiel 45°) zu der zu messenden Oberfläche **115** ausgerichtet ist.

Unter einem zweiten vorbestimmten Winkel (hier ebenfalls 45°) ist eine zweite optische Einrichtung angeordnet, die eine Linseneinrichtung **121**, eine Blendeneinrichtung **122** und einen senkrecht zur optischen Achse ausgerichteten Meßsensor **123** aufweist, welcher in diesem Ausführungsbeispiel als Farb-CCD-Chip ausgeführt ist.

An einer dritten optischen Einrichtung **130** sind drei lichtausstrahlende Elemente **132**, **133** und **134** angeordnet, die hier als LEDs ausgeführt sind, welche jeweils eine unterschiedliche spektrale Charakteristik aufweisen, d. h., welche Licht mit unterschiedlichen Farbcharakteristiken ausstrahlen.

In dem hier dargestellten Ausführungsbeispiel fällt das von den LEDs ausgestrahlte Licht im wesentlichen senkrecht auf die zu untersuchende Oberfläche.

Das von der ersten und der dritten optischen Einrichtung ausgestrahlte Licht wird an der zu untersuchenden Oberfläche **115** reflektiert und fällt zum Teil auf den FOTOSensor bzw. Farb-CCD-Chip **125**.

Da die lichtausstrahlenden Elemente **132-134** Licht mit unterschiedlichen Farbcharakteristiken im sichtbaren Bereich des Spektrums ausstrahlen, ist es möglich, einen Farbkennwert für die zu untersuchende Oberfläche zu bestimmen.

Patentansprüche

1. Vorrichtung zur quantifizierten Bestimmung der Qualität strukturierter Oberflächen mit:
einer ersten optischen Einrichtung mit wenigstens einer Lichtquelle, deren Licht in einem vorbestimmten Winkel auf eine Meßfläche, die Teil der zu messenden Oberfläche ist, gerichtet ist,
einer zweiten optischen Einrichtung, welche in einem vorbestimmten Winkel zu dieser Meßfläche ausgerichtet ist, und welche das von der Meßfläche reflektierte Licht aufnimmt, wobei diese zweite optische Einrichtung wenigstens einen FOTOSensor mit wenigstens einer lichtempfindlichen Fläche aufweist, wobei dieser wenigstens eine FOTOSensor ein elektrisches Meßsignal ausgibt, welches für das reflektierte Licht charakteristisch ist;
einer Auswerteeinrichtung, die zur Steuerung des Meßablaufs vorgesehen ist und die wenigstens eine Prozessoreinrichtung und wenigstens eine Speichereinrichtung aufweist;
einer Ausgabe-Einrichtung;
wobei diese erste optische Einrichtung und diese zweite optische Einrichtung so beschaffen sind, daß dieses reflektierte Licht durch die Struktur dieser Meßfläche beeinflußt wird, und
wobei diese Auswerteeinrichtung dieses reflektierte Licht auswertet und daraus wenigstens eine Strukturkennzahl ableitet, welche wenigstens eine strukturebedingte Eigenschaft dieser Oberfläche charakterisiert.
2. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung über ein in dieser Speichereinrichtung gespeichertes Programm diese Meßsignale auswertet und/oder in dieser Speichereinrichtung speichert.
3. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese zweite optische Einrichtung eine Vielzahl von FOTOSensoren beinhaltet, welche in Reihen und/oder Spalten angeordnet sind.
4. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein erster Teil dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts ein Lichtmuster aufweist.
5. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieses von dieser ersten optischen Einrichtung ausgestrahlte Lichtmuster wenigstens eine Hell-/Dunkelkante oder eine Vielzahl von Hell-/Dunkelkanten aufweist.
6. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Teil dieser Vielzahl von Hell-/Dunkelkan-

ten wenigstens abschnittsweise parallel zueinander verläuft.

7. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Teil dieser Vielzahl von Hell-/Dunkelkanten gitter- oder kreuzgitter- oder kreisförmig angeordnet sind.

8. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieser erste Teil dieses von dieser ersten optischen Einrichtung ausgestrahlten Lichts dieses Lichtmuster aufweist und daß ein zweiter Teil ohne Lichtmuster auf diese Meßfläche gerichtet ist.

9. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung für wenigstens einen Teil dieser Vielzahl von Fotosensoren eine Steigung des elektrischen und/oder digitalen Meßsignals bestimmt, indem eine Differenz des elektrischen und/oder digitalen Meßsignals dieses Fotosensors mit diesem elektrischen und/oder digitalen Meßsignals des nächsten Fotosensors in dieser Reihe und/oder Spalte gebildet wird.

10. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung für wenigstens einen Teil dieser Vielzahl von Fotosensoren die Steigungen des elektrischen und/oder digitalen Meßsignals bestimmt, indem die Differenzen des elektrischen und/oder digitalen Meßsignals dieser Fotosensoren mit den elektrischen und/oder digitalen Meßsignalen aller benachbarten Fotosensoren gebildet werden.

11. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung wenigstens einen Durchschnitt wenigstens eines Teils der Steigungen bestimmt und daraus diese charakteristische Strukturkennzahl für diese wenigstens eine strukturbedingte Eigenschaft dieser Oberfläche bestimmt.

12. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß eine, zwei, drei oder mehr charakteristische optische Kenngrößen dieser Meßfläche bestimmt werden.

13. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese charakteristische optische Kenngröße der Glanz oder der Glanzschleier oder die Abbildungsschärfe (DOT) der Meßfläche ist.

14. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese charakteristische optische Kenngröße ein repräsentatives Maß für die typische Wellenlänge und deren Amplitude (Orange peel) der Topologie der Schichtdicke dieser Meßfläche in einem vorbestimmten Wellenlängenintervall ist, wobei diese Auswertung auch in zwei oder mehr Wellenlängenbereichen erfolgen kann.

15. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Vorrichtung eine dritte optische Einrichtung mit wenigstens einer Lichtquelle aufweist und Licht mit einer vorgegebenen spektralen Charakteristik ausstrahlt, welches in einem vorbestimmten Winkel auf diese Meßfläche gerichtet ist.

16. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieser vorbestimmte Winkel, in welchem das von der ersten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist, und/oder daß dieser vorbestimmte Winkel, in welchem die zweite optische

Einrichtung das von dieser Meßfläche reflektierte Licht aufnimmt, und/oder daß dieser vorbestimmte Winkel, in welchem das von der dritten optischen Einrichtung ausgestrahlte Licht zu dieser Meßfläche ausgerichtet ist, einstellbar ist/sind und insbesondere die Winkel 5°, 10°, 15°, 20°, 30°, 45°, 60°, 75°, 80° und 85° umfaßt.

17. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß dieses von dieser dritten optischen Einrichtung ausgestrahlte Licht in einem derartigen Winkel auf die Oberfläche gerichtet ist, daß das unmittelbar von der Meßfläche gemäß der Fresnel'schen Reflexion gerichtet reflektierte Licht gegenüber dieser Meßfläche einen anderen Winkel aufweist, als der Winkel zwischen dieser Meßfläche und dem von dieser Meßfläche gerichtet reflektierten Licht, welches von dieser ersten optischen Einrichtung ausgestrahlt wird.

18. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese wenigstens eine Lichtquelle dieser dritten optischen Einrichtung wenigstens zwei, vorzugsweise drei lichtausstrahlende Elemente aufweist, deren Licht sich in der emittierten spektralen Charakteristik derart unterscheidet, daß die emittierten Wellenlängenbereiche unterschiedlich sind.

19. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß sich diese emittierten Wellenlängenbereiche dieser lichtausstrahlenden Elemente im sichtbaren Bereich des Lichtspektrums zumindest teilweise überlappen, und daß diese emittierten spektralen Charakteristiken dieser lichtausstrahlenden Elemente voneinander linear unabhängig sind.

20. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Fotosensor wenigstens zwei, vorzugsweise drei oder mehr fotosensitive Elemente aufweist, deren elektrische Ausgangssignale einzeln erfaßbar sind und die sich in ihrer spektralen Charakteristik unterscheiden, so daß als optische Kenngröße dieser Meßfläche die Farbe des reflektierten Lichtes erfaßbar ist.

21. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese wenigstens eine Lichtquelle dieser ersten optischen und/oder dieser dritten optischen Einrichtung eine oder mehr Leuchtdioden und/oder Laserlichtquellen aufweist.

22. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste und/oder diese dritte optische Einrichtung im wesentlichen paralleles Licht ausstrahlt.

23. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste und/oder dritte optische Einrichtung im wesentlichen divergentes oder konvergentes Licht ausstrahlt.

24. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste und/oder dritte optische Einrichtung wenigstens einen Lichtfleck mit vorbestimmten Durchmesser senkrecht zur Ausbreitungsrichtung oder wenigstens einen Lichtstreifen mit vorbestimmter Länge und Breite senkrecht zur Ausbreitungsrichtung ausstrahlt.

25. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens eine dieser wenigstens einen optischen Kenngrößen aus den elektrischen und/oder digitalen Meßsignalen dieser Fotosensoren bestimmt wird, welche diesen von der Meßfläche reflektierten zweiten Teil dieses

von dieser ersten optischen Einrichtung ausgestrahlten Lichts aufnehmen.

26. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß in möglichst unmittelbarer Nähe dieser Lichtquelle(n) dieser ersten optischen Einrichtung und/oder dieser Lichtquelle(n) dieser dritten optischen Einrichtung und/oder dieses wenigstens einen Fotosensors wenigstens eine Temperaturmeß-Einrichtung angeordnet ist, welche zur Bestimmung der charakteristischen Temperatur der jeweiligen Lichtquelle(n) und/oder der jeweiligen Fotosensoren vorgesehen ist, damit eine temperaturkorrigierte Bestimmung dieses wenigstens einen Strukturkennwerts und/oder dieser wenigstens einen optischen Kenngröße erfolgen kann.

27. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Teil des Verlaufs des Abbilds dieser wenigstens einen Hell-/Dunkelkante auf dieser Vielzahl von Fotosensoren bestimmt wird und durch eine Abweichung vom idealen Verlauf von diesem gemessenen Verlauf ein charakteristischer Profilhöhen-Kennwert dieser Meßfläche bestimmt wird.

28. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Strukturkennzahl mittels eines Triangulationsverfahrens bestimmt wird.

29. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens einer dieser wenigstens einen Sensors die Intensität des von dieser ersten optischen Einrichtung ausgestrahlten und von der zu untersuchenden Oberfläche reflektierten Lichts bestimmt, und daß diese Auswerteeinheit einen charakteristischen Orts-Kennwert für den Ort der Messung in dieser Speichereinrichtung speichert und daß diese Auswerteeinheit aus dieser gemessenen Intensität einen charakteristischen Abstands-Kennwert für den Abstand der zu untersuchenden Oberfläche von dieser Vorrichtung bestimmt und in dieser Speichereinrichtung speichert.

30. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese erste optische Einrichtung eine Scan-Einrichtung aufweist, welche diese zu untersuchende Oberfläche abtastet.

31. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerte-Einrichtung mit diesen Abstands-Kennwerten und diesen Orts-Kennwerten wenigstens eine dieser wenigstens einen Strukturkennzahl bestimmt.

32. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung wenigstens einen quantitativen Steilheitskennwert für die Steilheit der Struktur der zu untersuchenden Oberfläche bestimmt.

33. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerteeinrichtung wenigstens einen quantitativen Schärfenkennwert für die Schärfe der Kanten der Struktur der zu untersuchenden Oberfläche bestimmt.

34. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens eine Bewertungs-Strukturkennzahl aus dieser wenigstens einen Strukturkennzahl und wenigstens einer der genannten charakteristischen optischen Größen (Glanz, Schleierglanz, Abbildungsschärfe, Orange Peel) gebildet wird, oder daß für jede der bestimmten charakteristischen optischen Kenngrößen (Glanz,

Schleierglanz, Abbildungsschärfe, Orange Peel) wenigstens eine Bewertungs-Strukturkennzahl bestimmt wird.

35. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß jeweils wenigstens ein unterer und/oder wenigstens ein oberer Grenzwert für diese wenigstens eine Strukturkennzahl und/oder diese wenigstens eine charakteristische optische Kenngröße und /oder diese wenigstens eine Bewertungs-Strukturkennzahl festlegbar ist, bei dessen Unter- bzw. Überschreitung eine Nachricht auf dieser Ausgabe-Einrichtung oder ein Alarmsignal ausgegeben wird.

36. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens einer dieser wenigstens einen Grenzwerts durch Messung wenigstens einer Referenzoberfläche automatisch bestimmt werden.

37. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß eine Eingabe-Einrichtung angeordnet ist, in welche der Benutzer ein Typkennzeichen für den zu messenden Oberflächentyp eingeben kann, und daß die nachfolgenden Messungen mit Bezug auf dieses Typkennzeichen in dieser Speichereinrichtung dauerhaft abgelegt werden.

38. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß zusätzlich ein Ein-/Ausschalter angeordnet ist, und daß der Benutzer beim Einschalten durch ein akustisches und/ oder optisches Signal aufgefordert wird, in diese Eingabe-Einrichtung ein Typkennzeichen für den zu messenden Oberflächentyp einzugeben.

39. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß der Benutzer durch Eingabe eines Typänderungszeichens oder durch Betätigen eines Typänderungsschalters diese Auswerte-Einrichtung in einen Typänderungsmodus umschaltet, und daß nach Eingabe eines weiteren Typkennzeichens die Ergebnisse nachfolgender Messungen mit Bezug auf dieses weitere Typkennzeichen in dieser Speichereinrichtung dauerhaft abgelegt werden.

40. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß der Benutzer nach jeder Messung durch Eingabe einer Benutzer-Beurteilungszahl in diese Eingabe-Einrichtung diese Meßfläche bewerten kann, und daß diese Benutzer-Beurteilungszahl zusammen mit wenigstens einer dieser Kenngrößen und -werte und dieser wenigstens einen Bewertungs-Strukturkennzahl in dieser Speichereinrichtung dauerhaft abgelegt wird, wobei diese Benutzer-Beurteilungszahl ein subjektives Maß des Benutzers für die Qualität dieser Meßfläche ist, und wobei der Wertebereich dieser Benutzer-Beurteilungszahl wenigstens zwei unterschiedliche Werte, die "gut" und "schlecht" entsprechen, umfaßt.

41. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Auswerte-Einrichtung wenigstens ein Teil dieser in dieser Speichereinrichtung abgelegten Benutzer-Beurteilungszahlen und zugehörigen Kenngrößen und -werten zur automatischen Bestimmung wenigstens eines dieses wenigstens einen Grenzwerts für dieses Typkennzeichen verwendet wird.

42. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß diese Vorrichtung relativ zur Meßfläche in einer im wesentli-

chen abstandsgleichen Richtung verschiebbar ist, und daß eine Wegstreckenmeßeinrichtung vorgesehen ist, welche diese relative Verschiebung quantitativ erfaßt, und daß weiterhin eine Speichereinrichtung vorgesehen ist, in welcher die entlang vorgegebener Meßstellen auf der Oberfläche gemessenen Strukturkennwerte und/oder optischen Kenngrößen abgespeichert werden.

43. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens ein Meßrad vorgesehen ist, welches während der Messung auf der zu messenden Oberfläche aufgesetzt ist, und sich während der Relativbewegung zwischen der Vorrichtung und der zu messenden Oberfläche dreht.

44. Vorrichtung nach mindestens einem der vorhergehenden Ansprüche, dadurch gekennzeichnet, daß wenigstens eines dieses wenigstens einen Meßrades mit einem Drehwinkel-Geber verbunden ist, der ein elektrisches Drehwinkelsignal ausgibt, welches für den vom Meßrad zurückgelegten Drehwinkel repräsentativ ist.

45. Verfahren zur quantifizierten Bestimmung der Qualität strukturierter Oberflächen insbesondere unter Verwendung einer Vorrichtung gemäß mindestens einem vorhergehenden Ansprüche, bei welchem eine erste optische Einrichtung mit einer ersten Lichtquelle vorgesehen ist, um Licht unter einem vorbestimmten Winkel auf eine zu messende Oberfläche zu richten, und

eine zweite optische Einrichtung vorgesehen ist, welche wenigstens einen Fotosensor aufweist, welche in einem zweiten vorbestimmten Winkel zu dieser Meßfläche ausgerichtet ist, um das von der Meßfläche reflektierte Licht aufzunehmen, wobei dieser wenigstens eine Fotosensor ein elektrisches Meßsignal ausgibt, welches für das aufgenommene Licht charakteristisch ist, und

eine Auswerteeinrichtung angeordnet ist, die den Meßablauf steuert und die wenigstens eine Prozessoreinrichtung beinhaltet und welche diese Meßsignale in einer Speichereinrichtung speichert, und eine Ausgabe-Einrichtung die Meßergebnisse ausgibt, und diese Auswerteeinrichtung dieses reflektierte Licht auswertet und daraus wenigstens eine Strukturkennzahl ableitet, welche strukturbedingte Eigenschaften der Oberfläche charakterisiert.

Hierzu 4 Seite(n) Zeichnungen

50

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60

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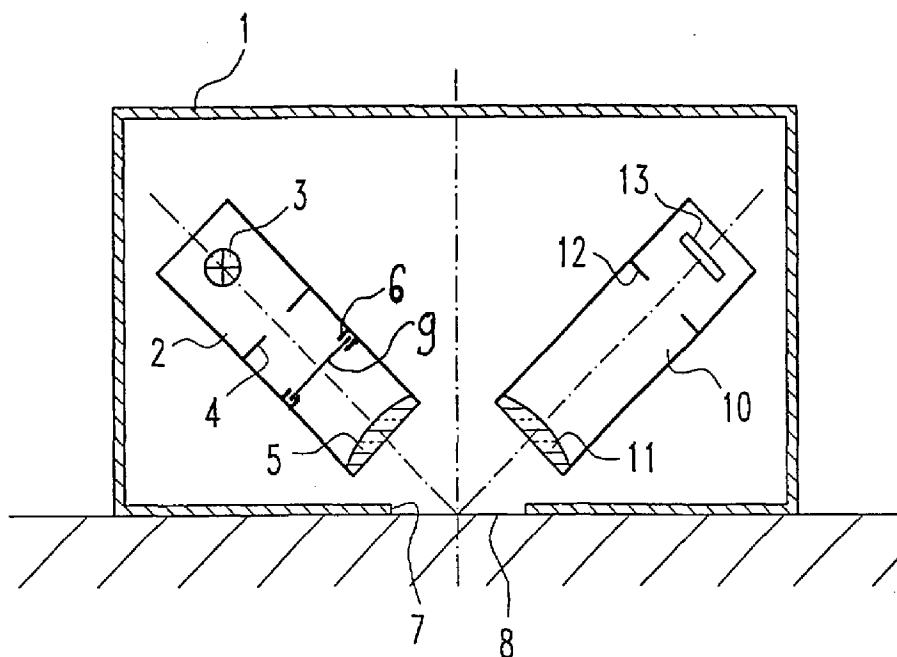


Fig.1

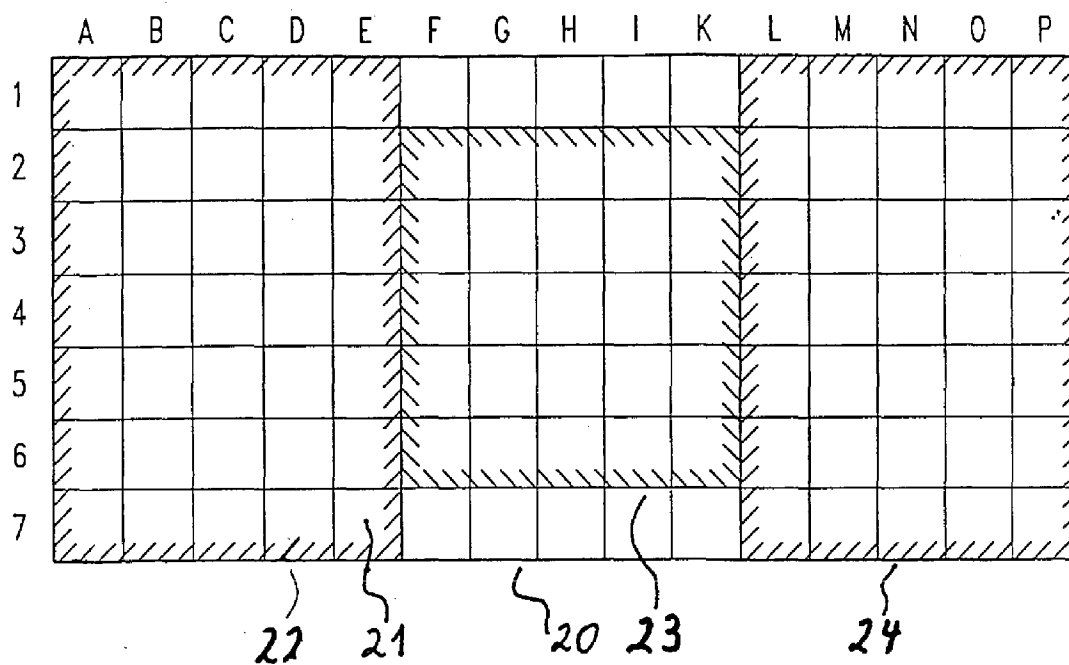


Fig. 2

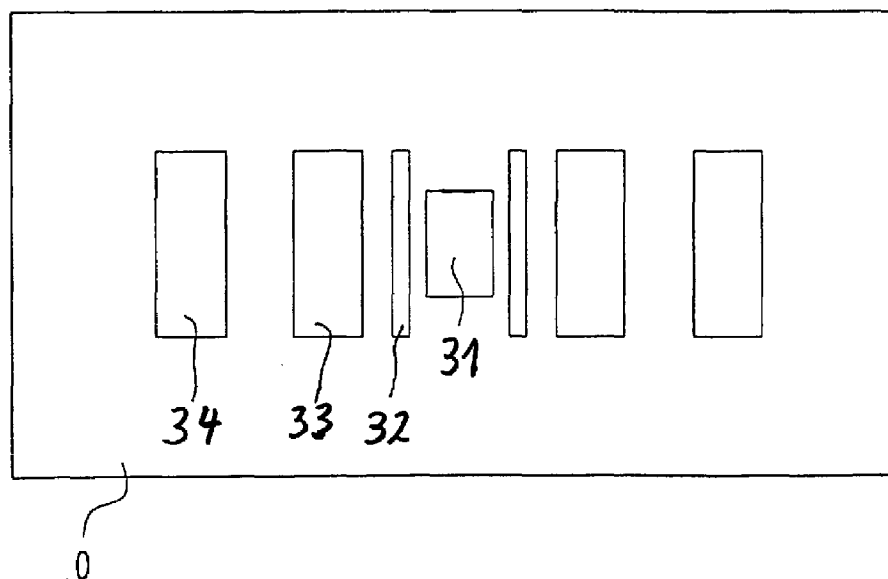


Fig. 3

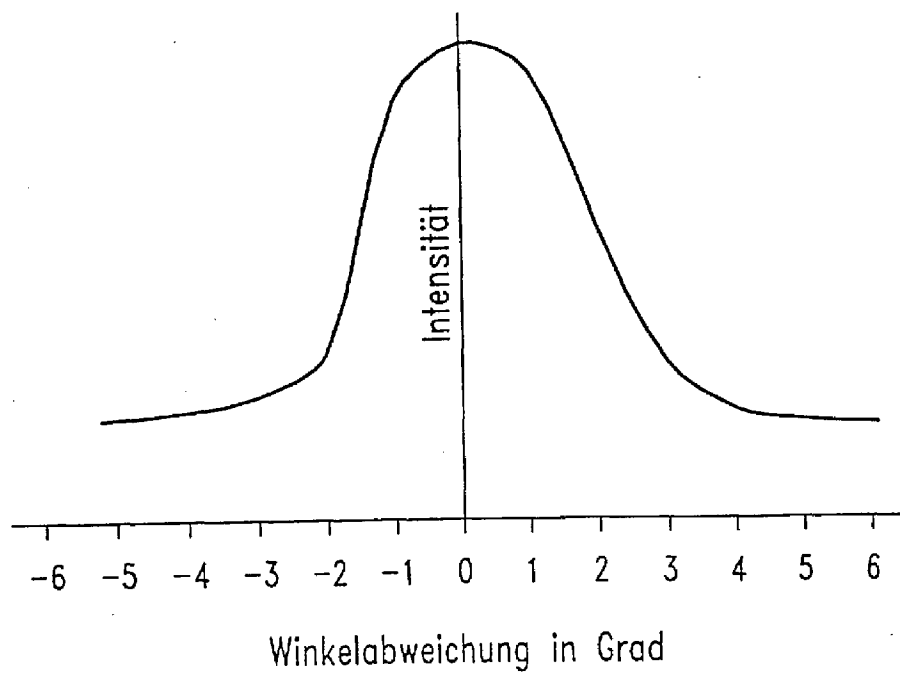


Fig. 4

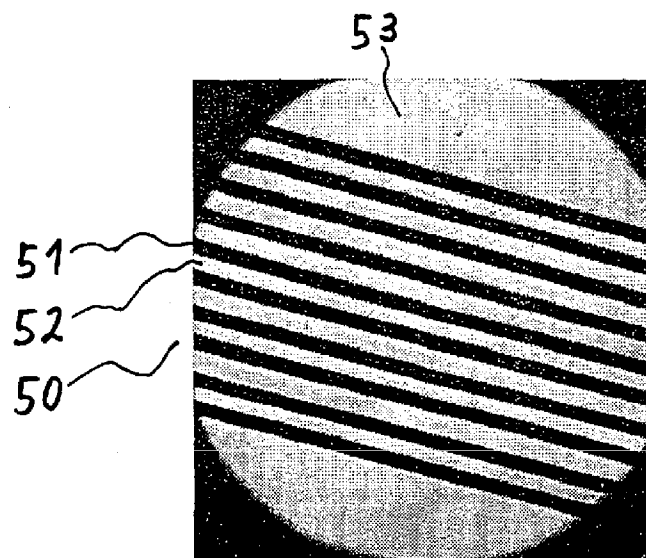


Fig. 5

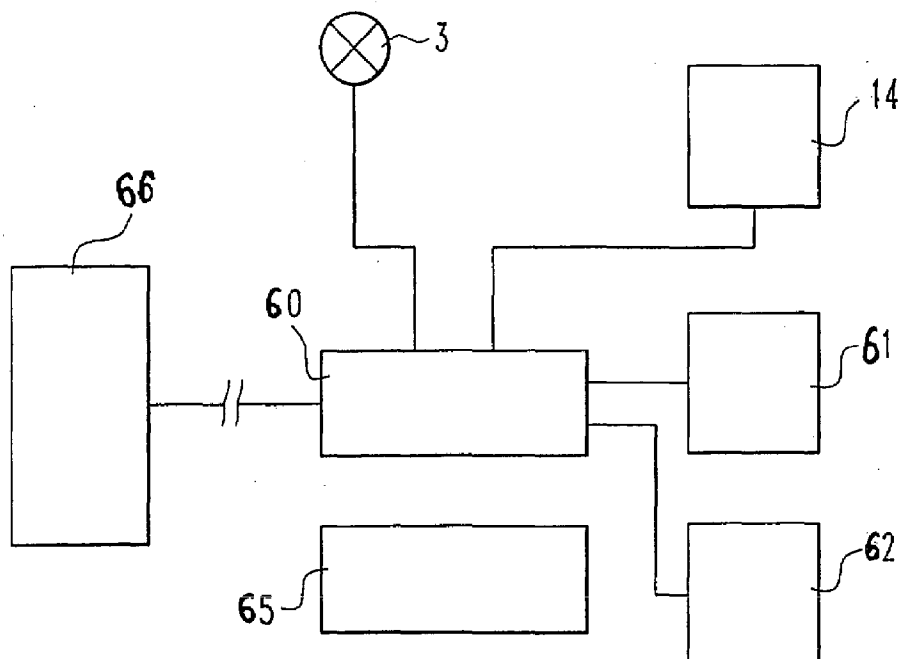


Fig. 6

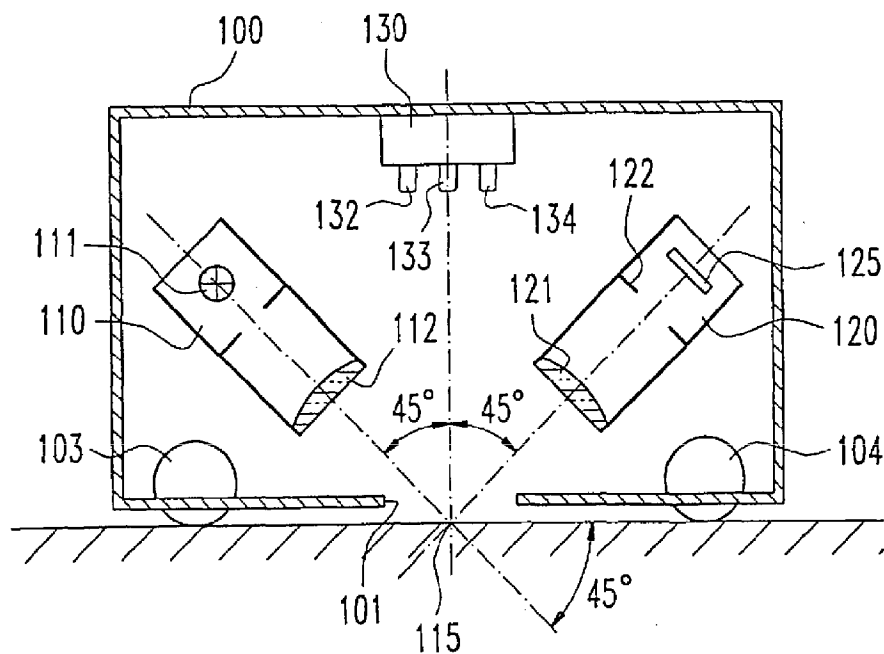
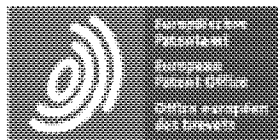


Fig. 7



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DESCRIPTION DE19846619A1

[0001]

¹³ The present invention relates to a device and a method for determining the quality of structured surfaces.

[0002]

¹⁸ The quality of a surface or a structured surface should be understood here to mean the physical properties of a surface that determine the appearance of a surface for the human observer.

[0003]

²³ In particular, properties such as structure, color, brightness of the color, gloss, sharpness of image

²⁵ DOI), glossy veils, surface textures and surface waviness (orange peel) etc. characterize structured surfaces.

[0004]

³⁰ The nature of the visible surfaces is an essential property of objects of everyday life, such as furnishings and everyday objects such as cars and the like.

[0005]

³⁵ In the field of automotive engineering, great importance is attached to the visual impression of the visible surfaces.

³⁷ The technical problems that arise in the design of surfaces in the motor vehicle sector are

explained in more detail below, without, however, restricting the application of the present invention in any way.

[0006]

⁴³ Motor vehicle bodies are usually provided with a high-gloss or metallic finish, the reflectivity or gloss characteristic of which the corresponding values of other surfaces, such as.

⁴⁵ B. furniture, is far superior.

⁴⁶ The high gloss of the paints used and the relatively large, flat surfaces require extremely careful preparation of the surfaces to be painted and particularly careful application of the paint.

[0007]

⁵¹ In the prior art, various methods, devices and apparatus are known, such as.

⁵² B. in DE 44 34 203 A1, with which the visual properties and especially the reflection behavior of surfaces can be determined.

[0008]

⁵⁷ Furthermore, a large number of structured surfaces are processed in the automotive industry in the interior of motor vehicles, which are made of natural materials, such as.

⁵⁹ B. consist of wood or plastic.

[0009]

⁶³ The visual impression of structured surfaces depends essentially on the distinctiveness of the structure of the surface, but is also determined by the other optical parameters mentioned above.

[0010]

⁶⁹ The automobile manufacturers now employ a large number of testers and use a large number of complex, large-scale measuring devices that check the quality of the surfaces visually or automatically in order to identify quality defects in surfaces during production.

⁷² However, this method has a number of disadvantages.

[0011]

⁷⁶ The work of the visual inspectors is very strenuous and requires work rooms whose lighting conditions are always precisely defined.

⁷⁸ Nevertheless, large differences were found in the evaluation of the same surfaces by different testers, because on the one hand the respective physiological impression differs from testers to

testers, and on the other hand the eyesight of the individual testers also depends on their respective physical condition.

[0012]

⁸⁵ It is therefore very difficult to define a lower quality limit, below which it can no longer be accepted.

[0013]

⁹⁰ Furthermore, it is difficult even for experienced testers to determine what causes an inadequate visual impression.

⁹² In the case of structured surfaces in particular, the visual impression is determined by a combination of different parameters, so that it is difficult to determine e.g.

⁹⁴ B. to change control values of an automatic painting device to improve the quality.

[0014]

⁹⁸ Structured plastic surfaces are often produced using an embossing process in which the shapes or shapes used are

¹⁰⁰ Embossing rollers etc. are subject to wear.

¹⁰¹ With an increasing number of embossed surfaces, the structure of the embossing mold and thus also the structure of the plastic surface becomes smaller and smaller.

¹⁰³ By regularly exchanging the embossing tools, production can be optimized in terms of quality and quantity; However, the service life of the tools used differs in the event of fluctuations in the material composition or fluctuations in the ambient conditions, so that the tools are generally already exchanged, although their quality is still sufficient for the production of finished structured surfaces.

[0015]

¹¹¹ Are automatically working devices used to check the quality of structured surfaces, e.g.

¹¹² B. to determine the gloss or the sharpness of the image, quantitative numerical values can be obtained.

¹¹⁴ However, a decisive disadvantage in the visual and also in the automatic tests is that although values are determined for the optical properties examined, the user has no information whatsoever about the cause.

[0016]

¹²⁰ Another disadvantage of conventional measuring apparatus is that they require a considerable amount of space and therefore cannot be taken along by the user.

[0017]

¹²⁵ It is therefore the object of the present invention to provide a method and a device of the type mentioned above, so that a reproducible assessment of the quality of surfaces and in particular structured surfaces can be carried out and that such physical quantities of the surface to be measured can be determined that a characteristic structure-related property of this surface is determined.

[0018]

¹³³ Another aspect of the object is to create a device which is compact and lightweight and is designed in such a way that it can be easily taken along by an operator and used to evaluate a surface without further aids.

[0019]

¹³⁹ Another aspect of the object is to create a device for measuring visual properties in which, despite the compact design according to the previous aspect of the object of the invention, the possibilities for measurement are considerably expanded compared to the devices known in the prior art.

[0020]

¹⁴⁶ Another aspect of the task is to specify a method which enables an advantageous detection of a characteristic structure-related property of surfaces.

[0021]

¹⁵¹ According to the invention, this object is achieved by a device as defined in claim 1.

¹⁵² The method according to the invention is the subject of claim 45.

[0022]

¹⁵⁶ Preferred developments of the invention are the subject matter of the subclaims.

[0023]

¹⁶⁰ In the device according to the invention, a first optical device, which has one or more light sources, directs light at a predetermined angle onto a measuring surface which is part of the surface to be measured.

[0024]

¹⁶⁶ The light reflected from the measuring surface is recorded with a second optical device which is oriented at a second, predetermined angle to this measuring surface and has one or more light-sensitive photosensors.

[0025]

¹⁷² This photosensor or photosensors, which have at least one light-sensitive surface, emit an electrical measurement signal which is characteristic of the reflected light, the first and second optical devices being designed so that the reflected light passes through the structure this measuring surface is influenced.

[0026]

¹⁷⁹ An evaluation device is arranged in the device according to the invention, which is provided for controlling the measurement sequence and which has at least one processor device and at least one memory device and which evaluates the reflected light and derives at least one structural code from it, which characterizes at least one of the structural properties of this surface.

[0027]

¹⁸⁷ An output device which is arranged in the device according to the invention is used to output the measured values.

[0028]

¹⁹² The evaluation device evaluates the measurement signals via a program stored in the memory device and, in a preferred development of the device according to the invention, stores the measurement signals and / or the measured values in this memory device.

[0029]

¹⁹⁸ In a further preferred development of the device according to the invention, a multiplicity of photo sensors are arranged in the second optical device, which photo sensors can be arranged in rows and / or columns.

[0030]

²⁰⁴ In contrast to conventional devices, in the device according to the invention the photosensors can be arranged on a common substrate.

206 The individual photosensors can also be formed by interconnecting photosensitive surfaces on a common substrate, the surface of each photosensitive sensor being selected or selectable such that it corresponds to a predetermined angular range of the angle of reflection in the measuring device.

[0031]

213 The device according to the invention has considerable advantages: An essential prerequisite for controlling the production sequence in the production of structured and non-structured surfaces is the determination of the optical quality or the visual impression of the surface.

216 By analyzing the light reflected on the surface, a structural index is derived that describes the quality of the surface.

218 A key figure for the structure of structured surfaces is not determined by conventional visual or automatic measuring methods.

[0032]

223 This is particularly advantageous since the profile of the structure of a structured surface makes a significant contribution to the overall visual impression of this surface.

[0033]

228 When embossing, casting or other processes are used, the molds used are subject to wear, including depends on the number of structured surfaces produced.

230 In addition to the number of manufactured components with structured surfaces, environmental conditions such as air humidity, material and air temperature also play a decisive role in the quality of the surface to be manufactured.

[0034]

236 In conventional manufacturing processes, the molds used must be replaced after a certain number of embossed or cast components or after a certain period of time in order to ensure a consistently high quality of the components produced.

[0035]

242 This method leads to an unnecessarily high effort, since tools of sufficient quality are often exchanged, while on the other hand it also happens that the tools are worn out before the intended change time due to environmental influences or material inhomogeneities or differences.

[0036]

²⁴⁹ One way of avoiding excessive rejects is to check them regularly.

²⁵⁰ For this purpose, the production process must be stopped to check the tools, so that costly downtimes result.

[0037]

²⁵⁵ With the device according to the invention it is possible to determine the quality of structured surfaces easily, quickly and reliably during ongoing production, so that impermissible wear of the tools can be detected early and unnecessary downtimes of the production plant can be avoided.

[0038]

²⁶² Another advantage of using the device according to the invention is that unnecessarily frequent changes of the tools, molds and stamping rollers and the like used can be omitted, which leads to lower production costs.

[0039]

²⁶⁸ Another important advantage is that the device according to the invention is compact and can therefore be taken along by the user at any time.

[0040]

²⁷³ Therefore, the device according to the invention is also suitable for the final inspection in poorly accessible places or in interiors of z.

²⁷⁵ B. motor vehicles, where a variety of structured surfaces is used.

[0041]

²⁷⁹ A further advantage of such a small device is that the quality can be determined simply and reliably even on concavely or convexly curved surfaces.

[0042]

²⁸⁴ It is also advantageous that the device according to the invention can be used to measure a wide variety of points on a surface to be examined quickly and easily and thus the quality distribution over a surface or also the reproducibility can be tested.

[0043]

290 In a preferred development of the device according to the invention, at least a first part of this light emitted by this first optical device has a light pattern.

292 The first optical device may include a pattern device.

293 The light emitted by this at least one light source of this first optical device is at least partially radiated onto this pattern device in this development of the device according to the invention.

295 The light transmitted through this pattern device is influenced in amplitude and / or phase in accordance with the pattern of the pattern device.

[0044]

300 In a further preferred development of the device according to the invention, the light pattern emitted by this pattern device has at least one light / dark edge, or a multiplicity of light / dark edges, in a further development of the device according to the invention at least part of this multiplicity of Light / dark edges run parallel to one another at least in sections.

[0045]

307 The pattern of this pattern device can be arranged symmetrically to the optical axis.

308 In the event that the emitted light pattern has a plurality of light / dark edges, the individual light / dark edges can have a straight, circular or sinusoidal course.

[0046]

313 In a preferred development, at least a part of this plurality of light / dark edges is arranged in a grid, cross-grid or circular shape.

[0047]

318 The projection of a light pattern onto the surface to be examined is particularly advantageous since the evaluation of the light pattern reflected from the surface to be examined or the measuring surface allows conclusions to be drawn about the quality and the structure of the measuring surface to be examined.

[0048]

325 It is possible that the light pattern does not have any sharp transitions from light to dark, but rather influences the intensity of the projected light in a sinusoidal or sawtooth shape.

[0049]

330 In a further, preferred development of the device according to the invention, a first part of a light emitted by this first optical device has this light pattern, while a second part without a light

pattern is directed onto the measuring surface.

[0050]

³³⁶ However, it is also possible that the pattern device can be switched on in the device according to the invention, so that an alternating measurement with a light pattern and without a light pattern is possible.

[0051]

³⁴² For this purpose, it is possible that this pattern device z.

³⁴³ B. has an LCD device in which individual areas or individual pixels can be connected to different light patterns as required, so that the light pattern can be quickly and easily adapted to the given requirements or switched off.

[0052]

³⁴⁹ By evaluating the signal from this at least one photosensor, which records at least part of the reflected light pattern, this evaluation device can determine a structural code number which is characteristic of at least one structure-related property of this surface.

[0053]

³⁵⁵ In a further, preferred development of the device according to the invention, this evaluation device determines a slope of the electrical and / or digital measurement signal for at least some of this plurality of photosensors by adding a difference between the electrical and / or digital measurement signal of the photosensor and this electrical and / or digital measurement signal of the next photosensor in this row and / or column is formed, or by forming differences in the electrical and / or digital measurement signal of this photosensor with this electrical and / or digital measurement signal of at least a part of the directly or diagonally adjacent photosensors.

[0054]

³⁶⁶ This is particularly advantageous since the gradients of the measurement signals and in particular the gradients of the measurement signals when irradiated with a light pattern are suitable for deriving a structural code which is characteristic of at least one structure-related property of the measurement surface to be examined.

[0055]

³⁷³ In a further preferred development of the device according to the invention, this evaluation

device determines at least an average of at least part of these slopes of the measurement signals of these photosensors and from this at least one average determines this characteristic structural number for this at least one structure-related property of this surface.

[0056]

380 From these slopes, however, a histogram can also be determined, in which the slope over the number of these slopes in the corresponding intervals or ranges for predetermined and / or selectable areas.

383 Areas is applied.

384 With the formation of the center of gravity and / or the mean value of the gradients, a characteristic structural number can be derived in this evaluation device, which characterizes the structure-related properties of this surface.

[0057]

390 This preferred development of the device according to the invention is particularly advantageous because a characteristic structural number for this measuring surface can be determined quickly and reliably by determining at least some of the slopes and by forming the average slope.

[0058]

397 Rounded surface structures, as they are e.g.

398 B. when using worn or worn tools in the production of structured surfaces, reflect light projected onto this surface more diffusely than surfaces with a sharper profile.

[0059]

403 By using such a development of the device according to the invention, it is possible to determine the structure-related properties.

[0060]

408 In a further preferred development of the device according to the invention, in addition to this characteristic structural number, at least one, two, three or more characteristic, optical parameters of this measuring surface are determined as optical parameters.

[0061]

414 In preferred developments of the invention, the gloss or haze or the distinction of image DOI or a representative measure of the typical wavelength and its amplitude (orange peel) is used

The topology of the layer thickness of this measuring surface is determined.

[0062]

⁴²⁰ The orange peel is determined in a predetermined wavelength interval; however, the evaluation to determine the orange peel can also be carried out in two or more wavelength ranges.

⁴²² It is also possible to determine the color of the measuring surface or the brightness of the color.

[0063]

⁴²⁶ In a further preferred development of this invention, in addition to the structural code, further of the above-mentioned characteristic optical parameters are determined.

[0064]

⁴³¹ Such a development of the device according to the invention is particularly advantageous because this small device u.

⁴³³ U. all interesting, visual and physical properties of this measuring surface can be determined.

[0065]

⁴³⁷ In an advantageous further development of all the exemplary embodiments described above, a third optical device is arranged which has at least one light source and emits light with a predetermined spectral characteristic which is directed at this measuring surface at a predetermined angle.

[0066]

⁴⁴⁴ In the previously mentioned preferred developments of the device according to the invention, the predetermined angle at which the light emitted by the first optical device is aligned with this measuring surface and / or the predetermined angle at which the second optical device receives the light reflected from this measuring surface, and / or the predetermined angle at which the light emitted by the third optical device is aligned with this measuring surface can be freely set and includes in particular the angles 5 °, 10 °, 15 °, 20 °, 30 °, 45 °, 60 °, 75 °, 80 ° and 85 ° between the normal of the measuring surface and the corresponding angle.

⁴⁵¹ It should be noted, however, that angles other than those mentioned here are also possible.

[0067]

⁴⁵⁵ In a further preferred development of the invention, this light emitted by this third optical device is directed onto the surface at an angle such that the light reflected directly from the measuring surface according to the Fresnel reflection has a different angle to this measuring surface than

the angle between this measuring surface and the light reflected by this measuring surface, which is emitted by this first optical device, so that the second optical device or the photosensors essentially receive the light emitted by the third optical device and light reflected diffusely from this measuring surface.

[0068]

⁴⁶⁵ This at least one light source of these third optical devices can have one, two and particularly preferably three light-emitting elements, the light of which differs in the emitted spectral characteristic in such a way that the emitted wavelength ranges are different.

[0069]

⁴⁷¹ The emitted wavelength ranges of these light-emitting elements can at least partially overlap in the visible range of the light spectrum, and in a further preferred development of the device according to the invention, the emitted, spectral characteristics of these light-emitting elements are linearly independent of one another.

[0070]

⁴⁷⁸ In a further preferred development of the device according to the invention, at least two, preferably three or more photosensors are designed in such a way that they differ in their spectral characteristics, so that the color of the reflected light can be detected as the optical parameter of this measuring surface.

[0071]

⁴⁸⁵ In another preferred development of the device according to the invention, at least one photosensor has at least two, preferably three or more photosensitive elements, the electrical output signals of which can be detected individually and which differ in their spectral characteristics, so that the color of the light reflected from this measuring surface can be detected is.

[0072]

⁴⁹³ A further development of the device according to the invention, which makes it possible to detect the color of the light reflected from this measuring surface, is particularly advantageous since the color and the brightness of the color are essential features of surfaces and also of structured surfaces.

[0073]

500 In further preferred developments of the device according to the invention, one or more light-emitting diodes or laser light sources are arranged in this first and / or this third optical device.

[0074]

505 At least part of the light emitted by this first optical device and / or this third optical device is essentially parallel, divergent or convergent.

[0075]

510 In a further, preferred development of the device according to the invention, this first and / or third optical device emits at least one light spot with a predetermined diameter perpendicular to the direction of propagation of the emitted light or at least one light strip with a predetermined length and width perpendicular to the direction of propagation of the emitted light.

[0076]

517 This light pattern projected by the first optical device can also consist of different light spots or light strips.

[0077]

522 It should be noted that this at least one optical parameter or this structural code can also be determined from the electrical and / or digital measurement signals of these photosensors, which record this second part reflected by this measurement surface without a light pattern of this light emitted by this first optical device.

[0078]

529 In a further preferred development of the device according to the invention, at least one light source of this first optical device and / or this at least one light source or this at least one light-emitting element of this third optical device and / or this at least one photo sensor or this at least one is in the immediate vicinity Photosensitive element arranged at least one temperature measuring device, which is provided for determining the characteristic temperature of the respective light source, light emitting elements, photosensors or light-sensitive elements so that a temperature-corrected determination of this at least one structural parameter and / or this at least one optical parameter can take place.

[0079]

540 The determination of the characteristic temperatures of the light sources and photosensors is particularly advantageous because these electrical devices work as a function of temperature

and the reproducibility of measurements is increased by a temperature-corrected determination of the parameters and parameters; because with increasing measurement duration or with changing measuring apparatus, the temperature and its distribution within the device according to the invention can change.

[0080]

549 In a further, preferred development of the invention, at least part of the course of the light pattern imaged on this plurality of photo sensors is determined.

551 A characteristic profile height parameter of this measuring surface can be determined by a deviation of the measured profile from the ideal profile.

[0081]

556 In a particularly preferred development, the course of at least one light / dark edge is determined in sections on this plurality of photo sensors or on the CCD chip.

[0082]

561 If a light or dark line is projected onto a surface to be examined, which has a rectangular profile with a profile height h , the reflection of this line is spatially separated on the CCD chip of the second optical device.

[0083]

567 The parts of the dark or light line that are reflected by the lower edges of the rectangular profile are reflected spatially offset parallel to those that are reflected at the upper edges of the rectangular profile.

[0084]

573 The reflected image of an ideal, rectangular structured surface are two dashed light or dark lines on the CCD chip.

[0085]

578 With the known angle of incidence and reflection and the distance between these two dashed lines on the CCD chip, the height of the profile can be determined at any location on the measuring surface.

[0086]

584 When measuring non-rectangular surface profiles, a profile height characteristic value of this measuring surface can be determined analogously with the corresponding geometric relationships.

[0087]

590 The determination of a profile height parameter is particularly advantageous because when the manufacturing tools wear out, the profile height of the manufactured components can also decrease.

[0088]

596 In a further preferred development of the invention, this structural code number is determined by means of a triangulation method, it being possible for this structural code number to be determined exclusively or additionally by means of this triangulation method.

599 It should be pointed out that one of the common triangulation methods, as they have become known in the prior art, can be used.

[0089]

604 As an example, reference should be made to the triangulation methods shown in the American patents US Pat. No. 5,546,189 and US Pat. No. 5,744,793, which hereby become part of the disclosure of the present invention.

[0090]

610 In a further preferred development of the device according to the invention, in contrast to the method produced in US Pat. No. 5,546,189, not only a gray value is determined which is representative of the height of the measuring surface at the measuring location, but the surface height at the measuring location is as large as possible Precision determined.

[0091]

617 In a further, preferred development of the invention, an additional optical device with at least one photo sensor is provided, which is oriented at a predetermined angle to this measuring surface.

620 In a particularly preferred development, this additional optical device is arranged in the plane spanned by the measuring point, the second optical device and the third optical device, and it is particularly preferred that the angle at which this additional optical device absorbs the reflected light is equal to the angle at which the second optical device receives the reflected light, so that the second and the additional optical device are arranged symmetrically to the light emitted by the first optical device or symmetrically to the light emitted by the third optical

device.

[0092]

⁶³⁰ In a further, preferred development of the invention, the intensity and the location of the point of impact of the reflected intensity on this second optical device and this additional optical device is determined, and a characteristic location value for the location of the measurement and the measured intensity is particularly preferably provided by this evaluation unit stored in this memory device, and this evaluation unit determines a characteristic distance value for the distance to the measuring point on the surface to be examined and stores this in this memory device.

[0093]

⁶⁴⁰ In a further preferred development of the invention, the first optical device has a scanning device which scans the surface to be examined in such a way that the light emitted by this first optical device scans the surface to be examined one after the other.

[0094]

⁶⁴⁶ Particularly preferably, this evaluation device uses these distance parameters and these location parameters to determine at least one of these at least one structure code.

[0095]

⁶⁵¹ In a further, preferred development of the device according to the invention, at least one quantitative steepness parameter is determined for the steepness of the structure of the surface to be examined.

[0096]

⁶⁵⁷ In a further, preferred development of the invention, this evaluation device determines at least one quantitative sharpness parameter for the sharpness of the edges of the surface to be examined.

[0097]

⁶⁶³ These developments are particularly advantageous because with the wear of the tools, the structure of the surface of the components to be manufactured decreases and, in particular, the steepness of the structure decreases due to wear and tear and, in particular, the sharpness of the edges decreases, since the profile of the tool rounds off with increasing wear.

[0098]

⁶⁷⁰ In a further, preferred development, an evaluation structure index is derived from this at least one structural index and at least one of the aforementioned characteristic optical quantities (gloss, haze, image sharpness, orange peel), which is characteristic of at least some of the visual properties of this surface.

[0099]

⁶⁷⁷ When evaluating surfaces and, in particular, structured surfaces, the visual impression depends not only on the development of the structure, but also on the other characteristic, optical parameters mentioned.

⁶⁸⁰ For example, in the case of particularly smooth and highly reflective surfaces, the demands on the manufacturing process are particularly high, since the human eye perceives even the smallest disturbances, unevenness and changes on such surfaces.

⁶⁸³ In motor vehicles with so-called "metallic" paintwork, it is almost impossible to touch up paint damage over a large area in such a way that this damage remains invisible to the viewer.

[0100]

⁶⁸⁸ In contrast to this, the requirements for the coating quality of the paint and the underlying surface are considerably lower for matt and weakly reflective surfaces.

[0101]

⁶⁹³ While surface roughness and orange peel disturb the physiological perception of a viewer even to a small extent on high-gloss surfaces, far greater roughness and surface waviness on weakly reflective and matt surfaces have little or no effect on the viewer's visual impression.

⁶⁹⁶ In the case of structured surfaces, the visual impression due to the structure is less dependent on the surface roughness, the gloss or the other optical parameters.

[0102]

⁷⁰¹ According to this preferred development of the invention, therefore, at least one evaluation structure index and at least one of the named characteristic variables are formed, which represents a measure of the quality of the measuring surface.

[0103]

⁷⁰⁷ In a preferred development of the invention, this characteristic value for the gloss of the measuring surface and this structural characteristic number is used to form an evaluation structure characteristic number in such a way that a sufficient or satisfactory evaluation

structure characteristic number is formed for areas with low gloss even if the structure characteristic number is not ideal.

[0104]

715 In a further, preferred development it is provided that this evaluation structure code is formed such that, for.

717 B. a higher gloss compensates for a poorer structural index, so that this evaluation structural index achieves a qualitatively sufficient result.

[0105]

722 In a further, preferred development it is provided that an evaluation structure index is formed for at least two or all of the mentioned optical variables gloss, haze, image sharpness, orange peel or that from two or more of the mentioned characteristic variables (gloss, haze, image sharpness, orange peel) and from this structural index an evaluation structure index is derived, which is characteristic of the visual impression of this measuring surface.

[0106]

730 The formation of at least one evaluation structure index from this at least one structural index and at least one of the characteristic optical variables, such as gloss, etc., is particularly advantageous since a particularly simple and automatic or automated determination of the quality of the surface to be examined is possible.

[0107]

737 In a further, preferred development of the device according to the invention, at least one lower or upper limit value can be determined for this at least one structural code number and / or this at least one characteristic optical parameter and / or this at least one evaluation structure code number.

741 If a message is exceeded on this output device or an alarm signal of an acoustic or optical type is output.

[0108]

746 In a further, preferred development of the invention, this at least one limit value is automatically determined by measuring at least one reference surface.

748 It is Z.

749 B. possible that one, two or more evaluated reference surfaces of different quality are measured, wherein advantageously at least one of these reference surfaces has a high quality and at least one other has an inadequate quality.

[0109]

⁷⁵⁵ Such a partially automated calibration of this device enables different measuring points to be carried out quickly and reliably on a surface or on different surfaces to be examined.

[0110]

⁷⁶⁰ In a further, preferred development, an entry or

⁷⁶¹ Arranged on-off switch.

[0111]

⁷⁶⁵ Furthermore, an input device can be provided on this device according to the invention, this being inter alia. it can be provided that the user can enter a type identifier for the type of surface to be measured, with subsequent measurements being evaluated with reference to this type identifier by this evaluation device and stored in this memory device.

[0112]

⁷⁷² In a further, preferred development of the invention, the user will be prompted by an acoustic or optical signal when switching on to enter a type identifier for the surface type to be measured in this input device, whereby in a particularly preferred development the user can enter another type identifier at any time Can change the surface type to which the following measurements refer.

[0113]

⁷⁸⁰ Entering or selecting a type identifier before measuring a specific surface type is particularly advantageous because the individual characteristic values have a different meaning when evaluating different surface types and a different weighting when forming an evaluation structure characteristic number.

[0114]

⁷⁸⁷ In a further, preferred development of the invention, the user can evaluate this measurement area according to his physiological impression during each measurement by entering the user assessment number in this input device.

⁷⁹⁰ This user assessment number is particularly preferably stored permanently in this memory device together with this at least one characteristic, optical parameter and this at least one structure code and this at least one assessment structure code, the range of values of this user assessment number at least two different quality values, the "good" and " bad "includes.

[0115]

⁷⁹⁷ In a further, preferred development of the invention, at least some of these user assessment numbers stored in this memory device with associated parameters and parameters of the corresponding measurements are used to automatically determine at least one of this at least one limit value for this type code of the surface to be examined.

[0116]

⁸⁰⁴ In a further, preferred development of the device according to the invention, this evaluation device takes into account essentially all of the user assessment numbers stored in this memory device for this type code of the surface to be measured and the corresponding parameters and parameters and structural parameters for determining this at least one limit value and for determining the weighting factors which are used to determine the evaluation structure index, so that these weighting factors and this at least one limit value are adapted with an increasing number of measurements in such a way that this evaluation structure index is adapted to the visual impression of the user.

[0117]

⁸¹⁵ This preferred development of the invention is particularly advantageous since this device can learn when determining these parameters and parameters and this evaluation structure code and, on the basis of an expert system, the reliability in determining this at least one evaluation structure code increases with the number of measurements.

[0118]

⁸²² In this preferred development of the invention, it is also provided that when determining these limit values and weighting factors, not all measured and stored values are used for this type identifier, since the physiological condition of the user affects the evaluation of this measurement area, and because of erroneous input incorrect user rating numbers, undesired limit values and weighting factors can be determined.

[0119]

⁸³⁰ It is possible e.g.

⁸³¹ B. that a certain percentage of z.

⁸³² B. 5% or 10% remain unconsidered in the formation of these weighting factors and these limit values, the selection of the unconsidered values is made according to known statistical methods.

[0120]

⁸³⁸ This development of the invention is particularly advantageous because even if incorrect user assessment numbers are entered by mistake, they are essentially ignored, so that the learning capability of this evaluation device increases the reliability of the automatically determined evaluation structure parameters with an increasing number of measurements.

[0121]

⁸⁴⁵ In an advantageous further development of all of the exemplary embodiments described above, this device can be displaced relative to the measuring surface in a direction essentially equidistant from one another, and a distance measuring device is provided which quantitatively detects this relative displacement.

⁸⁴⁹ The structural parameters and / or characteristic optical parameters and / or evaluation structure parameters measured along predetermined measurement thresholds on this surface are stored in this memory device.

[0122]

⁸⁵⁵ In a further, preferred development, at least one measuring wheel is provided which is placed on the surface to be measured during the measurement and rotates during the relative movement between the device and the surface to be measured, whereby this at least one measuring wheel can be connected to a rotary encoder, which outputs an electrical angle of rotation signal which is representative of the angle of rotation covered by the measuring wheel.

[0123]

⁸⁶³ This preferred development of the invention is particularly advantageous because surfaces to be examined z.

⁸⁶⁵ B. can be measured systematically over the entire surface.

[0124]

⁸⁶⁹ In the preferred development of the invention, in which a plurality of photo sensors are arranged, a CCD chip is particularly preferably used, which in a further preferred development of the invention can be designed as a color CCD chip.

[0125]

⁸⁷⁵ When using a two-dimensional CCD chip, a two-dimensional image of this measuring surface is imaged on the CCD chip in the second optical device.

[0126]

880 In a preferred development of the invention, a dark image is recorded before the start of the measurements or before the start of each individual measurement, and the values are averaged.

883 During the measurement, the image is recorded with illumination, also averaged and the mean value of the dark image subtracted from this.

885 A value is determined for the contrast of this image.

886 The brightness of the measured image is normalized with the mean values of the dark image, the illuminated image and the value for the contrast.

[0127]

891 The further evaluation takes place with the normalized values or with the normalized image.

[0128]

895 In a further, preferred development of the invention, the evaluation takes place via one or more Fourier transformations.

897 The spatial frequency coefficients are particularly preferably determined for each line from a Fourier transformation of this line and the spatial frequency coefficients for the entire image are determined from the mean value of the frequencies of all lines.

[0129]

903 In a further, preferred development of the device according to the invention, this structural parameter is determined using a conventional method for measuring the roughness of surfaces, such as, for.

906 B. in DIN 4768 (edition 1990-05), ISO 13565-1 (edition 1996-12) or ISO 13565-2 (edition 1996-12), which thus become part of the disclosure.

[0130]

911 It should be pointed out that the interaction of the individual features according to the invention in any combination is preferred.

913 In particular, the combinations of features disclosed by the independent and dependent claims are also preferred, with one or more features being omitted.

915 It is evident to the person skilled in the art that, in addition to the exemplary embodiments shown here, a large number of further modifications and designs are conceivable which are covered by the invention.

918 In particular, the invention is not limited to the embodiments shown here.

[0131]

⁹²² Further advantages, features and possible applications of the present invention emerge from the following description of exemplary embodiments in conjunction with the drawings.

[0132]

⁹²⁷ Show in it:

[0133]

⁹³¹ 1 shows a section of a first embodiment according to the present invention;

[0134]

⁹³⁵ 2 shows a further exemplary embodiment of a sensor element in which the areas used for light measurement can be varied;

[0135]

⁹⁴⁰ 3 shows a further exemplary embodiment of a sensor element in which seven photosensors are formed with the light-sensitive surfaces;

[0136]

⁹⁴⁵ 4 shows a diagram which shows the reflection behavior when illuminated without a light pattern, the measured light intensity being plotted on the ordinate and the angular deviation with respect to the ideal reflection angle being plotted on the abscissa;

[0137]

⁹⁵¹ 5 shows an image of the light pattern projected onto the measuring surface,

[0138]

⁹⁵⁵ 6 shows the basic circuit design of a measuring device as is to be used in the exemplary embodiments according to FIGS. 1-5; and

[0139]

⁹⁶⁰ 7 shows a further exemplary embodiment of the invention, which is also suitable for further measurements.

[0140]

965 A first embodiment of the invention will now be described with reference to FIG.

[0141]

969 The measuring device shown schematically in the figure for determining the quality of structured surfaces has a housing in which a first exposure tube 2 is arranged.

[0142]

974 In this exposure tube, indicated schematically, a light source 3, a diaphragm 4, a light pattern device 6 and a lens device 5 are arranged.

[0143]

979 The light emitted by the point-shaped light source 3 is limited in its aperture by the diaphragm 4 and strikes the light pattern device 6.

981 Depending on the embodiment, a partially transparent light pattern plate 9 can be arranged in the light pattern device 6, in which the phase and / or the amplitude of the incident light is influenced on part of the illuminated surface in such a way that the transmitted light has a characteristic light pattern.

985 In this exemplary embodiment, the amplitude of the light emitted by the light source 3 is influenced by light / dark edges on the light pattern plate 9.

[0144]

990 Furthermore, the emitted light is bundled by the lens 5 and impinges on the measuring surface 8 through the opening 7.

992 The light is reflected from the measuring surface 8 and enters the measuring tube 10, which also has a lens 11, a diaphragm 12 and the actual sensor 13.

[0145]

997 The reflection measuring device also has a control device (not shown) by which the operation of the device is controlled and a display device, also not shown, by which the measured structural parameters and characteristic optical parameters are displayed.

[0146]

1003 In Fig. 2, the sensor of a further embodiment is shown.

1004 With the exception of the sensor, this exemplary embodiment is constructed in the same way as in FIG. 1.

1006 FIG. 2 shows the sensor device 20 in which a multiplicity of photosensitive elements 21 are arranged in rows and columns.

[0147]

1011 A CCD chip is used in this example.

1012 On the surface of the CCD chip, individual surface elements are interconnected to form different photosensors 22, 23, 24, so that only the signals from these photosensors 22, 23, 24 are present during the measurement.

[0148]

1018 It should be pointed out that the most varied of measurement geometries can be implemented by interconnecting individual surface elements to form photosensors.

1020 Furthermore, not all surface elements have to be used for the measurement.

[0149]

1024 In Fig. 3, a further embodiment of the sensor device is shown.

1025 The further construction and operation of this device takes place in the same way as in FIG. 1.

[0150]

1029 In FIG. 3, the sensor device 30 is shown, which, as in the exemplary embodiment according to FIG. 2, is designed as a CCD chip.

[0151]

1034 Different photo sensors 31, 32, 33 and 34 are formed on the surface of the CCD chip by interconnecting individual surface elements.

1036 By interconnecting individual surface elements to form photosensors, different measuring standards, such as.

1038 B. the American ASTM E 430 can be measured.

[0152]

1042 In a further exemplary embodiment, the basic structure is the same as that of the exemplary embodiment according to FIG. 1, but no light pattern is projected onto the measuring surface.

1044 A CCD chip is used as the sensor device, with which all surface elements are measured individually.

[0153]

- 1049 In the diagram according to FIG. 4, the measured light intensity is plotted on the ordinate and the angular deviation with respect to the ideal reflection angle is plotted on the abscissa.
- 1051 The measured intensity is highest in the area of the ideal reflection angle and then decreases with increasing angular distance.
- 1053 From the curve recorded in this way, the reflection behavior on the surface can be assessed in a simple manner.

[0154]

- 1058 It is also possible to derive the characteristic values from the curve, which characterize the reflection behavior of surfaces according to the various standards.

[0155]

- 1063 In FIG. 5, the image reflected from a measuring surface onto the sensor device is shown.
- 1064 The light pattern 50 has dark edges 51 and light lines 52.
- 1065 A second part of the reflex 53 has no light pattern.

[0156]

- 1069 With increasing roughness, the contrast in the light pattern decreases, while a poorer gloss affects the intensity.
- 1071 Orange peel leads to a distortion of the individual dark and light lines, and in the case of a rectangular profile, a structured surface leads to the displacement of the individual lines in the area of the depressions.
- 1074 In the case of sawtooth or triangular profiles, correlating inclinations of the lines can be determined at the corresponding sections.

[0157]

- 1079 The intensity and contrast of the recorded image are evaluated and the gradients in the area of the light pattern are determined.
- 1081 A characteristic measure for the structure of the measuring surface is determined by averaging the gradients.

[0158]

- 1086 In Fig. 7, the basic, cleavage-related structure of a measuring device, as is to be used in the embodiments according to FIGS. 1-6, is shown.

[0159]

1091 The basic measurement setup is the same for all shown exemplary embodiments; only the programming differs depending on the type of sensor used.

[0160]

1096 The measuring device overall has a control device 60 which contains a commercially available microprocessor which is controlled by a program which is stored in a memory 61.

1098 The communication between the control device 60 and the user is provided by an input device 62 which contains a number of switches in order to start the operation of the control device and to be able to switch between individual operating modes (in the corresponding exemplary embodiments).

1102 Furthermore, the user can enter a type identifier for the structured surface to be measured into the input device 61, to which subsequent measurements refer.

[0161]

1107 In addition to the microprocessor, the control device has input / output devices which serve to connect the control device to the individual components of the device.

[0162]

1112 The control device is connected to the light source 3 and the sensor 14.

1113 The results of the measurement are displayed in a display 65, which is preferably designed as an LCD display.

1115 A connection to an external computer 76 is provided for further evaluation of the measurement; the measurement results are preferably also stored in the memory device 61.

[0163]

1120 The measuring device is powered by a battery (not shown).

[0164]

1124 The measuring device as a whole is preferably arranged in a housing 1 which has approximately the dimensions of a pocket book.

[0165]

1129 Another embodiment of the present invention will now be described with reference to FIG.

[0166]

1133 In this exemplary embodiment, the device is arranged as a whole in a housing 100 which has an opening 101 with which the device is placed on the surface to be measured.

[0167]

1138 In contrast to the previous exemplary embodiments, however, the device is not placed directly on the surface, but by means of (schematically indicated) at least two rubber rollers 103, 104 or at least four rubber wheels 103, 104, which are rotatably mounted (not shown) in the housing 100.

[0168]

1145 At least one of the rubber wheels or rollers is provided with a distance measuring device (not shown) which detects the angular movements of the rubber wheels 103, 104 and outputs a representative electrical signal.

[0169]

1151 The device also has a first optical device 110 in which a point light source 111 and a lens 112 are arranged.

1153 This first optical device 110 is aligned such that the optical axis is aligned at a predetermined angle (45 ° in the example shown) to the surface 115 to be measured.

[0170]

1158 A second optical device is arranged at a second predetermined angle (here also 45 °), which has a lens device 121, a diaphragm device 122 and a measuring sensor 123 oriented perpendicular to the optical axis, which in this embodiment is designed as a color CCD chip.

[0171]

1164 Arranged on a third optical device 130 are three light-emitting elements 132, 133 and 134, which are embodied here as LEDs, each of which has a different spectral characteristic, i. that is, which light emit with different color characteristics.

[0172]

1170 In the exemplary embodiment shown here, the light emitted by the LEDs falls essentially perpendicularly onto the surface to be examined.

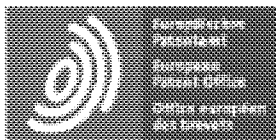
[0173]

¹¹⁷⁵ The light emitted by the first and the third optical device is reflected on the surface 115 to be examined and is partly incident on the photosensor or the photo sensor.

¹¹⁷⁷ Color CCD chip 125.

[0174]

¹¹⁸¹ Since the light-emitting elements 132-134 emit light with different color characteristics in the visible region of the spectrum, it is possible to determine a color characteristic for the surface to be examined.



Notice

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CLAIMS DE19846619A1

1.

¹³ Apparatus for the quantified determination of the quality of structured surfaces with: a first optical device with at least one light source, the light of which is directed at a predetermined angle onto a measuring surface which is part of the surface to be measured, a second optical device which is directed at a predetermined angle is aligned with this measuring surface and which receives the light reflected from the measuring surface, this second optical device having at least one photosensor with at least one light-sensitive surface, this at least one photosensor outputting an electrical measurement signal which is characteristic of the reflected light; an evaluation device which is provided for controlling the measurement sequence and which has at least one processor device and at least one memory device; an output device; This first optical device and this second optical device are designed so that this reflected light is influenced by the structure of this measuring surface, and this evaluation device evaluates this reflected light and derives at least one structural code from it, which characterizes at least one structure-related property of this surface.

2.

²⁹ Device according to at least one of the preceding claims, characterized in that this evaluation device evaluates these measurement signals via a program stored in this memory device and / or stores them in this memory device.

3.

³⁵ Device according to at least one of the preceding claims, characterized in that this second optical device contains a plurality of photo sensors which are arranged in rows and / or columns.

4.

⁴¹ Device according to at least one of the preceding claims, characterized in that at least a first part of this light emitted by this first optical device has a light pattern.

5.

⁴⁶ Device according to at least one of the preceding claims, characterized in that this light pattern emitted by this first optical device has at least one light / dark edge or a plurality of light / dark edges.

6.

⁵² Device according to at least one of the preceding claims, characterized in that at least some of this plurality of light / dark edges run parallel to one another at least in sections.

7.

⁵⁷ Device according to at least one of the preceding claims, characterized in that at least some of this multiplicity of light / dark edges are arranged in a grid or cross grid or circular manner.

8.

⁶² Device according to at least one of the preceding claims, characterized in that this first part of this light emitted by this first optical device has this light pattern and that a second part without a light pattern is directed onto this measuring surface.

9.

⁶⁸ Device according to at least one of the preceding claims, characterized in that this evaluation device determines a slope of the electrical and / or digital measurement signal for at least some of this plurality of photosensors by comparing a difference between the electrical and / or digital measurement signal of this photosensor with this electrical and / or or digital measurement signal of the next photo sensor in this row and / or column is formed.

10.

⁷⁶ Device according to at least one of the preceding claims, characterized in that this evaluation device determines the gradients of the electrical and / or digital measurement signal for at least some of this plurality of photosensors by comparing the differences in the electrical and / or digital measurement signals of these photosensors with the electrical and / or or digital

measurement signals from all neighboring photo sensors are formed.

11.

⁸⁴ Device according to at least one of the preceding claims, characterized in that this evaluation device determines at least an average of at least part of the slopes and from this determines this characteristic structural number for this at least one structure-related property of this surface.

12.

⁹¹ Device according to at least one of the preceding claims, characterized in that one, two, three or more characteristic optical parameters of this measuring surface are determined.

13.

⁹⁶ Device according to at least one of the preceding claims, characterized in that this characteristic optical parameter is the gloss or the gloss haze or the image sharpness (DOI) of the measuring surface.

14.

¹⁰² Device according to at least one of the preceding claims, characterized in that this characteristic optical parameter is a representative measure for the typical wavelength and its amplitude (orange peel) of the topology of the layer thickness of this measuring surface in a predetermined wavelength interval, this evaluation also in two or more Wavelength ranges can be done.

15.

¹¹⁰ Device according to at least one of the preceding claims, characterized in that this device has a third optical device with at least one light source and emits light with a predetermined spectral characteristic which is directed at this measuring surface at a predetermined angle.

16.

¹¹⁶ Device according to at least one of the preceding claims, characterized in that this predetermined angle at which the light emitted by the first optical device is aligned with this measuring surface, and / or that this predetermined angle at which the second optical device is directed from this measuring surface absorbs reflected light, and / or that this predetermined angle at which the light emitted by the third optical device is aligned with this measuring surface is / are adjustable and in particular the angles 5 °, 10 °, 15 °, 20 °, 30 °, 45 °, 60 °, 75

°, 80 ° and 85 °.

17.

¹²⁶ Device according to at least one of the preceding claims, characterized in that this light emitted by this third optical device is directed onto the surface at an angle such that the light reflected directly from the measuring surface according to the Fresnel reflection is different from this measuring surface Has angle, as the angle between this measuring surface and the directed reflected light from this measuring surface, which is emitted by this first optical device.

18.

¹³⁴ Device according to at least one of the preceding claims, characterized in that this at least one light source of this third optical device has at least two, preferably three light-emitting elements, the light of which differs in the emitted spectral characteristic in such a way that the emitted wavelength ranges are different.

19.

¹⁴¹ Device according to at least one of the preceding claims, characterized in that these emitted wavelength ranges of these light-emitting elements at least partially overlap in the visible range of the light spectrum, and that these emitted spectral characteristics of these light-emitting elements are linearly independent of one another.

20.

¹⁴⁸ Device according to at least one of the preceding claims, characterized in that at least one photosensor has at least two, preferably three or more photosensitive elements, the electrical output signals of which can be recorded individually and which differ in their spectral characteristics, so that the optical characteristic of this measuring surface is the color of the reflected light can be detected.

21.

¹⁵⁶ Device according to at least one of the preceding claims, characterized in that this at least one light source of this first optical and / or this third optical device has one or more light-emitting diodes and / or laser light sources.

22.

¹⁶² Device according to at least one of the preceding claims, characterized in that this first and / or this third optical device emits substantially parallel light.

23.

¹⁶⁷ Device according to at least one of the preceding claims, characterized in that this first and / or third optical device emits essentially divergent or convergent light.

24.

¹⁷² Device according to at least one of the preceding claims, characterized in that this first and / or third optical device emits at least one light spot with a predetermined diameter perpendicular to the direction of propagation or at least one light strip with a predetermined length and width perpendicular to the direction of propagation.

25.

¹⁷⁹ Device according to at least one of the preceding claims, characterized in that at least one of these at least one optical parameters is determined from the electrical and / or digital measurement signals of these photosensors, which pick up this second part of this light emitted by this first optical device, reflected by the measurement surface.

26.

¹⁸⁶ Device according to at least one of the preceding claims, characterized in that at least one temperature measuring device is arranged in the immediate vicinity of this light source (s), this first optical device and / or this light source (s), this third optical device and / or this at least one photosensor which is provided for determining the characteristic temperature of the respective light source (s) and / or the respective photosensors so that a temperature-corrected determination of this at least one structural parameter and / or this at least one optical parameter can take place.

27.

¹⁹⁶ Device according to at least one of the preceding claims, characterized in that at least part of the course of the image of this at least one light / dark edge is determined on this plurality of photosensors and a characteristic profile height parameter is determined by a deviation from the ideal course of this measured course Measurement area is determined.

28.

²⁰³ Device according to at least one of the preceding claims, characterized in that this structural code number is determined by means of a triangulation method.

29.

²⁰⁸ Device according to at least one of the preceding claims, characterized in that at least one of said at least one sensor determines the intensity of the light emitted by this first optical device and reflected by the surface to be examined, and that this evaluation unit has a characteristic location value for the location of the The measurement is stored in this memory device and that this evaluation unit determines a characteristic distance value for the distance of the surface to be examined from this device from this measured intensity and stores it in this memory device.

30.

²¹⁸ Device according to at least one of the preceding claims, characterized in that this first optical device has a scanning device which scans this surface to be examined.

31.

²²³ Device according to at least one of the preceding claims, characterized in that this evaluation device uses these distance parameters and these location parameters to determine at least one of these at least one structure code.

32.

²²⁹ Device according to at least one of the preceding claims, characterized in that this evaluation device determines at least one quantitative steepness characteristic value for the steepness of the structure of the surface to be examined.

33.

²³⁵ Device according to at least one of the preceding claims, characterized in that this evaluation device determines at least one quantitative sharpness parameter for the sharpness of the edges of the structure of the surface to be examined.

34.

²⁴¹ Device according to at least one of the preceding claims, characterized in that at least one evaluation structural code is formed from this at least one structural code and at least one of the said characteristic optical variables (gloss, haze, image sharpness, orange peel), or that for each of the determined characteristic optical parameters (gloss, haze, image sharpness, orange peel) at least one evaluation structure index is determined.

35.

²⁴⁹ Device according to at least one of the preceding claims, characterized in that in each case at least one lower and / or at least one upper limit value can be determined for this at least one structural code and / or this at least one characteristic optical parameter and / or this at least one evaluation structural code its lower resp.

²⁵³ Exceeding a message on this output device or an alarm signal is issued.

36.

²⁵⁷ Device according to at least one of the preceding claims, characterized in that at least one of these at least one limit value is automatically determined by measuring at least one reference surface.

37.

²⁶³ Device according to at least one of the preceding claims, characterized in that an input device is arranged into which the user can enter a type identifier for the surface type to be measured, and that the subsequent measurements with reference to this type identifier are permanently stored in this memory device.

38.

²⁷⁰ Device according to at least one of the preceding claims, characterized in that an on / off switch is additionally arranged, and that the user is prompted by an acoustic and / or optical signal when switching on to enter a type identifier for the type of surface to be measured in this input device to enter.

39.

²⁷⁷ Device according to at least one of the preceding claims, characterized in that the user switches this evaluation device to a type change mode by entering a type change code or by actuating a type change switch, and that after entering a further type code the results of subsequent measurements with reference to this further type code in stored permanently in this storage device.

40.

²⁸⁵ Device according to at least one of the preceding claims, characterized in that the user can evaluate this measuring surface after each measurement by entering a user assessment number into this input device, and that this user assessment number together with at least one of these parameters and values and this at least one structural evaluation number is permanently stored in this memory device, this user evaluation number being a subjective

measure of the user for the quality of this measurement area, and the range of values of this user evaluation number at least two different values, the "good" and "bad" correspond, includes.

41.

²⁹⁶ Device according to at least one of the preceding claims, characterized in that this evaluation device uses at least part of these user assessment numbers and associated parameters and values stored in this memory device for the automatic determination of at least one of said at least one limit value for this type identifier.

42.

³⁰³ Device according to at least one of the preceding claims, characterized in that this device is displaceable relative to the measuring surface in a substantially equidistant direction, and that a distance measuring device is provided which quantitatively detects this relative displacement, and that a storage device is also provided in which the structural parameters and / or optical parameters measured along predetermined measuring points on the surface are stored.

43.

³¹² Device according to at least one of the preceding claims, characterized in that at least one measuring wheel is provided which is placed on the surface to be measured during the measurement and rotates during the relative movement between the device and the surface to be measured.

44.

³¹⁹ Device according to at least one of the preceding claims, characterized in that at least one of this at least one measuring wheel is connected to a rotation angle transmitter which outputs an electrical rotation angle signal which is representative of the rotation angle covered by the measuring wheel.

45.

³²⁶ Method for the quantified determination of the quality of structured surfaces, in particular using a device according to at least one of the preceding claims, in which a first optical device with a first light source is provided to direct light at a predetermined angle onto a surface to be measured, and a second optical device Device is provided which has at least one photo sensor which is aligned at a second predetermined angle to this measuring surface in order to receive the light reflected from the measuring surface, this at least one photo sensor emitting

an electrical measurement signal which is characteristic of the recorded light, and an evaluation device is arranged which controls the measurement sequence and which contains at least one processor device and which stores these measurement signals in a memory device, and an output device outputs the measurement results, and this evaluation device The device evaluates this reflected light and derives at least one structural number from it, which characterizes the structure-related properties of the surface.



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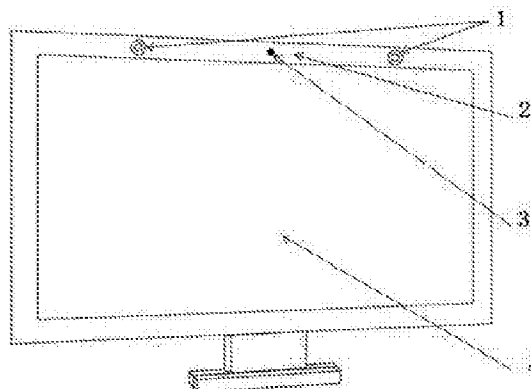
权利要求书2页 说明书5页 附图1页

(54)发明名称

一种基于双目结构光的三维手势控制智能电视的方法

(57)摘要

本发明公开了一种基于双目结构光的三维手势控制智能电视的方法,包括以下步骤:首先通过双目摄像头同步采集图像,根据获得的左右视图重建出具有深度信息的三维图像,并进行预处理,分割出三维手势后,再对手势动作进行识别,将识别出的手势动作转换为智能电视的操作指令,并执行。本发明采用双目视觉和结构光结合的方式,可以有效地提高环境抗干扰能力、降低光照条件的便会对指尖定位的影响;满足控制智能电视的实时性和有效性。



1. 一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,包括如下步骤:

步骤一,采集图像:通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片;

步骤二,重建图像三维:根据双目匹配算法,找出左视图、右视图之间各像素点的一一对应关系,根据他们之间的像素位置的横坐标差得到视差图,然后根据投影模型与三角测量原理,获得原始图像的深度信息;

步骤三,预处理深度图像:对深度图像进行中值滤波去除噪声点,再对深度图像中的内部空洞进行形态学处理;

步骤四,检测手势图像:对获得的图像深度数据进行分析,将人和背景分离开来,只选取人体的像素信息,将其余像素信息全部置零,获得人体的深度图像,最后获得手势的轮廓;

步骤五,识别三维手势:根据获得的手势轮廓,对手势轮廓提取中心距特征,中心距是指图形中必点到各边缘点的距离;设点 $O(x_0, y_0)$ 为手势的几何中心点,点集 $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ 为手势的边缘轮廓,不妨把手势的边缘轮廓看成有一个个边缘点连接而成,由此,几何中心点 $O(x_0, y_0)$ 和中心距 d 可由如下公式计算:

$$x_0 = \frac{1}{N} \sum_{i=1}^N x_i;$$

$$y_0 = \frac{1}{N} \sum_{i=1}^N y_i;$$

$$d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2};$$

再结合三维信息,利用基于贝叶斯网络的分类器进行识别;

步骤六,执行操作指令:根据在双目摄像头前做出的手势,智能电视执行相应的预先设定的操作指令,完成人机交互。

2. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,步骤一,采集图像:通过红外探测器探测操作环境的光照情况,若红外光过多则打开红外激光器;再通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片。

3. 根据权利要求2所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,若红外探测器探测操作环境的红外辐射强度小于 $0.2\text{W}/\text{m}^2 \cdot 850\text{nm}$,则打开红外激光器。

4. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,上述步骤三中的形态学处理即对图像边缘存在的毛刺和图像的内部空洞进行开、闭运算。

5. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,上述步骤四中获得手势的轮廓的具体步骤为:通过像素的灰度值反映改点距离镜头的远近,统计出人体的深度图像的灰度分布直方图,灰度直方图较小的峰是手部区域的灰度分布,较大的峰是人体躯干的灰度分布,从而分割出人体的手部图像,再对手部图像使用Canny边缘算了进行边缘检测,从而获得手势的轮廓。

6. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法, 其特征在于, 上述步骤五中利用基于贝叶斯网络的分类器进行识别的具体步骤如下:

步骤a, 计算不同手势类别下所有特征的高斯分布信息, 即是计算所有的均值与方差;

步骤b, 对于测试的每一个输入样本, 求出它在手势1类别下, 各个特征的概率, 由于每个特征是相互独立的, 因而特征下的概率分布也满足独立性, 由此便可以得出该输入样本在手势1类别下的总概率; 以此类推, 求出该样本在所有手势类别下的不同概率;

步骤c, 比较输入样本在不同类别下得到概率值, 找出其最大概率值所属的类别, 便预测该类别就是输入样本所属的类别。

一种基于双目结构光的三维手势控制智能电视的方法

技术领域

[0001] 本发明涉及人机交互领域,特别是一种手势控制的方法。

背景技术

[0002] 随着互联网技术的发展,智能电视——一种具备开放式操作系统与芯片,拥有开放式应用平台,可实现双向人机交互功能,集影音、娱乐、数据等多种功能于一体,以满足用户多样化和个性化需求的电视产品成为了电视的潮流趋势。

[0003] 传统的按键式遥控器主要是由集成电路电板和用来产生不同讯息的按钮所组成。内装有一个中央处理器CPU,CPU在制造时就将电视机各种菜单密码信息输入其中,电视机的遥控发射器只要发出与之对应的密码就可以实现电视机的遥控了。

[0004] 智能电视的发展对电视的交互操作提出了更多的要求。首先便是需要简洁方便的文字输入、网页浏览和频道选取,传统的遥控器显然无法满足新的交互需求。如果在智能电视上外接鼠标和键盘,用户则需要配置多个控制终端,且无法在远距离(3~5米)和复杂环境下(鼠标、键盘需要相对平坦的桌面)完成交互,给用户操作带来了很多的障碍。这也是目前电视的人机交互落后于计算机和手持设备的原因。

[0005] 日前,基于视觉的电视人机交互成为一个主流技术,而手势识别是其中非常重要的研究领域。手势语言是一种依靠动作和视觉进行交际的特殊语言,它是包含信息量最多的人体语言。而手势识别的本质就是根据用户的手势,识别出所表达的意义。然而目前的手势检测、指尖定位等图像处理技术大部分是基于二维的,适合近距离、简单背景的,不符合智能电视的交互需求。而三维手势识别能够判断出三维空间内的手势变化,以更为自然、直观的交互方式得到广泛关注,但是日前来讲,一些三维手势识别设备使用了数据手套,虽然其识别率高,但也失去了手势交互的自然性,费用也很昂贵;一些利用飞行时间的设备,对硬件要求高、费用高昂、分辨率也较低;基于双目视觉的设备视角较小、在昏暗环境下以及特征不明显的情况下效果较差;而基于结构光的设备非常容易受到光照影响。可以说现有的手势识别设备都存在不同程度的缺陷。

发明内容

[0006] 为解决现有技术的不足,本发明的目的在于提供一种基于双目结构光的三维手势控制智能电视的方法,本发明通过双目摄像头和红外结构光结合的方式获得优质的双手的三维信息,然后进行实时手势分割、检测以及指尖定位,完成三维手势动作与电视指令的实时转换,满足电视人机交互操作实时性的要求。

[0007] 为了实现上述目标,本发明采用如下的技术方案:

[0008] 一种基于双目结构光的三维手势控制智能电视的方法,包括如下步骤:

[0009] 步骤一,采集图像:通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片;

[0010] 步骤二,重建图像三维:根据双目匹配算法,找出左视图、右视图之间各像素点的

一一对应关系,根据他们之间的像素位置的横坐标差得到视差图,然后根据投影模型与三角测量原理,获得原始图像的深度信息;

[0011] 步骤三,预处理深度图像;对深度图像进行中值滤波去除噪声点,再对深度图像中的内部空洞进行形态学处理;

[0012] 步骤四,检测手势图像;对获得的图像深度数据进行分析,将人和背景分离开来,只选取人体的像素信息,将其余像素信息全部置零,获得人体的深度图像,最后获得手势的轮廓;

[0013] 步骤五,识别三维手势;根据获得的手势轮廓,对手势轮廓提取中心距特征,中心距是指图形中必点到各边缘点的距离;设点 $O(x_0, y_0)$ 为手势的几何中心点,点集 $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ 为手势的边缘轮廓,不妨把手势的边缘轮廓看成有一个个边缘点连接而成,由此,几何中心点 $O(x_0, y_0)$ 和中心距 d 可由如下公式计算:

$$[0014] \quad x_0 = \frac{1}{N} \sum_{i=1}^N x_i;$$

$$[0015] \quad y_0 = \frac{1}{N} \sum_{i=1}^N y_i;$$

$$[0016] \quad d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2};$$

[0017] 再结合三维信息,利用基于贝叶斯网络的分类器进行识别;

[0018] 步骤六,执行操作指令;根据在双目摄像头前做出的手势,智能电视执行相应的预先设定的操作指令,完成人机交互。

[0019] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤一,采集图像;通过红外探测器探测操作环境的光照情况,若红外光过多则打开红外激光器;再通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片。

[0020] 前述的一种基于双目结构光的三维手势控制智能电视的方法,若红外探测器探测操作环境的红外辐射强度小于 $0.2W/m^2 \cdot 850nm$,则打开红外激光器。

[0021] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤三中的形态学处理即对图像边缘存在的毛刺和图像的内部空洞进行开、闭运算。

[0022] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤四中获得手势的轮廓的具体步骤为:通过像素的灰度值反映改点距离镜头的远近,统计出人体的深度图像的灰度分布直方图,灰度直方图较小的峰是手部区域的灰度分布,较大的峰是人体躯干的灰度分布,从而分割出人体的手部图像,再对手部图像使用Canny边缘算子进行边缘检测,从而获得手势的轮廓。

[0023] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤五中利用基于贝叶斯网络的分类器进行识别的具体步骤如下:

[0024] 步骤a,计算不同手势类别下所有特征的高斯分布信息,即是计算所有的均值与方差;

[0025] 步骤b,对于测试的每一个输入样本,求出它在手势 l 类别下,各个特征的概率,由于每个特征是相互独立的,因而特征下的概率分布也满足独立性,由此便可以得出该输入

样本在手势1类别下的总概率;以此类推,求出该样本在所有手势类别下的不同概率;

[0026] 步骤c,比较输入样本在不同类别下得到概率值,找出其最大概率值所属的类别,便预测该类别就是输入样本所属的类别。

[0027] 本发明的有益之处在于:本发明提供一种基于双目结构光的三维手势控制智能电视的方法,本发明通过双目摄像头和红外结构光结合的方式获得优质的双手的三维信息,然后进行实时手势分割、检测以及指尖定位,完成三维手势动作与电视指令的实时转换,满足电视人机交互操作实时性的要求;

[0028] 本发明因为采用了双目视觉和结构光结合的方式,可以有效地提高环境抗干扰能力、降低光照条件的便会对指尖定位的影响,当在室内或环境昏暗时采用结构光测量深度信息,当太阳光照较强时则转为纯双目的方式。而且双目结构光的标定与激光器无关,当激光器损坏时替换较为方便;

[0029] 本发明通过对手势图像的前景分割减少了手势检测的遍历范围,可以满足电视人机交互操作实时性的要求。

附图说明

[0030] 图1是本发明的一种实施例的结构示意图;

[0031] 图中附图标记的含义:

[0032] 1红外摄像头,2红外探测器,3红外激光器,4电视。

具体实施方式

[0033] 以下结合附图和具体实施例对本发明作具体的介绍。

[0034] 如图1所示,本发明的所需的所有模块都集成在智能电视4当中。水平安装在智能电视4左右两边的是已经标定好的红外摄像头1,用于获取左右视角的图像。在两个红外摄像头1中间的是红外激光器3,用来发出红外条纹光,增加表面纹理,便于双目匹配,方便在昏暗条件下依旧能够准确识别手势。激光器旁边的是红外探测器2,用来监测操作环境的红外情况,以选择是否打开红外激光器3。而后续的图像三维重建、深度图像预处理、手势图像检测、三维手势识别以及人机交互则由智能电视4内的处理器来完成。

[0035] 一种基于双目结构光的三维手势控制智能电视4的方法,包括如下步骤:

[0036] 步骤一,采集图像;

[0037] 先通过红外探测器2探测操作环境的红外辐射情况,若红外辐射强度大于 $0.2\text{W}/\text{m}^2 \times 850\text{nm}$,激光器发出的编码光斑容易被淹没,则选用纯双目的方式;若红外辐射强度小于 $0.2\text{W}/\text{m}^2 \times 850\text{nm}$,编码光斑不容易被干扰,则打开红外激光器3,以提高双目匹配精度。

[0038] 再打开水平安装在智能电视4机内已经标定好的两个相同型号的红外摄像头1,双目摄像头以25帧每秒捕获摄像头前方图像,采集到的图像大小为 640×480 的RGB彩色空间的数字图像,并将双目摄像头拍摄的的左视图、右视图分别储存在智能电视4里两个独立的存储空间,再由处理器分别处理储存器内的两张图片。

[0039] 步骤二,重建图像三维;根据双目匹配算法,找出左视图、右视图之间各像素点的一一对应关系,根据他们之间的像素位置的横坐标差得到视差图,然后根据投影模型与三角测量原理,获得原始图像的深度信息;

[0040] 步骤三, 预处理深度图像; 由于获得的深度图像中手势边缘的深度值不稳定, 有许多0值噪声点, 直接处理会对算法的效果造成影响, 因此采用中值滤波去除噪声点, 降低目标边缘的模糊程度。再对深度图像中的内部空洞进行形态学处理; 形态学处理即对图像边缘存在的毛刺和图像的内部空洞进行开、闭运算。

[0041] 步骤四, 检测手势图像; 对获得的图像深度数据进行分析, 将人和背景分离开来, 只选取人体的像素信息, 将其余像素信息全部置零, 获得人体的深度图像, 最后获得手势的轮廓; 获得手势的轮廓的具体步骤为: 通过像素的灰度值反映改点距离镜头的远近, 统计出人体的深度图像的灰度分布直方图, 灰度直方图较小的峰是手部区域的灰度分布, 较大的峰是人体躯干的灰度分布, 从而分割出人体的手部图像, 再对手部图像使用Canny边缘算子进行边缘检测, 从而获得手势的轮廓。

[0042] 步骤五, 识别三维手势; 根据获得的手势轮廓, 对手势轮廓提取中心距特征(SCCD), 中心距是指图形中必点到各边缘点的距离; 设点 $O(x_0, y_0)$ 为手势的几何中心点, 点集 $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ 为手势的边缘轮廓, 不妨把手势的边缘轮廓看成有一个个边缘点连接而成, 由此, 几何中心点 $O(x_0, y_0)$ 和中心距 d 可由如下公式计算:

$$[0043] \quad x_0 = \frac{1}{N} \sum_{i=1}^N x_i;$$

$$[0044] \quad y_0 = \frac{1}{N} \sum_{i=1}^N y_i;$$

$$[0045] \quad d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2};$$

[0046] 再结合三维信息, 利用基于贝叶斯网络的分类器进行识别;

[0047] 利用基于贝叶斯网络的分类器进行识别的具体步骤如下:

[0048] 步骤a, 计算不同手势类别下所有特征的高斯分布信息, 即是计算所有的均值与方差;

[0049] 步骤b, 对于测试的每一个输入样本, 求出它在手势 l 类别下, 各个特征的概率, 由于每个特征是相互独立的, 因而特征下的概率分布也满足独立性, 由此便可以得出该输入样本在手势 l 类别下的总概率; 以此类推, 求出该样本在所有手势类别下的不同概率;

[0050] 步骤c, 比较输入样本在不同类别下得到概率值, 找出其最大概率值所属的类别, 便预测该类别就是输入样本所属的类别。

[0051] 步骤六, 执行操作指令; 根据在双目摄像头前做出的手势, 智能电视4执行相应的预先设定的操作指令, 完成人机交互。

[0052] 本发明提供一种基于双目结构光的三维手势控制智能电视4的方法, 本发明通过双目摄像头和红外结构光结合的方式获得优质的双手的三维信息, 然后进行实时手势分割、检测以及指尖定位, 完成三维手势动作与电视4指令的实时转换, 满足电视4人机交互操作实时性的要求;

[0053] 本发明因为采用了双目视觉和结构光结合的方式, 可以有效地提高环境抗干扰能力、降低光照条件的便会对指尖定位的影响, 当在室内或环境昏暗时采用结构光测量深度信息, 当太阳光照较强时则转为纯双日的方式。而且双目结构光的标定与激光器无关, 当激光器损坏时替换较为方便;

[0054] 本发明通过对手势图像的前景分割减少了手势检测的遍历范围,可以满足电视4人机交互操作实时性的要求。

[0055] 以上显示和描述了本发明的基本原理、主要特征和优点。本行业的技术人员应该了解,上述实施例不以任何形式限制本发明,凡采用等同替换或等效变换的方式所获得的技术方案,均落在本发明的保护范围内。

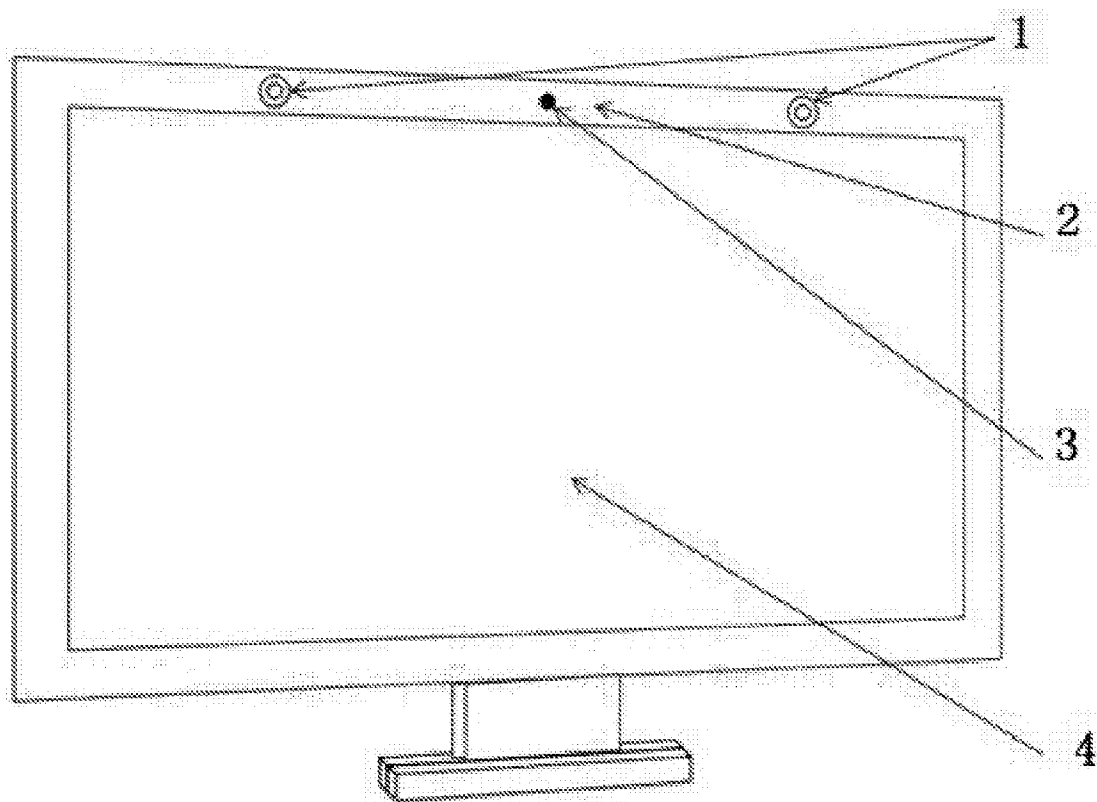
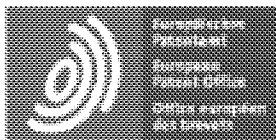


图1



Notice

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DESCRIPTION CN108363482A

¹⁰ Method for controlling smart TV with three-dimensional gesture based on binocular structured light

[0001]

¹⁴ Technical field

[0002]

¹⁸ The invention relates to the field of human-computer interaction, in particular to a method of gesture control.

[0003]

²² Background technique

[0004]

²⁶ With the development of Internet technology, smart TV-an open operating system and chip, with an open application platform, can achieve two-way human-computer interaction, and integrate multiple functions such as audio and video, entertainment, and data to meet TV products with diversified and individual needs of users have become the trend of TV.

[0005]

³³ The traditional push-button remote control is mainly composed of an integrated circuit board and buttons used to generate different messages.

³⁵ It is equipped with a central processing unit CPU, and the CPU inputs various menu password information of the TV when it is manufactured, and the remote control transmitter of the TV can realize the remote control of the TV as long as the corresponding password is issued.

[0006]

⁴¹ The development of smart TV puts forward more requirements on the interactive operation of the TV.

⁴² The first is the need for simple and convenient text input, web browsing and channel selection. The traditional remote control obviously cannot meet the new interactive needs.

⁴⁴ If an external mouse and keyboard are connected to the smart TV, the user needs to configure multiple control terminals, and it is not possible to complete the interaction in a long distance (3 to 5 meters) and in a complex environment (mouse and keyboard require a relatively flat desktop) to operate for the user Brings a lot of obstacles.

⁴⁸ This is also the reason why the current human-computer interaction of TV lags behind computers and handheld devices.

[0007]

⁵³ At present, vision-based television human-computer interaction has become a mainstream technology, and gesture recognition is a very important research field.

⁵⁵ Gesture language is a special language that relies on movement and vision to communicate. It is the human body language that contains the most information.

⁵⁷ The essence of gesture recognition is to recognize the meaning expressed according to the user's gestures.

However, most of the current image processing technologies such as gesture detection and fingertip positioning are based on two-dimensional, suitable for short-distance, simple background, and do not meet the interactive requirements of smart TVs. The three-dimensional gesture recognition can determine the gesture changes in the three-dimensional space, and has attracted widespread attention in a more natural and intuitive way of interaction. However, at present, some three-dimensional gesture recognition devices use data gloves. Although its recognition rate is high, it is also The naturalness of gesture interaction is lost, and the cost is also very expensive; some devices that use time of flight have high hardware requirements, high costs, and low resolution; devices based on binocular vision have small viewing angles, in dim environments, and features The effect is poor when it is not obvious; and equipment based on structured light is very susceptible to light. It can be said that the existing gesture recognition devices have various degrees of defects.

[0008]

⁷¹ Summary of the invention

[0009]

⁷⁵ In order to solve the shortcomings of the prior art, the purpose of the present invention is to provide a method for controlling a smart TV with a three-dimensional gesture based on binocular structured light. The present invention obtains high-quality three-dimensional information of both hands through a combination of binocular camera and infrared structured light. , And then perform real-time gesture segmentation, detection, and fingertip positioning to complete the real-time conversion of three-dimensional gesture actions and TV

commands to meet the real-time requirements of TV human-computer interactive operations.

[0010]

⁸⁴ In order to achieve the above objectives, the present invention adopts the following technical solutions:

[0011]

⁸⁸ A method for controlling a smart TV with a three-dimensional gesture based on binocular structured light includes the following steps:

[0012]

⁹³ Step one, capture images; capture images in front of the camera through the binocular camera set in the smart TV, and store the collected left view and right view into the memory respectively, and then the processor processes the two images in the memory respectively;

[0013]

⁹⁹ Step two, reconstruct the image in three dimensions; according to the binocular matching algorithm, find the one-to-one correspondence between the pixels of the left view and the right view, and obtain the disparity map according to the abscissa difference of the pixel positions between them, and then according to the projection model With the principle of triangulation, the depth information of the original image is obtained;

[0014]

¹⁰⁶ Step three, preprocess the depth image; perform median filtering on the depth image to remove noise points, and then perform morphological processing on the internal holes in the depth image;

[0015]

¹¹¹ Step 4: Detect the gesture image; analyze the obtained image depth data, separate the person from the background, select only the pixel information of the human body, and set all other pixel information to zero to obtain the depth image of the human body, and finally obtain the contour of the gesture ;

[0016]

¹¹⁷ Step 5: Recognize the three-dimensional gesture; extract the center distance feature of the gesture contour according to the obtained gesture contour. The center distance refers to the distance between the necessary point in the graph and each edge point; set the point $O(x_0, y_0)$ as the geometric center point of the gesture , The point set $C=\{(x_i, y_i)|i=1, 2, \dots, N\}$ is the edge contour of the gesture. It may be considered that the edge contour of the gesture is formed by connecting edge points. Therefore, the geometric center The point $O(x_0,$

y0) and the center distance d can be calculated by the following formula:

[0020]

¹²⁶ Combined with three-dimensional information, use the classifier based on Bayesian network for recognition;

[0021]

¹³⁰ Step 6, execute the operation instruction; according to the gesture made in front of the binocular camera, the smart TV executes the corresponding preset operation instruction to complete the human-computer interaction.

[0022]

¹³⁶ The aforementioned method for controlling a smart TV with three-dimensional gestures based on binocular structured light. Step one is to collect images; detect the lighting conditions of the operating environment through an infrared detector; if there is too much infrared light, turn on the infrared laser; The binocular camera in the TV captures the image in front of the camera, and the collected left view and right view are respectively stored in the memory, and then the two pictures in the memory are processed by the processor.

[0023]

¹⁴⁴ In the aforementioned method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, if the infrared detector detects that the infrared radiation intensity of the operating environment is less than $0.2W/m^2 \cdot 850nm$, the infrared laser is turned on.

[0024]

¹⁵⁰ In the aforementioned method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, the morphological processing in step 3 is to open and close the burr existing on the edge of the image and the internal cavity of the image.

[0025]

¹⁵⁶ In the aforementioned method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, the specific steps for obtaining the outline of the gesture in step 4 are: the gray value of the pixel reflects the distance of the changed point from the lens, and the depth image of the human body is calculated. The grayscale distribution histogram of the grayscale histogram, the smaller peak of the grayscale histogram is the gray distribution of the hand area, and the larger peak is the gray distribution of the human torso, so as to segment the human hand image, and then use the hand image. The Canny edge operator performs edge detection to obtain the contour of the gesture.

[0026]

¹⁶⁶ In the foregoing method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, the specific steps of using a Bayesian network-based classifier for recognition in step 5 are as follows:

[0027]

¹⁷² Step a: Calculate the Gaussian distribution information of all features in different gesture categories, that is, calculate all the mean and variance;

[0028]

¹⁷⁷ Step b: For each input sample tested, find the probability of each feature in the gesture 1 category. Since each feature is independent of each other, the probability distribution under the feature also satisfies the independence, which can be obtained Find the total probability of the input sample in the gesture 1 category; and so on, find the different probabilities of the sample in all gesture categories;

[0029]

¹⁸⁴ Step c: Compare the probability values of the input samples under different categories, find the category to which the maximum probability value belongs, and predict that the category is the category to which the input sample belongs.

[0030]

¹⁹⁰ The advantage of the present invention is that the present invention provides a method for controlling a smart TV with a three-dimensional gesture based on binocular structured light. The present invention obtains high-quality three-dimensional information of both hands through a combination of binocular camera and infrared structured light, and then performs Real-time gesture segmentation, detection, and fingertip positioning, complete the real-time conversion of three-dimensional gesture actions and TV commands, and meet the real-time requirements of TV man-machine interactive operations;

[0031]

¹⁹⁹ Because the invention adopts the combination of binocular vision and structured light, it can effectively improve the environment's anti-interference ability and reduce the influence of light conditions on fingertip positioning. When indoors or when the environment is dim, the structured light is used to measure depth information , When the sun is strong, it turns to a pure binocular way.

²⁰³ Moreover, the calibration of binocular structured light has nothing to do with the laser, and it is more convenient to replace when the laser is damaged;

[0032]

²⁰⁸ The invention reduces the traversal range of the gesture detection by segmenting the foreground of the gesture image, and can meet the real-time requirements of the television human-computer interactive operation.

[0033]

²¹³ Description of the drawings

[0034]

²¹⁷ Figure 1 is a schematic structural diagram of an embodiment of the present invention;

[0035]

²²¹ The meaning of the reference signs in the figure:

[0036]

²²⁵ 1 Infrared camera, 2 infrared detectors, 3 infrared lasers, 4 TVs.

[0037]

²²⁹ Detailed ways

[0038]

²³³ Hereinafter, the present invention will be described in detail with reference to the drawings and specific embodiments.

[0039]

²³⁸ As shown in FIG. 1, all the modules required by the present invention are integrated in the smart TV 4.

²³⁹ Horizontally installed on the left and right sides of the smart TV 4 are calibrated infrared cameras 1 for obtaining images with left and right viewing angles.

²⁴¹ In the middle of the two infrared cameras 1 is an infrared laser 3, which is used to emit infrared fringe light, increase the surface texture, facilitate binocular matching, and facilitate accurate recognition of gestures in dim conditions.

²⁴⁴ Next to the laser is an infrared detector 2, which is used to monitor the infrared situation of the operating environment to choose whether to turn on the infrared laser 3.

²⁴⁶ The subsequent image 3D reconstruction, depth image preprocessing, gesture image detection, 3D gesture recognition, and human-computer interaction are completed by the processor in the smart TV 4.

[0040]

251 A method for controlling a smart TV 4 with a three-dimensional gesture based on binocular structured light includes the following steps:

[0041]

256 Step one, collect images;

[0042]

260 First detect the infrared radiation of the operating environment through the infrared detector 2. If the infrared radiation intensity is greater than $0.2W/m^2 \cdot 850nm$, the coded spot emitted by the laser is likely to be submerged, and the pure binocular method is used; if the infrared radiation intensity is less than $0.2W/m^2 \cdot 850nm$, the coded spot is not easy to be disturbed, then turn on the infrared laser 3 to improve the accuracy of binocular matching.

[0043]

268 Then turn on the two infrared cameras 1 of the same model that have been calibrated horizontally installed in the smart TV 4. The binocular camera captures the front image of the camera at 25 frames per second, and the collected image size is 640×480 in the RGB color space. Digital images, and the left view and right view taken by the binocular camera are respectively stored in two independent storage spaces in the smart TV 4, and then the two pictures in the storage are processed by the processor.

[0044]

276 Step two, reconstruct the image in three dimensions; according to the binocular matching algorithm, find the one-to-one correspondence between the pixels of the left view and the right view, and obtain the disparity map according to the abscissa difference of the pixel positions between them, and then according to the projection model With the principle of triangulation, the depth information of the original image is obtained;

[0045]

283 Step three, preprocess the depth image; because the depth value of the gesture edge in the obtained depth image is unstable, there are many zero-value noise points, direct processing will affect the effect of the algorithm, so the median filter is used to remove the noise points and reduce the target The degree of blurring of the edges.

287 Morphological processing is performed on the internal cavity in the depth image; the morphological processing is to open and close the burr existing on the edge of the image and the internal cavity of the image.

[0046]

²⁹² Step 4: Detect the gesture image; analyze the obtained image depth data, separate the person from the background, select only the pixel information of the human body, and set all other pixel information to zero to obtain the depth image of the human body, and finally obtain the contour of the gesture ; The specific steps to obtain the contour of the gesture are; the gray value of the pixel reflects the distance of the changed point from the lens, and the gray distribution histogram of the depth image of the human body is calculated. The smaller peak of the gray histogram is the hand area Gray-scale distribution, the larger peak is the gray-scale distribution of the human torso, thereby segmenting the human hand image, and then using the Canny edge operator to perform edge detection on the hand image to obtain the contour of the gesture.

[0047]

³⁰³ Step 5: Recognize the three-dimensional gesture; extract the center distance feature (SCCD) of the gesture contour according to the obtained gesture contour. The center distance refers to the distance from the edge point to the edge point in the figure; set point $O(x_0, y_0)$ as the gesture The geometric center point, point set $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ is the edge contour of the gesture. It may be considered that the edge contour of the gesture is formed by connecting edge points. , The geometric center point $O(x_0, y_0)$ and the center distance d can be calculated by the following formula:

[0051]

³¹² Combined with three-dimensional information, use the classifier based on Bayesian network for recognition;

[0052]

³¹⁶ The specific steps of using a Bayesian network-based classifier for recognition are as follows:

[0053]

³²⁰ Step a: Calculate the Gaussian distribution information of all features in different gesture categories, that is, calculate all the mean and variance;

[0054]

³²⁵ Step b: For each input sample tested, find the probability of each feature in the gesture 1 category. Since each feature is independent of each other, the probability distribution under the feature also satisfies the independence, which can be obtained Find the total probability of the input sample in the gesture 1 category; and so on, find the different probabilities of the sample in all gesture categories;

[0055]

332 Step c: Compare the probability values of the input samples under different categories, find the category to which the maximum probability value belongs, and predict that the category is the category to which the input sample belongs.

[0056]

338 Step 6, execute the operation instruction; according to the gesture made in front of the binocular camera, the smart TV 4 executes the corresponding preset operation instruction to complete the human-computer interaction.

[0057]

344 The present invention provides a three-dimensional gesture control method for smart TV 4 based on binocular structured light. The present invention obtains high-quality three-dimensional information of both hands through the combination of binocular camera and infrared structured light, and then performs real-time gesture segmentation, detection and pointing. Pointer positioning, complete the real-time conversion of three-dimensional gesture actions and TV 4 instructions, and meet the real-time requirements of TV 4 man-machine interactive operation;

[0058]

353 Because the invention adopts the combination of binocular vision and structured light, it can effectively improve the environment's anti-interference ability and reduce the influence of light conditions on fingertip positioning. When indoors or when the environment is dim, the structured light is used to measure depth information , When the sun is strong, it turns to a pure binocular way.

357 Moreover, the calibration of binocular structured light has nothing to do with the laser, and it is more convenient to replace when the laser is damaged;

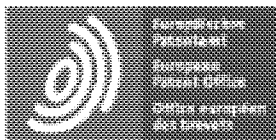
[0059]

362 The invention reduces the traversal range of the gesture detection by segmenting the foreground of the gesture image, and can meet the real-time requirements of the human-computer interaction operation of the TV 4.

[0060]

367 The basic principles, main features and advantages of the present invention have been shown and described above.

369 Those skilled in the industry should understand that the foregoing embodiments do not limit the present invention in any form, and all technical solutions obtained by equivalent substitutions or equivalent transformations fall within the protection scope of the present invention.



Notice

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CLAIMS CN108363482A

1.

¹³ A method for controlling a smart TV with a three-dimensional gesture based on binocular structured light is characterized in that it comprises the following steps:

¹⁵ Step one, capture images; capture images in front of the camera through the binocular camera set in the smart TV, and store the collected left view and right view into the memory respectively, and then the processor processes the two images in the memory respectively;

¹⁸ Step two, reconstruct the image in three dimensions; according to the binocular matching algorithm, find the one-to-one correspondence between the pixels of the left view and the right view, and obtain the disparity map according to the abscissa difference of the pixel positions between them, and then according to the projection model With the principle of triangulation, the depth information of the original image is obtained;

²² Step three, preprocess the depth image; perform median filtering on the depth image to remove noise points, and then perform morphological processing on the internal holes in the depth image;

²⁴ Step 4: Detect the gesture image; analyze the obtained image depth data, separate the person from the background, select only the pixel information of the human body, and set all other pixel information to zero to obtain the depth image of the human body, and finally obtain the contour of the gesture ;

²⁷ Step 5: Recognize the three-dimensional gesture; extract the center distance feature of the gesture contour according to the obtained gesture contour. The center distance refers to the distance between the necessary point in the graph and each edge point; set the point $O(x_0, y_0)$ as the geometric center point of the gesture , The point set $C=\{(x_i, y_i)|i=1, 2, \dots, N\}$ is the edge contour of the gesture. It may be considered that the edge contour of the gesture is formed by connecting edge points. Therefore, the geometric center The point $O(x_0, y_0)$ and the center distance d can be calculated by the following formula:

³³ Combined with three-dimensional information, use the classifier based on Bayesian network for recognition;

³⁴ Step 6, execute the operation instruction; according to the gesture made in front of the binocular camera, the smart TV executes the corresponding preset operation instruction to complete the human-computer interaction.

2.

⁴⁰ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, characterized in that, in step one, an image is acquired; Turn on the infrared laser; then capture the image in front of the camera through the binocular camera set in the smart TV, and store the collected left view and right view into the memory respectively, and then the processor processes the two images in the memory respectively.

3.

⁴⁸ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 2, wherein if the infrared radiation intensity of the operating environment detected by the infrared detector is less than $0.2\text{W}/\text{m}^2 \cdot 850\text{nm}$, the infrared laser is turned on .

4.

⁵⁴ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, wherein the morphological processing in the third step is to open the burr existing on the edge of the image and the internal cavity of the image. Close operation.

5.

⁶⁰ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, wherein the specific steps of obtaining the outline of the gesture in the above step 4 are: reflecting the change point distance through the gray value of the pixel The distance of the lens is to calculate the gray distribution histogram of the depth image of the human body. The smaller peak of the gray histogram is the gray distribution of the hand area, and the larger peak is the gray distribution of the human torso, so as to segment the human body. Then use Canny edge operator to perform edge detection on the hand image of the hand image to obtain the contour of the gesture.

6.

⁷⁰ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, wherein the specific steps of using a Bayesian network-based classifier to perform recognition in the above step 5 are as follows:

⁷³ Step a: Calculate the Gaussian distribution information of all features in different gesture categories, that is, calculate all the mean and variance;

⁷⁵ Step b: For each input sample tested, find the probability of each feature in the gesture 1 category. Since each feature is independent of each other, the probability distribution under the feature also satisfies the independence, which can be obtained Find the total probability of the input sample in the gesture 1 category; and so on, find the different probabilities of the sample in all gesture categories;

⁷⁹ Step c: Compare the probability values of the input samples under different categories, find the category to

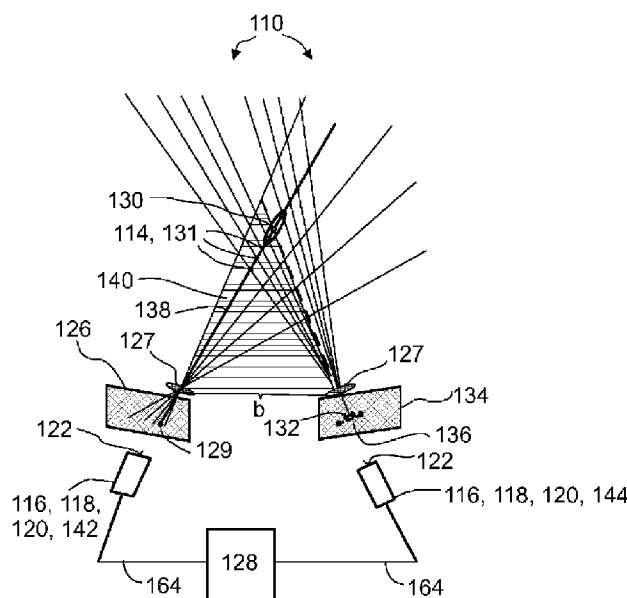
which the maximum probability value belongs, and predict that the category is the category to which the input sample belongs.



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(54) Title: DETECTOR FOR DETERMINING A POSITION OF AT LEAST ONE OBJECT

FIG.2



(57) Abstract: A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises -at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein each optical sensor (120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (122) by a reflection light beam (182) propagating from the object (112) to the detector (110), wherein the sensor element (116) is adapted to determine at least one reflection image (126); -at least one evaluation device (128), wherein the evaluation device (128) is adapted to select at least one reflection feature of the reflection image (126), wherein the evaluation device (128) is configured for determining at least one longitudinal region (130) of the selected reflection feature of the reflection image (134) by evaluating a combined signal Q from the sensor signals, wherein the evaluation device (128) is adapted to determine at least one

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displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130), wherein the evaluation device (128) is adapted to match the selected reflection feature with at least one reference feature within the displacement region (132).

Detector for determining a position of at least one object

Description

5 Field of the invention

The invention relates to a detector and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

Prior art

Optical 3D sensing methods generally may determine unreliable results in case of environment causing multiple reflections, with biasing light sources, or reflective measurement objects. Furthermore, 3D sensing methods, such as triangulation methods using structured light or stereo cameras, with imaging capabilities often suffer from a high demand of computational power to solve correspondence problems. Necessary computational power may result in high costs for processors or Field Programmable Gate Arrays (FPGAs), for heat removal in view of ventilation requirements or water proof housing difficulties, for electrical power consumption, in particular in mobile devices, and, in addition, in measurement uncertainties. The high demand in computational power may be prohibitive to realize real-time applications, high framerates or even standards video speed frame rates of 25 frames per second.

A large number of optical devices are known from the prior art using triangulation imaging methods. For example, structured light methods or stereo methods are known. For example, passive stereo methods using two cameras in a fixed relative orientation or active stereo technologies, where an additional light projector is used. Another example is the structures light approach, where one light projector and one camera in a fixed relative orientation are used. In order to determine a depth image via triangulation, the correspondence problem has to be solved first. Therefore, in passive stereo camera techniques enough corresponding feature points have to be identified in both camera views. In structured light approaches correspondences between pre-stored and projected pseudo random light patterns have to be determined. For a robust solution of these correspondence problems computational imaging algorithms such as algorithms scaling approximately quadratically with the number of points in the projected point pattern have to be employed. In structured light methods, for example, using a stereo system comprising two detectors having a fixed relative distance, a light source projects a pat-

tern such as points, pseudo-random, random, non-periodic or irregular point patterns, or the like. Each of the detectors generates an image of a reflection pattern and an image analysis task is to identify corresponding features in the two images. Due to fixed relative position a corresponding feature point selected in one of the two images lies along an epipolar line in the other image. However, solving the so-called correspondence problem may be difficult. In stereo and triangulation systems, the distance of all feature points along the epipolar line have to have reasonable correspondence among each other. A correspondence decision cannot be made one after another. If one correspondence is wrong, this has implications for other feature points, such as invisibility. This usually yields quadratic scaling evaluation algorithms.

For example, US 2008/0240502 A1 and US 2010/0118123 A1 describe an apparatus for mapping an object including an illumination assembly, which includes a single transparency containing a fixed pattern of spots. A light source transilluminates the single transparency with optical radiation so as to project the pattern onto the object. An image capture assembly captures an image of the pattern that is projected onto the object using the single transparency. A processor processes the image captured by the image capture assembly as to reconstruct a three-dimensional map of the object.

US 2008/118143 A1 describes a method and apparatus for obtaining an image to determine a three dimensional shape of a stationary or moving object using a bi dimensional coded light pattern having a plurality of distinct identifiable feature types. The coded light pattern is projected on the object such that each of the identifiable feature types appears at most once on predefined sections of distinguishable epipolar lines. An image of the object is captured and the reflected feature types are extracted along with their location on known epipolar lines in the captured image. Displacements of the reflected feature types along their epipolar lines from reference coordinates thereupon determine corresponding three dimensional coordinates in space and thus a 3D mapping or model of the shape of the object at any point in time.

US 2009/066929 A1 describes establishing of the correspondence of the points for creation of a range image with stereo cameras. The scene is illuminated twice; thereof at least once with a random or pseudo random pattern. For both cameras, an image is taken for each of the illuminations and the quotient of brightnesses is calculated pixelwise. The correspondence is established on the basis of a comparison of the quotient of pixels on epipolar lines of different cameras. The illumination pattern is preferably highly modulated along the epipolar line; transversally or diagonally to it, it is not or only slightly modulated. For illumination, a projection unit is used which in a preferred arrangement comprises two superposed grating patterns which have a distance to each other, with at least one varying in a pseudo random manner, and with two closely neighboring light sources which shine through the gratings and thereby generate different pseudo random patterns, especially moiré patterns.

Despite the advantages implied by the above-mentioned devices and detectors, several technical challenges remain. Thus, the evaluation algorithms used require high computation power which is a severe cost driver. Furthermore, due to energy consumption and heat production of

the required computational resources, the computational demand limits use of such 3D sensor methods in outdoor and mobile applications.

Problem addressed by the invention

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It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

10

Summary of the invention

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This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

20

As used in the following, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

25

Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, notwithstanding the fact that the respective feature or element may be present once or more than once.

35

Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without

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any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for determining a position of at least one object is disclosed. As used herein, the term "object" refers to a point or region emitting at least one light beam. The light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is adapted to determine at least one reflection image;
- at least one evaluation device, wherein the evaluation device is adapted to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals, wherein the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as

a combination of several devices. As further used herein, the term "matrix" generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

10 The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane.

15 As used herein, an "optical sensor" generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a "light-sensitive area" generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term "the optical sensors each having at least one light sensitive area" refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. Thus, the term "optical sensor" furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one CCD and/or CMOS device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible. Thus, as an example, an optical device comprising two, three, four or more than four light-sensitive areas may be used which is regarded as two, three, four or more than four optical sensors in the context of the present invention. As outlined above, the sensor element comprises a matrix of optical sensors. Thus, as an example, the optical sensors may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

Specifically, the optical sensors may be or may comprise inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge: Au photodiode, a Ge: Hg photodiode, a Ge: Cu photodiode, a Ge: Zn photodiode, a Si: Ga photodiode, a Si: As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The matrix may be composed of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term "essentially perpendicular", reference may be made to the definition given above. Thus, as an example, tolerances of less than 20°, specifically less than 10° or even less than 5°, may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Preferably, the sensor element may be oriented essentially perpendicular to an optical axis of the detector. Again, with respect to the term "essentially perpendicular", reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The reflection light beam may propagate from the object towards the detector. As will be outlined in further detail below, the reflection light beam may originate from the object or may originate from an illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. The illumination source may be or may comprise at least one multiple beam light source. For example, the illumination source may comprise at least one laser source and one or more diffractive optical elements (DOEs). Additionally or alternatively, the detector may use radiation already present in the scene such as from at least one ambient light source.

The reflection light beam specifically may fully illuminate the at least one optical sensor from which the center signal is generated, such that the at least one optical sensor from which the center signal arises is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the at least one optical sensor from which the sensor signal arises. Contrarily, preferably, the reflection light beam specifically may create a light spot on the entire matrix which is smaller than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further de-

tail below. As further used herein, a “light spot” generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range

from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may
5 be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a
10 CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as com-
15 mercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The detector may be adapted to determine at least one distance information of the object by using triangulation and/or structured light techniques. In known 3D sensing devices, such as
20 devices using triangulation or structured light techniques, due to correspondence problems regular, constant or periodic pattern are not suitable since each measured point has to be assigned to one reference point of a reference pattern.

As used herein, the term "reflection image" refers to an image determined by the optical sensor comprising at least one reflection feature. As used herein, the term "reflection feature" refers to
25 a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. The reflection image may comprise the at least one reflection pattern comprising the at least one reflection feature. As used herein, the term "illumination feature" refers to at least one arbitrary shaped feature generated by at least one ambient light source or at least one illumination source adapted to illuminate the object. As used herein, the
30 term "determining at least one reflection image" refers to one or more of imaging, recording and generating of the reflection image. As used herein, the term "select at least one reflection feature" refers to one or more of identifying, determining and choosing at least one reflection feature of the reflection image. The detector may be adapted to determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection image from the com-
35 bined signal. Thus, the detector may be adapted to pre-classify the at least one reflection feature of the reflection image. This allows using illumination patterns comprising regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal patterns may be preferred.

40 The evaluation device may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image

processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals. The evaluation device may be adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q, wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \varepsilon$. As used herein, the term "longitudinal region" refers to at least one uncertainty interval defined by the longitudinal coordinate z and the measurement uncertainty $\pm \varepsilon$ of the determination of the longitudinal coordinate from the combined signal Q. Error ε may depend on measurement uncertainty of the optical sensor. The measurement uncertainty of the optical sensors may be pre-determined and/or estimated and/or may be deposited in at least one data storage unit of the evaluation device. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$. As used herein, the term "combined signal Q" refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region. For example, the evaluation device may be configured for deriving the combined

$$\text{signal Q by } Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. As used herein, the term "beam profile" relates to a spatial distribution, in particular in at least one plane per-

pendicular to the propagation of the light beam, of an intensity of the light beam. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_0)$ and beam shape $S(x,y;z_0)$, $E(x,y;z_0)=L \cdot S$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_0 independent from the object size. Thus, the combined signal allows determination of the distance z_0 independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q . The light-sensitive areas may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term "essentially edge information" generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts

of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

- 5 Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area
10 may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam
15 profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 25 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
30 cuts may result in improved distance information.

In one embodiment, the light beam propagating from the object to the detector may illuminate the sensor element with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The
35 central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

For example, the light beam propagating from the object to the detector may illuminate the sensor element with at least one point pattern. The point pattern may be generated by the object, for example in response to an illumination by the at least one light source with an illumination pattern comprising the at least one point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The light beam propagating from the object to the detector may illuminate the sensor element with a reflection pattern comprising both point patterns and line patterns. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

The evaluation device may be configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

For example, the evaluation device may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;

- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the at least one longitudinal coordinate z of the selected feature by evaluating the combined signal.

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As explained, e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot may be characterized by two measurement variables: a measurement signal measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their respective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming the combined signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the quotient signal for deriving the longitudinal coordinate.

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The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus, may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.

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The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, as will be outlined in further detail below, the detector according to the present

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invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

Consequently, in accordance with the present invention, the term "center signal" generally refers to the at least one sensor signal comprising essentially center information of the beam profile. For example, the center signal may be the signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. As used herein, the term "highest sensor signal" refers to one or both of a local maximum or a maximum in a region of interest. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an in-

tegral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

- 5 Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix.
- 10 When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of
- 15 edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.
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Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment

25 will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

- 30 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
- 35 cuts may result in an improved distance information.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which

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contains at least one item of information on a comparison between the center signal and the sum signal.

The evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the object by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

Raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations, however, are feasible.

As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be per-

formed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a

quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the combined signal Q . This determining may be a one-step process, such as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined sig-

nal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

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The evaluation device may be configured for using at least one predetermined relationship between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

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The combined signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider. The divider may fully or partially be integrated into the sensor element answers or may fully or partially be embodied independent from the sensor element.

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The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term "transfer device", also denoted as "transfer system", may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term "focal length" of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a "focus" which may also be denoted as "focal point". Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer

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device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term "optical axis of the transfer device" generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information

of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

As outlined above, the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region. As used herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. As outlined above, the evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. Relative position of the reference image and reflection image may be known. For example, relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

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As used herein, the term "displacement region" refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the refer-

ence feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term "reference feature" refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \varepsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance measurement using the combined signal Q may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device may determine a displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm \varepsilon$ from the combined signal Q to determine the displacement region along the epipolar line corresponding to $z \pm \varepsilon$. The evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region. As used herein, the term "matching" refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be adapted to match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement

region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be configured to perform the following steps:

- 5 - Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
- 10 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

- 15 Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ϵ -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or
- 20 reference feature in shorter distance and/or ϵ -weighted distance to the reference feature and/or reflection feature.

- Preferably, the detector may be adapted to pre-classify the selected reflection feature using the combined signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference image may have a relative distance to each other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line. For
- 25 example, the illumination pattern may comprise at least one hexagonal pattern. Preferably, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Preferably, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point.
- 30 The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other.

- 40 The evaluation device may be adapted to determine a displacement of the matched reference feature and the selected reflection feature. As used herein, the term "displacement" refers to difference between a position in the reference image to a position in the reflection image. The evaluation device may be adapted to determine a longitudinal information of the matched fea-

ture using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term "longitudinal information" refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a distance value. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship. The evaluation device may be adapted to store parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector. The evaluation device may be adapted to generate the parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector such as by performing a Tsai camera calibration. The evaluation device may be adapted to compute and/or estimate parameters such as the focal length of the transfer device, the radial lens distortion coefficient, the coordinates of the center of radial lens distortion, scale factors to account for any uncertainty due to imperfections in hardware timing for scanning and digitization, rotation angles for the transformation between the world and camera coordinates, translation components for the transformation between the world and camera coordinates, aperture angles, image sensor format, principal point, skew coefficients, camera center, camera heading, baseline, rotation or translation parameters between camera and/or illumination source, apertures, focal distance, or the like.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce the possible number of solutions along the epipolar line significantly. The number of possible solutions may even be reduced to one. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature

may enhance accuracy and/or speed and/or may lower computational demand in comparison to conventional triangulation systems. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be reduced to increase the light intensity in each illumination point such as to compete with ambient light while complying to eye safety regulations. A reduced number of illumination features in a conventional triangulation system might increase the difficulty to match reflection features and reference features. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be increased, such as to increase the resolution of the distance measurement, such as to increase the resolution of the obtained depth map without increasing the processing power of the evaluation device such as in a mobile application.

The detector may further comprise an illumination source. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source may be adapted to generate at least one illumination pattern for illumination of the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "illumination pattern" refers to a pattern which illuminates the object. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern

such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that a hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination device may be attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-

frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

Specifically, the illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.

The illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The sensor element may be adapted to determine at least one reflection pattern. As used herein, the term "reflection pattern" refers to a response pattern generated by reflection or scattering of light at the surface of the object, in particular generated by the object in response to illumination by the illumination pattern. As outlined above, the illumination pattern comprises at least one feature adapted to illuminate the object. As outlined above, the illumination feature may be generated by ambient light or by the at least one illumination source. The reflection pattern may comprise at least one feature corresponding to at least one feature of the illumination pattern.

The reflection pattern may comprise, in comparison to the illumination pattern, at least one distorted pattern, wherein the distortion depends on the distance of the object, such as surface properties of the object. The evaluation device may be adapted to select at least one feature of the reflection pattern and to determine the longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals, as described above.

For example, the reference image may be an image of the illumination pattern at an image plane at a position of the illumination source. The evaluation device may be adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The illumination source and the sensor element may be separated by a fixed distance.

For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to

- determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be adapted to select at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The evaluation device may be adapted to select the at least one reflection feature in the reflection pattern and to determine the longitudinal region of the selected feature by evaluating the combined signal Q from the sensor signals. The evaluation device may be adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region.
- 15 As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

- In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein,
5 the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

10 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment
15 function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system
20 according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

25 In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

30 In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further
35 comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further
40 used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different

from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for
5 scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the
10 scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to
15 emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order
20 to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.
25

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic
30 and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other
35 properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector,
40 such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least

one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in

order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one
5 or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array
10 of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an im-
15 age by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to
20 measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applica-
25 tions, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as
30 mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning sys-
35 tem and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical sys-
40 tems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as

such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a drill, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a

wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one in-

candescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is pro-

duced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used. The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method comprises the following method steps:

- generating at least two sensor signals in response to an illumination of at least one sensor element having a matrix of optical sensors by at least one reflection light beam propagating from the object to the detector, the optical sensors each having a light-sensitive area;
- determining at least one reflection image by using the sensor element;
- selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal region of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
- determining at least one displacement region in at least one reference image corresponding to the longitudinal region;
- matching the selected reflection feature with at least one reference feature within the displacement region.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

The method may further comprise determining a displacement of the matched reference feature and the selected reflection feature. A longitudinal information of the matched feature may be determined using a predetermined relationship between a longitudinal coordinate and the displacement.

The deriving of the combined signal Q may comprise one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

5 In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

15 The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

25 As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for house-

hold applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

5 The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with
10 mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus,
15 specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or
20 other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance,
25 scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head-mounted displays.

35 Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention
40 may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries,

obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in

combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

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Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

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Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems, or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present

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invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.

5 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. 10 Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, 20 such as in the windows like rear window, side window or front window, on the bumpers or on the lights. 25

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions. 30

35 Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used 40 for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endo-

scopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric surgery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment. Further, the devices according to the present invention may be used in wearable robots such as in exoskeletons or prosthesis or the like. Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe. In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

Further, the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

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Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

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Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

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The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

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The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

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Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detec-

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tion etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore, multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection, such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections, inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft when landing or starting. Therein, fixed, known active targets may be used for precise

guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or accelerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racket, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and/or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both

from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the present invention may be used in drones or multicopters to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concertized movements of several drones or recharging or refueling in the air or the like.

Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as dog or cat litter robots, charging robot for electrical vehicles, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, robotic shopping carts, luggage carrying robots, line following robots, laundry robots, ironing robots, window washing robots, toy robots, patient monitoring robots, baby monitoring robots, elderly monitoring robots, children monitoring robots, transport robots, telepresence robots, professional service robots, programmable toy robots, pathfinder robots, social robots providing company to less mobile people, following robots, smart card following robots, psychotherapy robots, or robots translating speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the

present invention may be used in humanoid robots, especially in the context of using humanoid hands to pick up or hold or place objects. Further, the devices according to the present invention may be used in combination with audio interfaces especially in combination with household robots which may serve as a digital assistant with interfaces to online or offline computer applications. Further, the devices according to the present invention may be used in robots that can control switches and buttons in industrial and household purposes. Further, the devices according to the present invention may be used in smart home robots such as Mayfield's Kuri. Further the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate insecure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

Further, devices according to the present invention may be used for the inspection of adhesive beads, sealing beads, or the like, such as to recognize disruptions, slubs, contractions, asymmetries, local defects, or the like. Further, devices according to the present invention may be used to count objects such as dry fruits on a conveyer belt, such as in difficult situations, such as when fruit of similar color and shape may be in direct contact with each other. Further, devices according to the present invention may be used in quality control of die cast or injection molded parts such as to ensure flawless casting or molding, recognize surface damages, worn out toolings or the like. Further, devices according to the present invention may be used for laserscribing such as for quality control and positioning of the laser. Further, devices according to the present invention may be used for sorting systems, such as to detect position, rotation, and shape of an object, compare it to a database of objects, and classify the object. Further, devices according to the present invention may be used for stamping part inspection, packaging inspection, such as food and pharma packaging inspection, filament inspection, or the like.

Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that are can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.

Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition devices, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the

presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.

- 5 Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.
- 10 The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may
- 15 be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or greenhouse or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.
- 20 Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific
- 25 groups of animals such as bees.

Further, devices according to the present invention may be used to guide users during a shaving, hair cutting, or cosmetics procedure, or the like. Further, devices according to the present invention may be used to record or monitor what is played on an instrument, such as a violin.

- 30 Further, devices according to the present invention may be used in smart household appliances such as a smart refrigerator, such as to monitor the contents of the refrigerator and transmit notifications depending on the contents. Further, devices according to the present invention may be used for monitoring or tracking populations of humans, animals, or plants, such as deer or tree populations in forests. Further, devices according to the present invention may be used
- 35 in harvesting machines, such as for harvesting crops, flowers or fruits, such as grapes, corn, hops, apples, grains, rice, strawberries, asparagus, tulips, roses, soy beans, or the like. Further, devices according to the present invention may be used to monitor the growth of plants, animals, algae, fish, or the like, such as in breeding, food production, agriculture or research applications, to control irrigation, fertilization, humidity, temperature, use of herbicides, insecticides, fungicides, rodenticides, or the like. Further, devices according to the present invention may be
- 40 used in feeding machines for animals or pets, such as for cows, pigs, cats, dogs, birds, fish, or the like. Further, devices according to the present invention may be used in animal product production processes, such as for collecting milk, eggs, fur, meat, or the like, such as in automated

milking or butchering processes. Further, devices according to the present invention may be used for automated seeding machines, or sowing machines, or planting machines such as for planting corn, garlic, trees, salad or the like. Further, devices according to the present invention may be used to assess or monitor weather phenomena, such as clouds, fog, or the like, or to warn from danger of avalanches, tsunamis, gales, earthquakes, thunder storms, or the like. Further, devices according to the present invention may be used to measure motions, shocks, concussions, or the like such as to monitor earthquake risk. Further, devices according to the present invention may be used in traffic technology to monitor dangerous crossings, to control traffic lights depending on traffic, to monitor public spaces, to monitor roads, gyms, stadiums, ski resorts, public events, or the like. Further, devices according to the present invention may be used in medical applications such as to monitor or analyze tissues, medical or biological assays, changes in tissues such as in moles or melanoma or the like, to count bacteria, blood cells, cells, algae, or the like, for retina scans, breath or pulse measurements, gastroscopy, patient surveillance, or the like. Further, devices according to the present invention may be used to monitor the shape, size, or circumference of drops, streams, jets, or the like or to analyze, assess, or monitor profiles or gas or liquid currents such as in a wind channel, or the like. Further, devices according to the present invention may be used to warn drivers such as car or train drivers when they are getting sick or tired or the like. Further, devices according to the present invention may be used in material testing to recognize strains or tensions or fissures, or the like. Further, devices according to the present invention may be used in sailing to monitor and optimize sail positions such as automatically. Further, devices according to the present invention may be used for fuel level gauges.

Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or viruses or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further, the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location.

- 5 Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location specific information, such as the current position on a map, or information on goods sold, or the like.
- 10 Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.
- 15 Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices according to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.
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- As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further,
- 25 the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, devices according to the present invention may be recalibrated using bar codes, QR-codes or prerecorded symbols of known size by using the at least one image matrix and comparing the prerecorded size with a measured property of the recorded image of
- 30 the bar code, QR code or prerecorded symbol, such as by comparing the width or height of the symbol with a prerecorded value. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention
- 35 may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to
- 40 the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the

like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

5 Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least
10 one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally
15 refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital
20 video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at
25 least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

30 As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively,
35 however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field
40 of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device adapted for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based algorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared. Further, the devices according to the present invention may be used in the context of gesture recognition. In this context, gesture recognition in combination with devices according to the present invention may, in particular, be used as a human-machine interface for transmitting information via motion of a body, of body parts or of objects to a machine. Herein, the information may, preferably, be transmitted via a motion of hands or hand parts, such as fingers, in particular, by pointing at objects, applying sign language, such as for deaf people, making signs for numbers, approval, disapproval, or the like, by waving the hand, such as when asking someone to approach, to leave, or to greet a person, to press an object, to take an object, or, in the field of sports or music, in a hand or finger exercise, such as a warm-up exercise. Further, the information may be transmitted by motion of arms or legs, such as rotating, kicking, grabbing, twisting, rotating, scrolling, browsing, pushing, bending, punching, shaking, arms, legs, both arms, or both legs, or a combination of arms and legs, such as for a purpose of sports or music, such as for entertainment, exercise, or training function of a machine. Further, the information may be transmitted by motion of the

whole body or major parts thereof, such as jumping, rotating, or making complex signs, such as sign language used at airports or by traffic police in order to transmit information, such as “turn right”, “turn left”, “proceed”, “slow down”, “stop”, or “stop engines”, or by pretending to swim, to dive, to run, to shoot, or the like, or by making complex motions or body positions such as in
5 yoga, pilates, judo, karate, dancing, or ballet. Further, the information may be transmitted by using a real or mock-up device for controlling a virtual device corresponding to the mock-up device, such as using a mock-up guitar for controlling a virtual guitar function in a computer program, using a real guitar for controlling a virtual guitar function in a computer program, using
10 a real or a mock-up book for reading an e-book or moving pages or browsing through in a virtual document, using a real or mock-up pen for drawing in a computer program, or the like. Further, the transmission of the information may be coupled to a feedback to the user, such as a sound, a vibration, or a motion.

In the context of music and/or instruments, devices according to the present invention in combination with gesture recognition may be used for exercising purposes, control of instruments,
15 recording of instruments, playing or recording of music via use of a mock-up instrument or by only pretending to have a instrument present such as playing air guitar, such as to avoid noise or make recordings, or, for conducting of a virtual orchestra, ensemble, band, big band, choir, or the like, for practicing, exercising, recording or entertainment purposes or the like.

20 Further, in the context of safety and surveillance, devices according to the present invention in combination with gesture recognition may be used to recognize motion profiles of persons, such as recognizing a person by the way of walking or moving the body, or to use hand signs or movements or signs or movements of body parts or the whole body as access or identification
25 control such as a personal identification sign or a personal identification movement.

Further, in the context of smart home applications or internet of things, devices according to the present invention in combination with gesture recognition may be used for central or non-central
30 control of household devices which may be part of an interconnecting network of home appliances and/or household devices, such as refrigerators, central heating, air condition, microwave ovens, ice cube makers, or water boilers, or entertainment devices, such as television sets, smart phones, game consoles, video recorders, DVD players, personal computers, laptops, tablets, or combinations thereof, or a combination of household devices and entertainment devices.

35 Further, in the context of virtual reality or of augmented reality, devices according to the present invention in combination with gesture recognition may be used to control movements or function of the virtual reality application or of the augmented reality application, such as playing or controlling a game using signs, gestures, body movements or body part movements or the like,
40 moving through a virtual world, manipulating virtual objects, practicing, exercising or playing sports, arts, crafts, music or games using virtual objects such as a ball, chess figures, go stones, instruments, tools, brushes.

Further, in the context of medicine, devices according to the present invention in combination with gesture recognition may be used to support rehabilitation training, remote diagnostics, or to monitor or survey surgery or treatment, to overlay and display medical images with positions of medical devices, or to overlay display prerecorded medical images such as from magnetic resonance tomography or x-ray or the like with images from endoscopes or ultra sound or the like that are recorded during a surgery or treatment.

Further, in the context of manufacturing and process automation, devices according to the present invention in combination with gesture recognition may be used to control, teach, or program robots, drones, unmanned autonomous vehicles, service robots, movable objects, or the like, such as for programming, controlling, manufacturing, manipulating, repairing, or teaching purposes, or for remote manipulating of objects or areas, such as for safety reasons, or for maintenance purposes.

Further, in the context of business intelligence metrics, devices according to the present invention in combination with gesture recognition may be used for people counting, surveying customer movements, areas where customers spend time, objects, customers test, take, probe, or the like.

Further, devices according to the present invention may be used in the context of do-it-yourself or professional tools, especially electric or motor driven tools or power tools, such as drilling machines, saws, chisels, hammers, wrenches, staple guns, disc cutters, metals shears and nibblers, angle grinders, die grinders, drills, hammer drills, heat guns, wrenches, sanders, engravers, nailers, jig saws, biscuit joiners, wood routers, planers, polishers, tile cutters, washers, rollers, wall chasers, lathes, impact drivers, jointers, paint rollers, spray guns, morticers, or welders, in particular, to support precision in manufacturing, keeping a minimum or maximum distance, or for safety measures.

Further, the devices according to the present invention may be used to aid visually impaired persons. Further, devices according to the present invention may be used in touch screen such as to avoid direct contact such as for hygienic reasons, which may be used in retail environments, in medical applications, in production environments, or the like. Further, devices according to the present invention may be used in agricultural production environments such as in stable cleaning robots, egg collecting machines, milking machines, harvesting machines, farm machinery, harvesters, forwarders, combine harvesters, tractors, cultivators, ploughs, destoners, harrows, strip tills, broadcast seeders, planters such as potato planters, manure spreaders, sprayers, sprinkler systems, swathers, balers, loaders, forklifts, mowers, or the like.

Further, devices according to the present invention may be used for selection and/or adaption of clothing, shoes, glasses, hats, prosthesis, dental braces, for persons or animals with limited communication skills or possibilities, such as children or impaired persons, or the like. Further, devices according to the present invention may be used in the context of warehouses, logistics, distribution, shipping, loading, unloading, smart manufacturing, industry 4.0, or the like. Further,

in a manufacturing context, devices according to the present invention may be used in the context of processing, dispensing, bending, material handling, or the like.

5 The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as
10 one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

15 The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.
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The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the
25 optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted
30 information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks
35 such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

40 The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue

interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces
5 ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or
10 more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication
15 device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may
20 be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display
30 device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design
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may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

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As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

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The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

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The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

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The human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

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The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance

with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller
5 might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally,
10 the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may
15 fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as
20 at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.
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The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object
30 itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of:
35 a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
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As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The proposed devices and methods provide a large number of advantages over known detectors of this kind. Thus, the detector generally may avoid the shortcomings of the known prior art systems disclosed above. Specifically, the detector may avoid the use of FiP sensors, thereby allowing for e.g. using simple and cheap and commercially available semiconductor sensors such as silicon photodiodes. These photodiodes generally do not show a luminance dependency, and the method disclosed above is generally independent from the brightness of the scenery and/or the brightness of the light spot on the light beam. Consequently, a range of measurement in terms of luminance or total power of the light beam entering the detector is generally larger in the present invention as compared to many of the devices disclosed above. Further, the measurement by using the detector according to the present invention is generally independent from the target spot size, i.e. from either the size of the object, the size of a light spot projected onto the object or the size of a beacon device being one or more of attached to the object, integrated into the object or held by the object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional option of measuring one or more transversal coordinates, thereby integrating the functionality of a PSD.

The result of the measurement, i.e. the position determined by the detector such as the at least one longitudinal coordinate, may be rendered widely independent from the brightness of the scenery and/or the brightness of the object, the brightness of the at least one beacon device, the brightness of the at least one illumination source or the total power of the light beam propagating from the object to the detector. Further, due to this independence and wide range of measurement in terms of brightness, reflective objects or non-reflective objects may be used.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention, such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the quotient signal, as mentioned above, contrarily, provides a wide range of

continuously and monotonously decreasing or increasing functions which may be used to determine the longitudinal coordinate from the quotient signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the quotient signal from the total power of the light beam, at least as long as no saturation of one or both of the optical sensors is reached, this also provides a very wide range of measurement in terms of brightness, i.e. in terms of total power of the light beam.

The light beam, within the detector, generally may propagate along an optical axis of the detector. The light beam, however, may also propagate in other ways than along the optical axis. As an example, an illumination light beam may be generated which propagates along the optical axis or which propagates parallel to the optical axis or at an angle to the optical axis which is different from 0° , such as an angle of 1° to 20° . Other embodiments are feasible.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector comprising

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is adapted to determine at least one reflection image;
- at least one evaluation device, wherein the evaluation device is adapted to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals, wherein the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region.

Embodiment 2: The detector according to the preceding embodiment, wherein the evaluation device is adapted to determine a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region.

Embodiment 5: The detector according to any one of the preceding embodiments, wherein the evaluation device is adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q , wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm\epsilon$.

Embodiment 6: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image.

Embodiment 7: The detector according to the preceding embodiment, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 8: The detector according to the preceding embodiment, wherein the evaluation device is configured to perform the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is adapted to match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z .

Embodiment 10: The detector according to the preceding embodiments, wherein the evaluation algorithm is a linear scaling algorithm.

Embodiment 11: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o.

Embodiment 12: The detector according to the preceding embodiment, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the reflection light beam.

Embodiment 13: The detector according to any of the two preceding embodiments, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

Embodiment 14: The detector according to any one of the three preceding embodiments, wherein the light sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile.

Embodiment 16: The detector according to any one of the two preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.

Embodiment 17: The detector according to any one of the two preceding embodiments, wherein the beam profile is segmented by a secant or chord passing the center of the beam profile by a given distance, whereas the first segment comprises the center information, whereas the second segment comprises essentially edge information.

Embodiment 18: The detector according to the preceding embodiment, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

Embodiment 19: The detector according to any one of the two preceding embodiments, wherein the evaluation device is configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the at least one evaluation device is configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the longitudinal region by evaluating the combined signal.

Embodiment 21: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor

signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.

5 Embodiment 23: The detector according to any one of the three preceding embodiments, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.

10 Embodiment 24: The detector according to any one of the preceding embodiment, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

15 Embodiment 25: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one optical element, wherein the optical element is at least one element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element.

20 Embodiment 26: The detector according to any one of the six preceding embodiments, wherein the evaluation device is further configured for determining at least one transversal coordinate of the object by evaluating a transversal position of the at least one optical sensor having the highest sensor signal.

25 Embodiment 27: The detector according to any one of the preceding embodiments, wherein the sensor element comprises at least one CMOS sensor.

30 Embodiment 28: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object.

35 Embodiment 29: The detector according to preceding embodiment, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

40 Embodiment 30: The detector according to any one of the two preceding embodiments, wherein the sensor element is adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the

longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals.

5 Embodiment 31: The detector according to the preceding embodiment, wherein the reference image is an image of the illumination pattern at an image plane at a position of the illumination source, wherein the evaluation device is adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern, wherein the evaluation device is adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region.

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Embodiment 32: The detector according to any one of the four preceding embodiments, wherein the illumination source and the sensor element are separated by a fixed distance.

15 Embodiment 33: The detector according to any one of the five preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors.

20 Embodiment 34: The detector according to the preceding embodiment, where at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein a relative distance between the first sensor element and the second element is fixed.

25 Embodiment 35: The detector according to any one of the two preceding embodiments, wherein the at least one first sensor element is adapted to determine at least one first reflection pattern and the at least one second sensor element is adapted to determine at least one second reflection pattern, wherein the evaluation device is adapted to select at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

30 Embodiment 36: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

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40 Embodiment 37: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 38: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein
5 the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 39: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding
10 embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 40: A scanning system for determining a depth profile of a scenery, the scanning
15 system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 41: A camera for imaging at least one object, the camera comprising at least one
20 detector according to any one of the preceding embodiments referring to a detector.

Embodiment 42: A method for determining a position of at least one object by using at least one detector according to any one of the preceding embodiments referring to a detector, the method comprising the following steps:

- 25 - generating at least two sensor signals in response to an illumination of at least one sensor element having a matrix of optical sensors by at least one reflection light beam propagating from the object to the detector, the optical sensors each having a light-sensitive area;
- determining at least one reflection image by using the sensor element;
- 30 - selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal region of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
- determining at least one displacement region in at least one reference image corresponding to the longitudinal region;
- 35 - matching the selected reflection feature with at least one reference feature within the displacement region.

Embodiment 43: The method according to the preceding embodiment, wherein a displacement
40 of the matched reference feature and the selected reflection feature is determined, wherein a longitudinal information of the matched feature is determined using a predetermined relationship between a longitudinal coordinate and the displacement.

Embodiment 44: The method according to any one of the preceding embodiments referring to a method, wherein the deriving of the combined signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

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Embodiment 45: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; an outdoor application; a mobile application; a communication application; a tracking application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

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Brief description of the figures

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Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary

embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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Figure 1 shows a first embodiment of a detector according to the present invention;

Figure 2 shows a further embodiment of the detector according to the present invention;

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Figure 3 shows an embodiment of a detector system, camera, entertainment device, tracking system;

Figure 4 shows an embodiment of determining a longitudinal coordinate z using a line pattern;

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Figures 5A to C show three embodiments of a hexagonal illumination pattern; and

Figure 6 shows an embodiment of a scanning system.

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Detailed description of the embodiments:

Fig. 1 shows in a highly schematic fashion a first embodiment of a detector 110 for determining a position of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.

The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is

given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

In the embodiment shown in Figure 1, the detector 110 further comprises an illumination source 124. As an example, the illumination source 124 may be configured for generating an illuminating light beam for illuminating the object 112. The detector 110 may be configured such that the illuminating light beam propagates from the detector 110 towards the object 112 along an optical axis of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source 124 may be adapted to generate at least one illumination pattern for illumination of the object 112. Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; at least one pattern comprising at least one pre-known feature. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points. The illumination source 124 may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source 124 may comprise at least one light source adapted to generate the illumination pattern directly. The illumination source 124 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source 124 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source 124.

In the embodiment shown in Figure 1, the illumination source 124 may illuminate the at least one object 112 with the illumination pattern. The illumination pattern may comprise a plurality of points as image features. These points are illustrated as light beams 115 emerging from the illumination source 124.

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Each optical sensor 120 is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam propagating from the object 112 to the detector 110.

10 Furthermore, the sensor element 116 is adapted to determine at least one reflection image 126. The matrix 118 may comprise the reflection image 126. The reflection image 126 may comprise points as reflection features. These points result from reflection light beams 125 originating from the at least one object 112. Crossings 131 of the illumination light beams and the reflection light beams 125 correspond to possible positions 114 of the at least one object 112.

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The detector 127 may comprise at least one transfer device 127 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam de-
20 flection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 127 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

25 The detector 110 comprises at least one evaluation device 128. The evaluation device 126 is adapted to select at least one reflection feature of the reflection image 126, for example selected reflection feature 129. The evaluation device 128 may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image
30 analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency de-
35 composition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a blob analysis; applying an edge filter; applying a low-pass filter; applying a Fourier transformation; apply-
40 ing a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be deter-

mined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

If the selected reflection feature 129 is matched to a corresponding reference feature, the corresponding light beams are matched and the distance of the object 112 can be calculated. For a selected reflection feature 129 and a corresponding reflection light beam, the number of corresponding illumination light beams, and thus the number of corresponding reference features 112, is reduced after determining a combined signal Q.

- 5 The evaluation device 128 is configured for determining at least one longitudinal region 130 of the selected reflection feature 129 of the reflection image by evaluating the combined signal Q from the sensor signals. The evaluation device 128 may be adapted to determine at least one longitudinal coordinate z of the selected reflection feature 129 by evaluating the combined signal Q, wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \varepsilon$. Error ε may depend on measurement uncertainty of the optical sensor. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$. The evaluation device 128 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device 128 may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region. For example, the evaluation device 128 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance L(z_o) and beam shape S(x,y;z_o), E(x,y;z_o)=L•S. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. The light-sensitive areas 122 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device 128 may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably, the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1. Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 128 may be adapted for determining an area integral of the beam profile. The evaluation device 128 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 128 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be

replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 5 Additionally or alternatively, the evaluation device 128 may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
10 cuts may result in improved distance information.

For example, the evaluation device 128 may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor 120 having the highest sensor signal and forming at least one center signal;
- 15 b) evaluating the sensor signals of the optical sensors 120 of the matrix 118 and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the at least one longitudinal coordinate z of the selected feature by evaluating
20 the combined signal.

- For example, the center signal may be the signal of the at least one optical sensor 118 having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors 120 of the entire matrix 118 or of a region of interest within the matrix 118, wherein the region of
25 interest may be predetermined or determinable within an image generated by the optical sensors of the matrix 118. The center signal may arise from a single optical sensor 120 or from a group of optical sensors 120, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors 120 may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors 120 from which the center signal arises
30 may be a group of neighboring optical sensors, such as optical sensors 120 having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors 120 from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The
35 evaluation device 128 may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal.

- The determination of the center signal may be performed electronically, since a comparison of
40 sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. The detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be performed fully or partially electronically or fully or partially by using one or more software

algorithms. Specifically, the evaluation device 128 may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at
5 least one sensor element and/or may fully or partially be embodied independently from the sensor element.

Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance
10 from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors 120; a sum of sensor signals from a group of optical sensors 120 containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors 120; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors 120 containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.
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For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix 118 or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable
25 within an image generated by the optical sensors 120 of the matrix 118. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors 120 from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device 128 may be adapted to determine the sum signal by integrating signals of the entire matrix 118, or of the region of interest within the matrix 118. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by
30 geometric considerations.
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The sum signal may be derived from all sensor signals of the matrix 118, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors 120 contributing to the center signal excluded. In every case, a reliable
40 sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix 118; a sum of all sensor signals of the matrix 118; an integral of all sensor signals of the matrix 118; an average over all

sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; a sum of all sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; an integral of all sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; a sum of sensor signals of optical sensors 120 within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor 120 having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor 120 having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors 120 being located within a predetermined range from the optical sensor 120 having the highest sensor signal. Other options, however, exist. The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device 128 may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal. The evaluation device 128 may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device 128 may be configured to determine the at least one coordinate z of the by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

The comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a

quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device 128 may be configured for forming the one or more quotient signals. The evaluation device 128 may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient
5 signal.

The evaluation device 128 specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and
10 due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is
15 due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The
20 quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum
25 signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by
30 analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

The evaluation device 128 may be configured for using at least one predetermined relationship
40 between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 128 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The sensor element 116 may be adapted to determine at least one reflection pattern. The reflection pattern may comprise at least one feature corresponding to at least one feature of the illumination pattern. The reflection pattern may comprise, in comparison to the illumination pattern, at least one distorted pattern, wherein the distortion depends on the distance of the object, such as surface properties of the object. The evaluation device 128 may be adapted to select at least one feature of the reflection pattern and to determine the longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals, as described above.

The evaluation device 128 is adapted to determine at least one displacement region 132 in at least one reference image 134 corresponding to the longitudinal region 130. The longitudinal region corresponds to the displacement region 132. The reference image 134 and the reflection image 126 may be images of the object 112 determined at different spatial positions having a fixed distance b, also called baseline. The evaluation device 128 may be adapted to determine an epipolar line 136 in the reference image 134. Relative position of the reference image 134 and reflection image 126 may be known. For example, relative position of the reference image 134 and reflection image 126 may be stored within at least one storage unit of the evaluation device 128. The evaluation device 128 may be adapted to determine a straight line 138 extending from the selected reflection feature 129 of the reflection image 126. The straight line 138 may comprise possible object positions 114. The straight line 138 and the baseline b span an epipolar plane 140. As the reference image 134 is determined at a different relative position from the reflection image, the corresponding reference features may be imaged on the epipolar line 136 in the reference image 134.

For example, in the embodiment shown in Figure 1, the reference image 134 may be an image of the illumination pattern at an image plane at a position of the illumination source 124. The evaluation device 128 may be adapted to determine the displacement region 132 in the reference image 134 corresponding to the longitudinal region 130 of the selected feature 129 of the reflection pattern. The evaluation device 128 may be adapted to match the selected feature 129 of the reflection pattern with at least one feature of the reference pattern within the displacement region 132. The illumination source 124 and the sensor element 116 may be separated by a fixed distance.

The displacement region 132 may extend along the epipolar line 136. The evaluation device 128 may be adapted to determine the reference feature along the epipolar line 136 corresponding to the longitudinal coordinate z and to determine an extent of the displacement region 132 along the epipolar line 136 corresponding to the error interval $\pm\epsilon$. The evaluation device 128 is adapted to match the selected reflection feature 129 with at least one reference feature within the displacement region 132. The evaluation device 128 may be adapted to match the selected feature of the reflection image with the reference feature within the displacement region 132 by using at least one evaluation algorithm considering the determined longitudinal coordinate z. The evaluation algorithm may be a linear scaling algorithm.

The evaluation device 128 may be adapted to determine a displacement of the matched reference feature and the selected reflection feature 129. The evaluation device 128 may be adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The evaluation device 128 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature 129 in the reflection image 126 and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, a longitudinal coordinate of the corresponding object feature 114 may be determined by triangulation. Thus, the evaluation device 128 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. For example, Figure 1 shows possible projection lines resulting in reflection features in the reflection image 126 and corresponding to reference features in the reference image 134. In addition, intersection points of these lines are shown corresponding to object positions. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device 128 may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

In Figure 2, an embodiment of the detector 110 is shown, wherein the detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element 142 and at least one second sensor element 144 may be positioned at different spatial positions. A relative distance b between the first sensor element 142 and the second element 144 may be fixed. The at least one first sensor element 142 may be adapted to determine at least one first reflection pattern and the at least one second sensor element 144 may be adapted to determine at least one second reflection pattern. The evaluation device 128 may be adapted to select at least one image determined by the first sensor element 142 or the second sensor element 144 as reflection image and to select at least one image determined by the other one of the first sensor element 142 or the second sensor element 144 as reference image. In Figure 2, the image of the first sensor element 142 is selected as reflection image 126, whereas the image of the second sensor element 144 is selected as reference image 134. The evaluation device 128 may be adapted to select the at least one reflection feature in the reflection pattern and to determine the longitudinal region 130 of the selected feature by evaluating the combined signal Q from the sensor signals, as outlined with respect to Figure 1. The evaluation device 128 may be adapted to determine the displacement region 132 in the reference image 134 corresponding to the longitudinal region 120 of the selected feature 129 of the reflection pattern. The evaluation device 128 may be adapted to match the selected feature 129

of the reflection pattern with at least one feature of the reference pattern within the displacement region 132, as outlined with respect to Figure 1.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region 132 corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce possible number of solutions along the epipolar line 136 significantly. The number of solutions may even be reduced to one. Subsequent matching of the selected reflection feature and reference feature may enhance accuracy. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature 129 and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices, where battery life or ventilation or heat management is an issue. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature 129 and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may enhance accuracy in comparison the triangulation systems. Furthermore, it may be possible to use denser pattern in order to enhance further resolution or to use a less dense pattern in order to increase power per point.

Figure 3 shows, in a highly schematic illustration, an exemplary embodiment of the detector 110, for example according to the embodiment shown in Figure 2. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 3 further shows an embodiment of a detector system 148, which, besides the at least one detector 110, comprises one or more beacon devices 150, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 3 further shows an exemplary embodiment of a human-machine interface 152, which comprises the at least one detector system 148 and, further, an entertainment device 154, which comprises the human-machine interface 152. The figure further shows an embodiment of a tracking system 156 for tracking a position of the object 112, which comprises the detector system 148. The components of the devices and systems shall be explained in further detail below.

Fig. 3 further shows an exemplary embodiment of a scanning system 158 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 158 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well as, optionally, at least one further illumination source, not depicted here. The illumination source

124, generally, is configured to emit at least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 150 and/or on a surface of the object 112. The scanning system 158 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 158, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 3 is shown in Fig. 2. Thus, the detector 110, besides the optical sensors 120 comprises at least one evaluation device 128, having e.g. at least one divider 160 and/or the at least one position evaluation device 162, as symbolically depicted in Fig. 3. The components of the evaluation device 128 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 128 may be interconnected by one or more connectors 164 and/or by one or more interfaces, as symbolically depicted in Fig. 3. Further, the one or more connectors 164 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 164, the evaluation device 128 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 166 of the detector 110. Additionally or alternatively, the evaluation device 128 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 168, the position of which may be manipulated by a user 170. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 170 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 166. Further, the detector 110 may comprise the at least one transfer device 127 such as one or more optical systems, preferably comprising one or more lenses. An opening 172 inside the housing 166, which, preferably, is located concentrically with regard to an optical axis 174 of the detector 110, preferably defines a direction of view 176 of the detector 110. A coordinate system 178 may be defined, in which a direction parallel or anti-parallel to the optical axis 174 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 174 may be defined as transversal directions. In the coordinate system 178, symbolically depicted in Fig. 3, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

One or more light beams 180, such as light beams of the illumination pattern, are propagating from the object 112 and/or from one or more of the beacon devices 150, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. In case no illumination source 124 is used, the beacon devices 150 and/or at least one of these
5 beacon devices 150 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. Alternatively, ambient light sources may be used.

The light beam 180, for example generated by the illumination source 124 such as the at least one illumination pattern, may be directed towards the object by using one or more optical elements. The object 112 may in response to the illumination generate the at least one reflection
10 light beam 182. For details of the evaluation, reference may be made to Fig. 1 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 152, in order to
15 provide at least one item of information to a machine 184. In the embodiments schematically depicted in Fig. 3, the machine 184 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 128 may even be fully or partially integrated into the machine 184, such as into the computer.

As outlined above, Fig. 3 also depicts an example of a tracking system 156, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 156 comprises the detector 110 and at least one track controller 186. The track controller 186 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 186 may be an independent device and/or may be fully or partially integrated
20 into the machine 184, specifically the computer, as indicated in Fig. 3 and/or into the evaluation device 128.

Similarly, as outlined above, the human-machine interface 152 may form part of an entertainment device 154. The machine 184, specifically the computer, may also form part of the entertainment device 154. Thus, by means of the user 170 functioning as the object 112 and/or by
30 means of the user 170 handling a control device functioning as the object 112, the user 170 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Referring to Figure 4, the detector 110 such as the detector as described with respect to Figures 1 to 3 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 120. For example, the detector 110 may comprise a plurality of optical sensors arranged in the matrix 118. The detector 110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector 110 may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. As outlined above, this principle is
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called Distance by Photon Ratio (DPR). In one embodiment, the light beam 180 may illuminate the sensor element 116 with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 128 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 4, the reflection light beam 182 propagating from the object 112 to the detector 110 may illuminate the matrix 118 with at least one line pattern 188. The line pattern 188 may be generated by the object 112 for example in response to an illumination by the at least one illumination source 124 with an illumination pattern comprising at least one illumination line pattern. A1 may correspond to an area with a full line width of the line pattern 188 in the matrix 118. The line pattern 188 in the matrix 118 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a line width in the matrix 118 is increased. In particular, the line width of the line pattern 188 in the matrix 118 may change from one column to another column. A2 may be a central area of the line pattern 188 in the matrix 118. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern 188 may be segmented in the matrix 118. Each of the columns may comprise center information of intensity in the central area of the line pattern 188 and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern 188.

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Figures 5A to C show three embodiments of a hexagonal illumination pattern. The illumination source 124 may be adapted to generate at least one illumination pattern for illuminating the object 112. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. Specifically, the illumination source 124 may comprise at least one laser

and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination pattern may comprise at least one feature such as a point or symbol. The illumination pattern may comprise a plurality of features. The illumination pattern may com-
5 prise an arrangement of periodic or non-periodic features. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one qua-
10 siperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to
15 project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature
20 may depend on the circle of confusion in the image.

The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on an epipolar line. As shown in Figure 5A, the illumination pattern may comprise at least one hexagonal pattern, wherein the individual points are positioned on
25 epipolar lines 190. As shown in Figure 5B, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Such a positioning of the illumination features allows enhancing distance between the individual points on each epipolar line. For example as shown in Figure 5C, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are dis-
30 placed by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other

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Figure 6 shows an embodiment of a scanning system 158. The scanning system 158 may be adapted as a line scanning device. In particular, the scanning system 158 may comprise at least one sensor line or row of optical sensors 120. Furthermore, the scanning system 158 may comprise the at least one transfer device 127 and the at least one illumination source 124. Triangu-
40 lation systems require a sufficient baseline, however due to the baseline in the near field no detection may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-

zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. Further, these near field and far field problems can be overcome by using the scanning system 158 of Figure 6. The scanning system 158 may be adapted to detect a plurality of light beams 182 propagating from the object 112 to the scanning system 158 on the CMOS line. The light beams 182 may be generated at different position on the object 112 or by movement of the object 112. The scanning system 158 may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the combined signal Q as described above.

List of reference numbers

110	Detector
112	Object
114	object positions
116	sensor element
118	Matrix
120	optical sensor
122	light sensitive area
124	illumination source
126	reflection image
127	transfer device
128	evaluation device
129	selected reflection feature
130	longitudinal region
131	Crossing
132	displacement region
134	reference image
136	epipolar line
138	straight line
140	epipolar plane
142	first sensor element
144	second sensor element
146	Camera
148	detector system
150	beacon devices
152	human-machine-interface
154	entertainment device
156	tracking system
158	scanning system
160	Divider
162	position evaluation device
164	Connector
166	Housing
168	control device
170	User
172	Opening
174	optical axis
176	direction of view
178	coordinate system
182	reflection light beam
184	Machine
186	track controller

188 Line pattern
190 Epipolar line

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Patent claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) comprising
 - 5 - at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein each optical sensor (120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (122) by a reflection light beam (182) propagating from the object (112) to the detector (110), wherein the sensor element (116) is adapted to determine at least one reflection image (126);
 - 10 - at least one evaluation device (128), wherein the evaluation device (128) is adapted to select at least one reflection feature of the reflection image (126), wherein the evaluation device (128) is configured for determining at least one longitudinal region (130) of the selected reflection feature of the reflection image (134) by evaluating a combined signal Q from the sensor signals, wherein the evaluation device (128) is adapted to determine at least one displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130), wherein the evaluation device (128) is adapted to match the selected reflection feature with at least one reference feature within the displacement region (132).
 - 15
 - 20
2. The detector (110) according to the preceding claim, wherein the evaluation device (128) is adapted to determine a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device (128) is adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement.
- 25
3. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.
- 30
4. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region.
- 35
5. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q, wherein the longitudinal region (130) is given by the longitudinal coordinate z and an error interval $\pm \epsilon$.
- 40
6. The detector (110) according any one of the preceding claims, wherein the reference image (134) and the reflection image (126) are images of the object determined at different spatial

positions having a fixed distance, wherein the evaluation device (128) is adapted to determine an epipolar line (136) in the reference image (134).

7. The detector (110) according to the two preceding claims, wherein the displacement region (132) extends along the epipolar line (136), wherein the evaluation device (128) is adapted to determine the reference feature along the epipolar line (136) corresponding to the longitudinal coordinate z and to determine an extent of the displacement region (132) along the epipolar line (136) corresponding to the error interval $\pm\epsilon$.
8. The detector (110) according to the preceding claim, wherein the evaluation device (128) is configured to perform the following steps:
 - Determining the displacement region for the image position of each reflection feature;
 - Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.
9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is adapted to match the selected feature of the reflection image (126) with the reference feature within the displacement region (132) by using at least one evaluation algorithm considering the determined longitudinal coordinate z , wherein the evaluation algorithm is a linear scaling algorithm.
10. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are different areas of at least one beam profile of the reflection light beam (182) at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the reflection light beam.

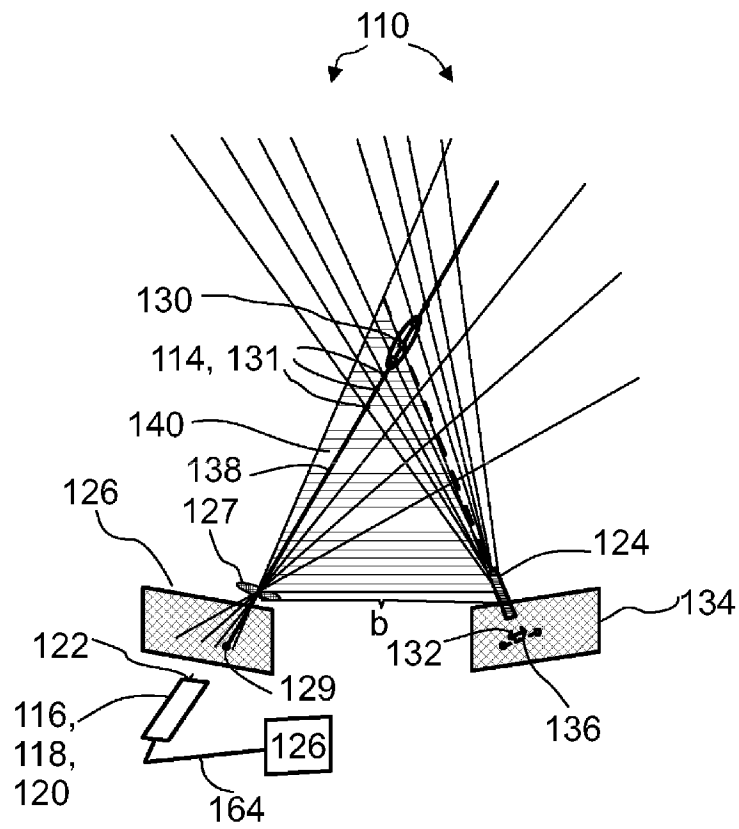
11. The detector (110) according to the preceding claim, wherein the light-sensitive areas (122) are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.
12. The detector (110) according to the preceding claim, wherein the evaluation device (128) is configured to determine the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
13. The detector (110) according to any one of the preceding claims, wherein the at least one evaluation device (128) is configured for evaluating the sensor signals, by
- a) determining at least one optical sensor (120) having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors (120) of the matrix (118) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining the longitudinal region by evaluating the combined signal.
14. The detector (110) according to the preceding claim, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.
15. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is adapted to generate at least one illumination pattern for illumination of the object (112), wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; at least one pattern comprising at least one pre-known feature.
16. The detector (110) according to the preceding claim, wherein the sensor element (116) is adapted to determine at least one reflection pattern, wherein the evaluation device (128) is adapted to select at least one feature of the reflection pattern and to determine the longitudinal region (130) of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals.

17. The detector (110) according to the preceding claim, wherein the reference image (134) is an image of the illumination pattern at an image plane at a position of the illumination source (124), wherein the evaluation device (128) is adapted to determine the displacement region (132) in the reference image (134) corresponding to the longitudinal region (130) of the selected feature of the reflection pattern, wherein the evaluation device (128) is adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region (132).
18. The detector (110) according to any one of the three preceding claims, wherein the detector (110) comprises at least two sensor elements (116) each having a matrix (118) of optical sensors (120), where at least one first sensor element (142) and at least one second sensor element (144) are positioned at different spatial positions, wherein a relative distance between the first sensor element (142) and the second element (144) is fixed, wherein the at least one first sensor element (142) is adapted to determine at least one first reflection pattern and the at least one second sensor element (144) is adapted to determine at least one second reflection pattern, wherein the evaluation device (128) is adapted to select at least one image determined by the first sensor element (142) or the second sensor element (144) as reflection image (126) and to select at least one image determined by the other one of the first sensor element (142) or the second sensor element (144) as reference image (134).
19. A detector system (148) for determining a position of at least one object (112), the detector system (148) comprising at least one detector (110) according to any one of the preceding claims, the detector system (148) further comprising at least one beacon device (150) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (150) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
20. A human-machine interface (152) for exchanging at least one item of information between a user (170) and a machine (184), wherein the human-machine interface (152) comprises at least one detector system (148) according to the preceding claim, wherein the at least one beacon device (150) is adapted to be at least one of directly or indirectly attached to the user (170) and held by the user (170), wherein the human-machine interface (152) is designed to determine at least one position of the user (170) by means of the detector system (148), wherein the human-machine interface (152) is designed to assign to the position at least one item of information.
21. An entertainment device (154) for carrying out at least one entertainment function, wherein the entertainment device (154) comprises at least one human-machine interface (152) according to the preceding claim, wherein the entertainment device (154) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (152), wherein the entertainment device (154) is designed to vary the entertainment function in accordance with the information.

22. A tracking system (156) for tracking a position of at least one movable object (112), the tracking system (156) comprising at least one detector system (148) according to any one of the preceding claims referring to a detector system (148), the tracking system (156) further comprising at least one track controller (186), wherein the track controller (186) is adapted to track a series of positions of the object (112) at specific points in time.
23. A scanning system (158) for determining a depth profile of a scenery, the scanning system comprising at least one detector (110) according to any of the preceding claims referring to a detector, the scanning system (158) further comprising at least one illumination source (124) adapted to scan the scenery with at least one light beam.
24. A camera (146) for imaging at least one object (112), the camera (146) comprising at least one detector (110) according to any one of the preceding claims referring to a detector.
25. A method for determining a position of at least one object (112) by using at least one detector (110) according to any one of the preceding claims referring to a detector, the method comprising the following steps:
- generating at least two sensor signals in response to an illumination of at least one sensor element (116) having a matrix (118) of optical sensors (120) by at least one reflection light beam (182) propagating from the object (112) to the detector (110), the optical sensors (120) each having a light-sensitive area (122);
 - determining at least one reflection image (126) by using the sensor element (116);
 - selecting at least one reflection feature of the reflection image (126) and evaluating the sensor signals by using at least one evaluation device (128), thereby, determining at least one longitudinal region (130) of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
 - determining at least one displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130);
 - matching the selected reflection feature with at least one reference feature within the displacement region (132).
26. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application;

1/6

FIG.1



2/6

FIG.2

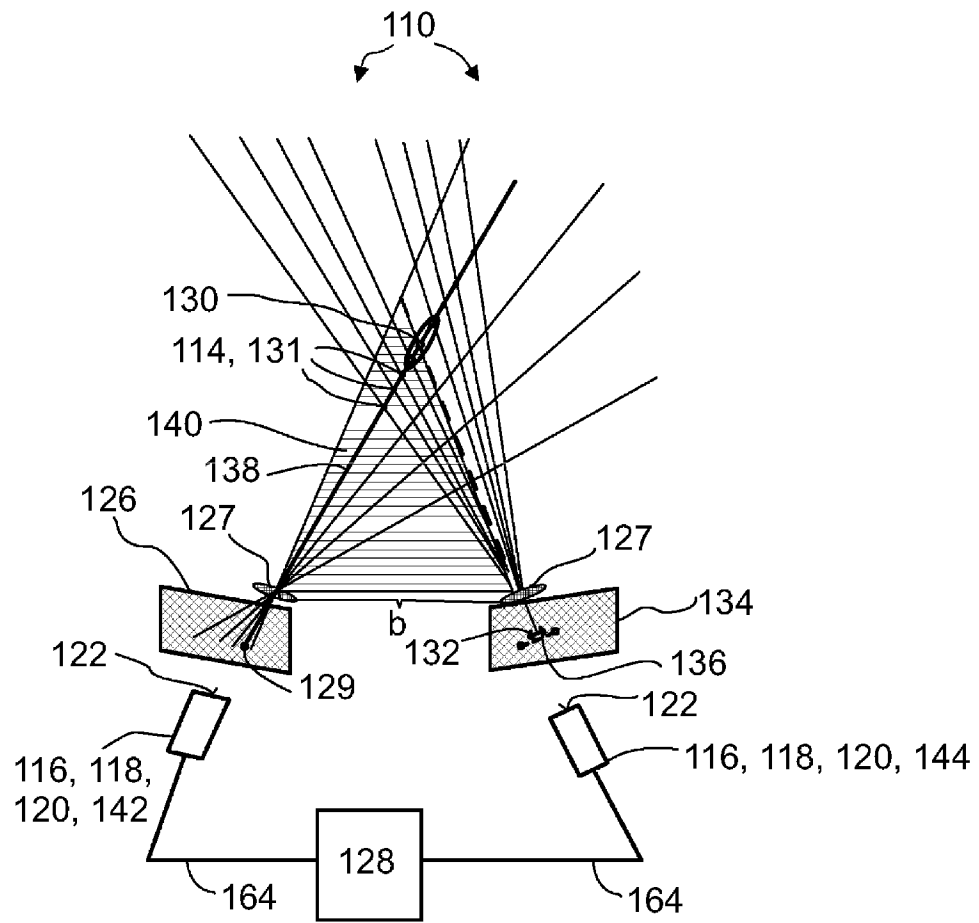
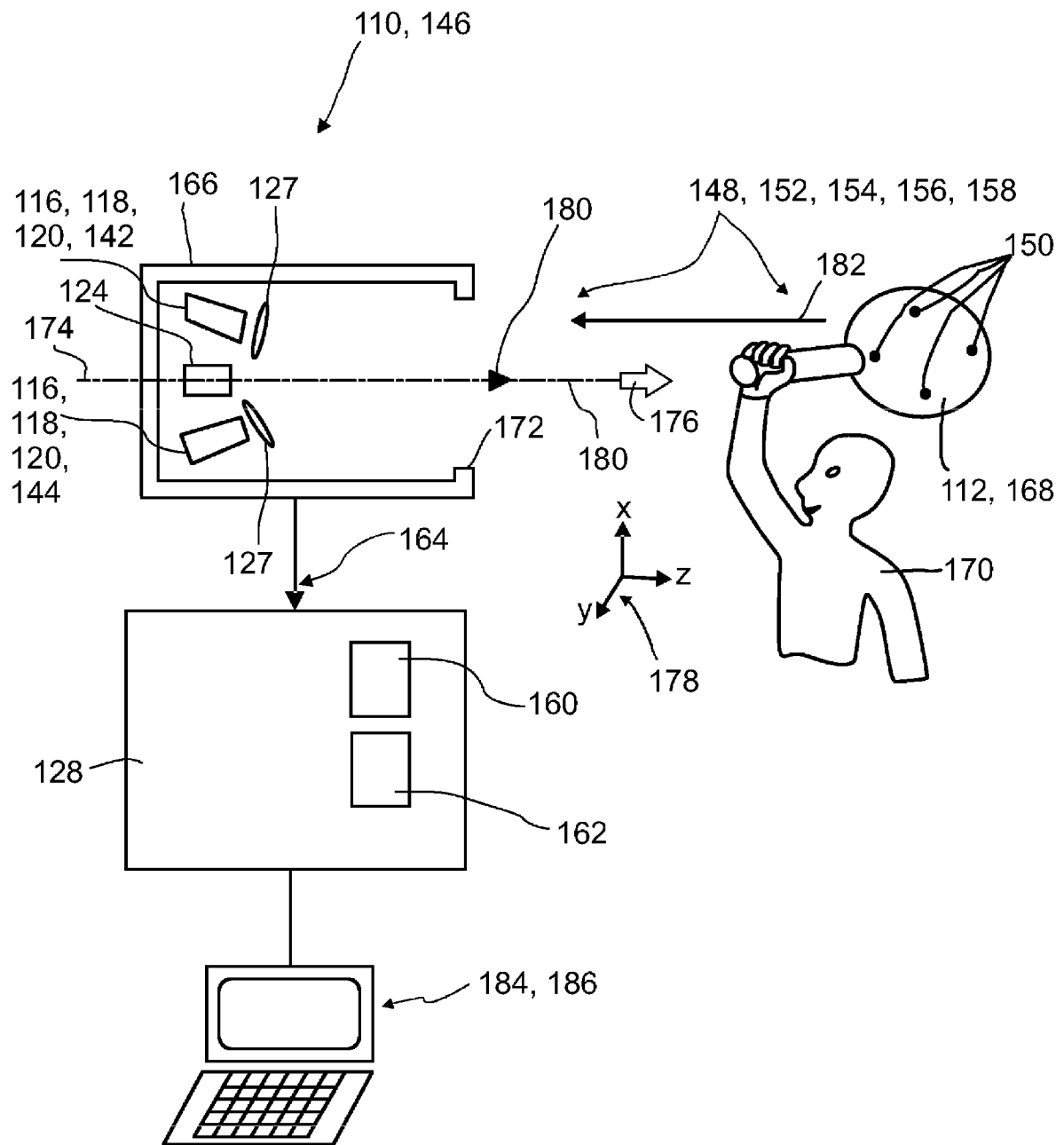


FIG.3



4/6

FIG.4

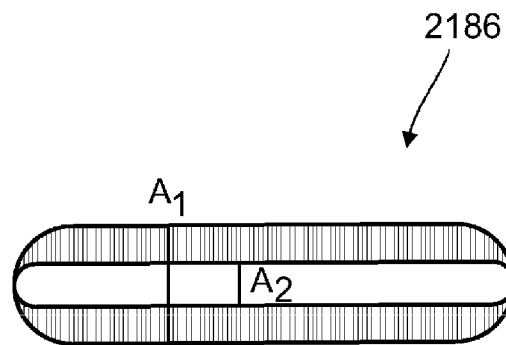


FIG.5A

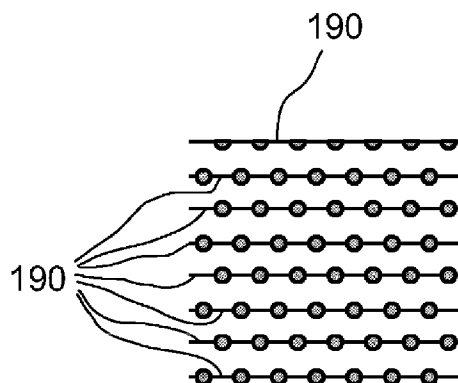


FIG.5B

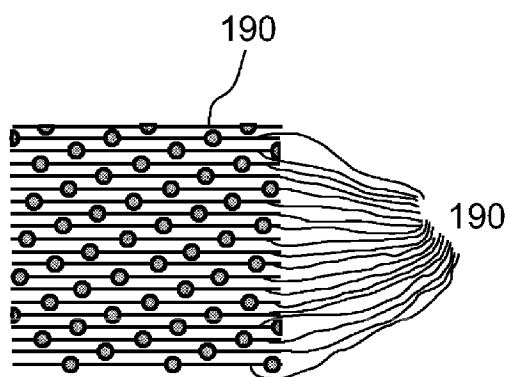


FIG.5C

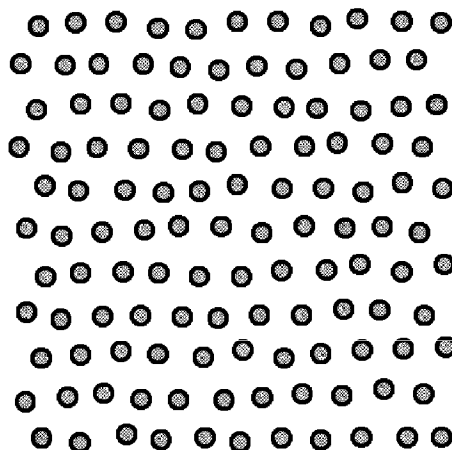
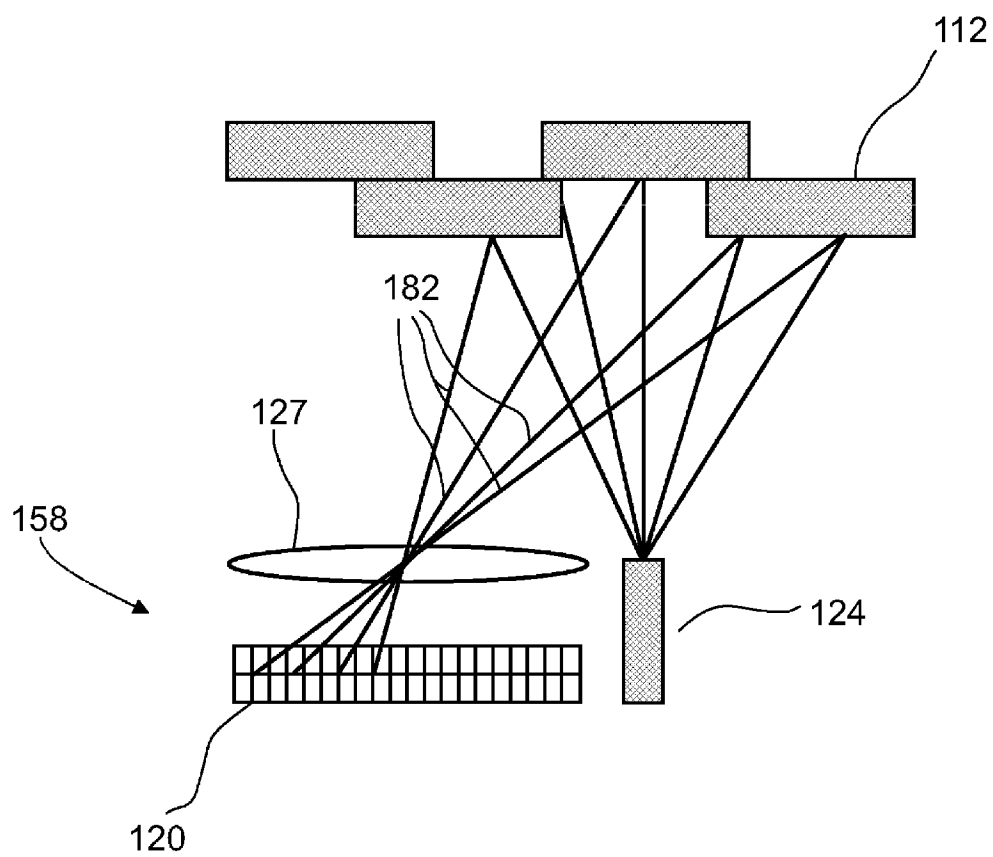


FIG.6



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2018/073067

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S7/48 G01S17/46
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, INSPEC, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2008/118143 A1 (GORDON EYAL [IL] ET AL) 22 May 2008 (2008-05-22) abstract; figures 1-25 paragraph [0309] - paragraph [0374] -----	1-26
X	US 2009/066929 A1 (TROPF HERMANN [DE]) 12 March 2009 (2009-03-12) abstract paragraph [0018] - paragraph [0110] -----	1,13, 19-26



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

15 November 2018

Date of mailing of the international search report

26/11/2018

Name and mailing address of the ISA/

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NL - 2280 HV Rijswijk
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Fax: (+31-70) 340-3016

Authorized officer

López de Valle, J

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2018/073067

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2008118143 A1	22-05-2008	US 2008118143 A1	22-05-2008
		US 2012063672 A1	15-03-2012

US 2009066929 A1	12-03-2009	EP 1971820 A1	24-09-2008
		US 2009066929 A1	12-03-2009
		WO 2007079805 A1	19-07-2007

PATENT COOPERATION TREATY

PCT

INTERNATIONAL SEARCH REPORT

(PCT Article 18 and Rules 43 and 44)

Applicant's or agent's file reference B182037PC	FOR FURTHER ACTION see Form PCT/ISA/220 as well as, where applicable, item 5 below.	
International application No. PCT/EP2020/056759	International filing date (<i>day/month/year</i>) 13 March 2020 (13-03-2020)	(Earliest) Priority Date (<i>day/month/year</i>) 15 March 2019 (15-03-2019)
Applicant TRINAMIX GMBH		

This international search report has been prepared by this International Searching Authority and is transmitted to the applicant according to Article 18. A copy is being transmitted to the International Bureau.

This international search report consists of a total of 4 sheets.

☒ It is also accompanied by a copy of each prior art document cited in this report.

1. Basis of the report

a. With regard to the **language**, the international search was carried out on the basis of:

- ☒ the international application in the language in which it was filed
☐ a translation of the international application into _____, which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b))

b. ☐ This international search report has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43.6**bis**(a)).

c. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, see Box No. I.

2. ☐ **Certain claims were found unsearchable** (See Box No. II)

3. ☐ **Unity of invention is lacking** (see Box No. III)

4. With regard to the **title**,

- ☒ the text is approved as submitted by the applicant
☐ the text has been established by this Authority to read as follows:

5. With regard to the **abstract**,

- ☒ the text is approved as submitted by the applicant
☐ the text has been established, according to Rule 38.2, by this Authority as it appears in Box No. IV. The applicant may, within one month from the date of mailing of this international search report, submit comments to this Authority

6. With regard to the **drawings**,

- a. the figure of the **drawings** to be published with the abstract is Figure No. 1
☒ as suggested by the applicant
☐ as selected by this Authority, because the applicant failed to suggest a figure
☐ as selected by this Authority, because this figure better characterizes the invention
b. ☐ none of the figures is to be published with the abstract

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2020/056759

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S7/48 G01N21/47 G01S17/46 G06K9/00 G06T7/00
G06T7/521

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S G01N G06T G06K G01B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data, COMPENDEX, INSPEC

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	DE 198 46 619 A1 (BYK GARDNER GMBH [DE]) 27 April 2000 (2000-04-27)	1-7, 9-16, 23-25
Y	column 13, line 45 - column 14, line 66; claims 1,29,7 column 2, lines 31-38 -----	17-20
X	CN 108 363 482 A (JIANGSU SIDIANLING ROBOT CO LTD) 3 August 2018 (2018-08-03)	1-7, 9-14,16, 21,23-25
Y	claim 1 -----	17-20
X	US 2018/033146 A1 (BLEYER MICHAEL [US] ET AL) 1 February 2018 (2018-02-01)	1-14,16, 22-25
Y	paragraphs [0013], [0036], [0037]; claims 1,4,6 -----	17-20
	-/-	



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

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"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

13 May 2020

Date of mailing of the international search report

29/05/2020

Name and mailing address of the ISA/

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Authorized officer

Grübl, Alexander

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2020/056759

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
Y	WO 2019/042956 A1 (TRINAMIX GMBH [DE]) 7 March 2019 (2019-03-07) claims 19-22,26 -----	17-20

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2020/056759

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
DE 19846619 A1	27-04-2000	DE 19846619 A1 JP 2000131226 A	27-04-2000 12-05-2000
-----	-----	-----	-----
CN 108363482 A	03-08-2018	NONE	
-----	-----	-----	-----
US 2018033146 A1	01-02-2018	CN 109477710 A EP 3491332 A1 US 2018033146 A1 WO 2018022386 A1	15-03-2019 05-06-2019 01-02-2018 01-02-2018
-----	-----	-----	-----
WO 2019042956 A1	07-03-2019	KR 20200040780 A WO 2019042956 A1	20-04-2020 07-03-2019
-----	-----	-----	-----

PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To:

see form PCT/ISA/220

PCT

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY (PCT Rule 43bis.1)

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/EP2020/056759

International filing date (day/month/year)
13.03.2020

Priority date (day/month/year)
15.03.2019

International Patent Classification (IPC) or both national classification and IPC
INV. G01S7/48 G01N21/47 G01S17/46 G06K9/00 G06T7/00 G06T7/521

Applicant
TRINAMIX GMBH

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☐ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA:



European Patent Office
D-80298 Munich
Tel. +49 89 2399 - 0
Fax: +49 89 2399 - 4465

Date of completion of
this opinion

see form
PCT/ISA/210

Authorized Officer

Grübl, Alexander

Telephone No. +49 89 2399-0



Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43bis.1(a))
3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:
 - a. ☐ forming part of the international application as filed:
 - ☐ in the form of an Annex C/ST.25 text file.
 - ☐ on paper or in the form of an image file.
 - b. ☐ furnished together with the international application under PCT Rule 13ter.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.
 - c. ☐ furnished subsequent to the international filing date for the purposes of international search only:
 - ☐ in the form of an Annex C/ST.25 text file (Rule 13ter.1(a)).
 - ☐ on paper or in the form of an image file (Rule 13ter.1(b) and Administrative Instructions, Section 713).
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Yes: Claims	<u>4, 9-14, 16-20, 23</u>
	No: Claims	<u>1-3, 5-8, 15, 21, 22, 24, 25</u>
Inventive step (IS)	Yes: Claims	
	No: Claims	<u>1-25</u>
Industrial applicability (IA)	Yes: Claims	<u>1-25</u>
	No: Claims	

2. Citations and explanations

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

Re Item V

Reasoned statement with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

Reference is made to the following documents:

- D1 DE 198 46 619 A1 (BYK GARDNER GMBH [DE]) 27 April 2000 (2000-04-27)
- D2 CN 108 363 482 A (JIANGSU SIDIANLING ROBOT CO LTD) 3 August 2018 (2018-08-03)
- D3 US 2018/033146 A1 (BLEYER MICHAEL [US] ET AL) 1 February 2018 (2018-02-01)
- D4 WO 2019/042956 A1 (TRINAMIX GMBH [DE]) 7 March 2019 (2019-03-07)

- 1 The present application does not meet the criteria of Article 33(1) PCT, because the subject-matter of independent claims 1, 24 and 25 is not new in the sense of Article 33(2) PCT.

Claim 1

A detector for identifying at least one material property comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object (D1: claim 1; D2: claim 1 - camera; D3: claim 1, claim 11 - imaging system)

- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image (D1: claim 1; D2: claim 1 - processor, depth information; D3: claim 1, claim 11 - intensity profile determination),

wherein the evaluation device is configured for determining at least one distance feature by applying at least one distance dependent image filter to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof, being one of the

depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, (D1: claim 29; D2: claim 1 - triangulation; D3: claims 1, 4 - depth map, triangulation)

wherein the evaluation device is configured for determining at least one material feature by applying at least one material dependent image filter to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Mother which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof (D1: claim 1, col. 14, l. 38-66; D2: claim 1 - median filter; D3: claim 6, [0036]),

wherein the evaluation device is configured for determining a longitudinal coordinate and the material property by evaluating the distance feature and the material feature (D1: claims 1 and 29; D2: claim 1 - median filtering, processing depth image; D3: claim 6, [0037]).

The method according to **claim 24** is disclosed in the same manner.

Claim 25

A use of the detector according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography

application; a machine vision application; a robotics application; a quality control application; a manufacturing application (D1: col 2, l. 31-38; D2: claim 1; D3: [0013]).

- 2 The present application does not meet the criteria of Article 33(1) PCT, because the subject-matter of independent **claim 23** does not involve an inventive step in the sense of Article 33(3) PCT.

Inertial measurement units are generally known, being adapted to receive data of a sensor as defined in claim 1, a material property detector, does not allow to establish a technical effect resulting from this feature.

- 3 The dependent claims do not contain particular features which appear suitable for rendering the independent claims clear, new and defining inventive subject matter:

Claims 2, 3, 5-7 relate to further vague definitions of the filters, not adding any more substance.

Claim 4 relates to material properties usually determined: e.g. D1: column 14, lines 38-45; D3: claim 1.

Claim 8: see D3, claim 6

Claims 9-11: Recursive, iterative or simultaneous application of filters and applying a further filter are considered usual alternatives.

Claims 12, 13: Using combined signals from sensor elements or a convolution based algorithm are considered common measures.

Claim 14: A CMOS sensor is a common matrix sensor.

Claim 15: D1, claim 7

Claim 16: A diffractive optical element is a usual means for producing such patterns.

The features of claims 17-20 are known from D4, claims 19-22, which the skilled person would turn to in order to find implementations.

Claim 21: D2, claim 1

Claim 22: D3, claim 1

Re Item VIII

Certain observations on the international application

The claims are not clear (Article 6 PCT) and the disclosure of the invention defined therein is not sufficient for enabling the skilled person to carry it out (Article 5 PCT).

The independent claims 1 and 24 relate to determining a longitudinal coordinate and a material property. It is neither clear how it is arrived at the desired distance feature and material feature nor which "further" filters are covered by the claim. The vague indication of correlation with one of the listed filters is an infinite definition of filters which does not allow to put into practice the invention or determine what is covered by the claims. The list of filters contains filters which are not commonly known and lack a definition in the claims.

The sign φ_{1m} in claim 10 is not defined.

Claim 23: As the detector as defined in claim 1 determines a material property, it is not clear how it can be used in an inertial measurement unit. A disclosure in this respect is also missing in the description.

Electronic Patent Application Fee Transmittal

Application Number:

Filing Date:

Title of Invention:

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

First Named Inventor/Applicant Name:

Friedrich Schick

Filer:

CHRISTOPHER M GOFF/Donna McGraw

Attorney Docket Number:

27843-1809 182037US01

Filed as Large Entity

Filing Fees for U.S. National Stage under 35 USC 371

Description

Fee Code

Quantity

Amount

**Sub-Total in
USD(\$)**

Basic Filing:

NATIONAL STAGE FEE

1631

1

320

320

NATL STAGE SEARCH FEE - REPORT PROVIDED

1642

1

540

540

NATIONAL STAGE EXAM - ALL OTHER CASES

1633

1

800

800

Pages:

NATL STAGE APPL SZ FEE PER 50 PGS >100

1681

1

420

420

Claims:

CLAIMS IN EXCESS OF 20

1615

5

100

500

Miscellaneous-Filing:

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				2580

Electronic Acknowledgement Receipt

EFS ID:	43769869
Application Number:	17439492
International Application Number:	PCT/EP2020/056759
Confirmation Number:	9002
Title of Invention:	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY
First Named Inventor/Applicant Name:	Friedrich Schick
Customer Number:	76656
Filer:	CHRISTOPHER M GOFF/Donna McGraw
Filer Authorized By:	CHRISTOPHER M GOFF
Attorney Docket Number:	27843-1809 182037US01
Receipt Date:	15-SEP-2021
Filing Date:	
Time Stamp:	12:07:17
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 2580
RAM confirmation Number	E20219EC08010885
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1	Transmittal of New Application	2021-09-15_371_Transmittal_27843-1809_.pdf	1562649	no	4
			d9a361d56aad880d0d50317bef2642041d08c1c4		
Warnings:					
Information:					
2		2021-09-15_PRIORITY_27843-1809_.pdf	977644	yes	139
			eb957a3f2804526ce45596cf0dd1ccc1a75857f7		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Abstract		1	2	
	Specification		3	132	
	Claims		133	137	
	Drawings-other than black and white line drawings		138	139	
Warnings:					
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3	Oath or Declaration filed	2021-09-15_Declaration_27843-1809_.pdf	1133725	no	7
			9c76a14a3e190805108b097caa8a4273c0c1fc80		
Warnings:					
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4		2021-09-15_Preliminary_Amendment_27843-1809_.pdf	238509	yes	13
			9869f86cb0f7cb5dba06b4ec9b7d4c51f79f0cc2		
	Multipart Description/PDF files in .zip description				
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	Preliminary Amendment		1	1
	Specification		2	2
	Abstract		3	4
	Claims		5	12
	Abstract		13	13

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5	Power of Attorney	2021-09-15_POA_27843-1809_.pdf	375728	no	2
			f6aace714e1de56acc5a620deaacf7605d383326		

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6	Application Data Sheet	2021-09-15_ADS_27843-1809_.pdf	1375768	no	12
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			fde7a115ada782abbb3d1fde60aba027067364e		

Warnings:
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8	Foreign Reference	DE19846619A1_EngTrans_.pdf	10442747	no	54
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9	Foreign Reference	CN108363482A_EngTrans_.pdf	3701835	no	21
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10	Foreign Reference	WO2019042956_.pdf	22837593 8ea499f7f90fd831d3fb9c166d179db359f0f83e	no	107
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11	Other Reference-Patent/App/Search documents	ISR_WO_.pdf	1533676 433b762d83d6402f1fe408adbeafe09abf0aab3a	no	11
Warnings:					
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12	Non Patent Literature	R_A_Street_NPL_.pdf	2578288 e43cdfefb1400b414d99aa31023bdd68a7a229b97	no	5
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	Filing Date		2021-09-15
	First Named Inventor	Schick, Friedrich	
	Art Unit		
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Application Number		17439492
Filing Date		2021-09-15
First Named Inventor	Schick, Friedrich	
Art Unit		
Examiner Name		
Attorney Docket Number	27843-1809 182037US01	

3	2018091649	WO	A1	2018-05-24	TRINAMIX GMBH		
4	2018091638	WO	A1	2018-05-24	TRINAMIX GMBH		
5	2012110924	WO	A1	2012-08-23	BASF SE et al.		
6	2014198629	WO	A1	2014-12-18	BASF SE		

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Preliminary Report on Patentability for corresponding PCT/EP2020/056759 mailed 30 Sept. 2021, 9 Pages	
	2	X. Jiang, H. Bunke: "Dreidimensionales Computersehen" Springer, Berlin Heidelberg, 1997, Chapter 2. 32 pages. Cited in specification, no English translation available.	
	3	EICHLER et al., "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 171 to 266, Springer Verlag, ISBN 0939-0979. Cited in specification, no English translation available.	

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Application Number	17439492
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Application Number	17439492
Filing Date	2021-09-15
First Named Inventor	Schick, Friedrich
Art Unit	
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Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2021-10-05
Name/Print	Christopher M. Goff	Registration Number	41785

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G01C 3/06 (2006.01)(21) International Application Number:
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61/739,173 19 December 2012 (19.12.2012) US
61/749,964 8 January 2013 (08.01.2013) US
61/867,169 19 August 2013 (19.08.2013) US(71) Applicant: **BASF SE** [DE/DE]; 67056 Ludwigshafen (DE).(71) Applicant (for MN only): **BASF (CHINA) COMPANY LIMITED** [CN/CN]; 300 Jiangxinsha Road, Shanghai 200137 (CN).(72) Inventors: **BRUDER, Ingmar**; Neustadter Str. 30, 67112 Mutterstadt (DE). **SPILLER, Simone Christina**; L12, 1, 68161 Mannheim (DE). **THIEL, Erwin**; Schöntalstr. 4, 57076 Siegen (DE). **IRLE, Stephan**; Theodorstr. 3, 57074 Siegen (DE). **SEND, Robert**; Luisenstr. 25, 76137 Karls-ruhe (DE). **WONNEBERGER, Henrike**; Richard-Wagner-Str. 15, 68165 Mannheim (DE).(74) Agent: **STÖBEL, Matthias**; Herzog Fiesser & Partner, Dudenstrasse 46, 68167 Mannheim (DE).

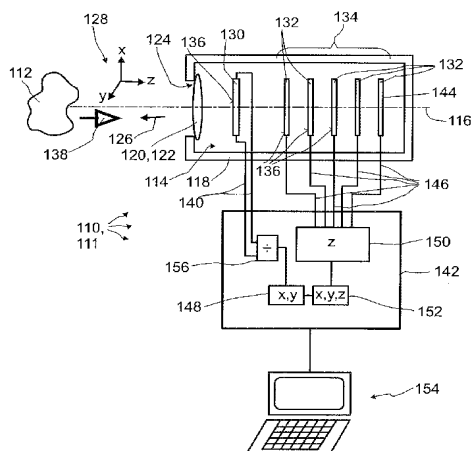
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[Continued on next page]

(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.1A



(57) Abstract: A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises: - at least one transversal optical sensor (130), the transversal optical sensor (130) being adapted to determine a transversal position of at least one light beam (138) traveling from the object (112) to the detector (110), the transversal position being a position in at least one dimension perpendicular to an optical axis (116) of the detector (110), the transversal optical sensor (130) being adapted to generate at least one transversal sensor signal; - at least one longitudinal optical sensor (132), wherein the longitudinal optical sensor (132) has at least one sensor region (136), wherein the longitudinal optical sensor (132) is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region (136) by the light beam (138), wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam (138) in the sensor region (136); - at least one evaluation device (142), wherein the evaluation device (142) is designed to generate at least one item of information on a transversal position of the object (112) by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object (112) by evaluating the longitudinal sensor signal.



Published:

— with international search report (Art. 21(3))

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Detector for optically detecting at least one objectDescription5 Field of the invention

The invention relates to a detector for determining a position of at least one object. Furthermore, the invention relates to a human-machine interface, an entertainment device, a tracking system and a camera. Furthermore, the invention relates to a method for optically detecting a position
10 of at least one object and to various uses of the detector. Such devices, methods and uses can be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, medical technology or in the sciences. Additionally or alternatively, the application may be applied in the field of mapping of spaces, such as for generating maps of one or more rooms, one or more buildings or one or more streets. However,
15 other applications are also possible in principle.

Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While
20 photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultraviolet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and for reasons of large-area processing, sensors comprising at least one organic sensor material
30 are being used, as described for example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO 2009/013282 A1.

As a further example, WO 2013/144177 A1 discloses quinolinium dyes having a fluorinated counter anion, an electrode layer which comprises a porous film made of oxide semiconductor fine particles sensitized with these kinds of quinolinium dyes having a fluorinated counter anion, a photoelectric conversion device which comprises such a kind of electrode layer, and a dye sensitized solar cell which comprises such a photoelectric conversion device.
35

A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or
40

microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for optically detecting objects are triangulation systems, by means of which distance measurements can likewise be carried out.

Proceeding from such known detectors and methods for optically detecting objects, it can be ascertained that in many cases a considerable technical outlay has to be implemented in order to carry out this object detection with sufficient precision.

By way of example, in microscopy a considerable outlay in respect of apparatus is required in order to obtain correct focusing of a light beam and/or in order to obtain depth information about the sample to be imaged.

Another example for recording depth detection are active sensors which usually function by emitting a signal and measuring reflections of the signal caused by the at least one object. The active sensors have found a wide dissemination, particularly in applications related to computer vision. M.R. Andersen, T. Jensen, P. Lisouski, A.K. Mortensen, M.K. Hansen, T. Gregersen, and P. Ahrendt, *Kinect Depth Sensor Evaluation for Computer Vision Applications*, Technical report ECE-TR-6, 2012, Dept. of Engineering, Aarhus University, Denmark, describe a depth sensor which projects an infrared (IR) point pattern onto at least one object and measures its reflection pattern by use of an IR camera. According to their study, the authors could observe that this depth sensor required a settling time of approximately 30 seconds from the time the device was powered until a reliable depth signal was acquired, in particular owing to a large amount of computing power being required for performing this kind of procedure. Equivalently, a similar settling time could be observed when the depth sensor was rotated and quickly pointed towards a different scene.

A further problem arises when more than one active sensor is employed for acquiring depth information, particularly for extending a measurement range. As soon as signals which originate from one or more different or identical active sensors overlap, the origin of the signals becomes unidentifiable, so that in the region where the signals overlap no depth information can be recorded. A similar problem arises when two or more active sensors point towards each other. Particularly, since the measurement volume of an optical sensor can be described as an approximate cone, a large measurement volume can only be covered by a number of active sensors which, however, results in overlapping volumes and, thus, uncovered regions.

Distance measurements, by contrast, are based in many cases on technically inadequate assumptions such as, for example, the assumption of a specific size of an object in an image evaluation. Other methods are based in turn on complex pulse sequences, such as, for

example, distance measurements by means of laser pulses. Yet other methods are based on the use of a plurality of detectors such as, for example, triangulation methods.

In WO 2005/106965 A1, a setup of an organic solar cell is disclosed. A photo current is generated in response to incident light. Further, a method for producing the organic solar cell is disclosed. Therein, reference is made to the fact that defects or traps may diminish the efficiency of the organic solar cell.

Various position detectors are known in the art. Thus, in JP 8-159714 A, a distance measurement device is disclosed. Therein, by using a detector and a shadow-forming element, a distance between an object and the detector is determined based on the fact that shadow formation of objects depends on the distance. In US 2008/0259310 A1, an optical position detector is disclosed. The position of a transmission system is determined by using a variety of known distances and measured angles. In US 2005/0184301 A1, a distance measurement device is disclosed. The measurement device makes use of a plurality of light-emitting diodes having different wavelengths. In CN 101650173 A, a position detector is disclosed which is based on the use of geometric principles. Further, in JP 10-221064 A, a complex optical setup is disclosed which is similar to optical setups used in holography.

In US 4,767,211, a device and a method for optical measurement and imaging are disclosed. Therein, a ratio of reflected light traveling along an optical axis and reflected light traveling off-axis is determined by using different photo-detectors and a divider. By using this principal, depressions in a sample may be detected.

In US 4,647,193, a range of a target object is determined by using a detector having multiple components. The detector is placed away from a focal plane of a lens. The size of a light spot of light from the object varies with the range of the object and, thus, is dependent on the range of the object. By using different photo-detectors, the size of the light spot and, thus, the range of the object may be determined by comparing signals generated by the photo-detectors.

In US 6,995,445 and US 2007/0176165 A1, a position sensitive organic detector is disclosed. Therein, a resistive bottom electrode, is used which is electrically contacted by using at least two electrical contacts. By forming a current ratio of the currents from the electric contacts, a position of a light spot on the organic detector may be detected.

In US 2007/0080925 A1, a low power consumption display device is disclosed. Therein, photoactive layers are utilized which both respond to electrical energy to allow a display device to display information and which generate electrical energy in response to incident radiation. Display pixels of a single display device may be divided into displaying and generating pixels. The displaying pixels may display information and the generating pixels may generate electrical energy. The generated electrical energy may be used to provide power to drive an image.

US provisional applications 61/739,173, filed on December 19, 2012, and 61/749,964, filed on January 8, 2013, the full content of which is herewith included by reference, disclose a method and a detector for determining a position of at least one object, by using at least one transversal optical sensor and at least one optical sensor. Specifically, the use of sensor stacks is disclosed, in order to determine a longitudinal position of the object with a high degree of accuracy and without ambiguity.

European patent application number EP 13171898.3, filed on June 13, 2013, the full content of which is herewith included by reference, discloses an optical detector comprising an optical sensor having a substrate and at least one photosensitive layer setup disposed thereon. The photosensitive layer setup has at least one first electrode at least one second electrode and at least one photovoltaic material sandwiched in between the first electrode and the second electrode. The photovoltaic material comprises at least one organic material. The first electrode comprises a plurality of first electrode stripes, and the second electrode comprises a plurality of second electrode stripes, wherein the first electrode stripes and the second electrode stripes intersect such that a matrix of pixels is formed at intersections of the first electrode stripes and the second electrode stripes. The optical detector further comprises at least one readout device, the readout device comprising a plurality of electrical measurement devices being connected to the second electrode stripes and a switching device for subsequently connecting the first electrode stripes to the electrical measurement devices.

European patent application number EP 13171900.7, also filed on June 13, 2013, the full content of which is herewith included by reference, discloses a detector device for determining an orientation of at least one object, comprising at least two beacon devices being adapted to be at least one of attached to the object, held by the object and integrated into the object, the beacon devices each being adapted to direct light beams towards a detector, and the beacon devices having predetermined coordinates in a coordinate system of the object. The detector device further comprises at least one detector adapted to detect the light beams traveling from the beacon devices towards the detector and at least one evaluation device, the evaluation device being adapted to determine longitudinal coordinates of each of the beacon devices in a coordinate system of the detector. The evaluation device is further adapted to determine an orientation of the object in the coordinate system of the detector by using the longitudinal coordinates of the beacon devices.

European patent application number EP 13171901.5, filed on June 13, 2013, the full content of which is herewith included by reference, discloses a detector for determining a position of at least one object. The detector comprises at least one optical sensor being adapted to detect a light beam traveling from the object towards the detector, the optical sensor having at least one matrix of pixels. The detector further comprises at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam. The evaluation device is further adapted to determine at least one

longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

In WO 2012/110924 A1, on which the present invention is based and the content of which is herewith included by reference, a detector for optically detecting at least one object is proposed. The detector comprises at least one optical sensor. The optical sensor has at least one sensor region. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area. The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

Despite the advantages implied by the above-mentioned devices and detectors specifically by the detector disclosed in WO 2012/110924 A1, there still is a need for a simple, cost-efficient and, still, reliable spatial detector. Thus, an improved spatial resolution of an object in conjunction with the possibility of tracking the object in space is desirable.

Problem addressed by the invention

Therefore, a problem addressed by the present invention is that of specifying devices and methods for optically detecting at least one object which at least substantially avoid the disadvantages of known devices and methods of this type. In particular, an improved detector for determining the position of an object in space and, preferably, for reliably tracking the object in space is desirable.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used herein, the expressions "have", "comprise" and "contain" as well as grammatical variations thereof are used in a non-exclusive way. Thus, the expression "A has B" as well as the expression "A comprises B" or "A contains B" may both refer to the fact that, besides B, A contains one or more further components and/or constituents, and to the case in which, besides B, no other components, constituents or elements are present in A.

In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed.

The object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

As used herein, a position generally refers to an arbitrary item of information on a location and/or orientation of the object in space. For this purpose, as an example, one or more coordinate systems may be used, and the position of the object may be determined by using one, two, three or more coordinates. As an example, one or more Cartesian coordinate systems and/or other types of coordinate systems may be used. In one example, the coordinate system may be a coordinate system of the detector in which the detector has a predetermined position and/or orientation. As will be outlined in further detail below, the detector may have an optical axis, which may constitute a main direction of view of the detector. The optical axis may form an axis of the coordinate system, such as a z-axis. Further, one or more additional axes may be provided, preferably perpendicular to the z-axis.

Thus, as an example, the detector may constitute a coordinate system in which the optical axis forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As used herein, a detector for determining a position of at least one object generally is a device adapted for providing at least one item of information on the position of the at least one object. The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The detector may be adapted to provide the at least one item of information on the position of the at least one object in any feasible way. Thus, the information may e.g. be provided electronically, visually, acoustically or in any arbitrary combination thereof. The information may
5 further be stored in a data storage of the detector or a separate device and/or may be provided via at least one interface, such as a wireless interface and/or a wire-bound interface.

The detector comprises:

- at least one transversal optical sensor, the transversal optical sensor being adapted to
10 determine a transversal position of at least one light beam traveling from the object to the detector, the transversal position being a position in at least one dimension perpendicular an optical axis of the detector, the transversal optical sensor being adapted to generate at least one transversal sensor signal;
- at least one longitudinal optical sensor, wherein the longitudinal optical sensor has at
15 least one sensor region, wherein the longitudinal optical sensor is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region;
- at least one evaluation device, wherein the evaluation device is designed to generate at
20 least one item of information on a transversal position of the object by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

As will be outlined in further detail below, the components listed above may be separate
25 components. Alternatively, two or more of the components listed above may be integrated into one component. Thus, the at least one transversal optical sensor and the at least one longitudinal optical sensor may at least partially be integrated into one optical sensor. Alternatively, at least one longitudinal optical sensor may be provided which is separate from at least one transversal optical sensor. Further, the at least one evaluation device may be formed
30 as a separate evaluation device independent from the at least one transversal optical sensor and the at least one longitudinal optical sensor, but may preferably be connected to the at least one transversal optical sensor and the at least one longitudinal optical sensor in order to receive the transversal sensor signal and the longitudinal sensor signal. Alternatively, the at least one evaluation device may fully or partially be integrated into the at least one transversal optical
35 sensor and/or the at least one longitudinal optical sensor.

As used herein, the term transversal optical sensor generally refers to a device which is adapted to determine a transversal position of at least one light beam traveling from the object to the detector. With regard to the term transversal position, reference may be made to the
40 definition given above. Thus, preferably, the transversal position may be or may comprise at least one coordinate in at least one dimension perpendicular to an optical axis of the detector. As an example, the transversal position may be a position of a light spot generated by the light

beam in a plane perpendicular to the optical axis, such as on a light-sensitive sensor surface of the transversal optical sensor. As an example, the position in the plane may be given in Cartesian coordinates and/or polar coordinates. Other embodiments are feasible.

- 5 For potential embodiments of the transversal optical sensor, reference may be made to the position sensitive organic detector as disclosed in US 6,995,445 and US 2007/0176165 A1. However, other embodiments are feasible and will be outlined in further detail below.

10 The at least one transversal sensor signal generally may be an arbitrary signal indicative of the transversal position. As an example, the transversal sensor signal may be or may comprise a digital and/or an analog signal. As an example, the transversal sensor signal may be or may comprise a voltage signal and/or a current signal. Additionally or alternatively, the transversal sensor signal may be or may comprise digital data. The transversal sensor signal may comprise a single signal value and/or a series of signal values. The transversal sensor signal may further
15 comprise an arbitrary signal which is derived by combining two or more individual signals, such as by averaging two or more signals and/or by forming a quotient of two or more signals, as will be outlined in further detail below.

20 As used herein, a longitudinal optical sensor generally is a device which is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region. For potential embodiments of the longitudinal optical sensor, reference may be made to the optical sensor as disclosed in WO 2012/110924 A1. Preferably, however, as will be outlined in
25 further detail below, the detector according to the present invention comprises a plurality of optical sensors, such as a plurality of optical sensors as disclosed in WO 2012/110924 A1, preferably as a sensor stack.

30 Thus, as an example, the detector according to the present invention may comprise a stack of optical sensors as disclosed in WO 2012/110924 A1, in combination with one or more transversal optical sensors. As an example, one or more transversal optical sensors may be disposed on a side of the stack of longitudinal optical sensors facing towards the object. Alternatively or additionally, one or more transversal optical sensors may be disposed on a side of the stack of longitudinal optical sensors facing away from the object. Again, additionally or
35 alternatively, one or more transversal article sensors may be interposed in between the longitudinal optical sensors of the stack.

40 As will further be outlined below, preferably, both the at least one transversal optical sensor longitudinal optical sensor and the at least one longitudinal optical sensor may comprise one or more photo detectors, preferably one or more organic photodetectors and, most preferably, one or more dye-sensitized organic solar cells (DSCs, also referred to as dye solar cells), such as one or more solid dye-sensitized organic solar cells (s-DSCs). Thus, preferably, the detector

may comprise one or more DSCs (such as one or more sDSCs) acting as the at least one transversal optical sensor and one or more DSCs (such as one or more sDSCs) acting as the at least one longitudinal optical sensor, preferably a stack of a plurality of DSCs (preferably a stack of a plurality of sDSCs) acting as the at least one longitudinal optical sensor.

5

As used herein, the term evaluation device generally refers to an arbitrary device designed to generate the at least one item of information on the transversal position of the object and the longitudinal position of the object. As an example, the evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated
10 circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the transversal sensor signal and/or the longitudinal sensor signal, such as one or more AD-converters and/or one or more
15 filters. Further, the evaluation device may comprise one or more data storage devices. Further, as outlined above, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

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The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program performing or supporting the step of generating the at least one item of information on the transversal position and/or the step of generating the at least one item of information on the longitudinal position. As an example, one or more algorithms may be implemented which, by using the transversal sensor signal and/or the longitudinal sensor signal as input variables, may perform a predetermined transformation into
25 the transversal position and/or the longitudinal position of the object.

30

As outlined above, preferably, the transversal optical sensor is a photo detector having at least one first electrode, at least one second electrode and at least one photovoltaic material, wherein the photovoltaic material is embedded in between the first electrode and the second electrode.
As used herein, a photovoltaic material generally is a material or combination of materials adapted to generate electric charges in response to an illumination of the photovoltaic material with light.

35

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared (IR) spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1000 μm , preferably in the range of 780 nm to 3.0 μm . The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the
40 range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally refers to an amount of light emitted into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beam may be or may comprise one or more Gaussian light beams which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

Preferably, the second electrode of the transversal optical sensor may be a split electrode having at least two partial electrodes, wherein the transversal optical sensor has a sensor area, wherein the at least one transversal sensor signal indicates a position of the light beam in the sensor area. Thus, as outlined above, the transversal optical sensor may be or may comprise one or more photo detectors, preferably one or more organic photo detectors, more preferably one or more DSCs or sDSCs. The sensor area may be a surface of the photo detector facing towards the object. The sensor area preferably may be oriented perpendicular to the optical axis. Thus, the transversal sensor signal may indicate a position of a light spot generated by the light beam in a plane of the sensor area of the transversal optical sensor.

Generally, as used herein, the term partial electrode refers to an electrode out of a plurality of electrodes, adapted for measuring at least one current and/or voltage signal, preferably independent from other partial electrodes. Thus, in case a plurality of partial electrodes is provided, the second electrode is adapted to provide a plurality of electric potentials and/or electric currents and/or voltages via the at least two partial electrodes, which may be measured and/or used independently.

When using at least one transversal optical sensor having at least one split electrode having two or more partial electrodes as a second electrode, currents through the partial electrodes may be dependent on a position of the light beam in the sensor area. This may generally be due to the fact that Ohmic losses or resistive losses may occur on the way from a location of generation of electrical charges due to the impinging light to the partial electrodes. Thus, besides the partial electrodes, the second electrode may comprise one or more additional electrode materials connected to the partial electrodes, wherein the one or more additional electrode materials provide an electrical resistance. Thus, due to the Ohmic losses on the way from the location of generation of the electric charges to the partial electrodes through with the one or more additional electrode materials, the currents through the partial electrodes depend on the location of the generation of the electric charges and, thus, to the position of the light beam in the sensor area. For details of this principle of determining the position of the light beam in the sensor area, reference may be made to the preferred embodiments below and/or to the physical principles and device options as disclosed e.g. in US 6,995,445 and/or US 2007/0176165 A1.

The transversal optical sensor may further be adapted to generate the transversal sensor signal in accordance with the electrical currents through the partial electrodes. Thus, a ratio of electric currents through two horizontal partial electrodes may be formed, thereby generating an x-coordinate, and/or a ratio of electric currents through two vertical partial electrodes may be formed, thereby generating a y-coordinate. The detector, preferably the transversal optical sensor and/or the evaluation device, may be adapted to derive the information on the transversal position of the object from at least one ratio of the currents through the partial electrodes. Other ways of generating position coordinates by comparing currents through the partial electrodes are feasible.

The partial electrodes generally may be defined in various ways, in order to determine a position of the light beam in the sensor area. Thus, two or more horizontal partial electrodes may be provided in order to determine a horizontal coordinate or x-coordinate, and two or more vertical partial electrodes may be provided in order to determine a vertical coordinate or y-coordinate. Thus, the partial electrodes may be provided at a rim of the sensor area, wherein an interior space of the sensor area remains free and may be covered by one or more additional electrode materials. As will be outlined in further detail below, the additional electrode material preferably may be a transparent additional electrode material, such as a transparent metal and/or a transparent conductive oxide and/or, most preferably, a transparent conductive polymer.

Further preferred embodiments may refer to the photovoltaic material. Thus, the photovoltaic material of the transversal optical sensor may comprise at least one organic photovoltaic material. Thus, generally, the transversal optical sensor may be an organic photo detector.

Preferably, the organic photo detector may be a dye-sensitized solar cell. The dye-sensitized solar cell preferably may be a solid dye-sensitized solar cell, comprising a layer setup embedded in between the first electrode and the second electrode, the layer setup comprising at least one n-semiconducting metal oxide, at least one dye, and at least one solid p-semiconducting organic material. Further details and optional embodiments of the dye-sensitized solar cell (DSC) will be disclosed below.

The at least one first electrode of the transversal optical sensor preferably is transparent. As used in the present invention, the term transparent generally refers to the fact that the intensity of light after transmission through the transparent object equals to or exceeds 10%, preferably 40% and, more preferably, 60% of the intensity of light before transmission through the transparent object. More preferably, the at least one first electrode of the transversal optical sensor may fully or partially be made of at least one transparent conductive oxide (TCO). As an example, indium-doped tin oxide (ITO) and/or fluorine-doped tin oxide (FTO) may be named. Further examples will be given below.

Further, the at least one second electrode of the transversal optical sensor preferably may fully or partially be transparent. Thus, specifically, the at least one second electrode may comprise

two or more partial electrodes and at least one additional electrode material contacting the two or more partial electrodes. The two or more partial electrodes may be intransparent. As an example, the two or more partial electrodes may fully or partially be made of a metal. Thus, the two or more partial electrodes preferably are located at a rim of the sensor area. The two or more partial electrodes, however, may electrically be connected by the at least one additional electrode material which, preferably, is transparent. Thus, the second electrode may comprise an intransparent rim having the two or more partial electrodes and a transparent inner area having the at least one transparent additional electrode material. More preferably, the at least one second electrode of the transversal optical sensor, such as the above-mentioned at least one additional electrode material, may fully or partially be made of at least one conductive polymer, preferably a transparent conductive polymer. As an example, conductive polymers having an electrical conductivity of at least 0.01 S/cm may be used, preferably of at least 0.1 S/cm or, more preferably, of at least 1 S/cm or even at least 10 S/cm or at least 100 S/cm. As an example, the at least one conductive polymer may be selected from the group consisting of: a poly-3,4-ethylenedioxythiophene (PEDOT), preferably PEDOT being electrically doped with at least one counter ion, more preferably PEDOT doped with sodium polystyrene sulfonate (PEDOT:PSS); a polyaniline (PANI); a polythiophene.

As outlined above, the conductive polymer may provide an electrical connection between the at least two partial electrodes. The conductive polymer may provide an Ohmic resistivity, allowing for determining the position of charge generation. Preferably, the conductive polymer provides an electric resistivity of 0.1 – 20 k Ω between the partial electrodes, preferably an electric resistivity of 0.5 – 5.0 k Ω and, more preferably, an electric resistivity of 1.0 – 3.0 k Ω .

Generally, as used herein, a conductive material may be a material which have a specific electrical resistance of less than 10⁴, less than 10³, less than 10², or of less than 10 Ω m. Preferably, the conductive material has a specific electrical resistance of less than 10⁻¹, less than 10⁻², less than 10⁻³, less than 10⁻⁵, or less than 10⁻⁶ Ω m. Most preferably, the specific electrical resistance of the conductive material is less than 5 x 10⁻⁷ Ω m or is less than 1 x 10⁻⁷ Ω m, particularly in the range of the specific electrical resistance of aluminum.

As outlined above, preferably, at least one of the transversal optical sensor and the longitudinal optical sensor is a transparent optical sensor. Thus, the at least one transversal optical sensor may be a transparent transversal optical sensor and/or may comprise at least one transparent transversal optical sensor. Additionally or alternatively, the at least one longitudinal optical sensor may be a transparent longitudinal optical sensor and/or may comprise at least one transparent longitudinal optical sensor. In case a plurality of longitudinal optical sensors is provided, such as a stack of longitudinal optical sensors, preferably all longitudinal optical sensors of the plurality and/or the stack or all longitudinal optical sensors of the plurality and/or the stack but one longitudinal optical sensor are transparent. As an example, in case a stack of longitudinal optical sensors is provided, wherein the longitudinal optical sensors are arranged along the optical axis of the detector, preferably all longitudinal optical sensors but the last

longitudinal optical sensor facing away from the object may be transparent longitudinal optical sensors. The last longitudinal optical sensor, i.e. the longitudinal optical sensor on the side of the stack facing away from the object, may be a transparent longitudinal optical sensor or an intransparent longitudinal optical sensor. Exemplary embodiments will be given below.

5 In case one of the transversal optical sensor and the longitudinal optical sensor is a transparent optical sensor or comprises at least one transparent optical sensor, the light beam may pass through the transparent optical sensor before impinging on the other one of the transversal optical sensor and the longitudinal optical sensor. Thus, the light beam from the object may
10 subsequently reach the transversal optical sensor and the longitudinal optical sensor or vice versa.

Further embodiments refer to the relationship between the transversal optical sensor and the longitudinal optical sensor. Thus, in principle, the transversal optical sensor and the longitudinal
15 optical sensor at least partially may be identical, as outlined above. Preferably, however, the transversal optical sensor and the longitudinal optical sensor at least partially may be independent optical sensors, such as independent photo detectors and, more preferably, independent DSCs or sDSCs.

20 As outlined above, the transversal optical sensor and the longitudinal optical sensor preferably may be stacked along the optical axis. Thus, a light beam travelling along the optical axis may both impinge on the transversal optical sensor and on the longitudinal optical sensor, preferably subsequently. Thus, the light beam may subsequently pass through the transversal optical sensor and the longitudinal optical sensor or vice versa.

25 Further embodiments of the present invention referred to the nature of the light beam which propagates from the object to the detector. The light beam might be admitted by the object itself, i.e. might originate from the object. Additionally or alternatively, another origin of the light beam is feasible. Thus, as will be outlined in further detail below, one or more illumination
30 sources might be provided which illuminate the object, such as by using one or more primary rays or beams, such as one or more primary rays or beams having a predetermined characteristic. In the latter case, the light beam propagating from the object to the detector might be a light beam which is reflected by the object and/or a reflection device connected to the object.

35 As outlined above, the at least one longitudinal sensor signal, given the same total power of the illumination by the light beam, is dependent on a beam cross-section of the light beam in the sensor region of the at least one longitudinal optical sensor. As used herein, the term beam cross-section generally refers to a lateral extension of the light beam or a light spot generated
40 by the light beam at a specific location. In case a circular light spot is generated, a radius, a diameter or a Gaussian beam waist or twice the Gaussian beam waist may function as a measure of the beam cross-section. In case non-circular light-spots are generated, the cross-

section may be determined in any other feasible way, such as by determining the cross-section of a circle having the same area as the non-circular light spot, which is also referred to as the equivalent beam cross-section.

5 Thus, given the same total power of the illumination of the sensor region by the light beam, a light beam having a first beam diameter or beam cross-section may generate a first longitudinal sensor signal, whereas a light beam having a second beam diameter or beam-cross section being different from the first beam diameter or beam cross-section generates a second longitudinal sensor signal being different from the first longitudinal sensor signal. Thus, by
10 comparing the longitudinal sensor signals, an information or at least one item of information on the beam cross-section, specifically on the beam diameter, may be generated. For details of this effect, reference may be made to WO 2012/110924 A1. Specifically in case one or more beam properties of the light beam propagating from the object to the detector are known, the at least one item of information on the longitudinal position of the object may thus be derived from
15 a known relationship between the at least one longitudinal sensor signal and a longitudinal position of the object. The known relationship may be stored in the evaluation device as an algorithm and/or as one or more calibration curves. As an example, specifically for Gaussian beams, a relationship between a beam diameter or beam waist and a position of the object may easily be derived by using the Gaussian relationship between the beam waist and a longitudinal
20 coordinate.

The above-mentioned effect, which is also referred to as the FiP-effect (alluding to the effect that the beam cross section ϕ influences the electric power P generated by the longitudinal optical sensor) may depend on or may be emphasized by an appropriate modulation of the light
25 beam, as disclosed in WO 2012/110924 A1. Thus, preferably, the detector may furthermore have at least one modulation device for modulating the illumination. The detector may be designed to detect at least two longitudinal sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies. In this case, the evaluation device may be designed to generate the at least one item of information on
30 the longitudinal position of the object by evaluating the at least two longitudinal sensor signals.

Generally, the longitudinal optical sensor may be designed in such a way that the at least one longitudinal sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination. Further details and exemplary
35 embodiments will be given below. This property of frequency dependency is specifically provided in DSCs and, more preferably, in sDSCs. However, other types of optical sensors, preferably photo detectors and, more preferably, organic photo detectors may exhibit this effect.

Preferably, the transversal optical sensor and the longitudinal optical sensor both are thin film
40 devices, having a layer setup of layer including electrode and photovoltaic material, the layer setup having a thickness of preferably no more than 1 mm, more preferably of at most 500 μm or even less. Thus, the sensor region of the transversal optical sensor and/or the sensor region

of the longitudinal optical sensor preferably each may be or may comprise a sensor area, which may be formed by a surface of the respective device, wherein the surface may face towards the object or may face away from the object. Hereby, it may further be feasible to arrange the at least one transversal optical sensor and the at least one longitudinal optical sensor in a way that some surfaces comprising the sensor regions may face towards the object where other surfaces may face away from the object. Such a kind of arrangement of the respective devices, which might be helpful to optimize the path of the light beam through the stack and/or to reduce reflections within the light path, may, for any reason or purpose, be implemented in an alternating manner, such as with one, two, three or more devices where the sensor regions may face towards the object alternating with one, two, three or more other devices where the sensor regions may face away from the object.

Preferably, the sensor region of the transversal optical sensor and/or the sensor region of the longitudinal optical sensor may be formed by one continuous sensor region, such as one continuous sensor area or sensor surface per device. Thus, preferably, the sensor region of the longitudinal optical sensor or, in case a plurality of longitudinal optical sensors is provided (such as a stack of longitudinal optical sensors), each sensor region of the longitudinal optical sensor, may be formed by exactly one continuous sensor region. The longitudinal sensor signal preferably is a uniform sensor signal for the entire sensor region of the longitudinal optical sensor or, in case a plurality of longitudinal optical sensors is provided, is a uniform sensor signal for each sensor region of each longitudinal optical sensor.

The at least one transversal optical sensor and/or the at least one longitudinal optical sensor each, independently, may have a sensor region providing a sensitive area, also referred to as a sensor area, of at least 1 mm², preferably of at least 5 mm², such as a sensor area of 5 mm² to 1000 cm², preferably a sensor area of 7 mm² to 100 cm², more preferably a sensor area of 1 cm². The sensor area preferably has a rectangular geometry, such as a square geometry. However, other geometries and/or sensor areas are feasible.

The longitudinal sensor signal preferably may be selected from the group consisting of a current (such as a photocurrent) and a voltage (such as a photo voltage). Similarly, the transversal sensor signal preferably may be selected from the group consisting of a current (such as a photocurrent) and a voltage (such as a photo voltage) or any signal derived thereof, such as a quotient of currents and/or voltages. Further, longitudinal sensor signal and/or the transversal sensor signal may be preprocessed, in order to derive refined sensor signals from raw sensor signals, such as by averaging and/or filtering.

Generally, the longitudinal optical sensor may comprise at least one semiconductor detector, in particular an organic semiconductor detector comprising at least one organic material, preferably an organic solar cell and particularly preferably a dye solar cell or dye-sensitized solar cell, in particular a solid dye solar cell or a solid dye-sensitized solar cell. Preferably, the longitudinal optical sensor is or comprises a DSC or sDSC. Thus, preferably, the longitudinal

optical sensor comprises at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode. In a preferred embodiment, the longitudinal optical sensor comprises at least one DSC or, more preferably, at least one sDSC. As outlined above, preferably, the at least one longitudinal optical sensor is a transparent longitudinal optical sensor or comprises at least one transparent longitudinal optical sensor. Thus, preferably, both the first electrode and the second electrode are transparent or, in case a plurality of longitudinal optical sensors is provided, at least one of the longitudinal optical sensors is designed such that both the first electrode and the second electrode are transparent.

As outlined above, in case a stack of longitudinal optical sensors is provided, preferably some or even all longitudinal optical sensors of the stack are transparent but the last longitudinal optical sensor of the stack. The last longitudinal optical sensor of the stack, i.e. the longitudinal optical sensor of the stack furthest away from the object, may be transparent or intransparent.

The stack may, besides the at least one transversal optical sensor and the at least one longitudinal optical sensor, comprise one or more further optical sensors which may function as one or more of a transversal optical sensor, a longitudinal optical sensor (also referred to as an imaging device) and an imaging sensor.

Thus, it may be possible to locate an imaging device in the optical path of the light beam in a manner that the light beam travels through the stack of the transparent longitudinal optical sensors until it impinges on the imaging device.

Thus, generally, the detector may further comprise at least one imaging device, i.e. a device capable of acquiring at least one image. The imaging device can be embodied in various ways. Thus, the imaging device can be for example part of the detector in a detector housing. Alternatively or additionally, however, the imaging device can also be arranged outside the detector housing, for example as a separate imaging device. Alternatively or additionally, the imaging device can also be connected to the detector or even be part of the detector. In a preferred arrangement, the stack of the transparent longitudinal optical sensors and the imaging device are aligned along a common optical axis along which the light beam travels. However, other arrangements are possible.

In addition, the detector may comprise at least one transfer device, such as an optical lens, which will be described later in more detail, and which may further be arranged along the common optical axis. By way of example, the light beam which emerges from the object may in this case travel first through the at least one transfer device and thereafter through the stack of the transparent longitudinal optical sensors until it finally impinges on the imaging device.

As used herein, an imaging device is generally understood as a device which can generate a one-dimensional, a two-dimensional, or a three-dimensional image of the object or of a part thereof. In particular, the detector, with or without the at least one optional imaging device, can

be completely or partly used as a camera, such as an IR camera, or an RGB camera, i.e. a camera which is designed to deliver three basic colors which are designated as red, green, and blue, on three separate connections. Thus, as an example, the at least one imaging device may be or may comprise at least one imaging device selected from the group consisting of: a
5 pixelated organic camera element, preferably a pixelated organic camera chip; a pixelated inorganic camera element, preferably a pixelated inorganic camera chip, more preferably a CCD- or CMOS-chip; a monochrome camera element, preferably a monochrome camera chip; a multicolor camera element, preferably a multicolor camera chip; a full-color camera element, preferably a full-color camera chip. The imaging device may be or may comprise at least one
10 device selected from the group consisting of a monochrome imaging device, a multi-chrome imaging device and at least one full color imaging device. A multi-chrome imaging device and/or a full color imaging device may be generated by using filter techniques and/or by using intrinsic color sensitivity or other techniques, as the skilled person will recognize. Other embodiments of the imaging device are also possible.

15 The imaging device may be designed to image a plurality of partial regions of the object successively and/or simultaneously. By way of example, a partial region of the object can be a one-dimensional, a two-dimensional, or a three-dimensional region of the object which is delimited for example by a resolution limit of the imaging device and from which electromagnetic
20 radiation emerges. In this context, imaging should be understood to mean that the electromagnetic radiation which emerges from the respective partial region of the object is fed into the imaging device, for example by means of the at least one optional transfer device of the detector. The electromagnetic rays can be generated by the object itself, for example in the form of a luminescent radiation. Alternatively or additionally, the at least one detector may
25 comprise at least one illumination source for illuminating the object.

In particular, the imaging device can be designed to image sequentially, for example by means of a scanning method, in particular using at least one row scan and/or line scan, the plurality of partial regions sequentially. However, other embodiments are also possible, for example
30 embodiments in which a plurality of partial regions is simultaneously imaged. The imaging device is designed to generate, during this imaging of the partial regions of the object, signals, preferably electronic signals, associated with the partial regions. The signal may be an analogue and/or a digital signal. By way of example, an electronic signal can be associated with each partial region. The electronic signals can accordingly be generated simultaneously or else
35 in a temporally staggered manner. By way of example, during a row scan or line scan, it is possible to generate a sequence of electronic signals which correspond to the partial regions of the object, which are strung together in a line, for example. Further, the imaging device may comprise one or more signal processing devices, such as one or more filters and/or analogue-digital-converters for processing and/or preprocessing the electronic signals.

40 As outlined above, the at least one longitudinal optical sensor may be transparent or intransparent, or may comprise at least one transparent longitudinal optical sensor. A

combination of at least one transparent and at least one intransparent longitudinal optical sensor is possible.

5 In a further preferred embodiment, the last longitudinal optical sensor may be intransparent. For this purpose, at least those parts of the last longitudinal optical sensor which are subject to be illuminated by the light beam which travels from the object and which may impinge on the last longitudinal optical sensor may comprise an optical sensor material, preferably an inorganic optical sensor material, and/or an organic optical sensor material, and/or a hybrid organic – inorganic optical sensor material, which exhibits intransparent optical properties. Intransparency
10 may also be achieved by using at least one intransparent electrode. In this embodiment, the last longitudinal optical sensor may be designed such that its electrode facing towards the object is transparent, whereas its electrode facing away from the object may be intransparent, or vice-versa. Additionally or alternatively, a respective material which exhibits intransparent optical properties may be selected for at least one of the n-semiconducting metal oxides, for at least
15 one of the dyes, and/or for at least one of the p-semiconducting organic materials which may comprise the last longitudinal optical sensor.

As outlined above, the detector may comprise at least one imaging device. The imaging device may fully or partially be embodied as an independent imaging device, independent from the at
20 least one transversal optical sensor and the at least one longitudinal optical sensor. Additionally or alternatively, the at least one optional imaging device may fully or partially be integrated into one or both of the at least one transversal optical sensor and the at least one longitudinal optical sensor. Thus, as an example, the imaging device may be used for determining a transversal position of a light spot and, thus, may be used as a transversal optical sensor or as a part
25 thereof.

As outlined above, the detector may comprise a stack of at least two optical sensors, the at least two optical sensors comprising the at least one transversal optical sensor and the at least one longitudinal optical sensor and, optionally, the at least one imaging device. Thus, as an
30 example, the stack may comprise the at least one transversal optical sensor, the at least one longitudinal optical sensor (preferably at least one transparent longitudinal optical sensor) and, optionally, in a position furthest away from the object, at least one imaging device, preferably at least one intransparent imaging device such as a CCD or CMOS chip.

35 The stack of at least two optical sensors optionally may partially or fully be immersed in an oil, in a liquid and/or in a solid material in order to avoid and/or decrease reflections at interfaces. Hereby, the oil, the liquid, and/or the solid material may preferably be transparent, preferentially to a high degree, at least over a part of the ultraviolet, visible, and/or infrared spectral range. In a preferred embodiment, the solid material may be generated by inserting at least one curable
40 substance into a region between at least two optical sensors and treating the curable substance with a treatment, such as by incident light, particularly with light within the ultraviolet range, and/or by an application of a temperature above or below room temperature, by which treatment

the curable substance might be cured, preferentially by hardening the curable substance into the solid material. Alternatively, at least two different curable substances may be inserted into a region between at least two optical sensors, whereby the two different curable substances are selected in a manner that they begin to set into the solid material with or without the treatment as indicated above. However, further treatments and/or other procedures of providing the transparent solid material may be possible. Thus, at least one of the optical sensors of the stack may fully or partially be immersed in the oil and/or the liquid and/or covered with the solid material.

- 10 Alternatively or additionally, the region between the at least two optical sensors may be partially or fully filled with a substance, such as the oil, the liquid and/or the solid material. Hereby, the substance may preferably exhibit a refractive index with a value which may differ from that of the optical sensors adjoining to the substance on one or both sides of the region. However, inserting the additional substance in the regions may require the optical sensors within the stack to observe a minimum spacing between them.

- 15 In case a stack of at least two optical sensors is used, the last optical sensor of the stack may be transparent or intransparent. Thus, an intransparent inorganic optical sensor may be used in a position furthest away from the object. As an example, the last optical sensor of the stack may be or may comprise the at least one optional imaging device, such as at least one CCD or CMOS chip, preferably a full-color CCD or CMOS chip.

- 20 Thus, the intransparent last optical sensor may be used as an imaging device, wherein the imaging device is impinged by the light beam after the light beam has traveled before through the stack of the transparent optical sensors until it impinges the imaging device. In particular, the imaging device can be completely or partly used as a camera, such as an IR camera, or an RGB camera, as described above. Hereby, the intransparent last optical sensor can be embodied in various ways as imaging device. Thus, the intransparent last optical sensor can be for example part of the detector in a detector housing. Alternatively or additionally, however, the intransparent last optical sensor can also be arranged outside the detector housing, for example as a separate imaging device.

- 25 The stack comprising the at least one transversal optical sensor, the at least one longitudinal optical sensor and the optional at least one imaging device may be designed such that the elements of the stack are arranged along an optical axis of the detector. The last element of the stack may be an intransparent optical sensor, preferably selected from the group consisting of an intransparent transversal optical sensor, an intransparent longitudinal optical sensor and an intransparent imaging device such as an intransparent CCD or CMOS chip.

- 30 In a preferred arrangement, the stack comprising the at least one transversal optical sensor, the at least one longitudinal optical sensor and, optionally, the at least one imaging device may be arranged along a common optical axis of the detector, along which the light beam may travel. In

case the stack contains a plurality of optical sensors, the optical sensors comprising the at least one transversal optical sensor, the at least one longitudinal optical sensor and, optionally, the at least one imaging device, wherein at least one of the optical sensors is a transparent optical sensor and wherein at least one of the optical sensors is an intransparent optical sensor, the transparent optical sensor and the intransparent optical sensor, the latter preferably being positioned furthest away from the object, may be arranged along the optical axis of the detector. However, other arrangements are possible.

In a further preferred embodiment, the intransparent last optical sensor is having at least one matrix of pixels, wherein a 'matrix' generally refers to an arrangement of a plurality of the pixels in space, which may be a linear arrangement or an areal arrangement. Generally, the matrix may, thus, preferably be selected from the group consisting of a one-dimensional matrix and a two-dimensional matrix. As an example, the matrix may comprise 100 to 100 000 000 pixels, preferably 1 000 to 1 000 000 pixels and, more preferably, 10 000 to 500 000 pixels. Most preferably, the matrix is a rectangular matrix having pixels arranged in rows and columns.

As further used herein, a pixel generally refers to a light-sensitive element of an optical sensor, such as a minimum uniform unit of the optical sensor adapted to generate a light signal. As an example, each pixel may have a light-sensitive area of $1\ \mu\text{m}^2$ to $5\ 000\ 000\ \mu\text{m}^2$, preferably $100\ \mu\text{m}^2$ to $4\ 000\ 000\ \mu\text{m}^2$, preferably $1\ 000\ \mu\text{m}^2$ to $1\ 000\ 000\ \mu\text{m}^2$ and more preferably $2\ 500\ \mu\text{m}^2$ to $50\ 000\ \mu\text{m}^2$. Still, other embodiments are feasible. The intransparent last optical sensor may be adapted to generate at least one signal which indicates an intensity of illumination for each of the pixels. Thus, as an example, the intransparent last optical sensor may be adapted to generate at least one electronic signal for each of the pixels, whereby each signal indicates the intensity of illumination for the respective pixel. The signal may be an analogue and/or a digital signal. Further, the detector may comprise one or more signal processing devices, such as one or more filters and/or analogue-digital-converters for processing and/or preprocessing the at least one signal.

The intransparent last optical sensor having a matrix of pixels can be selected from the group consisting of: an inorganic semiconductor sensor device such as a CCD chip and/or a CMOS chip; an organic semiconductor sensor device. In the latter case, as an example, the optical sensor may, for example, comprise at least one organic photovoltaic device having a matrix of pixels. As used herein, an organic photovoltaic device generally refers to a device having at least one organic photosensitive element and/or at least one organic layer. Therein, generally, any type of organic photovoltaic device may be used, such as organic solar cells and/or an arbitrary device having at least one organic photosensitive layer. As an example, an organic solar cell and/or a dye-sensitized organic solar cell may be comprised. Further, a hybrid device may be used, such as inorganic – organic photovoltaic devices.

Further preferred embodiments refer to the evaluation device. Thus, the evaluation device may be designed to generate the at least one item of information on the longitudinal position of the

object from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object with respect to the detector, preferably taking account of a known power of the illumination and optionally taking account of a modulation frequency with which the illumination is modulated.

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In a further preferred embodiment, the detector furthermore may comprise at least one transfer device, wherein the transfer device is designed to feed light emerging from the object to the transversal optical sensor and the longitudinal optical sensor, preferably subsequently. Details and preferred embodiments will be given below.

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As outlined above, the light beam propagating from the object of the detector may originate from the object or may originate from any other source. Thus, the object itself may emit the light beam. Additionally or alternatively, the object might be illuminated by using an illumination source generating primary light, wherein the object elastically or inelastically reflects the primary light, thereby generating the light beam propagating to the detector. The illumination source itself may be part of the detector. Thus, the detector may comprise at least one illumination source. The illumination source generally may be selected from: an illumination source, which is at least partly connected to the object and/or is at least partly identical to the object; an illumination source which is designed to at least partly illuminate the object with a primary radiation, preferably primary light, wherein the light beam preferably is generated by a reflection of the primary radiation on the object and/or by light emission by the object itself, stimulated by the primary radiation.

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As outlined above, the detector preferably has a plurality of longitudinal optical sensors. More preferably, the plurality of longitudinal optical sensors is stacked, such as along the optical axis of the detector. Thus, the longitudinal optical sensors may form a longitudinal optical sensor stack. The longitudinal optical sensor stack preferably may be oriented such that the sensor regions of the longitudinal optical sensors are oriented perpendicular to the optical axis. Thus, as an example, sensor areas or sensor surfaces of the single longitudinal optical sensors may be oriented in parallel, wherein slight angular tolerances might be tolerable, such as angular tolerances of no more than 10° , preferably of no more than 5° .

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In case stacked longitudinal optical sensors are provided, the at least one transversal optical sensor preferably fully or partially is located on a side of the stacked longitudinal optical sensors facing the object. However, other embodiments are feasible. Thus, embodiments in which the at least one transversal optical sensor is fully or partially located on a side of the transversal optical sensor stack facing away from the object. Again, additionally or alternatively, embodiments are feasible in which the at least one transversal optical sensor is located fully or partially in between the longitudinal optical sensor stack.

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The longitudinal optical sensors preferably are arranged such that a light beam from the object illuminates all longitudinal optical sensors, preferably sequentially. Specifically in this case,

preferably, at least one longitudinal sensor signal is generated by each longitudinal optical sensor. This embodiment is specifically preferred since the stacked setup of the longitudinal optical sensors allows for an easy and efficient normalization of the signals, even if an overall power or intensity of the light beam is unknown. Thus, the single longitudinal sensor signals may be known to be generated by one and the same light beam. Thus, the evaluation device may be adapted to normalize the longitudinal sensor signals and to generate the information on the longitudinal position of the object independent from an intensity of the light beam. For this purpose, use may be made of the fact that, in case the single longitudinal sensor signals are generated by one and the same light beam, differences in the single longitudinal sensor signals are only due to differences in the cross-sections of the light beam at the location of the respective sensor regions of the single longitudinal optical sensors. Thus, by comparing the single longitudinal sensor signals, information on a beam cross-section may be generated even if the overall power of the light beam is unknown. From the beam cross-section, information regarding the longitudinal position of the object may be gained, specifically by making use of a known relationship between the cross-section of the light beam and the longitudinal position of the object.

Further, the above-mentioned stacking of the longitudinal optical sensors and the generation of a plurality of longitudinal sensor signals by these stacked longitudinal optical sensors may be used by the evaluation device in order to resolve an ambiguity in a known relationship between a beam cross-section of the light beam and the longitudinal position of the object. Thus, even if the beam properties of the light beam propagating from the object to the detector are known fully or partially, it is known that, in many beams, the beam cross-section narrows before reaching a focal point and, afterwards, widens again. Thus, before and often as a focal point in which the light beam has the narrowest beam cross-section, positions along the axis of propagation of the light beam occur in which the light beam has the same cross-section. Thus, as an example, at a distance z_0 before and after the focal point, the cross-section of the light beam is identical. Thus, in case only one longitudinal optical sensor is used, a specific cross-section of the light beam might be determined, in case the overall power or intensity of the light beam is known. By using this information, the distance z_0 of the respective longitudinal optical sensor from the focal point might be determined. However, in order to determine whether the respective longitudinal optical sensor is located before or behind the focal point, additional information is required, such as a history of movement of the object and/or the detector and/or information on whether the detector is located before or behind the focal point. In typical situations, this additional information may not be provided. Therefore, by using a plurality of longitudinal optical sensors, additional information may be gained in order to resolve the above-mentioned ambiguity. Thus, in case the evaluation device, by evaluating the longitudinal sensor signals, recognizes that the beam cross-section of the light beam on a first longitudinal optical sensor is larger than the beam cross-section of the light beam on a second longitudinal optical sensor, wherein the second longitudinal optical sensor is located behind the first longitudinal optical sensor, the evaluation device may determine that the light beam is still narrowing and that the location of the first longitudinal optical sensor is situated before the focal point of the

light beam. Contrarily, in case the beam cross-section of the light beam on the first longitudinal optical sensor is smaller than the beam cross-section of the light beam on the second longitudinal optical sensor, the evaluation device may determine that the light beam is widening and that the location of the second longitudinal optical sensor is situated behind the focal point.

5 Thus, generally, the evaluation device may be adapted to recognize whether the light beam widens or narrows, by comparing the longitudinal sensor signals of different longitudinal sensors.

10 In addition to the at least one longitudinal coordinate of the object, at least one transversal coordinate of the object may be determined. Thus, generally, the evaluation device may further be adapted to determine at least one transversal coordinate of the object by determining a position of the light beam on the at least one transversal optical sensor, which may be a pixelated, a segmented or a large-area transversal optical sensor, as will be outlined in further detail below.

15 Thus, in case a pixelated transversal optical sensor is used and/or in case the at least one transversal optical sensor comprises at least one pixelated optical sensor having a matrix of pixels, the evaluation device may be adapted to determine a center of illumination of the at least one matrix by the light beam, wherein the at least one transversal coordinate of the object is
20 determined by evaluating at least one coordinate of the center of illumination. Thus, the coordinate of the center of illumination may be a pixel coordinate of the center of illumination. As an example, the matrix may comprise rows and columns of pixels, wherein the row number of the light beam and/or the center of the light beam within the matrix may provide an x-coordinate, and wherein the column number of the light beam and/or the center of the light
25 beam within the matrix may provide a y-coordinate.

The detector, as outlined above, may comprise at least one stack of optical sensors, the optical sensors comprising the at least one transversal optical sensor and the at least one longitudinal optical sensor, and, optionally, the at least one imaging device. The stack of optical sensors
30 may comprise at least one longitudinal optical sensor stack, being a stack of longitudinal optical sensors, having at least two longitudinal optical sensors in a stacked fashion. The longitudinal optical sensor stack preferably may comprise at least three longitudinal optical sensors, more preferably at least four longitudinal optical sensors, even more preferably at least five longitudinal optical sensors or even at least six longitudinal optical sensors. By tracking the
35 longitudinal sensor signals of the longitudinal optical sensors, even a beam profile of the light beam might be evaluated.

In case a plurality of the longitudinal optical sensors is used, wherein the plurality of the optical sensors may be arranged in the stacked fashion and/or in another arrangement, the longitudinal
40 optical sensors may have identical spectral sensitivities or may provide different spectral sensitivities. Thus, as an example, at least two of the longitudinal optical sensors may have a differing spectral sensitivity. As used herein, the term spectral sensitivity generally refers to the

fact that the sensor signal of the optical sensor, for the same power of the light beam, may vary with the wavelength of the light beam. Thus, generally, at least two of the optical sensors may differ with regard to their spectral properties. This embodiment generally may be achieved by using different types of absorbing materials for the optical sensors, such as different types of dyes or other absorbing materials.

Preferably, the at least one transversal optical sensor uses at least one transparent substrate. Similarly, preferably, the at least one longitudinal optical sensor uses at least one transparent substrate. In case a plurality of longitudinal optical sensors is used, such as a stack of longitudinal optical sensors, preferably, at least one of these longitudinal optical sensors uses a transparent substrate. Herein, the substrates employed for the plurality of optical sensors may exhibit identical properties or may differ from each other, in particular, with regard to a geometrical quantity and/or a material quantity related to the substrates, such as the thickness, the shape, and/or the refractive index of each substrate. Thus, identical planar glass plates may be used for the plurality of optical sensors within the stack. On the other hand, different substrates may be employed for some optical sensors or for each optical sensor within a plurality of optical sensors, in particular, for a purpose of optimizing the light path within the stack, especially for guiding the light path along regions on the optical axis which might particularly be suited for exploiting the FiP-effect as described elsewhere in this application. Within this regard, the thickness of some substrates or of each substrate, which may be defined by the light path as traversed by the light beam travelling through the respective substrate, may thus be varied, in particular, for reducing or else increasing or even maximizing reflections of the light beam.

In addition or alternatively, the substrates employed for the plurality of optical sensors may differ by exhibiting a different shape which may be selected from the group comprising a planar, a planar-convex, a planar-concave, a biconvex, a biconcave or any other form which may be employed for optical purposes, such as lenses or prisms. Herein, the substrates may be rigid or else flexible. Suitable substrates are, as well as metal foils, in particular plastic sheets or films and especially glass sheets or glass films. Shape-changing materials, such as shape-changing polymers, constitute an example of materials which may preferentially be employed as flexible substrates. Furthermore, the substrate may be covered or coated, in particular, for the purpose of reducing and/or modifying reflections of the incident light beam. As an example, the substrate may be shaped in a manner that it might exhibit a mirror effect, such as that of a dichroic mirror, which might particularly be useful in a setup where a splitting of the optical axis behind the substrate may be required for any purpose.

Additionally or alternatively, differing spectral properties of the optical sensors may be generated by other means implemented into the optical sensors and/or into the detector, such as by using one or more wavelength-selective elements, such as one or more filters (such as color filters) in front of the optical sensors and/or by using one or more prisms and/or by using one or more dichroitic mirrors and/or by using one or more color conversion elements. Thus, in

case a plurality of longitudinal optical sensors is provided, at least one of the longitudinal optical sensors may comprise a wavelength-selective element such as a color filter, having a specific transmission or reflection characteristic, thereby generating differing spectral properties of the optical sensors. Further, the longitudinal optical sensors may all be organic optical sensors, may all be inorganic optical sensors, may all be hybrid organic – inorganic optical sensors or may comprise an arbitrary combination of at least two optical sensors selected from the group consisting of organic optical sensors, inorganic optical sensors and hybrid organic – inorganic optical sensors.

In case the plurality of the longitudinal optical sensors is used, wherein at least two of the longitudinal optical sensors differ with regard to their respective spectral sensitivity, the evaluation device generally may be adapted to determine a color of the light beam by comparing sensor signals of the longitudinal optical sensors having the differing spectral sensitivity. As used herein, the expression “determine a color” generally refers to the step of generating at least one item of spectral information about the light beam. The at least one item of spectral information may be selected from the group consisting of a wavelength, specifically a peak wavelength; color coordinates, such as CIE coordinates. As further used herein, a “color” of the light beam generally refers to a spectral composition of the light beam. Specifically, the color of the light beam may be given in any arbitrary color coordinate system and/or in spectral units such as by giving a wavelength of a dominant peak of a spectrum of the light. Other embodiments are feasible. In case the light beam is a narrow-band light beam such as a laser light beam and/or a light beam generated by a semiconductor device such as a light-emitting diode, the peak wavelength of the light beam may be given to characterize the color of the light beam. The determination of the color of the light beam may be performed in various ways which are generally known to the skilled person. Thus, the spectral sensitivities of the longitudinal optical sensors may span a coordinate system in color space, and the signals provided by the optical sensors may provide a coordinate in this color space, as known to the skilled person for example from the way of determining CIE coordinates. As an example, the detector may comprise two, three or more longitudinal optical sensors in a stack. Thereof, at least two, preferably at least three, of the optical sensors may have differing spectral sensitivities, whereby three different longitudinal optical sensors with maximum absorption wavelengths in a spectral range between 600 nm and 780 nm (red), between 490 nm and 600 nm (green), and between 380 nm and 490 nm (blue) are generally preferred. Further, the evaluation device may be adapted to generate at least one item of color information for the light beam by evaluating the signals of the longitudinal optical sensors having differing spectral sensitivities.

The evaluation device may be adapted to generate at least two color coordinates, preferably at least three color coordinates, wherein each of the color coordinates is determined by dividing a signal of one of the spectrally sensitive optical sensors by a normalization value. As an example, the normalization value may contain a sum of the signals of all spectrally sensitive optical sensors. Additionally or alternatively, the normalization value may contain a detector signal of a white detector. The at least one item of color information may contain the color

coordinates. The at least one item of color information may, as an example, contain CIE coordinates.

Furthermore, in addition to the preferred at least two, more preferably at least three, spectrally sensitive longitudinal optical sensors, wherein at least two of the optical sensors differ with regard to their respective spectral sensitivity, particularly wherein the spectral sensitivities of the at least two longitudinal optical sensors may span a coordinate system in color space, the stack may comprise a last longitudinal optical sensor, wherein the last longitudinal optical sensor may be intransparent. An intransparency of the last longitudinal optical sensor may be achieved by selecting an optically intransparent material from the group consisting of: organic optical sensor materials, inorganic optical sensor materials and hybrid organic – inorganic optical sensor materials.

In a preferred embodiment, the intransparent last longitudinal optical sensor may be configured to exhibit an absorption spectrum which either does not vary at all or varies only to minor extent over the spectral range of the at least two different optical sensors. Whereas each of the at least two different optical sensors displays a specific spectral sensitivity which may, as described above, vary considerably over their spectral range allowing them to be sensitive to a specific color, the intransparent last longitudinal optical sensor may, thus, absorb substantially all colors over the spectral ranges of the at least two of the longitudinal optical sensors have a differing spectral sensitivity. By this property, the last longitudinal optical sensor may be described as a white detector, wherein the white detector may be adapted to absorb light in an absorption range covering the visible spectral range. This arrangement has the advantage that, irrespective of the specific color, each beam which propagates through the at least two different optical sensors until it impinges the intransparent last longitudinal optical sensor, will be recorded by at least two different optical sensors, i.e. by at least one of the at least two different optical sensors which is sensitive to the color as the first longitudinal optical sensor and by the intransparent last longitudinal optical sensor as the second longitudinal optical sensor. As described above, by recording at least one object by both the first longitudinal optical sensor and the second longitudinal optical sensor it will be possible to resolve the ambiguity in the known relationship between the beam cross-section of the light beam and the longitudinal position of the at least one object, wherein the resolving of the ambiguity may work separately for each color which may be recorded. In addition, it may be advantageous to set a first signal recorded by the first longitudinal optical sensor in relationship to a second signal recorded by the last or second longitudinal optical sensor, such as by calculating a quotient or another related relationship. A result obtained by creating such a relationship may particularly facilitate the recognition of a certain color by this arrangement.

As outlined above, the intransparent last longitudinal optical sensor may be a large-area sensor having a single sensitive area or may comprise at least one matrix of pixels, i.e. may be a pixelated optical sensor or imaging sensor. In a further preferred embodiment, the pixels themselves may be equipped with different spectral sensitivity which may allow them to become

sensitive to a specific color. Hereby, the specific colors may be distributed over the area of the last longitudinal optical sensor in any fashion, for example in a random fashion. However, an arrangement is preferred wherein pixels with sensitivity to a specific color are placed in an alternating fashion over the area of the intransparent last longitudinal optical sensor. By way of example, two, three, or four pixels with different sensitivity, e.g. three pixels with a specific sensitivity to red, green, and blue, alternate each other in a one-dimensional or, preferably, in a two-dimensional manner, over the area of the last longitudinal optical sensor. This arrangement may further be interrupted by white pixels, i.e. pixels which exhibit a spectral sensitivity over a large spectral range, e.g. over the visible spectrum, or over a spectrum which may include infrared or ultraviolet shares.

In a further preferred embodiment, the detector may be adapted, such as by using at least one optically sensitive element. As further used herein, the "optically sensitive element" may be considered as an arbitrary optical element which may be sensitive to a specific value of an optical property or to a specific range of values of the optical property of the light beam which may impinge on the optically sensitive element in a manner that the specific value of the optical property or to the specific range of values of the optical property is preferred with regard to other values of the light beam. The optical property as employed here may be selected from the group referring to the light beam and consisting of: a wavelength, a phase, and a polarization. Thus, the optically sensitive element may be designated as a wavelength-sensitive element, a phase-sensitive element, and/or a polarization-sensitive element, respectively. Examples for wavelength-sensitive elements may comprise one or more of a prism, a grating, a dichroitic mirror, a color wheel, or a color drum.

By using the optically sensitive element, the light beam may be influenced by an optical effect related to the at least one optical property immediately before an interaction, during the interaction and/or immediately after the interaction of the light beam with the optically sensitive element. Consequently, an expression like "passing through the optically sensitive element" may refer to a time period during which the light beam may interact with the respective element. In general, the optically sensitive element may induce the optical effect on the light beam impinging on the optically sensitive element, such as by altering a transmission or a reflectance of the impinging light beam.

Within this particular embodiment, for sequentially detecting detector signals for the different optical properties, such as different colors, different phases, and/or different polarizations, the at least one optically sensitive element may be adapted to sequentially influence the light beam. As an example for a sequential process, a rotating filter wheel may be used having filter segments of different transmission properties, for periodically influencing the light beam. Thus, each cycle of rotation of the filter wheel may be split into time segments, wherein each segment may correspond to a different optical property, such as to a different color, a different phase and/or a different polarization. With regard to a filter wheel wherein each segment corresponds to a different polarization, the filter wheel may preferably exhibit an elliptical polarization, in

particular a circular polarization, which may be changing along the circumference of the filter wheel, for example in a discrete or in a continuous manner. However, other embodiments, such as filter wheels which may employ different orientations of a linear polarization within different segments, may be feasible.

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In general, at least one optical sensor may be placed behind the filter wheel, in order to generate at least one combined detector signal. By evaluating the at least one combined detector signal in a time-resolved fashion, such as by using a phase-sensitive detection, the combined detector signal may be split into partial detector signals corresponding to the different
10 time segments and, thus, corresponding to the different colors of the light beam. Thereby, detector signals for each color, each phase and/or each polarization may be generated, which may correspond to the light beams impinging on the detector. Collecting the data for different optical properties, i.e. different colors or phases or polarizations, from the broadly absorbing stack of optical sensors, may result in acquiring the overall distribution. Consequently, the use
15 of a filter wheel may, therefore, allow determining simultaneously the respective optical property, such as color, phase, or polarization, and intensity and depth by using the detector according to the present invention, thereby avoiding the previous necessity to employ solar cells exhibiting different absorption spectra.

20 Generally, as outlined above, the evaluation device may be adapted to generate the at least one item of information on the longitudinal position of the object by determining a diameter of the light beam from the at least one longitudinal sensor signal. As used herein and as used in the following, the diameter of the light beam or, equivalently, a beam waist of the light beam might be used to characterize the beam cross-section of the light beam at a specific location. As
25 outlined above, a known relationship might be used between the longitudinal position of the object and the beam cross-section in order to determine the longitudinal position of the object by evaluating the at least one longitudinal sensor signal. As an example, as outlined above, a Gaussian relationship might be used, assuming that the light beam propagates at least approximately in a Gaussian Manner. For this purpose, the light beam might be shaped
30 appropriately, such as by using an illumination source generating a light beam having known propagation properties, such as a known Gaussian profile. For this purpose, the illumination source itself may generate the light beam having the known properties, which, for example, is the case for many types of lasers, as the skilled person knows. Additionally or alternatively, the illumination source and/or the detector may have one or more beam-shaping elements, such as
35 one or more lenses and/or one or more diaphragms, in order to provide a light beam having known properties, as the skilled person will recognize. Thus, as an example, one or more transfer elements may be provided, such as one or more transfer elements having known beam-shaping properties. Additionally or alternatively, the illumination source and/or the detector, such as the at least one optional transfer element, may have one or more wavelength-
40 selective elements, such as one or more filters, such as one or more filter elements for filtering out wavelengths outside an excitation maximum of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor.

Thus, generally, the evaluation device may be adapted to compare the beam cross-section and/or the diameter of the light beam with known beam properties of the light beam in order to determine the at least one item of information on the longitudinal position of the object,
5 preferably from a known dependency of a beam diameter of the light beam on at least one propagation coordinate in a direction of propagation of the light beam and/or from a known Gaussian profile of the light beam.

In a specific embodiment, the evaluation device is adapted to determine a number N of pixels of
10 at least one pixelated optical sensor of the detector, such as of the last longitudinal optical sensor, which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam. Therefore, the evaluation device may be adapted to compare, for each of the pixels, the signal to at least one threshold in order to determine
15 whether the pixel is an illuminated pixel or not. This at least one threshold may be an individual threshold for each of the pixels or may be a threshold which is a uniform threshold for the whole matrix. In case a plurality of optical sensors is provided, at least one threshold may be provided for each of the optical sensors and/or for a group comprising at least two of the optical sensors, wherein, for two optical sensors, their respective thresholds may be identical or different. Thus,
20 for each of the optical sensors, an individual threshold may be provided. The threshold may be predetermined and/or fixed. Alternatively, the at least one threshold may be variable. Thus, the at least one threshold may be determined individually for each measurement or groups of measurements. Thus, at least one algorithm may be provided adapted to determine the threshold.

25 The evaluation device may generally be adapted to determine at least one pixel having the highest illumination out of the pixels by comparing the signals of the pixels. Thus, the detector generally may be adapted to determine one or more pixels and/or an area or region of the matrix having the highest intensity of the illumination by the light beam. As an example, in this
30 way, a center of illumination by the light beam may be determined. The highest illumination and/or the information about the at least one area or region of highest illumination may be used in various ways. Thus, as outlined above, the at least one above-mentioned threshold may be a variable threshold. As an example, the evaluation device may be adapted to choose the above-mentioned at least one threshold as a fraction of the signal of the at least one pixel having the
35 highest illumination. Thus, the evaluation device may be adapted to choose the threshold by multiplying the signal of the at least one pixel having the highest illumination with a factor of $1/e^2$. This option is particularly preferred in case Gaussian propagation properties are assumed for the at least one light beam, since the threshold $1/e^2$ generally determines the borders of a light spot having a beam radius or beam waist generated by a Gaussian light beam on the
40 optical sensor.

A further aspect of the present invention makes use of at least two detectors according to the present invention, wherein each of such detectors may be selected as of at least one detector according to one or more of the embodiments disclosed above or disclosed in further detail below. Thus, for optional embodiments of the method, reference might be made to the
5 respective embodiments of the detector.

In a preferred embodiment, the at least one object might be illuminated by using at least one illumination source which generates primary light, wherein the at least one object elastically or inelastically reflects the primary light, thereby generating a plurality of light beams which
10 propagate to one of the at least two detectors. The at least one illumination source may form or may not form a constituent part of each of the at least two detectors which. Thus, the at least one illumination source may be formed independently of the at least two detectors and may, therefore, particularly be located in at least one position which is separated from the at least two detectors. By way of example, the at least one illumination source itself may be or may
15 comprise an ambient light source and/or may be or may comprise an artificial illumination source. This embodiment is preferably suited for an application in which at least two detectors, preferentially two identical detectors, are employed for acquiring depth information, in particular, for the purpose to providing a measurement volume which extends the inherent measurement volume of a single detector.

The inherent measurement volume of a single detector may, in many cases, be described as an approximate half-cone, wherein a first object which is located within the inherent measurement volume may be detected by the single detector whereas a second object which is located outside the inherent measurement volume can principally not be detected by the single
20 detector. The conical surface of the approximate half-cone can be considered to be formed by a virtual reverse light beam which would be emitted by the at least one optical sensor. The virtual reverse light beam would, thus, emerge from the surface of the at least one optical sensor which, however, does not constitute a point source but rather an extended area. From simple geometrical considerations it can be deduced that no virtual reverse light beam emitted by the
25 at least one optical sensor in this manner will be able to reach all positions in all directions in the volume which surrounds the at least one optical sensor. However, the locations which may be virtually hit by the virtual reverse light beam form an approximate half-cone which can be described as the inherent measurement volume of the single detector.

Therefore, in order to be able to cover a large measurement volume which exceeds the inherent measurement volume of a single detector, at least two detectors may be used, wherein the at least two detectors may be identical or may differ from each other with respect to at least one certain technical property as described elsewhere. In general, the large measurement volume comprises an overlapping volume, which denotes a region in space, which may be coherent or
30 not, wherein double or even multiple detection may occur, i.e. a specific object may be detected by two or more detectors independently, at the same time or at different times. Even in case when two or more detectors, particularly two or more identical detectors, are employed, the
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double or multiple detection of a specific object does not destroy a reliable acquisition of the depth information about the specific object in the overlapping volume. Since the at least one illumination source may be formed independently of the at least two detectors, typically, no relationship exists between a specific illumination source and a specific detector. Therefore, a reliable acquisition of the depth information will also be possible when at least two of the detectors point towards each other. As a result of this disconnection between the specific illumination source and the specific detector, the recording of the depth information about the specific object will not be impaired when the specific object may be located in the overlapping volume. On the contrary, the depth information in the overlapping volume which concerns a specific object may independently been acquired by more than one detector at the same time and can, thus, be used to improve the accuracy of the depth measurement for the specific object. By way of example, this improvement may be achieved by comparing respective depth values which are recorded simultaneously or successively for the same object by at least two separate single detectors.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is proposed. The human-machine interface as proposed may make use of the fact that the above-mentioned detector in one or more of the embodiments mentioned above or as mentioned in further detail below may be used by one or more users for providing information and/or commands to a machine. Thus, preferably, the human-machine interface may be used for inputting control commands.

The human-machine interface comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments as disclosed in further detail below, wherein the human-machine interface is designed to generate at least one item of geometrical information of the user by means of the detector wherein the human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command.

Generally, as used herein, the at least one item of geometrical information of the user may imply one or more items of information on a transversal position and/or on a longitudinal position of the user and/or one or more body parts of the user. Thus, preferably, the geometrical information of the user may imply one or more items of information on a transversal position and/or a longitudinal position as provided by the evaluation device of the detector. The user, a body part of the user or a plurality of body parts of the user may be regarded as one or more objects which may be detected by the at least one detector. Therein, precisely one detector may be provided, or a combination of a plurality of detectors may be provided. As an example, a plurality of detectors may be provided for determining positions of a plurality of body parts of the user and/or for determining an orientation of at least one body part of the user. The human-machine interface may comprise one or more detectors, wherein, in case a plurality of detectors is provided, the detectors may be identical or may differ. Herein, in case a plurality of detectors

is used, the plurality of detectors, particularly the plurality of identical detectors, still allows for a reliable acquisition of depth information about the at least one object in an overlapping volume which may be recorded, as described above, by the plurality of detectors.

- 5 Thus, preferably, the at least one item of geometrical information of the user is selected from the group consisting of: a position of a body of the user; a position of at least one body part of the user; an orientation of a body of the user; an orientation of at least one body part of the user.
- 10 The human-machine interface may further comprise at least one beacon device connectable to the user. As used herein, a beacon device generally is an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the
- 15 detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user. The attachment may take place
- 20 by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

The human-machine interface may be adapted such that the detector may generate an

25 information on the position of the at least one beacon device. Specifically in case a manner of attachment of the at least one beacon device to the user is known, from the at least one item of information on the position of the at least one beacon device at least one item of information regarding a position and/or an orientation of the user or one or more body parts of the user may be gained.

30 The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be

35 transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

40 The beacon device preferably may comprise at least one of: a garment to be worn by the user, preferably a garment selected from the group consisting of a glove, a jacket, a hat, shoes,

trousers and a suit; a stick that may be held by hand; a bat; a club; a racket; a cane; a toy, such as a toy gun.

In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Additionally or alternatively, the entertainment device may also be used for other purposes, such as for exercising, sports, physical therapy or motion tracking in general. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device.

The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one of a movement, the orientation and a position of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

In a further embodiment of the present invention, the entertainment device may be part of an equipment, the equipment being a mobile piece or, in particular, an immobile piece, wherein the equipment may at least partially incorporate the entertainment device. The equipment may

include a single, separate piece being located at a position, either a fixed position or a position at least intermittently subject to a variation, but the equipment may also comprise at least two pieces, preferably two to ten pieces, such as three, four, five, or six pieces, wherein the at least two pieces may be distributed over at least two positions differing from each other within an area, such as a room or a part thereof. Hereby, the entertainment device may be part of the equipment, wherein preferably some or each piece of the equipment may exhibit a part of the entertainment device, e.g. in such a manner that some or each piece of the equipment may comprise at least one detector according to the present invention or a part thereof, such as a sensor. As used herein, "immobile equipment" may include an immobile electronic article, in particular, designated as consumer electronics, wherein "consumer electronics" comprises electronic articles preferentially intended for everyday use, mainly in entertainment, communications and office matters, such as radio receivers, monitors, television sets, audio players, video players, personal computers and/or telephones. A particular example constituting the immobile equipment may be a surround system, which may for example be formed by a number of two, three, four, five, six or more separate pieces of the equipment, such as individual monitors or audio players, including loudspeakers, which may be distributed over an area, preferentially in a specific fashion, such as forming an arc-shaped assembly enclosing a room or a part thereof.

Additionally or alternatively, the entertainment device or a part thereof, such as one, some or each piece of the equipment, may further be equipped with one or more of the following devices: a photographic device, such as a camera, particularly a 2D camera, a picture analysis software, particularly a 2D picture analysis software, and a reference object, particularly a geometrically symmetric reference object, such as a book or a specially formed toy. Hereby, the reference object may also be a part of the immobile equipment and fulfill a further function of the entertainment device as described above and/or below, wherein the reference object may further comprise a detector, a 2D camera or another photographic device. Preferably, a constructive interaction of the photographic device, the picture analysis software and the particularly symmetric reference object, may facilitate an alignment of a 2D picture of an object in question as recorded by the photographic device with the 3D position of the same object as determined by the at least one detector.

In a further embodiment of the present invention, the object which may constitute a target of the at least one detector comprised within the at least one human-machine interface of the entertainment device, may be part of a controller as comprised within mobile equipment, wherein the mobile equipment may be configured to control another mobile equipment or immobile equipment. As used herein, "mobile equipment" may, thus, include mobile electronic articles, in particular, designated as consumer electronics, such as mobile phones, radio receivers, video recorders, audio players, digital cameras, camcorders, mobile computers, video game consoles and/or other devices adapted for remote control. This embodiment may particularly allow controlling immobile equipment with any kind of mobile equipment, preferably with a lesser number of pieces of equipment. As a non-limiting example, it may, thus, be

possible to simultaneously control for example both a game console and a television set by using a mobile phone.

5 Additionally or alternatively, the object which may constitute the target of the detector may further be equipped with an additional sensor (apart from the sensors as comprised within the detector) particularly configured for determining a physical and/or chemical quantity related to the object, such as an inertia sensor for measuring the inertial motion of the object, or an acceleration sensor for determining the acceleration of the object. However, besides these preferred examples, other kinds of sensors adapted for acquiring further parameter related to
10 the object, such as a vibrational sensor for determining vibrations of the object, a temperature sensor for recording the temperature of the object, or a humidity sensor for recording the humidity of the object may be employed. An application of the additional sensor within the object may allow improving the quality and/or the scope of the detection of the position of the object. As a non-limiting example, the additional inertia sensor and/or acceleration sensor may
15 particularly be configured to record additional movements of the object, such as a rotation of the object, which may particularly be employed for increasing the accuracy of the object detection. Moreover, the additional inertia sensor and/or acceleration sensor may preferentially still be addressed in a case where the object being equipped with at least one of these sensors may leave a visual range of the detector comprised within the human-machine interface of the
20 entertainment device. In this case, it might, nevertheless, be possible to follow the object after the object may have left the visual range of the detector by still being able to recording signals emitted from at least one of these sensors and using these signals for determining the location of the object by taking into account its actual inertia and acceleration values and calculating the position therefrom.

25 Additionally or alternatively, the object which may constitute the target of the detector may further be equipped with a further feature allowing both to simulate and/or to stimulate motions, for example by simulating motions of the object, wherein the object may be virtual or real and wherein the object may be controlled by a controller, and/or by stimulating motions of the object
30 by applying the controller accordingly. This feature may particularly be employed for providing a more realistic entertainment experience for a user. As an illustrating example, a steering wheel as employed in an entertainment device may vibrate, wherein the amplitude of the vibrations may depend on a nature of a ground on which a virtual car may be driving on. As a further embodiment, the motion of the object may be stimulated by employing a gyroscope, which may
35 be used for example for a stabilization of flying vehicles as described under the address en.wikipedia.org/wiki/gyroscope.

In a further embodiment of the present invention, the object which may constitute the target of the at least one detector may be equipped with at least one modulation device for modulating
40 the illumination, in particular for a periodic modulation. Preferably, the object may comprise at least one illumination source, which may be part of the object or, alternatively or additionally, be held by the object or attached thereto and which may act as a beacon in a manner as described

elsewhere in this application. The illumination may be adapted to generate at least one light beam to be transmitted to the detector, whereby the illumination source comprises a modulation device for modulating the illumination and/or the modulation device may be a separate device configured for controlling the emission of the illumination source. According to this embodiment, apart from the fundamental modulation of the illumination source, the modulation device may generate additional modulation frequencies, also be denoted as “overtones”, which might be used for transmitting any item of additional information or data from the object to the detector. This embodiment, which might also be designated as a “modulating retro-reflector”, may open the way to employ the object equipped with the modulation device configured for generating both the fundamental and the additional modulation frequencies as remote control. Further, an already existing remote control may be replaced by the object equipped with the described modulation device. Against this background, this kind of object and a piece configured for remote control may interchangeably be used within an arrangement incorporating a detector according to the present invention.

In a further embodiment of the present invention, the entertainment device may further be equipped with additional items such as items commonly used within such an environment. Glasses or other devices being configured for creating a 3D vision within the mind of a player may constitute a particular example.

In a further embodiment of the present invention, the entertainment device may further be equipped with an augmented reality application. As further used herein, “augmented reality” may describe a live perception of reality comprising elements which may be modified by computer-generated data primarily related to physical phenomena such as sound, images, or others. An example may be vision glasses which are particularly adapted for augmented reality applications. Another example may include an arrangement comprising at least two detectors, preferably a multitude thereof, wherein the detectors may particularly be arranged to cover an area within a room or a preferably large part thereof, a photographic device, such as a camera, particularly a 2D camera, and an augmented reality application, wherein the arrangement might be employed to transform a real area into a playing field, which may also be designated as an entertainment field.

In a further aspect of the present invention, a tracking system for tracking the position of at least one movable object is provided. As used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object or at least one part of an object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The tracking system may comprise one, two or more detectors, particularly two or more identical detectors, which allow for a reliable acquisition of depth information about the at least one object in an overlapping volume between the two or more detectors. The track controller is adapted to track a series of positions of the object, each position comprising at least one item of information on a transversal position of the object at a specific point in time and at least one item of information on a longitudinal position of the object at a specific point in time.

The tracking system may further comprise at least one beacon device connectable to the object. For a potential definition of the beacon device, reference may be made to the disclosure above. The tracking system preferably is adapted such that the detector may generate an information on the position of the object of the at least one beacon device. For potential embodiments of the beacon device, reference may be made to the disclosure above. Thus, again, the beacon device may fully or partially be embodied as an active beacon device and/or as a passive beacon device. As an example, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in

communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to the present invention, such as disclosed in one or more of the embodiments given above or given in further detail below.

Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term "photography" generally refers to the technology of acquiring image information of at least one object. As further used herein, a "camera" generally is a device adapted for performing photography. As further used herein, the term "digital photography" generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term "3D photography" generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

In a further aspect of the present invention, a method for determining a position of at least one object is disclosed. The method preferably may make use of at least one detector according to

the present invention, such as of at least one detector according to one or more of the embodiments disclosed above or disclosed in further detail below. Thus, for optional embodiments of the method, reference might be made to the embodiments of the detector.

- 5 The method comprises the following steps, which may be performed in the given order or in a different order. Further, additional method steps might be provided which are not listed. Further, two or more or even all of the method steps might be performed at least partially simultaneously. Further, two or more or even all of the method steps might be performed twice or even more than twice, repeatedly.

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In a first method step, which might also be referred to as a step of determining at least one transversal position, at least one transversal optical sensor is used. The transversal optical sensor determines a transversal position of at least one light beam traveling from the object to the detector, wherein the transversal position is a position in at least one dimension
15 perpendicular to an optical axis of the detector. The transversal optical sensor generates at least one transversal sensor signal.

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In a further method step, which might also be referred to as a step of determining at least one longitudinal position, at least one longitudinal optical sensor is used. The longitudinal optical sensor has at least one sensor region. The longitudinal optical sensor generates at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam. The longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region.

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In a further method step, which might also be referred to as an evaluation step, at least one evaluation device is used. The evaluation device generates at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and wherein the evaluation device further generates at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

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In a further aspect of the present invention, a use of a detector according to the present invention is disclosed. Therein, a use of the detector for a purpose of use is proposed, selected from the group consisting of: a distance measurement, in particular in traffic technology; a position measurement, in particular in traffic technology; an entertainment application; a security
35 application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space.

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In the following, some additional remarks regarding potential embodiments of the detector, the human-machine interface, the tracking system and the method according to the present invention are given. As outlined above, preferably, for potential details of the setups of the at least one transversal optical detector and the at least one longitudinal optical detector,

reference may be made to WO 2012/110924 A1, specifically with regard to potential electrode materials, organic materials, inorganic materials, layer setups and further details.

The object can generally be a living or else inanimate object. Examples of objects which can be detected completely or partly by means of the detector are described in even greater detail below.

Further, with regard to potential embodiments of the optional transfer device, reference may be made to WO 2012/110924 A1. Thus, this optional transfer device can comprise for example at least one beam path. The transfer device can for example comprise one or a plurality of mirrors and/or beam splitters and/or beam deflecting elements in order to influence a direction of the electromagnetic radiation. Alternatively or additionally, the transfer device can comprise one or a plurality of imaging elements which can have the effect of a converging lens and/or a diverging lens. By way of example, the optional transfer device can have one or a plurality of lenses and/or one or a plurality of convex and/or concave mirrors. Once again alternatively or additionally, the transfer device can have at least one wavelength-selective element, for example at least one optical filter. Once again alternatively or additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the optional transfer device can, in principle, be realized individually or in any desired combination.

Further, generally, it shall be noted that, in the context of the present invention, an optical sensor may refer to an arbitrary element which is designed to convert at least one optical signal into a different signal form, preferably into at least one electrical signal, for example a voltage signal and/or a current signal. In particular the optical sensor can comprise at least one optical-electrical converter element, preferably at least one photodiode and/or at least one solar cell. As is explained in even greater detail below, in the context of the present invention, preference is attached particularly to a use of at least one organic optical sensor, that is to say an optical sensor which comprises at least one organic material, for example at least one organic semiconductor material.

In the context of the present invention, a sensor region should be understood to mean a two-dimensional or three-dimensional region which preferably, but not necessarily, is continuous and can form a continuous region, wherein the sensor region is designed to vary at least one measurable property, in a manner dependent on the illumination. By way of example, said at least one property can comprise an electrical property, for example, by the sensor region being designed to generate, solely or in interaction with other elements of the optical sensor, a photo voltage and/or a photocurrent and/or some other type of signal. In particular, the sensor region can be embodied in such a way that it generates a uniform, preferably a single, signal in a manner dependent on the illumination of the sensor region. The sensor region can thus be the smallest unit of the optical sensor for which a uniform signal, for example, an electrical signal, is

generated, which preferably can no longer be subdivided to partial signals, for example for partial regions of the sensor region. The transversal optical sensor and/or the longitudinal optical sensor each can have one or else a plurality of such sensor regions, the latter case for example by a plurality of such sensor regions being arranged in a two-dimensional and/or three-dimensional matrix arrangement.

The at least one sensor region can comprise for example at least one sensor area, that is to say a sensor region whose lateral extent considerably exceeds the thickness of the sensor region, for example by at least a factor of 10, preferably by at least a factor of 100 and particularly preferably by at least a factor of 1000. Examples of such sensor areas can be found in organic or inorganic photovoltaic elements, for example, in accordance with the prior art described above, or else in accordance with the exemplary embodiments described in even greater detail below. The detector can have one or a plurality of such optical sensors and/or sensor regions. By way of example, a plurality of optical sensors can be arranged linearly in a spaced-apart manner or in a two-dimensional arrangement or else in a three-dimensional arrangement, for example by a stack of photovoltaic elements being used, preferably organic photovoltaic elements, preferably a stack in which the sensor areas of the photovoltaic elements are arranged parallel to one another. Other embodiments are also possible.

The optional transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the transversal optical sensor and/or the longitudinal optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the transversal and/or longitudinal optical sensor. The optional transfer device can also, as explained in even greater detail below, be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient light source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a

plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one
5 illumination source can also be integrated into the object or connected or spatially coupled to the object.

The light emerging from the object can accordingly, alternatively or additionally from the option that said light originates in the object itself emerge from the illumination source and/or be
10 excited by the illumination source. By way of example, the electromagnetic light emerging from the object can be emitted by the object itself and/or be reflected by the object and/or be scattered by the object before it is fed to the optical sensor. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift
15 can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary light source, for example by the object or a partial region of the object being excited to effect luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in
20 particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

25 The at least one illumination source of the detector can generally be adapted to the emission and/or reflective properties of the object, for example in terms of its wavelength. Various embodiments are possible.

30 The at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Most preferably, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm
35 to 750 nm or at 690 nm to 700 nm.

The feeding of the light to the transversal and/or longitudinal optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the transversal and/or
40 longitudinal optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example

in the case of an object arranged within a visual range of the detector, is arranged completely on the sensor region, in particular the sensor area. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

5 The at least one longitudinal optical sensor, as outlined above, can be designed for example in such a way that the longitudinal sensor signal, given the same power of the illumination, that is to say for example given the same integral over the intensity of the illumination on the sensor area, is dependent on the geometry of the illumination, that is to say for example on the diameter and/or the equivalent diameter for the sensor spot. By way of example, the longitudinal
10 optical sensor can be designed in such a way that upon a doubling of the beam cross section given the same total power, a signal variation occurs by at least a factor of 3, preferably by at least a factor of 4, in particular a factor of 5 or even a factor of 10. This condition can hold true for example for a specific focusing range, for example for at least one specific beam cross section. Thus, by way of example, the longitudinal sensor signal can have, between at least one
15 optimum focusing at which the signal can have for example at least one global or local maximum and a focusing outside said at least one optimum focusing, a signal difference by at least a factor of 3, preferably by at least a factor of 4, in particular a factor of 5 or even a factor of 10. In particular, the longitudinal sensor signal can have as a function of the geometry of the illumination, for example of the diameter or equivalent diameter of a light spot, at least one
20 pronounced maximum, for example with a boost by at least a factor of 3, particularly preferably by at least a factor of 4 and particularly preferably by at least a factor of 10. Consequently, the longitudinal optical sensor may be based on the above-mentioned FiP-effect, which is disclosed in great detail in WO 2012/110924 A1. Thus, specifically in sDSCs, the focusing of the light beam may play a decisive role, i.e. the cross-section or cross-sectional area on which a certain
25 number of auf photons (nph) is incident. The more tightly the light beam is focused, i.e. the smaller its cross-section, the higher the photo current may be. The term 'FiP' expresses the relationship between the cross-section ϕ (Fi) of the incident beam and the solar cell's power (P).

30 The at least one longitudinal optical sensor is combined with at least one transversal optical sensor in order to preferably provide appropriate position information of the object.

Such effects of the dependence of the at least one longitudinal sensor signal on a beam geometry, preferably a beam cross-section of the at least one light beam, were observed in the context of the investigations leading to the present invention in particular in the case of organic
35 photovoltaic components, that is to say photovoltaic components, for example, solar cells, which comprise at least one organic material, for example at least one organic p-semiconducting material and/or at least one organic dye. By way of example, such effects, as is explained in even greater detail below by way of example, were observed in the case of dye solar cells, that is to say components which have at least one first electrode, at least one n-semiconducting
40 metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid organic p-type semiconductor, and at least one second electrode. Such dye solar cells, preferably solid dye solar cells (solid dye sensitized solar cells, sDSC), are known in principle in

numerous variations from the literature. The described effect of the dependence of the sensor signal on a geometry of the illumination on the sensor area and a use of this effect have not, however, been described heretofore.

- 5 In particular, the at least one longitudinal optical sensor can be designed in such a way that the sensor signal, given the same total power of the illumination, is substantially independent of a size of the sensor region, in particular of a size of the sensor area, in particular as long as the light spot of the illumination lies completely within the sensor region, in particular the sensor area. Consequently, the longitudinal sensor signal can be dependent exclusively on a focusing
10 of the electromagnetic rays on the sensor area. In particular the sensor signal can be embodied in such a way that a photocurrent and/or a photo voltage per sensor area have/has the same values given the same illumination, for example the same values given the same size of the light spot.
- 15 The evaluation device can comprise in particular at least one data processing device, in particular an electronic data processing device, which can be designed to generate the at least one item of information on the transversal position of the object by evaluating the at least one transversal sensor signal and to generate the at least one item of information on the longitudinal position of the object by evaluating the at least one longitudinal sensor signal. Thus, the
20 evaluation device is designed to use the at least one transversal sensor signal and the at least one longitudinal sensor signal as input variables and to generate the items of information on the transversal position and the longitudinal position of the object by processing these input variables. The processing can be done in parallel, subsequently or even in a combined manner. The evaluation device may use an arbitrary process for generating these items of information,
25 such as by calculation and/or using at least one stored and/or known relationship. Besides the at least one transversal sensor signal and at least one longitudinal sensor signal, one or a plurality of further parameters and/or items of information can influence said relationship, for example at least one item of information about a modulation frequency. The relationship can be determined or determinable empirically, analytically or else semi-empirically. Particularly
30 preferably, the relationship comprises at least one calibration curve, at least one set of calibration curves, at least one function or a combination of the possibilities mentioned. One or a plurality of calibration curves can be stored for example in the form of a set of values and the associated function values thereof, for example in a data storage device and/or a table. Alternatively or additionally, however, the at least one calibration curve can also be stored for
35 example in parameterized form and/or as a functional equation. Separate relationships for processing the at least one transversal sensor signal into the at least one item of information on the transversal position and for processing the at least one longitudinal sensor signal into the at least one item of information on the longitudinal position may be used. Alternatively, at least one combined relationship for processing the sensor signals is feasible. Various possibilities are
40 conceivable and can also be combined.

By way of example, the evaluation device can be designed in terms of programming for the purpose of determining the items of information. The evaluation device can comprise in particular at least one computer, for example at least one microcomputer. Furthermore, the evaluation device can comprise one or a plurality of volatile or nonvolatile data memories. As an
5 alternative or in addition to a data processing device, in particular at least one computer, the evaluation device can comprise one or a plurality of further electronic components which are designed for determining the items of information, for example an electronic table and in particular at least one look-up table and/or at least one application-specific integrated circuit (ASIC).

10 The combination of the at least one item of information on the transversal position and the at least one item of information on the longitudinal position allows for a multiplicity of possible uses of the detector, which will be described by way of example hereinafter. As outlined in great detail in WO 2012/110924 A1, a cross-section of the light beam, resulting in a specific diameter
15 or equivalent diameter of a light spot on the sensor region of the at least one longitudinal optical sensor, can be dependent on a distance between the object and the detector and/or on the optional transfer device of the detector, for example at least one detector lens. By way of example, a variation of the distance between the object and a lens of the optional transfer device can lead to a defocusing of the illumination on the sensor region, accompanied by a
20 change in the geometry of the illumination, for example a widening of a light spot, which can result in a correspondingly altered longitudinal sensor signal or a multiplicity of altered longitudinal sensor signals, in case a plurality of longitudinal optical sensors is used. Even without a transfer device, by way of example, from a known beam profile from the sensor signal and/or a variation thereof, for example, by means of a known beam profile and/or a known
25 propagation of the light beam, it is possible to deduce a defocusing and/or the geometrical information. By way of example, given a known total power of the illumination, it is thus possible to deduce from the longitudinal sensor signal of the longitudinal optical sensor a geometry of the illumination and therefrom in turn the geometrical information, in particular at least one item of location information of the object.

30 Similarly, the at least one transversal optical sensor allows for an easy detection of a transversal position of the object. For this purpose, use may be made of the fact that a change in a transversal position of the object generally will lead to a change in a transversal position of the light beam in the sensor region of the at least one transversal optical sensor. Thus, for
35 example by detecting a transversal position of a light spot generated by the light beam impinging on a sensor region, such as a sensor area, of the transversal optical sensor, the transversal position or at least one item of information on the transversal position of the object may be generated. Thus, the position of the light spot may be determined by comparing currents and/or voltage signals of partial electrodes of the transversal optical sensor, such as by
40 forming at least one ratio of at least two currents through at least two different partial electrodes. For this measurement principle, as an example, reference may be made to US 6,995,445 and/or US 2007/0176165 A1. The above-mentioned at least one relationship between the at

least one transversal sensor signal and the at least one item of information on the transversal position of the object may include a known relationship between a transversal position of the light spot on a sensor region of the transversal optical sensor and a transversal position of the object. For this purpose, known imaging properties of the detector, specifically known imaging properties of at least one transfer device of the detector, may be used. Thus, as an example, the transfer device may include at least one lens, and the known imaging properties may make use of a known lens equation of the lens, thereby allowing for a transformation of at least one transversal coordinate of a light spot into at least one transversal coordinate of the object, as the skilled person will recognize. Therein, the known relationship may also make use of additional information, such as the at least one longitudinal sensor signal and/or the at least one item of information on the longitudinal position of the object. Thus, by using the at least one longitudinal sensor signal, the evaluation device may, as an example, first determine at least one item of information on a lateral position of the object, such as at least one distance between the object and the detector, specifically the transfer device and, more preferably, at least one lens of the transfer device. This item of information on the longitudinal position of the object may then be used in the lens equation in order to transform the at least one transversal sensor signal into the at least one item of information on the transversal position of the object, such as by transforming the at least one transversal coordinate of the light spot in the sensor region of the at least one transversal optical sensor into at least one transversal coordinate of the object. Other algorithms are feasible.

As outlined above, the total intensity of total power of the light beam is often unknown, since this total power e.g. may depend on the properties of the object, such as reflecting properties, and/or may depend on a total power of an illumination source and/or may depend on a large number of environmental conditions. Since the above-mentioned known relationship between the at least one longitudinal optical sensor signal and a beam cross-section of the light beam in the at least one sensor region of the at least one longitudinal optical sensor and, thus, a known relationship between the at least one longitudinal optical sensor signal and the at least one item of information on the longitudinal position of the object may depend on the total power of total intensity of the light beam, various ways of overcoming this uncertainty are feasible. Thus, as outlined in great detail in WO 2012/110924 A1, a plurality of longitudinal sensor signals may be detected by the same longitudinal optical sensor, such as by using different modulation frequencies of an illumination of the object. Thus, at least two longitudinal sensor signals may be acquired at different frequencies of a modulation of the illumination, wherein, from the at least two sensor signals, for example by comparison with corresponding calibration curves, it is possible to deduce the total power and/or the geometry of the illumination, and/or therefrom, directly or indirectly, to deduce the at least one item of information on the longitudinal position of the object.

Additionally or alternatively, however, as outlined above, the detector may comprise a plurality of longitudinal optical sensors, each longitudinal optical sensor being adapted to generate at least one longitudinal sensor signal. The longitudinal sensor signals generated by the

longitudinal optical sensors may be compared, in order to gain information on the total power and/or intensity of the light beam and/or in order to normalize the longitudinal sensor signals and/or the at least one item of information on the longitudinal position of the object for the total power and/or total intensity of the light beam. Thus, as an example, a maximum value of the longitudinal optical sensor signals may be detected, and all longitudinal sensor signals may be divided by this maximum value, thereby generating normalized longitudinal optical sensor signals, which, then, may be transformed by using the above-mentioned known relationship, into the at least one item of longitudinal information on the object. Other ways of normalization are feasible, such as a normalization using a mean value of the longitudinal sensor signals and dividing all longitudinal sensor signals by the mean value. Other options are possible. Each of these options is suited to render the transformation independent from the total power and/or intensity of the light beam. In addition, information on the total power and/or intensity of the light beam might be generated.

The detector described can advantageously be developed in various ways. Thus, the detector can furthermore have at least one modulation device for modulating the illumination, in particular for periodic modulation, in particular a periodic beam interrupting device. A modulation of the illumination should be understood to mean a process in which a total power of the illumination is varied, preferably periodically, in particular with one or a plurality of modulation frequencies. In particular, a periodic modulation can be effected between a maximum value and a minimum value of the total power of the illumination. The minimum value can be 0, but can also be > 0 , such that, by way of example, complete modulation does not have to be effected. The modulation can be effected for example in a beam path between the object and the optical sensor, for example by the at least one modulation device being arranged in said beam path. Alternatively or additionally, however, the modulation can also be effected in a beam path between an optional illumination source - described in even greater detail below - for illuminating the object and the object, for example by the at least one modulation device being arranged in said beam path. A combination of these possibilities is also conceivable. The at least one modulation device can comprise for example a beam chopper or some other type of periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates at constant speed and which can thus periodically interrupt the illumination. Alternatively or additionally, however, it is also possible to use one or a plurality of different types of modulation devices, for example modulation devices based on an electro-optical effect and/or an acousto-optical effect. Once again alternatively or additionally, the at least one optional illumination source itself can also be designed to generate a modulated illumination, for example by said illumination source itself having a modulated intensity and/or total power, for example a periodically modulated total power, and/or by said illumination source being embodied as a pulsed illumination source, for example as a pulsed laser. Thus, by way of example, the at least one modulation device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

The detector can be designed in particular to detect at least two sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies. The evaluation device can be designed to generate the geometrical information from the at least two sensor signals. As described above, in this way, by way of example, it is possible to resolve ambiguities and/or it is possible to take account of the fact that, for example, a total power of the illumination is generally unknown.

Further possible embodiments of the detector relate to the embodiment of the at least one optional transfer device. As explained above, said at least one transfer device can have imaging properties or else can be embodied as a pure non-imaging transfer device, which has no influence on a focusing of the illumination. It is particularly preferred, however, if the transfer device has at least one imaging element, for example at least one lens and/or at least one curved mirror, since, in the case of such imaging elements, for example, a geometry of the illumination on the sensor region can be dependent on a relative positioning, for example a distance, between the transfer device and the object. Generally, it is particularly preferred if the transfer device is designed in such a way that the electromagnetic radiation which emerges from the object is transferred completely to the sensor region, for example is focused completely onto the sensor region, in particular the sensor area, in particular if the object is arranged in a visual range of the detector.

As explained above, the optical sensor can furthermore be designed in such a way that the sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination. The detector can be embodied, in particular, as explained above, in such a way that sensor signals at different modulation frequencies are picked up, for example in order to generate one or a plurality of further items of information about the object. As described above, by way of example, a sensor signal at at least two different modulation frequencies can, in each case, be picked up, wherein, by way of example, in this way, a lack of information about a total power of the illumination can be supplemented. By way of example, by comparing the at least two sensor signals picked up at different modulation frequencies with one or a plurality of calibration curves, which can be stored for example in a data storage device of the detector, even in the case of an unknown total power of the illumination, it is possible to deduce a geometry of the illumination, for example a diameter or an equivalent diameter of a light spot on the sensor area. For this purpose, by way of example, it is possible to use the at least one evaluation device described above, for example at least one data processing data, which can be designed to control such picking-up of sensor signals at different frequencies and which can be designed to compare said sensor signals with the at least one calibration curve in order to generate therefrom the geometrical information, for example information about a geometry of the illumination, for example information about a diameter or equivalent diameter of a light spot of the illumination on a sensor area of the optical sensor. Furthermore, as is explained in even greater detail below, the evaluation device can alternatively or additionally be designed to generate at least one item of geometrical information about the object, for example at least one item of location information. This generation of the at

least one item of geometrical information, as explained above, can be effected for example taking account of at least one known relationship between a positioning of the object relative to the detector and/or the transfer device or a part thereof and a size of a light spot, for example empirically, semi-empirically or analytically using corresponding imaging equations.

5 In contrast to known detectors, in which a spatial resolution and/or imaging of objects is also generally tied to the fact that the smallest possible sensor areas are used, for example the smallest possible pixels in the case of CCD chips, the sensor region of the proposed detector can be embodied in a very large fashion, in principle, since for example the geometrical
10 information, in particular the at least one item of location information, about the object can be generated from a known relationship for example between the geometry of the illumination and the sensor signal. Accordingly, the sensor region can have for example a sensor area, for example an optical sensor area, which is at least 0.001 mm^2 , in particular at least 0.01 mm^2 , preferably at least 0.1 mm^2 , more preferably at least 1 mm^2 , more preferably at least 5 mm^2 ,
15 more preferably at least 10 mm^2 , in particular at least 100 mm^2 or at least 1000 mm^2 or even at least $10\,000 \text{ mm}^2$. In particular, sensor areas of 100 cm^2 or more can be used. The sensor area can generally be adapted to the application. In particular, the sensor area should be chosen in such a way that, at least if the object is situated within a visual range of the detector, preferably within a predefined viewing angle and/or a predefined distance from the detector, the light spot
20 is always arranged within the sensor area. In this way, it can be ensured that the light spot is not trimmed by the limits of the sensor region, as a result of which signal corruption could occur.

As described above, the sensor region can be in particular a continuous sensor region, in particular a continuous sensor area, which can preferably generate a uniform, in particular a
25 single, sensor signal. Consequently, the sensor signal can be in particular a uniform sensor signal for the entire sensor region, that is to say a sensor signal to which each partial region of the sensor region contributes, for example additively. The sensor signal can generally, as explained above, in particular be selected from the group consisting of a photocurrent and a photo voltage.

30 The optical sensor can comprise in particular at least one semiconductor detector and/or be at least one semiconductor detector. In particular, the optical sensor can comprise at least one organic semiconductor detector or be at least one organic semiconductor detector, that is to say a semiconductor detector comprising at least one organic semiconducting material and/or at
35 least one organic sensor material, for example at least one organic dye. Preferably, the organic semiconductor detector can comprise at least one organic solar cell and particularly preferably a dye solar cell, in particular a solid dye solar cell. Exemplary embodiments of such preferred solid dye solar cells are explained in even greater detail below.

40 In particular, the optical sensor can comprise at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably at least one solid p-semiconducting organic material, and at least one second

electrode. Generally, however, it is pointed out that the described effect in which the sensor signal, given a constant total power, is dependent on a geometry of the illumination of the sensor region is with high probability not restricted to organic solar cells and in particular not to dye solar cells. Without intending to restrict the scope of protection of the invention by this theory, and without the invention being bound to the correctness of this theory, it is supposed that generally photovoltaic elements are suitable as optical sensors in which at least one semiconducting material having trap states is used. Consequently, the optical sensor can comprise at least one n-semiconducting material and/or at least one p-semiconducting material which can have for example a conduction band and a valence band, wherein, in the case of organic materials, conduction band and valence band should correspondingly be replaced by LUMO (lowest unoccupied molecular orbital) and HOMO (highest occupied molecular orbital). Trap states should be understood to mean energetically possible states which are disposed between the conduction band (or LUMO) and the valence band (or HOMO) and which can be occupied by charge carriers. By way of example, it is possible to provide trap states for hole conduction which are disposed at at least one distance ΔE_h above the valence band (or HOMO) and/or trap states for electron conduction which are disposed at at least one distance ΔE_e below the conduction band (or LUMO). Such traps can be achieved for example by impurities and/or defects, which can optionally also be introduced in a targeted manner, or can be present intrinsically. By way of example, in the case of a low intensity, that is to say for example in the case of a light spot having a large diameter, only a low current can flow, since firstly the trap states are occupied before holes in the conduction band or electrons in the valence band contribute to a photocurrent. It is only starting from a higher intensity, that is to say for example starting from a more intense focusing of the light spot in the sensor region, that a considerable photocurrent can then flow. The described frequency dependence can be explained for example by the fact that charge carriers leave the traps again after a residence duration τ , such that the described effect occurs only in the case of modulated illumination with a high modulation frequency.

By way of example, the detector can be designed to bring about a modulation of the illumination of the object and/or at least one sensor region of the detector, such as at least one sensor region of the at least one longitudinal optical sensor, with a frequency of 0.05 Hz to 1 MHz, such as 0.1 Hz to 10 kHz. As outlined above, for this purpose, the detector may comprise at least one modulation device, which may be integrated into the at least one optional illumination source and/or may be independent from the illumination source. Thus, at least one illumination source might, by itself, be adapted to generate the above-mentioned modulation of the illumination, and/or at least one independent modulation device may be present, such as at least one chopper and/or at least one device having a modulated transmissibility, such as at least one electro-optical device and/or at least one acousto-optical device.

The above-mentioned trap states can be present for example with a density of 10^{-5} to 10^{-1} , relative to the n-semiconducting material and/or the p-semiconducting material and/or the dye.

The energy differences ΔE with respect to the conduction band and with respect to the valence band can be in particular 0.05-0.3 eV.

The detector has, as described above, at least one evaluation device. In particular, the at least one evaluation device can also be designed to completely or partly control or drive the detector, for example by the evaluation device being designed to control one or a plurality of modulation devices of the detector and/or to control at least one illumination source of the detector. The evaluation device can be designed, in particular, to carry out at least one measurement cycle in which one or a plurality of sensor signals, such as a plurality of longitudinal sensor signals, are picked up, for example a plurality of sensor signals of successively at different modulation frequencies of the illumination.

However, unlike in conventional semiconductor devices in which, e.g. the absorption of incident light may occur in a p-type conductor, the absorption of incident light in the present detector may be spatially separated from the movement of the charge carriers in dye sensitized solar cells (DSCs), where incident light may cause the light-absorbing organic dye to switch into an excited state such as Frenkel excitons, i.e. excited, strongly bound electron-hole pairs. As long as the energy levels of both the p-type conductor and the n-type conductor are well-matched to the energy states of the excited light-absorbing organic dye, the excitons may be separated and, thus, the electrons and the holes may travel through the n-type and the p-type conductor, respectively, to the appropriate contact electrodes. Hereby, the moving charge carriers may be majority charge carriers, i.e. in the n-type conductor electrons, in the p-type conductor holes are traveling. Since the light-absorbing organic dye itself is a non-conducting substance, an effective charge transport may depend on an extent to which molecules of the light-absorbing organic dye are in close contact with both the p-type conductor and the n-type conductor. For potential details of DSCs, reference may be made e.g. to U. Bach, M. Grätzel, D. Lupo, P. Comte, J. E. Moser, F. Weissörtel, J. Salbeck, and H. Spreitzer. "Solid-state dye-sensitized mesoporous TiO₂ solar cells with high proton-to-electron conversion efficiencies". *Nature*, Vol. 395, no. 6702, pp. 583-585, 1998.

As described above, when light falls on the cell, it may be absorbed by the light-absorbing organic dye and an exciton may be created. If the energy of the absorbed photons is greater than an energy gap between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) of the light-absorbing organic dye, an electron of the HOMO may be raised to the LUMO of the photoexcited dye and charge separation may take place at the boundary between the dye and the semiconductor which nanoporous titanium dioxide. From there, the electron may travel within femtoseconds to picoseconds further into the conduction band of nanoporous titanium dioxide. Preferably, the energy level of the excited state is matched to the lower limit of the conduction band of the titanium dioxide in order to minimize energy losses during electron transition and the LUMO level should extend sufficiently beyond the lower limit of the conduction band of the titanium dioxide. The oxidation potential of the hole conductor should extend over the HOMO level of the dye allowing it to transport away

the holes of the excited dye. If a load is connected in an external circuit, a current may flow across the titanium dioxide and the anode. The reduced dye may be regenerated through electron donation by the organic p-type conductor to the dye, which may prevent recombination of the electrons from the conduction band of the titanium dioxide with the oxidised dye. The p-type conductor, in turn, may be regenerated through the counter-electrode, which may ensure a constant conversion of energy from incident light into electrical energy without requiring a permanent chemical change.

The evaluation device is designed, as described above, to generate at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal. Said position of the object can be static or may even comprise at least one movement of the object, for example a relative movement between the detector or parts thereof and the object or parts thereof. In this case, a relative movement can generally comprise at least one linear movement and/or at least one rotational movement. Items of movement information can for example also be obtained by comparison of at least two items of information picked up at different times, such that for example at least one item of location information can also comprise at least one item of velocity information and/or at least one item of acceleration information, for example at least one item of information about at least one relative velocity between the object or parts thereof and the detector or parts thereof. In particular, the at least one item of location information can generally be selected from: an item of information about a distance between the object or parts thereof and the detector or parts thereof, in particular an optical path length; an item of information about a distance or an optical distance between the object or parts thereof and the optional transfer device or parts thereof; an item of information about a positioning of the object or parts thereof relative to the detector or parts thereof; an item of information about an orientation of the object and/or parts thereof relative to the detector or parts thereof; an item of information about a relative movement between the object or parts thereof and the detector or parts thereof; an item of information about a two-dimensional or three-dimensional spatial configuration of the object or of parts thereof, in particular a geometry or form of the object. Generally, the at least one item of location information can therefore be selected for example from the group consisting of: an item of information about at least one location of the object or at least one part thereof; information about at least one orientation of the object or a part thereof; an item of information about a geometry or form of the object or of a part thereof, an item of information about a velocity of the object or of a part thereof, an item of information about an acceleration of the object or of a part thereof, an item of information about a presence or absence of the object or of a part thereof in a visual range of the detector.

The at least one item of location information can be specified for example in at least one coordinate system, for example a coordinate system in which the detector or parts thereof rest. Alternatively or additionally, the location information can also simply comprise for example a

distance between the detector or parts thereof and the object or parts thereof. Combinations of the possibilities mentioned are also conceivable.

As outlined above, the detector may comprise at least one illumination source. The illumination source can be embodied in various ways. Thus, the illumination source can be for example part of the detector in a detector housing. Alternatively or additionally, however, the at least one illumination source can also be arranged outside a detector housing, for example as a separate light source. The illumination source can be arranged separately from the object and illuminate the object from a distance. Alternatively or additionally, the illumination source can also be connected to the object or even be part of the object, such that, by way of example, the electromagnetic radiation emerging from the object can also be generated directly by the illumination source. By way of example, at least one illumination source can be arranged on and/or in the object and directly generate the electromagnetic radiation by means of which the sensor region is illuminated. By way of example, at least one infrared emitter and/or at least one emitter for visible light and/or at least one emitter for ultraviolet light can be arranged on the object. By way of example, at least one light emitting diode and/or at least one laser diode can be arranged on and/or in the object. The illumination source can comprise in particular one or a plurality of the following illumination sources: a laser, in particular a laser diode, although in principle, alternatively or additionally, other types of lasers can also be used; a light emitting diode; an incandescent lamp; an organic light source, in particular an organic light emitting diode. Alternatively or additionally, other illumination sources can also be used. It is particularly preferred if the illumination source is designed to generate one or more light beams having a Gaussian beam profile, as is at least approximately the case for example in many lasers. However, other embodiments are also possible, in principle.

As outlined above, a further aspect of the present invention proposes a human-machine interface for exchanging at least one item of information between a user and a machine. A human-machine interface should generally be understood to mean a device by means of which such information can be exchanged. The machine can comprise in particular a data processing device. The at least one item of information can generally comprise for example data and/or control commands. Thus, the human-machine interface can be designed in particular for the inputting of control commands by the user.

The human-machine interface has at least one detector in accordance with one or a plurality of the embodiments described above. The human-machine interface is designed to generate at least one item of geometrical information of the user by means of the detector wherein the human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command. By way of example, said at least one item of geometrical information can be or comprise an item of location information and/or position information and/or orientation information about a body and/or at least one body part of the user, for example an item of location information about a hand posture and/or a posture of some other body part of the user.

In this case, the term user should be interpreted broadly and can for example also encompass one or a plurality of articles directly influenced by the user. Thus, the user can for example also wear one or a plurality of gloves and/or other garments, wherein the geometrical information is at least one item of geometrical information of this at least one garment. By way of example, such garments can be embodied as reflective to a primary radiation emerging from at least one illumination source, for example by the use of one or a plurality of reflectors. Once again alternatively or additionally, the user can for example spatially move one or a plurality of articles whose geometrical information can be detected, which is likewise also intended to be subsumable under generation of at least one item of geometrical information of the user. By way of example, the user can move at least one reflective rod and/or some other type of article, for example by means of said user's hand.

The at least one item of geometrical information can be static, that is to say can for example once again comprise a snapshot, but can also for example once again comprise a series of sequential items of geometrical information and/or at least one movement. By way of example, at least two items of geometrical information picked up at different times can be compared, such that, by way of example, the at least one item of geometrical information can also comprise at least one item of information about a velocity and/or an acceleration of a movement. Accordingly, the at least one item of geometrical information can for example comprise at least one item of information about at least one body posture and/or about at least one movement of the user.

The human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command. As explained above, the term information should in this case be interpreted broadly and can comprise for example data and/or control commands. By way of example, the human-machine interface can be designed to assign the at least one item of information to the at least one item of geometrical information, for example by means of a corresponding assignment algorithm and/or a stored assignment specification. By way of example, a unique assignment between a set of items of geometrical information and corresponding items of information can be stored. In this way, for example by means of a corresponding body posture and/or movement of the user, an inputting of at least one item of information can be effected.

Such human-machine interfaces can generally be used in the machine control or else for example in virtual reality. By way of example, industrial controllers, manufacturing controllers, machine controllers in general, robot controllers, vehicle controllers or similar controllers can be made possible by means of the human-machine interface having the one or the plurality of detectors. However, the use of such a human-machine interface in consumer electronics is particularly preferred.

Accordingly, as outlined above, a further aspect of the present invention proposes an entertainment device for carrying out at least one entertainment function, in particular a game. The entertainment function can comprise in particular at least one game function. By way of example, one or a plurality of games can be stored which can be influencable by a user, who in this context is also called a player hereinafter. By way of example, the entertainment device can comprise at least one display device, for example at least one screen and/or at least one projector and/or at least one set of display spectacles.

The entertainment device furthermore comprises at least one human-machine interface in accordance with one or more of the embodiments described above. The entertainment device is designed to enable at least one item of information of a player to be input by means of the human-machine interface. By way of example, the player, as described above, can adopt or alter one or a plurality of body postures for this purpose. This includes the possibility of the player for example using corresponding articles for this purpose, for example garments such as e.g. gloves, for example garments which are equipped with one or a plurality of reflectors for reflecting the electromagnetic radiation of the detector. The at least one item of information can comprise for example, as explained above, one or a plurality of control commands. By way of example, in this way, changes in direction can be performed, inputs can be confirmed, a selection can be made from a menu, specific game options can be initiated, movements can be influenced in a virtual space or similar instances of influencing or altering the entertainment function can be performed.

The above-described detector, the method, the human-machine interface and the entertainment device and also the proposed uses have considerable advantages over the prior art. Thus, generally, a simple and, still, efficient detector for determining a position of at least one object in space may be provided. Therein, as an example, three-dimensional coordinates of an object or part of an object may be determined in a fast and efficient way. Specifically the combination of the at least one transversal optical sensor and the at least one longitudinal optical sensor, which each may be designed in a cost-efficient way, may lead to a compact, cost-efficient and, still, highly precise device. Both the at least one transversal optical sensor and the at least one longitudinal optical sensor preferably may fully or partially be designed as organic photovoltaic devices, such as by using dye-sensitized solar cells for each of these optical sensors, preferably sDSCs.

As compared to devices known in the art, which of them are based on complex triangulation methods, the detector as proposed provides a high degree of simplicity, specifically with regard to an optical setup of the detector. Thus, in principle, a simple combination of one, two or more sDSCs, preferably in combination with a suited transfer device, specifically a suited lens, and in conjunction with an appropriate evaluation device, is sufficient for high precision position detection.

This high degree of simplicity, in combination with the possibility of high precision measurements, is specifically suited for machine control, such as in human-machine interfaces and, more preferably, in gaming. Thus, cost-efficient entertainment devices may be provided which may be used for a large number of gaming purposes.

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Thus, as for the optical detectors and devices disclosed in WO 2012/110924 A1 or in US provisional applications 61/739,173, filed on December 19, 2012, and 61/749,964, filed on January 8, 2013, the optical detector, the detector system, the human-machine interface, the entertainment device, the tracking system or the camera according to the present invention (in
10 the following simply referred to as “the devices according to the present invention” or “Fip-devices”) may be used for a plurality of application purposes, such as one or more of the purposes disclosed in further detail in the following.

Thus, firstly, Fip-devices may be used in mobile phones, tablet computers, laptops, smart
15 panels or other stationary or mobile computer or communication applications. Thus, Fip-devices may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, Fip-devices may be used as cameras and/or sensors, such as in combination with mobile software for scanning environment, objects and living beings. Fip-devices may even be
20 combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. Fip-devices may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with gesture recognition. Thus, specifically, Fip-devices acting as human-machine interfaces, also referred to as FiP input devices, may be used in mobile applications, such as for controlling other electronic devices or
25 components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one Fip-device may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

Further, Fip-devices may be used in webcams or other peripheral devices for computing
30 applications. Thus, as an example, Fip-devices may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, Fip-devices are particularly useful for giving commands by facial expressions and/or body expressions. Fip-devices can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, etc. Further,
35 Fip-devices may be used in applications for gaming, such as by using a webcam. Further, Fip-devices may be used in virtual training applications and/or video conferences.

Further, Fip-devices may be used in mobile audio devices, television devices and gaming
40 devices, as partially explained above. Specifically, Fip-devices may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, Fip-devices may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications.

Further, Fip-devices may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of Fip-devices in mobile applications such as mobile phones, as disclosed above.

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Further, Fip-devices may be used for security and surveillance applications. Specifically, Fip-devices may be used for optical encryption. FIP-based detection can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. Fip-devices may further be combined with an active infrared light source to allow detection in low light surroundings. Fip-devices such as FIP-based sensors are generally advantageous as compared to active detector systems, specifically since Fip-devices avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector device is. Thus, generally, Fip-devices may be used for an unrecognized and undetectable tracking of moving objects. Additionally, Fip-devices generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using Fip-devices, Fip-devices generally may be used for facial, body and person recognition and identification. Therein, Fip-devices may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, Fip-devices may be used in security devices and other personalized applications.

Further, Fip-devices may be used as 3D-Barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, Fip-devices generally can be used for surveillance and monitoring of spaces and areas. Thus, Fip-devices may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, Fip-devices may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in combination with image intensifiers or image enhancement devices and/or photomultipliers.

Further, Fip-devices may advantageously be applied in camera applications such as video and camcorder applications. Thus, Fip-devices may be used for motion capture and 3D-movie recording. Therein, Fip-devices generally provide a large number of advantages over conventional optical devices. Thus, Fip-devices generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing Fip-devices having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are

voluminous, such as due to the general need for voluminous beam-splitters. Further, Fip-devices generally may be used for focus/autofocus devices, such as autofocus cameras. Further, Fip-devices may also be used in optical microscopy, especially in confocal microscopy.

5 Further, Fip-devices generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, Fip-devices may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, rear cross traffic alert, and other automotive and traffic applications. Therein, Fip-devices may be used as standalone devices or
10 in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, Fip-devices may be used for autonomous driving and safety issues. Further, in these applications, Fip-devices may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. Advantage, in these applications, the generally passive nature of typical Fip-devices is
15 advantageous. Thus, since Fip-devices generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. Fip-devices specifically may be used in combination with recognition software, such as standard image recognition software. Thus, signals and data as provide by Fip-devices typically are readily processable and, therefore, generally require lower calculation power than established
20 stereovision systems such as LIDAR. Given the low space demand, Fip-devices such as cameras using the FiP-effect may be placed at virtually any place in a vehicle, such as on a window screen, on a front hood, on bumpers, on lights, on mirrors or other places the like. Various detectors based on the FiP-effect can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts.
25 Thus, various FiP-based sensors may be combined with other Fip-based sensors and/or conventional sensors, such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of a FiP-sensor with one or more rain detection sensors is also possible. This is
30 due to the fact that Fip-devices generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one Fip-device with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions.

35 Further, Fip-devices generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, Fip-devices may be used for detecting free parking spaces in parking lots.

40 Further, Fip-devices may be used is the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endoscopes, may be named, since, as outlined above, Fip-devices may require a low volume only and may be integrated into other

devices. Specifically, Fip-devices having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, Fip-devices may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. These applications are specifically valuable e.g. in medical treatments and long-distance diagnosis and tele-medicine.

Further, Fip-devices may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, Fip-devices may be applied in the field of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing etc. Fip-devices can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

FIP devices may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

Fip-devices further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the Fip-devices may also be applied for distance diagnostics.

Further, Fip-devices may be applied in the field of machine vision. Thus, one or more Fip-devices may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, Fip-devices may allow for autonomous movement and/or autonomous detection of failures in parts. Fip-devices may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. Given the passive nature of Fip-devices, Fip-devices may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of Fip-devices is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, Fip-devices generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. Fip-devices can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, Fip-devices may be used for assessment of surface quality, such as for surveying the surface evenness of a

product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible.

Further, Fip-devices may be used in the polls, airplanes, ships, spacecrafts and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircrafts, vehicles and the like may be named. Detection devices based on the FiP-effect for monitoring the speed and/or the direction of moving objects are feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one Fip-detector specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one Fip-device can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. Fip-devices generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. Fip-devices are particularly useful but not limited to e.g. speed control and air traffic control devices.

Fip-devices generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircrafts at landing or starting. Wherein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving in dangerous but well defined routes, such as mining vehicles.

Further, as outlined above, Fip-devices may be used in the field of gaming. Thus, Fip-devices can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of Fip-devices for giving commands are feasible, such as by using one or more Fip-devices for gesture or facial recognition. Fip-devices may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more Fip-devices with one or more IR or VIS light sources is possible, such as with a detection device based on the FiP effect. A combination of a FiP-based detector with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racquet, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, Fip-devices generally may be used in the field of building, construction and cartography. Thus, generally, FiP-based devices may be used in order to measure and/or monitor environmental areas, e.g. country side or buildings. Therein, one or more Fip-devices may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. FiP-devices can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like.

FiP-devices may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support.

FiP devices may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, FiP detectors may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices.

FiP-based devices can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of Fip-devices, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, Fip-devices may be used in inspections and maintenance, such as pipeline inspection gauges.

As outlined above, Fip-devices may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, Fip-devices may be used in logistics applications. Thus, Fip-devices may be used for optimized loading or packing containers or vehicles. Further, Fip-devices may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, Fip-devices may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, Fip-devices may be used for process control in production, e.g. for observing filling level of tanks. Further, Fip-devices may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, Fip-devices may be used for analyzing 3D-quality marks. Further, Fip-devices may be used in manufacturing tailor-made goods such as tooth inlays, dental braces,

prosthesis, clothes or the like. Fip-devices may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, Fip-devices may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

As outlined above, preferably, the at least one transversal optical sensor and/or the at least one longitudinal optical sensor both may comprise at least one organic semiconductor detector, particularly preferably at least one dye solar cell, DSC or sDSC. In particular, the transversal optical sensor and/or the longitudinal optical sensor each may comprise at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material and at least one second electrode, preferably in the stated order. The stated elements can be present as layers in a layer construction, for example. The layer construction can be applied for example to a substrate, preferably a transparent substrate, for example a glass substrate.

Preferred embodiments of the above-mentioned elements of the preferred optical sensor are described below by way of example, wherein these embodiments can be used in any desired combination. However, numerous other configurations are also possible, in principle, wherein reference can be made for example to WO 2012/110924 A1, US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 and WO 2009/013282 A1 cited above.

As outlined above, the at least one transversal optical sensor may be designed as a dye-sensitized solar cell (DSC), preferably a solid dye-sensitized solar cell (sDSC). Similarly, the at least one longitudinal optical sensor may be designed as at least one dye-sensitized solar cell (DSC) or may comprise at least one dye-sensitized solar cell (DSC), preferably a solid dye-sensitized solar cell (sDSC). More preferably, the at least one longitudinal optical sensor comprises a stack of DSCs, preferably a stack of sDSCs. Preferred components of the DSCs or sDSCs will be disclosed in the following. However, it shall be understood that other embodiments are feasible.

First electrode and n-semiconductive metal oxide

Generally, for preferred embodiments of the first electrode and the n-semiconductive metal oxide, which may be used in a layer setup of the transversal optical sensor and/or the longitudinal optical sensor, reference may be made to WO 2012/110924 A1. The n-semiconductive metal oxide used in the dye solar cell of the transversal optical sensor and/or the longitudinal optical sensor may be a single metal oxide or a mixture of different oxides. It is also possible to use mixed oxides. The n-semiconductive metal oxide may especially be porous and/or be used in the form of a nanoparticulate oxide, nanoparticles in this context being understood to mean particles which have an average particle size of less than 0.1 micrometer. A nanoparticulate oxide is typically applied to a conductive substrate (i.e. a carrier with a

conductive layer as the first electrode) by a sintering process as a thin porous film with large surface area.

Preferably, the at least one transversal optical sensor uses at least one transparent substrate.

- 5 Similarly, preferably, the at least one longitudinal optical sensor uses at least one transparent substrate. In case a plurality of longitudinal optical sensors is used, such as a stack of longitudinal optical sensors, preferably, at least one of these longitudinal optical sensors uses a transparent substrate. Thus, as an example, all longitudinal optical sensors but the last longitudinal optical sensor facing away from the object, each may use a transparent substrate.
- 10 The last longitudinal optical sensor may either use a transparent or an intransparent substrate.

Similarly, the at least one transversal optical sensor uses at least one transparent first electrode. Further, the at least one longitudinal optical sensor may use at least one transparent first electrode. In case a plurality of longitudinal optical sensors is used, such as a stack of

15 longitudinal optical sensors, preferably, at least one of these longitudinal optical sensors uses a transparent first electrode. Thus, as an example, all longitudinal optical sensors but the last longitudinal optical sensor facing away from the object each may use a transparent first electrode. The last longitudinal optical sensor may either use a transparent or an intransparent first electrode.

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The substrate may be rigid or else flexible. Suitable substrates (also referred to hereinafter as carriers) are, as well as metal foils, in particular plastic sheets or films and especially glass sheets or glass films. Particularly suitable electrode materials, especially for the first electrode according to the above-described, preferred structure, are conductive materials, for example

25 transparent conductive oxides (TCOs), for example fluorine- and/or indium-doped tin oxide (FTO or ITO) and/or aluminum-doped zinc oxide (AZO), carbon nanotubes or metal films. Alternatively or additionally, it would, however, also be possible to use thin metal films which still have a sufficient transparency. In case an intransparent first electrode is desired and used, thick metal films may be used.

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The substrate can be covered or coated with these conductive materials. Since generally, only a single substrate is required in the structure proposed, the formation of flexible cells is also possible. This enables a multitude of end uses which would be achievable only with difficulty, if at all, with rigid substrates, for example use in bank cards, garments, etc.

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The first electrode, especially the TCO layer, may additionally be covered or coated with a solid metal oxide buffer layer (for example of thickness 10 to 200 nm), in order to prevent direct contact of the p-type semiconductor with the TCO layer (see Peng *et al.*, Coord. Chem. Rev. 248, 1479 (2004)). The inventive use of solid p-semiconducting electrolytes, in the case of

40 which contact of the electrolyte with the first electrode is greatly reduced compared to liquid or gel-form electrolytes, however, makes this buffer layer unnecessary in many cases, such that it is possible in many cases to dispense with this layer, which also has a current-limiting effect

and can also worsen the contact of the n-semiconducting metal oxide with the first electrode. This enhances the efficiency of the components. On the other hand, such a buffer layer can in turn be utilized in a controlled manner in order to match the current component of the dye solar cell to the current component of the organic solar cell. In addition, in the case of cells in which the buffer layer has been dispensed with, especially in solid cells, problems frequently occur with unwanted recombinations of charge carriers. In this respect, buffer layers are advantageous in many cases specifically in solid cells.

As is well known, thin layers or films of metal oxides are generally inexpensive solid semiconductor materials (n-type semiconductors), but the absorption thereof, due to large band gaps, is typically not within the visible region of the electromagnetic spectrum, but rather usually in the ultraviolet spectral region. For use in solar cells, the metal oxides therefore generally, as is the case in the dye solar cells, have to be combined with a dye as a photosensitizer, which absorbs in the wavelength range of sunlight, i.e. at 300 to 2000 nm, and, in the electronically excited state, injects electrons into the conduction band of the semiconductor. With the aid of a solid p-type semiconductor used additionally in the cell as an electrolyte, which is in turn reduced at the counter electrode, electrons can be recycled to the sensitizer, such that it is regenerated.

Of particular interest for use in organic solar cells are semiconductors zinc oxide, tin dioxide, titanium dioxide or mixtures of these metal oxides. The metal oxides can be used in the form of nanocrystalline porous layers. These layers have a large surface area which is coated with the dye as a sensitizer, such that a high absorption of sunlight is achieved. Metal oxide layers which are structured, for example nanorods, give advantages such as higher electron mobilities or improved pore filling by the dye.

The metal oxide semiconductors can be used alone or in the form of mixtures. It is also possible to coat a metal oxide with one or more other metal oxides. In addition, the metal oxides may also be applied as a coating to another semiconductor, for example GaP, ZnP or ZnS.

Particularly preferred semiconductors are zinc oxide and titanium dioxide in the anatase polymorph, which is preferably used in nanocrystalline form.

In addition, the sensitizers can advantageously be combined with all n-type semiconductors which typically find use in these solar cells. Preferred examples include metal oxides used in ceramics, such as titanium dioxide, zinc oxide, tin(IV) oxide, tungsten(VI) oxide, tantalum(V) oxide, niobium(V) oxide, cesium oxide, strontium titanate, zinc stannate, complex oxides of the perovskite type, for example barium titanate, and binary and ternary iron oxides, which may also be present in nanocrystalline or amorphous form.

Due to the strong absorption that customary organic dyes and phthalocyanines and porphyrins have, even thin layers or films of the n-semiconducting metal oxide are sufficient to absorb the

required amount of dye. Thin metal oxide films in turn have the advantage that the probability of unwanted recombination processes falls and that the internal resistance of the dye subcell is reduced. For the n-semiconducting metal oxide, it is possible with preference to use layer thicknesses of 100 nm up to 20 micrometers, more preferably in the range between 500 nm and approx. 3 micrometers.

Dye

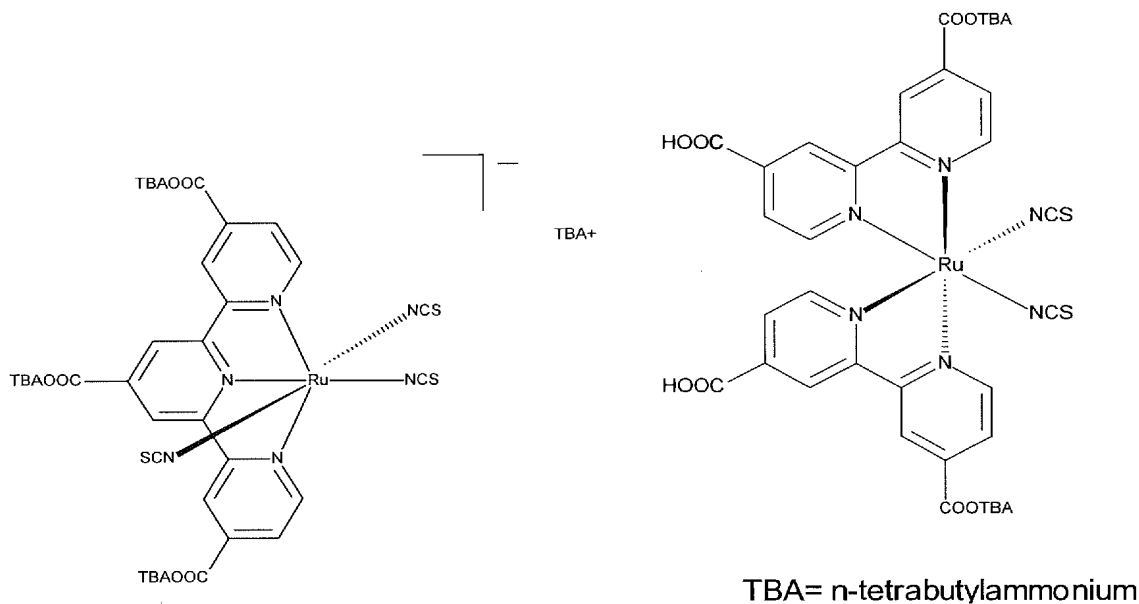
In the context of the present invention, as usual in particular for DSCs, the terms "dye", "sensitizer dye" and "sensitizer" are used essentially synonymously without any restriction of possible configurations. Numerous dyes which are usable in the context of the present invention are known from the prior art, and so, for possible material examples, reference may also be made to the above description of the prior art regarding dye solar cells. As a preferred example, one or more of the dyes disclosed in WO 2012/110924 A1 may be used.

Additionally or alternatively, one or more quinolinium dyes with fluorinated counter anion may be used in the detector according to the present invention, such as one or more of the dyes as disclosed in WO 2013/144177 A1. Specifically, one or more of the dyes disclosed in the following may be used in the at least one longitudinal optical sensor and/or in the at least one transversal optical sensor. Details of these dyes and details of the disclosure of these unpublished applications will be given below. Specifically, dye D-5, which will be described below in more detail, may be used. However, one or more other dyes may be used in addition or alternatively.

All dyes listed and claimed may in principle also be present as pigments. Dye-sensitized solar cells based on titanium dioxide as a semiconductor material are described, for example, in US 4 927 721 A, Nature 353, p. 737-740 (1991) and US 5 350 644 A, and also Nature 395, p. 583-585 (1998) and EP 1 176 646 A1. The dyes described in these documents can in principle also be used advantageously in the context of the present invention. These dye solar cells preferably comprise monomolecular films of transition metal complexes, especially ruthenium complexes, which are bonded to the titanium dioxide layer via acid groups as sensitizers.

Dyes for use in dye-sensitized solar cells which may comprise ruthenium complexes have been rather of academic interest so far, particularly due to the high costs of ruthenium. However, a dye-sensitized solar cell which may be used in the detector according to the present invention would only require such a small amount of ruthenium that the cost argument may easily be overruled by its attractive features for use within the present method for determining a position of at least one object, particularly in a case where at least one light beam which travels from the object might involve a spectral range which may at least partially include a part of the infrared (IR) region, i.e. a part of the electromagnetic spectrum ranging from approximately 750 nm to 1000 μ m, preferentially a part thereof generally denoted as near infrared (NIR) region, which is usually considered to range from approximately 750 nm to 1.5 μ m. Examples of known

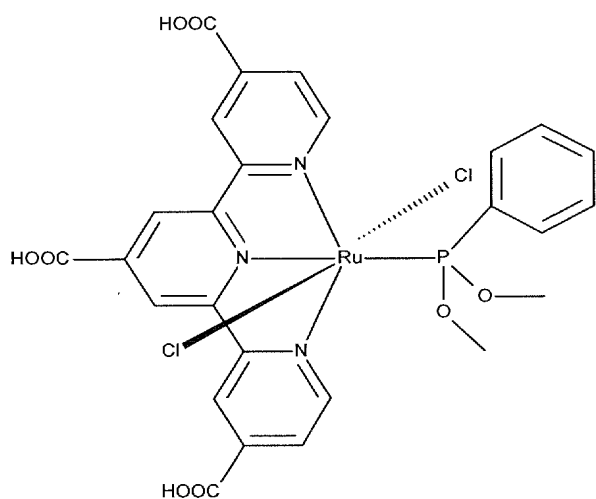
ruthenium complexes which might be suitable for an application within the detector according to the present invention are:



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Another example may be found in T. Kinoshita, J.T. Dy, S. Uchida, T. Kubo, and H. Segawa, *Wideband dye-sensitized solar cells employing a phosphine-coordinated ruthenium sensitizer*, Nature Photonics, 7, 535-539 (2013), wherein a phosphine-coordinated ruthenium complex is described, which exhibits a strong absorption in the NIR, particularly within the range from 750 nm to 950 nm, which may thus, yield a dye-sensitized solar cell with a promising efficiency:

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15 Owing to weak absorption properties of most known dyes within the IR region, including the NIR region, dyes comprising ruthenium complexes may, thus, be able to extend the scope of the

detector according to the present invention into the IR region, in particular into the NIR region, e.g. as being used as active depth sensors, particularly in applications related to computer vision, wherein IR light may play an important role, as described elsewhere in this application.

5 Many sensitizers which have been proposed include metal-free organic dyes, which are likewise also usable in the context of the present invention. High efficiencies of more than 4%, especially in solid dye solar cells, can be achieved, for example, with indoline dyes (see, for example, Schmidt-Mende *et al.*, Adv. Mater. 2005, 17, 813). US-A-6 359 211 describes the use, also implementable in the context of the present invention, of cyanine, oxazine, thiazine and acridine
10 dyes which have carboxyl groups bonded via an alkylene radical for fixing to the titanium dioxide semiconductor.

Organic dyes now achieve efficiencies of almost 12.1% in liquid cells (see, for example, P. Wang *et al.*, ACS. Nano 2010). Pyridinium-containing dyes have also been reported, can be
15 used in the context of the present invention and exhibit promising efficiencies.

Particularly preferred sensitizer dyes in the dye solar cell proposed are the perylene derivatives, terrylene derivatives and quaterrylene derivatives described in DE 10 2005 053 995 A1 or WO 2007/054470 A1. The use of these dyes, which is also possible in the context of the
20 present invention, leads to photovoltaic elements with high efficiencies and simultaneously high stabilities.

The rylenes exhibit strong absorption in the wavelength range of sunlight and can, depending on the length of the conjugated system, cover a range from about 400 nm (perylene derivatives I from DE 10 2005 053 995 A1) up to about 900 nm (quaterrylene derivatives I from
25 DE 10 2005 053 995 A1). Rylene derivatives I based on terrylene absorb, according to the composition thereof, in the solid state adsorbed onto titanium dioxide, within a range from about 400 to 800 nm. In order to achieve very substantial utilization of the incident sunlight from the visible into the near infrared region, it is advantageous to use mixtures of different rylene
30 derivatives I. Occasionally, it may also be advisable also to use different rylene homologs.

The rylene derivatives I can be fixed easily and in a permanent manner to the n-semiconducting metal oxide film. The bonding is effected via the anhydride function (x1) or the carboxyl groups –COOH or –COO– formed in situ, or via the acid groups A present in the imide or condensate
35 radicals ((x2) or (x3)). The rylene derivatives I described in DE 10 2005 053 995 A1 have good suitability for use in dye-sensitized solar cells in the context of the present invention.

It is particularly preferred when the dyes, at one end of the molecule, have an anchor group which enables the fixing thereof to the n-type semiconductor film. At the other end of the
40 molecule, the dyes preferably comprise electron donors Y which facilitate the regeneration of the dye after the electron release to the n-type semiconductor, and also prevent recombination with electrons already released to the semiconductor.

For further details regarding the possible selection of a suitable dye, it is possible, for example, again to refer to DE 10 2005 053 995 A1. By way of example, it is possible especially to use ruthenium complexes, porphyrins, other organic sensitizers, and preferably rylenes.

The dyes can be fixed onto or into the n-semiconducting metal oxide films in a simple manner. For example, the n-semiconducting metal oxide films can be contacted in the freshly sintered (still warm) state over a sufficient period (for example about 0.5 to 24 h) with a solution or suspension of the dye in a suitable organic solvent. This can be accomplished, for example, by immersing the metal oxide-coated substrate into the solution of the dye.

If combinations of different dyes are to be used, they may, for example, be applied successively from one or more solutions or suspensions which comprise one or more of the dyes. It is also possible to use two dyes which are separated by a layer of, for example, CuSCN (on this subject see, for example, Tennakone, K.J., Phys. Chem. B. 2003, 107, 13758). The most convenient method can be determined comparatively easily in the individual case.

In the selection of the dye and of the size of the oxide particles of the n-semiconducting metal oxide, the organic solar cell should be configured such that a maximum amount of light is absorbed. The oxide layers should be structured such that the solid p-type semiconductor can efficiently fill the pores. For instance, smaller particles have greater surface areas and are therefore capable of adsorbing a greater amount of dyes. On the other hand, larger particles generally have larger pores which enable better penetration through the p-conductor.

p-semiconducting organic material

As described above, the at least one DSC or sDSC of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor can comprise in particular at least one p-semiconducting organic material, preferably at least one solid p-semiconducting material, which is also designated hereinafter as p-type semiconductor or p-type conductor. Hereinafter, a description is given of a series of preferred examples of such organic p-type semiconductors which can be used individually or else in any desired combination, for example in a combination of a plurality of layers with a respective p-type semiconductor, and/or in a combination of a plurality of p-type semiconductors in one layer.

In order to prevent recombination of the electrons in the n-semiconducting metal oxide with the solid p-conductor, it is possible to use, between the n-semiconducting metal oxide and the p-type semiconductor, at least one passivating layer which has a passivating material. This layer should be very thin and should as far as possible cover only the as yet uncovered sites of the n-semiconducting metal oxide. The passivation material may, under some circumstances, also be applied to the metal oxide before the dye. Preferred passivation materials are especially one or more of the following substances: Al_2O_3 ; silanes, for example CH_3SiCl_3 ; Al^{3+} ; 4-tert-butylpyridine

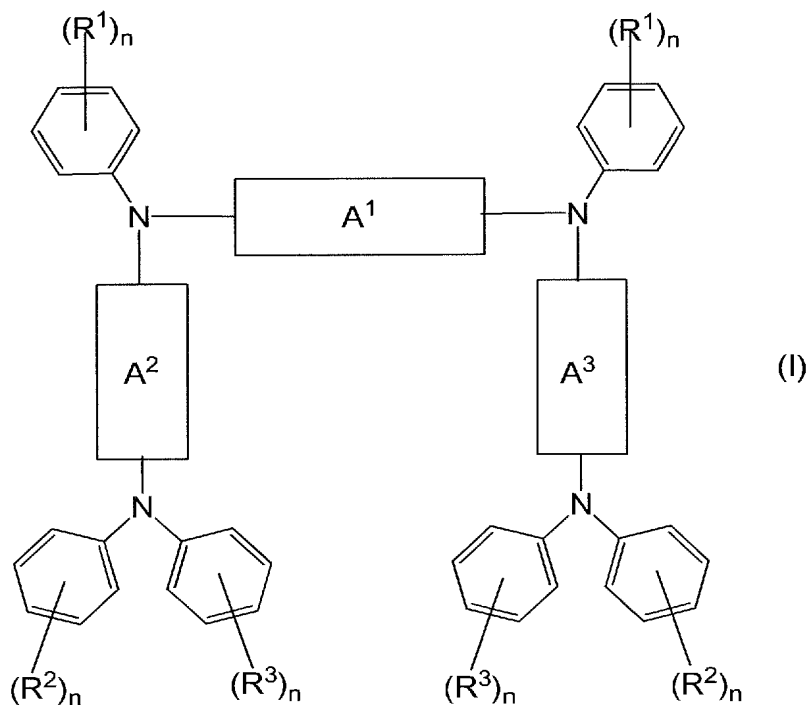
(TBP); MgO; GBA (4-guanidinobutyric acid) and similar derivatives; alkyl acids; hexadecylmalonic acid (HDMA).

As described above, in the context of the organic solar cell, preferably one or more solid organic p-type semiconductors are used – alone or else in combination with one or more further p-type semiconductors which are organic or inorganic in nature. In the context of the present invention, a p-type semiconductor is generally understood to mean a material, especially an organic material, which is capable of conducting holes, that is to say positive charge carriers. More particularly, it may be an organic material with an extensive π -electron system which can be oxidized stably at least once, for example to form what is called a free-radical cation. For example, the p-type semiconductor may comprise at least one organic matrix material which has the properties mentioned. Furthermore, the p-type semiconductor can optionally comprise one or a plurality of dopants which intensify the p-semiconducting properties. A significant parameter influencing the selection of the p-type semiconductor is the hole mobility, since this partly determines the hole diffusion length (cf. Kumara, G., Langmuir, 2002, 18, 10493-10495). A comparison of charge carrier mobilities in different spiro compounds can be found, for example, in T. Saragi, Adv. Funct. Mater. 2006, 16, 966-974.

Preferably, in the context of the present invention, organic semiconductors are used (i.e. low molecular weight, oligomeric or polymeric semiconductors or mixtures of such semiconductors). Particular preference is given to p-type semiconductors which can be processed from a liquid phase. Examples here are p-type semiconductors based on polymers such as polythiophene and polyarylamines, or on amorphous, reversibly oxidizable, nonpolymeric organic compounds, such as the spirobifluorenes mentioned at the outset (cf., for example, US 2006/0049397 and the spiro compounds disclosed therein as p-type semiconductors, which are also usable in the context of the present invention). Preference is given to using low molecular weight organic semiconductors, such as the low molecular weight p-type semiconducting materials as disclosed in WO 2012/110924 A1, preferably spiro-MeOTAD, and/or one or more of the p-type semiconducting materials disclosed in Leijtens et al., ACS Nano, VOL. 6, NO. 2, 1455-1462 (2012). Additionally or alternatively, one or more of the p-type semiconducting materials as disclosed in WO 2010/094636 A1 may be used, the full content of which is herewith included by reference. In addition, reference may also be made to the remarks regarding the p-semiconducting materials and dopants from the above description of the prior art.

The p-type semiconductor is preferably producible or produced by applying at least one p-conducting organic material to at least one carrier element, wherein the application is effected for example by deposition from a liquid phase comprising the at least one p-conducting organic material. The deposition can in this case once again be effected, in principle, by any desired deposition process, for example by spin-coating, knife-coating, printing or combinations of the stated and/or other deposition methods.

The organic p-type semiconductor may especially comprise at least one spiro compound and/or especially be selected from: a spiro compound, especially spiro-MeOTAD; a compound with the structural formula:



in which

A¹, A², A³ are each independently optionally substituted aryl groups or heteroaryl groups,

R¹, R², R³ are each independently selected from the group consisting of the substituents -R, -OR, -NR₂, -A⁴-OR and -A⁴-NR₂,

where R is selected from the group consisting of alkyl, aryl and heteroaryl,

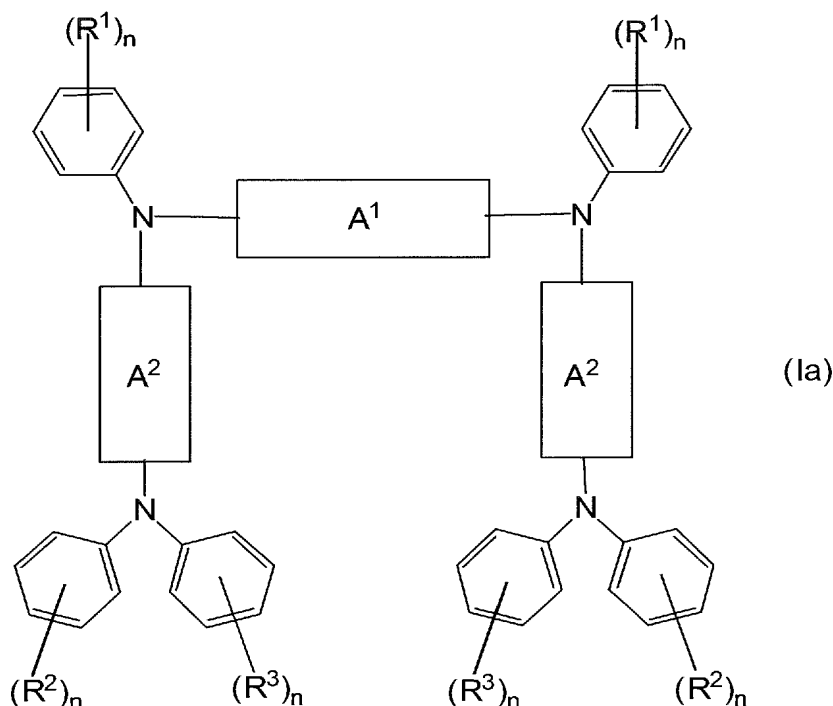
and

where A⁴ is an aryl group or heteroaryl group, and

where n at each instance in formula I is independently a value of 0, 1, 2 or 3,

with the proviso that the sum of the individual n values is at least 2 and at least two of the R¹, R² and R³ radicals are -OR and/or -NR₂.

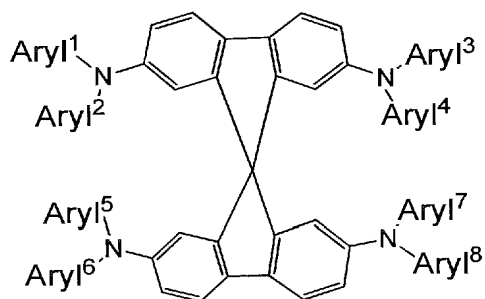
Preferably, A² and A³ are the same; accordingly, the compound of the formula (I) preferably has the following structure (Ia)



- 5 More particularly, as explained above, the p-type semiconductor may thus have at least one low molecular weight organic p-type semiconductor. A low molecular weight material is generally understood to mean a material which is present in monomeric, nonpolymerized or nonoligomerized form. The term "low molecular weight" as used in the present context preferably means that the p-type semiconductor has molecular weights in the range from 100 to 25 000 g/mol. Preferably, the low molecular weight substances have molecular weights of 500 to 2000 g/mol.

In general, in the context of the present invention, p-semiconducting properties are understood to mean the property of materials, especially of organic molecules, to form holes and to transport these holes and/or to pass them on to adjacent molecules. More particularly, stable oxidation of these molecules should be possible. In addition, the low molecular weight organic p-type semiconductors mentioned may especially have an extensive π -electron system. More particularly, the at least one low molecular weight p-type semiconductor may be processable from a solution. The low molecular weight p-type semiconductor may especially comprise at least one triphenylamine. It is particularly preferred when the low molecular weight organic p-type semiconductor comprises at least one spiro compound. A spiro compound is understood to mean polycyclic organic compounds whose rings are joined only at one atom, which is also referred to as the spiro atom. More particularly, the spiro atom may be sp^3 -hybridized, such that the constituents of the spiro compound connected to one another via the spiro atom are, for example, arranged in different planes with respect to one another.

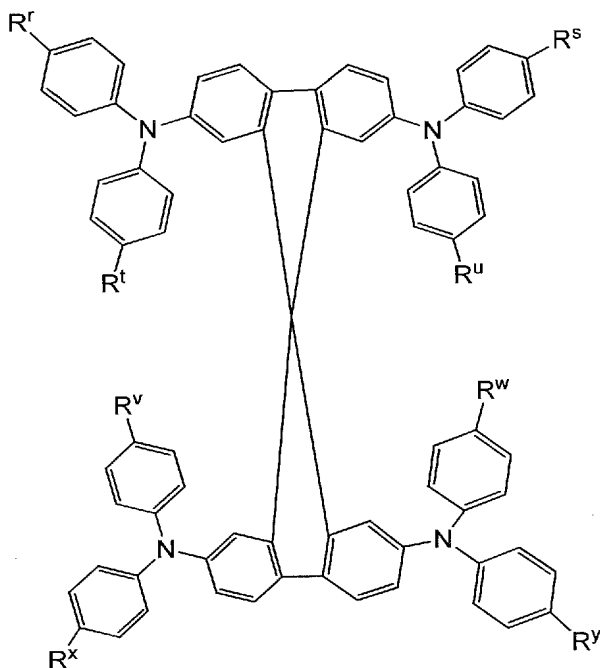
More preferably, the spiro compound has a structure of the following formula:



- 5 where the aryl¹, aryl², aryl³, aryl⁴, aryl⁵, aryl⁶, aryl⁷ and aryl⁸ radicals are each independently selected from substituted aryl radicals and heteroaryl radicals, especially from substituted phenyl radicals, where the aryl radicals and heteroaryl radicals, preferably the phenyl radicals, are each independently substituted, preferably in each case by one or more substituents selected from the group consisting of -O-alkyl, -OH, -F, -Cl, -Br and -I, where alkyl is preferably
 10 methyl, ethyl, propyl or isopropyl. More preferably, the phenyl radicals are each independently substituted, in each case by one or more substituents selected from the group consisting of -O-Me, -OH, -F, -Cl, -Br and -I.

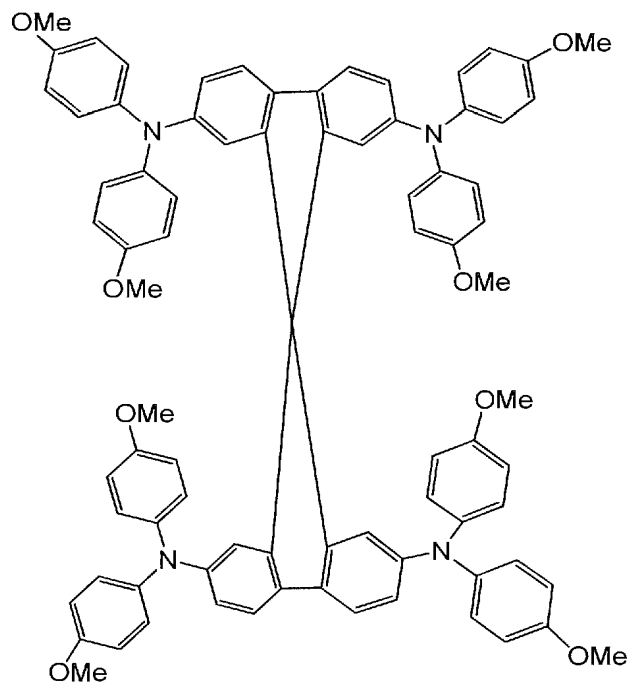
Further preferably, the spiro compound is a compound of the following formula:

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- where Rʳ, Rˢ, Rᵗ, Rᵁ, Rᵛ, Rʷ, Rˣ and Rʸ are each independently selected from the group consisting of -O-alkyl, -OH, -F, -Cl, -Br and -I, where alkyl is preferably methyl, ethyl, propyl or
 20 isopropyl. More preferably, Rʳ, Rˢ, Rᵗ, Rᵁ, Rᵛ, Rʷ, Rˣ and Rʸ are each independently selected from the group consisting of -O-Me, -OH, -F, -Cl, -Br and -I.

More particularly, the p-type semiconductor may comprise spiro-MeOTAD or consist of spiro-MeOTAD, i.e. a compound of the formula below, commercially available, for example, from Merck KGaA, Darmstadt, Germany:



Alternatively or additionally, it is also possible to use other p-semiconducting compounds, especially low molecular weight and/or oligomeric and/or polymeric p-semiconducting compounds.

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In an alternative embodiment, the low molecular weight organic p-type semiconductor comprises one or more compounds of the above-mentioned general formula I, for which reference may be made, for example, to PCT application number PCT/EP2010/051826, which will be published after the priority date of the present application. The p-type semiconductor may comprise the at least one compound of the above-mentioned general formula I additionally or alternatively to the spiro compound described above.

15

The term "alkyl" or "alkyl group" or "alkyl radical" as used in the context of the present invention is understood to mean substituted or unsubstituted C₁-C₂₀-alkyl radicals in general. Preference is given to C₁- to C₁₀-alkyl radicals, particular preference to C₁- to C₈-alkyl radicals. The alkyl radicals may be either straight-chain or branched. In addition, the alkyl radicals may be substituted by one or more substituents selected from the group consisting of C₁-C₂₀-alkoxy, halogen, preferably F, and C₆-C₃₀-aryl which may in turn be substituted or unsubstituted.

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Examples of suitable alkyl groups are methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl and octyl, and also isopropyl, isobutyl, isopentyl, sec-butyl, tert-butyl, neopentyl, 3,3-dimethylbutyl, 2-

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ethylhexyl, and also derivatives of the alkyl groups mentioned substituted by C₆-C₃₀-aryl, C₁-C₂₀-alkoxy and/or halogen, especially F, for example CF₃.

The term "aryl" or "aryl group" or "aryl radical" as used in the context of the present invention is understood to mean optionally substituted C₆-C₃₀-aryl radicals which are derived from monocyclic, bicyclic, tricyclic or else multicyclic aromatic rings, where the aromatic rings do not comprise any ring heteroatoms. The aryl radical preferably comprises 5- and/or 6-membered aromatic rings. When the aryls are not monocyclic systems, in the case of the term "aryl" for the second ring, the saturated form (perhydro form) or the partly unsaturated form (for example the dihydro form or tetrahydro form), provided the particular forms are known and stable, is also possible. The term "aryl" in the context of the present invention thus comprises, for example, also bicyclic or tricyclic radicals in which either both or all three radicals are aromatic, and also bicyclic or tricyclic radicals in which only one ring is aromatic, and also tricyclic radicals in which two rings are aromatic. Examples of aryl are: phenyl, naphthyl, indanyl, 1,2-dihydronaphthenyl, 1,4-dihydronaphthenyl, fluorenyl, indenyl, anthracenyl, phenanthrenyl or 1,2,3,4-tetrahydronaphthyl. Particular preference is given to C₆-C₁₀-aryl radicals, for example phenyl or naphthyl, very particular preference to C₆-aryl radicals, for example phenyl. In addition, the term "aryl" also comprises ring systems comprising at least two monocyclic, bicyclic or multicyclic aromatic rings joined to one another via single or double bonds. One example is that of biphenyl groups.

The term "heteroaryl" or "heteroaryl group" or "heteroaryl radical" as used in the context of the present invention is understood to mean optionally substituted 5- or 6-membered aromatic rings and multicyclic rings, for example bicyclic and tricyclic compounds having at least one heteroatom in at least one ring. The heteroaryls in the context of the invention preferably comprise 5 to 30 ring atoms. They may be monocyclic, bicyclic or tricyclic, and some can be derived from the afore-mentioned aryl by replacing at least one carbon atom in the aryl base skeleton with a heteroatom. Preferred heteroatoms are N, O and S. The heteroaryl radicals more preferably have 5 to 13 ring atoms. The base skeleton of the heteroaryl radicals is especially preferably selected from systems such as pyridine and five-membered heteroaromatics such as thiophene, pyrrole, imidazole or furan. These base skeletons may optionally be fused to one or two six-membered aromatic radicals. In addition, the term "heteroaryl" also comprises ring systems comprising at least two monocyclic, bicyclic or multicyclic aromatic rings joined to one another via single or double bonds, where at least one ring comprises a heteroatom. When the heteroaryls are not monocyclic systems, in the case of the term "heteroaryl" for at least one ring, the saturated form (perhydro form) or the partly unsaturated form (for example the dihydro form or tetrahydro form), provided the particular forms are known and stable, is also possible. The term "heteroaryl" in the context of the present invention thus comprises, for example, also bicyclic or tricyclic radicals in which either both or all three radicals are aromatic, and also bicyclic or tricyclic radicals in which only one ring is aromatic, and also tricyclic radicals in which two rings are aromatic, where at least one of the rings, i.e. at least one aromatic or one nonaromatic ring has a heteroatom. Suitable fused heteroaromatics are, for example,

carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl. The base skeleton may be substituted at one, more than one or all substitutable positions, suitable substituents being the same as have already been specified under the definition of C₆-C₃₀-aryl. However, the hetaryl radicals are preferably unsubstituted. Suitable hetaryl radicals are, for example, pyridin-2-yl, pyridin-3-yl, pyridin-4-yl, thiophen-2-yl, thiophen-3-yl, pyrrol-2-yl, pyrrol-3-yl, furan-2-yl, furan-3-yl and imidazol-2-yl and the corresponding benzofused radicals, especially carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl.

In the context of the invention the term "optionally substituted" refers to radicals in which at least one hydrogen radical of an alkyl group, aryl group or heteroaryl group has been replaced by a substituent. With regard to the type of this substituent, preference is given to alkyl radicals, for example methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl and octyl, and also isopropyl, isobutyl, isopentyl, sec-butyl, tert-butyl, neopentyl, 3,3-dimethylbutyl and 2-ethylhexyl, aryl radicals, for example C₆-C₁₀-aryl radicals, especially phenyl or naphthyl, most preferably C₆-aryl radicals, for example phenyl, and hetaryl radicals, for example pyridin-2-yl, pyridin-3-yl, pyridin-4-yl, thiophen-2-yl, thiophen-3-yl, pyrrol-2-yl, pyrrol-3-yl, furan-2-yl, furan-3-yl and imidazol-2-yl, and also the corresponding benzofused radicals, especially carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl. Further examples include the following substituents: alkenyl, alkynyl, halogen, hydroxyl.

The degree of substitution here may vary from monosubstitution up to the maximum number of possible substituents.

Preferred compounds of the formula I for use in accordance with the invention are notable in that at least two of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. The at least two radicals here may be only -OR radicals, only -NR₂ radicals, or at least one -OR and at least one -NR₂ radical.

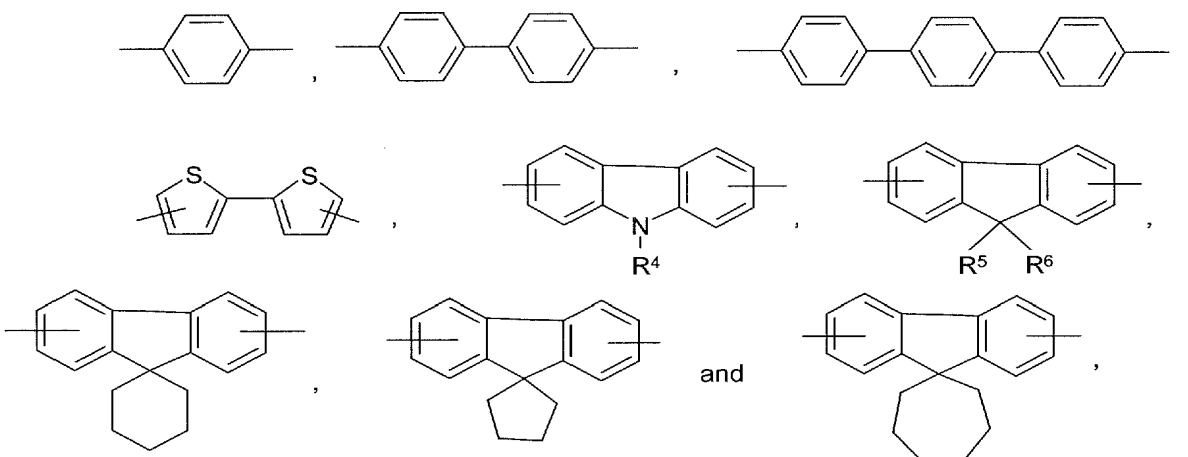
Particularly preferred compounds of the formula I for use in accordance with the invention are notable in that at least four of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. The at least four radicals here may be only -OR radicals, only -NR₂ radicals or a mixture of -OR and -NR₂ radicals.

Very particularly preferred compounds of the formula I for use in accordance with the invention are notable in that all of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. They may be only -OR radicals, only -NR₂ radicals or a mixture of -OR and -NR₂ radicals.

In all cases, the two R in the -NR₂ radicals may be different from one another, but they are preferably the same.

Preferably, A¹, A² and A³ are each independently selected from the group consisting of

76



in which

5

m is an integer from 1 to 18,

R^4 is alkyl, aryl or heteroaryl, where R^4 is preferably an aryl radical, more preferably a phenyl radical,

10

R^5, R^6 are each independently H, alkyl, aryl or heteroaryl,

where the aromatic and heteroaromatic rings of the structures shown may optionally have further substitution. The degree of substitution of the aromatic and heteroaromatic rings here may vary from monosubstitution up to the maximum number of possible substituents.

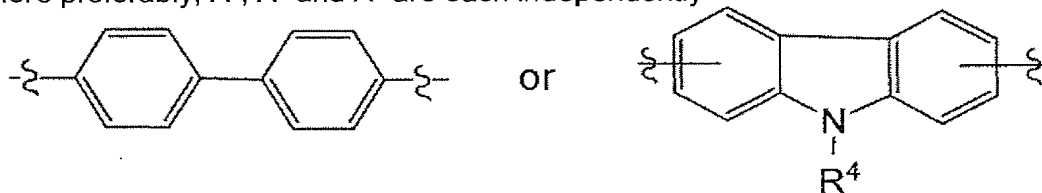
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Preferred substituents in the case of further substitution of the aromatic and heteroaromatic rings include the substituents already mentioned above for the one, two or three optionally substituted aromatic or heteroaromatic groups.

20

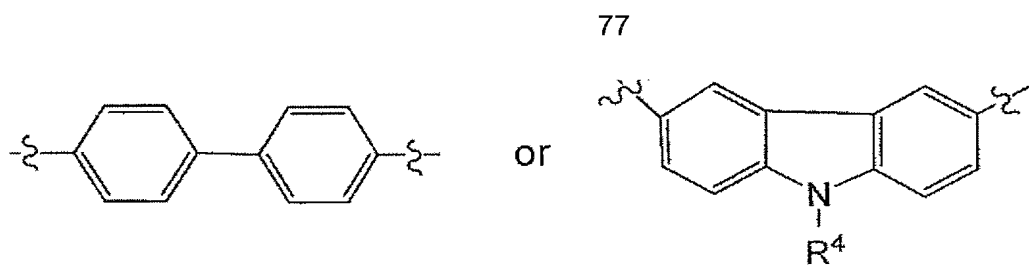
Preferably, the aromatic and heteroaromatic rings of the structures shown do not have further substitution.

More preferably, A^1, A^2 and A^3 are each independently



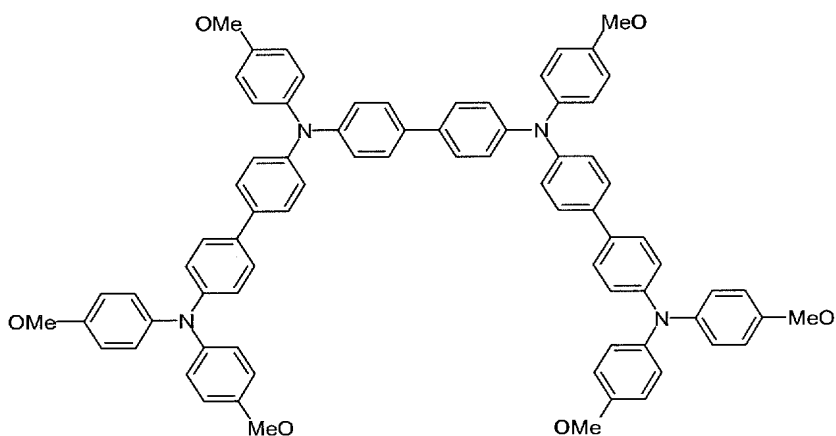
25

more preferably



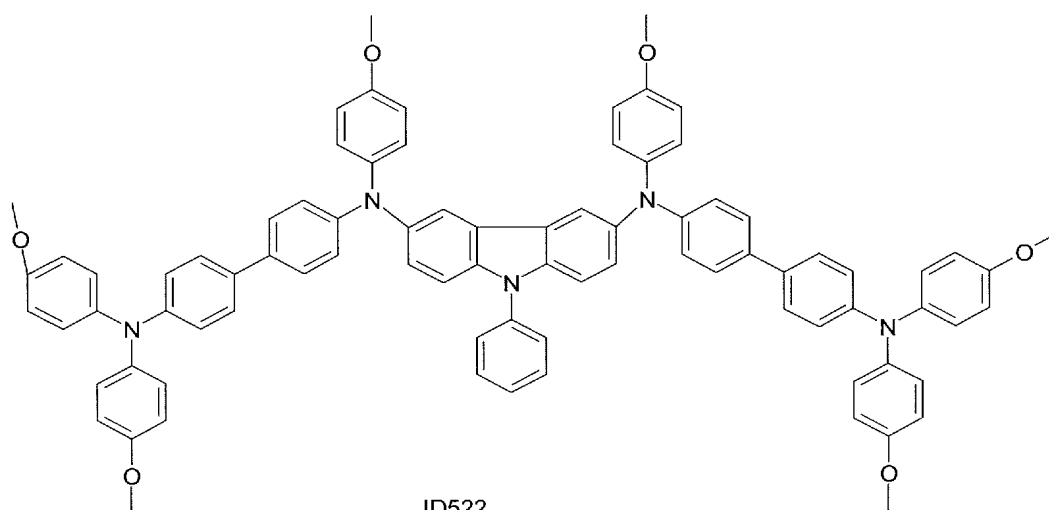
More preferably, the at least one compound of the formula (I) has one of the following structures:

5



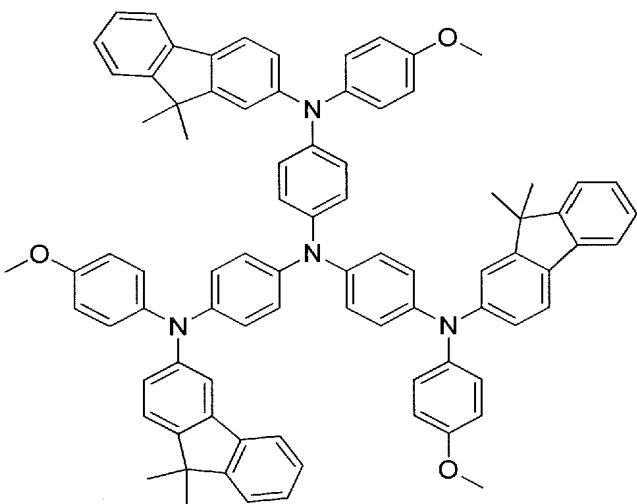
ID367

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ID522

In an alternative embodiment, the organic p-type semiconductor comprises a compound of the type ID322 having the following structure:



The compounds for use in accordance with the invention can be prepared by customary methods of organic synthesis known to those skilled in the art. References to relevant (patent) literature can additionally be found in the synthesis examples adduced below.

Second electrode

a) General remarks

The second electrode may be a bottom electrode facing the substrate or else a top electrode facing away from the substrate. As outlined above, the second electrode may be fully or partially transparent or, else, may be intransparent. As used herein, the term partially transparent refers to the fact that the second electrode may comprise transparent regions and intransparent regions.

In case the second electrode is fully or partially transparent, the second electrode may comprise at least one transparent conductive electrode material, which may be selected from the group consisting of: an inorganic transparent conductive material; an organic transparent conductive material. As an example of an inorganic conductive transparent material, a metal oxide may be used, such as ITO and/or FTO. As an example of an organic transparent conductive material, one or more electrically conductive polymer materials may be used. As used herein, the term "transparent" refers to the actual layer or layer setup of the second electrode. Thus, the transparency may be generated by using thin layers, such as layers having a thickness of less than 100 nm, more preferably less than 50 nm.

One or more materials of the following group of materials may be used: at least one metallic material, preferably a metallic material selected from the group consisting of aluminum, silver, platinum, gold; at least one nonmetallic inorganic material, preferably LiF; at least one organic conductive material, preferably at least one electrically conductive polymer and, more preferably, at least one transparent electrically conductive polymer.

The second electrode may comprise one or more metals in a pure form and/or may comprise one or more metal alloys. The second electrode may further comprise a single layer and/or may comprise a layer setup of two or more layers, wherein, preferably at least one layer is a metal layer comprising one or more metals or metal alloys. As an example, the second electrode may comprise at least one metal selected from the group listed in the preceding paragraph in a pure form and/or as a component of an alloy. As an example, the second electrode may comprise at least one alloy selected from the group consisting of: a molybdenum alloy; a niobium alloy; a neodymium alloy; an aluminum alloy. Most preferably, the second electrode may comprise at least one alloy selected from the group consisting of: MoNb; AlNd; MoNb. As an example, a layer setup comprising two or more layers of two or more of the named alloys may be used, such as a layer setup comprising the following layers: MoNb / AlNd / MoNb. As an example, the following layer thicknesses may be used: MoNb 30nm / AlNd 100 nm / MoNb 30nm. Additionally or alternatively, however, other setups and/or other layer thicknesses may be used.

The second electrode may comprise at least one metal electrode, wherein one or more metals in pure form or as a mixture/alloy, such as especially aluminum or silver may be used.

Additionally or alternatively, nonmetallic materials may be used, such as inorganic materials and/or organic materials, both alone and in combination with metal electrodes. As an example, the use of inorganic/organic mixed electrodes or multilayer electrodes is possible, for example the use of LiF/Al electrodes. Additionally or alternatively, conductive polymers may be used. Thus, the second electrode of the at least one transversal optical sensor and/or the second electrode of the at least one longitudinal optical sensor preferably may comprise one or more conductive polymers.

As an example, one or more electrically conductive polymers may be used, selected from the group consisting of: polyaniline (PANI) and/or its chemical relatives; a polythiophene and/or its chemical relatives, such as poly(3-hexylthiophene) (P3HT) and/or PEDOT:PSS (poly(3,4-ethylenedioxythiophene) poly(styrenesulfonate)). Additionally or alternatively, one or more of the conductive polymers as disclosed in EP2507286 A2, EP2205657 A1 or EP2220141 A1

In addition or alternatively, inorganic conductive materials may be used, such as inorganic conductive carbon materials, such as carbon materials selected from the group consisting of: graphite, graphene, carbon nano-tubes, carbon nano-wires.

In addition, it is also possible to use electrode designs in which the quantum efficiency of the components is increased by virtue of the photons being forced, by means of appropriate reflections, to pass through the absorbing layers at least twice. Such layer structures are also referred to as "concentrators" and are likewise described, for example, in WO 02/101838 (especially pages 23-24).

The second electrode may be identical for the at least one transversal optical sensor and the at least one longitudinal optical sensor. Still, different setups of the second electrode for the transversal optical sensor and the longitudinal optical sensor may be used.

5 b) Second electrode of the transversal sensor device

Preferably, the second electrode for the at least one transversal sensor device is at least partially transparent. As an example, the second electrode of the transversal sensor device may comprise at least one transparent electrode layer which covers a sensor region, preferably a
10 sensor area, of the transversal optical sensor. As outlined above, the at least one transparent electrode layer preferably may comprise at least one layer of an electrically conductive polymer, preferably a transparent electrically conductive polymer.

Additionally, the second electrode of the transversal sensor device may comprise two or more
15 partial electrodes, which, preferably, may be made of one or more metals, such as one or more of the metals and/or metal alloys listed above. As an example, the two or more partial electrodes may form a frame which surrounds the sensor region, preferably the sensor area, of the transversal optical sensor. The frame may have a polygonal shape, such as a rectangular or, preferably, a square shape. Preferably, on each side of the polygon, preferably the rectangle
20 or square, one partial electrodes is provided, such as a partial electrode being formed as a bar fully or partially extending along the side.

The at least one electrically conductive polymer may have an electrical conductivity of at least one order of magnitude below electrical conductivity of the material of the partial electrodes,
25 preferably of at least two orders of magnitude below. The at least one electrically conductive polymer may electrically interconnect the partial electrodes. Thus, as outlined above, the partial electrodes may form a frame surrounding a sensor region, preferably a sensor area, of the transversal optical sensor. The at least one layer of the electrically conductive polymer may form a transparent, electrically conductive layer which fully or partially covers the sensor region
30 and which electrically contacts the partial electrodes. As an example, the partial electrodes may comprise metal strips or metal bars along the sides of a rectangle, wherein an inner region of the rectangle forms the sensor region, wherein the at least one layer of the electrically conductive polymer forms one or more transparent electrode layers fully or partially covering the inner region of the rectangle and electrically contacting the metal strips or bars.

35 In case two or more partial electrodes are used which, preferably, are electrically interconnected by at least one layer of an electrically conductive polymer, each of the partial electrodes may be contacted individually, such as by one or more electrical leads or contact pads. Thus, by electrically contacting the partial electrodes, an electrical current through each of the partial
40 electrodes may be measured individually, such as by using an individual current measurement device and/or by using a sequential measurement scheme for individually detecting the currents through the partial electrodes. The detector, for the purpose of measuring the currents through

the partial electrodes, may provide an appropriate measurement setup comprising one or more current measurement devices.

c) Second electrode of the longitudinal sensor device

Generally, with regard to the at least one second electrode of the at least one longitudinal sensor device, the above-mentioned details regarding the transversal sensor device may apply mutatis mutandis. Again, the second electrode of the at least one longitudinal sensor device preferably is transparent. In case a plurality of longitudinal sensor devices is provided, such as in a stack, preferably all second electrodes of the longitudinal sensor devices are transparent but the second electrode of the last longitudinal sensor device facing away from the object. The second electrode of the last longitudinal sensor device may be transparent or intransparent.

With regard to materials which may be used for the second electrode of the longitudinal sensor device, reference may be made to the above-mentioned materials, which may be selected from metallic materials, nonmetallic inorganic materials and electrically conductive organic materials.

Again, the second electrode of the longitudinal optical sensor or, in case a plurality of longitudinal optical sensors is provided, of at least one of the longitudinal optical sensors may optionally be subdivided into partial electrodes which may be contacted individually. However, since for the purpose of the at least one longitudinal optical sensor generally only one individual longitudinal sensor signal is required per longitudinal optical sensor, the second electrode of the at least one longitudinal optical sensor may as well be designed to provide a single sensor signal and, thus, may provide a single electrode contact only.

The second electrode of the longitudinal optical sensor, again, preferably may comprise one or more layers of an electrically conductive polymer, such as one or more of the polymers mentioned above. The at least one layer of the electrically conductive polymer, which preferably is transparent, may fully or partially cover the sensor region, preferably a sensor area, of the longitudinal optical sensor. In addition, one or more contact pads may be provided which electrically contact the at least one electrically conductive polymer layer. This at least one contact pad for the second electrode of the longitudinal optical sensor preferably may be made of at least one metal, such as at least one of the methods mentioned above, and/or may fully or partially be made of at least one inorganic conductive material, such as one or more transparent conductive oxides, such as one or more of the conductive oxides mentioned above with regard to the first electrode.

Encapsulation

The at least one transversal optical sensor and/or the at least one longitudinal optical sensor may further be encapsulated and/or packaged, in order to provide protection against

environmental influences, such as oxygen and/or humidity. Thereby, an increased long-term stability may be provided.

Therein, each of the optical sensors may be encapsulated individually. Thus, an individual
5 encapsulation may be provided for each of the optical sensors, such as an encapsulation for the transversal optical sensor or each of the transversal optical sensors, and an individual encapsulation for the longitudinal optical sensor or for each of the longitudinal optical sensors. Additionally or alternatively, a plurality of optical sensors may be encapsulated as a group. Thus, an encapsulation may be provided which encapsulates more than one optical sensor,
10 such as a plurality of transversal optical sensors, a plurality of longitudinal optical sensors, or at least one transversal optical sensor and at least one longitudinal optical sensor.

For the purpose of encapsulation, various techniques may be used. Thus, the detector may comprise an airtight housing which protects the optical sensors. Additionally or alternatively,
15 specifically in case organic photodetectors and, more preferably, DSCs or sDSCs are used, an encapsulation by one or more lids interacting with a substrate of the optical sensors may be used. Thus, a lid made of a metal, a ceramic material or a glass material may be glued to a substrate of the optical sensors, wherein a layer setup is located in inner space of the lid. Two or more contact leads for contacting the at least one first electrode and the at least one second
20 electrode may be provided which may be contacted from the outside of the lid.

Various other encapsulation techniques may be used alternatively or in addition. Thus, encapsulation by one or more encapsulation layers may be provided. The at least one encapsulation layer may be deposited on top of a layer setup of the devices. Thus, one or more
25 organic and/or inorganic encapsulation materials such as one or more barrier materials may be used.

Synthesis examples:

30 Syntheses of various compounds which can be used in dye solar cells in the context of the present invention, in particular as p-type semiconductors, are listed by way of example in WO 2012/110924 A1, the content of which is herewith included by reference.

35 Overall, in the context of the present invention, the following embodiments are regarded as particularly preferred:

Embodiment 1: A detector for determining a position of at least one object, comprising:

- 40
- at least one transversal optical sensor, the transversal optical sensor being adapted to determine a transversal position of at least one light beam traveling from the object to the detector, the transversal position being a position in at least one dimension

perpendicular to an optical axis of the detector, the transversal optical sensor being adapted to generate at least one transversal sensor signal;

- at least one longitudinal optical sensor, wherein the longitudinal optical sensor has at least one sensor region, wherein the longitudinal optical sensor is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region;
- at least one evaluation device, wherein the evaluation device is designed to generate at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

Embodiment 2: The detector according to the preceding embodiment, wherein the transversal optical sensor is a photo detector having at least one first electrode, at least one second electrode and at least one photovoltaic material, wherein the photovoltaic material is embedded in between the first electrode and the second electrode, wherein the photovoltaic material is adapted to generate electric charges in response to an illumination of the photovoltaic material with light, wherein the second electrode is a split electrode having at least two partial electrodes, wherein the transversal optical sensor has a sensor region, wherein the at least one transversal sensor signal indicates a position of the light beam in the sensor region, preferably a sensor area.

Embodiment 3: The detector according to the preceding embodiment, wherein electrical currents through the partial electrodes are dependent on a position of the light beam in the sensor region.

Embodiment 4: The detector according to the preceding embodiment, wherein the transversal optical sensor is adapted to generate the transversal sensor signal in accordance with the electrical currents through the partial electrodes.

Embodiment 5: The detector according to any of the two preceding embodiments, wherein the detector, preferably the transversal optical sensor and/or the evaluation device, is adapted to derive the information on the transversal position of the object from at least one ratio of the currents through the partial electrodes.

Embodiment 6: The detector according to any of the four preceding embodiments, wherein at least four partial electrodes are provided.

Embodiment 7: The detector according to any of the five preceding embodiments, wherein the photovoltaic material comprises at least one organic photovoltaic material and wherein the transversal optical sensor is an organic photo detector.

Embodiment 8: The detector according to any of the six preceding embodiments, wherein the organic photo detector is a dye-sensitized solar cell.

5 Embodiment 9: The detector according to the preceding embodiment, wherein the dye-sensitized solar cell is a solid dye-sensitized solar cell, comprising a layer setup embedded in between the first electrode and the second electrode, the layer setup comprising at least one n-semiconducting metal oxide, at least one dye, and at least one solid p-semiconducting organic material.

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Embodiment 10: The detector according to any of the eight preceding embodiments, wherein the first electrode at least partially is made of at least one transparent conductive oxide, wherein the second electrode at least partially is made of an electrically conductive polymer, preferably a transparent electrically conductive polymer.

15

Embodiment 11: The detector according to the preceding embodiment, wherein the conductive polymer is selected from the group consisting of: a poly-3,4-ethylenedioxythiophene (PEDOT), preferably PEDOT being electrically doped with at least one counter ion, more preferably PEDOT doped with sodium polystyrene sulfonate (PEDOT:PSS); a polyaniline (PANI); a polythiophene.

20

Embodiment 12: The detector according to any of the two preceding embodiments, wherein the conductive polymer provides an electric resistivity of 0.1 – 20 k Ω between the partial electrodes, preferably an electric resistivity of 0.5 – 5.0 k Ω and, more preferably, an electric resistivity of 1.0 – 3.0 k Ω .

25

Embodiment 13: The detector according to any of the preceding embodiments, wherein at least one of the transversal optical sensor and the longitudinal optical sensor is a transparent optical sensor.

30

Embodiment 14: The detector according to the preceding embodiment, wherein the light beam passes through the transparent optical sensor before impinging on the other one of the transversal optical sensor and the longitudinal optical sensor.

35 Embodiment 15: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one imaging device.

Embodiment 16: The detector according to the preceding claim, wherein the detector comprises a stack of optical sensors, the optical sensors comprising the at least one transversal optical sensor and the at least one longitudinal optical sensor, the stack further comprising the imaging device.

40

Embodiment 17: The detector according to the preceding claim, wherein the imaging device is located in a position of the stack furthest away from the object.

Embodiment 18: The detector according to any of the three preceding embodiments, wherein the light beam passes through the at least one longitudinal optical sensor before illuminating the imaging device.

Embodiment 19: The detector according to any of the four preceding embodiments, wherein the imaging device comprises a camera.

Embodiment 20: The detector according to any of the five preceding embodiments, wherein the imaging device comprises at least one of: an inorganic camera; a monochrome camera; a multichrome camera; a full-color camera; a pixelated inorganic chip; a pixelated organic camera; a CCD chip, preferably a multi-color CCD chip or a full-color CCD chip; a CMOS chip; an IR camera; an RGB camera.

Embodiment 21: The detector according to any of the preceding embodiments, wherein the transversal optical sensor and the longitudinal optical sensor at least partially are identical.

Embodiment 22: The detector according to any of the preceding embodiments, wherein the transversal optical sensor and the longitudinal optical sensor at least partially are independent optical sensors.

Embodiment 23: The detector according to any of the preceding embodiments, wherein the transversal optical sensor and the longitudinal optical sensor are stacked along the optical axis such that a light beam travelling along the optical axis both impinges on the transversal optical sensor and on the longitudinal optical sensor.

Embodiment 24: The detector according to the preceding embodiment, wherein the light beam subsequently passes through the transversal optical sensor and the longitudinal optical sensor or vice versa.

Embodiment 25: The detector according to any of the preceding embodiments, wherein the detector furthermore has at least one modulation device for modulating the illumination.

Embodiment 26: The detector according to the preceding embodiment, wherein the detector is designed to detect at least two longitudinal sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies, wherein the evaluation device is designed to generate the at least one item of information on the longitudinal position of the object by evaluating the at least two longitudinal sensor signals.

Embodiment 27: The detector according to any of the preceding embodiments, wherein the longitudinal optical sensor is furthermore designed in such a way that the longitudinal sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination.

5

Embodiment 28: The detector according to any of the preceding embodiments, wherein the sensor region of the longitudinal optical sensor is exactly one continuous sensor region, wherein the longitudinal sensor signal is a uniform sensor signal for the entire sensor region.

10

Embodiment 29: The detector according to any of the preceding embodiments, wherein the sensor region of the transversal optical sensor and/or the sensor region of the longitudinal optical sensor is or comprises a sensor area, the sensor area being formed by a surface of the respective device, wherein the surface faces towards the object or faces away from the object.

15

Embodiment 30: The detector according to any of the preceding embodiments, wherein the longitudinal sensor signal is selected from the group consisting of a current and a voltage.

20

Embodiment 31: The detector according to any of the preceding embodiments, wherein the transversal sensor signal is selected from the group consisting of a current and a voltage or any signal derived thereof.

25

Embodiment 32: The detector according to any of the preceding embodiments, wherein the longitudinal optical sensor comprises at least one semiconductor detector, in particular an organic semiconductor detector comprising at least one organic material, preferably an organic solar cell and particularly preferably a dye solar cell or dye-sensitized solar cell, in particular a solid dye solar cell or a solid dye-sensitized solar cell.

30

Embodiment 33: The detector according to the preceding embodiment, wherein the longitudinal optical sensor comprises at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode.

35

Embodiment 34: The detector according to the preceding embodiment, wherein both the first electrode and the second electrode are transparent.

40

Embodiment 35: The detector according to any of the preceding embodiments, wherein the evaluation device is designed to generate the at least one item of information on the longitudinal position of the object from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object with respect to the detector, preferably taking account of a known power of the illumination and optionally taking account of a modulation frequency with which the illumination is modulated.

Embodiment 36: The detector according to any of the preceding embodiments, furthermore comprising at least one transfer device, wherein the transfer device is designed to feed light emerging from the object to the transversal optical sensor and the longitudinal optical sensor.

- 5 Embodiment 37: The detector according to any of the preceding embodiments, furthermore comprising at least one illumination source.

10 Embodiment 38: The detector according to the preceding embodiment, wherein the illumination source is selected from: an illumination source, which is at least partly connected to the object and/or is at least partly identical to the object; an illumination source which is designed to at least partly illuminate the object with a primary radiation, wherein the light beam preferably is generated by a reflection of the primary radiation on the object and/or by light emission by the object itself, stimulated by the primary radiation.

- 15 Embodiment 39: The detector according to any of the preceding embodiments, wherein the detector has a plurality of longitudinal optical sensors, wherein the longitudinal optical sensors are stacked.

20 Embodiment 40: The detector according to the preceding embodiment, wherein the longitudinal optical sensors are stacked along the optical axis.

25 Embodiment 41: The detector according to any of the two preceding embodiments, wherein the longitudinal optical sensors form a longitudinal optical sensor stack, wherein the sensor regions of the longitudinal optical sensors are oriented perpendicular to the optical axis.

Embodiment 42: The detector according to any of the three preceding embodiments, wherein the transversal optical sensor is located on a side of the stacked longitudinal optical sensors facing the object.

30 Embodiment 43: The detector according to any of the four preceding embodiments, wherein the longitudinal optical sensors are arranged such that a light beam from the object illuminates all longitudinal optical sensors, preferably sequentially, wherein at least one longitudinal sensor signal is generated by each longitudinal optical sensor.

35 Embodiment 44: The detector according to any of the five preceding embodiments, wherein a last longitudinal optical sensor is arranged in a manner that the light beam first illuminates all other longitudinal optical sensors apart from the last longitudinal optical sensor, until the light beam finally illuminates the last longitudinal optical sensor.

40 Embodiment 45: The detector according to the preceding embodiment, wherein the last longitudinal optical sensor is intransparent with respect to the light beam.

Embodiment 46: The detector according to any of the seven preceding embodiments, wherein at least two of the longitudinal optical sensors have a differing spectral sensitivities.

Embodiment 47: The detector according to the preceding embodiment, wherein the differing spectral sensitivities are arranged over a spectral range allowing each of the at least two of the longitudinal optical sensors to be sensitive to a specific color.

Embodiment 48: The detector according to the preceding embodiment, wherein the longitudinal optical sensors comprise at least one first longitudinal optical sensor absorbing light in a first spectral range, wherein the longitudinal optical sensors further comprise at least one second longitudinal optical sensor absorbing light in a second spectral range being different from the first spectral range, wherein the longitudinal optical sensors further comprise at least one third longitudinal optical sensor absorbing light in a third spectral range which includes both the first spectral range and the second spectral range.

Embodiment 49: The detector according to any of the six preceding embodiments, wherein the evaluation device is adapted to normalize the longitudinal sensor signals and to generate the information on the longitudinal position of the object independent from an intensity of the light beam.

Embodiment 50: The detector according to any of the seven preceding embodiments, wherein the evaluation device is adapted to recognize whether the light beam widens or narrows, by comparing the longitudinal sensor signals of different longitudinal sensors.

Embodiment 51: The detector according to any of the preceding embodiments, wherein the stack of at least two optical sensors is partially or fully be immersed in an oil, in a liquid and/or in a solid material.

Embodiment 52: The detector according to the preceding embodiment, wherein the oil, the liquid, and/or the solid material are transparent at least over a part of the ultraviolet, visible, and/or infrared spectral range.

Embodiment 53: The detector according to any of the two preceding embodiments, wherein the solid material is generable by using at least one curable substance and treating the curable substance with a treatment, by which treatment the curable substance is cured into the solid material.

Embodiment 54: The detector according to any of the three preceding embodiments, wherein a region between the at least two optical sensors is partially or fully filled with a substance.

Embodiment 55: The detector according to the preceding embodiment, wherein the substance exhibits a refractive index differing from that of the optical sensors adjoining to the substance on one or both sides of the region.

- 5 Embodiment 56: The detector according to any of the preceding embodiments, wherein the at least one transversal optical sensor and/or the at least one longitudinal optical sensor uses at least two transparent substrates.

- 10 Embodiment 57: The detector according to the preceding embodiment, wherein the substrates exhibit identical properties.

Embodiment 58: The detector according to any of the two preceding embodiments, wherein the substrates differ from each other with regard to a geometrical quantity and/or to a material quantity related to the substrates.

- 15 Embodiment 59: The detector according to the preceding embodiment, wherein the substrates differ from each other by the thickness.

- 20 Embodiment 60: The detector according to any of the two preceding embodiments, wherein the substrates differ from each other by shape.

- 25 Embodiment 61: The detector according to the preceding embodiment, wherein the shape is selected from the group comprising a planar, a planar-convex, a planar-concave, a biconvex, a biconcave or any other form employed for optical purposes.

Embodiment 62: The detector according to any of the six preceding embodiments, wherein the substrates are rigid or flexible.

- 30 Embodiment 63: The detector according to any of the seven preceding embodiments, wherein the substrates are covered or coated.

Embodiment 63: The detector according to any of the eight preceding embodiments, wherein the substrate is shaped in a manner that it exhibits a mirror effect.

- 35 Embodiment 64: The detector according to the preceding embodiment, wherein the substrate is shaped in a manner that it exhibits the effect of a dichroic mirror.

- 40 Embodiment 65: The detector according to any of the preceding embodiments, wherein the evaluation device is adapted to generate the at least one item of information on the longitudinal position of the object by determining a diameter of the light beam from the at least one longitudinal sensor signal.

Embodiment 66: The detector according to the preceding embodiment, wherein the evaluation device is adapted to compare the diameter of the light beam with known beam properties of the light beam in order to determine the at least one item of information on the longitudinal position of the object, preferably from a known dependency of a beam diameter of the light beam on at least one propagation coordinate in a direction of propagation of the light beam and/or from a known Gaussian profile of the light beam.

Embodiment 67: The detector according to any of the preceding embodiments, wherein the detector comprises an optically sensitive element.

Embodiment 68: The detector according to the preceding embodiment, wherein the optically sensitive element is located between the transfer device and the optical sensor.

Embodiment 69: The detector according to any of the two preceding embodiments, wherein the optically sensitive element comprises a wavelength-sensitive element, a phase-sensitive element, and/or a polarization-sensitive element.

Embodiment 70: The detector according to the preceding embodiment, wherein the wavelength-sensitive element comprises one or more of a prism, a grating, a dichroitic mirror, a color wheel, or a color drum.

Embodiment 71: The detector according to the preceding embodiment, wherein the color wheel comprises a sequential color recapture wheel.

Embodiment 72: The detector according to any of the two preceding embodiments, wherein the color wheel or the color drum comprises two or more of at least one red, green, blue, white, cyan, yellow, or magenta segment.

Embodiment 73: The detector according to embodiment 54, wherein the polarization-sensitive element comprises a filter wheel using an elliptically polarizing filter.

Embodiment 74: The detector according to the preceding embodiment, wherein the polarization-sensitive element comprises a filter wheel using a circular polarizing filter.

Embodiment 75: The detector according to any of the two preceding embodiments, wherein the optically sensitive element comprises at least two wheels, at least one first wheel and at least one second wheel, wherein the first wheel constitutes a color wheel and wherein the second wheel constitutes an elliptically polarizing filter.

Embodiment 76: An arrangement comprising at least two detectors according to any of the preceding embodiments.

Embodiment 77: An arrangement according to the preceding embodiment, wherein the at least two detectors have identical optical properties.

5 Embodiment 78: The arrangement according to any of the two preceding embodiments, wherein the arrangement further comprises at least one illumination source.

10 Embodiment 79: A human-machine interface for exchanging at least one item of information between a user and a machine, in particular for inputting control commands, wherein the human-machine interface comprises at least one detector according to any of the preceding
embodiments relating to a detector, wherein the human-machine interface is designed to generate at least one item of geometrical information of the user by means of the detector wherein the human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command.

15 Embodiment 80: The human-machine interface according to the preceding embodiment, wherein the at least one item of geometrical information of the user is selected from the group consisting of: a position of a body of the user; a position of at least one body part of the user; an orientation of a body of the user; an orientation of at least one body part of the user.

20 Embodiment 81: The human-machine interface according to any of the two preceding embodiments, wherein the human-machine interface further comprises at least one beacon device connectable to the user, wherein the human-machine interface is adapted such that the detector may generate an information on the position of the at least one beacon device.

25 Embodiment 82: The human-machine interface according to the preceding embodiment, wherein the beacon device is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user.

30 Embodiment 83: The human-machine interface according to the preceding embodiment, wherein the beacon device comprises at least one illumination source adapted to generate at least one light beam to be transmitted to the detector.

35 Embodiment 84: The human-machine interface according to any of the two preceding embodiments, wherein the beacon device comprises at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

40 Embodiment 85: The human-machine interface according to any of the three preceding embodiments, wherein the beacon device comprises at least one of: a garment to be worn by the user, preferably a garment selected from the group consisting of a glove, a jacket, a hat, shoes, trousers and a suit; a stick that may be held by hand; a bat; a club; a racket; a cane; a toy, such as a toy gun.

Embodiment 86: An entertainment device for carrying out at least one entertainment function, in particular a game, wherein the entertainment device comprises at least one human-machine interface according to any of the preceding embodiments referring to a human-machine interface, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 87: A tracking system for tracking the position of at least one movable object, the tracking system comprising at least one detector according to any of the preceding embodiments referring to a detector, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object, each position comprising at least one item of information on a transversal position of the object at a specific point in time and at least one item of information on a longitudinal position of the object at a specific point in time.

Embodiment 88: The tracking system according to the preceding embodiment, wherein the tracking system further comprises at least one beacon device connectable to the object, wherein the tracking system is adapted such that the detector may generate an information on the position of the object of the at least one beacon device.

Embodiment 89: The tracking system according to the preceding embodiment, wherein the beacon device comprises at least one illumination source adapted to generate at least one light beam to be transmitted to the detector.

Embodiment 90: The tracking system according to any of the two preceding embodiments, wherein the beacon device comprises at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

Embodiment 91: The tracking system according to any of the preceding embodiments referring to a tracking system, wherein the track controller is adapted to initiate at least one action in accordance with an actual position of the object.

Embodiment 92: The tracking system according to the preceding embodiment, wherein the action is selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

Embodiment 93: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 94: A method for determining a position of at least one object, in particular using a detector according to any of the preceding embodiments relating to a detector,

- wherein at least one transversal optical sensor of a detector is used, wherein the transversal optical sensor determines a transversal position of at least one light beam traveling from the object to the detector, the transversal position being a position in at least one dimension perpendicular to an optical axis of the detector, wherein the transversal optical sensor generates at least one transversal sensor signal;
- wherein at least one longitudinal optical sensor of the detector is used, wherein the longitudinal optical sensor has at least one sensor region, wherein the longitudinal optical sensor generates at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region;
- wherein at least one evaluation device is used, wherein the evaluation device generates at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and wherein the evaluation device further generates at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

Embodiment 95: The use of a detector according to any of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a distance measurement, in particular in traffic technology; a position measurement, in particular in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented alone or with several in combination. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1A shows an exemplary embodiment of a detector according to the present invention;

- Figure 1B shows a further exemplary embodiment of a detector according to the present invention;
- 5 Figure 1C shows a further exemplary embodiment of a detector according to the present invention;
- Figure 2A and 2B show different views of an embodiment of a transversal detector which may be used in the detector of the present invention;
- 10 Figures 3A to 3D show principles of generating transversal sensor signals and deriving information on a transversal position of an object;
- Figures 4A to 4C show different views of embodiments of a longitudinal optical sensor which may be used in the detector according to the present invention;
- 15 Figures 5A to 5E show the principle of generating longitudinal sensor signals and deriving information on a longitudinal position of an object; and
- 20 Figure 6 shows a schematic embodiment of a human-machine interface and of an entertainment device according to the present invention.

Exemplary embodiments

25 Detector

Figure 1A illustrates, in a highly schematic illustration, an exemplary embodiment of a detector 110 according to the invention, for determining a position of at least one object 112. The detector 110 preferably may form a camera 111 or may be part of a camera 111. Other
30 embodiments are feasible.

The detector 110 comprises a plurality of optical sensors 114, which, in the specific embodiment, are all stacked along an optical axis 116 of the detector 110. Specifically, the optical axis 116 may be an axis of symmetry and/or rotation of the setup of the optical sensors
35 114. The optical sensors 114 may be located inside a housing 118 of the detector 110. Further, at least one transfer device 120 may be comprised, such as one or more optical systems, preferably comprising one or more lenses 122. An opening 124 in the housing 118, which, preferably, is located concentrically with regard to the optical axis 116, preferably defines a direction of view 126 of the detector 110. A coordinate system 128 may be defined, in which a
40 direction parallel or antiparallel to the optical axis 116 is defined as a longitudinal direction, whereas directions perpendicular to the optical axis 116 may be defined as transversal directions. In the coordinate system 128, symbolically depicted in Fig. 1A, a longitudinal

direction is denoted by z and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 128 are feasible.

The optical sensors 114 comprise at least one transversal optical sensor 130 and, in this embodiment, a plurality of longitudinal optical sensors 132. The longitudinal optical sensors 132 form a longitudinal optical sensor stack 134. In the embodiment shown in Fig. 1A, five longitudinal sensors 132 are depicted. It shall be noted, however, that embodiments having a different number of longitudinal optical sensors 132 are feasible.

The transversal optical sensor 132 comprises a sensor region 136, which, preferably, is transparent to a light beam 138 travelling from the object 112 to the detector 110. The transversal optical sensor 130 is adapted to determine a transversal position of the light beam 138 in one or more transversal directions, such as in direction x and/or in direction y. Therein, embodiments are feasible in which a transversal position in only one transversal direction is determined, embodiments in which transversal positions in more than one transversal direction are determined by one and the same transversal optical sensor 130, and embodiments in which a transversal position in a first transversal direction is determined by a first transversal optical sensor and wherein at least one further transversal position in at least one further transversal direction is determined by at least one further transversal optical sensor.

The at least one transversal optical sensor 130 is adapted to generate at least one transversal sensor signal. This transversal sensor signal may be transmitted by one or more transversal signal leads 140 to at least one evaluation device 142 of the detector 110, which will be explained in further detail below.

The longitudinal optical sensors 132 each comprise at least one sensor region 136, too. Preferably, one, more or all of the longitudinal optical sensors 132 are transparent, but the last longitudinal optical sensor 144 of the longitudinal optical sensor stack 134, i.e. the longitudinal optical sensor 132 on the side of the stack 134 facing away from the object 112. This last longitudinal sensor 144 may fully or partially be intransparent.

Each of the longitudinal optical sensors 132 is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the respective sensor region 136 by the light beam 138. The longitudinal sensor signals, given the same total power of the illumination, is dependent on a beam cross-section of the light beam 138 in the respective sensor region 136, as will be outlined in further detail below. Via one or more longitudinal signal leads 146, the longitudinal sensor signals may be transmitted to the evaluation device 142. As will be outlined in further detail below, the evaluation device may be designed to generate at least one item of information on at least one transversal position of the object 112 by evaluating the at least one transversal sensor signal and to generate at least one item of information on at least one longitudinal position of the object 112 by evaluating the longitudinal sensor signal. For this purpose, the evaluation device 142 may comprise one or more electronic devices and/or

one or more software components, in order to evaluate the sensor signals, which is symbolically denoted by transversal evaluation unit 148 (denoted by "xy") and longitudinal evaluation unit 150 (denoted by "z"). By combining results derived by these evolution units 148, 150, a position information 152, preferably a three-dimensional position information, may be generated
5 (denoted by "x, y, z").

The evaluation device 142 may be part of a data processing device 154 and/or may comprise one or more data processing devices 154. The evaluation device 142 may be fully or partially integrated into the housing 118 and/or may fully or partially be embodied as a separate device
10 which is electrically connected in a wireless or wire-bound fashion to the optical sensors 114. The evaluation device 142 may further comprise one or more additional components, such as one or more electronic hardware components and/or one or more software components, such as one or more measurement units (not depicted in Fig. 1A) and/or one or more transformation units 156. Symbolically, in Fig. 1A, one optional transformation unit 156 is depicted which may
15 be adapted to transform at least two transversal sensor signals into a common signal or common information.

In the following, embodiments of the transversal optical sensor 130 and the at least one longitudinal optical sensor 132 are disclosed. It shall be noted, however, that other
20 embodiments are feasible. Thus, in the embodiments disclosed hereinafter, the optical sensors 114 are all designed as solid dye-sensitized solar cells (sDSCs). It shall be noted, however, that other embodiments are feasible.

Fig. 1B illustrates, in a highly schematic illustration, a further exemplary embodiment of the detector 110 according to the present invention, for determining a position of the at least one object 112. In this particular embodiment, the detector 110 may comprise one or more
25 illumination sources 192, which may include an ambient light source and/or an artificial light source, and/or may comprise one or more reflective elements which may, for example, be connected to the object 112 for reflecting one or more primary light beams 206, as indicated in Fig. 1B. In addition or alternatively, the light beam 138 which emerges from the object 112 can
30 fully or partially be generated by the object 112 itself, for example in the form of a luminescent radiation.

The optical sensors 114 comprise at least one transversal optical sensor 130 and, in this
35 embodiment, a plurality of longitudinal optical sensors 132. The longitudinal optical sensors 132 form a longitudinal optical sensor stack 134. In the embodiment shown in Fig. 1B, five longitudinal optical sensors 132 are depicted, wherein the last longitudinal optical sensor 144 comprises a single longitudinal optical sensor 132 out of the plurality of the longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 which faces away from the object
40 112. In this embodiment, it is particularly preferred to arrange the last longitudinal optical sensor 144 in the longitudinal optical sensor stack 134 in a manner that a light beam 138 first travels

through the plurality of the longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 until it impinges the last longitudinal optical sensor 144.

The last longitudinal optical sensor 144 may be configured in various ways. Thus, the last longitudinal optical sensor 144 can for example be part of the detector 110 within the detector housing 118. Alternatively, the last longitudinal optical sensor 144 may be arranged outside the detector housing 118. It shall be emphasized that embodiments having a different number of longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 are feasible.

Whereas the plurality of the longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 with the exception of the last longitudinal optical sensor 144 preferably are at least partially transparent, particularly to enabling a high relative intensity at each of these longitudinal optical sensors 132, the last longitudinal optical sensor 144 may either be transparent or intransparent. In case the last longitudinal optical sensor 144 is transparent, it may be feasible to further place an additional optical sensor behind the longitudinal optical sensor stack 134, such as a separate imaging device 157 in a manner that a light beam 138 first travels through the plurality of the longitudinal optical sensors 132 including the last longitudinal optical sensor 144 until it impinges on the imaging device 157.

The imaging device 157 may be configured in various ways. Thus, the imaging device 157 can for example be part of the detector 110 within the detector housing 118. Alternatively, the imaging device 157 may be separately located outside the detector housing 118. The imaging device 157 may be fully or partially transparent or intransparent. The imaging device 157 may be or may comprise an organic imaging device or an inorganic imaging device. Preferably, the imaging device 157 may comprise at least one matrix of pixels, wherein the matrix of pixels is particularly selected from the group consisting of: an inorganic semiconductor sensor device such as a CCD chip and/or a CMOS chip; an organic semiconductor sensor device. The imaging device signal may be transmitted by one or more imaging device signal leads 159 to at least one evaluation device 142 of the detector 110, which will be explained in further detail below.

In a further preferred embodiment, at least two of the longitudinal optical sensors 132 exhibit a differing spectral sensitivity. Preferably, a last longitudinal optical sensor 144, which preferably is intransparent, is configured to absorb all over the spectral ranges of the at least two of the longitudinal optical sensors have the differing spectral sensitivity. By way of example, the differing spectral sensitivity of at least two of the longitudinal optical sensors 132 varies over spectral range in a manner that the at least two of the longitudinal optical sensors 132 are sensitive to a specific color whereas the intransparent last longitudinal optical sensor 144 is adapted to be sensitive to all colors of the visible spectra range. In a specific embodiment, the longitudinal optical sensors 132 with the exception of the last longitudinal optical sensor 144 are each sensitive to a different specific color with little overlap. This feature particularly allows reliably acquiring depth information for a colored object 112 irrespective of the specific color of

the object 112, for which information at least two of the longitudinal optical sensors 132 are required, as described above. This feature is realized by the fact that for each color, absorption of the light beam 118 always occurs at least in two of the longitudinal optical sensors 132, i.e. in the one of the longitudinal optical sensor 132 which is sensitive to the specific color of the object 112, as well as in the last longitudinal optical sensor 144 being sensitive to all colors.

With respect to the other features as presented in an exemplary fashion in Fig. 1B, reference is made to the above description of Fig. 1A.

A further exemplary embodiment of the detector 110 according to the present invention is shown in Fig. 1C in a highly schematic manner. In this particular embodiment, the detector 110 may comprise one or more optically sensitive elements 161 which may particularly be placed along the optical axis 116 of the detector 110. As depicted herein, the optically sensitive element 161 may preferably be located between the transfer device 120, which here comprises one lens 122, and the optical sensor 114, which, in this embodiment, comprises at least one transversal optical sensor 130 and a plurality of longitudinal optical sensors 132 forming a longitudinal optical sensor stack 134. The optically sensitive element 161 may be configured in a number of ways, such as a wavelength-sensitive element, a phase-sensitive element, or a polarization-sensitive element.

As a non-limiting example, a wavelength-sensitive element comprising a color wheel 163 may be employed here as the optically sensitive element 161. Alternative wavelength-sensitive elements may comprise one or more of a prism, a grating, and/or a dichroitic mirror. The color-wheel 163 may especially comprise a circle around which at least two, preferably two to eight, separate segments are arranged which may differ from each other by their optical effect, particularly a grade of transmission, to the impinging light beam 138. Herein, the separate segments may each act as a filter which may allow determining a contribution of the colors red (R), green (G), and blue (B), also designated as "RGB color wheel". Alternatively, the color wheel 163 commonly designated as a "RGB-W color wheel" and additionally comprising a white (W) segment may be employed. As a further alternative, the color wheel 163 may comprise clear or transparent segments which may be used to increase luminous efficiency. Furthermore, the color wheel 163 may, alternatively or additionally, comprise one or more of the colors cyan, yellow, and magenta. A further alternative may comprise a so-called "color drum", wherein the segments are arranged on the inner surface of a drum where the incident light beam may be directed through in order to impinge an optical sensor.

Moreover, a sequential color recapture wheel (SCR wheel) may be generated from RGB dichroic coatings arranged in an Archimedes spiral pattern. Herein, the Archimedes spiral may exhibit the property that the boundaries between two segments may move at a constant speed in radial direction. Furthermore, it may be possible to include also here white or clear segments. Further examples for rotating color wheels, such as known from video beamer devices, and for color drums may be found under the address www.hcinema.de/farbrad.htm.

With respect to the other features as presented in an exemplary fashion in Fig. 1C, reference is made to the above descriptions concerning Figs. 1A and/or 1B.

5 In Figs. 2A and 2B, different views of a potential embodiment of a transversal optical sensor 130 are depicted. Therein, Fig. 2A shows a top-view on a layer setup of the transversal optical sensor 130, whereas Fig. 2B shows a partial cross-sectional view of the layer setup in a schematic setup. For alternative embodiments of the layer setup, reference may be made to the disclosure above.

10 The transversal optical sensor 130 comprises a transparent substrate 158, such as a substrate made of glass and/or a transparent plastic material. The setup further comprises a first electrode 160, an optical blocking layer 162, at least one n-semiconducting metal oxide 164, sensitized with at least one dye 166, at least one p-semiconducting organic material 168 and at
15 least one second electrode 170. These elements are depicted in Fig. 2B. The setup may further comprise at least one encapsulation 172 which is not depicted in Fig. 2B and which is symbolically depicted in the top-view of Fig. 2A, which may cover a sensor region 136 of the transversal optical sensor 130.

20 As an exemplary embodiment, the substrate 158 may be made of glass, the first electrode 160 may fully or partially be made of fluorine-doped tin oxide (FTO), the blocking layer 162 may be made of dense titanium dioxide (TiO₂), the n-semiconducting metal oxide 164 may be made of nonporous titanium dioxide, the p-semiconducting organic material 168 may be made of spiro-MiOTAD, and the second electrode 170 may comprise PEDOT:PSS. Further, dye ID504, as
25 e.g. disclosed in WO 2012/110924 A1, may be used. Other embodiments are feasible.

As depicted in Figs. 2A and 2B, the first electrode 160 may be a large-area electrode, which may be contacted by a single electrode contact 174. As depicted in the top-view in Fig. 2A, the electrode contacts 174 of the first electrode 160 may be located in corners of the transversal
30 optical sensor 130. By providing more than one electrode contact 174, a redundancy may be generated, and resistive losses over the first electrode 160 might be eliminated, thereby generating a common signal for the first electrode 160.

Contrarily, the second electrode 170 comprises at least two partial electrodes 176. As can be
35 seen in the top-view in Fig. 2A, the second electrode 170 may comprise at least two partial electrodes 178 for an x-direction, and at least two partial electrodes 180 for a y-direction via contact leads 182, these partial electrodes 176 may be contacted electrically through the encapsulation 172.

40 The partial electrodes 176, in this specific embodiment, form a frame which surrounds the sensor region 136. As an example, a rectangular or, more preferably, a square frame may be formed. By using appropriate current measurement devices, electrode currents through the

partial electrodes 176 may be determined individually, such as by current measurement devices implemented into the evaluation device 142. By comparing e.g. electrode currents through the two single x-partial electrodes 178, and by comparing the electrode currents through the individual y-partial electrodes 180, x- and y-coordinates of a light spot 184 generated by the light beam 138 in the sensor region 136 may be determined, as for the outlined with respect to Figs. 3A to 3D below.

In Figs. 3A to 3D, two different situations of a positioning of the object 112 are depicted. Thus, Fig. 3A and Fig. 3B show a situation in which the object 112 is located on the optical axis 116 of the detector 110. Therein, Fig. 3A shows a side-view and Fig. 3B shows a top-view onto the sensor region 136 of the transversal optical sensor 130. The longitudinal optical sensors 132 are not depicted in this setup.

In Fig. 3C and 3D, the setup of Figs. 3A and 3B is depicted in analogous views with the object 112 shifted in a transversal direction, to an off-axis position.

It shall be noted that, in Figs. 3A and 3C, the object 112 is depicted as the source of one or more light beams 138. As will be outlined in further detail below, specifically with respect the embodiment in Fig. 6, the detector 110 may as well comprise one or more illumination sources which may be connected to the object 112 and, thus, which may emit the light beams 138, and/or which might be adapted to illuminate the object 112 and, by the object 112 reflecting primary light beams, generate the light beams 138 by reflection and/or diffusion.

According to well-known imaging equations, the object 112 is imaged onto the sensor region 136 of the transversal optical sensor 130, thereby generating an image 186 of the object 112 on the sensor region 136, which, in the following, will be considered a light spot 184 and/or a plurality of light spots 184.

As can be seen in the partial images 3B and 3D, the light spot 184 on the sensor region 136 will lead, by generating charges in the layer setup of the sDSC, electrode currents, which, in each case, are denoted by i_1 to i_4 . Therein, electrode currents i_1, i_2 denote electrode currents through partial electrodes 180 in y-direction and electrode currents i_3, i_4 denote electrode currents through partial electrodes 178 in x-direction. These electrode currents may be measured by one or more appropriate electrode measurement devices simultaneously or sequentially. By evaluating these electrode currents, x- and y-coordinates may be determined. Thus, the following equations may be used:

$$x_0 = f\left(\frac{i_3 - i_4}{i_3 + i_4}\right) \text{ and } y_0 = f\left(\frac{i_1 - i_2}{i_1 + i_2}\right).$$

Therein, f might be an arbitrary known function, such as a simple multiplication of the quotient of the currents with a known stretch factor and/or an addition of an offset. Thus, generally, the

electrode currents i_1 to i_4 might form transversal sensor signals generated by the transversal optical sensor 130, whereas the evaluation device 142 might be adapted to generate information on a transversal position, such as at least one x-coordinate and/or at least one y-coordinate, by transforming the transversal sensor signals by using a predetermined or determinable transformation algorithm and/or a known relationship.

In Figs. 4A to 4C, various views of longitudinal optical sensors 132 are shown. Therein, Fig. 4A shows a cross-sectional view of a potential layer setup, and Figs. 4B and 4C show top-view of two embodiments of potential longitudinal optical sensors 132. Therein, Fig. 4C shows a potential embodiment of the last longitudinal optical sensor 144, wherein Fig. 4B shows potential embodiments of the remaining longitudinal optical sensors 132 of the longitudinal optical sensor stack 134. Thus, the embodiment in Fig. 4B may form a transparent longitudinal optical sensor 132, whereas the embodiment in Fig. 4C may be an intransparent longitudinal optical sensor 132. Other embodiments are feasible. Thus, the last longitudinal optical sensor 144, alternatively, might also be embodied as a transparent longitudinal optical sensor 132.

As can be seen in the schematic cross-sectional view in Fig. 4A, the longitudinal optical sensor 132 again, might be embodied as an organic photo-detector, preferably as an sDSC. Thus, similarly to the setup of Fig. 2B, a layer setup using a substrate 158, a first electrode 160, a blocking layer 162, an n-semiconducting metal oxide 164 being sensitized with a dye 166, a p-semiconducting organic material 168 and a second electrode 170 may be used. Additionally, an encapsulation 172 may be provided. For potential materials of the layers, reference may be made to Fig. 2B above. Additionally or alternatively, other types of materials may be used.

It shall be noted that, in Fig. 2B, an illumination from the top is symbolically depicted, i.e. an illumination by the light beam 138 from the side of the second electrode 170. Alternatively, an illumination from the bottom, i.e. from the side of the substrate 158 and through the substrate 158, may be used. The same holds true for the setup of Fig. 4A.

However, as depicted in Fig. 4A, in a preferred orientation of the longitudinal optical sensor 132, an illumination by the light beam 138 preferably takes place from the bottom, i.e. through the transparent substrate 158. This is due to the fact that the first electrode 160 may easily be embodied as a transparent electrode, such as by using a transparent conductive oxide, such as FTO. The second electrode 170, as will be outlined in further detail below may either be transparent or, specifically, for the last longitudinal optical sensor 144, intransparent.

In Fig. 4B and Fig. 4C, different setups of the second electrode 170 are depicted. Therein, in Fig. 4B, corresponding to the cross-sectional view of Fig. 4A, the first electrode 160 may be contacted by one or more electrode contacts 174, which, as an example, may comprise one or more metal pads, similar to the setup in Fig. 2B. These electrode contacts 174 may be located in the corners of the substrate 158. Other embodiments are feasible.

The second electrode 170, however, in the setup of Fig. 4B may comprise one or more layers of a transparent electrically conductive polymer 188. As an example, similar to the setup of Figs. 2A and 2B, PEDOT:PSS may be used. Further, one or more top contacts 190 may be provided, which may be made of a metallic material, such as aluminum and/or silver. By using one or more contact leads 182, leading through the encapsulation 172, this top contact 190 may be electrically contacted.

In the exemplary embodiment shown in Fig. 4B, the top contact 190 forms a closed opened frame surrounding the sensor region 136. Thus, as opposed to the partial electrodes 176 in Figs. 2A and 2B, only one top contact 190 is required. However, the longitudinal optical sensor 132 and the transversal optical sensor 130 may be combined in one single device, such as by providing partial electrodes in the setup of Figs. 4A to 4C. Thus, in addition to the FiP effect which will be outlined in further detail below, transversal sensor signals may be generated with the longitudinal optical sensor 132. Thereby, a combined transversal and longitudinal optical sensor may be provided.

The use of the transparent electrically conductive polymer 188 allows for an embodiment of the longitudinal optical sensor 132 in which both the first electrode 160 and the second electrode 170 are at least partially transparent. The same, preferably, holds true for the transversal optical sensor 130. In Fig. 4C, however, a setup of the longitudinal optical sensor 132 is disclosed which uses an intransparent second electrode 170. Thus, as an example, the second electrode 170 may be embodied by using one or more metal layers, such as aluminum and/or silver, instead of or in addition to the at least one electrically conductive polymer 188. Thus, as an example, the electrically conductive polymer 188 may be replaced or may be reinforced by one or more metal layers which, preferably, may cover the full sensor region 136.

In Figs. 5A to 5E, the above-mentioned FiP effect shall be explained. Therein, Fig. 5A shows a side-view of a part of a detector 110, in a plain parallel to the optical axis 116, similar to the setup in Figs. 1, 3A and 3C. Of the detector 110, only the longitudinal optical sensors 132 and the transfer device 120 are depicted. Not shown is at least one transversal optical sensor 130. This transversal optical sensor 130 may be embodied as a separate optical sensor 114 and/or may be combined with one or more of the longitudinal optical sensors 132.

Again, the measurement starts with an emission and/or reflection of one or more light beams 138 by at least one object 112. The object 112 may comprise an illumination source 192, which may be considered as a part of the detector 110. Additionally or alternatively, a separate illumination source 192 may be used.

Due to a characteristic of the light beam 138 itself and/or due to beam shaping characteristics of the transfer device 120, preferably of the at least one lens 122, the beam properties of the light beam 138 in the region of the longitudinal optical sensors 132 at least partially are known. Thus,

as depicted in Fig. 5A, one or more focal points 194 might occur. In the focal point 194, a beam waist or a cross-section of the light beam 138 may assume a minimum value.

In Fig. 5B, in a top-view onto the sensor regions 136 of the longitudinal optical sensors 132 in Fig. 5A, a development of the light spots 184 generated by the light beam 138 impinging on the sensor regions 136 is depicted. As can be seen, close to the focal point 194, the cross-section of the light spot 184 assumes a minimum value.

In Fig. 5C, a photo current I of the longitudinal optical sensors 132 is given for the five cross-sections of the light spot 184 in Fig. 5B, in case longitudinal optical sensors 132 exhibiting the above-mentioned FiP effect are used. Thus, as an exemplary embodiment, five different photo currents I for the spot cross-sections as shown in Figs. 5B are shown for typical DSC devices preferably sDSC devices. The photo current I is depicted as a function of the area A of the light spot 184, which is a measure of the cross-section of the light spots 184.

As can be seen in Fig. 5C, the photo current I , even if all longitudinal optical sensors 132 are illuminated with the same total power of the illumination, the photo current I is dependent on the cross-section of the light beam 138, such as by providing a strong dependency on the cross-sectional area A and/or the beam waist of the light spot 184. Thus, the photo current is a function both of the power of the light beam 138 and of the cross-section of the light beam 138:

$$I=f(n,a).$$

Therein, I denotes the photo current provided by each longitudinal optical sensor 132, such as a photo current measured in arbitrary units, as a voltage over at least one measurement resistor and/or in amps. n denotes the overall number of photons impinging on the sensor region 136 and/or the overall power of the light beam in the sensor region 136. A denotes the beam cross-section of the light beam 138, provided in arbitrary units, as a beam waist, as a beam diameter of beam radius or as an area of the light spot 134. As an example, the beam cross-section may be calculated by the $1/e^2$ diameter of the light spot 184, i.e. a cross-sectional distance from a first point on a first side of a maximum intensity having an intensity of $1/e^2$ as compared to the maximum intensity of the light spot 184, to a point on the other side of the maximum having the same intensity. Other options of quantifying the beam cross-section are feasible.

The setup in Fig. 5C shows the photo current of a longitudinal optical sensor 132 according to the present invention which may be used in the detector 110 according to the present invention, showing the above-mentioned FiP effect. Contrarily, Fig. 5D in a diagram corresponding to the diagram of Fig. 5C, photo currents of traditional optical sensors are shown, for the same setup as depicted in Fig. 5A. As an example, silicon photo detectors may be used for this measurement. As can be seen, in these traditional measurements, the photo current or photo signal of the detectors is independent from the beam cross-section A .

Thus, by evaluating the photo currents and/or other types of longitudinal sensor signals of the longitudinal optical sensors 132 of the detector 110, the light beam 138 may be characterized. Since the optical characteristics of the light beam 138 depend on the distance of the object 112 from the detector 110, by evaluating these longitudinal sensor signals, a position of the object 112 along the optical axis 116, i.e. a z-position, may be determined. For this purpose, the photo currents of the longitudinal optical sensors 132 may be transformed, such as by using one or more known relationships between the photo current I and the position of the object 112, into at least one item of information on a longitudinal position of the object 112, i.e. a z-position. Thus, as an example, the position of the focal point 194 may be determined by evaluating the sensor signals, and a correlation between the focal point 194 and a position of the object 112 in the z-direction may be used for generating the above-mentioned information. Additionally or alternatively, a widening and/or narrowing of the light beam 138 may be evaluated by comparing the sensor signals of the longitudinal sensors 132. As an example, known beam properties may be assumed, such as a beam propagation of the light beam 138 according to Gaussian laws, using one or more Gaussian beam parameters.

Further, the use of a plurality of longitudinal optical sensors 132 provides additional advantages as opposed to the use of a single longitudinal optical sensor 132. Thus, as outlined above, the overall power of the light beam 138 generally might be unknown. By normalizing the longitudinal sensor signals, such as to a maximum value, the longitudinal sensor signals might be rendered independent from the overall power of the light beam 138, and a relationship

$$I_n = g(A)$$

may be used by using normalized photo currents and/or normalized longitudinal sensor signals, which is independent from the overall power of the light beam 138.

Additionally, by using the plurality of longitudinal optical sensors 132, an ambiguity of the longitudinal sensor signals may be resolved. Thus, as can be seen by comparing the first and the last image in Fig. 5B and/or by comparing the second and the fourth image in Fig. 5B, and/or by comparing the corresponding photo currents in Fig. 5C, longitudinal optical sensors 132 being positioned at a specific distance before or behind the focal point 194 may lead to the same longitudinal sensor signals. A similar ambiguity might arise in case the light beam 138 weakens during propagations along the optical axis 116, which might generally be corrected empirically and/or by calculation. In order to resolve this ambiguity in the z-position, the plurality of longitudinal sensor signals clearly shows the position of the focal point and of the maximum. Thus, by e.g. comparing with one or more neighboring longitudinal sensor signals, it may be determined whether a specific longitudinal optical sensor 132 is located before or beyond a focal point on the longitudinal axis.

In Figure 5E, a longitudinal sensor signal for a typical example of an sDSC is depicted, in order to demonstrate the possibility of the longitudinal sensor signal and the above-mentioned FiP

effect being dependent on a modulation frequency. In this figure, a short-circuit current I_{sc} is given as the longitudinal sensor signal on the vertical axis, in arbitrary units, for a variety of modulation frequencies f . On the horizontal axis, a longitudinal coordinate z is depicted. The longitudinal coordinate z , given in micrometers, is chosen such that a position of a focus of the light beam on the z -axis is denoted by position 0, such that all longitudinal coordinates z on the horizontal axis are given as a distance to the focal point of the light beam. Consequently, since the beam cross-section of the light beam depends on the distance from the focal point, the longitudinal coordinate in Figure 5E denotes the beam cross-section in arbitrary units. As an example, a Gaussian light beam may be assumed, with known or determinable beam parameters, in order to transform the longitudinal coordinate into a specific beam waist or beam cross-section.

In this experiment, longitudinal sensor signals are provided for a variety of modulation frequencies of the light beam, for 0 Hz (no modulation), 7 Hz, 377 Hz and 777 Hz. As can be seen in the figure, for modulation frequency 0Hz, no FiP effect or only a very small FiP effect, which may not easily be distinguished from the noise of the longitudinal sensor signal, may be detected. For higher modulation frequencies, a pronounced dependency of the longitudinal sensor signal on the cross section of the light beam may be observed. Typically, modulation frequencies in the range of 0.1 Hz to 10 kHz may be used for the detector according to the present invention, such as modulation frequencies of 0.3 Hz.

Human-machine interface, entertainment device and tracking system:

In Fig. 6, an exemplary embodiment of a human-machine interface 196 according to the invention, which can simultaneously also be embodied as an exemplary embodiment of an entertainment device 198 according to the invention or which can be a constituent part of such an entertainment device 198, is depicted. Further, the human-machine interface 196 and/or the entertainment device 198 may also form an exemplary embodiment of a tracking system 199 adapted for tracking a user 200 and/or one or more body parts of the user 200. Thus, a motion of one or more of the body parts of the user 200 may be tracked.

By way of example, at least one detector 110, according to the present invention can once again be provided, for example, in accordance with one or more of the embodiments described above, with one or a plurality of optical sensors 114, which may comprise one or more transversal optical sensors 130 and one or more longitudinal optical sensors 132. Further elements of the detector 110 can be provided, which are not illustrated in Fig. 6, such as, for example, elements of an optional transfer device 120. For a potential embodiment, reference may be made to Fig. 1A and/or B. Furthermore, one or a plurality of illumination sources 192 may be provided. Generally, with regard to these possible embodiments of the detector 110, reference can be made for example to the description above.

The human-machine interface 196 can be designed to enable an exchange of at least one item of information between a user 200 and a machine 202, which is merely indicated in Fig. 6. For example, in exchange of control commands, and/or information may be performed by using the human-machine interface 196. The machine 202 can comprise, in principle, any desired device
5 having at least one function which can be controlled and/or influenced in some way. At least one evaluation device 142 of the at least one detector 110 and/or a part thereof can, as indicated in Fig. 6, be wholly or partially integrated into said machine 201, but can, in principle, also be formed fully or partially separately from the machine 202.

The human-machine interface 196 can be designed for example to generate, by means of the detector 110, at least one item of geometrical information of the user 200 by means of the detector 110 and can assign the geometrical information at least to one item of information, in particular at least one control command. For this purpose, by way of examples, by means of the detector 110, a movement and/or a change in posture of the user 200 can be identified. For
15 example, as indicated in Fig. 6, a hand movement and/or a specific hand posture of the user 200 may be detected. Additionally or alternatively, other types of geometrical information of the user 200 may be detected by the one or more detectors 110. For this purpose, one or more positions and/or one or more position information regarding the user 200 and/or one or more body parts of the user 200 may be identified by the at least one detector 110. It is then possible
20 to recognize, for example by comparison with a corresponding command list, that the user 200 would like to effect a specific input, for example would like to give the machine 202 a control command. As an alternative or an addition to direct diametrical information about the actual user 200, it is also possible, for example, to generate at least one item of geometrical
25 information about at least one beacon device 204 attached to the user 200, such as at least one item of geometrical information about a garment of the user 200 and/or a glove and/or an article moved by the user 200, such as a stick, a bat, a club, a racket, a cane, a toy, such as a toy gun. One or more beacon devices 204 may be used. The beacon device 204 may be embodied as an active beacon device and/or as a passive beacon device. Thus, the beacon device 204 may
30 comprise one or more illumination sources 192 and/or may comprise one or more reflective elements for reflecting one or more primary light beams 206, as indicated in Fig. 6.

The machine 202 can furthermore comprise one or a plurality of further human-machine interfaces, which need not necessarily be embodied according to the invention, for example, as indicated in Fig. 6, at least one display 208 and/or at least one keyboard 210. Additionally or
35 alternatively, other types of human-machine interfaces may be provided. The machine 202 can, in principle, be any desired type of machine or combination of machines, such as a personal computer.

The at least one evaluation device 142 and/or parts thereof may further function as a track
40 controller 201 of the tracking system 199. Additionally or alternatively, one or more additional track controllers 201 may be provided, such as one or more additional data evaluation devices. The track controller 201 may be or may comprise one or more data memories, such as one or

more volatile and/or non-volatile memories. In this at least one data memory, a plurality of subsequent positions and/or orientation of one or more objects of parts of an object may be stored, in order to allow for storing a past trajectory. Additionally or alternatively, a future trajectory of the object and/or parts thereof may be predicted, such as by calculation, extrapolation or any other suitable algorithm. As an example, a past trajectory of an object or a part thereof may be extrapolated to future values, in order to predict at least one of a future position, a future orientation and a future trajectory of the object or a part thereof.

In the context of an entertainment device 198, said machine 202 can be designed for example to carry out at least one entertainment function, for example at least one game, in particular with at least one graphical display on the display 208 and, optionally, a corresponding audio output. The user 200 can input at least one item of information, for example via the human-machine interface 196 and/or one or more other interfaces, wherein the entertainment device 198 is designed to alter the entertainment function in accordance with the information. By way of example, specific movements or one or more virtual articles, for example of virtual persons in a game and/or movements of virtual vehicles in a game, may be controlled by means of corresponding movements of the user 200 and/or one or more body parts of the user 200, which, in turn, may be recognized by the detector 110. Other types of control of at least one entertainment function by the user 200, by means of the at least one detector 110, are also possible.

Exemplary embodiments of sDSCs for a 3-D position sensor:

The practical implementation of the FiP-effect of the sDSCs in the form of a 3-D sensor, and achieving good spatial resolution both in the x,y- and in the z-direction, typically may require the cells to have an active area of approximately 1 cm x 1 cm and meet certain requirements. Therefore, in the following, preferred requirements for the individual cells of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor are given. It shall be noted, however, that other embodiments are feasible.

Optical properties of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor:

As can be seen in Figures 5A to 5C, one particular current signal can imply two different spatial points (in front and behind the focus). In order to obtain unambiguous depth information on the z-axis, therefore, preferably at least two cells need to be arranged one behind the other. Unambiguous information is then derived from the ratio between the current signals of the two cells. For the sake of precise z-information, this sensor should have six cells stacked behind each other. This requires the cells to be transparent, i.e. the back electrode that normally consists of vapor-deposited silver across its entire area needs to be replaced by a transparent conducting material.

To ensure that sufficient illumination reaches the last cell and it supplies a useful current signal, the front five cells may have only low absorption at the excitation wavelength. The wavelength used for excitation should be around 700 nm.

5 Cross-resistance of the transversal optical sensors:

To achieve precise x,y resolution, there has to be a sufficient potential difference between each pair of opposite sides in this square cell. Fig. 2A shows such a transparent cell with which the x,y resolution is possible.

10

Even without a silver back electrode, sufficiently good electron transport from the p-type conductor into the oxidized dye has to be ensured across the entire surface area of the cells so that the dye is regenerated rapidly by a supply of electrons. Since the p-type conductor itself has very low conductance (10^{-5} S/cm), a conducting layer needs to be coated onto the p-type

15 conductor. Thanks to this additional layer, a defined cross-resistance R is to be achieved between the opposite sides of this square cell.

Transparency of the transversal optical sensors:

20

Due to their good conductance, normal solar cells have back electrodes (second electrodes) made of silver. The cells being developed here, however, have to be transparent, which is why the 1 cm² cell area typically requires a transparent back electrode. The material preferably used for this purpose is the conducting polymer poly(3,4-ethylenedioxythiophene)-poly(styrenesulfonate) (PEDOT:PSS) in aqueous dispersion. The conjugated polymer

25 PEDOT:PSS is highly transparent; it absorbs in the blue-green region (450-550 nm) only at considerable layer thicknesses, and only minimally in the red spectral range.

30

The additional PEDOT layer makes possible good electron transport in the p-type conductor. To improve the conductance of this layer and provide contacts, four silver electrodes 1 cm in length are vapor-deposited around the square cell. The arrangement of the silver electrodes is shown in Fig. 3.3a. Fig. 3.3b shows a cell with a transparent PEDOT back electrode.

Extinction of the cells of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor:

35

It is not only the back electrode that has to be transparent, but the entire cell. To ensure that a sufficient amount of light still reaches the last cell in the stack, the extinction of the front five cells should be as low as possible. This is determined first and foremost by the dye's absorption. The extinction of a solar cell, i.e. the absorption of light by the dye, has a decisive

40 effect on the cell's output current. Typically, the wavelength-dependent absorption spectrum has a maximum – the wavelength of maximum absorption is characteristic of the particular dye used. The more dye has been adsorbed in the np TiO₂ layer, the higher the cell's absorption.

The more dye molecules are adsorbed, the more electrons can reach the cb of the TiO₂ through optical excitation and the higher the current. A cell with higher extinction will, therefore, have a higher output current than one with low extinction.

- 5 The objective here is to obtain the maximum total current from the complete cell arrangement – which in the ideal case is divided equally among all the cells. Since the intensity of the light is attenuated by absorption in the cells, the ones located further back in the stack receive less and less light. In order, nonetheless, to obtain similar output currents from all six cells, it would make sense for the front cells to have lower extinction than the ones at the back. As a result, they will
- 10 stop less of the light reaching the following cells, which in turn will absorb a larger proportion of the already weakened light. Through optimal adjustment of the extinctions at the positions of the cells in the stack, in this way theoretically one could obtain the same current from all the cells. The extinction of a solar cell can be adjusted by staining with a dye and by controlling the thickness of the np TiO₂ layer.

15

Optimizing the extinction and the output current of the cells of the longitudinal optical sensor stack:

- 20 The last cell in the stack preferably should absorb almost all the incident light. For this reason, this cell should have maximum extinction. Starting with the current obtained under maximum extinction at the last cell, the extinctions of the front cells need to be so adjusted that all cells together supply a maximum total current, one that is distributed as uniformly as possible across all the cells.

- 25 Optimizing the stack's output currents is carried out as follows:

- The choice of dye
- Maximum extinction/maximum output current of the last cell
- Dye concentration for staining the last cell

30

- Staining time of the last cell
- Optimum thickness of the last cell's np TiO₂ layer
- Maximum output current of the complete stack
- Optimum thickness of the np TiO₂ layers of the front five cells

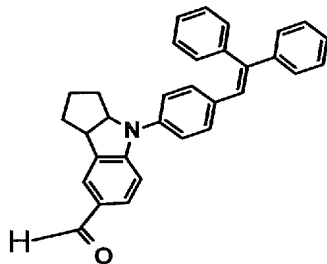
- 35 The extinction was measured with a Zeiss spectrometer MCS 501 UV-NIR using Zeiss lamp MCS 500. The results were evaluated with an Aspect Plus software program.

Choice of dye:

- 40 To begin with, a dye should be found that absorbs sufficiently at the excitation wavelength of approximately 700 nm. The ideal dye for solar cells typically has a broad absorption spectrum and should absorb completely the incident light below a wavelength of ca. 920 nm. In actual

fact, most dyes have an absorption maximum in the wavelength range between 450-600 nm; above 650 nm they usually absorb weakly or not at all.

The dye with which the first experiments were performed was ID504, as e.g. disclosed in WO 2012/110924 A1. This dye, however, turned out to exhibit only a low absorption in the range of 700 nm. Therefore, for the stack, the dye D-5 (also referred to as ID 1338) was used:



Preparation and properties of the Dye D-5 are disclosed in WO 2013/144177 A1.

Additionally or alternatively, however, other dyes may be used. The staining time, i.e. the duration of staining of the TiO₂-layer with the respective dye, turned out to have an influence on the absorption properties. The tests were performed with cells having np TiO₂ layers having a thickness of 1.3 microns. The absorption maximum of D-5 is around 550-560 nm which exhibits an extinction $\epsilon \approx 59000$ at this maximum.

In this experimental series, the dye concentration was 0.3 mM and the staining time was increased so as to be 10 to 30 minutes. A pronounced increase in extinction at longer staining times was observed, so finally staining times of 30 min. were used for D-5.

Still, even after optimizing the staining time, the absorption was determined to be rather very low. Therefore, generally, the absorption will have to be maximized by increasing the dye concentration, the staining time and the thickness of the np TiO₂ layer.

Dye concentration and staining time of the last cell in the longitudinal optical sensor stack:

Several experiments regarding the staining time and the dye concentration were performed. The standard concentration of the dye solution for a layer thickness of the TiO₂-layers of 1-2 microns was 0.5 mM. At these concentrations, the dye should already be present in excess. Here the dye concentration was increased to 0.7 mM. In order to prevent inhomogeneities across the cell's area, the dye solution was cleaned by removing undissolved dye particles and other impurities, using a 0.2 micron syringe filter, before placing the cells in it.

If the dye is present in excess, then after a 1 hour staining time a dye monolayer dye should have been adsorbed to the surface of the np TiO₂ layer, which leads to maximum absorption by

the dye being used. The maximum staining time tested here was 75 minutes, which was finally used for the cells.

Finally, cells having a layer thickness of the TiO₂-layer of 1.3 microns, a dye concentration of 0.7 mM and a staining time of 75 min was used. The cell's extinction turned out to be 0.4 at 700 nm.

The np TiO₂ layer thickness of the last cell of the longitudinal optical sensor stack:

Ultimately, the thickness of the nanoporous (np) layer and thus the TiO₂ surface area available for dye adsorption may be an important factor influencing the absorption behavior and therefore the cells' output current. So far, maximizing the extinction was done in cells with np TiO₂ layers whose thickness was 1.3 microns. Since more dye can be adsorbed in thicker np TiO₂ layers, the thickness of the TiO₂ layer was increased in steps to 3 microns, and the thickness at which the greatest output current occurred was determined.

The nanoporous TiO₂ layers were applied by spin coating. Spin coating is suitable for applying low-volatility substances dissolved in a high-volatility solvent (here: terpineol). As a starting product, TiO₂ paste made by Dyesol (DSL 18 NR-T) was used. This paste was mixed with terpineol, which decreases the viscosity of the paste. Depending on the composition ratio of the paste : terpineol mixture, and at a constant spin velocity of 4500 1/min, np TiO₂ layers of varying thicknesses are obtained. The higher the terpineol proportion, the lower the viscosity of the diluted paste and the thinner the cells will be.

The diluted TiO₂ paste was also cleaned using a 1.2 micron syringe filter to remove larger particles, before applying the paste the next day by spin coating on the cells coated with blocking layer.

When varying the np TiO₂ layer thickness, it should be noted that the concentration of the p-type conductor dissolved in chlorobenzene needs to be adjusted. Thicker np layers have a larger cavity volume that has to be filled with p-type conductor. For this reason, in the case of thicker np layers the amount of supernatant p-type conductor solution on top of the np layer is smaller. To ensure that the solid p-type conductor layer remaining on the np TiO₂ layer after spin coating has a constant thickness (the solvent evaporates during spin coating), higher p-type conductor concentrations are needed for thick np TiO₂ layers than for thin ones. The optimal p-type conductor concentrations are not known for all the TiO₂ layer thicknesses tested here. For this reason, the p-type conductor concentration is varied for the unknown layer thicknesses and the output currents compared for equal layer thicknesses but different p-type conductor concentrations.

The chosen starting value for the layer thickness variation was a 1.3 microns for the np TiO₂ layer. 1.3 microns corresponds to a TiO₂ paste : terpineol mass composition of 5 g : 5 g. A test

series with cells whose np TiO₂ layers are thicker than 1.3 microns will show at which layer thickness the largest output current is obtained from the last cell in the stack.

These cells were stained with the aforementioned optimized parameters for maximum extinction (D-5; c = 0.7 mM; staining time: 75 minutes). The extinction of these cells was found to be approximately 0.6 at 700 nm.

Since the last cell generally does not have to be transparent, the back electrode was vapor-deposited on the whole 1 cm² area directly onto the p-type conductor – without PEDOT.

The measurement results indicated that, as expected, the output currents of the cells with a whole-area back electrode (second electrode) are much higher. Highest output currents were obtained with a TiO₂ : terpineol mass ratio of 5 : 3. This corresponds to a TiO₂ layer thickness of 2-3 microns.

Therefore, in subsequent experiments, a TiO₂ paste : terpineol composition of 5 : 3 was used for the last cell in the stack. The back electrode was vapor-deposited across the entire 1 cm² cell area.

The np TiO₂ layer thickness of the front cells of the longitudinal optical sensor stack:

Starting from the maximum output current obtained with the last cell, the thicknesses of the front cells' np TiO₂ layers are to be so adjusted that every cell in the stack generates the maximum possible output current. This requires low extinction values in the front cells.

During the experiments, it turned out that, in practical terms, it is rather difficult to obtain reproducible low extinctions through the dye concentration and staining time parameters. In order to make cells with low, reproducible extinction, therefore, it makes sense to fabricate cells with thin np TiO₂ layers and to keep them in the dye solution for the time needed to ensure dye saturation of the np TiO₂ surface. The terpineol proportion in the TiO₂ layers was increased in a stepwise fashion. All the cells were stained under the same conditions. Since their extinction is intended to be decreased significantly, the dye concentration here was 0.5 mM, and the staining time was 60 minutes.

Surprisingly, in this series, the cell's output voltages turned out to begin with an increase in output current with decreasing np TiO₂ layer thickness. The optimum out of the tested TiO₂ paste dilutions turned out to be 5 : 6. At higher dilutions and therefore thinner np TiO₂ layers, the output current tends to decrease. The reason for the exception to this trend at a dilution of 5 : 9 is likely to be optimal adjustment of the p-type conductor concentration of 100 mg/ml for this layer thickness.

If, however, one considers the decrease in extinction relative to that of the output current, it makes sense to accept the lower output current in order to ensure that the following cells receive very much more light than would be the case with 5 : 6 dilution. Photographs of cells with TiO₂ : terpineol mixtures of 5 : 4.1, 5 : 6 and 5 : 10 were taken, which illustrated this effect.

- 5 Effects of inhomogeneity were observed. In order to achieve a homogeneous layer within a 1 cm² cell, the TiO₂ area was increased for later cells such that the region in which the TiO₂ banks up during spin coating lies outside the silver electrodes and thus outside the cell.

- 10 The construction of the cell stack with regard to the TiO₂ layer's thickness in the cells and their positioning in the stack, was carried out by testing various arrangements of cells with np TiO₂ layers of various thicknesses.

Preparation and the properties of a dye sensitized solar cell (DSC) prepared with the Dye D-5

- 15 FTO (tin oxide doped with fluorine) glass substrates (< 12 ohms/sq, A11DU80, supplied by AGC Fabritech Co., Ltd.) were used as the base material, which were successively treated with glass cleaner, Semico Clean (Furuuchi Chemical Corporation), fully deionized water and acetone, in each case for 5 min in an ultrasonic bath, then baked for 10 minutes in isopropanol and dried in a nitrogen flow.

- 20 A spray pyrolysis method was used to produce the solid TiO₂ buffer layer. Titanium oxide paste (PST-18NR, supplied by Catalysts & Chemicals Ind. Co., Ltd.) was applied onto the FTO glass substrate by screen printing method. After being dried for 5 minutes at 120°C, a working electrode layer having a thickness of 1.6 µm was obtained by applying heat treatment in air at 25 450°C for 30 minutes and 500°C for 30 minutes. Obtained working electrode is then treated with TiCl₄, as described by M. Grätzel et al., for example, in Grätzel M. et al., Adv. Mater. 2006, 18, 1202. After sintering the sample was cooled to 60 to 80°C. The sample was then treated with an additive as disclosed in WO 2012/001628 A1. 5 mM of the additive in ethanol was prepared and the intermediate was immersed for 17 hours, washed in a bath of pure ethanol, briefly dried in a 30 nitrogen stream and subsequently immersed in a 0.5 mM solution of dye D-5 in a mixture solvent of acetonitrile + t-butyl alcohol (1:1) for 2 hours so as to adsorb the dye. After removal from the solution, the specimen was subsequently washed in acetonitrile and dried in a nitrogen flow.

- 35 A p-type semiconductor solution was spin-coated on next. To this end a 0.165M 2,2',7,7'-tetrakis(N,N-di-p-methoxyphenyl-amine)-9,9'-spirobifluorene (spiro-MeOTAD) and 20mM LiN(SO₂CF₃)₂ (Wako Pure Chemical Industries, Ltd.) solution in chlorobenzene was employed. 20 µl/cm² of this solution was applied onto the specimen and allowed to act for 60 s. The supernatant solution was then spun off for 30 s at 2000 revolutions per minute. The substrate 40 was stored overnight under ambient conditions. Thus, the HTM was oxidized and for this reason the conductivity increased.

As the metal back electrode, Ag was evaporated by thermal metal evaporation in a vacuum at a rate of 0.5 nm/s in a pressure of 1×10^{-5} mbar, so that an approximately 100 nm thick Ag layer was obtained.

- 5 In order to determine the photo-electric power conversion efficiency η of the above photoelectric conversion device, the respective current/voltage characteristic such as short-circuit current density J_{sc} , open-circuit voltage V_{oc} and fill factor FF was obtained with a Source Meter Model 2400 (Keithley Instruments Inc.) under the illumination of an artificial sunlight (AM 1.5, 100 mW/cm² intensity) generated by a solar simulator (Peccell Technologies, Inc). As a result, the
- 10 DSC prepared with the Dye D-5 exhibited the following parameters:

J_{sc} [mA/cm ²]	V_{oc} [mV]	FF [%]	η [%]
10.5	721	59	4.5

Results of the optimized output currents for the longitudinal optical sensor stack:

- 15 The best result in terms of the cell stack's output currents were obtained when all five transparent cells of the longitudinal optical sensors step have an np TiO₂ layer thickness of 0.45 micrometers (i.e. a TiO₂ paste dilution of 5 : 10). These cells with a 0.45 micrometers np TiO₂ layer were stained for 60 minutes in a 0.5 mM dye solution. Only the last cell had a np TiO₂ layer just under 3 micrometers and was stained for 75 minutes (0.7 mM). Since the last cell
- 20 does not have to be transparent, the back electrode (second electrode) of the last cell was a vapor-deposited silver layer across the whole 1 cm² area so as to be able to pick up the maximum possible current. The following photo currents were observed with this stack, in order from the first cell to the last cell of the stack :

25 Current [μ A]: 37 9.7 7.6 4.0 1.6 1.9

- The first five cells were made identically. The last cell had a thicker np TiO₂ layer and a silver back electrode vapor-deposited across the whole cell area. One can see that the current of the second cell has already dropped to 1/4 of the first cell's. Even in these five highly transparent
- 30 cells, the last cell's current is only a fraction of the current in the first cell. The cells were excited with a red laser (690 nm, 1 mW) directed at the centre of the cell area.

- The current obtained with cells with a TiO₂ : terpineol dilution of 5 : 9, 5 : 8 or 5 : 7 (i.e. thicker cells) was at most 10 microamps larger than the current of a cell with 5 : 10 dilution of the TiO₂ paste. These cells, however, exhibited a significantly higher extinction, as a result of which the
- 35 output current of the following cells decreases considerably.

- The cells with a TiO₂ dilution of 5 : 6, in which – compared with the TiO₂ dilutions of 5 : 9; 5 : 8 and 5 : 7 – a significantly higher current was obtained, nevertheless absorbed so much light that
- 40 no more light reaches the last cell of the stack. Even when placing just one of these cells in

position 5 with four preceding 450 nm thin cells, the output current of the last cell decreased considerably, such that the last cell supplies practically no more current.

It needs mentioning that each of these cells in the test stack was sealed with an additional glass plate for protection against ambient effects. This, however, created many additional interfaces at which the light beam of the 690 nm laser (1 mW) can be reflected and scattered, as a result of which the extinction of such a sealed cell is higher. In the later device the cell stack was kept in nitrogen, which is why the sealing becomes unnecessary and the cells lie directly on top of each other. This decreased the stack's extinction, since the losses resulting from scattering at the cover glasses no longer occur.

Cross-resistance of the transversal optical sensors:

A defined cross-resistance between the opposite sides of the square cell makes precise x,y resolution possible. The principle of x,y resolution is illustrated in Figures 3A to 3D. The cross-resistance across the area of the cell is determined by the PEDOT layer present between the p-type conductor and the silver electrodes bordering the cell. In the undoped state, PEDOT is a semiconductor. Conductivity is made possible by the system of conjugated double bonds extending across the entire molecule in combination with the doping with a negatively charged counter-ion. The PEDOTs used which were used for the present experiments were all doped with the negatively charged polymer polystyrene sulfonate (PSS). PEDOT:PSS is available in a wide range of embodiments as regards conductance, solid content, ionisation potential (IP), viscosity and pH.

Factors affecting the cross-resistance:

The PEDOT was also applied to the cells by spin coating. During the spinning process, the solvents ethanol and isopropanol evaporate whereas the low-volatility PEDOT remains on the substrate in the form of a film. The resistance of this layer depends on the conductance of the PEDOT being used and on the thickness of the layer:

$$R = \rho \cdot \frac{l}{A}$$

where ρ is the resistivity, l the distance across which the resistance is measured and A the cross-sectional area through which the charge carriers flow (A is a function of the PEDOT layer's thickness).

According to known principles of spin-coating, the layer thickness d to be expected when coating a non-Newtonian fluid can be determined by:

$$d = \sqrt[3]{\frac{3 \cdot x_s^3 \cdot u_k \cdot e}{2 \cdot (1 - x_s) \cdot \omega^2}}$$

$$\sim \sqrt[3]{\frac{x_s^3 \cdot u_k}{(1 - x_s) \cdot \omega^{3/2}}}$$

where x_s is the PEDOT percentage in the mixed diluted solution, u_k is the kinematic viscosity, e the evaporation rate of the solvent(s) and ω the angular velocity during spin coating. The evaporation rate is proportional to $\omega^{1/2}$.

The thickness of the PEDOT layer can, therefore, be affected by various parameters: the angular velocity, the viscosity of the PEDOT solution and the percentage of PEDOT in the solution. The angular velocity can be varied directly. The viscosity and the percentage of PEDOT in the solution can only be affected indirectly, i.e. via the ratio at which the PEDOT is mixed with ethanol and isopropanol.

The following parameters, therefore, can be used in order to adjust the cross-resistance, and they will be optimized in due course:

- The choice of PEDOT
- The layer thickness of the PEDOT
- The PEDOT/solvent ratio
- The spin speed during the PEDOT's spin coating
- The number of PEDOT layers
- The time interval Δt between applying and spin coating the PEDOT

Optimizing the cross-resistance:

The PEDOT solution was mixed with ethanol and isopropanol in a standard volumetric ratio of 1: 1 : 1, and larger particles removed with a 0.45 micrometer syringe filter. The entire cell was covered with this diluted PEDOT solution (approximately 900 microliters were needed per substrate) and spin coated at a speed of 2000 1/s. At this speed, 30 s turned out to suffice to remove and evaporate the solvents ethanol and isopropanol.

The aforementioned parameters were then varied systematically, with the objective of obtaining a cross-resistance of ca. 2 k Ω between the opposite electrodes of the square cell.

Choice of PEDOT:

The greatest impact on the PEDOT layer's cross-resistance turned out to come from the conductance of the PEDOT solution used. In order to obtain a first impression of the order of magnitude of the resistances of such a PEDOT layer across 1 cm, three PEDOT products with very different conductances were tested:

- Clevios™ PVP AI 4083 from Heraeus
- Clevios™ PH 1000 from Heraeus
- Orgacon™ N-1005 from Sigma Aldrich

The relevant parameters of dynamic viscosity η_d , ionisation potential IP and resistivity ρ are summarised in Table 1. The IP is an important selection criterion for the PEDOT. The PEDOT's IP generally should be less than 5 eV in order to ensure good functionality of the cell.

	AI 4083	PH 1000	N-1005
η_d [mPas]	5 - 12	15 - 50	7 - 12
IP [eV]	5.2	4.8	N/A
ρ [Ωcm]	500 - 5000	0.0012	75 - 120

Table 1: Relevant parameters for various PEDOTs.

For these first tests, 1.3 micrometer np TiO₂ layers were coated onto FTO-free glass substrates. In this first experimental series, only 300 microliters of each of the three prepared PEDOT solutions were coated directly on np TiO₂ layers – without the staining or p-type conductor coating steps. For each PEDOT solution, three substrates with 1, 2 and 3 PEDOT layers were made. The resistance was measured by applying a layer of conducting silver paint at 1 cm spacing at several positions on each substrate.

It is to be expected that the resistance of the substrate fabricated in this way, will be smaller than the resistance resulting from applying the PEDOT on a smooth p-type conductor layer.

As expected, the experiments showed a decrease in cross-resistance with an increasing number of layers and therefore increasing total thickness of the PEDOT. The cross-resistance of AI 4083 is in the M Ω range even with three layers, therefore it was not used in further tests. PH 1000 with two applied layers was in the required range. The cross-resistance of N 1005 is also in the k Ω range, and could be decreased through optimization. Since, however, it can be assumed that when applying the PEDOT on the smooth p-type conductor's surface the resistance will be higher than when applying it directly on the np TiO₂ layer as in this test series, further optimizations will focus on PH 1000.

Applying several PEDOT layers:

A further option for increasing the total thickness of the PEDOT layer is to apply several PEDOT layers consecutively. Tests were conducted with 1 and 2 applied PEDOT layers. PH 1000 was mixed with ethanol and isopropanol at a 1 : 1 : 1 volumetric ratio. The cells were covered completely with 900 microliters of the PEDOT solution and the excess solution was removed by spin coating at an RPM of 2000 1/min.

Unlike the first experimental series, these tests were conducted on 'complete' cells, i.e. stained cells coated with p-type conductor. The cross-resistance was measured between two circular vapour-deposited electrodes ca. 2 mm apart, in order to exclude errors resulting from different contact resistances of PEDOT/conducting silver paint and PEDOT/vapour-deposited silver. In addition, cell efficiency measurements can be automated with this electrode arrangement. These test cells are very much simpler and quicker to make than the square transparent ones, but they suffice for the requirements of these tests (for one thing, the cross-resistance of the PEDOT layer(s) is to be measured here across a defined section; for another, the cells are to be tested for functionality, i.e. whether there is good contact between the p-type conductor and the PEDOT and the IP of the PEDOT matches the energy level of the p-type conductor). The cross-resistance across 1 cm is calculated using equation 3.1 – assuming equal layer thickness and therefore equal area A – through multiplication by the factor 5 (the resistivity of the solution is constant).

Table 2 shows the results of the resistance measurement between the two circular vapor-deposited back electrodes and the calculated cross-resistances across 1 cm for 1 and 2 PEDOT layers. The last column shows the efficiency of the cells. The smallest and largest measurements obtained in several tests are shown in each case.

Mixture ratio	Number of layers	RPM [1/min]	$R_{l=2mm}$ [k Ω]	$R_{l=1cm}$ [k Ω]	η [%]
1 : 1 : 1	1	2000	6 - 19	30 - 90	1.5 - 2.5
	2		0.18 - 0.27	0.9 - 1.35	0.4 - 0.6

Table 2: Results of resistance measurements for one and two PEDOT layers.

The difference in cross-resistances between one and two applied PEDOT layers can be seen clearly. For one PEDOT layer it is significantly higher than the required 2 k Ω , for two layers it is very much lower. It is also evident, however, that the efficiency of the cells with two applied PEDOT layers is very much smaller, which means that the contact between the two PEDOT layers is poor. It should be noted that the efficiency being measured here refers to the circular cell, i.e. to a cell with a back electrode vapor-deposited across the whole surface. The efficiency of the square transparent cells, therefore, will be much lower still, which is why the idea of several PEDOT layers applied consecutively has to be discarded. Further experiments will attempt to minimize the cross-resistance of one PEDOT layer only.

It is noticeable that the resistance with two applied PEDOT layers is smaller than in the first experimental series, in which the PEDOT was applied directly onto the np TiO₂ layer. Presumably, the reason for this discrepancy is that in the first experiments the layers were applied immediately after each other, even though the first PEDOT layer had not yet dried out completely. In this experiment, the cell was placed on the hotplate at 60 °C between applying the two layers.

As expected, in this experimental series – in which the PEDOT was applied onto the p-type conductor – the cross-resistance with only one applied PEDOT layer was higher than when the PEDOT was applied onto the rough surface of the np TiO₂.

Increasing the PEDOT concentration of the solution:

As mentioned above, usually, the PEDOT solution is mixed with ethanol and isopropanol at a volume ratio of 1 : 1 : 1 in order to decrease the viscosity of the solution and obtain homogeneous layers through spin coating. When the PEDOT proportion in the mixture is increased, the solution's viscosity rises. Due to the higher viscosity, an increase in the thickness of the PEDOT layer remaining on the cell after spin coating is expected (for comparison: η_d , ethanol, 20°C = 1.19 mPas; η_d , isopropanol, 20°C = 2.43 mPas; η_d , PEDOT = 5-50 mPas).

In order to investigate the practical effect of the PEDOT solution's viscosity and the amount of substance it contains on layer thickness and therefore on the cross-resistance, the PEDOT proportion was increased a little to begin with, and then significantly. The volumetric mixture ratios ethanol : isopropanol : PEDOT tested here were as follows:

1 : 1 : 1
1 : 1 : 2
1 : 1 : 5
1 : 1 : 10
2 : 2 : 1

Since a preliminary experiment showed that small variations in PEDOT concentration do not result in significant differences as far as the resistance is concerned, the PEDOT proportion in the mixed solutions was increased considerably. This was the first experimental series in which the construction of the cells and the arrangement of the electrodes corresponded to those of the actual square cells.

The thickness of the cells' np TiO₂ layer was 1.3 micrometers. Each time, a PEDOT layer with a different proportion of PH 1000 was applied. The PEDOT solution was spun for 90 seconds at 2000 or 1500 1/min. Then the PEDOT layer was dried for ca. 1 minute with the hot air blower, before the silver electrodes (ca. 2 micrometers thick) were vapor-deposited.

Mixture	Number	RPM	R
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ratio	of layers	[1/min]	[k Ω]
1 : 1 : 1	1	2000	22 - 39
1 : 1 : 5			31 - 51
1 : 1 : 10			29 - 36
1 : 1 : 1	1	1500	21 - 27
1 : 1 : 5			21 - 32
1 : 1 : 10			26 - 37

Table 3: Cross-resistances for various mixture ratios and spin speeds of PEDOT layers.

As can be seen in Table 3, the cross-resistance does not decrease as expected with the increasing PEDOT proportion in the applied solution. Both at an angular velocity of 2000 and of 1500 1/min, the cross-resistance increases with the increasing PEDOT concentration in the solution. However, it is also noticeable that the resistance tends to decrease with decreasing RPM for the same PEDOT proportion, but is still 10-15 orders of magnitude too high.

Adjusting the time interval between application and spin coating of the PEDOT (Δt) and minimizing the RPM during spin coating:

The classic method for increasing layer thickness during spin coating is to decrease the angular velocity. In this way, layer thickness can easily be increased and the cross-resistance decreased. In the experimental series thus far, this was the only variation that led to a sensible result. The angular velocity during spin coating cannot, however, be decreased to an arbitrary value, since at excessively low RPM the solvent no longer evaporates sufficiently quickly which results in inhomogeneous PEDOT layers.

Tests have shown, however, that the time interval between applying the PEDOT solution to the substrate and starting the spin coating (and thus, removing excess solution from the substrate) has a significant effect on the cross-resistance. Therefore, subsequently, the cross-resistance was minimized through iterative optimization of the two parameters Δt and angular velocity during spin coating.

Therefore, over several tests series, the time interval Δt between applying the PEDOT solution to the cell and the start of spin coating was increased in steps from 30 seconds to 2 minutes, later in combination with RPM optimization from 1 to 3 minutes and finally from 3.5 to 5 minutes. This involves decreasing the RPM from 2000 1/min to 350 1/min. When decreasing the RPM to under 1000 1/min, 30 seconds turned out to no longer suffice for complete evaporation of the solvent. This time was, therefore, extended to 2 minutes in every case. Thereafter, the cells were dried with the hot air blower for approximately 1 minute before the electrodes were vapor-deposited.

RPM [1/min] 2000 1000 750 600 500 450 400 350

The results of the optimization are summarized in Tables 4 to 7.

- 5 In the first experimental series of the final optimization (Table 4), in which to begin with the time interval Δt between applying the PEDOT solution on the cell and spinning at constant angular velocity was increased, at an RPM of 1000 1/min there seems to be an optimum at $\Delta t = 60$ s (4.1 - 4.2 k Ω). For this time interval and a further decrease in RPM, a new minimum was obtained for 600 1/min (2.6 - 2.7 k Ω).

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	1000	30	8.5 - 8.7
			60	4.1 - 4.2
			90	4.7 - 4.8
			120	5.4 - 5.8
		750	60	9.0 - 9.2
		600	60	2.6 - 2.7
		500	60	3.7 - 4.0

Table 4: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 1.

5

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	600	60	15.8 - 17.6
			90	11.0 - 11.6
			120	11.2 - 11.4
		500	60	6.0 - 6.8
			90	5.0
			120	6.4 - 6.8
			180	3.1 - 4.1

Table 5: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 2.

10

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	500	180	0.4
		450	180	0.6
		400	180	1.1 - 1.3
		350	180	2.0 - 2.7
		350	210	1.8 - 2.5
		350	240	1.6
		350	270	1.5 - 1.8
		350	300	1.8 - 1.9

Table 6: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 3.

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	500	180	1.3 - 2.4
		450	180	1.0 - 2.0

Table 7: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 4.

Since, however, the results do not differ substantially between 600 and 500 1/min, in the next experimental series the time interval Δt was increased further stepwise for both RPM values. The results are shown in Table 5.

A further decrease in RPM and increase in Δt did not exhibit further improvement. In fact, the cross-resistance even increased again at an RPM < 450 1/min (see Table 6).

Since the values for 500 and 450 1/min lie very close together, a last comparative test was performed (see Table 7).

It was shown that the cross-resistance at an RPM of 450 1/min is slightly smaller than at 500 1/min. Since, however, no significant is achieved thereby and since PEDOT layers coated at an excessively low RPM are no longer homogeneous, 500 1/min was chosen as the optimal RPM. The time interval Δt is then 180 s.

Generally speaking, it is noticeable that the resistance values in the last set of experimental series, where there was a time interval between applying the PEDOT solution to the cell and starting the spinning, do not fluctuate as much within one series for constant parameters as

before. In the last experimental series (Table 7), the two cross-resistances were measured on each of four cells (left-right and top-bottom) and the results varied only by ca. 1 k Ω . The fact that in some cases the results from different series vary significantly for the same experimental parameters, is likely to be due to the production of the PEDOT solution since the individual experimental series in themselves provide coherent results.

The spin coater's open lid during the time interval Δt may be an important interference factor in these experiments. In one experimental series, it was not closed immediately after applying the PEDOT solution to the cell but only before spin coating. The cross-resistances measured in this experimental series are very much higher and there is a considerable variation in their values between the substrates – but not on any one substrate. It could not be determined exactly why the cross-resistance is so strongly influenced by the time interval Δt before the PEDOT solution's spin coating. Perhaps some of the PEDOT solution dries and adheres to the cell during this time, resulting in a thicker PEDOT layer.

Results of the optimized parameters:

The minimum cross-resistance obtained in this way lies between 1 and 3 k Ω . The parameters that bring about a minimum cross-resistance here were:

- PEDOT: Clevios[®] PH 1000 from Heraeus
- Number of layers: 1
- PEDOT : ethanol : isopropanol ratio = 1 : 1 : 1
- Time interval between applying and spin coating the PEDOT: $\Delta t = 180$ s
- RPM during PEDOT spin coating: $n = 500$ 1/min ($t = 120$ s)

The final cells used in the experiments:

The cells used during the optimization process so far were made on thick 2.5 mm TEC 8 glass carriers, with the FTO layer already applied during the fabrication process. They have a highly homogeneous FTO layer, onto which the application of homogeneous np TiO₂ layers is possible. This makes possible the fabrication of cells that appear homogeneous to the human eye.

For the technical realization of the sensor stack, however, the cells were made on thin 1 mm special glass carriers made of quartz glass, which were coated subsequently with FTO. Therein, loss carriers with beveled edges were used. The beveled edges served as bases for cells contacts. The silver contacts were vapor-deposited up to the edge of the bevels. This made it possible to contact the cells directly adjacent to each other in the stack individually with pins.

The subsequently applied FTO layer on these special carriers exhibited in part inhomogeneities arising from the fabrication process. The fabrication of homogeneous cells on these carriers

turned out to be very difficult, as was shown by generating current diagrams of the final cells. Even in the first cell of the stack which supplied a homogeneous current signal across its entire area, four locations were identified that supplied lower currents due to inhomogeneities. Current diagrams were obtained by exciting the cells with a laser at a wavelength 690 of nm. The laser
5 scanned the cells at 1 mm intervals. The cells were scanned in their final arrangement as a stack, i.e. the current diagram of the last cells was recorded with five 'thin' cells located in front of the last cell.

At the excitation wavelength of 690 nm, the developed cells had an extinction of 0.13. At the
10 maximum (at ca. 550 nm) the extinction of these cells was ca. 0.4. Despite this low absorption by the cells and the fact that the back electrode consists of a poorly conducting – unlike silver – transparent layer, the efficiency of such a cell is still ~0.3% (AM 1.5*), that of the last cell ~2%.

Figures 2A, 4B and 4C show the final cells on the 1 mm special glass carriers. The first cell in
15 the stack, which acts as the transversal optical sensor, requires the special electrode arrangement for x,y-resolution. For cells 2-5, which form the longitudinal optical sensor stack, only the total current is needed for z-resolution, which is why the contacting silver electrodes are combined here into one electrode surrounding the cell. Otherwise, however, the first five cells were made identically.

20 The last cell of the longitudinal optical sensor stack is intended to absorb the remaining light preferably completely, which is why it was chosen to have a significantly higher extinction than the front cells. In addition, it has a back electrode covering its entire area in order to supply maximum output current.

25 The cell shown in Fig. 2A, which forms the transversal optical sensor, in this experiment, was used only once for x,y resolution at position 1 in the stack. The cells in Fig. 4B were used four times in the entire stack of the optical sensors, i.e. for positions 2-5 of the entire stack. The last cell, which is depicted in Fig. 4C, was used at position 6 of the entire stack of optical sensors.
30 Thus, generally, a stack of optical sensors was formed with the first optical sensor being the transversal optical sensor (Fig. 2A), followed by four transparent longitudinal optical sensors (Fig. 4B) and a last longitudinal optical sensor having the set-top of Fig. 4C.

35 When illuminating one of these single transparent final cells with a red laser (690 nm, 1 mW), this cell supplied a current of 30-40 microamps. The last longitudinal optical sensor provided a current of approximately 70 μ A. The cross-resistance between any two opposite electrodes in the first cell on these special glass carriers turned out to be 0.1 and 0.3 k Ω .

40 Since the fabrication of transparent cells on the special glass carriers was problematic due to the poor FTO coating, these cells had to be fabricated in large numbers. The cells were screened, and only selected cells were used for the final setup of the detector forming a prototype 3-D sensor. For this screening procedure, specifically for the transversal optical

sensor, the cells were excited by a laser beam (690 nm, 1 mW) at the center of the cell. If the cells are homogenous, the currents at all four contacts are equal ($I_1 = I_2 = I_3 = I_4$). By comparing the currents, specific cells were selected for use in the prototype.

- 5 The x,y resolution achieved with the detector of the set-top turned out to be approximately 1 mm at a distance of 3 m. The z-resolution of this detective setup turned out to be approximately 1 cm.

List of reference numbers

	110	detector
5	111	camera
	112	object
	114	optical sensors
	116	optical axis
	118	housing
10	120	transfer device
	122	lens
	124	opening
	126	direction of view
	128	coordinate system
15	130	transversal optical sensor
	132	longitudinal optical sensor
	134	longitudinal optical sensor stack
	136	sensor region
	138	light beam
20	140	transversal signal lead
	142	evaluation device
	144	last longitudinal optical sensor
	146	longitudinal signal leads
	148	transversal evaluation unit
25	150	longitudinal evaluation unit
	152	position information
	154	data processing device
	156	transformation unit
	157	imaging device
30	158	substrate
	159	imaging device signal lead
	160	first electrode
	161	optically sensitive element
	162	blocking layer
35	163	color wheel
	164	n-semiconducting metal oxide
	166	dye
	168	p-semiconducting organic material
	170	second electrode
40	172	encapsulation
	174	electrode contact
	176	partial electrode

178 partial electrode, x
180 partial electrode, y
182 contact leads
184 light spot
5 186 image
188 electrically conductive polymer
190 top contact
192 illumination source
194 focal point
10 196 human-machine interface
198 entertainment device
199 tracking system
200 user
201 track controller
15 202 machine
204 beacon device
206 primary light beam
208 display
210 keyboard
20

Patent claims

1. A detector (110) for determining a position of at least one object (112), comprising:
 - 5 - at least one transversal optical sensor (130), the transversal optical sensor (130) being adapted to determine a transversal position of at least one light beam (138) traveling from the object (112) to the detector (110), the transversal position being a position in at least one dimension perpendicular an optical axis (116) of the detector (110), the transversal optical sensor (130) being adapted to generate at least one transversal sensor signal;
10
 - at least one longitudinal optical sensor (132), wherein the longitudinal optical sensor (132) has at least one sensor region (136), wherein the longitudinal optical sensor (132) is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region (136) by the light beam (138), wherein the
15 longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam (138) in the sensor region (136);
 - at least one evaluation device (142), wherein the evaluation device (142) is designed to generate at least one item of information on a transversal position of the object (112) by evaluating the transversal sensor signal and to generate at least one item of information
20 on a longitudinal position of the object (112) by evaluating the longitudinal sensor signal.
2. The detector (110) according to the preceding claim, wherein the transversal optical sensor (130) is a photo detector having at least one first electrode (160), at least one second electrode (170) and at least one photovoltaic material (164, 166, 168), wherein the
25 photovoltaic material (164, 166, 168) is embedded in between the first electrode (160) and the second electrode (170), wherein the photovoltaic material (164, 166, 168) is adapted to generate electric charges in response to an illumination of the photovoltaic material (164, 166, 168) with light, wherein the second electrode (170) is a split electrode having at least two partial electrodes (176), wherein the transversal optical sensor (130) has a
30 sensor region (136), wherein the at least one transversal sensor signal indicates a position of the light beam (138) in the sensor region (136).
3. The detector (110) according to the preceding claim, wherein electrical currents through the partial electrodes (176) are dependent on a position of the light beam (138) in the
35 sensor region (136), wherein the transversal optical sensor (130) is adapted to generate the transversal sensor signal in accordance with the electrical currents through the partial electrodes (176).
4. The detector (110) according to the preceding claim, wherein the detector (110) is
40 adapted to derive the information on the transversal position of the object (112) from at least one ratio of the currents through the partial electrodes (176).

5. The detector (110) according to any of the three preceding claims, wherein the photo detector is a dye-sensitized solar cell.
- 5 6. The detector (110) according to any of the four preceding claims, wherein the first electrode (160) at least partially is made of at least one transparent conductive oxide, wherein the second electrode (170) at least partially is made of an electrically conductive polymer (188), preferably a transparent electrically conductive polymer (188).
- 10 7. The detector (110) according to any of the preceding claims, wherein at least one of the transversal optical sensor (130) and the longitudinal optical sensor (132) is a transparent optical sensor.
- 15 8. The detector (110) according to any of the preceding claims, wherein the transversal optical sensor (130) and the longitudinal optical sensor (132) are stacked along the optical axis (116) such that a light beam (138) travelling along the optical axis (116) both impinges on the transversal optical sensor (130) and on the longitudinal optical sensor (132).
- 20 9. The detector (110) according to any of the preceding claims, wherein the longitudinal optical sensor (132) comprises at least one dye-sensitized solar cell.
- 25 10. The detector (110) according to the preceding claim, wherein the longitudinal optical sensor (132) comprises at least one first electrode (160), at least one n-semiconducting metal oxide (164), at least one dye (166), at least one p-semiconducting organic material (168), preferably a solid p-semiconducting organic material (168), and at least one second electrode (170).
- 30 11. The detector (110) according to the preceding claim, wherein both the first electrode (160) and the second electrode (170) are transparent.
- 35 12. The detector (110) according to any of the preceding claims, wherein the evaluation device (142) is designed to generate the at least one item of information on the longitudinal position of the object (112) from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object (112) with respect to the detector (110).
13. The detector (110) according to any of the preceding claims, furthermore comprising at least one illumination source (192).
- 40 14. The detector (110) according to any of the preceding claims, wherein the detector (110) has a plurality of longitudinal optical sensors (132), wherein the longitudinal optical sensors (132) are stacked.

15. The detector (110) according to the preceding claim, wherein the longitudinal optical sensors (132) are arranged such that a light beam (138) from the object (112) illuminates all longitudinal optical sensors (132), wherein at least one longitudinal sensor signal is generated by each longitudinal optical sensor (132), wherein the evaluation device (142) is adapted to normalize the longitudinal sensor signals and to generate the information on the longitudinal position of the object (112) independent from an intensity of the light beam (138).
16. The detector (110) according to any of the two preceding claims, wherein a last longitudinal optical sensor (144) is arranged in a manner that the light beam (138) illuminates all other longitudinal optical sensors (132) apart from the last longitudinal optical sensor (144), until the light beam (138) impinges on the last longitudinal optical sensor (144), wherein the last longitudinal optical sensor (144) is intransparent with respect to the light beam (138).
17. The detector (110) according to any of the three preceding claims, wherein the stack of at least two optical sensors is partially or fully be immersed in an oil, in a liquid and/or in a solid material.
18. The detector (110) according to any of the preceding claims, wherein the at least one transversal optical sensor and/or the at least one longitudinal optical sensor uses at least two different transparent substrates.
19. The detector (110) according to any of the preceding claims, wherein the detector (110) further comprises at least one imaging device (157).
20. The detector (110) according to any of the preceding claims, wherein the detector (110) further comprises at least one optically sensitive element (159).
21. The detector (110) according to any of the preceding claims, wherein the optically sensitive element (161) comprises a color wheel (163), a color drum, and/or a filter wheel using an elliptically polarizing filter.
22. The detector (110) according to any of the preceding claims, wherein the evaluation device (142) is adapted to generate the at least one item of information on the longitudinal position of the object (112) by determining a diameter of the light beam (138) from the at least one longitudinal sensor signal.
23. The detector (110) according to the preceding claim, wherein the evaluation device (142) is adapted to compare the diameter of the light beam (138) with known beam properties of

the light beam (138) in order to determine the at least one item of information on the longitudinal position of the object (112).

24. The detector (110) according to any of the preceding claims, wherein the longitudinal optical sensor (132) is furthermore designed in such a way that the longitudinal sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination.
25. A human-machine interface (196) for exchanging at least one item of information between a user (200) and a machine, wherein the human-machine interface (196) comprises at least one detector (110) according to any of the preceding claims relating to a detector (110), wherein the human-machine interface (196) is designed to generate at least one item of geometrical information of the user (200) by means of the detector (110) wherein the human-machine interface (196) is designed to assign to the geometrical information at least one item of information.
26. An entertainment device (198) for carrying out at least one entertainment function, wherein the entertainment device (198) comprises at least one human-machine interface (196) according to any of the preceding claims referring to a human-machine interface (196), wherein the entertainment device (198) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (196), wherein the entertainment device (198) is designed to vary the entertainment function in accordance with the information.
27. A tracking system (199) for tracking the position of at least one movable object (112), the tracking system (199) comprising at least one detector (110) according to any of the preceding claims referring to a detector (110), the tracking system (199) further comprising at least one track controller (201), wherein the track controller (201) is adapted to track a series of positions of the object (112), each position comprising at least one item of information on a transversal position of the object (112) at a specific point in time and at least one item of information on a longitudinal position of the object (112) at a specific point in time.
28. A camera (111) for imaging at least one object (112), the camera (111) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
29. A method for determining a position of at least one object (112),
- wherein at least one transversal optical sensor (130) of a detector (110) is used, wherein the transversal optical sensor (130) determines a transversal position of at least one light beam (138) traveling from the object (112) to the detector (110), the transversal position being a position in at least one dimension perpendicular to an optical axis (116) of the

detector (110), wherein the transversal optical sensor (130) generates at least one transversal sensor signal;

- wherein at least one longitudinal optical sensor (132) of the detector (110) is used, wherein the longitudinal optical sensor (132) has at least one sensor region (136), wherein the longitudinal optical sensor (132) generates at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region (136) by the light beam (138), wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam (138) in the sensor region (136);
- wherein at least one evaluation device (142) is used, wherein the evaluation device (142) generates at least one item of information on a transversal position of the object (112) by evaluating the transversal sensor signal and wherein the evaluation device (142) further generates at least one item of information on a longitudinal position of the object (112) by evaluating the longitudinal sensor signal.

30. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use, selected from the group consisting of: a distance measurement, in particular in traffic technology; a position measurement, in particular in traffic technology; a tracking application, in particular in traffic technology.

31. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as an entertainment application.

32. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as a camera (111), in particular in a security application.

33. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as a human-machine interface (196) application.

34. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as a mapping application, in particular for generating maps of at least one space.

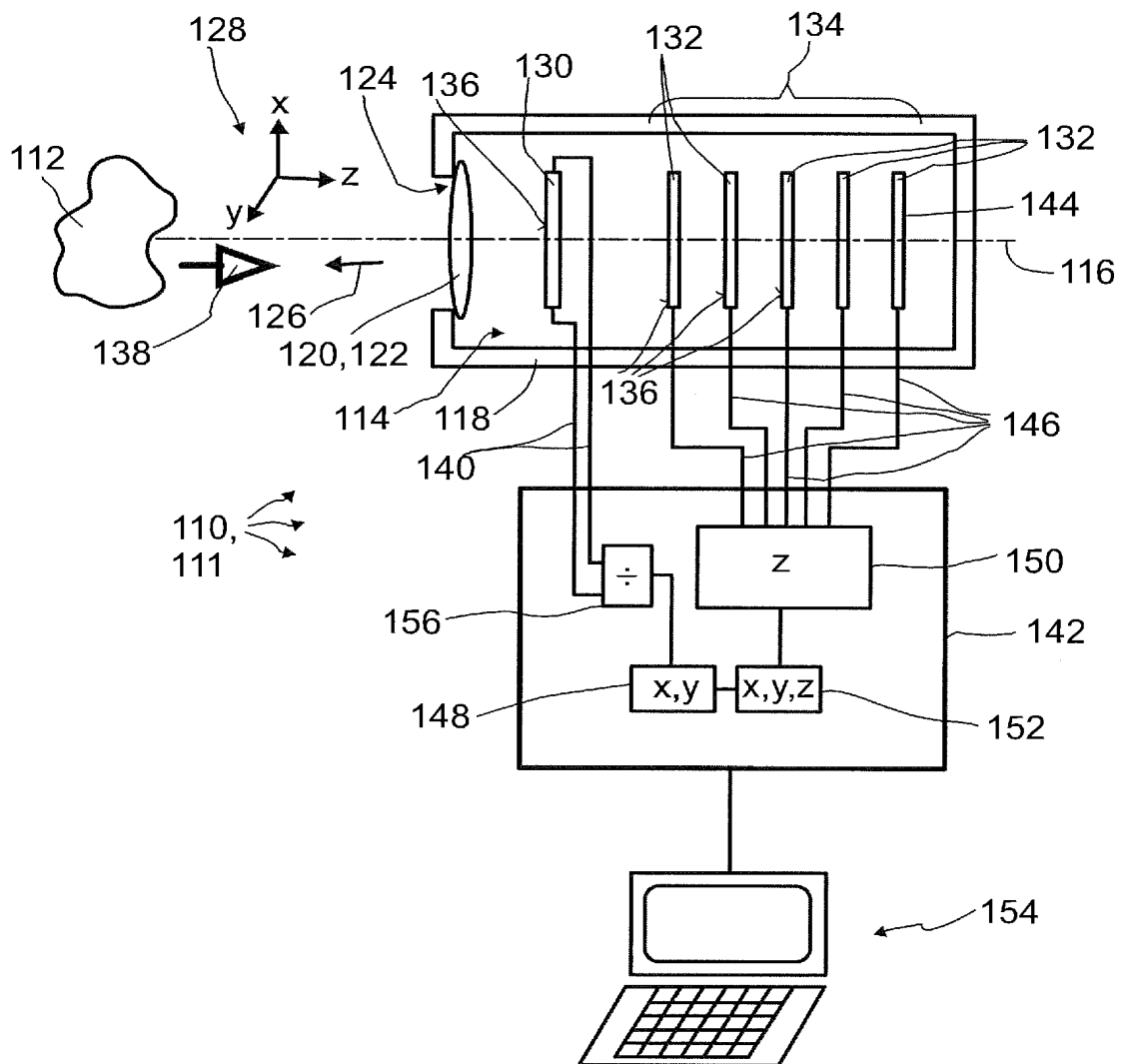
35. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in automated machine processes, selected from the group consisting of: a distance measurement; a position measurement; a tracking application.

36. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in high-precision metrology, in particular in analytics.

37. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in modeling of manufacturing parts.
 38. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in medical operations, in particular in endoscopic methods.
- 5

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FIG.1A



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FIG.1B

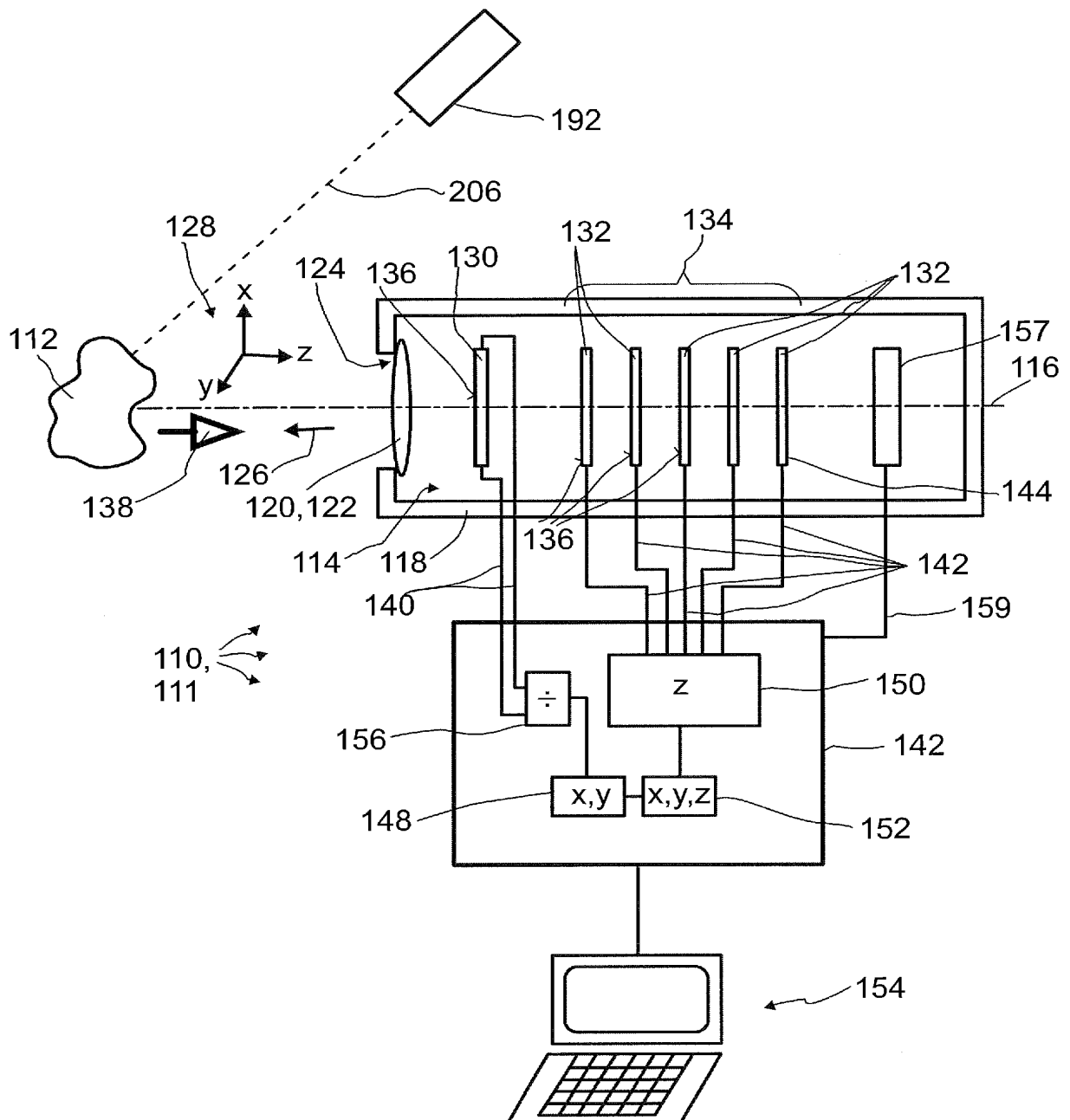
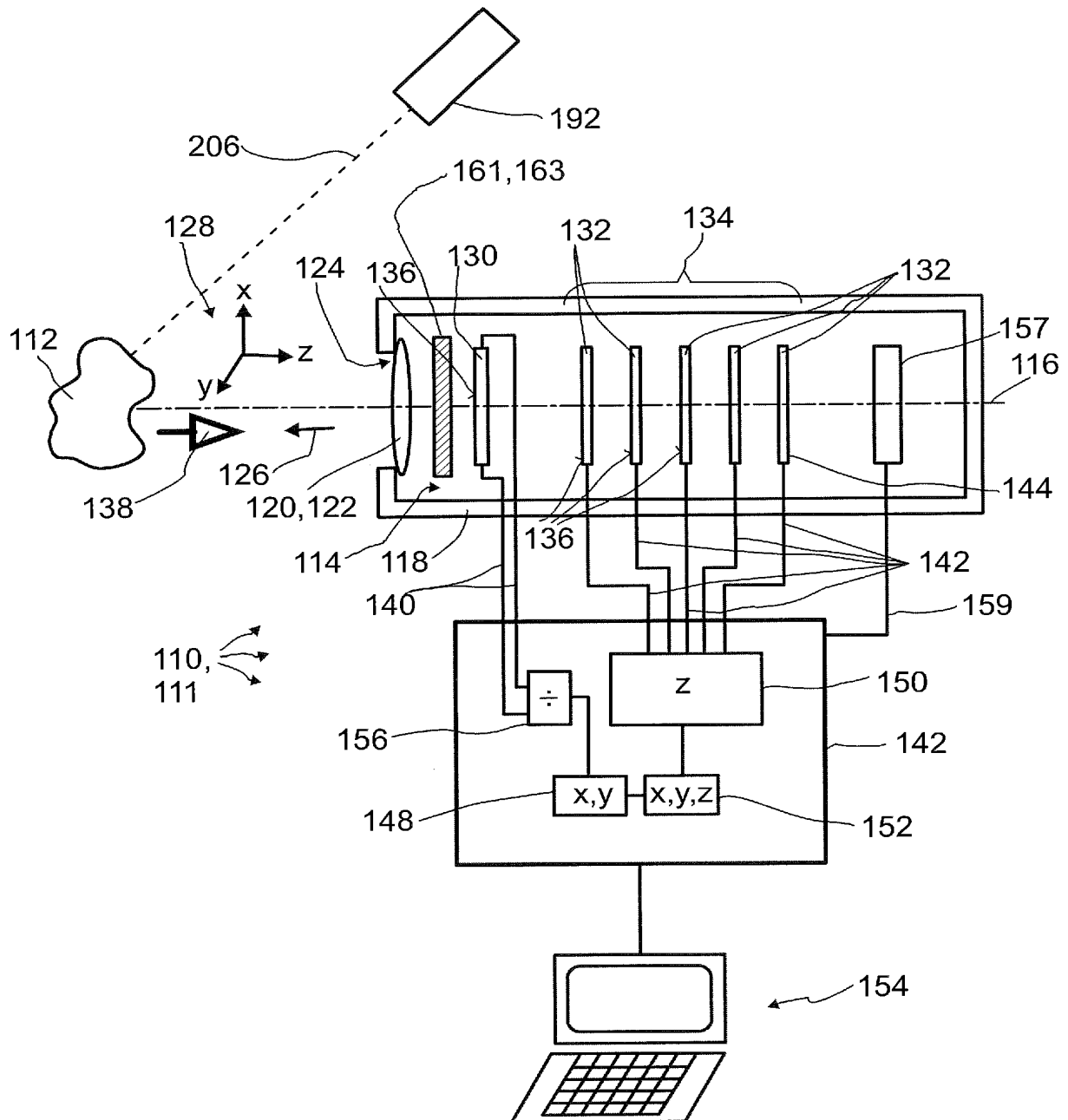


FIG.1C



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FIG.2A

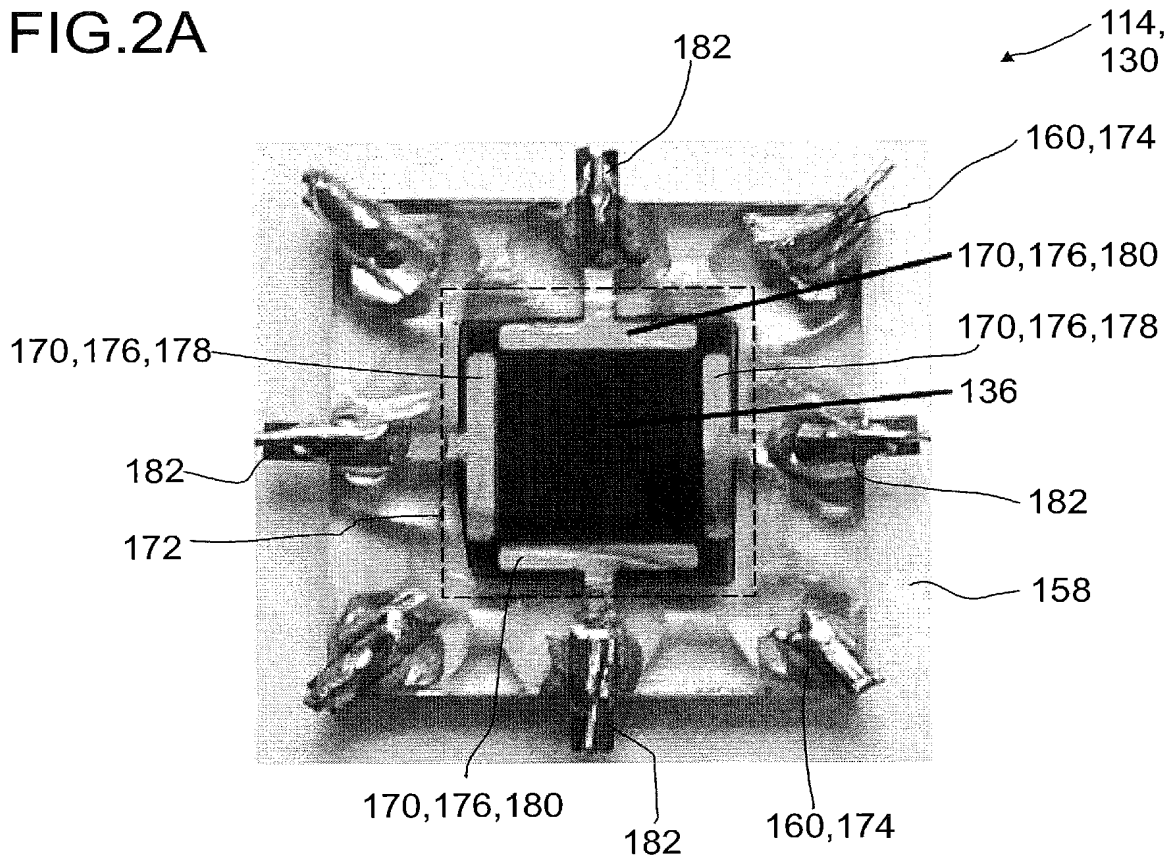


FIG.2B

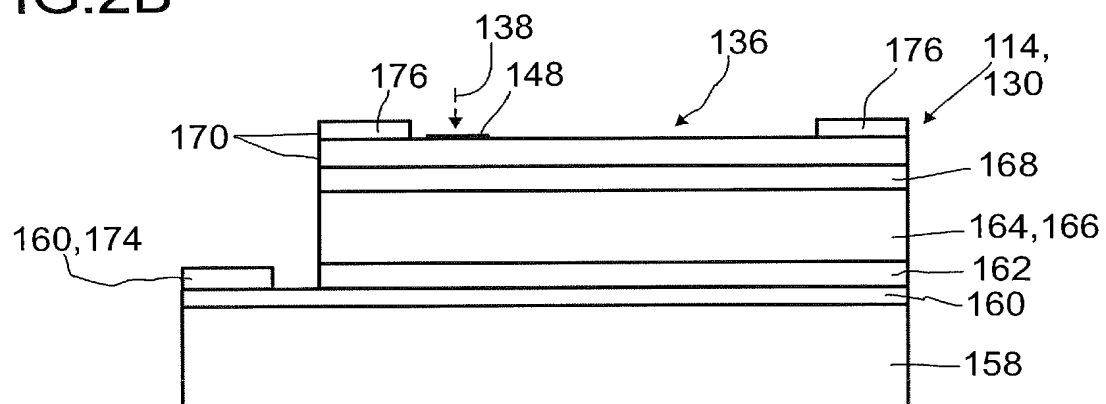


FIG.3A

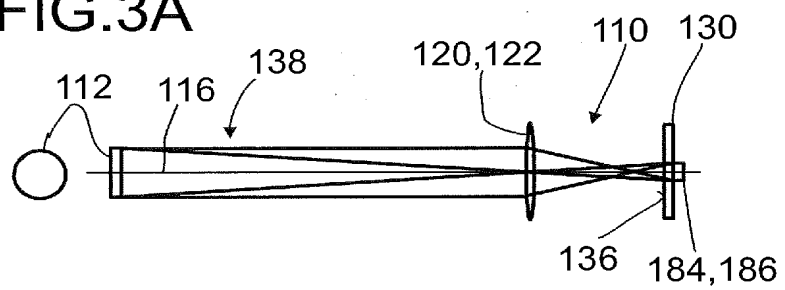


FIG.3B

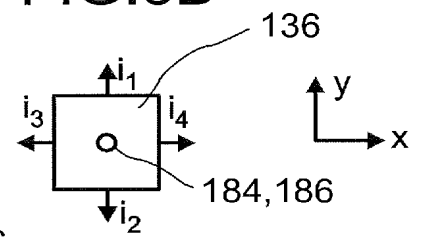


FIG.3C

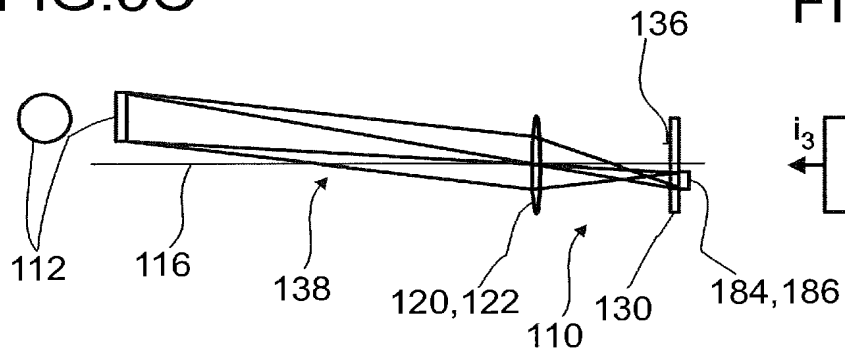
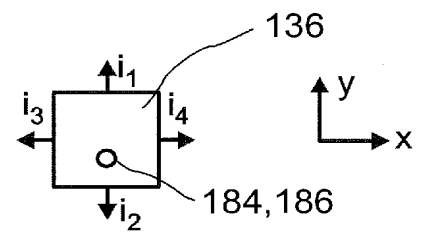


FIG.3D



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FIG.4A

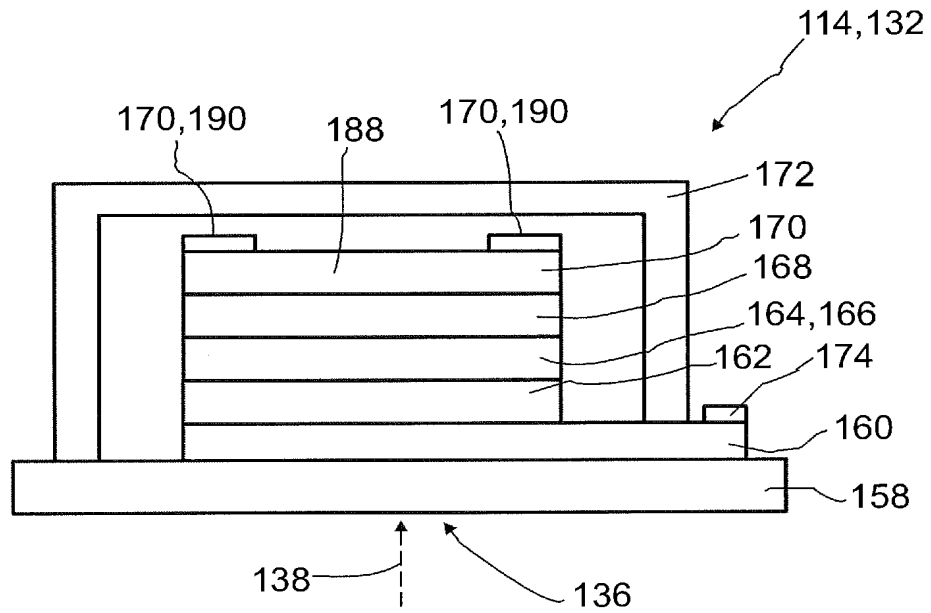


FIG.4B

FIG.4C

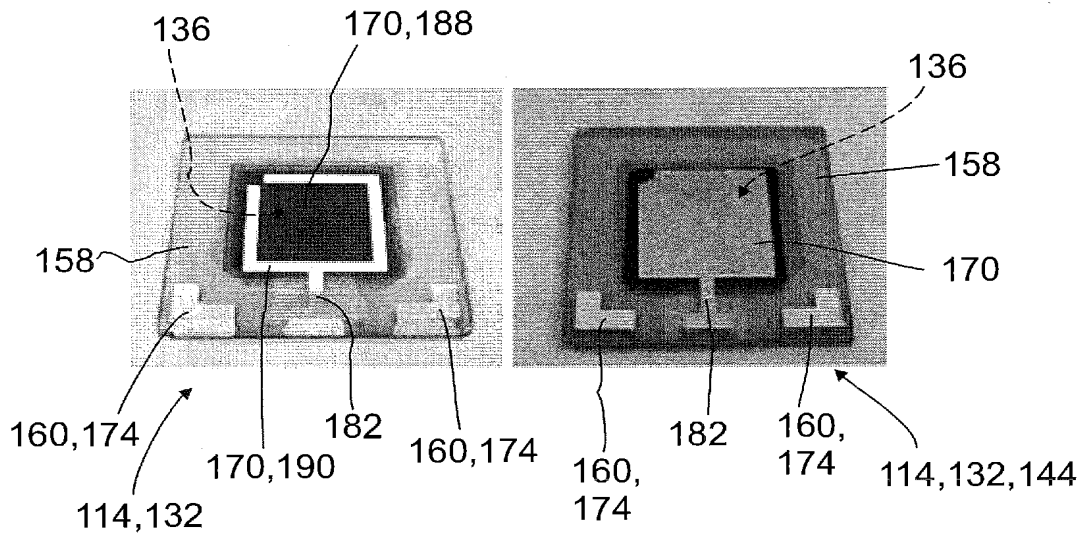


FIG.5A

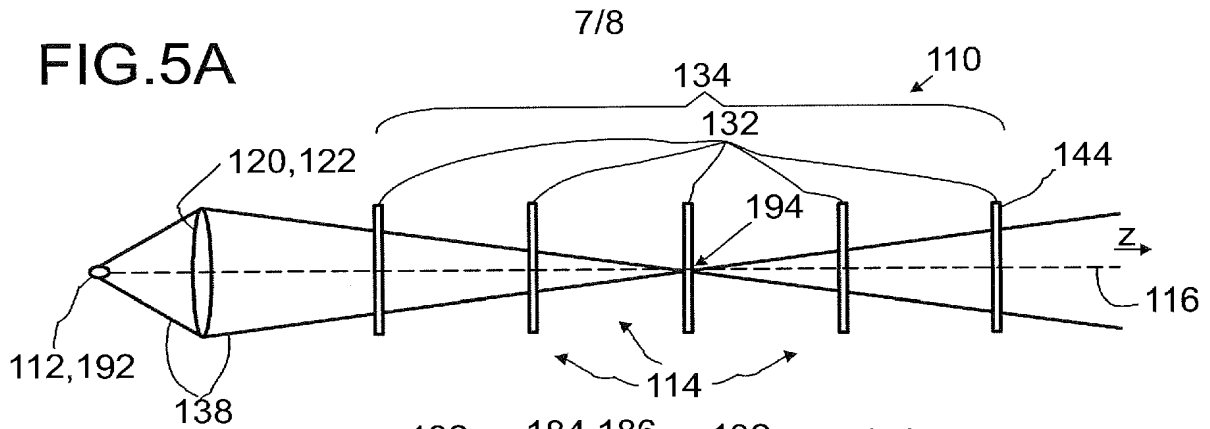


FIG. 5B

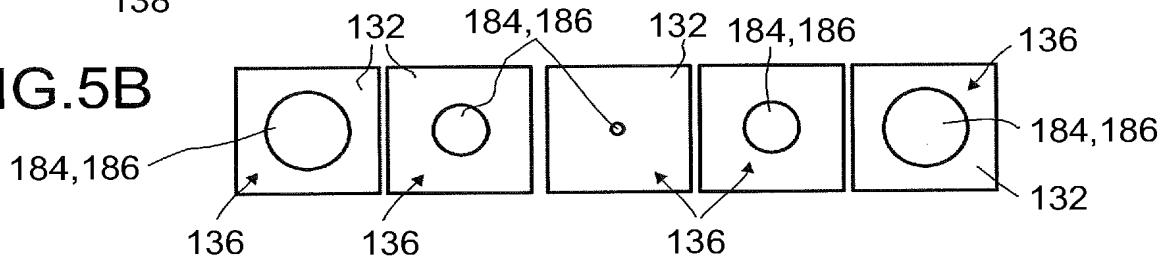


FIG.5C



FIG.5D

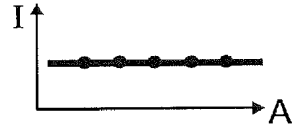
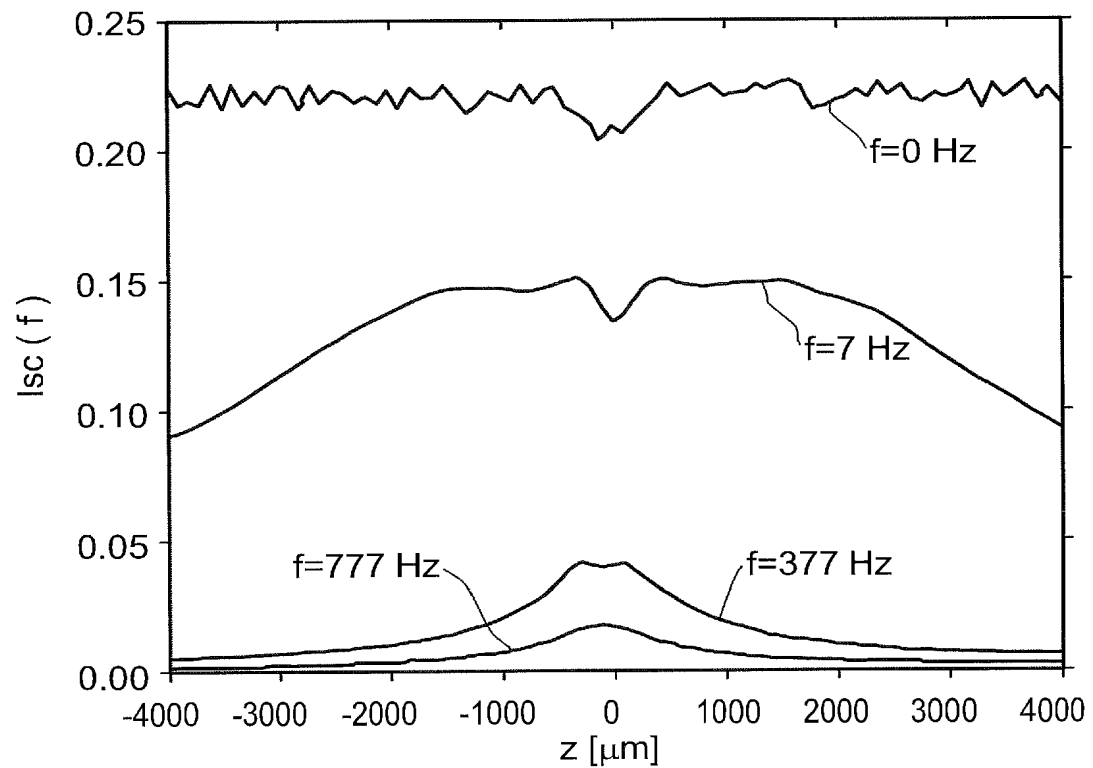
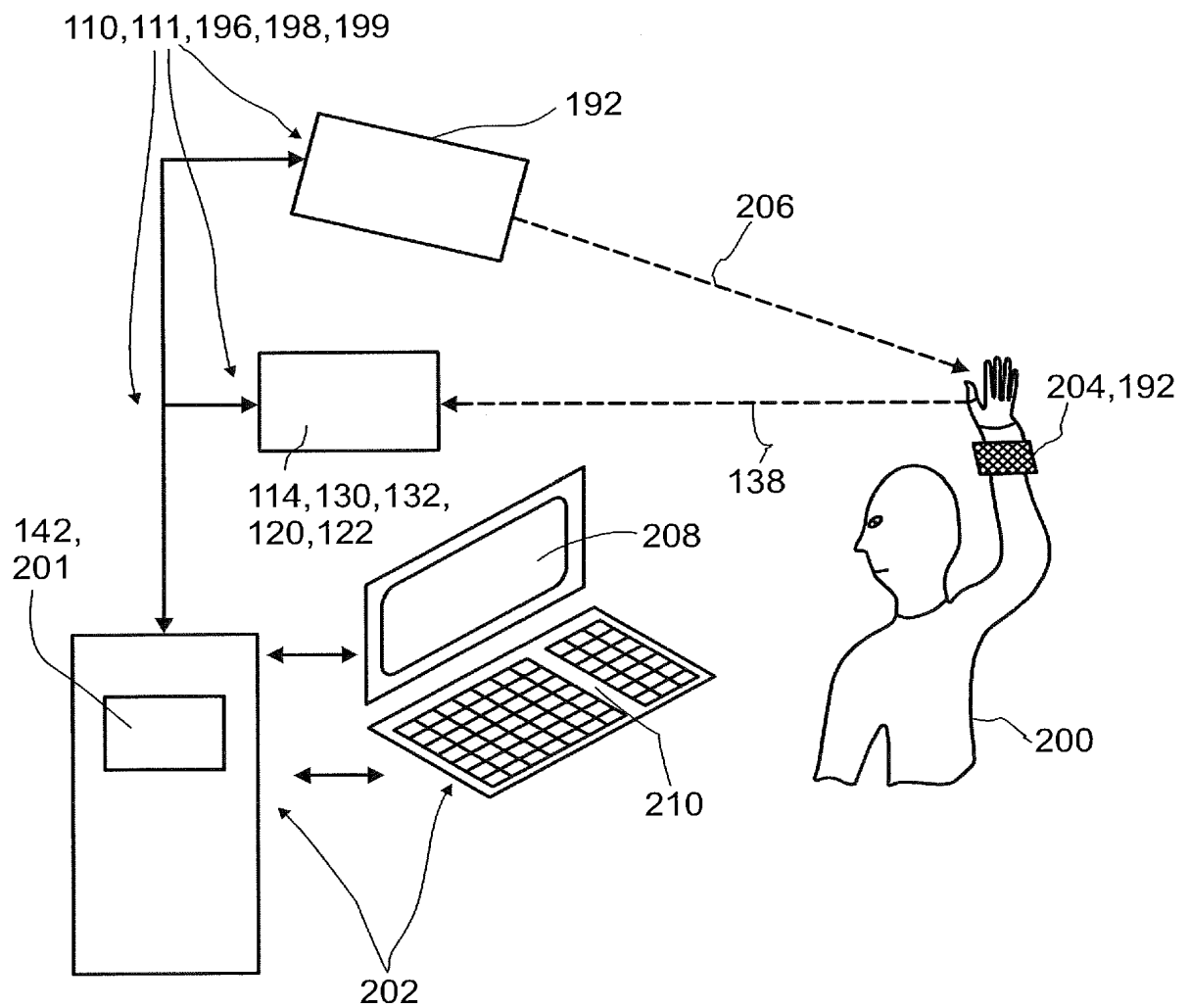


FIG.5E



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FIG. 6



INTERNATIONAL SEARCH REPORT

International application No.

PCT/IB2013/061095

A. CLASSIFICATION OF SUBJECT MATTER

G01C 3/06(2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

WPI;EPODOC;CNKI;IEEE;CNPAT:measur+, positon, coordinate?, depth, distance, transver+, cross, longitudinal, length+ w direction, sensor?, beam, optical, light, luminous, spot?, speckle?, facula?, focus+

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US 2008259310A1 ((WADA, HIDEO)) 23 October 2008 (2008-10-23) description, paragraphs [0085]-[0086], [0105]-[0108]	1-38
A	CN 1896686A ((TOPCON CORPORATION)) 17 January 2007 (2007-01-17) the whole document	1-38
A	CN 102549381A ((ROBERT BOSCH GMBH)) 04 July 2012 (2012-07-04) the whole document	1-38
A	US 2009097010A1 ((YAMAGUCHI, AKIFUMI)) 16 April 2009 (2009-04-16) the whole document	1-38
A	US 2011194097A1 ((YAMAGUCHI, AKIFUMI ET AL.)) 11 August 2011 (2011-08-11) the whole document	1-38
A	JP 2005241340A ((SHARP K.K.)) 08 September 2005 (2005-09-08) the whole document	1-38



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents:

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“E”	earlier application or patent but published on or after the international filing date	“X”	document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
“L”	document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)	“Y”	document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
“O”	document referring to an oral disclosure, use, exhibition or other means	“&”	document member of the same patent family
“P”	document published prior to the international filing date but later than the priority date claimed		

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Date of mailing of the international search report

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No.

PCT/IB2013/061095

Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
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		JP 2008267898A	06 November 2008
CN 1896686A	17 January 2007	EP 1744122A2	17 January 2007
		JP 2007017406A	25 January 2007
		EP 1760430A2	07 March 2007
		US 2011096319A1	28 April 2011
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		CN 102192723A	21 September 2011
		TW 201140111A	16 November 2011
JP 2005241340A	08 September 2005	CN 1677053A	05 October 2005
		US 2005184301A1	25 August 2005



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G01S 17/46 (2006.01) *G01S* 7/48 (2006.01)
G01S 3/783 (2006.01) *G01S* 11/12 (2006.01)

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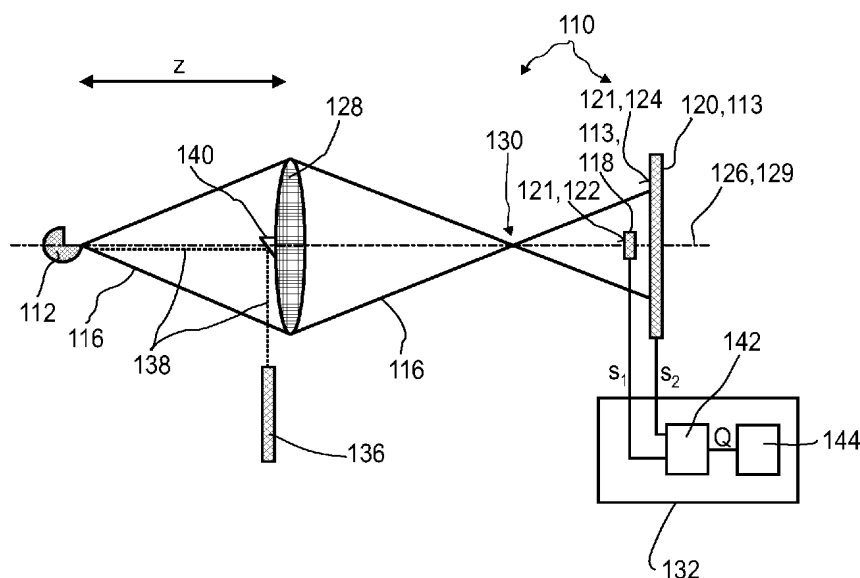
16199397.7 17 November 2016 (17.11.2016) EP
16199398.5 17 November 2016 (17.11.2016) EP
16199399.3 17 November 2016 (17.11.2016) EP
17161334.2 16 March 2017 (16.03.2017) EP

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(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.2



(57) **Abstract:** A detector (110, 1110, 2110) for determining a position of at least one object (112) is proposed. The detector (110, 1110, 2110) comprises: at least one transfer device (128, 1128), wherein the transfer device (128, 1128) has at least one focal length in response to at least one incident light beam (116, 1116) propagating from the object (112, 1112) to the detector (110, 1110, 2110); at least two optical sensors (113, 1118, 1120), wherein each optical sensor (113, 1118, 1120) has at least one light sensitive area (121, 1122, 1124), wherein each optical sensor (113, 1118, 1120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam (116, 1116), at least one evaluation device (132, 1132) being configured for determining at least one longitudinal coordinate z of the object (112, 1112) by evaluating a quotient signal Q from the sensor signals. The detector is adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from

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(81) **Designated States** (*unless otherwise indicated, for every kind of national protection available*): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

(84) **Designated States** (*unless otherwise indicated, for every kind of regional protection available*): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:

- *without international search report and to be republished upon receipt of that report (Rule 48.2(g))*

Detector for optically detecting at least one object

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, systems, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

20 Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultra-violet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1, US 2007/0176165 A1, WO 2009/013282 A1, WO 2012/110924 A1, WO 2014/097181 A1, WO 2015/024871 A1, US 4,767,211, WO 2014/198629 A1, WO 2014/198626 A1, WO 2014/198629 A1 and WO 2014/198625 A1, the full content of which is herewith included by reference. Further, referring to potential materials and optical sensors which may also be employed in the context of the present invention, reference may be made to European patent applications EP 15 153 215.7, filed on January 30, 2015, EP 15 157 363.1, filed on March 3, 2015, EP 15 164 653.6, filed on April 22, 2015, EP 15177275.3, filed on July 17, 2015, EP 15180354.1 and EP 15180353.3, both filed on August 10, 2015, and EP 15 185 005.4, filed on September 14, 2015, EP 15 196 238.8 and EP 15 196 239.6, both filed on November 25, 2015, EP 15 197 744.4, filed on December 3, 2015, the full content of all of which is herewith also included by reference.

Further, reference may be made to detector concepts comparing signals of at least two different sources for determining a position of an object. Thus, as an example, reference may be made to EP 16155834.1, EP 16155835.8 or EP 16155845.7, all filed on February 16, 2016, the full

disclosure of which is herewith included by reference. Further, P. Bartu, R. Koeppe, N. Arnold, A. Neulinger, L. Fallon, and S. Bauer, Conformable large-area position-sensitive photodetectors based on luminescence collecting silicone waveguides, J. Appl. Phys. 107, 123101 (2010), describe a kind of position sensitive detector (PSD) device which might be suitable for large areas and on curved surfaces.

For example, US 5,323,222 A describes a distance measuring device contains a light-projecting section for projecting luminous flux onto a subject. A light-receiving section is placed the base length away from the light-projecting section, receives the reflected rays of light from the subject, and produces a photoelectric current according to the amount of light received. The light-receiving section is divided into at least four light-receiving elements by a first dividing line that crosses the base length at right angles and by a second dividing line that crosses the first dividing line obliquely. A first ratio calculating section calculates the ratio of the sum of the photoelectric currents generated by the light-receiving elements on one side of the first dividing line to the sum of the photoelectric currents produced by the light-receiving elements on the other side. A second ratio calculating section calculates the ratio of the sum of the photoelectric currents generated by the light-receiving elements on one side of the second dividing line to the sum of the photoelectric currents produced by the light-receiving elements on the other side. A distance-determining section calculates the sum of the outputs of the first and second ratio calculating sections, and based on the resulting value, determines the distance from the device to the subject.

US 4,675,517 A describes a photosensitive element having first and second sensitive regions arranged to receive a light spot in such a way that as the light spot lies astride the boundary there between, the ratio of the areas of the two parts of the spot light which are received by the first and second sensitive regions, respectively, varies depending on the received position of the light spot, whereby the received position of the light spot is detected from the variation of the ratio of the outputs of the first and second sensitive regions, wherein an error of variation of the ratio of the outputs of the first and second sensitive regions for the received position of the light spot, which takes place when the diameter of the received spot light is so large as to overrun the first and second sensitive regions, is corrected by providing a supplementary sensitive region surrounding the first or second sensitive region.

WO 2015/024871 A1 describes an optical detector. The optical detector comprising: - at least one spatial light modulator being adapted to modify at least one property of a light beam in a spatially resolved fashion, having a matrix of pixels, each pixel being controllable to individually modify the at least one optical property of a portion of the light beam passing the pixel; - at least one optical sensor adapted to detect the light beam after passing the matrix of pixels of the spatial light modulator and to generate at least one sensor signal; - at least one modulator device adapted for periodically controlling at least two of the pixels with different modulation frequencies; and - at least one evaluation device adapted for performing a frequency analysis in order to determine signal components of the sensor signal for the modulation frequencies.

However, determination of distance to an object using these known methods and devices depends on the object size. In particular, the optical sensors may show a luminance dependency which renders the evaluation of the measurement result more difficult, and, additionally, may dependent on a target spot size. Dependency on object size may result in susceptibility of distance determination on manufacturing tolerances and environmental conditions. Furthermore, these methods and devices require a large baseline, and thus, large dimensions in order to reach a reliable distance value.

10 Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

20 This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

25 As used in the following, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

35 Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with op-

tional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for determining a position of at least one object is disclosed. As used herein, the term "object" refers to a point or region emitting at least one light beam. The light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least one transfer device, wherein the transfer device has at least one focal length in response to at least one incident light beam propagating from the object to the detector;
- at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam,
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q from the sensor signals.

The detector may be adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from the object size in an object plane.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term “at least two optical sensors each having at least one light sensitive area” refers to configurations with two single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having at least two light sensitive areas. Thus, the term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one CCD and/or CMOS device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible. Thus, as an example, an optical device comprising two, three, four or more than four light-sensitive areas may be used which is regarded as two, three, four or more than four optical sensors in the context of the present invention. As an example, the optical device may comprise a matrix of light-sensitive areas. Thus, as an example, the optical sensors may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces

of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to an optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The light beam may propagate from the object towards the detector. As will be outlined in further detail below, the light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source, as an example, may be or may comprise one or more of an external illumination source, an illumination source integrated into the detector or an illumination source integrated into a beacon device being one or more of attached to the object, integrated into the object or held by the object. Thus, the detector may be used in active and/or passive illumination scenarios. For example, the illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. Additionally or alternatively, the object may be adapted to generate and/or to emit the at least one light beam. The light source may be or may comprise at least one multiple beam light source. For example, the light source may comprise at least one laser source and one or more diffractive optical elements (DOEs).

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. As used herein, the term “beam profile” generally refers to a transverse intensity profile of the light beam. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The light beam may be a monochromatic light beam. The light beam may comprise a narrow band of wavelengths, preferably the light beam may comprise a single wavelength. The at least one light source may be adapted to generate at least one monochromatic light beam and/or the detector may comprise at least one filter element adapted to filter a narrow band of wavelength, such as a monochromator.

The illumination source may be adapted to generate at least one illumination pattern for illuminating the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "illumination pattern" refers to a pattern which illuminates the object. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines; at least one fringe pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that a hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may com-

prise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing
5 lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination
10 source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source may be adapted to generate pulsed illumination. The illumination source
15 may be adapted to generate at least one light pulse. As used herein, the term "light pulse" or "pulsed illumination" refers to a light beam limited in time. The light pulse may have a pre-defined length or time duration, for example in the nanoseconds range. For example, the illumination source may be adapted to generate pulses with a pulse length of less than a nanosecond, such as a tenth of a nanosecond, up to a tenth of a second. The illumination source may
20 be adapted to periodically generate the light pulse. For example, the illumination source may be adapted to generate the light pulse with a frequency of 10 Hz to 10 GHz.

The illumination source may be adapted to generate a pulsed light beam. For example, the illumination source may be adapted to generate a continuous illumination light beam and the de-
25 tector may comprise at least one interruption device adapted to interrupt the illumination, in particular periodically. The interruption device may comprise at least one shutter and/or a beam chopper or some other type of mechanical or electronical periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates at constant speed and which can thus periodically interrupt the illumination. By way of example,
30 the at least one interruption device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range
35 from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 microme-
40 ters. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a

CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes. As further used herein, the term "photosensitive element" generally refers to an element which is sensitive against illumination in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the photosensitive element may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used.

In view of the technical challenges involved in the prior art documents discussed above, specifically in view of the technical effort which is required for generating the so called FiP effect as described in WO 2015/024871 it has to be noted that the present invention specifically may be realized by using non-FiP optical sensors. In fact, since optical sensors having the FiP characteristic typically exhibit a strong peak in the respective sensor signals at a focal point, the range of measurement of a detector according to the present invention using FiP sensors as optical sensors may be limited to a range in between the two positions and which the first and second optical sensors are in focus of the light beam. When using linear optical sensors, however, i.e. optical sensors not exhibiting the FiP effect, this problem, with the setup of the present invention, generally may be avoided. Consequently, the first and second optical sensor may each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals may be dependent on the total power of illumination of the respective optical sensor and may be independent from a diameter of a light spot of the illumination. It shall be noted, however, that other embodiments are feasible, too.

The first and second optical sensors each specifically may be semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes. Thus, as opposed to complex and expensive FiP sensors, the present invention simply may be realized by using commercially available inorganic photodiodes, i.e. one small photodiode and one large area photodiode. Thus, the setup of the present invention may be realized in a cheap and inexpensive fashion.

However, embodiments are feasible in which the detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871, as will be outlined in detail below.

The detector may further comprise at least one illumination source for illuminating the object. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector. For this

purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source, as outlined above, specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the optical sensors may be arranged in a twisted arrangement.

The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. As an example, the illuminating light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illuminating light beam may be on the optical axis or off the optical axis. As an example, the illuminating light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

The detector comprises at least one transfer device. The term "transfer device", also denoted as "transfer system", may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one aspheric lens; at least one diffractive optical

element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

- 5 As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of
10 example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimat-
15 ed light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a pre-defined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the
20 transfer device can, in principle, be realized individually or in any desired combination.

- The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The
25 optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or
30 tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the
35 detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis and the condition noted above generally may refer to each beam path independently.. The optical sensors may be located in one
40 and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths. In case the optical sensors are distributed over different partial beam paths, the above-mentioned condition may be described such that at least one first optical sensor is located in at least one first partial beam path, being offset from the

optical axis of the first partial beam path by a first spatial offset, and at least one second optical sensor is located in at least one second partial beam path, being offset from the optical axis of the second partial beam path by at least one second spatial offset, wherein the first spatial offset and the second spatial offset are different.

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The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

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The optical sensors may be positioned off focus. As used herein, the term “focus” generally refers to one or both of a minimum extend of a circle of confusion of the light beam, in particular of at least one light beam emitted from one point of the object, caused by the transfer device or a focal length of the transfer device. As used herein, the term “circle of confusion” refers to a light spot caused by a cone of light rays of the light beam focused by the transfer device. The circle of confusion may depend on a focal length f of the transfer device, a longitudinal distance from the object to the transfer device, a diameter of an exit pupil of the transfer device, a longitudinal distance from the transfer device to the light sensitive area, a distance from the transfer device to an image of the object. For example, for Gaussian beams, a diameter of the circle of confusion may be a width of the Gaussian beam. In particular, for a point like object situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from the object into a focus point at the focal length of the transfer device. For non-point like objects situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from at least one point of the object into a focus plane at the focal length of the transfer device. For point like objects not situated or placed at infinite distance from the detector, the circle of confusion may have a minimum extend at least at one longitudinal coordinate. For non-point like objects not situated or placed at infinite distance from the detector, the circle of confusion of the light beam from at least one point of the object may have a minimum extend at least at one longitudinal coordinate. As used herein, the term “positioned off focus” generally refers to a position other than the minimum extent of a circle of confusion of the light beam caused by the transfer device or a focal length of the transfer device. In particular, the focal point or minimum extend of the circle of confusion may be at a longitudinal coordinate l_{focus} , whereas the position of each of the optical sensors may have a longitudinal coordinate l_{sensor} different from l_{focus} . For example, the longitudinal coordinate l_{sensor} may be, in a longitudinal direction, arranged closer to the position of the transfer device than the longitudinal coordinate l_{focus} or may be arranged further away from the position of the transfer device than the longitudinal coordinate l_{focus} . Thus, the longitudinal coordinate l_{sensor} and the longitudinal coordinate l_{focus} may be situated at different distances from the transfer device. For example, the optical sensors may be spaced apart from the minimum extend of the circle of confusion in longitu-

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dinal direction by $\pm 2\%$ of focal length, preferably by $\pm 10\%$ of focal length, most preferably $\pm 20\%$ of focal length. For example, at a focal length of the transfer device may be 20mm and the longitudinal coordinate l_{sensor} may be 19,5 mm, i.e. the sensors may be positioned at 97,5% focal length, such that l_{sensor} is spaced apart from the focus by 2,5% of focal length.

- 5 The optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.

Each light-sensitive area may have a geometrical center. As used herein, the term “geometrical center” of an area generally may refer to a center of gravity of the area. As an example, if an
10 arbitrary point inside or outside the area is chosen, and if an integral is formed over the vectors interconnecting this arbitrary point with each and every point of the area, the integral is a function of the position of the arbitrary point. When the arbitrary point is located in the geometrical center of the area, the integral of the absolute value of the integral is minimized. Thus, in other
15 words, the geometrical center may be a point inside or outside the area with a minimum overall or sum distance from all points of the area.

For example, each geometrical center of each light-sensitive area may be arranged at a longitudinal coordinate $l_{\text{center},i}$, wherein i denotes the number of the respective optical sensor. In the case of the detector comprising precisely two optical sensors and in the case of the detector
20 comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor, wherein the first optical sensor, in particular the geometrical center, being arranged at a first longitudinal coordinate $l_{\text{center},1}$, and at least one second optical sensor, wherein the second optical sensor, in particular the geometrical center, being at a second longitudinal coordinate $l_{\text{center},2}$, wherein the first longitudinal coordinate and the second longitudinal coordi-
25 nate differ. For example, the first optical sensor and the second optical sensor may be located in different planes which are offset in a direction of the optical axis. The first optical sensor may be arranged in front of the second optical sensor. Thus, as an example, the first optical sensor may simply be placed on the surface of the second optical sensor. Additionally or alternatively, the first optical sensor may be spaced apart from the second optical sensor, for example, by no
30 more than five times the square root of a surface area of the first light-sensitive area. Additionally or alternatively, the first optical sensor may be arranged in front of the second optical sensor and may be spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm. Relative distance of the first optical sensor and second optical sensor may depend, for example, on focal length or object distance.

35 The longitudinal coordinates of the optical sensors may also be identical, as long as one of the above-mentioned conditions is fulfilled. For example, the longitudinal coordinates of the optical sensors may be identical, but the light-sensitive areas may be spaced apart from the optical axis and/or the surface areas differ.

40 Each geometrical center of each light-sensitive area may be spaced apart from the optical axis of the transfer device, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located. The distance, in particular in transversal direc-

tion, between the geometrical center and the optical axis is denoted by the term "spatial offset". In the case of the detector comprising precisely two optical sensors and in the case of the detector comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor being spaced apart from the optical axis by a first spatial offset and at least one second optical sensor being spaced apart from the optical axis by a second spatial offset, wherein the first spatial offset and the second spatial offset differ. The first and second spatial offsets, as an example, may differ by at least a factor of 1.2, more preferably by at least a factor of 1.5, more preferably by at least a factor of two. The spatial offsets may also be zero or may assume negative values, as long as one of the above-mentioned conditions is fulfilled.

As used herein, the term "surface area" generally refers to both of a shape and a content of at least one light-sensitive area. In the case of the detector comprising precisely two optical sensors and in the case of the detector comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor having a first surface area and at least one second optical sensor having a second surface area. In the case of the detector comprising more than two optical sensors, e.g. a sensor element comprising a matrix of optical sensors, a first group of optical sensors or at least one of the optical sensors of the matrix may form a first surface area, wherein a second group of optical sensors or at least one other optical sensor of the matrix may form a second surface area. The first surface area and the second surface area may differ. In particular, the first surface area and the second surface area are not congruent. Thus, the surface area of the first optical sensor and the second optical sensor may differ in one or more of the shape or content. For example, the first surface area may be smaller than the second surface area. As an example, both the first surface area and the second surface area may have the shape of a square or of a rectangle, wherein side lengths of the square or rectangle of the first surface area are smaller than corresponding side lengths of the square or rectangle of the second surface area. Alternatively, as an example, both the first surface area and the second surface area may have the shape of a circle, wherein a diameter of the first surface area is smaller than a diameter of the second surface area. Again, alternatively, as an example, the first surface area may have a first equivalent diameter, and the second surface area may have a second equivalent diameter, wherein the first equivalent diameter is smaller than the second equivalent diameter. The surface areas may be congruent, as long as one of the above-mentioned conditions is fulfilled.

The optical sensors, in particular the light-sensitive areas, may overlap or may be arranged such that no overlap between the optical sensors is given.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware el-

ements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

5 The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Field Programmable Gate Arrays (FPGAs), or digital
10 signal processors (DSPs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

15 The evaluation device is configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q of the sensor signals A as used herein, the term "quotient signal Q " refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. In the following, the quotient signal is also denoted as combined signal or combined sensor signal.

20 The object size may be a priori unknown. An object image size, i.e. the size of the object in the image plane, may be distance dependent. However, different materials may have different scattering properties such that deviations from Lambertian distribution may occur. In particular, a luminance dependence may deviate significantly from a $1/z^2$. In addition, due type of scattering
25 and/or due to reflective properties and/or scattering properties of the object, for example depending on a material of the object, the object image size, in particular diameter of the spot, may vary. Thus, methods and devices using the spot size, for example by counting pixels, or absolute photon numbers for distance determination may be not suitable and may require calibrations depending on the material of the object. Methods and devices adapted for determination
30 of the distance independently from the object size are necessary in order to allow robust distance determination for any materials with one calibration, only. Further, the object size may be altered by the light source itself. For example, the properties of the optics may be altered by one or more of dirt, raindrops, scratches or manufacturing precision of the light source and/or its optics. Thus, the light beam emitted by the light source may be changed in one or more of its
35 properties such as diameter, beam profile, divergence or the like. Thus, methods and devices adapted for determination of the distance independently from the object size are necessary in order to allow robust distance determination for higher manufacturing tolerances or harsher environmental conditions or the like with one calibration, only. Further, object size may be altered depending on temperature as the optical properties of the detector and or the light source may
40 change such as due to a temperature dependent change of the distance between emitter and optics of the light source or due to a temperature dependent change of the distance between sensor and optics of the detector or due to a temperature dependent change of the refracting properties of the lens or the like. Object size independence is, thus, important for measure-

ments with little or no a priori knowledge on one or more of the measured object, the measurement environment, the temperature of the detector and its environment, the manufacturing quality of the sensor and/or the light source. Further, object size independence is, thus, important for measurements that require high flexibility concerning one or more of the measured object, the measurement environment, the temperature of the detector and its environment, the manufacturing quality of the sensor and/or the light source. Further, object size independence is important, when diverging or converging illumination light beams shall be used, such as when diverging or converging laser light beams are used where the size of the illuminated spot changes with distance. Further, object size independence is, thus, important for measurements that require a high robustness in one or more of measurement quality independent of the measured object, the measurement environment, the temperature of the detector and its environment, the manufacturing quality of the sensor and/or the light source.

Known 3D sensing methods are dependent on the object size. For example US 4,675,517 and US 5,323,222 describe devices and methods exhibiting dependence on object size. Specifically, US 4,675,517 describes with respect to Figures 3A to 3C, column 2, line 16 that the distance information changes with spot size. In particular, it is described that in actual practice, with the use of such a system, as the diameter of the image of the reflection of the projected image spot P changes at random, or the optical position deviates, the defined object distance boundaries are caused to shift largely. In particular, problems due to too large spot sizes are addressed. In case the spot size is too large, the sensor is outshined and the resulting ratio is wrong. US 4,675,517 describes to correct this spot size dependence by enlarging the sensor area. However, the spot size dependence is not eliminated. In particular, problems due to too small spot sizes are not addressed. Furthermore, specifically, US 5,323,222 describes that the ratio used for determining the distance depends on a chip size of the light source. In particular, in column 2, line 11 it is outlined that solid lines I1 and I2 in Fig 14A show the relationship between the reciprocal of the distance a and the calculation output $I1/(I1+I2)$ for the light-projecting chip sizes t1 and t2 ($t1 < t2$), respectively. Furthermore, it is described that if the distance between the projection lens and light emitting element is fT, the base length L, and the chip size of the light emitting element t, then the distance measuring range S will be: $S = \infty$ through $fT \cdot L/t$. However, US 5,323,222 does not further address eliminating impact of object size dependence.

As used herein, the term “object size”, generally refers to a size of the object in an object plane, in particular to the size of a spot and/or region and/or the entire object emitting the at least one light beam. As further used herein, the terms “independent from the object size” and “object size independence” refer to the fact that variations of the object size do not influence the determination of the longitudinal coordinate z. As will be outlined in detail below, the quotient signal Q may be used for determining the longitudinal coordinate. Thus, object size independence furthermore refers to the fact that the quotient signal is essentially independent from the object size such that, in case of identical object distance, a first quotient signal determined at a first object size and a second quotient signal determined at a different object size are identical, with a tolerance of $\pm 20\%$ or less, preferably a tolerance of $\pm 10\%$ or less, more preferably a tolerance of $\pm 5\%$ or less. Requirement or condition of object size independence can be stated as follows:

the circle of confusion of at least one light beam emitted from at least one point of the object is greater than an image size of the object, i.e. a size of the object in an image plane. The detector may be adapted to determine the longitudinal coordinate z of the object independent from the object size if the circle of confusion of the at least one light beam emitted from the at least one point of the object is greater than an image size of the object. This condition may be calculated by $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$, wherein z_0 is a longitudinal distance from the object to the transfer device; E_x is a diameter of an exit pupil of the transfer device; z_s is a longitudinal distance from the transfer device to the light sensitive area, z_i is a distance from the transfer device to an image of the object; O_{size} is an object size of the object in the object plane. As used herein, the term “diameter of an exit pupil” refers to an aperture of the transfer device. Given a measurement range the condition can be fulfilled by for example varying the sensor position, the diameter of the exit pupil or the image of the object, for example by varying the focal length. The detector may be adapted to determine the longitudinal coordinate z of the object independent from the object size if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for at least one distance z_0 . For example, the detector may be adapted to determine the longitudinal coordinate z at the at least one distance z_0 if the object size varies and/or changes within a large range, e.g. by more than 100% of the object size. The detector may be adapted to determine the longitudinal coordinate z at distances, at which the above-mentioned condition $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ is fulfilled only weakly, if the object size varies and/or changes within a smaller range, in particular if the object size varies and/or changes by $\pm 20\%$ or less, preferably by $\pm 10\%$ or less, more preferably by $\pm 5\%$ or less. The detector may be adapted to determine the longitudinal coordinate z over the whole measurement range if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for the at least one distance z_0 and if the object size and if the object size varies and/or changes by $\pm 20\%$ or less, preferably by $\pm 10\%$ or less, more preferably by $\pm 5\%$ or less. As used herein, the term “holds true for at least one distance” refers to that at the at least one distance the longitudinal coordinate z may be determined independent from the object size and that the detector is adapted to determine the longitudinal coordinate z at other distances, in particular within the whole or entire measurement range, if the object size varies and/or changes by $\pm 20\%$ or less, preferably by $\pm 10\%$ or less, more preferably by $\pm 5\%$ or less.

As used herein, the term “measurement range” generally refers to an arbitrary range in which the determination of the longitudinal coordinate z is performed. The measurement range may be adjustable by adjusting one or more parameters selected from the group consisting of: the longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; the diameter of the exit pupil of the transfer device E_x ; the longitudinal distance from the transfer device to the light sensitive area z_s , the distance from the transfer device to an image of the object z_i ; the object size O_{size} of the object in the object plane. For example, if the object size does not exceed an object-size limit, a unique relation of the quotient signal Q to the longitudinal distance from the object to the transfer device exists. The object-size limit may depend on the longitudinal distance from the transfer device to the light sensitive area, the longitudinal distance from the object to the transfer device, and an F-number of the transfer device $F\#$, i.e. a ratio of the focal length of the transfer device and the diameter of an exit pupil of the transfer device. For example, the measurement range may be adapted by selecting and/or choosing one or more of the longitudinal distance from the transfer device to the light sensitive area z_s , the dis-

tance from the transfer device to an image of the object z_i , the longitudinal distance from the transfer device to the light sensitive area, the focal length and the F-number of the transfer device. For example, the focal length and/or F-number may be adjusted by using a zoom objective. In particular, the focal length may be between 10 and 200 mm, preferably between 20 and 150 mm. The F-number may be between 1 and 10, preferably between 1,5 and 6. The longitudinal distance from the transfer device may be as short as possible. The longitudinal distance from the transfer device to the light sensitive area may be between 0 and 200 mm, preferably between 20 and 50 mm. For a given system setup (f , $F\#$, z_s) a unique value of the object-size limit can be calculated. For example, in case the focal length is 3,5 mm and the F-number is 2,0, the object size may be smaller than 1,75 mm. Preferably a lower object size limit may be 0.5 μm or more, more preferably 1 μm or more and most preferably 10 μm or more. The lower object size limit refers to a minimum object size using an active measurement system, in particular to a laser spot size.

The evaluation device may be configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate. For example, the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are areas of the beam profile at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A_1 and area A_2 may differ. In particular, A_1 and A_2 are not congruent. Thus, A_1 and A_2 may differ in one or more of the shape or content. As used herein, the term “beam profile” relates to a spatial distribution, in particular in at least one plane perpendicular to the propagation of the light beam, of an intensity of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_o)$ and beam shape $S(x, y; z_o)$, $E(x, y; z_o) = L \cdot S$. Thus, by deriving the quotient signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the quotient signal allows determination of the distance z_o independent from the object size. Thus, the quotient signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like. Further-

more, even at very large spot sizes, for example at an outshining, reliable determination of the longitudinal coordinate of the object may be possible.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the quotient signal Q. The light-sensitive regions may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term "essentially edge information" generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, the center signal may be a signal corresponding to a rectangular stripe comprising the center of the light spot. The evaluation device may be adapted to form a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. In par-

5 ticular, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile. The first and the second area, as an example, may be adjacent, with a minimum separation from one another. However, the separation between adjacent areas may also be varied such as enlarged to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent areas may be increased by not evaluating the area between the first and the second area. This may reduce the light on one of the areas relative and/or absolute as compared to the adjacent area which will may increase a quotient of the signal of the two
10 areas. Further, one or both of the areas may consist of separated subareas which may be adjacent to each other and/or which may be separated by areas that are not evaluated and/or which may be evaluated as part of a different quotient. Further, the first and the second area may consist of a linear combination of subareas, whereas the signal contributed by each subarea may be weighted differently when forming the center signal and/or the sum signal. This may further
15 be beneficial for increasing the dynamic range of the quotient system.

The extent of the circle of confusion may be larger than the extent of the optical sensor. For example, the optical sensor may be positioned as such that the circle of confusion extends beyond the optical sensor. The optical sensor may thus only partially evaluate the beam profile of
20 the light beam. The evaluation device may be adapted to extrapolate a beam profile such as due to symmetry considerations or due to comparing the partial beam profile to prerecorded beam profiles or the like. Further, the evaluation device may be adapted to evaluate a partial sum signal and a partial center signal of a partial beam profile and convert it into a sum signal and a center signal of an extrapolated and/or fitted and/or prerecorded beam profile or the like.

25 The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the
30 second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by
35 equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

40 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot

may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The detector may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors. For example, the detector may comprise a plurality of optical sensors arranged in a matrix. The detector may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, a defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. This principle is called Distance by Photon Ratio (DPR).

The DPR principle can be applied, for example, to many sub-regions within a segmented image profile, such as a segmented image of at least one feature generated by the at least one light beam on the matrix of optical sensors, which is demonstrated hereafter. The quotient $Q_A(z)$ in the two-dimensional case can be written as

$$Q_A(z) = \frac{\iint_{-r_{out}}^{r_{out}} P(x,y,z) dx dy}{\iint_{-r_{in}}^{r_{in}} P(x,y,z) dx dy},$$

where $P(x, y, z)$ is a two-dimensional beam profile and r_{in} and r_{out} are inner and outer circle radii, respectively. For line-segmented quotients $Q_y(z)$ along the y -dimension, this can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{out}}^{r_{out}} P(x,z) dx}{\int_{-r_{in}}^{r_{in}} P(x,z) dx}.$$

Without wishing to be bound by this theory, the overall defocused beam profile $P(x, z)$ can be regarded as a superposition of defocused image point profiles $p(x, z)$ along the image width $b(x, z)$. This relationship can be modelled as a convolution given by

$$P(x, z) = b(x, z) * p(x, z),$$

wherein $p(x, z)$ represents the Point-Spread-Function (PSF) of a lens in defocus, which in the field of paraxial optics is also known as Circle of Confusion (CoC). By inserting $P(x, z)$ into $Q_y(z)$ the above described quotient $Q_y(z)$ can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{out}}^{r_{out}} b(x,z) * p(x,z) dx}{\int_{-r_{in}}^{r_{in}} b(x,z) * p(x,z) dx} = \frac{\int_{-r_{out}}^{r_{out}} b(x,z) dx \cdot \int_{-r_{out}}^{r_{out}} p(x,z) dx}{\int_{-r_{in}}^{r_{in}} b(x,z) dx \cdot \int_{-r_{in}}^{r_{in}} p(x,z) dx}.$$

The CoC radius may be r_c and r_o may be an image spot radius of the feature on the matrix of optical sensors, then $r_{out} = r_c + r_o \Rightarrow r_{out} \geq r_c$ and $r_{out} \geq r_o$, which yields to

$$Q_y(z) = \frac{\int_{-r_o}^{r_o} b(x,z)dx \cdot \int_{-r_c}^{r_c} p(x,z)dx}{\int_{-r_{in}}^{r_{in}} b(x,z)dx \cdot \int_{-r_{in}}^{r_{in}} p(x,z)dx}.$$

In case of $r_o \leq r_{in} < r_c$,

$$Q_y(z) = \frac{\int_{-r_o}^{r_o} b(x,z)dx \cdot \int_{-r_c}^{r_c} p(x,z)dx}{\int_{-r_o}^{r_o} b(x,z)dx \cdot \int_{-r_{in}}^{r_{in}} p(x,z)dx} = \frac{\int_{-r_c}^{r_c} p(x,z)dx}{\int_{-r_{in}}^{r_{in}} p(x,z)dx},$$

which indicates that the quotient is independent of the object size on condition that $r_o < r_c$.

The Object Size Independence (OSI) in DPR applies as long as the image width i remains below the Circle of Confusion (CoC) width denoted as c . In the one-dimensional case, this is represented by diameters $d_o < d_c$. Using paraxial optics, these diameters can be substituted, such that

$$\frac{i}{a} o_s < \frac{D_x}{i} |d - i|,$$

with a as the object distance, i as the image distance, o_s as the object size, D_x as the lens exit pupil diameter and d as the sensor position with respect to image-side principal plane. After re-arranging, this can be written as

$$o_s < \frac{a D_x}{i^2} |d - i|.$$

When focusing a single lens to infinity, this simplifies to

$$o_s < \frac{f}{F\#} \left| 1 - \frac{f}{a} \right|,$$

where f is the focal length and $F\#$ the F-number of the lens.

Furthermore, with a matrix of optical sensors a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates. Accordingly, the parameterization of a single line may be given by

$$\omega = x \cos(\theta) + y \sin(\theta).$$

The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega,\theta)}\{f(x,y)\} = \iint_{-\infty}^{\infty} f(x,y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy,$$

where δ denotes the Dirac delta function and $f(x,y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is given by

$$R = \frac{\Re_{(\omega,\theta)}\{f'(x,y)\}}{\Re_{(\omega,\theta)}\{f(x,y)\}},$$

with $f'(x,y)$ as the inner region. The variation of θ may yield different ratios R for skewed object surfaces at a certain distance. It is sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

In one embodiment, the light beam propagating from the object to the detector may illuminate the sensor element with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at

least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

For example, the light beam propagating from the object to the detector may illuminate the sensor element with at least one point pattern. The point pattern may be generated by the object, for example in response to an illumination by the at least one light source with an illumination pattern comprising the at least one point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The light beam propagating from the object to the detector may illuminate the sensor element with a reflection pattern comprising both point patterns and line patterns. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

For example, the optical sensors may comprise a first optical sensor having a first light sensitive area and a second optical sensor having a second light sensitive area, wherein the first and the second light sensitive areas are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$

is satisfied, wherein “a” is a ratio of photons hitting both an inner region of a plane perpendicular to the optical axis intersecting the optical axis at a distance equal to half of a focal length of the transfer device and the first light sensitive area, “b” is a ratio of photons hitting both the inner region of the plane and the second light sensitive area, “c” is a ratio of photons hitting both an outer region of the plane and the first light-sensitive area, “d” is a ratio of the photons hitting both the outer region of the plane and the second light sensitive area. The first light sensitive area and the second light sensitive areas may be arranged such that the inner region may have an area with a geometrical center point on the optical axis and an extension such that half of the photons hit the plane within the inner region and the other half hit the plane outside the inner region. The first light sensitive area and the second light sensitive areas may be arranged such that the inner region may be designed as a circle with a center point on the optical axis and a radius r which is chosen such that half of the photons hit the plane within the circle and the other half hit the plane outside the circle.

As outlined above, the detector may comprise at least one illumination source. A distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors may be small. The distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. The illumination source and the optical axis may be separated by a small baseline. As used herein, the term “baseline”, also denoted as basis line, refers to a distance, for example in a xy -plane, between the optical axis and the illumination source, in particular a distance between the optical axis and a z-component of the illumination light beam. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and of the at least one optional transfer device, which will be described in detail below. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device.

At least one of the optical sensors may be adapted to generate at least one sensor signal dependent on a time-of-flight (TOF) the illumination light beam has traveled from the illumination source to the object and the reflection light beam has traveled from the object to the light sensitive area of the optical sensor. The evaluation device may be configured for determining at least one TOF-longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. The optical sensor adapted to generate the TOF sensor signal may be designed as time-of-flight detector. The time-of-flight detector may be selected from the group consisting of: at least one pulsed time-of-flight detector; at least one phase modulated time-of-flight detector; at least one direct time-of-flight detector; at least one indirect time-of-flight detector. For example, the pulsed time-of-flight detector may be at least one range gated imager and/or at least one direct time-of-flight imager. For example the phase modulated time-of-flight detector may be at least one RF-

modulated light source with at least one phase detector. The optical sensor may be adapted to determine a time delay between emission of the illumination light beam by the illumination source and receipt of the reflection light beam.

- 5 For example, the optical sensor adapted to generate the TOF sensor signal may be designed as pulsed time-of-flight detector. The detector may comprise the at least one interruption device, such as at least one shutter element, adapted to generate a pulsed light beam. The optical sensor may be adapted to store the TOF sensor signal dependent on the receiving time of the reflection light beam in a plurality of time windows, in particular subsequent time windows. The
- 10 optical sensor may be adapted to store dependent on receiving time of the reflection light beam the generated TOF sensor signal in at least one first time window and/or in at least one second time window. The first and second time windows may be correlated with the opening and closure of the interruption device. Duration of first and second time windows may be pre-defined. For example, the TOF sensor signal may be stored in the first time window during opening of
- 15 the interruption device, whereas during closure of the interruption device the TOF sensor signal may be stored in the second time window. Other durations of time windows are thinkable. The first and the second time window may comprise information about background, signal height and signal shift.
- 20 For example, the optical sensor adapted to generate the TOF sensor signal may be designed as direct time-of-flight imager. The direct time-of-flight imager may comprise the at least one illumination source adapted to generate at least one single laser pulse. The single laser pulse may be reflected back from the object onto the optical sensor. The optical sensor may comprise at least one photodiode, for example at least one Avalanche Photo Diode (APD), such as at
- 25 least one Si APD, or such as at least one InGaAs APD, or at least one PIN photo detector array, or at least one Single-Photon Avalanche Photodiode (SPAD), adapted to image the reflection light beam. The direct time-of flight imager may be adapted to image at least one image comprising spatial and temporal data.
- 30 For example, the optical sensor adapted to generate the TOF sensor signal may be designed as phase modulated time-of-flight modulator. The phase modulated time-of-flight modulator may be adapted to measure a difference in phase, in particular a phase shift, by determining a correlated signal, for example by multiplying a received signal, i.e. of the reflection light beam, with the emitted signal, i.e. the illumination light beam. A DC component of the correlated signal may
- 35 comprise an information about the difference in phase. The evaluation device may be adapted to determine the second longitudinal coordinate of the object from the phase difference. For example, the illumination source and the optical sensor adapted to generate the TOF sensor signal may be designed as RF-modulated light source with at least one phase detector. The illumination source may comprise, for example, at least one LED and/or at least one laser. The
- 40 illumination source may comprise at least one modulation device adapted to modulate the light beam having a pre-defined phase shift. For example, the modulation device may comprise at least one radio frequency module. The radio frequency module may be adapted to modulate the

illumination beam with an RF carrier. The optical sensor may be adapted to determine a phase shift of the reflective light beam impinging on the optical sensor.

The optical sensor may be designed as and/or may comprise at least one time-of-flight pixel.

- 5 Preferably, the detector may comprise at least two optical sensors, wherein each optical sensor is designed as and/or comprises at least one TOF pixel. For example, the detector, in particular the optical sensor may comprise a quadrant diode adapted to generate the TOF sensor signal. For example, the detector, in particular the optical sensor, may comprise at least one pixelated TOF-imager.

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The evaluation device may be configured for determining the at least one TOF longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. As used herein, the term "TOF longitudinal coordinate" refers to a longitudinal coordinate derived from the TOF sensor signal.

- 15 As described above, the illumination light source may be adapted to periodically generate at least one light pulse. The detector may be adapted to generate the first longitudinal sensor signal for each period. The evaluation device may be adapted to determine in which pulse period the TOF sensor signal was generated using the second longitudinal coordinate. The detector may be adapted to uniquely assign to which period the ToF-signal refers to by using the combined sensor signal. Both, the TOF sensor signal and the combined sensor signal, may be non-monotonous functions of the longitudinal coordinate z_{real} . Thus, the longitudinal coordinate may not be uniquely determined from one of the TOF sensor signal or the combined sensor signal alone and a measurement range may not be restricted to a longitudinal range in which the signals are unique functions of z_{real} . The term "measurement range" generally refers to a range from the object to the detector in which determination of the longitudinal coordinate is possible.

- 25 The term "longitudinal range" generally refers to a range from the object to the detector in which unique determination of the longitudinal coordinate is possible. Below and/or above certain distances from the object to the detector a determination of the longitudinal coordinate may not be possible. For example, time-of-flight measurements are not possible below a certain distance from the object to the detector, possibly due to the minimum measurement time of the internal clock. Furthermore, in time-of-flight measurements the sensor signal may be unique within a longitudinal period, but the sensor signal may be the same in case integer multiples of the longitudinal period are added such that the determined longitudinal coordinate may be non-unique. Thus, the same TOF sensor signal will be obtained for a distance z_1 and a distance $z_1 + n \cdot z_{1p}$, where n is an integer denoting the longitudinal period and z_{1p} is the longitudinal period of the TOF sensor signal, wherein the distances z_1 and $z_1 + n \cdot z_{1p}$ are within the measurement range.
- 35 As used herein, the term "longitudinal period" refers to partitions of a period, in particular a distance range, in which the longitudinal coordinate can be unambiguously determined from the TOF sensor signal. Non-unique longitudinal coordinates may be denoted as relative longitudinal coordinates and the unique longitudinal coordinates may be denoted as absolute longitudinal coordinates.
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If both the TOF sensor signal $F1$ and the combined sensor signals $F2$ are available, it may be possible to uniquely determine the longitudinal coordinate and to extend the longitudinal range,

as long as each signal pair (F1,F2) corresponds to a unique distance and vice versa. In particular, if a unique signal pair (F1,F2) exists for each longitudinal coordinate and vice versa, the evaluation device may be adapted to determine the unique combined longitudinal coordinate by

- (1) selecting at least one first selected signal such as the TOF sensor signal and/or the combined sensor signal and determining non-unique first longitudinal coordinates;
- (2) selecting a second selected signal such as the combined signal Q and/or the TOF sensor signal, which was not selected in step (1), and determining non-unique second longitudinal coordinates;
- (3) determining whether one of the non-unique first longitudinal coordinates and the non-unique second longitudinal coordinates match up to a predetermined tolerance threshold;
- (4) setting a combined unique longitudinal coordinate to be a matching longitudinal coordinate.

In step (1) and step (2), signals may be selected in the given order or may be performed in a different order. For example, in step (1) the TOF sensor signal may be selected and in step (2) the combined signal Q may be selected. In another example, in step (1) the combined sensor signal may be selected, and the non-unique first longitudinal sensor signal may be determined therefrom. In step (2), the TOF sensor signal may be selected.

Additionally or alternatively to step (4), the evaluation device may be adapted to output an error signal in case no matching coordinates are found and/or to output an error signal in case more than one matching coordinates are found. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be stored in a look-up table. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be approximated or described by an analytical function which is evaluated to find the longitudinal coordinate corresponding to a given signal pair.

The evaluation device may comprise at least two memory elements. As used herein, the term “memory element” refers to a device adapted to store information. The evaluation device may be adapted to receive and to store information provided by the optical sensors, for example the at least one first sensor signal. Such information may comprise raw sensor data and/or processed sensor data. For example, the memory element may be adapted to store information for further evaluation by the evaluation device. The memory element may be a volatile or non-volatile memory element.

As described above, the optical sensor may be designed as and/or may comprise at least one ToF pixel. The detector may comprise at least two switches. Each of the switches may be connected to the optical sensor adapted to generate the first sensor signal, for example by at least one connector. In particular, each of the switches may be connected to the ToF pixel. The switches are adapted to provide the TOF sensor signal to one of the memory elements. In particular, the switches may be adapted to let, dependent on receiving time of the reflection light beam, the generated TOF sensor signal pass through one of the switches. For example, the TOF sensor signal may pass one of the switches during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may pass the other

switch. Each of the switches may be controlled by a control signal having a pulse length identical to a pulse length of a light pulse generated by the illumination source. The control signal of one of the switches may be delayed. For example, the delay may correspond to the pulse length of the light pulse. The evaluation device may be adapted to sample and/or store depending on the delay a first part, or fraction, of the TOF sensor signal through a first switch in a first memory element and the other, second part, or fraction, of the TOF sensor signal through a second switch in a second memory element. The evaluation device may be adapted to determine the first longitudinal coordinate by evaluating the first part and second part of the TOF sensor signal. The evaluation device may be adapted to determine the first longitudinal coordinate z_1 by

$$z_1 = \frac{1}{2} \cdot c \cdot t_0 \cdot \frac{S_{12}}{S_{11} + S_{12}} + z_0;$$

wherein c is the speed of light, t_0 is the pulse length of the illumination light beam, z_0 is a distance offset often determined by the resolution of the time measurement, and S_{11} and S_{12} are the first part and second part of the TOF sensor signal, respectively.

As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

In a first preferred embodiment of the present invention, the detector may comprise:

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor may be configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
- at least one second optical sensor having a second light-sensitive area, wherein the second optical sensor may be configured to generate at least one second sensor signal in response to an illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area may be smaller than the second light-sensitive area; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

In this first preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their longitudinal coordinate and/or their surface areas.

As will further be outlined in detail below, each optical sensor, preferably, may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, preferably, each optical sensor is a single area optical sensor rather than a pixelated optical sensor like e.g. a sensor matrix. Other embodiments, however, are feasible. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and effi-

cient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the setup, one with a small light-sensitive area and one with a larger light-sensitive area, as will be explained in further detail below.

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The first and second light-sensitive areas specifically may be oriented towards the object.

The light beam propagating from the object to the detector specifically may fully illuminate the first light-sensitive area, such that the first light-sensitive area is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the first optical sensor. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the second light-sensitive area, which is smaller than the second light-sensitive area, such that the light spot is fully located within the second light-sensitive area. Within the light spot on the second light-sensitive area, a shadow created by the first optical sensor may be located. Thus, generally, the first optical sensor, having the smaller first light-sensitive area, may be located in front of the second optical sensor, as seen from the object, with the first light-sensitive area being fully located within the light beam and with the light beam generating a light spot on the second light-sensitive area being smaller than the second light-sensitive area, and with further a shadow created by the first optical sensor within the light spot. The situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a light spot generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

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As outlined above, the first light-sensitive area is smaller than the second light-sensitive area. As used therein, the term "is smaller than" refers to the fact that the surface area of the first light-sensitive area is smaller than the surface area of the second light-sensitive area, such as by at least a factor of 0.9, e.g. at least a factor of 0.7 or even by at least a factor of 0.5. As an example, both the first light-sensitive area and the second light-sensitive area may have the shape of a square or of a rectangle, wherein side lengths of the square or rectangle of the first light-sensitive area are smaller than corresponding side lengths of the square or rectangle of the second light-sensitive area. Alternatively, as an example, both the first light-sensitive area and the second light-sensitive area may have the shape of a circle, wherein a diameter of the first light-sensitive area is smaller than a diameter of the second light-sensitive area. Again, alternatively, as an example, the first light-sensitive area may have a first equivalent diameter, and the second light-sensitive area may have a second equivalent diameter, wherein the first equivalent diameter is smaller than the second equivalent diameter.

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The first light-sensitive area specifically may overlap with the second light-sensitive area in a direction of propagation of the light beam. The light beam may illuminate both the first light-sensitive area and, fully or partially, the second light-sensitive area. Thus, as an example, as seen from an object located on an optical axis of the detector, the first light-sensitive area may

be located in front of the second light-sensitive area, such that the first light-sensitive area, as seen from the object, is fully located within the second light-sensitive area. When the light beam from this object propagates towards the first and second light-sensitive areas, as outlined above, the light beam may fully illuminate the first light-sensitive area and may create a light spot on the second light-sensitive area, wherein a shadow created by the first optical sensor is located within the light spot. It shall be noted, however, that other embodiments are feasible.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between first and second sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the object by using at least one known, determinable or predetermined relationship between the quotient signal derived from the first and second sensor signals and the longitudinal coordinate.

Thus, the evaluation device specifically may be configured for deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. As an example, Q may simply be determined as

$$Q = s_1/s_2$$

or

$$Q = s_2/s_1,$$

with s_1 denoting the first sensor signal and s_2 denoting the second sensor signal. Additionally or alternatively, Q may be determined as

$$Q = a \cdot s_1 / b \cdot s_2$$

or

$$Q = b \cdot s_2 / a \cdot s_1,$$

with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as

$$Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2),$$

with a , b , c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as

$$Q = s_1 / (s_1 + s_2).$$

Other quotient signals are feasible.

Typically, in the setup described above, Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q=s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the first signal s_1 and the second signal s_2 decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the first signal s_1 decreases more rapidly than the second signal s_2 , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the first and second sensor signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

The evaluation device is configured for determining the longitudinal coordinate by evaluating the quotient signal Q . The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The quotient signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

The first and second optical sensors specifically may be arranged linearly in one and the same beam path of the detector. As used herein, the term “linearly” generally refers to that the sensors are arranged along one axis. Thus, as an example, the first and second optical sensors both may be located on an optical axis of the detector. Specifically, the first and second optical sensors may be arranged concentrically with respect to an optical axis of the detector.

The first optical sensor may be arranged in front of the second optical sensor. Thus, as an example, the first optical sensor may simply be placed on the surface of the second optical sensor. Additionally or alternatively, the first optical sensor may be spaced apart from the second optical sensor by no more than five times the square root of a surface area of the first light-sensitive area. Additionally or alternatively, the first optical sensor may be arranged in front of the second optical sensor and may be spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm.

As outlined above, the second light-sensitive area is larger than the first light-sensitive area. Thus, as an example, the second light-sensitive area may be larger by at least a factor of two, more preferably by at least a factor of three and most preferably by at least a factor of five and the first light-sensitive area.

The first light-sensitive area specifically may be a small light-sensitive area, such that, preferably, the light beam fully illuminates this light-sensitive area. Thus, as an example which may be applicable to typical optical configurations, the first light-sensitive area may have a surface area of 1 mm² to 150 mm², more preferably a surface area of 10 mm² to 100 mm².

The second light-sensitive area specifically may be a large area. Thus, preferably, within a measurement range of the detector, light spots generated by a light beam propagating from the object of the detector may fully be located within the second light-sensitive area, such that the light spot is fully located within the borders of the second light-sensitive area. As an example, which is applicable e.g. in typical optical setups, the second light-sensitive area may have a surface area of 160 mm² to 1000 mm², more preferably a surface area of 200 mm² to 600 mm².

Infrared optical sensors which may be used for the first optical sensor, for the second optical sensor or for both the first and second optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany. Thus, as an example, the first optical sensor, the second optical sensor or both the first and the second optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the first optical sensor, the second optical sensor or both the first and the second optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor pho-

todiode selected from the group consisting of: a Ge:Ga photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the first optical sensor, the second optical sensor or both the first and the second optical sensor may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The first and second optical sensors each, independently, may be opaque, transparent or semi-transparent. For the sake of simplicity, however, opaque sensors which are not transparent for the light beam, may be used, since these opaque sensors generally are widely commercially available.

The first and second optical sensors each specifically may be uniform sensors having a single light-sensitive area each. Thus, the first and second optical sensors specifically may be non-pixelated optical sensors.

As outlined above, by evaluating the first and second sensor signals, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more additional transversal sensors may be used for determining at least one transversal coordinate of the object. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object.

In a further embodiment, the detector may comprise:

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor may be configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;

- at least one second optical sensor having a second light-sensitive area, wherein the second optical sensor may be configured to generate at least one second sensor signal in response to an illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area and the second light sensitive area may be arranged such that the light-sensitive areas differ in their longitudinal coordinate; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

With respect to definitions and embodiments reference can be made to the description of the first preferred embodiment. In particular, the first light-sensitive area may be smaller than the second light-sensitive area.

In a further preferred embodiment of the present invention, the detector may comprise:

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor may be configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;
- at least one evaluation device may be configured for evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

In this further preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in spatial offset and/or surface areas.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. As further used herein, the term "matrix" generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane.

As explained in more detail in e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot, may be characterized by two measurement variables: a measurement signal measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their respective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming the quotient signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the quotient signal for deriving the longitudinal coordinate.

The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus, may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.

The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or

alternatively, as will be outlined in further detail below, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

5 Consequently, in accordance with the present invention, the term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals
10 generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from
15 which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from
20 which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the
25 trapezoid, in particular of a plateau of the trapezoid.

Similarly, the term “sum signal” generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the
30 entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation
35 device may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of
40 the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

Similarly, the term “combined signal”, as used herein, generally refers to a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

The light beam propagating from the object to the detector specifically may fully illuminate the at least one optical sensor from which the center signal is generated, such that the at least one optical sensor from which the center signal arises is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the at least one optical sensor from which the sensor signal arises. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the entire matrix which is smaller than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a “light spot” generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal

may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

As outlined above, raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition.; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or maybe determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by

the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations, however, are feasible.

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As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be performed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

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As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

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The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the quotient signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined sig-

nal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the quotient signal Q such as the combined signal. This determining may be a one-step process, such as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined signal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

The matrix may be composed of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

As further outlined above, the matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term “essentially perpendicular”, reference may be made to the definition given above. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10

rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, as outlined above, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

As further outlined above, preferably, the sensor element may be oriented essentially perpendicular to an optical axis of the detector. Again, with respect to the term “essentially perpendicular”, reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The detector may be adapted to determine at least one distance information of the object by using triangulation and/or structured light techniques. In known 3D sensing devices, such as devices using triangulation or structured light techniques, due to correspondence problems regular, constant or periodic pattern are not suitable since each measured point has to be assigned to one reference point of a reference pattern. The at least one light beam propagating from the object to the detector may be adapted to generate at least one reflection pattern on the matrix of optical sensors. The term “reflection pattern” refers to at least one image determined by the optical sensors in response to illumination of the light sensitive areas by the light beam propagating from the object to the detector which was generated in response to illumination by the illumination pattern. The reflection pattern may comprise at least one reflection feature depending on the corresponding illumination feature of the illumination pattern. The detector may be adapted to determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection pattern from the quotient signal. Thus, the detector may be adapted to pre-classify the at least one reflection feature of the reflection pattern. This allows using illumination patterns comprising regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal patterns may be preferred.

The evaluation device may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor

signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device may be adapted to determine at least one reference feature in at least one reference pattern corresponding to the at least one reflection feature. The term "reference pattern" refers to an image of comparison at a specific spatial position comprising at least one reference feature. As outlined above, the evaluation device may be adapted to perform an image analysis and to identify features of the reflection pattern. The evaluation device may be adapted to identify at least one reference feature in the reference pattern having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference pattern and the reflection pattern may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The distance may be a relative distance, also denoted as baseline. The evaluation device may be adapted to determine an epipolar line in the reference pattern. The relative position of the reference pattern and reflection pattern may be known. For example, the relative position of the reference pattern and the reflection pattern may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature to the real world feature from which it originates. Thus, the straight line may comprise possible object features corresponding to the selected reflection feature. The straight line and the baseline span an epipolar plane. As the reference pattern is determined at a different relative constellation from the reflection pattern, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference pattern. The epipolar line may be the intersection of the epipolar plane and the reference pattern. Thus, a feature of the reference pattern corresponding to the selected feature of the reflection pattern lies on the epipolar line.

Depending on the distance to the object, the reference feature corresponding to the image position of the reflection feature may be displaced within the reference pattern compared to the image position in the reflection pattern. The reference pattern may comprise at least one displacement region in which the reference feature corresponding to the selected reflection feature may be imaged. The displacement region may comprise only one reference feature. The dis-

placement region may extend along the epipolar line. The evaluation device may be adapted to determine the reference feature along the epipolar line. The evaluation device may be adapted to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm \varepsilon$ from the quotient signal Q to determine a displacement region along the epipolar line corresponding to $z \pm \varepsilon$. The evaluation device may be adapted to match the selected reflection feature with at least one reference feature within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating the corresponding reference and reflection features. The evaluation device may be adapted to match the selected feature of the reflection pattern with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. Preferably, the detector may be adapted to pre-classify the selected reflection feature using the quotient signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference pattern may have a relative distance to each other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line. For example, the illumination pattern may comprise at least one hexagonal pattern. Preferably, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Preferably, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other.

The evaluation device may be adapted to determine a displacement of the reference feature and the reflection feature. The evaluation device may be adapted to determine the displacement of the matched reference feature and the selected reflection feature. As used herein, the term “displacement” refers to the difference between an image position in the reference image to an image position in the reflection image. The evaluation device may be adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate z_{triang} . For example, the longitudinal information may be a distance value. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case the position of the selected reflection feature in the reflection pattern and the position of the matched reference feature and/or the relative displacement of the selected reflection feature and the matched reference feature are known, the longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. The displacement and the corresponding distance value may be stored in at least one storage

device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

The detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, at least one optical sensors of the matrix of optical sensors may be adapted to generate a so called FiP signal. For example, each of the optical sensors of the matrix may be adapted to generate the at least one FiP signal. For example, the matrix of optical sensors may be a pixelated FiP-sensor. Additionally or alternatively, the FiP signal may be determined, such as extracted and/or simulated, from the sensor signals of the matrix and the quotient as outlined above may be determined from the sensor signals.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image

may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

The illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or object recognition method.

Preferably, the illumination source may be a movable and/or mobile illumination source. For example, the illumination source may be moved during the measurement such as to illuminate the object from different positions and/or angles. However, embodiments are feasible in which the illumination source may be positioned in at least one fixed position, for example during a complete measurement time. The evaluation device may be adapted to illumination sources with unclear position such as due to high manufacturing tolerances and/or user interaction and/or user assembly or the like.

The illumination source may illuminate the object with a convergent and/or divergent and/or collimated light beam.

In a further preferred embodiment the detector may comprise:

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

10 In this further preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas

The light-sensitive areas of the optical sensors may overlap, as visible from the object, or may not overlap, i.e. may be placed next to each other without overlap. The light-sensitive areas may be spaced apart from each other or may directly be adjacent.

The detector may comprise more than two optical sensors. In any case, i.e. in the case of the detector comprising precisely two optical sensors and in the case of the detector comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor being spaced apart from the optical axis by a first spatial offset and at least one second optical sensor being spaced apart from the optical axis by a second spatial offset, wherein the first spatial offset and the second spatial offset differ. In case further optical sensors are provided, besides the first and second optical sensors, these additional optical sensors may also fulfill the condition or, alternatively, may be spaced apart from the optical axis by the first spatial offset, by the second spatial offset or by a different spatial offset. The first and second spatial offsets, as an example, may differ by at least a factor of 1.2, more preferably by at least a factor of 1.5, more preferably by at least a factor of two. The spatial offsets may also be zero or may assume negative values, as long as the above-mentioned conditions are fulfilled.

30 As outlined above, each light-sensitive area has a geometrical center. Each geometrical center of each light-sensitive area may be spaced apart from the optical axis of the detector, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located.

35 As outlined above, the optical sensors specifically may be located in one and the same plane, which, preferably, is a plane perpendicular to the optical axis. Other configurations, however, are possible. Thus, two or more of the optical sensors may also be spaced apart in a direction parallel to the optical axis.

40 For example, the optical sensors may be partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector. As used herein, the term "partial diode" may comprise several diodes that are connected in series or in parallel. This example is rather simple and cost-efficiently realizable. Thus, as an example, bi-cell diodes

or quadrant diodes are widely commercially available at low cost, and driving schemes for these bi-cell diodes or quadrant diodes are generally known. As used herein, the term "bi-cell diode" generally refers to a diode having two partial diodes in one packaging. Bi-cell and quadrant diodes may have two or four separate light sensitive areas, in particular two or four active areas.

5 As an example, the bi-cell diodes may each form independent diodes having the full functionality of a diode. As an example, each of the bi-cell diodes may have a square or rectangular shape, and the two diodes may be placed in one plane such that the two partial diodes, in total, form a 1 x 2 or 2 x 1 matrix having a rectangular shape. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes and quadrant diode is proposed, as will be outlined in further detail below. Generally, however, the optical sensors specifically may be partial diodes of a quadrant diode, with a center of the quadrant diode being off-centered from the optical axis of the detector. As used herein, the term "quadrant diode" generally refers to a diode having four partial diodes in one packaging. As an example, the four partial diodes may each form independent diodes having the full functionality of a diode. As an example, the four partial diodes may each have a square or rectangular shape, and the four partial diodes may be placed in one plane such that the four partial diodes, in total, form a 2 x 2 matrix having a rectangular or square shape. In a further example, the four partial diodes, in total, may form a 2 x 2 matrix having a circular or elliptical shape. The partial diodes, as an example, may be adjacent, with a minimum separation from one another. However, the separation between adjacent diodes may also be varied to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent diodes may be increased by using an opaque mask. This may reduce the light on one of the diodes relative and/or absolute as compared to the adjacent diode which will may increase a quotient of the signal of the two diodes.

25 In case a quadrant diode is used, having a 2x2 matrix of partial diodes, the center of the quadrant diode specifically may be off-centered or offset from the optical axis. Thus, as an example, the center of the quadrant diodes, which may be an intersection of the geometrical centers of the optical sensors of the quadrant diode, may be off-centered from the optical axis by at least 0.2 mm, more preferably by at least 0.5 mm, more preferably by at least 1.0 mm or even 2.0 mm. Similarly, when using other types of optical sensors setups having a plurality of optical sensors, an overall center of the optical sensors may be offset from the optical axis by the same distance.

35 As outlined above, bi-cell and quadrant diodes may have two or four separate light sensitive areas, in particular two or four active areas. The light sensitive areas may be separated by a dividing line. The dividing line, for example, of two Si-diodes, can be arranged parallel to the baseline of the detector, in particular parallel to a spot movement on the optical sensors. However, other arrangements are feasible. For example, the quadrant diode may comprise two dividing lines. The dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other. In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device may be adapted to determine a second quotient using at least three or all four sensor signals of the

quadrant diode. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotient signals allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient signal may be determined by dividing the sensor signals of two top quadrants, also called top segment, by the sensor signals of two bottom quadrants, also called bottom segment. Using the quotient signal of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between the top and bottom segment is parallel to the baseline, the quotient signal determined from top segment divided by bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient signal by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals of the quadrants such that distance measurement is possible over a wide range with a large resolution.

Generally, the light-sensitive areas of the optical sensors may have an arbitrary surface area or size. Preferably, however, specifically in view of a simplified evaluation of the sensor signals, the light-sensitive areas of the optical sensors are substantially equal, such as within a tolerance of less than 10%, preferably less than 5% or even less than 1%. This, specifically, is the case in typical commercially available quadrant diodes.

The detector may comprise at least one quadrant diode comprising four quadrants. For example, an upper left quadrant of the quadrant diode may be denoted "ol", a lower left quadrant "ul", an upper right quadrant may be denoted "or" and a lower right quadrant may be denoted "ur".

The light spot may move from the left in a small distance range to the left in a far distance range. The detector may comprise at least one neutral density filter. The quadrants of the small distance range may be covered by the neutral density filter. In the small distance range the quotient ol/ul may be used to determine the longitudinal coordinate z . In the far distance range a quotient or/ur and/or a quotient from a combination of all quadrants may be used. The neutral density filter allows that objects in the small distance range are adapted to illuminate the optical sensors brighter by constant amplification without saturating the amplifier. At the same time it is possible to obtain in the far distance range more of the signal. Thus, the neutral density filter allows to enhance dynamic of the detector.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between sensor signals and/or any secondary signal derived thereof and the longitudinal coordinate. Thus, the evaluation device may be config-

ured for determining at least one combined sensor signal out of the at least two sensor signals, i.e. of the at least one sensor signal of at least one first optical sensor and out of the at least one sensor signal of at least one second optical sensor.

5 The detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the bi-cell or quadrant diode may be adapted to generate a so called FiP signal. As outlined e.g. in WO
2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information
10 over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. According to the FiP-effect, the longitudinal sensor signal, given the same total power, may exhibit at least one pronounced maximum for one or a plurality of focusings and/or
15 for one or a plurality of specific sizes of the light spot on the sensor region or within the sensor region. For purposes of comparison, an observation of a maximum of the longitudinal sensor signal in a condition in which the corresponding material is impinged by a light beam with the
20 smallest possible cross-section, such as when the material may be located at or near a focal point as affected by an optical lens, may be denominated as a "positive FiP-effect". The negative FiP effect describes an observation of a minimum of the longitudinal sensor signal under a condition in which the corresponding material is impinged by a light beam with the smallest
25 available beam cross-section, in particular, when the material may be located at or near a focal point as effected by an optical lens. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

25 As outlined above, the detector may comprise the at least one illumination source. A distance perpendicular to an optical axis of the detector between the illumination source and the bi-cell or quadrant diode may be small. The distance perpendicular to the optical axis of the detector between the illumination source and the bi-cell or quadrant diode may be less than 0,1 m, preferably
30 less than 0,05 m, more preferably less than 0,025 m. The illumination source and the optical axis may be separated by a small baseline. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the bi-cell or quadrant diode and of the at least one optional transfer device. The baseline may be less than 0,1 m, preferably
35 less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device.

40 As generally used herein, the term "combine" generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least

one comparison signal or comparison result. As will be outlined in further detail below, the combined sensor signal or secondary signal may be or may comprise at least one quotient signal.

In case more than two sensor signals are provided, as an example, more than one quotient signal may be generated, such as by forming quotient signals of more than one pair of sensor signals.

Thus, as an example, in case more than two optical sensors are provided, the above-mentioned quotient formation may take place between two of the sensor signals generated by these optical sensors or may take place between more than two of the sensor signals. Thus, instead of using the first one of the sensor signals and the second one of the sensor signals in the formulae given above, combined sensor signals may be used for quotient formation.

The quotient signal Q generally is an example for an asymmetry parameter denoting an asymmetry or an asymmetric distribution of the light spot generated by the light beam on the light-sensitive areas. The quotient of the two or more optical sensors, such as the two or more photodiodes, may provide a combined signal which typically is monotonously dependent on the distance between the detector and the object from which the light beam travels towards the detector, as will be shown by experimental data below. In addition or as an alternative to the quotient signal, other types of combined functions implementing the sensor signals of two or more sensors in the setup of the present invention may be used, which also may show a dependency on the distance between the object and the detector. The asymmetry or the asymmetry parameter of the light spot, as an example, may be an indication of the width of a light beam. If this asymmetry parameter depends on the distance only, the measurement can be used to determine the distance.

In typical setups, commercially available quadrant diodes such as quadrant photodiodes are used for positioning, i.e. for adjusting and/or measuring a transversal coordinate of a light spot in the plane of the quadrant photodiode. Thus, as an example, laser beam positioning by using quadrant photodiodes is well known. According to a typical prejudice, however, quadrant photodiodes are used for xy-positioning, only. According to this assumption, quadrant photodiodes are not suitable for measuring distances. The above-mentioned findings, however, using an off-centered quadrant photodiode with regard to an optical axis of the detector, show otherwise, as will be shown in further measurements below. Thus, as indicated above, in quadrant photodiodes, the asymmetry of the spot can be measured by shifting the quadrant diode slightly off-axis, such as by the above-mentioned offset. Thereby, a monotonously z-dependent function may be generated, such as by forming the quotient signal Q of two or more of the sensor signals of two or more partial photodiodes, i.e. quadrants, of the quadrant photodiode. Therein, in principle, only two photodiodes are necessary for the measurement. The other two diodes may be used for noise cancellation or to obtain a more precise measurement.

In addition or as an alternative to using a quadrant diode or quadrant photodiode, other types of optical sensors may be used. Thus, as will be shown in further detail below, staggered optical sensors may be used.

- 5 The use of quadrant diodes provides a large number of advantages over known optical detectors. Thus, quadrant diodes are used in a large number of applications in combination with LEDs or active targets and are widely commercially available at very low price, with various optical properties such as spectral sensitivities and in various sizes. No specific manufacturing process has to be established, since commercially available products may be implemented into
10 the detector according to the present invention.

The detector according to the present invention specifically may be used in multilayer optical storage discs, such as disclosed by international patent application number PCT/IB2015/052233, filed on March 26, 2015. Measurements performed by using the detector
15 according to the present invention specifically may be used in order to optimize the focus position in optical storage discs.

As will be outlined in further detail below, the distance measurement by using the detector according to the present invention may be enhanced by implementing one or more additional distance measurement means into the detector and/or by combining the detector with other types
20 of distance measurement means. Thus, as an example, the detector may comprise or may be combined with at least one triangulation distance measurement device. Thus, the distance measurement can be enhanced by making use of a combination of the measurement principle discussed above and a triangulation type distance measurement. Further, means for measuring
25 one or more other coordinates, such as x- and/or y-coordinates, may be provided.

In case a quadrant diode is used, the quadrant diode may also be used for additional purposes. Thus, the quadrant diode may also be used for conventional x-y-measurements of a light spot, as generally known in the art of optoelectronics and laser physics. Thus, as an example, the
30 lens or detector position can be adjusted using the conventional xy-position information of the quadrant diode to optimize the position of the spot for the distance measurement. As a practical example, the light spot, initially, may be located right in the center of the quadrant diode, which typically does not allow for the above-mentioned distance measurement using the quotient function Q. Thus, firstly, conventional quadrant photodiode techniques may be used for off-centering
35 a position of the light spot on the quadrant photodiode, such that, e.g., the spot position on the quadrant diode is optimal for the measurement. Thus, as an example, the different off-centering of the optical sensors of the detector may simply be a starting point for movement of the optical sensors relative to the optical axis such that the light spot is off-centered with respect to the optical axis and with respect to a geometrical center of the array of the optical sensors.

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Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned quadrant diode. The case $m=2$, $n=2$ is the case of

the quadrant diode or quadrant optical sensor, which, for practical reasons, is one of the preferred cases, since quadrant photodiodes are widely available.

As a starting point, a geometrical center of the optical sensors within the array may be off-centered from the optical axis, such as by the above-mentioned offset. The sensor array specifically may be movable relative to the optical axis, for example along a gradient, preferably automatically, such as by moving the sensor array, e.g. in a plane perpendicular to the optical axis, and/or by moving the optical axis itself, e.g. shifting the optical axis in a parallel shift and/or tilting the optical axis. Thus, the sensor array may be shifted in order to adjust a position of a light spot generated by the light beam in the plane of the sensor array. Additionally or alternatively, the optical axis may be shifted and/or tilted by using appropriate elements, such as by using one or more deflection elements and/or one or more lenses. The movement, as an example, may take place by using one or more appropriate actuators, such as one or more piezo actuators and/or one or more electromagnetic actuators and/or one or more pneumatic or mechanical actuators, which, e.g., move and/or shift the array and/or move and/or shift and/or tilt one or more optical elements in the beam path in order to move the optical axis, such as parallel shifting the optical axis and/or tilting the optical axis. The evaluation device specifically may be adjusted to control a relative position of the sensor array to the optical axis, e.g. in the plane perpendicular to the optical axis. An adjustment procedure may take place in that the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly, moving the array relative to the optical axis, such as by moving the array and/or the optical axis, e.g. by moving the array in the plane to the optical axis until the light spot is off-centered and/or by tilting a lens until the light spot is off-centered. As used therein, a transversal position may be a position in a plane perpendicular to the optical axis, which may also be referred to as the x-y-plane. For the measurement of the transversal coordinate, as an example, the sensor signals of the optical sensors may be compared. As an example, in case the sensor signals are found to be equal and, thus, in case it is determined that the light spot is located symmetrically with respect to the optical sensors, such as in the center of the quadrant diodes, a shifting of the array and/or a tilting of a lens may take place, in order to off-center the light spot in the array. Thus, as outlined above, the off-centering of the array from the optical axis, such as by off-centering the center of the quadrant photodiode from the optical axis, may simply be a starting point in order to avoid the situation which is typical, in which the light spot is located on the optical axis and, thus, is centered. By off-centering the array relative to the optical axis, thus, the light spot should be off-centered. In case this is found not to be true, such that the light spot, incidentally, is located in the center of the array and equally illuminates all optical sensors, the above-mentioned shifting of the array relative to the optical axis may take place, preferably automatically, in order to off-center the light spot on the array. Thereby, a reliable distance measurement may take place.

Further, in a scanning system with a movable light source, the position of the light spot on the quadrant diode may not be fixed. This is still possible, but may necessitate that different calibrations are used, dependent on the xy-position of the spot in the diode.

Further, the use of the above-mentioned quotient signal Q is a very reliable method for distance measurements. Typically, Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q=s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, the sensor signals, such as the above-mentioned first sensor signal s_1 and the above-mentioned second sensor signal s_2 , decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, due to the off-centering, the one of the sensor signals decreases more rapidly than the other, since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. By spreading the light spot, however, the portion of the light illuminating the one or more optical sensors outside the center of the light spot increases, as compared to a situation of a very small light spot. Thus, the quotient of the sensor signals continuously changes, i.e. increases or decreases, with increasing diameter of the light beam or diameter of the light spot. The quotient, further, may further be rendered mainly independent from the total power of the light beam, since the total power of the light beam forms a factor in all sensor signals. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the sensor signals and the size or diameter of the light beam.

As outlined above, specifically, quadrant photodiodes may be used. As an example, commercially available quadrant photodiodes may be integrated in order to provide four optical sensors, such as one or more quadrant photodiodes available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany, such as quadrant Si PIN photodiodes of the type S4349, which are sensitive in the UV spectral range to the near IR spectral range. In case an array of optical sensors is used, the array may be a naked chip or may be an encapsulated array, such as encapsulated in a TO-5 metal package. Additionally or alternatively, a surface mounted device may be used, such as TT Electronics OPR5911 available from TT Electronics plc, Fourth Floor, St Andrews House, West Street Woking Surrey, GU21 6EB, England. It shall be noted that other optical sensors may also be used.

Further, it shall be noted that, besides the option of using precisely one quadrant photodiode, two or more quadrant photodiodes may also be used. Thus, as an example, a first quadrant photodiode may be used for the distance measurement, as described above, providing the two or more optical sensors. Another quadrant photodiode may be used, e.g. in a second partial beam path split off from the beam path of the first quadrant photodiode, for a transversal position measurement, such as for using at least one transversal coordinate x and/or y . The second quadrant photodiode, as an example, may be located on-axis with respect to the optical axis.

Further, it shall be noted that, besides the option of using one or more quadrant photodiodes, one or more quadrant photodiodes or further photodiode arrays may also be replaced or mim-

icked by separated photodiodes that are arranged or assembled close to each other, preferably in a symmetric shape such as a rectangular matrix, such as a 2 x 2 matrix. However further arrangements are feasible. In such an arrangement or assembly, the photodiodes may be arranged or assembled in a housing or mount, such as all photodiodes in a single housing or mount or groups of photodiodes in one housing or mount, or each of the photodiodes in a separate housing or mount. Further, the photodiodes may also be assembled directly on a circuit board. In such arrangements or assemblies, photodiodes may be arranged as such that the separation between the active area of the photodiodes, has a distinct value less than one centimeter, preferably less than one millimeter, more preferably as small as possible. Further, to avoid optical reflexes, distortions, or the like that may deteriorate the measurement, the space between the active areas may be either empty or filled with a material, preferably with a light absorbing material such as a black polymer, such as black silicon, black polyoxymethylene, or the like, more preferably optically absorbing and electrically insulating material, such as black ceramics or insulating black polymers such as black silicon, or the like. Further, the distinct value of the photodiode separation may also be realized by adding a distinct building block between the photodiodes such as a plastic separator. Further embodiments are feasible. The replacement of quadrant photodiodes by single diodes arranged in a similar setup such as in a 2 x 2 rectangular matrix with minimal distance between the active areas may further minimize the costs for the optical detector. Further, two or more diodes from a quadrant diode may be connected in parallel or in series to form a single light sensitive area.

The optical sensors each specifically may be uniform sensors having a single light-sensitive area each. Thus, the optical sensors specifically may be non-pixelated optical sensors.

In a further preferred embodiment the detector may comprise

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
- at least one second optical sensor having:
 - o at least one fluorescent waveguiding sheet forming a second light-sensitive area, wherein the fluorescent waveguiding sheet is oriented towards the object such that at least one light beam propagating from the object towards the detector generates at least one light spot in the second light-sensitive area, wherein the fluorescent waveguiding sheet contains at least one fluorescent material, wherein the fluorescent material is adapted to generate fluorescence light in response to the illumination by the light beam; and
 - o at least one photosensitive element located at at least one edge of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive element by the fluorescent waveguiding sheet and capable of generating at least one second sensor signal in response to the illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and

- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

In this further preferred embodiment the optical sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas. The first and second light-sensitive areas specifically may be oriented towards the object

The light beam propagating from the object to the detector specifically may fully illuminate the first light-sensitive area, such that the first light-sensitive area is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the first optical sensor. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the second light-sensitive area, which is smaller than the second light-sensitive area, such that the light spot is fully located within the second light-sensitive area. Within the light spot on the second light-sensitive area, a shadow created by the first optical sensor may be located. Thus, generally, the first optical sensor, having the smaller first light-sensitive area, may be located in front of the second optical sensor, as seen from the object, with the first light-sensitive area being fully located within the light beam and with the light beam generating a light spot on the second light-sensitive area being smaller than the second light-sensitive area, and with further a shadow created by the first optical sensor within the light spot. The situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a light spot generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

The first and second optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. The first and second optical sensors may be sensitive in one and the same spectral range or may have differing spectral sensitivities.

As further used herein, a fluorescent waveguiding sheet generally refers to an element with both waveguiding properties and fluorescent properties. Therein, waveguiding generally refers to the property or a plurality of an element to guide the light in one or more of the ultraviolet, visible or the infrared spectral range by internal reflection, specifically by internal total reflection. The term "fluorescence" generally refers to the property of an element or a material to emit secondary light, also referred to as fluorescence light, in one or more of the ultraviolet, visible or infrared spectral range, in response to excitation by electromagnetic radiation, also referred to as primary radiation or excitation radiation, such as primary light or excitation light. In most cases, the emitted light, fluorescence light or secondary light has a longer wavelength and a lower energy than the primary radiation. The primary radiation typically induces the presence of excited states within the fluorescent material, such as so-called excitons. Typically, excited state decay times for photon emissions with energies from the UV to near infrared are within the range of 0.5 to 20 nanoseconds. Other examples, however, are feasible within the present invention. Similarly, as used herein, the term "fluorescent material" generally refers to a material having fluorescence

properties. The term “fluorescence light” generally refers to the secondary light generated during the above-mentioned fluorescence process.

For potential embodiments of the fluorescent waveguiding sheet, generally, reference may be made to the so-called “Q-Foil Technology”, as disclosed e.g. in the above-mentioned article by P. Bartu, R. Koeppel, N. Arnold, A. Neuling, L. Fallon, and S. Bauer, Conformable large-area position-sensitive photodetectors based on luminescence collecting silicone waveguides, J. Appl. Phys. 107, 123101 (2010). Fluorescent waveguiding sheets and/or second optical sensors as disclosed therein may also be used in the context of the present invention, within the second optical sensor.

The fluorescent waveguiding sheet, as will be outlined in further detail below, specifically may be an element or may comprise an element which has a sheet-like shape or which is a sheet. As used herein, a “sheet” generally refers to an element which has a lateral extension, such as a diameter or an equivalent diameter, which significantly exceeds a thickness of the element, such as by at least a factor of 5, more preferably by at least a factor of 10 or even more preferably by at least a factor of 20, a factor of 50 or even a factor of 100. The sheet specifically may be flexible, deformable or rigid.

As will be outlined in further detail below, the fluorescent waveguiding sheet specifically may be or may comprise a transparent material, specifically a transparent sheet. The transparency may be a transparency of at least 50 to 70% in the visible spectral range or in a part thereof, such as in a range of 500 nm to 700 nm. Other embodiments are feasible.

As further used herein, the term “sensitive area” generally refers to a two-dimensional or three-dimensional region of an element, specifically of the fluorescent waveguiding sheet or the optical sensor, which is sensitive to external influences and, e.g., produces at least one reaction in response to an external stimulus. In this case, specifically, the sensitive area may be sensitive to an optical excitation. The sensitive area specifically may be a part of a surface or the volume of the fluorescent waveguiding sheet, such as the whole surface of the fluorescent waveguiding sheet or a part thereof.

As further used herein, the term “fluorescent waveguiding sheet is oriented towards the object” generally refers to the situation that the surface of the fluorescent waveguiding sheet or a part of this surface, specifically the sensitive area, is fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the sensitive area may form an angle with a surface element of the sensitive area or of the fluorescent waveguiding sheet which is different from 0°, such as an angle in the range of 20°-90°. Other embodiments, however, are feasible.

Most preferably, however, the at least one fluorescent waveguiding sheet, the sensitive area or a part thereof is oriented essentially perpendicular to an optical axis of the optical sensor and/or of the detector.

As further used herein, the term “edge” of the fluorescent waveguiding sheet generally refers to a boundary of the at least one fluorescent waveguiding sheet, such as a side boundary or side edge or a front or back face of the fluorescent waveguiding sheet. Thus, as the skilled person will recognize, the term “edge” of the fluorescent waveguiding sheet may generally refer to an interface or boundary between the fluorescent waveguiding sheet and a surrounding atmosphere, such as air. Specifically, the edge may be a border of a second light-sensitive area formed by the fluorescent waveguiding sheet. The term “located at” generally refers to the fact that the photosensitive elements are either located directly on the edge or in close proximity to the edge. As an example, at least 50% of the light emerging from the edge into the direction of the photosensitive element may be collected by the respective photosensitive element, without being lost by scattering, beam widening or other losses. As an example, the photosensitive element may be located at a position spaced apart from the edge of no more than 10 mm, more preferably of no more than 5 mm. It shall be noted, however, that other embodiments are feasible for connecting the fluorescence light. Most preferably, all photosensitive elements are located relative to their respective edges of the fluorescent waveguiding sheet in the same way, in order to provide similar measurement conditions for all photosensitive elements.

The at least one photosensitive element located at the at least one edge of the fluorescent waveguiding sheet may, as an example, be fully or partially located in the same plane as the fluorescent waveguiding sheet and/or may fully or partially be located in a different plane. In the latter case, as will be outlined in further detail below, as an example, an optical coupling between the edge of the fluorescent waveguiding sheet and the at least one photosensitive element may take place, by using at least one optical coupling element. Further, in case a plurality of photosensitive elements is provided in the second optical sensor, at least one of the photosensitive elements may be located in the same plane as the fluorescent waveguiding sheet and at least one of the photosensitive elements may be located outside the plane of the fluorescent waveguiding sheet. Further, a direction of view of the at least one photosensitive element may be parallel to the plane of the fluorescent waveguiding sheet or may be directed otherwise, such as perpendicular to the plane. Therein, when talking about a “plane” of the fluorescent waveguiding sheet, this term does not necessarily imply that the fluorescent waveguiding sheet is fully planar. Thus, as an example, the fluorescent waveguiding sheet may also be curved or bent, and the plane of the fluorescent waveguiding sheet at the location of the at least one photosensitive element may be a local tangential plane.

The term “edge” of the fluorescent waveguiding sheet, as used herein, may refer to a straight line or straight border area of the fluorescent waveguiding sheet, in the following also referred to as a “straight edge”, or may also refer to a non-straight line or non-straight border area of the fluorescent waveguiding sheet, such as a corner of the fluorescent waveguiding sheet. Thus, the at least one photosensitive element or, in case a plurality of photosensitive elements is provided, at least one of the photosensitive elements, may also be located at at least one corner of the fluorescent waveguiding sheet, such as at at least one corner of the second light-sensitive area formed by the fluorescent waveguiding sheet. Thus, specifically, the edge may comprise a

rim or a portion of a rim of the fluorescent waveguiding sheet, such as a corner and/or a straight rim portion. Additionally or alternatively, however, the edge may also comprise a flat surface of the fluorescent waveguiding sheet, such as a front side or back side.

- 5 As already outlined above, in order to improve feeding fluorescence light guided from the light spot towards the at least one photosensitive element into the at least one respective photosensitive element, at least one optical coupling may take place, by using at least one optical coupling element in between the fluorescent waveguiding sheet and the respective photosensitive element. Thus, the at least one photosensitive element, or, in case a plurality of photosensitive
- 10 elements is provided, at least one of the photosensitive elements may be optically coupled to the fluorescent waveguiding sheet by at least one optical coupling element configured for at least partially coupling the fluorescence light guided by the fluorescent waveguiding sheet out of the fluorescent waveguiding sheet and, preferably, at least partially into the photosensitive element. As used herein, the term "optical coupling element" generally refers to an arbitrary element
- 15 which is configured for one or more of disturbing, diminishing or interrupting an internal total reflection within the fluorescent waveguiding sheet which takes place during waveguiding within the fluorescent waveguiding sheet. Thus, as an example, the optical coupling element may be an arbitrary transparent element having an index of refraction in between an index of refraction of the fluorescent waveguiding sheet and the photosensitive element and/or the ambient atmosphere, such as air. Thus, as an example, in case an index of refraction of the fluorescent waveguiding sheet is denoted by n_1 , and an index of refraction of the photosensitive element is denoted by n_2 , an index of refraction n_3 of the optical coupling element may be
- 20 $n_1 < n_3 < n_2$ or $n_1 > n_3 > n_2$.
- 25 The optical coupling element may be in direct contact with the fluorescent waveguiding sheet, such as with at least one surface, such as a surface facing the object and/or a surface facing away from the object, of the fluorescent waveguiding sheet. Further, the optical coupling element may also be in direct contact with the at least one photosensitive element. Further, for each photosensitive element, an independent optical coupling element may be provided, or alternatively, a plurality of photosensitive elements may share a common optical coupling element, or, alternatively, a plurality of optical coupling elements may be coupled to one photosensitive element.
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- Various ways of optical coupling are generally known to the skilled person and may also be
- 35 used for coupling fluorescence light from the fluorescent waveguiding sheet into the photosensitive element. Thus, as an example, the at least one optical coupling element may comprise at least one element selected from the group consisting of: a portion of transparent adhesive attaching the photosensitive element to the fluorescent waveguiding sheet; an etched portion within the fluorescent waveguiding sheet, such as within a surface of the fluorescent waveguiding sheet, such as a surface facing the object and/or facing away from the object; a scratch in the fluorescent waveguiding sheet, such as a scratch in the surface of the fluorescent waveguiding sheet, such as a surface facing the object and/or facing away from the object; a prism. Additionally or alternatively, other optical coupling elements are generally known and may also
- 40

be used in the present invention. In a most simple case, the at least one photosensitive element may simply be adhered or glued to a surface of the fluorescent waveguiding sheet, such as by at least one transparent glue or adhesive, e.g. a transparent epoxy. Other ways of optical coupling are feasible.

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The first optical sensor and the second optical sensor specifically may have an essentially identical electrical capacitance. When referring to a capacitance of an optical sensor, generally, the capacity C as may be determined in an oscillating circuit is referred to, as the person skilled in the art of electronics generally will recognize. Thus, when implementing the first optical sensor or, alternatively, the second optical sensor into an oscillating circuit, the oscillating circuit, as an example, may have essentially the same resonance frequency. As used herein, the term “essentially identical” generally refers to the fact that, as an example, the capacitance of the first optical sensor and the capacitance of the second optical sensor differ by no more than a factor of 10, more preferably a by no more than a factor of 5 or even by no more than a factor of 2.

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Further, when referring to the capacitance of the second optical sensor, reference is made to the capacitance of the at least one photosensitive element of the second optical sensor, since the photosensitive element determines the electrical behavior and, specifically, the high-frequency behavior of the second optical sensor. In case a plurality of photosensitive elements are present in the second optical sensor, the capacitance of at least one of these photosensitive elements is meant.

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As outlined above, for example with respect to the first preferred embodiment, for evaluating the first and second sensor signals at least one predetermined or determinable relationship between the first and second sensor signals and the longitudinal coordinate may be used. With respect to evaluation of the first and second sensor signals reference can be made to the description of the first preferred embodiment. Thus, the evaluation device specifically may be configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. The evaluation device may be configured for determining the longitudinal coordinate by evaluating the quotient signal Q .

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Typically, in the setup described above, Q may be a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. This is due to the fact that, in the preferred setup described above, the first light-sensitive area is fully illuminated and preferably is located in the center of the light spot, such as on the optical axis. Consequently, with increasing diameter of the light spot, the peak intensity of the illumination, which mainly may be located in the first light-sensitive area, decreases and, thus, the first sensor signal decreases. The second sensor signal, however, may be more or less independent from the size of the light spot, as long as the light spot, except for the shadow created by the first optical sensor, is located on the second light-sensitive area. Consequently, the second sensor signal may remain more or less constant. The quotient of the first and second sensor

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signals, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the first and second light-sensitive areas. The quotient, further, is mainly independent from the total power of the light beam, since the total power of the light beam forms a factor both in the first sensor signal and in the second sensor signal. Consequently, the quotient

5 Q may form a secondary signal which provides a unique and unambiguous relationship between the first and second sensor signals and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship

10 between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal

15 coordinate of the object, or both.

The first optical sensor and the second optical sensor may be positioned in various ways. Thus, in a first example, the first optical sensor is positioned in front of the second optical sensor, such that the light beam, firstly, passes the first optical sensor and, secondly, thereafter, reaches the

20 second optical sensor. Alternatively, however, a reverse order is possible. Thus, as an alternative, the second optical sensor may be placed in front of the first optical sensor, such that the light beam, firstly, reaches the second optical sensor, fully or partially passes through the second optical sensor and, secondly, thereafter, reaches the first optical sensor. The latter alternative is mainly possible due to the fact that the fluorescent waveguiding sheet of the second optical

25 sensor may, in this embodiment or other embodiments, be embodied fully or partially transparent, specifically in the spectral range of the light beam. Thus, the smaller first optical sensor may be placed on a reverse side of the second optical sensor, from the perspective of the object.

30 The second optical sensor and, more specifically, the fluorescent waveguiding sheet of the second optical sensor, may be placed in various positions in one or more beam paths. Thus, as an example, the second optical sensor and/or the fluorescent waveguiding sheet of the second optical sensor may be placed close to a focal point within a beam path. Alternatively, however, specifically in order to improve the function of summing, the second optical sensor may also be

35 placed away from the focal point, such as by one, two or more than two Rayleigh wavelengths.

As outlined above, the second optical sensor comprises at least one photosensitive element, from which the at least one second sensor signal arises. Therein, precisely one photosensitive

40 element may be used, such as for collecting at least a part of the fluorescence light and for creating a second sensor signal in response. Alternatively, however, the second optical sensor may comprise two, three, four or more photosensitive elements which, preferably, are located at at least two edges of the fluorescent waveguiding sheet, such as at least two opposing portion

of a rim of the fluorescent waveguiding sheet. The photosensitive elements may be capable of detecting fluorescence light guided from the light spot towards the photosensitive elements by the fluorescent waveguiding sheet and capable of generating at least one sensor signal each. Thus, a plurality of second sensor signals may exist.

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In case at least two photosensitive elements are provided, being located at at least two edges of the fluorescent waveguiding sheet, a plurality of possibilities exists. Thus, as an example, the at least two photosensitive elements may be located at one or more of: at least two straight edges of the fluorescent waveguiding sheet, such as at least two opposing edges, such as opposing rim portions; at least two corners of the fluorescent waveguiding sheet, such as at least two opposing corners; at least one corner of the fluorescent waveguiding sheet and at least one straight edge, such as at least one straight rim portion, of the fluorescent waveguiding sheet. Other possibilities are generally given.

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The detector and, preferably, the evaluation device may specifically be embodied to combine at least two of these second sensor signals to form at least one common second sensor signal. This at least one common second sensor signal, specifically, may be used, in conjunction with the first second sensor signal, for determining the longitudinal coordinate. Additionally, at least one common second sensor signal may be used for acquiring additional information, such as at least one item of information on a transversal position of the object or a part thereof, as will be outlined in further detail below.

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Thus, for combining the at least two second sensor signals, the evaluation device may comprise at least one summing device configured for summing up the sensor signals of the at least two photosensitive elements, thereby forming a sum signal S. As an example, the sum signal may be derived by using the formula

$$S = \sum_i c_i s_i, \quad (1)$$

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with s_i denoting the second sensor signals, with $i = 1 \dots N$ with N being a positive integer indicating the number of the photosensitive elements or a smaller positive integer, and with c_i being respective calibration coefficients.

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As outlined above, the evaluation device specifically may be configured to determine the at least one longitudinal coordinate z of the object by using at least one predetermined relationship between the first sensor signal, the at least one second sensor signal and the longitudinal coordinate z. In case a plurality of second sensor signals is given, as outlined above, at least one combined second sensor signal may be used in this relationship. Thus, specifically, the sum signal S may be used. Thus, as an example, the evaluation device may be configured to determine the at least one longitudinal coordinate z by using at least one predetermined relationship between the first sensor signal, the sum signal S of the second sensor signals and the longitudinal coordinate z.

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In addition to the at least one longitudinal coordinate, the detector may further be configured for determining at least one transversal coordinate of the object and/or of a part thereof. For this purpose, the detector may comprise at least one transversal optical sensor. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or WO 2014/198629 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Thus, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Preferably, however, the at least one second optical sensor itself may be used as a transversal optical sensor, for determining at least one transversal coordinate of the object. This setup is specifically advantageous in case a plurality of photosensitive elements is used in the second optical sensor, such as in different positions and located at different edges of the fluorescent waveguiding sheet. The evaluation device may be configured to determine at least one transversal coordinate x , y of the object by evaluating the sensor signals of the photosensitive elements. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

Thus, generally, the evaluation device may further be configured to determine at least one transversal coordinate x , y of the object by evaluating the second sensor signals of the photosensitive elements. For the purpose of determining the at least one transversal coordinate in one or more directions, the sensor signals of the photosensitive elements may be compared. Thus, as is evident to the skilled person, the sensor signal of a respective photosensitive element, which represents the fluorescence light guided to the photosensitive elements by the fluorescent waveguiding sheet from the light spot and, thus, from the location of generation of the fluorescence light, depends on a distance between the light spot and the respective photosensitive element. Generally, with an increasing distance between the light spot and the photosensitive element, the sensor signal of the respective photosensitive element will decrease, such as due to losses during waveguiding and/or due to spreading of the fluorescence light. By comparing the sensor signals of the photosensitive elements located at different, known positions, the lateral or transversal position of the light spot on the fluorescent waveguiding sheet may be determined and, therefrom, by using e.g. a known or determinable relationship between the transversal position of the light spot and the transversal coordinate of the object, the transversal coordinate of the object. Again, empirical relationships and/or semiempirical relationships and/or analytical relationships may be used, such as the lens equation which is generally known to the skilled person.

Thus, the evaluation device specifically may be configured for evaluating at least two second sensor signals of at least two different photosensitive elements. As an example, these at least two second sensor signals may be compared in order to determine a transversal position of the light spot on the fluorescent waveguiding sheet and/or for directly deriving a transversal position of the object. Thus, as an example, a difference between at least two second sensor signals may be determined in order to derive at least one item of information on a transversal position of the object. Consequently, the evaluation device may comprise at least one subtracting device configured to form at least one difference signal D between second sensor signals generated by at least two of the photosensitive elements. The second sensor signals may comprise at least one second sensor signal s_{21} and at least one second sensor signal s_{22} , wherein the at least one difference signal D is proportional to $a \cdot s_{21} - b \cdot s_{22}$:

$$D \sim a \cdot s_{21} - b \cdot s_{22}, \quad (2)$$

with a, b being real number coefficients, preferably with $a=1$ and $b=1$. Additionally or alternatively, the at least one difference signal D may be derived according to the formula:

$$D = (a \cdot s_{21} - b \cdot s_{22}) / (a \cdot s_{21} + b \cdot s_{22}). \quad (3)$$

The subtracting device specifically may be configured to form at least one first difference signal D_x from which at least one first transversal coordinate x of the object is derived. Additionally or alternatively, the subtracting device may be further configured to form at least one second difference signal D_y from which at least one second transversal coordinate y of the object is derived. Thus, the first difference signal D_x may be generated from at least two second sensor signals s_{x1} , s_{x2} of at least two photosensitive elements located at opposing edges of the waveguiding sheet in a first dimension. The second difference signal D_y may be generated from at least two second sensor signals s_{y1} , s_{y2} of at least two photosensitive elements located at opposing edges of the waveguiding sheet in a second dimension. The at least one first difference signal D_x may be derived according to the formula

$$D_x = (a \cdot s_{x1} - b \cdot s_{x2}) / (a \cdot s_{x1} + b \cdot s_{x2}), \quad (4)$$

and the at least one second difference signal D_y may be derived according to the formula

$$D_y = (c \cdot s_{y1} - d \cdot s_{y2}) / (c \cdot s_{y1} + d \cdot s_{y2}), \quad (5)$$

with a, b, c, d being real number coefficients, preferably with $a=1$, $b=1$, $c=1$ and $d=1$.

As outlined above, in case the second optical sensor comprises a plurality of photosensitive elements, the second optical sensor specifically may comprise at least two photosensitive elements located at opposing edges of the fluorescent waveguiding sheet. The photosensitive elements may comprise at least one first pair of photosensitive elements located at opposing edges of the fluorescent waveguiding sheet in a first dimension of a coordinate system, and the

photosensitive elements may further comprise at least one second pair of photosensitive elements located at opposing edges of the fluorescent waveguiding sheet in a second dimension of the coordinate system.

- 5 Further optional details may refer to the at least one fluorescent waveguiding sheet and/or to the second light-sensitive area. Thus, as an example, the second light-sensitive area specifically may be a homogeneous light-sensitive area. Thus, the second light-sensitive area may not be subdivided physically into partial areas, such as pixels. Contrarily, the sensitive area may be one homogeneous area which forms a uniform fluorescence.

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The second light-sensitive area specifically may be a large sensitive area. Thus, as an example, the second light-sensitive area may have a surface of at least 5 mm², preferably of at least 10 mm², more preferably of at least 100 mm², more preferably of at least 400 mm². As an example, the sensitive area may have a surface of 5 mm² to 10,000 mm², such as 100 mm² to 2500 mm².

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The large-area design of the sensitive area is advantageous in many ways. Thus, specifically, by increasing the surface of the second light-sensitive area, a resolution of the determination of the transversal coordinates may be increased. Further, the field of view of the detector, e.g. the viewing angle, may be widened by using a large second light-sensitive area.

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The fluorescent waveguiding sheet specifically may comprise at least one planar sheet. Therein, however, slight curvatures still may be tolerated. In other embodiments, however, the fluorescent waveguiding sheet may also be embodied as a curved fluorescent waveguiding sheet, such as in order to provoke specific optical effects which might be desirable in certain applications. Thus, one of the advantages of the present detector may reside in the fact that the fluorescent waveguiding sheet specifically may be curved, flexible or having a specific geometry.

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The fluorescent waveguiding sheet may have a thickness of 10 µm to 3 mm, preferably a thickness of 100 µm to 1 mm, e.g. a thickness of 50 µm to 2 mm. The thickness of the waveguiding sheet specifically may be dimensioned of the waveguiding sheet along an optical axis of the detector. The thickness may be adapted to improve or optimize waveguiding properties of the fluorescence light.

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The fluorescent waveguiding sheet may fully or partially be rigid or, alternatively, may fully or partially be embodied flexible or deformable.

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The fluorescent waveguiding sheet may comprise at least one matrix material. As used herein, the term "matrix material" generally refers to a material which forms the main part of the fluorescent waveguiding sheet and which defines the main body of the fluorescent waveguiding sheet. As an example, the matrix material may be a material which is capable of receiving one or more additional materials therein, such as by intermixing, chemical bonding, dispersion or solving. Thus, the at least one fluorescent material may be one or more mixed into the matrix material, dispersed into the matrix material, chemically bound to the matrix material or dissolved in the matrix material.

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The matrix material specifically may be or may comprise at least one plastic material. The plastic material specifically may be or may comprise at least one polymer material. The plastic material, as an example, may be or may comprise at least one material selected from the group consisting of a polycarbonate, a poly(methyl methacrylate), a polystyrene, a polyurethane, a polypropylene, a polyethylene terephthalate, a polyvinylchloride. Other materials, however, are feasible.

The fluorescent material specifically may comprise an arbitrary fluorophore. Specifically, the at least one fluorescent material may comprise at least one fluorescent dye. A wide variety of fluorescent dyes is generally known to the skilled person. Thereof, as outlined above, most of these fluorescent dyes exhibit the above-mentioned saturation effects, thereby rendering the fluorescence being a nonlinear function of the excitation. As an example, the fluorescent dye may be capable of being saturated by the light beam, such that a total power of the fluorescence light generated by the fluorescent dye is a nonlinear function of an intensity of the light beam. Specifically, the total power of the fluorescence light may be sub-proportional to the intensity of the light beam.

The fluorescent dye specifically may comprise at least one organic fluorescent dye. Inorganic dyes, however, may be used additionally or alternatively. The fluorescent dye specifically may be selected from the group consisting of: a xanthene derivative, preferably one or more of fluorescein, rhodamine, oregon green, eosin, texas red, or a derivative of any component thereof; a cyanine derivative, preferably one or more of cyanine, indocarbocyanine, oxacarbocyanine, thiocarbocyanine, merocyanine, or a derivative of any component thereof; a squaraine derivative or a ring-substituted squaraine, preferably one or more of Seta, SeTau, and Square dyes, or a derivative of any component thereof; a naphthalene derivative, preferably one or more of a dansyl or a prodan derivative thereof; a coumarin derivative; a oxadiazole derivative, preferably one or more of pyridyloxazole, nitrobenzoxadiazole, benzoxadiazole, or a derivative of any component thereof; an anthracene derivative, preferably one or more of an anthraquinone, DRAQ5, DRAQ7, CyTRAK orange, or a derivative of any component thereof; a pyrene derivative, preferably cascade blue; an oxazine derivative, preferably one or more of nile red, nile blue, cresyl violet, oxazine 170, or a derivative of any component thereof; an acridine derivative, preferably one or more of proflavin, acridine orange, acridine yellow, or a derivative of any component thereof; an arylmethine derivative, preferably one or more of auramine, crystal violet, malachite green, or a derivative of any component thereof; a tetrapyrrole derivative, preferably one or more of porphin, phthalocyanine, bilirubin; a rylene dye or any derivative thereof, such as a perylene dye; a naphthalene imid or perylene imide; a naphthoilene benzimidazole dye such as published in WO 2012/168395 A1; or a derivative of any component of the listed substances. It shall be noted, however, that other dyes may be used additionally or alternatively.

Further optional details of the detector referred to the at least one photosensitive element or to the plurality of photosensitive elements. Thus, as an example, the at least one photosensitive element may comprise at least one photodiode, preferably at least one inorganic photodiode, such as a silicon photodiode. The at least one photosensitive element may comprise at least

one elongated photosensitive element extending along at least one segment of an edge or rim portion of the waveguiding sheet.

5 As outlined above, the fluorescent waveguiding sheet specifically may be a rectangular fluorescent waveguiding sheet, preferably a square waveguiding sheet. In the setup, as an example, the photosensitive elements may be located at each of the four edges, such as at each of the four straight rim sections or rim portions, of the waveguiding sheet. Other embodiments are feasible.

10 The first and second optical sensors each, independently, may be opaque, transparent or semi-transparent. For the sake of simplicity, however, opaque sensors which are not transparent for the light beam, may be used, since these opaque sensors generally are widely commercially available. The fluorescent waveguiding sheet, however, may fully or partially be transparent.

15 The first optical sensor specifically may be a uniform sensor having a uniform light-sensitive area. Thus, specifically, the first optical sensor specifically may be a non-pixelated optical sensor. Alternatively, however, the at least one first optical sensor may also comprise one or more pixelated sensors, each having two or more pixels. Other embodiments are feasible.

20 Similarly, as discussed in further detail below, the at least one second optical sensor may comprise precisely one second optical sensor or a plurality of optical sensors. Further, the at least one fluorescent waveguiding sheet of the second optical sensor may be precisely one coherent fluorescent waveguiding sheet. Alternatively, however, the at least one second optical sensor also may comprise a plurality of second optical sensors. The latter alternative, as an example,
25 may be realized by providing a plurality of fluorescent waveguiding sheets. Thus, a plurality of fluorescent waveguiding sheets may be provided, each having one or more photosensitive elements. Other embodiments are feasible.

The detector, such as of the second optical sensor, may further comprise at least one optical
30 filter element, preferably at least one optical short-pass filter. Thus, as an example, only photons which are capable of exciting fluorescence light may pass through to the fluorescent waveguiding sheet. Other embodiments, however, are feasible.

The first and second optical sensors specifically may be arranged linearly in one and the same
35 beam path of the detector. Thus, as an example, the first and second optical sensors may be arranged concentrically with respect to an optical axis of the detector, specifically with the first optical sensor being placed in front of the second optical sensor. It shall be noted, however, that other embodiments are feasible, such as embodiments with the first optical sensor and the second optical sensor being placed in separate and different beam paths, wherein the beam path of
40 the detector is split by one or more beam-splitting elements. The linear setup with the first and second optical sensors being placed in one and the same beam path, however, is preferred, due to the simple optical setup and the low volume required for the detector. Thus, the present invention specifically allows for a simple, efficient and inexpensive optical setup.

As outlined above, the first optical sensor specifically may be a semiconductor sensor, preferably an inorganic semiconductor sensor, more preferably a photodiode and most preferably a silicon photodiode. As an example, the first optical sensor may be a uniform sensor having a single light-sensitive area. As an example, a commercially available silicon photodiode may be used.

In a further preferred embodiment the detector may comprise

- at least one angle dependent optical element adapted to generate at least one light beam having at least one beam profile depending on an angle of incidence of an incident light beam propagating from the object towards the detector and illuminating the angle dependent optical element;
- at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element;
- the at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the quotient signal Q from the sensor signals.

The angle dependent optical element may comprise at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element, at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one prism; at least one lens; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter; at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.

At least one of the light-sensitive areas may be oriented towards the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. Additionally or alternatively, at least one of the light-sensitive areas may be oriented differing from an orientation towards the object. For example, at least one of the optical sensors may be oriented perpendicular or under an arbitrary angle to the optical axis and with respect to the object. The angle dependent optical element may be adapted to generate the light beam such that the light beam impinges on the light-sensitive areas. For example, in case at least one of the light-sensitive areas is oriented under the arbitrary angle with respect to the optical axis,

the angle dependent optical element may be adapted to guide the light beam onto the light-sensitive area.

As used herein, the term “angle dependent optical element” refers to an optical element adapted to generate the at least one light beam having at least one beam profile depending on the angle of incidence of the incident light beam propagating from the object towards the detector and illuminating the angle dependent optical element. In particular, the angle dependent optical element may be adapted to influence and/or change and/or adjust the beam profile of the incident light beam. For example, the angle dependent optical element may have one or more of angle dependent transmission properties, angle dependent reflection properties or angle dependent absorption properties. The light beam generated by the angle dependent optical element may comprise at least one transmission light beam and/or at least one reflection light beam. The angle of incidence may be measured with respect to an optical axis of the angle dependent optical element.

An electromagnetic wave impinging on a first side, for example a surface and/or an entrance, of the angle dependent optical element may be partly, depending on the properties of the angle dependent optical element, absorbed and/or reflected and/or transmitted. The term “absorption” refers to a reduction of power and/or intensity of the incident light beam by the angle dependent optical element. For example, the power and/or intensity of the incident light beam may be transformed by the angle dependent optical element to heat or another type of energy. As used herein, the term “transmission” refers to a part of the electromagnetic wave which is measurable outside the angle dependent optical element in a half-space with angles from 90° and higher with respect to the optical axis. For example, transmission may be a remaining part of the electromagnetic wave impinging on the first side of the angle dependent optical element, penetrating the angle dependent optical element and leaving the angle dependent optical element at a second side, for example an opposite side and/or an exit. The term “reflection” refers to a part of the electromagnetic wave which is measurable outside the angle dependent optical element in a half-space with angles below 90° with respect to the optical axis. For example, reflection may be a change in direction of a wavefront of the incident light beam due to interaction with the angle dependent optical element.

The total power of the electromagnetic wave impinging on the angle dependent optical element may be distributed by the angle dependent optical element in at least three components, i.e. an absorption component, a reflection component and a transmission component. A degree of transmission may be defined as power of the transmission component normalized by the total power of the electromagnetic wave impinging on the angle dependent optical element. A degree of absorption may be defined as power of the absorption component normalized by the total power of the electromagnetic wave impinging on the angle dependent optical element. A degree of reflection may be defined as power of the reflection component normalized by the total power of the electromagnetic wave impinging on the angle dependent optical element.

As used herein, "angle dependent transmission" refers to the fact that the degree of transmission depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element. As outlined above, the angle of incident may be measured with respect to an optical axis of the angle dependent optical element. The angle dependent optical element may be arranged in the direction of propagation behind at least one transfer device. The transfer device may, for example, comprise at least one collimating lens. The angle dependent optical element may be designed to weaken rays impinging with larger angles compared to rays impinging with a smaller angle. For example, the degree of transmission may be highest for light rays parallel to the optical axis, i.e. at 0°, and may decrease for higher angles. In particular, at at least one cut-off angle the degree of transmission may steeply fall to zero. Thus, light rays having a large angle of incidence may be cut-off.

As used herein, the term "angle dependent absorption" refers to the fact that the degree of absorption depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element. As used herein, the term "angle dependent absorption" refers to the fact that a degree of absorption depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element. For example, photon energy and/or intensity of the light beam propagating from the object to the detector may be reduced depending on the angle of incidence.

As used herein, the term "angle dependent reflection" refers to the fact that the degree of reflection depends on the angle of incidence at which the incident light beam propagating from the object towards the detector impinges on the angle dependent optical element.

For example, the angle dependent optical element comprises at least one optical fiber. The optical fiber may be designed such that the degree of transmission may be highest for incoming light rays parallel, i.e. at an angle of 0°, to the optical fiber, neglecting reflection effects. The optical fiber may be designed such that for higher angles, for example angles from 1° to 10°, the degree of transmission may decrease smoothly to around 80% of the degree of transmission for parallel light rays and may remain at this level constantly up to an acceptance angle of the optical fiber. As used herein, the term "acceptance angle" may refer to an angle above which total reflection within the optical fiber is not possible such that the light rays are reflected out of the optical fiber. The optical fiber may be designed that at the acceptance angle, the degree of transmission may steeply fall to zero. Light rays having a large angle of incidence may be cut-off.

The optical fiber may be adapted to transmit at least parts of the incident light beam which are not absorbed and/or reflected, between two ends of the optical fiber. The optical fiber may have a length and may be adapted to permit transmission over a distance. The optical fiber may comprise at least one material selected from the group consisting of: silica, aluminosilicate glass, germane silicate glass, fluorozirconate, rare earth doped glass, fluoride glass, chalco-

genide glasses, sapphire, doped variants, especially for silica glass, phosphate glass, PMMA, polystyrene, fluoropolymers such as poly(perfluoro-butenylvinyl ether), or the like. The optical fiber may be a single or multi-mode fiber. The optical fiber may be a step index fiber, a polarizing fiber, a polarization maintaining fiber, a plastic optical fiber or the like. The optical fiber may
5 comprise at least one fiber core which is surrounded by at least one fiber cladding having a lower index of refraction as the fiber core. The fiber cladding may also be a double or multiple cladding. The fiber cladding may comprise a so-called outer jacket. The fiber cladding may be coated by a so-called buffer adapted to protect the optical fiber from damages and moisture. The buffer may comprise at least one UV-cured urethane acrylate composite and/or at least one
10 polyimide material. In one embodiment, a refractive index of the fiber core may be higher than the refractive index of the fiber cladding material and the optical fiber may be adapted to guide the incoming light beam by total internal reflection below the angle of acceptance. In one embodiment, the optical fiber may comprise at least one hollow core fiber, also called photonic bandgap fiber. The hollow-core fiber may be adapted to guide the incoming light beam essentially
15 within a so-called hollow region, wherein a minor portion of the light beam is lost due to propagation into the fiber cladding material.

The optical fiber may comprise one or more fiber connectors at the end of the fiber. The optical fiber may comprise end caps such as coreless end caps. The optical fiber may comprise one or
20 more of a fiber coupler, a fiber Bragg grating, a fiber polarizer, a fiber amplifier, a fiber coupled diode laser, a fiber collimator, a fiber joint, a fiber splicing, a fiber connector, a mechanical splicing, a fusion splicing, or the like. The optical fiber may comprise a polymer coating.

The optical fiber may comprise at least two or more fibers. The optical fiber may be at least one
25 multifurcated optical fiber, in particular at least one bifurcated optical fiber. For example, the bifurcated optical fiber may comprise two fibers, in particular at least one first fiber and at least one second fiber. The first fiber and the second fiber may be arranged close to each other at an entrance end of the bifurcated optical fiber and may split into two legs separated by a distance at an exit end of the bifurcated optical fiber. The first and second fiber may be designed as fi-
30 bers having identical properties or may be fibers of different type. The first fiber may be adapted to generate at least one first transmission light beam and the second fiber may be adapted to generate at least one second transmission light beam. The bifurcated optical fiber may be arranged such that the incident light beam may impinge at a first angle of incidence into the first fiber and at a second angle of incidence, different from the first angle, into the second fiber,
35 such that the degree of transmission is different for the first transmission light beam and the second transmission light beam. One of the optical sensors may be arranged at an exit end of the first fiber and the other optical sensor may be arranged at an exit end of the second fiber. The optical fiber may comprise more than two fibers, for example three, four or more fibers. For example, the multifurcated may comprise multiple fibers wherein each fiber may comprise at
40 least one of a core, a cladding, a buffer, a jacket, and one or more fibers may partially or entirely be bundled by a further jacket such as a polymer hose to ensure that the fibers stay close to each other such as at one end of the fiber.

The detector may comprise a plurality of optical fibers, for example a plurality of single optical fibers or a plurality of multifurcated optical fibers. For example, the detector may comprise a plurality of single optical fibers, for example optical fibers having identical properties. The optical fibers, i.e. the single optical fibers or multifurcated optical fibers, may be arranged such that the incident light beam may impinge at different angles of incidence into each of the optical fibers such that the degree of transmission is different for each of the optical fibers. At the exit ends of each optical fiber at least one optical sensor may be arranged. Alternatively, at least two or more of the optical fibers may use the same optical sensor.

The illumination source may be adapted to illuminate the object through the angle dependent optical element. The optical fiber may comprise at least one illumination fiber adapted to transmit at least one incoming light beam generated by the illumination source such that it illuminates the object. The illumination source may be adapted to couple the at least one light beam generated by the illumination source into the illumination fiber. The detector may comprise at least one coupling element, for example at least one in-coupling element and/or at least one out-coupling element, which may be arranged in front of and/or behind the optical fiber. The coupling element may be or may comprise at least one transfer device.

The detector may comprise a further coupling element, in particular a further in-coupling element, adapted to couple the light beam travelling from the object to the detector into the illumination fiber. The further coupling element may be arranged in direction of propagation of the light beam travelling from the object to the detector in front of the optical fiber. The further coupling element may be or may comprise at least one transfer device.

In case the detector comprises a plurality of optical fibers, the detector may comprise one illumination source or a plurality of identical illumination sources and/or a plurality of different illumination sources. For example, the plurality of illumination sources may comprise at least two illumination sources generating light with different properties such as color or modulation frequencies. In one embodiment, at least two of the optical fibers of the plurality of the optical fibers may be illuminated using the same illumination source and/or two identical illumination sources. At the exit ends of each optical fiber illuminated by the same illumination source and/or by identical illumination sources at least one optical sensor may be arranged. Alternatively, at least two or more of the optical fibers illuminated by the same illumination source may use the same optical sensor. Such a detector, in which for a plurality of optical fibers the same illumination source and same optical sensors are used, may be used as proximity sensor. In one embodiment, at least two of the optical fibers of the plurality of optical fibers may be illuminated by using at least two different illumination sources. At the exit ends of each optical fiber at least one optical sensor may be arranged. Alternatively, at least two or more of the optical fibers may use the same optical sensor.

As outlined above, the distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors may be small. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may

be 21 mm. In particular in case the angle dependent optical element is an optical fiber, the baseline may even be smaller than a radius of the transfer device.

For example, the angle dependent optical element may be an optical fiber. The illumination
5 source may be guided through an optical fiber wherein the illumination beam may exit the optical fiber behind the transfer device within a baseline that is smaller than the radius of the transfer device. The illumination source may be adapted to generate at least one illumination light beam, wherein the optical fiber guiding the illumination beam may be attached to the transfer device such as by a polymer or glue or the like, to reduce reflections at interfaces with larger
10 differences in refractive index. For example, the detector may be a compact device without further optical elements, wherein the illumination source may be placed as close as possible to an edge of the transfer device. Thus, the baseline may be close to half a diameter of the transfer device, in particular the lens diameter and housings of lens and light source. For example, the detector may be an even more compact device, wherein a mirror, in particular a small mirror,
15 may be positioned in front of the transfer device, in particular in a center, for example a geometrical center, or close to the center of the transfer device, in order to couple the illumination beam into the beam path. Thus, the baseline may be less than half the diameter of the transfer device. The illumination source may be arranged such that the baseline is as small as possible. By arranging the illumination source such that the direction of propagation of the illumination light
20 beam is essentially parallel to the optical axis and that the illumination source and the optical axis are separated by the small baseline, very compact devices are possible. For example, a distance from the center of the transfer device to the illumination source, in particular along a connecting line from the center of the transfer device to the illumination source, may be preferably less than 2.5 times a distance from the center of the transfer device to an edge if the transfer
25 device, more preferably less than 1.5 times the distance center to edge of the transfer device, and most preferably less 1 times the distance center to edge of the transfer device. The transfer device may have an arbitrary shape, in particular a non-circular shapes are possible. At small distances an aperture of the illumination source may be small and the baseline may be small. At large distances an aperture of the illumination source may be large and the baseline may be
30 small. This is contrarily as in triangulation methods, wherein at large distances a large baseline is necessary. Further, triangulation based systems have a minimum detection range significantly greater than zero, for example such as 20 cm from the detector system, due to necessary spatial extend of the baseline. Such a large baseline may result in that the illuminated light scattered from the object may not reach the light sensitive area of the optical sensor behind the
35 transfer device. In addition, in triangulation based systems, using a small baseline would reduce the minimum detection range, however, at the same time would reduce a maximum detection range. Further, triangulation based systems require a plurality of light sensitive areas and sensor signals, for example sensor signals of at least one detector row. According to the invention, determination of the longitudinal coordinate z is possible with a reduced number of sensor signals, in particular with less than 20, preferably less than 10 and more preferably less than 5
40 sensor signals. The illumination source and the angle dependent optical element may be arranged in the direction of propagation of the light beam traveling from the object to the detector behind the transfer device, which will be described in detail below. The distance perpendicular

to the optical axis of the detector between the illumination source and the optical sensors may be less than the radius of the transfer device.

5 The illumination source and the optical sensors and/or the illumination source and the angle dependent optical element and/or the optical sensors may be arranged with a relative spatial offset from the optical axis of the detector. Such an arrangement may allow enhancing the tendency of the quotient, and thus, accuracy of the distance measurement. In particular, with increasing spatial offset a slope in a Q vs distance diagram increases and thus allows distinguishing similar distances more accurately. For example, one of the illumination source and the optical sensors may be arranged on the optical axis and the other one may be arranged spaced apart from the optical axis. For example, both of illumination source and the optical sensors may be arranged spaced apart from the optical axis by at least one different distance, in particular perpendicular to the optical axis. For example, the at least two fibers of a multifurcated optical fiber may be arranged at different distances from the optical axis. The angle dependent optical element is adapted to mimic a larger distance compared to the real distance perpendicular to an optical axis between the illumination source and the optical sensors without moving the illumination source and/or optical sensors.

20 By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

30 Alternatively to the linear arrangement of the two optical sensors, the optical sensors may be arranged in different beam paths of the detector. The angle dependent optical element may be adapted to generate the first light beam and the second light beam. The first light beam and the second light beam may be generated with different degree of transmission. The first optical sensor may be configured to generate the first sensor signal in response to the illumination of the first light-sensitive area by the first light beam generated by the angle dependent optical element. The second optical sensor may be configured to generate the second sensor signal in response to the illumination of the second light-sensitive area by the second light beam generated by the angle dependent optical element. For example, as outlined above, the angle dependent optical element may comprise at least one multifurcated optical fiber which may be arranged such that the incident light beam may impinge at the first angle of incidence into the first fiber and at the second angle of incidence, different from the first angle, into the second fiber, such that the degree of transmission is different for the first light beam, in this case a first

transmission light beam, and the second light beam, in this case a second transmission light beam. One of the first and second optical sensors may be arranged at the exit end of the first fiber and the other optical sensor may be arranged at the exit end of the second fiber.

- 5 At least one sensor element may have a matrix of optical sensors, wherein the optical sensors each having a light-sensitive area, wherein each optical sensor may be configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by the light beam generated by the angle dependent optical element.
- 10 For example, the detector may comprise two sensor elements, in particular at least one first sensor element and at least one second sensor element, arranged in different beam paths of the detector. The angle dependent optical element may be adapted to generate the first light beam and the second light beam. The first light beam and the second light beam may be generated with different degree of transmission. The first sensor element may be configured to
- 15 generate the first sensor signal in response to the illumination by the first light beam generated by the angle dependent optical element. The second sensor element may be configured to generate the second sensor signal in response to an illumination by the second light beam generated by the angle dependent optical element. For example, as outlined above, the angle dependent optical element may comprise at least one multifurcated optical fiber arranged such that the
- 20 incident light beam may impinge at the first angle of incidence into the first fiber and at the second angle of incidence, different from the first angle, into the second fiber, such that the degree of transmission is different for the first transmission light beam and the second transmission light beam. One of the first and second sensors elements may be arranged at the exit end of the first fiber and the other sensor element may be arranged at the exit end of the second fiber.
- 25 The detector may comprise at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in
- 30 response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element.

The evaluation device may be configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals. In this further embodiment the optical

35 sensors may be arranged such that the light-sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas

The detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. For example, in case the angle

40 dependent optical element comprises at least one multifurcated optical fiber, each of the fibers of the multifurcated optical fiber may constitute an independent beam path. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis, and the condi-

tion noted above generally may refer to each beam path independently. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths. In case the optical sensors are distributed over different partial beam paths, the above-mentioned condition may be described such that at least one first optical sensor is located in at least one first partial beam path, being offset from the optical axis of the first partial beam path by a first spatial offset, and at least one second optical sensor is located in at least one second partial beam path, being offset from the optical axis of the second partial beam path by at least one second spatial offset, wherein the first spatial offset and the second spatial offset are different.

As outlined above, each light-sensitive area has a geometrical center. Each geometrical center of each light-sensitive area may be spaced apart from the optical axis of the detector, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located.

As outlined above, the optical sensors specifically may be located in one and the same plane, which, preferably, is a plane perpendicular to the optical axis. Other configurations, however, are possible. Thus, two or more of the optical sensors may also be spaced apart in a direction parallel to the optical axis.

For example, the optical sensors may be partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector. As an example, bi-cell diodes or quadrant diodes are widely commercially available at low cost, and driving schemes for these bi-cell diodes or quadrant diodes are generally known. As an example, the bi-cell diodes may each form independent diodes having the full functionality of a diode. As an example, each of the bi-cell diodes may have a square or rectangular shape, and the two diodes may be placed in one plane such that the two partial diodes, in total, form a 1 x 2 or 2 x 1 matrix having a rectangular shape. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes and quadrant diode is proposed, as will be outlined in further detail below. Generally, however, the optical sensors specifically may be partial diodes of a quadrant diode, with a center of the quadrant diode being off-centered from the optical axis of the detector. As an example, the four partial diodes may each form independent diodes having the full functionality of a diode. As an example, the four partial diodes may each have a square or rectangular shape, and the four partial diodes may be placed in one plane such that the four partial diodes, in total, form a 2 x 2 matrix having a rectangular or square shape. In a further example, the four partial diodes, in total, may form a 2 x 2 matrix having a circular or elliptical shape. The partial diodes, as an example, may be adjacent, with a minimum separation from one another.

In case a quadrant diode is used, having a 2x2 matrix of partial diodes, the center of the quadrant diode specifically may be off-centered or offset from the optical axis. Thus, as an example, the center of the quadrant diodes, which may be an intersection of the geometrical centers of the optical sensors of the quadrant diode, may be off-centered from the optical axis by at least

0.2 mm, more preferably by at least 0.5 mm, more preferably by at least 1.0 mm or even 2.0 mm. Similarly, when using other types of optical sensors setups having a plurality of optical sensors, an overall center of the optical sensors may be offset from the optical axis by the same distance.

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In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector
10 system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more
15 illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the
20 one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein,
25 elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

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In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein,
35 the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

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In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is config-

ured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment function in accordance with the information.

- 5 In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

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- In a further aspect of the present invention, a readout device for optical storage media is proposed. The readout device comprises at least one detector according to any one of the preceding embodiments referring to a detector. As used therein, a readout device for optical storage media generally refers to a device which is capable of optically retrieving information stored in optical storage media such as optical storage discs, e.g. CCD, DVD or Blu-ray discs. Thus, the above-described measurement principle of the detector according to the present invention may be used for detecting data modules within an optical storage medium such as in optical storage discs. As an example, in case a reflective data module is present and reflects the illuminating light beam, the detector will not only detect the reflected light beam according to the above-mentioned measurement principle but will also detect a distance between the detector and the reflective data module, i.e. a depth of the reflective data module within the optical storage medium. Thus, as an example, the detector may be used for detecting different layers of information modules or data modules within the optical storage medium. Thereby, as an example, two layer discs or three layer discs or even discs having more than three layers may be generated and read out.

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- In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination

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light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning.

Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning.

The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-

related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object.

- 5 For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.
- 10 Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission
- 15 in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning device. In particular, the scanning system may, thus, use at
- 20 least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be
- 25 used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a
- 30 scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination
- 35 source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated
- 40 out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with a several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning device may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning device may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning

system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning device is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or drill at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning device.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon

photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses an inertial measurement unit for use in an electronic device. The electronic device may be a mobile electronic device. The electronic device may be a camera. The electronic device may be a mobile phone. The inertial measurement unit is adapted to receive data determined by at least one inertial measurement unit comprising at least one detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. As used herein, the term "data determined by the at least one detector" refers to at least one information on the at least one longitudinal coordinate z . The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. As used herein, the term "data determined by the at least one further sensor" refers to at least one information selected from the group consisting of: angle information; speed information; information on turn rate; information on inclination. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed. The inertial measurement unit may comprise at least one processor. The processor may be adapted to evaluate data recorded by the further sensor. In particular, the processor may be adapted to determine and/or calculate one or more of spatial position, spatial orientation, movement and velocity. The inertial measurement unit may comprise a plurality of further sensors. The inertial measurement unit may be adapted to fuse information determined from at least two of the further sensors. The inertial measurement unit may be adapted to fuse information of at least two further sensors by using at least one Kalman filter. As outlined

above, the detector may be adapted to provide an absolute measurement of the longitudinal coordinate z . The processor, e.g. the evaluation device as described above, may be adapted to fuse the information of the at least two further sensors considering the longitudinal coordinate z . The various sensor signals may be used within a Kalman filter or a linear quadratic estimated to take into account that each sensor signal is subject to measurement errors and inaccuracies. Fusion of these sensor signals within a Kalman filter may yield an improved estimate such as for the measurement of the longitudinal coordinate.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used. The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method comprises the following method steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by the light beam, wherein the detector has at least one transfer device, wherein the transfer device has at least one focal length in response to the light beam propagating from the object to the detector;
- illuminating each of light-sensitive areas of at least two optical sensors of the detector with a light beam propagating from the object to the detector, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving a quotient signal Q of the sensor signals.

The longitudinal coordinate z of the object may be determined in at least one measurement range independent from the object size in an object plane.

The deriving of the quotient signal Q may comprise one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

The method may comprise adjusting the measurement range by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the object in the object plane.

The transfer device may have an optical axis. With respect to embodiments of the transfer device reference is made to the embodiments of the transfer device described with respect to a first aspect above.

5 For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

10 In a first preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

15 The method steps may be as follows:

- illuminating at least one first light-sensitive area of at least one first optical sensor of the detector with a light beam propagating from the object to the detector and, thereby, generating at least one first sensor signal;
- illuminating at least one second light-sensitive area of at least one second optical
20 sensor of the detector with the light beam and, thereby, generating at least one second sensor signal, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
- evaluating the first and second sensor signals, thereby determining at least one longitudinal coordinate z of the object.

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Specifically, evaluating the first and second sensor signal may comprise deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. Further, the determining the longitudinal coordinate may comprise evaluating the quotient signal

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Q .

In a second preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed.

35 Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps may be as follows:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical
40 sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;
- evaluating the sensor signals, by

- determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - a) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - b) determining at least one combined signal by combining the center signal and the sum signal; and
 - c) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Specifically, the evaluation of the first and second sensor signal may comprise deriving the quotient signal Q by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Further, the determination of the longitudinal coordinate may comprise evaluating the quotient signal Q . The evaluation of the quotient signal specifically may comprise using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate, in order to determine the longitudinal coordinate.

In a third preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps may be as follows:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light;
- illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

Specifically, evaluating the first and second sensor signal may comprise deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q . Further, the determining of the longitudinal coordinate may comprise evaluating the quotient signal Q .

In a further preferred embodiment the method may comprise the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps may be as follows:

- illuminating at least one first light-sensitive area of at least one first optical sensor with a light beam propagating from the object to the detector and generating at least one first sensor signal by the first optical sensor, in response to the illumination of the first light-sensitive area;
 - illuminating at least one second light-sensitive area of at least one second optical sensor with the light beam and generating at least one second sensor signal by the second optical sensor, in response to the illumination of the second light-sensitive area, the second optical sensor having:
 - o at least one fluorescent waveguiding sheet forming a second light-sensitive area, wherein the fluorescent waveguiding sheet is oriented towards the object such that at least one light beam propagating from the object towards the detector generates at least one light spot in the second light-sensitive area, wherein the fluorescent waveguiding sheet contains at least one fluorescent material, wherein the fluorescent material is adapted to generate fluorescence light in response to the illumination by the light beam; and
 - o at least one photosensitive element located at at least one edge of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive element by the fluorescent waveguiding sheet and capable of generating the at least one second sensor signal in response to the illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
 - determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.
- Specifically, as discussed above, the evaluating the first and second sensor signals may comprise deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals. The determining the at least one longitudinal coordinate of the object may comprise evaluating the quotient signal Q . as discussed above, at least one predetermined relationship between the quotient signal and the longitudinal coordinate may be used for determining the longitudinal coordinate.

As also discussed above, the second optical sensor specifically may comprise at least two photosensitive elements. The method may further comprise determining at least one transversal coordinate x , y of the object by evaluating the sensor signals of the photosensitive elements.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to

the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further preferred embodiment, the method comprises the following steps:

- 5 - providing at least one angle dependent optical element and generating at least one light beam having at least one beam profile depending on an angle of incidence;
- providing at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light
- 10 beam generated by the angle dependent optical element;
- illuminating each of the light-sensitive areas of the at least two optical sensors of the detector with the light beam generated by the angle dependent optical element, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- 15 evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving the quotient signal Q of the sensor signals.

Specifically, evaluating the first and second sensor signal may comprise deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals.

20 Further, the determining the longitudinal coordinate may comprise evaluating the quotient signal Q .

As also discussed above, the second optical sensor specifically may comprise at least two photosensitive elements. The method may further comprise determining at least one transversal

25 coordinate x , y of the object by evaluating the sensor signals of the photosensitive elements.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or

30 given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position

35 measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; a machine vision application; a robotics application;

40 a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector sys-

tem. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

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As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

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Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for household applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

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The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus, specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling

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other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

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Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head-mounted displays.

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Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries, obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

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Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

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Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further

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be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general need for voluminous beam-splitters. Further, the devices according to the present invention

generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

- 5 Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems, or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.
- 30 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the

present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions.

Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endoscopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric surgery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment. Further, the devices according to the present invention may be used in wearable robots such as in exoskeletons or prosthesis or the like.

Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe.

In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

Further the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection, such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections,

inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, air-
5 planes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed
10 and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a
15 guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to
20 active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for air-
25 craft when landing or starting. Wherein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or ac-
30 celerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use
35 with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or
40 facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light

sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racket, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the present invention may be used in drones or multicopters to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concertized movements of several drones or recharging or refueling in the air or the like.

Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network)

to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as dog or cat litter robots, charging robot for electrical vehicles, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, robotic shopping carts, luggage carrying robots, line following robots, laundry robots, ironing robots, window washing robots, toy robots, , patient monitoring robots, baby monitoring robots, elderly monitoring robots, children monitoring robots, transport robots, telepresence robots, professional service robots, programmable toy robots, pathfinder robots, social robots providing company to less mobile people, following robots, smart card following robots, psychotherapy robots, or robots translating and speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the present invention may be used in humanoid robots, especially in the context of using humanoid hands to pick up or hold or place objects. Further, the devices according to the present invention may be used in combination with audio interfaces especially in combination with household robots which may serve as a digital assistant with interfaces to online or offline computer applications. Further, the devices according to the present invention may be used in robots that can control switches and buttons in industrial and household purposes. Further, the devices according to the present invention may be used in smart home robots such as Mayfield's Kuri. Further the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate insecure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

Further, devices according to the present invention may be used for the inspection of adhesive beads, sealing beads, or the like, such as to recognize disruptions, slubs, contractions, asy-

metries, local defects, or the like. Further, devices according to the present invention may be used to count objects such as dry fruits on a conveyer belt, such as in difficult situations, such as when fruit of similar color and shape may be in direct contact with each other. Further, devices according to the present invention may be used in quality control of die cast or injection molded parts such as to ensure flawless casting or molding, recognize surface damages, worn out toolings or the like. Further, devices according to the present invention may be used for laserscribing such as for quality control and positioning of the laser. Further, devices according to the present invention may be used for sorting systems, such as to detect position, rotation, and shape of an object, compare it to a database of objects, and classify the object. Further, devices according to the present invention may be used for stamping part inspection, packaging inspection, such as food and pharma packaging inspection, filament inspection, or the like.

Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that are can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.

Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition devices, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.

Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.

The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or green-

house or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.

Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific groups of animals such as bees.

Further, devices according to the present invention may be used to guide users during a shaving, hair cutting, or cosmetics procedure, or the like. Further, devices according to the present invention may be used to record or monitor what is played on an instrument, such as a violin. Further, devices according to the present invention may be used in smart household appliances such as a smart refrigerator, such as to monitor the contents of the refrigerator and transmit notifications depending on the contents. Further, devices according to the present invention may be used for monitoring or tracking populations of humans, animals, or plants, such as deer or tree populations in forests. Further, devices according to the present invention may be used in harvesting machines, such as for harvesting crops, flowers or fruits, such as grapes, corn, hops, apples, grains, rice, strawberries, asparagus, tulips, roses, soy beans, or the like. Further, devices according to the present invention may be used to monitor the growth of plants, animals, algae, fish, or the like, such as in breeding, food production, agriculture or research applications, to control irrigation, fertilization, humidity, temperature, use of herbicides, insecticides, fungicides, rodenticides, or the like. Further, devices according to the present invention may be used in feeding machines for animals or pets, such as for cows, pigs, cats, dogs, birds, fish, or the like. Further, devices according to the present invention may be used in animal product production processes, such as for collecting milk, eggs, fur, meat, or the like, such as in automated milking or butchering processes. Further, devices according to the present invention may be used for automated seeding machines, or sowing machines, or planting machines such as for planting corn, garlic, trees, salad or the like. Further, devices according to the present invention may be used to assess or monitor weather phenomena, such as clouds, fog, or the like, or to warn from danger of avalanches, tsunamis, gales, earthquakes, thunder storms, or the like. Further, devices according to the present invention may be used to measure motions, shocks, concussions, or the like such as to monitor earthquake risk. Further, devices according to the present invention may be used in traffic technology to monitor dangerous crossings, to control traffic lights depending on traffic, to monitor public spaces, to monitor roads, gyms, stadiums, ski resorts, public events, or the like. Further, devices according to the present invention may be used in medical applications such as to monitor or analyze tissues, medical or biological assays, changes in tissues such as in moles or melanoma or the like, to count bacteria, blood cells, cells, algae, or the like, for retina scans, breath or pulse measurements, gastroscopy, patient surveillance, or the like. Further, devices according to the present invention may be used to monitor the shape, size, or circumference of drops, streams, jets, or the like or to analyze, assess, or monitor profiles or gas or liquid currents such as in a wind channel, or the like. Further,

devices according to the present invention may be used to warn drivers such as car or train drivers when they are getting sick or tired or the like. Further, devices according to the present invention may be used in material testing to recognize strains or tensions or fissures, or the like. Further, devices according to the present invention may be used in sailing to monitor and optimize sail positions such as automatically. Further, devices according to the present invention may be used for fuel level gauges.

Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or viruses or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location. Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location specific information, such as the current position on a map, or information on goods sold, or the like.

Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.

Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices ac-

According to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.

5

As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate

along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

- 5 The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device adapted for detecting light beams, such as
10 the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The
15 detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the de-
20 tector are feasible.

The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as
25 simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based al-
30 gorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

35 The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying
40 distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor,

or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

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The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared. Further, the devices according to the present invention may be used in the context of gesture recognition. In this context, gesture
10 recognition in combination with devices according to the present invention may, in particular, be used as a human-machine interface for transmitting information via motion of a body, of body parts or of objects to a machine. Herein, the information may, preferably, be transmitted via a motion of hands or hand parts, such as fingers, in particular, by pointing at objects, applying sign language, such as for deaf people, making signs for numbers, approval, disapproval, or the
15 like, by waving the hand, such as when asking someone to approach, to leave, or to greet a person, to press an object, to take an object, or, in the field of sports or music, in a hand or finger exercise, such as a warm-up exercise. Further, the information may be transmitted by motion of arms or legs, such as rotating, kicking, grabbing, twisting, rotating, scrolling, browsing, pushing, bending, punching, shaking, arms, legs, both arms, or both legs, or a combination of
20 arms and legs, such as for a purpose of sports or music, such as for entertainment, exercise, or training function of a machine. Further, the information may be transmitted by motion of the whole body or major parts thereof, such as jumping, rotating, or making complex signs, such as sign language used at airports or by traffic police in order to transmit information, such as "turn right", "turn left", "proceed", "slow down", "stop", or "stop engines", or by pretending to swim, to
25 dive, to run, to shoot, or the like, or by making complex motions or body positions such as in yoga, pilates, judo, karate, dancing, or ballet. Further, the information may be transmitted by using a real or mock-up device for controlling a virtual device corresponding to the mock-up device, such as using a mock-up guitar for controlling a virtual guitar function in a computer program, using a real guitar for controlling a virtual guitar function in a computer program, using
30 a real or a mock-up book for reading an e-book or moving pages or browsing through in a virtual document, using a real or mock-up pen for drawing in a computer program, or the like. Further, the transmission of the information may be coupled to a feedback to the user, such as a sound, a vibration, or a motion.

35 In the context of music and/or instruments, devices according to the present invention in combination with gesture recognition may be used for exercising purposes, control of instruments, recording of instruments, playing or recording of music via use of a mock-up instrument or by only pretending to have a instrument present such as playing air guitar, such as to avoid noise or make recordings, or, for conducting of a virtual orchestra, ensemble, band, big band, choir, or
40 the like, for practicing, exercising, recording or entertainment purposes or the like.

Further, in the context of safety and surveillance, devices according to the present invention in combination with gesture recognition may be used to recognize motion profiles of persons, such

as recognizing a person by the way of walking or moving the body, or to use hand signs or movements or signs or movements of body parts or the whole body as access or identification control such as a personal identification sign or a personal identification movement.

- 5 Further, in the context of smart home applications or internet of things, devices according to the present invention in combination with gesture recognition may be used for central or non-central control of household devices which may be part of an interconnecting network of home appliances and/or household devices, such as refrigerators, central heating, air condition, microwave ovens, ice cube makers, or water boilers, or entertainment devices, such as television sets,
10 smart phones, game consoles, video recorders, DVD players, personal computers, laptops, tablets, or combinations thereof, or a combination of household devices and entertainment devices.

- 15 Further, in the context of virtual reality or of augmented reality, devices according to the present invention in combination with gesture recognition may be used to control movements or function of the virtual reality application or of the augmented reality application, such as playing or controlling a game using signs, gestures, body movements or body part movements or the like, moving through a virtual world, manipulating virtual objects, practicing, exercising or playing sports, arts, crafts, music or games using virtual objects such as a ball, chess figures, go
20 stones, instruments, tools, brushes.

- Further, in the context of medicine, devices according to the present invention in combination with gesture recognition may be used to support rehabilitation training, remote diagnostics, or to monitor or survey surgery or treatment, to overlay and display medical images with positions of
25 medical devices, or to overlay display prerecorded medical images such as from magnetic resonance tomography or x-ray or the like with images from endoscopes or ultra sound or the like that are recorded during an surgery or treatment.

- 30 Further, in the context of manufacturing and process automation, devices according to the present invention in combination with gesture recognition may be used to control, teach, or program robots, drones, unmanned autonomous vehicles, service robots, movable objects, or the like, such as for programming, controlling, manufacturing, manipulating, repairing, or teaching purposes, or for remote manipulating of objects or areas, such as for safety reasons, or for maintenance purposes.

- 35 Further, in the context of business intelligence metrics, devices according to the present invention in combination with gesture recognition may be used for people counting, surveying customer movements, areas where customers spend time, objects, customers test, take, probe, or the like.

- 40 Further, devices according to the present invention may be used in the context of do-it-yourself or professional tools, especially electric or motor driven tools or power tools, such as drilling machines, saws, chisels, hammers, wrenches, staple guns, disc cutters, metals shears and nib-

blers, angle grinders, die grinders, drills, hammer drills, heat guns, wrenches, sanders, engravers, nailers, jig saws, biscuit joiners, wood routers, planers, polishers, tile cutters, washers, rollers, wall chasers, lathes, impact drivers, jointers, paint rollers, spray guns, morticers, or welders, in particular, to support precision in manufacturing, keeping a minimum or maximum distance, or for safety measures.

Further, the devices according to the present invention may be used to aid visually impaired persons. Further, devices according to the present invention may be used in touch screen such as to avoid direct contact such as for hygienic reasons, which may be used in retail environments, in medical applications, in production environments, or the like. Further, devices according to the present invention may be used in agricultural production environments such as in stable cleaning robots, egg collecting machines, milking machines, harvesting machines, farm machinery, harvesters, forwarders, combine harvesters, tractors, cultivators, ploughs, destoners, harrows, strip tills, broadcast seeders, planters such as potato planters, manure spreaders, sprayers, sprinkler systems, swathers, balers, loaders, forklifts, mowers, or the like.

Further, devices according to the present invention may be used for selection and/or adaption of clothing, shoes, glasses, hats, prosthesis, dental braces, for persons or animals with limited communication skills or possibilities, such as children or impaired persons, or the like. Further, devices according to the present invention may be used in the context of warehouses, logistics, distribution, shipping, loading, unloading, smart manufacturing, industry 4.0, or the like. Further, in a manufacturing context, devices according to the present invention may be used in the context of processing, dispensing, bending, material handling, or the like.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which

will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

As outlined above, the human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device.

Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element,

the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a

computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

5 The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one
10 item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements
15 may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

20 The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which
25 might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

30 As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may
35 fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

40 The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the

object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.

- 5 The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly at-
10 tachable to or integratable into the object to be tracked.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other
15 types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detec-
20 tor.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial
25 applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication
30 systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional
35 option of measuring one or more transversal coordinates, thereby integrating the functionality of a PSD.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention,
40 such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the quotient signal, as mentioned above, contrarily, provides a wide range of a

continuously and monotonously decreasing or increasing functions which may be used to determine the longitudinal coordinate from the quotient signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the quotient signal from the total power of the light beam, at least as long as no saturation of one or both of the optical sensors is reached, also provides a very wide range of measurement in terms of brightness, i.e. in terms of total power of the light beam.

The light beam, within the detector, generally may propagate along an optical axis of the detector. The first and second optical sensors may be placed on the optical axis. The light beam, however, may also propagate in other ways than along the optical axis. As an example, an illumination light beam may be generated which propagates along the optical axis or which propagates parallel to the optical axis or at an angle to the optical axis which is different from 0° , such as an angle of 1° to 20° . Other embodiments are feasible.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector comprising:

- at least one transfer device, wherein the transfer device has at least one focal length in response to at least one incident light beam propagating from the object to the detector;
- at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam,
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q from the sensor signals.

Embodiment 2: The detector according to the preceding embodiment, wherein the detector is adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from the object size in an object plane.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the measurement range is adjustable by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the object in the object plane.

Embodiment 4: The detector according to the preceding embodiment, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object

size if a circle of confusion of at least one light beam emitted from at least one point of the object is greater than an image size of the object.

Embodiment 5: The detector according to the preceding embodiment, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object size if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for at least one distance.

Embodiment 6: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 7: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 8: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are areas of the beam profile at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o .

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the optical sensors are positioned off focus.

Embodiment 10: The detector according to any one of the preceding embodiments, wherein the transfer device has an optical axis, wherein the transfer device constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis, wherein the optical sensors are arranged such that the light-sensitive areas of the optical sensors differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.

Embodiment 11: The detector according to any one of the preceding embodiments, wherein the light beam may be a monochromatic light beam.

Embodiment 12: The detector according to any one of the preceding embodiments, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of at least one beam profile of the light beam.

Embodiment 13: The detector according to the preceding embodiment, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

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Embodiment 14: The detector according to any one of the two preceding embodiments, wherein the light-sensitive regions are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

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Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile.

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Embodiment 16: The detector according to any one of the four preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.

20

Embodiment 17: The detector according to any one of the preceding embodiments, wherein the edge information comprises an information relating to a number of photons in the first area of the beam profile and the center information comprises an information relating to a number of photons in the second area of the beam profile.

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Embodiment 18: The detector according to any one of the two the preceding embodiments, wherein the evaluation device is configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.

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Embodiment 19: The detector according to any one of the three preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensors with at least one line pattern, wherein A1 corresponds to an area with a full line width of the line pattern on the optical sensors, wherein A2 is a central area of the line pattern on the optical sensors.

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Embodiment 20: The detector according to any one of the four preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensor with at least one point pattern, wherein A1 corresponds to an area with a full radius of a point of the point pattern on the optical sensors, wherein A2 is a central area of the point in the point pattern on the optical sensors.

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Embodiment 21: The detector according to any one of the preceding embodiments, wherein the optical sensors comprise a first optical sensor having a first light sensitive area and a second optical sensor having a second light sensitive area, wherein the first and the second light sensitive areas are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$

- 5 is satisfied, wherein "a" is a ratio of photons hitting both an inner region of a plane perpendicular to the optical axis intersecting the optical axis at a distance equal to half of a focal length of the transfer device and the first light sensitive area, "b" is a ratio of photons hitting both the inner region of the plane and the second light sensitive area, "c" is a ratio of photons hitting both an outer region of the plane and the first light-sensitive area, "d" is a ratio of the photons hitting
10 both the outer region of the plane and the second light sensitive area.

Embodiment 22: The detector according to the preceding embodiment, wherein the first light sensitive area and the second light sensitive area are arranged such that the inner region has an area with a geometrical center point on the optical axis and an extension such that half of the
15 photons hit the plane within the inner region and the other half hit the plane outside the inner region.

Embodiment 23: The detector according to the preceding embodiment, wherein the first light sensitive area and the second light sensitive area are arranged such that the inner region is
20 designed as a circle with a center point on the optical axis and a radius r which is chosen such that half of the photons hit the plane within the circle and the other half hit the plane outside the circle.

- Embodiment 24: A detector for determining a position of at least one object, the detector having
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- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
 - at least one second optical sensor having a second light-sensitive area, wherein the
30 second optical sensor is configured to generate at least one second sensor signal in response to an illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
 - at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.
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Embodiment 25: The detector according to the preceding embodiment, wherein the evaluation device is configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device is configured for determining the longitudinal coordinate by evaluating the quotient signal Q.
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Embodiment 26: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.

- 5 Embodiment 27: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

- 10 Embodiment 28: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

- 15 Embodiment 29: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

Embodiment 30: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

- 20 Embodiment 31: The detector according to any one of the seven preceding embodiments, wherein the first and second optical sensors are arranged linearly in one and the same beam path of the detector.

- 25 Embodiment 32: The detector according to the preceding embodiment, wherein the first and second optical sensors are arranged concentrically with respect to an optical axis of the detector.

- 30 Embodiment 33: The detector according to any one of the eight preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than five times the square root of a surface area of the first light-sensitive area.

- 35 Embodiment 34: The detector according to any one of the nine preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm.

- 40 Embodiment 35: The detector according to any one of the ten preceding embodiments, wherein the second light-sensitive area is larger by at least a factor of two, more preferably by at least a factor of three and most preferably by at least a factor of five as compared to the first light-sensitive area.

Embodiment 36: The detector according to any one of the eleven preceding embodiments, wherein the first light-sensitive area has a surface area of 1 mm² to 150 mm², more preferably a surface area of 10 mm² to 100 mm².

- 5 Embodiment 37: The detector according to any one of the twelve preceding embodiments, specifically according to the preceding embodiment, wherein the second light-sensitive area has a surface area of 160 mm² to 1000 mm², more preferably a surface area of 200 mm² to 600 mm².

- 10 Embodiment 38: The detector according to any one of the thirteen preceding embodiments, wherein the first and second optical sensor each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

- 15 Embodiment 39: The detector according to any one of the fourteen preceding embodiments, wherein the first and second optical sensors each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

- 20 Embodiment 40: The detector according to any one of the fifteen preceding embodiments, wherein the first and second optical sensors each are uniform sensors having a single light-sensitive area each.

- 25 Embodiment 41: The detector according to any one of the sixteen preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

- 30 Embodiment 42: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

- 35 Embodiment 43: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

- 40 Embodiment 44: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pat-

tern comprising convex uniform tilings; at least one fringe pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

5 Embodiment 45: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the quotient signal Q .

10 Embodiment 46: The detector according to any one of the twenty-one preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

15 Embodiment 47: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

20 Embodiment 48: The detector according to any one of the twenty-three preceding embodiments, wherein the first light-sensitive area, in a direction of propagation of the light beam, overlaps with the second light-sensitive area.

25 Embodiment 49: A detector for determining a position of at least one object, the detector having - at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;

30 - at least one evaluation device configured for evaluating the sensor signals, by
a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one
35 sum signal;
c) determining at least one combined signal by combining the center signal and the sum signal; and
d) determining at least one longitudinal coordinate z of the object by evaluating the combined
40 signal.

Embodiment 50: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal;

an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Embodiment 51: The detector according to any one of the two preceding embodiments, wherein the detector, specifically the evaluation device, is configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals.

Embodiment 52: The detector according to the preceding embodiment, wherein the transforming of the sensor signals comprises at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and brightness channels, a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image.

Embodiment 53: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal.

Embodiment 54: The detector according to the preceding embodiment, wherein the center detector is fully or partially integrated into the sensor element.

Embodiment 55: The detector according to any one of the two preceding embodiments, wherein the center detector is fully or partially embodied in one or both of software or hardware.

Embodiment 56: The detector according to any one of the seven preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.

Embodiment 57: The detector according to the preceding embodiment, wherein the evaluation device comprises at least one summing device for forming the sum signal.

Embodiment 58: The detector according to the preceding embodiment, wherein the summing device is fully or partially integrated into the sensor element.

Embodiment 59: The detector according to any one of the two preceding embodiments, wherein the summing device is fully or partially embodied in one or both of software or hardware.

Embodiment 60: The detector according to any one of the eleven preceding embodiments, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

Embodiment 61: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 62: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

Embodiment 63: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

Embodiment 64: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

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Embodiment 65: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 66: The detector according to any one of the two preceding embodiments, wherein the divider is fully or partially integrated into the sensor element.

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Embodiment 67: The detector according to any one of the eighteen preceding embodiments, wherein the optical sensors are photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors.

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Embodiment 68: The detector according to any one of the nineteen preceding embodiments, wherein the optical sensors are sensitive in the infrared spectral range.

Embodiment 69: The detector according to any one of the twenty preceding embodiments, wherein the matrix comprises one or both of a CCD detector, preferably a CCD detector chip, or a CMOS detector, preferably a CMOS detector chip.

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Embodiment 70: The detector according to any one of the twenty-one preceding embodiments, wherein the matrix is a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns.

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Embodiment 71: The detector according to the preceding embodiment, wherein the rows and columns are oriented essentially perpendicular.

Embodiment 72: The detector according to any one of the two preceding embodiments, wherein the matrix has at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows.

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Embodiment 73: The detector according to any one of the three preceding embodiments, wherein the matrix has at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns.

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Embodiment 74: The detector according to any one of the four preceding embodiments, wherein the ratio of the number of rows and the number of columns is close to 1 preferable larger than 1:3, more preferably larger than 1:2.

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Embodiment 75: The detector according to any one of the twenty-six preceding embodiments, wherein the matrix comprises at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors.

Embodiment 76: The detector according to any one of the twenty-seven preceding embodiments, wherein the sensor element is oriented essentially perpendicular to an optical axis of the detector.

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Embodiment 77: The detector according to any one of the twenty-eight preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the respective sensor signals are dependent on the total power of illumination of the respective optical sensor.

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Embodiment 78: The detector according to any one of the twenty-nine preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

Embodiment 79: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

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Embodiment 80: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

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Embodiment 81: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one fringe pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

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Embodiment 82: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the quotient signal Q .

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Embodiment 83: The detector according to any one of the thirty-four preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

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Embodiment 84: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

Embodiment 85: The detector according to any one of the thirty-six preceding embodiments, wherein the evaluation device is further configured for determining at least one transversal coordinate of the object by evaluating a transversal position of the at least one optical sensor having the highest sensor signal.

Embodiment 86: The detector according to any one of the thirty-eight preceding embodiments, wherein the evaluation device is configured for distinguishing different light beams having different modulations.

Embodiment 87: A detector for determining a position of at least one object, the detector having

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

Embodiment 88: The detector according to the preceding embodiment, wherein the optical sensors are located in the same plane.

Embodiment 89: The detector according to any one of the two preceding embodiments, wherein the optical sensors are partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector.

Embodiment 90: The detector according to any one of the three preceding embodiments, wherein the optical sensors are part of a sensor array, such as an array comprising 2-4 optical sensors such as a quadrant photodiode.

Embodiment 91: The detector according to the preceding embodiment, wherein a geometrical center of the sensor array is offset from the optical axis.

Embodiment 92: The detector according to any one of the two preceding embodiments, wherein the sensor array is movable, preferably automatically, relative to the optical axis.

Embodiment 93: The detector according to the preceding embodiment, wherein the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly,
5 moving the sensor array relative to the optical axis until the light spot is off-centered.

Embodiment 94: The detector according to any one of the seven preceding embodiments, wherein the optical sensors are partial diodes of a quadrant diode, with a geometrical center of the quadrant diode being off-centered from the optical axis of the detector.
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Embodiment 95: The detector according to any one of the eight preceding embodiments, wherein the light-sensitive areas of the optical sensors are equal.

Embodiment 96: The detector according to any one of the nine preceding embodiments, wherein the detector comprises at least one lens, wherein the optical axis of the detector is an axis of symmetry of the lens.
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Embodiment 97: The detector according to any one of the ten preceding embodiments, wherein the evaluation device is configured for deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the evaluation device is further configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
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Embodiment 98: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.
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Embodiment 99: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.
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Embodiment 100: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.
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Embodiment 101: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

Embodiment 102: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.
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Embodiment 103: The detector according to any one of the sixteen preceding embodiments, wherein the optical sensors are arranged in one and the same beam path of the detector.

Embodiment 104: The detector according to any one of the seventeen preceding embodiments, wherein the optical sensors each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

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Embodiment 105: The detector according to any one of the eighteen preceding embodiments, wherein the optical sensors each are uniform sensors having a single light-sensitive area each.

Embodiment 106: The detector according to any one of the nineteen preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

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Embodiment 107: The detector according to any one of the twenty preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object, e.g. an illumination source comprising at least one laser, such as an illumination source providing at least one illuminating light beam being on the optical axis or being not on the optical axis.

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Embodiment 108: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

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Embodiment 109: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

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Embodiment 110: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one fringe pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

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Embodiment 111: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the quotient signal

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Q.

Embodiment 112: The detector according to any one of the twenty-five preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

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Embodiment 113: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example, at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one diffractive optical element; at least one Fresnel lens; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

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Embodiment 114: A detector for determining a position of at least one object, the detector having

- at least one first optical sensor having a first light-sensitive area, wherein the first optical sensor is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area by a light beam propagating from the object to the detector;
- at least one second optical sensor having:
 - o at least one fluorescent waveguiding sheet forming a second light-sensitive area, wherein the fluorescent waveguiding sheet is oriented towards the object such that at least one light beam propagating from the object towards the detector generates at least one light spot in the second light-sensitive area, wherein the fluorescent waveguiding sheet contains at least one fluorescent material, wherein the fluorescent material is adapted to generate fluorescence light in response to the illumination by the light beam; and
 - o at least one photosensitive element located at at least one edge of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive element by the fluorescent waveguiding sheet and capable of generating at least one second sensor signal in response to the illumination of the second light-sensitive area by the light beam, wherein the first light-sensitive area is smaller than the second light-sensitive area; and
- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating the first and second sensor signals.

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Embodiment 115: The detector according to the preceding embodiment, wherein the at least one photosensitive element is located at at least one of: a corner of the fluorescent waveguiding sheet; a straight edge, e.g. a straight rim portion, of the fluorescent waveguiding sheet.

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Embodiment 116: The detector according to any one of the two preceding embodiments, wherein the at least one photosensitive element is optically coupled to the fluorescent waveguiding sheet by at least one optical coupling element configured for at least partially coupling the fluo-

rescence light guided by the fluorescent waveguiding sheet out of the fluorescent waveguiding sheet.

5 Embodiment 117: The detector according to the preceding embodiment, wherein the optical coupling element is configured for at least partially coupling the fluorescence light coupled out of the fluorescent waveguiding sheet into the photosensitive element.

10 Embodiment 118: The detector according to any one of the two preceding embodiments, wherein the optical coupling element is selected from the group consisting of: a portion of transparent adhesive attaching the photosensitive element to the fluorescent waveguiding sheet; an etched portion within the fluorescent waveguiding sheet; a scratch in the fluorescent waveguiding sheet; a prism.

15 Embodiment 119: The detector according to any one of the five preceding embodiments, wherein the at least one first optical sensor and the at least one photosensitive element of the at least one second optical sensor have essentially identical electrical capacitance.

20 Embodiment 120: The detector according to any one of the six preceding embodiments, wherein the evaluation device is configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device is configured for determining the longitudinal coordinate by evaluating the quotient signal Q .

25 Embodiment 121: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.

30 Embodiment 122: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

35 Embodiment 123: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

Embodiment 124: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

40 Embodiment 125: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 126: The detector according to any one of the twelve preceding embodiments, wherein the first optical sensor is located in front of the second optical sensor, preferably such

that the light beam, in the given order, illuminates the first optical sensor and, afterwards, the second optical sensor.

Embodiment 127: The detector according to any one of the thirteen preceding embodiments, wherein the second optical sensor comprises at least two, preferably at least three and most preferably at least four photosensitive elements located at at least two edges of the fluorescent waveguiding sheet, capable of detecting fluorescence light guided from the light spot towards the photosensitive elements by the fluorescent waveguiding sheet and capable of generating at least one sensor signal each.

Embodiment 128: The detector according to the preceding embodiment, wherein the at least two photosensitive elements are located at one or more of: at least two straight edges, e.g. straight rim portions, of the fluorescent waveguiding sheet, at least two corners of the fluorescent waveguiding sheet, at least one corner of the fluorescent waveguiding sheet and at least one straight edge of the fluorescent waveguiding sheet.

Embodiment 129: The detector according to the preceding embodiment, wherein the evaluation device comprises at least one summing device configured for summing up the sensor signals of the at least two photosensitive elements, thereby forming a sum signal S.

Embodiment 130: The detector according to the preceding embodiment, wherein the sum signal is derived by using the formula $S = \sum_i c_i s_i$, with s_i being the sensor signals, with $i = 1 \dots N$ with N being a positive integer indicating the number of the photosensitive elements or a smaller positive integer, and with c_i being respective calibration coefficients.

Embodiment 131: The detector according to any one of the two preceding embodiments, wherein the evaluation device is configured to determine the at least one longitudinal coordinate z of the object by using at least one predetermined relationship between the first sensor signal, the sum signal S of the second sensor signals and the longitudinal coordinate z.

Embodiment 132: The detector according to any one of the four preceding embodiments, wherein the evaluation device is further configured to determine at least one transversal coordinate x, y of the object by evaluating the sensor signals of the photosensitive elements.

Embodiment 133: The detector according to the preceding embodiment, wherein the evaluation device comprises at least one subtracting device configured to form at least one difference signal D between second sensor signals generated by at least two of the photosensitive elements.

Embodiment 134: The detector according to the preceding embodiment, wherein the second sensor signals comprise at least one second sensor signal s_{21} and at least one second sensor signal s_{22} , wherein the at least one difference signal D is proportional to $a \cdot s_{21} - b \cdot s_{22}$, with a, b being real number coefficients, preferably with $a=1$ and $b=1$.

Embodiment 135: The detector according to the preceding embodiment, wherein the at least one difference signal D is derived according to the formula $D = (a \cdot s_{21} - b \cdot s_{22}) / (a \cdot s_{21} + b \cdot s_{22})$.

Embodiment 136: The detector according to any one of the three preceding embodiments, wherein the subtracting device is configured to form at least one first difference signal D_x from which at least one first transversal coordinate x of the object is derived, wherein the subtracting device is further configured to form at least one second difference signal D_y from which at least one second transversal coordinate y of the object is derived.

Embodiment 137: The detector according to the preceding embodiment, wherein the first difference signal D_x is generated from at least two second sensor signals s_{x1} , s_{x2} of at least two photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the waveguiding sheet in a first dimension, and wherein the second difference signal D_y is generated from at least two second sensor signals s_{y1} , s_{y2} of at least two photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the waveguiding sheet in a second dimension.

Embodiment 138: The detector according to the preceding embodiment, wherein the at least one first difference signal D_x is derived according to the formula $D_x = (a \cdot s_{x1} - b \cdot s_{x2}) / (a \cdot s_{x1} + b \cdot s_{x2})$, and wherein the at least one second difference signal D_y is derived according to the formula $D_y = (c \cdot s_{y1} - d \cdot s_{y2}) / (c \cdot s_{y1} + d \cdot s_{y2})$, with a , b , c , d being real number coefficients, preferably with $a=1$, $b=1$, $c=1$ and $d=1$.

Embodiment 139: The detector according to any one of the 11 preceding embodiments, wherein the photosensitive elements comprise at least two photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the fluorescent waveguiding sheet.

Embodiment 140: The detector according to any one of the 12 preceding embodiments, wherein the photosensitive elements comprise at least one first pair of photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the fluorescent waveguiding sheet in a first dimension of a coordinate system, wherein the photosensitive elements further comprise at least one second pair of photosensitive elements located at opposing edges, e.g. opposing rim portions and/or corners, of the fluorescent waveguiding sheet in a second dimension of the coordinate system.

Embodiment 141: The detector according to any one of the 27 preceding embodiments, wherein the second sensitive area is a homogeneous sensitive area.

Embodiment 142: The detector according to any one of the 28 preceding embodiments, wherein the second sensitive area has a surface area of at least 5 mm^2 , preferably of at least 10 mm^2 , more preferably of at least 100 mm^2 , more preferably of at least 400 mm^2 .

Embodiment 143: The detector according to any one of the 29 preceding embodiments, wherein the fluorescent waveguiding sheet comprises at least one planar sheet.

Embodiment 144: The detector according to any one of the 30 preceding embodiments, wherein the fluorescent waveguiding sheet has a thickness of 10 μm to 3 mm, preferably a thickness of 100 μm to 1 mm, such as a thickness of 50 μm to 2 mm.

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Embodiment 145: The detector according to any one of the 31 preceding embodiments, wherein the fluorescent waveguiding sheet is flexible or deformable.

Embodiment 146: The detector according to any one of the 32 preceding embodiments, wherein the fluorescent waveguiding sheet comprises at least one matrix material, wherein the at least one fluorescent material is one or more of mixed into the matrix material, dispersed into the matrix material, chemically bound to the matrix material or dissolved in the matrix material.

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Embodiment 147: The detector according to the preceding embodiment, wherein the matrix material comprises at least one plastic material.

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Embodiment 148: The detector according to the preceding embodiment, wherein the plastic material comprises at least one polymer material.

Embodiment 149: The detector according to any one of the two preceding embodiments, wherein the plastic material comprises at least one material selected from the group consisting of: a polycarbonate, a poly(methyl methacrylate), a polystyrene, a polyurethane, a polypropylene, a polyethylene terephthalate, a polyvinylchloride.

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Embodiment 150: The detector according to any one of the 36 preceding embodiments, wherein the fluorescent material comprises at least one fluorescent dye.

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Embodiment 151: The detector according to the preceding embodiment, wherein the fluorescent dye comprises at least one organic fluorescent dye.

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Embodiment 152: The detector according to any one of the two preceding embodiments, wherein the fluorescent dye is selected from the group consisting of: a xanthene derivative, preferably one or more of fluorescein, rhodamine, oregon green, eosin, texas red, or a derivative of any component thereof; a cyanine derivative, preferably one or more of cyanine, indocarbocyanine, oxacarbocyanine, thiocarbocyanine, merocyanine, or a derivative of any component thereof; a squaraine derivative or a ring-substituted squaraine, preferably one or more of Seta, SeTau, and Square dyes, or a derivative of any component thereof; a naphthalene derivative, preferably one or more of a dansyl or a prodan derivative thereof; a coumarin derivative; a oxadiazole derivative, preferably one or more of pyridyloxazole, nitrobenzoxadiazole, benzoxadiazole, or a derivative of any component thereof; an anthracene derivative, preferably one or more of an anthraquinone, DRAQ5, DRAQ7, CyTRAK orange, or a derivative of any component thereof; a pyrene derivative, preferably cascade blue; an oxazine derivative, preferably one or more of nile red, nile blue, cresyl violet, oxazine 170, or a derivative of any component thereof; an acridine derivative, preferably one or more of proflavin, acridine orange, acridine yellow, or a derivative

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of any component thereof; an arylmethine derivative, preferably one or more of auramine, crystal violet, malachite green, or a derivative of any component thereof; a tetrapyrrole derivative, preferably one or more of porphin, phthalocyanine, bilirubin; a rylene dye or any derivative thereof, such as a perylene dye; a naphthalene imide or perylene imide; a naphthoilene benzimidazole dye such as published in WO 2012/168395 A1; or a derivative of any component thereof.

Embodiment 153: The detector according to any one of the 39 preceding embodiments, wherein the at least one photosensitive element comprises at least one elongated photosensitive element extending along at least one segment of an edge of the fluorescent waveguiding sheet.

Embodiment 154: The detector according to any one of the 40 preceding embodiments, wherein the fluorescent waveguiding sheet is a rectangular fluorescent waveguiding sheet, preferably a square fluorescent waveguiding sheet, wherein photosensitive elements are located at each of the four edges, e.g. each of the four rim portions and/or corners, of the fluorescent waveguiding sheet.

Embodiment 155: The detector according to any one of the 41 preceding embodiments, wherein the detector further comprises at least one optical filter element, preferably at least one optical short-pass filter.

Embodiment 156: The detector according to any one of the 42 preceding embodiments, wherein the first and second optical sensors are arranged linearly in one and the same beam path of the detector.

Embodiment 157: The detector according to the preceding embodiment, wherein the first and second optical sensors are arranged concentrically with respect to an optical axis of the detector.

Embodiment 158: The detector according to any one of the 44 preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than five times the square root of a surface area of the first light-sensitive area.

Embodiment 159: The detector according to any one of the 45 preceding embodiments, wherein the first optical sensor is arranged in front of the second optical sensor and is spaced apart from the second optical sensor by no more than 50 mm, preferably by no more than 15 mm.

Embodiment 160: The detector according to any one of the 46 preceding embodiments, wherein the second light-sensitive area is larger by at least a factor of two, more preferably by at least a factor of three and most preferably by at least a factor of five as compared to the first light-sensitive area.

Embodiment 161: The detector according to any one of the 47 preceding embodiments, wherein the first light-sensitive area has a surface area of 1 mm² to 150 mm², more preferably a surface area of 10 mm² to 100 mm².

- 5 Embodiment 162: The detector according to any one of the 48 preceding embodiments, specifically according to the preceding embodiment, wherein the second light-sensitive area has a surface area of 160 mm² to 1000 mm², more preferably a surface area of 200 mm² to 600 mm².

- 10 Embodiment 163: The detector according to any one of the 49 preceding embodiments, wherein the first and second optical sensor each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

- 15 Embodiment 164: The detector according to any one of the 50 preceding embodiments, wherein the first optical sensor is a semiconductor sensor, preferably an inorganic semiconductor sensor, more preferably a photodiode and most preferably a silicon photodiode.

- 20 Embodiment 165: The detector according to any one of the 51 preceding embodiments, wherein the first optical sensor is a uniform sensor having a single light-sensitive area.

Embodiment 166: The detector according to any one of the 52 preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

- 25 Embodiment 167: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

- 30 Embodiment 168: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

- 35 Embodiment 169: The detector according to any one of the 55 preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

- 40 Embodiment 170: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, at least one focus-tunable lens, at least one spheric lens, at least one aspheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

Embodiment 171: The detector according to any one of the 57 preceding embodiments, wherein the first light-sensitive area, in a direction of propagation of the light beam, overlaps with the second light-sensitive area.

- 5 Embodiment 172: A detector for determining a position of at least one object, the detector comprising:
- at least one angle dependent optical element adapted to generate at least one transmission light beam having at least one beam profile depending on an angle of incidence of an incident light beam propagating from the object towards the detector and illuminating the angle dependent optical element;
 - 10 - at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element;
 - 15 - at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by evaluating a quotient signal Q from the sensor signals.

Embodiment 173: The detector according to the preceding embodiment, wherein the angle dependent optical element comprises at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element; at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter;

20 at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.

30 Embodiment 174: The detector according to any one of the two preceding embodiments, wherein the detector comprises at least one transfer device, wherein the transfer device has at least one focal length in response to at least one incident light beam propagating from the object to the detector.

35 Embodiment 175: The detector according to any one of the three preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

Embodiment 176: The detector according to the preceding embodiment, wherein the illumination source is adapted to illuminate the object through the angle dependent optical element.

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Embodiment 177: The detector according to any one of the two preceding embodiments, wherein the illumination source and the angle dependent optical element are arranged in the direction

of propagation of the light beam traveling from the object to the detector behind the transfer device.

Embodiment 178: The detector according to any one of the three preceding embodiments, wherein a distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors is small.

Embodiment 179: The detector according to the preceding embodiment, wherein the distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors is less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m.

Embodiment 180: The detector according to any one of the two preceding embodiments, wherein the angle dependent optical element is an optical fiber, wherein the illumination source is guided through the optical fiber, wherein the illumination source is adapted to generate at least one illumination beam, wherein the illumination beam exits the optical fiber behind the transfer device within a baseline that is smaller than a radius of the transfer device, wherein the optical fiber guiding the illumination beam may be attached to the transfer device to reduce reflections at interfaces with larger differences in refractive index.

Embodiment 181: The detector according to the preceding embodiment, wherein the optical fiber is attached to the transfer device by one or more of a polymer, a glue or other means for attachment.

Embodiment 182: The detector according to any one of the four preceding embodiments, wherein the distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors is less than the radius of the transfer device.

Embodiment 183: The detector according to any one of the eleven preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 184: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 185: The detector according to any one of the thirteen preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are areas of the beam profile at a sensor position of the optical sensors, and $E(x,y,z_0)$ denotes the beam profile given at the object distance z_0 .

- 5 Embodiment 186: The detector according to any one of the fourteen preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

- 10 Embodiment 187: The detector according to any one of the fifteen preceding embodiments, wherein the optical sensors are positioned off focus.

- 15 Embodiment 188: The detector according to any one of the sixteen preceding embodiments, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the beam profile of the light beam generated by the angle dependent optical element.

- 20 Embodiment 189: The detector according to the preceding embodiment, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

- 25 Embodiment 190: The detector according to any one of the two preceding embodiments, wherein the light-sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

- 30 Embodiment 191: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile.

- 35 Embodiment 192: The detector according to any one of the four preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.

- 40 Embodiment 193: The detector according to any one of the twenty preceding embodiments, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

- Embodiment 194: The detector according to any one of the two preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensors with at least one line pattern, wherein $A1$ corresponds to an area with a full line width of the line pat-

tern on the optical sensors, wherein A2 is a central area of the line pattern on the optical sensors.

Embodiment 195: The detector according to any one of the four preceding embodiments, wherein the light beam propagating from the object to the detector illuminates the optical sensors with at least one point pattern, wherein A1 corresponds to an area with a full radius of a point of the point pattern on the optical sensors, wherein A2 is a central area of the point in the point pattern on the optical sensors.

Embodiment 196: The detector according to any one of the two the preceding embodiments, wherein the evaluation device is configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center in-formation.

Embodiment 197: The detector according to any one of the 23 preceding embodiments, wherein the detector has at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by the light beam generated by the angle dependent optical element.

Embodiment 198: The detector according to the preceding embodiment, wherein the evaluation device is configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 199: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor

signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

- 5 Embodiment 200: The detector according to any one of the two preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for
10 sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a
15 sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.
- 20 Embodiment 201: The detector according to any one of the three preceding embodiments, wherein the combined signal is the quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a
25 quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

Embodiment 202: The detector according to any one of the 27 preceding embodiments, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets,
30 wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element.

35 Embodiment 204: The detector according to the preceding embodiment, wherein the optical sensors are part of a sensor array, wherein a geometrical center of the sensor array is offset from the optical axis.

Embodiment 205: The detector according to the preceding embodiment, wherein the sensor
40 array is movable relative to the optical axis.

Embodiment 206: The detector according to the preceding embodiment, wherein the evaluation device is configured for, firstly, determining the transversal position of a light spot generated by

the light beam on the sensor array by using the sensor signals and for, secondly, moving the sensor array relative to the optical axis until the light spot is off-centered.

5 Embodiment 207: The detector according to any one of the four preceding embodiments, wherein the optical sensors are partial diodes of a bi-cell or quadrant diode, with a geometrical center of the quadrant diode being off-centered from the optical axis of the detector.

10 Embodiment 208: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

15 Embodiment 209: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to
20 assign to the position at least one item of information.

Embodiment 210: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one
25 item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

30 Embodiment 211: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

35 Embodiment 212: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

40 Embodiment 213: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 214: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of the preceding embodiment referring to a detector, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

Embodiment 215: A readout device for optical storage media, the readout device comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 216: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by the light beam, wherein the detector has at least one transfer device, wherein the transfer device has at least one focal length in response to the light beam propagating from the object to the detector;
- illuminating each of the light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving a quotient signal Q of the sensor signals.

Embodiment 217: The method according to the preceding embodiment, wherein the longitudinal coordinate z of the object is determined in at least one measurement range independent from the object size in an object plane.

Embodiment 218: The method according to the preceding embodiment, wherein the deriving of the quotient signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 219: The method according to any one of the preceding embodiments referring to a method, wherein the method comprises adjusting the measurement range by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive

area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the object in the object plane.

Embodiment 220: The method according to any one of the preceding embodiments referring to a method, wherein the transfer device has an optical axis, wherein the transfer device constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis, wherein the method comprises arranging the optical sensors such that the light-sensitive areas of the optical sensors differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.

Embodiment 221: The method according to any one of the preceding embodiments referring to a method, wherein each of the sensor signals comprises at least one information of at least one beam profile of the light beam, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles, wherein the light-sensitive regions are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

Embodiment 222: The method according to the preceding embodiment, wherein the evaluating comprises determining the first area of the beam profile and the second area of the beam profile.

Embodiment 223: The method according to any one of the two preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises an information relating to a number of photons in the first area of the beam profile and the center information comprises an information relating to a number of photons in the second area of the beam profile, wherein the evaluating comprises deriving the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.

Embodiment 224: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- illuminating at least one first light-sensitive area of at least one first optical sensor of the detector with a light beam propagating from the object to the detector and, thereby, generating at least one first sensor signal;
- illuminating at least one second light-sensitive area of at least one second optical sensor of the detector with the light beam and, thereby, generating at least one sec-

ond sensor signal, wherein the first light-sensitive area is smaller than the second light-sensitive area; and

- evaluating the first and second sensor signals, thereby determining at least one longitudinal coordinate z of the object.

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Embodiment 225: The method according to the preceding embodiment, wherein the evaluating the first and second sensor signal comprises deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, and wherein the determining the longitudinal coordinate comprises evaluating the quotient signal Q .

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Embodiment 226: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;

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- evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;

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- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;

- c) determining at least one combined signal by combining the center signal and the sum signal; and

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- d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 227: The method according to the preceding embodiment, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

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Embodiment 228: The method according to the preceding embodiment, wherein determining the longitudinal coordinate comprises evaluating the quotient signal Q , wherein evaluating the quotient signal comprises using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

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Embodiment 229: The method according to any one of the three preceding method embodiments, wherein the method comprises using the detector according to any one of the preceding embodiments referring to a detector.

Embodiment 230: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam;
- illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

Embodiment 231: The method according to the preceding embodiment, wherein evaluating of the sensor signals comprises deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q .

Embodiment 232: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least one angle dependent optical element and generating at least one light beam having at least one beam profile depending on an angle of incidence;
- providing at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam generated by the angle dependent optical element;
- illuminating each of the light-sensitive areas of the at least two optical sensors of the detector with the light beam generated by the angle dependent optical element, wherein, thereby, each of the light-sensitive areas generates at least one sensor signal; and
- evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object, wherein the evaluating comprises deriving a combined signal Q of the sensor signals.

Embodiment 233: The method according to the preceding embodiment, wherein the deriving of the combined signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 234: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage appli-

cation; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use in combination with optical data storage and readout.

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Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this
10 context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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| 20 | Figures 1 and 2 | show different embodiments of a detector according to the present invention; |
| | Figure 3 | shows a quotient signal Q as a function of a longitudinal coordinate z of an object, for various intensities of a light source; |
| 25 | Figure 4 | shows a quotient signal Q as a function of a longitudinal coordinate z of an object for various target sizes; |
| 30 | Figure 5 | shows an exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera. |
| | Figures 6 and 7 | show further different embodiments of a detector according to the present invention; |
| 35 | Figure 8 | shows an example of a light spot on a matrix of optical sensors and the evaluation thereof; |
| 40 | Figure 9 | shows a further exemplary embodiment of a detector according to the present invention; a detector system, a human machine interface, an entertainment device, a tracking system, a scanning system and a camera; and |
| | Figure 10 | shows an exemplary embodiment of a quotient signal Q as a function of a longitudinal coordinate z , for various illumination intensities; |

- Figures 11 and 12 show further different embodiments of a detector according to the present invention;
- 5 Figure 13 shows a further exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera;
- 10 Figure 14 shows an off-centered position of the optical axis of the detector and of a light spot on a quadrant diode;
- Figure 15 shows an alternative embodiment of a detector with staggered optical sensors;
- 15 Figures 16 to 18 show a quotient signal Q of two sensor signals of two photodiodes of a quadrant photodiode as a function of a longitudinal coordinate z of an object under various measurement conditions;
- 20 Figures 19A and 19B show a modification of the embodiment of Figure 12, with an off-axis illumination light beam;
- Figures 19C to 19E shows a comparison of two experimental setups using a detector setup according to Figure 19A with a Bi-cell as optical sensors and experimental results;
- 25 Figures 20 and 21 show further different embodiments of a detector according to the present invention;
- Figures 22A and 22B show different views of an exemplary embodiment of a second optical sensor which may be used in the present invention;
- 30 Figure 23 shows a top view onto the light-sensitive area of the second optical sensor shown in Figures 22A and 22B, with a light spot generated by a light beam;
- 35 Figure 24 shows a further exemplary schematic setup of an evaluation device;
- Figure 25 shows a further exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera;
- 40 Figures 26A and 26B show an alternative embodiment of a second optical sensor which may be used in the present invention;

- Figures 27A and 27 B show further exemplary embodiments of a detector according to the present invention;
- 5 Figure 28 shows experimental results of a distance determination with a detector according to the present invention;
- Figure 29 shows a further exemplary embodiment of a detector according to the present invention;
- 10 Figure 30 shows an exemplary cloud of points impinging on a sensor element;
- Figures 31 A to O show further exemplary configurations of optical sensors according to the present invention;
- 15 Figure 32 shows experimental results of a determination of a longitudinal coordinate z for different object sizes;
- Figures 33A and B show an exemplary beam profile and determination of first area and second area of the beam profile;
- 20 Figure 34 shows a further exemplary embodiment of the detector;
- Figure 35 shows an exemplary embodiment of a detector according to the present invention;
- 25 Figure 36 shows an embodiment with a multifurcated optical fiber;
- Figure 37 shows a cut through the optical fiber of Figure 36;
- 30 Figures 38 visualizes angle dependent transmission of an angle dependent optical element;
- Figure 39 shows angle dependent transmission power of the optical fiber at constant irradiated power;
- 35 Figures 40A and B show experimental results of distance measurement;
- Figure 41 shows an exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera;
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- Figure 42 shows an embodiment of determining a longitudinal coordinate z using a line pattern;
- 5 Figure 43 shows a simulation testing of object size independence for a one-dimensional case using computational ray tracing;
- 10 Figures 44A and B show a defocused beam profile subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates and Radon transformation different image regions on a matrix of optical sensors;
- Figures 45A and B show further embodiments of the detector according to the present invention comprising at least one bi-cell;
- 15 Figure 46 shows spot diameter independence of the combined sensor signal;
- Figures 47A to C show three embodiments of a hexagonal illumination pattern;
- 20 Figure 48 shows an embodiment of a scanning device;

Detailed description of the embodiments:

- 25 In Fig. 1, a schematic view of a first embodiment of a detector 110 for determining a position of at least one object 112 is depicted. The detector 110 comprises at least two optical sensors 113, for example a first optical sensor 118 and a second optical sensor 120, each having at least one light-sensitive area 121. In this case, the object 112 comprises a beacon device 114, from which a light beam 116 propagates towards the first optical sensor 118 and the second
- 30 optical sensor 120. The first optical sensor 118 may comprise a first light-sensitive area 122, and the second optical sensor 120 may comprise a second light-sensitive area 124. The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110, further, comprises at least one transfer device 128, such as at least one lens or a lens system, specifically for beam shaping.
- 35 The transfer device 128 has at least one focal length in response to the incident light beam 116 propagating from the object 112 to the detector 110. The transfer device 128 has an optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. The transfer device 128 constitutes a coordinate system. A direction parallel or anti-parallel to the optical axis 126, 129 may be defined as a longitudinal direction, whereas direc-
- 40 tions perpendicular to the optical axis 126, 129 may be defined as transversal directions, wherein a longitudinal coordinate l is a coordinate along the optical axis 126, 129 and wherein d is a spatial offset from the optical axis 126, 129. Consequently, the light beam 116 is focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a

longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. The optical sensors 118, 120 may be positioned off focus. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

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In this first preferred embodiment the optical sensors 118, 120 may be arranged such that the light-sensitive areas 122, 124 differ in their longitudinal coordinate and/or their surface areas and/or their surface shapes. As can be seen in Fig. 1, the first optical sensor 118 is a small optical sensor, whereas the second optical sensor 120 is a large optical sensor. Thus, the width of the light beam 116 fully covers the first light-sensitive area 122, whereas, on the second light-sensitive area 124, a light spot is generated which is smaller than the second light-sensitive area 124, such that the light spot is fully located within the second light-sensitive area 124. As an example, the first light-sensitive area 122 may have a surface area of 1 mm^2 to 100 mm^2 , whereas the second light-sensitive area 124 may have a surface area of 50 to 600 mm^2 . Other embodiments, however, are feasible.

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The first optical sensor 118, in response to the illumination by the light beam 116, may generate a first sensor signal s_1 , whereas the second optical sensor 120 may generate a second sensor signal s_2 . Preferably, the optical sensors 118, 120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 116 or of the portion of the light beam 116 illuminating their respective light-sensitive areas 122, 124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination. In other words, preferably, the optical sensors 118, 120 do not exhibit the above-described FiP effect.

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The sensor signals s_1 and s_2 are provided to an evaluation device 132 of the detector 110. The evaluation device 132, as symbolically shown in Fig. 1, is embodied to derive a quotient signal Q , as explained above. From the quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 112 and/or the beacon device 114, from which the light beam 116 propagates towards the detector 110. For further details of this evaluation, reference is made to Fig. 3 and 4 below.

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The detector 110, in combination with the at least one beacon device 114, may be referred to as a detector system 134, as will be explained in further detail below with reference to Fig. 5.

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In Fig. 2, a modification of the embodiment of Fig. 1 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Fig. 1. Instead of using an active light source, i.e. a beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 comprises at least one illumination source 136. The illumination source 136, as an example, may comprise a laser, whereas, in Fig. 1, as an example, the beacon device 114 may comprise a light-emitting diode (LED). The illumination source 136 may be configured for generating at least one illumi-

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nation light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116.

- 5 As shown in Fig. 2, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too. In order to provide an on-axis illumination, as shown in Fig. 2, as an example, one or more reflective elements 140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

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Apart from these modifications, the setup of the embodiment in Fig. 2 corresponds to the setup in Fig. 1. Thus, again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q , and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the quotient signal Q . It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

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It shall further be noted that the embodiments shown in Fig. 1 and 2 simply provide embodiments for determining the longitudinal coordinate z of the object 112. It is also feasible, however, to modify the setups of Fig. 1 and 2 to provide additional information on a transversal coordinate of the object 112 and/or of parts thereof. As an example, e.g. in between the transfer device 128 and the optical sensors 118, 120, one or more parts of the light beam 116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 112 and/or of parts thereof. The transversal coordinate may be used to verify and/or enhance the quality of the distance information. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

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In Fig. 3 and 4 typical quotient signals Q are depicted, as a function of the longitudinal coordinate z of an object 112 in a test setup. Therein, a simple quotient s_1/s_2 is shown, for an exemplary setup of the detector 110. Fig. 3 and 4, each, show a bundle of experiments which are not resolved in these figures. Thus, in Fig. 3, various curves are given for the setup shown in Fig. 1, with an active beacon device 114 having an LED. The current of the LED target of the beacon device 114, in this experiment, is changed from 1000 mA to 25 mA. Basically, no difference in the quotient signal, as a function of the longitudinal coordinate z (given in mm) can be detected over the spatial measurement range of 250 mm to 2,250 mm. The experiment clearly shows that the setup of the detector 110 according to the present invention is independent from the total power of the light beam 116. Thus, no additional information on the total power of the light beam, and, thus, no additional illumination on the luminance is required in order to derive the

longitudinal coordinate. Thus, as shown in Fig. 3, as an example, a unique relationship between a quotient signal Q^* as measured in an experiment and a longitudinal coordinate z^* exists.

Thus, the curves as shown in Fig. 3, as an example, may be used as calibration curves for indicating a unique and predetermined or determinable relationship between the quotient signal Q and the longitudinal coordinate. The curves as shown in Fig. 3, as an example, may be stored in a data storage and/or in a lookup table. The calibration curves Q may simply be determined by calibration experiments. It is also feasible, however, to derive these curves by one or more of modelling, analytically, semi-empirically and empirically.

The experiment shown in Fig. 3 clearly demonstrates that the setup of the detector 110 according to the present invention provides a large range of measurement, both in terms of space (e.g. a measurement range from 270 to 2,250 mm) and in terms of brightness or total power of the light beam 116. In Fig. 4, an additional experiment is shown which demonstrates that the setup is widely independent from the target size, i.e. the lateral diameter of the beacon device 114. For this experiment, again, an LED beacon device 114 was used, similar to the setup shown in Fig. 1, wherein the size of the target, i.e. the visible part of the LED, was changed by using a diffuser and an adjustable aperture. Thereby, the aperture or the size of the target was varied from 1 mm to 25 mm in diameter. Without resolving the curves shown in Fig. 4 in detail, it is clearly visible that the quotient signal Q , again, is widely independent from the target size, in between a target size of 1 mm to 25 mm. Thus, again, a unique relationship between the quotient signal Q and the longitudinal coordinate z can be derived, for various target sizes, which may be used for evaluation.

The results shown in Fig. 3 and 4 were derived experimentally, by varying the named parameters and by measuring appropriate signals. The results, however, may also be derived analytically, semi-analytically or by modelling. Comparable results were obtained.

Fig. 5 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments shown in Fig. 1 or 2. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 5 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 5 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 148. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises the detector system 134. The components of the devices and systems shall be explained in further detail below.

Fig. 5 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136. The illumination source 136, generally, is configured to emit at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 5 is shown in Fig. 1 and 2. Thus, the detector 110, besides the optical sensors 118, 120, comprises at least one evaluation device 132, having e.g. the at least one divider 142 and/or the at least one position evaluation device 144, as symbolically depicted in Fig. 5. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 118, 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Fig. 5. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 118, 120 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the optical sensors 118, 120. The optical sensors 118, 120 may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. A coordinate system 168 may be defined, in which a direction parallel or anti-parallel to the optical axis 126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 126 may be defined as transversal directions. In the coordinate system 128, symbolically depicted in Fig. 5, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the optical sensors 118, 120 as well as, optionally, further optical sensors. The optical sensors 118, 120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 118 covers a portion of the second optical sensor 120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 112 and/or of parts thereof. Alternatively, however, the optical sensors 118, 120 may be located at the same longitudinal coordinate.

One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as explained above in the context of Fig. 1 to 4, the evaluation device 132 is configured to evaluate sensor signals provided by the optical sensors 118, 120. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 118, 120 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 136 is used, the beacon devices 114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflected beacon devices 114 having at least one reflective surface such as a mirror, retro reflector, reflective film, or the like. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the light-sensitive areas 122, 124 of the optical sensors 118, 120. For details of the evaluation, reference may be made to Fig. 1 to 4 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Fig. 5, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Fig. 5 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may be fully or partially integrated into the machine 170, specifically the computer, as indicated in Fig. 5 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Figure 6, a schematic view of a further embodiment of the detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises the at least one beacon device 114, from which the light beam 116 propagates towards at least one sensor element 115. The sensor element 115 comprises a matrix 117 of optical sensors 113, each optical sensor 113 having at least one light-sensitive area 121 facing the object 112. In this second preferred embodiment the optical sensors 118, 120 may be arranged such that the light-sensitive areas of the optical sensors 113 differ in spatial offset and/or surface areas. The light beam 116, as an example, may propagate along the optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110 comprises the at least one transfer device 128, such as at least one lens and/or at least one lens system, specifically for beam shaping. Consequently, the light beam 116 may be focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on the longitudinal coordinate z of the object 112, such as on the distance between the detector 110 and the beacon device 114 and/or the object 112. The transfer device 128 constitutes the optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. Consequently, the light beam 116 is focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. The optical sensors 118, 120 are positioned off focus. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Figure 6, the light beam 116 generates a light spot 131 on the matrix 117. In Figure 8, an exemplary view of the light spot 131 on the matrix 117 is shown. As can be seen, in this exemplary embodiment, the matrix 117 specifically may be a rectangular matrix, with rows numbered by "i", from 1 to n , and with columns, denoted by "j", from 1 to m , with n , m being integers. The center of the light spot 131, in this exemplary embodiment, is located in the sensor

element denoted by i^* , j^* . The optical sensors 113 may provide sensor signals s_{ij} to an evaluation device 132 which, out of the sensor signals, determines at least one center signal, denoted symbolically by $s_{i^*j^*}$. As outlined in further detail above, for generating the center signal, the evaluation device 132 may comprise at least one center detector 133. As an example, the center detector 133 simply may determine the maximum sensor signal out of the plurality of sensor signals generated by the optical sensors 113. Alternative methods are feasible. Thus, as an example, instead of determining a single maximum optical sensor signal, a plurality of sensor signals may be used for generating the center signal. Thus, as an example, neighboring optical sensors which are adjacent to the optical sensor i^* , j^* may contribute to the center signal, such as optical sensors with the coordinates $i^*-1, \dots, i^*+1$ and $j^*-1, \dots, j^*+1$. These coordinates, in the simple exemplary embodiment, may form a square around the optical sensor i^* , j^* . Instead of a square having a side length of 3, as in this embodiment, other environments around the optical sensor having the highest sensor signal may be used such as to optimize the signal to noise ratio of the detector signal and or of the distance information. Further, additionally or alternatively, the center signal may be generated by adding up and/or averaging over sensor signals within a certain range from the maximum sensor signal which may for example be beneficial to the measurement precision concerning noise such as pixel noise. Further, additionally or alternatively, for the determination of the center signal or sum signal, image processing techniques such as sub-pixel processing, interpolation, normalization or the like may be employed. Other alternatives are feasible. The evaluation device 132 may be adapted to determine the center signal by integrating of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device 132 may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid. Further, when trapezoid beam profiles may be assumed, the evaluation device 132 may be adopted to determine the edge and center signals by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations. Additionally or alternatively, the evaluation device 132 may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

Further, the evaluation device 132 is configured for determining at least one sum signal out of the sensor signals of the matrix 117. For this purpose, the evaluation device 132 may comprise at least one summing device 135. The summing device 135 may be configured for adding up, integrating or averaging over the sensor signals of the entire matrix 117, of a region of interest within the matrix 117, each option with or without the optical sensors from which the center signal is generated. Thus, in the exemplary embodiment shown in Figure 8, the summing device 135 is simply configured for summing over the sensor signals s_{ij} of the entire matrix 117, except for the center optical detector with the coordinates i^* , j^* . Other options, however, are feasible.

The evaluation device 132 may be adapted to determine the sum signal by integrating of signals of the entire matrix 117, of the region of interest within the matrix 117. For example, the beam profile may be a trapezoid beam profile and the evaluation device 132 may be adapted to determine an integral of the entire trapezoid.

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The evaluation device 132 may be adapted to determine at least one region of interest within the matrix, such as one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform at least one filtering, for example at least one object recognition method.

10 The region of interest may be determined manually by a user or maybe determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device 132 further is configured for forming at least one combined signal out of the center signal and the sum signal. For this purpose, the evaluation device 132, as an example, may comprise at least one combining device 137, such as at least one divider 142. As a very simple embodiment, a quotient Q may be formed, by dividing the center signal by the sum signal or vice versa. Other options are feasible and are given above.

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Finally, the evaluation device 132 is configured for determining at least one longitudinal coordinate z of the object by evaluating the combined signal. For this purpose, the evaluation device may comprise at least one further component, such as at least one evaluation component, for example a position evaluation device 144. It shall be noted that the components of the evaluation device 132 shown in Figure 8 may fully or partially be embodied in hardware and/or software. Further, the components may fully or partially be embodied as independent or separate components, and/or may fully or partially be embodied as components which are integrated into the sensor element 115.

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The embodiment of Figure 8 further shows that, in addition to the longitudinal coordinate z , at least one item of information on a transversal coordinate of the object 112 and/or the beacon device 114 may be generated. Thus, the coordinates i^* and j^* provide additional items of information on a transversal position of the object 112 and/or the beacon device 114. In the setup of Figure 6, the beacon device 114, for the sake of simplicity, is positioned in the center, i.e. on the optical axis 126, 129. In this case, the light spot 131 is likely to be centered in the middle of the matrix 117. In the embodiment shown in Figure 8, however, as can easily be detected, the light spot 131 is off-centered. This off-centering is characterized by the coordinates i^* , j^* . By using known optical relationships between this off-centering and a transversal position of the object 112 and/or the beacon device 114, such as by using the lens equation, at least one transversal coordinate of the object 112 and/or the beacon device 114 may be generated. This option is also shown in the exemplary embodiment of Figure 8.

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In Figure 7, a modification of the embodiment of Figure 6 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Figure 6. Instead of using an active beacon device 114 with light-emitting

properties for generating the light beam 116, however, the detector 110 itself comprises at least one illumination source 136. The illumination source 136, as an example, may comprise at least one laser, whereas, in Figure 6, as an example, the beacon device 114 may comprise a light-emitting diode (LED). Other embodiments, however, are feasible. The illumination source 136
5 may be configured for generating at least one illumination light beam 138 for fully or partially illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116.

As shown in Figure 7, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too. In order to provide an on-axis illumination, as shown in Figure 7, as an example, one or more reflective elements 140 may be used, such as one or more
10 prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Figure 7 corresponds to the setup in Figure 6. Thus, again, an evaluation device 132 may be used, having e.g. at least one divider 142 for forming the quotient signal Q, and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the quotient signal Q and/or another type of combined signal. It shall be noted that the evaluation device 132, again,
20 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 133, 135, 137, 142, 144 may fully or partially be embodied by appropriate software components and/or may fully or partially be embodied by hardware components.

The optical sensors 113 of the matrix 117, as an example, may be pixels of a pixelated optical sensor, such as a CCD and/or a CMOS sensor chip. Thus, as an example, the optical sensors 113 may have a side length and/or an equivalent diameter in the range of a few micrometers to several hundred micrometers. It shall be noted, however, that larger pixels or optical sensors 113 may be used. Further, instead of using an integrated sensor element 115 such as a CCD and/or CMOS sensor chip, non-integrated matrices may be used.
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Figure 9 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments in Figures 6 or 7. The detector 110, specifically, may be embodied as the camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or
35 image sequences such as digital video clips. Other embodiments are feasible.

Figure 9 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached to and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Figure 9 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 151. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises
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the detector system 134. The components of the devices and systems shall be explained in further detail below.

Figure 9 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the at least one object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110 and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136, which is not shown. The illumination source 136, generally, may be configured to emit the at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. It shall be noted, however, that an active beacon device, as e.g. shown in the setup of Figure 6, may also be used, and, thus, that setups with no integrated illumination source 136 are also feasible. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112 and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Figure 9 is shown in Figures 6 and 7. Thus, the detector 110, besides the sensor element 115, comprises the at least one evaluation device 132, having, e.g., the at least one center detector 133, the at least one summing device 135, the at least one combining device 140, the at least one divider 142, the at least one position evaluation device 144 and/or combinations thereof. These components, which may optionally be present, are symbolically depicted in Figure 9. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the components of the evaluation device 132 and one or more of the components of the sensor element 115 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Figure 9. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the sensor element 115 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the sensor element 115. The sensor 115, wherein one or more of the sensor elements 115 may be provided, may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

5 An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. In the coordinate system 168, symbolically depicted in Figure 9, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other
10 types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the sensor element 115 as well as, optionally, one or more further optical sensors. A non-branched beam path may be used or, alternatively, a branched beam path may be possible, with, e.g., additional optical sensors in one or more additional
15 beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining the transversal coordinates of the object 112 and/or parts thereof. As outlined above, however, in the context of Figure 8, the at least one transversal coordinate may also be determined by the sensor element 115 itself, such as by determining the transversal coordinates of the center of the light spot 131.

20 One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as outlined above in the context of Figures 6 to 8, the evaluation device 132 may be configured to evaluate the sensor signals
25 provided by the optical sensors 113 of the matrix 117 of the sensor element 115. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 113 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may comprise
30 active beacon devices with an integrated illumination source such as light-emitting diodes. In case the illumination source 136 is used, the beacon devices do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by
35 one or more lenses, illuminates the sensor element 118. For details of the evaluation, reference may be made to Figures 6 to 8 above.

40 As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Figure 9, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Figure 9 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may fully or partially be integrated into the machine 170, specifically the computer, as indicated in Figure 9, and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Figure 10, experimental data are shown which demonstrate an exemplary embodiment of the present invention in terms of measurement data. The figure shows a quotient signal Q as a function of a longitudinal coordinate z , given in millimeters, for various illumination intensities.

In order to gain the experimental data shown in Figure 10, an experimental setup was used with a sensor element 118 formed by a Basler AC 1920-40GC camera, with a transfer device 128 formed by a Nikkor 50 mm lens.

As a beacon device 114, a light-emitting diode (LED) was used, having a central nominal wavelength of 532 nm. A diffusor made from Teflon film was used in front of the LED and a diaphragm, in order to provide a well-defined light-emitting area having a diameter of 5 mm. The intensity of the LED was varied, by varying a drive current of the LED between 5 mA and 150 mA.

In the experiments, the distance z between the LED and the lens was varied from 300 mm to 1700 mm. The signal of the Basler AC 1920-40GC camera was evaluated by the following procedure. As a center signal, an accumulated signal of an inner circle having a radius of 15 pixels around the optical axis was determined, with the light spot centered at the optical axis. As a sum signal, the sum of the signals of all pixels of the camera within the light spot was generated. A quotient signal was formed, by dividing the sum signal by the center signal.

In Figure 10, the solid curves, overlapping, show the quotient signal curves for LED currents of 150 mA, 125 mA, 100 mA and 50 mA. As can be seen, there are basically no differences between these curves, within the whole measurement range. This experiment clearly shows that the quotient signal is widely independent on the total power of the light beam. Only at lower intensities, the curves may be distinguished. Thus, the dotted curve shows measurements at an

LED current of 25 mA, the dashed-dotted line at an LED current of 10 mA, and the dashed line at an LED current of 5 mA. Still, even at these low intensities, the measurement curves are very close to the solid curves, which shows the high tolerances of the measurement.

- 5 In Fig. 11, a schematic view of a further embodiment of a detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises the beacon device 114, from which the light beam 116 propagates towards the first optical sensor 118 and the second optical sensor 120. The first optical sensor 118 may comprise a first light-sensitive area 122, and the second optical sensor 120 comprises the second light-sensitive area 124.
- 10 The optical sensors 118, 120, as shown e.g. in Fig. 14, may be part of an array 174 of optical sensors 176, such as the first optical sensor 118 being the optical sensor 176 in the upper left corner of the array 174 and the second optical sensor 120 being the optical sensor 176 in the lower right corner of the array 174. Other choices are feasible. The array 174, as an example, may be a quadrant photodiode 178, and the optical sensors 176 may be partial diodes of the
- 15 quadrant photodiode 178.

The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110 comprises the at least one transfer device 128, such as at least one lens and/or at least one lens system, specifically

20 for beam shaping. Consequently, the light beam 116 may be focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on the longitudinal coordinate z of the object 112, such as on the distance between the detector 110 and the beacon device 114 and/or the object 112. The optical sensors 118, 120 are positioned off focus. In this third preferred embodiment the optical sensors 118, 120 may be arranged such that the light-

25 sensitive areas of the optical sensors differ in their spatial offset and/or their surface areas. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Fig. 14, the setup of the detector 110 is off-centered in various ways. Thus, a

30 geometrical center 180 of the array 174 may be off-centered from the optical axis 126, 129 by offset d_0 . Further, a geometrical center 182 of the first optical sensor 118 is off-centered from the optical axis 126 by offset d_1 , and a geometrical center 184 of the second optical sensor 120 is off-centered from the optical axis 126 by offset d_2 , wherein $d_1 \neq d_2$. In other words, a light spot 186 is formed, which is unequally distributed over the light-sensitive areas 122, 124.

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As will be shown in further detail below, the detector 110 may be configured for automatically establishing the off-centered position of the light spot 186 on the array 174. For this purpose, firstly, the detector 110 may be configured for determining whether the sensor signals generated by the optical sensors 176 of the array 174 are equal. If this should be the case, the detector

40 110 may be configured to determine that the light spot 186 is centered in the array 174 and, consequently, may shift the light spot 186 out of the geometrical center 180 of the array 174, such as by shifting the whole array 174 in a plane perpendicular to the optical axis 126, 129. For

this purpose, as will be shown in further detail below with respect to Fig. 13, one or more actuators may be provided in the detector 110.

Turning back to the setup of Fig. 11, the first optical sensor 118, in response to the illumination by the light beam 116, generates a first sensor signal s_1 , whereas the second optical sensor 120 generates a second sensor signal s_2 . Preferably, the optical sensors 118, 120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 116 or of the portion of the light beam 116 illuminating their respective light-sensitive areas 122, 124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination. In other words, preferably, the optical sensors 118, 120 do not exhibit the above-described FiP effect.

The sensor signals s_1 and s_2 are provided to an evaluation device 132 of the detector 110. The evaluation device 110, as symbolically shown in Fig. 1, may specifically be embodied to derive a quotient signal Q , as explained above. The quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 112 and/or the beacon device 114, from which the light beam 116 propagates towards the detector 110, as will be explained in further detail with reference to the corrosion signals shown in Fig. 16 to 18 below.

The detector 110, in combination with the at least one beacon device 114, may be referred to as the detector system 134, as will be explained in further detail below with reference to Fig. 13.

In Fig. 12, a modification of the embodiment of Fig. 11 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Fig. 11. Instead of using an active light source, i.e. a beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 comprises the at least one illumination source 136. The illumination source 136, as an example, may comprise a laser, whereas, in Fig. 11, as an example, the beacon device 114 may comprise a light-emitting diode (LED). The illumination source 136 may be configured for generating at least one illumination light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The illumination source 136, as an example, may comprise one or more diaphragms 190, such as an adjustable diaphragm 190, e.g. an adjustable iris diaphragm and/or a pin hole.

The setup shown in Fig. 12, as an example, may also be used in or as a readout device 192 for optical storage media. Thus, as an example, the object 112 may be an optical storage medium such as in optical storage discs, e.g. a CCD, DVD or Blu-ray disc. By measuring the presence or non-presence of data storage modules and the depth of the same within the object 112, by using the above-mentioned measurement principle, a data readout may take place.

The light beam 116, specifically, may travel along the optical axis 126 of the detector 110. As shown in Fig. 12, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too, as will be shown in the context of Fig. 19A and 19B below. In order to provide an on-axis illumination, as shown in Fig. 12, as an example, one or more reflective elements 140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Fig. 12 corresponds to the setup in Fig. 11. Thus, again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q , and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the at least one quotient signal Q . It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

It shall further be noted that the embodiments shown in Fig. 11 and 12 simply provide embodiments for determining the longitudinal coordinate z of the object 112. It is also feasible, however, to modify the setups of Fig. 11 and 12 to provide additional information on a transversal coordinate of the object 112 and/or of parts thereof. As an example, e.g. in between the transfer device 128 and the optical sensors 118, 120, one or more parts of the light beam 116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or additional quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 112 and/or of parts thereof. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

Fig. 13 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments shown in Fig. 11 or 12. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 13 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 13 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 148. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises the detector system 134. The components of the devices and systems shall be explained in further detail below.

Fig. 13 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136. The illumination source 136, generally, is configured to emit at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 13 is shown in Fig. 11 and 12 or will be shown, as an alternative embodiment, in Fig. 15 below. Thus, the detector 110, besides the optical sensors 118, 120, comprises at least one evaluation device 132, having e.g. the at least one divider 142 and/or the at least one position evaluation device 144, as symbolically depicted in Fig. 13. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 118, 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Fig. 15. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 118, 120 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In Fig. 13, as an example, one or more reflective elements 140 may be used, for example partially transparent, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

The detector 110 as symbolically shown in the exemplary embodiment of Fig. 13 may also comprise at least one actuator 188 for moving the array 174 of the optical sensors 176 relative to the optical axis 126. As outlined above, for providing this movement, the optical axis 126 may be moved in relation to the array 174 by moving the optical axis 126, by moving the array 174 or both. Thus, as an example, the optical axis may be moved by using one or more of deflecting elements and/or by using the transfer device 128. As a simple example, a lens of the transfer device 128 may be tilted, such as by using one or more actuators 188 (not depicted). Additionally or alternatively, the array 174 may be shifted by the one or more actuators 188, preferably in

a plane perpendicular to the optical axis 126. As an example, one or more electromechanical actuators may be used, such as one electromechanical actuator for an x-direction and another electromechanical actuator for a y-direction. Other embodiments are feasible. Thereby, the above-mentioned off-centering procedure may be implemented for establishing an off-centered situation as shown e.g. in Fig. 14.

In the exemplary embodiment shown in Fig. 13, further, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the optical sensors 176, including at least the first optical sensor 118 and the second optical sensor 120. The optical sensors 176 may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. In the coordinate system 128, symbolically depicted in Fig. 15, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the optical sensors 118, 120 as well as, optionally, further optical sensors. The optical sensors 118, 120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 118 covers a portion of the second optical sensor 120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 112 and/or of parts thereof.

One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as explained above in the context of Fig. 11, 12 and 14, the evaluation device 132 is configured to evaluate sensor signals provided by the optical sensors 118, 120. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 118, 120 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may be or may comprise active beacon devices with an inte-

grated illumination source such as a light-emitting diode. In case the illumination source 136 is used, the beacon devices 114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the light-sensitive areas 122, 124 of the optical sensors 118, 120. For details of the evaluation, reference may be made to Fig. 11, 12 and 14 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Fig. 13, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Fig. 13 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may be fully or partially integrated into the machine 170, specifically the computer, as indicated in Fig. 13 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In the setup of the detectors 110 as shown in Fig. 11, 12 and 13, the optical sensors 176 are part of an array 174, and all optical sensors 176 may be located in one and the same plane oriented essentially perpendicular to the optical axis 126. As noted in this context, when mentioning “perpendicular” or “essentially perpendicular”, preferably, a 90° orientation is given. However, tolerances may be present, such as angular tolerances of no more than 20°, preferably of no more than 10° or more preferably of no more than 5°. The optical sensors 176, however, not necessarily have to be located in an array 174 and not necessarily have to be located in one and the same plane, as is shown in an alternative setup of the detector 110 shown in Fig. 15. In this figure, only the optical components are shown. For other components, reference may be made to Fig. 11, 12 and 13 above.

As can be seen, in this alternative setup, two or more optical sensors 176 are present, comprising at least one first optical sensor 118 and at least one second optical sensor 120 located in different planes which are offset in a direction of the optical axis 126, also referred to as the z-direction. Thus, further, as can also be seen, the optical sensors 118, 120 may overlap, whereas in the previous embodiments, preferably, no overlap between the optical sensors 176 is given. Apart from these modifications, the functionality and the evaluation of the sensor signals generally corresponds to the embodiment of Fig. 11, 12 and 13 above.

As discussed above, for evaluating the at least two sensor signals of the at least two optical sensors 176 and for deriving an information on the longitudinal position of the object 112 thereof, such as a distance between the detector 110 and the object 112 and/or a z-coordinate of the object 112, preferably, at least one combined sensor signal is generated by the evaluation device 132. The combined sensor signal, as long as this combined sensor signal provides, at least over a measurement range, a unique function of the distance, may be used for deriving the longitudinal coordinate. As an example, the combined sensor signal may be or may comprise at least one quotient signal Q. In Fig. 16 to 18, quotient signals Q of two sensor signals of two optical sensors 176 are shown under various measurement conditions. In each case, the quotient signal Q is denoted on the vertical axis, as a function of the longitudinal coordinate z of the object 112 on the horizontal axis, the latter given in centimeters.

In all experiments, a setup as shown in Fig. 12 was used. As an illumination source 136, in the experiments of Fig. 16 and 17, a 980 nm Picotronic laser source was used, in conjunction with a lens having a focal length of 100 mm. In the experiment of Fig. 18, a Laser Components laser light source having a wavelength of 850 nm was used, in conjunction with a lens having a focal length of 79 mm. In all experiments, the laser beam was aligned on the optical axis 126 via a small prism in front of the lens 128, forming a reflective element 140. A diaphragm 190 in front of the laser source was used to vary the spot size. The quadrant diode 178 was used to measure the reflection of the laser source on different materials. In all experiments, the distance dependency is given by the quotient Q of two adjacent quadrant currents.

In Fig. 16, the laser power was varied during the experiment, from 8 nA laser current, denoted by the dotted line, to 106 nA, denoted by the solid line. Therein, since the laser current typically does not provide a measure for the laser intensity, the laser current indicated therein is a current of a silicon photodetector in a measurement setup in which the laser illuminates a white sheet of paper at a distance of 330 mm from the lens. As is clearly visible, the curves are nearly identical and, at least within this range of variation of the laser power, do not significantly depend on the laser power. This experiment shows that the quotient signal provides a reliable and monotonous function of the longitudinal coordinate, independent from the influence of the brightness of the illumination source.

In Fig. 17, a spot size of the illumination source 136 was varied, by varying the open diameter of the diaphragm 190 in front of the laser. The spot size was varied from 1.5 mm, denoted by the dotted line, to 3.5 mm, denoted by the solid line, in steps of 0.5 mm. As can be seen, up to a

distance of approximately 200 cm, the quotient signal Q does not depend on the spot size and, thus, again, is not negatively affected by this variation.

In Fig. 18, a material of the object 112 illuminated by the laser beam was varied. Therein, the dotted line denotes white paper, the dashed line with the smallest dashes denotes black paper, the dashed line with the medium dashes denotes wood, and the dashed line with the largest dashes denotes an aluminum plate. As can be seen, at least up to a measurement range of approximately 250 cm, the experiment does not strongly depend on the type of material used for the object 112.

The experiments shown in Fig. 16 to 18, thus, clearly demonstrate that the quotient signal Q provides a reliable function of the distance. At least within a range of measurement, the function monotonously rises with the distance. The function is not strongly influenced by the most significant variations which may occur in real life measurements, such as the brightness of the illumination source, the spot size of the illumination source or the material of the object 112. Thus, by evaluating the quotient signal Q of two or more optical sensors 176, reliable distance information may be generated. Thus, as an example, the curves shown in Fig. 16 to 18 directly may be used as calibration curves for the purpose of the evaluation device 132. Other evaluation methods, however, are feasible.

In Fig. 19A and 19B, an alternative embodiment of the detector 110 is shown which is a modification of the setup shown in Fig. 2. Thus, for most elements and optional details as well as further elements not shown in the schematic Figures 19A and 19B, reference may be made to the description of Fig. 12 above.

In Figure 12, the illumination light beam 138, as discussed above, preferably travels along the optical axis 126, i.e. parallel to the optical axis 126 or even on the optical axis 126. In the setup, the position of the center of the light spot 186 typically does not depend on the z-coordinate of the object 112, such as on a distance between the object 112 and the detector 110. In other words, the diameter or equivalent diameter of the light spot 186 changes with the distance between the object 112 and the detector 110 whereas, typically, the position of the light spot 186 on the array 174 does not.

Contrarily, in Fig. 19A and 19B, a setup of the detector 110 is shown in which an illumination light beam 138 travels off-axis, i.e. one or both of at an angle other than 0° with the optical axis 126 or parallel to the optical axis 126 but shifted from the optical axis 126. This embodiment, as will be discussed in further detail below, demonstrates that the method according to the present invention can be further enhanced by increasing the z-dependency of a combined sensor signal. Thus, in Fig. 19A, a side view is shown with two different positions of the object 112, i.e. a first position at z_1 , drawn in solid lines, and a second position at z_2 , drawn in dashed lines. As can be seen, the illumination light beam 138 which, as an example, propagates at an angle of 5° to 30° , e.g. 10° to 20° , with the optical axis 126, hits the object 112 in both cases at different positions. From these points of the object 112 illuminated by the illumination light beam 138,

light beams 116 propagate towards the detector 110, wherein, again, the light beam 116 for the object 112 being located at position z_1 is drawn in solid lines, wherein the light beam 116 for the object 112 being located at position z_2 is drawn in dashed lines.

5 In Fig. 19B, the array 174, e.g. a quadrant photodiode, is shown in an enlarged fashion. As can be seen in this setup, the position of the light spot 186 moves with the longitudinal position z of the object 112. Thus, not only is the size of the light spot 186 affected by the longitudinal position z but also is the position on the array 174 of the light spot 186 changed. In Fig. 19B, this movement of the light spot 186 is denoted by arrow z .

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Consequently, by this movement of the light spot 186, the z -dependency of a combined sensor signal taking into account at least two sensor signals of the optical sensors 176 may be increased. As an example, the four diodes of the array 174, in Fig. 19B, are denoted by D1-D4.

15 The quotient signal Q , as an example, may be formed as $Q = i(D1)/i(D4)$, with $i(D1)$ being the sensor signal of photodiode D1, and $i(D4)$ being the sensor signal of photodiode D4. As shown in Figure 19B, the quadrant diode may comprise two dividing lines. The dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other.

20 In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device 132 may be adapted to determine a second quotient using at least three or all four sensor signals of the quadrant diode. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotients allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient may be determined

25 by $Q = i(D1+D2)/i(D3+D4)$, wherein the sensor signals of the two top quadrants, also called top segment, are divided by the sensor signals of the two bottom quadrants, also called bottom segment. Using the quotient of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing

30 line between top and bottom segment is parallel to the baseline, the quotient signal determined from the top segment divided by the bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom

35 segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments, i.e. $Q = i(D1+D3)/i(D2+D4)$, may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient by dividing sensor

40 signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals $i(D1)$, $i(D2)$, $i(D3)$ and $i(D4)$ of the quadrants such that distance measurement is possible over a wide range with a large resolution.

In the situation shown in Fig. 12, the position of the light spot 186 does not depend on z . With a change in z , depending on the optical situation, the spot will become larger or smaller, such as by becoming more diffuse or more focused. In case the spot size increases and the spot becomes more diffuse, $i(D4)$ will increase more rapidly than $i(D1)$, such that the quotient signal Q decreases.

Contrarily, in the situation of Fig. 19A, both the size and the position of the light spot 186 are dependent on the z -coordinate. Thus, the tendency of the z -dependency of the combined sensor signal such as the quotient signal Q will be increased. In the situation of Figure 12, depending on the z -coordinate, the sensor signal of at least one sensor will increase and simultaneously the sensor signal of at least one different sensor will decrease, resulting in the z -dependent quotient signal Q . In the situation of Fig. 19A, the position dependency of the light spot 186 can result in three different situations depending on the relative position of light source, optical axis, and sensor: Firstly, the position dependency of the light spot 186 may result in a further decrease of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot 186 may result in a further increase of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 12. Secondly, the position dependency of the light spot 186 may result in a reduced decrease or even increase of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot 186 may result in a reduced increase or even decrease of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 12. Thirdly, the position dependency of the light spot 186 may be as such that the z -dependence of the sensor signals is largely unchanged compared to the situation in Figure 12. However, according to the present invention, object distance is not determined from the position of the light spot 186 on a sensor as done in triangulation methods. Instead, movement of the light spot 186 on the array 174 may be used to enhance dynamic of the sensor signals and or the resulting quotient signal Q which may result in an enhanced dynamic of the z -dependency. Furthermore, movement of the light spot 186 on the array 174 during measurement may be used to establish and/or to enhance object size independence for the whole measurement range by suitable relative positioning of the optical sensor 176 and the illumination source 136. Thus, movement of the light spot 186 may not be used for the purpose of triangulation but for the purpose of object size independence.

Additionally, as known from the prior art, the sensor signals $i(D1)$, $i(D2)$, $i(D3)$, $i(D4)$ may also be used for determining a transversal position x , y of the object 112. Further, the sensor signals may also be used for verifying the z -coordinate determined by the present invention.

Figure 19C shows a comparison of two experimental setups using a detector setup according to Figure 19A with a Bi-cell as optical sensors 176 with two light sensitive areas. In a first experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 186 may move in parallel to the linear boundary of the two optical sensors 176 of the Bi-cell along a direction of movement 210 in dependence of the object distance. Since the direction of movement 210 of the light spot 186 is in parallel to the linear

boundary of the two light sensitive areas in dependence of the object distance, the resulting sensor signals are identical to a situation with no movement of the light spot 186 depending on object distance as shown in Figure 12. In a second experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 186 may move as such that the distance of the center of the light spot 186 to the boundary of the two optical sensors 176 of the Bi-cell changes in dependence of the object distance such as a movement orthogonal to the boundary of the two optical sensors 176 such as a movement along a direction of movement 208 in dependence of the object distance. The detector setup allowing movement of the light spot 186 may be a modification of the setup shown in Figure 19A. Thus, for most elements and optional details as well as further elements, reference may be made to the description of Fig. 19A above. In Figure 19C, the optical sensors 176 may be a bi-cell diode.

Figure 19D shows experimental results of the comparison of the two experimental setups using a detector setup according to Figure 19A, allowing movement of the light spot 186 according to Figure 19C with movement of the light spot depending on the object distance along directions of movement 210 and 208. Curve 212 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup allowing movement of the light spot 186 along a direction of movement 210 as shown in Figure 19C which is in parallel to the boundary of the optical sensors of the Bi-Cell and, which is a situation equivalent to Figure 12 without a movement of the light spot depending on the object distance. Curve 214 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup according to Figure 19A and using a detector setup allowing movement of the light spot 186 with movement of the light spot 186 according to Figure 19C with movement of the light spot depending on the object distance along a direction of movement 208. The experimental setup was as follows: The optical sensors 176 may be a bi-cell diode, in particular a Si-Bi-Cell. The illumination source 136 may be a 950 nm laser with a spot size of 4 mm. The transfer device 128 may have a focal length of 20 mm, e.g. a lens available as Thorlabs Asphere, $f = 20$ mm. The distance of the object 112 was varied from 0 to 3000 mm. Determination of the longitudinal coordinate z may be possible without allowing movement of the light spot 186. In particular, according to the present invention, movement of the light spot may not be essential for determination of the longitudinal coordinate z . With the detector setup allowing movement of the light spot 186 along a direction 210 or without any movement determination of object distance is possible at very small distance, whereas with movement along a direction 208 determination of object distance is possible for object distance such as distances greater than 500mm.

Figure 19E shows object independence of the two experimental setups using a detector setup according to Figure 19A, allowing movement of the light spot 186 according to Figure 19C with movement of the light spot depending on the object distance along directions of movement 208 and 210. In addition, for both experimental setups, the object size was varied from 1mm (dashed line), 2mm (dotted line), 6mm (solid line) and 12mm (loosely dotted line) by varying the aperture of the laser illumination source. Set of curves 216 shows dependency of quotient Q on the longitudinal coordinate z for the experimental setup allowing movement of the light spot 186

along a direction 208. Set of curves 218 shows dependency of quotient Q on the longitudinal coordinate z for the experimental setup allowing movement of the light spot 186 along a direction 210 or without any movement. Set of curves 216 show only small deviations, in particular less than 5%, whereas set of curves 218 show larger deviations, in particular with increasing distance z . Thus, movement of the light spot 186 on the array 174 during measurement may be used to establish and/or to enhance object size independence for the whole measurement range by suitable relative positioning of the optical sensor 176 and the illumination source 136.

In Fig. 20, a schematic view of a further embodiment of a detector 1110 for determining a position of at least one object 1112 is depicted. In this case, the object 1112 may comprise a beacon device 1114, from which a light beam 1116 propagates towards a first optical sensor 1118 and a second optical sensor 1120. The first optical sensor 1118 comprises a first light-sensitive area 1122, and the second optical sensor 1120 comprises a second light-sensitive area 1124. Details of the second optical sensor 1120 and the second light-sensitive area 124 will be explained in further detail below, with reference to Fig. 22A, 22B and 23.

It shall be noted therein, that, in the embodiment shown in Fig. 20, the first optical sensor 1118 is positioned in front of the second optical sensor 1120, such that the light beam 1116 reaches the first optical sensor 1118 before the second optical sensor 1120. As discussed above, however, another order is feasible. Thus, as an example, the second optical sensor 1120 may be positioned in front of the first optical sensor 1118. The latter option, which is not depicted herein, is specifically possible in case the second light-sensitive area 1124 is fully or partially transparent, such as by providing a transparent fluorescent waveguiding sheet 1174, as will be outlined in further detail below.

The light beam 1116, as an example, may propagate along an optical axis 1126 of the detector 1110. Other embodiments, however, are feasible.

The detector 1110, further, may comprise at least one transfer device 1128, such as at least one lens or a lens system, specifically for beam shaping. Consequently, the light beam 1116 may be focused, such as in one or more focal points 1130, and a beam width of the light beam 1116 may depend on a longitudinal coordinate z of the object 1112, such as on a distance between the detector 1110 and the beacon device 1114 and/or the object 1112. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Fig. 20, the first optical sensor 1118 is a small optical sensor, whereas the second optical sensor 1120 is a large optical sensor. Thus, the width of the light beam 1116 fully may cover the first light-sensitive area 1122, whereas, on the second light-sensitive area 1124, a light spot is generated which is smaller than the light-sensitive area 1124, such that the light spot is fully located within the second light-sensitive area 1124. Possible embodiments will be explained below with reference to Fig. 23. Thus, as an example, the first light-sensitive area 1122 may have a surface area of 10 mm^2 to 100 mm^2 , whereas the second light-sensitive area

1124 may have a surface area of more than 100 mm², such as 200 mm² or more, e.g. 200 to 600 mm² or 500 mm² or more. Other embodiments, however, are feasible.

5 The first optical sensor 1118, in response to the illumination by the light beam 1116, may generate a first sensor signal s_1 , and the second optical sensor 1120 may generate at least one second sensor signal s_2 . As an example, the first optical sensor 1118 may be a linear optical sensor, i.e. the sensor signal s_1 is dependent on the total power of the light beam 1116 or on the portion of the light beam 1116 illuminating the first light-sensitive area 1122, whereas the sensor signal s_1 is independent from the actual size of the light spot of illumination. In other words, the
10 first optical sensor 1118, preferably, does not exhibit the above-described FiP effect.

The sensor signals s_1 and s_2 may be provided to an evaluation device 1132 of the detector 1110. The evaluation device 1110, as symbolically depicted in Fig. 20, may specifically be embodied to derive a quotient signal Q , as explained above. The quotient signal Q , derived by dividing e.g. the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used
15 for deriving at least one item of information on a longitudinal coordinate z of the object 1112 and/or the beacon device 1114, from which the light beam 1116 propagates towards the detector 1110. Thus, as an example, a unique evaluation curve may exist, in which, for each quotient signal Q , a longitudinal coordinate z is assigned.

20 The detector 1110, in combination with the at least one beacon device 1114, may be referred to as a detector system 1134, as will be explained in further detail below, with reference to Fig. 25.

In Fig. 21, a modification of the embodiment of Fig. 20 is shown, which forms an alternative detector 1110. The alternative embodiment of the detector 1110 widely corresponds to the embodiment shown in Fig. 20. Instead of using an active light source, i.e. a beacon device 1114 with light-emitting properties for generating the light beam 1116, however, the detector 1110 may comprise at least one illumination source 1136. The illumination source 1136, as an example, may comprise a laser, whereas, in Fig. 20, as an example, the beacon device 1114 may comprise a light-emitting diode (LED). The illumination source 1136 may be configured for generating at least one illumination light beam 1138 for illuminating the object 1112. The illumination light beam 1138 may fully or partially be reflected by the object 1112 and may travel back towards the detector 1110, thereby forming the light beam 1116.

35 As shown in Fig. 20, as an example, the illumination light beam 1138 may be parallel to the optical axis 1126 of the detector 1110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too. In order to provide an on-axis illumination, as shown in Fig. 21, as an example, one or more reflective elements 1140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

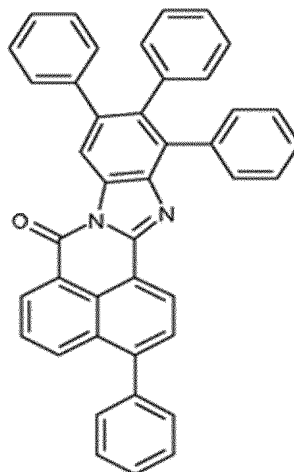
40 Apart from these modifications, the setup of the embodiment in Fig. 21 corresponds to the setup in Fig. 20. Thus, again, an evaluation device 1132 may be used, having, e.g., at least one divider 1142 for forming the quotient signal Q , and, as an example, at least one position evaluation

device 1144, for deriving the at least one longitudinal coordinate z from the quotient signal Q . It shall be noted that the evaluation device 1132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 1142, 1144 may be embodied by appropriate software components.

It shall be further noted that the embodiments shown in Fig. 20 and 21 simply provide embodiments for determining the longitudinal coordinate of the object 1112. As will be outlined in further detail below with reference to Fig. 22A and 22B as well as to Fig. 23, the detector 1110 may also be used for providing additional information on at least one transversal coordinate of the object 1112 and/or of parts thereof.

In Fig. 22A and 22B, a top view (Fig. 22A) and a cross-sectional view of the second optical sensor 1120, which may be used in the setups e.g. of Fig. 20 and/or 21, is shown. The second optical sensor 1120 may comprise a fluorescent waveguiding sheet 1174 which forms the second light-sensitive area 1124 facing towards the object 1112. The fluorescent waveguiding sheet 1174, in this exemplary embodiment, may be designed as a flat waveguiding sheet, in which, as symbolically depicted by the arrow 1176 in Fig. 22B, waveguiding by internal reflection may take place, specifically by internal total reflection, specifically a waveguiding of fluorescence light generated within the fluorescent waveguiding sheet 1174. The fluorescent waveguiding sheet 1174, as an example, may have a lateral extension of at least 25 mm², such as at least 100 mm², more preferably of at least 400 mm². As an example, a 10 mm x 10 mm square sheet, a 20 mm x 20 mm square sheet, a 50 mm x 50 mm square sheet or another dimension may be used. It shall be noted, however, that non-square geometries or even non-rectangular geometries may be used, such as circular or oval geometries or polygonal geometries.

The fluorescent waveguiding sheet 1174, as an example, may comprise a matrix material 1178 and at least one fluorescent material 1180 disposed therein, such as at least one fluorophore, e.g. a fluorescent dye. For exemplary embodiments, reference may be made to the above-mentioned materials, such as one or more of the materials listed in WO 2012/168395 A1. As an example, the following fluorescent material may be used:



This fluorescent material is disclosed as substance 34.2 in WO 2012/168395 A1, including potential synthesis methods. The material may be immersed in polystyrene, such as at a concentration of 0.001 - 0.5 wt.%. The fluorescent material 1180 may be designed to generate fluorescence light in response to an illumination by the light beam 1116.

The fluorescent material 1180 and/or the concentration of the fluorescent material 1180 within the matrix material 1178, specifically may be chosen to show linear properties, at least within a range of measurement, i.e. within a range of intensities, such that the total power of the fluorescence light generated in response to an excitation is a linear function of the intensity of the illumination by the excitation light, i.e. by the light beam 1116. As an example, the materials and/or intensities may be chosen such that saturation effects are avoided.

The second optical sensor 1120 further, in this embodiment, may comprise a plurality of photosensitive elements 1182, 1184, 1186, 1188, in Fig. 22A and 22B referred to as PD1-PD4, located at respective edges 190, 192, 194, 196 of the fluorescent waveguiding sheet 174, e.g. rim portions of the fluorescent waveguiding sheet 1174. In this exemplary embodiment, the fluorescent waveguiding sheet 1174 may have a rectangular shape, such that pairs of edges are opposing each other, such as the pair of edges 1190, 1192 and the pair of edges 1194, 1196. The sides of the rectangular shape of the fluorescent waveguiding sheet 174 may define a Cartesian coordinate system, with an x-dimension defined by an interconnection between edges 1190 and 192, and a y-dimension defined by an interconnection between edges 1196, 1194, as indicated in Fig. 22A. It shall be noted, however, that other coordinate systems are feasible.

The photosensitive elements 1182, 1184, 1186, 1188, as an example, may comprise photodiodes. Specifically, these photosensitive elements 1182, 1184, 1186, 1188 may have, each, a comparable, preferably an identical, electrical capacity as the first optical sensor 1118. It shall be noted, however, that other embodiments are feasible. The photosensitive elements 1182, 1184, 1186, 1188, as an example, may be or may comprise strip-shaped photodiodes covering, preferably, the full length of the respective edges 1190, 1192, 1194, 1196, or, preferably, covering at least 50 % or more preferably at least 70 % of the length of these respective edges 1190, 1192, 1194, 1196. Other embodiments, however, are feasible, such as embodiments in which more than one photosensitive element is located at a respective edge.

The photosensitive elements 1182, 1184, 1186, 1188 each produce at least one sensor signal, in response to the light, specifically the fluorescence light, detected by these photosensitive elements 1182, 1184, 1186, 1188. All of these sensor signals are referred to as second sensor signals, wherein, in the following, PD1 creates sensor signal $s_{2,1}$, PD2 creates sensor signal $s_{2,2}$, PD3 creates sensor signal $s_{2,3}$, and PD4 creates sensor signal $s_{2,4}$, with the first index 2 denoting the fact that these sensor signals are second sensor signals, and with the second index, from 1 to 4, indicating the respective photosensitive element 1182, 1184, 1186, 1188 from which the respective sensor signal originates.

As outlined above in Fig. 20 and 21, the at least one first sensor signal s_1 and the second sensor signals $s_{2,j}$ (with $j = 1, \dots, 4$) are provided to the evaluation device 1132 of the detector 1110, the function of which will be explained in further detail below, specifically with reference to Fig. 24.

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The evaluation device 1132 is configured to determine at least one longitudinal coordinate z of the object 1112, which is not depicted in these figures, and from which the light beam 1116 propagates towards the detector 1110, by evaluating the first and second sensor signals. Additionally as will be outlined in further detail below, at least one transversal coordinate x and/or y may be determined, as will also be outlined in further detail below, with reference to Fig. 3 and 24.

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The second optical sensor 1120, as depicted in Fig. 22B, may further optionally comprise at least one optical filter element 1198. The optical filter element 1198 may be placed in front of an optional reference photosensitive element 1200, which may further, with or without the optical filter element 1198, be present in the detector 1110. As an example, the reference photosensitive element 1200 may comprise a large area photodiode. Other setups, however, are feasible. Thus, it shall be noted, that the reference photosensitive element 1200 may also be left out in this embodiment, since the first optical sensor 1118 may also take over the functionality of the reference photosensitive element 1200. Specifically, in case a transparent fluorescent waveguiding sheet 1174 is used and in case the first optical sensor 1118 is placed behind the second optical sensor 1120, the first optical sensor 1118 may also take over the functionality of the reference photosensitive element 1200. It shall further be noted that one or both of the first optical sensor 1118 and the second optical sensor 1120 may be a uniform optical sensor, having a single light-sensitive area 1122, 1124, each, or that one or both of these optical sensors 1118, 1120 may be pixelated.

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As an example, the at least one optical filter element 1198 may be designed to prevent fluorescence light from entering the reference photosensitive element 1200 or, at least, may attenuate fluorescence light by at least 70 %, or, preferably, by at least 80 %.

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In Fig. 23, an illumination of the second light-sensitive area 1124 by the light beam 1116 is shown. Therein, two different situations are depicted, representing different distances between the object 1112 and from which the light beam 1116 propagates towards the detector 1110, and the detector 1110 itself, resulting in two different spot sizes of light spots generated by the light beam in the fluorescent waveguiding sheet 1174. Firstly, a small light spot 1202 and, secondly, a large light spot 1204. In both cases, the overall power of the light beam remains the same over light spots 1202, 1204. Further, a shadow 1206 is depicted, which is generated by the first optical sensor 1118 being placed in front of the second optical sensor 1120. In the following, it is assumed that the first optical sensor 1118 is still fully illuminated by the light beam 1116.

The illumination by the light beam 1116 induces fluorescence which, as depicted in Fig. 22B above, is fully or partially transported by waveguiding towards the photosensitive elements

1182, 1184, 1186, 1188. As indicated above, corresponding second sensor signals are generated by these photosensitive elements, and are provided to the evaluation device 1132, in conjunction with the first sensor signal and, optionally, further in conjunction with at least one reference sensor signal generated by the at least one reference photosensitive element 1200.

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The evaluation device 1132, as symbolically depicted in Fig. 24, is designed to evaluate the sensor signals which, therein, are represented as outlined above. The sensor signals may be evaluated by the evaluation device in various ways, in order to determine a location information and/or a geometrical information of the object 1112, such as at least one longitudinal coordinate z of the object 1112 and, optionally, one or more transversal coordinates of the object 1112.

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Firstly, the evaluation device 1132 may comprise at least one summing device 1208 configured to form a sum signal S of the sensor signals PD1 to PD4, such as according to formula (1) above, for the second sensor signals $s_{2,i}$, with $i = 1, \dots, 4$ (the first index, for the sake of simplicity, is left out in the above-mentioned formula (1)). This sum signal S may replace the second sensor signal s_2 in general and/or, for a part of the further evaluation, may be used as "the" second sensor signal of the second optical sensor 1120. This sum signal S may represent the total power of the fluorescence light generated by the light beam 1116. Even though, some losses may occur, since, generally, not all of the fluorescence light will actually reach the photosensitive elements 1182, 1184, 1186, 1188. Thus, as an example, losses in waveguiding may occur, or some of the fluorescence light may actually be emitted from the edges 1190, 1192, 1194, 1196, in a direction which is not covered by the photosensitive elements 1182, 1184, 1186, 1188. Still, the sum signal S provides a fairly good measure for the total power of the fluorescence generated within the fluorescent waveguiding sheet 1174.

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The evaluation device 1132 may comprise at least one divider 1142 which, as symbolically depicted in Fig. 24, may be part of a position evaluation device 1144 and which may be configured for forming at least one quotient signal out of the first and second sensor signals s_1 , s_2 , with s_2 , as an example, being the sum signal S of the respective second sensor signals, as outlined above. Thus, as an example, the divider 1142 may be configured for one or more of dividing the first and second sensor signals, dividing multiples of the first and second sensor signals or dividing linear combinations of the first and second sensor signals. The position evaluation device 1144 further may be configured for determining the at least one longitudinal coordinate z by evaluating the quotient signal Q , such as by using at least one predetermined or determinable relationship between the quotient signal Q and the longitudinal coordinate. As an example, calibration curves may be used. The divider 1142 and/or the position evaluation device 1144 may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the quotient signal, at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

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As outlined above, additional information may be derived from the second sensor signals $s_{2,1}$, $s_{2,2}$, $s_{2,3}$ and $s_{2,4}$, besides the at least one longitudinal coordinate z of the object. Thus, additionally, at least one transversal coordinate x , y may be derived. This is mainly due to the fact that the distances between a center of the light spots 1202, 1204 and the photosensitive elements 1182, 1184, 1186, 1188 are non-equal. Thus, the center of the light spot 1202, 1204 has a distance from the photosensitive element 1182 of l_1 , a distance from the photosensitive element 1184 of l_2 , from the photosensitive element 1186 of l_3 and from the photosensitive element 1188 of l_4 . Due to the differences in these distances between the location of the generation of the fluorescence light and the photosensitive elements detecting said fluorescence light, the sensor signals will differ. This is due to various effects. Firstly, again, internal losses will occur during waveguiding, since each internal total reflection implies a certain loss, such that the fluorescence light will be attenuated on its way, depending on the length of the path. The longer the distance of travel, the higher the attenuation and the higher the losses. Further, absorption effects will occur. Thirdly, a spreading of the light will have to be considered. The longer the distance between the light spot 1202, 1204 to the respective photosensitive element 1182, 1184, 1186, 1188, the higher the probability that a photon will be directed into a direction other than the photosensitive element. Consequently, by comparing the sensor signals of the photosensitive elements 1182, 1184, 1186, 1188, at least one item of information on a transversal coordinate of the light spot 1202, 1204 and, thus, of the object 1112 may be generated.

The comparison of the sensor signals may take place in various ways. Thus, generally, the evaluation device 1132 may be designed to compare the sensor signals in order to derive the at least one transversal coordinate of the object 1112 and/or of the light spot 1202, 1204. As an example, the evaluation device 1132 may comprise at least one subtracting device 1210 and/or any other device which provides a function which is dependent on at least one transversal coordinate, such as on the coordinates x , y , of the object 1112. For exemplary embodiments, the subtracting device 1210 and/or any other device may provide a function which is dependent on at least one transversal coordinate, such as on the coordinates x , y . For exemplary embodiments, the subtracting device 1210 may be designed to generate at least one difference signal, such as a signal according to formula (4) and/or (5) above, for one or each of dimensions x , y in Fig. 23. As an example, a simple difference between PD1 and PD2, such as $PD1 - PD2 / (PD1 + PD2)$ may be used, as a measure for the x -coordinate, and a difference between PD3 and PD4, such as $(PD3 - PD4) / (PD3 + PD4)$, may be used as measure for the y -coordinate. A transformation of the transversal coordinates of the light spot 1202, 1204 in the plane of the second light-sensitive area 1124, as an example, into transversal coordinates of the object from which the light beam 1116 propagates to the detector 1110, may simply be made by using the well-known lens equation. For further details, as an example, reference may be made to WO 2014/097181 A1.

It shall be noted, however, that other transformations or other algorithms for processing the sensor signals by evaluating device 1140 are feasible. Thus, besides subtractions or the linear combinations with positive or negative coefficients, non-linear transformations are generally feasible. As an example, for transforming the sensor signals into z -coordinates and/or x , y -

coordinates, one or more known or determinable relationships may be used, which, as an example, may be derived empirically, such as by calibrating experiments with the object placed at various distances from the detector 1110 and/or by calibrating experiments with the object placed at various transversal positions or three-dimensional positions, and by recording the respective sensor signals.

Fig. 25 shows, in a highly schematic illustration, an exemplary embodiment of a detector 1110, e.g. according to the embodiments shown in Fig. 20 or 21. The detector 1110 specifically may be embodied as a camera 1146 and/or may be part of a camera 1146. The camera 1146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 25 further shows an embodiment of a detector system 1134, which, besides the at least one detector 1110, comprises one or more beacon devices 1114, which, in this example, may be attached and/or integrated into an object 1112, the position of which shall be detected by using the detector 1110. Fig. 25 further shows an exemplary embodiment of a human-machine interface 1148, which comprises the at least one detector system 1134 and, further, an entertainment device 1150, which comprises the human-machine interface 1148. The figure further shows an embodiment of a tracking system 1152 for tracking a position of the object 1112, which comprises the detector system 1134. The components of the devices and systems shall be explained in further detail below.

Fig. 25 further shows an exemplary embodiment of a scanning system 1154 for scanning a scenery comprising the object 1112, such as for scanning the object 1112 and/or for determining at least one position of the at least one object 1112. The scanning system 1154 comprises the at least one detector 1110, and, further, optionally, the at least one illumination source 1136 as well as, optionally, at least one further illumination source 1136. The illumination source 1136, generally, is configured to emit at least one illumination light beam 1138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 1114 and/or on a surface of the object 1112. The scanning system 1154 may be designed to generate a profile of the scenery including the object 1112 and/or a profile of the object 1112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 1154, specifically the detector 1110, by using the at least one detector 1110. In Fig. 13, as an example, one or more reflective elements 1140 may be used, for example partially transparent, such as one or more prisms and

As outlined above, an exemplary embodiment of the detector 1110 which may be used in the setup of Fig. 25 is shown in Fig. 20 and 21. Thus, the detector 1110, besides the optical sensors 1118, 1120, comprises at least one evaluation device 1132, having e.g. the at least one divider 1142 and/or the at least one position evaluation device 1144, as symbolically depicted in Fig. 25. The components of the evaluation device 1132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 1110. Besides the possibility of fully or partially combining two or more components, one or

more of the optical sensors 1118, 1120 and one or more of the components of the evaluation device 1132 may be interconnected by one or more connectors 1156 and/or by one or more interfaces, as symbolically depicted in Fig. 25. Further, the one or more connectors 1156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 1156, the evaluation device 1132 may fully or partially be integrated into one or both of the optical sensors 1118, 1120 and/or into a housing 1158 of the detector 1110. Additionally or alternatively, the evaluation device 1132 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 1112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 1160, the position of which may be manipulated by a user 1162. As an example, the object 1112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 1112 are possible. Further, the user 1162 himself or herself may be considered as the object 1112, the position of which shall be detected.

As outlined above, the detector 1110 comprises at least the optical sensors 1118, 1120. The optical sensors 1118, 1120 may be located inside the housing 1158 of the detector 1110. Further, the at least one transfer device 1128 may be comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 1164 inside the housing 1158, which, preferably, is located concentrically with regard to the optical axis 1126 of the detector 1110, preferably defines a direction of view 1166 of the detector 1110. A coordinate system 1168 may be defined, in which a direction parallel or anti-parallel to the optical axis 1126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 1126 may be defined as transversal directions. In the coordinate system 1128, symbolically depicted in Fig. 25, a longitudinal direction is denoted by z , and transversal directions are denoted by x and y , respectively. Other types of coordinate systems 1168 are feasible, such as non-Cartesian coordinate systems.

The detector 1110 may comprise the optical sensors 1118, 1120 as well as, optionally, further optical sensors. The optical sensors 1118, 1120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 1118 covers a portion of the second optical sensor 1120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 1112 and/or of parts thereof.

One or more light beams 1116 are propagating from the object 1112 and/or from one or more of the beacon devices 1114, towards the detector 1110. The detector 1110 is configured for determining a position of the at least one object 1112. For this purpose, as explained above in the context of Fig. 20 to 23, the evaluation device 1132 is configured to evaluate sensor signals

provided by the optical sensors 1118, 1120. The detector 1110 is adapted to determine a position of the object 1112, and the optical sensors 1118, 1120 are adapted to detect the light beam 1116 propagating from the object 1112 towards the detector 1110, specifically from one or more of the beacon devices 1114. In case no illumination source 1136 is used, the beacon devices 1114 and/or at least one of these beacon devices 1114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 1136 is used, the beacon devices 1114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 1112 may be used, such as integrated reflected beacon devices 1114 having at least one reflective surface. The light beam 1116, directly and/or after being modified by the transfer device 1128, such as being focused by one or more lenses, illuminates the light-sensitive areas 1122, 1124 of the optical sensors 1118, 1120. For details of the evaluation, reference may be made to Fig. 20 to 23 above.

As outlined above, the determination of the position of the object 1112 and/or a part thereof by using the detector 1110 may be used for providing a human-machine interface 1148, in order to provide at least one item of information to a machine 1170. In the embodiments schematically depicted in Fig. 25, the machine 1170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 1132 may even be fully or partially integrated into the machine 1170, such as into the computer.

As outlined above, Fig. 25 also depicts an example of a tracking system 1152, configured for tracking the position of the at least one object 1112 and/or of parts thereof. The tracking system 1152 comprises the detector 1110 and at least one track controller 1172. The track controller 1172 may be adapted to track a series of positions of the object 1112 at specific points in time. The track controller 1172 may be an independent device and/or may be fully or partially integrated into the machine 1170, specifically the computer, as indicated in Fig. 25 and/or into the evaluation device 1132.

Similarly, as outlined above, the human-machine interface 1148 may form part of an entertainment device 1150. The machine 1170, specifically the computer, may also form part of the entertainment device 1150. Thus, by means of the user 1162 functioning as the object 1112 and/or by means of the user 1162 handling a control device 1160 functioning as the object 1112, the user 1162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Fig. 26A and 26B, an alternative embodiment of the second optical sensor 1120 is shown, in a top view (Fig. 26A) and in a cross-sectional view (Fig. 26B). For most of the details of the second optical sensor 1120, reference may be made to Fig. 22A and 22B above. The embodiment, however, shows various variations from the embodiment of Fig. 22A and 22B, which may be realized in an isolated fashion or in combination.

Thus, firstly, the embodiment shows variations of the placement of the photosensitive elements. Besides the photosensitive elements 1182, 1184, 1186, 1188 located at opposing edges 1190, 1192, 1194, 1196, which, in this embodiment, are straight edges, additional photosensitive elements 1212 are located at corners 1214 of the fluorescent waveguiding sheet 1174. The edges 1190, 1192, 1194, 1196 in combination may form a rim of the fluorescent waveguiding sheet 1174, such as a rectangular rim. The rim itself may be roughened or even blackened in order to avoid back reflections from the rim. The corners 1214 also are part of the edges of the fluorescent waveguiding sheet 1174. The photosensitive elements 1212 located at the corners 1214 may provide additional second sensor signals which may be evaluated in a similar fashion as shown e.g. in Fig. 24. They may provide an increased accuracy of the determination of the z-coordinate and/or of the x, y-coordinate. Thus, as an example, these additional sensor signals may be included in the sum signal, such as formed by using formula (1) above. Additionally or alternatively, these additional sensor signals may be implemented into the formation of difference signals, such as according to formulae (2) and/or (3) above. As an example, difference signals between two photosensitive elements 1212 located at opposing corners 1214 may be formed and/or difference signals between one photosensitive element 1212 located at a corner 1214 and one photosensitive element located at a straight edge, e.g. a straight rim portion, may be formed. The difference signal D, in each case, may denote a location of the light spot on an axis interconnecting the two photosensitive elements.

Further, the embodiment of Fig. 26A and 26B shows a variation of the placement of the photosensitive elements 1182, 1184, 1186, 1188, 1212 with respect to the fluorescent waveguiding sheet 1174. Thus, in the embodiment of Fig. 22A and 22B, the photosensitive elements 1182, 1184, 1186, 1188 may be located within the plane of the fluorescent waveguiding sheet 1174. Additionally or alternatively, as shown in the embodiment of Fig. 26A and 26B, some or even all of the photosensitive elements 1182, 1184, 1186, 1188, 1212 may be located outside the plane of the fluorescent waveguiding sheet 1174. Specifically, as shown in the cross-sectional view of figure 26B, as an example, the photosensitive elements 1182, 1184, 1186, 1188, 1212 may be optically coupled to the fluorescent waveguiding sheet 1174 by optical coupling elements 1216. As an example, the photosensitive elements 1182, 1184, 1186, 1188, 1212 simply may be glued to the fluorescent waveguiding sheet 1174 by using one or more transparent adhesives, such as an epoxy adhesive.

Further, the embodiment of Fig. 26A and 26B shows a variation of the size and shape of the photosensitive elements 1182, 1184, 1186, 1188, 1212. Thus, the photosensitive elements 1182, 1184, 1186, 1188, 1212 do not necessarily have to be strip-shaped photosensitive elements. As an example, very small photodiodes may be used, such as rectangular photodiodes or even point-like or spot-like photodiodes. As outlined above, a small size of the photodiodes generally may lead to a lower electrical capacitance and, thus, may lead to a faster response of the second optical sensor 1120.

Further, the embodiment of Fig. 26A and 26B shows that no reference photosensitive element 1200 is necessary. Thus, as discussed above, the sum signal itself may replace the function of

the reference photosensitive element 1200. Thus, the second optical sensor 1120 as shown in the embodiment of Fig. 26A and 26B provides a fully functional and, optionally, transparent PSD. No further PSDs are required.

5 Figures 27A and 27 B show a schematic view of a further exemplary embodiment of a detector 110 according to the present invention. In Figure 27 A, the detector 110 comprises at least two optical sensors 113, for example a first optical sensor 118 and a second optical sensor 120, each having at least one light-sensitive area 121. The optical detector 110, further, comprises at least one transfer device 128, such as at least one lens or a lens system, specifically for beam
10 shaping. The transfer device 128 has an optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. The detector 110 may comprise at least one illumination source 136. The illumination source 136, as an example, may comprise a laser source. The illumination source 136 may be arranged such that the illumination light beam 138 is one or both of non-parallel to the optical axis 126, but off-axis, or shifted
15 from the optical axis 126. The illumination source 136 may be configured for generating at least one illumination light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The light beam 116 propagates from the object 112 towards the first optical sensor 118 and the second optical sensor 120. The first optical sensor 118 may com-
20 prise a first light-sensitive area 122, and the second optical sensor 120 may comprise a second light-sensitive area 124. In this embodiment the optical sensors 118, 120 may be arranged such that the light-sensitive areas 122, 124 have identical surface areas. For example, the optical sensors 118, 120 may be identical. The detector 110 may further comprise the reflective element 140, such as at least one beam splitter, which is adapted to lead the light beam 116 from
25 the transfer device 128 to both of the optical sensors 118, 120. The first optical sensor 118 may have a distance db_1 from the beam splitter and the second optical sensor 120 may have a distance db_2 from the beam splitter, wherein $db_1 \neq db_2$. Again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q, and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordi-
30 nate z from the quotient signal Q. It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

In Figure 27 B, the detector 110 comprises at least two optical sensors 113, for example a first
35 optical sensor 118 and a second optical sensor 120, each having at least one light-sensitive area 121. The optical detector 110, further, may comprise at least one transfer device 128, such as at least one lens or a lens system. The transfer device 128 has an optical axis 129, wherein the transfer device 128 and the optical detector preferably may have a common optical axis. The detector 110 may comprise at least one illumination source 136. The illumination source
40 136, as an example, may comprise a laser source, for example with a 1550nm laser source. The illumination source 136 may be arranged such that the illumination light beam 138 is one or both of non-parallel to the optical axis 126, but off-axis, or shifted from the optical axis 126. The illumination source 136 may be configured for generating at least one illumination light beam

138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The light beam 116 propagates from the object 112 towards the first optical sensor 118 and the second optical sensor 120. The first optical sensor 118 may comprise the first light-sensitive area 122, and the second optical sensor 120 may comprise the second light-sensitive area 124. As can be seen in Fig. 27B, the first optical sensor 118 is a small optical sensor, whereas the second optical sensor 120 is a large optical sensor. The optical sensors 118, 120 may be Ge-sensors. The first optical sensor 118 may have a first distance from the transfer device 128 and the second optical sensor 120 may have a second distance from the transfer device 128. In Figure 27B, the first optical sensor 118 may be close to the transfer device 128, whereas the second optical sensor 120 may be arranged further away in direction to the focus. The first optical sensor 118 may be arranged such that, independent from a distance from the object, a sensor signal of the first optical sensor 118 may be proportional to the total power of the light beam passing the transfer device 128. Again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q , and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the quotient signal Q . It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

In Figure 28 experimental results of a distance determination with the detector 110 is shown. In this experimental setup, the transfer device 128 was a plano-convex lens having a focal length of 150 mm, a diameter of 75 mm and coated with an anti-reflective coating for a range of 1050-1700 nm, available as Thorlabs LA1002-C. The object 112, in this case a piece of carpet, was illuminated by a laser diode with 30mW CW-power output at a wavelength of 1550 nm, available as Schäfter+Kirchhoff 55cm-1550-30-Q04-T12-C-6. The illumination source 136 was placed laterally next to the transfer device and was operated at 367Hz with a 50:50 rectangle modulation. A second optical sensor 120, in this experimental setup, a Ge photodiode with dimensions of 10mm x 10mm, available as Thorlabs FDG1010, was arranged directly on the transfer device, and a first optical sensor 118 having a diameter of 5mm, available as Thorlabs FDG05 was placed with a distance of 0,85m from the transfer device 128. Figure 28 shows a dependency of distance d in m, corresponding to the longitudinal coordinate z of the object, of the determined quotient signal Q .

In Figure 29, a further exemplary embodiment of the detector 110 is depicted. For details of the optical sensor 113 reference is made to Figure 6 above. As in Figures 27A and B, the illumination source 136 may be positioned off-axis. The illumination source 136 may be adapted to generate and/or to project a cloud of points, for example the illumination source 136 may comprise one optical element 194, in particular one or more optical elements selected from the group consisting of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The sensor element 115 may comprise a matrix 117 of optical sensors 113, each optical sensor 113 having at least one light-

sensitive area 121 facing the object 112. The sensor element 115 may comprise at least one CMOS sensor. In Figure 30, schematically the cloud of points impinging on the sensor element 115 is depicted. Additionally, disturbances may be present on the matrix 117 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device 132 may be adapted to determine at least one region of interest 196, for example one or more pixels illuminated by the light beam 116 which are used for determination of the longitudinal coordinate of the object 112. In Figure 30, regions of interest 196 are shown exemplary as circular areas with dashed lines. For example, the evaluation device 132 may be adapted to perform a filtering method, for example, a blob-analysis and/or object recognition method.

Figure 31 A to O show further exemplary configurations of optical sensors according to the present invention, in particular top view in direction of propagation of the light beam 116. In Figure 31A, a top view of two rectangular optical sensors 113 is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120. The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset, in particular in a transversal direction y , from the optical axis 126. In Figures 31B and 31C, top view of a large rectangular optical sensor 120 is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120 having a triangle shaped (Figure 31B) or star-shaped (Figure 31C) light-sensitive area 121. In Figures 31 M to O, a top view of two rectangular optical sensors 113 is shown, wherein the first optical sensor 118 and the second optical sensor 120 are rectangular sensors with the same size. In Figures 31 M to O a mask 119 is arranged in front of the first and second optical sensors 118, 120. The mask 119 may be arranged with a different offset from the optical axis 126. The mask 119 may have an arbitrary size and shape, for example, the mask may be rectangular shaped (Figure 31M), triangle shaped (Figure 31N) or star-shaped (Figure 31O). However, other sizes and shapes are feasible. Mask 119 may be adapted to prevent light impinging on the light sensitive areas of the first and second optical sensors 118, 120. If used in a situation comparable to the situation illustrated in Figure 19A, the mask may result in a further z -dependent decrease of a decreasing sensor signal, resulting in an increased z -dependency of the resulting quotient signal Q .

The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset from the optical axis 126. Figure 31K shows two circular shaped optical sensors 113, wherein the first optical sensor 118 is a small optical sensor in front of the larger second optical sensor 120. In Figures 31D, the light sensitive area of the first optical sensor 118 is square-shaped, and the light sensitive area of the second optical sensor 120 is rectangular, such that the surface areas in x and y differ. In addition, a center of the first optical sensor 118 and a center of second optical sensor 120 may have different x coordinates such that the optical sensors 118, 120 may have different spatial offset in one or more of x and y direction from the optical axis. In Figure 31H, both the first optical sensor 118 and the second optical sensor 120 may be rectangular. The first optical sensor 118 and the second optical sensor 120 may be arranged such that the center of the first optical sensor 118 and the center of second optical sensor 120 may have different x coordinates and that the surface areas in x and y differ. The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset from the

optical axis 126. In Figure 31L, the first optical sensor 118 may have a deviating shape from the shape of the second optical sensor 120 such as a circular or semicircular shape. Figures 31 E, F, G, I, J show sensor element 115 having the matrix of pixels 117. In Figures 31 E, F, G the sensor element 115 has a rectangular shape, whereas in Figures 31 I and J the sensor element
 5 115 has a circular shape. Rows and columns may be arranged equidistant or non-equidistant. In case of equidistant rows and/or columns the sensor element 115 may be arranged with a spatial offset to the optical axis 126.

Figure 32 shows experimental results of a determination of a longitudinal coordinate z for different object sizes. The experimental setup was comparable to the setup shown in Figure 19A. In
 10 the measurement setup the object 112, a paper target, was illuminated by laser 136 with a wavelength of 905 nm, 1,6 mW and modulated with 23 Hz. Light reflected from the object 112 was led to a quadrant diode 178, available as OSI Optoelectronics, OSI Spot-4D. Between the object 112 and quadrant diode 178 a lens 128 having an aspherical effective focal length of
 15 20,0 mm, a diameter of 25,0 mm was placed, available as Thorlabs AL2520M-B. A distance from quadrant diode 178 to lens 128 was 19,7 mm and the quadrant diode 178 had an offset from the optical axis in $y = 0,5$ mm. Further, different from the situation in Figure 19A and not shown in Fig. 19A, in the situation of Figure 32, an iris diaphragm or a further lens was placed in front of the laser 136 between the laser 136 and the object 112, to modify the illumination light
 20 beam 138. The iris diaphragm was used to modify the width of the illumination light beam 138. The further lens was used to obtain a diverging illumination light beam 138 with a beam width decreasing with the distance from the laser 136. Figure 32 shows the quotient Q of two adjacent quadrant currents as a function of the distance, i.e. longitudinal coordinate of the object 112, z in mm. In a first experiment, a diameter of a illumination light beam 138 was varied by an iris
 25 diaphragm from 1 mm, solid line, to 3,5 mm, loosely dashed line, and to 5 mm, dash-dot line. In a second experiment, the diameter of the illumination light beam 138 was varied by the further lens such that the beam width of the illumination light beam 138 diverges with increasing distance from the further lens. To characterize the diverging illumination light beam 138, the beam width at 1m, 2m, and 3m from the lens 128 is given. The dashed line shows the quotient Q , wherein the beam width was 10 mm at 1 m distance, 16 mm at 2 m distance and 22 mm at 3 m
 30 distance from the lens 128. The dotted line shows the quotient Q , wherein the beam width was 15 mm at 1 m distance, 32 mm at 2 m distance and 49 mm at 3 m distance from the lens 128. Below $z = 2300$ mm all curves show the same dependency of Q from z and deviations below ± 5 % and thus independence from the beam width. In the situation of Figure 32 the beam width at the object 112 corresponds to the object size that is measured. The independence of the
 35 quotient Q from the beam width and thus from the object size clearly demonstrates the property of object size independence. In an application, the influence of the further lens leading to a diverging illumination light beam may be caused by a liquid drop, or rain, or dirt or the like such as on the laser module. Thus, object size independence is an important property for robust meas-
 40 urements.

Figures 33A and B show an exemplary beam profile and determination of first area 198 and second area 200 of the beam profile. In Figure 33A normalized intensity I_{norm} as a function of the

transversal coordinate x in mm is depicted. The object size was 20mm and the distance object to sensor was 1200mm. The first area 198 of the beam profile may comprise essentially edge information of the beam profile and the second area 200 of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, a maximum value of the beam profile and/or a center point of a plateau of the beam profile. In Figure 33A the center of the plateau may be at 500mm. The beam profile may further comprise falling edges extending from the plateau. The second area 200 may comprise inner regions of the cross section and the first area 198 may comprise outer regions of the cross section. At least one area of the beam profile may be determined and/or selected as first area 198 of the beam profile if it comprises at least parts of the falling edges of the cross section. In Figure 33A, the first area 198 at both sides from the center is depicted in dark grey. At least one area of the beam profile may be determined and/or selected as second area 200 of the beam profile if it is close or around the center and comprises essentially center information. In Figure 33A, the second area 200 is depicted in light grey. Figure 33B shows the corresponding light spot of the intensity distribution as shown in Figure 33A and the corresponding first area 198, and second area 200.

Figure 34 shows a further exemplary embodiment of the detector 110. The optical sensors 113 may comprise the first optical sensor 118 having the first light sensitive area 122 and the second optical sensor 120 having the second light sensitive area 124. The first light sensitive area 122 and the second light sensitive area 124 are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$

is satisfied. "a" is a ratio of photons hitting both an inner region 202 of a plane 204 perpendicular to the optical axis 126 intersecting the optical axis 126 at a distance equal to half of a focal length f of the transfer device 128 and the first light sensitive area 122. "b" is a ratio of photons hitting both the inner region 202 of the plane 204 and the second light sensitive area 124. "c" is a ratio of photons hitting both an outer region 206 of the plane 204 and the first light-sensitive area 122. "d" is a ratio of the photons hitting both the outer region 206 of the plane 204 and the second light sensitive area 124. The inner region 202 may have an area with a geometrical center point on the optical axis 126 and an extension such that half of the photons hit the plane 204 within the inner region 202 and the other half hit the plane outside the inner region 202. The inner region 202 may be designed as a circle with a center point on the optical axis 126 and a radius r which is chosen such that half of the photons hit the plane 204 within the circle and the other half hit the plane outside the circle.

In Fig. 25, a schematic view of an exemplary embodiment of a detector 2110 for determining a position of at least one object 2112 is depicted. In Figure 35, the object 2112 is depicted for two different object distances. The detector 2110 comprises at least two optical sensors 2113, for example a first optical sensor 2118 and a second optical sensor 2120, each having at least one light-sensitive area 2121. The object 2112 may comprise at least one beacon device 2114, from which a light beam 2116, also denoted as incident light beam, propagates towards the detector 2110. Additionally or alternatively, the detector may comprise at least one illumination source 2115 for illuminating the object 2112. As an example, the illumination source 2115 may be con-

figured for generating an illuminating light beam for illuminating the object 2112. Specifically, the illumination source 2115 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source 2115 may
5 comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source 2115 may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the
10 range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. Further, the illumination source 2115 may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources 2115 is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below,
15 later on may be used for distinguishing the light beams.

The first optical sensor 2118 may comprise a first light-sensitive area 2122, and the second optical sensor 2120 may comprise a second light-sensitive area 2124. The light beam 2116, as an example, may propagate along an optical axis 2126 of the detector 2110. Other embodiments,
20 however, are feasible. The first light-sensitive area 2122 and the second light-sensitive area may be oriented towards the object 2112. The optical detector 2110, further, may comprise at least one transfer device 2128, such as at least one lens or a lens system, specifically for beam shaping. The transfer device 2128 may have at least one focal length in response to the incident light beam 2116 propagating from the object 2112 to the detector 2110. The transfer device 2128 may have an optical axis 2129, wherein the transfer device 2128 and the optical detector preferably may have a common optical axis. The transfer device 2128 may constitute a coordinate system. A direction parallel or anti-parallel to the optical axis 2126, 2129 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 2126, 2129 may be defined as transversal directions, wherein a longitudinal coordinate l is a coordinate along the optical axis 2126, 2129 and wherein d is a spatial offset from the optical axis 2126, 2129. Consequently, the light beam 2116 is focused, such as in one or more focal points, and a beam width of the light beam 2116 may depend on a longitudinal coordinate z of the object 2112, such as on a distance between the detector 2110 and the beacon device 2114 and/or the object 2112. The optical sensors 2118, 2120 may be positioned off focus. For details of this
30 beam width dependency on the longitudinal coordinate, reference may be made to one or more of the WO 2012/110924 A1 and/or WO 2014/097181 A1.

The detector comprises at least one angle dependent optical element 2130 adapted to generate at least one light beam 2131 having at least one beam profile depending on an angle of incidence of an incident light beam propagating from the object 2112 towards the detector 2110 and illuminating the angle dependent optical element 2130. The angle dependent optical element 2130 may have angle dependent transmission properties such that an electromagnetic wave impinging on a first side 2132, for example a surface and/or an entrance, of the angle de-
40

pendent optical element 2130 may be partly, depending on the properties of the angle dependent optical element, absorbed and/or reflected and/or transmitted. A degree of transmission may be defined as quotient of transmitted power of the electromagnetic wave, i.e. the power behind the angle dependent optical element 2130, and the incident power of the electromagnetic wave, i.e. the power before impinging on the angle dependent optical element 2130. The angle dependent optical element 2130 may be designed such that the degree of transmission depends on an angle of incidence at which the incident light beam propagating from the object towards the detector 2116 impinges on the angle dependent optical element 2130. The angle of incident may be measured with respect to an optical axis of the angle dependent optical element 2130.

The angle dependent optical element 2130 may be arranged in the direction of propagation behind the transfer device 2128. The transfer device may, for example, comprise at least one collimating lens. The angle dependent optical element 2130 may be designed to weaken rays impinging with larger angles compared to rays impinging with a smaller angle. For example, the degree of transmission may be highest for light rays parallel to the optical axis, i.e. at 0° , and may decrease for higher angles. In particular, at at least one cut-off angle the degree of transmission may steeply fall to zero. Thus, light rays having a large angle of incidence may be cut-off.

The angle dependent optical element 2130 may comprise at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element; at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one prism; at least one lens; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter; at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.

The first optical sensor 2118, in response to the illumination by the light beam 2131, may generate a first sensor signal s_1 , whereas the second optical sensor 2120 may generate a second sensor signal s_2 . Preferably, the optical sensors 2118, 2120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 131 or of the portion of the light beam 2131 illuminating their respective light-sensitive areas 2122, 2124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination.

The sensor signals s_1 and s_2 are provided to an evaluation device 2133 of the detector 2110. The evaluation device 2133 is embodied to derive a quotient signal Q , as explained above.

From the quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 2112 and/or the beacon device 2114, from which the light beam 2116 propagates towards the detector 2110. The evaluation device 2133 may have at least one di-

vider 2134 for forming the combined signal Q, and, as an example, at least one position evaluation device 2136, for deriving the at least one longitudinal coordinate z from the combined signal Q. It shall be noted that the evaluation device 2133 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 2134, 2136 may be embodied by appropriate software components.

In Fig. 36, a modification of the embodiment of Fig. 35 is shown, which forms an alternative detector 2110. The alternative embodiment of the detector 2110 widely corresponds to the embodiment shown in Fig. 35. In Figure 36, the angle dependent optical element 2130 may comprise at least one optical fiber 2138. The optical fiber 2138 may be adapted to transmit at least parts of incident light beam 2116 which are not absorbed and/or reflected, between two ends of the optical fiber. The optical fiber 2138 may have a length and may be adapted to permit transmission over a distance. The optical fiber 2138 may comprise at least one fiber core which is surrounded by at least one fiber cladding having a lower index of refraction as the fiber core. Below the angle of acceptance, the optical fiber 2138 may be adapted to guide the incoming light beam by total internal reflection.

The optical fiber 2138 may be designed such that the degree of transmission may be highest for incoming light rays parallel, i.e. at an angle of 0° , to the optical fiber, neglecting reflection effects. The optical fiber 2130 may be designed such that for higher angles, for example angles from 1° to 10° , the degree of transmission may decrease smoothly to around 80% of the degree of transmission for parallel light rays and may remain at this level constantly up to an acceptance angle of the optical fiber 2138. The optical fiber 2138 may be designed such that above the acceptance angle total reflection within the optical fiber 2138 is not possible such that the light rays are reflected out of the optical fiber 2138. The optical fiber 2138 may be designed that at the acceptance angle, the degree of transmission may steeply fall to zero. Light rays having a large angle of incidence may be cut-off.

As shown in Figure 36, the illumination source 2115 may be adapted to illuminate the object 2112 through the angle dependent optical element 2130. The optical fiber 2138 may comprise at least one illumination fiber 2140 adapted to transmit the light beam 2142 generated by the illumination source 2115 such that it illuminates the object 2112. The illumination source 2115 may be adapted to couple the at least one light beam 2142 generated by the illumination source 2115 into the illumination fiber 2140.

The optical fiber 2138 may comprise at least two or more fibers. The optical fiber 2138 may be at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber. In the embodiment of Figure 36, and as shown in the cut through in Figure 37, the optical fiber 2138 may comprise four fibers. In particular the optical fiber may comprise the illumination fiber 2138 and at least two fibers each for generating at least one light beam 2131, denoted as first fiber 2144 and second fiber 2146. As shown in Figure 37, the first fiber 2144 and the second fiber 2146 may be arranged close to each other at an entrance end 2148 of the optical fiber 2138 and may split into legs separated by a distance at an exit end 2150 of the optical fiber 2138.

The first fiber 2144 and second fiber 2146 may be designed as fibers having identical properties or may be fibers of different type. The first fiber 2144 may be adapted to generate at least one first light beam 2152 and the second fiber 2146 may be adapted to generate at least one second light beam 2154. The optical fiber 138 may be arranged such that the incident light beam 2116 may impinge at a first angle of incidence into the first fiber 2144 and at a second angle of incidence, different from the first angle, into the second fiber 2146, such that the degree of transmission is different for the first light beam 2152 and the second light beam 2154. One of the optical sensors 2113 may be arranged at an exit end of the first fiber 2144 and the other optical sensor 2113 may be arranged at an exit end of the second fiber 2146. The optical fiber may comprise more than three fibers, for example four fibers as depicted in Figure 37.

It shall further be noted that the embodiments shown in Fig. 35 and 36 simply provide embodiments for determining the longitudinal coordinate z of the object 2112. It is also feasible, however, to modify the setups of Fig. 35 and 36 to provide additional information on a transversal coordinate of the object 2112 and/or of parts thereof. As an example, e.g. in between the transfer device 2128 and the optical sensors 2118, 2120, one or more parts of the light beam 2116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 2112 and/or of parts thereof. The transversal coordinate may be used to verify and/or enhance the quality of the distance information. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

Figures 38 visualizes angle dependent transmission of an angle dependent optical element 2130. The angle dependent optical element 2130 may be designed such that the degree of transmission depends on an angle of incidence at which the incident light beam propagating from the object towards the detector 2116 impinges on the angle dependent optical element 2130. The angle dependent optical element 2130 may be designed to weaken rays impinging with larger angles compared to rays impinging with a smaller angle. In particular, at the cutoff angle the degree of transmission may steeply fall to zero and the light rays having a large angle of incidence may be cut-off. As shown in Figure 38 regions of the incident light beam 2116 are cut-off by the angle dependent optical element 2130 in the generated light beam 2131.

Figure 39 shows a dependency of the transmission power P in W of the optical fiber at constant irradiated power as a function of angle of incidence A in degree. The acceptance angle is shown as vertical line. The degree of transmission may be highest for incoming light rays parallel, i.e. at an angle of 0° , to the optical fiber, neglecting reflection effects. For higher angles, for example angles from 1° to 10° , the degree of transmission may decrease smoothly to around 80% of the degree of transmission for parallel light rays and may remain at this level constantly up to an acceptance angle of the optical fiber 2138. At the acceptance angle, the degree of transmission may steeply fall to zero. Light rays having a large angle of incidence may be cut-off.

Figures 40A and 40B show experimental results of distance measurements. In The determined distance z_{meas} in mm is shown as a function of the object distance z_{obj} in mm. As illumination source 115 a Laser was used having a wavelength of 980 nm and average power of 2,4 mW available under Flexpoint® Laser components module. Two Si-photodetectors were used as optical sensors 113. As optical fiber 2138 and transfer device 2128 available under Thorlabs Fixed Focus Collimation package F220SMA-980 was used. In Figure 40A, the solid line indicates where $z_{\text{meas}} = z_{\text{obj}}$. For the measurement the object distance was varied and two different types of object were used, in particular a black paper object, curve 2156 (dotted line), and a white paper object, curve 2158 (dashed line). The determined object distance is in agreement with the real distance within 2% for small and medium distances and within 10% for large distances. In Figure 40B, the combined signal Q determined by dividing the signals of the two photodetectors as a function of the distance z_{obj} in mm is shown for the black paper object (dotted line) and the white paper object (dashed line). The determined quotient for both object types is in agreement within 2% for small and medium distances and within 10% for large distances.

Figure 41 shows, in a highly schematic illustration, an exemplary embodiment of a detector 2110, for example according to the embodiments shown in Figure 35 or 36. The detector 2110 specifically may be embodied as a camera 2156 and/or may be part of a camera 2156. The camera 156 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 41 further shows an embodiment of a detector system 2158, which, besides the at least one detector 2110, comprises one or more beacon devices 2114, which, in this example, may be attached and/or integrated into an object 2112, the position of which shall be detected by using the detector 2110. Fig. 41 further shows an exemplary embodiment of a human-machine interface 2160, which comprises the at least one detector system 2158 and, further, an entertainment device 2162, which comprises the human-machine interface 2160. The figure further shows an embodiment of a tracking system 2164 for tracking a position of the object 2112, which comprises the detector system 2158. The components of the devices and systems shall be explained in further detail below.

Fig. 41 further shows an exemplary embodiment of a scanning system 2166 for scanning a scenery comprising the object 2112, such as for scanning the object 2112 and/or for determining at least one position of the at least one object 2112. The scanning system 2166 comprises the at least one detector 2110, and, further, optionally, the at least one illumination source 2115 as well as, optionally, at least one further illumination source 2115. The illumination source 2115, generally, is configured to emit at least one illumination light beam 2142, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 2114 and/or on a surface of the object 2112. The scanning system 2166 may be designed to generate a profile of the scenery including the object 2112 and/or a profile of the object 2112, and/or may be designed to generate at least one item of information about the dis-

tance between the at least one dot and the scanning system 2166, specifically the detector 2110, by using the at least one detector 2110.

As outlined above, an exemplary embodiment of the detector 2110 which may be used in the setup of Fig. 41 is shown in Fig. 35 and 36. Thus, the detector 2110, besides the optical sensors 2118, 2120, comprises at least one evaluation device 2133, having e.g. the at least one divider 2134 and/or the at least one position evaluation device 2136, as symbolically depicted in Fig. 41. The components of the evaluation device 2133 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 2110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 2118, 2120 and one or more of the components of the evaluation device 2133 may be interconnected by one or more connectors 2168 and/or by one or more interfaces, as symbolically depicted in Fig. 41. Further, the one or more connectors 2168 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 2168, the evaluation device 2133 may fully or partially be integrated into one or both of the optical sensors 2118, 2120 and/or into a housing 2170 of the detector 2110. Additionally or alternatively, the evaluation device 2133 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 2112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 2172, the position of which may be manipulated by a user 2174. As an example, the object 2112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 2112 are possible. Further, the user 2174 himself or herself may be considered as the object 2112, the position of which shall be detected.

As outlined above, the detector 2110 comprises at least the optical sensors 2118, 2120. The optical sensors 2118, 2120 may be located inside the housing 2170 of the detector 2110. Further, the at least one transfer device 2128 is comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 2176 inside the housing 2170, which, preferably, is located concentrically with regard to the optical axis 2126 of the detector 2110, preferably defines a direction of view 2178 of the detector 2110. A coordinate system 2180 may be defined, in which a direction parallel or anti-parallel to the optical axis 2126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 126 may be defined as transversal directions. In the coordinate system 2180, symbolically depicted in Fig. 41, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 2180 are feasible, such as non-Cartesian coordinate systems.

The detector 2110 may comprise the optical sensors 2118, 2120 as well as, optionally, further optical sensors. The optical sensors 2118, 2120 may be located in one and the same beam

path, for example one behind the other, such that the first optical sensor 2118 covers a portion of the second optical sensor 2120. Alternatively, however, a branched beam path may be possible, for example using a multifurcated optical fiber. The branched beam path may comprise additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 2112 and/or of parts thereof. Alternatively, however, the optical sensors 2118, 2120 may be located at the same longitudinal coordinate.

One or more light beams 2116 are propagating from the object 2112 and/or from one or more of the beacon devices 2114, towards the detector 2110. The detector 2110 is configured for determining a position of the at least one object 2112. For this purpose, as explained above in the context of Fig. 35 to 40, the evaluation device 2133 is configured to evaluate sensor signals provided by the optical sensors 2118, 2120. The detector 2110 is adapted to determine a position of the object 2112, and the optical sensors 2118, 2120 are adapted to detect the light beam 2131. In case no illumination source 2115 is used, the beacon devices 2114 and/or at least one of these beacon devices 2114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 2115 is used, the beacon devices 2114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 2112 may be used, such as integrated reflected beacon devices 2114 having at least one reflective surface such as a mirror, retro reflector, reflective film, or the like. The light beam 2116, directly and/or after being modified by the transfer device 2128, such as being focused by one or more lenses, impinges on the angle dependent element 2130 which generates the at least one light beam which illuminates the light-sensitive areas 2122, 2124 of the optical sensors 2118, 2120. For details of the evaluation, reference may be made to Fig. 35 to 40 above.

As outlined above, the determination of the position of the object 2112 and/or a part thereof by using the detector 2110 may be used for providing a human-machine interface 2160, in order to provide at least one item of information to a machine 2182. In the embodiments schematically depicted in Fig. 41, the machine 2182 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 2133 may even be fully or partially integrated into the machine 2182, such as into the computer.

As outlined above, Fig. 41 also depicts an example of a tracking system 2164, configured for tracking the position of the at least one object 2112 and/or of parts thereof. The tracking system 2164 comprises the detector 2110 and at least one track controller 2184. The track controller 2184 may be adapted to track a series of positions of the object 2112 at specific points in time. The track controller 2184 may be an independent device and/or may be fully or partially integrated into the machine 2182, specifically the computer, as indicated in Fig. 41 and/or into the evaluation device 2133.

Similarly, as outlined above, the human-machine interface 2160 may form part of an entertainment device 2162. The machine 2182, specifically the computer, may also form part of the en-

entertainment device 2162. Thus, by means of the user 2174 functioning as the object 2112 and/or by means of the user 2174 handling a control device 2172 functioning as the object 2112, the user 2174 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Referring to Figure 42, the detector 110, 1110, 2110, such as the detector as described with respect to Figures 1 to 41 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 113, 1118, 1120, 2113. For example, the detector 110, 1110, 2110 may comprise a plurality of optical sensors arranged in the matrix 117. The detector 110, 1110, 2110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector 110, 1110, 2110 may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. As outlined above, this principle is called Distance by Photon Ratio (DPR). In one embodiment, the light beam 116, 1116 may illuminate the sensor element with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 132 may be configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 42, the light beam 116, 1116 propagating from the object 112, 1112 to the detector 110, 1110, 2110 may illuminate the matrix 117 with at least one line pattern 2186. The line pattern 2186 may be generated by the object 112, 1112, for example in response to an illumination by the at least one illumination source 136 with an illumination pattern comprising at least one illumination line pattern. A1 may correspond to an area with a full line width of the line pattern 2186 in the matrix 117. The line pattern 2186 in the matrix 117 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a

line width in the matrix 117 is increased. In particular, the line width of the line pattern 2186 in the matrix 117 may change from one column to another column. A2 may be a central area of the line pattern 2186 in the matrix 117. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern 2186 may be segmented in the matrix 117. Each of the columns may comprise center information of intensity in the central area of the line pattern 2186 and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern 2186.

Figure 43 shows a simulation testing of object size independence for a one-dimensional case using computational ray tracing. In a simulation testing, an aspheric lens with $f=10$ mm, 10 mm pupil diameter in a distance range of 100 mm to 600 mm was used. Using this specification, object size independence of about 10 mm was obtained, such that the spot size was varied from 1 mm to 25 mm. In Fig. 43, the quotient $Q(z)$ over the longitudinal coordinate z is shown for 1 mm variation (curve 2188), 2 mm variation (curve 2190), 5 mm variation (curve 2192), 15 mm variation (curve 2194), 20 mm variation (curve 2196) and 25 mm variation (curve 2198). It can be seen that the quotient for object sizes above 10 mm deviates, whereas that of object sizes smaller than 10 mm yield identical ratios. This feature reduces calibration efforts for targets of varying size and is inherent to DPR analysis.

Referring to Figures 44A and B, as outlined above, the detector 110, 1110, 2110 may comprise the at least one matrix 117 of optical sensors 113, 1118, 1120, 2113. With the aid of such a pixelated imaging device, a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates as shown in Figure 44A. Accordingly, the parameterization of a single line would be given by $\omega = x \cos(\theta) + y \sin(\theta)$. The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega,\theta)}\{f(x,y)\} = \iint_{-\infty}^{\infty} f(x,y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy$$

where δ denotes the Dirac delta function and $f(x,y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is then given by

$$R = \frac{\Re_{(\omega,\theta)}\{f'(x,y)\}}{\Re_{(\omega,\theta)}\{f(x,y)\}}$$

with $f'(x,y)$ as the overshined image region highlighted in Fig. 44B. It is expected that the variation of θ yields different ratios R for skewed object surfaces. It may be sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

Figures 45A and B show further embodiments of the detector 110 according to the present invention comprising at least one bi-cell. The illumination source 136, such as a laser source, may generate the light beam 138 illuminating object 112. The reflected light beam 116 may propagate from the object 112 to the transfer device 128 and may impinge on the bi-cell of optical

sensors 176. In Figure 45A a side view is shown and in Figure 45B a front view is shown. The detector 110 may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the bi-cell in Figures 45 A and B may be adapted to generate a so called FiP signal. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

Figure 46 shows experimental results, in particular spot diameter independence and luminance independence of the combined sensor signal, determined using the detector setup shown in Figure 46. In particular, the bi-cell was a PbS-Bi-cell and a 1550 nm laser was used with a laser spot size of 4 mm. The baseline was 12.5 mm. The transfer device was a Thorlabs Asphere lens with focal length of $f = 20$ mm and diameter of $D = 25$ mm. Figure 46 shows quotient Q over the longitudinal coordinate z for different luminance and spot diameter, in particular for luminance of 2.6 mW and spot diameter of 12 mm (curve 2200), 2.4 mW and 6 mm (curve 2202) and 1.2 mW and spot diameter of 3 mm (curve 2204). All curves show identical curve shape and thus, spot diameter independence.

Figures 47A to C show three embodiments of a hexagonal illumination pattern. The illumination source 136 may be adapted to generate at least one illumination pattern for illuminating the object 112. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. Specifically, the illumination source 136 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination pattern may comprise at least one feature such as a point or symbol. The illumination pattern may comprise a plurality of features. The illumination pattern may comprise an arrangement of periodic or non-periodic features. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasi-periodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as

possible features per area such that hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

- 5 The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on an epipolar line. As shown in Figure 47A, the illumination pattern may comprise at least one hexagonal pattern, wherein the individual points are positioned on epipolar lines 2206. As shown in Figure 47B, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Such a positioning
- 10 of the illumination features allows enhancing distance between the individual points on each epipolar line. For example as shown in Figure 47C, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the
- 15 distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other. Such a positioning allows to enhance the number of possible features per area.
- 20 Figure 48 shows an embodiment of a scanning device 154. The scanning device may be adapted as a line scanning device. In particular, the scanning device 154 may comprise at least one sensor line or row of optical sensors 113. Furthermore, the scanning device may comprise the at least one transfer device 128 and the at least one illumination source 136. Triangulation systems require a sufficient baseline, however due to the baseline in the near field no detection
- 25 may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. Further, these near field and far field
- 30 problems can be overcome by using the scanning device 154 of Figure 48. The scanning device 154 may be adapted to detect a plurality of light beams 116 propagating from the object 112 to the scanning device 154 on the CMOS line. The light beams 116 may be generated at different position on the object 112 or by movement of the object 112. The scanning device 154 may be
- 35 adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above.

List of reference numbers

110	detector
112	object
113	optical sensors
114	beacon device
115	sensor element
116	light beam
117	matrix
118	first optical sensor
119	mask
120	second optical sensor
121	light-sensitive area
122	first light-sensitive area
124	second light-sensitive area
126	optical axis of the detector
128	transfer device
129	optical axis of the transfer device
130	focal point
131	light spot
132	evaluation device
133	center detector
134	detector system
135	summing device
136	illumination source
137	combining device
138	illumination light beam
140	reflective element
142	divider
144	position evaluation device
146	camera
148	human-machine interface
150	entertainment device
152	tracking system
154	scanning system
156	connector
158	housing

160 control device
162 user
164 opening
166 direction of view
168 coordinate system
170 machine
172 track controller
174 array
176 optical sensor
178 quadrant photodiode
180 geometrical center of every
182 geometrical center of first optical sensor
184 geometrical center of second optical sensor
186 light spot
188 actuator
190 diaphragm
192 readout device for optical storage media
194 optical element
196 region of interest
198 first area
200 second area
202 inner region
204 plane
206 outer region
208 direction of movement
210 direction of movement
212 curve
214 curve
216 set of curves
218 set of curves
1110 detector
1112 object
1114 beacon device
1116 light beam
1118 first optical sensor
1120 second optical sensor
1122 first light-sensitive area

1124 second light-sensitive area
1126 optical axis
1128 transfer device
1130 focal point
1132 evaluation device
1134 detector system
1136 illumination source
1138 illumination light beam
1140 reflective element
1142 divider
1144 position evaluation device
1146 camera
1148 human-machine interface
1150 entertainment device
1152 tracking system
1154 scanning system
1156 connector
1158 housing
1160 control device
1162 user
1164 opening
1166 direction of view
1168 coordinate system
1170 machine
1172 track controller
1174 fluorescent waveguiding sheet
1176 waveguiding
1178 matrix material
1180 fluorescent material
1182 photosensitive element
1184 photosensitive element
1186 photosensitive element
1188 photosensitive element
1190 edge
1192 edge
1194 edge
1196 edge

1198 optical filter element
1200 reference photosensitive element
1202 small light spot
1204 large light spot
1206 shadow
1208 summing device
1210 subtracting device
1212 photosensitive element
1214 corner
1216 optical coupling element
2110 detector
2112 object
2113 optical sensors
2114 beacon device
2115 illumination source
2116 light beam
2118 first optical sensor
2120 second optical sensor
2121 light-sensitive area
2122 first light-sensitive area
2124 second light-sensitive area
2126 optical axis of the detector
2128 transfer device
2129 optical axis of the transfer device
2130 angle dependent optical element
2131 light beam
2132 first side
2133 evaluation device
2134 divider
2136 position evaluation device
2138 Optical fiber
2140 illumination fiber
2142 Light beam
2144 First fiber
2146 Second fiber
2148 entrance end
2150 exit end

2152 first light beam
2154 Second light beam
2156 camera
2158 Detector system
2160 Human-machine interface
2162 Entertainment device
2164 Tracking system
2166 Scanning system
2168 connector
2170 housing
2172 Control device
2174 user
2176 opening
2178 Direction of view
2180 Coordinate system
2182 machine
2184 Track controller
2186 Line pattern
2188 curve
2190 curve
2192 curve
2194 curve
2196 curve
2198 curve
2200 curve
2202 curve
2204 curve
2206 Epipolar line

Patent claims

1. A detector (110, 1110, 2110) for determining a position of at least one object (112, 1112), the detector (110, 1110, 2110) comprising:
 - 5 - at least one transfer device (128, 1128, 2128), wherein the transfer device (128, 1128, 2128) has at least one focal length in response to at least one incident light beam (116, 1116) propagating from the object (112, 1112, 2112) to the detector (110, 1110, 2110);
 - 10 - at least two optical sensors (113, 1118, 1120, 2113), wherein each optical sensor (113, 1118, 1120) has at least one light sensitive area (121, 1122, 1124, 2121), wherein each optical sensor (113, 1118, 1120, 2113) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam (116, 1116, 2131),
 - 15 - at least one evaluation device (132, 1132, 2133) being configured for determining at least one longitudinal coordinate z of the object (112, 1112, 2112) by evaluating a quotient signal Q from the sensor signals;
 wherein the detector is adapted to determine the longitudinal coordinate z of the object in at least one measurement range independent from the object size in an object plane.
- 20 2. The detector (110, 1110, 2110) according to the preceding claim, wherein the measurement range is adjustable by adjusting one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to
 25 an image of the object z_i ; an object size O_{size} of the object in the object plane.
3. The detector (110, 1110, 2110) according to the preceding claim, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object size if a circle of confusion of at least one light beam emitted from at least one point of the
 30 object is greater than an image size of the object.
4. The detector (110, 1110, 2110) according to the preceding claim, wherein the detector is adapted to determine the longitudinal coordinate z of the object independent from the object size if $O_{size} < \frac{z_0 E_x}{z_i^2} |z_s - z_i|$ holds true for at least one distance.
- 35 5. The detector (110, 1110, 2110) according to the preceding claim, wherein a lower object size limit is preferably 0.5 μm or more, more preferably 1 μm or more and most preferably 10 μm or more.
- 40 6. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for deriving the quotient signal Q by one or more of di-

viding the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

7. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.
8. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o.

9. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the optical sensors (113) are positioned off focus.
10. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the transfer device (128) has an optical axis (129), wherein the transfer device (128) constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis (129) and wherein d is a spatial offset from the optical axis (129), wherein the optical sensors (113) are arranged such that the light-sensitive areas (121) of the optical sensors (113) differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.
11. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of at least one beam profile of the light beam.
12. The detector (110, 1110, 2110) according to the preceding claim, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.
13. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the light-sensitive regions are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a

second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

14. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured to determine the first area of the beam profile and the second area of the beam profile.
15. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.
16. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
17. The detector (110, 1110, 2110) according to any one of the two the preceding claims, wherein the evaluation device (132) is configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.
18. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the optical sensors (113) comprise a first optical sensor (118) having a first light sensitive area (122) and a second optical sensor (120) having a second light sensitive area (124), wherein the first and the second light sensitive areas (122, 124) are arranged such that a condition

$$\frac{a}{c} \neq \frac{b}{d}$$
 is satisfied, wherein a is a ratio of photons hitting both an inner region of a plane perpendicular to the optical axis (129) intersecting the optical axis (129) at a distance equal to half of a focal length of the transfer device (129) and the first light sensitive area (122), b is a ratio of photons hitting both the inner region of the plane and the second light sensitive area (124), c is a ratio of photons hitting both an outer region of the plane and the first light-sensitive area (122), d is a ratio of the photons hitting both the outer region of the plane and the second light sensitive area (124).
19. The detector (110, 1110, 2110) according to the preceding claim, wherein the first light sensitive area (118) and the second light sensitive area (122) are arranged such that the inner region has an area with a geometrical center point on the optical axis (129) and an ex-

tension such that half of the photons hit the plane within the inner region and the other half hit the plane outside the inner region.

20. The detector (110, 1110, 2110) according to the preceding claim, wherein the first light sensitive area (118) and the second light sensitive area (122) are arranged such that the inner region is designed as a circle with a center point on the optical axis and a radius r which is chosen such that half of the photons hit the plane within the circle and the other half hit the plane outside the circle.
21. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the optical sensors (113) comprise a first optical sensor (118) having a first light sensitive area (122) and a second optical sensor (120) having a second light sensitive area (124),
- wherein the first optical sensor (118) is configured to generate at least one first sensor signal in response to the illumination of the first light-sensitive area (122) by the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110);
 - wherein the second optical sensor (120) is configured to generate at least one second sensor signal in response to the illumination of the second light-sensitive area (124) by the light beam (116), wherein the first light-sensitive area (122) is smaller than the second light-sensitive area (124).
22. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured for deriving the quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device (132) is configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
23. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the first and second optical sensors (118, 120) are arranged linearly in one and the same beam path of the detector (110, 1110, 2110).
24. The detector (110, 1110, 2110) according to the preceding claim, wherein the first and second optical sensors (118, 120) are arranged concentrically with respect to an optical axis (126) of the detector (110, 1110, 2110).
25. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the first and second optical sensors (118, 120) each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

26. The detector (110, 1110, 2110) according to any one of the five preceding claims, wherein the first and second optical sensors (118, 120) each are uniform sensors having a single light-sensitive area each.
- 5 27. The detector (110, 1110, 2110) according to any one of the six preceding claims, wherein the detector (110, 1110, 2110) further comprises an illumination source (136) for illuminating the object (112).
- 10 28. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein the detector (110, 1110, 2110) has
- at least one sensor element (115) having a matrix (117) of optical sensors (113), the optical sensors (113) each having a light-sensitive area (121), wherein each optical sensor (113) is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area (121) by at least one light beam (116) propagating from the object (112) to the detector (110, 1110, 2110);
 - 15 - the at least one evaluation device (132) configured for evaluating the sensor signals, by
 - a) determining at least one optical sensor (113) having the highest sensor signal and forming at least one center signal;
 - 20 b) evaluating the sensor signals of the optical sensors (113) of the matrix (117) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - 25 d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.
29. The detector (110, 1110, 2110) according to the preceding claim, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors (113) containing the optical sensor (113) having the highest sensor signal and a predetermined group of neighboring optical sensors (113); a sum of sensor signals from a group of optical sensors (113) containing the optical sensor (113) having the highest sensor signal and a predetermined group of neighboring optical sensors (113); a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors (113) containing the optical sensor (113) having the highest sensor signal and a predetermined group of neighboring optical sensors (113); an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.
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30. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix (117); a sum of all sensor signals of the matrix (117); an integral of all sensor signals of the matrix (117); an average over all sensor signals of the matrix (117) except for sensor signals from those optical sensors (113) contributing to the center signal; a sum of all sensor signals of the matrix (117) except for sensor signals from those optical sensors (113) contributing to the center signal; an integral of all sensor signals of the matrix (117) except for sensor signals from those optical sensors (113) contributing to the center signal; a sum of sensor signals of optical sensors (113) within a predetermined range from the optical sensor (113) having the highest sensor signal; an integral of sensor signals of optical sensors (113) within a predetermined range from the optical sensor (113) having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors (113) being located within a predetermined range from the optical sensor (113) having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors (113) being located within a predetermined range from the optical sensor (113) having the highest sensor signal.
31. The detector (110, 1110, 2110) according to any one of the three preceding claims, wherein the combined signal is the quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.
32. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate.
33. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the evaluation device (132) comprises at least one divider (142), wherein the divider (142) is configured for deriving the quotient signal.
34. The detector (110, 1110, 2110) according to any one of the six preceding claims, wherein the evaluation device (132) is further configured for determining at least one transversal coordinate of the object (112) by evaluating a transversal position of the at least one optical sensor (113) having the highest sensor signal.
35. The detector (110, 1110, 2110) according to any one of the preceding claims, wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical

axis (126) of the detector (110, 1110, 2110) by different spatial offsets, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light beam (116) propagating from the object (112) to the detector (110, 1110, 2110).

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36. The detector (110, 1110, 2110) according to the preceding claim, wherein the optical sensors (118, 120, 176) are part of a sensor array (174), wherein a geometrical center (180) of the sensor array (174) is offset from the optical axis (126).

10 37. The detector (110, 1110, 2110) according to the preceding claim, wherein the sensor array (174) is movable relative to the optical axis (126).

38. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (132) is configured for, firstly, determining the transversal position of a light spot (186) generated by the light beam (116) on the sensor array (174) by using the sensor signals and for, secondly, moving the sensor array (174) relative to the optical axis (126) until the light spot (186) is off-centered.

15 39. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the optical sensors (118, 120, 176) are partial diodes of a quadrant diode (178), with a geometrical center (180) of the quadrant diode (178) being off-centered from the optical axis (126) of the detector (110, 1110, 2110).

20 40. The detector (110, 1110, 2110) according to any one of the preceding claims, the detector (110, 1110, 2110) having

- at least one first optical sensor (1118) having a first light-sensitive area (1122), wherein the first optical sensor (1118) is configured to generate at least one first sensor signal in response to an illumination of the first light-sensitive area (122) by a light beam (1116) propagating from the object (1112) to the detector (1110);
- 30 - at least one second optical sensor (1120) having:
 - at least one fluorescent waveguiding sheet (1174) forming a second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) is oriented towards the object (1112) such that at least one light beam (1116) propagating from the object (1112) towards the detector (1110) generates at least one light spot (1202, 1204) in the second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) contains at least one fluorescent material (1180), wherein the fluorescent material (1180) is adapted to generate fluorescence light in response to the illumination by the light beam (1116); and
 - 35 ○ at least one photosensitive element (1182, 1184, 1186, 1188, 1212) located at at least one edge (1190, 1192, 1194, 1196) of the fluorescent waveguiding sheet (1174), capable of detecting fluorescence light guided from the light spot (1202, 1204) towards the photosensitive element (1182, 1184, 1186, 1188, 1212) by the fluorescent waveguiding sheet (1174) and capable of generating at least one second sensor signal in response to the illumination of the second light-sensitive
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area (1124) by the light beam (1116), wherein the first light-sensitive area (1122) is smaller than the second light-sensitive area (1124); and

- at least one evaluation device (1132) being configured for determining at least one longitudinal coordinate z of the object (1112) by evaluating the first and second sensor signals.

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41. The detector (110, 1110, 2110) according to the preceding claim, wherein the at least one photosensitive element (1182, 1184, 1186, 1188, 1212) is optically coupled to the fluorescent waveguiding sheet (1174) by at least one optical coupling element (1216) configured for at least partially coupling the fluorescence light guided by the fluorescent waveguiding sheet (1174) out of the fluorescent waveguiding sheet (1174).
42. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the evaluation device (1132) is configured for deriving a quotient signal Q by dividing the first and second sensor signals, by dividing multiples of the first and second sensor signals or by dividing linear combinations of the first and second sensor signals, wherein the evaluation device (1132) is configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
43. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (1132) is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.
44. The detector (110, 1110, 2110) according to any one of the four preceding claims, wherein the first optical sensor (1118) is located in front of the second optical sensor (1120), preferably such that the light beam (1116), in the given order, illuminates the first optical sensor (1118) and, afterwards, the second optical sensor (1120).
45. The detector (110, 1110, 2110) according to any one of the five preceding claims, wherein the second optical sensor (1120) comprises at least two photosensitive elements (1182, 1184, 1186, 1188, 1212) located at at least two edges (1190, 1192, 1194, 1196) of the fluorescent waveguiding sheet (1174), capable of detecting fluorescence light guided from the light spot (1202, 1204) towards the photosensitive elements (1182, 1184, 1186, 1188, 1212) by the fluorescent waveguiding sheet (1174) and capable of generating at least one sensor signal each.
46. The detector (110, 1110, 2110) according to the preceding claim, wherein the evaluation device (1132) comprises at least one summing device (1208) configured for summing up the sensor signals of the at least two photosensitive elements (1182, 1184, 1186, 1188, 1212), thereby forming a sum signal S , wherein the evaluation device (1132) is configured to determine the at least one longitudinal coordinate z of the object (1112) by using at least one predetermined relationship between the first sensor signal, the sum signal S of the second sensor signals and the longitudinal coordinate z .

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47. The detector (110, 1110, 2110) according to any one of the two preceding claims, wherein the evaluation device (1132) is further configured to determine at least one transversal coordinate x, y of the object (1112) by evaluating the sensor signals of the photosensitive elements (1182, 1184, 1186, 1188, 1212).
- 5 48. The detector (110, 1110, 2110) according to any one of the eight preceding claims, wherein the second light-sensitive area (1124) is larger by at least a factor of two than the first light-sensitive area (1122).
- 10 49. The detector (110, 1110, 2110) according to any one of the nine preceding claims, wherein the detector (110, 1110, 2110) further comprises an illumination source (1136) for illuminating the object (1112).
- 15 50. The detector (110, 1110, 2110) according to any one of the preceding claims, the detector (2110) comprising:
- at least one angle dependent optical element (2130) adapted to generate at least one light beam (2131) having at least one beam profile depending on an angle of incidence of an incident light beam (2116) propagating from the object (2112) towards the detector (2110) and illuminating the angle dependent optical element (2130);
 - 20 - at least two optical sensors (2113), wherein each optical sensor (2113) has at least one light sensitive area (2121), wherein each optical sensor (2113) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by the light beam (2131) generated by the angle dependent optical element (130);
 - 25 - at least one evaluation device (2133) being configured for determining at least one longitudinal coordinate z of the object (2112) by evaluating the quotient signal Q from the sensor signals.
- 30 51. The detector (2110) according to the preceding claim, wherein the angle dependent optical element (2130) comprises at least one optical element selected from the group consisting of: at least one optical fiber, in particular at least one multifurcated optical fiber, in particular at least one bifurcated optical fiber; at least one diffractive optical element; at least one angle dependent reflective element, at least one diffractive grating element, in particular a blaze grating element; at least one aperture stop; at least one prism; at least one lens; at least one lens array, in particular at least one microlens array; at least one optical filter; at least one polarization filter; at least one bandpass filter; at least one liquid crystal filter, in particular a liquid crystal tunable filter; at least one short-pass filter; at least one long-pass filter; at least one notch filter; at least one interference filter; at least one transmission grating; at least one nonlinear optical element, in particular one birefringent optical element.
- 35 52. A detector system (134) for determining a position of at least one object (112), the detector system (134) comprising at least one detector (110, 1110, 2110) according to any one of the preceding claims, the detector system (134) further comprising at least one beacon device (114) adapted to direct at least one light beam (116) towards the detector (110, 1110, 2110),
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wherein the beacon device (114) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).

53. A human-machine interface (148) for exchanging at least one item of information between a user (162) and a machine (170), wherein the human-machine interface (148) comprises at least one detector system (134) according to the preceding claim, wherein the at least one beacon device (114) is adapted to be at least one of directly or indirectly attached to the user (162) and held by the user (162), wherein the human-machine interface (148) is designed to determine at least one position of the user (162) by means of the detector system (134), wherein the human-machine interface (148) is designed to assign to the position at least one item of information.
54. An entertainment device (150) for carrying out at least one entertainment function, wherein the entertainment device (150) comprises at least one human-machine interface (148) according to the preceding claim, wherein the entertainment device (150) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (148), wherein the entertainment device (150) is designed to vary the entertainment function in accordance with the information.
55. A tracking system (152) for tracking a position of at least one movable object (112), the tracking system (152) comprising at least one detector system (134) according to any one of the preceding claims referring to a detector system (134), the tracking system (152) further comprising at least one track controller (172), wherein the track controller (172) is adapted to track a series of positions of the object (112) at specific points in time.
56. A scanning system (154) for determining a depth profile of a scenery, the scanning system (154) comprising at least one detector (110, 1110, 2110) according to any of the preceding claims referring to a detector (110, 1110, 2110), the scanning system (154) further comprising at least one illumination source (136) adapted to scan the scenery with at least one light beam (116).
57. A camera (146) for imaging at least one object (112), the camera (146) comprising at least one detector (110, 1110, 2110) according to any one of the preceding claims referring to a detector (110, 1110, 2110).
58. A readout device (192) for optical storage media, the readout device (192) comprising at least one detector (110, 1110, 2110) according to any one of the preceding claims referring to a detector (110, 1110, 2110).
59. A method for determining a position of at least one object (112) by using at least one detector (110, 1110, 2110), the method comprising the following steps:
- providing at least two optical sensors (113), each optical sensor (113) having a light-sensitive area (121), wherein each optical sensor (113) is configured to generate a sen-

- sor signal in response to an illumination of its respective light-sensitive area by a light beam (116), wherein the detector has at least one transfer device (128), wherein the transfer device (128) has at least one focal length in response to the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110);
- 5 - illuminating each of the light-sensitive areas (121) of the at least two optical sensors (113) of the detector with the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110), wherein, thereby, each of the light-sensitive areas (121) generates at least one sensor signal; and
- 10 - evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object (112), wherein the evaluating comprises deriving a quotient signal Q of the sensor signals,
wherein the longitudinal coordinate z of the object is determined in at least one measurement range independent from the object size in an object plane.
- 15 60. The method according to the preceding claim, wherein the deriving of the quotient signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.
- 20 61. The method according to any one of the preceding claims referring to a method, wherein the method comprises adjusting the measurement range by adjusting on one or more parameters selected from the group consisting of: a longitudinal distance from the object to the transfer device z_0 ; the focal length of the transfer device f ; a diameter of an exit pupil of the transfer device E_x ; a longitudinal distance from the transfer device to the light sensitive area z_s , a distance from the transfer device to an image of the object z_i ; an object size O_{size} of the object in the object plane.
- 25 62. The method according to any one of the preceding claims referring to a method, wherein the transfer device (128) has an optical axis (129), wherein the transfer device (128) constitutes a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis (129) and wherein d is a spatial offset from the optical axis (129), wherein the method comprises arranging the optical sensors (113) such that the light-sensitive areas (121) of the optical sensors (113) differ in at least one of: their longitudinal coordinate, their spatial offset, or their surface areas.
- 30 63. The method according to any one of the preceding claims referring to a method, wherein each of the sensor signals comprises at least one information of at least one beam profile of the light beam, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles, wherein the light-sensitive regions are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.
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64. The method according to the preceding claim, wherein the evaluating comprises determining the first area of the beam profile and the second area of the beam profile.
- 5 65. The method according to any one of the two preceding claims, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises an information relating to a number of photons in the first area of the beam profile and the center information comprises an information relating to a number of photons in the second area of the beam profile, wherein the evaluating comprises deriving the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.
- 10
- 15 66. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- illuminating at least one first light-sensitive area (122) of at least one first optical sensor (118) of the detector (110, 1110, 2110) with the light beam (116) propagating from the object (112) to the detector (110, 1110, 2110) and, thereby, generating at least one first sensor signal;
 - illuminating at least one second light-sensitive area (124) of at least one second optical sensor (120) of the detector (110, 1110, 2110) with the light beam (116) and, thereby, generating at least one second sensor signal, wherein the first light-sensitive area (122) is smaller than the second light-sensitive area (124).
- 20
- 25
67. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- illuminating at least one sensor element (115) of the detector (110, 1110, 2110) with the at least one light beam (116) propagating from the object (112) to the detector (110, 1110, 2110), the detector (110, 1110, 2110) having a matrix (117) of optical sensors (113), the optical sensors (113) each having a light-sensitive area (121), wherein each optical sensor (113) generates at least one sensor signal in response to the illumination;
 - evaluating the sensor signals, by
 - a) determining at least one optical sensor (113) having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors (113) of the matrix (117) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.
- 30
- 35
- 40

68. The method according to the preceding claim, wherein the combined signal is the quotient signal Q, derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.
69. The method according to any one of the preceding claims referring to a method, wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110, 1110, 2110) by different separations.
70. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- illuminating at least one first light-sensitive area (1122) of at least one first optical sensor (1118) with a light beam (1116) propagating from the object (1112) to the detector (110, 1110, 2110) and generating at least one first sensor signal by the first optical sensor (1118), in response to the illumination of the first light-sensitive area (1122);
 - illuminating at least one second light-sensitive area (1124) of at least one second optical sensor (1120) with the light beam (1116) and generating at least one second sensor signal by the second optical sensor (1120), in response to the illumination of the second light-sensitive area (1124), the second optical sensor (1120) having:
 - o at least one fluorescent waveguiding sheet (1174) forming a second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) is oriented towards the object (1112) such that at least one light beam (1116) propagating from the object (1112) towards the detector (110, 1110, 2110) generates at least one light spot (1202, 1204) in the second light-sensitive area (1124), wherein the fluorescent waveguiding sheet (1174) contains at least one fluorescent material (1180), wherein the fluorescent material (1180) is adapted to generate fluorescence light in response to the illumination by the light beam (1116); and
 - o at least one photosensitive element (1182, 1184, 1186, 1188, 1212) located at at least one edge (1190, 1192, 1194, 1196) of the fluorescent waveguiding sheet (1174), capable of detecting fluorescence light guided from the light spot (1202, 1204) towards the photosensitive element (1182, 1184, 1186, 1188, 1212) by the fluorescent waveguiding sheet (1174) and capable of generating the at least one second sensor signal in response to the illumination of the second light-sensitive area (1124) by the light beam (1116), wherein the first light-sensitive area (1122) is smaller than the second light-sensitive area (1124); and
 - determining at least one longitudinal coordinate z of the object (1112) by evaluating the first and second sensor signals.

71. The method according to any one of the preceding claims referring to a method, the method comprising the following steps:
- providing at least one angle dependent optical element (130) and generating at least one light beam (131) having at least one beam profile depending on an angle of incidence;
 - providing at least two optical sensors (113), wherein each optical sensor (113) has at least one light sensitive area (121), wherein each optical sensor (113) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (121) by the light beam (131) generated by the angle dependent optical element (130);
 - illuminating each of the light-sensitive areas (131) of the at least two optical sensors (113) of the detector (110) with the light beam (131) generated by the angle dependent optical element (130), wherein, thereby, each of the light-sensitive areas (121) generates at least one sensor signal; and
 - evaluating the sensor signals, thereby, determining at least one longitudinal coordinate z of the object (112), wherein the evaluating comprises deriving the quotient signal Q of the sensor signals.
72. A use of the detector (110, 1110, 2110) according to any one of the preceding claims relating to a detector (110, 1110, 2110), for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface (148) application; a logistics application; a tracking application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use in combination with optical data storage and readout.

FIG.1

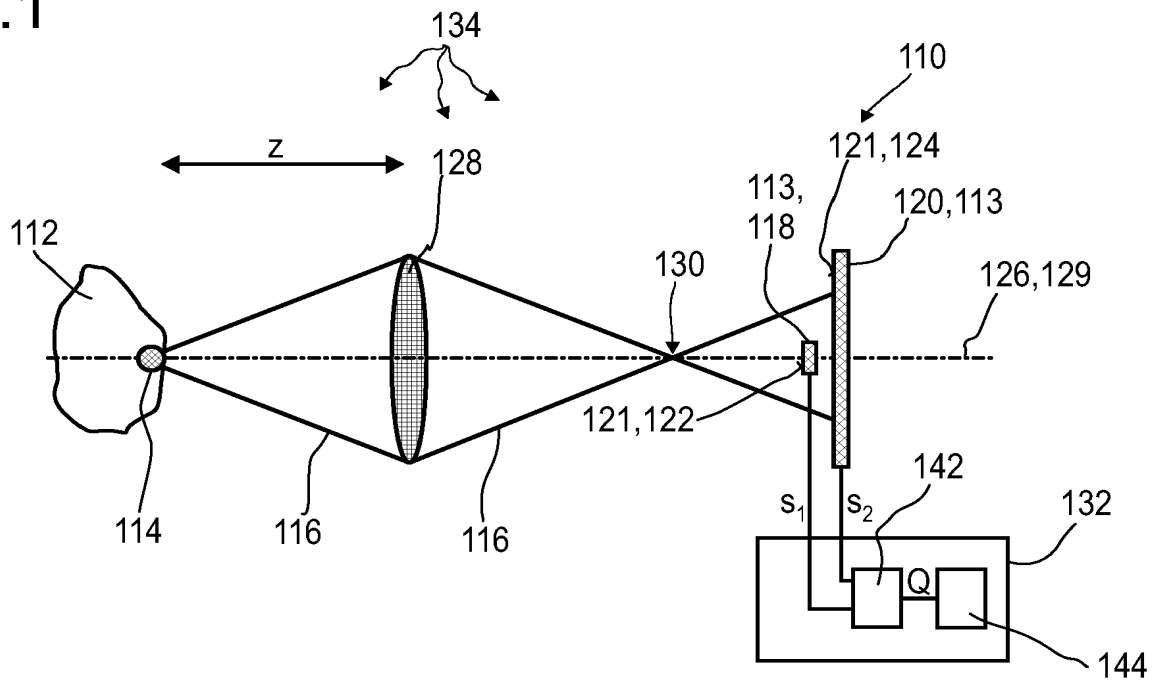


FIG.2

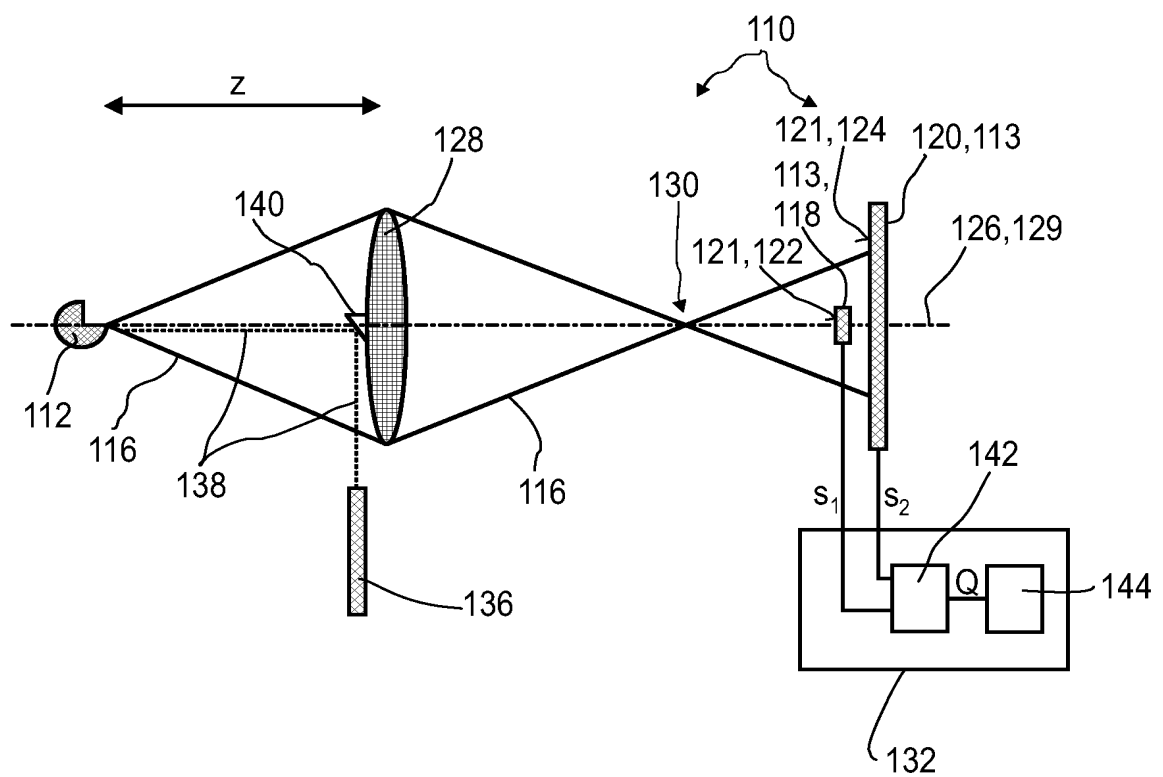


FIG.3

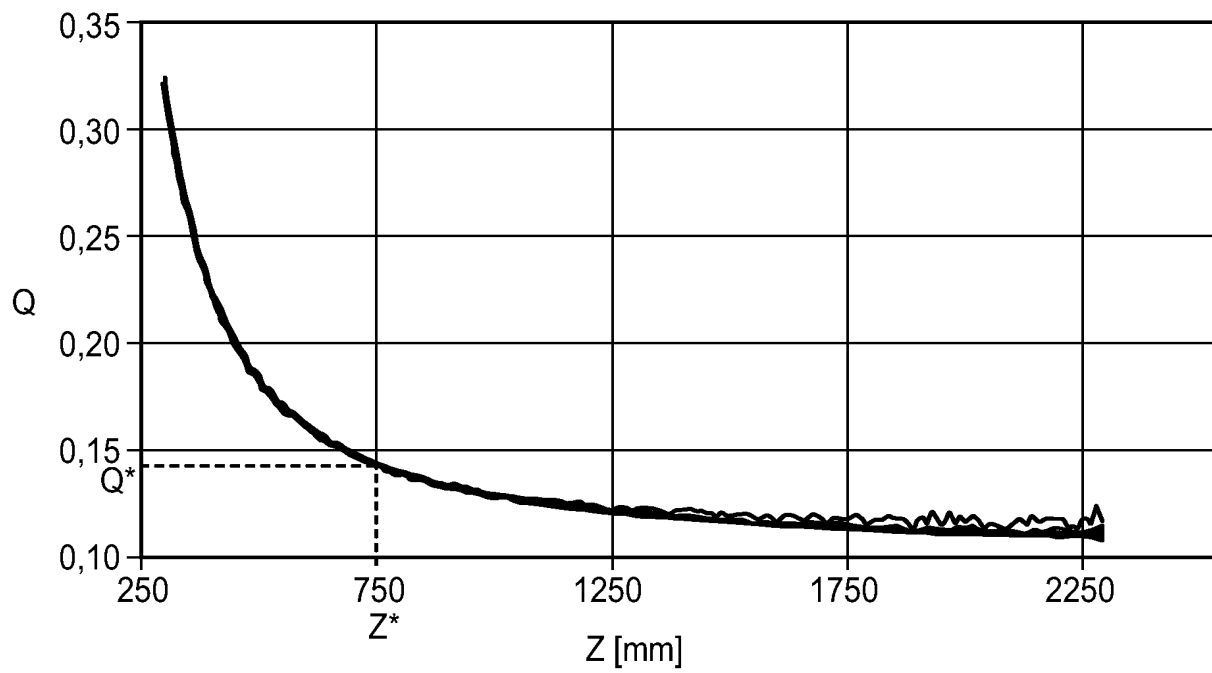


FIG.4

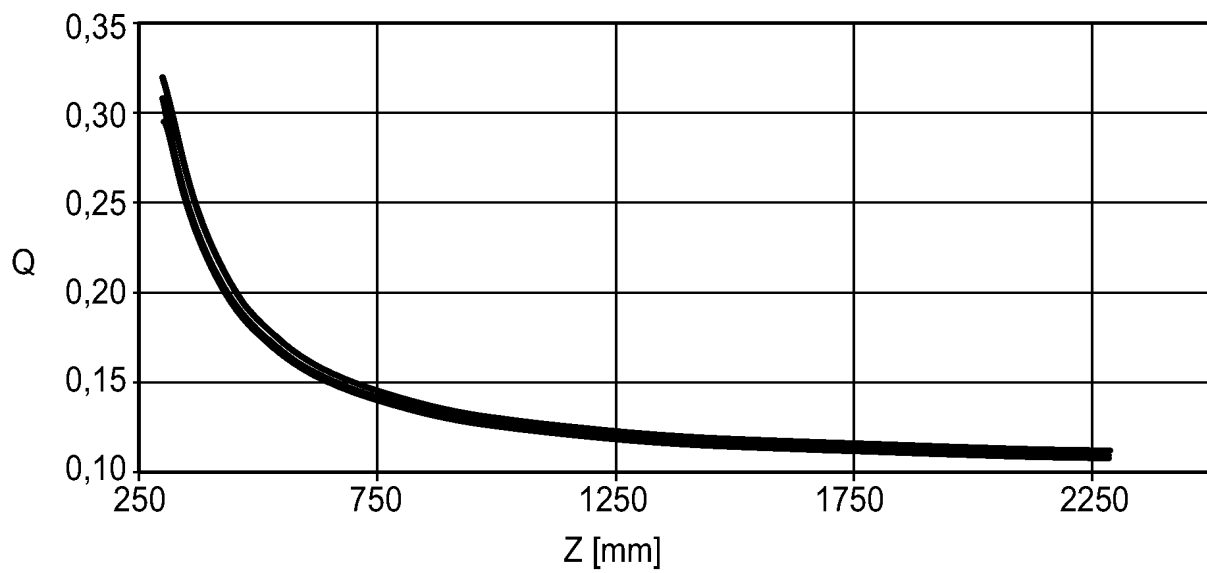


FIG.5

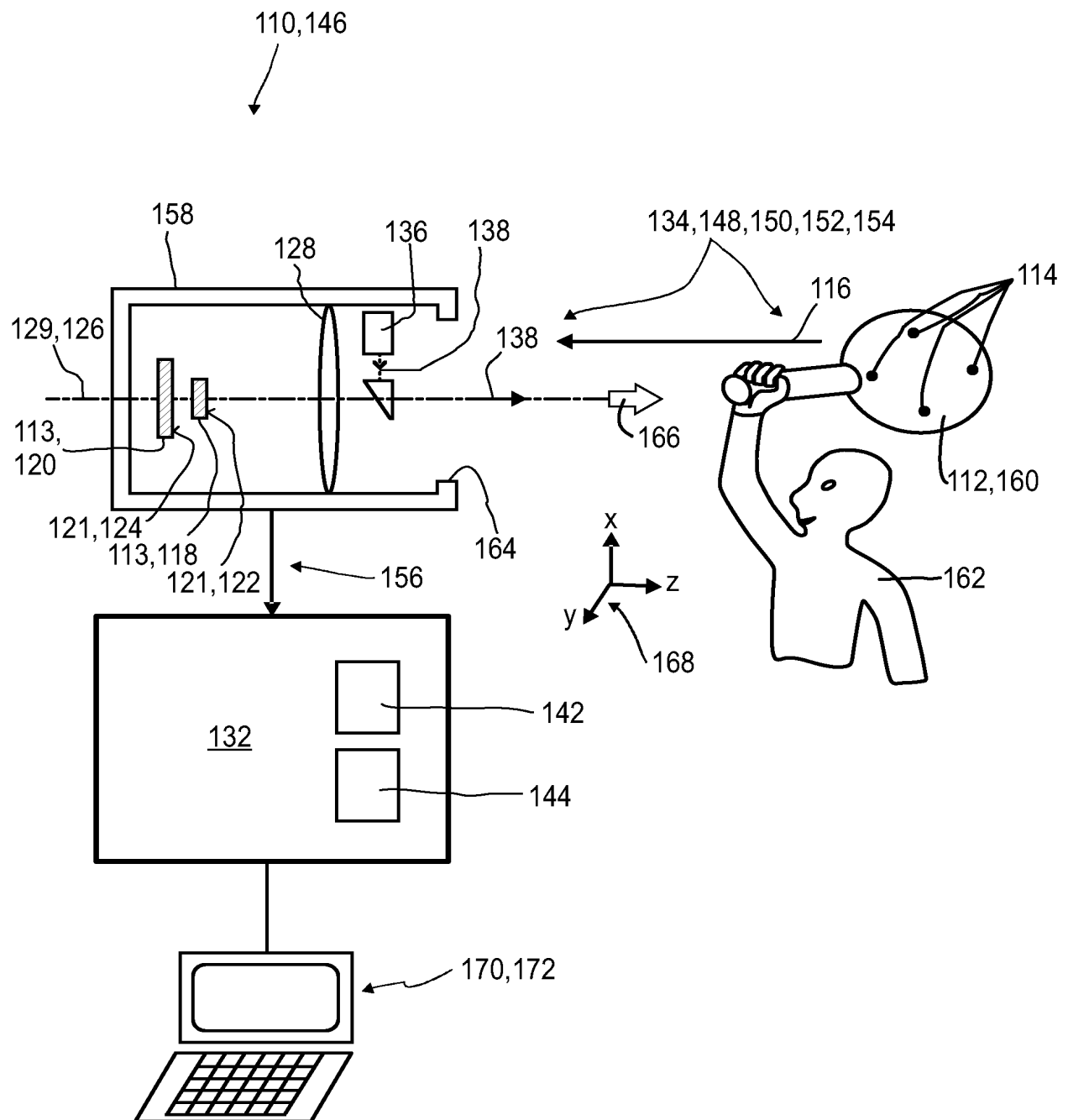


FIG.6

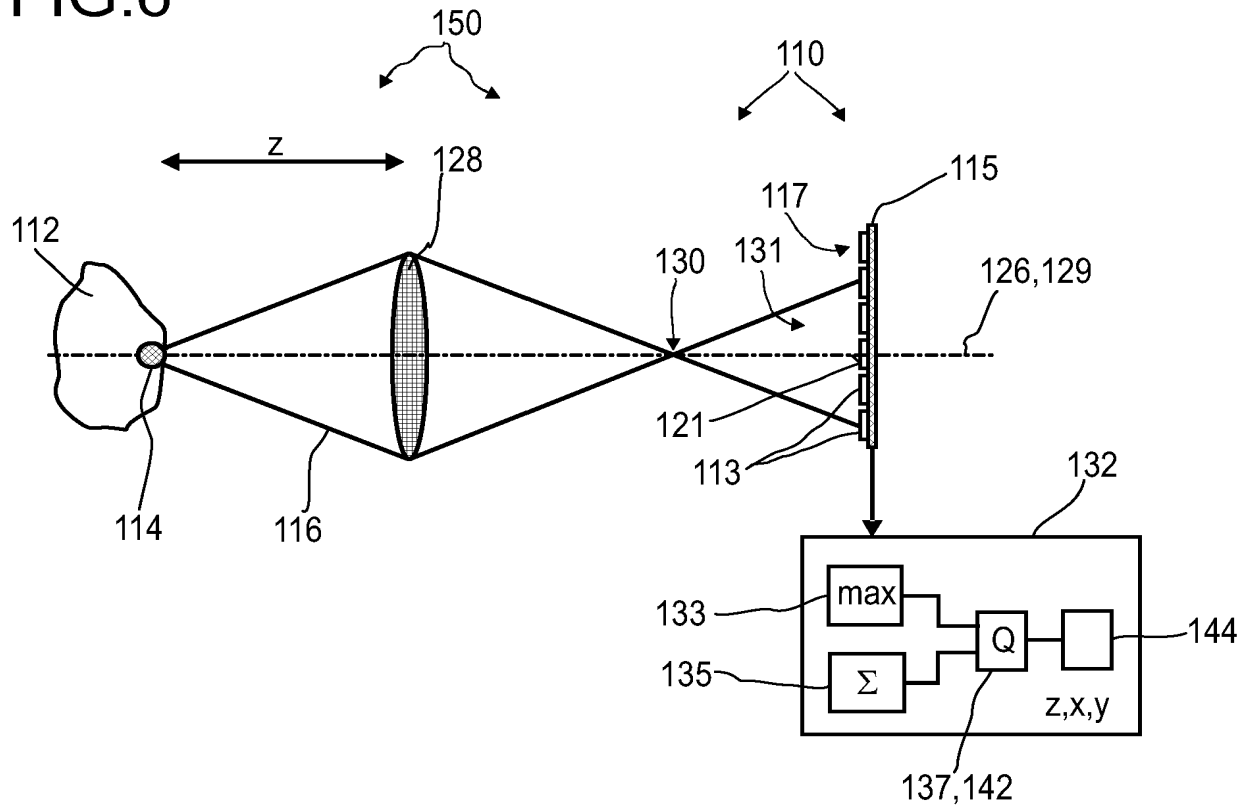


FIG.7

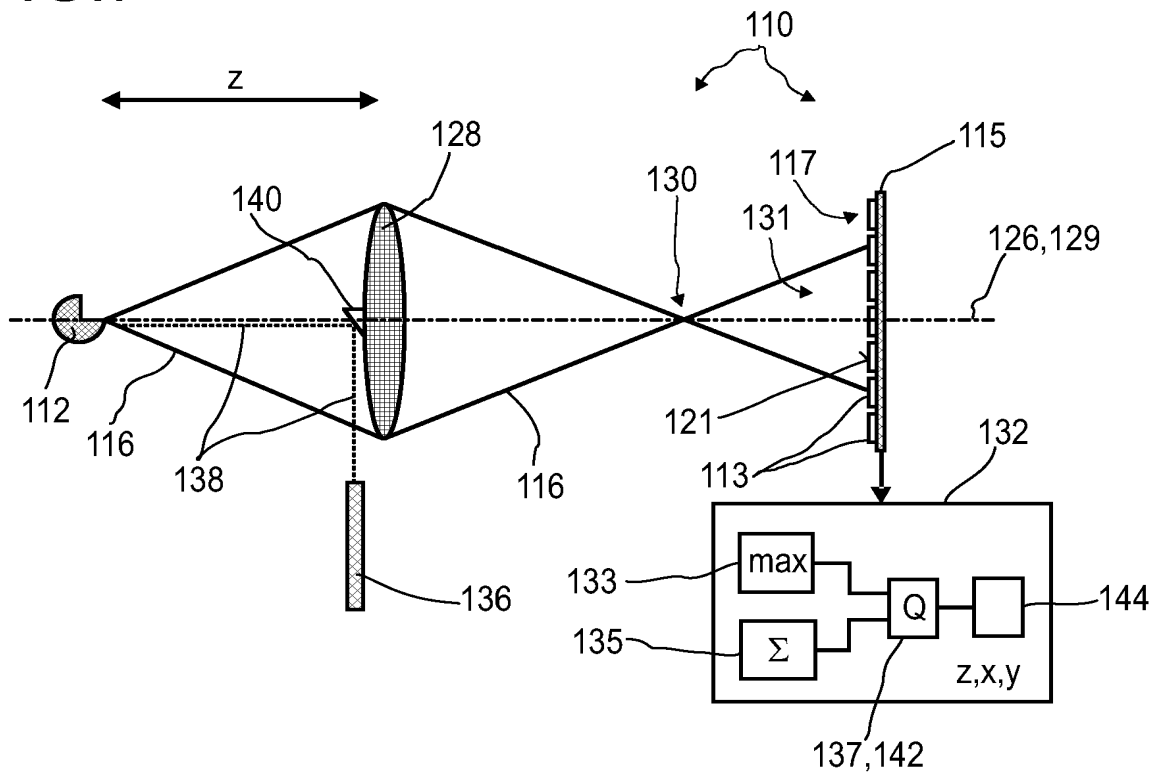


FIG.8

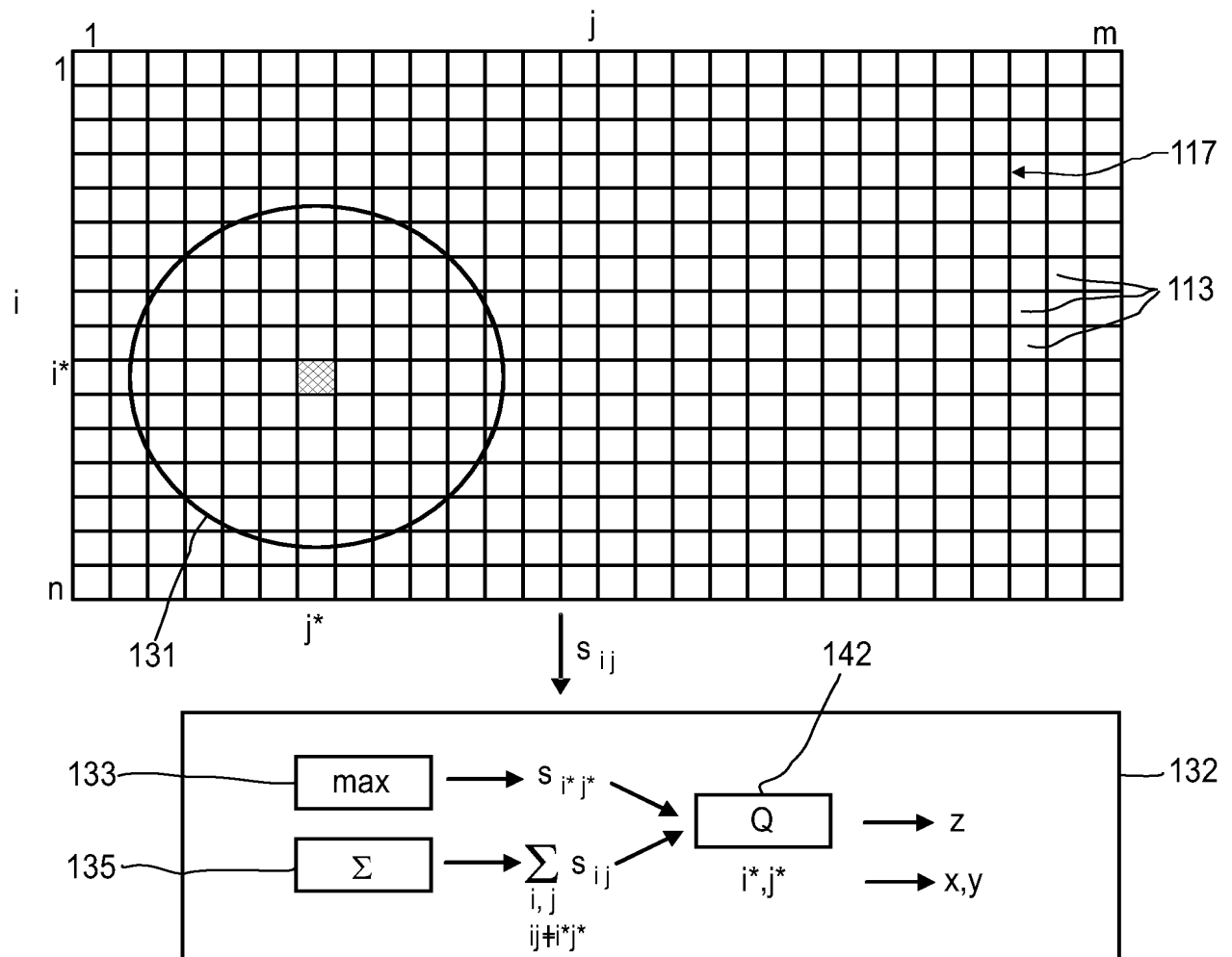


FIG.9

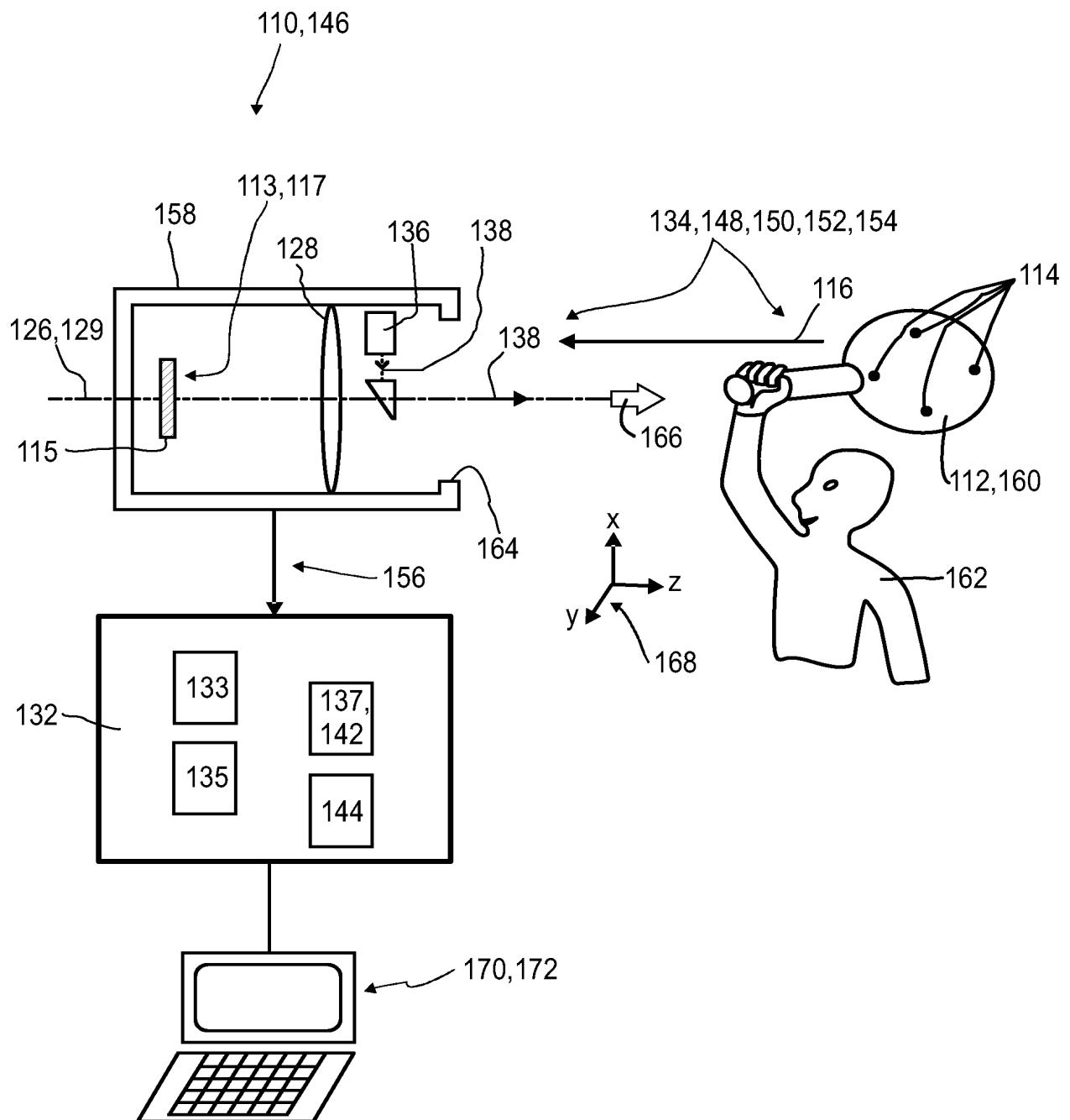


FIG.10

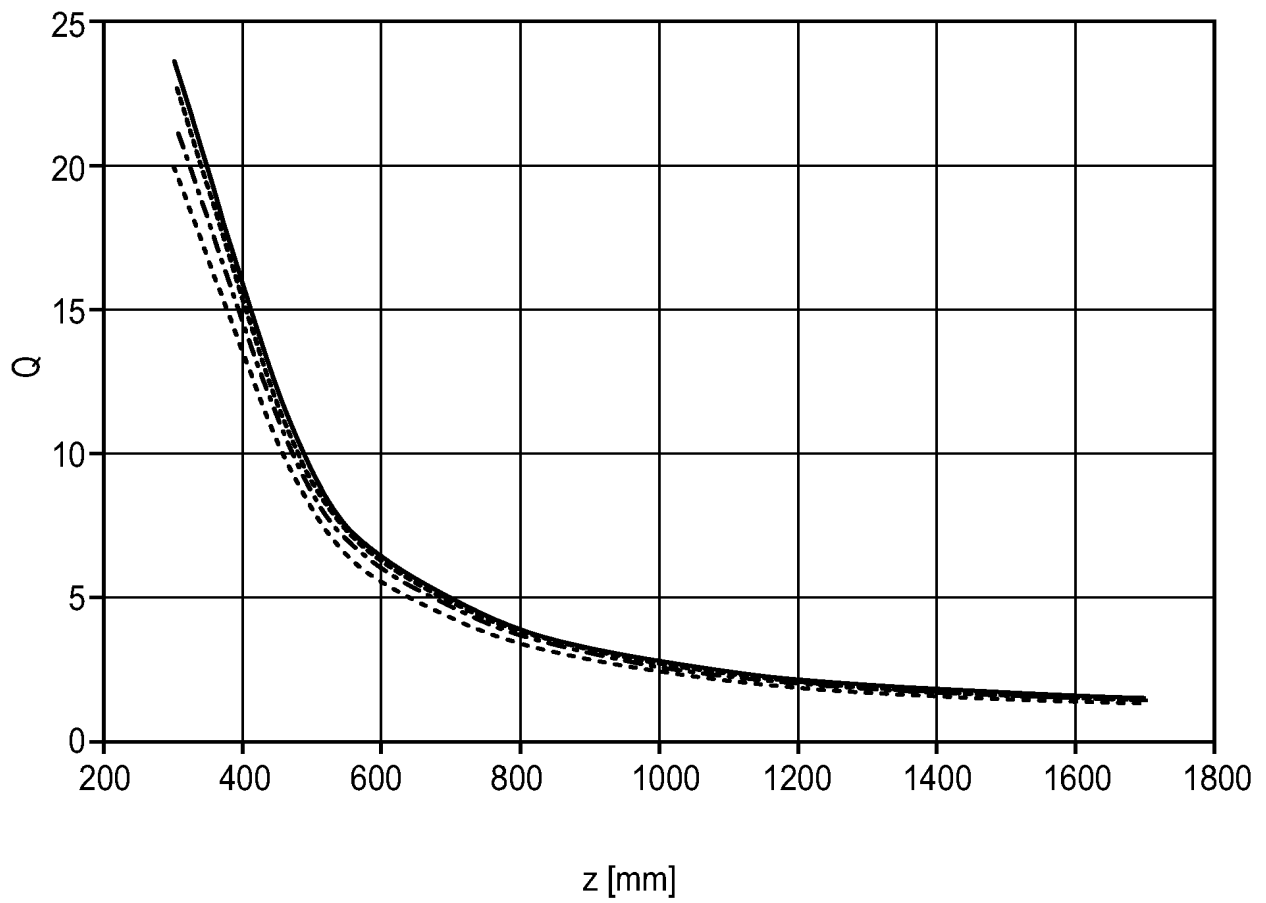


FIG.11

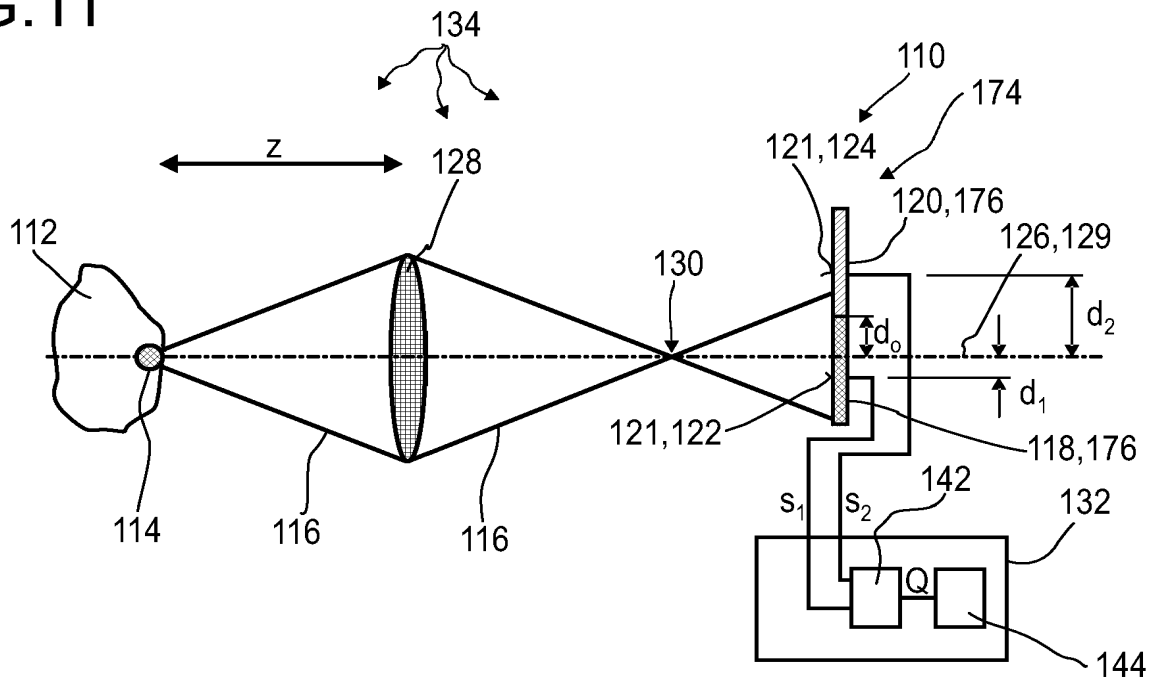


FIG.12

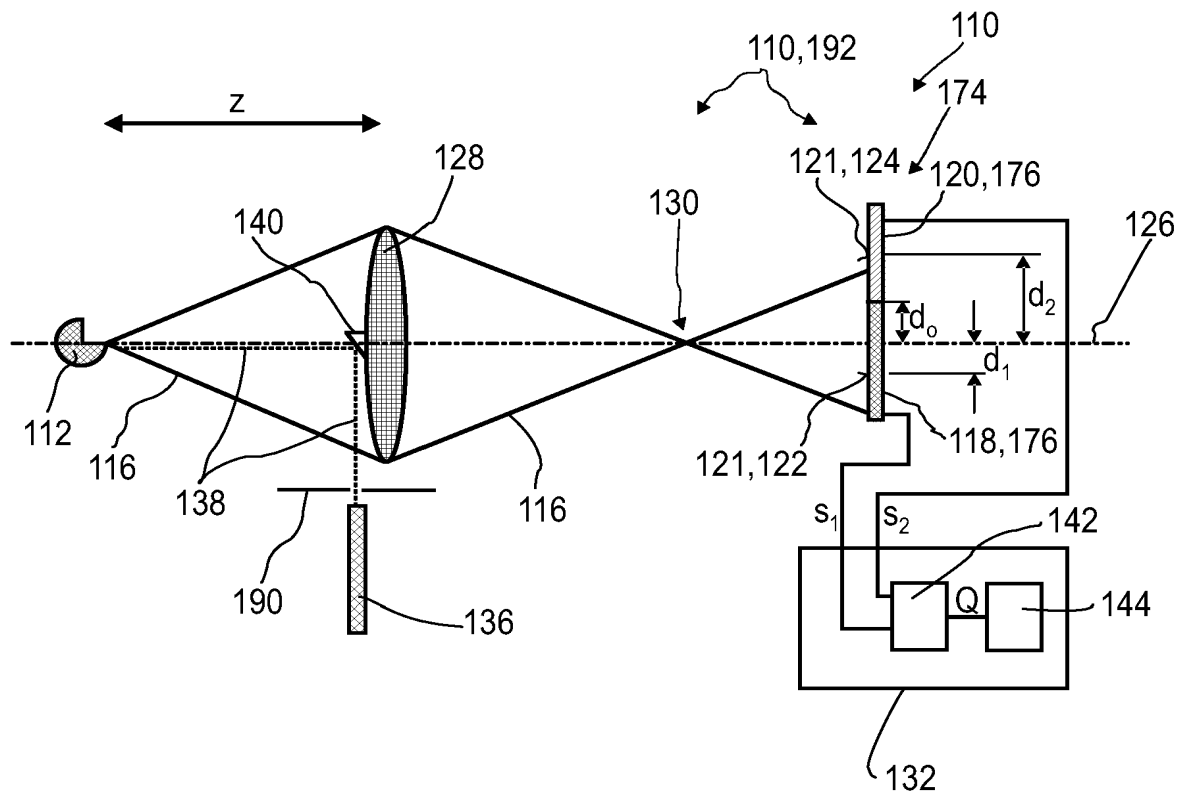


FIG.13

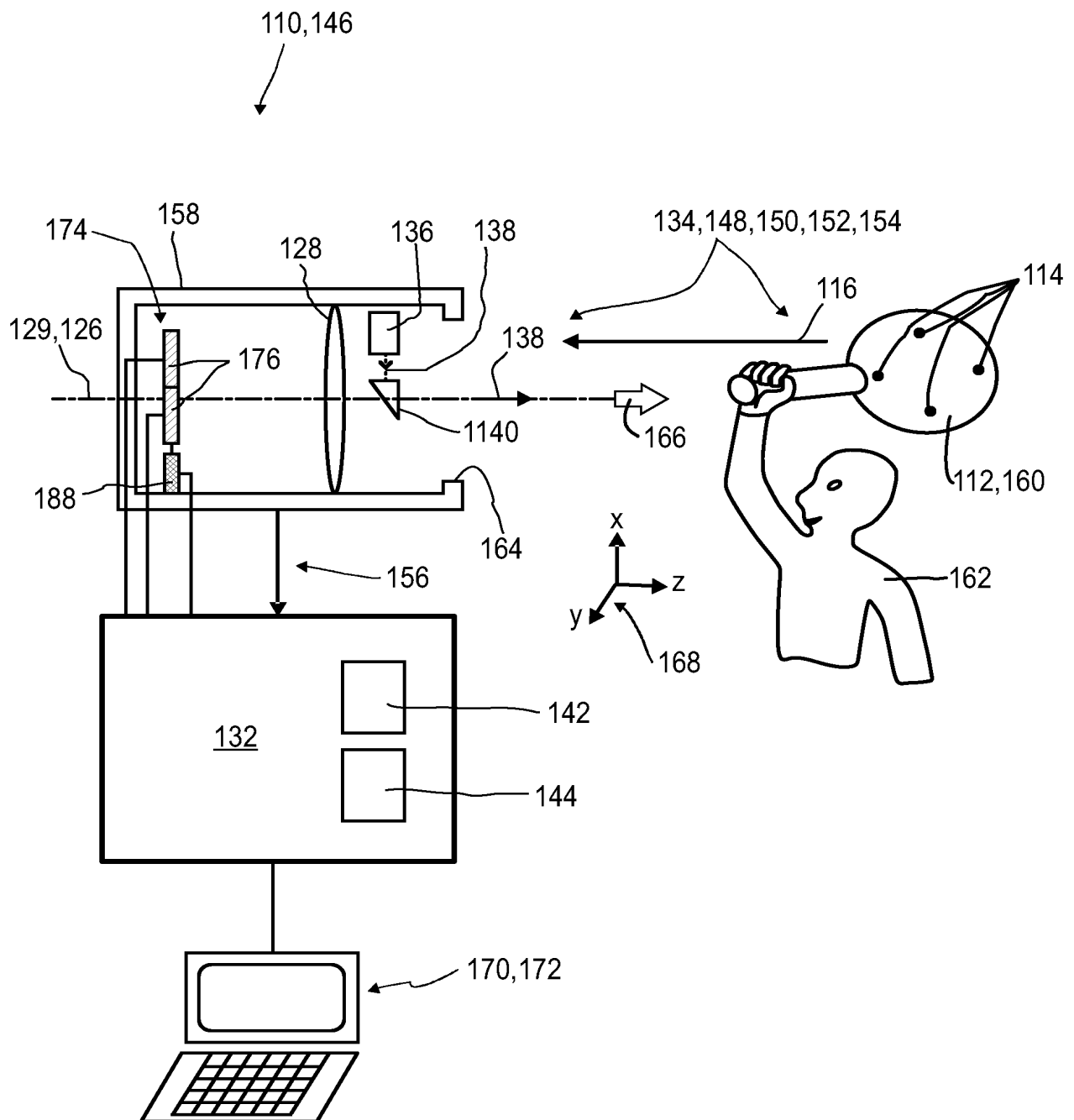


FIG.14

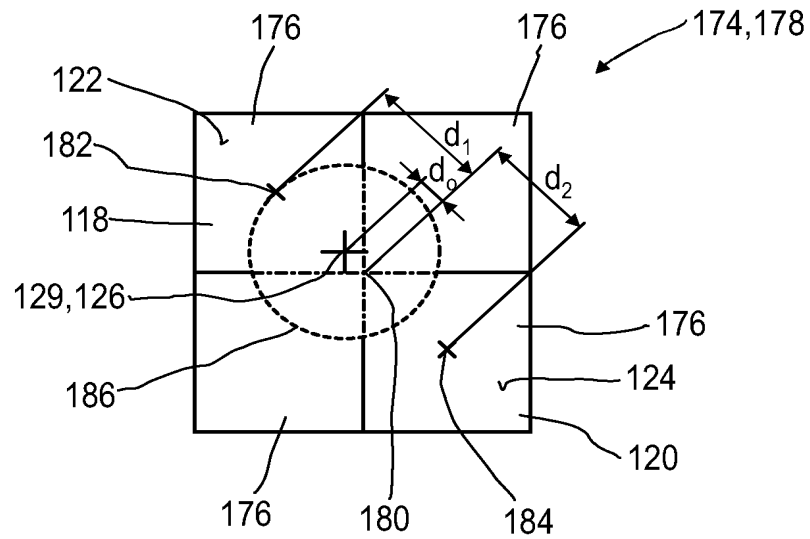


FIG.15

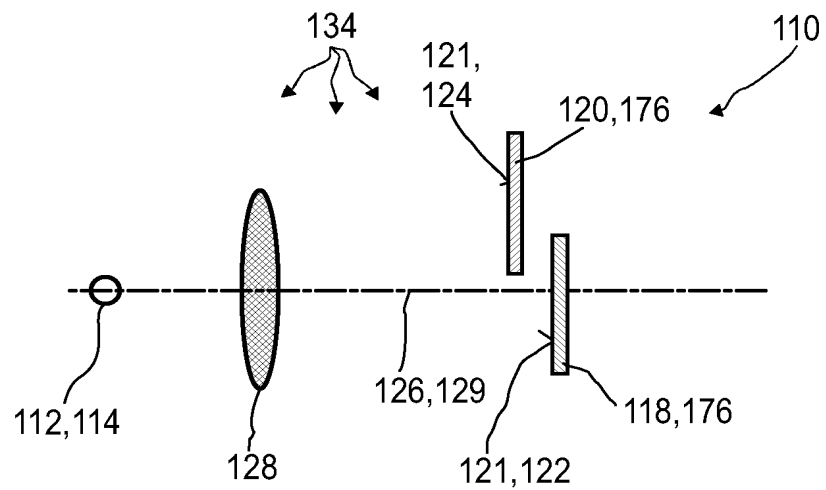


FIG.16

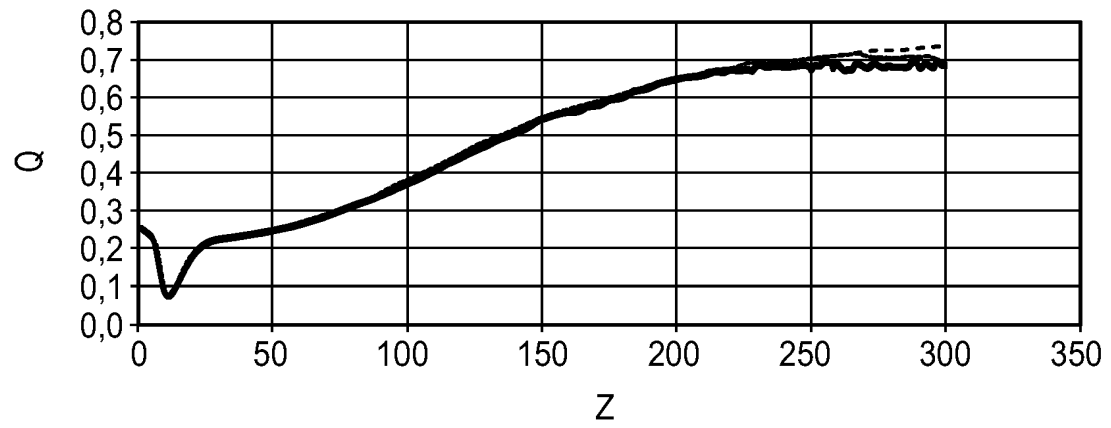


FIG.17

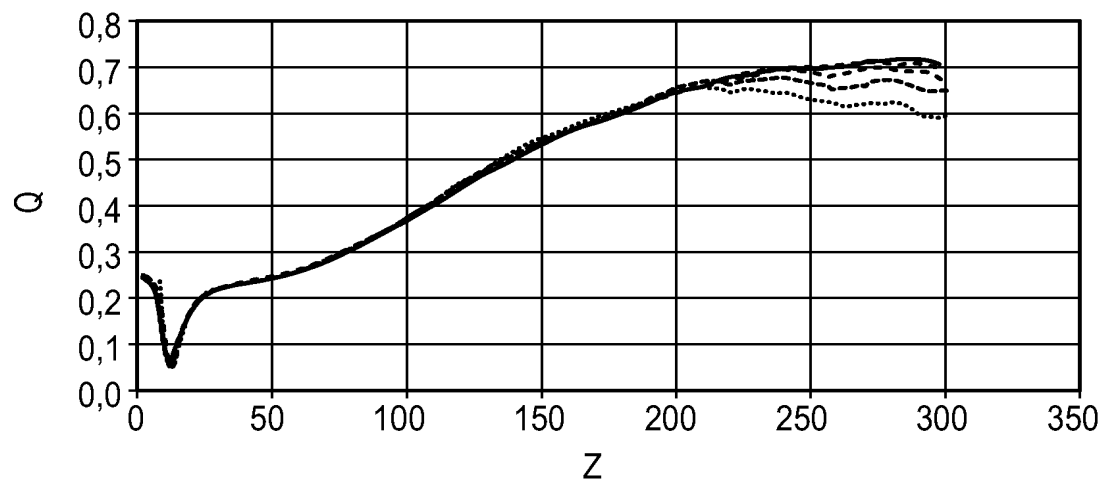


FIG.18

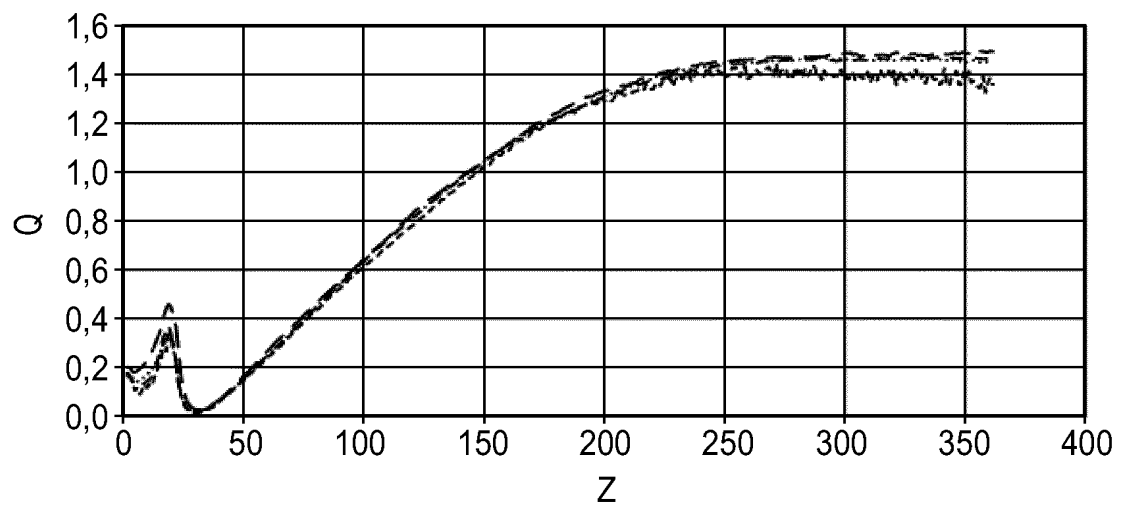


FIG.19A

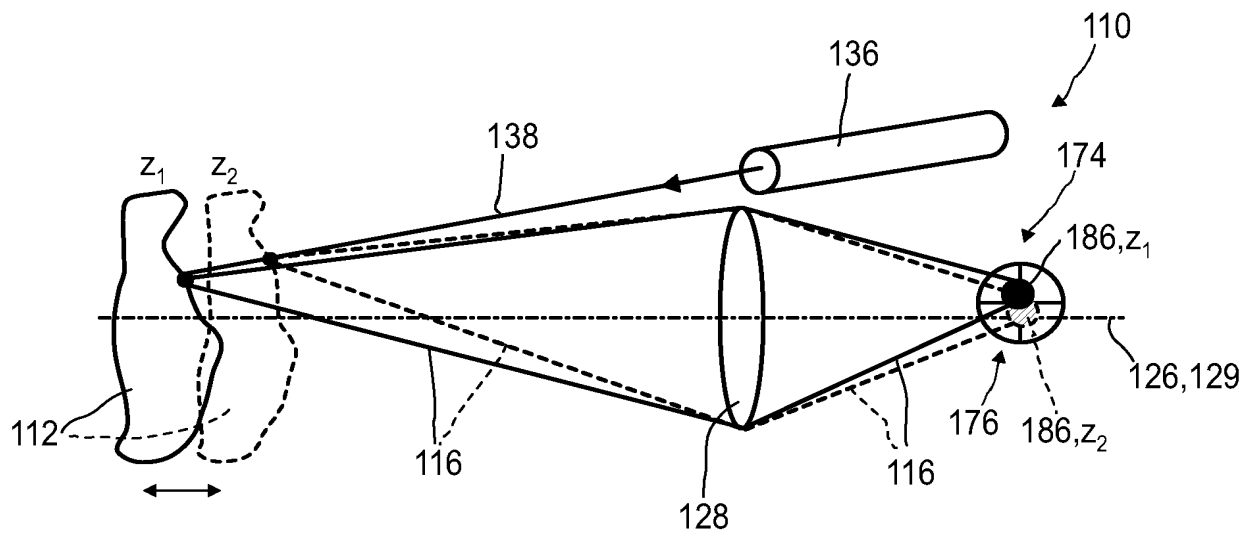


FIG.19B

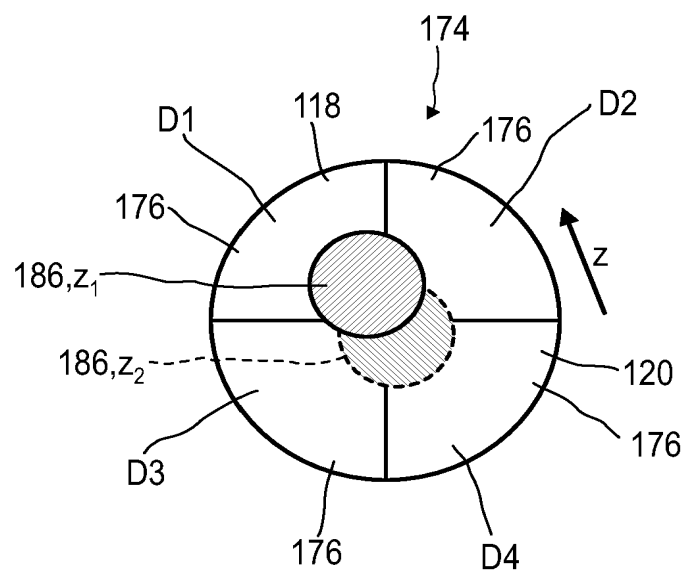


FIG.19C

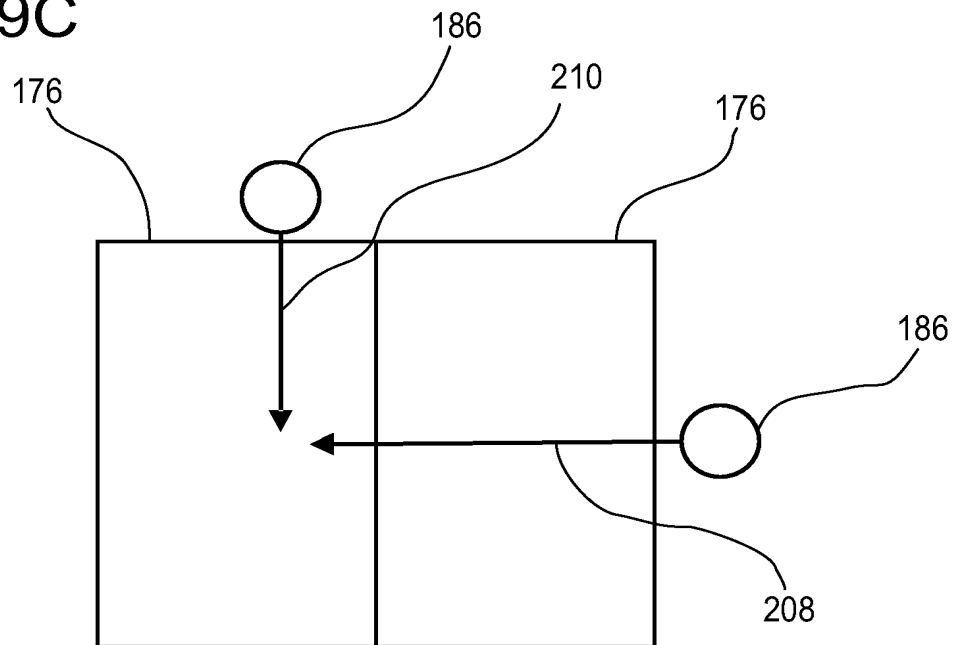


FIG.19D

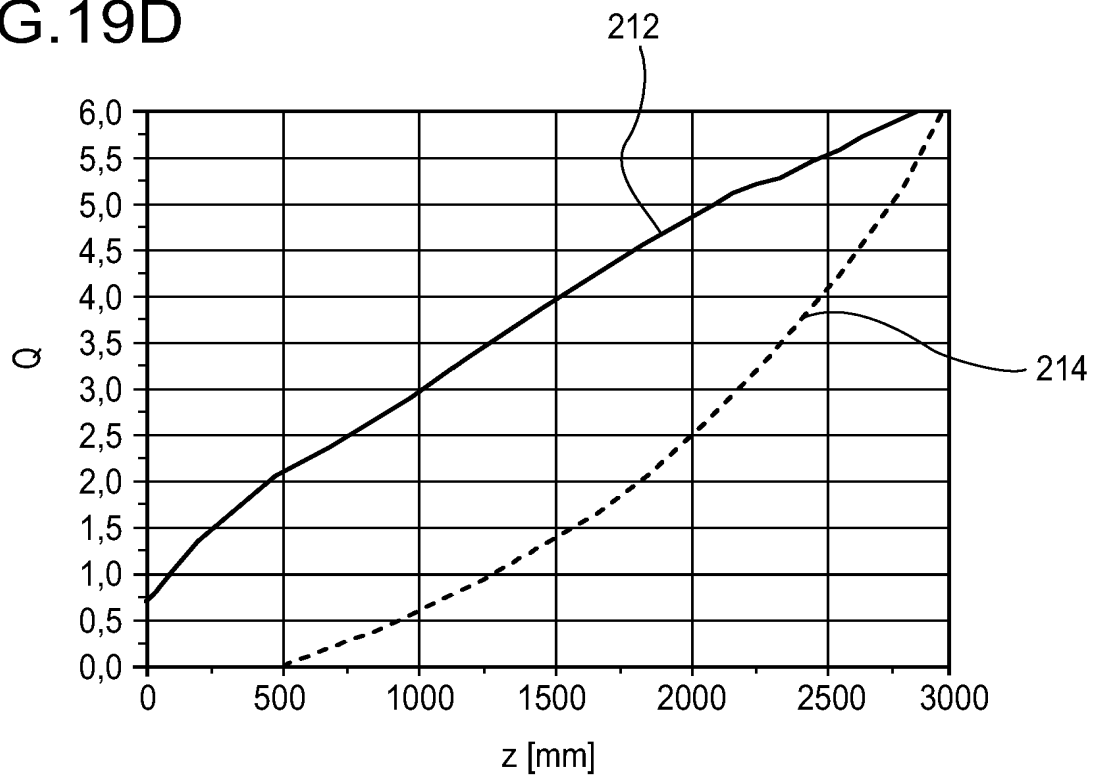


FIG.19E

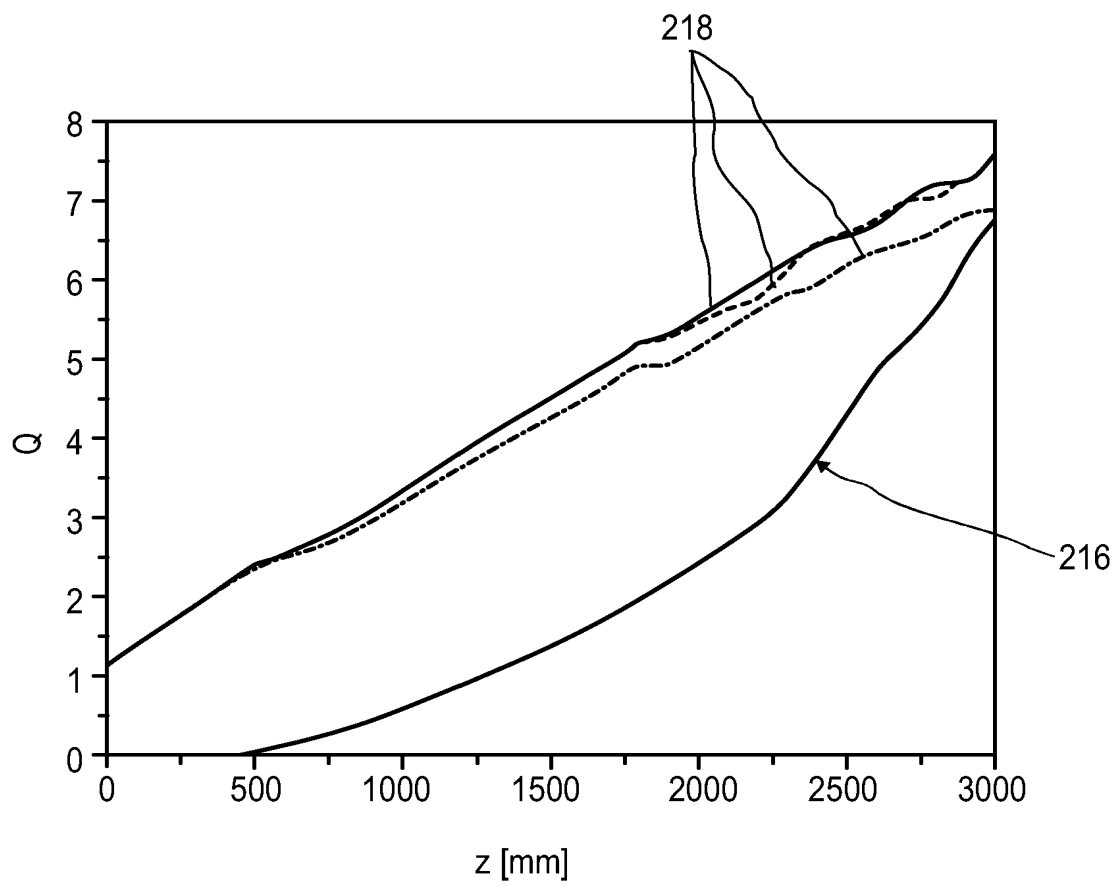


FIG.20

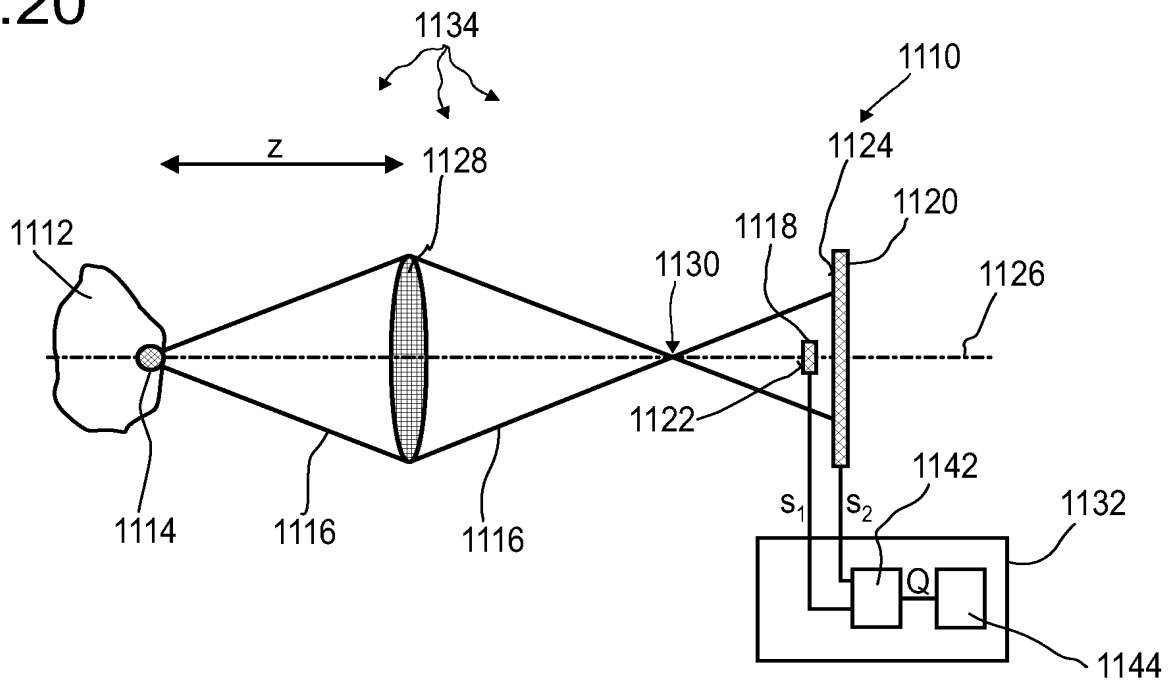


FIG.21

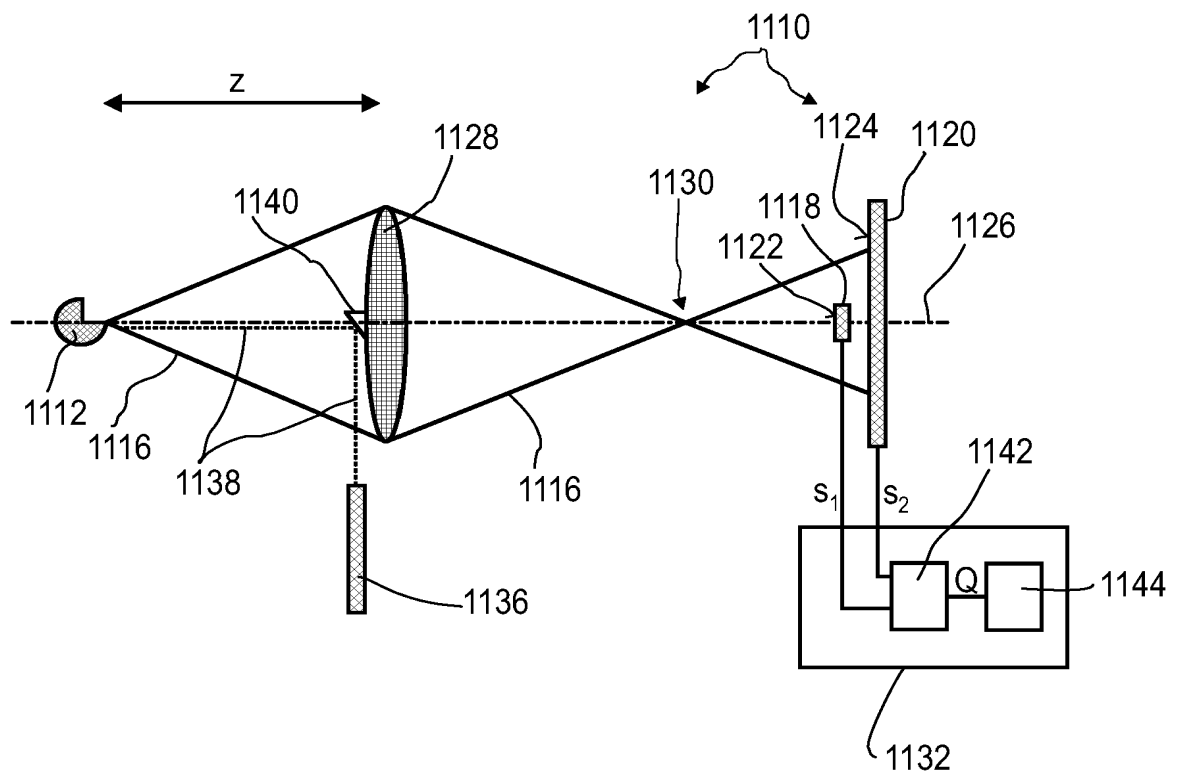


FIG.22A

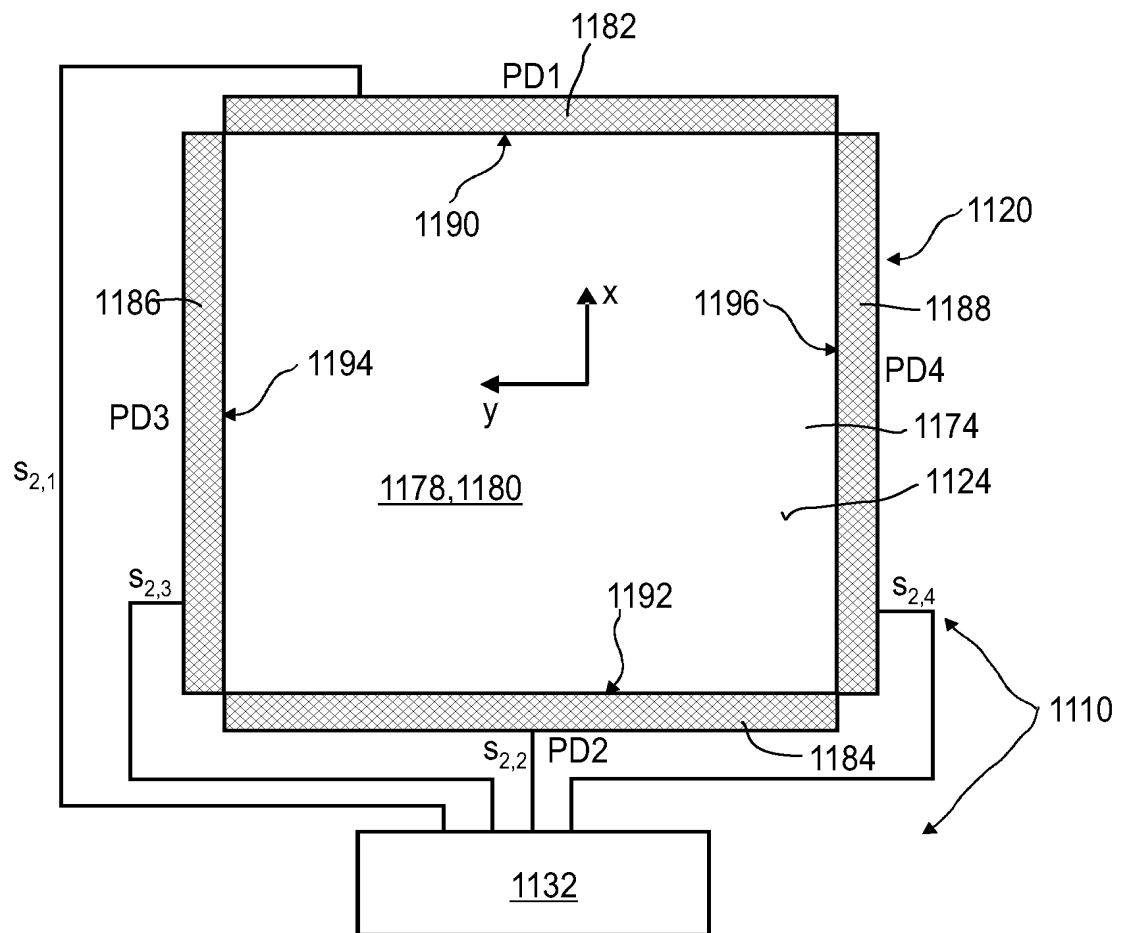


FIG.22B

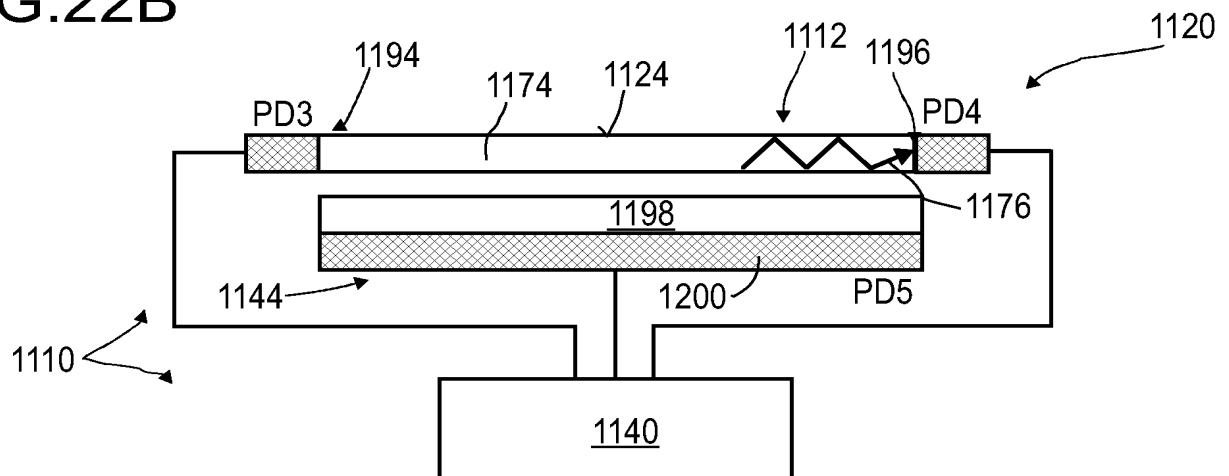


FIG.23

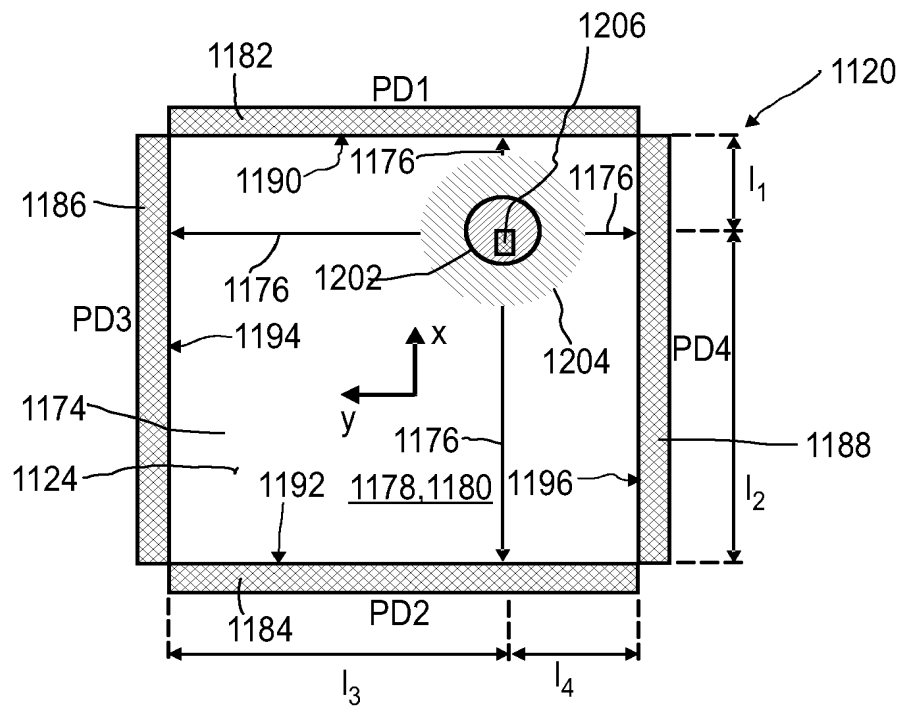


FIG.24

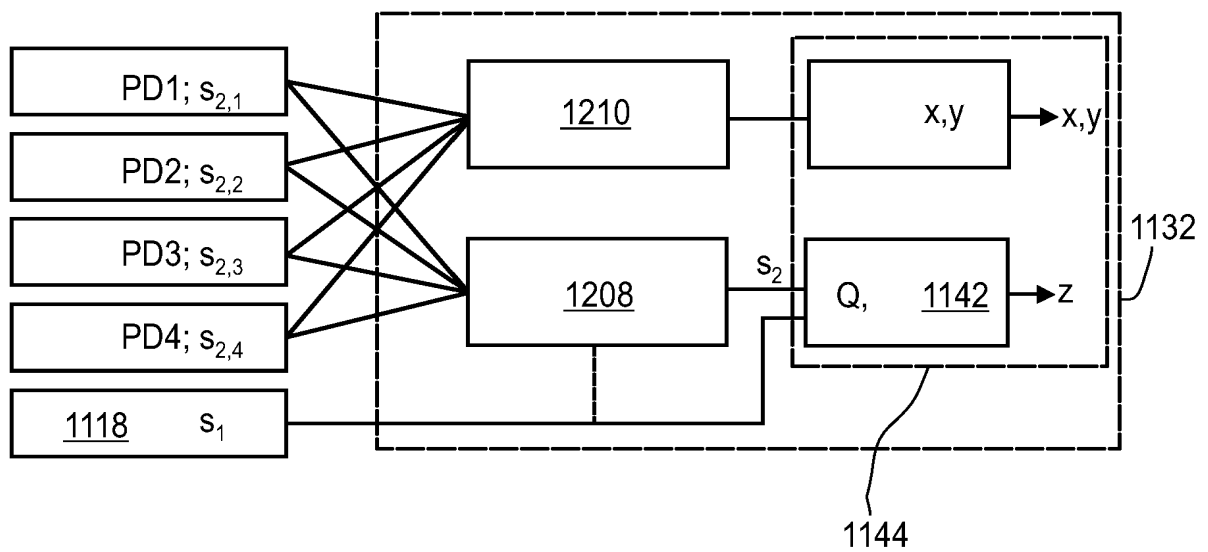


FIG.25

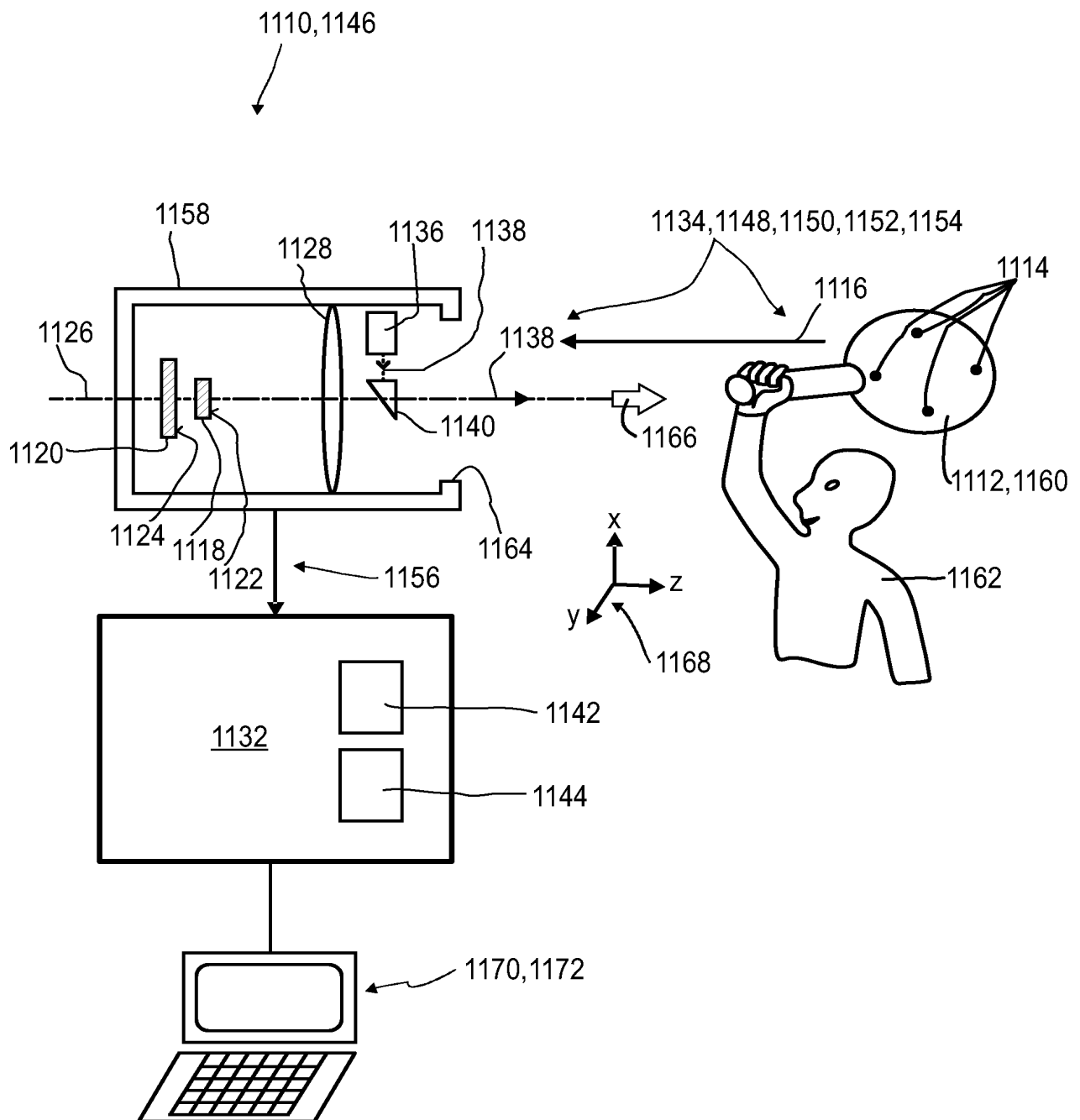


FIG.26A

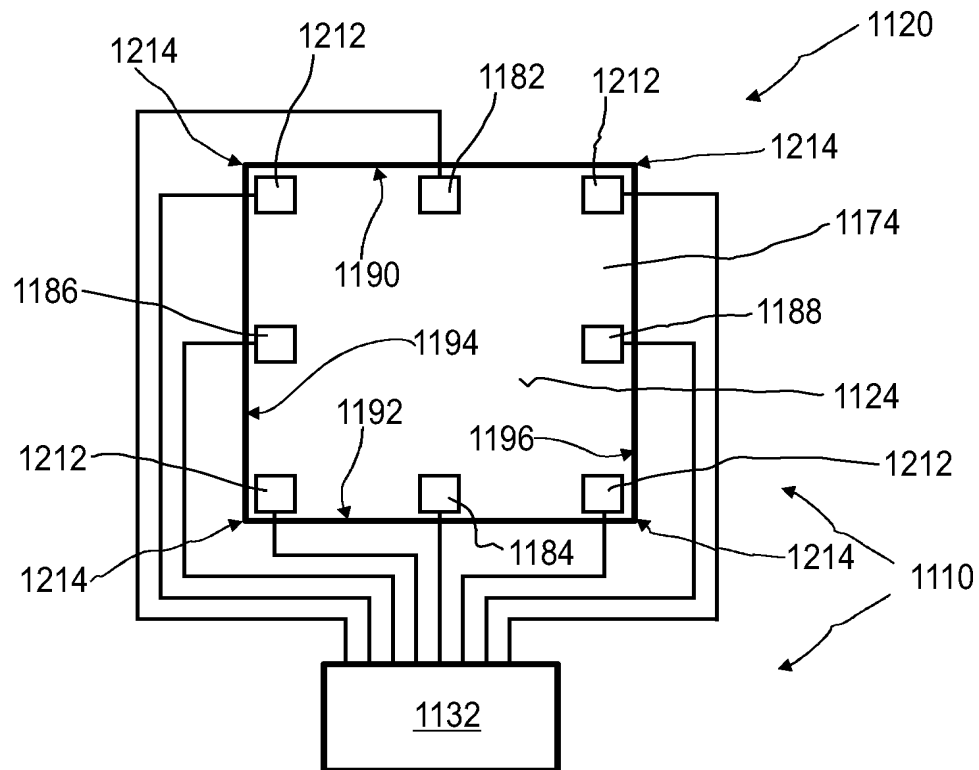


FIG.26B

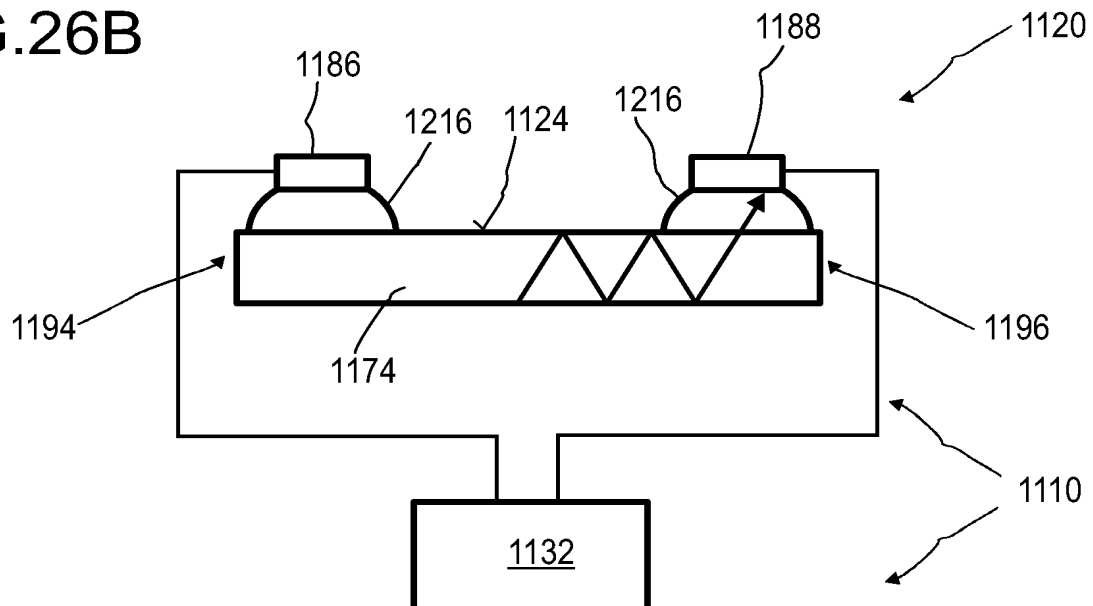


FIG.27A

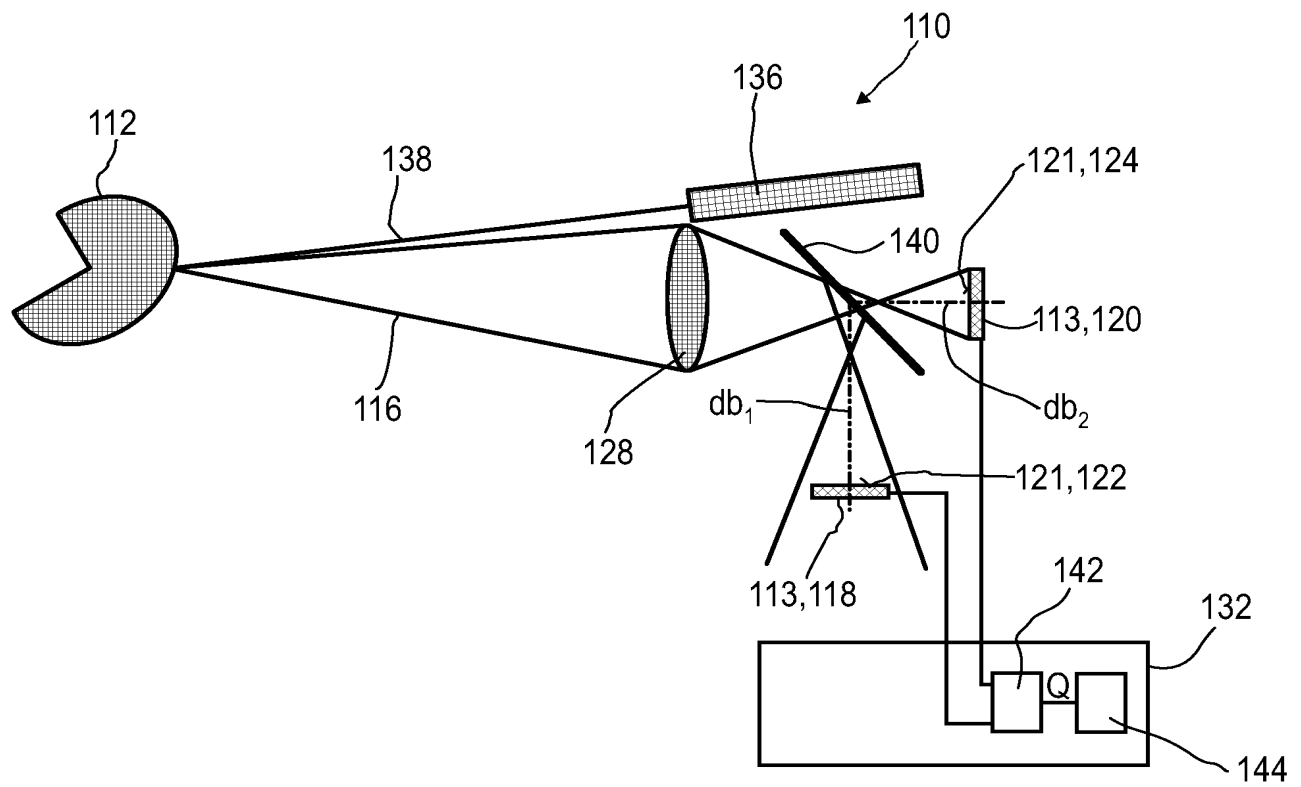


FIG.27B

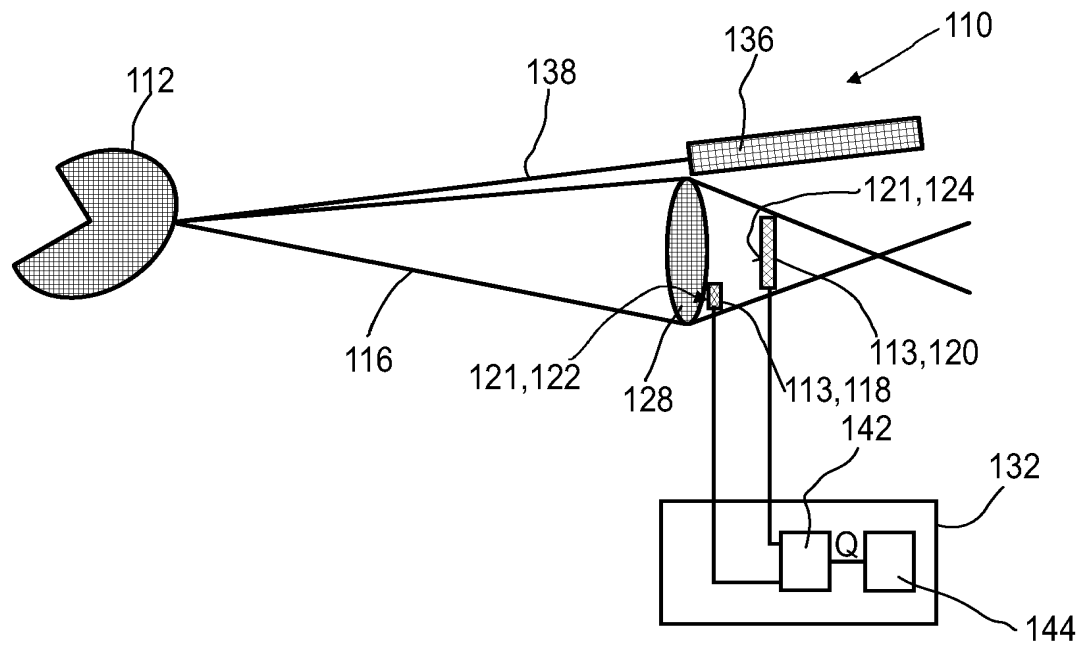


FIG.28

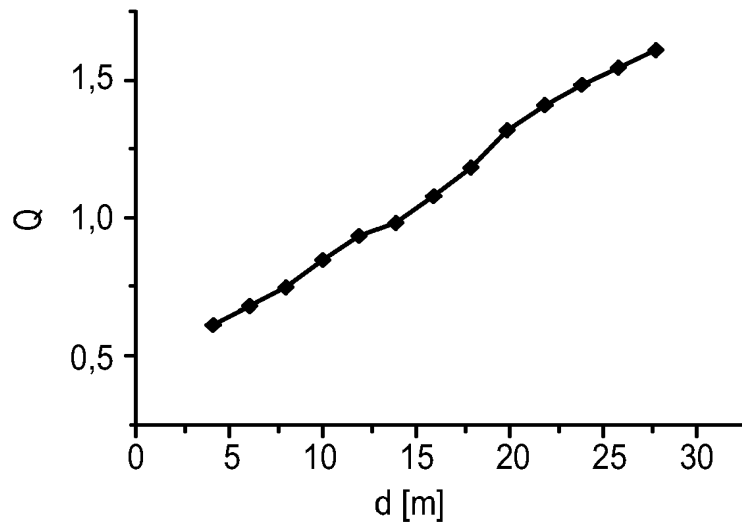


FIG.29

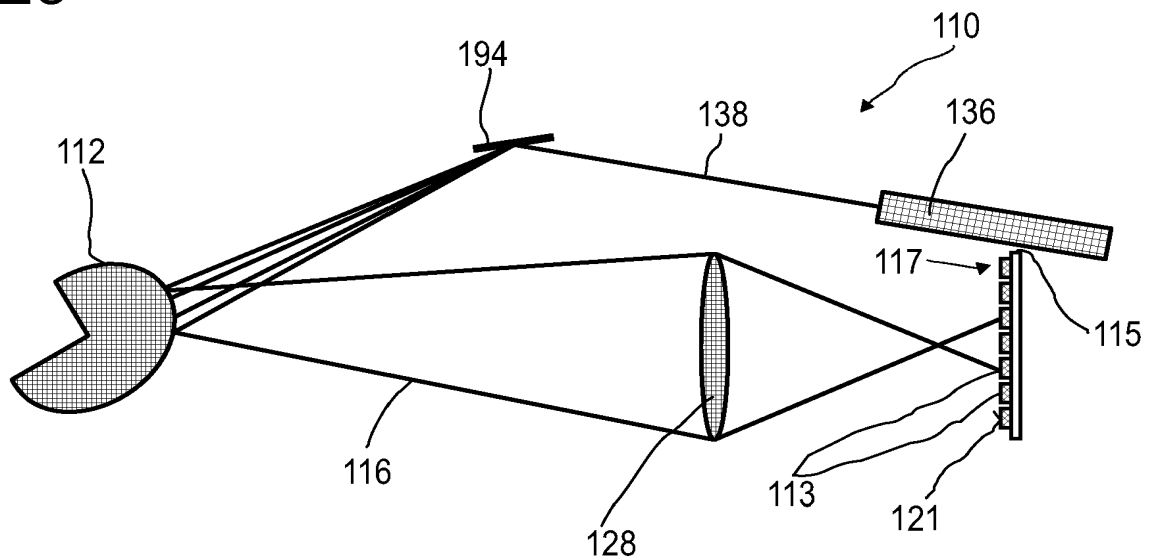


FIG.30

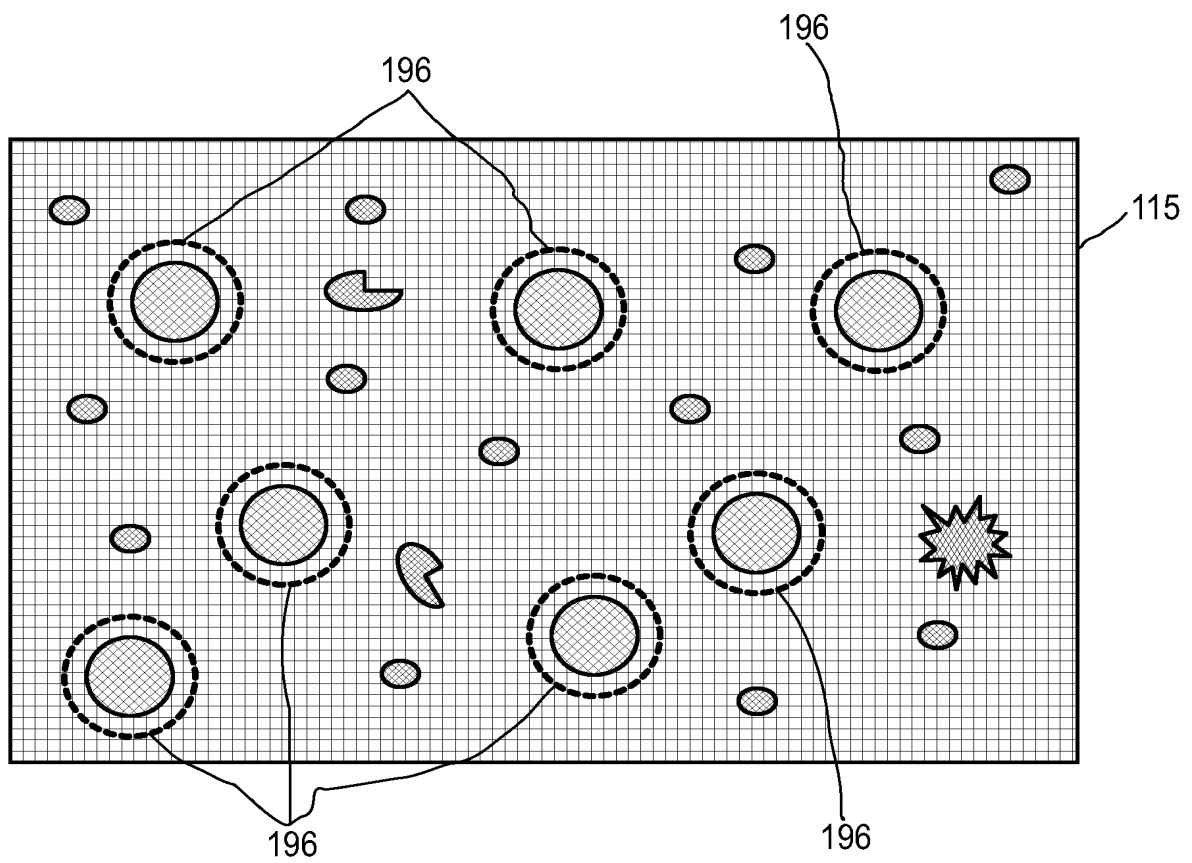


FIG.31

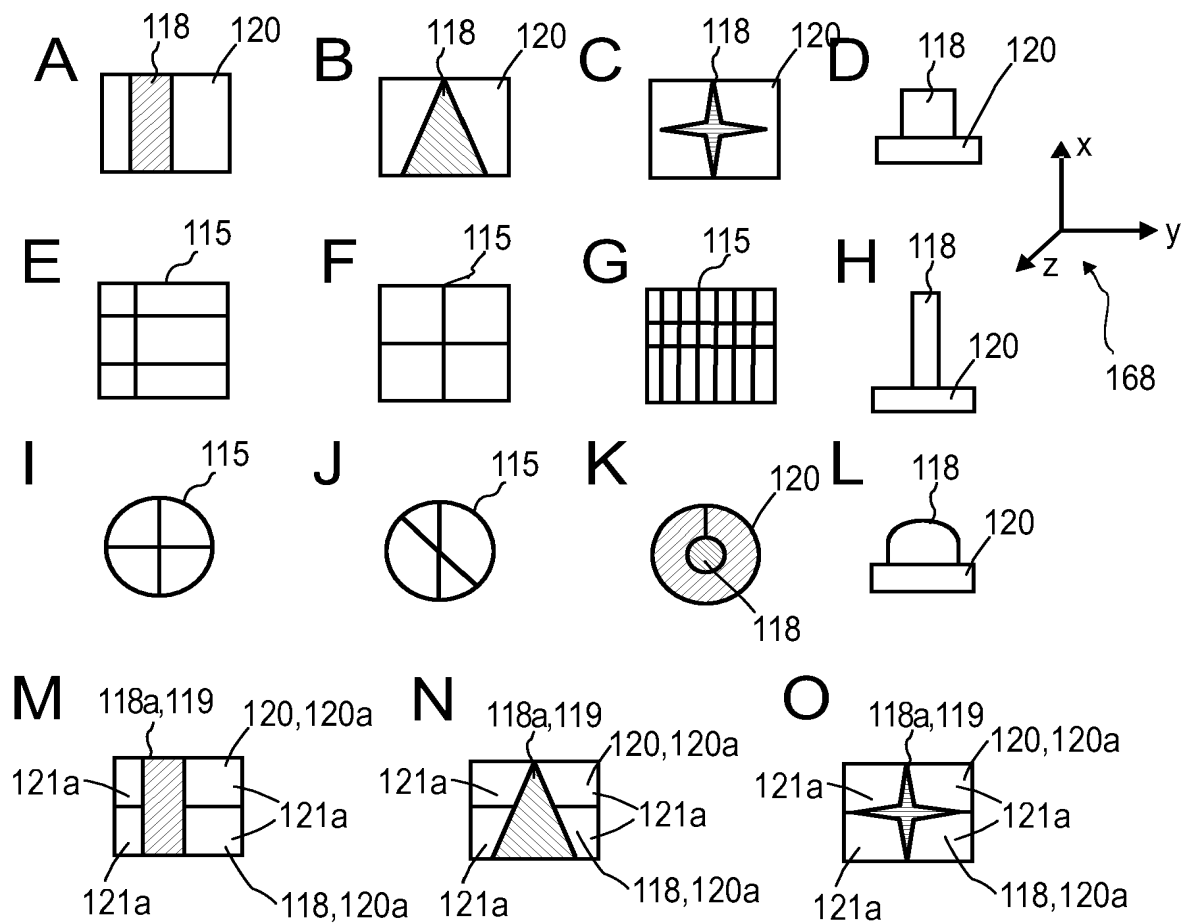


FIG.32

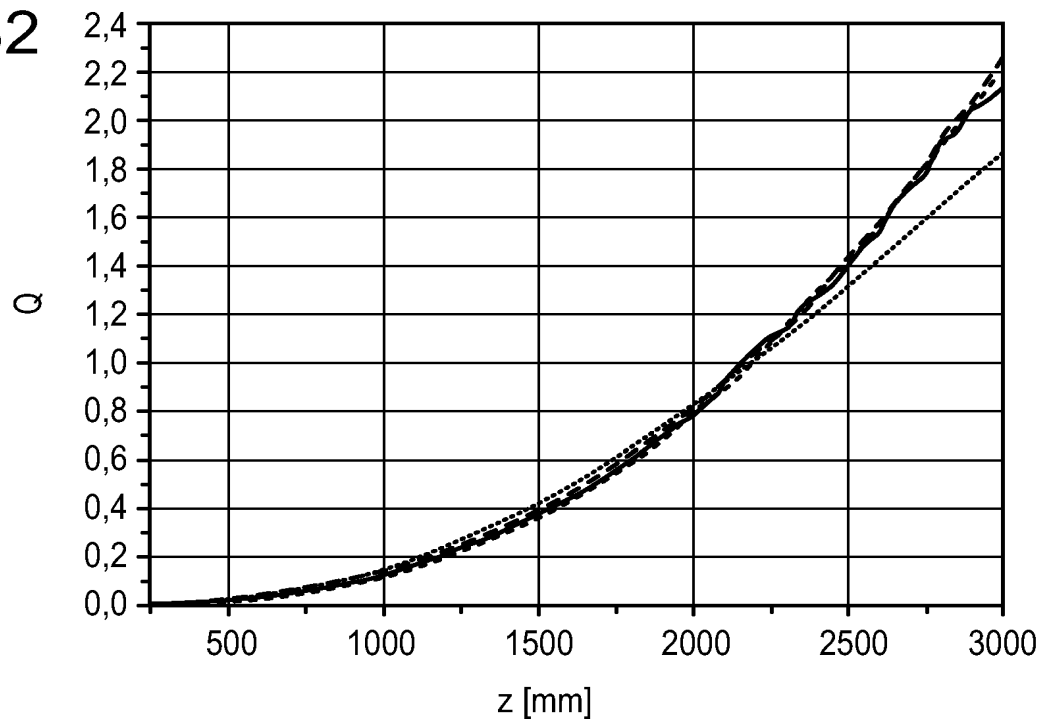


FIG.33A

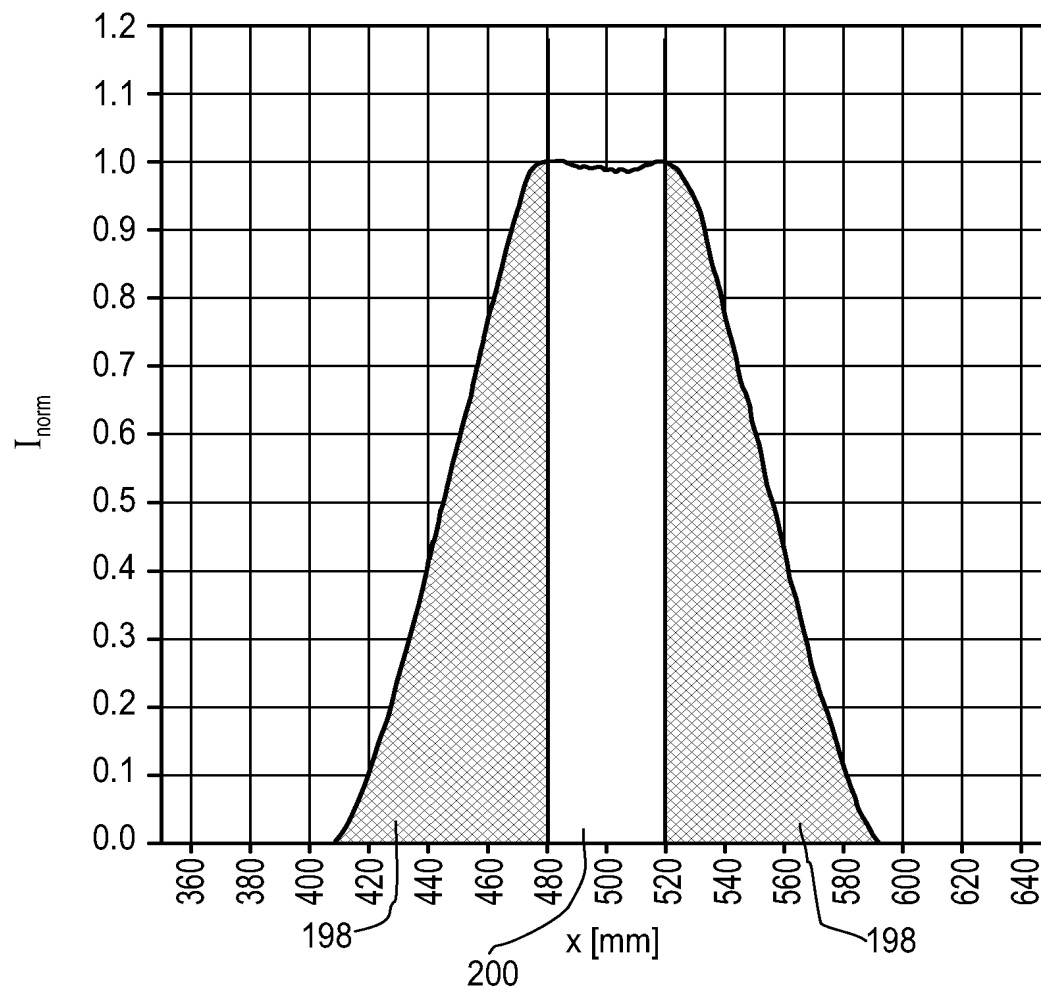


FIG.33B

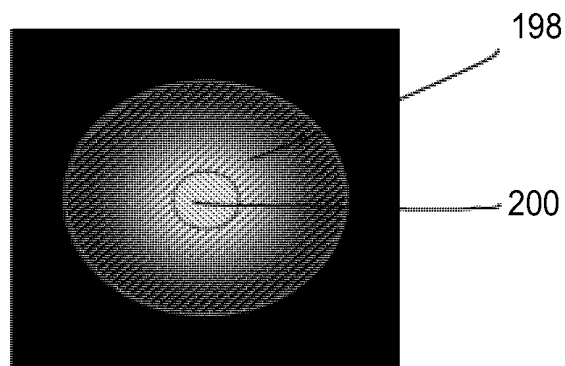


FIG.34

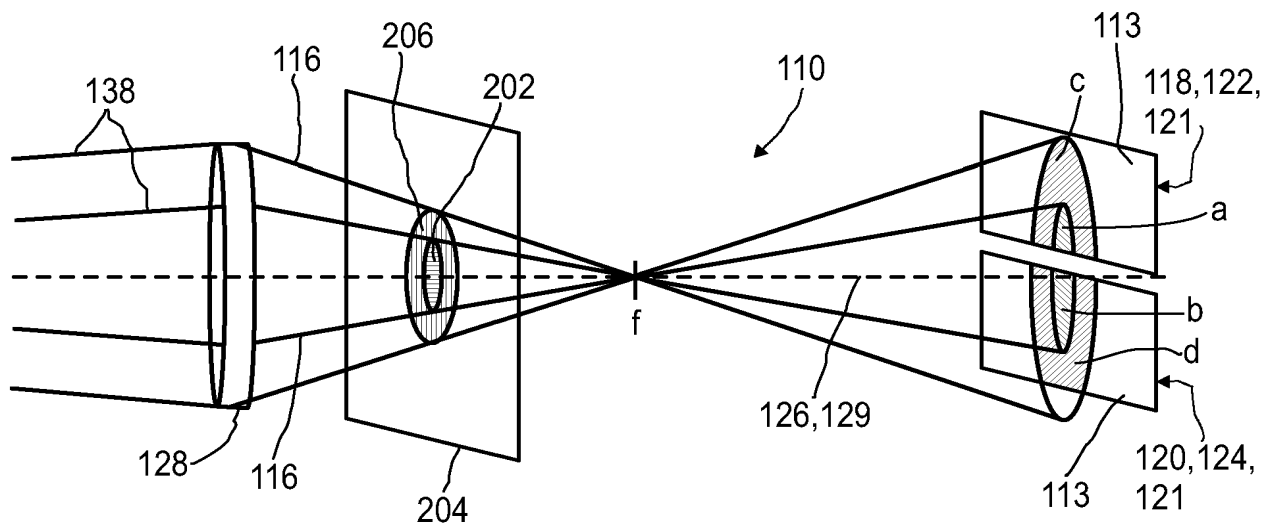


FIG.35

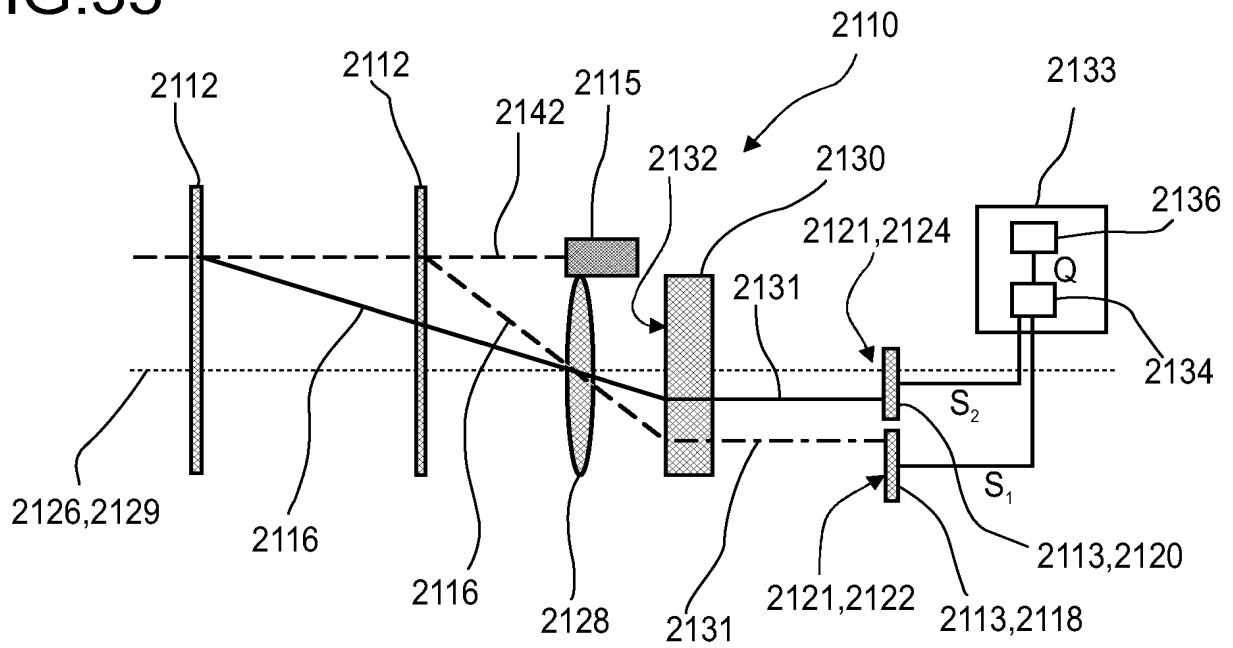


FIG.36

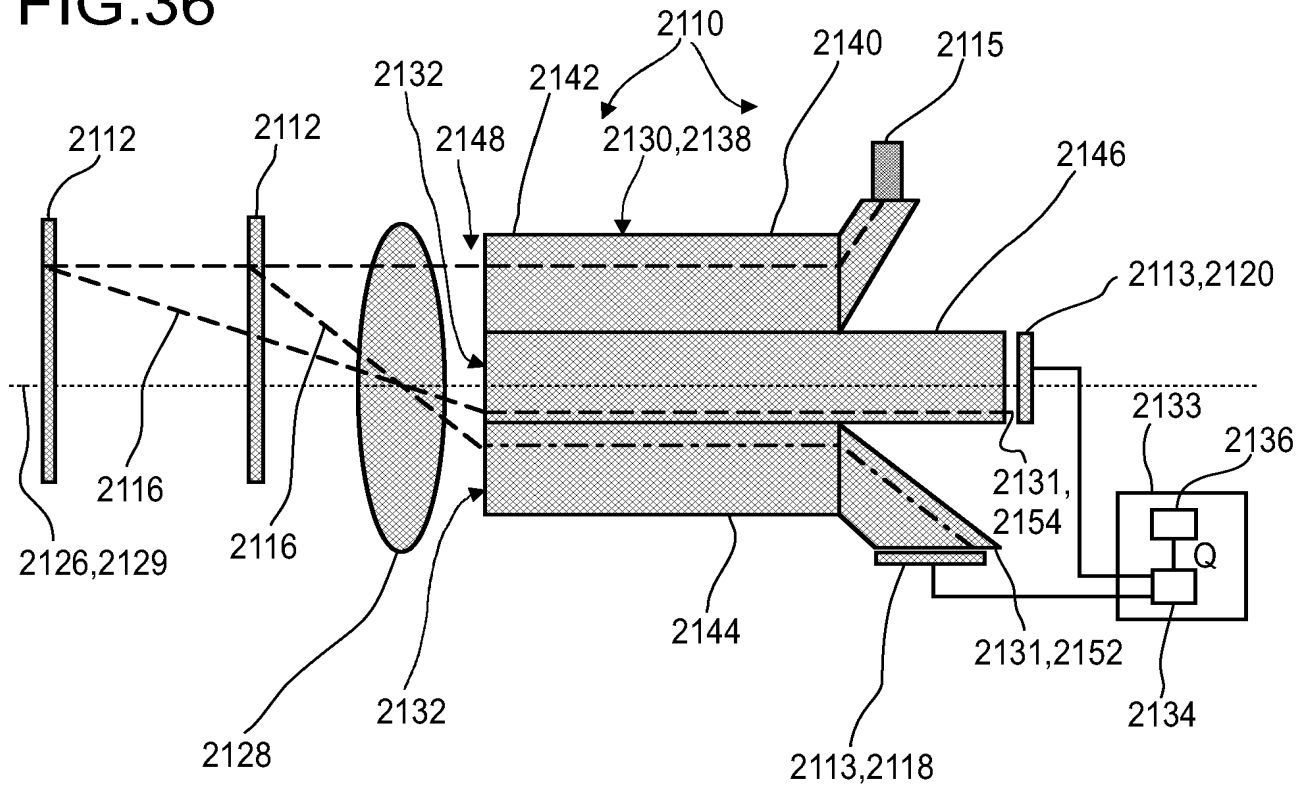


FIG.37

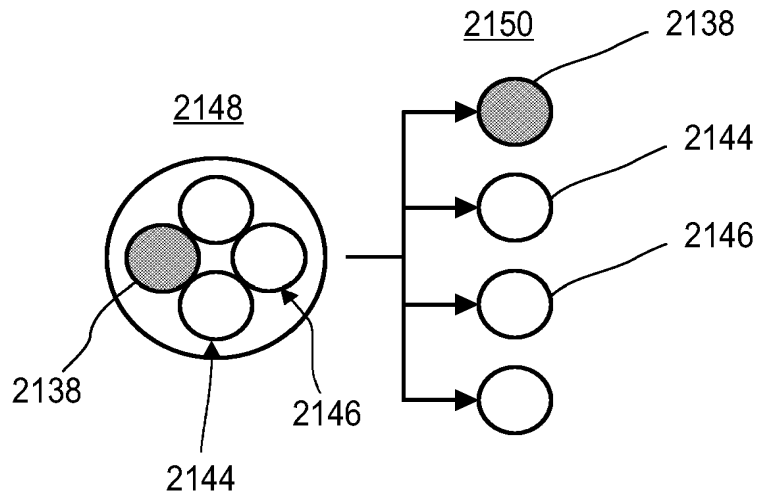


FIG.38

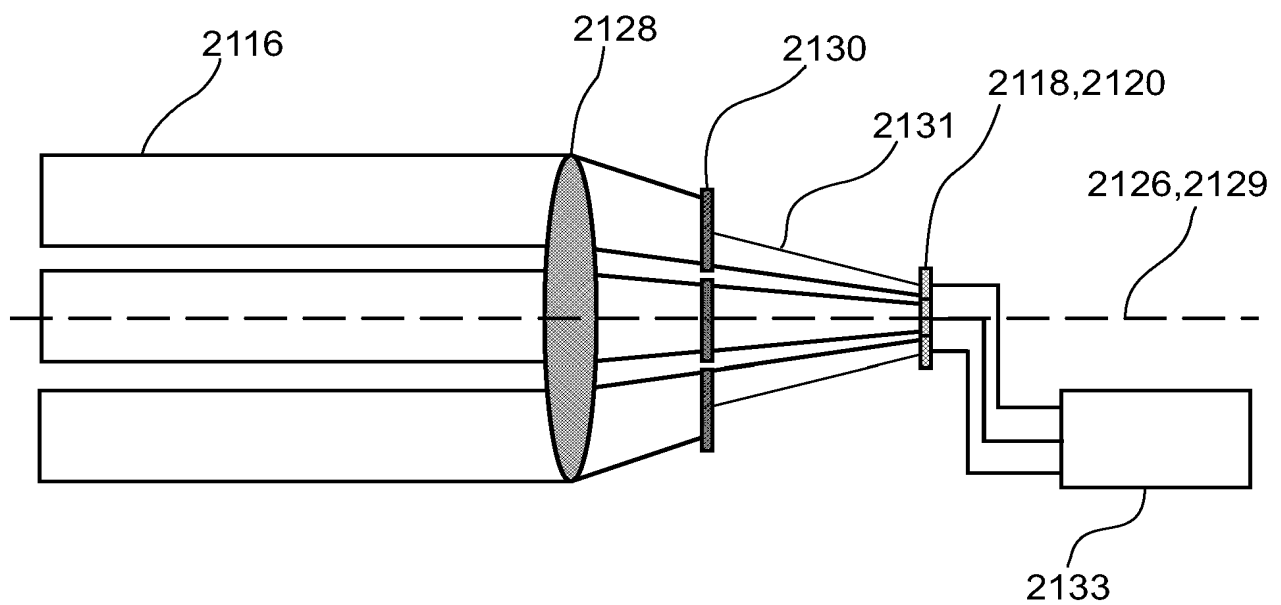


FIG.39

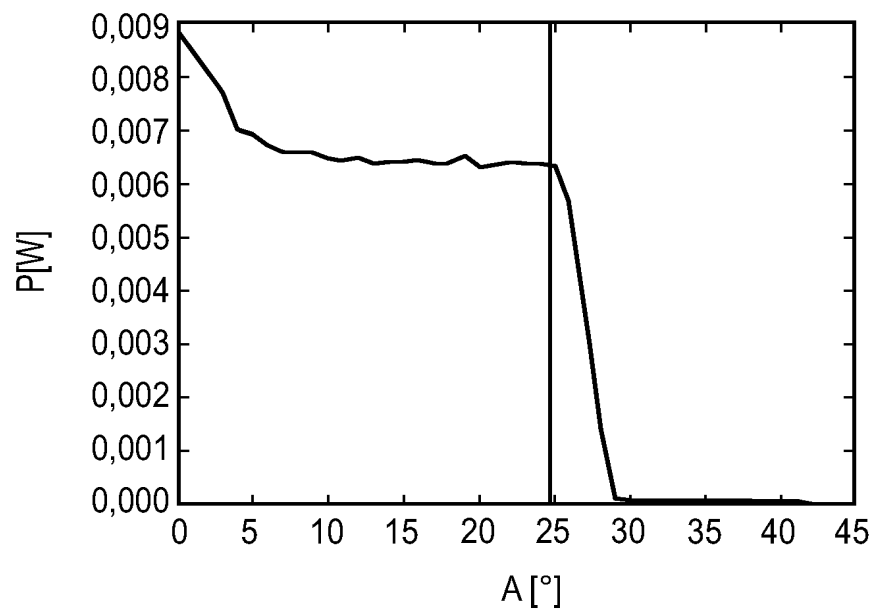


FIG.40A

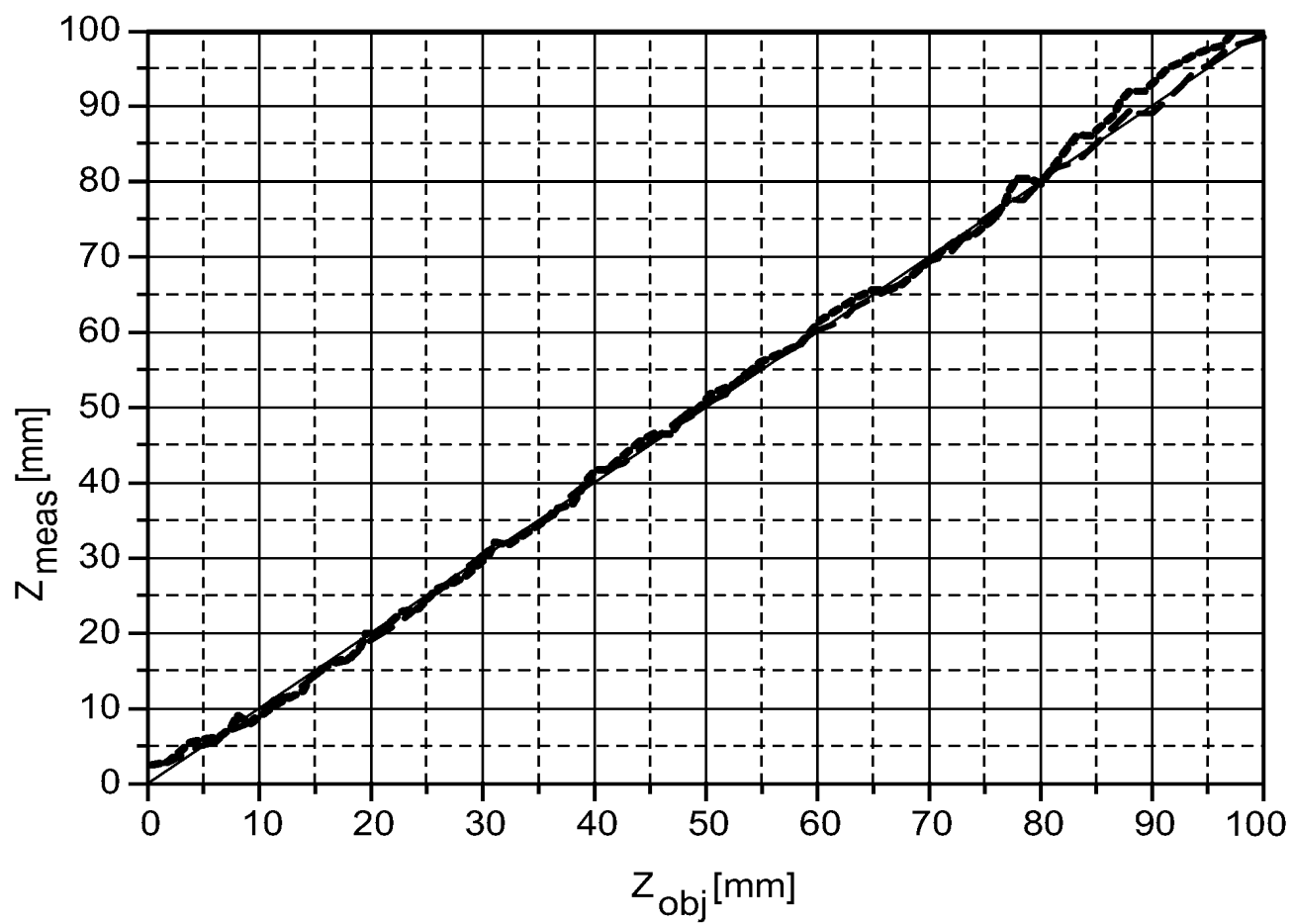


FIG.40B

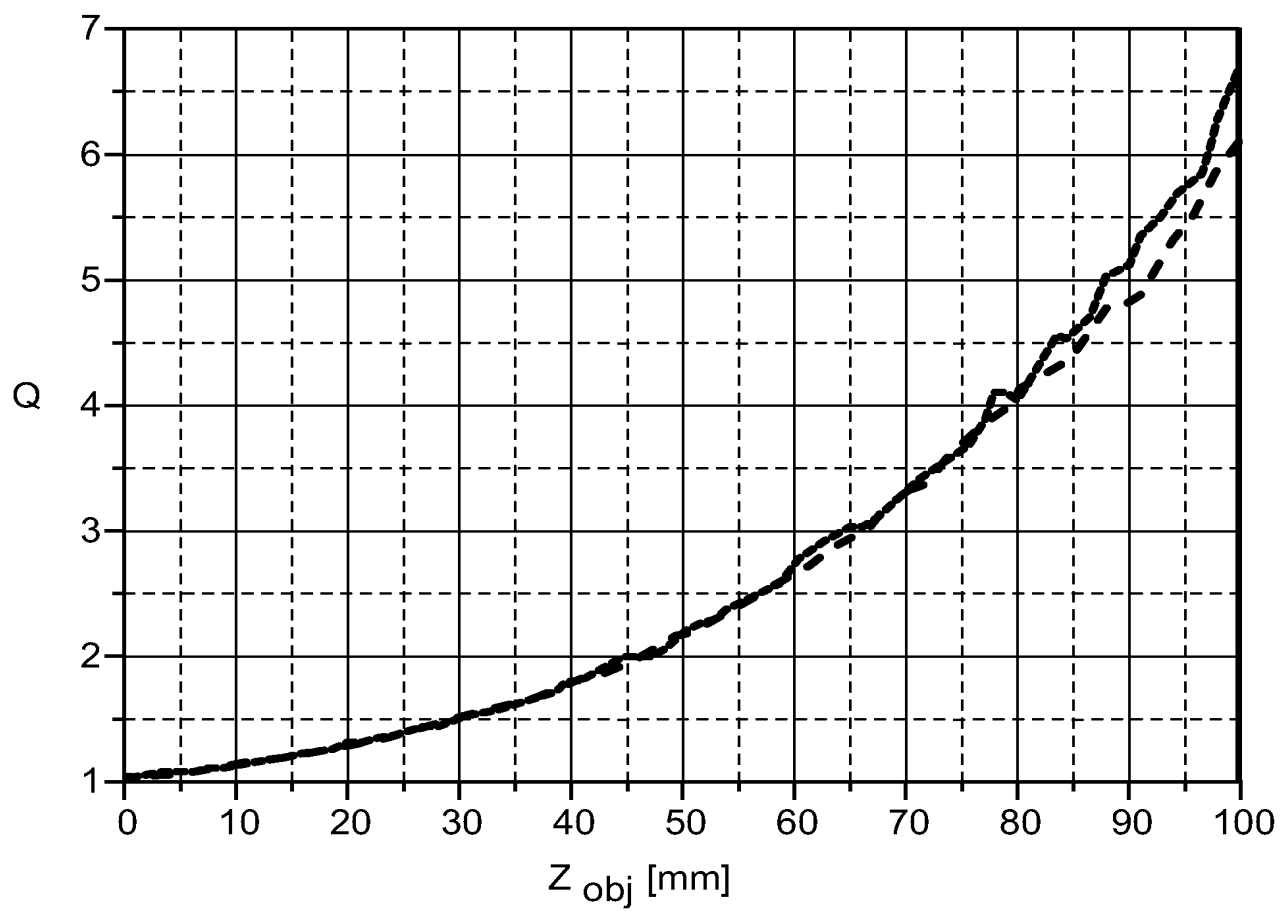


FIG.41

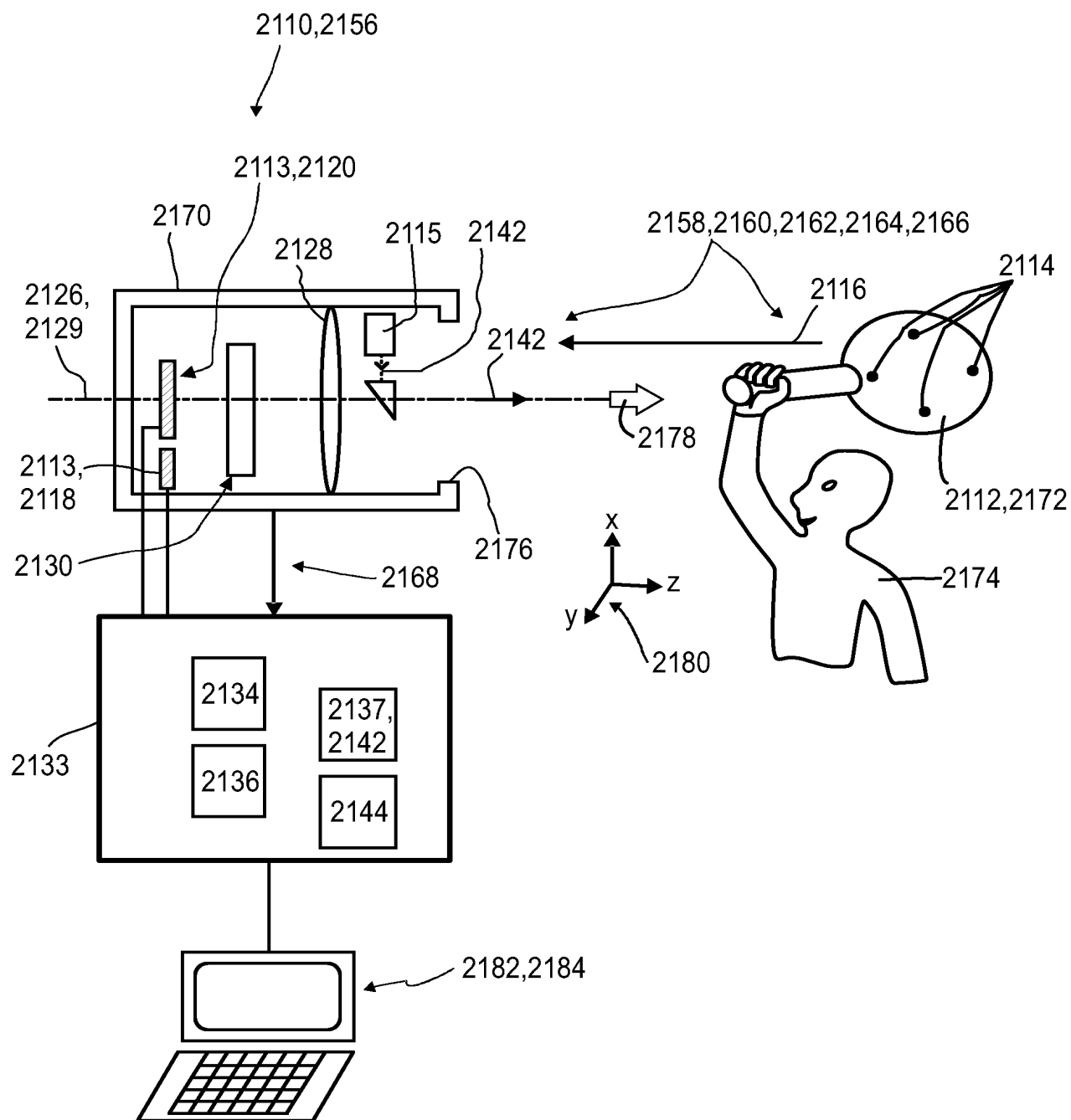


FIG.42

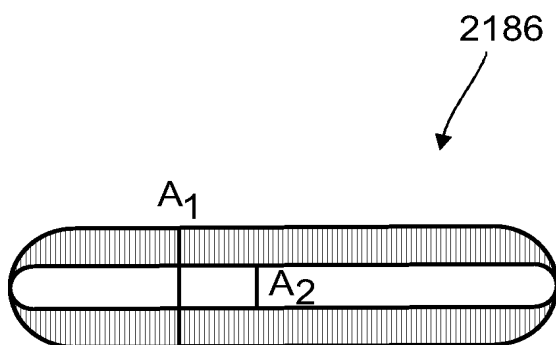


FIG.43

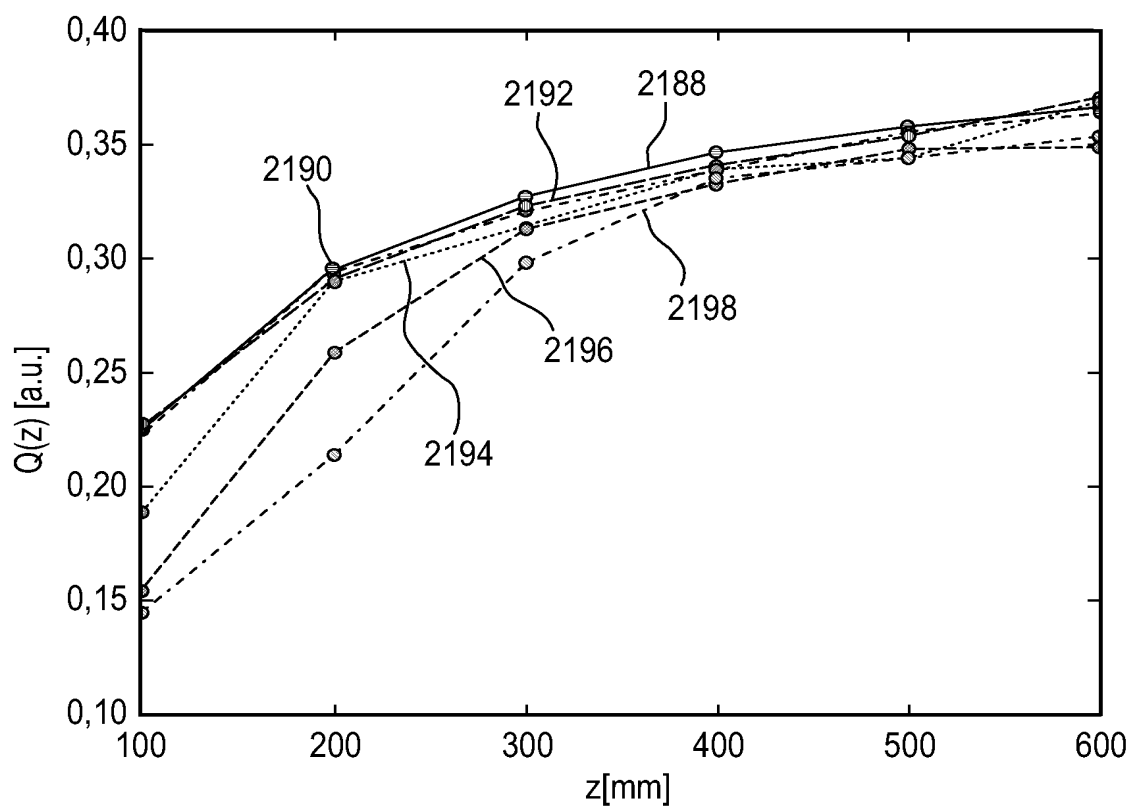


FIG.44A

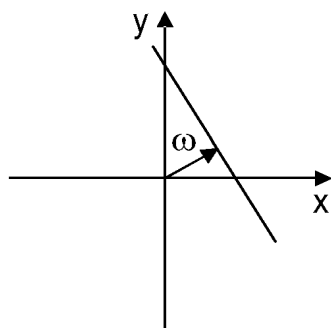


FIG.44B

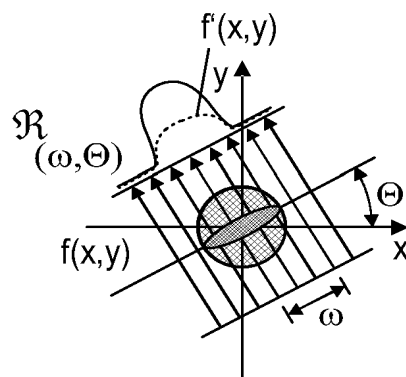


FIG.45A

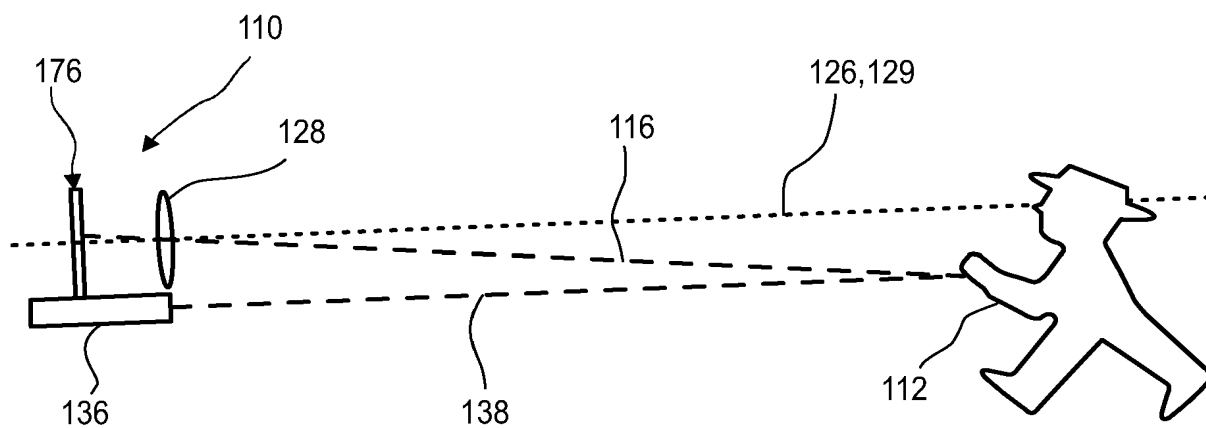


FIG.45B

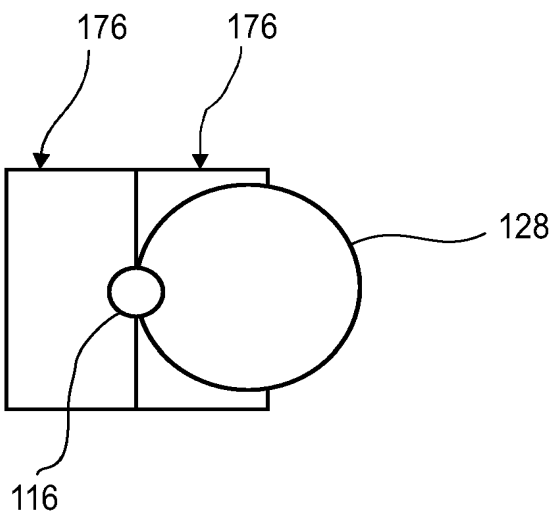


FIG.46

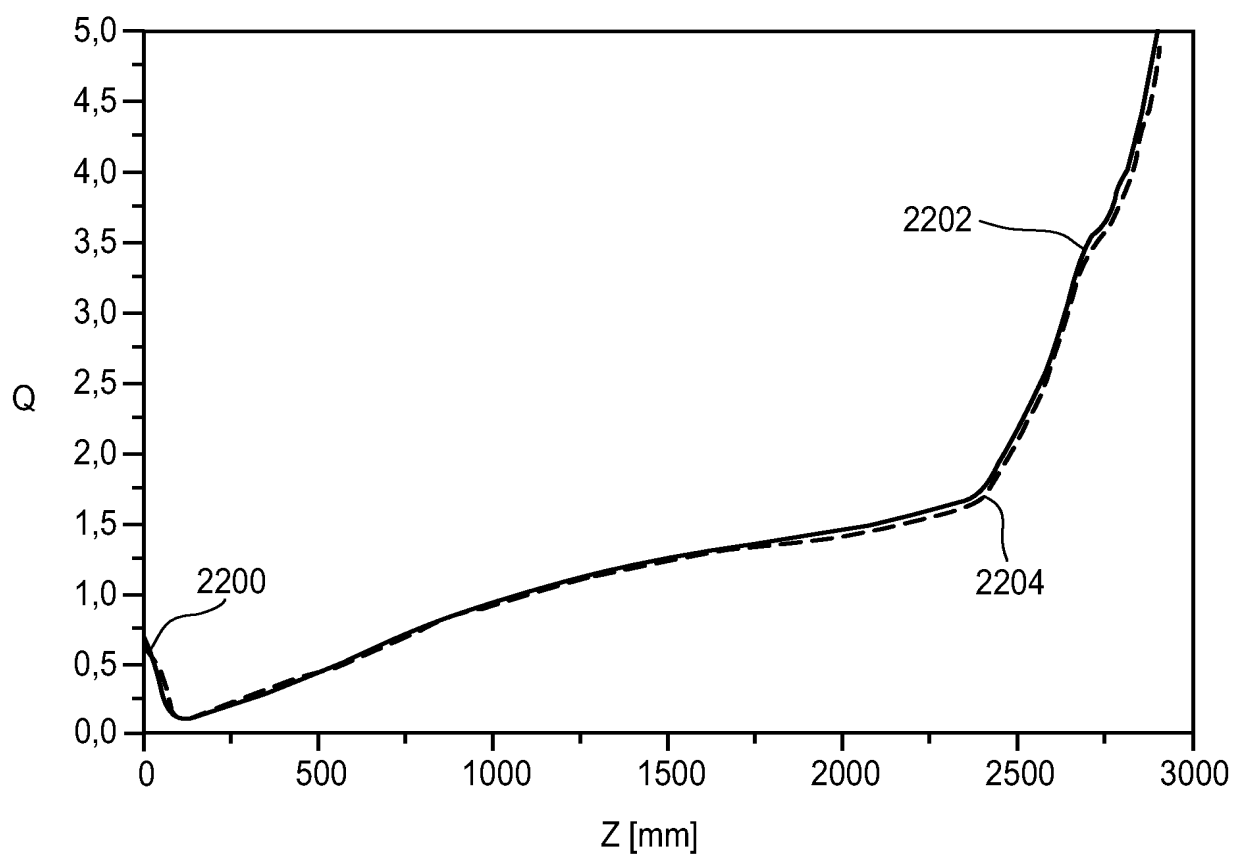


FIG.47A

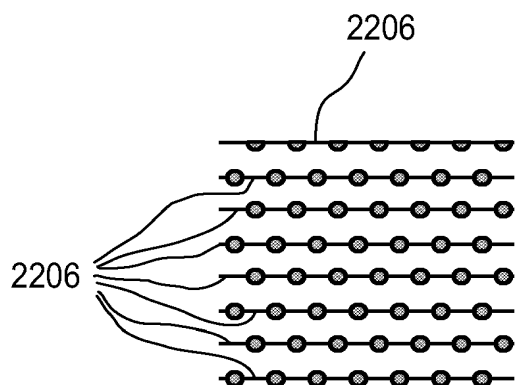


FIG.47B

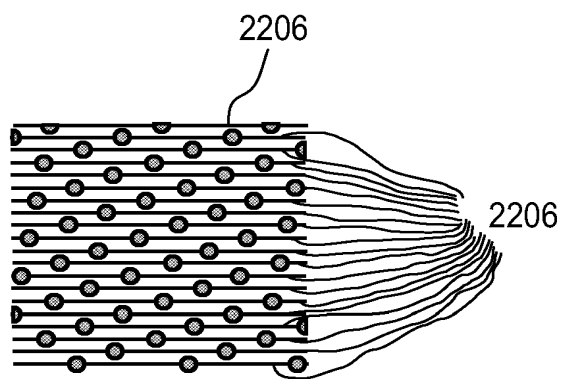


FIG.47C

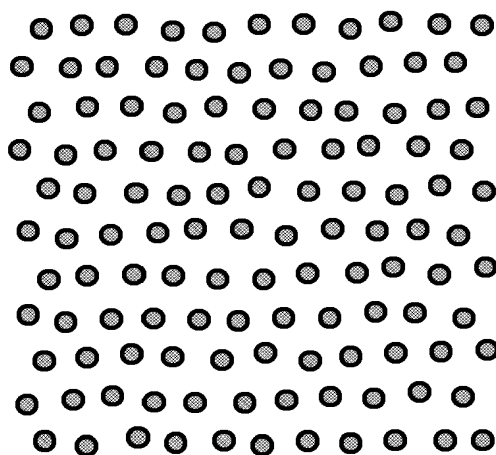
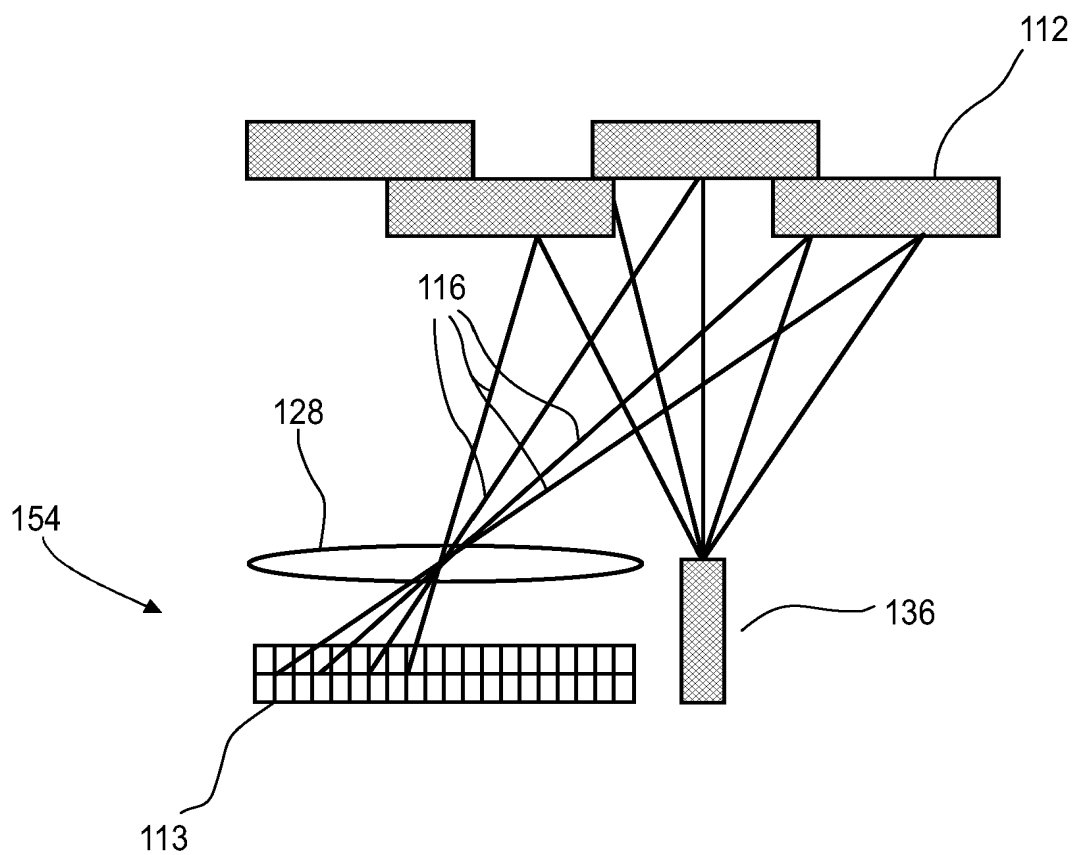


FIG.48





(51) International Patent Classification:

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G01S 7/481 (2006.01) G01S 11/12 (2006.01)
G01S 5/16 (2006.01) G01S 7/48 (2006.01)

17195427.4 09 October 2017 (09.10.2017) EP
17195398.7 09 October 2017 (09.10.2017) EP

(21) International Application Number:

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(22) International Filing Date:

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(25) Filing Language:

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(26) Publication Language:

English

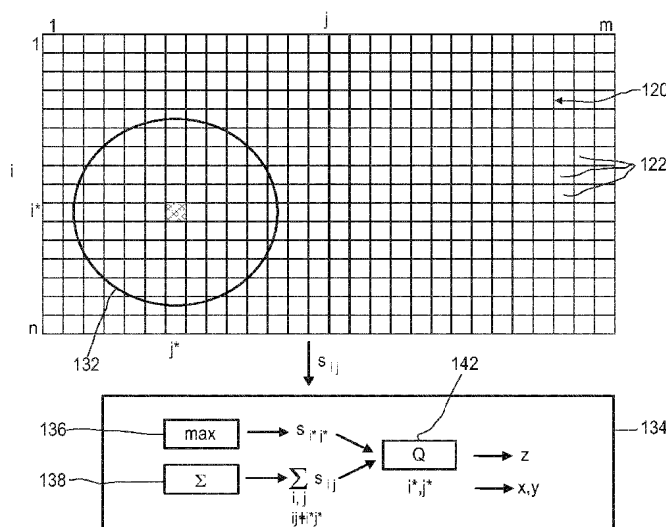
(30) Priority Data:

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16199399.3 17 November 2016 (17.11.2016) EP
16199397.7 17 November 2016 (17.11.2016) EP
17161334.2 16 March 2017 (16.03.2017) EP

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(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.3



(57) **Abstract:** A detector (110) for determining a position of at least one object (112) is disclosed. The detector (110) comprises: - at least one sensor element (118) having a matrix (120) of optical sensors (122), the optical sensors (122) each having a light-sensitive area (124), wherein each optical sensor (122) is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area (124) by at least one light beam (116) propagating from the object (112) to the detector (110); - at least one evaluation device (134) configured for evaluating the sensor signals, by a) determining at least one optical sensor (122) having the highest sensor signal and forming at least one center signal; b) evaluating the sensor signals of the optical sensors (122) of the matrix (120) and forming at least one sum signal; c) determining at least one combined signal by combining the center signal and the sum signal; and d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.

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Detector for optically detecting at least one object

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The
10 devices, systems, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the
15 invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

20 Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultra-violet, visible or infrared light, into electrical signals or electrical energy, optical detectors are
25 generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed
30 in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and for reasons of large-area processing, sensors comprising at least one organic sensor material are being used, as described for example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO
35 2009/013282 A1.

A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or
40 microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for opti-

cally detecting objects are triangulation systems, by means of which distance measurements can likewise be carried out. Kurt Konolige et al., A Low-Cost Laser Distance Sensor, 2008 IEEE International Conference on Robotics and Automation, Pasadena, CA, USA, May 19-23, 2008, discuss competing technologies using triangulation for planar laser distance sensor (LDS).

5 Structured line devices use a light stripe laser and offset camera to determine range to a set of points. Because the laser energy is spread over a line, it is difficult to achieve accurate range, especially in the presence of ambient light, or with darker objects. Point scan devices for 3D scanning of small objects typically use a scanning mirror to direct a point laser beam and redirect the laser return to an optical receiver. Such devices cannot be miniaturized, and their cost and mechanical fragility will remain high. Centroid point modules typically use position-sensitive devices (PSD). These devices measure the centroid of all light impinging on their surface. Although modulation techniques can be used to offset some of the effects of ambient light, PSDs do not perform well unless the laser spot has a very strong reflection, limiting their use to ranges of a meter or less. Pixel-based point modules search the pixel with maximum signal intensity to determine the position of the light spot on the sensor. Typically CMOS line arrays are used for detection. Konolige et al. introduce a low-cost version of a Pixel-based point module.

US 5 235 377 A describes a measuring apparatus for a camera, wherein light is projected to an object from a projection section. Light reflected on the object is received at a first light receiving section and a second light receiving section arranged to be adjacent to each other. In the first light receiving section, a first and second signals relating to a light receiving position and a quantity of received light are output, and in the second light receiving section, a third signal relating to a quantity of light of the reflected light is output. In a discriminating and calculating section, if it is discriminated that the reflected light is positioned at a place, which is nearer than a predetermined distance, a measuring distance can be calculated from an output ratio of the first signal to the second signal by a measuring calculation circuit. On the other hand, if it is discriminated that the reflected light is positioned at a place, which is farther than the predetermined distance, a measuring distance can be calculated from the output ratio of the sum of the first and second signals to the third signal by the measuring calculation circuit.

US 5 512 997 A describes a distance measuring device having an infrared light-emitting diode by which an infrared light is emitted towards a subject, and first and second linear line sensors on which an image corresponding to light reflected from the subject is formed. The first and second linear line sensors are disposed adjacently and in parallel to each other, and offset by an amount corresponding to half of the predetermined A/D converting step. Analog distance data outputted from the linear line sensors are converted to digital distance data using the predetermined A/D converting step, and then, the digital distance data are added together to obtain a subject distance with a high accuracy.

DE 42 11 875 A1 describes an optical rangefinder with electronic correction for spot eccentricity. The optical rangefinder has an evaluation circuit, which determines if an output distance measurement value is valid or erroneous due to e.g. inclination of object surface. The distance signal is determined by evaluation a position a light spot on a first detector. A second control

signal, which characterizes inclination of the object surface, is determined by evaluating eccentricity of a light spot on a second detector, arranged in a different plane.

5 WO 2016/051323 A1 describes a detector and a method for optically determining a position of at least one object. The detector comprises at least one optical sensor for determining a position of at least one light beam and at least one evaluation device for generating at least one item of information on a transversal position of the object and at least one item of information on a longitudinal position of the object. The sensor has at least a first electrode and a second electrode. At least one photovoltaic material is embedded in between the first electrode and the
10 second electrode. The first electrode or the second electrode is a split electrode having at least three partial electrodes. The detector and the method can determine three-dimensional coordinates of an object in a fast and efficient way.

15 In WO 2012/110924 A1, the content of which is herewith included by reference, a detector for optically detecting at least one object is proposed. The detector comprises at least one optical sensor. The optical sensor has at least one sensor region. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor
20 area. The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

25 WO 2014/097181 A1, the full content of which is herewith included by reference, discloses a method and a detector for determining a position of at least one object, by using at least one transversal optical sensor and at least one optical sensor. Specifically, the use of sensor stacks is disclosed, in order to determine a longitudinal position of the object with a high degree of accuracy and without ambiguity.

30 WO 2015/024871 A1 and US 2015/286340 A1, the full content of which is herewith included by reference, discloses an optical detector, comprising:

- at least one spatial light modulator being adapted to modify at least one property of a light beam in a spatially resolved fashion, having a matrix of pixels, each pixel being controllable to individually modify the at least one optical property of a portion of the light beam passing the
35 pixel;
- at least one optical sensor adapted to detect the light beam after passing the matrix of pixels of the spatial light modulator and to generate at least one sensor signal;
- at least one modulator device adapted for periodically controlling at least two of the pixels with different modulation frequencies; and
- 40 – at least one evaluation device adapted for performing a frequency analysis in order to determine signal components of the sensor signal for the modulation frequencies.

US 4,767,211 discloses an apparatus for and a method of measuring a boundary surface of a sample in which a ratio of the light quantity of a part of reflected light from a sample which travels in the vicinity of the optical axis of the reflected light, to the light quantity of another part of the reflected light which is directed to a position deviating from the optical axis by a predetermined distance is used to accurately measure a boundary surface of a sample. Since the accuracy of measurement is increased by using the above ratio, light capable of passing through the sample can be used as incident light. Thus, a deep hole in the surface of the sample and a void such as an air bubble in a living being sample, which cannot be measured by the prior art, can be measured very accurately.

WO 2014/198629 A1, the full content of which is herewith included by reference, discloses a detector for determining a position of at least one object, comprising:

- at least one optical sensor, the optical sensor being adapted to detect a light beam propagating from the object towards the detector, the optical sensor having at least one matrix of pixels; and
- at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

Further, generally, for various other detector concepts, reference may be made to WO 2014/198626 A1, WO 2014/198629 A1 and WO 2014/198625 A1, the full content of which is herewith included by reference. Further, referring to potential materials and optical sensors which may also be employed in the context of the present invention, reference may be made to European patent applications EP 15 153 215.7, filed on January 30, 2015, EP 15 157 363.1, filed on March 3, 2015, EP 15 164 653.6, filed on April 22, 2015, EP 15177275.3, filed on July 17, 2015, EP 15180354.1 and EP 15180353.3, both filed on August 10, 2015, and EP 15 185 005.4, filed on September 14, 2015, EP 15 196 238.8 and EP 15 196 239.6, both filed on November 25, 2015, EP 15 197 744.4, filed on December 3, 2015, the full content of all of which is herewith also included by reference.

US 2006/044546 A1 describes a ranging apparatus capable of ranging simultaneously to a three dimensional scene. An illumination means illuminates a scene with a two dimensional array of spots. A detector is located near to the illumination means and arranged to look toward the scene. A processor responds to the output from the detector and, from the location of a spot in the image of the scene, determines the range to that spot.

US 2008/13005 A1 describes driving schemes in which rows are selected one at a time and column data voltages are inverted to provide inversion schemes for display devices comprising pixels arranged in rows and columns. The order in which rows are selected is such that a first group of first polarity rows is selected in a first order, a first group of second polarity rows is selected in a second order, a second group of first polarity rows is selected in the second order,

and a second group of second polarity rows is selected in the first order, the first order being one of ascending or descending row number order, and the second order being the other of ascending or descending row number order.

- 5 Despite the advantages implied by the above-mentioned devices and detectors, several technical challenges remain. Thus, generally, a need exists for detectors for detecting a position of an object in space which is both reliable and may be manufactured at low cost. Specifically, a need exists for 3D-sensing concepts. Various known concepts are at least partially based on using so-called FiP sensors, such as several of the above-mentioned concepts. Therein, as an
- 10 example, large area sensors may be used, in which the individual sensor pixels are significantly larger than the light spot and which are fixed to a specific size. Still, large area sensors in many cases are inherently limited in the use of the FiP measurement principle, specifically in case more than one light spot is to be investigated simultaneously.
- 15 A further challenge using FiP detectors is the detector area or active area. Typically, for distance measurements, a large active area of the detector is used or even is required. This area, however, may cause noise problems, specifically when the tetralateral conductivity concept is employed to build a PSD. This often results in poor signal-to-noise-ratios and slow detector response times due to the large capacitance in conjunction with the series resistance of the detector. A further challenge using FiP detectors is cost of manufacturing. Thus, in many cases, typical FiP sensors are expensive, as compared to e.g. conventional Si-based photodiodes. Further, the evaluation of the measurement results of measurements using FiP-sensors remains an
- 20 issue, specifically in case the total power of a light beam is unknown. In case a plurality of FiP-sensors is used, located at different positions along an axis of propagation of the light beam, a range of measurement typically is limited to the range in between the two positions of the sensors. Further, many FiP-detectors show a luminance dependency which renders the evaluation of the measurement result more difficult, and, additionally, FiP-measurements in many cases are dependent on a target spot size.

30 Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a

35 position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

- 40 This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by “in an embodiment of the invention” or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed. As used herein, the term “position” refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;
- at least one evaluation device configured for evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

As used herein, the term “sensor element” generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices.

As further used herein, the term “matrix” generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

As further used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane. As will further be outlined in detail below, each optical sensor, preferably, may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-

sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, preferably, each optical sensor is a single area optical sensor rather than a pixelated optical sensor like e.g. a sensor matrix. Other embodiments, however, are feasible. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photosensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the setup, one with a small light-sensitive area and one with a larger light-sensitive area, as will be explained in further detail below.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. The sensor signal may comprise at least one information of at least one beam profile of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The present invention may implement a general idea for determining the at least one longitudinal coordinate of the object, which might be characterized as follows. Thus, as explained in more detail in one or more of the above-mentioned prior art documents, e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot, may

be characterized by two measurement variables: a measurement signal measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their respective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming a combined signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the combined signal for deriving the longitudinal coordinate.

The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus, may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.

The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, as will be outlined in further detail below, the detector according to the present invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

Consequently, in accordance with the present invention, the term "center signal" generally refers to the at least one sensor signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. As used herein, the term "highest sensor signal" refers to one or both of a local maximum or a maximum in a

region of interest. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range.

The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

Similarly, the term "sum signal" generally refers to a signal derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be adapted to determine the sum signal by integrating of signals of the entire matrix, of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of sum and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of the center signal or sum signal from at least one slice or cut of the light spot. Slices or cuts may be realized for example by using one or more specific rows and/or columns. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

Similarly, the term "combined signal", as used herein, generally refers to a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an

arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

5 The light beam propagating from the object to the detector specifically may fully illuminate the at least one optical sensor from which the center signal is generated, such that the at least one optical sensor from which the center signal arises is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the at least one optical sensor from which the sensor signal arises. Contrarily, preferably, the light beam propagating from the object to the detector specifically may create a light spot on the entire matrix which is small-
10 er than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further detail below. As further used herein, a "light spot" generally refers to a visible or detectable round or non-round illumination of an
15 article, an area or object by a light beam.

The light beam may propagate from the object towards the detector. As will be outlined in further detail below, the light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may
20 originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source, as an example, may be or may comprise one or more of an external illumination source, and illumination source integrated into the detector or an illumination source integrated into a beacon device
25 being one or more of attached to the object, integrated into the object or held by the object.

As used herein, the term "ray" generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term "beam" generally refers to a collection of rays. In the following, the terms "ray" and "beam" will be used as synonyms. As
30 further used herein, the term "light beam" generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a
35 conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. As used herein, the term "beam profile" generally refers to a transverse intensity profile of the light beam. In particular, the term "beam profile" relates to a spatial distribution, in particular in at least one plane perpendicular to the propagation of the light beam, of an intensity of the light beam. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments
40 are feasible, however. The detector may comprise at least one transfer device which may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The optical sensors of the sensor element may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. The optical sensors, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers, and/or in the visible spectral range, specifically, in the range of 380 nm to 780 nm. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm.

Specifically, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the combined signal and/or any secondary signal derived thereof and the longitudinal coordinate. The combined signal, specifically, as will be outlined in further detail below, may be a quotient signal.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the opti-

cal sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold;
5 an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

10 As outlined above, raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating
15 secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by
20 using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image
25 created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition.; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a
30 Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or maybe determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations,
40 however, are feasible.

As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be per-

formed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more combined signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa;

forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one combined signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sum signal and the center signal, dividing multiples of the sum signal and the center

signal, dividing linear combinations of the sum signal and the center signal. For example, the evaluation device may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 5 wherein x and y are transversal coordinates, A1 and A2 are areas of at least one beam profile at the position of the sensor element, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The center signal may be a sensor signal comprising essentially center information of the beam profile. The sum signal may be a
- 10 signal comprising essentially edge information of the beam profile. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.
- 15 Each of the center signal and the sum signal may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the position of the sensor element used for determining the combined signal Q. The light-sensitive regions may be arranged such that the sum signal signal comprises information of a first area of the beam profile and the
- 20 center signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.
- 25 The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the
- 30 beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e.
- 35 proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information com-

pared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, the center signal may be a signal corresponding to a rectangular stripe comprising the center of the light spot. The evaluation device may be adapted to form a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal. In particular, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile. The first and the second area, as an example, may be adjacent, with a minimum separation from one another. However, the separation between adjacent areas may also be varied such as enlarged to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent areas may be increased by not evaluating the area between the first and the second area. This may reduce the light on one of the areas relative and/or absolute as compared to the adjacent area which will may increase a quotient of the signal of the two areas. Further, one or both of the areas may consist of separated subareas which may be adjacent to each other and/or which may be separated by areas that are not evaluated and/or which may be evaluated as part of a different quotient. Further, the first and the second area may consist of a linear combination of subareas, whereas the signal contributed by each subarea may be weighted differently when forming the center signal and/or the sum signal. This may further be beneficial for increasing the dynamic range of the quotient system.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determi-

nation of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by one or more line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

The evaluation device may be configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

The detector may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors. For example, the detector may comprise a plurality of optical sensors arranged in a matrix. The detector may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, a defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector may be adapted to determine the depth information using the radiance ratio. This principle is called Distance by Photon Ratio (DPR).

The DPR principle can be applied, for example, to many sub-regions within a segmented image profile, such as a segmented image of at least one feature generated by the at least one light beam on the matrix of optical sensors, which is demonstrated hereafter. The quotient $Q_A(z)$ in the two-dimensional case can be written as

$$Q_A(z) = \frac{\iint_{-r_{out}}^{r_{out}} P(x,y,z) dx dy}{\iint_{-r_{in}}^{r_{in}} P(x,y,z) dx dy},$$

where $P(x, y, z)$ is a two-dimensional beam profile and r_{in} and r_{out} are inner and outer circle radii, respectively. For line-segmented quotients $Q_y(z)$ along the y-dimension, this can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{out}}^{r_{out}} P(x,z) dx}{\int_{-r_{in}}^{r_{in}} P(x,z) dx}.$$

Without wishing to be bound by this theory, the overall defocused beam profile $P(x, z)$ can be regarded as a superposition of defocused image point profiles $p(x, z)$ along the image width $b(x, z)$. This relationship can be modelled as a convolution given by

$$P(x, z) = b(x, z) * p(x, z),$$

wherein $p(x, z)$ represents the Point-Spread-Function (PSF) of a lens in defocus, which in the field of paraxial optics is also known as Circle of Confusion (CoC). By inserting $P(x, z)$ into $Q_y(z)$ the above described quotient $Q_y(z)$ can be rewritten as

$$Q_y(z) = \frac{\int_{-r_{out}}^{r_{out}} b(x, z) \cdot p(x, z) dx}{\int_{-r_{in}}^{r_{in}} b(x, z) \cdot p(x, z) dx} = \frac{\int_{-r_{out}}^{r_{out}} b(x, z) dx \cdot \int_{-r_{out}}^{r_{out}} p(x, z) dx}{\int_{-r_{in}}^{r_{in}} b(x, z) dx \cdot \int_{-r_{in}}^{r_{in}} p(x, z) dx}.$$

Furthermore, with the matrix of optical sensors a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates. Accordingly, the parameterization of a single line may be given by

$$\omega = x \cos(\theta) + y \sin(\theta).$$

The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega, \theta)}\{f(x, y)\} = \iint_{-\infty}^{\infty} f(x, y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy,$$

where δ denotes the Dirac delta function and $f(x, y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is given by

$$R = \frac{\Re_{(\omega, \theta)}\{f'(x, y)\}}{\Re_{(\omega, \theta)}\{f(x, y)\}},$$

with $f'(x, y)$ as the inner region. The variation of θ may yield different ratios R for skewed object surfaces at a certain distance. It is sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

In one embodiment, the light beam propagating from the object to the detector may illuminate the sensor element with at least one pattern comprising at least one feature point. As used herein, the term “feature point” refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central

area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

For example, the light beam propagating from the object to the detector may illuminate the sensor element with at least one point pattern. The point pattern may be generated by the object, for example in response to an illumination by the at least one light source with an illumination pattern comprising the at least one point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The light beam propagating from the object to the detector may illuminate the sensor element with a reflection pattern comprising both point patterns and line patterns. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the combined signal such as the combined signal Q. This determining may be a one-step process, such as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined signal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

The evaluation device may be configured for using at least one predetermined relationship between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The combined signal Q may be determined by using various means. As an example, a software means for deriving the combined signal, a hardware means for deriving the combined signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the combined signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider. The divider may fully or partially be integrated into the

sensor element answers or may fully or partially be embodied independent from the sensor element.

As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

Specifically, the optical sensors may be or may comprise inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The matrix may be composed of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1, 4:3, 16:9$ or similar. As an example, the array may be a square array,

having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

As further outlined above, the matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term “essentially perpendicular”, reference may be made to the definition given above. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, as outlined above, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

As further outlined above, preferably, the sensor element may be oriented essentially perpendicular to an optical axis of the detector. Again, with respect to the term “essentially perpendicular”, reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The matrix may comprise two or four separate light sensitive areas and/or may be sub-divided into two or more segments, in particular two or four active areas. The light sensitive areas and/or segments may be separated by a dividing line. The dividing line, for example, of two Si-diodes, can be arranged parallel to the baseline of the detector, in particular parallel to a spot movement on the optical sensors. However, other arrangements are feasible. For example, the quadrant diode may comprise at least two dividing lines. The at least two dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the combined signal for near field and far field applications independently from each other. In addition to determining the combined signal of sum signal and center signal, the evaluation device may be adapted to determine a second quotient using other signals of the matrix. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotient signals allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient signal may be determined by dividing the sensor signals of two top quadrants, also called top segment, by the sensor signals of two bottom quadrants, also called bottom segment. Using the quotient signal of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient

without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between the top and bottom segment is parallel to the baseline, the quotient signal determined from top segment divided by bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient signal by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals of the quadrants such that distance measurement is possible over a wide range with a large resolution.

In view of the technical challenges involved in the prior art documents discussed above, specifically in view of the technical effort which is required for generating the FiP effect, it has to be noted that the present invention specifically may be realized by using non-FiP optical sensors. In fact, since optical sensors having the FiP characteristic typically exhibit a strong peak in the respective sensor signals at a focal point, the range of measurement of a detector according to the present invention using FiP sensors as optical sensors may be limited to a range in between the positions and which the optical sensors are in focus of the light beam. When using linear optical sensors, however, i.e. optical sensors not exhibiting the FiP effect, this problem, with the setup of the present invention, generally may be avoided. Consequently, the optical sensors each may have, at least within a range of measurement, a linear signal characteristic such that the respective sensor signals are dependent on the total power of illumination of the respective optical sensor.

The detector may further comprise an illumination source for illuminating the object. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

As outlined above, the illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source, as outlined above, specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at

least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

5 The detector may comprise the at least one illumination source. The illumination source may be adapted to illuminate the object with at least one illumination light beam. The illumination light beam may be fully or partially reflected by the object and may travel back towards the detector. The illumination source may be arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis. As used herein, the term “essentially parallel to the optical axis” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. The illumination source may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis do not intersect in space. In particular, the illumination may be on-axis. In order to provide such an on-axis illumination, as an example, one or more reflective elements may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms. The reflected light beam may be essentially parallel to the optical axis. Thus, according to the present invention, the object distance is not determined from an angle of the light beam, i.e. a position of the light spot on a sensor, as done in triangulation methods. Instead, the object distance may be determined using light beams essential parallel to the optical axis. Determination of the object distance by a non-triangulation method but by combining the at least two sensor signals, allows arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis.

25 The light beam may be a monochromatic light beam. The light beam may comprise a narrow band of wavelengths, preferably the light beam may comprise a single wavelength. The at least one light source may be adapted to generate at least one monochromatic light beam and/or the detector may comprise at least one filter element adapted to filter a narrow band of wavelength, such as a monochromator.

30 The illumination source may be adapted to generate at least one illumination pattern for illuminating the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term “pattern” refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term “illumination pattern” refers to a pat-

tern which illuminates the object. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines; at least one fringe pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that a hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source may be adapted to generate pulsed illumination. The illumination source may be adapted to generate at least one light pulse. As used herein, the term "light pulse" or "pulsed illumination" refers to a light beam limited in time. The light pulse may have a predefined length or time duration, for example in the nanoseconds range. For example, the illumination source may be adapted to generate pulses with a pulse length of less than a nanosecond, such as a tenth of a nanosecond, up to a tenth of a second. The illumination source may

be adapted to periodically generate the light pulse. For example, the illumination source may be adapted to generate the light pulse with a frequency of 10 Hz to 10 GHz.

5 The illumination source may be adapted to generate a pulsed light beam. For example, the illumination source may be adapted to generate a continuous illumination light beam and the detector may comprise at least one interruption device adapted to interrupt the illumination, in particular periodically. The interruption device may comprise at least one shutter and/or a beam chopper or some other type of mechanical or electronical periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates
10 at constant speed and which can thus periodically interrupt the illumination. By way of example, the at least one interruption device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

15 Preferably, the illumination source may be a movable and/or mobile illumination source. For example, the illumination source may be moved during the measurement such as to illuminate the object from different positions and/or angles. However, embodiments are feasible in which the illumination source may be positioned in at least one fixed position, for example during a complete measurement time. The evaluation device may be adapted to illumination sources with unclear position such as due to high manufacturing tolerances and/or user interaction
20 and/or user assembly or the like.

The illumination source may illuminate the object with a convergent and/or divergent and/or collimated light beam.

25 The illumination source and the sensor element may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the optical sensors may be arranged in a twisted arrangement.

30 The illumination source and the optical axis may be separated by a small baseline. As used herein, the term "baseline", also denoted as basis line, refers to a distance, for example in a xy - plane, between the optical axis and the illumination source, in particular a distance between the optical axis and a z-component of the illumination light beam. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical
35 sensors and transfer device. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. For example, the detector may be a compact device without further optical elements, wherein the illumination source may be placed as close as possible to an edge of the transfer device. Thus, the baseline
40 may be close to half a diameter of the transfer device, in particular the lens diameter and housings of lens and light source. For example, the detector may be an even more compact device, wherein a mirror, in particular a small mirror, may be positioned in front of the transfer device, in particular in a center, for example a geometrical center, or close to the center of the transfer

device, in order to couple the illumination beam into the beam path. Thus, the baseline may be less than half the diameter of the transfer device. The illumination source may be arranged such that the baseline is as small as possible. By arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis and that the illumination source and the optical axis are separated by the small baseline, very compact devices are possible. For example, a distance from the center of the transfer device to the illumination source, in particular along a connecting line from the center of the transfer device to the illumination source, may be preferably less than 2.5 times a distance from the center of the transfer device to an edge if the transfer device, more preferably less than 1.5 times the distance center to edge of the transfer device, and most preferably less 1 times the distance center to edge of the transfer device. The transfer device may have an arbitrary shape, in particular a non-circular shapes are possible. At small distances an aperture of the illumination source may be small and the baseline may be small. At large distances an aperture of the illumination source may be large and the baseline may be small. This is contrarily as in triangulation methods, wherein at large distances a large baseline is necessary. Further, triangulation based systems have a minimum detection range significantly greater than zero, for example such as 20 cm from the detector system, due to necessary spatial extent of the baseline. Such a large baseline may result in that the illuminated light scattered from the object may not reach the light sensitive area of the optical sensor behind the transfer device. In addition, in triangulation based systems, using a small baseline would reduce the minimum detection range, however, at the same time would reduce a maximum detection range. Further, triangulation based systems require a plurality of light sensitive areas and sensor signals, for example sensor signals of at least one detector row. According to the invention, determination of the longitudinal coordinate z is possible with a reduced number of sensor signals, in particular with less than 20, preferably less than 10 and more preferably less than 5 sensor signals.

The detector may further comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

The optical sensor may be positioned off focus. As used herein, the term "focus" generally refers to one or both of a minimum extent of a circle of confusion of the light beam, in particular of at least one light beam emitted from one point of the object, caused by the transfer device or a focal length of the transfer device. As used herein, the term "circle of confusion" refers to a light spot caused by a cone of light rays of the light beam focused by the transfer device. The circle of confusion may depend on a focal length f of the transfer device, a longitudinal distance from the object to the transfer device, a diameter of an exit pupil of the transfer device, a longitudinal distance from the transfer device to the light sensitive area, a distance from the transfer device

to an image of the object. For example, for Gaussian beams, a diameter of the circle of confusion may be a width of the Gaussian beam. In particular, for a point like object situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from the object into a focus point at the focal length of the transfer device. For non-point like objects situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from at least one point of the object into a focus plane at the focal length of the transfer device. For point like objects not situated or placed at infinite distance from the detector, the circle of confusion may have a minimum extent at least at one longitudinal coordinate. For non-point like objects not situated or placed at infinite distance from the detector, the circle of confusion of the light beam from at least one point of the object may have a minimum extent at least at one longitudinal coordinate. As used herein, the term "positioned off focus" generally refers to a position other than the minimum extent of a circle of confusion of the light beam caused by the transfer device or a focal length of the transfer device. In particular, the focal point or minimum extent of the circle of confusion may be at a longitudinal coordinate l_{focus} , whereas the position of the optical sensor may have a longitudinal coordinate l_{sensor} different from l_{focus} . For example, the longitudinal coordinate l_{sensor} may be, in a longitudinal direction, arranged closer to the position of the transfer device than the longitudinal coordinate l_{focus} or may be arranged further away from the position of the transfer device than the longitudinal coordinate l_{focus} . Thus, the longitudinal coordinate l_{sensor} and the longitudinal coordinate l_{focus} may be situated at different distances from the transfer device. For example, the optical sensors may be spaced apart from the minimum extent of the circle of confusion in longitudinal direction by $\pm 2\%$ focal length, preferably by $\pm 10\%$ focal length, most preferably $\pm 20\%$ focal length. For example, at a focal length of the transfer device may be 20mm and the longitudinal coordinate l_{sensor} may be 19,5 mm, i.e. the optical sensor may be positioned at 97,5% focal length, such that l_{sensor} is spaced apart from the focus by 2,5% of focal length. The extent of the circle of confusion may be larger than the extent of the optical sensor. For example, the optical sensor may be positioned as such that the circle of confusion extends beyond the optical sensor. The optical sensor may thus only partially evaluate the beam profile of the light beam. The evaluation device may be adapted to extrapolate a beam profile such as due to symmetry considerations or due to comparing the partial beam profile to prerecorded beam profiles or the like. Further, the evaluation device may be adapted to evaluate a partial sum signal and a partial center signal of a partial beam profile and convert it into a sum signal and a center signal of an extrapolated and/or fitted and/or prerecorded beam profile or the like.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the

light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

The illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or object recognition method.

As outlined above, the detector may comprise the at least one illumination source. A distance perpendicular to an optical axis of the detector between the illumination source and the optical sensors may be small. The distance perpendicular to the optical axis of the detector between the illumination source and the optical sensors may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. The illumination source and the optical axis may be separated by a small baseline. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and of the at least one optional transfer device, which will be described in detail below. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device.

At least one of the optical sensors may be adapted to generate at least one sensor signal dependent on a time-of-flight (TOF) the illumination light beam has traveled from the illumination source to the object and the reflection light beam has traveled from the object to the light sensitive area of the optical sensor. The evaluation device may be configured for determining at least one TOF-longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. The optical sensor adapted to generate the TOF sensor signal may be designed as time-of-flight detector. The time-of-flight detector may be selected from the group consisting of: at least one pulsed time-of-flight detector; at least one phase modulated time-of-flight detector; at least one direct time-of-flight detector; at least one indirect time-of-flight detector. For example, the pulsed time-of-flight detector may be at least one range gated imager and/or at least one direct time-of-flight imager. For example the phase modulated time-of-flight detector may be at least one RF-modulated light source with at least one phase detector. The optical sensor may be adapted to

determine a time delay between emission of the illumination light beam by the illumination source and receipt of the reflection light beam.

For example, the optical sensor adapted to generate the TOF sensor signal may be designed as pulsed time-of-flight detector. The detector may comprise the at least one interruption device, such as at least one shutter element, adapted to generate a pulsed light beam. The optical sensor may be adapted to store the TOF sensor signal dependent on the receiving time of the reflection light beam in a plurality of time windows, in particular subsequent time windows. The optical sensor may be adapted to store dependent on receiving time of the reflection light beam the generated TOF sensor signal in at least one first time window and/or in at least one second time window. The first and second time windows may be correlated with the opening and closure of the interruption device. Duration of first and second time windows may be pre-defined. For example, the TOF sensor signal may be stored in the first time window during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may be stored in the second time window. Other durations of time windows are thinkable. The first and the second time window may comprise information about background, signal height and signal shift.

For example, the optical sensor adapted to generate the TOF sensor signal may be designed as direct time-of-flight imager. The direct time-of-flight imager may comprise the at least one illumination source adapted to generate at least one single laser pulse. The single laser pulse may be reflected back from the object onto the optical sensor. The optical sensor may comprise at least one photodiode, for example at least one Avalanche Photo Diode (APD), such as at least one Si APD, or such as at least one InGaAs APD, or at least one PIN photo detector array, or at least one Single-Photon Avalanche Photodiode (SPAD), adapted to image the reflection light beam. The direct time-of flight imager may be adapted to image at least one image comprising spatial and temporal data.

For example, the optical sensor adapted to generate the TOF sensor signal may be designed as phase modulated time-of-flight modulator. The phase modulated time-of-flight modulator may be adapted to measure a difference in phase, in particular a phase shift, by determining a correlated signal, for example by multiplying a received signal, i.e. of the reflection light beam, with the emitted signal, i.e. the illumination light beam. A DC component of the correlated signal may comprise an information about the difference in phase. The evaluation device may be adapted to determine the second longitudinal coordinate of the object from the phase difference. For example, the illumination source and the optical sensor adapted to generate the TOF sensor signal may be designed as RF-modulated light source with at least one phase detector. The illumination source may comprise, for example, at least one LED and/or at least one laser. The illumination source may comprise at least one modulation device adapted to modulate the light beam having a pre-defined phase shift. For example, the modulation device may comprise at least one radio frequency module. The radio frequency module may be adapted to modulate the illumination beam with an RF carrier. The optical sensor may be adapted to determine a phase shift of the reflective light beam impinging on the optical sensor.

The optical sensor may be designed as and/or may comprise at least one time-of-flight pixel. Preferably, the detector may comprise at least two optical sensors, wherein each optical sensor is designed as and/or comprises at least one TOF pixel. For example, the detector, in particular the optical sensor may comprise a quadrant diode adapted to generate the TOF sensor signal.

5 For example, the detector, in particular the optical sensor, may comprise at least one pixelated TOF-imager.

The evaluation device may be configured for determining the at least one TOF longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. As used herein, the term “TOF longitudinal coordinate” refers to a longitudinal coordinate derived from the TOF sensor signal. As described above, the illumination light source may be adapted to periodically generate at least one light pulse. The detector may be adapted to generate the first longitudinal sensor signal for each period. The evaluation device may be adapted to determine in which pulse period the TOF sensor signal was generated using the second longitudinal coordinate. The detector
10 may be adapted to uniquely assign to which period the ToF-signal refers to by using the combined sensor signal. Both, the TOF sensor signal and the combined sensor signal, may be non-monotonous functions of the longitudinal coordinate z_{real} . Thus, the longitudinal coordinate may not be uniquely determined from one of the TOF sensor signal or the combined sensor signal alone and a measurement range may not be restricted to a longitudinal range in which the signals are unique functions of z_{real} . The term “measurement range” generally refers to a range
20 from the object to the detector in which determination of the longitudinal coordinate is possible. The term “longitudinal range” generally refers to a range from the object to the detector in which unique determination of the longitudinal coordinate is possible. Below and/or above certain distances from the object to the detector a determination of the longitudinal coordinate may not be possible. For example, time-of-flight measurements are not possible below a certain distance from the object to the detector, possibly due to the minimum measurement time of the internal clock. Furthermore, in time-of-flight measurements the sensor signal may be unique within a longitudinal period, but the sensor signal may be the same in case integer multiples of the longitudinal period are added such that the determined longitudinal coordinate may be non-unique.
30 Thus, the same TOF sensor signal will be obtained for a distance z_1 and a distance $z_1 + n \cdot z_{1p}$, where n is an integer denoting the longitudinal period and z_{1p} is the longitudinal period of the TOF sensor signal, wherein the distances z_1 and $z_1 + n \cdot z_{1p}$ are within the measurement range. As used herein, the term “longitudinal period” refers to partitions of a period, in particular a distance range, in which the longitudinal coordinate can be unambiguously determined from the
35 TOF sensor signal. Non-unique longitudinal coordinates may be denoted as relative longitudinal coordinates and the unique longitudinal coordinates may be denoted as absolute longitudinal coordinates.

If both the TOF sensor signal $F1$ and the combined sensor signals $F2$ are available, it may be
40 possible to uniquely determine the longitudinal coordinate and to extend the longitudinal range, as long as each signal pair ($F1, F2$) corresponds to a unique distance and vice versa. In particular, if a unique signal pair ($F1, F2$) exists for each longitudinal coordinate and vice versa, the evaluation device may be adapted to determine the unique combined longitudinal coordinate by

(1) selecting at least one first selected signal such as the TOF sensor signal and/or the combined sensor signal and determining non-unique first longitudinal coordinates;

(2) selecting a second selected signal such as the combined signal Q and/or the TOF sensor signal, which was not selected in step (1), and determining non-unique second longitudinal coordinates;

(3) determining whether one of the non-unique first longitudinal coordinates and the non-unique second longitudinal coordinates match up to a predetermined tolerance threshold;

(4) setting a combined unique longitudinal coordinate to be a matching longitudinal coordinate.

In step (1) and step (2), signals may be selected in the given order or may be performed in a different order. For example, in step (1) the TOF sensor signal may be selected and in step (2) the combined signal Q may be selected. In another example, in step (1) the combined sensor signal may be selected, and the non-unique first longitudinal sensor signal may be determined therefrom. In step (2), the TOF sensor signal may be selected.

Additionally or alternatively to step (4), the evaluation device may be adapted to output an error signal in case no matching coordinates are found and/or to output an error signal in case more than one matching coordinates are found. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be stored in a look-up table. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be approximated or described by an analytical function which is evaluated to find the longitudinal coordinate corresponding to a given signal pair.

The evaluation device may comprise at least two memory elements. As used herein, the term “memory element” refers to a device adapted to store information. The evaluation device may be adapted to receive and to store information provided by the optical sensors, for example the at least one first sensor signal. Such information may comprise raw sensor data and/or processed sensor data. For example, the memory element may be adapted to store information for further evaluation by the evaluation device. The memory element may be a volatile or non-volatile memory element.

As described above, the optical sensor may be designed as and/or may comprise at least one ToF pixel. The detector may comprise at least two switches. Each of the switches may be connected to the optical sensor adapted to generate the first sensor signal, for example by at least one connector. In particular, each of the switches may be connected to the ToF pixel. The switches are adapted to provide the TOF sensor signal to one of the memory elements. In particular, the switches may be adapted to let, dependent on receiving time of the reflection light beam, the generated TOF sensor signal pass through one of the switches. For example, the TOF sensor signal may pass one of the switches during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may pass the other switch. Each of the switches may be controlled by a control signal having a pulse length identical to a pulse length of a light pulse generated by the illumination source. The control signal of one of the switches may be delayed. For example, the delay may correspond to the pulse

length of the light pulse. The evaluation device may be adapted to sample and/or store depending on the delay a first part, or fraction, of the TOF sensor signal through a first switch in a first memory element and the other, second part, or fraction, of the TOF sensor signal through a second switch in a second memory element. The evaluation device may be adapted to determine the first longitudinal coordinate by evaluating the first part and second part of the TOF sensor signal. The evaluation device may be adapted to determine the first longitudinal coordinate z_1 by

$$z_1 = \frac{1}{2} \cdot c \cdot t_0 \cdot \frac{S_{12}}{S_{11} + S_{12}} + z_0;$$

wherein c is the speed of light, t_0 is the pulse length of the illumination light beam, z_0 is a distance offset often determined by the resolution of the time measurement, and S_{11} and S_{12} are the first part and second part of the TOF sensor signal, respectively.

The detector may be adapted to determine at least one distance information of the object by using triangulation and/or structured light techniques. In known 3D sensing devices, such as devices using triangulation or structured light techniques, due to correspondence problems regular, constant or periodic pattern are not suitable since each measured point has to be assigned to one reference point of a reference pattern. The at least one light beam propagating from the object to the detector may be adapted to generate at least one reflection pattern on the matrix of optical sensors. The term "reflection pattern" refers to at least one image determined by the optical sensors in response to illumination of the light sensitive areas by the light beam propagating from the object to the detector which was generated in response to illumination by the illumination pattern. The reflection pattern may comprise at least one reflection feature depending on the corresponding illumination feature of the illumination pattern. The detector may be adapted to determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection pattern from the quotient signal. Thus, the detector may be adapted to pre-classify the at least one reflection feature of the reflection pattern. This allows using illumination patterns comprising regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal patterns may be preferred.

The evaluation device may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator;

applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device may be adapted to determine at least one reference feature in at least one reference pattern corresponding to the at least one reflection feature. The term "reference pattern" refers to an image of comparison at a specific spatial position comprising at least one reference feature. As outlined above, the evaluation device may be adapted to perform an image analysis and to identify features of the reflection pattern. The evaluation device may be adapted to identify at least one reference feature in the reference pattern having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference pattern and the reflection pattern may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The distance may be a relative distance, also denoted as baseline. The evaluation device may be adapted to determine an epipolar line in the reference pattern. The relative position of the reference pattern and reflection pattern may be known. For example, the relative position of the reference pattern and the reflection pattern may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature to the real world feature from which it originates. Thus, the straight line may comprise possible object features corresponding to the selected reflection feature. The straight line and the baseline span an epipolar plane. As the reference pattern is determined at a different relative constellation from the reflection pattern, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference pattern. The epipolar line may be the intersection of the epipolar plane and the reference pattern. Thus, a feature of the reference pattern corresponding to the selected feature of the reflection pattern lies on the epipolar line.

Depending on the distance to the object, the reference feature corresponding to the image position of the reflection feature may be displaced within the reference pattern compared to the image position in the reflection pattern. The reference pattern may comprise at least one displacement region in which the reference feature corresponding to the selected reflection feature may be imaged. The displacement region may comprise only one reference feature. The displacement region may extend along the epipolar line. The evaluation device may be adapted to determine the reference feature along the epipolar line. The evaluation device may be adapted to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm \varepsilon$ from the quotient signal Q to determine a displacement region along the epipolar line corresponding to $z \pm \varepsilon$. The evaluation device may be adapted to match the selected reflection feature with at

least one reference feature within the displacement region. As used herein, the term “matching” refers to determining and/or evaluating the corresponding reference and reflection features. The evaluation device may be adapted to match the selected feature of the reflection pattern with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. Preferably, the detector may be adapted to pre-classify the selected reflection feature using the quotient signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference pattern may have a relative distance to each other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line. For example, the illumination pattern may comprise at least one hexagonal pattern. Preferably, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Preferably, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other.

The evaluation device may be adapted to determine a displacement of the reference feature and the reflection feature. The evaluation device may be adapted to determine the displacement of the matched reference feature and the selected reflection feature. As used herein, the term “displacement” refers to the difference between an image position in the reference image to an image position in the reflection image. The evaluation device may be adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term “longitudinal information” refers to information relating to the longitudinal coordinate z_{triang} . For example, the longitudinal information may be a distance value. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case the position of the selected reflection feature in the reflection pattern and the position of the matched reference feature and/or the relative displacement of the selected reflection feature and the matched reference feature are known, the longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. The displacement and the corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one

data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

5 The detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, at least one optical sensors of the matrix of optical sensors may be adapted to generate a so called FiP signal. For example, each of the optical sensors of the matrix may be adapted to generate the at least one FiP signal. For example, the matrix of optical sensors may be a pixelated FiP-sensor. Additionally or alternatively, the FiP signal may be determined, such as extracted and/or simulated, from
10 the sensor signals of the matrix and the quotient as outlined above may be determined from the sensor signals. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. According to the FiP-effect, the longitudinal sensor signal, given the same total power, may exhibit at least one pronounced maximum for one or a plurality of focusings and/or for one or a plurality of specific sizes of the light spot on the sensor region or within the sensor region. For purposes of comparison, an observation of a maximum of the longitudinal sensor signal in a condition in which the corresponding material is impinged by a light beam with the smallest possible cross-section, such as when the material may be located at or near a focal point as affected by an optical lens, may be denominated as a "positive FiP-effect". The negative FiP effect describes an observation of a minimum of the longitudinal sensor signal under a condition in which the corresponding material is impinged by a light beam with the smallest available beam cross-section, in particular, when the material may be located at or near a focal point as effected by an optical lens. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

As outlined above, the detector may further comprise one or more additional elements such as
30 one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing. The detector specifically may comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensor. The transfer device may comprise one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at
35 least one multi-lens system. Therein, generally, the term "transfer device" may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical
40 sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of , preferably at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection

element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

5 In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

30 In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

35 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source.

Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined.

Alternatively, the illumination source of the scanning system may emit two or more individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning device. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as

mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with a several corners and surfaces. One or more flexibly movable laser sources may be attached to the scanning system. The one or more laser sources may be moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system may be measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information may be stored in a mobile phone application. The laser sources may be moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and/or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning device may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning device may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of

optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning device is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning device.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The optional transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The optional

transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. Most preferably, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the sensor element can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the sensor element. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses an inertial measurement unit for use in an electronic device. The electronic device may be a mobile electronic device. The electronic device may be a camera. The electronic device may be a mobile phone. The inertial measurement unit is adapted to receive data determined by at least one inertial measurement unit comprising at least one detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. As used herein, the term "data determined by the at least one detector" refers to at least one information on the at least one longitudinal coordinate z . The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. As used herein, the term "data determined by the at least one further sensor" refers to at least one information selected from the group consisting of: angle information; speed information; information on turn rate; information on inclination. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed. The inertial measurement unit may comprise at least one processor. The processor may be adapted to evaluate data recorded by the further sensor. In particular, the processor may be adapted to determine and/or calculate one or more of spatial position, spatial orientation, movement and velocity. The inertial measurement unit may comprise a plurality of further sensors. The inertial measurement unit may be adapted to fuse information determined from at least two of the further sensors. The inertial measurement unit may be adapted to fuse information of at least two further sensors by using at least one Kalman filter. As outlined

above, the detector may be adapted to provide an absolute measurement of the longitudinal coordinate z . The processor, e.g. the evaluation device as described above, may be adapted to fuse the information of the at least two further sensors considering the longitudinal coordinate z . The various sensor signals may be used within a Kalman filter or a linear quadratic estimated to take into account that each sensor signal is subject to measurement errors and inaccuracies. Fusion of these sensor signals within a Kalman filter may yield an improved estimate such as for the measurement of the longitudinal coordinate.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used.

The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps are as follows:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;
- evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Specifically, the evaluation of the first and second sensor signal may comprise deriving a combined signal Q by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Further, the determination of the longitudinal coordinate may comprise evaluating the combined signal Q . The evaluation of the combined signal specifically may comprise using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the longitudinal coordinate.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

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In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application;.

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The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

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As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

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Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially land-mark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for house-

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hold applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus, specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head-mounted displays.

Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries,

obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in

combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems, or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present

invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.

5 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. 10 Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, 20 such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions. 30

Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used 40 for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endo-

scopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric surgery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment.

Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe. In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

Further, the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a

surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

5 Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation
10 etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

15 Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

20 The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

25 The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

30 Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to
35 avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the
40 devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the

same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D
5 imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially
10 useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection,
15 such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections, inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

20 Further, the devices according to the present invention may be used in vehicles, trains, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed
25 and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a
30 guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to
35 active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

40 The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft when landing or starting. Therein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to

detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or accelerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

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Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racquet, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light-emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the

present invention may be used in drones or multicopters to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concertized movements of several drones or recharging or refueling in the air or the like.

Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as cat litter robots, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, window washing robots, toy robots, telepresence robots, social robots providing company to less mobile people, or robots translating and speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate inse-

cure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

5 Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that are can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.

10 Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition device, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.

15 Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.

20 The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or greenhouse or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.

25 Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific groups of animals such as bees.

30 Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or virus-

es or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location. Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location-specific information, such as the current position on a map, or information on goods sold, or the like.

Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.

Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices according to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.

As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, the devices according to the present invention may be used in logistics applications. Thus, the

5 devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

20 Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

35 Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes

the position. The detector generally may be a device adapted for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device.

- 5 Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

- 10 The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

- 15 The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside a camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based algorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

- 30 The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

- 40 The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared.

5 The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

15 The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

20 The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

40 The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The

evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and/or processing 3D information.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device

and/or the data processing device may be especially suitable for such integration into automotive design.

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

As outlined above, the human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon

device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device

may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller

might include one or more data processing devices, such as one or more computers and/or microcontrollers.

- As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.
- The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.
- The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.
- The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
- As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the

article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The proposed devices and methods provide a large number of advantages over known detectors of this kind. Thus, the detector generally may avoid the shortcomings of the known prior art systems disclosed above. Specifically, the detector may avoid the use of FiP sensors, thereby allowing for e.g. using simple and cheap and commercially available semiconductor sensors such as silicon photodiodes. These photodiodes generally do not show a luminance dependency, and the method disclosed above is generally independent from the brightness of the scenery and/or the brightness of the light spot on the light beam. Consequently, a range of measurement in terms of luminance or total power of the light beam entering the detector is generally larger in the present invention as compared to many of the devices disclosed above. Further, the measurement by using the detector according to the present invention is generally independent from the target spot size, i.e. from either the size of the object, the size of a light spot projected on to the object or the size of a beacon device being one or more of attached to the object, integrated into the object or held by the object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional option of measuring one or more transversal coordinates, thereby integrating the functionality of a PSD.

The detector specifically may use linear photodetectors, such as the near silicon photodiodes. The use of a beam splitter is optional but is not generally necessary, which allows for using a simple optical setup.

The result of the measurement, i.e. the position determined by the detector such as the at least one longitudinal coordinate, may be rendered widely independent from the brightness of the scenery and/or the brightness of the object, the brightness of the at least one beacon device, the brightness of the at least one illumination source or the total power of the light beam propagating from the object to the detector. Further, due to this independence and wide range of measurement in terms of brightness, reflective objects or non-reflective objects may be used.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention, such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the combined signal, as mentioned above, contrarily, provides a wide range of

a continuously and monotonously decreasing or increasing functions which may be used to determine the longitudinal coordinate from the combined signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the combined signal from the total power of the light beam, at least as long as no saturation of the optical sensors is reached, the present invention also provides a very wide range of measurement in terms of brightness, i.e. in terms of total power of the light beam.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector having

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area by at least one light beam propagating from the object to the detector;

- at least one evaluation device configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;

- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;

- c) determining at least one combined signal by combining the center signal and the sum signal; and

- d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 2: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the detector, specifically the evaluation device, is configured for transforming the sensor signals of

the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals.

5 Embodiment 4: The detector according to the preceding embodiment, wherein the transforming of the sensor signals comprises at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a
10 decomposition into color channels; a decomposition into hue, saturation, and brightness channels, a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; ap-
15 plying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image.

Embodiment 5: The detector according to any one of the preceding embodiments, wherein the
20 evaluation device comprises at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal.

Embodiment 6: The detector according to the preceding embodiment, wherein the center detec-
25 tor is fully or partially integrated into the sensor element.

Embodiment 7: The detector according to any one of the two preceding embodiments, wherein the center detector is fully or partially embodied in one or both of software or hardware.

Embodiment 8: The detector according to any one of the preceding embodiments, wherein the
30 sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all
35 sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a prede-
40 termined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.

Embodiment 9: The detector according to the preceding embodiment, wherein the evaluation
45 device comprises at least one summing device for forming the sum signal.

Embodiment 10: The detector according to the preceding embodiment, wherein the summing device is fully or partially integrated into the sensor element.

- 5 Embodiment 11: The detector according to any one of the two preceding embodiments, wherein the summing device is fully or partially embodied in one or both of software or hardware.

Embodiment 12: The detector according to any one of the preceding embodiments, wherein the combined signal is a combined signal Q, derived by one or more of: forming a quotient of the
 10 center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

- 15 Embodiment 13: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- 20 wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o.

Embodiment 14: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for using at least one predetermined relationship between the
 25 combined signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 15: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

30 Embodiment 16: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

35 Embodiment 17: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the combined signal.

40 Embodiment 18: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 19: The detector according to any one of the two preceding embodiments, wherein the divider is fully or partially integrated into the sensor element.

5 Embodiment 20: The detector according to any one of the preceding embodiments, wherein the optical sensors are photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors.

10 Embodiment 21: The detector according to any one of the preceding embodiments, wherein the optical sensors are sensitive in the infrared spectral range.

Embodiment 22: The detector according to any one of the preceding embodiments, wherein the matrix comprises one or both of a CCD detector, preferably a CCD detector chip, or a CMOS detector, preferably a CMOS detector chip.

15 Embodiment 23: The detector according to any one of the preceding embodiments, wherein the matrix is a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns.

20 Embodiment 24: The detector according to the preceding embodiment, wherein the rows and columns are oriented essentially perpendicular.

Embodiment 25: The detector according to any one of the two preceding embodiments, wherein the matrix has at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows.

25 Embodiment 26: The detector according to any one of the three preceding embodiments, wherein the matrix has at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns.

30 Embodiment 27: The detector according to any one of the preceding embodiments, wherein the matrix comprises at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors.

35 Embodiment 28: The detector according to any one of the four preceding embodiments, wherein the ratio of the number of rows and the number of columns is close to 1 preferable larger than 1:3, more preferably larger than 1:2.

Embodiment 29: The detector according to any one of the preceding embodiments, wherein the sensor element is oriented essentially perpendicular to an optical axis of the detector.

40 Embodiment 30: The detector according to any one of the preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the respective sensor signals are dependent on the total power of illumination of the respective optical sensor.

45 Embodiment 31: The detector according to any one of the preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object.

Embodiment 32: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

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Embodiment 33: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

10 Embodiment 34: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at
15 least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings.

20 Embodiment 35: The detector according to the preceding embodiment, wherein the optical sensors are adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the longitudinal coordinate z of the selected feature of the reflection pattern by evaluating the combined signal Q .

25 Embodiment 36: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

30 Embodiment 37: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one
35 beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

Embodiment 38: The detector according to any one of the preceding embodiments, wherein the evaluation device is further configured for determining at least one transversal coordinate of the
40 object by evaluating a transversal position of the at least one optical sensor having the highest sensor signal.

Embodiment 39: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for distinguishing different light beams having different modulations.
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Embodiment 40: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is adapted to illuminate the object with at least one illumination light beam, wherein the illumination source is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis.

Embodiment 41: The detector according to any one of the preceding embodiments, wherein the detector comprises the at least one illumination source, wherein the illumination source and the optical axis are separated by a small baseline.

Embodiment 42: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

Embodiment 43: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 44: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 45: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 46: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 47: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 48: An inertial measurement unit for use in an electronic device, wherein the inertial measurement unit is adapted to receive data determined by at least one detector according to any one of the preceding embodiment referring to a detector, wherein the inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion, rotation, acceleration, orientation, angle position, inclination, turn rate, speed.

Embodiment 49: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- illuminating at least one sensor element of the detector with at least one light beam propagating from the object to the detector, the detector having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor generates at least one sensor signal in response to the illumination;
- evaluating the sensor signals, by
 - a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining at least one longitudinal coordinate z of the object by evaluating the combined signal.

Embodiment 50: The method according to the preceding embodiment, wherein the combined signal is a combined signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

Embodiment 51: The method according to the preceding embodiment, wherein the determining the longitudinal coordinate comprises evaluating the combined signal Q , wherein the evaluating the combined signal comprises using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

Embodiment 52: The method according to any one of the preceding method embodiments, wherein the method comprises using the detector according to any one of the preceding claims referring to a detector.

Embodiment 53: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figures 1 and 2 show different embodiments of a detector according to the present invention;

Figure 3 shows an example of a light spot on a matrix of optical sensors and the evaluation thereof;

Figure 4 shows an exemplary embodiment of a detector according to the present invention; a detector system, a human machine interface, an entertainment device, a tracking system, a scanning system and a camera;

Figure 5 shows an exemplary embodiment of a combined signal Q as a function of a longitudinal coordinate z , for various illumination intensities; and

Figures 6A and 6B show a modification of the embodiment of Figure 2, with an off-axis illumination light beam;

Figures 6C and 6D show a comparison of two experimental setups using a detector setup according to Figure 6A with a Bi-cell as optical sensors and experimental results;

Figures 7A and 7B show a further exemplary embodiment of a detector according to the present invention with a small baseline;

Figure 8 shows an embodiment of determining a longitudinal coordinate z using a line pattern;

Figures 9A and B show a defocused beam profile subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates and Radon transformation different image regions on a matrix of optical sensors;

Figures 10A to C show three embodiments of a hexagonal illumination pattern; and

Figure 11 shows an embodiment of a scanning device.

10 Detailed description of the embodiments

In Figure 1, a schematic view of an embodiment of a detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises a beacon device 114, from which a light beam 116 propagates towards a sensor element 118. The sensor element 118 comprises a matrix 120 of optical sensors 122, each optical sensor 122 having a light-sensitive area 124 facing the object 112. The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110, further, may comprise at least one transfer device 128, such as at least one lens and/or at least one lens system, specifically for beam shaping. Consequently, the light beam 116 may be focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of the above-mentioned prior art documents, such as to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Figure 1, the light beam 116 generates a light spot 132 on the matrix 120. In Figure 3, an exemplary view of the light spot 132 on the matrix 120 is shown. As can be seen, in this exemplary embodiment, the matrix 120 specifically may be a rectangular matrix, with rows numbered by "i", from 1 to n , and with columns, denoted by "j", from 1 to m , with n , m being integers. The center of the light spot 132, in this exemplary embodiment, is located in the sensor element denoted by i^* , j^* . The optical sensors 122 may provide sensor signals s_{ij} to an evaluation device 134 which, out of the sensor signals, determines at least one center signal, denoted symbolically by s_{ij^*} . As outlined in further detail above, for generating the center signal, the evaluation device 134 may comprise at least one center detector 136. As an example, the center detector simply may determine the maximum sensor signal out of the plurality of sensor signals generated by the optical sensors 122. Alternative methods are feasible. Thus, as an example, instead of determining a single maximum optical sensor signal, a plurality of sensor signals may be used for generating the center signal. Thus, as an example, neighboring optical sensors which are adjacent to the optical sensor i^* , j^* may contribute to the center signal, such as optical sensors with the coordinates $i^*-1, \dots, i^*+1$ and $j^*-1, \dots, j^*+1$. These coordinates, in the simple exemplary embodiment, may form a square around the optical sensor i^* , j^* . Instead of a square having a side length of 3, as in this embodiment, other environments around the optical sensor

having the highest sensor signal may be used. Further, additionally or alternatively, the center signal may be generated by adding up and/or averaging over sensor signals within a certain range from the maximum sensor signal which may for example be beneficial to the measurement precision concerning noise such as pixel noise. Further, additionally or alternatively, for the determination of the center signal or sum signal, image processing techniques such as sub-pixel processing, interpolation, normalization or the like may be employed. Other alternatives are feasible. For example, the beam profile may be a trapezoid beam profile and the evaluation device 134 may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid. Further, when trapezoid beam profiles may be assumed, the evaluation device 134 may be adopted to determine the sum and center signals by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Further, the evaluation device 134 is configured for determining at least one sum signal out of the sensor signals of the matrix 120. For this purpose, the evaluation device 134 may comprise at least one summing device 138. The summing device 138 may be configured for adding up, integrating or averaging over the sensor signals of the entire matrix 120, of a region of interest within the matrix 120, each option with or without the optical sensors from which the center signal is generated. Thus, in the exemplary embodiment shown in Figure 3, the summing device is simply configured for summing over the sensor signals s_{ij} of the entire matrix 120, except for the center optical detector with the coordinates i^* , j^* . Other options, however, are feasible.

The evaluation device 134 further is configured for forming at least one combined signal out of the center signal and the sum signal. For this purpose, the evaluation device 134, as an example, may comprise at least one combining device 140, such as at least one divider 142. As a very simple embodiment, a quotient Q may be formed, by dividing the center signal by the sum signal or vice versa. Other options are feasible and are given above.

Finally, the evaluation device 134 is configured for determining at least one longitudinal coordinate z of the object by evaluating the combined signal. For this purpose, the evaluation device may comprise at least one further component, such as at least one evaluation component, which is not depicted in the figures. It shall be noted that the components of the evaluation device 134 shown in Figure 3 may fully or partially be embodied in hardware and/or software. Further, the components may fully or partially be embodied as independent or separate components, and/or may fully or partially be embodied as components which are integrated into the sensor element 118.

The embodiment of Figure 3 further shows that, in addition to the longitudinal coordinate z , at least one item of information on a transversal coordinate of the object 112 and/or the beacon device 114 may be generated. Thus, the coordinates i^* and j^* provide additional items of information on a transversal position of the object 112 and/or the beacon device 114. In the setup of Figure 1, the beacon device 114, for the sake of simplicity, is positioned in the center, i.e. on the

optical axis 126. In this case, the light spot 132 is likely to be centered in the middle of the matrix 120. In the embodiment shown in Figure 3, however, as can easily be detected, the light spot 132 is off-centered. This off-centering is characterized by the coordinates i^* , j^* . By using known optical relationships between this off-centering and a transversal position of the object 112 and/or the beacon device 114, such as by using the lens equation, at least one transversal coordinate of the object 112 and/or the beacon device 114 may be generated. This option is also shown in the exemplary embodiment of Figure 3.

In Figure 2, a modification of the embodiment of Figure 1 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Figure 1. Instead of using an active beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 itself comprises at least one illumination source 144. The illumination source 144, as an example, may comprise at least one laser, whereas, in Figure 1, as an example, the beacon device 114 may comprise a light-emitting diode (LED). Other embodiments, however, are feasible. The illumination source 144 may be configured for generating at least one illumination light beam 146 for fully or partially illuminating the object 112. The illumination light beam 146 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116.

As shown in Figure 2, as an example, the illumination light beam 146 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too, as will be shown in the context of Fig. 6A and 6B below. In order to provide an on-axis illumination, as shown in Figure 2, as an example, one or more reflective elements 148 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Figure 2 corresponds to the setup in Figure 1. Thus, again, an evaluation device 134 may be used, having e.g. at least one divider 142 for forming the combined signal Q, and, as an example, at least one position evaluation device 143, for deriving the at least one longitudinal coordinate z from the combined signal Q and/or another type of combined signal. It shall be noted that the evaluation device 134, again, may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 136, 138, 140, 142, 143 may fully or partially be embodied by appropriate software components and/or may fully or partially be embodied by hardware components.

The optical sensors 122 of the matrix 120, as an example, may be pixels of a pixelated optical sensor, such as a CCD and/or a CMOS sensor chip. Thus, as an example, the optical sensors 122 may have a side length and/or an equivalent diameter in the range of a few micrometers to several hundred micrometers. It shall be noted, however, that larger pixels or optical sensors 122 may be used. Further, instead of using an integrated sensor element 118 such as a CCD and/or CMOS sensor chip, non-integrated matrices may be used.

Figure 4 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments in Figures 1 or 2. The detector 110, specifically, may be embodied as a camera 149 and/or may be part of a camera 149. The camera 149 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Figure 4 further shows an embodiment of a detector system 150, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached to and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Figure 4 further shows an exemplary embodiment of a human-machine interface 151, which comprises the at least one detector system 150 and, further, an entertainment device 152, which comprises the human-machine interface 151. The figure further shows an embodiment of a tracking system 153 for tracking a position of the object 112, which comprises the detector system 150. The components of the devices and systems shall be explained in further detail below.

Figure 4 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the at least one object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110 and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136, which is not shown. The illumination source 136, generally, may be configured to emit the at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. It shall be noted, however, that an active beacon device, as e.g. shown in the setup of Figure 1, may also be used, and, thus, that setups with no integrated illumination source 136 are also feasible. The scanning system 134 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112 and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Figure 4 is shown in Figures 1 and 2. Thus, the detector 110, besides the sensor element 118, comprises the at least one evaluation device 134, having, e.g., the at least one center detector 136, the at least one summing device 138, the at least one combining device 140, the at least one divider 142, the at least one position evaluation device 143 and/or combinations thereof. These components, which may optionally be present, are symbolically depicted in Figure 4. The components of the evaluation device 134 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the components of the evaluation device 134 and one or more of the components of the sensor element 118 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Figure 4. Further, the one or more connectors 156 may

comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 134 may fully or partially be integrated into one or both of the sensor element 118 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 134 may
5 fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object
10 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the sensor element 118. The sensor
15 118, wherein one or more of the sensor elements 118 may be provided, may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 may be comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard
20 to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. A coordinate system 168 may be defined, in which a direction parallel or antiparallel to the optical axis 126, such as parallel or antiparallel to the direction of view 166, may be defined as a transversal direction. In the coordinate system 168, symbolically depicted in Figure 4,
25 a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the sensor element 118 as well as, optionally, one or more further optical sensors. A non-branched beam path may be used or, alternatively, a branched
30 beam path may be possible, with, e.g., additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining the transversal coordinates of the object 112 and/or parts thereof. As outlined above, however, in the context of Figure 3, the at least one transversal coordinate may also be determined by the sensor element 118 itself, such as by determining
35 the transversal coordinates of the center of the light spot 132.

One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as outlined above in the context
40 of Figures 1 to 3, the evaluation device 134 may be configured to evaluate the sensor signals provided by the optical sensors 122 of the matrix 120 of the sensor element 118. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 122 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110,

specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least one of these beacon devices 114 may comprise active beacon devices with an integrated illumination source such as light-emitting diodes. In case the illumination source 136 is used, the beacon devices do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the sensor element 118. For details of the evaluation, reference may be made to Figures 1 to 3 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 151, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Figure 4, the machine 170 may be a computer and/or may comprise a computer.

Other embodiments are feasible. The evaluation device 134 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Figure 4 also depicts an example of a tracking system 153, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 153 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may fully or partially be integrated into the machine 170, specifically the computer, as indicated in Figure 4, and/or into the evaluation device 134.

Similarly, as outlined above, the human-machine interface 151 may form part of an entertainment device 152. The machine 170, specifically the computer, may also form part of the entertainment device 152. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In Figure 5, experimental data are shown which demonstrate an exemplary embodiment of the present invention in terms of measurement data. The figure shows a combined signal Q as a function of a longitudinal coordinate z , given in millimeters, for various illumination intensities.

In order to gain the experimental data shown in Figure 5, an experimental setup was used with a sensor element 118 formed by a Basler AC 1920-40GC camera, with a transfer device 128 formed by a Nikkor 50 mm lens.

As a beacon device 114, a light-emitting diode (LED) was used, having a central nominal wavelength of 532 nm. A diffusor was used in front of the LED and a diaphragm, in order to provide a

well-defined light-emitting area having a diameter of 5 mm. The intensity of the LED was varied, by varying a drive current of the LED between 5 mA and 150 mA.

In the experiments, the distance z between the LED and the lens was varied from 300 mm to 1700 mm. The signal of the Basler AC 1920-40GC camera was evaluated by the following procedure. As a center signal, an accumulated signal of an inner circle having a radius of 15 pixels around the optical axis was determined, with the light spot centered at the optical axis. As a sum signal, the sum of the signals of all pixels of the camera within the light spot was generated. A combined signal was formed, by dividing the sum signal by the center signal.

In Figure 5, the solid curves, overlapping, show the combined signal curves for LED currents of 150 mA, 125 mA, 100 mA and 50 mA. As can be seen, there are basically no differences between these curves, within the whole measurement range. This experiment clearly shows that the combined signal is widely independent on the total power of the light beam. Only at lower intensities, the curves may be distinguished. Thus, the dotted curve shows measurements at an LED current of 25 mA, the dashed-dotted line at an LED current of 10 mA, and the dashed line at an LED current of 5 mA. Still, even at these low intensities, the measurement curves are very close to the solid curves, which shows the high tolerances of the measurement.

In Fig. 6A and 6B, an alternative embodiment of the detector 110 is shown which is a modification of the setup shown in Fig. 2. Thus, for most elements and optional details as well as further elements not shown in the schematic Figures 6A and 6B, reference may be made to the description of Fig. 2 above.

In Figure 2, the illumination light beam 146, as discussed above, preferably travels along the optical axis 126, i.e. parallel to the optical axis 126 or even on the optical axis 126. In the setup, the position of the center of the light spot 132 typically does not depend on the z -coordinate of the object 112, such as on a distance between the object 112 and the detector 110. In other words, the diameter or equivalent diameter of the light spot 132 changes with the distance between the object 112 and the detector 110 whereas, typically, the position of the light spot 132 on the matrix 120 does not.

Contrarily, in Fig. 6A and 6B, a setup of the detector 110 is shown in which an illumination light beam 146 travels off-axis, i.e. one or both of at an angle other than 0° with the optical axis 126 or parallel to the optical axis 126 but shifted from the optical axis 126. This embodiment, as will be discussed in further detail below, demonstrates that the method according to the present invention can be further enhanced by increasing the z -dependency of a combined sensor signal. Thus, in Fig. 6A, a side view is shown with two different positions of the object 112, i.e. a first position at z_1 , drawn in solid lines, and a second position at z_2 , drawn in dashed lines. As can be seen, the illumination light beam 146 which, as an example, propagates at an angle of 5° to 30° , e.g. 10° to 20° , with the optical axis 126, hits the object 112 in both cases at different positions. From these points of the object 112 illuminated by the illumination light beam 146, light beams 116 propagate towards the detector 110, wherein, again, the light beam 116 for the

object 112 being located at position z_1 is drawn in solid lines, wherein the light beam 116 for the object 112 being located at position z_2 is drawn in dashed lines.

In Fig. 6B, the matrix 120 is shown in an enlarged fashion. As can be seen in this setup, the position of the light spot 132 moves with the longitudinal position z of the object 112. Thus, not only is the size of the light spot 132 affected by the longitudinal position z but also is the position on the matrix 120 of the light spot 132 changed. In Fig. 6B, this movement of the light spot 132 is denoted by arrow z .

Consequently, by this movement of the light spot 132, the z -dependency of a combined signal taking into account at least two sensor signals of the optical sensor 122 may be increased.

In the situation shown in Fig. 2, the position of the light spot 132 does not depend on z . With a change in z , depending on the optical situation, the spot will become larger or smaller, such as by becoming more diffuse or more focused. In case the spot size increases and the spot becomes more diffuse such that the combined signal $Q = s_{\text{center}}/s_{\text{sum}}$ decreases.

Contrarily, in the situation of Fig. 6A, both the size and the position of the light spot 132 are dependent on the z -coordinate. Thus, the tendency of the z -dependency of the combined signal such as the combined signal Q will be increased. In the situation of Figure 2, depending on the z -coordinate, the sensor signal of at least one optical sensor of the matrix will increase and simultaneously the sensor signal of at least one different sensor will decrease, resulting in the z -dependent combined signal Q . In the situation of Fig. 6A, the position dependency of the light spot 132 can result in three different situations depending on the relative position of light source, optical axis, and sensor, especially in case the center signal and sum signal are determined from one or more optical sensors, whereas the optical sensors used for determining the center signal and sum signal are kept fixed, independent of the z -coordinate and/or position of the light spot: Firstly, the position dependency of the light spot may result in a further decrease of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a further increase of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Secondly, the position dependency of the light spot may result in a reduced decrease or even increase of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a reduced increase or even decrease of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Thirdly, the position dependency of the light spot may be as such that the z -dependence of the sensor signals is largely unchanged compared to the situation in Figure 2. However, according to the present invention, object distance is not determined from the position of the light spot on a sensor as done in triangulation methods. Instead, movement of the light spot 132 on the matrix 120 may be used to enhance dynamic of the sensor signals and or the resulting combined signal Q which may result in an enhanced dynamic of the z -dependency.

Figure 6C shows a comparison of two experimental setups using a detector setup according to Figure 6A with a Bi-cell as optical sensors 120 with two light sensitive areas. In a first experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 132 may move in parallel to the linear boundary of the two optical sensors 120 of the Bi-cell along a direction of movement 174 in dependence of the object distance. Since the direction of movement 174 of the light spot 132 is in parallel to the linear boundary of the two light sensitive areas in dependence of the object distance, the resulting sensor signals are identical to a situation with no movement of the light spot 132 depending on object distance as shown in Figure 2. In a second experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 132 may move as such that the distance of the center of the light spot 132 to the boundary of the two optical sensors 120 of the Bi-cell changes in dependence of the object distance such as a movement orthogonal to the boundary of the two optical sensors 120 such as a movement along a direction of movement 176 in dependence of the object distance. The detector setup allowing movement of the light spot 132 may be a modification of the setup shown in Figure 6A. Thus, for most elements and optional details as well as further elements, reference may be made to the description of Fig. 6A above. In Figure 6C, the optical sensors 120 may be 1x2 matrix.

Figure 6D shows experimental results of the comparison of the two experimental setups using a detector setup according to Figure 6A, allowing movement of the light spot 132 according to Figure 6C with movement of the light spot depending on the object distance along directions of movement 174 and 176. Curve 178 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup allowing movement of the light spot 132 along a direction of movement 174 as shown in Figure 6C which is in parallel to the boundary of the optical sensors of the Bi-Cell and, which is a situation equivalent to Figure 2 without a movement of the light spot depending on the object distance. Curve 180 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup according to Figure 6A and using a detector setup allowing movement of the light spot 186 with movement of the light spot 132 according to Figure 6C with movement of the light spot depending on the object distance along a direction of movement 176. The experimental setup was as follows: The optical sensors 122 may be a 1x2 matrix, in particular a Si-Bi-Cell. The illumination source 144 may be a 950 nm laser with a spot size of 4 mm. The transfer device 128 may have a focal length of 20 mm, e.g. a lens available as Thorlabs Asphere, f= 20 mm. The distance of the object 112 was varied from 0 to 3000 mm. Determination of the longitudinal coordinate z may be possible without allowing movement of the light spot 132. In particular, according to the present invention, movement of the light spot may not be essential for determination of the longitudinal coordinate z. With the detector setup allowing movement of the light spot 132 along a direction 174 or without any movement determination of object distance is possible at very small distance, whereas with movement along a direction 176 determination of object distance is possible for object distance such as distances greater than 500mm.

Figures 7A and 7B show an exemplary embodiment of a detector 110 with a small baseline 182. Figure 7A shows a side view of the detector 110. The detector 110 may comprise the at least one illumination source 144. The illumination source 144 may be adapted to illuminate the object 112 with at least one illumination light beam 146. The illumination light beam 146 may be fully or partially reflected by the object and may travel back towards the optical sensor 122. The illumination source 144 may be arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis 126. The illumination source 144 may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis 126 do not intersect in space. The illumination source 144 and the optical axis 126 may be separated by a small baseline 182. The illumination source 144 may be spaced apart from the optical axis 126 by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and transfer device. Figure 7B shows a front view on the illumination light beam 144 and the transfer device 128. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the detector 110 may be a compact device without further optical elements, wherein the illumination source 144 may be placed as close as possible to an edge of the transfer device 128. Thus, the baseline 182 may be close to half a diameter of the transfer device 128, in particular the lens diameter and housings of lens and light source. The illumination source 144 may be arranged such that the baseline 182 is as small as possible. For example, a distance from the center of the transfer device 128 to the illumination source 144, in particular along a connecting line from the center of the transfer device 128 to the illumination source 144, may be preferably less than 2.5 times a distance from the center of the transfer device 128 to an edge if the transfer device 128, more preferably less than 1.5 times the distance center to edge of the transfer device 128, and most preferably less 1 times the distance center to edge of the transfer device 128.

Referring to Figure 8, the detector 110 such as the detector as described with respect to Figures 1 to 7 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 122. For example, the detector 110 may comprise a plurality of optical sensors arranged in the matrix 120. The detector 110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector 110 may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. As outlined above, this principle is called Distance by Photon Ratio (DPR). In one embodiment, the light beam 116 may illuminate the sensor element with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 134 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are areas of the beam profile at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . A_1 may correspond to a full or complete area of a feature point on the optical sensors. A_2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 8, the light beam 116 propagating from the object 112 to the detector 110 may illuminate the matrix 120 with at least one line pattern 184. The line pattern 184 may be generated by the object 112 for example in response to an illumination by the at least one illumination source 144 with an illumination pattern comprising at least one illumination line pattern. A_1 may correspond to an area with a full line width of the line pattern 184 in the matrix 120. The line pattern 184 in the matrix 120 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a line width in the matrix 120 is increased. In particular, the line width of the line pattern 184 in the matrix 120 may change from one column to another column. A_2 may be a central area of the line pattern 184 in the matrix 120. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern 184 may be segmented in the matrix 120. Each of the columns may comprise center information of intensity in the central area of the line pattern 184 and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern 184.

Referring to Figures 9A and 9B, as outlined above, the detector 110 may comprise the at least one matrix 120 of optical sensors 122. With the aid of such a pixelated imaging device, a defocused beam profile may be subdivided into cross-sections along lines of a certain angle θ and with a distance ω from the origin of ordinates as shown in Figure 9A. Accordingly, the parameterization of a single line would be given by $\omega = x \cos(\theta) + y \sin(\theta)$. The integration of the intensity along parallel lines can be mathematically described by an integral projection $\Re\{\cdot\}$ of the well-known Radon transform which reads

$$\Re_{(\omega, \theta)}\{f(x, y)\} = \iint_{-\infty}^{\infty} f(x, y) \delta(x \cos(\theta) + y \sin(\theta) - \omega) dx dy$$

where δ denotes the Dirac delta function and $f(x, y)$ is the intensity of an enclosed defocused beam profile. The photon ratio R for a given angle θ and projection width ω is then given by

$$R = \frac{\Re_{(\omega, \theta)}\{f'(x, y)\}}{\Re_{(\omega, \theta)}\{f(x, y)\}}$$

with $f'(x, y)$ as the overshined image region highlighted in Fig. 9. The variation of θ may yield different ratios R for skewed object surfaces at a certain distance. It is sufficient to let θ vary in the following interval $\{\theta \in \mathbb{R}_+, \theta < \pi\}$.

- 5 Figures 10A to 10C show three embodiments of a hexagonal illumination pattern. The illumination source 144 may be adapted to generate at least one illumination pattern for illuminating the object 112. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. Specifically, the illumination source 144 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor
- 10 lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination pattern may comprise at least one feature such as a point or symbol. The illumination pattern may comprise a plurality of features. The illumination pattern may comprise an arrangement of periodic or non-periodic features. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one
- 15 illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasi-periodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines; at least one fringe pattern. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern
- 20 may comprise as much as possible features per area such that hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.
- 25
- 30 The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on an epipolar line. As shown in Figure 10A, the illumination pattern may comprise at least one hexagonal pattern, wherein the individual points are positioned on epipolar lines 186. As shown in Figure 10B, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Such a positioning
- 35 of the illumination features allows enhancing distance between the individual points on each epipolar line. For example as shown in Figure 10C, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the
- 40 distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other. Such a positioning allows to enhance the number of possible features per area.

Figure 11 shows an embodiment of a scanning system 154. The scanning system 154 may be adapted as a line scanning device. In particular, the scanning system 154 may comprise at least one sensor line or row of optical sensors 113. Furthermore, the scanning system may comprise the at least one transfer device 128 and the at least one illumination source 144. Triangulation systems require a sufficient baseline, however due to the baseline in the near field no detection may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. Further, these near field and far field problems can be overcome by using the scanning system 154 of Figure 11. The scanning system 154 may be adapted to detect a plurality of light beams 116 propagating from the object 112 to the scanning device 154 on the CMOS line. The light beams 116 may be generated at different position on the object 112 or by movement of the object 112. The scanning system 154 may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the combined signal Q as described above.

List of reference numbers

110	detector
112	object
114	beacon device
116	light beam
118	sensor element
120	matrix
122	optical sensor
124	light-sensitive area
126	optical axis
128	transfer device
130	focal point
132	light spot
134	evaluation device
136	center detector
138	summing device
140	combining device
142	divider
143	position evaluation device
144	illumination source
146	illumination light beam
148	reflective element
149	camera
150	detector system
151	human-machine interface
152	entertainment device
153	tracking system
154	scanning system
156	connector
158	housing
160	control device
162	user
164	opening
166	direction of view
168	coordinate system
170	machine
172	track controller
174	direction of movement
176	direction of movement
178	curve
180	curve
182	baseline

- 184 Line pattern
- 186 Epipolar line

Patent claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) having
- 5 - at least one sensor element (118) having a matrix (120) of optical sensors (122), the optical sensors (122) each having a light-sensitive area (124), wherein each optical sensor (122) is configured to generate at least one sensor signal in response to an illumination of the light-sensitive area (124) by at least one light beam (116) propagating from the object (112) to the detector (110);
- 10 - at least one evaluation device (134) configured for evaluating the sensor signals, by
- a) determining at least one optical sensor (122) having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors (122) of the matrix (120) and forming at least one sum signal;
- 15 c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.
- 20 2. The detector (110) according to the preceding claim, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors (122) containing the optical sensor (122) having the highest sensor signal and a predetermined group of neighboring optical sensors (122); a
- 25 sum of sensor signals from a group of optical sensors (122) containing the optical sensor (122) having the highest sensor signal and a predetermined group of neighboring optical sensors (122); a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an inte-
- 30 gral of sensor signals from a group of optical sensors (122) containing the optical sensor (122) having the highest sensor signal and a predetermined group of neighboring optical sensors (122); an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a pre-
- 35 determined threshold.
3. The detector (110) according to any one of the preceding claims, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix (120); a sum of all sensor signals of the matrix (120); an integral of all sensor signals of the matrix (120); an average over all sensor signals of the matrix (120) except for sensor signals from those
- 40 optical sensors (122) contributing to the center signal; a sum of all sensor signals of the matrix (120) except for sensor signals from those optical sensors (122) contributing to the center signal; an integral of all sensor signals of the matrix (120) except for sensor signals from those optical sensors (122) contributing to the center signal; a sum of sensor signals of optical sensors (122) within a predetermined range from the optical sensor (122) having the highest sensor

signal; an integral of sensor signals of optical sensors (122) within a predetermined range from the optical sensor (122) having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors (122) being located within a predetermined range from the optical sensor (122) having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors (122) being located within a predetermined range from the optical sensor (122) having the highest sensor signal.

4. The detector (110) according to any one of the preceding claims, wherein the combined signal is a combined signal Q, derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of a first linear combination of the center signal and the sum signal and a second linear combination of the center signal and the sum signal.

5. The detector (110) according to the preceding claim, wherein the evaluation device (134) is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate for determining the longitudinal coordinate.

6. The detector (110) according to any one of the two preceding claims, wherein the evaluation device (134) comprises at least one divider (142), wherein the divider (142) is configured for deriving the combined signal.

7. The detector (110) according to any one of the preceding claims, wherein the evaluation device (134) is further configured for determining at least one transversal coordinate of the object (112) by evaluating a transversal position of the at least one optical sensor (122) having the highest sensor signal.

8. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (144), wherein the illumination source (144) is adapted to illuminate the object (112) with at least one illumination light beam, wherein the illumination source (144) is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis (126).

9. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises the at least one illumination source (144), wherein the illumination source (144) and the optical axis (126) are separated by a small baseline (182).

10. A detector system (150) for determining a position of at least one object (112), the detector system (150) comprising at least one detector (110) according to any one of the preceding claims, the detector system (150) further comprising at least one beacon device (114) adapted to direct at least one light beam (116) towards the detector (110), wherein the beacon device (114) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).

11. A human-machine interface (151) for exchanging at least one item of information between a user (162) and a machine (170), wherein the human-machine interface (151) comprises at least one detector system (150) according to the preceding claim, wherein the at least one beacon device (114) is adapted to be at least one of directly or indirectly attached to the user (162) and held by the user (162), wherein the human-machine interface (151) is designed to determine at least one position of the user (162) by means of the detector system (150), wherein the human-machine interface (151) is designed to assign to the position at least one item of information.
12. An entertainment device (152) for carrying out at least one entertainment function, wherein the entertainment device (152) comprises at least one human-machine interface (151) according to the preceding claim, wherein the entertainment device (152) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (151), wherein the entertainment device (152) is designed to vary the entertainment function in accordance with the information.
13. A tracking system (153) for tracking a position of at least one movable object (112), the tracking system (153) comprising at least one detector system (150) according to any one of the preceding claims referring to a detector system (150), the tracking system (153) further comprising at least one track controller (172), wherein the track controller (172) is adapted to track a series of positions of the object (112) at specific points in time.
14. A scanning system (154) for determining a depth profile of a scenery, the scanning system (154) comprising at least one detector (110) according to any of the preceding claims referring to a detector (110), the scanning system (154) further comprising at least one illumination source (144) adapted to scan the scenery with at least one light beam (146, 116).
15. A camera (149) for imaging at least one object (112), the camera (149) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
16. A method for determining a position of at least one object (112) by using at least one detector (110), the method comprising the following steps:
- illuminating at least one sensor element (118) of the detector (110) with at least one light beam (116) propagating from the object (112) to the detector (110), the detector (110) having a matrix (120) of optical sensors (122), the optical sensors (122) each having a light-sensitive area (124), wherein each optical sensor (122) generates at least one sensor signal in response to the illumination;
 - evaluating the sensor signals, by
 - a) determining at least one optical sensor (122) having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors (122) of the matrix (120) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and

d) determining at least one longitudinal coordinate z of the object (112) by evaluating the combined signal.

- 5 17. The method according to the preceding claim, wherein the combined signal is a combined signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.
- 10 18. A use of the detector (110) according to any one of the preceding claims relating to a detector (110), for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface (151) application; a tracking application; a photography application; a logistics application; a machine vision application; a
- 15 robotics application; a quality control application; a manufacturing application.

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FIG.1

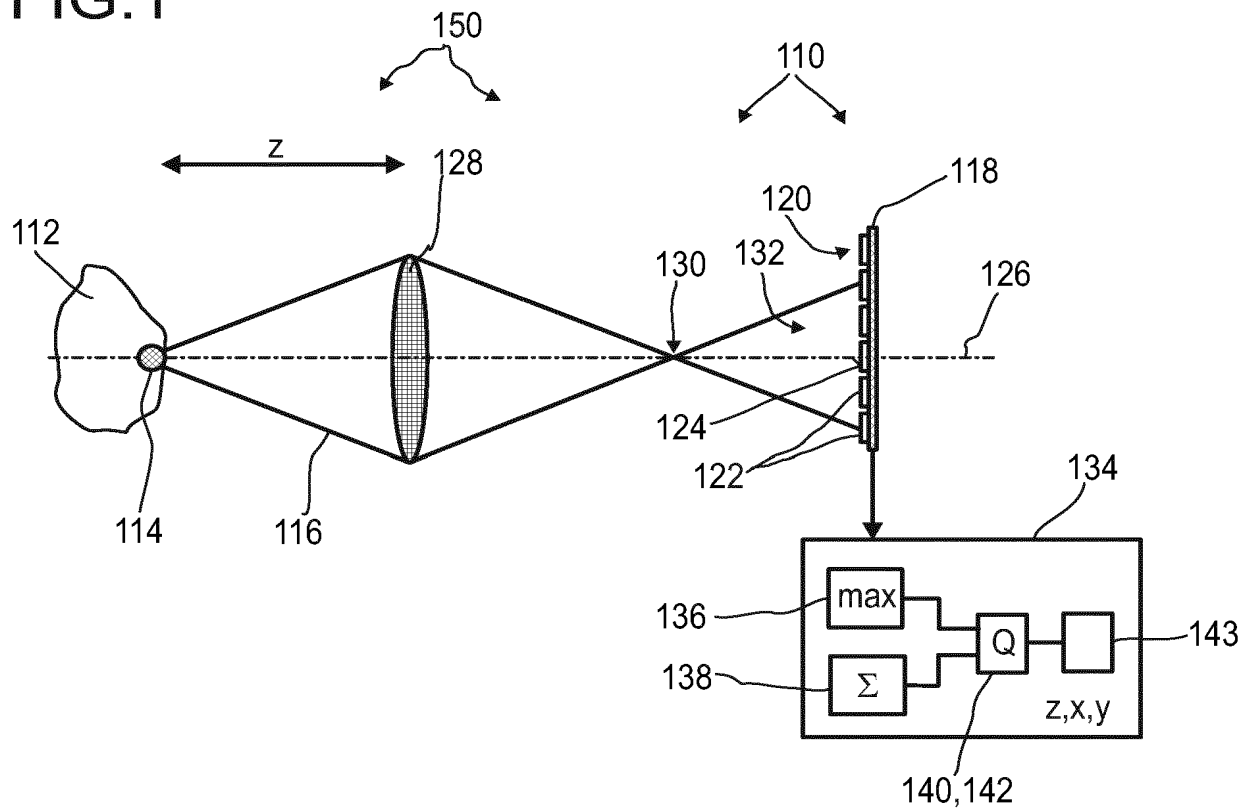


FIG.2

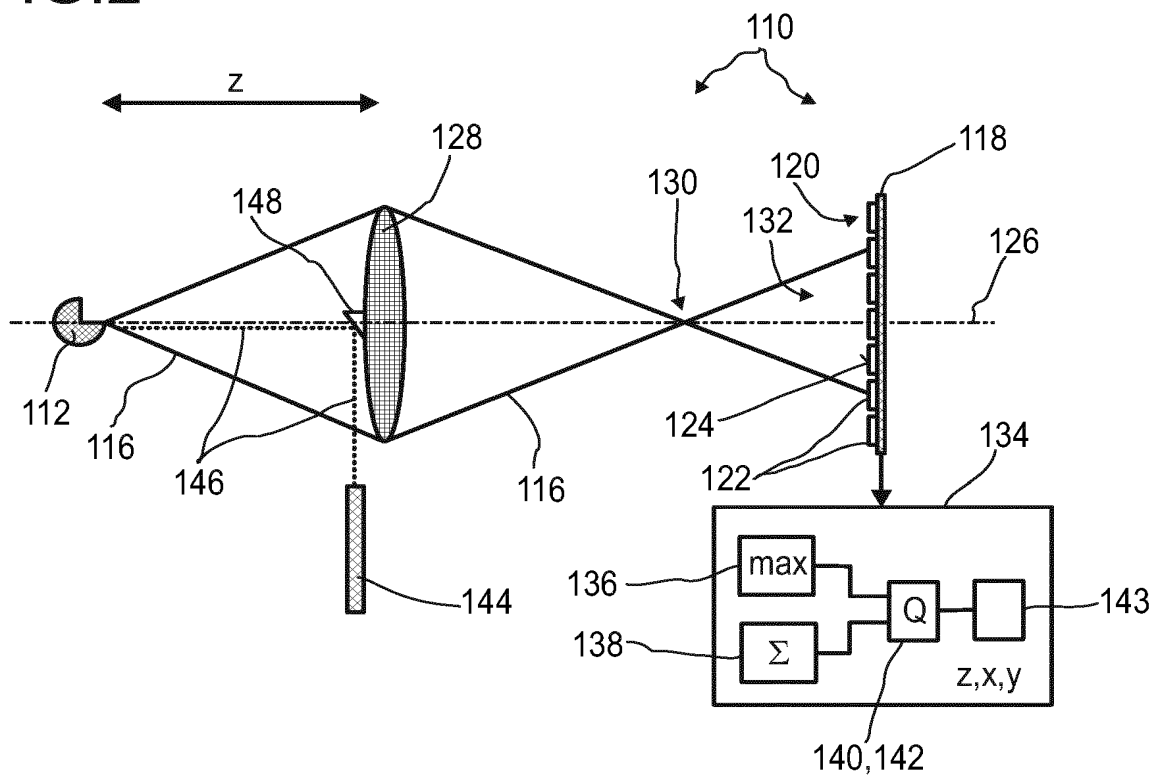


FIG.3

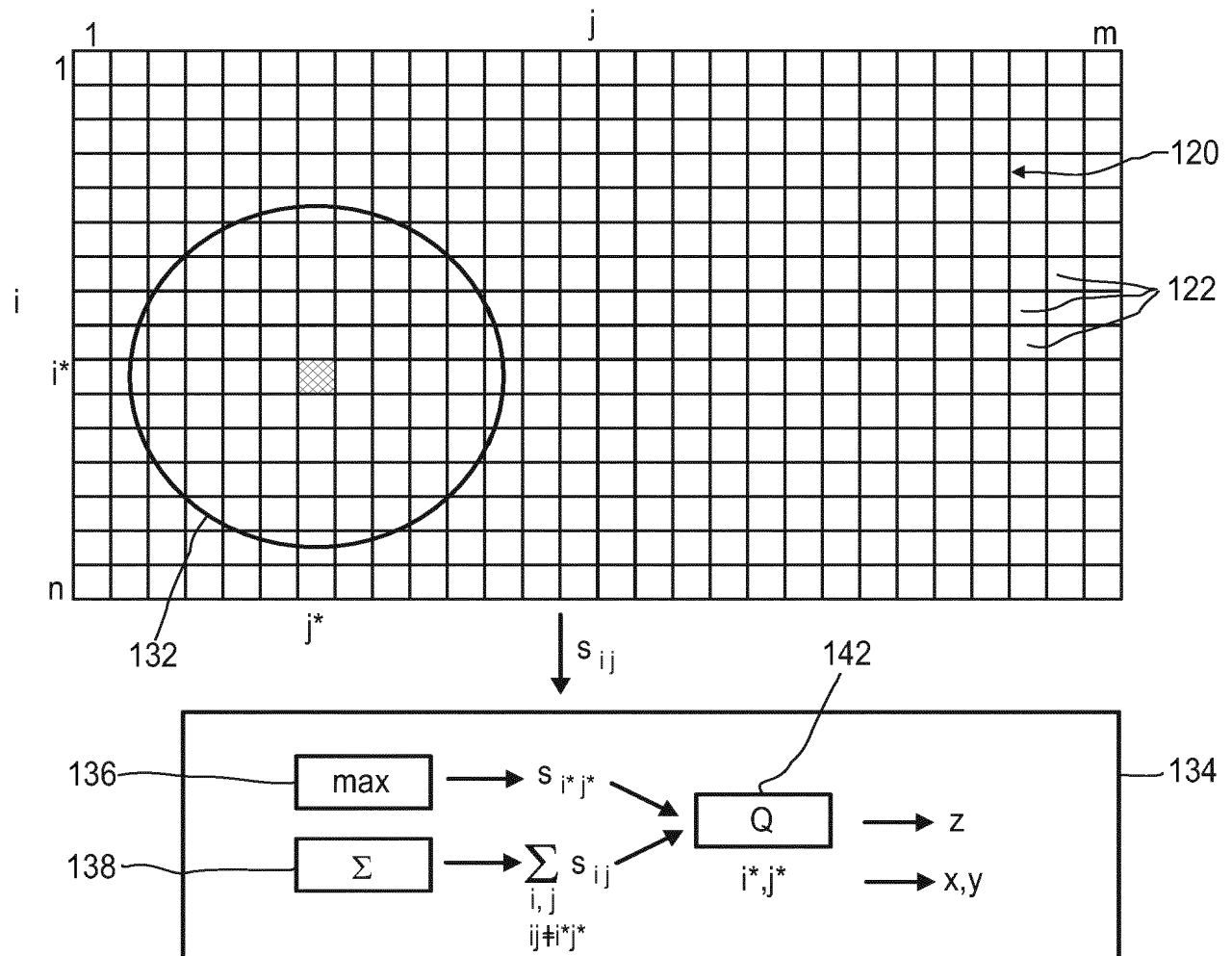
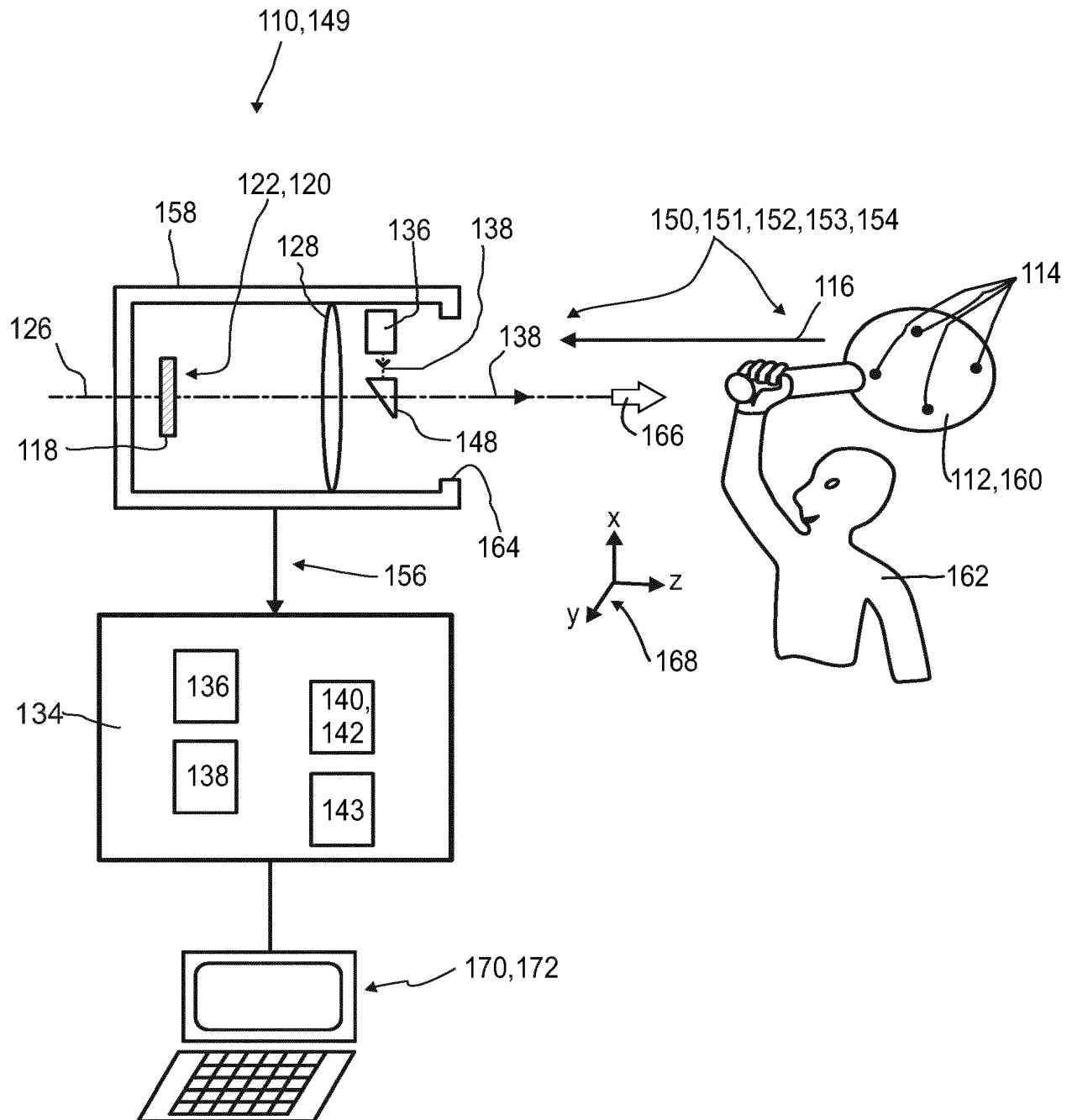


FIG.4



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FIG.5

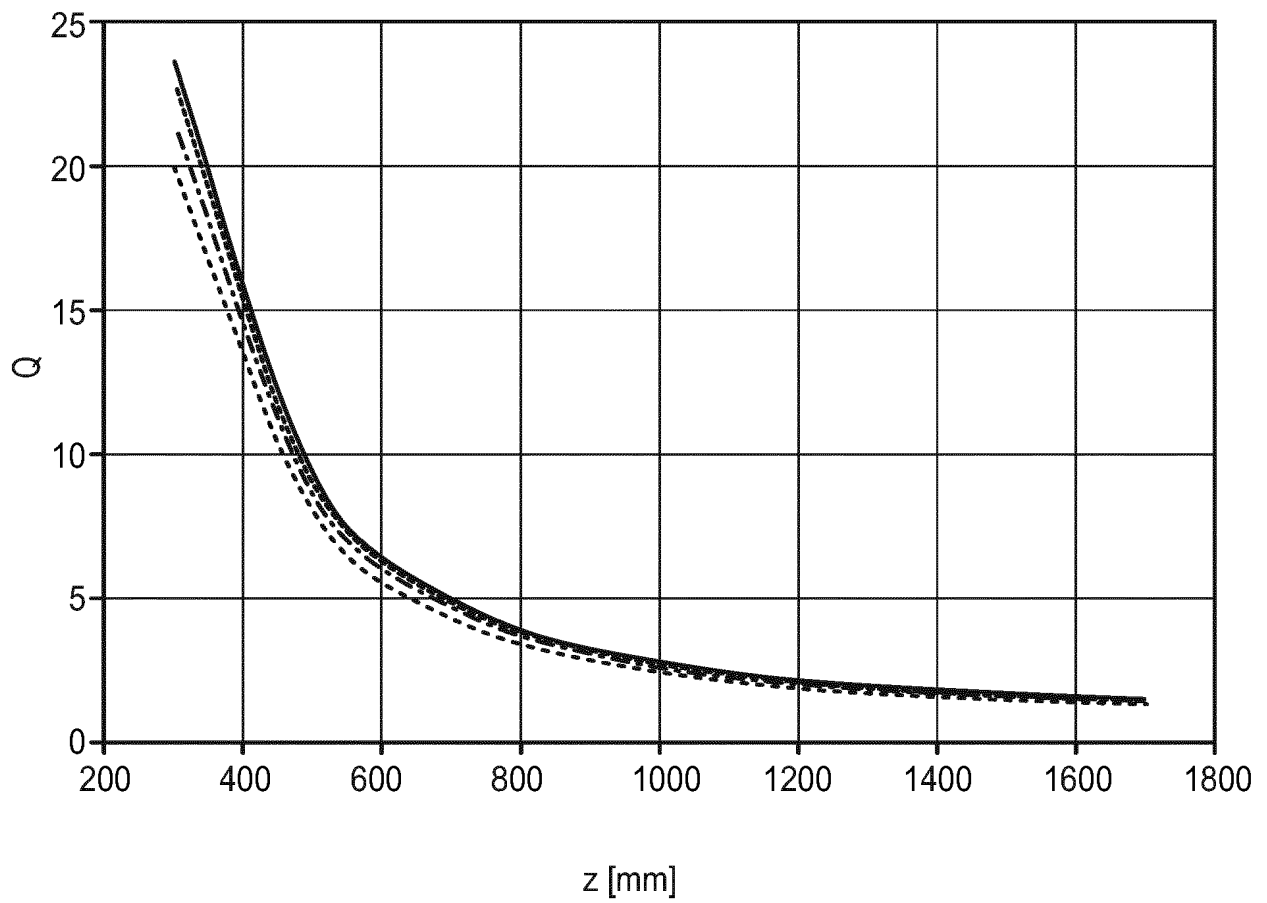


FIG.6A

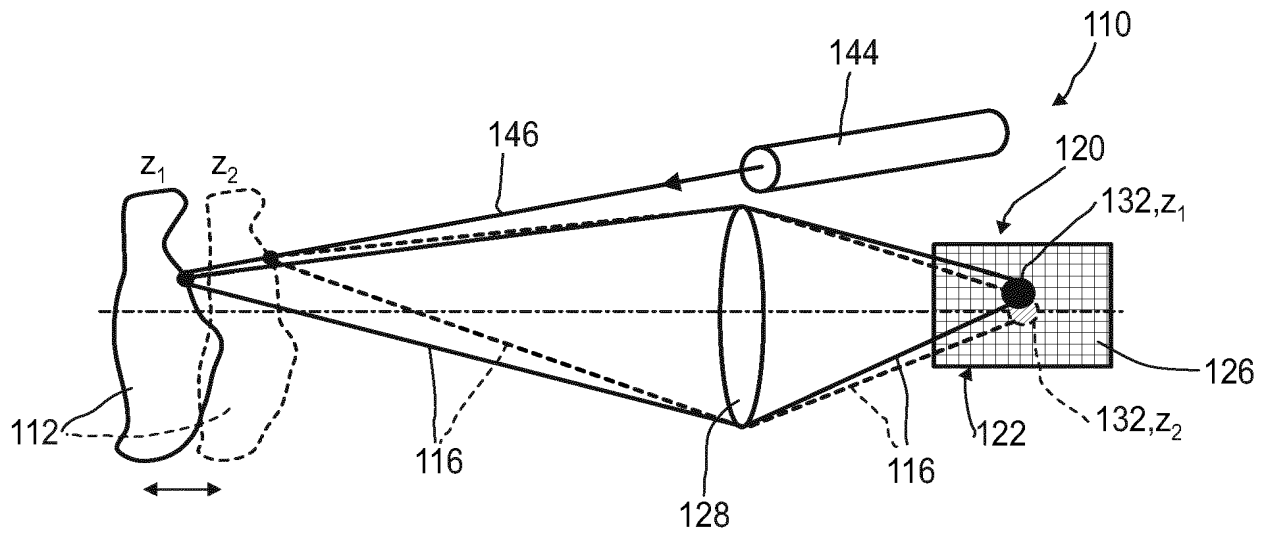
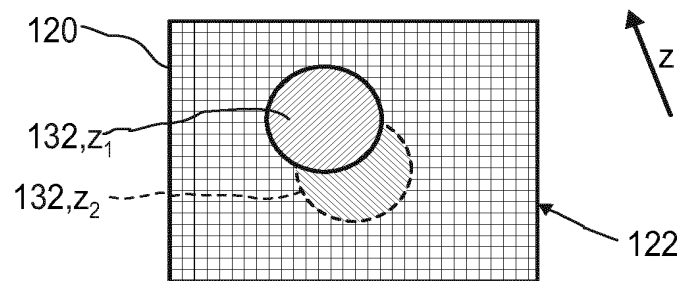


FIG.6B



6/10

FIG.6C

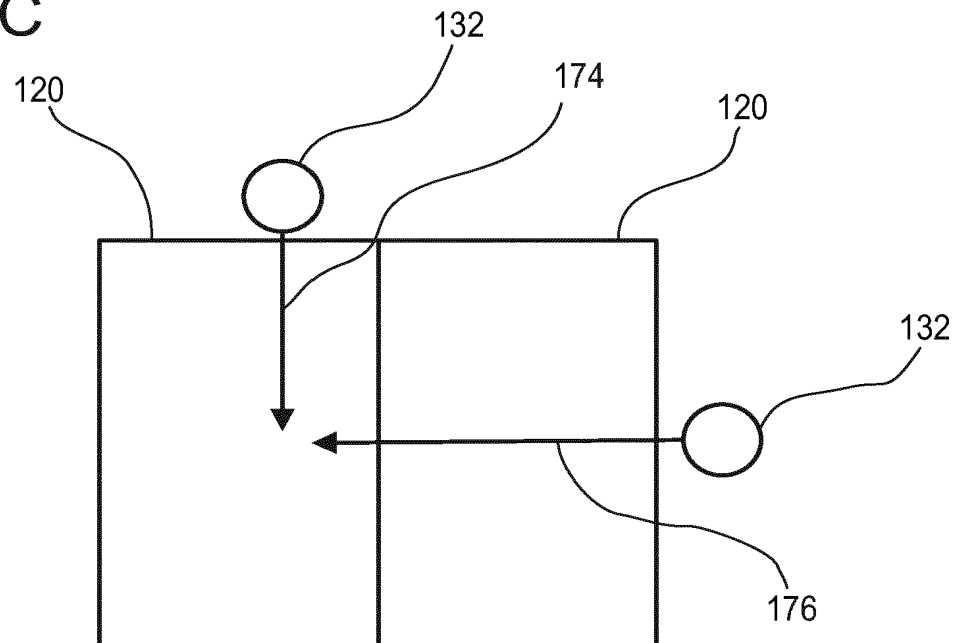


FIG.6D

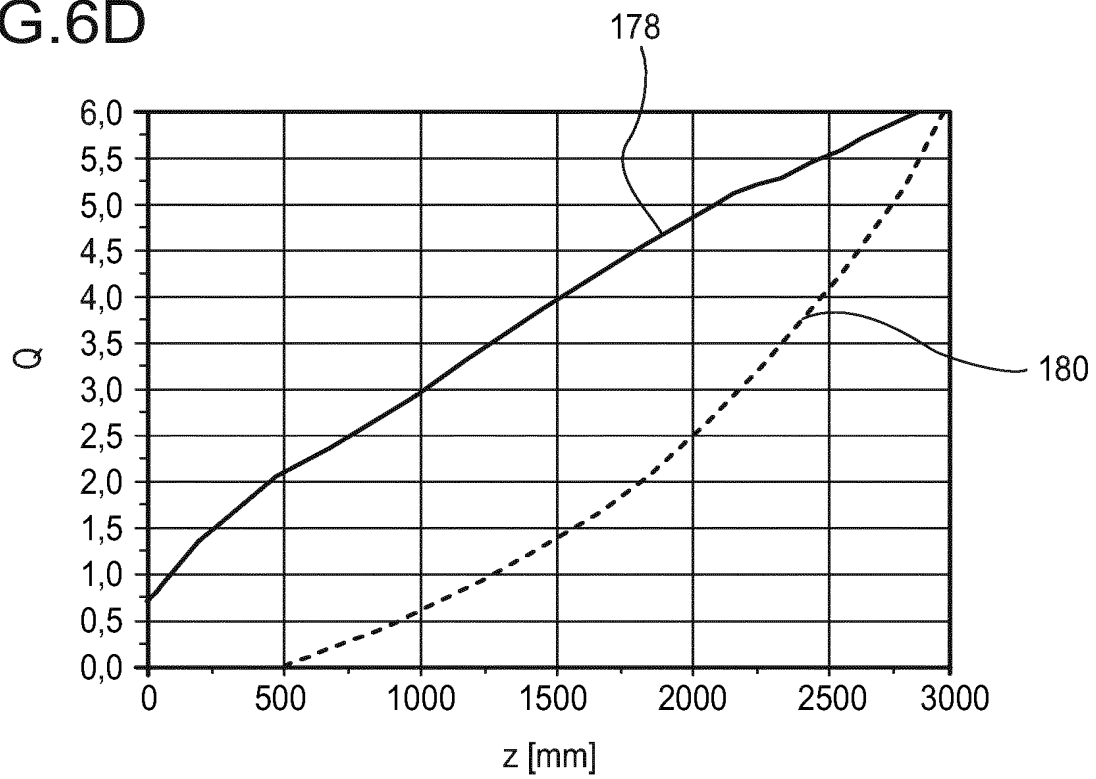


FIG.7A

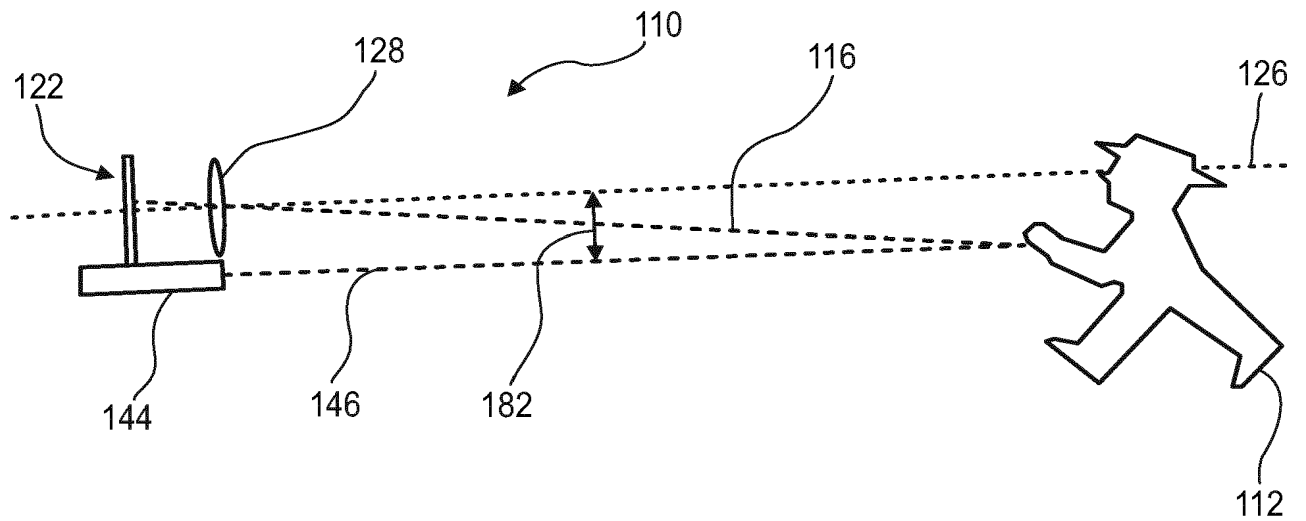


FIG.7B

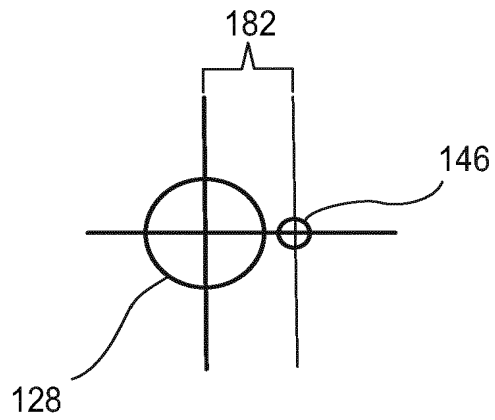


FIG.8

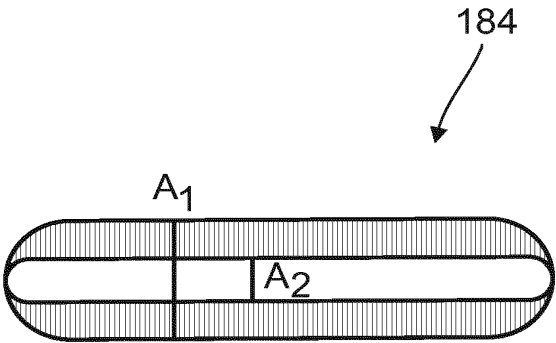


FIG.9A

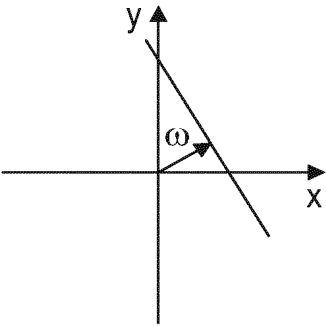


FIG.9B

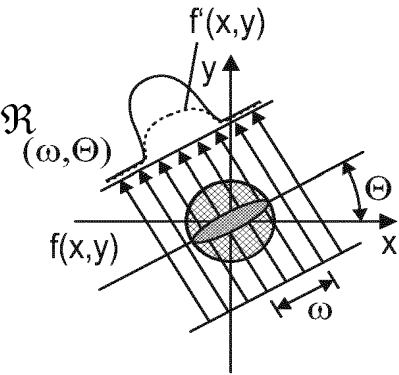


FIG.10A

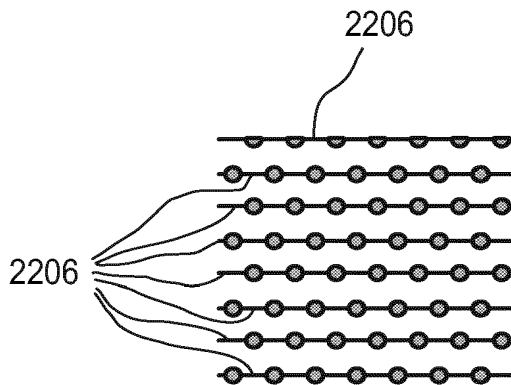


FIG.10B

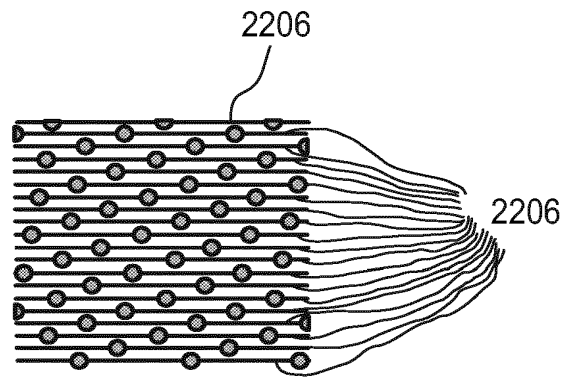


FIG.10C

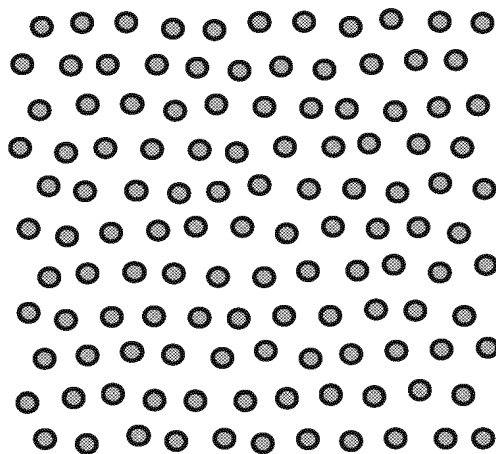
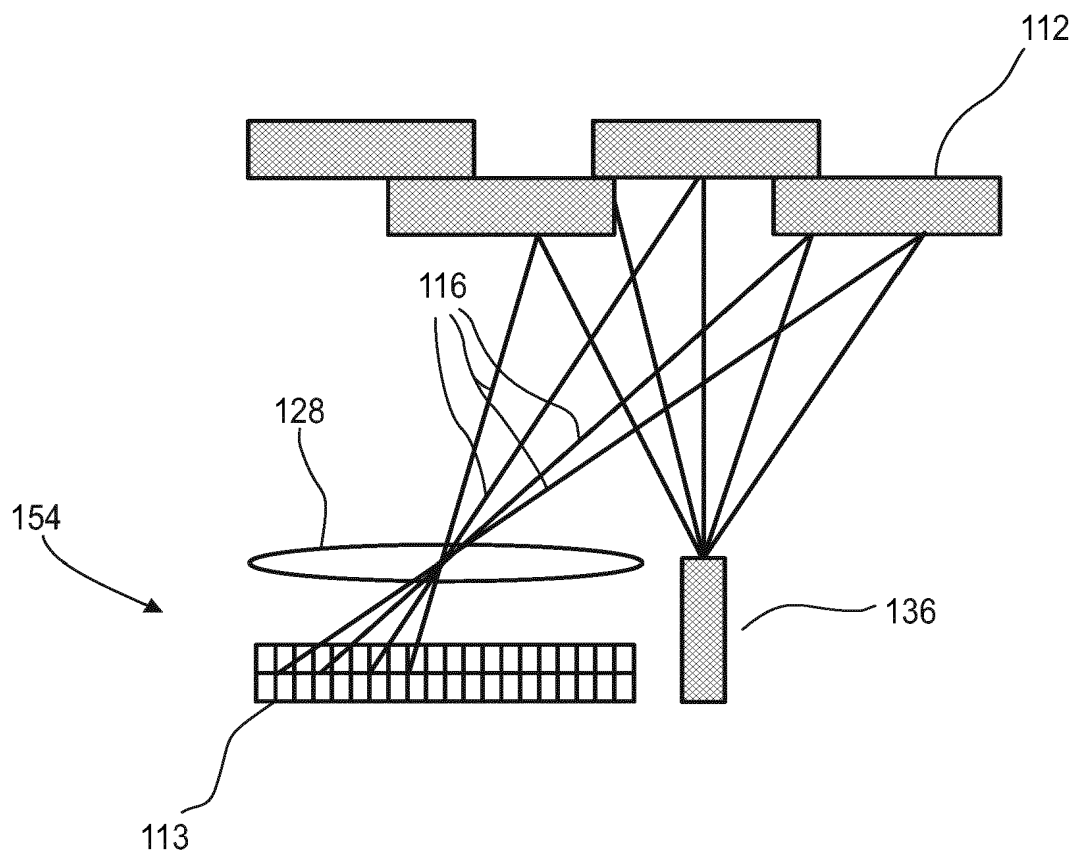


FIG.11



INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2017/079577

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S17/46 G01S7/481 G01S5/16 G01S7/491 G01S11/12
G01S7/48

ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data, INSPEC

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2006/044546 A1 (LEWIN ANDREW C [GB] ET AL) 2 March 2006 (2006-03-02)	1-18
Y	paragraphs [0001], [0030], [0093], [0110], [0118], [0119], [0121] - paragraphs [0123], [0129]; figures 2a, 2b	10
X	US 2008/130005 A1 (WASLOWSKI KAI [DE] ET AL) 5 June 2008 (2008-06-05)	1-3,16
	paragraphs [0011], [0014] - paragraphs [0016], [0030], [0037], [0041], [0049], [0051], [0054], [0063], [0086], [0091]; figure 1	
Y	US 2015/286340 A1 (SEND ROBERT [DE] ET AL) 8 October 2015 (2015-10-08)	10
	cited in the application paragraph [0519]	



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

2 February 2018

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Beer, Mark

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2017/079577

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		US 2015286340 A1 08-10-2015	
		US 2017082486 A1 23-03-2017	
		WO 2015024871 A1 26-02-2015	



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17195427.4	09 October 2017 (09.10.2017)	EP

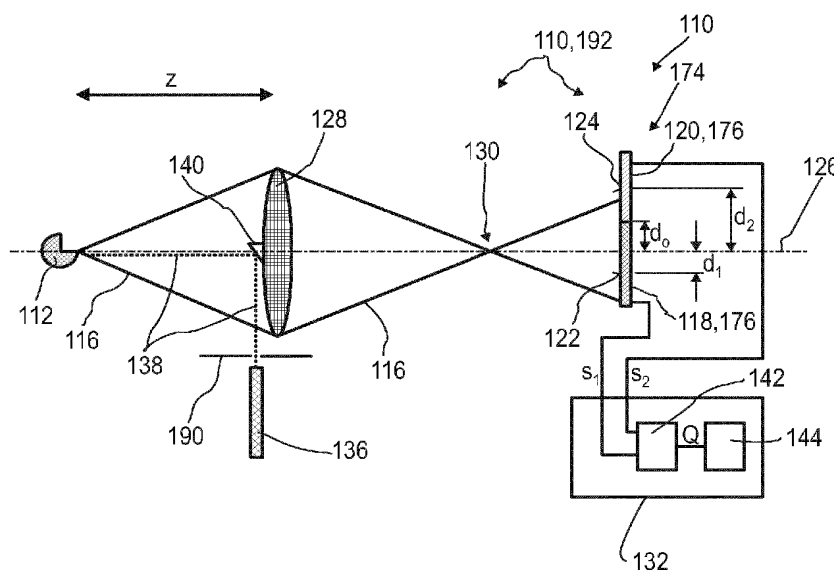
17195398.7 09 October 2017 (09.10.2017) EP

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(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.2



(57) **Abstract:** A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises: -at least two optical sensors (118, 120, 176), each optical sensor (118, 120, 176) having a light-sensitive area (122, 124), wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110) by different spatial offsets, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light beam (116) propagating from the object (112) to the detector (110); and -at least one evaluation device (132) being configured for determining at least one longitudinal coordinate z of the object (112) by combining the at least two sensor signals.

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(81) **Designated States** (*unless otherwise indicated, for every kind of national protection available*): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IR, IS, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, SM, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, ZA, ZM, ZW.

(84) **Designated States** (*unless otherwise indicated, for every kind of regional protection available*): ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:

- with international search report (Art. 21(3))
- with information concerning incorporation by reference of missing parts and/or elements (Rule 20.6)

Detector for optically detecting at least one object

Description

5 Field of the invention

The invention relates to a detector, a detector system and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system, a readout device for optical storage media and various uses of the detector device. The devices, systems, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

20 Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultraviolet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and for reasons of large-area processing, sensors comprising at least one organic sensor material are being used, as described for example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO 2009/013282 A1.

A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for opti-

cally detecting objects are triangulation systems, by means of which distance measurements can likewise be carried out. Kurt Konolige et al., A Low-Cost Laser Distance Sensor, 2008 IEEE International Conference on Robotics and Automation, Pasadena, CA, USA, May 19-23, 2008, discuss competing technologies using triangulation for planar laser distance sensor (LDS).

5 Structured line devices use a light stripe laser and offset camera to determine range to a set of points. Because the laser energy is spread over a line, it is difficult to achieve accurate range, especially in the presence of ambient light, or with darker objects. Point scan devices for 3D scanning of small objects typically use a scanning mirror to direct a point laser beam and redirect the laser return to an optical receiver. Such devices cannot be miniaturized, and their cost and mechanical fragility will remain high. Centroid point modules typically use position-sensitive devices (PSD). These devices measure the centroid of all light impinging on their surface. Although modulation techniques can be used to offset some of the effects of ambient light, PSDs do not perform well unless the laser spot has a very strong reflection, limiting their use to ranges of a meter or less. Pixel-based point modules search the pixel with maximum signal intensity to determine the position of the light spot on the sensor. Typically CMOS line arrays are used for detection. Konolige et al. introduce a low-cost version of a Pixel-based point module.

US 5 235 377 A describes a measuring apparatus for a camera, wherein light is projected to an object from a projection section. Light reflected on the object is received at a first light receiving section and a second light receiving section arranged to be adjacent to each other. In the first light receiving section, a first and second signals relating to a light receiving position and a quantity of received light are output, and in the second light receiving section, a third signal relating to a quantity of light of the reflected light is output. In a discriminating and calculating section, if it is discriminated that the reflected light is positioned at a place, which is nearer than a predetermined distance, a measuring distance can be calculated from an output ratio of the first signal to the second signal by a measuring calculation circuit. On the other hand, if it is discriminated that the reflected light is positioned at a place, which is farther than the predetermined distance, a measuring distance can be calculated from the output ratio of the sum of the first and second signals to the third signal by the measuring calculation circuit.

US 5 512 997 A describes a distance measuring device having an infrared light-emitting diode by which an infrared light is emitted towards a subject, and first and second linear line sensors on which an image corresponding to light reflected from the subject is formed. The first and second linear line sensors are disposed adjacently and in parallel to each other, and offset by an amount corresponding to half of the predetermined A/D converting step. Analog distance data outputted from the linear line sensors are converted to digital distance data using the predetermined A/D converting step, and then, the digital distance data are added together to obtain a subject distance with a high accuracy.

DE 42 11 875 A1 describes an optical rangefinder with electronic correction for spot eccentricity. The optical rangefinder has an evaluation circuit, which determines if an output distance measurement value is valid or erroneous due to e.g. inclination of object surface. The distance signal is determined by evaluation a position a light spot on a first detector. A second control

signal, which characterizes inclination of the object surface, is determined by evaluating eccentricity of a light spot on a second detector, arranged in a different plane.

WO 2016/051323 A1 describes a detector and a method for optically determining a position of at least one object. The detector comprises at least one optical sensor for determining a position of at least one light beam and at least one evaluation device for generating at least one item of information on a transversal position of the object and at least one item of information on a longitudinal position of the object. The sensor has at least a first electrode and a second electrode. At least one photovoltaic material is embedded in between the first electrode and the second electrode. The first electrode or the second electrode is a split electrode having at least three partial electrodes. The detector and the method can determine three-dimensional coordinates of an object in a fast and efficient way.

In WO 2012/110924 A1, the content of which is herewith included by reference, a detector for optically detecting at least one object is proposed. The detector comprises at least one optical sensor. The optical sensor has at least one sensor region. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area. The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

WO 2014/097181 A1, the full content of which is herewith included by reference, discloses a method and a detector for determining a position of at least one object, by using at least one transversal optical sensor and at least one optical sensor. Specifically, the use of sensor stacks is disclosed, in order to determine a longitudinal position of the object with a high degree of accuracy and without ambiguity.

WO 2015/024871 A1, the full content of which is herewith included by reference, discloses an optical detector, comprising:

- at least one spatial light modulator being adapted to modify at least one property of a light beam in a spatially resolved fashion, having a matrix of pixels, each pixel being controllable to individually modify the at least one optical property of a portion of the light beam passing the pixel;
- at least one optical sensor adapted to detect the light beam after passing the matrix of pixels of the spatial light modulator and to generate at least one sensor signal;
- at least one modulator device adapted for periodically controlling at least two of the pixels with different modulation frequencies; and
- at least one evaluation device adapted for performing a frequency analysis in order to determine signal components of the sensor signal for the modulation frequencies.

US 4,767,211 discloses an apparatus for and a method of measuring a boundary surface of a sample in which a ratio of the light quantity of a part of reflected light from a sample which travels in the vicinity of the optical axis of the reflected light, to the light quantity of another part of the reflected light which is directed to a position deviating from the optical axis by a predetermined distance is used to accurately measure a boundary surface of a sample. Since the accuracy of measurement is increased by using the above ratio, light capable of passing through the sample can be used as incident light. Thus, a deep hole in the surface of the sample and a void such as an air bubble in a living being sample, which cannot be measured by the prior art, can be measured very accurately.

WO 2014/198629 A1, the full content of which is herewith included by reference, discloses a detector for determining a position of at least one object, comprising:

- at least one optical sensor, the optical sensor being adapted to detect a light beam propagating from the object towards the detector, the optical sensor having at least one matrix of pixels; and
- at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

Further, generally, for various other detector concepts, reference may be made to WO 2014/198626 A1, WO 2014/198629 A1 and WO 2014/198625 A1, the full content of which is herewith included by reference. Further, referring to potential materials and optical sensors which may also be employed in the context of the present invention, reference may be made to European patent applications EP 15 153 215.7, filed on January 30, 2015, EP 15 157 363.1, filed on March 3, 2015, EP 15 164 653.6, filed on April 22, 2015, EP 15177275.3, filed on July 17, 2015, EP 15180354.1 and EP 15180353.3, both filed on August 10, 2015, and EP 15 185 005.4, filed on September 14, 2015, EP 15 196 238.8 and EP 15 196 239.6, both filed on November 25, 2015, EP 15 197 744.4, filed on December 3, 2015, the full content of all of which is herewith also included by reference.

Further, reference may be made to detector concepts comparing signals of at least two different sources for determining a position of an object. Thus, as an example, reference may be made to EP 16155834.1, EP 16155835.8 or EP 16155845.7, all filed on February 16, 2016, the full disclosure of which is herewith included by reference.

US 5,235,377 A describes a measuring apparatus for a camera. Light is projected to an object from a projection section, and light reflected on the object is received at a first light receiving section and a second light receiving section arranged to be adjacent to each other. In the first light receiving section, a first and second signals relating to a light receiving position and a quantity of received light are output, and in the second light receiving section, a third signal relating to a quantity of light of the reflected light is output. In a discriminating and calculating sec-

tion, if it is discriminated that the reflected light is positioned at a place, which is nearer than a predetermined distance, a measuring distance can be calculated from an output ratio of the first signal to the second signal by a measuring calculation circuit. On the other hand, if it is discriminated that the reflected light is positioned at a place, which is farther than the predetermined distance, a measuring distance can be calculated from the output ratio of the sum of the first and second signals to the third signal by the measuring calculation circuit.

US 5,512,997 A describes a distance measuring device having an infrared light-emitting diode by which an infrared light is emitted towards a subject, and first and second linear line sensors on which an image corresponding to light reflected from the subject is formed. The first and second linear line sensors are disposed adjacently and in parallel to each other, and offset by an amount corresponding to half of the predetermined A/D converting step. Analog distance data outputted from the linear line sensors are converted to digital distance data using the predetermined A/D converting step, and then, the digital distance data are added together.

DE 42 11 875 A1 describes that a beam from e.g. a collimated laser diode passes through crossed cylinder lenses which focus the rays in both sectional planes and adapt the diameter of the beam to the numerical aperture of an objective lens. A mirror section cuts off half of the incident beam and deflects light scattered from the object to a differential photodiode detector. A beam splitter reflects part of the deflected beam on to a four-quadrant photodetector in an orthogonal plane. A first electronic circuit produces a signal corresponding to the position of the light spot on the detector. A second circuit produces a monitoring or correction signal derived from the eccentricity of the spot position.

US 4,767,211 A describes an apparatus for and a method of measuring a boundary surface of a sample are disclosed in which a ratio of the light quantity of a part of reflected light from a sample which travels in the vicinity of the optical axis of the reflected light, to the light quantity of another part of the reflected light which is directed to a position deviating from the optical axis by a predetermined distance is used to measure a boundary surface of a sample.

Despite the advantages implied by the above-mentioned devices and detectors, several technical challenges remain. Thus, generally, a need exists for detectors for detecting a position of an object in space which is both reliable and may be manufactured at low cost. Specifically, a need exists for 3D-sensing concepts. Various known concepts are at least partially based on using so-called FiP sensors, such as several of the above-mentioned concepts. Therein, as an example, large area sensors may be used, in which the individual sensor pixels are significantly larger than the light spot and which are fixed to a specific size. Still, large area sensors in many cases are inherently limited in the use of the FiP measurement principle, specifically in case more than one light spot is to be investigated simultaneously.

A further challenge using FiP detectors is detector area or active area. Typically, for distance measurements, a large active area of the detector is used or even is required. This area, however, may cause noise problems, specifically when the tetralateral conductivity concept is em-

ployed to build a PSD. This often results in poor signal-to-noise-ratios and slow detector response times due to the large capacitance in conjunction with the series resistance of the detector. A further challenge using FiP detectors is cost of manufacturing. Thus, in many cases, typical FiP sensors are expensive, as compared to e.g. conventional Si-based photodiodes. Further, the evaluation of the measurement results of measurements using FiP-sensors remains an issue, specifically in case the total power of a light beam is unknown. In case a plurality of FiP-sensors is used, located at different positions along an axis of propagation of the light beam, a range of measurement typically is limited to the range in between the two positions of the sensors. Further, many FiP-detectors show a luminance dependency which renders the evaluation of the measurement result more difficult, and, additionally, FiP-measurements in many cases are dependent on a target spot size.

Referring to concepts comparing sensor signals generated by at least two different sources such as at least two different optical sensors, one technical challenge remains a reduction of cost of the overall system and, specifically, of the optical sensors. Thus, in many concepts, special optical sensors have to be used, having a dedicated design and placement, which typically requires setting up an expensive optoelectronic semiconductor manufacturing process and a complex assembly scheme.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide cost-efficient devices and methods which reliably may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and

- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term “at least two optical sensors, each having at least one light sensitive area” refers to configurations with two single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having at least two light sensitive areas. Thus, the term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one bi-cell and/or quadrant diode device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor, may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. Other embodiments, however, are feasible. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photosensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the setup.

As further used herein, a “sensor signal” generally refers to a single signal or a plurality of sensor signals such as a series of sensor signals or a continuous sensor signal, generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the

light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

The light beam may propagate from the object towards the detector. As will be outlined in further detail below, the light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The illumination source, as an example, may be or may comprise one or more of an external illumination source, an illumination source integrated into the detector or an illumination source integrated into a beacon device being one or more of attached to the object, integrated into the object or held by the object.

As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam specifically may be a Gaussian light beam, as will be outlined in further detail below. Other embodiments are feasible, however.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The light-sensitive areas of the optical sensors may overlap, as visible from the object, or may not overlap, i.e. may be placed next to each other without overlap. The light-sensitive areas may be spaced apart from each other or may directly be adjacent.

The detector has an optical axis. As used herein, the term “optical axis” generally refers to a line of symmetry of the optical setup of the detector. In case the detector has a lens or lens system, such as a lens or lens system being part of a transfer system which will be explained in further detail below, the optical axis, as an example, may be an axis of mirror symmetry or rotational symmetry of the lens or lens system. Thus, as will be outlined in further detail below, the detector may comprise at least one transfer system, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with

the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis, and the condition noted above generally may refer to each beam path independently. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths. In case the optical sensors are distributed over different partial beam paths, the above-mentioned condition may be described such that at least one first optical sensor is located in at least one first partial beam path, being offset from the optical axis of the first partial beam path by a first spatial offset, and at least one second optical sensor is located in at least one second partial beam path, being offset from the optical axis of the second partial beam path by at least one second spatial offset, wherein the first spatial offset and the second spatial offset are different.

The detector may comprise more than two optical sensors. In any case, i.e. in the case of the detector comprising precisely two optical sensors and in the case of the detector comprising more than two optical sensors, the optical sensors may comprise at least one first optical sensor being spaced apart from the optical axis by a first spatial offset and at least one second optical sensor being spaced apart from the optical axis by a second spatial offset, wherein the first spatial offset and the second spatial offset differ. In case further optical sensors are provided, besides the first and second optical sensors, these additional optical sensors may also fulfill the condition or, alternatively, may be spaced apart from the optical axis by the first spatial offset, by the second spatial offset or by a different spatial offset. The first and second spatial offsets, as an example, may differ by at least a factor of 1.2, more preferably by at least a factor of 1.5, more preferably by at least a factor of two. The spatial offsets may also be zero or may assume negative values, as long as the above-mentioned conditions are fulfilled.

As outlined above, each light-sensitive area has a geometrical center. As used herein, the term "geometrical center" of an area generally may refer to a center of gravity of the area. As an example, if an arbitrary point inside or outside the area is chosen, and if an integral is formed over the vectors interconnecting this arbitrary point with each and every point of the area, the integral is a function of the position of the arbitrary point. When the arbitrary point is located in the geometrical center of the area, the integral of the absolute value of the integral is minimized. Thus, in other words, the geometrical center may be a point inside or outside the area with a minimum overall or sum distance from all points of the area.

Each geometrical center of each light-sensitive area is spaced apart from the optical axis of the detector, such as the optical axis of the beam path or the respective beam path in which the respective optical sensor is located. The distance between the geometrical center and the optical axis is denoted by the term "spatial offset".

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As outlined above, the optical sensors specifically may be located in one and the same plane, which, preferably, is a plane perpendicular to the optical axis. Other configurations, however, are possible. Thus, two or more of the optical sensors may also be spaced apart in a direction parallel to the optical axis.

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In a preferred embodiment, which is rather simple and cost-efficiently realizable, the optical sensors are partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector. As used herein, the term "partial diode" may comprise several diodes that are connected in series or in parallel. This example is rather simple and cost-efficiently realizable. Thus, as an example, bi-cell diodes or quadrant diodes are widely commercially available at low cost, and driving schemes for these quadrant diodes are generally known. As used herein, the term "bi-cell diode" generally refers to a diode having two partial diodes in one packaging. Bi-cell and quadrant diodes may have two or four separate light sensitive areas, in particular two or four active areas. As an example, the bi-cell diodes may each form independent diodes having the full functionality of a diode. As an example, each of the bi-cell diodes may have a square or rectangular shape, and the two diodes may be placed in one plane such that the two partial diodes, in total, form a 1 x 2 or 2 x 1 matrix having a rectangular shape. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes or quadrant diode is proposed, as will be outlined in further detail below. In the present invention, however, a new scheme for evaluating the sensor signals of the bi-cell diodes and the quadrant diode is proposed, as will be outlined in further detail below. Generally, however, the optical sensors specifically may be partial diodes of a quadrant diode, with a center of the quadrant diode being off-centered from the optical axis of the detector. As used herein, the term "quadrant diode" generally refers to a diode having four partial diodes in one packaging. As an example, the four partial diodes may each form independent diodes having the full functionality of a diode. As an example, the four partial diodes may each have a square or rectangular shape, and the four partial diodes may be placed in one plane such that the four partial diodes, in total, form a 2 x 2 matrix having a rectangular or square shape. In a further example, the four partial diodes, in total, may form a 2 x 2 matrix having a circular or elliptical shape. The partial diodes, as an example, may be adjacent, with a minimum separation from one another. However, the separation between adjacent diodes may also be varied to improve the dynamic range of the quotient signal. As an example, the separation between two adjacent diodes may be increased by using an opaque mask. This may reduce the light on one of the diodes relative and/or absolute as compared to the adjacent diode which may increase a quotient of the signal of the two diodes. The rows and columns of the matrix specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the partial diodes are arranged in concentric circles or ellipses about a center

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point, hexagonal arrangements, triangular arrangements, polygonal arrangements or combinations thereof are also feasible, wherein the partial diodes are arranged as regular or irregular polygons such as in hexagons or the like.

- 5 Further, arrangements may be feasible, wherein at least one of the partial diodes of a polygonal pattern may be arranged as star shaped element in a triangular pattern and/or striped element in a rectangular pattern, or the like. Further, arrangements may be feasible, wherein the partial diodes of a polygonal pattern are summarized to a larger, unseparated partial diode, such as star shaped partial diodes in a triangular pattern, striped partial diodes in a rectangular pattern, 10 or the like. Further, separated partial diodes may be summarized or connected to a single partial diode such as the corner partial diodes forming a first partial diode and the non-corner partial diodes forming a second partial diode. Further, for example, the matrix may be a single row of partial diodes. Further, arrangements, where the optical sensors and/or at least one of partial diodes may have a star shape and/or circular shape, and/or cross shape may be feasible. For 15 example, at least two of the partial diodes may be arranged in star shaped manner or cross shaped manner, whereas the other partial diodes may be arranged as such that they complete a square or rectangular shape of the optical sensor, so that a square shaped or rectangular shaped substrate can be employed. Other arrangements are feasible.
- 20 In case a quadrant diode is used, having a 2×2 matrix of partial diodes, the center of the quadrant diode specifically may be off-centered or offset from the optical axis. Thus, as an example, the center of the quadrant diodes, which may be an intersection of the geometrical centers of the optical sensors of the quadrant diode, may be off-centered from the optical axis by at least 0.2 mm, more preferably by at least 0.5 mm, more preferably by at least 1.0 mm or even 2.0 25 mm. Similarly, when using other types of optical sensors setups having a plurality of optical sensors, an overall center of the optical sensors may be offset from the optical axis by the same distance.

- As outlined above, bi-cell and quadrant diodes may have two or four separate light sensitive 30 areas, in particular two or four active areas. The light sensitive areas may be separated by a dividing line. The dividing line, for example, of two Si-diodes, can be arranged parallel to the baseline of the detector, in particular parallel to a spot movement on the optical sensors. However, other arrangements are feasible. For example, the quadrant diode may comprise two dividing lines. The dividing lines may be arranged orthogonal to each other. The orthogonal ar- 35 rangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other. In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device may be adapted to determine a second quotient using at least three or all four sensor signals of the quadrant diode. The two quotients can be formed such that two distinct distance ranges are 40 covered. The two quotient signals for the near field and far field may have an overlap region in which both quotient signals allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient signal may be determined by dividing the sensor signals of two top quadrants, also called top segment, by the sensor signals of two bottom quadrants, also

called bottom segment. Using the quotient signal of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between the top and bottom segment is parallel to the baseline, the quotient signal determined from top segment divided by bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient signal by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals of the quadrants such that distance measurement is possible over a wide range with a large resolution.

The detector may comprise at least one quadrant diode comprising four quadrants. For example, an upper left quadrant of the quadrant diode may be denoted "ol", a lower left quadrant "ul", an upper right quadrant may be denoted "or" and a lower right quadrant may be denoted "ur".

The light spot may move from the left in a small distance range to the left in a far distance range. The detector may comprise at least one neutral density filter. The quadrants of the small distance range may be covered by the neutral density filter. In the small distance range the quotient ol/ul may be used to determine the longitudinal coordinate z. In the far distance range a quotient or/ur and/or a quotient from a combination of all quadrants may be used. The neutral density filter allows that objects in the small distance range are adapted to illuminate the optical sensors brighter by constant amplification without saturating the amplifier. At the same time it is possible to obtain in the far distance range more of the signal. Thus, the neutral density filter allows to enhance dynamic of the detector.

Generally, the light-sensitive areas of the optical sensors may have an arbitrary surface area or size. Preferably, however, specifically in view of a simplified evaluation of the sensor signals, the light-sensitive areas of the optical sensors are substantially equal, such as within a tolerance of less than 10%, preferably less than 5% or even less than 1%. This, specifically, is the case in typical commercially available quadrant diodes.

As further used herein, the term evaluation device generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

Specifically, as will be outlined in further detail below, the evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between sensor signals and/or any secondary signal derived thereof and the longitudinal coordinate. Thus, the evaluation device may be configured for determining at least one combined sensor signal out of the at least two sensor signals, i.e. of the at least one sensor signal of at least one first optical sensor and out of the at least one sensor signal of at least one second optical sensor. As generally used herein, the term “combine” generally may refer to an arbitrary operation in which two or more components such as signals are one or more of mathematically merged in order to form at least one merged combined signal and/or compared in order to form at least one comparison signal or comparison result. As will be outlined in further detail below, the combined sensor signal or secondary signal may be or may comprise at least one quotient signal.

The evaluation device specifically may be configured for deriving at least one quotient signal Q by one or more of: dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. The evaluation device may further be configured for determining the longitudinal coordinate by evaluating the quotient signal Q . In case more than two sensor signals are provided, as an example, more than one quotient signal may be generated, such as by forming quotient signals of more than one pair of sensor signals.

As an example, Q may simply be determined as

$$Q = s_1/s_2$$

or

$$Q = s_2/s_1,$$

with s_1 denoting a first one of the sensor signals and s_2 denoting a second one of the sensor signals. Additionally or alternatively, Q may be determined as

$$Q = a \cdot s_1 / b \cdot s_2$$

or

$$Q = b \cdot s_2 / a \cdot s_1,$$

with a and b being real numbers which, as an example, may be predetermined or determinable. Additionally or alternatively, Q may be determined as

$$Q = (a \cdot s_1 + b \cdot s_2) / (c \cdot s_1 + d \cdot s_2),$$

with a, b, c and d being real numbers which, as an example, may be predetermined or determinable. As a simple example for the latter, Q may be determined as

$$Q = s_1 / (s_1 + s_2),$$

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or, as a further example, Q may be determined as

$$Q = (s_1 - s_2) / (s_1 + s_2).$$

10 Other quotient signals are feasible. Thus, as an example, in case more than two optical sensors are provided, the above-mentioned quotient formation may take place between two of the sensor signals generated by these optical sensors or may take place between more than two of the sensor signals. Thus, instead of using the first one of the sensor signals and the second one of the sensor signals in the formulae given above, combined sensor signals may be used for quotient formation.

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The quotient signal Q generally is an example for an asymmetry parameter denoting an asymmetry or an asymmetric distribution of the light spot generated by the light beam on the light-sensitive areas. The quotient of the two or more optical sensors, such as the two or more photodiodes, may provide a combined signal which typically is monotonously dependent on the distance between the detector and the object from which the light beam travels towards the detector, as will be shown by experimental data below. In addition or as an alternative to the quotient signal, other types of combined functions implementing the sensor signals of two or more sensors in the setup of the present invention may be used, which also may show a dependency on the distance between the object and the detector. The asymmetry or the asymmetry parameter of the light spot, as an example, may be an indication of the width of a light beam. If this asymmetry parameter depends on the distance only, the measurement can be used to determine the distance.

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30 The evaluation device may be configured for deriving the quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate for determining the longitudinal coordinate. For example, the evaluation device is configured for deriving the quotient signal Q by

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$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. As used herein, the term “beam profile” relates to a spa-

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tial distribution, in particular in at least one plane perpendicular to the propagation of the light beam, of an intensity of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_0)$ and beam shape $S(x,y;z_0)$,
 $E(x,y;z_0) = L \cdot S$.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the quotient signal Q . The light-sensitive regions may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term "essentially edge information" generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may com-

prise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

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The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

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Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient signal Q by a line integrals along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in an improved distance information.

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The evaluation device may be configured to derive the quotient signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

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In one embodiment, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of the optical sensors are arranged as a sensor element having at least two columns, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern. For example, the detector may comprise at least two rows of optical sensors, wherein the rows are arranged on top of each other. The line pattern may be imaged parallel to the rows such that the center of the line pattern is fixed or moves only little relative to a dividing line of the rows. The A1 may correspond to an area above the dividing line and A2 may be an area below the dividing line.

The light beam may be a monochromatic light beam. The light beam may comprise a narrow band of wavelengths, preferably the light beam may comprise a single wavelength. The at least one light source may be adapted to generate at least one monochromatic light beam and/or the detector may comprise at least one filter element adapted to filter a narrow band of wavelength, such as a monochromator.

The illumination source may illuminate the object with a convergent and/or divergent and/or collimated light beam.

The illumination source and the sensor element may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the optical sensors may be arranged in a twisted arrangement.

In typical setups, commercially available quadrant diodes such as quadrant photodiodes are used for positioning, i.e. for adjusting and/or measuring a transversal coordinate of a light spot in the plane of the quadrant photodiode. Thus, as an example, laser beam positioning by using quadrant photodiodes is well known. According to a typical prejudice, however, quadrant photodiodes are used for xy-positioning, only. According to this assumption, quadrant photodiodes are not suitable for measuring distances. The above-mentioned findings, however, using an off-

centered quadrant photodiode with regard to an optical axis of the detector, show otherwise, as will be shown in further measurements below. Thus, as indicated above, in quadrant photodiodes, the asymmetry of the spot can be measured by shifting the quadrant diode slightly off-axis, such as by the above-mentioned offset. Thereby, a monotonously z-dependent function may be generated, such as by forming the quotient signal Q of two or more of the sensor signals of two or more partial photodiodes, i.e. quadrants, of the quadrant photodiode. Therein, in principle, only two photodiodes are necessary for the measurement. The other two diodes may be used for noise cancellation or to obtain a more precise measurement.

In addition or as an alternative to using a quadrant diode or quadrant photodiode, other types of optical sensors may be used. Thus, as will be shown in further detail below, staggered optical sensors may be used.

The use of quadrant diodes provides a large number of advantages over known optical detectors. Thus, quadrant diodes are used in a large number of applications in combination with LEDs or active targets and are widely commercially available at very low price, with various optical properties such as spectral sensitivities and in various sizes. No specific manufacturing process has to be established, since commercially available products may be implemented into the detector according to the present invention.

The detector according to the present invention specifically may be used in multilayer optical storage discs, such as disclosed by international patent application number PCT/IB2015/052233, filed on March 26, 2015. Measurements performed by using the detector according to the present invention specifically may be used in order to optimize the focus position in optical storage discs.

As will be outlined in further detail below, the distance measurement by using the detector according to the present invention may be enhanced by implementing one or more additional distance measurement means into the detector and/or by combining the detector with other types of distance measurement means. Thus, as an example, the detector may comprise or may be combined with at least one triangulation distance measurement device. Thus, the distance measurement can be enhanced by making use of a combination of the measurement principle discussed above and a triangulation type distance measurement. Further, means for measuring one or more other coordinates, such as x- and/or y-coordinates, may be provided.

In case a quadrant diode is used, the quadrant diode may also be used for additional purposes. Thus, the quadrant diode may also be used for conventional x-y-measurements of a light spot, as generally known in the art of optoelectronics and laser physics. Thus, as an example, the lens or detector position can be adjusted using the conventional xy-position information of the quadrant diode to optimize the position of the spot for the distance measurement. As a practical example, the light spot, initially, may be located right in the center of the quadrant diode, which typically does not allow for the above-mentioned distance measurement using the quotient function Q. Thus, firstly, conventional quadrant photodiode techniques may be used for off-centering

a position of the light spot on the quadrant photodiode, such that, e.g., the spot position on the quadrant diode is optimal for the measurement. Thus, as an example, the different off-centering of the optical sensors of the detector may simply be a starting point for movement of the optical sensors relative to the optical axis such that the light spot is off-centered with respect to the optical axis and with respect to a geometrical center of the array of the optical sensors.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned quadrant diode. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like. The case $m=2$, $n=2$ is the case of the quadrant diode or quadrant optical sensor, which, for practical reasons, is one of the preferred cases, since quadrant photodiodes are widely available.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned quadrant diode. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like. The case $m=2$, $n=2$ is the case of the quadrant diode or quadrant optical sensor, which, for practical reasons, is one of the preferred cases, since quadrant photodiodes are widely available.

As a starting point, a geometrical center of the optical sensors within the array may be off-centered from the optical axis, such as by the above-mentioned offset. The sensor array specifically may be movable relative to the optical axis, for example along a gradient, preferably automatically, such as by moving the sensor array, e.g. in a plane perpendicular to the optical axis, and/or by moving the optical axis itself, e.g. shifting the optical axis in a parallel shift and/or tilting the optical axis. Thus, the sensor array may be shifted in order to adjust a position of a light spot generated by the light beam in the plane of the sensor array. Additionally or alternatively, the optical axis may be shifted and/or tilted by using appropriate elements, such as by using one or more deflection elements and/or one or more lenses. The movement, as an example, may take place by using one or more appropriate actuators, such as one or more piezo actuators and/or one or more electromagnetic actuators and/or one or more pneumatic or me-

chanical actuators, which, e.g., move and/or shift the array and/or move and/or shift and/or tilt one or more optical elements in the beam path in order to move the optical axis, such as parallel shifting the optical axis and/or tilting the optical axis. The evaluation device specifically may be adjusted to control a relative position of the sensor array to the optical axis, e.g. in the plane perpendicular to the optical axis. An adjustment procedure may take place in that the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly, moving the array relative to the optical axis, such as by moving the array and/or the optical axis, e.g. by moving the array in the plane to the optical axis until the light spot is off-centered and/or by tilting a lens until the light spot is off-centered. As used therein, a transversal position may be a position in a plane perpendicular to the optical axis, which may also be referred to as the x-y-plane. For the measurement of the transversal coordinate, as an example, the sensor signals of the optical sensors may be compared. As an example, in case the sensor signals are found to be equal and, thus, in case it is determined that the light spot is located symmetrically with respect to the optical sensors, such as in the center of the quadrant diodes, a shifting of the array and/or a tilting of a lens may take place, in order to off-center the light spot in the array. Thus, as outlined above, the off-centering of the array from the optical axis, such as by off-centering the center of the quadrant photodiode from the optical axis, may simply be a starting point in order to avoid the situation which is typical, in which the light spot is located on the optical axis and, thus, is centered. By off-centering the array relative to the optical axis, thus, the light spot should be off-centered. In case this is found not to be true, such that the light spot, incidentally, is located in the center of the array and equally illuminates all optical sensors, the above-mentioned shifting of the array relative to the optical axis may take place, preferably automatically, in order to off-center the light spot on the array. Thereby, a reliable distance measurement may take place.

Further, in a scanning system with a movable light source, the position of the light spot on the quadrant diode may not be fixed. This is still possible, but may necessitate that different calibrations are used, dependent on the xy-position of the spot in the diode.

Further, the use of the above-mentioned quotient signal Q is a very reliable method for distance measurements. Typically, Q is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, the quotient $Q = s_1/s_2$ is a monotonously decreasing function of the size of the light spot. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, the sensor signals, such as the above-mentioned first sensor signal s_1 and the above-mentioned second sensor signal s_2 , decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, due to the off-centering, the one of the sensor signals decreases more rapidly than the other, since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. By spreading the light spot, however, the portion of the light illuminating the one or more optical sensors outside the center of the light

spot increases, as compared to a situation of a very small light spot. Thus, the quotient of the sensor signals continuously changes, i.e. increases or decreases, with increasing diameter of the light beam or diameter of the light spot. The quotient, further, may further be rendered mainly independent from the total power of the light beam, since the total power of the light beam forms a factor in all sensor signals. Consequently, the quotient Q may form a secondary signal which provides a unique and unambiguous relationship between the sensor signals and the size or diameter of the light beam.

Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the first and second sensor signals and the longitudinal coordinate may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the first and second sensor signals or a secondary signal derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the quotient signal Q. The evaluation device may be configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The quotient signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider.

In view of the technical challenges involved in the prior art documents discussed above, specifically in view of the technical effort which is required for generating the FiP effect, it has to be noted that the present invention specifically may be realized by using non-FiP optical sensors. In fact, since optical sensors having the FiP characteristic typically exhibit a strong peak in the respective sensor signals at a focal point, the range of measurement of a detector according to the present invention using FiP sensors as optical sensors may be limited to a range in between the two positions and which the optical sensors are in focus of the light beam. When using linear optical sensors, however, i.e. optical sensors not exhibiting the FiP effect, this problem, with the setup of the present invention, generally may be avoided. Consequently, the optical sensors

may each have, at least within a range of measurement, a linear signal characteristic such that the respective first and second sensor signals may be dependent on the total power of illumination of the respective optical sensor and may be independent from a diameter of a light spot of the illumination. It shall be noted, however, that other embodiments are feasible, too.

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The optical sensors each specifically may be semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes. Thus, as opposed to complex and expensive FiP sensors, the present invention simply may be realized by using commercially available inorganic photodiodes, i.e. one small photodiode and
10 one large area photodiode. Thus, the setup of the present invention may be realized in a cheap and inexpensive fashion.

In one embodiment, the detector may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the
15 bi-cell or quadrant diode may be adapted to generate a so called FiP signal. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. According to the FiP-effect, the longitudinal sensor signal, given the same total power, may exhibit at least one pronounced maximum for one or a plurality of focusings
20 and/or for one or a plurality of specific sizes of the light spot on the sensor region or within the sensor region. For purposes of comparison, an observation of a maximum of the longitudinal sensor signal in a condition in which the corresponding material is impinged by a light beam with the smallest possible cross-section, such as when the material may be located at or near a focal point as affected by an optical lens, may be denominated as a "positive FiP-effect". The
25 negative FiP effect describes an observation of a minimum of the longitudinal sensor signal under a condition in which the corresponding material is impinged by a light beam with the smallest available beam cross-section, in particular, when the material may be located at or near a focal point as effected by an optical lens. The negative FiP effect may be used to tune small image effects at high distances. Image changes such as position, size, shape, sharpness, etc.
30 may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

At least one of the optical sensors may be adapted to generate at least one sensor signal dependent on a time-of-flight (TOF) the illumination light beam has traveled from the illumination
35 source to the object and the reflection light beam has traveled from the object to the light sensitive area of the optical sensor. The evaluation device may be configured for determining at least one TOF-longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. The optical sensor adapted to generate the TOF sensor signal may be designed as time-of-flight
40 detector. The time-of-flight detector may be selected from the group consisting of: at least one pulsed time-of-flight detector; at least one phase modulated time-of-flight detector; at least one direct time-of-flight detector; at least one indirect time-of-flight detector. For example, the pulsed time-of-flight detector may be at least one range gated imager and/or at least one direct time-of-

flight imager. For example the phase modulated time-of-flight detector may be at least one RF-modulated light source with at least one phase detector. The optical sensor may be adapted to determine a time delay between emission of the illumination light beam by the illumination source and receipt of the reflection light beam.

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For example, the optical sensor adapted to generate the TOF sensor signal may be designed as pulsed time-of-flight detector. The detector may comprise the at least one interruption device, such as at least one shutter element, adapted to generate a pulsed light beam. The optical sensor may be adapted to store the TOF sensor signal dependent on the receiving time of the reflection light beam in a plurality of time windows, in particular subsequent time windows. The optical sensor may be adapted to store dependent on receiving time of the reflection light beam the generated TOF sensor signal in at least one first time window and/or in at least one second time window. The first and second time windows may be correlated with the opening and closure of the interruption device. Duration of first and second time windows may be pre-defined.

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For example, the TOF sensor signal may be stored in the first time window during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may be stored in the second time window. Other durations of time windows are thinkable. The first and the second time window may comprise information about background, signal height and signal shift.

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For example, the optical sensor adapted to generate the TOF sensor signal may be designed as direct time-of-flight imager. The direct time-of-flight imager may comprise the at least one illumination source adapted to generate at least one single laser pulse. The single laser pulse may be reflected back from the object onto the optical sensor. The optical sensor may comprise at least one photodiode, for example at least one Avalanche Photo Diode (APD), such as at least one Si APD, or such as at least one InGaAs APD, or at least one PIN photo detector array, or at least one Single-Photon Avalanche Photodiode (SPAD), adapted to image the reflection light beam. The direct time-of flight imager may be adapted to image at least one image comprising spatial and temporal data.

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For example, the optical sensor adapted to generate the TOF sensor signal may be designed as phase modulated time-of-flight modulator. The phase modulated time-of-flight modulator may be adapted to measure a difference in phase, in particular a phase shift, by determining a correlated signal, for example by multiplying a received signal, i.e. of the reflection light beam, with the emitted signal, i.e. the illumination light beam. A DC component of the correlated signal may comprise an information about the difference in phase. The evaluation device may be adapted to determine the second longitudinal coordinate of the object from the phase difference. For example, the illumination source and the optical sensor adapted to generate the TOF sensor signal may be designed as RF-modulated light source with at least one phase detector. The illumination source may comprise, for example, at least one LED and/or at least one laser. The illumination source may comprise at least one modulation device adapted to modulate the light beam having a pre-defined phase shift. For example, the modulation device may comprise at least one radio frequency module. The radio frequency module may be adapted to modulate the

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illumination beam with an RF carrier. The optical sensor may be adapted to determine a phase shift of the reflective light beam impinging on the optical sensor.

The optical sensor may be designed as and/or may comprise at least one time-of-flight pixel.

5 Preferably, the detector may comprise at least two optical sensors, wherein each optical sensor is designed as and/or comprises at least one TOF pixel. For example, the detector, in particular the optical sensor may comprise a quadrant diode adapted to generate the TOF sensor signal. For example, the detector, in particular the optical sensor, may comprise at least one pixelated TOF-imager.

10 The evaluation device may be configured for determining the at least one TOF longitudinal coordinate z_{TOF} of the object by evaluating the TOF sensor signal. As used herein, the term "TOF longitudinal coordinate" refers to a longitudinal coordinate derived from the TOF sensor signal. As described above, the illumination light source may be adapted to periodically generate at least one light pulse. The detector may be adapted to generate the first longitudinal sensor signal for each period. The evaluation device may be adapted to determine in which pulse period the TOF sensor signal was generated using the second longitudinal coordinate. The detector may be adapted to uniquely assign to which period the ToF-signal refers to by using the combined sensor signal. Both, the TOF sensor signal and the combined sensor signal, may be non-monotonous functions of the longitudinal coordinate z_{real} . Thus, the longitudinal coordinate may not be uniquely determined from one of the TOF sensor signal or the combined sensor signal alone and a measurement range may not be restricted to a longitudinal range in which the signals are unique functions of z_{real} . The term "measurement range" generally refers to a range from the object to the detector in which determination of the longitudinal coordinate is possible.

25 The term "longitudinal range" generally refers to a range from the object to the detector in which unique determination of the longitudinal coordinate is possible. Below and/or above certain distances from the object to the detector a determination of the longitudinal coordinate may not be possible. For example, time-of-flight measurements are not possible below a certain distance from the object to the detector, possibly due to the minimum measurement time of the internal clock. Furthermore, in time-of-flight measurements the sensor signal may be unique within a longitudinal period, but the sensor signal may be the same in case integer multiples of the longitudinal period are added such that the determined longitudinal coordinate may be non-unique. Thus, the same TOF sensor signal will be obtained for a distance z_1 and a distance $z_1 + n \cdot z_{1p}$, where n is an integer denoting the longitudinal period and z_{1p} is the longitudinal period of the TOF sensor signal, wherein the distances z_1 and $z_1 + n \cdot z_{1p}$ are within the measurement range. As used herein, the term "longitudinal period" refers to partitions of a period, in particular a distance range, in which the longitudinal coordinate can be unambiguously determined from the TOF sensor signal. Non-unique longitudinal coordinates may be denoted as relative longitudinal coordinates and the unique longitudinal coordinates may be denoted as absolute longitudinal coordinates.

If both the TOF sensor signal $F1$ and the combined sensor signals $F2$ are available, it may be possible to uniquely determine the longitudinal coordinate and to extend the longitudinal range,

as long as each signal pair (F1,F2) corresponds to a unique distance and vice versa. In particular, if a unique signal pair (F1,F2) exists for each longitudinal coordinate and vice versa, the evaluation device may be adapted to determine the unique combined longitudinal coordinate by

(1) selecting at least one first selected signal such as the TOF sensor signal and/or the combined sensor signal and determining non-unique first longitudinal coordinates;

(2) selecting a second selected signal such as the combined signal Q and/or the TOF sensor signal, which was not selected in step (1), and determining non-unique second longitudinal coordinates;

(3) determining whether one of the non-unique first longitudinal coordinates and the non-unique second longitudinal coordinates match up to a predetermined tolerance threshold;

(4) setting a combined unique longitudinal coordinate to be a matching longitudinal coordinate.

In step (1) and step (2), signals may be selected in the given order or may be performed in a different order. For example, in step (1) the TOF sensor signal may be selected and in step (2) the combined signal Q may be selected. In another example, in step (1) the combined sensor signal may be selected, and the non-unique first longitudinal sensor signal may be determined therefrom. In step (2), the TOF sensor signal may be selected.

Additionally or alternatively to step (4), the evaluation device may be adapted to output an error signal in case no matching coordinates are found and/or to output an error signal in case more than one matching coordinates are found. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be stored in a look-up table. Additionally or alternatively, the signal pairs and their corresponding longitudinal coordinates may be approximated or described by an analytical function which is evaluated to find the longitudinal coordinate corresponding to a given signal pair.

The evaluation device may comprise at least two memory elements. As used herein, the term "memory element" refers to a device adapted to store information. The evaluation device may be adapted to receive and to store information provided by the optical sensors, for example the at least one first sensor signal. Such information may comprise raw sensor data and/or processed sensor data. For example, the memory element may be adapted to store information for further evaluation by the evaluation device. The memory element may be a volatile or non-volatile memory element.

As described above, the optical sensor may be designed as and/or may comprise at least one ToF pixel. The detector may comprise at least two switches. Each of the switches may be connected to the optical sensor adapted to generate the first sensor signal, for example by at least one connector. In particular, each of the switches may be connected to the ToF pixel. The switches are adapted to provide the TOF sensor signal to one of the memory elements. In particular, the switches may be adapted to let, dependent on receiving time of the reflection light beam, the generated TOF sensor signal pass through one of the switches. For example, the TOF sensor signal may pass one of the switches during opening of the interruption device, whereas during closure of the interruption device the TOF sensor signal may pass the other

switch. Each of the switches may be controlled by a control signal having a pulse length identical to a pulse length of a light pulse generated by the illumination source. The control signal of one of the switches may be delayed. For example, the delay may correspond to the pulse length of the light pulse. The evaluation device may be adapted to sample and/or store depending on the delay a first part, or fraction, of the TOF sensor signal through a first switch in a first memory element and the other, second part, or fraction, of the TOF sensor signal through a second switch in a second memory element. The evaluation device may be adapted to determine the first longitudinal coordinate by evaluating the first part and second part of the TOF sensor signal. The evaluation device may be adapted to determine the first longitudinal coordinate z_1 by

$$z_1 = \frac{1}{2} \cdot c \cdot t_0 \cdot \frac{S_{12}}{S_{11} + S_{12}} + z_0;$$

wherein c is the speed of light, t_0 is the pulse length of the illumination light beam, z_0 is a distance offset often determined by the resolution of the time measurement, and S_{11} and S_{12} are the first part and second part of the TOF sensor signal, respectively.

As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

As outlined above, specifically, quadrant photodiodes may be used. As an example, commercially available quadrant photodiodes may be integrated in order to provide four optical sensors, such as one or more quadrant photodiodes available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany, such as quadrant Si PIN photodiodes of the type S4349, which are sensitive in the UV spectral range to the near IR spectral range. In case an array of optical sensors is used, the array may be a naked chip or may be an encapsulated array, such as encapsulated in a TO-5 metal package. Additionally or alternatively, a surface mounted device may be used, such as TT Electronics OPR5911 available from TT Electronics plc, Fourth Floor, St Andrews House, West Street Woking Surrey, GU21 6EB, England. It shall be noted that other optical sensors may also be used.

Further, it shall be noted that, besides the option of using precisely one quadrant photodiode, two or more quadrant photodiodes may also be used. Thus, as an example, a first quadrant photodiode may be used for the distance measurement, as described above, providing the two or more optical sensors. Another quadrant photodiode may be used, e.g. in a second partial beam path split off from the beam path of the first quadrant photodiode, for a transversal position measurement, such as for using at least one transversal coordinate x and/or y . The second quadrant photodiode, as an example, may be located on-axis with respect to the optical axis.

Further, it shall be noted that, besides the option of using one or more quadrant photodiodes, one or more quadrant photodiodes or further photodiode arrays may also be replaced or mimicked by separated photodiodes that are arranged or assembled close to each other, preferably in a symmetric shape such as a rectangular matrix, such as a 2×2 matrix. However further arrangements are feasible. In such an arrangement or assembly, the photodiodes may be ar-

5 ranged or assembled in a housing or mount, such as all photodiodes in a single housing or mount or groups of photodiodes in one housing or mount, or each of the photodiodes in a separate housing or mount. Further, the photodiodes may also be assembled directly on a circuit board. In such arrangements or assemblies, photodiodes may be arranged as such that the separation between the active area of the photodiodes, has a distinct value less than one centimeter, preferably less than one millimeter, more preferably as small as possible. Further, to avoid optical reflexes, distortions, or the like that may deteriorate the measurement, the space between the active areas may be either empty or filled with a material, preferably with a light absorbing material such as a black polymer, such as black silicon, black polyoxymethylene, or the like, more preferably optically absorbing and electrically insulating material, such as black ceramics or insulating black polymers such as black silicon, or the like. Further, the distinct value of the photodiode separation may also be realized by adding a distinct building block between the photodiodes such as a plastic separator. Further embodiments are feasible. The replacement of quadrant photodiodes by single diodes arranged in a similar setup such as in a 2 x 2 rectangular matrix with minimal distance between the active areas may further minimize the costs for the optical detector. Further, two or more diodes from a quadrant diode may be connected in parallel or in series to form a single light sensitive area.

20 Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer. The optical sensors may comprise differing materials or photovoltaic concepts. As an example, at least one first optical sensor may be a Ge photodiode, whereas at least one second optical sensor may be an InGaAs photodiode. As a further example, the first and second optical sensor may comprise the same material with a differing doping such as a differing doping concentration or doping material.

35 The optical sensors each, independently, may be opaque, transparent or semitransparent. For the sake of simplicity, however, opaque sensors which are not transparent for the light beam, may be used, since these opaque sensors generally are widely commercially available.

The optical sensors each specifically may be uniform sensors having a single light-sensitive area each. Thus, the optical sensors specifically may be non-pixelated optical sensors.

40 The detector may further comprise an illumination source for illuminating the object. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam

propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

The illuminating light beam generally may be parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. As an example, the illuminating light beam, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the illuminating light beam may be on the optical axis or off the optical axis. As an example, the illuminating light beam may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

The detector may comprise the at least one illumination source. The illumination source may be adapted to illuminate the object with at least one illumination light beam. The illumination light beam may be fully or partially reflected by the object and may travel back towards the detector. The illumination source may be arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis. As used herein, the term "essentially parallel to the optical axis" refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. The illumination source may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis do not intersect in space. In particular, the illumination may be on-axis. In order to provide such an on-axis illumination, as an example, one or more reflective elements may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms. The reflected light beam may be essentially parallel to the optical axis. Thus, according to the present invention, the object distance is not determined from an angle of the light beam, i.e. a position of the light spot on a sensor, as done in triangulation methods. Instead, the object distance may be determined using light beams essential parallel to the optical axis. Determination of the object distance by a non-triangulation method but by combining the at least two sensor signals, allows arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis.

The illumination source and the optical axis may be separated by a small baseline. As used herein, the term "baseline", also denoted as basis line, refers to a distance, for example in a xy-plane, between the optical axis and the illumination source, in particular a distance between the

optical axis and a z-component of the illumination light beam. The illumination source may be spaced apart from the optical axis by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and transfer device. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the baseline may be 21 mm. Preferably, the illumination source may be arranged right next to the transfer device. For example, the transfer device may be flattened such that the illumination source can be positioned even closer to the optical axis. The illumination source may be arranged behind the transfer device. Triangulation systems require a sufficient baseline, however due to the baseline in the near field no detection may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. For example, the detector may be a compact device without further optical elements, wherein the illumination source may be placed as close as possible to an edge of the transfer device. Thus, the baseline may be close to half a diameter of the transfer device, in particular the lens diameter and housings of lens and light source. For example, the detector may be an even more compact device, wherein a mirror, in particular a small mirror, may be positioned in front of the transfer device, in particular in a center, for example a geometrical center, or close to the center of the transfer device, in order to couple the illumination beam into the beam path. Thus, the baseline may be less than half the diameter of the transfer device. The illumination source may be arranged such that the baseline is as small as possible. By arranging the illumination source such that the direction of propagation of the illumination light beam is essentially parallel to the optical axis and that the illumination source and the optical axis are separated by the small baseline, very compact devices are possible. For example, a distance from the center of the transfer device to the illumination source, in particular along a connecting line from the center of the transfer device to the illumination source, may be preferably less than 2.5 times a distance from the center of the transfer device to an edge of the transfer device, more preferably less than 1.5 times the distance center to edge of the transfer device, and most preferably less 1 times the distance center to edge of the transfer device. The transfer device may have an arbitrary shape, in particular a non-circular shapes are possible. At small distances an aperture of the illumination source may be small and the baseline may be small. At large distances an aperture of the illumination source may be large and the baseline may be small. This is contrarily as in triangulation methods, wherein at large distances a large baseline is necessary. Further, triangulation based systems have a minimum detection range significantly greater than zero, for example such as 20 cm from the detector system, due to necessary spatial extend of the baseline. Such a large baseline may result in that the illuminated light scattered from the object may not reach the light sensitive area of the optical sensor behind the transfer device. In addition, in triangulation based systems, using a small baseline would reduce the minimum detection range, however, at the same time would reduce a maximum detection range. Further, triangulation based systems require a plurality of light sensitive areas and sensor signals, for example sensor signals of at least one detector row.

According to the invention, determination of the longitudinal coordinate z is possible with a reduced number of sensor signals, in particular with less than 20, preferably less than 10 and more preferably less than 5 sensor signals.

5 The illumination source may be adapted to generate at least one illumination pattern for illuminating the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the
10 detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term "pattern"
15 refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "illumination pattern" refers to a pattern which illuminates the object. The illumination pattern may be generated by ambient light,
20 such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

25 The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light
30 source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which
35 may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate at least one point, e.g. a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source.

40 The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination source may be attached

to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

5 The illumination source may be adapted to generate pulsed illumination. The illumination source may be adapted to generate at least one light pulse. As used herein, the term “light pulse” or “pulsed illumination” refers to a light beam limited in time. The light pulse may have a pre-defined length or time duration, for example in the nanoseconds range. For example, the illumination source may be adapted to generate pulses with a pulse length of less than a nanosecond, such as a tenth of a nanosecond, up to a tenth of a second. The illumination source may
10 be adapted to periodically generate the light pulse. For example, the illumination source may be adapted to generate the light pulse with a frequency of 10 Hz to 10 GHz.

The illumination source may be adapted to generate a pulsed light beam. For example, the illumination source may be adapted to generate a continuous illumination light beam and the detector
15 may comprise at least one interruption device adapted to interrupt the illumination, in particular periodically. The interruption device may comprise at least one shutter and/or a beam chopper or some other type of mechanical or electronical periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates at constant speed and which can thus periodically interrupt the illumination. By way of example,
20 the at least one interruption device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

The detector may further comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors. The transfer device specifically may
25 comprise one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one diffractive optical element; at least one Fresnel lens; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

30 The optical sensors may be positioned off focus. As used herein, the term “focus” generally refers to one or both of a minimum extend of a circle of confusion of the light beam, in particular of at least one light beam emitted from one point of the object, caused by the transfer device or a focal length of the transfer device. As used herein, the term “circle of confusion” refers to a light spot caused by a cone of light rays of the light beam focused by the transfer device. The circle
35 of confusion may depend on a focal length f of the transfer device, a longitudinal distance from the object to the transfer device, a diameter of an exit pupil of the transfer device, a longitudinal distance from the transfer device to the light sensitive area, a distance from the transfer device to an image of the object. For example, for Gaussian beams, a diameter of the circle of confusion may be a width of the Gaussian beam. In particular, for a point like object situated or
40 placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from the object into a focus point at the focal length of the transfer device. For non-point like objects situated or placed at infinite distance from the detector the transfer device may be adapted to focus the light beam from at least one point of the object into a focus plane at the

focal length of the transfer device. For point like objects not situated or placed at infinite distance from the detector, the circle of confusion may have a minimum extend at least at one longitudinal coordinate. For non-point like objects not situated or placed at infinite distance from the detector, the circle of confusion of the light beam from at least one point of the object may have a minimum extend at least at one longitudinal coordinate. As used herein, the term “positioned off focus” generally refers to a position other than the minimum extend of a circle of confusion of the light beam caused by the transfer device or a focal length of the transfer device. In particular, the focal point or minimum extend of the circle of confusion may be at a longitudinal coordinate l_{focus} , whereas the position of each of the optical sensors may have a longitudinal coordinate l_{sensor} different from l_{focus} . For example, the longitudinal coordinate l_{sensor} may be, in a longitudinal direction, arranged closer to the position of the transfer device than the longitudinal coordinate l_{focus} or may be arranged further away from the position of the transfer device than the longitudinal coordinate l_{focus} . Thus, the longitudinal coordinate l_{sensor} and the longitudinal coordinate l_{focus} may be situated at different distances from the transfer device. For example, the optical sensors may be spaced apart from the minimum extend of the circle of confusion in longitudinal direction by $\pm 2\%$ focal length, preferably by $\pm 10\%$ focal length, most preferably $\pm 20\%$ focal length. For example, at a focal length of the transfer device may be 20mm and the longitudinal coordinate l_{sensor} may be 19,5 mm, i.e. the sensors may be positioned at 97,5% focal length, such that l_{sensor} is spaced apart from the focus by 2,5% of focal length. The extent of the circle of confusion may be larger than the extent of the optical sensor. For example, the optical sensor may be positioned as such that the circle of confusion extends beyond the optical sensor. The optical sensor may thus only partially evaluate the beam profile of the light beam.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Field Programmable Gate Arrays (FPGAs), or digital signal processors (DSPs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As outlined above, by evaluating the sensor signals, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more additional transversal sensors may be used for determining at least one transversal coordinate of the object. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. These devices may generally also be implemented into the detector according to the present invention.

As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing. The detector specifically may comprise at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensor. The transfer device may comprise one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. Further, if applicable, Fresnel lenses or diffractive optical elements may be used. Therein, generally, the term "transfer device" may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of, preferably at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one bea-

con device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, the at least one beacon device may be adapted to reflect one
5 or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties
10 of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine inter-
15 face comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein, the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface
20 is designed to assign to the position at least one item of information.

In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-
25 machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at spe-
30 cific points in time.

In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

40 In a further aspect of the present invention, a readout device for optical storage media is proposed. The readout device comprises at least one detector according to any one of the preceding embodiments referring to a detector. As used therein, a readout device for optical storage

media generally refers to a device which is capable of optically retrieving information stored in optical storage media such as optical storage discs, e.g. CCD, DVD or Blu-ray discs. Thus, the above-described measurement principle of the detector according to the present invention may be used for detecting data modules within an optical storage medium such as in optical storage
5 discs. As an example, in case a reflective data module is present and reflects the illuminating light beam, the detector will not only detect the reflected light beam according to the above-mentioned measurement principle but will also detect a distance between the detector and the reflective data module, i.e. a depth of the reflective data module within the optical storage medium. Thus, as an example, the detector may be used for detecting different layers of information
10 modules or data modules within the optical storage medium. Thereby, as an example, two layer discs or three layer discs or even discs having more than three layers may be generated and read out.

In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further comprises at least one illumination source adapted to scan the scenery with at least one light
20 beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of
30 information, such as at least one longitudinal coordinate, for each region. As an example, for scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers

to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector, such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention, at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as

a front edge or a back edge of the housing, and the single dot may, alternatively, be determined.

Alternatively, the illumination source of the scanning system may emit two individual laser
5 beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions,
10 such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to
15 the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in
20 particular a regular pitch, with respect to each other and which may be arranged in a manner in order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one
25 or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array
30 of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an im-
35 age by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning device. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning
40 devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics

applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with a several corners and surfaces. One or more flexibly movable laser sources may be attached to the scanning system. The one or more laser sources may be moved as such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system may be measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information may be stored in a mobile phone application. The laser sources may be moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and/or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning device may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a driller, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a wood plank while the scanning device is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning device.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The optional transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The optional transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor.

5 The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least
10 one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination
15 source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively
20 or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electro-
25 magnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence
30 and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be
35 regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the
40 visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. Most preferably, the at least one illumination source is adapted to emit light in the visible spectral

range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm.

5 The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of
10 example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

15 In a further aspect, the present invention discloses an inertial measurement unit for use in an electronic device. The electronic device may be a mobile electronic device. The electronic device may be a camera. The electronic device may be a mobile phone. The inertial measurement unit is adapted to receive data determined by at least one inertial measurement unit comprising at least one detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. As used herein, the term "data determined by the at least one detector" refers to at least
20 one information on the at least one longitudinal coordinate z. The inertial measurement unit further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer; a gyroscope. As used herein, the term "data determined by the at least one further sensor" refers to at least one information selected from the group consisting of: angle information; speed information; information on turn rate; information on inclination. The inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position
30 in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, speed. The inertial measurement unit may comprise at least one processor. The processor may be adapted to evaluate data recorded by the further sensor. In particular, the processor may be adapted to determine and/or calculate one or more of spatial position, spatial orientation, movement and velocity. The inertial measurement unit may comprise a plurality of further sensors. The inertial measurement unit may be adapted to fuse information determined from at least two of the further sensors. The inertial measurement unit may be adapted to fuse information of at least two further sensors by using at least one Kalman filter. As outlined above, the detector may be adapted to provide an absolute measurement of the longitudinal
40 coordinate z. The processor, e.g. the evaluation device as described above, may be adapted to fuse the information of the at least two further sensors considering the longitudinal coordinate z. The various sensor signals may be used within a Kalman filter or a linear quadratic estimated to take into account that each sensor signal is subject to measurement errors and inaccuracies.

Fusion of these sensor signals within a Kalman filter may yield an improved estimate such as for the measurement of the longitudinal coordinate.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used.

The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method steps are as follows:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light;
- illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

Specifically, evaluating the first and second sensor signal may comprise deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q . Further, the determining of the longitudinal coordinate may comprise evaluating the quotient signal Q .

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine

interface application; a tracking application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for household applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as con-

ventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus, specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head mounted displays.

Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries, obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an

object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of

lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general
5 need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

10 Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic
15 alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems,
20 or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present
25 invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located
30 as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.

35 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors.
40 In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal

sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions.

Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endoscopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric sur-

gery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment.

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Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

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Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe.

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In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

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Further the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

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Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support

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the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

5 Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

10 The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

15 The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

20 Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and/or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to
25 avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the
30 devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the
35 present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such
40 as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-

dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection, such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections, inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft when landing or starting. Therein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or accelerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, ap-

plications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racket, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the present invention may be used in drones or multicopters or the like to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used to recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization

and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concerted movements of several drones or recharging or refueling in the air or the like.

5 Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet-related appliances, child-related appliances, child surveillance, appliance related surveillance, support or service to elderly or ill persons, home
10 security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic
15 robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as cat litter robots, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, window washing robots, toy robots, telepresence robots, social robots providing company to less mobile people, or robots translating and speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots
30 or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate insecure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

35 Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.
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- Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition device, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.
- Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.
- The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or greenhouse or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.
- Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific groups of animals such as bees.
- Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or viruses or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.
- One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present inven-

tion may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further, the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location. Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location-specific information, such as the current position on a map, or information on goods sold, or the like.

Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.

Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices according to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.

As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the

present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or
5 may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an
10 optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction,
15 and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar
20 coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle
25 may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes
30 the position. The detector generally may be a device adapted for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device.
35 Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a
40 stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside the camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based algorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predeter-

mined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

As outlined above, the human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will

be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device.

Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club,

a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least

one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

5 The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.
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The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.
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The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of:
20 a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
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As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.
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The proposed devices and methods provide a large number of advantages over known detectors of this kind. Thus, the detector generally may avoid the shortcomings of the known prior art systems disclosed above. Specifically, the detector may avoid the use of FiP sensors, thereby allowing for e.g. using simple and cheap and commercially available semiconductor sensors such as silicon photodiodes. These photodiodes generally do not show a luminance dependen-
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cy, and the method disclosed above is generally independent from the brightness of the scenery and/or the brightness of the light spot on the light beam. Consequently, a range of measurement in terms of luminance or total power of the light beam entering the detector is generally larger in the present invention as compared to many of the devices disclosed above. Further, the measurement by using the detector according to the present invention is generally independent from the target spot size, i.e. from either the size of the object, the size of a light spot projected onto the object or the size of a beacon device being one or more of attached to the object, integrated into the object or held by the object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional option of measuring transversal coordinates, thereby integrating the functionality of a PSD.

The result of the measurement, i.e. the position determined by the detector such as the at least one longitudinal coordinate, may be rendered widely independent from the brightness of the scenery and/or the brightness of the object, the brightness of the at least one Beacon device, the brightness of the at least one illumination source or the total power of the light beam propagating from the object to the detector. Further, due to this independence and wide range of measurement in terms of brightness, reflective objects or non-reflective objects may be used.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention, such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the quotient signal, as mentioned above, contrarily, provides a wide range of a continuously and monotonously decreasing or increasing function which may be used to determine the longitudinal coordinate from the quotient signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the quotient signal from the total power of the light beam, at least as long as no saturation of one or both of the optical sensors is reached, also a very wide range of measurement in terms of brightness may be provided, i.e. in terms of total power of the light beam.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector having

- at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different spatial offsets, wherein each optical sensor is configured to generate a sensor signal

in response to an illumination of its respective light-sensitive area by a light beam propagating from the object to the detector; and

- at least one evaluation device being configured for determining at least one longitudinal coordinate z of the object by combining the at least two sensor signals.

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Embodiment 2: The detector according to the preceding embodiment, wherein the optical sensors are located in the same plane.

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Embodiment 3: The detector according to any one of the preceding embodiments, wherein the optical sensors are partial diodes of a segmented diode, with a center of the segmented diode being off-centered from the optical axis of the detector.

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Embodiment 4: The detector according to any one of the preceding embodiments, wherein the optical sensors are part of a sensor array, such as an array comprising 2-4 optical sensors such as a quadrant photodiode.

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Embodiment 5: The detector according to the preceding embodiment, wherein a geometrical center of the sensor array is offset from the optical axis.

Embodiment 6: The detector according to any one of the two preceding embodiments, wherein the sensor array is movable, preferably automatically, relative to the optical axis.

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Embodiment 7: The detector according to the preceding embodiment, wherein the evaluation device is configured for, firstly, determining the at least one transversal position of a light spot generated by the light beam on the sensor array by using the sensor signals and for, secondly, moving the sensor array relative to the optical axis until the light spot is off-centered.

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Embodiment 8: The detector according to any one of the preceding embodiments, wherein the optical sensors are partial diodes of a quadrant diode, with a geometrical center of the quadrant diode being off-centered from the optical axis of the detector.

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the light-sensitive areas of the optical sensors are equal.

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Embodiment 10: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one lens, wherein the optical axis of the detector is an axis of symmetry of the lens.

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Embodiment 11: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are areas of the beam profile at the sensor position, and $E(x,y,z_o)$ denotes the beam profile given at the object distance z_o .

Embodiment 12: The detector according to the preceding embodiment, wherein the light-

5 sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

10 Embodiment 13: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a
15 number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

Embodiment 14: The detector according to any one of the three preceding embodiments,

20 wherein the light beam propagating from the object to the detector illuminates the optical sensors with at least one line pattern, wherein $A1$ corresponds to an area with a full line width of the line pattern on the optical sensors, wherein $A2$ is a central area of the line pattern on the optical sensors.

Embodiment 15: The detector according to any one of the three preceding embodiments,

25 wherein the light beam propagating from the object to the detector illuminates the sensor element with at least one point pattern, wherein $A1$ corresponds to an area with a full radius of a point of the point pattern on the optical sensors, wherein $A2$ is a central area of the point on the point pattern on the optical sensors.

30 Embodiment 16: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the evaluation device is further configured for determining the longitudinal coordinate by evaluating the quotient signal Q .

35 Embodiment 17: The detector according to the preceding embodiment, wherein the evaluation device is configured for using at least one predetermined relationship between the quotient signal Q and the longitudinal coordinate.

40 Embodiment 18: The detector according to the preceding embodiment, wherein the predetermined relationship is one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship.

Embodiment 19: The detector according to any one of the two preceding embodiments, wherein the evaluation device comprises at least one data storage device for storing the predetermined relationship.

- 5 Embodiment 20: The detector according to any one of the four preceding embodiments, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

- 10 Embodiment 21: The detector according to the preceding embodiment, wherein the divider is fully or partially embodied as one or both of a software divider or a hardware divider.

Embodiment 22: The detector according to any one of the preceding embodiments, wherein the optical sensors are arranged in one and the same beam path of the detector.

- 15 Embodiment 23: The detector according to any one of the preceding embodiments, wherein the optical sensors each are semiconductor sensors, preferably inorganic semiconductor sensors, more preferably photodiodes and most preferably silicon photodiodes.

- 20 Embodiment 24: The detector according to any one of the preceding embodiments, wherein the optical sensors each are uniform sensors having a single light-sensitive area each.

- 25 Embodiment 25: The detector according to any one of the preceding embodiments, wherein the optical sensors each have, at least within a range of measurement, a linear signal characteristic such that the sensor signals are dependent on the total power of illumination of the respective optical sensor and are independent from a diameter of a light spot of the illumination.

- 30 Embodiment 26: The detector according to any one of the preceding embodiments, wherein the detector further comprises an illumination source for illuminating the object, e.g., an illumination source comprising at least one laser, such as an illumination source providing at least one illuminating light beam being on the optical axis or being not on the optical axis.

- 35 Embodiment 27: The detector according to the preceding embodiment, wherein the illumination source is configured for generating an illuminating light beam for illuminating the object, wherein the detector is configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector.

- 40 Embodiment 28: The detector according to the preceding embodiment, wherein the detector comprises at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

- 45 Embodiment 29: The detector according to any one of the three preceding embodiments, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern; at least one pattern comprising at least one pre-known feature; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

Embodiment 30: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensors.

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Embodiment 31: The detector according to the preceding embodiment, wherein the transfer device comprises one or more of: at least one lens, preferably at least one focus-tunable lens; at least one beam deflection element, preferably at least one mirror; at least one diffractive optical element; at least one Fresnel lens; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system.

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Embodiment 32: The detector according to any one of the preceding embodiments, wherein the detector comprises the at least one illumination source, wherein the illumination source is adapted to illuminate the object with at least one illumination light beam, wherein the illumination source is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis.

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Embodiment 33: The detector according to any one of the preceding embodiments, wherein the detector comprises the at least one illumination source, wherein the illumination source and the optical axis are separated by a small baseline.

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Embodiment 34: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

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Embodiment 35: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

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Embodiment 36: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

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Embodiment 37: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding

embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

5 Embodiment 38: A scanning system for determining a depth profile of a scenery, the scanning system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

10 Embodiment 39: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 40: A readout device for optical storage media, the readout device comprising at least one detector according to any one of the preceding embodiments referring to a detector.

15 Embodiment 41: A method for determining a position of at least one object by using at least one detector, the method comprising the following steps:

- providing at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light-sensitive area has a geometrical center, wherein the geometrical centers of the optical sensors are spaced apart from an optical axis of the detector by different separations, wherein each optical sensor is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area by a light beam;
- 20 - illuminating the at least two light-sensitive areas of the at least two optical sensors of the detector with a light beam propagating from the object to the detector and, thereby, generating at least two sensor signals; and
- 25 - evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object.

30 Embodiment 42: The method according to the preceding embodiment, wherein evaluating of the sensor signals comprises deriving a quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining of the longitudinal coordinate comprises evaluating the quotient signal Q .

35 Embodiment 43: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use for calibration of a distance measurement; a use in combination with optical data storage and readout.

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Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

- | | | |
|----|---------------------|---|
| 15 | Figures 1 and 2 | show different embodiments of a detector according to the present invention; |
| 20 | Figure 3 | shows an exemplary embodiment of a detector according to the present invention, a detector system, a human-machine interface, an entertainment device, a tracking system, a scanning system and a camera; |
| | Figure 4 | shows an off-centered position of the optical axis of the detector and of a light spot on a quadrant diode; |
| 25 | Figure 5 | shows an alternative embodiment of a detector with staggered optical sensors; |
| 30 | Figures 6 to 8 | show a quotient signal Q of two sensor signals of two photodiodes of a quadrant photodiode as a function of a longitudinal coordinate z of an object under various measurement conditions; |
| | Figures 9A and 9B | show a modification of the embodiment of Figure 2, with an off-axis illumination light beam; |
| 35 | Figures 9C and 9D | show a comparison of two experimental setups using a detector setup according to Figure 9A with a Bi-cell as optical sensors and experimental results; |
| 40 | Figures 10A and 10B | show a further exemplary embodiment of a detector according to the present invention with a small baseline; |
| | Figure 11 A to 11 O | show further exemplary configurations of optical sensors according to the present invention; |

Figures 12A and B show embodiments of determining a longitudinal coordinate z using a line pattern; and

Figures 13A and B show further embodiments of the detector according to the present invention comprising at least one bi-cell.

Detailed description of the embodiments

In Fig. 1, a schematic view of a first embodiment of a detector 110 for determining a position of at least one object 112 is depicted. In this case, the object 112 comprises a beacon device 114, from which a light beam 116 propagates towards a first optical sensor 118 and a second optical sensor 120. The first optical sensor 118 comprises a first light-sensitive area 122, and the second optical sensor 120 comprises a second light-sensitive area 124. The optical sensors 118, 120, as shown e.g. in Fig. 4, may be part of an array 174 of optical sensors 176, such as the first optical sensor 118 being the optical sensor 176 in the upper left corner of the array 174 and the second optical sensor 120 being the optical sensor 176 in the lower right corner of the array 174. Other choices are feasible. The array 174, as an example, may be a quadrant photodiode 178, and the optical sensors 176 may be partial diodes of the quadrant photodiode 178.

The light beam 116, as an example, may propagate along an optical axis 126 of the detector 110. Other embodiments, however, are feasible. The optical detector 110, further, may comprise at least one transfer device 128, such as at least one lens or a lens system, specifically for beam shaping. Consequently, the light beam 116 is focused, such as in one or more focal points 130, and a beam width of the light beam 116 may depend on a longitudinal coordinate z of the object 112, such as on a distance between the detector 110 and the beacon device 114 and/or the object 112. For details of this beam width dependency on the longitudinal coordinate, reference may be made to one or more of the above-mentioned prior art documents, such as to one or more of WO 2012/110924 A1 and/or WO 2014/097181 A1.

As can be seen in Fig. 4, the setup of the detector 110 is off-centered in various ways. Thus, a geometrical center 180 of the array 174 may be off-centered from the optical axis 126 by offset d_0 . Further, a geometrical center 182 of the first optical sensor 118 is off-centered from the optical axis 126 by offset d_1 , and a geometrical center 184 of the second optical sensor 120 is off-centered from the optical axis 126 by offset d_2 , wherein $d_1 \neq d_2$. In other words, a light spot 186 is formed, which is unequally distributed over the light-sensitive areas 122, 124.

As will be shown in further detail below, the detector 110 may be configured for automatically establishing the off-centered position of the light spot 186 on the array 174. For this purpose, firstly, the detector 110 may be configured for determining whether the sensor signals generated by the optical sensors 176 of the array 174 are equal. If this should be the case, the detector 110 may be configured to determine that the light spot 186 is centered in the array 174 and, consequently, may shift the light spot 186 out of the geometrical center 180 of the array 174, such as by shifting the whole array 174 in a plane perpendicular to the optical axis 126. For this

purpose, as will be shown in further detail below with respect to Fig. 3, one or more actuators may be provided in the detector 110.

Turning back to the setup of Fig. 1, the first optical sensor 118, in response to the illumination by the light beam 116, generates a first sensor signal s_1 , whereas the second optical sensor 120 generates a second sensor signal s_2 . Preferably, the optical sensors 118, 120 are linear optical sensors, i.e. the sensor signals s_1 and s_2 each are solely dependent on the total power of the light beam 116 or of the portion of the light beam 116 illuminating their respective light-sensitive areas 122, 124, whereas these sensor signals s_1 and s_2 are independent from the actual size of the light spot of illumination. In other words, preferably, the optical sensors 118, 120 do not exhibit the above-described FiP effect.

The sensor signals s_1 and s_2 are provided to an evaluation device 132 of the detector 110. The evaluation device 110, as symbolically shown in Fig. 1, may specifically be embodied to derive a quotient signal Q , as explained above. The quotient signal Q , derived by dividing the sensor signals s_1 and s_2 or multiples or linear combinations thereof, may be used for deriving at least one item of information on a longitudinal coordinate z of the object 112 and/or the beacon device 114, from which the light beam 116 propagates towards the detector 110, as will be explained in further detail with reference to the corrosion signals shown in Fig. 6 to 8 below.

The detector 110, in combination with the at least one beacon device 114, may be referred to as a detector system 134, as will be explained in further detail below with reference to Fig. 3.

In Fig. 2, a modification of the embodiment of Fig. 1 is shown, which forms an alternative detector 110. The alternative embodiment of the detector 110 widely corresponds to the embodiment shown in Fig. 1. Instead of using an active light source, i.e. a beacon device 114 with light-emitting properties for generating the light beam 116, however, the detector 110 comprises at least one illumination source 136. The illumination source 136, as an example, may comprise a laser, whereas, in Fig. 1, as an example, the beacon device 114 may comprise a light-emitting diode (LED). The illumination source 136 may be configured for generating at least one illumination light beam 138 for illuminating the object 112. The illumination light beam 138 is fully or partially reflected by the object 112 and travels back towards the detector 110, thereby forming the light beam 116. The illumination source 136, as an example, may comprise one or more diaphragms 190, such as an adjustable diaphragm 190, e.g. an adjustable iris diaphragm and/or a pin hole.

The setup shown in Fig. 2, as an example, may also be used in or as a readout device 192 for optical storage media. Thus, as an example, the object 112 may be an optical storage medium such as in optical storage discs, e.g. a CCD, DVD or Blu-ray disc. By measuring the presence or non-presence of data storage modules and the depth of the same within the object 112, by using the above-mentioned measurement principle, a data readout may take place.

The light beam 116, specifically, may travel along the optical axis 126 of the detector 110. As shown in Fig. 2, as an example, the illumination light beam 138 may be parallel to the optical axis 126 of the detector 110. Other embodiments, i.e. off-axis illumination and/or illumination at an angle, are feasible, too, as will be shown in the context of Fig. 9A and 9B below. In order to provide an on-axis illumination, as shown in Fig. 2, as an example, one or more reflective elements 140 may be used, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

Apart from these modifications, the setup of the embodiment in Fig. 2 corresponds to the setup in Fig. 1. Thus, again, an evaluation device 132 may be used, having, e.g., at least one divider 142 for forming the quotient signal Q, and, as an example, at least one position evaluation device 144, for deriving the at least one longitudinal coordinate z from the at least one quotient signal Q. It shall be noted that the evaluation device 132 may fully or partially be embodied in hardware and/or software. Thus, as an example, one or more of components 142, 144 may be embodied by appropriate software components.

It shall further be noted that the embodiments shown in Fig. 1 and 2 simply provide embodiments for determining the longitudinal coordinate z of the object 112. It is also feasible, however, to modify the setups of Fig. 1 and 2 to provide additional information on a transversal coordinate of the object 112 and/or of parts thereof. As an example, e.g. in between the transfer device 128 and the optical sensors 118, 120, one or more parts of the light beam 116 may be branched off, and may be guided to a position-sensitive device such as one or more CCD and/or CMOS pixelated sensors and/or additional quadrant detectors and/or other position sensitive devices, which, from a transversal position of a light spot generated thereon, may derive a transversal coordinate of the object 112 and/or of parts thereof. For further details, as an example, reference may be made to one or more of the above-mentioned prior art documents which provide for potential solutions of transversal sensors.

Fig. 3 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, e.g. according to the embodiments shown in Fig. 1 or 2. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible.

Fig. 3 further shows an embodiment of a detector system 134, which, besides the at least one detector 110, comprises one or more beacon devices 114, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 3 further shows an exemplary embodiment of a human-machine interface 148, which comprises the at least one detector system 134 and, further, an entertainment device 150, which comprises the human-machine interface 148. The figure further shows an embodiment of a tracking system 152 for tracking a position of the object 112, which comprises the detector system 134. The components of the devices and systems shall be explained in further detail below.

Fig. 3 further shows an exemplary embodiment of a scanning system 154 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 154 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 136 as well as, optionally, at least one further illumination source 136. The illumination source 136, generally, is configured to emit at least one illumination light beam 138, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 114 and/or on a surface of the object 112. The scanning system 154 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 154, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 3 is shown in Fig. 1 and 2 or will be shown, as an alternative embodiment, in Fig. 5 below. Thus, the detector 110, besides the optical sensors 118, 120, comprises at least one evaluation device 132, having e.g. the at least one divider 142 and/or the at least one position evaluation device 144, as symbolically depicted in Fig. 3. The components of the evaluation device 132 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 118, 120 and one or more of the components of the evaluation device 132 may be interconnected by one or more connectors 156 and/or by one or more interfaces, as symbolically depicted in Fig. 5. Further, the one or more connectors 156 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 156, the evaluation device 132 may fully or partially be integrated into one or both of the optical sensors 118, 120 and/or into a housing 158 of the detector 110. Additionally or alternatively, the evaluation device 132 may fully or partially be designed as a separate device.

In Fig. 3, as an example, one or more reflective elements 140 may be used, for example partially transparent, such as one or more prisms and/or mirrors, such as dichroitic mirrors, such as movable mirrors or movable prisms.

The detector 110 as symbolically shown in the exemplary embodiment of Fig. 3 may also comprise at least one actuator 188 for moving the array 174 of the optical sensors 176 relative to the optical axis 126. As outlined above, for providing this movement, the optical axis 126 may be moved in relation to the array 174 by moving the optical axis 126, by moving the array 174 or both. Thus, as an example, the optical axis may be moved by using one or more of deflecting elements and/or by using the transfer device 128. As a simple example, a lens of the transfer device 128 may be tilted, such as by using one or more actuators 188 (not depicted). Additionally or alternatively, the array 174 may be shifted by the one or more actuators 188, preferably in a plane perpendicular to the optical axis 126. As an example, one or more electromechanical actuators may be used, such as one electromechanical actuator for an x-direction and another

electromechanical actuator for a y-direction. Other embodiments are feasible. Thereby, the above-mentioned off-centering procedure may be implemented for establishing an off-centered situation as shown e.g. in Fig. 4.

5 In the exemplary embodiment shown in Fig. 3, further, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 160, the position of which may be manipulated by a user 162. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 162 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises at least the optical sensors 176, including at least the first optical sensor 118 and the second optical sensor 120. The optical sensors 176
15 may be located inside the housing 158 of the detector 110. Further, the at least one transfer device 128 may be comprised, such as one or more optical systems, preferably comprising one or more lenses.

An opening 164 inside the housing 158, which, preferably, is located concentrically with regard to the optical axis 126 of the detector 110, preferably defines a direction of view 166 of the detector 110. A coordinate system 168 may be defined, in which a direction parallel or anti-parallel to the optical axis 126 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 126 may be defined as transversal directions. In the coordinate system 128, symbolically depicted in Fig. 5, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 168 are feasible, such as non-Cartesian coordinate systems.

The detector 110 may comprise the optical sensors 118, 120 as well as, optionally, further optical sensors. The optical sensors 118, 120 preferably are located in one and the same beam path, one behind the other, such that the first optical sensor 118 covers a portion of the second optical sensor 120. Alternatively, however, a branched beam path may be possible, with additional optical sensors in one or more additional beam paths, such as by branching off a beam path for at least one transversal detector or transversal sensor for determining transversal coordinates of the object 112 and/or of parts thereof.

35 One or more light beams 116 are propagating from the object 112 and/or from one or more of the beacon devices 114, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. For this purpose, as explained above in the context of Fig. 1, 2 and 4, the evaluation device 132 is configured to evaluate sensor signals provided
40 by the optical sensors 118, 120. The detector 110 is adapted to determine a position of the object 112, and the optical sensors 118, 120 are adapted to detect the light beam 116 propagating from the object 112 towards the detector 110, specifically from one or more of the beacon devices 114. In case no illumination source 136 is used, the beacon devices 114 and/or at least

one of these beacon devices 114 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. In case the illumination source 136 is used, the beacon devices 114 do not necessarily have to be active beacon devices. Contrarily, a reflective surface of the object 112 may be used, such as integrated reflective beacon devices 114 having at least one reflective surface. The light beam 116, directly and/or after being modified by the transfer device 128, such as being focused by one or more lenses, illuminates the light-sensitive areas 122, 124 of the optical sensors 118, 120. For details of the evaluation, reference may be made to Fig. 1, 2 and 4 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 148, in order to provide at least one item of information to a machine 170. In the embodiments schematically depicted in Fig. 3, the machine 170 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 132 may even be fully or partially integrated into the machine 170, such as into the computer.

As outlined above, Fig. 3 also depicts an example of a tracking system 152, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 152 comprises the detector 110 and at least one track controller 172. The track controller 172 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 172 may be an independent device and/or may be fully or partially integrated into the machine 170, specifically the computer, as indicated in Fig. 3 and/or into the evaluation device 132.

Similarly, as outlined above, the human-machine interface 148 may form part of an entertainment device 150. The machine 170, specifically the computer, may also form part of the entertainment device 150. Thus, by means of the user 162 functioning as the object 112 and/or by means of the user 162 handling a control device 160 functioning as the object 112, the user 162 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer game.

In the setup of the detectors 110 as shown in Fig. 1, 2 and 3, the optical sensors 176 are part of an array 174, and all optical sensors 176 may be located in one and the same plane oriented essentially perpendicular to the optical axis 126. As noted in this context, when mentioning “perpendicular” or “essentially perpendicular”, preferably, a 90° orientation is given. However, tolerances may be present, such as angular tolerances of no more than 20°, preferably of no more than 10° or more preferably of no more than 5°. The optical sensors 176, however, not necessarily have to be located in an array 174 and not necessarily have to be located in one and the same plane, as is shown in an alternative setup of the detector 110 shown in Fig. 5. In this figure, only the optical components are shown. For other components, reference may be made to Fig. 1, 2 and 3 above.

As can be seen, in this alternative setup, two or more optical sensors 176 are present, comprising at least one first optical sensor 118 and at least one second optical sensor 120 located in different planes which are offset in a direction of the optical axis 126, also referred to as the z-direction. Further, as can also be seen, the optical sensors 118, 120 may overlap, whereas in the previous embodiments, preferably, no overlap between the optical sensors 176 is given. Apart from these modifications, the functionality and the evaluation of the sensor signals generally corresponds to the embodiment of Fig. 1, 2 and 3 above.

As discussed above, for evaluating the at least two sensor signals of the at least two optical sensors 176 and for deriving an information on the longitudinal position of the object 112 thereof, such as a distance between the detector 110 and the object 112 and/or a z-coordinate of the object 112, preferably, at least one combined sensor signal is generated by the evaluation device 132. The combined sensor signal, as long as this combined sensor signal provides, at least over a measurement range, a unique function of the distance, may be used for deriving the longitudinal coordinate. As an example, the combined sensor signal may be or may comprise at least one quotient signal Q . In Fig. 6 to 8, quotient signals Q of two sensor signals of two optical sensors 176 are shown under various measurement conditions. In each case, the quotient signal Q is denoted on the vertical axis, as a function of the longitudinal coordinate z of the object 112 on the horizontal axis, the latter given in centimeters.

In all experiments, a setup as shown in Fig. 2 was used. As an illumination source 136, in the experiments of Fig. 6 and 7, a 980 nm Picotronic laser source was used, in conjunction with a lens having a focal length of 100 mm. In the experiment of Fig. 8, a Laser Components laser light source having a wavelength of 850 nm was used, in conjunction with a lens having a focal length of 79 mm. In all experiments, the laser beam was aligned on the optical axis 126 via a small prism in front of the lens 128, forming a reflective element 140. A diaphragm 190 in front of the laser source was used to vary the spot size. The quadrant diode 178 was used to measure the reflection of the laser source on different materials. In all experiments, the distance dependency is given by the quotient Q of two adjacent quadrant currents.

In Fig. 6, the laser power was varied during the experiment, from 8 nA laser current, denoted by the dotted line, to 106 nA, denoted by the solid line. Therein, since the laser current typically does not provide a measure for the laser intensity, the laser current indicated therein is a current of a silicon photodetector in a measurement setup in which the laser illuminates a white sheet of paper at a distance of 330 mm from the lens. As is clearly visible, the curves are nearly identical and, at least within this range of variation of the laser power, do not significantly depend on the laser power. This experiment shows that the quotient signal provides a reliable and monotonous function of the longitudinal coordinate, independent from the influence of the brightness of the illumination source.

In Fig. 7, a spot size of the illumination source 136 was varied, by varying the open diameter of the diaphragm 190 in front of the laser. The spot size was varied from 1.5 mm, denoted by the dotted line, to 3.5 mm, denoted by the solid line, in steps of 0.5 mm. As can be seen, up to a

distance of approximately 200 cm, the quotient signal Q does not depend on the spot size and, thus, again, is not negatively affected by this variation.

In Fig. 8, a material of the object 112 illuminated by the laser beam was varied. Therein, the dotted line denotes white paper, the dashed line with the smallest dashes denotes black paper, the dashed line with the medium dashes denotes wood, and the dashed line with the largest dashes denotes an aluminum plate. As can be seen, at least up to a measurement range of approximately 250 cm, the experiment does not strongly depend on the type of material used for the object 112.

The experiments shown in Fig. 6 to 8, thus, clearly demonstrate that the quotient signal Q provides a reliable function of the distance. At least within a range of measurement, the function monotonously rises with the distance. The function is not strongly influenced by the most significant variations which may occur in real life measurements, such as the brightness of the illumination source, the spot size of the illumination source or the material of the object 112. Thus, by evaluating the quotient signal Q of two or more optical sensors 176, reliable distance information may be generated. Thus, as an example, the curves shown in Fig. 6 to 8 directly may be used as calibration curves for the purpose of the evaluation device 132. Other evaluation methods, however, are feasible.

In Fig. 9A and 9B, an alternative embodiment of the detector 110 is shown which is a modification of the setup shown in Fig. 2. Thus, for most elements and optional details as well as further elements not shown in the schematic Figures 9A and 9B, reference may be made to the description of Fig. 2 above.

In Figure 2, the illumination light beam 138, as discussed above, preferably travels along the optical axis 126, i.e. parallel to the optical axis 126 or even on the optical axis 126. In the setup, the position of the center of the light spot 186 typically does not depend on the z-coordinate of the object 112, such as on a distance between the object 112 and the detector 110. In other words, the diameter or equivalent diameter of the light spot 186 changes with the distance between the object 112 and the detector 110 whereas, typically, the position of the light spot 186 on the array 174 does not.

Contrarily, in Fig. 9A and 9B, a setup of the detector 110 is shown in which an illumination light beam 138 travels off-axis, i.e. one or both of at an angle other than 0° with the optical axis 126 or parallel to the optical axis 126 but shifted from the optical axis 126. This embodiment, as will be discussed in further detail below, demonstrates that the method according to the present invention can be further enhanced by increasing the z-dependency of a combined sensor signal. Thus, in Fig. 9A, a side view is shown with two different positions of the object 112, i.e. a first position at z_1 , drawn in solid lines, and a second position at z_2 , drawn in dashed lines. As can be seen, the illumination light beam 138 which, as an example, propagates at an angle of 5° to 30° , e.g. 10° to 20° , with the optical axis 126, hits the object 112 in both cases at different positions. From these points of the object 112 illuminated by the illumination light beam 138,

light beams 116 propagate towards the detector 110, wherein, again, the light beam 116 for the object 112 being located at position z_1 is drawn in solid lines, wherein the light beam 116 for the object 112 being located at position z_2 is drawn in dashed lines.

In Fig. 9B, the array 174, e.g. a quadrant photodiode, is shown in an enlarged fashion. As can be seen in this setup, the position of the light spot 186 moves with the longitudinal position z of the object 112. Thus, not only is the size of the light spot 186 affected by the longitudinal position z but also is the position on the array 174 of the light spot 186 changed. In Fig. 9B, this movement of the light spot 186 is denoted by arrow z .

Consequently, by this movement of the light spot 186, the z -dependency of a combined sensor signal taking into account at least two sensor signals of the optical sensors 176 may be increased. As an example, the four diodes of the array 174, in Fig. 9B, are denoted by D1-D4. The quotient signal Q , as an example, may be formed as $Q = i(D1)/i(D4)$, with $i(D1)$ being the sensor signal of photodiode D1, and $i(D4)$ being the sensor signal of photodiode D4.

As shown in Figure 9B, the quadrant diode may comprise two dividing lines. The dividing lines may be arranged orthogonal to each other. The orthogonal arrangement of the dividing lines allows adjusting of the quotient signal for near field and far field applications independently from each other. In addition to determining the quotient signal of sensor signals of two optical sensors of the quadrant diode, the evaluation device 132 may be adapted to determine a second quotient using at least three or all four sensor signals of the quadrant diode. The two quotients can be formed such that two distinct distance ranges are covered. The two quotient signals for the near field and far field may have an overlap region in which both quotients allow obtaining reasonable determination of the longitudinal distance z . For example, the quotient may be determined by $Q = i(D1+D2)/i(D3+D4)$, wherein the sensor signals of the two top quadrants, also called top segment, are divided by the sensor signals of the two bottom quadrants, also called bottom segment. Using the quotient of sensor signals determined by two sensor areas which have a dividing line parallel to the baseline of the detector may allow determining of the quotient without any distance dependent movement of the light spot. In particular, as an example, if the dividing line between top and bottom segment is parallel to the baseline, the quotient signal determined from the top segment divided by the bottom segment may be used in the near field, wherein the light spot may illuminate only one of a left or right segment of the quadrant diode. In this case determining the quotient signal by dividing sensor signals of the left and right segments may not be possible. However, determining the quotient by dividing the sensor signals of top and bottom segments may provide a reasonable distance measurement. The quotient signal determined by dividing sensor signals of the left and right segments, i.e. $Q = i(D1+D3)/i(D2+D4)$, may be used for far field measurement, wherein the light spot illuminates both left and right segments. Furthermore, the evaluation device may be adapted to determine the quotient by dividing sensor signals of opposing segments or neighboring segments. The evaluation device may be adapted to combine the acquired sensor signals $i(D1)$, $i(D2)$, $i(D3)$ and $i(D4)$ of the quadrants such that distance measurement is possible over a wide range with a large resolution.

In the situation shown in Fig. 2, the position of the light spot 186 does not depend on z . With a change in z , depending on the optical situation, the spot will become larger or smaller, such as by becoming more diffuse or more focused. In case the spot size increases and the spot be-
comes more diffuse, $i(D4)$ will increase more rapidly than $i(D1)$, such that the quotient signal Q decreases.

Contrarily, in the situation of Fig. 9A, both the size and the position of the light spot 186 are dependent on the z -coordinate. Thus, the tendency of the z -dependency of the combined sensor signal such as the quotient signal Q will be increased. In the situation of Figure 2, depending on the z -coordinate, the sensor signal of at least one sensor will increase and simultaneously the sensor signal of at least one different sensor will decrease, resulting in the z -dependent quotient signal Q . In the situation of Fig. 9A, the position dependency of the light spot 186 can result in three different situations depending on the relative position of light source, optical axis, and sensor: Firstly, the position dependency of the light spot may result in a further decrease of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a further increase of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Secondly, the position dependency of the light spot may result in a reduced decrease or even increase of the at least one decreasing sensor signal depending on the z -coordinate, while, simultaneously, the position dependency of the light spot may result in a reduced increase or even decrease of the at least one decreasing sensor signal depending on the z -coordinate compared to the situation in Figure 2. Thirdly, the position dependency of the light spot may be as such that the z -dependence of the sensor signals is largely unchanged compared to the situation in Figure 2. However, according to the present invention, object distance is not determined from the position of the light spot on a sensor as done in triangulation methods. Instead, movement of the light spot 186 on the array 174 may be used to enhance dynamic of the sensor signals and or the resulting quotient signal Q which may result in an enhanced dynamic of the z -dependency.

Additionally, as known from the prior art, the sensor signals $i(D1)$, $i(D2)$, $i(D3)$, $i(D4)$ may also be used for determining a transversal position x , y of the object 112. Further, the sensor signals may also be used for verifying the z -coordinate determined by the present invention.

Figure 9C shows a comparison of two experimental setups using a detector setup according to Figure 9A with a Bi-cell as optical sensors 176 with two light sensitive areas. In a first experimental setup, depending on the relative position of the illumination light source, the optical axis and the sensor, the light spot 186 may move in parallel to the linear boundary of the two optical sensors 176 of the Bi-cell along a direction of movement 194 in dependence of the object distance. Since the direction of movement 194 of the light spot 186 is in parallel to the linear boundary of the two light sensitive areas in dependence of the object distance, the resulting sensor signals are identical to a situation with no movement of the light spot 186 depending on object distance as shown in Figure 2. In a second experimental setup, depending on the relative

position of the illumination light source, the optical axis and the sensor, the light spot 186 may move as such that the distance of the center of the light spot 186 to the boundary of the two optical sensors 176 of the Bi-cell changes in dependence of the object distance such as a movement orthogonal to the boundary of the two optical sensors 176 such as a movement
5 along a direction of movement 196 in dependence of the object distance. The detector setup allowing movement of the light spot 186 may be a modification of the setup shown in Figure 9A. Thus, for most elements and optional details as well as further elements, reference may be made to the description of Fig. 9A above. In Figure 9C, the optical sensors 176 may be a bi-cell diode.

10 Figure 9D shows experimental results of the comparison of the two experimental setups using a detector setup according to Figure 9A, allowing movement of the light spot 186 according to Figure 9C with movement of the light spot depending on the object distance along directions of movement 194 and 196. Curve 198 shows the dependency of quotient Q on the longitudinal
15 coordinate z for the detector setup allowing movement of the light spot 186 along a direction of movement 194 as shown in Figure 9C which is in parallel to the boundary of the optical sensors of the Bi-Cell and, which is a situation equivalent to Figure 2 without a movement of the light spot depending on the object distance. Curve 200 shows the dependency of quotient Q on the longitudinal coordinate z for the detector setup according to Figure 9A and using a detector set-
20 up allowing movement of the light spot 186 with movement of the light spot 186 according to Figure 9C with movement of the light spot depending on the object distance along a direction of movement 196. The experimental setup was as follows: The optical sensors 176 may be a bi-cell diode, in particular a Si-Bi-Cell. The illumination source 136 may be a 950 nm laser with a spot size of 4 mm. The transfer device 128 may have a focal length of 20 mm, e.g. a lens avail-
25 able as Thorlabs Asphere, f= 20 mm. The distance of the object 112 was varied from 0 to 3000 mm. Determination of the longitudinal coordinate z may be possible without allowing movement of the light spot 186. In particular, according to the present invention, movement of the light spot may not be essential for determination of the longitudinal coordinate z. With the detector setup allowing movement of the light spot 186 along a direction 194 or without any movement deter-
30 mination of object distance is possible at very small distance, whereas with movement along a direction 196 determination of object distance is possible for object distance such as distances greater than 500mm.

35 Figures 10A and 10B show an exemplary embodiment of a detector 110 with a small baseline 202. Figure 10A shows side view of the detector 110. The detector 110 may comprise the at least one illumination source 136. The illumination source 136 may be adapted to illuminate the object 112 with at least one illumination light beam 138. The illumination light beam 138 may be fully or partially reflected by the object and may travel back towards the optical sensor 176. The illumination source 136 may be arranged such that a direction of propagation of the illumination
40 light beam is essentially parallel to the optical axis 126. The illumination source 136 may be arranged such that a line running in the direction of propagation of the illumination light beam and the optical axis 126 do not intersect in space. The illumination source 136 and the optical axis 126 may be separated by a small baseline 202. The illumination source 136 may be

spaced apart from the optical axis 126 by a minimum distance. The minimum distance from the optical axis may be defined by further detector elements such as size and position of the optical sensors and transfer device. Figure 10B shows a front view on the illumination light beam 138 and the transfer device 128. The baseline may be less than 0,1 m, preferably less than 0,05 m, more preferably less than 0,025 m. For example, the detector 110 may be a compact device without further optical elements, wherein the illumination source 136 may be placed as close as possible to an edge of the transfer device 128. Thus, the baseline 202 may be close to half a diameter of the transfer device 128, in particular the lens diameter and housings of lens and light source. The illumination source 136 may be arranged such that the baseline 202 is as small as possible. For example, a distance from the center of the transfer device 128 to the illumination source 136, in particular along a connecting line from the center of the transfer device 128 to the illumination source 136, may be preferably less than 2.5 times a distance from the center of the transfer device 128 to an edge if the transfer device 128, more preferably less than 1.5 times the distance center to edge of the transfer device 128, and most preferably less 1 times the distance center to edge of the transfer device 128.

Figure 11 A to O show further exemplary configurations of optical sensors according to the present invention, in particular top view in direction of propagation of the light beam 116. In Figure 11A, a top view of two rectangular optical sensors is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120. The first optical sensor 118 and the second optical sensor 120 may be arranged with a different offset, in particular in a transversal direction y, from the optical axis 126. In Figures 11B and 11C, top view of a large rectangular optical sensor 120 is shown, wherein the first optical sensor 118 is a small optical sensor in front of a larger second optical sensor 120 having a triangle shaped (Figure 11B) or star-shaped (Figure 11C) light-sensitive area. Alternatively, the first optical sensor 118 and the second optical sensor 120 may be located at the same longitudinal coordinate, such as in the same plane, such as as separated light sensitive elements.

In Figures 11 M to O, a top view of two rectangular optical sensors is shown, wherein the first optical sensor 118 and the second optical sensor 120 are rectangular sensors with the same size. In Figures 11 M to O a mask 119 is arranged in front of the first and second optical sensors 118, 120. The mask 119 may be arranged with a different offset from the optical axis 126. The mask 119 may have an arbitrary size and shape, for example, the mask may be rectangular shaped (Figure 11M), triangle shaped (Figure 11N) or star-shaped (Figure 11O). However, other sizes and shapes are feasible. Mask 119 may be adapted to prevent light impinging on the light sensitive areas of the first and second optical sensors 118, 120.

Alternatively to using a mask 119, Figures 11 M to O may be understood as a first optical sensor 118a having rectangular shape (Figure 11M), triangle shape (Figure 11N) or cross-shape or star-shape (Figure 11O) and yielding a first sensor signal and a second optical sensor 120a yielding a second sensor signal, wherein the second optical sensor 120a may be separated into several light sensitive areas, for example four light sensitive areas 121a.

The geometrical centers of the first optical sensor 118 and the second optical sensor 120 are spaced apart from the optical axis 126 of the detector 110. Figure 11K shows two circular shaped optical sensors, wherein the first optical sensor 118 is a small optical sensor in front of the larger second optical sensor 120. Alternatively, the first optical sensor 118 and the second optical sensor 120 may be located at the same longitudinal coordinate, such as in the same plane, such as as separated light sensitive elements. In Figures 11D, the light sensitive area of the first optical sensor 118 is square-shaped, and the light sensitive area of the second optical sensor 120 is rectangular, such that the surface areas in x and y differ. In addition, a center of the first optical sensor 118 and a center of second optical sensor 120 may have different x coordinates such that the optical sensors 118, 120 may have different spatial offset in one or more of x and y direction from the optical axis. In Figure 11H, both the first optical sensor 118 and the second optical sensor 120 may be rectangular. The first optical sensor 118 and the second optical sensor 120 may be arranged such that the center of the first optical sensor 118 and the center of second optical sensor 120 may have different x coordinates and that the surface areas in x and y differ. In Figure 11L, the first optical sensor 118 may have a deviating shape from the shape of the second optical sensor 120 such as a circular or semicircular shape. Figures 11 E, F, G, I, J show a matrix 115 comprising a plurality of partial diodes. In Figures 11 E, F, G the matrix 115 has a rectangular shape, whereas in Figures 11 I and J the matrix 115 has a circular shape. Rows and columns may be arranged equidistant or non-equidistant. In case of equidistant rows and/or columns the matrix 115 may be arranged with a spatial offset to the optical axis 126. The sensor signals of the light sensitive areas of the matrix 115 in Figure 11 E, F, G, I, J may be summarized such as added to form a first sensor signal of a first optical sensor and a second sensor signal of a second optical sensor. As an example, the sensor signals of the light sensitive areas in the corners of the sensor element 115 in Figure 11 G may be added to form a first sensor signal, whereas the remaining light sensitive areas of the sensor element 115 may be added to form a second sensor signal.

Referring to Figures 12A and B, the detector 110 such as the detector as described with respect to Figures 1 to 11 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 118, 120. For example, the detector 110 may comprise a plurality of optical sensors. The detector 110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by the at least two optical sensors 118, 120. In one embodiment, the light beam 116 may illuminate the at least two optical sensors 118, 120 with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 132 may be configured for deriving the quotient signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors 118, 120. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 41 A, the light beam 116 propagating from the object 112 to the detector 110 may illuminate the at least two optical sensors 118, 120 with at least one line pattern 2186. The line pattern 204 may be generated by the object 112, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising at least one illumination line pattern. A1 may correspond to an area with a full line width of the line pattern 204. The line pattern 204 on the at least two optical sensors 118, 120 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a line width on the at least two optical sensors 118, 120 is increased. A2 may be a central area of the line pattern 204 on the at least two optical sensors 118, 120. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. Figure 12B show an embodiment, wherein the detector 110 may comprise at least two rows of optical sensors 118, 120, wherein the rows are arranged on top of each other. The line pattern 204 may be imaged parallel to the rows such that the center of the line pattern is fixed or moves only little relative to a dividing line 206 of the rows. The A1 may correspond to an area above the dividing line 206 and A2 may be an area below the dividing line.

Figures 13A and 13B show further embodiments of the detector 110 according to the present invention comprising at least one bi-cell. The illumination source 136, such as a laser source, may generate the illumination light beam 138 illuminating object 112. The reflected light beam 116 may propagate from the object 112 to the transfer device 128 and may impinge on the bi-cell of optical sensors 118, 120. In Figure 13A a side view is shown and in Figure 13B a front view is shown. The detector 110 may comprise at least one FiP sensor adapted for generating the so called FiP effect as described in WO 2015/024871 or WO2016/120392. For example, the bi-cell in Figures 45 A and B may be adapted to generate a so called FiP signal. As outlined e.g. in WO 2015/024871 or WO2016/120392, the FiP signal can be used to determine depth information over a wide distance range. The FiP sensor may be adapted to exhibit a positive and/or a negative FiP effect. The negative FiP effect may be used to tune small image effects at high

distances. Image changes such as position, size, shape, sharpness, etc. may vanish at high distances while the negative FiP effect increases. Furthermore, no luminance dependence may be introduced since both cells are at the same longitudinal position and thus receive identical photon density.

List of reference numbers

110	Detector
112	Object
114	beacon device
115	matrix
116	light beam
118	first optical sensor
118a	first optical sensor
119	mask
120	second optical sensor
120a	second optical sensor
121a	light sensitive area
122	first light-sensitive area
124	second light-sensitive area
126	optical axis
128	transfer device
130	focal point
132	evaluation device
134	detector system
136	illumination source
138	illumination light beam
140	reflective element
142	divider
144	position evaluation device
146	camera
148	human-machine interface
150	entertainment device
152	tracking system
154	scanning system
156	connector
158	housing
160	control device
162	User
164	opening
166	direction of view
168	coordinate system
170	machine
172	track controller
174	Array
176	optical sensor
178	quadrant photodiode
180	geometrical center of every

182	geometrical center of first optical sensor
184	geometrical center of second optical sensor
186	light spot
188	actuator
190	diaphragm
192	readout device for optical storage media
194	direction of movement
196	direction of movement
198	curve
200	curve
202	baseline
204	Line pattern
206	Dividing line

Patent claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) having
 - 5 - at least two optical sensors (118, 120, 176), each optical sensor (118, 120, 176) having a light-sensitive area (122, 124), wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110) by different spatial offsets, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light beam (116) propagating from the object (112) to the detector (110); and
 - at least one evaluation device (132) being configured for determining at least one longitudinal coordinate z of the object (112) by combining the at least two sensor signals.
- 15 2. The detector (110) according to the preceding claim, wherein the optical sensors (118, 120, 176) are part of a sensor array (174), wherein a geometrical center (180) of the sensor array (174) is offset from the optical axis (126).
- 20 3. The detector (110) according to the preceding claim, wherein the sensor array (174) is movable relative to the optical axis (126).
4. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for, firstly, determining the transversal position of a light spot (186) generated by the light beam (116) on the sensor array (174) by using the sensor signals and for, secondly, moving the sensor array (174) relative to the optical axis (126) until the light spot (186) is off-centered.
- 25 5. The detector (110) according to any one of the preceding claims, wherein the optical sensors (118, 120, 176) are partial diodes of a quadrant diode (178), with a geometrical center (180) of the quadrant diode (178) being off-centered from the optical axis (126) of the detector (110).
- 30 6. The detector (110) according to any one of the preceding claims, wherein the evaluation device (132) is configured for deriving at least one quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the evaluation device (132) is further configured for determining the longitudinal coordinate by evaluating the quotient signal Q .
- 35 7. The detector (110) according to the preceding claim, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the at least one quotient signal Q and the longitudinal coordinate.
- 40

- 5 8. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (136), wherein the illumination source (136) is adapted to illuminate the object (112) with at least one illumination light beam, wherein the illumination source (136) is arranged such that a direction of propagation of the illumination light beam is essentially parallel to the optical axis (126).
- 10 9. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises the at least one illumination source (136), wherein the illumination source (136) and the optical axis (126) are separated by a small baseline (202).
- 15 10. A detector system (134) for determining a position of at least one object (112), the detector system (134) comprising at least one detector (110) according to any one of the preceding claims, the detector system (134) further comprising at least one beacon device (114) adapted to direct at least one light beam (116) towards the detector (110), wherein the beacon device (114) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
- 20 11. A human-machine interface (148) for exchanging at least one item of information between a user (162) and a machine (170), wherein the human-machine interface (148) comprises at least one detector system (134) according to the preceding claim, wherein the at least one beacon device (114) is adapted to be at least one of directly or indirectly attached to the user (162) and held by the user (162), wherein the human-machine interface (148) is designed to determine at least one position of the user (162) by means of the detector system (134), wherein the human-machine interface (148) is designed to assign to the position at least one item of information.
- 25 12. An entertainment device (150) for carrying out at least one entertainment function, wherein the entertainment device (150) comprises at least one human-machine interface (148) according to the preceding claim, wherein the entertainment device (150) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (148), wherein the entertainment device (150) is designed to vary the entertainment function in accordance with the information.
- 30 13. A tracking system (152) for tracking a position of at least one movable object (112), the tracking system (152) comprising at least one detector system (134) according to any one of the preceding claims referring to a detector system (134), the tracking system (152) further comprising at least one track controller (172), wherein the track controller (172) is adapted to track a series of positions of the object (112) at specific points in time.
- 35 14. A scanning system (154) for determining a depth profile of a scenery, the scanning system (154) comprising at least one detector (110) according to any of the preceding claims referring to a detector (110), the scanning system (154) further comprising at least one illumination source (136) adapted to scan the scenery with at least one light beam (116).
- 40

15. A camera (146) for imaging at least one object (112), the camera (146) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
16. A readout device (192) for optical storage media, the readout device (192) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
17. A method for determining a position of at least one object (112) by using at least one detector (110), the method comprising the following steps:
- providing at least two optical sensors (118, 120, 176), each optical sensor (118, 120, 176) having a light-sensitive area (122, 124), wherein each light-sensitive area (122, 124) has a geometrical center (182, 184), wherein the geometrical centers (182, 184) of the optical sensors (118, 120, 176) are spaced apart from an optical axis (126) of the detector (110) by different separations, wherein each optical sensor (118, 120, 176) is configured to generate a sensor signal in response to an illumination of its respective light-sensitive area (122, 124) by a light;
 - illuminating the at least two light-sensitive areas (122, 124) of the at least two optical sensors (118, 120, 176) of the detector (110) with a light beam (116) propagating from the object (112) to the detector (110) and, thereby, generating at least two sensor signals; and
 - evaluating the sensor signals by combining the at least two sensor signals, thereby determining at least one longitudinal coordinate z of the object (112).
18. The method according to the preceding claim, wherein the evaluating the sensor signals comprises deriving at least one quotient signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals, and wherein the determining the longitudinal coordinate comprises evaluating the quotient signal Q .
19. A use of the detector (110) according to any one of the preceding claims relating to a detector (110), for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; an optical data storage application; a security application; a surveillance application; a safety application; a human-machine interface (148) application; a tracking application; a photography application; a logistics application; a machine vision application; a robotics application; a quality control application; a manufacturing application; a use for calibration of a distance measurement; a use in combination with optical data storage and readout.

FIG.1

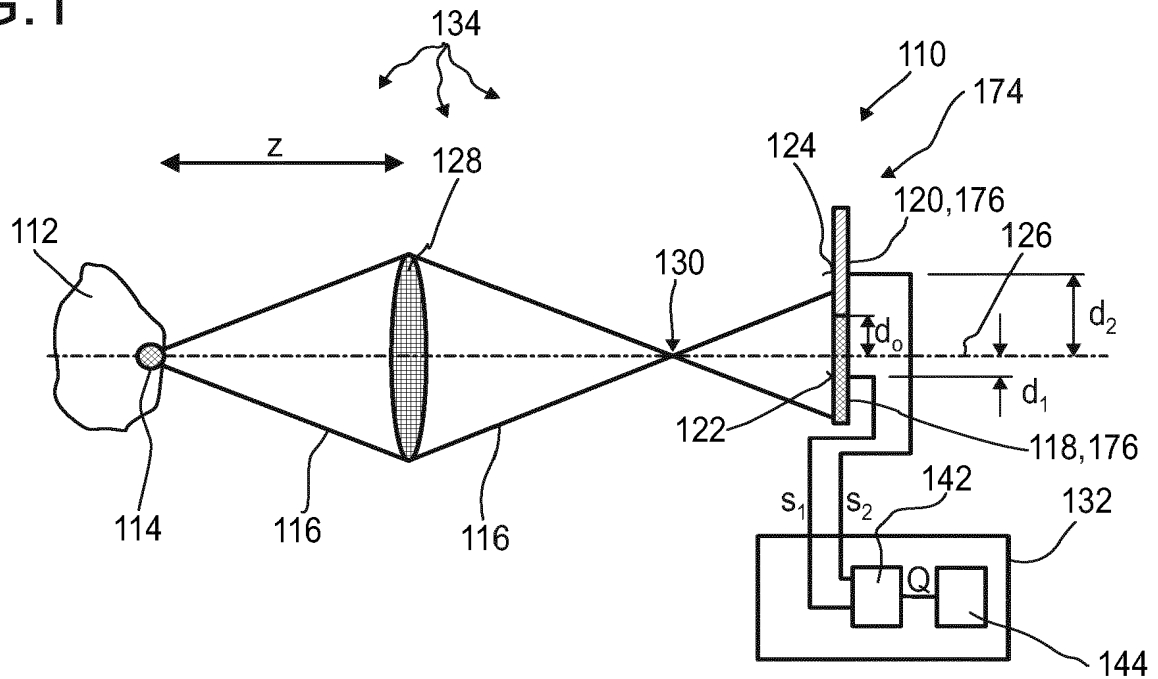


FIG.2

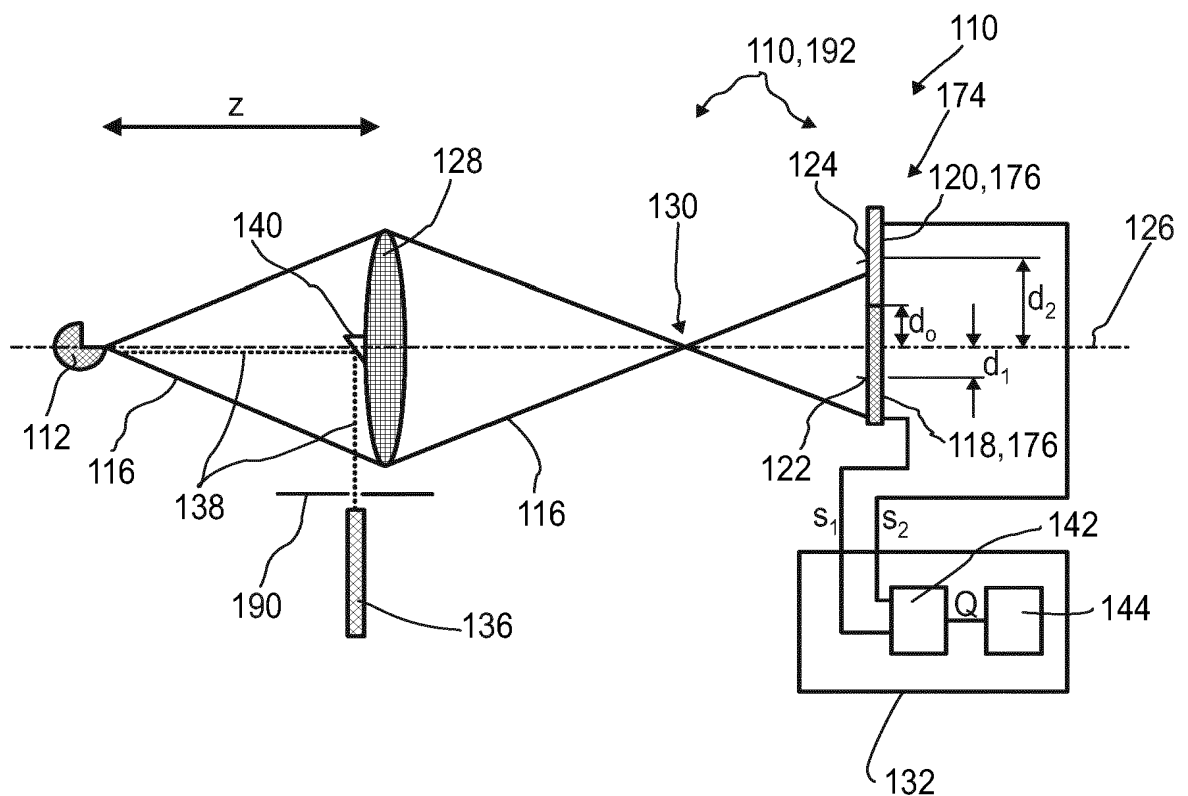


FIG.3

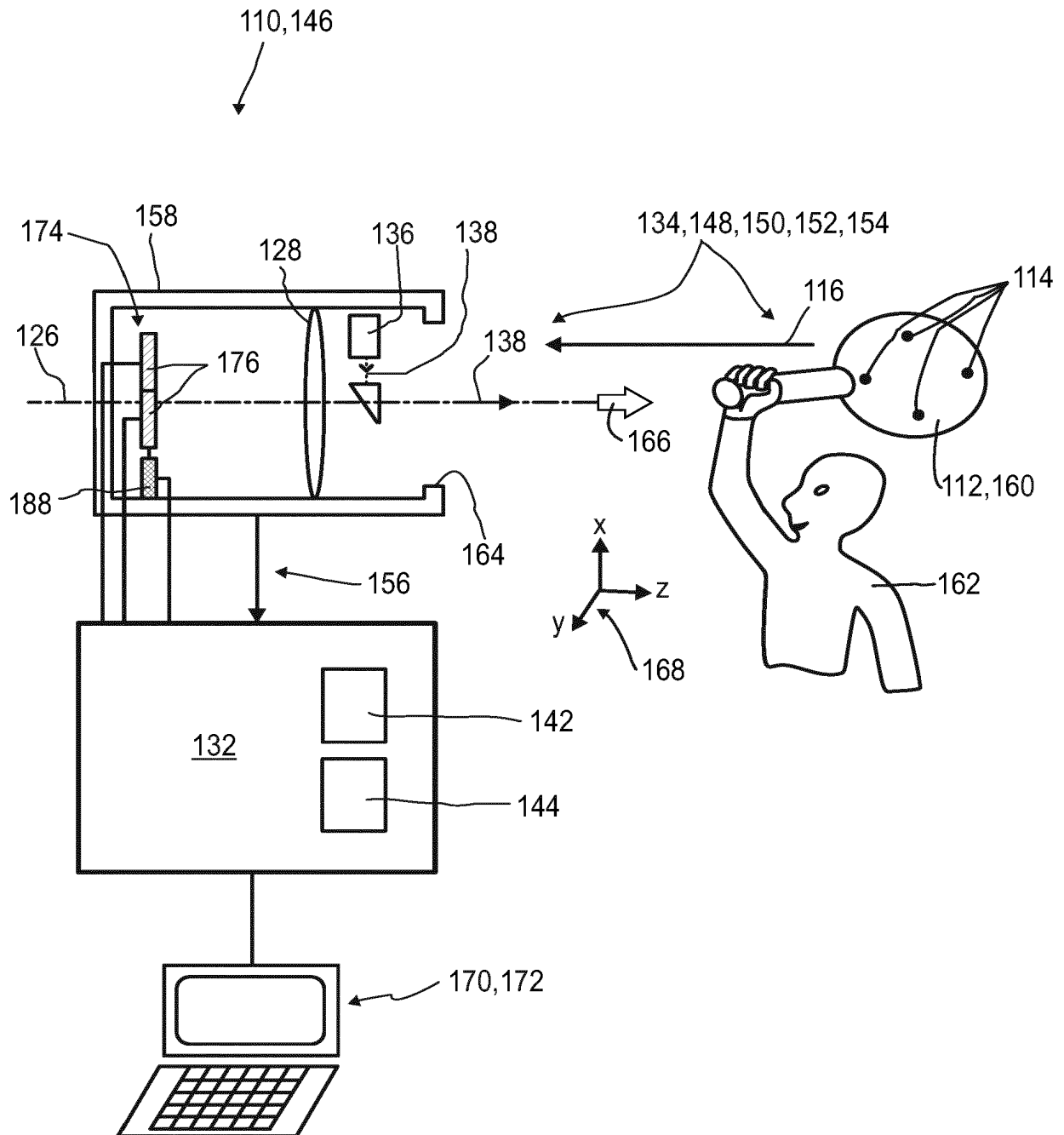


FIG.4

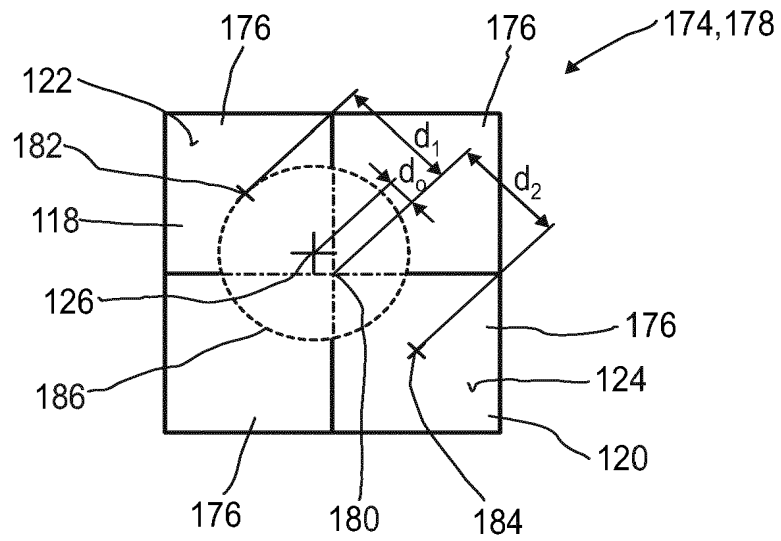


FIG.5

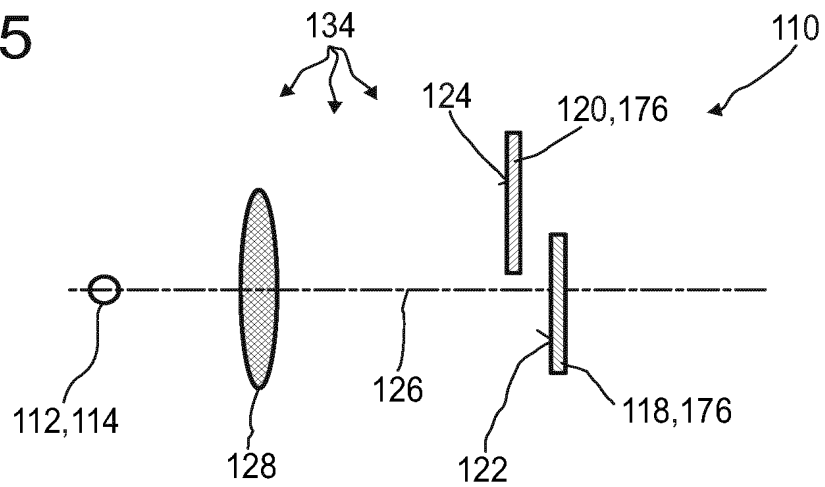


FIG.6

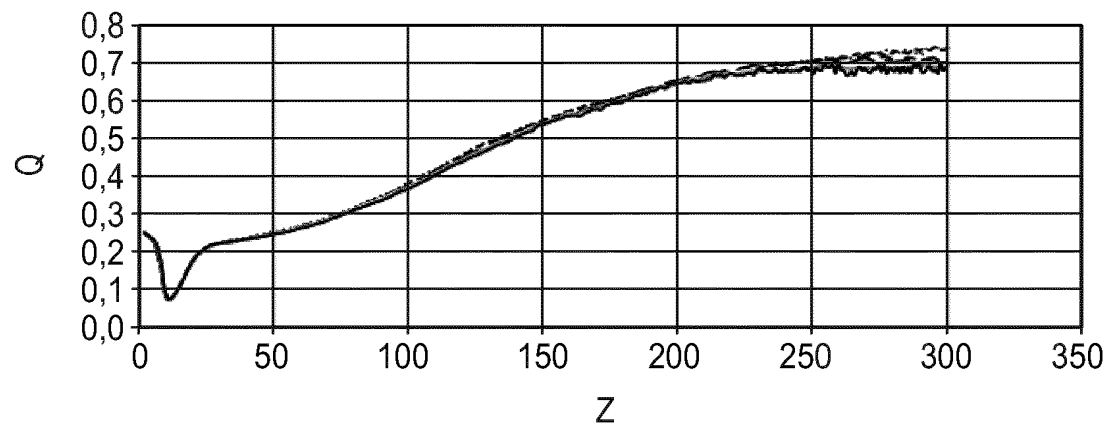


FIG.7

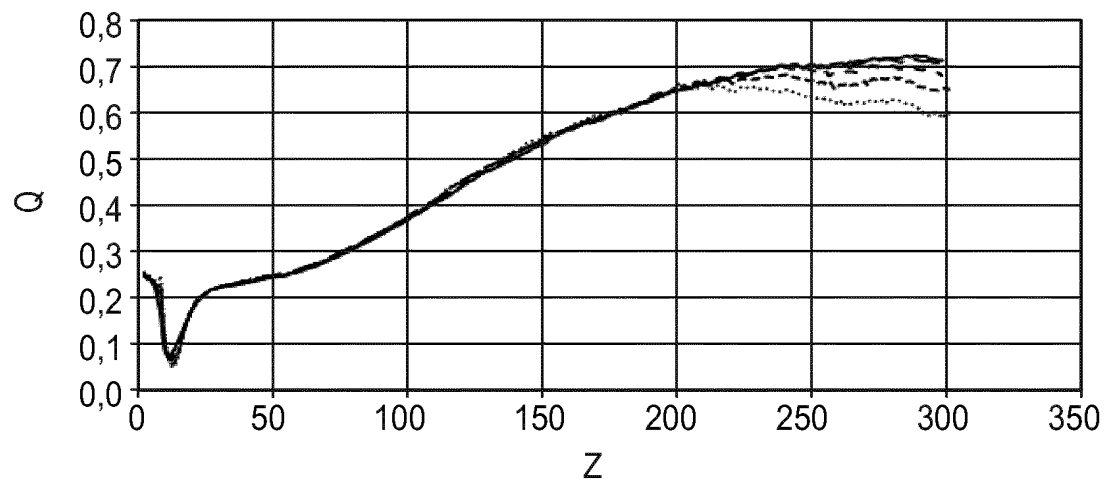


FIG.8

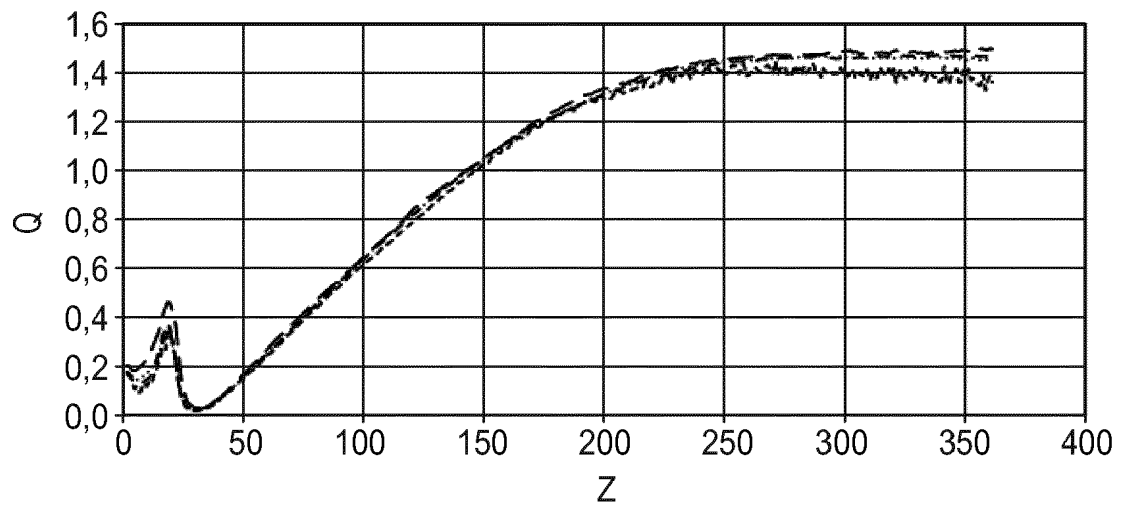


FIG. 9A

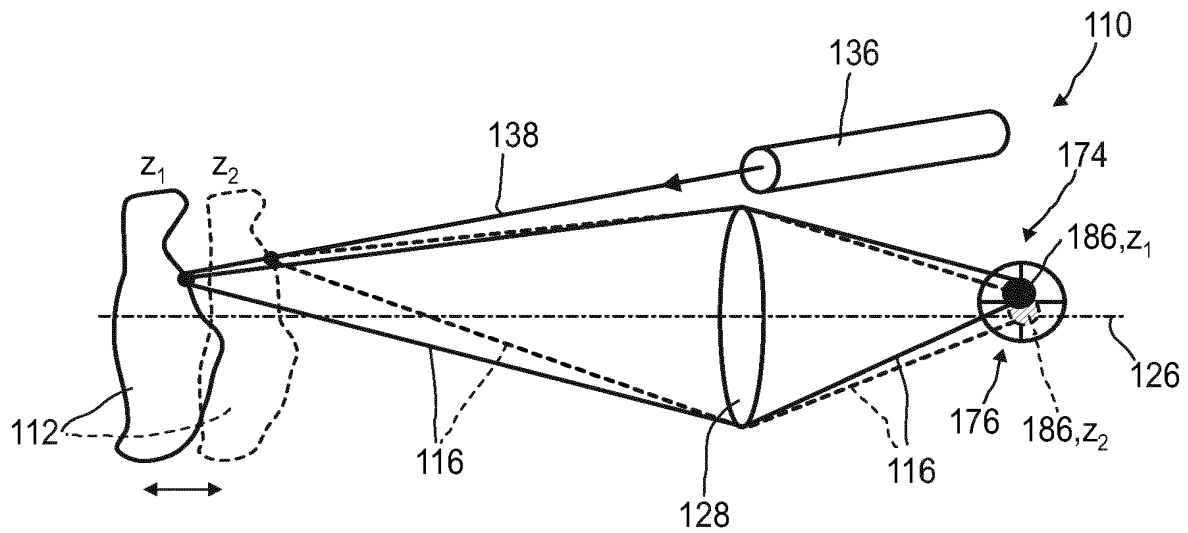


FIG.9B

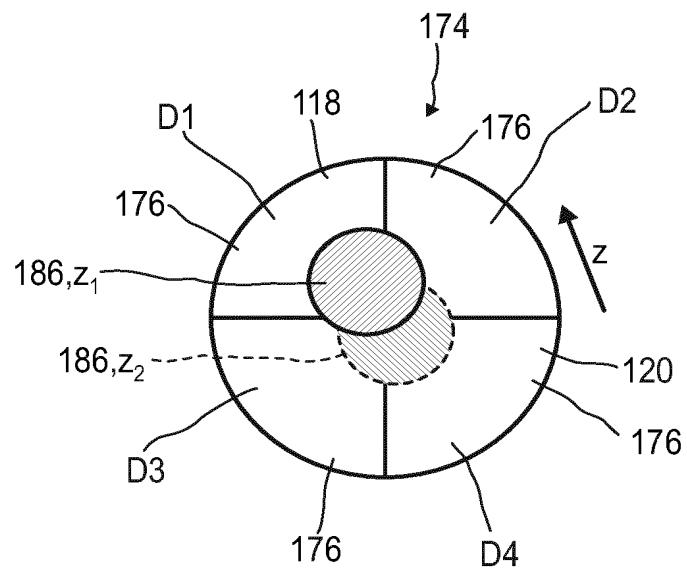


FIG.9C

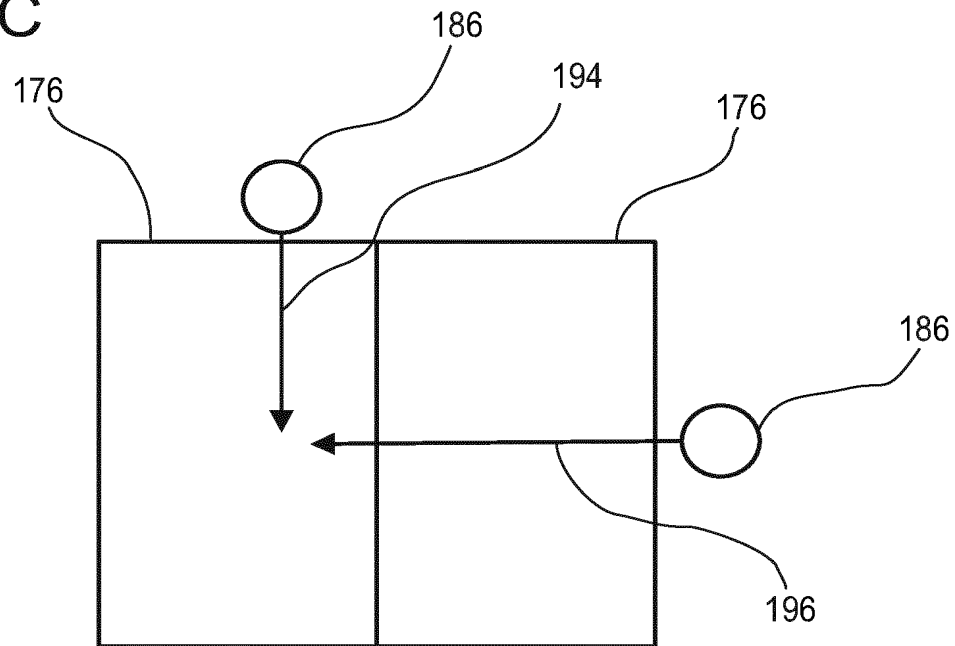


FIG.9D

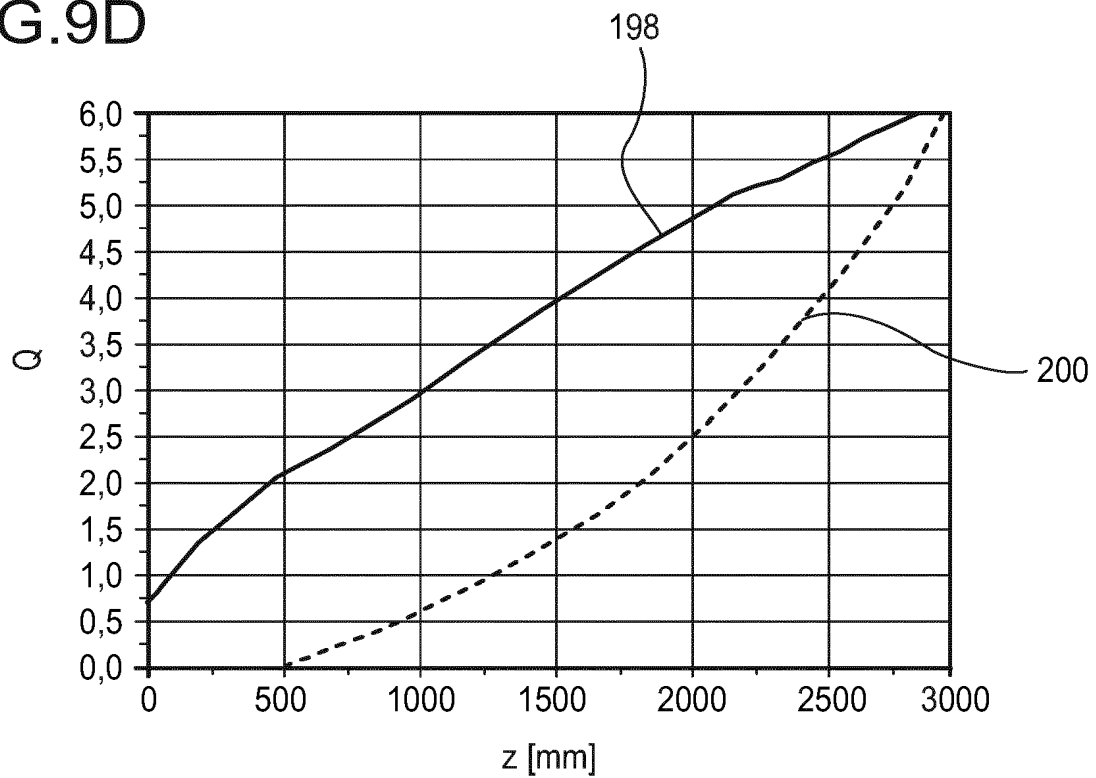


FIG.10A

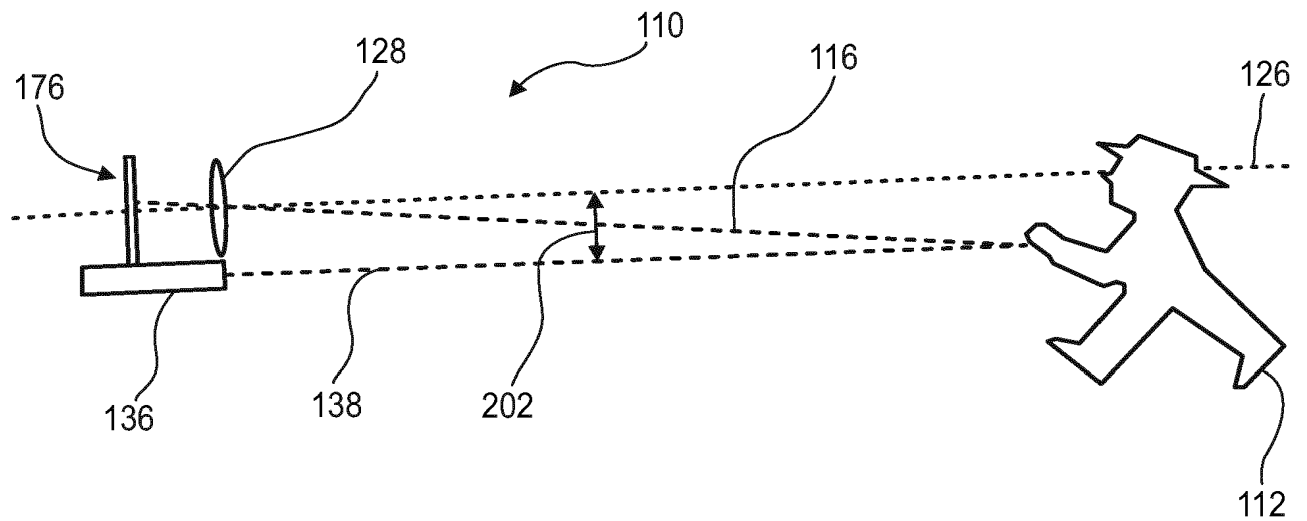


FIG.10B

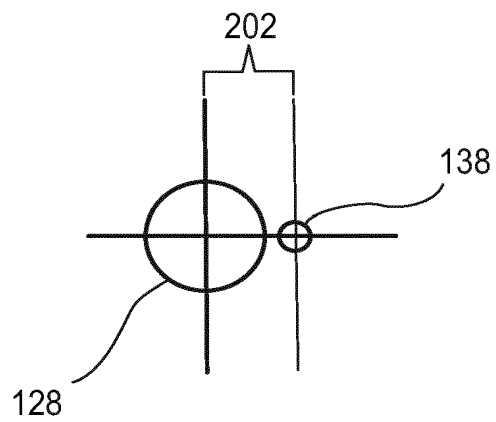


FIG.11

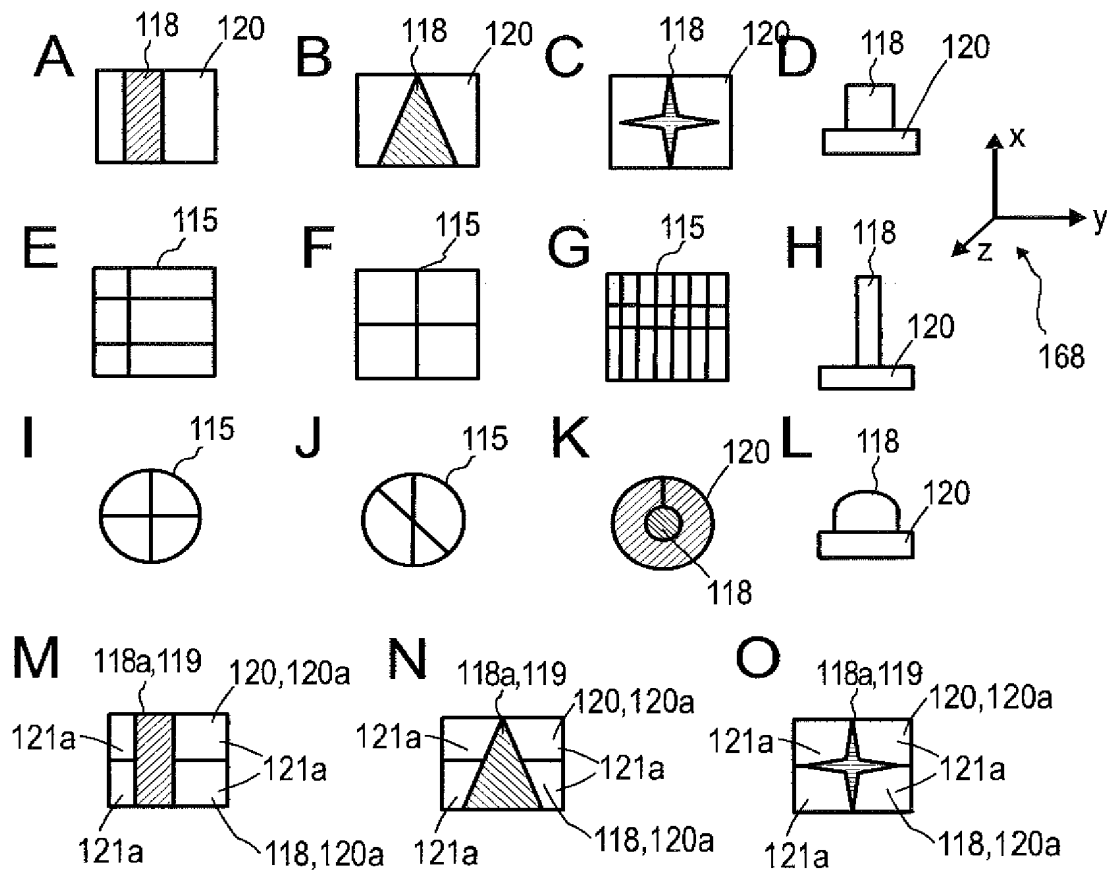


FIG.12A

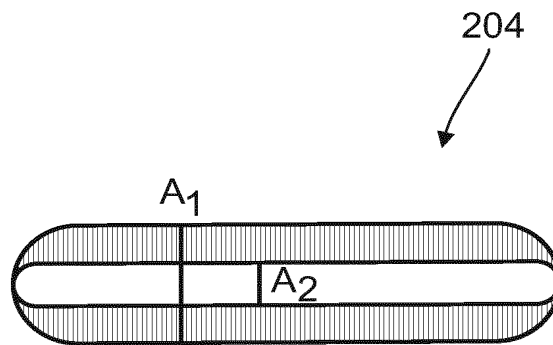


FIG.12B

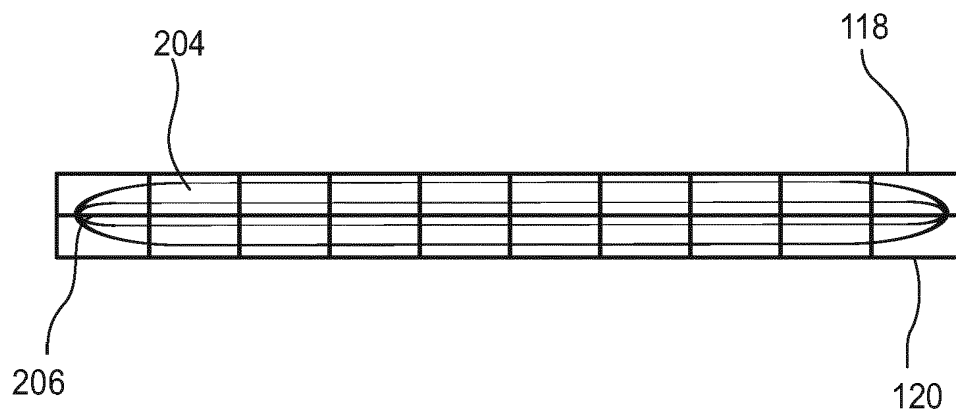


FIG.13A

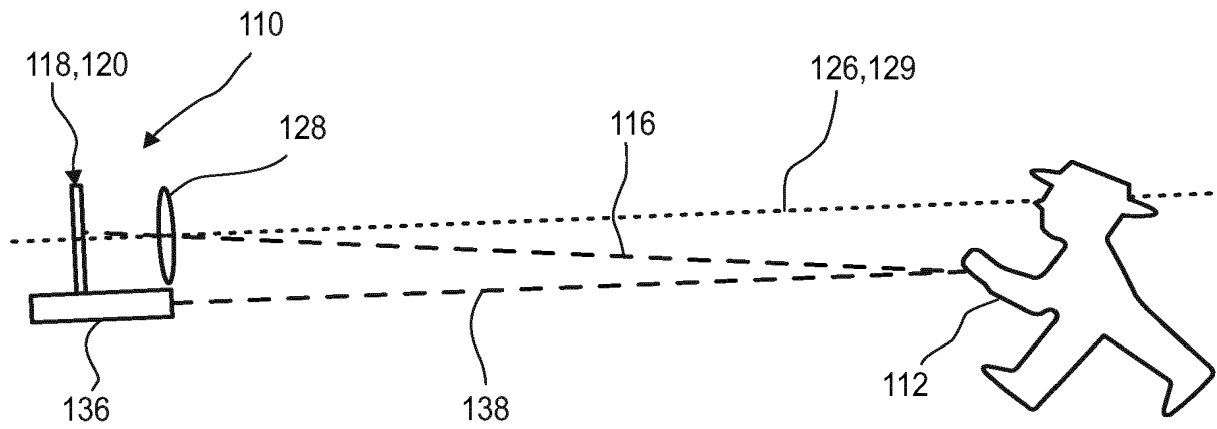
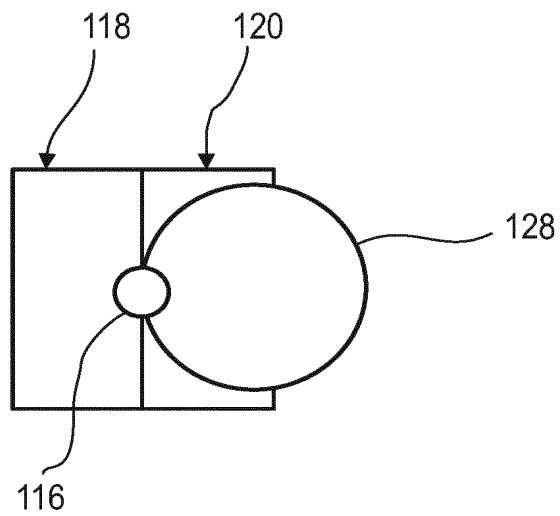


FIG.13B



INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2017/079558

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S7/481 G01S7/48 G01S5/16 G01S11/12
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data, INSPEC

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	EP 2 040 097 A1 (SICK AG [DE]) 25 March 2009 (2009-03-25) paragraphs [0001] - [0003] paragraph [0008] paragraphs [0019] - [0030] figures 1, 2	1-19
A	----- WO 2016/092454 A1 (BASF SE [DE]; BASF CHINA CO LTD [CN]) 16 June 2016 (2016-06-16) page 3, lines 1-13 page 54, line 31 - page 55, line 3 page 58, line 32 - page 61, line 25 page 66, line 41 - page 67, line 20 page 71, lines 24-32 -----	10-16,19



Further documents are listed in the continuation of Box C.



See patent family annex.

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"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

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"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

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Date of the actual completion of the international search

7 February 2018

Date of mailing of the international search report

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2017/079558

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			US 2017363465 A1	21-12-2017
			WO 2016092454 A1	16-06-2016



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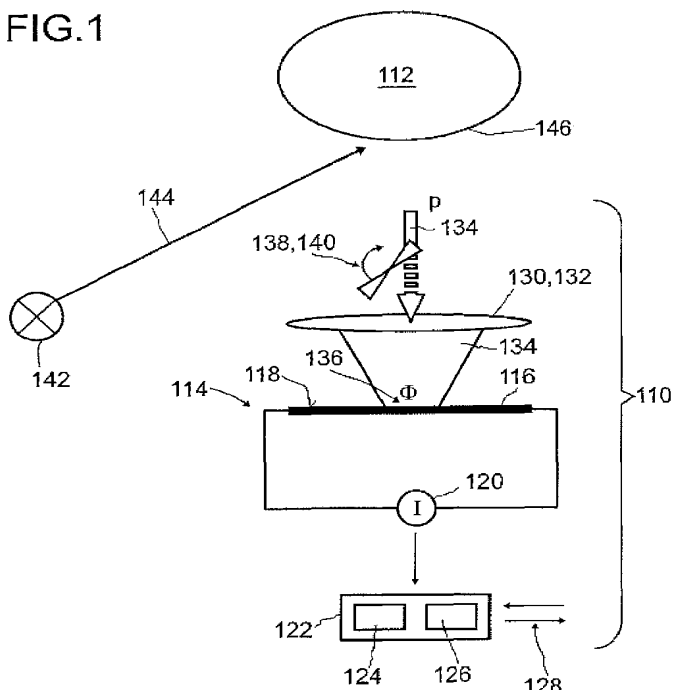
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(54) **Title:** DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.1



(57) **Abstract:** A detector (110) for optically detecting at least one object (112) is proposed. The detector (110) comprises at least one optical sensor (114). The optical sensor (114) has at least one sensor region (116). The optical sensor (114) is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region (116). The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area (118). The detector (110) furthermore has at least one evaluation device (122). The evaluation device (122) is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object (112).



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Detector for optically detecting at least one object

Description

5 Field of the invention

The invention relates to a detector for optically detecting at least one object. Furthermore, the invention relates to a distance measuring device, an imaging device, a human-machine interface, an entertainment device and a security device. Furthermore,
10 the invention relates to a method for optically detecting at least one object and to a use of an organic solar cell as optical sensor. Such devices, methods and uses can be employed for example in various areas of daily life, traffic technology, production technology, security technology, medical technology or in the sciences. However, other applications are also possible in principle.

15

Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for
20 example, ultraviolet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors
25 are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and for reasons of large-area processing, sensors comprising at least one organic sensor material are being used, as described for
30 example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO 2009/013282 A1.

A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on
35 the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring
40 devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for optically detecting

objects are triangulation systems, by means of which distance measurements can likewise be carried out.

Proceeding from such known detectors and methods for optically detecting objects, it can be ascertained that in many cases a considerable technical outlay has to be implemented in order to carry out this object detection with sufficient precision.

By way of example, in microscopy a considerable outlay in respect of apparatus is required in order to obtain correct focusing of a light beam and/or in order to obtain depth information about the sample to be imaged.

Distance measurements, by contrast, are based in many cases on technically inadequate assumptions such as, for example, the assumption of a specific size of an object in an image evaluation. Other methods are based in turn on complex pulse sequences, such as, for example, distance measurements by means of laser pulses. Yet other methods are based on the use of a plurality of detectors such as, for example, triangulation methods.

Problem addressed by the invention

Therefore, a problem addressed by the present invention is that of specifying devices and methods for optically detecting at least one object which at least substantially avoid the disadvantages of known devices and methods of this type. In particular, the proposed devices and methods are intended to make it possible to simplify an optical detection of at least one object in respect of apparatus.

Disclosure of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims.

A detector for optically detecting at least one object is proposed in a first aspect of the present invention.

In the context of the present invention, an optical detection should generally be understood to mean a process in which at least one item of information about the detected object is obtained. In this case, the term information should be interpreted broadly. The at least one item of information can comprise in particular one or more of the following items of information: an item of information about the fact that the object is present or not for example in a measurement range and/or visual range of the detector;

an item of information about at least one optical property of the detector, for example, at least one brightness and/or at least one radiation property, for example a luminescence property; an item of location information of the object, for example, an item of information about a distance between the object and the detector or a part of the detector and/or a relative orientation of the object with respect to the detector or a part of the detector and/or an item of position information of the object in at least one coordinate system which is determined by the detector and/or the object; an item of information about a movement state of the object, for example a one-dimensional, two-dimensional or three-dimensional velocity of the object and/or an acceleration of the object. Alternatively or additionally, other items of information can also be obtained during the detection of the object. The invention is described below substantially with reference to obtaining an item of location information about the object, without restriction of further embodiments of the at least one item of geometrical information which can be realized alternatively or additionally. Said geometrical information, in particular the location information, can relate to the entire object or else only a part of the object, for example a point, an area or a region of the object, which is detected by means of the detector. Said point, said area or said region can be arranged on a surface of the object or else at least partly within the object.

The object can generally be a living or else inanimate object. Examples of objects which can be detected completely or partly by means of the detector are described in even greater detail below.

The detector comprises at least one optical sensor. The optical sensor has at least one sensor region, in particular at least one sensor region comprising at least one sensor area. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination in the sensor region, in particular on the sensor area.

The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

The detector can furthermore comprise at least one transfer device. The transfer device can be designed to feed electromagnetic radiation emerging from the object to the optical sensor and to illuminate the sensor region in the process. In the context of the present invention, a transfer device should be understood to mean a device which is embodied in any desired manner, in principle, and which is designed to feed the electromagnetic

radiation emerging from the object to the optical sensor and there in particular to the sensor region or preferably the sensor area. This feeding can be embodied in imaging fashion or else in non-imaging fashion. Thus, this optional transfer device can comprise for example at least one beam path. The transfer device can for example comprise one or a plurality of mirrors and/or beam splitters and/or beam deflecting elements in order to influence a direction of the electromagnetic radiation. Alternatively or additionally, the transfer device can comprise one or a plurality of imaging elements which can have the effect of a converging lens and/or a diverging lens. By way of example, the optional transfer device can have one or a plurality of lenses and/or one or a plurality of convex and/or concave mirrors. Once again alternatively or additionally, the transfer device can have at least one wavelength-selective element, for example at least one optical filter. Once again alternatively or additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the optional transfer device can, in principle, be realized individually or in any desired combination.

In the context of the present invention, an optical sensor should be understood generally to mean an element which is designed to convert at least one optical signal into a different signal form, preferably into at least one electrical signal, for example a voltage signal and/or a current signal. In particular the optical sensor can comprise at least one optical-electrical converter element, preferably at least one photodiode and/or at least one solar cell. As is explained in even greater detail below, in the context of the present invention, preference is attached particularly to a use of at least one organic optical sensor, that is to say an optical sensor which comprises at least one organic material, for example at least one organic semiconductor material.

In the context of the present invention, a sensor region should be understood to mean a two-dimensional or three-dimensional region which preferably, but not necessarily, is continuous and can form a continuous region, wherein the sensor region is designed to vary at least one measurable property, in a manner dependent on the illumination. By way of example, said at least one property can comprise an electrical property, for example, by the sensor region being designed to generate, solely or in interaction with other elements of the optical sensor, a photovoltage and/or a photocurrent and/or some other type of signal. In particular the sensor region can be embodied in such a way that it generates a uniform, preferably a single, signal in a manner dependent on the illumination of the sensor region. The sensor region can thus be the smallest unit of the optical sensor for which a uniform signal, for example, an electrical signal, is generated, which preferably can no longer be subdivided to partial signals, for example for partial regions of the sensor region. The optical sensor can have one or else a plurality of such

sensor regions, the latter case for example by a plurality of such sensor regions being arranged in a two-dimensional and/or three-dimensional matrix arrangement.

5 The at least one sensor region can comprise for example at least one sensor area, that is to say a sensor region whose lateral extent considerably exceeds the thickness of the sensor region, for example by at least a factor of 10, preferably by at least a factor of 100 and particularly preferably by at least a factor of 1000. Examples of such sensor areas can be found in organic or inorganic photovoltaic elements, for example, in accordance with the prior art described above, or else in accordance with the exemplary
10 embodiments described in even greater detail below. The detector can have one or a plurality of such optical sensors and/or sensor regions. By way of example, a plurality of optical sensors can be arranged linearly in a spaced-apart manner or in a two-dimensional arrangement or else in a three-dimensional arrangement, for example by a stack of photovoltaic elements being used, preferably organic photovoltaic elements,
15 preferably a stack in which the sensor areas of the photovoltaic elements are arranged parallel to one another. Other embodiments are also possible.

The optional transfer device can, as explained above, be designed to feed
20 electromagnetic radiation emerging from the object to the optical sensor. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The optional transfer device can also, as explained in even greater detail below, be wholly or partly a constituent part of at least one optional illumination source, for
25 example by the illumination source being designed to provide electromagnetic radiation having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

30 The electromagnetic radiation can be in particular light in one or more of the following spectral ranges: in the ultraviolet spectral range, in the visible spectral range, in the infrared spectral range. The ultraviolet spectral range can be considered to be for example a range with a wavelength of 50 nm to 400 nm, the visible spectral range a range of 400 nm to 800 nm, and the infrared spectral range a range of 800 nm to
35 100 000 nm.

The electromagnetic radiation emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the optical sensor and the sensor region. The latter case can
40 be effected for example by at least one illumination source being used. This illumination source can be for example ambient light or else an artificial illumination source. By way

of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the object or connected or spatially coupled to the object.

The electromagnetic radiation emerging from the object can accordingly, alternatively or additionally from the option that said radiation originates in the object itself emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic radiation emerging from the object can be emitted by the object itself and/or be reflected by the object and/or be scattered by the object before it is fed to the optical sensor. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of radiation can be excited, for example, by a primary light source, for example by the object or a partial region of the object being excited to effect luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector. The at least one illumination source of the detector can generally be adapted to the emission and/or reflective properties of the object, for example in terms of its wavelength. Various embodiments are possible.

The feeding of the electromagnetic radiation to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on the sensor region, in

particular the sensor area. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

As described above, the optical sensor is designed to generate at least one sensor
5 signal in a manner dependent on the illumination of the sensor region. By way of
example, this can be an electrical signal, in particular a voltage signal and/or current
signal, for example a photovoltage and/or a photocurrent. Hereinafter, without restricting
possible further embodiments, reference is made in particular to optical sensors which, in
10 a manner dependent on the illumination generate at least one photocurrent. The optical
signal can be temporally constant or vary temporally and can be embodied, in principle,
in analog fashion or else in digital fashion, wherein analog signals are preferred
hereinafter. The sensor signal can be used as a raw signal, but can also be subjected to
one or a plurality of processing operations, for example one or a plurality of filtering
15 operations or similar processing operations. No distinction is drawn hereinafter between
these options and the use of a raw signal, and the term sensor signal is used uniformly.
Furthermore, without restricting further possible embodiments, it is assumed that the
sensor signal has positive values, and the term of a maximum should for example, also
be understood in this respect. If one or a plurality of sensor signals having a negative
20 sign are detected, which is likewise possible, then hereinafter either the expression of the
maximum should be replaced by the term of a minimum or generally of an extremum, or
the actual sensor signal should be replaced, for example by its absolute value, as will
readily be evident to the person skilled in the art.

As explained above, the optical sensor is designed to generate at least one sensor signal
25 in a manner dependent on the illumination of the sensor region, wherein the sensor
signal, given the same total power of the illumination, is dependent on a geometry of the
illumination, in particular on a beam cross section of the illumination on the sensor area.

A geometry of the illumination can generally summarize at least one property of the
30 illumination which characterizes a two-dimensional and/or three-dimensional
embodiment of a region of the sensor region which is exposed to the electromagnetic
radiation emerging from the object. By way of example, the geometry of the illumination
can be characterized by a beam cross section of the illumination on or in the sensor
35 region, for example on a sensor surface, for example by a diameter or equivalent
diameter of the illumination, for example of a light spot. As described above, a light spot
which can have for example a diameter or an equivalent diameter can be produced for
example on the sensor area. Said diameter or equivalent diameter can completely or
partly characterize for example the geometry of the illumination. A light spot can be
40 understood to mean for example an illuminated area, in particular in demarcation relative
to an unilluminated area. By way of example, light spot can be understood to mean an

area of the sensor region within which an intensity of the illumination is at least 10% of a maximum intensity. However, other definitions of a light spot are also possible, in principle, for example by other limit values being set instead of 10%, since an edge of the illumination in practice, will not be sharply defined by virtue of the intensity falling abruptly
5 to zero.

The optical sensor can be designed for example in such a way that the sensor signal, given the same power of the illumination, that is to say for example given the same integral over the intensity of the illumination on the sensor area, is dependent on the
10 geometry of the illumination, that is to say for example on the diameter and/or the equivalent diameter for the sensor spot. By way of example, the sensor can be designed in such a way that upon a doubling of the beam cross section given the same total power, a signal variation occurs by at least a factor of 3, preferably by at least a factor of 4, in particular a factor of 5 or even a factor of 10. This condition can hold true for
15 example for a specific focusing range, for example for at least one specific beam cross section. Thus, by way of example, the signal can have, between at least one optimum focusing at which the signal can have for example at least one global or local maximum and a focusing outside said at least one optimum focusing, a signal difference by at least a factor of 3, preferably by at least a factor of 4, in particular a factor of 5 or even a factor
20 of 10. In particular, the sensor signal can have as a function of the geometry of the illumination, for example of the diameter or equivalent diameter of a light spot, at least one pronounced maximum, for example with a boost by at least a factor of 3, particularly preferably by at least a factor of 4 and particularly preferably by at least a factor of 10.

Consequently, the invention is based generally on the hitherto unreported and surprising insight that specific optical sensors exist whose sensor signal is not only dependent on a total light power of the illumination of the sensor region, for example of the sensor area, of these sensors but in which a pronounced signal dependence on a geometry of the illumination, for example a size of a light spot of the illumination on the sensor region, for
30 example the sensor area, also exists. This is generally not the case for most conventional optical sensors, in particular for most inorganic semiconductor sensors, since here the sensor signal is generally dependent only on a total power of the illumination, that is to say an integral over the intensity over the entire light spot which is generally independent of the size of the light spot, that is to say the geometry of the
35 illumination, as long as the light spot lies within the limits of the sensor region. It has surprisingly been discovered, however, that in specific optical sensors, for example organic optical sensors, such a dependence of the sensor signal occurs in which the sensor signal on the one hand rises with the total power of the illumination, but on the other hand, even given a constant total power, is dependent on a geometry of the
40 illumination. Examples of such optical sensors are explained in even greater detail below. By way of example, the sensor signal, given the same total power, can have at

least one pronounced maximum for one or a plurality of focusings and/or for one or a plurality of specific sizes of the light spot on the sensor area or within the sensor region. This effect can additionally be dependent on, or intensified by, a frequency of the illumination by virtue of the fact that the electromagnetic radiation with which the sensor region is illuminated is not incident continuously on the sensor region, but rather is interrupted, for example interrupted periodically with a frequency f . The described optical sensors which have the stated effect of the dependence of the sensor signal, given the same total power of the illumination on a geometry of the illumination and optionally on a frequency of the illumination, are also designated hereinafter as fip sensors since, given the same total power p , the sensor signal can be dependent on the intensity i and optionally the frequency f or since the sensor signal, given the same total power p , can be dependent on an optical flux density Φ . By way of example, the sensor signal can comprise a photocurrent and/or a photovoltage. By way of example, the photocurrent can thus be a function of the total power p , of the flux Φ and/or of the geometry of the illumination (e.g. of a diameter or an equivalent diameter of a light spot) and optionally of the frequency, or for example a function of the total power p , of the intensity (for example of a maximum intensity) and of the frequency.

Such effects of the dependence of the sensor signal on a beam geometry were observed in the context of the investigations leading to the present invention in particular in the case of organic photovoltaic components, that is to say photovoltaic components, for example, solar cells, which comprise at least one organic material, for example at least one organic p-semiconducting material and/or at least one organic dye. By way of example, such effects, as is explained in even greater detail below by way of example, were observed in the case of dye solar cells, that is to say components which have at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid organic p-type semiconductor, and at least one second electrode. Such dye solar cells, preferably solid dye solar cells (solid dye sensitized solar cells, sDSC), are known in principle in numerous variations from the literature. The described effect of the dependence of the sensor signal on a geometry of the illumination on the sensor area and a use of this effect have not, however, been described heretofore.

In particular, the optical sensor can be designed in such a way that the sensor signal, given the same total power of the illumination, is substantially independent of a size of the sensor region, in particular of a size of the sensor area, in particular as long as the light spot of the illumination lies completely within the sensor region, in particular the sensor area. Consequently, the sensor signal can be dependent exclusively on a focusing of the electromagnetic rays on the sensor area. In particular the sensor signal can be embodied in such a way that a photocurrent and/or a photovoltage per sensor

area have/has the same values given the same illumination, for example the same values given the same size of the light spot.

Consequently, by means of the optical detector, for example by a suitable calibration and/or by a suitable analysis of the sensor signal of the optical sensor, at least one additional item of information can be obtained, which is designated hereinafter as geometrical information. The detector comprises at least one evaluation device which is designed to generate the at least one item of geometrical information from the sensor signal.

An item of geometrical information should be understood to mean, in principle, any desired item of information which can be derived directly or indirectly from the abovementioned effect that the sensor signal, given the same total power of the illumination, is dependent on the geometry of the illumination. The geometrical information can comprise, in particular, at least one item of information about the illumination, in particular a geometry of the illumination, and/or at least one item of geometrical information about the object. The geometrical information can, in particular, go beyond an item of information about the light power alone. The geometrical information can preferably comprise at least one item of information about the geometry of the illumination and/or at least one item of information about at least one influencing variable which influences the geometry of the illumination for example an item of information about a distance of the object.

Particularly preferably the geometrical information comprises at least one item of information, selected from the group consisting of: an item of information about the fact that the object is present or not for example in a measurement range and/or visual range of the detector; an item of information about at least one optical property of the detector, for example, at least one brightness and/or at least one radiation property, for example a luminescence property; an item of location information of the object, for example, an item of information about a distance between the object and the detector or a part of the detector and/or a relative orientation of the object with respect to the detector or a part of the detector and/or an item of position information of the object in at least one coordinate system which is determined by the detector and/or the object; an item of information about a movement state of the object, for example a one-dimensional, two-dimensional or three-dimensional velocity of the object and/or an acceleration of the object; an item of information about a geometry of the illumination of the sensor region; an item of information about the fact that an illumination has taken place or is taking place with a specific geometry, in particular an inhomogeneous illumination of the sensor region, in particular a focused illumination, for example an illumination in which at least one light spot is produced on a sensor area. Alternatively or additionally, the geometrical information can also comprise one or a plurality of other items of information. The

invention is described below substantially with reference to obtaining an item of geometrical information in the form of at least one item of location information about the object. Said location information can relate to the entire object or else only a part of the object, for example a point, an area or a region of the object which is detected by means
5 of the detector. This point, this area or this region can be arranged on a surface of the object or else at least partly within the object.

The geometrical information can be generated in any desired form, in principle. Preferably, the geometrical information is generated in a machine-readable form and/or a
10 form that can be used by a machine. By way of example, the geometrical information can be generated in the form of at least one electrical and/or optical signal. Alternatively or additionally, the geometrical information can also be generated in a form that can be read and/or detected by a human, for example by printout on paper, by display on a screen, by an output in visual form, by an output in acoustic form, by an output in haptic
15 form or by a combination of two or more of the stated and/or other output forms that can be detected by a human. The geometrical information may have been stored or can be stored in particular on at least one volatile or nonvolatile data memory which can be for example wholly or partly a constituent part of the evaluation device and/or of some other device. Alternatively or additionally, the at least one item of geometrical information can
20 also be provided and/or transferred by means of at least one interface, for example by means of at least one output device.

An evaluation device should generally be understood to mean a device which is designed to generate the at least one item of geometrical information from the at least
25 one sensor signal, using the above-described effect that the sensor signal, given the same total power of the illumination, is dependent on the geometry of the illumination. In particular, in this case it is possible to use a known relationship between the geometry of the illumination and the geometrical function, as is explained in even greater detail below by way of example.

The evaluation device can comprise in particular at least one data processing device, in particular an electronic data processing device, which can be designed to use the at least one sensor signal as at least one input variable and to generate the at least
35 geometrical information using said input variable, for example by calculation and/or using at least one stored and/or known relationship. Besides the at least one sensor signal, one or a plurality of further parameters and/or items of information can influence said relationship, for example at least one item of information about a modulation frequency. In this case, the relationship can be determined or determinable empirically, analytically or else semi-empirically. Particularly preferably, the relationship comprises at least one
40 calibration curve, at least one set of calibration curves, at least one function or a combination of the possibilities mentioned. One or a plurality of calibration curves can be

stored for example in the form of a set of values and the associated function values thereof, for example in a data storage device and/or a table. Alternatively or additionally, however, the at least one calibration curve can also be stored for example in parameterized form and/or as a functional equation. Various possibilities are conceivable and can also be combined.

By way of example, the evaluation device can be designed in terms of programming for the purpose of determining the at least one item of geometrical information. The evaluation device can comprise in particular at least one computer, for example at least one microcomputer. Furthermore, the evaluation device can comprise one or a plurality of volatile or nonvolatile data memories. As an alternative or in addition to a data processing device, in particular at least one computer, the evaluation device can comprise one or a plurality of further electronic components which are designed for determining the at least one item of geometrical information using the at least one sensor signal, for example an electronic table and in particular at least one look-up table and/or at least one application-specific integrated circuit (ASIC).

The geometrical information allows a multiplicity of possible uses of such detectors, which will be described by way of example hereinafter. By way of example, as is explained in even greater detail below, at least one item of location information of the object can be generated from said geometrical information, or the geometrical information can comprise the at least one item of location information, since for example a geometry of the illumination, for example a diameter or equivalent diameter of the light spot on the sensor area, can be dependent on a distance between the object and the detector and/or the optional transfer device of the detector, for example at least one detector lens. By way of example, a variation of the distance between the object and a lens of the optional transfer device can lead to a defocusing of the illumination on the sensor region, accompanied by a change in the geometry of the illumination, for example a widening of a light spot, which can result in a correspondingly altered sensor signal. Even without a transfer device, by way of example, from a known beam profile from the sensor signal and/or a variation thereof, for example, by means of a known beam profile and/or a known propagation of the electromagnetic rays, it is possible to deduce a defocusing and/or the geometrical information. By way of example, given a known total power of the illumination, it is thus possible to deduce from the sensor signal of the optical sensor a geometry of the illumination and therefrom in turn the geometrical information, in particular at least one item of location information of the object.

Preferably, at least two sensor signals are detected. If the total power of the illumination is not known, for example, then for example at least two sensor signals can be generated, for example at least two sensor signals at different frequencies of a modulation of the illumination, wherein, from the at least two sensor signals, for example

by comparison with corresponding calibration curves, it is possible to deduce the total power and/or the geometry of the illumination, and/or therefrom or directly the at least one item of geometrical information, in particular the at least one item of location information, of the object. In any case, the detector can thus be designed to generate, in particular on the basis of the effect described, the at least one item of geometrical information, which preferably goes beyond pure information about the total power of the illumination.

The detector described can advantageously be developed in various ways. Thus, the detector can furthermore have at least one modulation device for modulating the illumination, in particular for periodic modulation, in particular a periodic beam interrupting device. A modulation of the illumination should be understood to mean a process in which a total power of the illumination is varied, preferably periodically, in particular with one or a plurality of modulation frequencies. In particular, a periodic modulation can be effected between a maximum value and a minimum value of the total power of the illumination. The minimum value can be 0, but can also be > 0 , such that, by way of example, complete modulation does not have to be effected. The modulation can be effected for example in a beam path between the object and the optical sensor, for example by the at least one modulation device being arranged in said beam path. Alternatively or additionally, however, the modulation can also be effected in a beam path between an optional illumination source - described in even greater detail below - for illuminating the object and the object, for example by the at least one modulation device being arranged in said beam path. A combination of these possibilities is also conceivable. The at least one modulation device can comprise for example a beam chopper or some other type of periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates at constant speed and which can thus periodically interrupt the illumination. Alternatively or additionally, however, it is also possible to use one or a plurality of different types of modulation devices, for example modulation devices based on an electro-optical effect and/or an acousto-optical effect. Once again alternatively or additionally, the at least one optional illumination source itself can also be designed to generate a modulated illumination, for example by said illumination source itself having a modulated intensity and/or total power, for example a periodically modulated total power, and/or by said illumination source being embodied as a pulsed illumination source, for example as a pulsed laser. Thus, by way of example, the at least one modulation device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

The detector can be designed in particular to detect at least two sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies. The evaluation device can be designed to generate the

geometrical information from the at least two sensor signals. As described above, in this way, by way of example, it is possible to resolve ambiguities and/or it is possible to take account of the fact that, for example, a total power of the illumination is generally unknown.

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Further possible embodiments of the detector relate to the embodiment of the at least one optional transfer device. As explained above, said at least one transfer device can have imaging properties or else can be embodied as a pure non-imaging transfer device, which has no influence on a focusing of the illumination. It is particularly preferred, however, if the transfer device has at least one imaging element, for example at least one lens and/or at least one curved mirror, since, in the case of such imaging elements, for example, a geometry of the illumination on the sensor region can be dependent on a relative positioning, for example a distance, between the transfer device and the object. Generally, it is particularly preferred if the transfer device is designed in such a way that the electromagnetic radiation which emerges from the object is transferred completely to the sensor region, for example is focused completely onto the sensor region, in particular the sensor area, in particular if the object is arranged in a visual range of the detector.

As explained above, the optical sensor can furthermore be designed in such a way that the sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination. The detector can be embodied, in particular, as explained above, in such a way that sensor signals at different modulation frequencies are picked up, for example in order to generate one or a plurality of further items of information about the object. As described above, by way of example, a sensor signal at at least two different modulation frequencies can in each case be picked up, wherein, by way of example, in this way, a lack of information about a total power of the illumination can be supplemented. By way of example, by comparing the at least two sensor signals picked up at different modulation frequencies with one or a plurality of calibration curves, which can be stored for example in a data storage device of the detector, even in the case of an unknown total power of the illumination, it is possible to deduce a geometry of the illumination, for example a diameter or an equivalent diameter of a light spot on the sensor area. For this purpose, by way of example, it is possible to use the at least one evaluation device described above, for example at least one data processing data, which can be designed to control such picking-up of sensor signals at different frequencies and which can be designed to compare said sensor signals with the at least one calibration curve in order to generate therefrom the geometrical information, for example information about a geometry of the illumination, for example information about a diameter or equivalent diameter of a light spot of the illumination on a sensor area of the optical sensor. Furthermore, as is explained in even greater detail below, the evaluation device can alternatively or additionally be designed to generate at least one item of geometrical information about

the object, for example at least one item of location information. This generation of the at least one item of geometrical information, as explained above, can be effected for example taking account of at least one known relationship between a positioning of the object relative to the detector and/or the transfer device or a part thereof and a size of a light spot, for example empirically, semi-empirically or analytically using corresponding imaging equations.

In contrast to known detectors, in which a spatial resolution and/or imaging of objects is also generally tied to the fact that the smallest possible sensor areas are used, for example the smallest possible pixels in the case of CCD chips, the sensor region of the proposed detector can be embodied in a very large fashion, in principle, since for example the geometrical information, in particular the at least one item of location information, about the object can be generated from a known relationship for example between the geometry of the illumination and the sensor signal. Accordingly, the sensor region can have for example a sensor area, for example an optical sensor area, which is at least 0.001 mm^2 , in particular at least 0.01 mm^2 , preferably at least 0.1 mm^2 , more preferably at least 1 mm^2 , more preferably at least 5 mm^2 , more preferably at least 10 mm^2 , in particular at least 100 mm^2 or at least 1000 mm^2 or even at least $10\,000 \text{ mm}^2$. In particular, sensor areas of 100 cm^2 or more can be used. The sensor area can generally be adapted to the application. In particular, the sensor area should be chosen in such a way that, at least if the object is situated within a visual range of the detector, preferably within a predefined viewing angle and/or a predefined distance from the detector, the light spot is always arranged within the sensor area. In this way, it can be ensured that the light spot is not trimmed by the limits of the sensor region, as a result of which signal corruption could occur.

As described above, the sensor region can be in particular a continuous sensor region, in particular a continuous sensor area, which can preferably generate a uniform, in particular a single, sensor signal. Consequently, the sensor signal can be in particular a uniform sensor signal for the entire sensor region, that is to say a sensor signal to which each partial region of the sensor region contributes, for example additively. The sensor signal can generally, as explained above, in particular be selected from the group consisting of a photocurrent and a photovoltage.

The optical sensor can comprise in particular at least one semiconductor detector and/or be at least one semiconductor detector. In particular, the optical sensor can comprise at least one organic semiconductor detector or be at least one organic semiconductor detector, that is to say a semiconductor detector comprising at least one organic semiconducting material and/or at least one organic sensor material, for example at least one organic dye. Preferably, the organic semiconductor detector can comprise at least one organic solar cell and particularly preferably a dye solar cell, in particular a solid dye

solar cell. Exemplary embodiments of such preferred solid dye solar cells are explained in even greater detail below.

In particular, the optical sensor can comprise at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably at least one solid p-semiconducting organic material, and at least one second electrode. Generally, however, it is pointed out that the described effect in which the sensor signal, given a constant total power, is dependent on a geometry of the illumination of the sensor region is with high probability not restricted to organic solar cells and in particular not to dye solar cells. Without intending to restrict the scope of protection of the invention by this theory, and without the invention being bound to the correctness of this theory, it is supposed that generally photovoltaic elements are suitable as optical sensors in which at least one semiconducting material having trap states is used. Consequently, the optical sensor can comprise at least one n-semiconducting material and/or at least one p-semiconducting material which can have for example a conduction band and a valence band, wherein, in the case of organic materials, conduction band and valence band should correspondingly be replaced by LUMO (lowest unoccupied molecular orbital) and HOMO (highest occupied molecular orbital). Trap states should be understood to mean energetically possible states which are disposed between the conduction band (or LUMO) and the valence band (or HOMO) and which can be occupied by charge carriers. By way of example, it is possible to provide trap states for hole conduction which are disposed at at least one distance ΔE_h above the valence band (or HOMO) and/or trap states for electron conduction which are disposed at at least one distance ΔE_e below the conduction band (or LUMO). Such traps can be achieved for example by impurities and/or defects, which can optionally also be introduced in a targeted manner, or can be present intrinsically. By way of example, in the case of a low intensity, that is to say for example in the case of a light spot having a large diameter, only a low current can flow, since firstly the trap states are occupied before holes in the conduction band or electrons in the valence band contribute to a photocurrent. It is only starting from a higher intensity, that is to say for example starting from a more intense focusing of the light spot in the sensor region, that a considerable photocurrent can then flow. The described frequency dependence can be explained for example by the fact that charge carriers leave the traps again after a residence duration τ , such that the described effect occurs only in the case of modulated illumination with a high modulation frequency. By way of example, the detector can be designed to bring about a modulation of the illumination of the sensor region with a frequency of at least 1 Hz, preferably at least 10 Hz, in particular at least 100 Hz and particularly preferably at least 1 kHz. The trap states can be present for example with a density of 10^{-5} to 10^{-1} , relative to the n-semiconducting material and/or the p-semiconducting material and/or the dye. The energy differences ΔE with respect to the conduction band and with respect to the valence band can be in particular 0.05-0.3 eV.

The detector has, as described above, at least one evaluation device. In particular, the at least one evaluation device can also be designed to completely or partly control or drive the detector, for example by the evaluation device being designed to control one or a plurality of modulation devices of the detector and/or to control at least one illumination source of the detector. The evaluation device can be designed, in particular, to carry out at least one measurement cycle in which one or a plurality of sensor signals are picked up, for example a plurality of sensor signals successively at different modulation frequencies of the illumination.

The evaluation device is designed, as described above, to generate the at least one item of geometrical information from the at least one sensor signal. The at least one item of geometrical information can, in particular, as explained above, comprise at least one item of location information of the object and/or can be at least one item of location information of the object. In this case, an item of location information should generally be understood to mean an item of information which is adapted and/or suited to characterize at least one location and/or at least one orientation of the object or of a part of the object, for example of a region of the object, from which the electromagnetic radiation emerges. Said at least one item of location information can be static, that is to say can comprise for example a single item of location information or a plurality of items of location information which are picked up at the same time, but can, alternatively or additionally, also comprise a plurality of items of location information which are picked up at different times. By way of example, in this way, the at least one item of location information can also comprise at least one item of information about at least one movement, for example a relative movement between the detector or parts thereof and the object or parts thereof. In this case, a relative movement can generally comprise at least one linear movement and/or at least one rotational movement. Items of movement information can for example also be obtained by comparison of at least two items of location information picked up at different times, such that for example at least one item of location information can also comprise at least one item of velocity information and/or at least one item of acceleration information, for example at least one item of information about at least one relative velocity between the object or parts thereof and the detector or parts thereof. In particular, the at least one item of location information can generally be selected from: an item of information about a distance between the object or parts thereof and the detector or parts thereof, in particular an optical path length; an item of information about a distance or an optical distance between the object or parts thereof and the optional transfer device or parts thereof; an item of information about a positioning of the object or parts thereof relative to the detector or parts thereof; an item of information about an orientation of the object and/or parts thereof relative to the detector or parts thereof; an item of information about a relative movement between the object or parts thereof and the detector or parts thereof; an item of information about a two-dimensional or three-

dimensional spatial configuration of the object or of parts thereof, in particular a geometry or form of the object. Generally, the at least one item of location information can therefore be selected for example from the group consisting of: an item of information about at least one location of the object or at least one part thereof; information about at least one orientation of the object or a part thereof; an item of information about a geometry or form of the object or of a part thereof, an item of information about a velocity of the object or of a part thereof, an item of information about an acceleration of the object or of a part thereof, an item of information about a presence or absence of the object or of a part thereof in a visual range of the detector.

The at least one item of location information can be specified for example in at least one coordinate system, for example a coordinate system in which the detector or parts thereof rest. Alternatively or additionally, the location information can also simply comprise for example a distance between the detector or parts thereof and the object or parts thereof. Combinations of the possibilities mentioned are also conceivable.

The evaluation device can, as explained above, in particular be designed to determine the geometrical information from at least one predefined relationship between the geometry of the illumination, for example a diameter or equivalent diameter of a luminous spot on the sensor region and/or the sensor area, and a relative positioning of the object with respect to the detector, for example a distance and/or an optical path length between object and detector, preferably taking account of a known power of the illumination and optionally taking account of a modulation frequency with which the illumination is modulated. By way of example, the evaluation device can comprise at least one data storage device in which is stored a predefined relationship between a distance between detector and object and a geometry of the illumination, for example a diameter or equivalent diameter of a luminous spot on the sensor region. Said relationship can be stored for example in discrete fashion or continuously or else in the form of a function. By way of example, said relationship can comprise at least one calibration function. The relationship can for example also be stored in an electronic table, for example a so-called look-up table. If a power of the illumination is not known, then, as explained above, for example the evaluation device can be designed to pick up sensor signals at at least two different modulation frequencies of the illumination. Generally, the evaluation device can be designed for example to deduce in one or a plurality of steps, firstly, for example, from the at least one sensor signal, the geometry of the illumination, for example a diameter or equivalent diameter of a luminous spot on the sensor region or the sensor area. Furthermore, the evaluation device can be designed to deduce the at least one item of geometrical information from this determined geometry, preferably taking account of the modulation frequency. These two deductions can also be combined in one step, such that, by way of example, the evaluation device can be designed to directly deduce the geometrical information, in particular the location

information, from the at least one sensor signal. Examples of such relationships by means of which the evaluation device can generate the at least one item of geometrical information are explained in even greater detail below, including possible calibration curves. In particular in the case of a relationship between the geometry of the illumination and the geometrical information, in this case it is also possible to use analytical or semi-empirical models, for example imaging equations. Thus, by way of example, the illumination can be effected by means of one or a plurality of Gaussian beams, wherein for example the at least one item of geometrical information can be generated by the beam geometry and the Gaussian beam parameters of the illumination, for example by means of corresponding optical matrix calculations. By way of example, imaging parameters of the optional transfer device can be known, for example one or a plurality of focal lengths of one or a plurality of lenses and/or curved mirrors of the transfer device, such that, by way of example, a relationship between a positioning of the object relative to the transfer device and a geometry of the illumination of the sensor region, for example a diameter or equivalent diameter of the luminous spot, can be calculated. By way of example, Gaussian matrix optics and/or some other form of an imaging equation can be used for this purpose.

As explained above, the detector can furthermore comprise at least one illumination source for generating electromagnetic rays. The illumination source can be designed in particular to bring about the illumination of the sensor region. Thus, the illumination source can be designed for example to generate a primary radiation with which the object or a part of the object is irradiated, whereupon the electromagnetic radiation emerges from the object and is transferred to the optical sensor and the sensor region thereof, for example by means of the at least one optional transfer device. As explained above, the electromagnetic radiation by means of which the sensor region is illuminated can comprise for example the primary radiation in reflected or scattered form. Alternatively or additionally, however, for example an influencing of the primary radiation can also be effected, for example a spectral shift and/or a process in which the primary radiation emitted by the illumination source excites the object or a part thereof to emit the electromagnetic radiation, for example by excitation of luminescence.

The illumination source can be embodied in various ways. Thus, the illumination source can be for example part of the detector in a detector housing. Alternatively or additionally, however, the at least one illumination source can also be arranged outside a detector housing, for example as a separate light source. The illumination source can be arranged separately from the object and illuminate the object from a distance. Alternatively or additionally, the illumination source can also be connected to the object or even be part of the object, such that, by way of example, the electromagnetic radiation emerging from the object can also be generated directly by the illumination source. By way of example, at least one illumination source can be arranged on and/or in the object

and directly generate the electromagnetic radiation by means of which the sensor region is illuminated. By way of example, at least one infrared emitter and/or at least one emitter for visible light and/or at least one emitter for ultraviolet light can be arranged on the object. By way of example, at least one light emitting diode and/or at least one laser diode can be arranged on and/or in the object. The illumination source can comprise in particular one or a plurality of the following illumination sources: a laser, in particular a laser diode, although in principle, alternatively or additionally, other types of lasers can also be used; a light emitting diode; an incandescent lamp; an organic light source, in particular an organic light emitting diode. Alternatively or additionally, other illumination sources can also be used. It is particularly preferred if the illumination source is designed to generate electromagnetic beams having a Gaussian beam profile, as is at least approximately the case for example in many lasers. However, other embodiments are also possible, in principle.

Furthermore, it should be noted that the optional transfer device described above can be embodied independently of the optional at least one illumination source. Alternatively or additionally, however, the at least one transfer device can also already be wholly or partly integrated into the illumination source or be wholly or partly identical to said illumination source. Thus, the illumination source itself can already be designed to feed the electromagnetic radiation to the sensor region, for example by a corresponding orientation toward the sensor region and/or by a focusing and/or a corresponding beam profile. By way of example, the illumination source can comprise at least one laser which can already generate at least one laser beam having a known beam profile, for example a Gaussian beam profile, such that, by way of example, it is possible to dispense with beam shaping by one or a plurality of lens systems, since, by way of example, from corresponding equations a propagation of Gaussian beams, from a known beam geometry upon identification of a geometry of the illumination on the sensor region, for example a diameter or equivalent diameter of a luminous spot on the sensor region, a distance between the illumination source and the detector and/or the object and the detector can be deduced. Alternatively or additionally, however, the transfer device can comprise one or a plurality of additional imaging elements, for example one or a plurality of lenses and/or objectives. Alternatively or additionally, by way of example, one or a plurality of deflection elements can be included, for example one or a plurality of mirrors and/or one or a plurality of prisms.

Generally, therefore, as already mentioned above, the illumination source can in particular be selected in particular from an illumination source which is at least partly connected to the object and/or is at least partly identical to the object, and an illumination source which is designed to at least partly illuminate the object. Various other embodiments are possible and are described in even greater detail below by way of example.

The detector can therefore comprise at least one illumination source. The illumination source can be designed in particular to illuminate the object with at least one primary radiation. Said at least one primary radiation can comprise electromagnetic radiation, for example light, but can, alternatively or additionally, also comprise at least one radiation of a different type, for example a particle radiation. The electromagnetic radiation which emerges from the object and which is fed to the optical sensor and illuminates the sensor region in the process can, as already described in part above, comprise in particular a reflected radiation, wherein the reflected radiation can comprise at least part of the primary radiation after reflection at the object. Alternatively or additionally, the electromagnetic radiation emerging from the object can also comprise at least one scattered radiation, wherein the scattered radiation comprises at least one part of the primary radiation after scattering at the object. As explained above, this scattering can be effected without spectral properties being influenced or else with spectral properties being influenced. Once again alternatively or additionally, the electromagnetic radiation emerging from the object can also comprise at least one luminescent radiation which is excited by the primary radiation. Said luminescent radiation can comprise for example a fluorescent radiation and/or a phosphorus radiation. Other possibilities for the embodiment of the electromagnetic radiation which emerges from the object, or combinations of the stated and/or other possibilities, are also conceivable.

A further aspect of the present invention proposes a distance measuring device, in particular for use in a motor vehicle. In this case, a distance measuring device should be understood to mean a device which is designed to generate at least one item of location information of at least one object, for example relative to the distance measuring device itself. As explained above, said at least one item of location information can comprise for example a simple distance between the distance measuring device and the object, but can also generally comprise at least one item of information about a positioning of the object relative to the distance measuring device, for example a spatial positioning in at least one coordinate system and/or an orientation of the object in at least one coordinate system. For further embodiments of the at least one item of location information, reference can be made to the above description.

The proposed distance measuring device accordingly comprises at least one detector in accordance with one or a plurality of the embodiments described above, wherein the detector is designed to determine at least one item of geometrical information of at least one object, wherein the geometrical information comprises at least one item of location information of the object, in particular a distance between a motor vehicle and at least one object and preferably a distance between the motor vehicle and at least one object selected from the group consisting of a further motor vehicle, an obstacle, a cyclist and a pedestrian. In particular, the detector can be completely or partly integrated in one or a

plurality of motor vehicles and can be designed to determine a distance between a motor vehicle and at least one object, for example a distance between two motor vehicles and/or a distance between the motor vehicle and at least one object selected from the group consisting of a further motor vehicle, an obstacle, a cyclist, a pedestrian or some other kind of traffic participant. The detector can be for example completely integrated into the motor vehicle, but can also be arranged for example in a manner distributed over a plurality of motor vehicles. By way of example, as is explained in even greater detail below by way of example, the distance measuring device can comprise at least one detector and at least one illumination source, wherein, by way of example, the at least one illumination source is arranged on a rear side of a first motor vehicle, and the detector is arranged on a front side of at least one second motor vehicle, such that, by way of example, a distance between a front side of the second motor vehicle and a rear side of the first motor vehicle can be determined. Alternatively or additionally, at least one illumination source and at least one detector can also be integrated into one and the same motor vehicle, for example on a front side and/or a rear side of the motor vehicle. Thus, by way of example, the illumination source can be designed to illuminate a motor vehicle ahead and/or a following motor vehicle with primary radiation, and the detector can be designed to detect the electromagnetic radiation emerging from said motor vehicle, as described above, and to generate the location information therefrom.

A further aspect of the present invention proposes an imaging device for imaging at least one sample. In this case, an imaging device should generally be understood to mean a device which can generate a one-dimensional, a two-dimensional or a three-dimensional image of the sample or of a part of said sample.

In particular, the imaging device can be completely or partly used as a microscope. Preferably, the image device is designed for confocal imaging, that is to say has a confocal construction, or is designed as a confocal microscope. Other embodiments of the imaging device are also possible in principle, however, and are described in even greater detail below by way of example.

The imaging device has at least one detector in accordance with one or more of the embodiments described above. The imaging device is furthermore designed to image a plurality of partial regions of the sample successively or simultaneously onto the at least one sensor region of the detector. In this case, by way of example, a partial region of the sample can be a one-dimensional, two-dimensional or three-dimensional region of the sample which is delimited for example by a resolution limit of the imaging device and from which electromagnetic radiation emerges, which, on the sensor region of the detector, leads to an illumination, for example a common luminous spot, in particular on a sensor area. The plurality of partial regions can be imaged successively and/or simultaneously onto the at least one sensor region. In this context, imaging should be

understood to mean that electromagnetic radiation of the sample emerging from the respective partial region is fed to the optical sensor, for example by means of the at least one optional transfer device of the detector.

- 5 With regard to the possible embodiments of the optional transfer device, reference can be made to the above description. In particular, the transfer device can have imaging properties and can comprise for example at least one imaging element, for example at least one lens and/or at least one curved mirror. In particular, the imaging device can be designed to image sequentially, for example by means of a scanning method, in
10 particular using at least one row scan and/or line scan, the plurality of partial regions sequentially onto the sensor region. However, other embodiments are also possible, in principle, for example embodiments in which a plurality of partial regions are simultaneously imaged onto the at least one sensor region, for example by at least one sensor region being assigned to each partial region. By way of example, it is possible to
15 use a detector having a plurality of sensor regions, for example in accordance with one or more of the embodiments described above.

The imaging device is designed to generate, during this imaging of the partial regions of the sample, sensor signals assigned to the partial regions. By way of example, a sensor
20 signal can be assigned to each partial region. The sensor signals can accordingly be generated simultaneously or else in a temporally staggered manner. By way of example, during a row scan or line scan, it is possible to generate a sequence of sensor signals which correspond to the partial regions of the sample, which are strung together in a line, for example. The imaging device is designed to generate from the sensor signals items
25 of geometrical information of the respective partial regions, wherein the items of geometrical information comprise items of location information.

With regard to the embodiment of the items of location information and the various possibilities for generating these items of location information, reference can be made to
30 the above description. The electromagnetic rays emerging from the sample can once again be generated by the sample itself, for example in the form of a luminescent radiation. Alternatively or additionally, the imaging device or the at least one detector can also once again comprise at least one illumination source for illuminating the sample. For further possible embodiments of the imaging device, in particular for use in microscopy,
35 reference can be made to the exemplary embodiments below.

A further aspect of the present invention proposes a human-machine interface for exchanging at least one item of information between a user and a machine. A human-machine interface should generally be understood to mean a device by means of which
40 such information can be exchanged. The machine can comprise in particular a data processing device. The at least one item of information can generally comprise for

example data and/or control commands. Thus, the human-machine interface can be designed in particular for the inputting of control commands by the user.

The human-machine interface has at least one detector in accordance with one or a plurality of the embodiments described above. The human-machine interface is designed to generate at least one item of geometrical information, in particular at least one item of location information, of the user by means of the detector. By way of example, said at least one item of geometrical information can be or comprise an item of location information about a body part of the user, for example an item of location information about a hand posture and/or a posture of some other body part of the user.

In this case, the term user should be interpreted broadly and can for example also encompass one or a plurality of articles directly influenced by the user. Thus, the user can for example also wear one or a plurality of gloves and/or other garments, wherein the geometrical information is at least one item of geometrical information of this at least one garment. By way of example, such garments can be embodied as reflective to a primary radiation emerging from at least one illumination source, for example by the use of one or a plurality of reflectors. Once again alternatively or additionally, the user can for example spatially move one or a plurality of articles whose geometrical information can be detected, which is likewise also intended to be subsumable under generation of at least one item of geometrical information of the user. By way of example, the user can move at least one reflective rod and/or some other type of article, for example by means of said user's hand.

With regard to the embodiment of the at least one item of geometrical information, in particular the at least one item of location information, reference can likewise once again be made to the above description. The at least one item of geometrical information can be static, that is to say can for example once again comprise a snapshot, but can also for example once again comprise a series of sequential items of geometrical information and/or at least one movement. By way of example, at least two items of geometrical information picked up at different times can be compared, such that, by way of example, the at least one item of geometrical information can also comprise at least one item of information about a velocity and/or an acceleration of a movement. Accordingly, the at least one item of geometrical information can for example comprise at least one item of information about at least one body posture and/or about at least one movement of the user.

The human-machine interface is designed to assign to the at least one item of geometrical information at least one item of information, in particular at least one control command. As explained above, the term information should in this case be interpreted broadly and can comprise for example data and/or control commands. By way of

example, the human-machine interface can be designed to assign the at least one item of information to the at least one item of geometrical information, for example by means of a corresponding assignment algorithm and/or a stored assignment specification. By way of example, a unique assignment between a set of items of geometrical information and corresponding items of information can be stored. In this way, for example by means of a corresponding body posture and/or movement of the user, an inputting of at least one item of information can be effected.

Such human-machine interfaces can generally be used in the machine control or else for example in virtual reality. By way of example, robot controllers, vehicle controllers or similar controllers can be made possible by means of the human-machine interface having the one or the plurality of detectors. However, the use of such a human-machine interface in consumer electronics is particularly preferred. Accordingly, a further aspect of the present invention proposes an entertainment device for carrying out at least one entertainment function, in particular a game. The entertainment function can comprise in particular at least one game function. By way of example, one or a plurality of games can be stored which can be influencable by a user, who in this context is also called a player hereinafter. By way of example, the entertainment device can comprise at least one display device, for example at least one screen and/or at least one projector and/or at least one set of display spectacles.

The entertainment device furthermore comprises at least one human-machine interface in accordance with one or more of the embodiments described above. The entertainment device is designed to enable at least one item of information of a player to be input by means of the human-machine interface. By way of example, the player, as described above, can adopt or alter one or a plurality of body postures for this purpose. This includes the possibility of the player for example using corresponding articles for this purpose, for example garments such as e.g. gloves, for example garments which are equipped with one or a plurality of reflectors for reflecting the electromagnetic radiation of the detector. The at least one item of information can comprise for example, as explained above, one or a plurality of control commands. By way of example, in this way, changes in direction can be performed, inputs can be confirmed, a selection can be made from a menu, specific game options can be initiated, movements can be influenced in a virtual space or similar instances of influencing or altering the entertainment function can be performed.

A further aspect of the present invention proposes using the detector in a security device. As described above, by means of the detector, in particular the at least one evaluation device, it is possible to generate at least one item of geometrical information which preferably goes beyond a pure item of information about the total power of the illumination. Said at least one item of geometrical information can, inter alia, also be or

comprise an item of information about the fact that an illumination has taken place or is taking place with a specific geometry, in particular an inhomogeneous illumination of the sensor region, in particular a focused illumination, for example an illumination in which at least one light spot is produced on a sensor area. The proposed security device is therefore based for example on the insight that, by means of the proposed detector, it is possible to identify if intensive electromagnetic radiation, for example a focused light beam and in particular a laser beam, impinges on an article comprising the detector. By way of example, in this way it can be identified that a read-out of optically readable data of an optical data storage device has been effected, for example of a CD-ROM, of a bar code or of some other type of data storage device.

Accordingly, a security device for carrying out at least one security application is proposed. A security device should generally be understood to mean a device which fulfils at least one security function, for example a function which recognizes an access to data and can optionally implement one or a plurality of corresponding measures according to the situation. A security function can generally be a function which recognizes an access to an article or data and optionally prevents it or at least makes it more difficult. The at least one security application can accordingly comprise in particular at least one application in the field of data protection. In particular, the at least one security application can be an application in which an access, in particular an unauthorized access, to data of an optical data storage device is recognized and/or avoided. The optical data storage device can comprise, in principle, any desired type of optical data storage device, for example one or a plurality of the optical data storage devices mentioned above.

The security device furthermore comprises at least one detector in accordance with one or more of the above-described claims relating to a detector. In this respect, it should be pointed out that, in accordance with the above description, the detector can optionally comprise the at least one transfer device. In the security device, such a transfer device is not absolutely necessary, since, by way of example, as described above, the electromagnetic radiation can also be focused by a laser and/or an external imaging device, for example at least one external lens.

Accordingly, the security device or the detector of said security device can also be embodied without such a transfer device, or the illumination source or parts thereof can be regarded as a transfer device or parts thereof. On the other hand, the transfer device can also be wholly or partly integrated into an illumination source, such that the security device can also be embodied in a multipartite fashion. By way of example, the transfer device, as described above, can be wholly or partly integrated into an illumination source, which can likewise be regarded as a constituent part of the security device, but

which need not be formed integrally with the other parts of the security device, for example with the optical sensor of the detector of the security device.

The security device is furthermore designed to recognize, by means of the detector, 5 impingement of focused electromagnetic rays onto the security device, in particular impingement of one or a plurality of laser beams, in particular focused laser beams. By way of example, the security device can comprise the at least one evaluation device for this purpose, as described above. By way of example, this recognition can be effected by virtue of the fact that the at least one sensor signal of the at least one detector is 10 compared with at least one threshold value. While non-focused light can generate for example a sensor signal below said threshold value, for example focused light having the same total power which impinges on the sensor region can generate a sensor signal which reaches or exceeds the at least one threshold value. By way of example, a photocurrent or an absolute value of the photocurrent can be compared with said at least 15 one threshold value. In this way, it is possible for example to recognize if at least one reading beam, for example a weak laser beam, is radiated in so as to read out optically readable data. Even if said reading beam is chosen, in terms of its total power, to be so weak that overall there impinges on the sensor region a total power which does not exceed for example the total power of ambient light, for example assuming an ambient 20 light intensity of 1 to $1 \cdot 10^5$ W/m², such a reading beam can be recognized by means of the at least one detector according to the invention. Focused electromagnetic rays can be understood to mean electromagnetic rays by means of which only a partial region of the sensor region of the detector is irradiated with a boosted intensity, for example only a light spot of less than 10 mm², in particular of less than 1 mm² and particularly preferably 25 of less than 0.1 mm², although the focusing can be dependent on the application. For the definition of a light spot which can be for example a round, oval or else differently shaped light spot, reference can be made to the above description.

The security device can furthermore be designed, if impingement of such focused 30 electromagnetic rays is recognized, to perform at least one security function or further security functions. By way of example, the security device, in particular the evaluation device, can be designed to generate at least one warning signal. Said at least one warning signal can comprise for example at least one signal, in particular selected from the group consisting of an acoustic signal, an electrical signal, an optical signal and a 35 haptic signal. Alternatively or additionally, the warning signal can also comprise at least one variation of at least one state of at least one data storage device of the security device. By way of example, it is possible to use at least one data storage device with at least one read-out bit, wherein the at least one read-out bit can be changed over from a state "not read out" to a state "read out" by means of the at least one warning signal. 40 Alternatively or additionally, however, the warning signal can also be designed for example to bring about, after recognition of impingement of focused electromagnetic

rays, destruction of the security device and/or of the data storage device, such that only a single use is possible. The warning signal can for example also be output to a user and/or some other device.

5 The security device mentioned can be designed or applied in various ways. By way of example, the security device can be embodied as a security label, for example as a security label having at least one optical data storage device and the at least one detector. By way of example, in this way it is possible to prevent multiple uses of access
10 authorizations, for example tickets for events, for example by virtue of a first read-out being identified and/or by virtue of especially a focus light beam being radiated onto the detector. Alternatively or additionally, however, the security device can also be integrated for example into an optical data storage device, such that it is possible to embody an optical data storage device with a security device of this type. The at least one security
15 device can for example be embodied in such a way that the at least one sensor region of the at least one detector is arranged in direct spatial proximity to at least one optically readable data storage device, for example at least one data field of at least one optical ROM, and/or in direct proximity to a bar code and/or a holographic data storage device and/or as a constituent part thereof. In this way, it can be ensured, for example, that a
20 reading beam by means of which data are read out optically from the data storage device inevitably also impinges on the sensor region of the detector. By way of example, a spatial distance between the sensor region and the data storage device region can be not more than 5 mm, in particular not more than 1 mm and particularly preferably not more than 0.1 mm.

25 A further aspect of the present invention proposes a method for optically detecting at least one object. The method can be effected in particular using a detector in accordance with one or more of the embodiments described above, such that, with regard to optional embodiments of the method, reference can be made to the above description of the detector. In the method, at least one optical sensor is used, wherein the optical sensor
30 has at least one sensor region. Electromagnetic radiation emerging from the object is fed to the sensor, wherein the sensor region is illuminated. The optical sensor generates at least one sensor signal in a manner dependent on the illumination of the sensor region, wherein the sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination.

35 For further possible embodiments of the method, reference can be made to the above description. In particular, from the sensor signal it is possible to generate at least one item of geometrical information of the object, preferably at least one item of location information of the object, which preferably goes beyond a pure item of information about
40 the total power of the illumination. For possible embodiments of this geometrical information, reference can be made to the above description. In particular, this

geometrical information can comprise at least one item of location information of the object. With regard to possible embodiments of this location information, reference can likewise be made to the above description. The geometrical information can be determined in particular using at least one predefined relationship between the geometry of the illumination and a location of the object, preferably taking account of a known power of the illumination and/or taking account of a modulation frequency with which the illumination is modulated. In particular, therefore, the method can be embodied in such a way that the illumination is modulated, in particular is modulated periodically, with a modulation frequency which can be constant or else variable.

A further aspect of the present invention proposes a use of a detector in accordance with one or more of the embodiments described above for a purpose of use, which is selected from the group consisting of: distance measurement, in particular in traffic technology; imaging, in particular in microscopy; an entertainment application; a human-machine interface application; a security application. However, other uses of the detector are also possible, in principle.

A further aspect of the present invention proposes the use of an organic solar cell, in particular a dye solar cell and preferable a solid dye solar cell, as optical sensor, for example in a detector in accordance with one or more of the embodiments described above. In the use, at least one sensor signal is generated, wherein the sensor signal, given the same total power of an illumination of at least one sensor region of the optical sensor, is dependent on a geometry of the illumination on the organic solar cell, in particular a sensor area of the organic solar cell, wherein at least one item of geometrical information of at least one object is generated from the sensor signal in the use. For further possible embodiments of this use, reference can be made to the above description.

The above-described detector, the method, the distance measuring device, the imaging device, the human-machine interface, the entertainment device and the security device and also the proposed uses have considerable advantages over the prior art. Thus, in particular the outlay for object detection can be distinctly reduced by means of the proposed detector and by means of the proposed method. For distance measurements for example or other types of object detection, by way of example, technically complex propagation time measurements in which pulse propagation times of laser pulses are detected, for example, can be at least substantially avoided. In contrast to conventional imaging methods, it is possible in a simple manner, for example with erroneous assumptions being avoided, to generate items of geometrical information, in particular items of location information, about a detected object, without for example detectors having a high spatial resolution, for example a high number of pixels, being absolutely necessary for this purpose. By way of example, it is possible to use a detector having a

comparatively large sensor area, for example in accordance with the embodiment described above, on which a light spot is produced by electromagnetic radiation emerging from the object. The sensor signal can be, in particular, independent of the position of said light spot on the sensor region, for example the sensor area, as long as said light spot is arranged completely within the sensor region. However, the detector can be designed to deduce from the focusing of the electromagnetic rays, for example from a diameter or equivalent diameter of the light spot on the sensor area, the at least one item of geometrical information, for example the distance between the object and the detector. Such an application can be realized technically in an extremely simple manner. Furthermore, it is possible to use cost-effective optical sensors, in particular large-area optical sensors, which can be producible for example using one or a plurality of organic materials. In particular, it is possible to use solar cells, for example organic solar cells and in particular dye solar cells, which are produced as mass-produced products in photovoltaics. In this way, for the detector proposed, a multiplicity of new fields of application open up which were withheld from previous detectors in many cases on account of their high costs, such as, for example, applications in disposable articles or data storage devices.

As explained above, the optical sensor can comprise in particular an organic semiconductor detector, particularly preferably a dye solar cell. In particular, the optical sensor can comprise at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material and at least one second electrode, preferably in the stated order. The stated elements can be present as layers in a layer construction, for example. The layer construction can be applied for example to a substrate, preferably a transparent substrate, for example a glass substrate.

Preferred embodiments of the abovementioned elements of the preferred optical sensor are described below by way of example, wherein these embodiments can be used in any desired combination. However, numerous other configurations are also possible, in principle, wherein reference can be made for example to US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 and WO 2009/013282 A1 cited above.

First electrode and n-semiconductive metal oxide

The n-semiconductive metal oxide used in the dye solar cell may be a single metal oxide or a mixture of different oxides. It is also possible to use mixed oxides. The n-semiconductive metal oxide may especially be porous and/or be used in the form of a nanoparticulate oxide, nanoparticles in this context being understood to mean particles which have an average particle size of less than 0.1 micrometer. A nanoparticulate oxide

is typically applied to a conductive substrate (i.e. a carrier with a conductive layer as the first electrode) by a sintering process as a thin porous film with large surface area.

5 The substrate may be rigid or else flexible. Suitable substrates (also referred to hereinafter as carriers) are, as well as metal foils, in particular plastic sheets or films and especially glass sheets or glass films. Particularly suitable electrode materials, especially for the first electrode according to the above-described, preferred structure, are conductive materials, for example transparent conductive oxides (TCOs), for example
10 fluorine- and/or indium-doped tin oxide (FTO or ITO) and/or aluminum-doped zinc oxide (AZO), carbon nanotubes or metal films. Alternatively or additionally, it would, however, also be possible to use thin metal films which still have a sufficient transparency. The substrate can be covered or coated with these conductive materials. Since generally only a single substrate is required in the structure proposed, the formation of flexible cells is also possible. This enables a multitude of end uses which would be achievable only with
15 difficulty, if at all, with rigid substrates, for example use in bank cards, garments, etc.

The first electrode, especially the TCO layer, may additionally be covered or coated with a solid metal oxide buffer layer (for example of thickness 10 to 200 nm), in order to prevent direct contact of the p-type semiconductor with the TCO layer (see Peng *et al.*,
20 Coord. Chem. Rev. 248, 1479 (2004)). The inventive use of solid p-semiconducting electrolytes, in the case of which contact of the electrolyte with the first electrode is greatly reduced compared to liquid or gel-form electrolytes, however, makes this buffer layer unnecessary in many cases, such that it is possible in many cases to dispense with this layer, which also has a current-limiting effect and can also worsen the contact of the
25 n-semiconducting metal oxide with the first electrode. This enhances the efficiency of the components. On the other hand, such a buffer layer can in turn be utilized in a controlled manner in order to match the current component of the dye solar cell to the current component of the organic solar cell. In addition, in the case of cells in which the buffer layer has been dispensed with, especially in solid cells, problems frequently occur with
30 unwanted recombinations of charge carriers. In this respect, buffer layers are advantageous in many cases specifically in solid cells.

As is well known, thin layers or films of metal oxides are generally inexpensive solid semiconductor materials (n-type semiconductors), but the absorption thereof, due to
35 large bandgaps, is typically not within the visible region of the electromagnetic spectrum, but rather usually in the ultraviolet spectral region. For use in solar cells, the metal oxides therefore generally, as is the case in the dye solar cells, have to be combined with a dye as a photosensitizer, which absorbs in the wavelength range of sunlight, i.e. at 300 to 2000 nm, and, in the electronically excited state, injects electrons into the conduction
40 band of the semiconductor. With the aid of a solid p-type semiconductor used

additionally in the cell as an electrolyte, which is in turn reduced at the counterelectrode, electrons can be recycled to the sensitizer, such that it is regenerated.

Of particular interest for use in organic solar cells are the semiconductors zinc oxide, tin dioxide, titanium dioxide or mixtures of these metal oxides. The metal oxides can be used in the form of nanocrystalline porous layers. These layers have a large surface area which is coated with the dye as a sensitizer, such that a high absorption of sunlight is achieved. Metal oxide layers which are structured, for example nanorods, give advantages such as higher electron mobilities or improved pore filling by the dye.

The metal oxide semiconductors can be used alone or in the form of mixtures. It is also possible to coat a metal oxide with one or more other metal oxides. In addition, the metal oxides may also be applied as a coating to another semiconductor, for example GaP, ZnP or ZnS.

Particularly preferred semiconductors are zinc oxide and titanium dioxide in the anatase polymorph, which is preferably used in nanocrystalline form.

In addition, the sensitizers can advantageously be combined with all n-type semiconductors which typically find use in these solar cells. Preferred examples include metal oxides used in ceramics, such as titanium dioxide, zinc oxide, tin(IV) oxide, tungsten(VI) oxide, tantalum(V) oxide, niobium(V) oxide, cesium oxide, strontium titanate, zinc stannate, complex oxides of the perovskite type, for example barium titanate, and binary and ternary iron oxides, which may also be present in nanocrystalline or amorphous form.

Due to the strong absorption that customary organic dyes and phthalocyanines and porphyrins have, even thin layers or films of the n-semiconducting metal oxide are sufficient to absorb the required amount of dye. Thin metal oxide films in turn have the advantage that the probability of unwanted recombination processes falls and that the internal resistance of the dye subcell is reduced. For the n-semiconducting metal oxide, it is possible with preference to use layer thicknesses of 100 nm up to 20 micrometers, more preferably in the range between 500 nm and approx. 3 micrometers.

Dye

In the context of the present invention, as usual in particular for DSCs, the terms "dye", "sensitizer dye" and "sensitizer" are used essentially synonymously without any restriction of possible configurations. Numerous dyes which are usable in the context of the present invention are known from the prior art, and so, for possible material examples, reference may also be made to the above description of the prior art regarding

dye solar cells. All dyes listed and claimed may in principle also be present as pigments. Dye-sensitized solar cells based on titanium dioxide as a semiconductor material are described, for example, in US-A-4 927 721, Nature 353, p. 737-740 (1991) and US-A-5 350 644, and also Nature 395, p. 583-585 (1998) and EP-A-1 176 646. The dyes
5 described in these documents can in principle also be used advantageously in the context of the present invention. These dye solar cells preferably comprise monomolecular films of transition metal complexes, especially ruthenium complexes, which are bonded to the titanium dioxide layer via acid groups as sensitizers.

10 Not least for reasons of cost, sensitizers which have been proposed repeatedly include metal-free organic dyes, which are likewise also usable in the context of the present invention. High efficiencies of more than 4%, especially in solid dye solar cells, can be achieved, for example, with indoline dyes (see, for example, Schmidt-Mende *et al.*, Adv. Mater. 2005, 17, 813). US-A-6 359 211 describes the use, also implementable in the
15 context of the present invention, of cyanine, oxazine, thiazine and acridine dyes which have carboxyl groups bonded via an alkylene radical for fixing to the titanium dioxide semiconductor.

Organic dyes now achieve efficiencies of almost 12.1% in liquid cells (see, for example,
20 P. Wang *et al.*, ACS. Nano 2010). Pyridinium-containing dyes have also been reported, can be used in the context of the present invention and exhibit promising efficiencies.

Particularly preferred sensitizer dyes in the dye solar cell proposed are the perylene derivatives, terrylene derivatives and quaterrylene derivatives described in
25 DE 10 2005 053 995 A1 or WO 2007/054470 A1. The use of these dyes, which is also possible in the context of the present invention, leads to photovoltaic elements with high efficiencies and simultaneously high stabilities.

The rylenes exhibit strong absorption in the wavelength range of sunlight and can,
30 depending on the length of the conjugated system, cover a range from about 400 nm (perylene derivatives I from DE 10 2005 053 995 A1) up to about 900 nm (quaterrylene derivatives I from DE 10 2005 053 995 A1). Rylene derivatives I based on terrylene absorb, according to the composition thereof, in the solid state adsorbed onto titanium dioxide, within a range from about 400 to 800 nm. In order to achieve very substantial
35 utilization of the incident sunlight from the visible into the near infrared region, it is advantageous to use mixtures of different rylene derivatives I. Occasionally, it may also be advisable also to use different rylene homologs.

The rylene derivatives I can be fixed easily and in a permanent manner to the n-semiconducting metal oxide film. The bonding is effected via the anhydride function (x1)
40 or the carboxyl groups –COOH or –COO– formed in situ, or via the acid groups A present

in the imide or condensate radicals ((x2) or (x3)). The rylene derivatives I described in DE 10 2005 053 995 A1 have good suitability for use in dye-sensitized solar cells in the context of the present invention.

5 It is particularly preferred when the dyes, at one end of the molecule, have an anchor group which enables the fixing thereof to the n-type semiconductor film. At the other end of the molecule, the dyes preferably comprise electron donors Y which facilitate the regeneration of the dye after the electron release to the n-type semiconductor, and also prevent recombination with electrons already released to the semiconductor.

10 For further details regarding the possible selection of a suitable dye, it is possible, for example, again to refer to DE 10 2005 053 995 A1. By way of example, it is possible especially to use ruthenium complexes, porphyrins, other organic sensitizers, and preferably rylenes.

15 The dyes can be fixed onto or into the n-semiconducting metal oxide films in a simple manner. For example, the n-semiconducting metal oxide films can be contacted in the freshly sintered (still warm) state over a sufficient period (for example about 0.5 to 24 h) with a solution or suspension of the dye in a suitable organic solvent. This can be
20 accomplished, for example, by immersing the metal oxide-coated substrate into the solution of the dye.

If combinations of different dyes are to be used, they may, for example, be applied successively from one or more solutions or suspensions which comprise one or more of
25 the dyes. It is also possible to use two dyes which are separated by a layer of, for example, CuSCN (on this subject see, for example, Tennakone, K.J., Phys. Chem. B. 2003, 107, 13758). The most convenient method can be determined comparatively easily in the individual case.

30 In the selection of the dye and of the size of the oxide particles of the n-semiconducting metal oxide, the organic solar cell should be configured such that a maximum amount of light is absorbed. The oxide layers should be structured such that the solid p-type semiconductor can efficiently fill the pores. For instance, smaller particles have greater surface areas and are therefore capable of adsorbing a greater amount of dyes. On the
35 other hand, larger particles generally have larger pores which enable better penetration through the p-conductor.

p-semiconducting organic material

40 As described above, the optical sensor can comprise in particular at least one p-semiconducting organic material, preferably at least one solid p-semiconducting material,

which is also designated hereinafter as p-type semiconductor or p-type conductor. Hereinafter a description is given of a series of preferred examples of such organic p-type semiconductors which can be used individually or else in any desired combination, for example in a combination of a plurality of layers with a respective p-type semiconductor, and/or in a combination of a plurality of p-type semiconductors in one layer.

In order to prevent recombination of the electrons in the n-semiconducting metal oxide with the solid p-conductor, it is possible to use, between the n-semiconducting metal oxide and the p-type semiconductor, at least one passivating layer which has a passivating material. This layer should be very thin and should as far as possible cover only the as yet uncovered sites of the n-semiconducting metal oxide. The passivation material may, under some circumstances, also be applied to the metal oxide before the dye. Preferred passivation materials are especially one or more of the following substances: Al_2O_3 ; silanes, for example CH_3SiCl_3 ; Al^{3+} ; 4-tert-butylpyridine (TBP); MgO ; GBA (4-guanidinobutyric acid) and similar derivatives; alkyl acids; hexadecylmalonic acid (HDMA).

As described above, in the context of the organic solar cell, preferably one or more solid organic p-type semiconductors are used – alone or else in combination with one or more further p-type semiconductors which are organic or inorganic in nature. In the context of the present invention, a p-type semiconductor is generally understood to mean a material, especially an organic material, which is capable of conducting holes, that is to say positive charge carriers. More particularly, it may be an organic material with an extensive π -electron system which can be oxidized stably at least once, for example to form what is called a free-radical cation. For example, the p-type semiconductor may comprise at least one organic matrix material which has the properties mentioned. Furthermore, the p-type semiconductor can optionally comprise one or a plurality of dopants which intensify the p-semiconducting properties. A significant parameter influencing the selection of the p-type semiconductor is the hole mobility, since this partly determines the hole diffusion length (cf. Kumara, G., *Langmuir*, 2002, 18, 10493-10495). A comparison of charge carrier mobilities in different spiro compounds can be found, for example, in T. Saragi, *Adv. Funct. Mater.* 2006, 16, 966-974.

Preferably, in the context of the present invention, organic semiconductors are used (i.e. low molecular weight, oligomeric or polymeric semiconductors or mixtures of such semiconductors). Particular preference is given to p-type semiconductors which can be processed from a liquid phase. Examples here are p-type semiconductors based on polymers such as polythiophene and polyarylamines, or on amorphous, reversibly oxidizable, nonpolymeric organic compounds, such as the spirobifluorenes mentioned at the outset (cf., for example, US 2006/0049397 and the spiro compounds disclosed

therein as p-type semiconductors, which are also usable in the context of the present invention). Preference is given to using low molecular weight organic semiconductors. In addition, reference may also be made to the remarks regarding the p-semiconducting materials and dopants from the above description of the prior art.

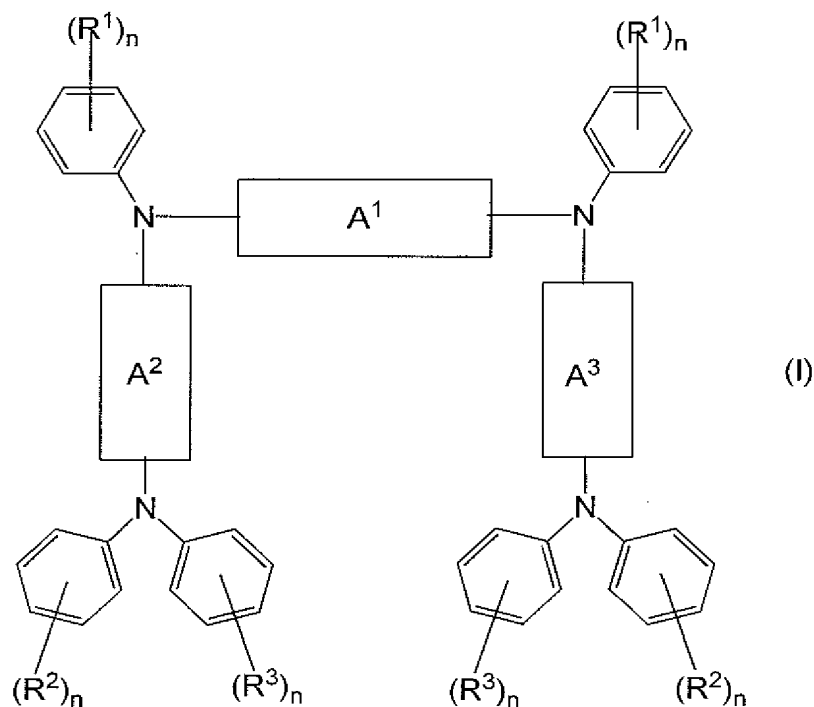
5

The p-type semiconductor is preferably producible or produced by applying at least one p-conducting organic material to at least one carrier element, wherein the application is effected for example by deposition from a liquid phase comprising the at least one p-conducting organic material. The deposition can in this case once again be effected, in principle, by any desired deposition process, for example by spin-coating, knife-coating, printing or combinations of the stated and/or other deposition methods.

10

The organic p-type semiconductor may especially comprise at least one spiro compound and/or especially be selected from: a spiro compound, especially spiro-MeOTAD; a compound with the structural formula:

15



in which

20

A^1 , A^2 , A^3 are each independently optionally substituted aryl groups or heteroaryl groups,

R^1 , R^2 , R^3 are each independently selected from the group consisting of the substituents $-R$, $-OR$, $-NR_2$, $-A^4-OR$ and $-A^4-NR_2$,

25

where R is selected from the group consisting of alkyl, aryl and heteroaryl,

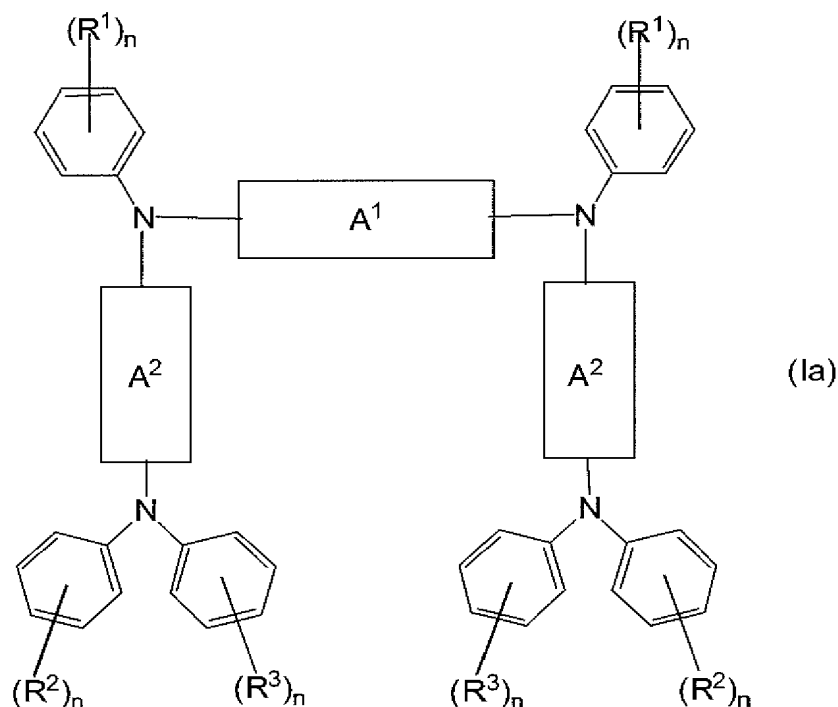
and

where A^4 is an aryl group or heteroaryl group, and

where n at each instance in formula I is independently a value of 0, 1, 2 or 3,

with the proviso that the sum of the individual n values is at least 2 and at least two of the R^1 , R^2 and R^3 radicals are -OR and/or -NR₂.

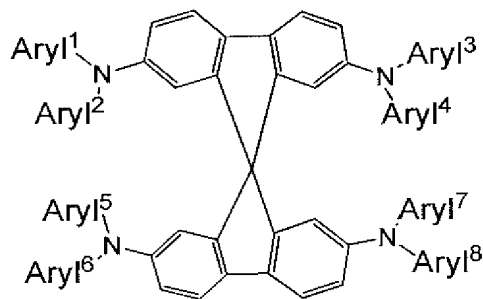
Preferably, A^2 and A^3 are the same; accordingly, the compound of the formula (I) preferably has the following structure (Ia)



More particularly, as explained above, the p-type semiconductor may thus have at least one low molecular weight organic p-type semiconductor. A low molecular weight material is generally understood to mean a material which is present in monomeric, nonpolymerized or nonoligomerized form. The term "low molecular weight" as used in the present context preferably means that the p-type semiconductor has molecular weights in the range from 100 to 25 000 g/mol. Preferably, the low molecular weight substances have molecular weights of 500 to 2000 g/mol.

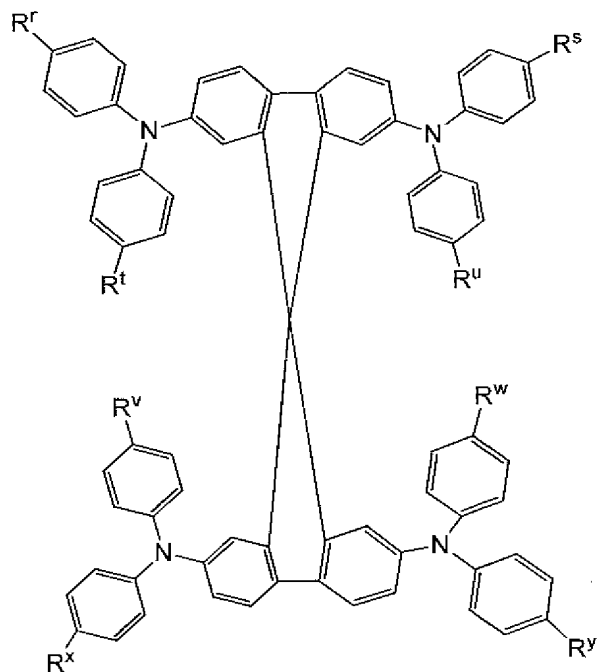
In general, in the context of the present invention, p-semiconducting properties are understood to mean the property of materials, especially of organic molecules, to form holes and to transport these holes and/or to pass them on to adjacent molecules. More particularly, stable oxidation of these molecules should be possible. In addition, the low molecular weight organic p-type semiconductors mentioned may especially have an extensive π -electron system. More particularly, the at least one low molecular weight p-type semiconductor may be processable from a solution. The low molecular weight p-type semiconductor may especially comprise at least one triphenylamine. It is particularly preferred when the low molecular weight organic p-type semiconductor comprises at least one spiro compound. A spiro compound is understood to mean polycyclic organic compounds whose rings are joined only at one atom, which is also referred to as the spiro atom. More particularly, the spiro atom may be sp^3 -hybridized, such that the constituents of the spiro compound connected to one another via the spiro atom are, for example, arranged in different planes with respect to one another.

More preferably, the spiro compound has a structure of the following formula:



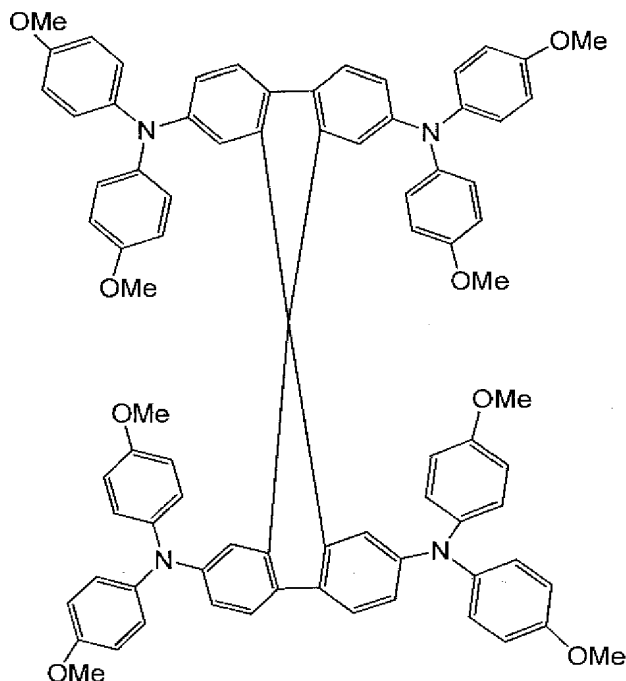
where the aryl¹, aryl², aryl³, aryl⁴, aryl⁵, aryl⁶, aryl⁷ and aryl⁸ radicals are each independently selected from substituted aryl radicals and heteroaryl radicals, especially from substituted phenyl radicals, where the aryl radicals and heteroaryl radicals, preferably the phenyl radicals, are each independently substituted, preferably in each case by one or more substituents selected from the group consisting of -O-alkyl, -OH, -F, -Cl, -Br and -I, where alkyl is preferably methyl, ethyl, propyl or isopropyl. More preferably, the phenyl radicals are each independently substituted, in each case by one or more substituents selected from the group consisting of -O-Me, -OH, -F, -Cl, -Br and -I.

Further preferably, the spiro compound is a compound of the following formula:



where R^r , R^s , R^t , R^u , R^v , R^w , R^x and R^y are each independently selected from the group consisting of -O-alkyl, -OH, -F, -Cl, -Br and -I, where alkyl is preferably methyl, ethyl, propyl or isopropyl. More preferably, R^r , R^s , R^t , R^u , R^v , R^w , R^x and R^y are each independently selected from the group consisting of -O-Me, -OH, -F, -Cl, -Br and -I.

More particularly, the p-type semiconductor may comprise spiro-MeOTAD or consist of spiro-MeOTAD, i.e. a compound of the formula below, commercially available, for example, from Merck KGaA, Darmstadt, Germany:



Alternatively or additionally, it is also possible to use other p-semiconducting compounds, especially low molecular weight and/or oligomeric and/or polymeric p-semiconducting compounds.

In an alternative embodiment, the low molecular weight organic p-type semiconductor comprises one or more compounds of the abovementioned general formula I, for which reference may be made, for example, to PCT application number PCT/EP2010/051826, which will be published after the priority date of the present application. The p-type semiconductor may comprise the at least one compound of the abovementioned general formula I additionally or alternatively to the spiro compound described above.

The term "alkyl" or "alkyl group" or "alkyl radical" as used in the context of the present invention is understood to mean substituted or unsubstituted C₁-C₂₀-alkyl radicals in general. Preference is given to C₁- to C₁₀-alkyl radicals, particular preference to C₁- to C₈-alkyl radicals. The alkyl radicals may be either straight-chain or branched. In addition, the alkyl radicals may be substituted by one or more substituents selected from the group consisting of C₁-C₂₀-alkoxy, halogen, preferably F, and C₆-C₃₀-aryl which may in turn be substituted or unsubstituted. Examples of suitable alkyl groups are methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl and octyl, and also isopropyl, isobutyl, isopentyl, sec-butyl, tert-butyl, neopentyl, 3,3-dimethylbutyl, 2-ethylhexyl, and also derivatives of the alkyl groups mentioned substituted by C₆-C₃₀-aryl, C₁-C₂₀-alkoxy and/or halogen, especially F, for example CF₃.

The term "aryl" or "aryl group" or "aryl radical" as used in the context of the present invention is understood to mean optionally substituted C₆-C₃₀-aryl radicals which are derived from monocyclic, bicyclic, tricyclic or else multicyclic aromatic rings, where the aromatic rings do not comprise any ring heteroatoms. The aryl radical preferably comprises 5- and/or 6-membered aromatic rings. When the aryls are not monocyclic systems, in the case of the term "aryl" for the second ring, the saturated form (perhydro form) or the partly unsaturated form (for example the dihydro form or tetrahydro form), provided the particular forms are known and stable, is also possible. The term "aryl" in the context of the present invention thus comprises, for example, also bicyclic or tricyclic radicals in which either both or all three radicals are aromatic, and also bicyclic or tricyclic radicals in which only one ring is aromatic, and also tricyclic radicals in which two rings are aromatic. Examples of aryl are: phenyl, naphthyl, indanyl, 1,2-dihydronaphthenyl, 1,4-dihydronaphthenyl, fluorenyl, indenyl, anthracenyl, phenanthrenyl or 1,2,3,4-tetrahydronaphthyl. Particular preference is given to C₆-C₁₀-aryl radicals, for example phenyl or naphthyl, very particular preference to C₆-aryl radicals, for example phenyl. In addition, the term "aryl" also comprises ring systems comprising at least two monocyclic, bicyclic or multicyclic aromatic rings joined to one another via single or double bonds. One example is that of biphenyl groups.

The term "heteroaryl" or "heteroaryl group" or "heteroaryl radical" as used in the context of the present invention is understood to mean optionally substituted 5- or 6-membered aromatic rings and multicyclic rings, for example bicyclic and tricyclic compounds having at least one heteroatom in at least one ring. The heteroaryls in the context of the invention preferably comprise 5 to 30 ring atoms. They may be monocyclic, bicyclic or tricyclic, and some can be derived from the aforementioned aryl by replacing at least one carbon atom in the aryl base skeleton with a heteroatom. Preferred heteroatoms are N, O and S. The hetaryl radicals more preferably have 5 to 13 ring atoms. The base skeleton of the heteroaryl radicals is especially preferably selected from systems such as pyridine and five-membered heteroaromatics such as thiophene, pyrrole, imidazole or furan. These base skeletons may optionally be fused to one or two six-membered aromatic radicals. In addition, the term "heteroaryl" also comprises ring systems comprising at least two monocyclic, bicyclic or multicyclic aromatic rings joined to one another via single or double bonds, where at least one ring comprises a heteroatom. When the heteroaryls are not monocyclic systems, in the case of the term "heteroaryl" for at least one ring, the saturated form (perhydro form) or the partly unsaturated form (for example the dihydro form or tetrahydro form), provided the particular forms are known and stable, is also possible. The term "heteroaryl" in the context of the present invention thus comprises, for example, also bicyclic or tricyclic radicals in which either both or all three radicals are aromatic, and also bicyclic or tricyclic radicals in which only one ring is aromatic, and also tricyclic radicals in which two rings are aromatic, where at least one of the rings, i.e. at least one aromatic or one nonaromatic ring has a heteroatom. Suitable

fused heteroaromatics are, for example, carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl. The base skeleton may be substituted at one, more than one or all substitutable positions, suitable substituents being the same as have already been specified under the definition of C₆-C₃₀-aryl. However, the hetaryl radicals are preferably unsubstituted. Suitable hetaryl radicals are, for example, pyridin-2-yl, pyridin-3-yl, pyridin-4-yl, thiophen-2-yl, thiophen-3-yl, pyrrol-2-yl, pyrrol-3-yl, furan-2-yl, furan-3-yl and imidazol-2-yl and the corresponding benzofused radicals, especially carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl.

In the context of the invention the term "optionally substituted" refers to radicals in which at least one hydrogen radical of an alkyl group, aryl group or heteroaryl group has been replaced by a substituent. With regard to the type of this substituent, preference is given to alkyl radicals, for example methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl and octyl, and also isopropyl, isobutyl, isopentyl, sec-butyl, tert-butyl, neopentyl, 3,3-dimethylbutyl and 2-ethylhexyl, aryl radicals, for example C₆-C₁₀-aryl radicals, especially phenyl or naphthyl, most preferably C₆-aryl radicals, for example phenyl, and hetaryl radicals, for example pyridin-2-yl, pyridin-3-yl, pyridin-4-yl, thiophen-2-yl, thiophen-3-yl, pyrrol-2-yl, pyrrol-3-yl, furan-2-yl, furan-3-yl and imidazol-2-yl, and also the corresponding benzofused radicals, especially carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl. Further examples include the following substituents: alkenyl, alkynyl, halogen, hydroxyl.

The degree of substitution here may vary from monosubstitution up to the maximum number of possible substituents.

Preferred compounds of the formula I for use in accordance with the invention are notable in that at least two of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. The at least two radicals here may be only -OR radicals, only -NR₂ radicals, or at least one -OR and at least one -NR₂ radical.

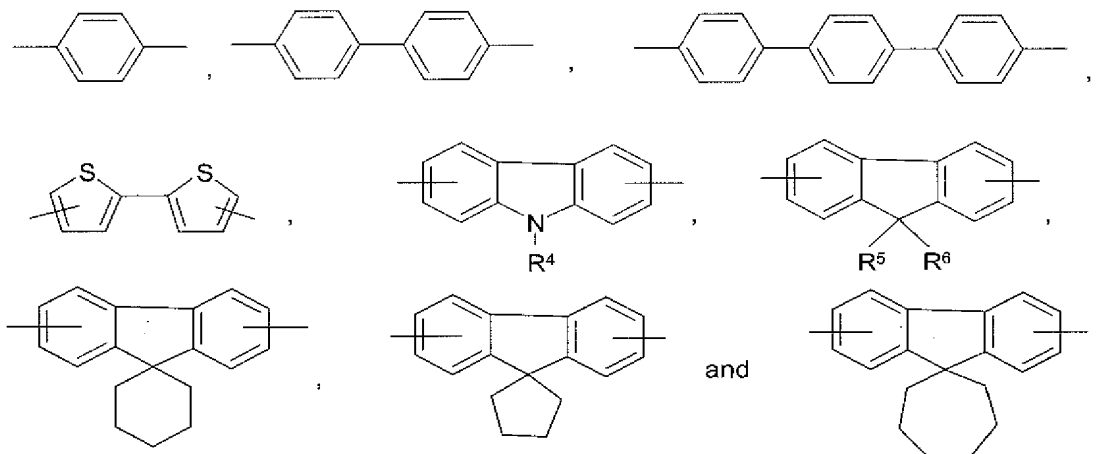
Particularly preferred compounds of the formula I for use in accordance with the invention are notable in that at least four of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. The at least four radicals here may be only -OR radicals, only -NR₂ radicals or a mixture of -OR and -NR₂ radicals.

Very particularly preferred compounds of the formula I for use in accordance with the invention are notable in that all of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. They may be only -OR radicals, only -NR₂ radicals or a mixture of -OR and -NR₂ radicals.

In all cases, the two R in the $-NR_2$ radicals may be different from one another, but they are preferably the same.

Preferably, A^1 , A^2 and A^3 are each independently selected from the group consisting of

5



in which

10

m is an integer from 1 to 18,

R^4 is alkyl, aryl or heteroaryl, where R^4 is preferably an aryl radical, more preferably a phenyl radical,

15

R^5 , R^6 are each independently H, alkyl, aryl or heteroaryl,

where the aromatic and heteroaromatic rings of the structures shown may optionally have further substitution. The degree of substitution of the aromatic and heteroaromatic rings here may vary from monosubstitution up to the maximum number of possible substituents.

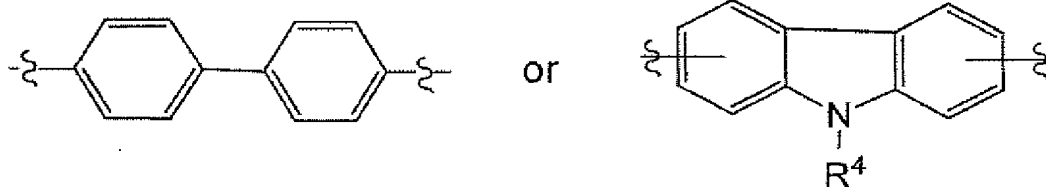
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Preferred substituents in the case of further substitution of the aromatic and heteroaromatic rings include the substituents already mentioned above for the one, two or three optionally substituted aromatic or heteroaromatic groups.

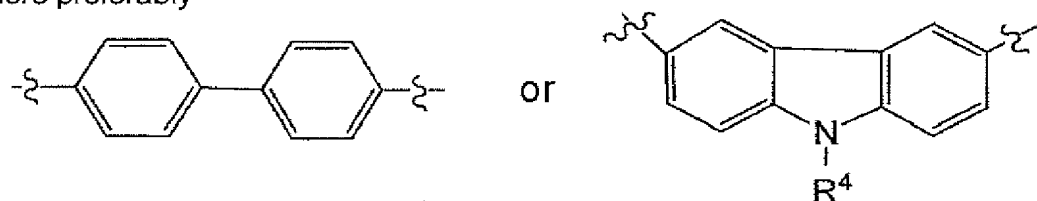
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Preferably, the aromatic and heteroaromatic rings of the structures shown do not have further substitution.

More preferably, A¹, A² and A³ are each independently

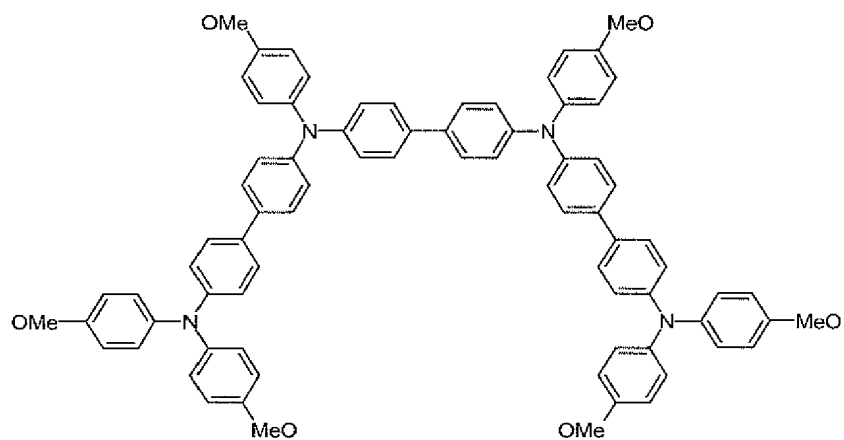


more preferably



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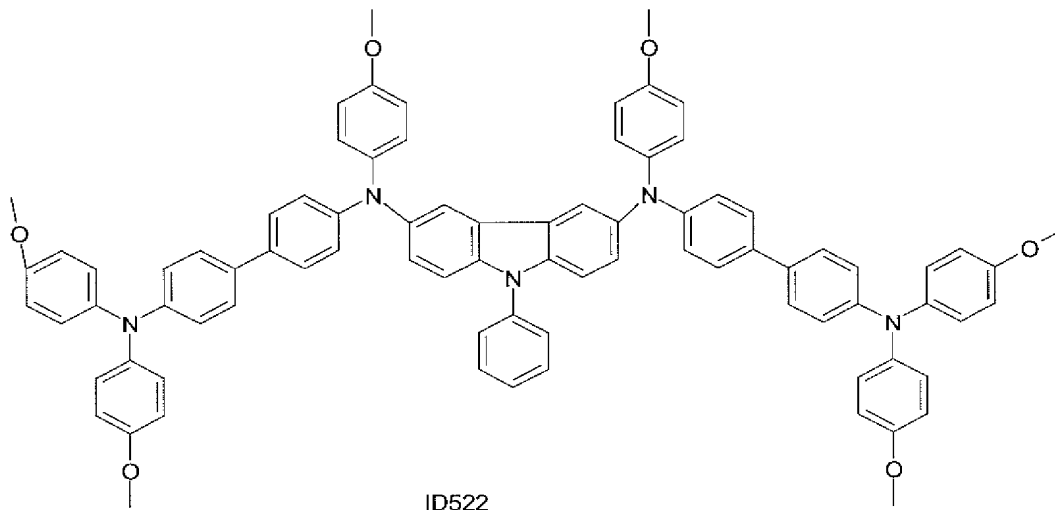
More preferably, the at least one compound of the formula (I) has one of the following structures:



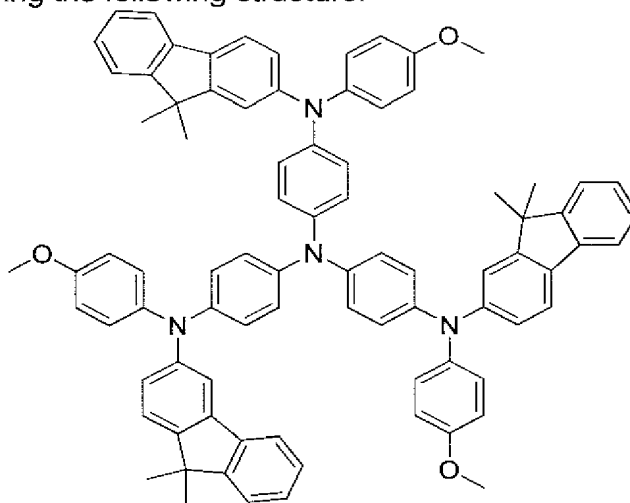
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ID367

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In an alternative embodiment, the organic p-type semiconductor comprises a compound of the type ID322 having the following structure:



5

The compounds for use in accordance with the invention can be prepared by customary methods of organic synthesis known to those skilled in the art. References to relevant (patent) literature can additionally be found in the synthesis examples adduced below.

10

Second electrode

The second electrode may be a bottom electrode facing the substrate or else a top electrode facing away from the substrate. The second electrode used can be especially metal electrodes which may have one or more metals in pure form or as a mixture/alloy, such as especially aluminum or silver. The use of inorganic/organic mixed electrodes or multilayer electrodes is also possible, for example the use of LiF/Al electrodes.

15

In addition, it is also possible to use electrode designs in which the quantum efficiency of the components is increased by virtue of the photons being forced, by means of appropriate reflections, to pass through the absorbing layers at least twice. Such layer structures are also referred to as "concentrators" and are likewise described, for example, in WO 02/101838 (especially pages 23-24).

The organic solar cell can furthermore comprise at least one encapsulation, wherein the encapsulation is designed to shield the organic solar cell, in particular the electrodes and/or the p-type semiconductor, from a surrounding atmosphere.

Overall, in the context of the present invention, the following embodiments are regarded as particularly preferred:

Embodiment 1: A detector for optically detecting at least one object, comprising at least one optical sensor, wherein the optical sensor has at least one sensor region, in particular at least one sensor region comprising at least one sensor area wherein the optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region, wherein the sensor signal, given the same total power of the illumination is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area, wherein the detector furthermore has at least one evaluation device, wherein the evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

Embodiment 2: The detector according to the preceding embodiment, wherein the detector furthermore has at least one modulation device for modulating the illumination.

Embodiment 3: The detector according to the preceding embodiment, wherein the detector is designed to detect at least two sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies, wherein the evaluation device is designed to generate the geometrical information from the at least two sensor signals.

Embodiment 4: The detector according to any of the preceding embodiments, wherein the optical sensor is furthermore designed in such a way that the sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination.

Embodiment 5: The detector according to any of the preceding embodiments, wherein the sensor region is exactly one continuous sensor region, wherein the sensor signal is a uniform sensor signal for the entire sensor region.

- 5 Embodiment 6: The detector according to any of the preceding embodiments, wherein the sensor signal is selected from the group consisting of a photocurrent and a photovoltage.

10 Embodiment 7: The detector according to any of the preceding embodiments, wherein the optical sensor comprises at least one semiconductor detector, in particular an organic semiconductor detector comprising at least one organic material, preferably an organic solar cell and particularly preferably a dye solar cell, in particular a solid dye solar cell.

15 Embodiment 8: The detector according to the preceding embodiment, wherein the optical sensor comprises at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode.

20 Embodiment 9: The detector according to any of the preceding embodiments, wherein the geometric information comprises at least one item of location information of the object.

25 Embodiment 10: The detector according to the preceding embodiment, wherein the evaluation device is designed to determine the geometrical information from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object with respect to the detector, preferably taking account of a known power of the illumination and optionally taking account of a modulation frequency with which the illumination is modulated.

30 Embodiment 11: The detector according to any of the preceding embodiments, furthermore comprising at least one transfer device, wherein the transfer device is designed to feed electromagnetic radiation emerging from the object to the optical sensor and thereby to illuminate the sensor region.

35 Embodiment 12: The detector according to any of the preceding embodiments, furthermore comprises at least one illumination source.

40 Embodiment 13: The detector according to the preceding embodiment, wherein the illumination source is selected from: an illumination source, which is at least partly

connected to the object and/or is at least partly identical to the object; an illumination source which is designed to at least partly illuminate the object with a primary radiation.

5 Embodiment 14: A distance measuring device, in particular for use in a motor vehicle, comprising at least one detector according to any of the preceding
embodiments, wherein the detector is designed to determine at least one item of
geometrical information of at least one object, wherein the geometrical information
comprises at least one item of location information of the object, in particular a distance
between a motor vehicle and at least one object and preferably a distance between the
10 motor vehicle and at least one object selected from the group consisting of a further
motor vehicle, an obstacle, a cyclist and a pedestrian.

Embodiment 15: An imaging device for imaging at least one sample, wherein the
imaging device comprises at least one detector according to any of the preceding
15 embodiments relating to a detector, wherein the imaging device is designed to image a
plurality of partial regions of the sample onto the sensor region and to thereby generate
sensor signals assigned to the partial regions, wherein the imaging device is designed to
generate items of geometrical information of the respective partial regions from the
sensor signals, wherein the items of geometrical information comprise items of location
20 information.

Embodiment 16: A human-machine interface for exchanging at least one item of
information between a user and a machine, in particular for inputting control commands,
wherein the human-machine interface comprises at least one detector according to any
25 of the preceding claims relating to a detector wherein the human-machine interface is
designed to generate at least one item of geometrical information of the user (218) by
means of the detector wherein the human-machine interface is designed to assign to the
geometrical information at least one item of information, in particular at least one control
command.

30 Embodiment 17: An entertainment device for carrying out at least one entertainment
function, in particular a game, wherein the entertainment device comprises at least one
human-machine interface according to the preceding embodiment, wherein the
entertainment device is designed to enable at least one item of information to be input by
35 a player by means of the human-machine interface, wherein the entertainment device is
designed to vary the entertainment function in accordance with the information.

Embodiment 18: A security device for carrying out at least one security application in
particular for identifying and/or avoiding an access to data of an optical data storage
40 device, wherein the security device comprises at least one detector according to any of
the preceding embodiments relating to a detector, wherein the security device is

designed to identify, by means of the detector, impingement of focused electromagnetic radiation, in particular laser beams, on the security device and preferably to generate at least one warning signal.

- 5 Embodiment 19: A method for optically detecting at least one object, in particular using a detector according to any of the preceding embodiments relating to a detector, wherein at least one optical sensor is used, wherein the optical sensor has at least one sensor region, wherein electromagnetic radiation emerging from the object is fed to the optical sensor and thereby the sensor region is illuminated, wherein the optical sensor
10 generates at least one sensor signal in a manner dependent on the illumination of the sensor region, wherein the sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination.

- Embodiment 20: The method according to the preceding embodiment, wherein at
15 least one item of geometrical information of the object is generated from the sensor signal, in particular at least one item of location information of the object.

- Embodiment 21: The use of a detector according to any of the preceding
embodiments relating to a detector for a purpose of use, selected from the group
20 consisting of: distance measurement, in particular in traffic technology; imaging, in particular in microscopy; an entertainment application; a security application; a human-machine interface application.

- Embodiment 22: The use of an organic solar cell, in particular a dye solar cell,
25 preferably a solid dye solar cell, as optical sensor, wherein at least one sensor signal is generated in the use, wherein the sensor signal, given the same total power of an illumination is dependent on a geometry of the illumination on the organic solar cell, wherein at least one item of geometrical information of at least one object is generated from the sensor signal in the use.

30

Brief description of the figures

- Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent
35 claims. In this context, the particular features may be implemented alone or with several in combination. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

40

Specifically, in the figures:

- 5 figure 1 shows an exemplary embodiment of a detector according to the invention for optically detecting at least one object;
- figure 2 shows an exemplary embodiment of an optical sensor for use in a detector according to the invention;
- 10 figure 3 shows an exemplary embodiment of possible energy levels of a layer structure of an optical sensor in accordance with figure 2;
- figure 4 shows an exemplary embodiment of a signal dependence of an optical sensor according to the present invention on a beam geometry;
- 15 figures 5 and 6 show measurements for demonstrating a geometry independence of the sensor signals of conventional semiconductor sensors;
- figure 7 shows a typical frequency dependence of the signal of optical sensors for use in a detector according to the present invention;
- 20 figure 8 shows a joint illustration of the dependence of a sensor signal of the detector according to the present invention on a modulation frequency f and a beam geometry;
- 25 figure 9 shows a schematic exemplary embodiment of a distance measuring device according to the invention;
- figure 10 shows a result of a distance measurement using a distance measuring device according to the invention;
- 30 figure 11 shows a use of a distance measuring device according to the invention in motor vehicles;
- 35 figures 12A and 12B show a comparison of a conventional imaging device (figure 12A) having a confocal beam path with an imaging device according to the invention (figure 12B);
- 40 figures 13 and 14 show further possible embodiments of imaging devices according to the invention;

figure 15 shows an exemplary embodiment of a human-machine interface and of an entertainment device according to the present invention; and

figure 16 shows an exemplary embodiment of a security device according to the invention.

Exemplary embodiments

Detector

Figure 1 illustrates, in a highly schematic illustration, an exemplary embodiment of a detector 110 according to the invention for optically detecting at least one object 112. The detector 110 comprises an optical sensor 114, having a sensor region 116, for example having a sensor area 118. The sensor is designed to generate at least one sensor signal which can comprise for example a photocurrent I. By way of example, the optical sensor 114 can comprise at least one measuring device 120 by means of which at least one physical property of the optical sensor 114 can be detected, for example by at least one photocurrent and/or at least one photovoltage being measured.

Furthermore, the detector 110 in accordance with the exemplary embodiment illustrated in figure 1 comprises an evaluation device 122, which, by way of example, can be connected to the optical sensor 114 and/or else can be wholly or partly integrated into the optical sensor 114, or vice-versa. Said evaluation device 122 can be designed, for example, to pick up the at least one sensor signal of the optical sensor 114 directly or indirectly. The evaluation device 122 can comprise for example at least one data processing device 124 and/or at least one data storage device 126. Furthermore, the evaluation device 122 can have for example a unidirectional or bidirectional interface 128, for example in order to be able to exchange data and/or control commands unidirectionally or bidirectionally with other devices.

Furthermore, in the exemplary embodiment illustrated in figure 1, the detector 110 optionally comprises at least one transfer device 130, for example having at least one lens 132 and/or other imaging or non-imaging elements. The optional transfer device 130 is designed to feed electromagnetic radiation 134 emerging from the object 112 to the optical sensor 114 and to illuminate the sensor region 116 in the process. By way of example, a light spot 136 having a diameter or equivalent diameter can arise on the sensor area 118 in this way. Said light spot 136 can have an area A, for example, wherein the total power of the electromagnetic radiation 134 is designated by p in figure 1, and the flux p/A is designated by Φ . It is pointed out that the optional transfer device

130 in figure 1 is only indicated schematically. Said transfer device can also be embodied in various other ways.

Furthermore, in the exemplary embodiment in accordance with figure 1, the detector 110 optionally comprises at least one modulation device 138 for modulating the electromagnetic rays 134. In particular, said modulation device 138 can comprise at least one beam interrupter 140, for example a so-called chopper wheel.

Furthermore, the detector 110 can optionally comprise at least one illumination source 142. Said illumination source 142 can be designed to irradiate the object 112 or a partial region of the object 112 with a primary radiation 144. By way of example, said primary radiation 144 can be reflected and/or scattered at the object 112. During this reflection and/or scattering, which can be supported for example by at least one optional reflective surface 146 of the object 112, the electromagnetic radiation 134 is optionally generated, which is then fed to the optical sensor 114, for example by means of the transfer device 130. As an alternative or in addition to reflection and/or scattering, however, the primary radiation 144 can also excite the object 112 or a part thereof to emit the electromagnetic radiation 134, for example in the form of an excitation of fluorescence and/or excitation of phosphorescence. In this case, the object 112 can comprise for example at least one fluorescent material and/or at least one phosphorescent material.

Optical sensor

As explained above, it is particularly preferred if the optical sensor 114 comprises at least one semiconductor detector 148, in particular an organic semiconductor detector 150, preferably an organic solar cell 152 and particularly preferably a dye solar cell 154. An exemplary embodiment of such optical sensors 114 is shown in figure 2 in a schematic sectional illustration through one possible layer structure. The dye solar cell 154 illustrated there can comprise for example a substrate 156, for example a glass substrate. However, other substrates 156 can also be used, as described above, for example a plastic substrate or else multilayer or laminate substrates. In the exemplary embodiment illustrated, at least one first electrode 158, which can also be designated as a working electrode and which can preferably be embodied in a transparent fashion, as described above, is applied on said substrate 156. At least one blocking layer 160 (also designated hereinafter as buffer layer) of an optional metal oxide can in turn optionally be applied on said first electrode 158, said layer preferably being nonporous and/or nonparticulate. An n-semiconducting metal oxide 162 is applied in turn on said layer, said metal oxide being sensitized with a dye 164. As explained above, this sensitization can be effected by applying a separate layer of the dye 164 and/or by complete or partial impregnation, wetting or mixing of the, preferably porous and/or particulate, n-semiconducting metal oxide 162 with the dye 164.

An organic p-type semiconductor 166 is applied to the dye 164. Said organic p-type semiconductor 166 is preferably solid in the finished state of the dye solar cell 154, that is to say embodied as a solid organic p-type semiconductor 166. A plurality of layers of this solid organic p-type semiconductor 166 can also be provided. A second electrode 168, which is also designated as a counterelectrode, is applied on the p-type semiconductor 166.

The layers illustrated in figure 2 together form a layer structure 170, which is preferably shielded from a surrounding atmosphere by an encapsulation 172, for example in order to shield the layer structure 170 from a surrounding atmosphere, for example in order to completely or partly protect the layer structure 170 from oxygen and/or moisture. One or both of the electrodes 158, 168 or connection contacts of said electrodes 158, 168 can be led out from the encapsulation 172, in order to be able to provide one or a plurality of contact-connection areas outside the encapsulation 172.

Figure 3 shows, in a highly schematic manner, one possible energy level diagram of the dye solar cell 154, for example according to figure 2. The illustration shows the Fermi levels 174 of the first electrode 158 and of the second electrode 168, and the HOMOs (Highest Occupied Molecular Orbitals) 176 and the LUMOs (Lowest Unoccupied Molecular Orbitals) 178 of the layers 160, 162 (which can comprise the same material, for example TiO_2) of the dye 164 (indicated by way of example, with a HOMO level of 5.7 eV) and of the p-type semiconductor 166 (also designated as HTL, Hole Transport Layer). FTO (fluorine-doped tin oxide) and silver are specified by way of example as materials for the first electrode 158 and the second electrode 168. It is pointed out that other electrode materials can also be used and that, for example, the order of the first electrode and the second electrode can also be reversed and that, for example, it is also possible to use a nontransparent substrate 156 in combination with a transparent second electrode 168 and optionally a transparent encapsulation 172, or an inverse layer structure. The organic semiconductor detector 150 can furthermore optionally comprise further elements that are not illustrated in figures 2 and 3.

Production of a dye solar cell

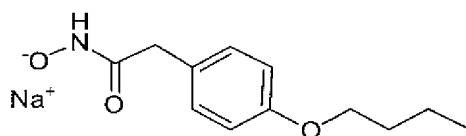
As an example of production of an organic solar cell 152, production of a dye solar cell 154 with a solid p-type semiconductor 166 is described below by way of example.

As the base material and substrate 156, glass plates which had been coated with fluorine-doped tin oxide (FTO) as the first electrode (working electrode) 158 and were of dimensions 25 mm × 25 mm × 3 mm (Hartford Glass) were used, which were treated successively in an ultrasound bath with glass cleaner (RBS 35), demineralized water and

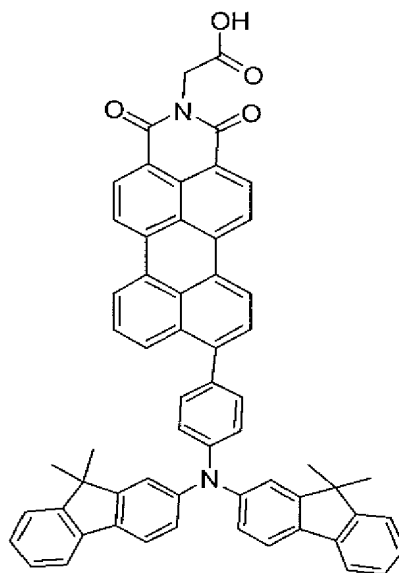
acetone, for 5 min in each case, then boiled in isopropanol for 10 min and dried in a nitrogen stream.

To produce an optional solid TiO₂ buffer layer 160, a spray pyrolysis process was used. Thereon, as an n-semiconducting metal oxide 162, a TiO₂ paste (Dyesol) which comprises TiO₂ particles with a diameter of 25 nm in a terpeneol/ethylcellulose dispersion was spun on with a spin-coater at 4500 rpm and dried at 90°C for 30 min. After heating to 450°C for 45 min and a sintering step at 450°C for 30 minutes, a TiO₂ layer thickness of approximately 1.8 μm was obtained.

After removal from the drying cabinet, the sample was cooled to 80°C and immersed into a 5 mM solution of an additive ID662 (obtainable according to example H below, for example) for 12 h and subsequently into a 0.5 mM solution of a dye 164 in dichloromethane for 1 h. The dye 164 used was the dye ID504 (obtainable according to example G below, for example), but a large number of other dyes can be used and the choice of dye generally has only little influence on the effect described above. In particular, the dye can be adapted to the respective purpose of use and the wavelengths of the electromagnetic radiation 134 used.



ID662



ID504

After removal from the solution, the sample was subsequently rinsed with the same solvent and dried in a nitrogen stream. The samples obtained in this way were subsequently dried at 40°C under reduced pressure.

Next, a p-type semiconductor 166 solution was spun on. For this purpose, a solution of 0.163 M spiro-MeOTAD (Merck) and 20 mM LiN(SO₂CF₃)₂ (Aldrich) in chlorobenzene

was made up. 125 μ l of this solution were applied to the sample and allowed to act for 60 s. Thereafter, the supernatant solution was spun off at 2000 rpm for 30 s.

Finally, a metal back electrode was applied as a second electrode 168 by thermal metal vaporization under reduced pressure. The metal used was Ag, which was vaporized at a rate of 3 $\text{\AA}/\text{s}$ at a pressure of approx. $2 \cdot 10^{-6}$ mbar, so as to give a layer thickness of about 200 nm.

Dependence of the sensor signal on the geometry of the illumination and the modulation frequency

With such dye solar cells 154 as optical sensor 114, firstly the method of operation of the detector 110 illustrated in figure 1 will be described. Figure 4 shows an example of a measurement of a photocurrent I (normalized to the maximum value) as a function of a position x (indicated in millimeters) of a lens 132, wherein the lens 132 focuses electromagnetic radiation 134 onto a sensor area 118 of the dye solar cell 154. The sensor area 118 can be for example a surface of the substrate 156 of the layer structure in figure 2, said surface facing away from the first electrode 158. Alternatively, however, the sensor area 118 can also be arranged in a layer plane of the organic layer structure, for example in the region of the dye 164. The exact position of said sensor area 118 within the organic semiconductor detector 150 is greatly dependent on the individual physical processes and is generally not of importance for the exact functioning of the detector 110. Alternatively, the entire organic layer structure or a part thereof can be regarded as a sensor region 116.

During the measurement in figure 4, the lens 132 is displaced relative to the optical sensor 114 in figure 1 in a direction perpendicular to the sensor area 118. As a result, the diameter or equivalent diameter of the light spot 136 on the sensor area 118 changes. By way of example, a lens having a focal length of 50 mm can be used, given a beam diameter of 25 mm, for example, which leads to a size of the light spot 136 of less than 2 mm, for example. A location of the lens 132 at which optimum focusing occurs is arbitrarily indicated as location $x = 0$ in figure 4. As an alternative or in addition to a displacement of the lens 132 or alteration of the optional transfer device 130, other measures could also be implemented in order to vary a focusing of the electromagnetic radiation 134 on the sensor area 118 or a diameter or equivalent diameter or some other geometry of the electromagnetic radiation 134 in the sensor region 116 and in particular on the sensor area 118.

It can clearly be discerned in figure 1 that the signal of the optical sensor 114, in this case the photocurrent, is greatly dependent on the geometry of the illumination. Outside a maximum at $x = 0$, the photocurrent falls to less than 10% of its maximum value.

In comparison therewith, figures 5 and 6 illustrate corresponding measurements on silicon diodes (figure 5) and germanium diodes (figure 6). In this case, diodes of the Hamamatsu S2386-8k type were used as silicon diodes, and diodes of the Hamamatsu J16-5SP-R03M type were used as germanium diodes. It can clearly be discerned that the signal of such semiconductor diodes, apart from marginal effects in figure 6, which may be attributable to a boundary of the sensor area, and apart from noise effects, which are below 10%, does not have the above-described geometry dependence of the sensor signals on the illumination of the sensor region 116. In other words, in the case of the optical sensors 114 of a conventional type that are used in figures 5 and 6, the sensor signal, given the same total power, is substantially independent of a geometry of the illumination of the sensor region 116 or the sensor area 118; wherein, by way of example, fluctuations of less than 10%, preferably of less than 5%, can still be accepted and should not yet be regarded as the effect according to the invention. In contrast thereto, the measurement by means of the detector 110 according to the invention and the optical sensor 114 in accordance with figure 2, for example, shows a pronounced geometry-dependent effect in the sensor signal.

This effect is generally dependent on a modulation frequency of the illumination of the optical sensor 114. The measurements in figures 4-6 were carried out using a modulation device 138 by means of which the electromagnetic radiation 134 was modulated before impinging on the sensor region 116 with a frequency of typically 30 Hz to 100 Hz. The modulation device 138 used was an electronic modulation device in the form of a pulsed current source having a duty cycle of 1:1. This is illustrated once again schematically in figure 7 for measurements analogous to the measurement in accordance with figure 4. Here the sensor signal I/I_0 , that is to say the sensor signal normalized to its maximum value I_0 , is plotted as a function of the spatial coordinate x of the lens 132 used for the focusing, but this can be exchanged, in principle, for any other parameter that characterizes the geometry of the illumination on the sensor region 116 or the sensor area 118. The illustration schematically shows a curve of the sensor signal at a modulation frequency $f = 0$ Hz and a modulation frequency f that is greater than a limiting frequency f_{Base} dependent on the type of optical sensor 114.

From the spatial coordinate x it is possible, by using imaging equations or else by simple observation, for example, to deduce the geometry of the illumination. On the other hand it is possible, for example given fixed positioning of the lens 132 or fixed embodiment of the transfer device 130, to deduce for example at least one spatial coordinate of the object 112 from the sensor signal and/or from the geometry of the illumination. For this purpose, calibration curves can be recorded and/or calculated, which can be stored for example in the evaluation device 122 and/or the data storage device 126. Figure 8 shows one possible example of such calibration curves in a three-dimensional

illustration. Here the photocurrent I is plotted as a function of the spatial coordinate x of the lens 132 and as a function of the modulation frequency f , specified in hertz. During this measurement, a photocurrent of 7 nA occurred as maximum photocurrent. In this case, the location of the maximum is chosen arbitrarily, and so in this case, without
5 restricting the usability of the results, the location $x = 0$ does not specify the location of the maximum. The curves clearly show that the sensor signal I is a distinct function of the spatial coordinate x , and hence the geometry of the illumination of the sensor region 116, and the frequency f .

10 In a manner similar to that illustrated in figure 8, a multiplicity of other calibration curves can be recorded. By way of example, it is possible to record calibration curves in which, instead of the spatial coordinate x , a coordinate is directly specified which characterizes at least one item of geometrical information of the object 112. Given a known modulation frequency f and measured intensity I , there is then only an ambiguity with regard to the
15 fact of on which side of the maximum the measurement result lies. This ambiguity can be resolved for example by prior knowledge of on which side of the maximum the measurement must lie, or by a plurality of measurements. The evaluation device 122 can be designed to resolve this ambiguity, for example by programming.

20 Distance measuring device

Figure 9 shows a highly schematic example of a distance measuring device 180, which in principle, can be embodied analogously to the detector 110 in accordance with figure 1 and/or can comprise such a detector 110. In this case, the at least one item of
25 geometrical information generated by means of the detector is embodied as location information or comprises at least one item of location information, and/or the distance measuring device 180 is designed to generate the at least one item of location information from the at least one item of geometrical information. The distance measuring device 180 therefore serves to generate at least one item of location
30 information about at least one object 112, which is illustrated symbolically as a pedestrian in figure 9. By way of example, this location information can comprise at least one distance d_1 or d_2 between the object 112 and the detector 110 or a part of the detector 110, for example a distance between the object 112 and a lens 132 of the detector 110.

35 The structure of the detector 110 in figure 9 can correspond, in principle, to the structure in accordance with figure 1. An optional modulation device 138, which can be arranged for example between object 112 and lens 132 and/or between lens 132 and optical sensor 114 and/or which can be integrated into the at least one illumination source 142, is not illustrated in figure 9.
40

Figure 9 illustrates by way of example two objects 112 at different distances d_1 , d_2 . These cause for example on the sensor region 116, and in particular the sensor area 118, different light spots 182, 184, with different geometries. Sensor signals which are picked up by the optical sensor 114 in accordance with the geometry of said light spots 182, 184 are correspondingly different. The evaluation device 122 can be designed, for example using calibration curves in accordance with figure 5 and/or other types of calibration curves, to generate from said sensor signals an item of geometrical information, in particular at least one item of location information, of the object 112, for example an item of information about the distance d_1 or d_2 between the objects 112 and the detector 110.

The detector 110 or distance measuring device 180 can be embodied for example in integral fashion or else in a multipartite fashion and can comprise for example at least one housing 186 into which, by way of example, the optical sensor 114 and/or the evaluation device 122 and optionally the at least one illumination source 142 can be integrated. Alternatively or additionally, however, it is also possible to provide multipartite embodiments, for example by one or a plurality of illumination sources 142 being arranged separately, for example in a separate housing. Overall, the distance measuring device 180 in accordance with figure 9 can have for example the form of a simple, camera-like arrangement and can for example also be embodied as a handheld device, that is to say as a device which can be carried by a user or can be taken along solely on the basis of muscle power. Various other configurations are possible.

The distance measuring device 180 in accordance with figure 9 can optionally comprise exactly one optical sensor 114. Alternatively, the distance measuring device 180 can also be embodied as a camera, for example by a plurality of optical sensors 114 and/or one or a plurality of optical sensors 114 having a plurality of sensor regions 116 being used, for example in a two-dimensional matrix arrangement and/or a three-dimensional matrix arrangement. One advantage of such cameras resides, in comparison with conventional recording techniques, in a shorter recording time, for example with a pixel clock of 10 MHz, corresponding to a time of 0.1 ms for 1000 pixels. Recording times for two-dimensional and three-dimensional recordings can generally be made equally short. In contrast thereto, conventional cameras, in order to generate three-dimensional images, generally have to record n images, which necessitates recording times in the Hz - kHz range. Furthermore, conventional cameras of the type mentioned are generally of comparatively low luminosity.

Evidence of the functioning of the arrangement illustrated in figure 9 is shown in figure 10. Here a distance x_M (corresponding, for example, to the spatial coordinates d_1 and d_2 in figure 9), specified in cm or 10 mm, measured by means of the distance measuring device 180 in accordance with figure 9 is plotted as a function of the actual distance x_s ,

likewise specified in cm or 10 mm. The measurements were carried out using a transfer device 130 in the form of an objective having an achromatic lens, an illumination source 142 in the form of a frequency-doubled neodymium laser and a dye solar cell 154 in accordance with the above description having a sensor area 118 of approximately 4 mm².

The measurements clearly show that, for example here in the case of distances of less than 1 m, the distance measurements by means of the distance measuring device 180 according to the invention correspond excellently to the actual distances. The distance measurement is readily also adaptable to other distances, for example by adapting the transfer device 130 and/or the illumination source 142 and/or the optical sensor 114. By way of example, microscopic distances can be identified by corresponding focusing of the primary radiation 144 and/or by corresponding embodiment of the optional transfer device 130. Greater distances than the distances shown in figure 10 can likewise be identified for example by using laser beams and/or corresponding optics in the transfer device 130. However, the measurement in figure 10 clearly shows that the measurement principle proposed can be realized.

This measurement principle of a distance measuring device 180 can be used very diversely, for example for determining at least one spatial coordinate of at least one object 112, of a plurality of objects, or for spatially detecting an entire environment, for example in a scanning method in which, within a visual range of the detector 110, one or a plurality of objects, a plurality of points of at least one object or contours of one or a plurality of objects are detected successively or simultaneously. In this respect, the distance measuring device 180 in the embodiment illustrated or another embodiment according to the invention can be used and embodied very diversely and can be used for example generally for one-dimensional, for two-dimensional or for three-dimensional detection of surroundings.

Figure 11 shows how a distance measuring device 180 of the type proposed can be used for example in motor vehicle technology, for example for detecting the surroundings. By way of example, a detector 110 or a part thereof can be arranged on a front side 188 of a first motor vehicle 190. An illumination source 142, for example in the form of an infrared transmitter, which directly emits the electromagnetic radiation 134 in the direction of the detector 110, can be arranged on a rear side 192 of a second motor vehicle 194, which in this case can function as the object 112. Said illumination source 142 can therefore be regarded as a constituent part of the detector 110 or else as a separate component. Alternatively or additionally, the at least one illumination source 142 or a part thereof could, however, also be arranged in the first motor vehicle 190 and emit primary radiation in the direction of the second motor vehicle 194, where it is for example reflected and/or scattered and/or excites luminescence, such that

electromagnetic radiation 134 in turn propagates back to the first motor vehicle 190 and the detector 110. Various embodiments are possible.

Generally, the illumination can be embodied in such a way that a geometry of an illumination of a sensor region 116 of the detector 110 changes with a distance d between the motor vehicles 190, 194. This can be effected for example once again by means of a corresponding embodiment of the illumination source 142 and/or an optional transfer device 130 of the detector 110. Consequently, a sensor signal of the detector 110 also changes, from which the distance d and/or some other geometrical information can in turn be deduced. This deduction can be effected for example once again by using one or a plurality of calibration functions, in particular once again by means of an evaluation device 122. The latter can be embodied separately, but can also be integrated for example wholly or partly into a device already present in the motor vehicle, for example an engine controller.

By way of example, a plurality of motor vehicles 190, 194 can each be equipped with such distance measuring devices 180. The at least one item of geometrical information, in particular the at least one item of location information, about the motor vehicle ahead and/or the following motor vehicle can for example be brought to the attention of a driver and/or used to generate warning signals and/or to intervene automatically in a driving behavior of the motor vehicle.

The structure shown in figure 11 can be modified in diverse ways in the context of the present invention. Thus, by way of example, the at least one illumination source 142 can also be integrated directly into the detector 110, such that, by way of example, as described above, the detector 110 can emit primary radiation 144 in the direction of, for example, another motor vehicle, in particular a motor vehicle ahead. Said radiation can then be for example reflected and/or scattered and/or used for excitation there and can be directed as electromagnetic radiation 134 back to the detector 110 again. Thus, by way of example, motor vehicles could also be equipped with one or a plurality of reflectors which can foster this reflection process. Furthermore, generally the at least one detector can be embodied for carrying out monitoring of the surroundings of a motor vehicle, for example by a viewing direction of the detector 110 and/or a radiation direction of the primary radiation 144 being made variable. Various other embodiments are conceivable.

In contrast for example to conventional triangulation methods, propagation time methods, image evaluation methods or similar methods for identifying distances, the detector 110 according to the invention and the distance measuring device 180 according to the invention afford the advantage of a simpler structure, in particular without the need for digital data analysis, a faster reaction and a more cost-effective embodiment.

Imaging device

Figures 12A-14 show that the detector 110 according to the invention can advantageously also be used in an imaging device 196 for imaging at least one sample 198. In this case, the sample 198 serves as the object 112. The measurement principle of the imaging device 196 can, in principle, be analogous, similar or even identical to the principle of the distance measuring device 180 described above, although (which is also possible, in principle, in the context of the distance measuring device 180) a plurality of points and/or regions of the at least one object 112 or of the sample 198 can be detected.

Figure 12A shows an imaging device 196 corresponding to the prior art in the form of a conventional confocal microscope having a confocal beam path. The confocal microscope has an illumination source 142, for example a laser, the focus of which is designated symbolically by the reference numeral 202 in figure 12A. Primary radiation 144 emerging from said illumination source 142 is directed by a beam splitter 204, for example a partly transmissive mirror and/or a beam splitter cube, onto the sample 198 and is focused for example by means of a lens 132 and/or an objective. Electromagnetic radiation 134 emerging from the sample 198 is directed by means of the beam splitter 204 onto a diaphragm 206, which is confocal with respect to the focus 202, and a sensor 208 lying behind said diaphragm. What is achieved by the confocal structure is that only light from the region of the image of the focus 202 can pass through the diaphragm 206 by means of the lens 132 in the sample 198 and can be detected by the sensor 208, for example a conventional semiconductor sensor. Light from other regions is at least substantially masked out by the diaphragm 206.

By contrast, figure 12B illustrates an exemplary embodiment of an imaging device 196 according to the invention. The structure can for example substantially correspond to the structure in accordance with figure 12A, but the diaphragm 206 can be obviated in that the sensor 208 can be replaced by an optical sensor 114 having the inventive properties of sensitivity relative to a geometry of the illumination of the sensor region 116.

From at least one known relationship between the sensor signal and at least one item of geometrical information (optionally by means of at least one intermediate step of determining a geometry of the illumination from the sensor signal and at least one optional intermediate step of determining the geometrical information from the geometry of the illumination), it is then possible to deduce from what depth of the sample 198 the electromagnetic radiation 134 originates. This can for example be effected once again by means of the evaluation device 122.

In this way, by way of example, an item of location information about the location of the origin of the electromagnetic radiation 134 can be generated, such that the sample 198 can be scanned, for example, and an item of information about a surface of the sample is in each case obtained, for example. Alternatively or additionally, it is also possible to obtain items of information from deeper regions of the sample 198. The optional transfer device 130 can comprise for example at least one scanning device, for example one or a plurality of movable mirrors, such that, for example, a line scan or else an area scan can be carried out on the sample 198. Such scanning devices are known for example from the field of confocal microscopy.

Furthermore, the optional transfer device 130 in the structure shown in figure 12B or else in other embodiments of devices and detectors 110 according to the invention can also comprise one or a plurality of wavelength-selective elements, for example wavelength-selective filters and/or wavelength-selective deflection elements such as, for example, one or a plurality of dichroic mirrors.

One advantage of the imaging device 196 illustrated in figure 12B in comparison with the conventional imaging device in accordance with figure 12A consists generally in a higher z-resolution, that is to say a higher resolution in the direction perpendicular to the sample 198, in a higher light intensity, since the need to use a diaphragm 206 is obviated, a higher insensitivity toward a transverse offset of an optical axis and the diaphragm 206, and a high intensity at an optimum z-resolution. Furthermore, the structure of the imaging device 196 according to the invention can be greatly simplified in comparison with conventional imaging devices 196, since generally, for example, by means of the detector 110, it is possible to generate not only an item of information about a total power of the electromagnetic radiation 134, but at least one item of geometrical information, for example at least one item of location information.

Figures 13 and 14 show modifications of the structure of the imaging device shown in figure 12B, once again in a highly schematic illustration. Thus, figure 13 shows a structure in which the detector 110 comprises a stack 210 of optical sensors 114. By way of example, the optical sensors 114 can be embodied such that they are fully or partly transparent. By means of these different sensors 114, which for example can in each case be aligned with the focus from a specific layer 212 or depth of the sample 198 or can be designed to receive signals corresponding to the electromagnetic radiation 134 emerging from said layer 212, a depth resolution in the z-direction perpendicular to the sample 198 can be automatically effected, without the optional transfer device 130 having to be designed for a movement in the z-direction. However, a scanning device parallel to the sample 198, which is designated by x, y in figures 13 and 14, can nevertheless still be provided, for example for carrying out a line or area scan.

Figure 14 shows a development of the exemplary embodiment in accordance with figure 13. This exemplary embodiment shows that in the detector 110 generally and in particular preferably in an imaging device 196, according to the invention, one or a plurality of sensors 114 can be provided, which can in each case have a plurality of sensor regions 116, for example in a matrix arrangement. This is merely indicated schematically in figure 14. For each of said sensor regions 116, by way of example, an evaluation method in accordance with the above description can be carried out, for example in order to generate at least one item of geometrical information, in particular at least one item of location information, about the at least one object 112 or a partial region of the object 112, for example that partial region from which the electromagnetic radiation 134 that impinges on the respective sensor region 116 originates. In order to ensure the spatial resolution, the optional transfer device 130 can have for example a plurality of lenses 132. By way of example, the transfer device 130, as illustrated in figure 14, can have at least one lens array in order, for example, in a targeted manner, to assign lateral regions of a layer of the sample 212 in each case to a sensor region 116 of an optical sensor 114. In this way, a lateral spatial resolution in the x, y direction parallel to the layers 212 can simultaneously be effected by evaluation of the sensor signals of the individual sensor regions 116 of the sensor array, and depth information can be effected by evaluation of the optical sensors 114 arranged one behind another. In this case, a three-dimensional region of the sample 198 can be imaged simultaneously.

Human-machine interface and entertainment device

Figure 15 shows an exemplary embodiment of a human-machine interface 214 according to the invention, which can simultaneously also be embodied as an exemplary embodiment of an entertainment device 216 or can be a constituent part of such an entertainment device 216.

By way of example, at least one detector 110 according to the present invention can once again be provided, for example in accordance with one or more of the embodiments described above, with one or a plurality of optical sensors 114. Further elements of the detector 110 can be provided, which are not illustrated in figure 15, such as, for example, elements of an optional transfer device 130. Furthermore, one or a plurality of illumination sources 142 can be provided. Generally, with regard to possible embodiments of the detector 110, reference can be made for example to the description above.

The human-machine interface 214 can be designed to make it possible to exchange at least one item of information between a user 218 of a machine 220, which is merely indicated in figure 15, and the machine 220, for example to exchange control commands. The machine 220 can comprise, in principle, any desired device having at least one

function which can be controlled and/or influenced in some way. At least one optional evaluation device 122 of the at least one detector 110 or a part thereof can, as indicated in figure 15, be wholly or partly integrated into said machine 220, but can, in principle, also be formed separately from the machine 220.

5

The human-machine interface can be designed for example to generate, by means of the detector 112, at least one item of geometrical information, in particular at least one item of location information, of the user and to assign to the geometrical information at least one item of information, in particular at least one control command. For this purpose, by way of example, by means of the detector 110, a movement and/or a change in posture of the user 218 can be identified, for example, as indicated in figure 15, a hand movement and/or a specific hand posture. If, by way of example, such a hand movement and/or hand posture of a specific type is identified, for example by evaluation of one or a plurality of items of geometrical information, in particular items of location information, of the user 218, then it is possible to recognize, for example by comparison with a corresponding command list, that the user 218 would like to effect a specific input, for example would like to give the machine 220 a control command. As an alternative or in addition to direct geometrical information about the actual user 218, it is also possible, for example, to generate at least one item of geometrical information about a garment of the user 218 and/or an article moved by the user 218.

The machine 220 can furthermore comprise one or a plurality of further human-machine interfaces, which need not necessarily be embodied according to the invention, for example, as indicated in figure 15, at least one display 222 and/or at least one keyboard 224. The machine 220 can be, in principle, any desired type of machine or combination of machines.

In the context of an entertainment device 216, said machine 220 can be designed for example to carry out at least one entertainment function, for example at least one game, in particular with at least one graphical display on the display 222 and optionally a corresponding audio output. The user 218 can input at least one item of information for example via the human-machine interface, wherein the entertainment device is designed to alter the entertainment function in accordance with the information. By way of example, specific movements of virtual articles, for example of virtual persons in a game and/or movements of virtual vehicles in a game, can be controlled by means of corresponding movements of the user 218, which are in turn recognized by the detector 110. Other types of control of at least one entertainment function by the user 218 by means of the at least one detector 110 are also possible.

40 Security device

Figure 16 shows an exemplary embodiment of a security device 226 according to the invention. Said security device 226 can be integrated for example into a memory device 228 having at least one optical data storage device 230. The security device 226 once again comprises at least one detector 110, for example of the type described above. The detector 110 once again comprises at least one optical sensor 114 and optionally at least one transfer device 130, for example at least one lens 132, for example in the form of a printed lens and/or a printed lens array. As explained above, however, the detector 110 of the security device 226 can also be embodied without a transfer device 130. Furthermore, the detector 110 once again comprises at least one evaluation device 122 for evaluating signals of the optical sensor 114.

The security device 226 is designed to carry out at least one security application. In particular, it can be designed to identify and/or avoid access to data of the optical data storage device 230. In general, one or a plurality of reading beams 232 are used for reading out the data of the optical data storage device 230. By way of example, this can involve focused reading beams in order to read individual information modules of the optical data storage device 230, the lateral resolution of which is typically in the range of a few hundred nanometers to a few tens or a few hundreds of micrometers. The security device 226 can be arranged so close to the optical data storage device 230 and/or be completely or partly integrated into the optical data storage device 230 such that the reading beam 232 or the reading beams 232 also impinge on the security device 226.

In contrast to impingement of diffuse light on the optical sensor 114 and the sensor region 116 thereof, which would lead to a comparatively low sensor signal, the impingement of focused light, for example of a focused reading beam 232, in particular of a focused laser beam, generally leads to a strongly boosted sensor signal, which can be identified by the evaluation device 122. This effect generally occurs - on account of the above-described geometry dependence of the sensor signal - even when the reading beam 232 has a very weak total power which does not exceed a total power of ambient light impinging on the sensor region 116.

Accordingly, the evaluation device 122 and/or other components of the memory device 228 or some other device which comprises the security device 226 can implement corresponding measures if a reading beam 232 is identified. In particular, the evaluation device 122 can output, via at least one interface 234, at least one warning signal which indicates, for example, that an attempt at reading by means of focused reading beams 232 has been effected. In accordance with this identification and the warning signal, a memory bit can be altered, for example, which indicates that the optical data storage device 230 has already been read once. Alternatively or additionally, other measures can be implemented, for example outputting of a corresponding warning to a user and/or

administrator and/or destruction of the optical data storage device 230, such that an optical data storage device 230 that can only be read once can be generated.

5 The security device 226 described can be integrated into various types of devices. Thus, it is possible, for example, to equip optical data storage devices 230 in the form of optical ROMs according to the invention. Optical data storage devices 230 in the form, for example, of holographic data storage devices and/or in the form of bar codes can also be embodied accordingly. In this way, it is possible for example to ensure, for example by the destruction of the optical data storage device 230 after the latter has been read once,
10 that the optical data storage device 230 and/or the memory device 228, for example an entry card and/or a ticket, can only be used once. Various other embodiments of the security device are conceivable.

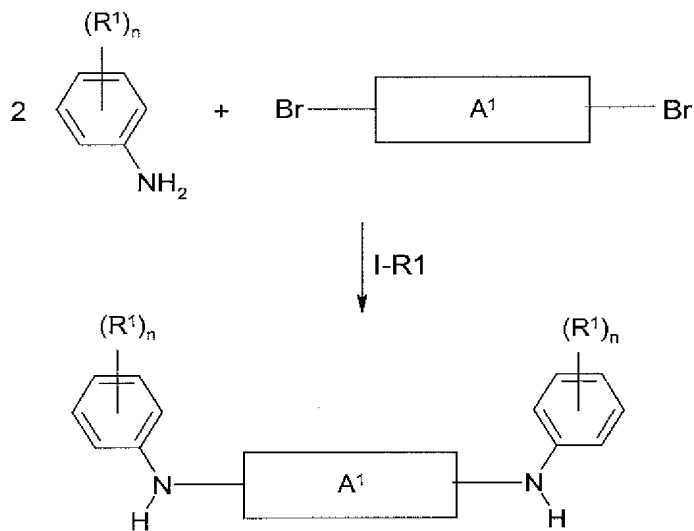
Synthesis examples:

Syntheses of various compounds which can be used in dye solar cells 154 in the context of the present invention, in particular as p-type semiconductors 166, are listed by way of example hereinafter. Possible syntheses of compounds of the formula (I) are described, for example:

(A) General synthesis schemes for preparation of compounds of the formula I:

10 (a) *Synthesis route I:*

(a1) *Synthesis step I-R1:*



15

The synthesis in synthesis step I-R1 was based on the references cited below:

a) Liu, Yunqi; Ma, Hong; Jen, Alex K-Y.; CHCOFS; Chem. Commun.; 24; 1998; 2747-2748,

20

b) Goodson, Felix E.; Hauck, Sheila; Hartwig, John F.; J. Am. Chem. Soc.; 121; 33; 1999; 7527 – 7539,

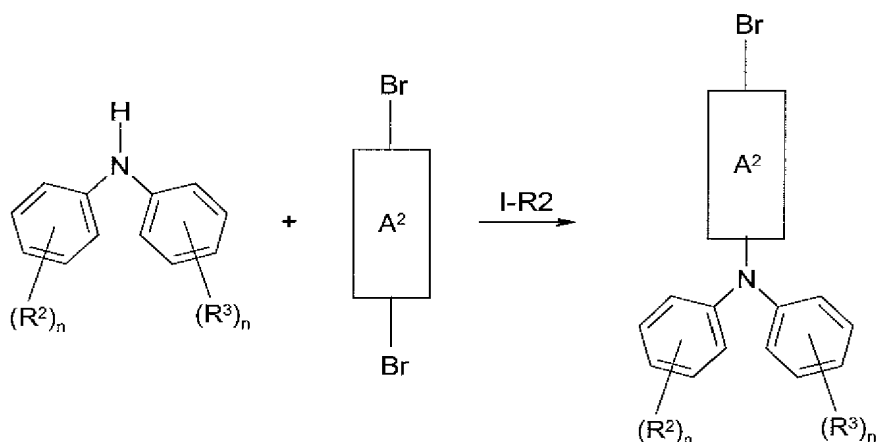
c) Shen, Jiun Yi; Lee, Chung Ying; Huang, Tai-Hsiang; Lin, Jiann T.; Tao, Yu-Tai; Chien, Chin-Hsiung; Tsai, Chiitang; J. Mater. Chem.; 15; 25; 2005; 2455 – 2463,

5 d) Huang, Ping-Hsin; Shen, Jiun-Yi; Pu, Shin-Chien; Wen, Yuh-Sheng; Lin, Jiann T.; Chou, Pi-Tai; Yeh, Ming-Chang P.; J. Mater. Chem.; 16; 9; 2006; 850 – 857,

e) Hirata, Narukuni; Kroeze, Jessica E.; Park, Taiho; Jones, David; Haque, Saif A.; Holmes, Andrew B.; Durrant, James R.; Chem. Commun.; 5; 2006; 535 - 537.

10

(a2) *Synthesis step I-R2:*



The synthesis in synthesis step I-R2 was based on the references cited below:

15

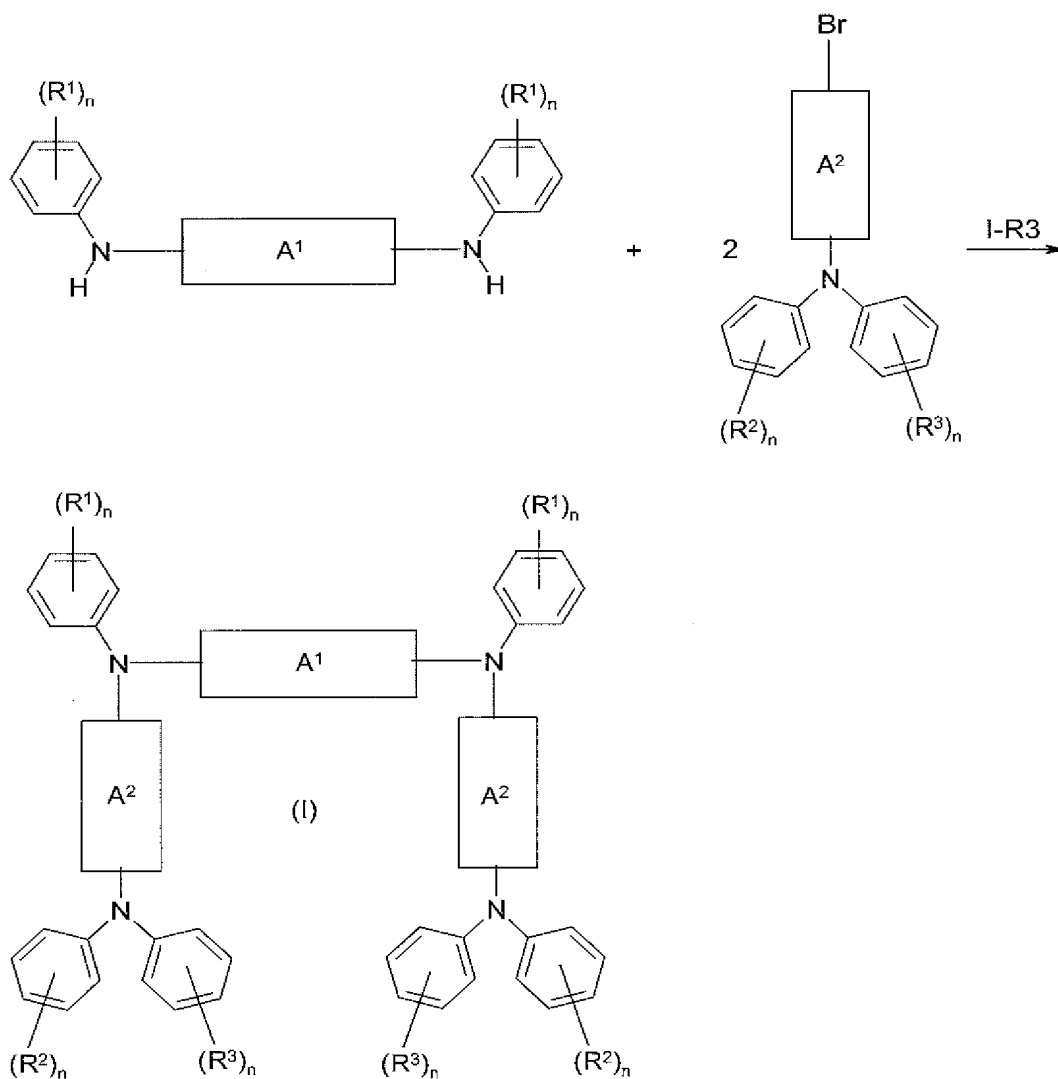
a) Huang, Qinglan; Evmenenko, Guennadi; Dutta, Pulak; Marks, Tobin J.; J. Am. Chem. Soc.; 125; 48; 2003; 14704 – 14705,

20 b) Bacher, Erwin; Bayerl, Michael; Rudati, Paula; Reckefuss, Nina; Mueller, C. David; Meerholz, Klaus; Nuyken, Oskar; Macromolecules; EN; 38; 5; 2005; 1640 – 1647,

c) Li, Zhong Hui; Wong, Man Shing; Tao, Ye; D'Iorio, Marie; J. Org. Chem.; EN; 69; 3; 2004; 921 - 927.

(a3) *Synthesis step I-R3:*

5



The synthesis in synthesis step I-R3 was based on the reference cited below:

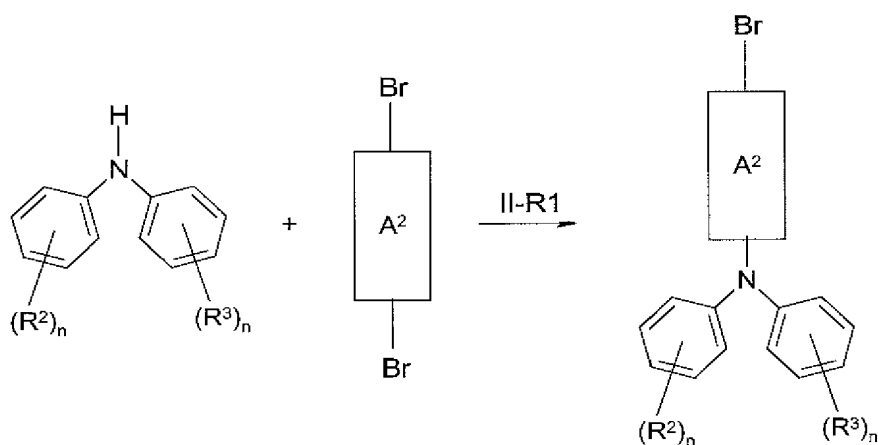
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J. Grazulevicius; J. of Photochem. and Photobio., A: Chemistry 2004 162(2-3), 249-252.

The compounds of the formula I can be prepared via the sequence of synthesis steps shown above in synthesis route I. In steps (I-R1) to (I-R3), the reactants can be coupled, for example, by Ullmann reaction with copper as a catalyst or under palladium catalysis.

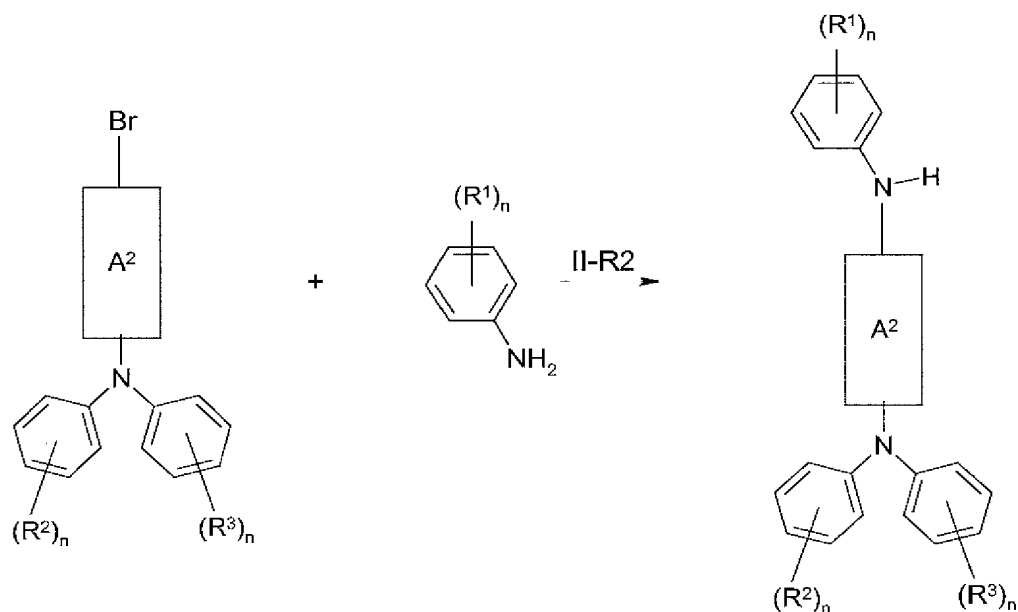
5 (b) *Synthesis route II:*

(b1) *Synthesis step II-R1:*



10 The synthesis in synthesis step II-R1 was based on the references cited under I-R2.

(b2) *Synthesis step II-R2:*

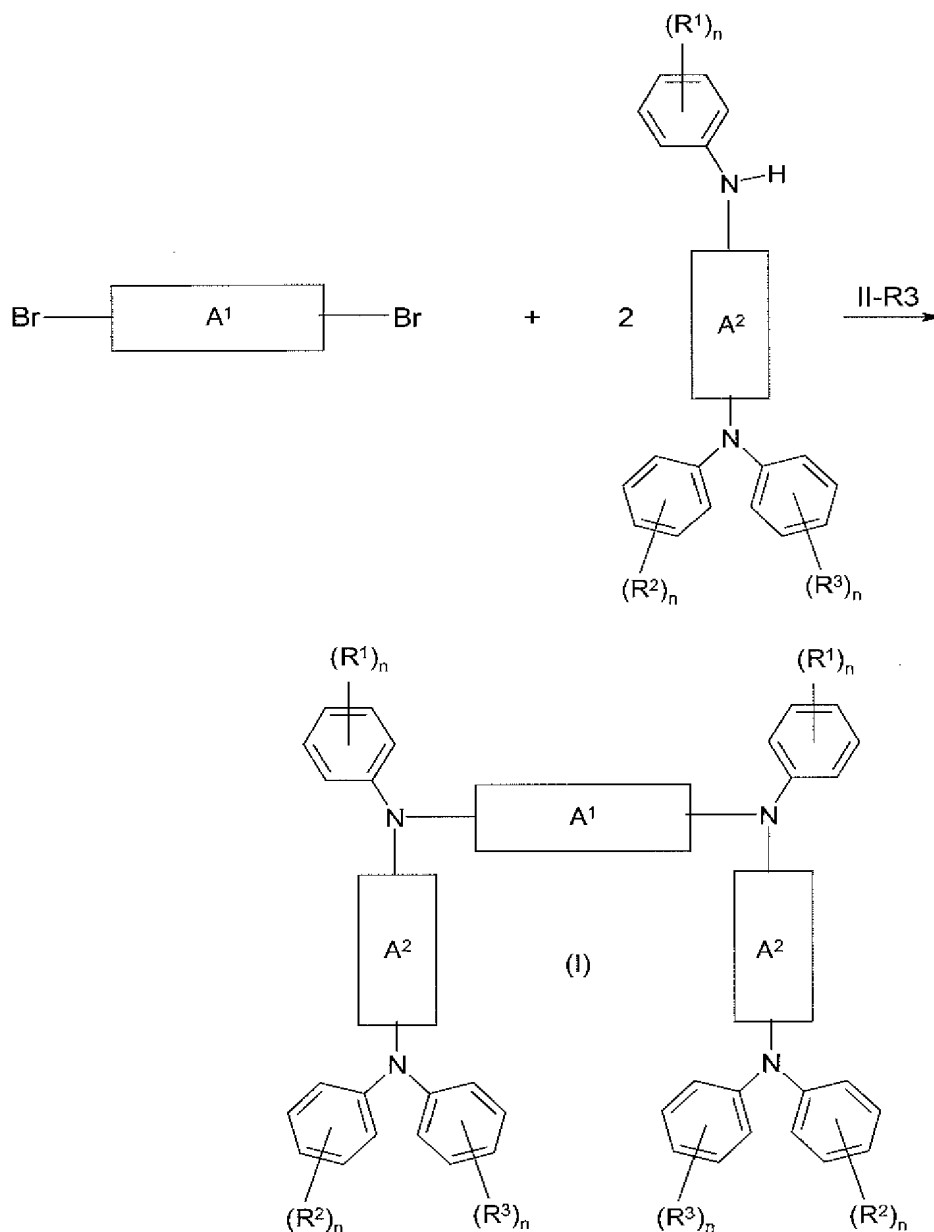


5 The synthesis in synthesis step II-R2 was based on the references cited below:

a) Bacher, Erwin; Bayerl, Michael; Rudati, Paula; Reckefuss, Nina; Müller, C. David; Meerholz, Klaus; Nuyken, Oskar; *Macromolecules*; 38; 5; 2005; 1640 – 1647,

10 b) Goodson, Felix E.; Hauck, Sheila; Hartwig, John F.; *J. Am. Chem. Soc.*; 121; 33; 1999; 7527 – 7539; Hauck, Sheila I.; Lakshmi, K. V.; Hartwig, John F.; *Org. Lett.*; 1; 13; 1999; 2057 – 2060.

(b3) *Synthesis step II-R3:*

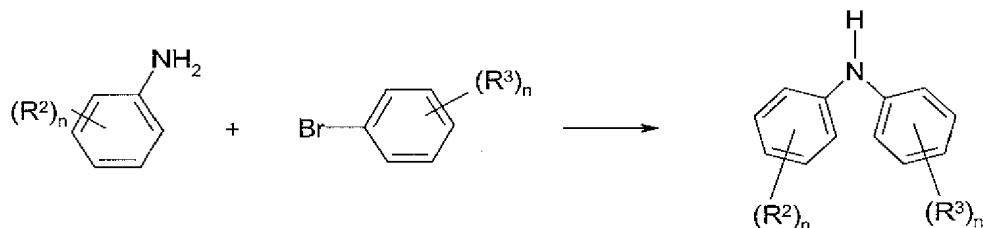


5

The compounds of the formula I can be prepared via the sequence of synthesis steps shown above in synthesis route II. In steps (II-R1) to (II-R3), the reactants can be coupled, as also in synthesis route I, for example, by Ullmann reaction with copper as a catalyst or under palladium catalysis.

(c) Preparation of the starting amines:

When the diarylamines in synthesis steps I-R2 and II-R1 of synthesis routes I and II are not commercially available, they can be prepared, for example, by Ullmann reaction with
 5 copper as a catalyst or under palladium catalysis, according to the following reaction:



The synthesis was based on the review articles listed below:

10

Palladium-catalyzed C-N coupling reactions:

a) Yang, Buchwald; J. Organomet. Chem. 1999, 576 (1-2), 125-146,

b) Wolfe, Marcoux, Buchwald; Acc. Chem. Res. 1998, 31, 805-818,

15

c) Hartwig; Angew. Chem. Int. Ed. Engl. 1998, 37, 2046-2067.

Copper-catalyzed C-N coupling reactions:

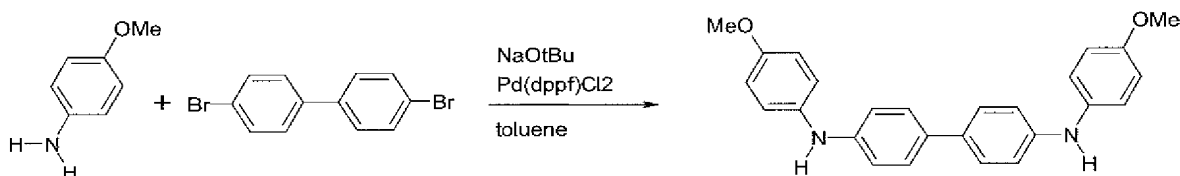
a) Goodbrand, Hu; Org. Chem. 1999, 64, 670-674,

20

b) Lindley; Tetrahedron 1984, 40, 1433-1456.

(B) Synthesis example 1: Synthesis of the compound ID367 (synthesis route I)

(B1): Synthesis step according to general synthesis scheme I-R1:

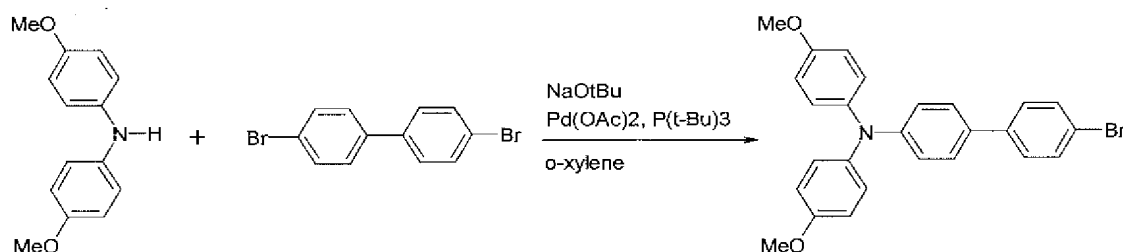


- 5 A mixture of 4,4'-dibromobiphenyl (93.6 g; 300 mmol), 4-methoxyaniline (133 g; 1.08 mol), Pd(dppf)Cl₂ (Pd(1,1'-bis(diphenylphosphino)ferrocene)Cl₂; 21.93 g; 30 mmol) and t-BuONa (sodium tert-butoxide; 109.06 g; 1.136 mol) in toluene (1500 ml) was stirred under a nitrogen atmosphere at 110°C for 24 hours. After cooling, the mixture was diluted with diethyl ether and filtered through a Celite® pad (from Carl Roth). The filter
10 bed was washed with 1500 ml each of ethyl acetate, methanol and methylene chloride. The product was obtained as a light brown solid (36 g; yield: 30%).

¹H NMR (400 MHz, DMSO): δ 7.81 (s, 2H), 7.34-7.32 (m, 4H), 6.99-6.97 (m, 4H), 6.90-6.88 (m, 4H), 6.81-6.79 (m, 4H), 3.64 (s, 6H).

15

(B2): Synthesis step according to general synthesis scheme I-R2:

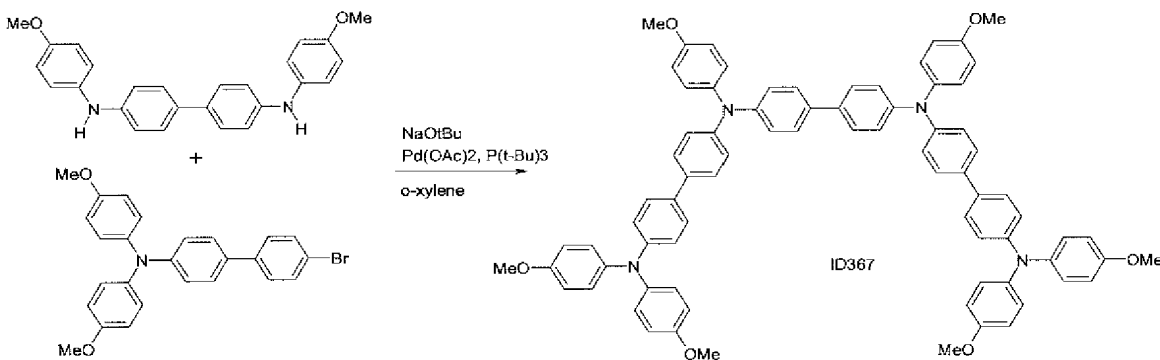


- 20 Nitrogen was passed for a period of 10 minutes through a solution of dppf (1,1'-bis(diphenylphosphino)ferrocene; 0.19 g; 0.34 mmol) and Pd₂(dba)₃ (tris(dibenzylideneacetone)dipalladium(0); 0.15 g; 0.17 mmol) in toluene (220 ml). Subsequently, t-BuONa (2.8 g; 29 mmol) was added and the reaction mixture was stirred for a further 15 minutes. 4,4'-Dibromobiphenyl (25 g; 80 mmol) and 4,4'-
25 dimethoxydiphenylamine (5.52 g; 20 mmol) were then added successively. The reaction mixture was heated at a temperature of 100°C under a nitrogen atmosphere for 7 hours.

After cooling to room temperature, the reaction mixture was quenched with ice-water, and the precipitated solid was filtered off and dissolved in ethyl acetate. The organic layer was washed with water, dried over sodium sulfate and purified by column chromatography (eluent: 5% ethyl acetate/hexane). A pale yellow solid was obtained (7.58 g, yield: 82%).

^1H NMR (300 MHz, DMSO- d_6): 7.60-7.49 (m, 6H), 7.07-7.04 (m, 4H), 6.94-6.91 (m, 4H), 6.83-6.80 (d, 2H), 3.75 (s, 6H).

(B3): Synthesis step according to general synthesis scheme I-R3:



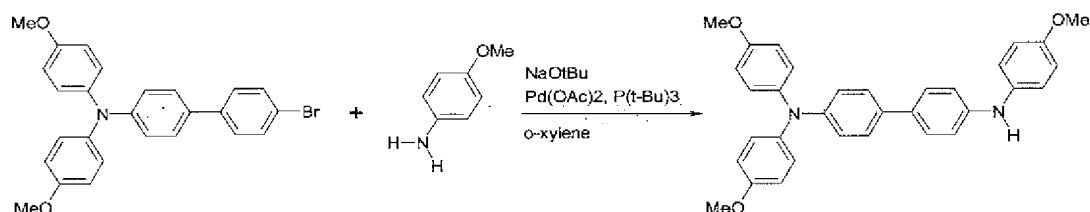
N⁴,N^{4'}-Bis(4-methoxyphenyl)biphenyl-4,4'-diamine (product from synthesis step I-R1; 0.4 g; 1.0 mmol) and product from synthesis step I-R2 (1.0 g; 2.2 mmol) were added under a nitrogen atmosphere to a solution of t-BuONa (0.32 g; 3.3 mmol) in o-xylene (25 ml). Subsequently, palladium acetate (0.03 g; 0.14 mmol) and a solution of 10% by weight of P(t-Bu)₃ (tris-t-butylphosphine) in hexane (0.3 ml; 0.1 mmol) were added to the reaction mixture which was stirred at 125°C for 7 hours. Thereafter, the reaction mixture was diluted with 150 ml of toluene and filtered through Celite®, and the organic layer was dried over Na₂SO₄. The solvent was removed and the crude product was reprecipitated three times from a mixture of tetrahydrofuran (THF)/methanol. The solid was purified by column chromatography (eluent: 20% ethyl acetate/hexane), followed by a precipitation with THF/methanol and an activated carbon purification. After removing the solvent, the product was obtained as a pale yellow solid (1.0 g, yield: 86%).

^1H NMR (400 MHz, DMSO- d_6): 7.52-7.40 (m, 8H), 6.88-7.10 (m, 32H), 6.79-6.81 (d, 4H), 3.75 (s, 6H), 3.73 (s, 12H).

(C) Synthesis example 2: Synthesis of the compound ID447 (synthesis route II)

5

(C1) Synthesis step according to general synthesis scheme II-R2:

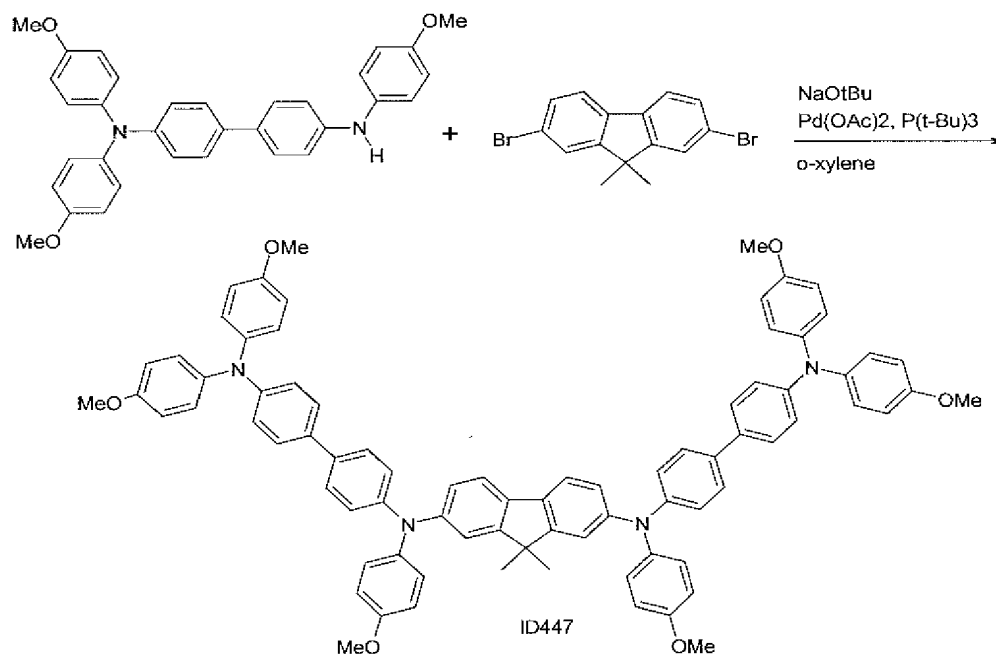


- 10 p-Anisidine (5.7 g, 46.1 mmol), t-BuONa (5.5 g, 57.7 mol) and P(t-Bu)₃ (0.62 ml, 0.31 mmol) were added to a solution of the product from synthesis step I-R2 (17.7 g, 38.4 mmol) in toluene (150 ml). After nitrogen had been passed through the reaction mixture for 20 minutes, Pd₂(dba)₃ (0.35 g, 0.38 mmol) was added. The resulting reaction mixture was left to stir under a nitrogen atmosphere at room temperature for 16 hours.
- 15 Subsequently, it was diluted with ethyl acetate and filtered through Celite®. The filtrate was washed twice with 150 ml each of water and saturated sodium chloride solution. After the organic phase had been dried over Na₂SO₄ and the solvent had been removed, a black solid was obtained. This solid was purified by column chromatography (eluent: 0–25% ethyl acetate/hexane). This afforded an orange solid (14 g, yield: 75%).

20

^1H NMR (300 MHz, DMSO): 7.91 (s, 1H), 7.43-7.40 (d, 4H), 7.08-6.81 (m, 16H), 3.74 (s, 6H), 3.72 (s, 3H).

(C1) Synthesis step according to general synthesis scheme II-R3:



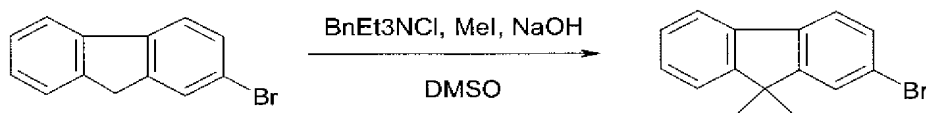
- 5 t-BuONa (686 mg; 7.14 mmol) was heated at 100°C under reduced pressure, then the reaction flask was purged with nitrogen and allowed to cool to room temperature. 2,7-Dibromo-9,9-dimethylfluorene (420 mg; 1.19 mmol), toluene (40 ml) and Pd[P(^tBu)₃]₂ (20 mg; 0.0714 mmol) were then added, and the reaction mixture was stirred at room temperature for 15 minutes. Subsequently, N,N,N'-p-trimethoxytriphenylbenzidine (1.5 g; 1.27 mmol) was added to the reaction mixture which was stirred at 120°C for 5 hours.
- 10 The mixture was filtered through a Celite®/MgSO₄ mixture and washed with toluene. The crude product was purified twice by column chromatography (eluent: 30% ethyl acetate/hexane) and, after twice reprecipitating from THF/methanol, a pale yellow solid was obtained (200 mg, yield: 13%).

15

¹H NMR: (400 MHz, DMSO-d₆): 7.60-7.37 (m, 8H), 7.02-6.99 (m, 16H), 6.92-6.87 (m, 20H), 6.80-6.77 (d, 2H), 3.73 (s, 6H), 3.71 (s, 12H), 1.25 (s, 6H)

(D) Synthesis example 3: Synthesis of the compound ID453 (synthesis route I)*(D1) Preparation of the starting amine:*

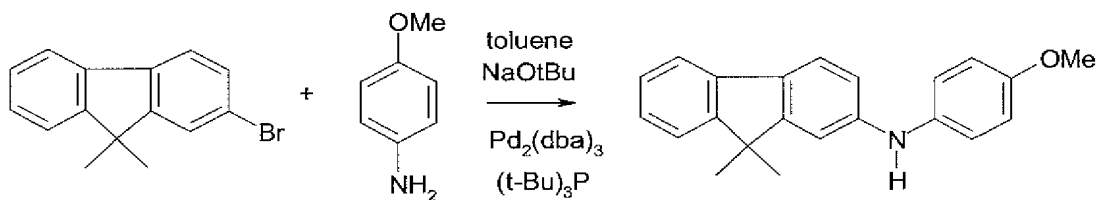
5 Step 1:



NaOH (78 g; 4 eq) was added to a mixture of 2-bromo-9H-fluorene (120 g; 1 eq) and
10 BnEt_3NCl (benzyltriethylammonium chloride; 5.9 g; 0.06 eq) in 580 ml of DMSO (dimethylsulfoxide). The mixture was cooled with ice-water, and methyl iodide (MeI) (160 g; 2.3 eq) was slowly added dropwise. The reaction mixture was left to stir overnight, then poured into water and subsequently extracted three times with ethyl acetate. The combined organic phases were washed with a saturated sodium chloride solution and
15 dried over Na_2SO_4 , and the solvent was removed. The crude product was purified by column chromatography using silica gel (eluent: petroleum ether). After washing with methanol, the product (2-bromo-9,9'-dimethyl-9H-fluorene) was obtained as a white solid (102 g).

20 ^1H NMR (400 MHz, CDCl_3): δ 1.46 (s, 6 H), 7.32 (m, 2 H), 7.43 (m, 2 H), 7.55 (m, 2 H), 7.68 (m, 1 H)

Step 2:

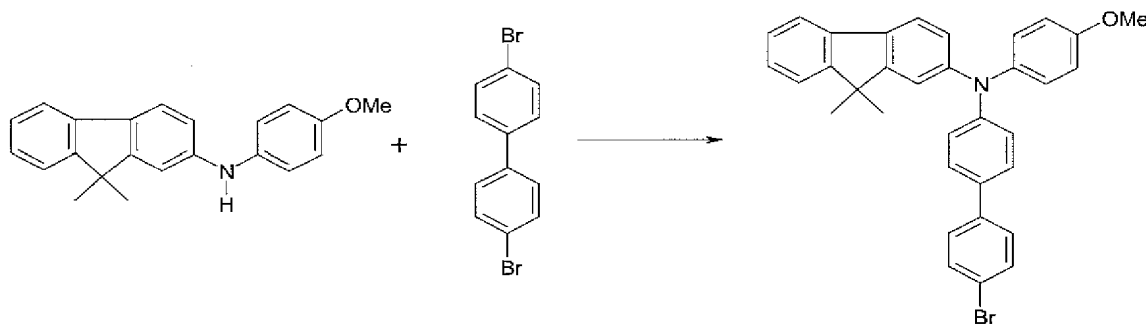


p-Anisidine (1.23 g; 10.0 mmol) and 2-bromo-9,9'-dimethyl-9H-fluorene (3.0 g; 11.0 mmol) were added under a nitrogen atmosphere to a solution of t-BuONa (1.44 g; 15.0 mmol) in 15 ml of toluene (15 ml). Pd₂(dba)₃ (92 mg; 0.1 mmol) and a 10% by weight solution of P(t-Bu)₃ in hexane (0.24 ml; 0.08 mmol) were added, and the reaction mixture was stirred at room temperature for 5 hours. Subsequently, the mixture was quenched with ice-water, and the precipitated solid was filtered off and dissolved in ethyl acetate. The organic phase was washed with water and dried over Na₂SO₄. After purifying the crude product by column chromatography (eluent: 10% ethyl acetate/hexane), a pale yellow solid was obtained (1.5 g, yield: 48%).

¹H NMR (300 MHz, C₆D₆): 7.59-7.55 (d, 1H), 7.53-7.50 (d, 1H), 7.27-7.22 (t, 2H), 7.19 (s, 1H), 6.99-6.95 (d, 2H), 6.84-6.77 (m, 4H), 4.99 (s, 1H), 3.35 (s, 3H), 1.37 (s, 6H).

(D2) Preparation of the compound ID453 for use in accordance with the invention

(D2.1): Synthesis step according to general synthesis scheme I-R2:

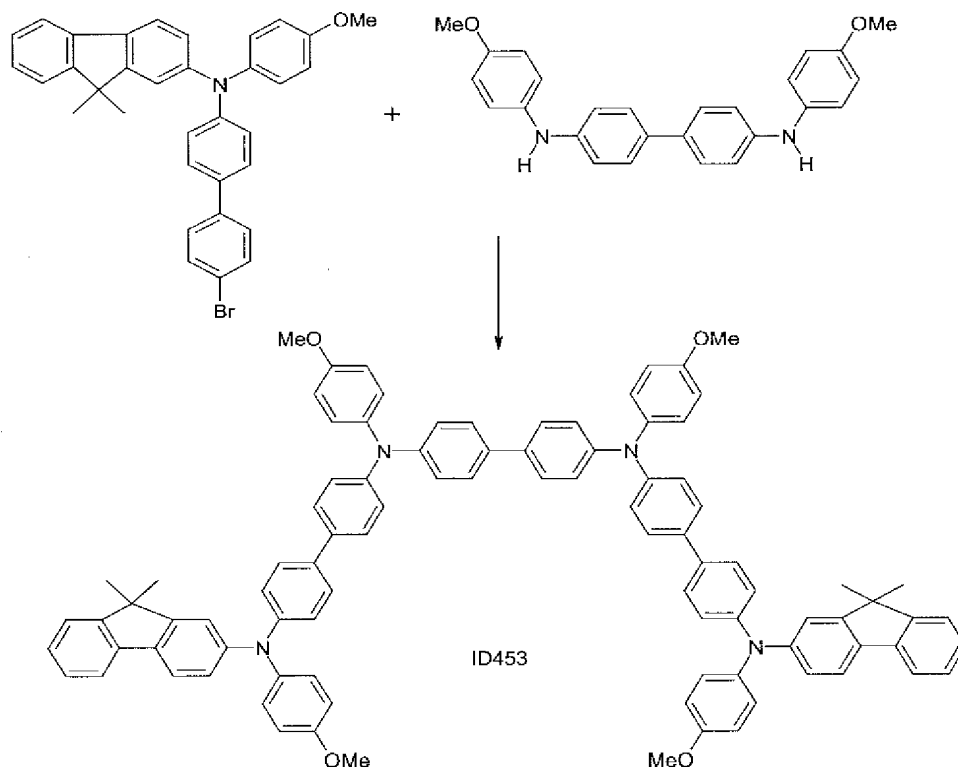


Product from a) (4.70 g; 10.0 mmol) and 4,4'-dibromobiphenyl (7.8 g; 25 mmol) were added to a solution of t-BuONa (1.15 g; 12 mmol) in 50 ml of toluene under nitrogen. Pd₂(dba)₃ (0.64 g; 0.7 mmol) and DPPF (0.78 g; 1.4 mmol) were added, and the reaction mixture was left to stir at 100°C for 7 hours. After the reaction mixture had been quenched with ice-water, the precipitated solid was filtered off and it was dissolved in ethyl acetate. The organic phase was washed with water and dried over Na₂SO₄. After purifying the crude product by column chromatography (eluent: 1% ethyl acetate/hexane), a pale yellow solid was obtained (4.5 g, yield: 82%).

¹H NMR (400 MHz, DMSO-d₆): 7.70-7.72 (d, 2H), 7.54-7.58 (m, 6H), 7.47-7.48 (d, 1H), 7.21-7.32 (m, 3H), 7.09-7.12 (m, 2H), 6.94-6.99 (m, 4H), 3.76 (s, 3H), 1.36 (s, 6H).

(D2.2) Synthesis step according to general synthesis scheme I-R3:

5



10 N⁴,N^{4'}-Bis(4-methoxyphenyl)biphenyl-4,4'-diamine (0.60 g; 1.5 mmol) and product from the preceding synthesis step I-R2 (1.89 g; 3.5 mmol) were added under nitrogen to a solution of t-BuONa (0.48 g; 5.0 mmol) in 30 ml of o-xylene. Palladium acetate (0.04 g; 0.18 mmol) and P(t-Bu)₃ in a 10% by weight solution in hexane (0.62 ml; 0.21 mmol) were added, and the reaction mixture was stirred at 125°C for 6 hours. Subsequently, the mixture was diluted with 100 ml of toluene and filtered through Celite®. The organic phase was dried over Na₂SO₄ and the resulting solid was purified by column chromatography (eluent: 10% ethyl acetate/hexane). This was followed by reprecipitation from THF/methanol to obtain a pale yellow solid (1.6 g, yield: 80%).

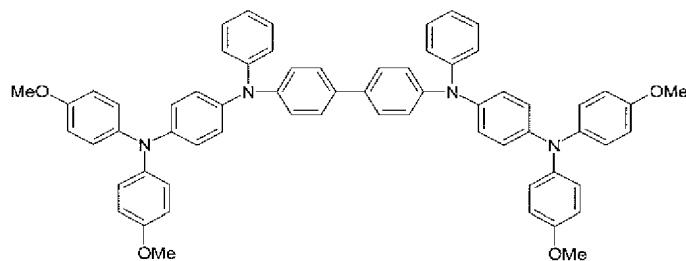
15

¹H NMR (400 MHz, DMSO-d₆): 7.67-7.70 (d, 4H), 7.46-7.53 (m, 14H), 7.21-7.31 (m, 4H), 7.17-7.18 (d, 2H), 7.06-7.11 (m, 8H), 6.91-7.01 (m, 22H), 3.75 (s, 12H), 1.35 (s, 12H).

(E) Further compounds of the formula I for use in accordance with the invention:

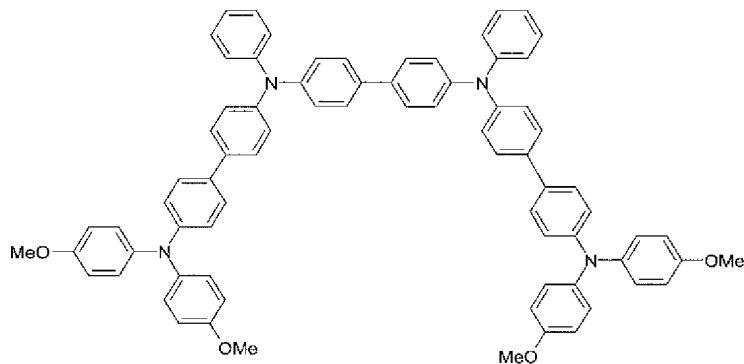
The compounds listed below were obtained analogously to the syntheses described above:

(E1) Synthesis example 4: Compound ID320



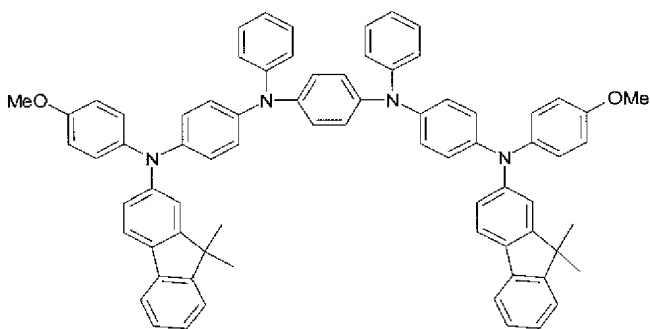
^1H NMR (300 MHz, THF- d_8): δ 7.43-7.46 (d, 4H), 7.18-7.23 (t, 4H), 7.00-7.08 (m, 16H), 6.81-6.96 (m, 18H), 3.74 (s, 12H)

(E2) Synthesis example 5: Compound ID321



^1H NMR (300 MHz, THF- d_8): δ 7.37-7.50 (t, 8H), 7.37-7.40 (d, 4H), 7.21-7.26 (d, 4H), 6.96-7.12 (m, 22H), 6.90-6.93 (d, 4H), 6.81-6.84 (d, 8H), 3.74 (s, 12H)

(E3) Synthesis example 6: Compound ID366

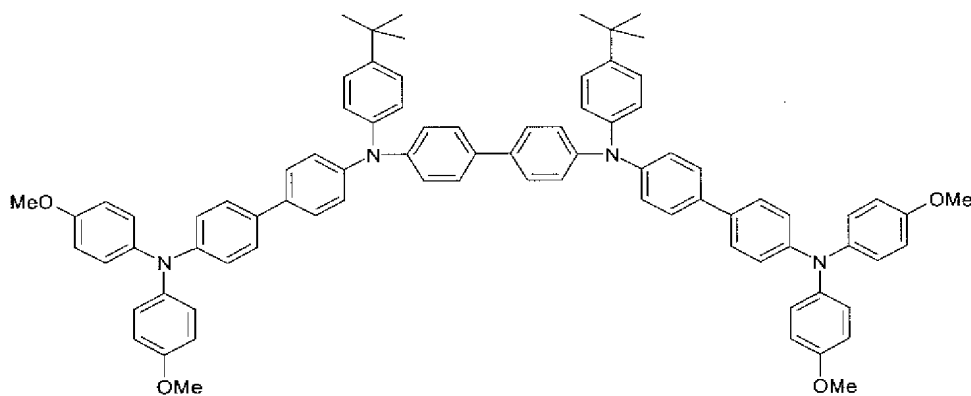


5

^1H NMR (400 MHz, DMSO- d_6): δ 7.60-7.70 (t, 4H), 7.40-7.55 (d, 2H), 7.17-7.29 (m, 8H), 7.07-7.09 (t, 4H), 7.06 (s, 2H), 6.86-7.00 (m, 24H), 3.73 (s, 6H), 1.31 (s, 12H)

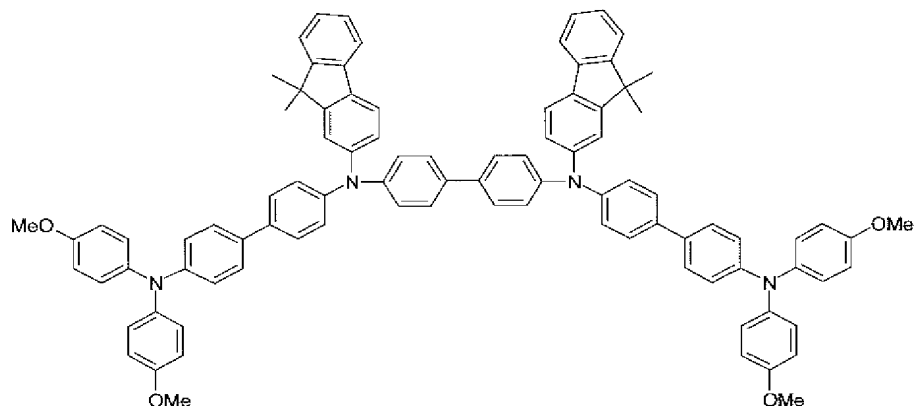
(E4) Synthesis example 7: Compound ID368

10

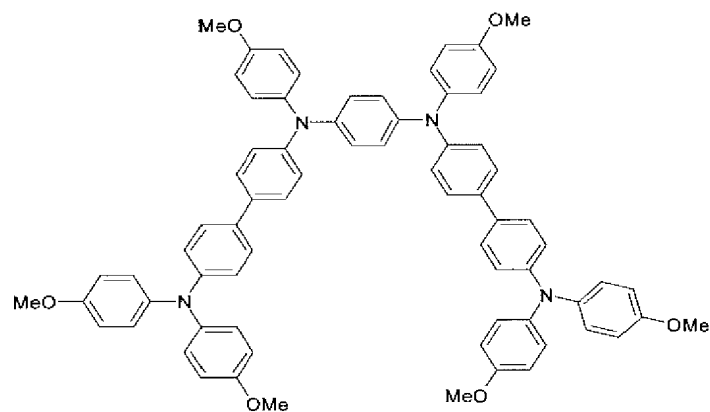


^1H NMR (400 MHz, DMSO- d_6): δ 7.48-7.55 (m, 8H), 7.42-7.46 (d, 4H), 7.33-7.28 (d, 4H), 6.98-7.06 (m, 20H), 6.88-6.94 (m, 8H), 6.78-6.84 (d, 4H), 3.73 (s, 12H), 1.27 (s, 18H)

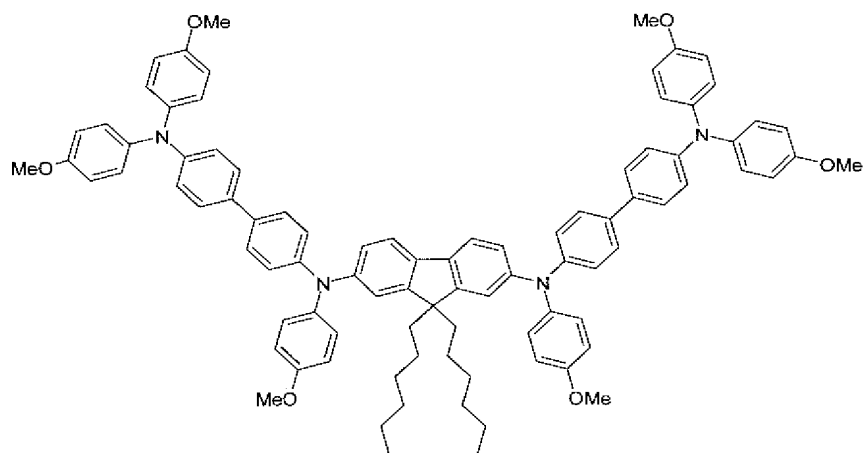
15

(E5) Synthesis example 8: Compound ID369

¹H NMR (400 MHz, THF-d₈): δ 7.60-7.70 (t, 4H), 7.57-7.54 (d, 4H), 7.48-7.51 (d, 4H),
 5 7.39-7.44 (t, 6H), 7.32-7.33 (d, 2H), 7.14-7.27 (m, 12H), 7.00-7.10 (m, 10H), 6.90-6.96
 (m, 4H), 6.80-6.87 (m, 8H), 3.75 (s, 12H), 1.42 (s, 12H)

(E6) Synthesis example 9: Compound ID446

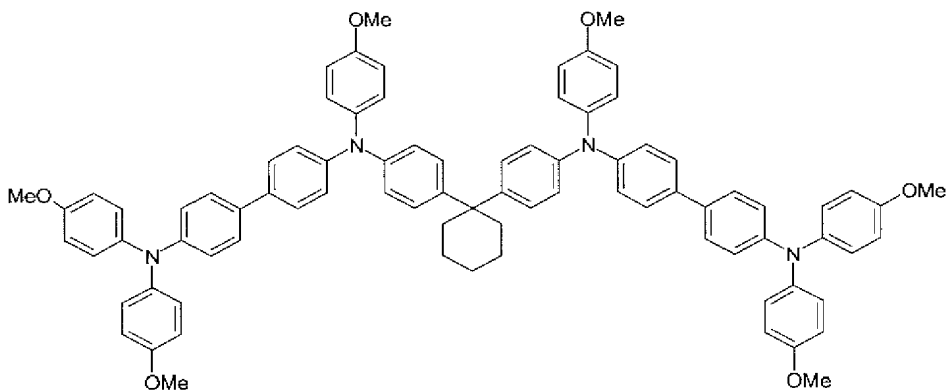
¹H NMR (400 MHz, dms_o-d₆): δ 7.39-7.44 (m, 8H), 7.00-7.07 (m, 13H), 6.89-6.94 (m,
 10 19H), 6.79-6.81 (d, 4H), 3.73 (s, 18H)

(E7) Synthesis example 10: Compound ID450

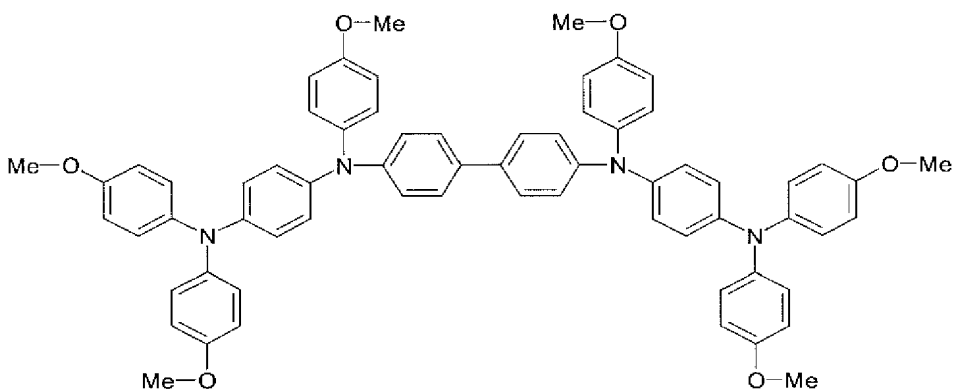
- 5 ^1H NMR (400 MHz, $\text{dms}\text{-d}_6$): δ 7.55-7.57 (d, 2H), 7.39-7.45 (m, 8H), 6.99-7.04 (m, 15H), 6.85-6.93 (m, 19H), 6.78-6.80 (d, 4H), 3.72 (s, 18H), 1.68-1.71 (m, 6H), 1.07 (m, 6H), 0.98-0.99 (m, 8H), 0.58 (m, 6H)

(E8) Synthesis example 11: Compound ID452

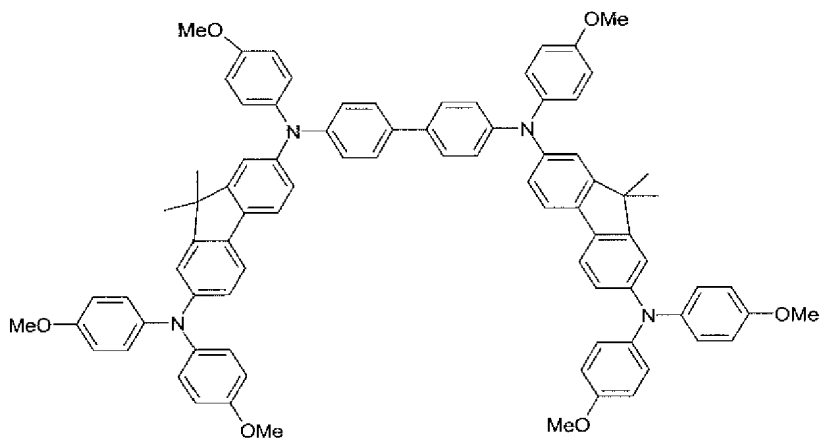
10



- 15 ^1H NMR (400 MHz, DMSO-d_6): δ 7.38-7.44 (m, 8H), 7.16-7.19 (d, 4H), 6.99-7.03 (m, 12H), 6.85-6.92 (m, 20H), 6.77-6.79 (d, 4H), 3.74 (s, 18H), 2.00-2.25 (m, 4H), 1.25-1.50 (m, 6H)

(E9) Synthesis example 12: Compound ID480

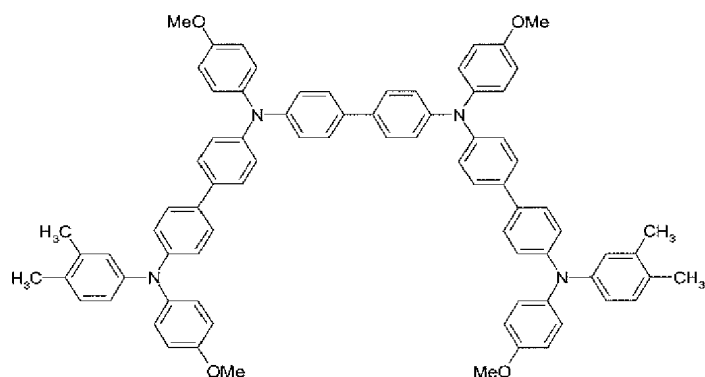
- 5 ^1H NMR (400 MHz, DMSO- d_6): δ 7.40-7.42 (d, 4H), 7.02-7.05 (d, 4H), 6.96-6.99 (m, 28H), 6.74-6.77 (d, 4H), 3.73 (s, 6H), 3.71 (s, 12H)

(E10) Synthesis example 13: Compound ID518

- 10 ^1H NMR (400 MHz, DMSO- d_6): 7.46-7.51 (m, 8H), 7.10-7.12 (d, 2H), 7.05-7.08 (d, 4H), 6.97-7.00 (d, 8H), 6.86-6.95 (m, 20H), 6.69-6.72 (m, 2H), 3.74 (s, 6H), 3.72 (s, 12H), 1.24 (t, 12H)

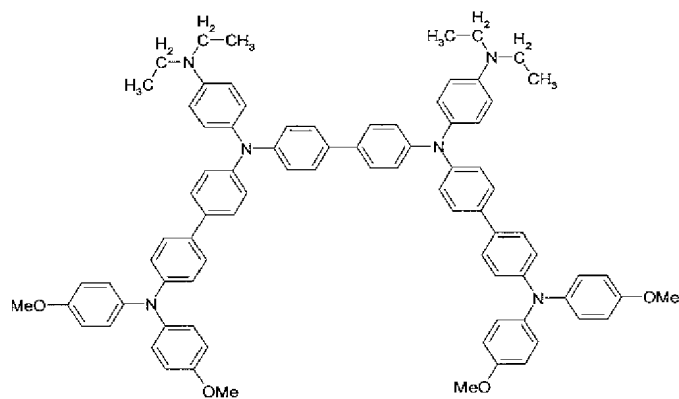
15

(E11) Synthesis example 14: Compound ID519

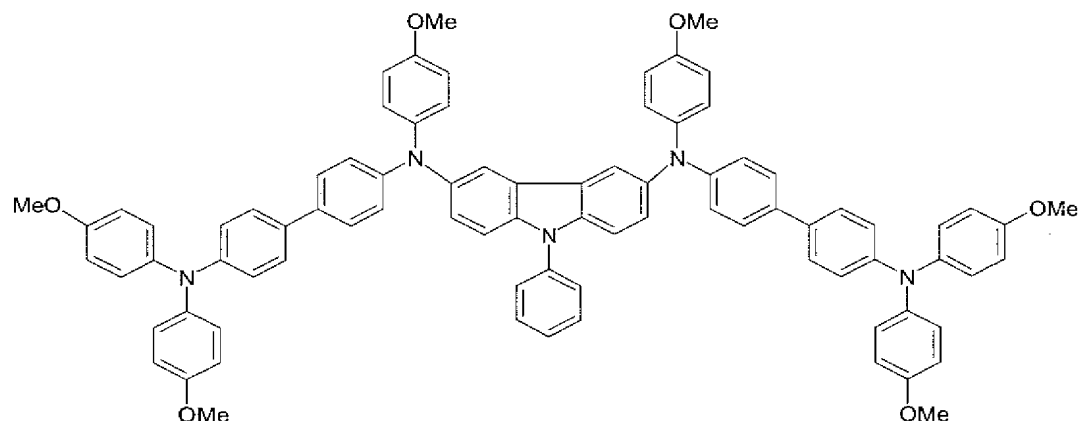


¹H NMR (400 MHz, DMSO-d₆): 7.44-7.53 (m, 12H), 6.84-7.11 (m, 32H), 6.74-6.77 (d, 2H), 3.76 (s, 6H), 3.74 (s, 6H), 2.17 (s, 6H), 2.13 (s, 6H)

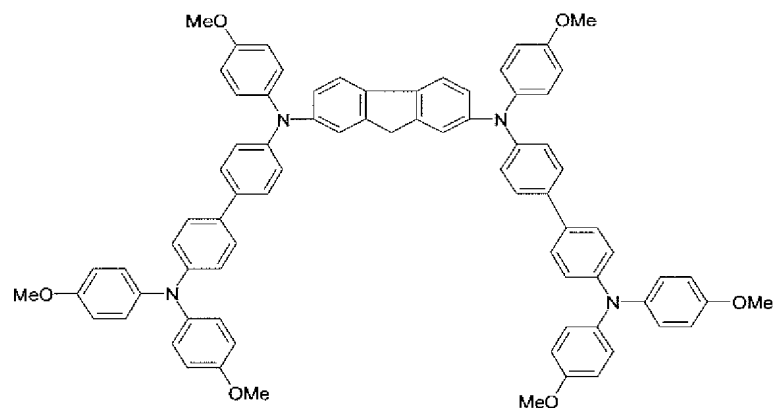
(E12) Synthesis example 15: Compound ID521



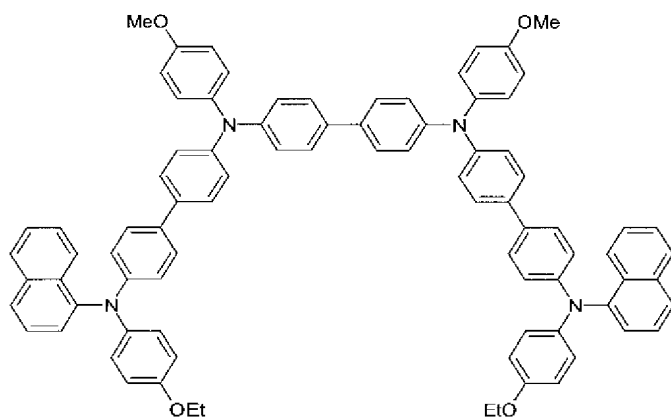
¹H NMR (400 MHz, THF-d₆): 7.36-7.42 (m, 12H), 6.99-7.07 (m, 20H), 6.90-6.92 (d, 4H), 6.81-6.84 (m, 8H), 6.66-6.69 (d, 4H), 3.74 (s, 12H), 3.36-3.38 (q, 8H), 1.41-1.17 (t, 12H)

(E13) Synthesis example 16: Compound ID522

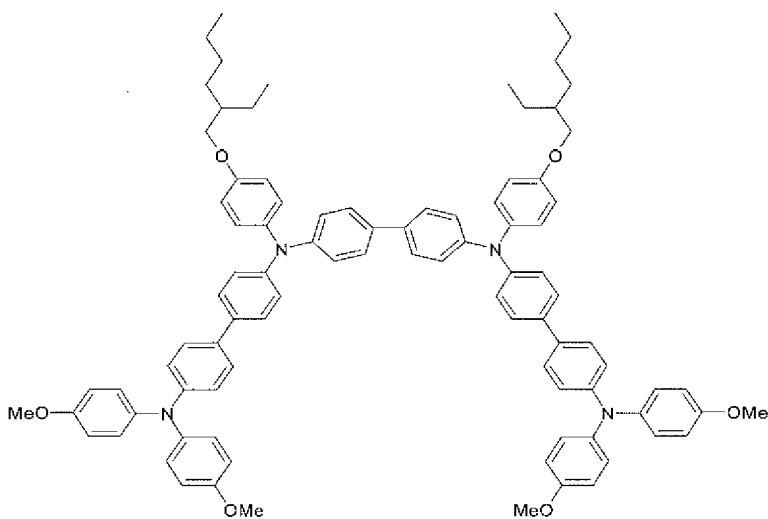
¹H NMR (400 MHz, DMSO-d₆): 7.65 (s, 2H), 7.52-7.56 (t, 2H), 7.44-7.47 (t, 1H), 7.37-7.39 (d, 2H), 7.20-7.22 (m, 10H), 7.05-7.08 (dd, 2H), 6.86-6.94 (m, 8H), 6.79-6.80-6.86 (m, 12H), 6.68-6.73, (dd, 8H), 6.60-6.62 (d, 4H), 3.68 (s, 12H), 3.62 (s, 6H)

(E14) Synthesis example 17: Compound ID523

¹H NMR (400 MHz, THF-d₈): 7.54-7.56 (d, 2H), 7.35-7.40 (dd, 8H), 7.18 (s, 2H), 7.00-7.08 (m, 18H), 6.90-6.92 (d, 4H), 6.81-6.86 (m, 12H), 3.75 (s, 6H), 3.74 (s, 12H), 3.69 (s, 2H)

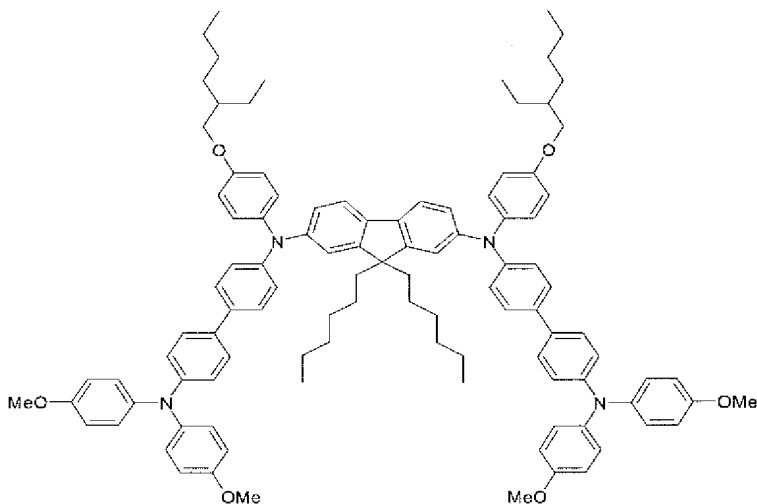
(E15) Synthesis example 18: Compound ID565

¹H NMR (400 MHz, THF-d₈): 7.97-8.00 (d, 2H), 7.86-7.89 (d, 2H), 7.73-7.76 (d, 2H),
 5 7.28-7.47 (m, 20H), 7.03-7.08 (m, 16H), 6.78-6.90 (m, 12H), 3.93-3.99 (q, 4H), 3.77 (s, 6H), 1.32-1.36 (s, 6H)

(E16) Synthesis example 19: Compound ID568

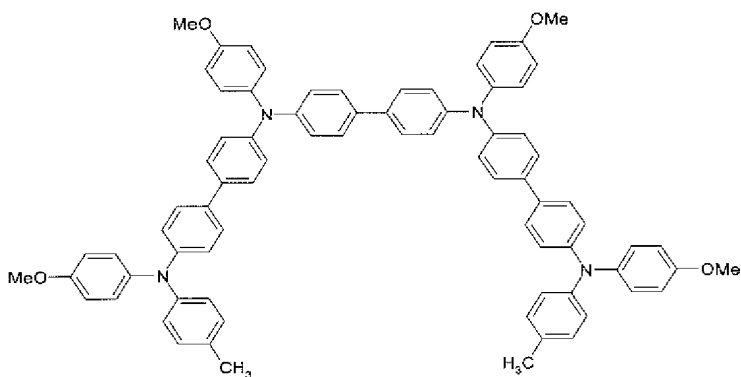
¹H NMR (400 MHz, DMSO-d₆): 7.41-7.51 (m, 12H), 6.78-7.06 (m, 36H), 3.82-3.84 (d, 4H), 3.79 (s, 12H), 1.60-1.80 (m, 2H), 0.60-1.60 (m, 28H)

15 *(E17) Synthesis example 20: Compound ID569*



¹H NMR (400 MHz, DMSO-d₆): 7.40-7.70 (m, 10H), 6.80-7.20 (m, 36H), 3.92-3.93 (d, 4H), 2.81 (s, 12H), 0.60-1.90 (m, 56H)

(E18) Synthesis example 21: Compound ID572



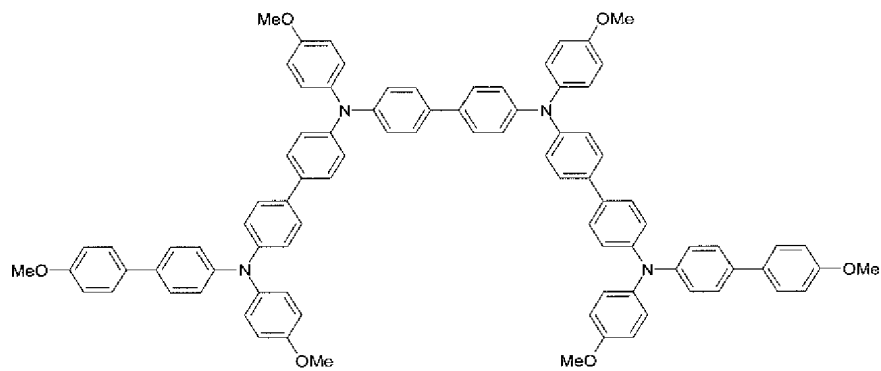
10

¹H NMR (400 MHz, THF-d₈): 7.39-7.47 (m, 12H), 7.03-7.11 (m, 20H), 6.39-6.99 (m, 8H), 6.83-6.90 (m, 8H), 3.78 (s, 6H), 3.76 (s, 6H), 2.27 (s, 6H)

(E19) Synthesis example 22: Compound ID573

15

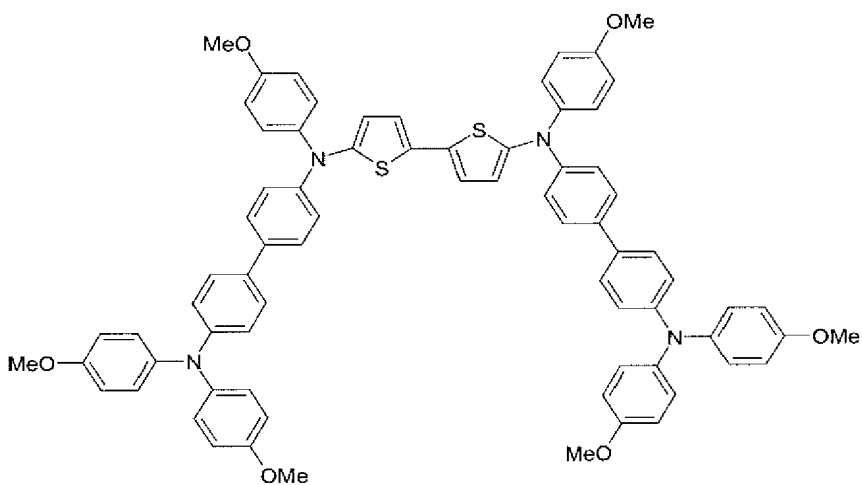
90



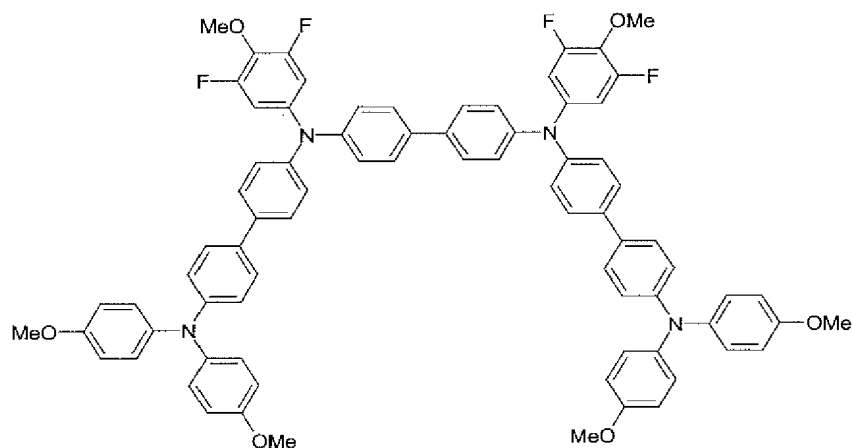
^1H NMR (400 MHz, THF- d_8): 7.43-7.51 (m, 20H), 7.05-7.12 (m, 24H), 6.87-6.95 (m, 12H), 3.79 (s, 6H), 3.78 (s, 12H)

5

(E20) Synthesis example 23: Compound ID575



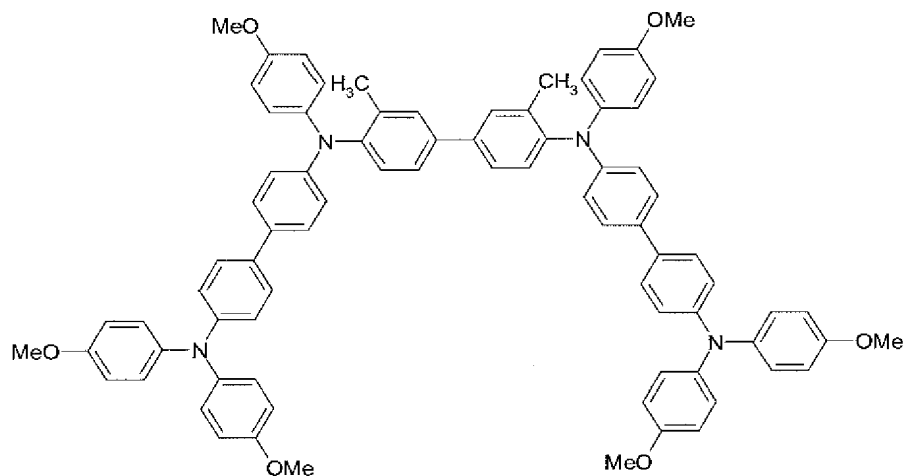
10 ^1H NMR (400 MHz, DMSO- d_6): 7.35-7.55 (m, 8H), 7.15-7.45 (m, 4H), 6.85-7.10 (m, 26H), 6.75-6.85 (d, 4H), 6.50-6.60 (d, 2H), 3.76 (s, 6H), 3.74 (s, 12H)

(E21) Synthesis example 24: Compound ID629

- 5 ^1H NMR (400 MHz, THF- d_8): 7.50-7.56 (dd, 8H), 7.38-7.41 (dd, 4H), 7.12-7.16 (d, 8H), 7.02-7.04 (dd, 8H), 6.91- 6.93 (d, 4H), 6.82-6.84 (dd, 8H), 6.65-6.68 (d, 4H) 3.87 (s, 6H), 3.74 (s, 12H)

(E22) Synthesis example 25: Compound ID631

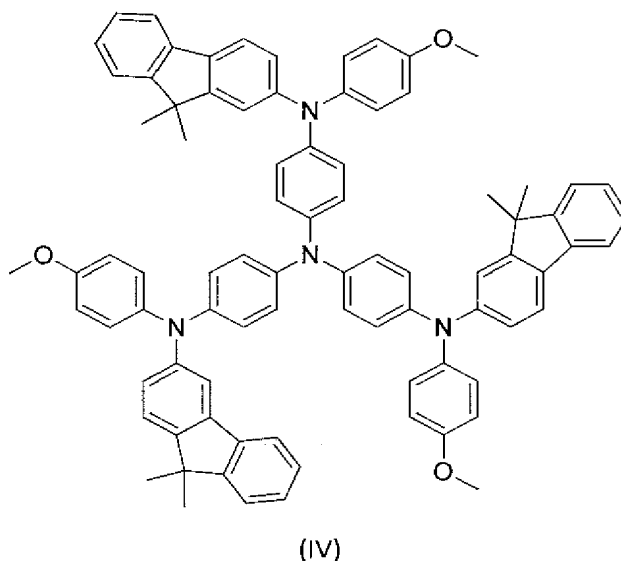
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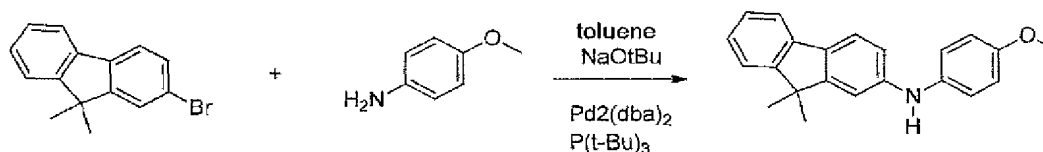
^1H NMR (400 MHz, THF- d_6): 7.52 (d, 2H), 7.43-7.47 (dd, 2H), 7.34-7.38 (m, 8H), 7.12-7.14 (d, 2H), 6.99-7.03 (m, 12H), 6.81-6.92 (m, 20H), 3.74 (s, 18H), 2.10 (s, 6H)

15

(F) Synthesis of compounds of the formula IV:



5

(a) Coupling of *p*-anisidine and 2-bromo-9,9-dimethyl-9H-fluorene

- 10 To 0.24 ml (0.08 mmol) of $P(t-Bu)_3$ ($d = 0.68$ g/ml) and 0.1 g of $Pd_2(dba)_2$ [=tris(dibenzylideneacetone)dipalladium(0)] (0.1 mmol) were added 10 ml to 15 ml of toluene (anhydrous, 99.8%), and the mixture was stirred at room temperature for 10 min. 1.44 g (15 mmol) of sodium tert-butoxide (97.0%) were added and the mixture was stirred at room temperature for a further 15 min. Subsequently, 2.73 g (11 mmol) of 2-bromo-9,9-dimethyl-9H-fluorene were added and the reaction mixture was stirred for a further 15 min. Finally, 1.23 g (10 mmol) of *p*-anisidine were added and the mixture was stirred at 90°C for 4 h.

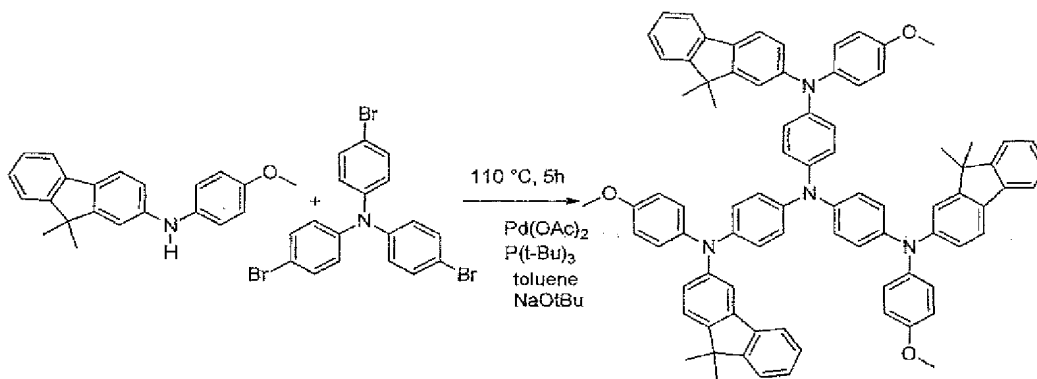
20 The reaction mixture was admixed with water and the product was precipitated from hexane. The aqueous phase was additionally extracted with ethyl acetate. The organic phase and the precipitated solid which had been filtered off were combined and purified by column chromatography on an SiO_2 phase (10:1 hexane:ethyl acetate).

1.5 g (yield: 47.6%) of a yellow solid were obtained.

^1H NMR (300 MHz, C_6D_6): 6.7-7.6 (m, 11H), 5.00 (s, 1H), 3.35 (s, 3H), 1.37 (s, 6H)

5

(b) Coupling of the product from (a) with tris(4-bromophenyl)amine



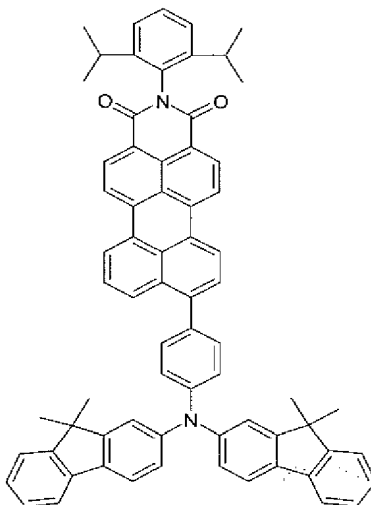
- 10 To 0.2 ml (0.07 mmol) of $\text{P}(\text{t-Bu})_3$ ($D = 0.68$ g/ml) and 0.02 g (0.1 mmol) of palladium acetate were added 25 ml of toluene (anhydrous), and the mixture was stirred at room temperature for 10 min. 0.4 g (1.2 mmol) of sodium tert-butoxide (97.0%) was added and the mixture was stirred at room temperature for a further 15 min. Subsequently, 0.63 g (1.3 mmol) of tris(4-bromophenyl)amine was added and the reaction mixture was stirred
- 15 for a further 15 min. Finally, 1.3 g (1.4 mmol) of the product from step (a) were added and the mixture was stirred at 90°C for 5 h.

- The reaction mixture was admixed with ice-cold water and extracted with ethyl acetate. The product was precipitated from a mixture of hexane/ethyl acetate and purified by
- 20 column chromatography on SiO_2 phase (9:1 → 5:1 hexane:ethyl acetate gradient).

0.7 g (yield: 45%) of a yellow product was obtained.

^1H NMR (300 MHz, C_6D_6): 6.6-7.6 (m, 45H), 3.28 (s, 9H), 1.26 (s, 18H)

25

(G) Synthesis of compounds ID504:

- 5 The preparation proceeded from (4-bromophenyl)bis(9,9-dimethyl-9H-fluoren-2-yl) (see Chemical Communications, 2004, 68-69), which was first reacted with 4,4,5,5,4',4',5',5'-octamethyl-[2,2']bi[[1,3,2]dioxaborolanyl] (step a). This was followed by coupling with 9Br-DIPP-PDCI (step b). This was followed by hydrolysis to give the anhydride (step c) and subsequent reaction with glycine to give the final compound (step d).

10

Step a:

- A mixture of 30 g (54 mmol) of (4-bromophenyl)bis(9,9-dimethyl-9H-fluoren-2-yl), 41 g (162 mmol) of 4,4,5,5,4',4',5',5'-octamethyl-[2,2']bi[[1,3,2]dioxaborolanyl], 1 g (1.4 mmol) of Pd(dppf)₂Cl₂, 15.9 g (162 mmol) of potassium acetate and 300 ml of dioxane was heated to 80°C and stirred for 36 h.

15

After cooling, the solvent was removed and the residue was dried at 50°C in a vacuum drying cabinet.

20

- Purification was effected by filtration through silica gel with the eluent 1:1 n-hexane:dichloromethane. After the removal of the reactant, the eluent was switched to dichloromethane. The product was isolated as a reddish and tacky residue. This was extracted by stirring with methanol at RT for 0.5 h. The light-colored precipitate was filtered off. After drying at 45°C in a vacuum drying cabinet, 24 g of a light-colored solid were obtained, which corresponds to a yield of 74%.

25

Analytical data

^1H NMR (500MHz, CD_2Cl_2 , 25°C): δ = 7.66-7.61 (m, 6H); 7.41-7.4 (m, 2H); 7.33-7.25 (m, 6H); 7.13-7.12 (m, 2H); 7.09-7.07 (m, 2H); 1.40 (s, 12H); 1.32 (s, 12H)

Step b:

17.8 g (32 mmol) of 9Br-DIPP-PDCl and 19 ml (95 mmol) of 5 molar NaOH were introduced into 500 ml of dioxane. This mixture was degassed with argon for 30 min. Then 570 mg (1.1 mmol) of $\text{Pd}[\text{P}(\text{tBu})_3]_2$ and 23 g (38 mmol) of stage a were added and the mixture was stirred at 85°C under argon for 17 h.

Purification was effected by column chromatography with the eluent 4:1 dichloromethane:toluene.

22.4 g of a violet solid were obtained, which corresponds to a yield of 74%.

Analytical data:

^1H NMR (500MHz, CH_2Cl_2 , 25°C): δ = 8.59-8.56 (m, 2H); 8.46-8.38 (m, 4H); 8.21-8.19 (d, 1H); 7.69-7.60 (m, 6H); 7.52-7.25 (m, 17H); 2.79-2.77 (m, 2H); 1.44 (s, 12H); 1.17-1.15 (d, 12H)

Step c:

22.4 g (23 mmol) of step b and 73 g (1.3 mol) of KOH were introduced into 200 ml of 2-methyl-2-butanol and the mixture was stirred at reflux for 17 h.

After cooling, the reaction mixture was added to 1 l of ice-water + 50 ml of concentrated acetic acid. The orange-brown solid was filtered through a frit and washed with water.

The solid was dissolved in dichloromethane and extracted with demineralized water. 10 ml of concentrated acetic acid were added to the organic phase, which was stirred at RT. The solvent was removed from the solution. The residue was extracted by stirring with methanol at RT for 30 min, filtered with suction through a frit and dried at 55°C in a vacuum drying cabinet.

This afforded 17.5 g of a violet solid, which corresponds to a yield of 94%. The product was used unpurified in the next step.

Step d:

17.5 g (22 mmol) of stage c, 16.4 g (220 mmol) of glycine and 4 g (22 mmol) of zinc acetate were introduced into 350 ml of N-methylpyrrolidone and the mixture was stirred at 130°C for 12 h.

- 5 After cooling, the reaction mixture was added to 1 l of demineralized water. The precipitate was filtered through a frit, washed with water and dried at 70°C in a vacuum drying cabinet.

- 10 Purification was effected by means of column chromatography with the eluent 3:1 dichloromethane:ethanol + 2% triethylamine. The isolated product was extracted by stirring at 60°C with 50% acetic acid. The solid was filtered off with suction through a frit, washed with water and dried at 80°C in a vacuum drying cabinet.

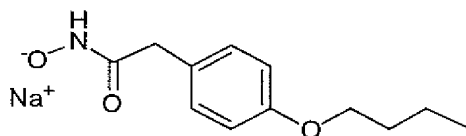
7.9 g of a violet solid were obtained, which corresponds to a yield of 42%.

15

Analytical data:

- 20 ¹H NMR (500MHz, THF, 25°C): δ = 8.37-8.34 (m, 2H); 8.25-8.18 (m, 4H); 8.12-8.10 (d, 1H); 7.74-7.70 (m, 4H); 7.59-7.53 (m, 4H); 7.45-7.43 (m, 4H); 7.39-7.37 (m, 2H); 7.32-7.22 (m, 6H); 4.82 (s, 2H); 1.46 (s, 12H)

(H) Synthesis of compounds ID662:



ID662

25

ID662 was prepared by reacting the corresponding commercially available hydroxamic acid [2-(4-butoxyphenyl)-N-hydroxyacetamide] with sodium hydroxide.

List of reference signs

110	Detector	188	Front side
112	Object	190	First motor vehicle
114	Optical sensor	192	Rear side
116	Sensor region	194	Second motor vehicle
118	Sensor area	196	Imaging device
120	Measuring device	198	Sample
122	Evaluation device	200	Confocal microscope
124	Data processing device	202	Focus
126	Data storage device	204	Beam splitter
128	Interface	206	Diaphragm
130	Transfer device	208	Sensor
132	Lens	210	Stack
134	Electromagnetic radiation	212	Layer
136	Light spot	214	Human-machine interface
138	Modulation device	216	Entertainment device
140	Beam interrupter	218	User
142	Illumination source	220	Machine
144	Primary radiation	222	Display
146	Reflective surface	224	Keyboard
148	Semiconductor detector	226	Security device
150	Organic semiconductor detector	228	Memory device
152	Organic solar cell	230	Optical data storage device
154	Dye solar cell	232	Reading beam
156	Substrate	234	Interface
158	First electrode		
160	Blocking layer, buffer layer		
162	n-semiconducting metal oxide		
164	Dye		
166	p-type semiconductor		
168	Second electrode		
170	Layer structure		
172	Encapsulation		
174	Fermi level		
176	HOMO		
178	LUMO		
180	Distance measuring device		
182	Light spot		
184	Light spot		
186	Housing		

Patent claims

1. A detector (110) for optically detecting at least one object (112), comprising at least one optical sensor (114), wherein the optical sensor (114) has at least one sensor region (116), in particular at least one sensor region (116) comprising at least one sensor area (118), wherein the optical sensor (114) is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region (116), wherein the sensor signal, given the same total power of the illumination is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area (118), wherein the detector (110) furthermore has at least one evaluation device (122), wherein the evaluation device (122) is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object (112).
2. The detector (110) according to the preceding claim, wherein the detector (110) furthermore has at least one modulation device (138) for modulating the illumination.
3. The detector (110) according to the preceding claim, wherein the detector (110) is designed to detect at least two sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies, wherein the evaluation device (122) is designed to generate the geometrical information from the at least two sensor signals.
4. The detector (110) according to any of the preceding claims, wherein the optical sensor (114) is furthermore designed in such a way that the sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination.
5. The detector (110) according to any of the preceding claims, wherein the sensor region (116) is exactly one continuous sensor region (116), wherein the sensor signal is a uniform sensor signal for the entire sensor region (116).
6. The detector (110) according to any of the preceding claims, wherein the sensor signal is selected from the group consisting of a photocurrent and a photovoltage.
7. The detector (110) according to any of the preceding claims, wherein the optical sensor (114) comprises at least one semiconductor detector (148), in particular an organic semiconductor detector (150) comprising at least one organic material,

preferably an organic solar cell (152) and particularly preferably a dye solar cell (154), in particular a solid dye solar cell.

- 5 8. The detector (110) according to the preceding claim, wherein the optical sensor (114) comprises at least one first electrode (158), at least one n-semiconducting metal oxide (162), at least one dye (164), at least one p-semiconducting organic material (166), preferably a solid p-semiconducting organic material, and at least one second electrode (168).
- 10 9. The detector (110) according to any of the preceding claims, wherein the geometrical information comprises at least one item of location information of the object (112).
- 15 10. The detector (110) according to the preceding claim, wherein the evaluation device (122) is designed to determine the geometrical information from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object (112) with respect to the detector (110), preferably taking account of a known power of the illumination and optionally taking account of a modulation frequency with which the illumination is modulated.
- 20 11. The detector (110) according to any of the preceding claims, furthermore comprising at least one transfer device (130), wherein the transfer device (130) is designed to feed electromagnetic radiation (134) emerging from the object (112) to the optical sensor (114) and thereby to illuminate the sensor region (116).
- 25 12. The detector (110) according to any of the preceding claims, furthermore comprising at least one illumination source (142).
- 30 13. The detector (110) according to the preceding claim, wherein the illumination source (142) is selected from: an illumination source (142), which is at least partly connected to the object (112) and/or is at least partly identical to the object (112); an illumination source (142) which is designed to at least partly illuminate the object (112) with a primary radiation (144).
- 35 14. A distance measuring device (180), in particular for use in a motor vehicle, comprising at least one detector (110) according to any of the preceding claims, wherein the detector (110) is designed to determine at least one item of geometrical information of at least one object (112), wherein the geometrical information comprises at least one item of location information of the object (112),
40 in particular a distance between a motor vehicle and at least one object (112) and preferably a distance between the motor vehicle and at least one object (112)

selected from the group consisting of a further motor vehicle, an obstacle, a cyclist and a pedestrian.

- 5 15. An imaging device (196) for imaging at least one sample (198), wherein the imaging device (196) comprises at least one detector (110) according to any of the preceding claims relating to a detector (110), wherein the imaging device (196) is designed to image a plurality of partial regions of the sample (198) onto the sensor region (116) and to thereby generate sensor signals assigned to the partial regions, wherein the imaging device (196) is designed to generate items of geometrical information of the respective partial regions from the sensor signals, wherein the items of geometrical information comprise items of location information.
- 10
- 15 16. A human-machine interface (214) for exchanging at least one item of information between a user (218) and a machine (220), in particular for inputting control commands, wherein the human-machine interface (214) comprises at least one detector (110) according to any of the preceding claims relating to a detector (110), wherein the human-machine interface (214) is designed to generate at least one item of geometrical information of the user (218) by means of the detector (110), wherein the human-machine interface (214) is designed to assign to the geometrical information at least one item of information, in particular at least one control command.
- 20
- 25 17. An entertainment device (216) for carrying out at least one entertainment function, in particular a game, wherein the entertainment device (216) comprises at least one human-machine interface (214) according to the preceding claim, wherein the entertainment device (216) is designed to enable at least one item of information to be input by a player (218) by means of the human-machine interface (214), wherein the entertainment device (216) is designed to vary the entertainment function in accordance with the information.
- 30
- 35 18. A security device (226) for carrying out at least one security application in particular for identifying and/or avoiding an access to data of an optical data storage device, wherein the security device (226) comprises at least one detector (110) according to any of the preceding claims relating to a detector (110), wherein the security device (226) is designed to identify, by means of the detector (110), impingement of focused electromagnetic radiation (134), in particular laser beams, on the security device (226) and preferably to generate at least one warning signal.

19. A method for optically detecting at least one object (112), in particular using a detector (110) according to any of the preceding claims relating to a detector (110), wherein at least one optical sensor (114) is used, wherein the optical sensor (114) has at least one sensor region (116), wherein electromagnetic radiation (134) emerging from the object is fed to the optical sensor (114) and thereby the sensor region (116) is illuminated, wherein the optical sensor (114) generates at least one sensor signal in a manner dependent on the illumination of the sensor region (116), wherein the sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination.
20. The method according to the preceding claim, wherein at least one item of geometrical information of the object (112) is generated from the sensor signal, in particular at least one item of location information of the object (112).
21. Use of a detector (110) according to any of the preceding claims relating to a detector (110) for a purpose of use, selected from the group consisting of: distance measurement, in particular in traffic technology; imaging, in particular in microscopy; an entertainment application; a security application; a human-machine interface application.
22. Use of an organic solar cell (152), in particular a dye solar cell (154), preferably a solid dye solar cell, as optical sensor (114), wherein at least one sensor signal is generated in the use, wherein the sensor signal, given the same total power of an illumination is dependent on a geometry of the illumination on the organic solar cell, wherein at least one item of geometrical information of at least one object (112) is generated from the sensor signal in the use.

FIG. 1

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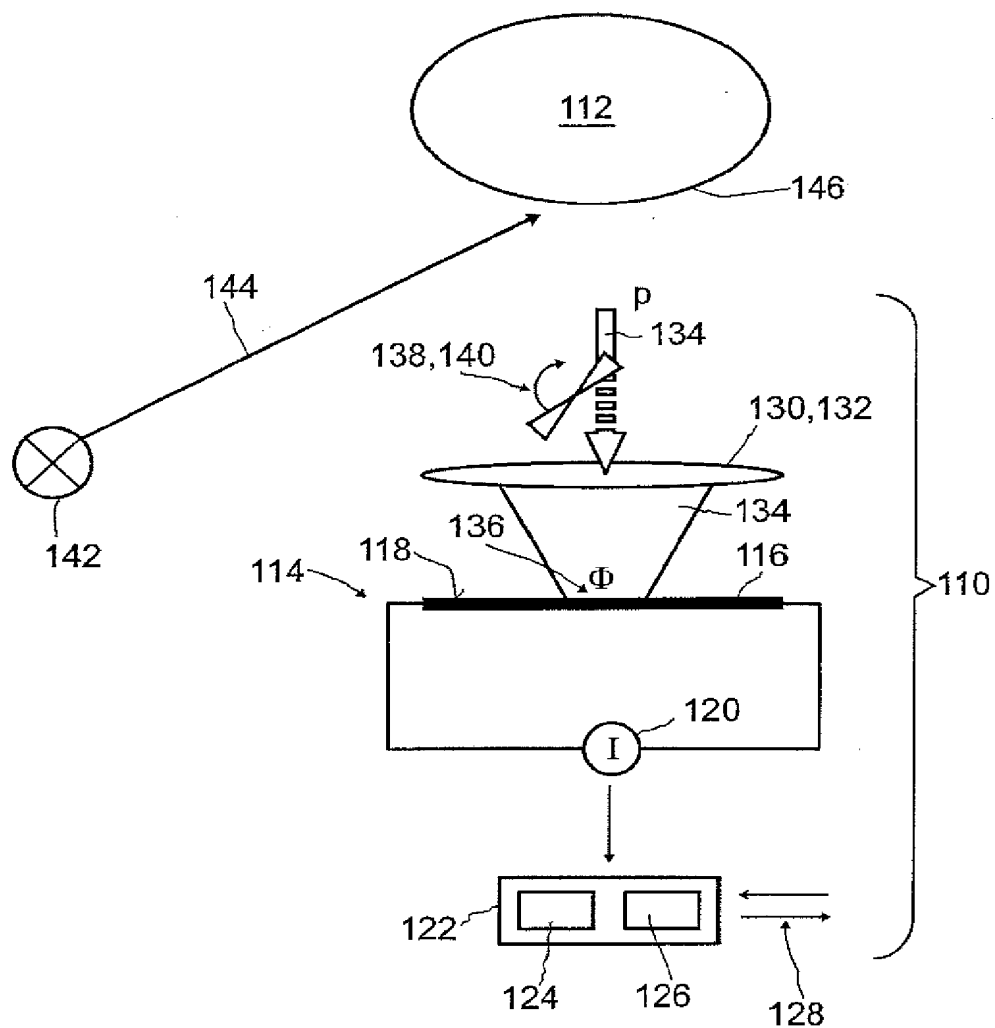


FIG.2

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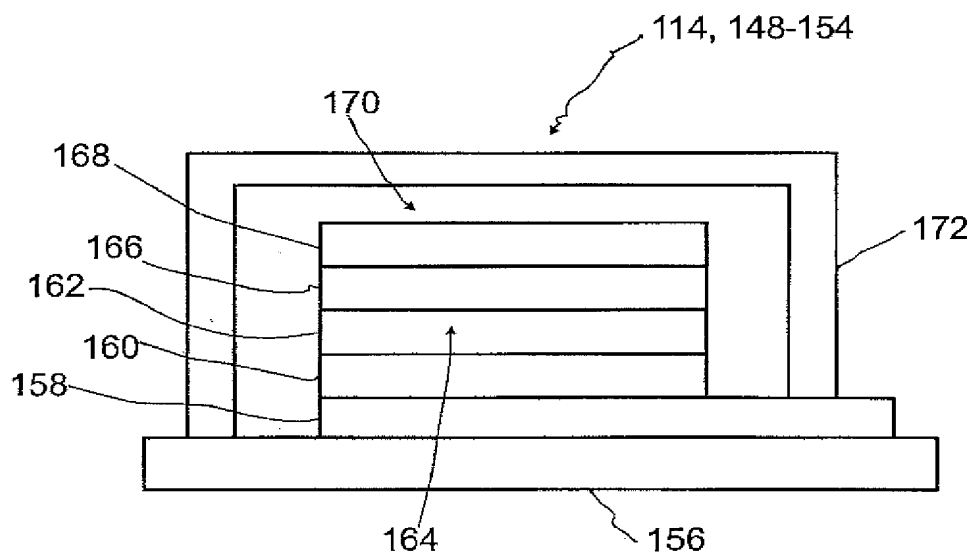


FIG.3

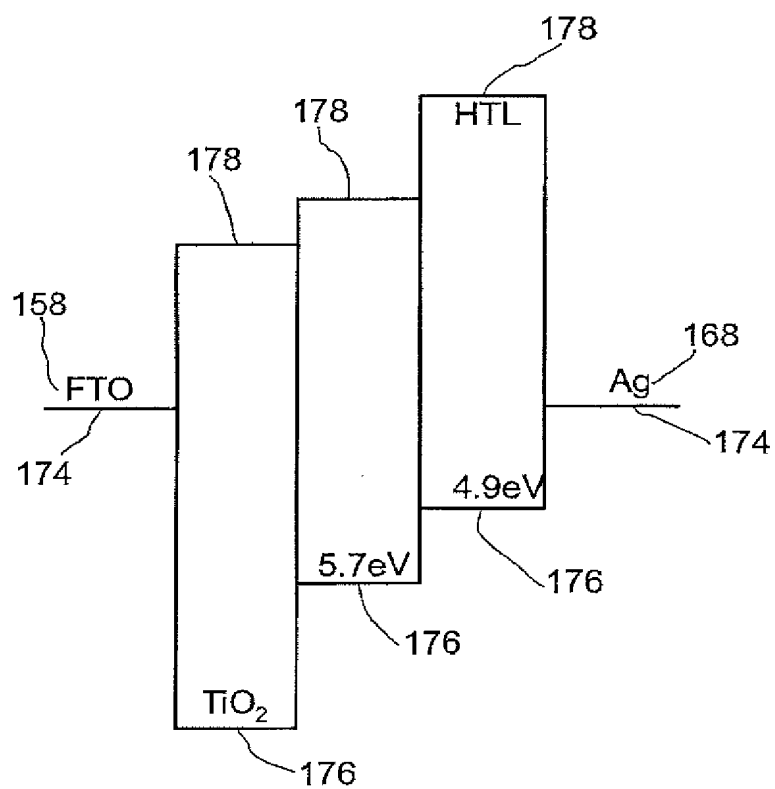


FIG.4

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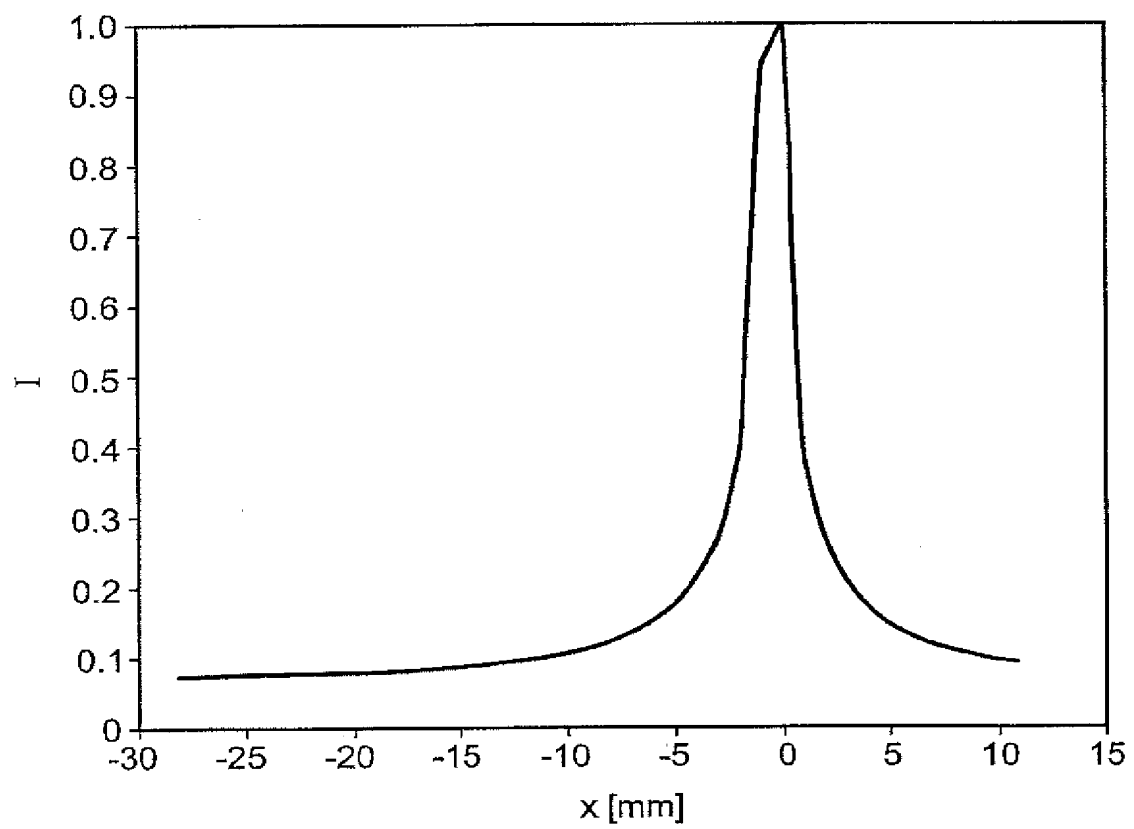


FIG.5

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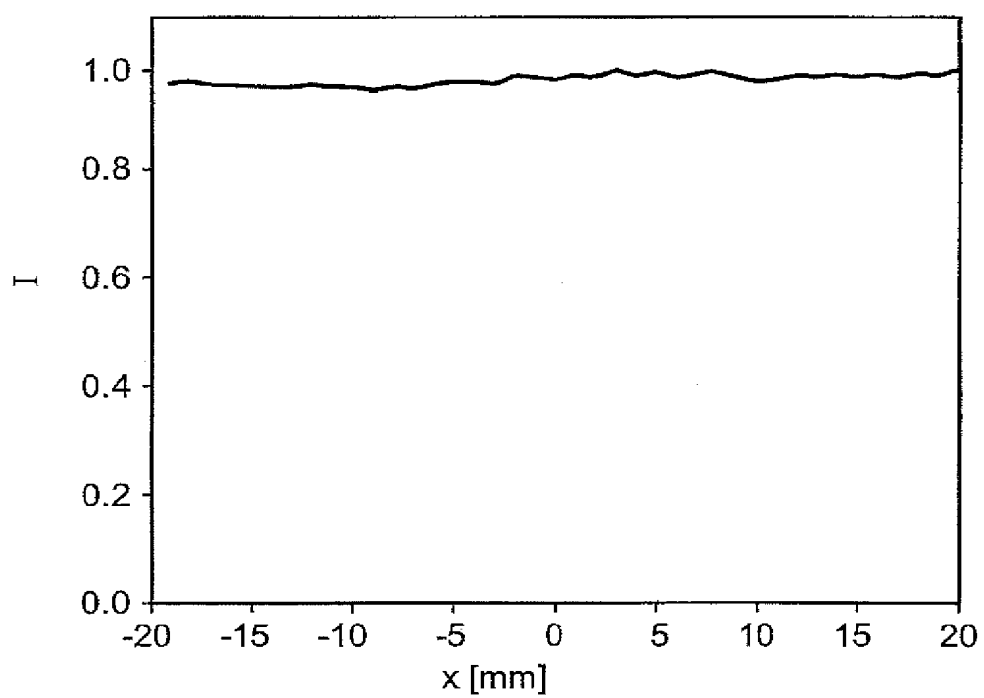


FIG.6

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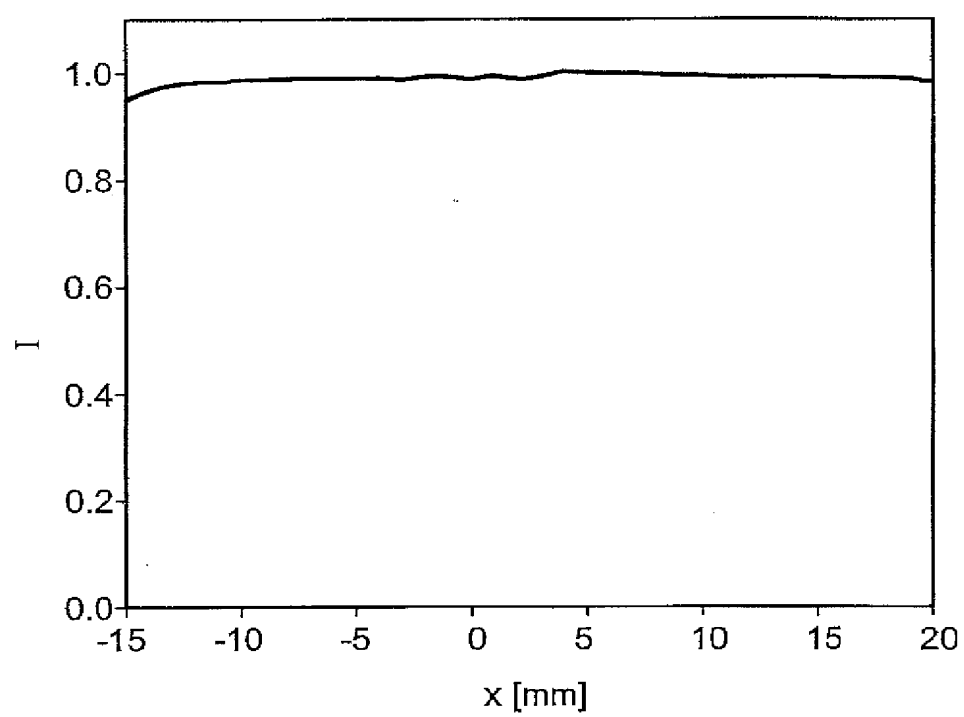


FIG.7

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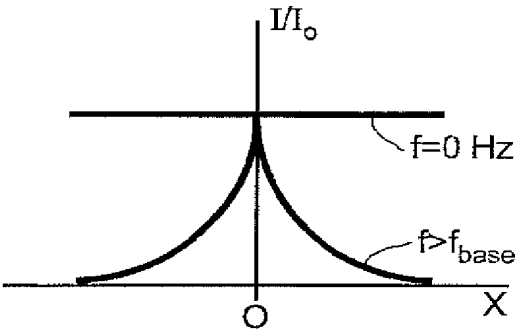


FIG.8

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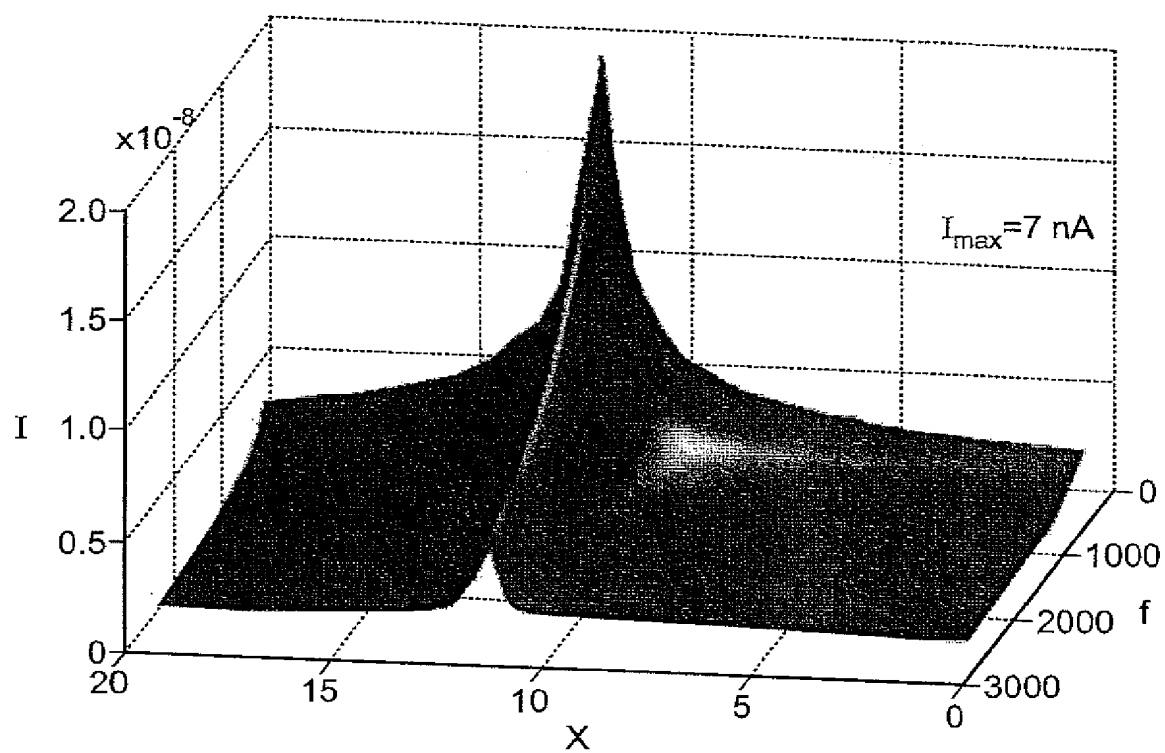


FIG.9

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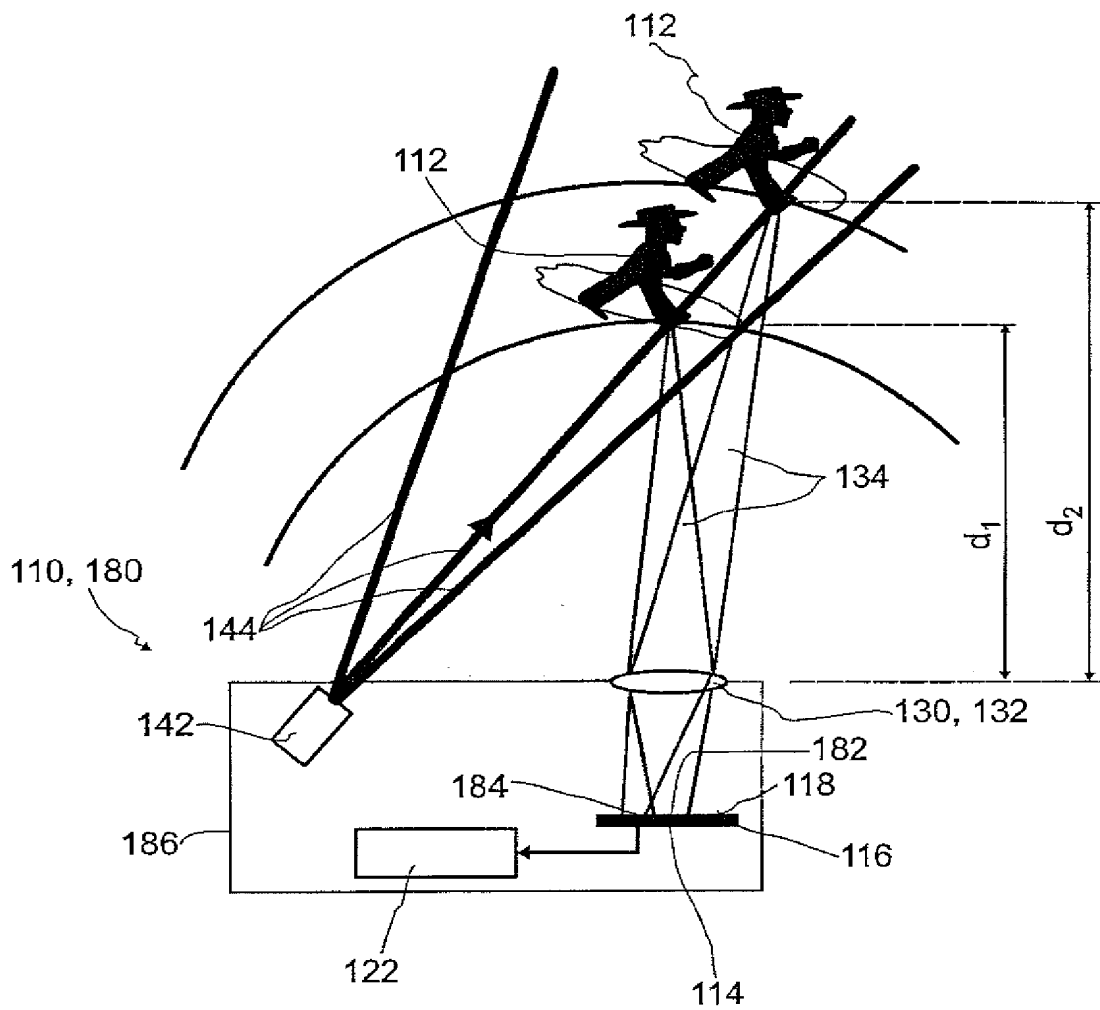


FIG.10

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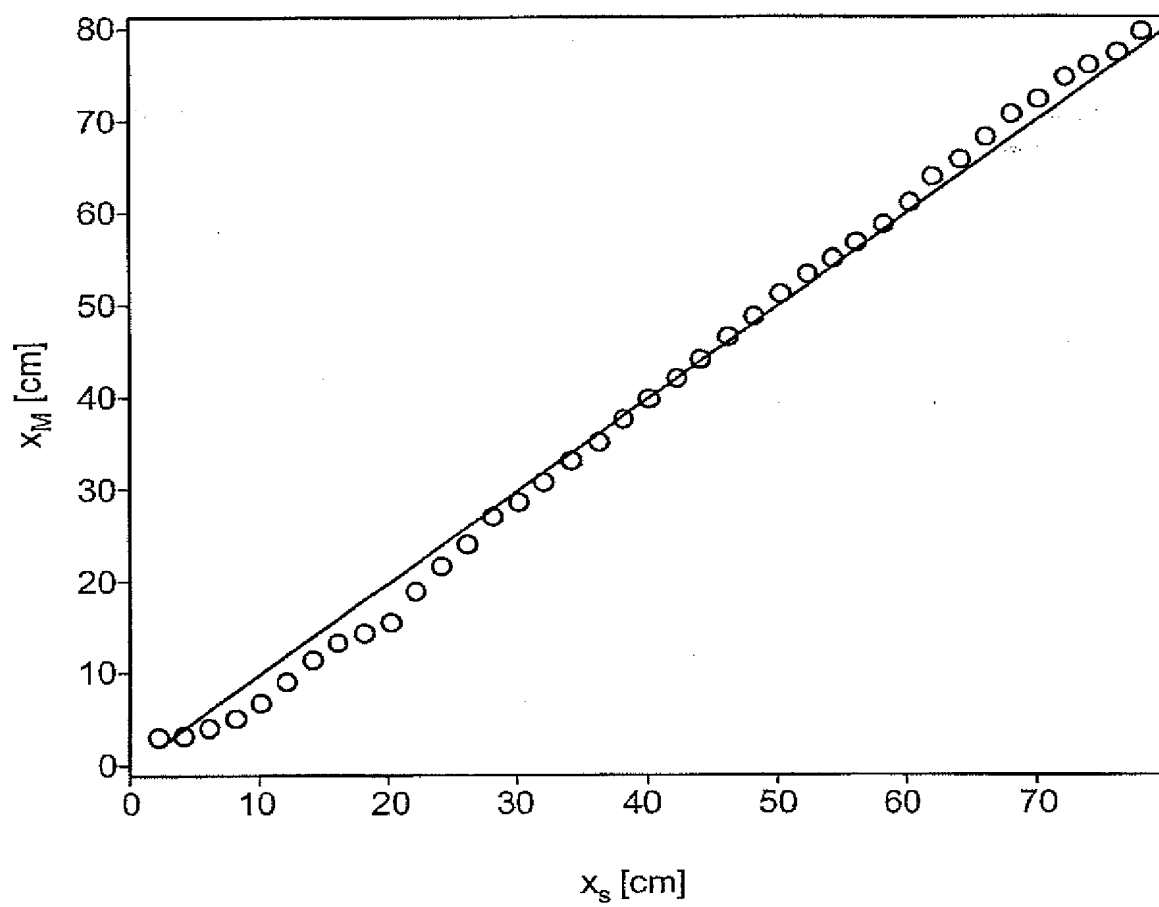


FIG.11

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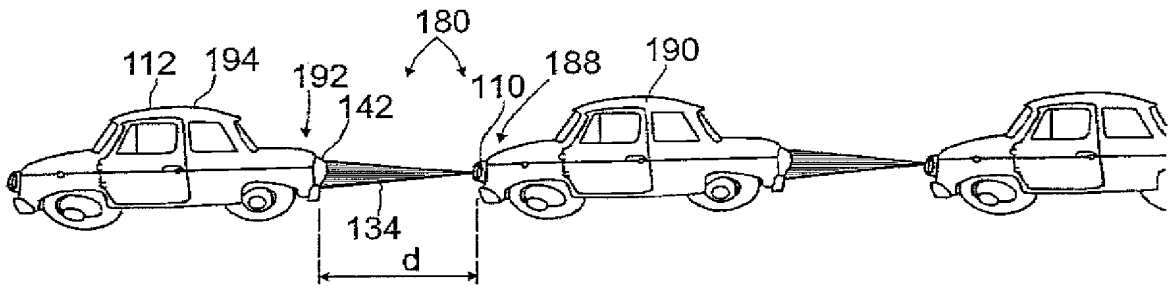


FIG.12A

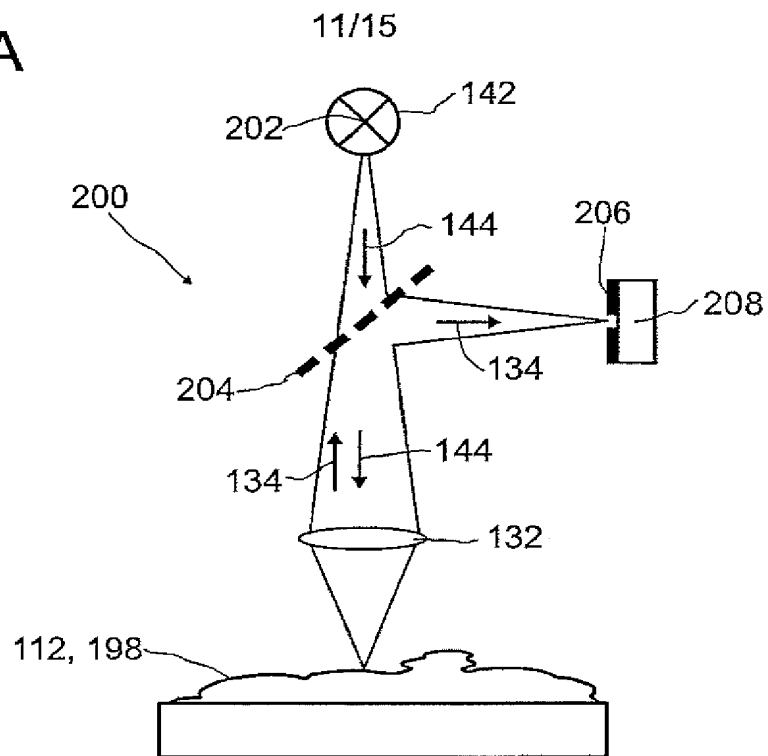


FIG.12B

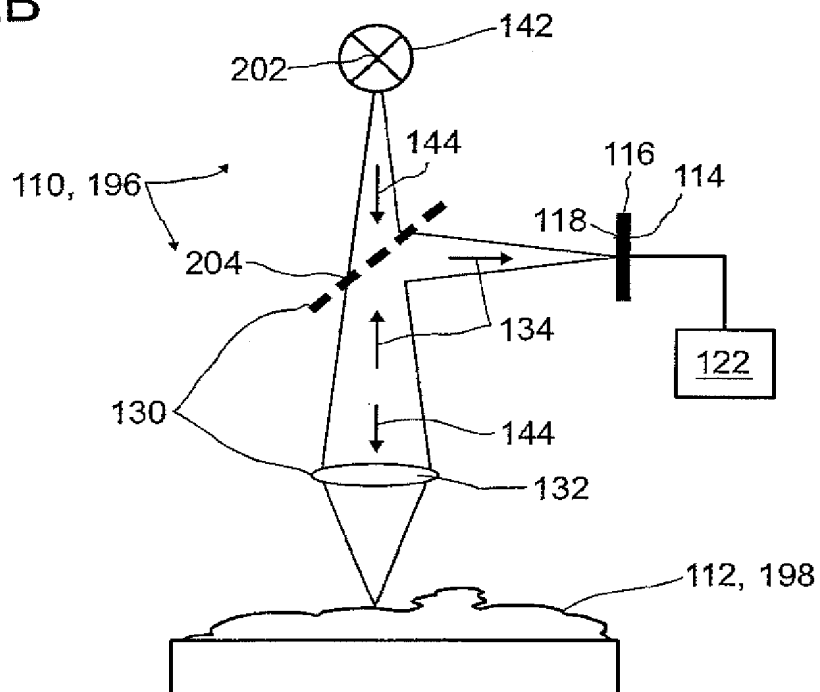


FIG.13

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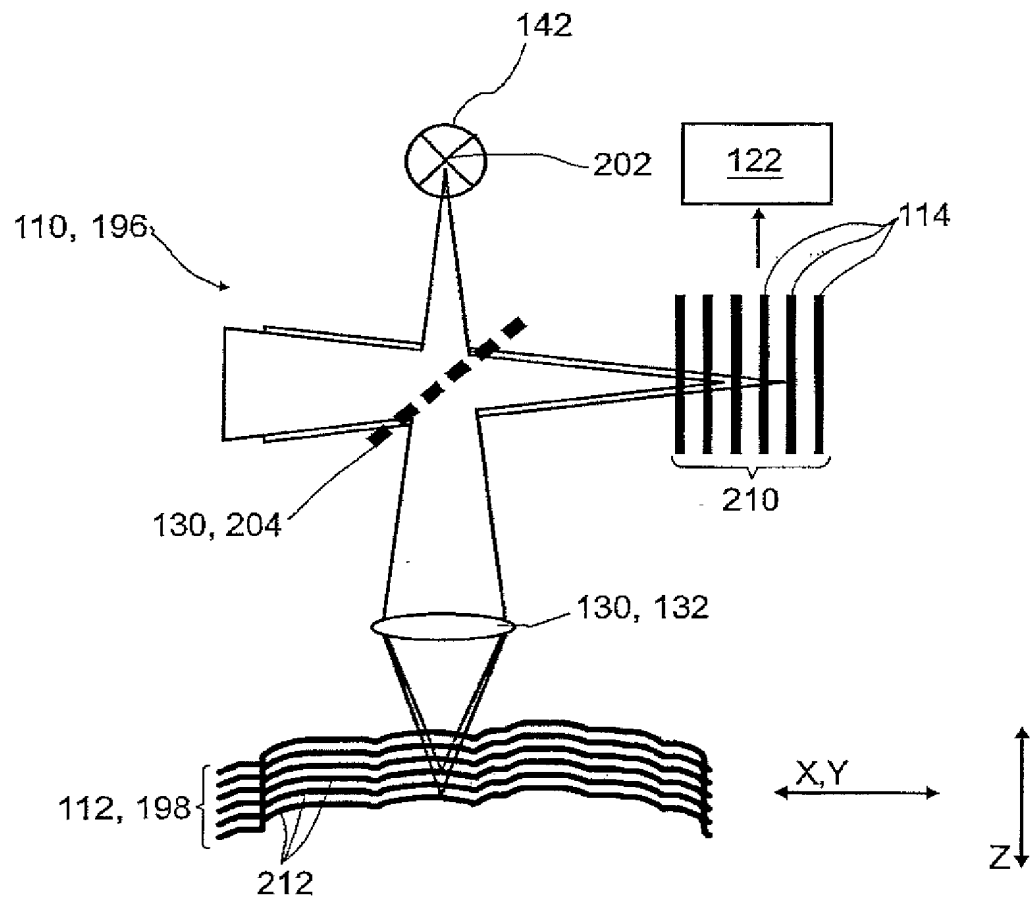


FIG.14

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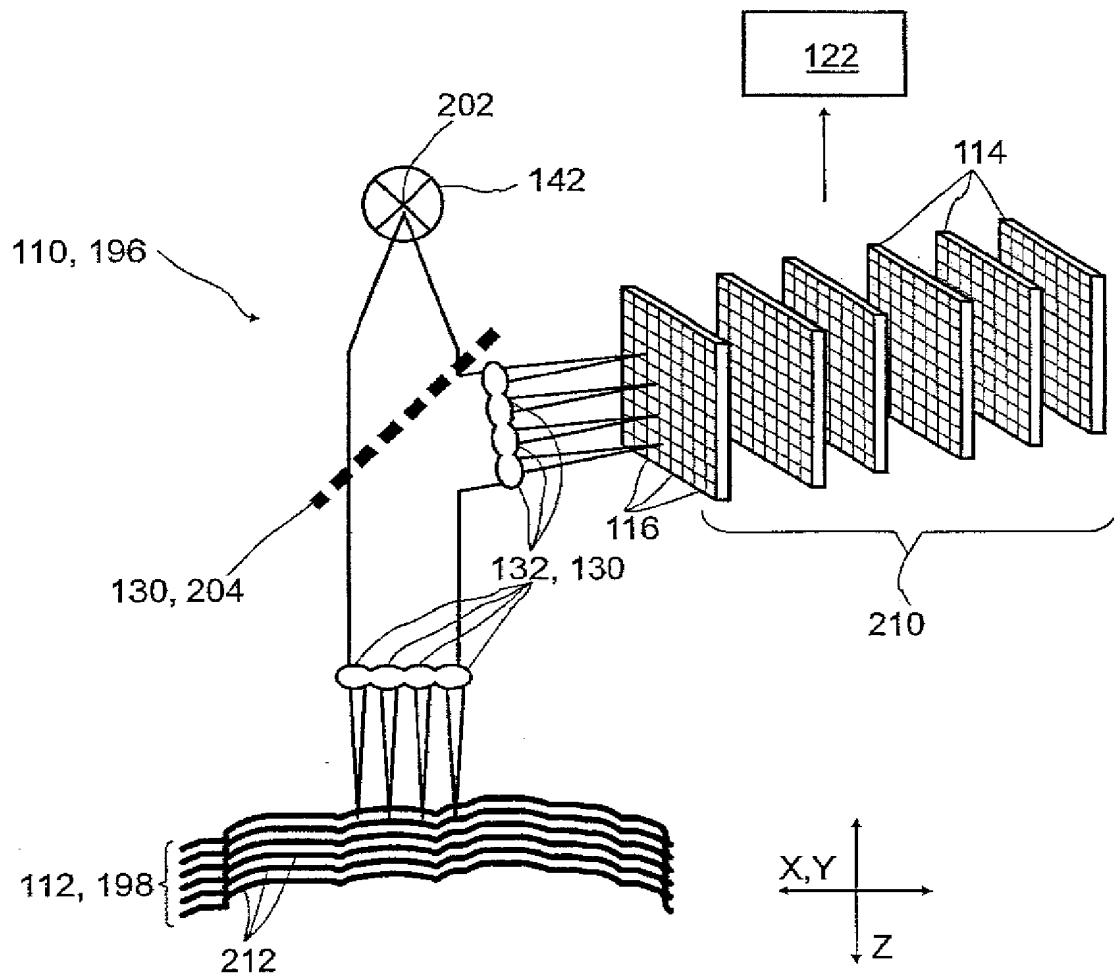


FIG.15

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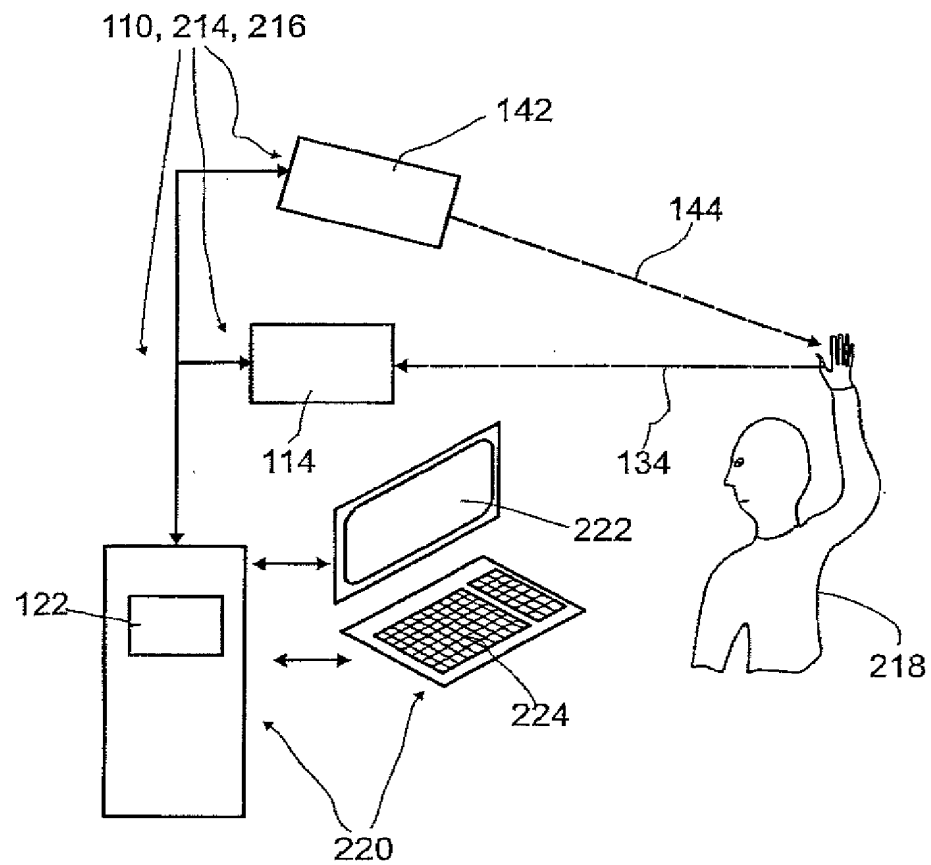
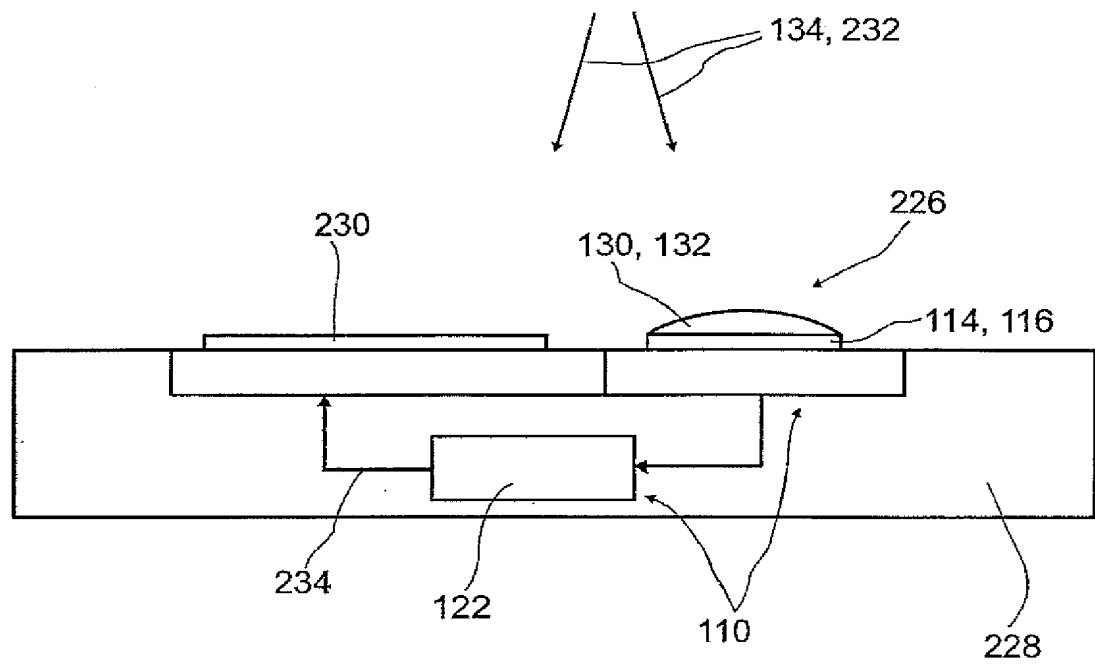


FIG.16

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INTERNATIONAL SEARCH REPORT

International application No.

PCT/IB2012/050592

A. CLASSIFICATION OF SUBJECT MATTER

G01C 3/00 (2006.01) i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

IPC: G01C3/-; G01B11/-; G01S11/-; G01C1/-

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

CNPAT,CNKI,WPI,EPODOC: position, distance, sensor, optical, light, geometry, area, solar cell

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	JP8-159714A (OMRON KK.) 21 Jun.1996 (21.06.1996) description paragraphs [0008]-[0010], figures 1-2	1, 2, 5, 6, 9, 11-22
A	CN101290348A (SHARP KK.) 22 Oct.2008 (22.10.2008) the whole document	1-22
A	CN1677053A (SHARP KK.) 05 Oct. 2005 (05.10.2005) the whole document	1-22
A	CN101650173A (UNIV. WUHAN TECHNOLOGY) 17 Feb.2010 (17.02.2010) the whole document	1-22
A	JP10-221064A (NIPPON JIDOSHA BUHIN SOGO) 21 Aug.1998 (21.08.1998) the whole document	1-22

☐ Further documents are listed in the continuation of Box C.☒ See patent family annex.

* Special categories of cited documents:

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“E” earlier application or patent but published on or after the international filing date

“L” document which may throw doubts on priority claim (S) or which is cited to establish the publication date of another citation or other special reason (as specified)

“O” document referring to an oral disclosure, use, exhibition or other means

“P” document published prior to the international filing date but later than the priority date claimed

“T” later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

“X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

“Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

“&” document member of the same patent family

Date of the actual completion of the international search
24 Apr.2012 (24.04.2012)Date of mailing of the international search report
31 May 2012 (31.05.2012)Name and mailing address of the ISA/CN
The State Intellectual Property Office, the P.R.China
6 Xitucheng Rd., Jimen Bridge, Haidian District, Beijing, China
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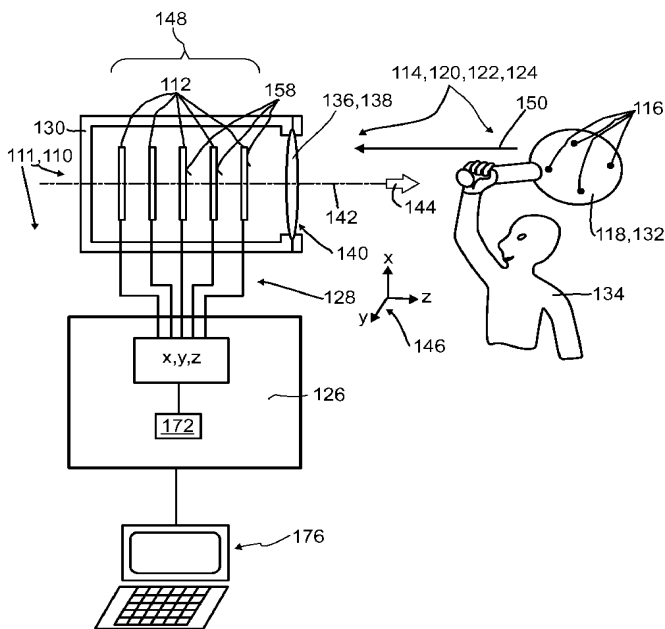
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[Continued on next page]

(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.1



(57) Abstract: A detector (110) for determining a position of at least one object (118) is disclosed. The detector (110) comprises: at least one optical sensor (112), the optical sensor (112) being adapted to detect a light beam (150) traveling from the object (118) towards the detector (110), the optical sensor (112) having at least one matrix (152) of pixels (154); and - at least one evaluation device (126), the evaluation device (126) being adapted to determine a number N of pixels (154) of the optical sensor (112) which are illuminated by the light beam (150), the evaluation device (126) further being adapted to determine at least one longitudinal coordinate of the object (118) by using the number N of pixels (154) which are illuminated by the light beam (150).



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Detector for optically detecting at least one object

Description

5 Field of the invention

The present invention is based on previous European patent application number 13171901.5, the full content of which is herewith included by reference. The invention relates to a detector, a detector system and a method for determining a position of at least one object. The invention
10 further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera and various uses of the detector device. The devices, systems, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as
15 digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. However, other applications are also possible.

Prior art

20 A large number of optical sensors and photovoltaic devices are known from the prior art. While photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultraviolet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

25 A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and
30 for reasons of large-area processing, sensors comprising at least one organic sensor material are being used, as described for example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO 2009/013282 A1.

35 A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological
40 samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for optically detecting objects are triangulation systems, by means of which distance measurements can likewise be carried out.

In US 2007/0080925 A1, a low power consumption display device is disclosed. Therein, photoactive layers are utilized that both respond to electrical energy to allow a display device to display information and that generate electrical energy in response to incident radiation. Display pixels of a single display device may be divided into displaying and generating pixels. The displaying pixels may display information and the generating pixels may generate electrical energy. The generated electrical energy may be used to provide power to drive an image.

In EP 1 667 246 A1, a sensor element capable of sensing more than one spectral band of electromagnetic radiation with the same spatial location is disclosed. The element consists of a stack of sub-elements each capable of sensing different spectral bands of electromagnetic radiation. The sub-elements each contain a non-silicon semiconductor where the non-silicon semiconductor in each sub-element is sensitive to and/or has been sensitized to be sensitive to different spectral bands of electromagnetic radiation.

In WO 2012/110924 A1, the content of which is herewith included by reference, a detector for optically detecting at least one object is proposed. The detector comprises at least one optical sensor. The optical sensor has at least one sensor region. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area. The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

US provisional applications 61/739,173, filed on December 19, 2012, 61/749,964, filed on January 8, 2013, and 61/867,169, filed on August 19, 2013, and international patent application PCT/IB2013/061095, filed on December 18, 2013, the full content of all of which is herewith included by reference, disclose a method and a detector for determining a position of at least one object, by using at least one transversal optical sensor and at least one optical sensor. Specifically, the use of sensor stacks is disclosed, in order to determine a longitudinal position of the object with a high degree of accuracy and without ambiguity.

Despite the advantages implied by the above-mentioned devices and detectors, specifically by the detectors disclosed in WO 2012/110924 A1, US 61/739,173 and 61/749,964, several technical challenges remain. Thus, generally, a need exists for detectors for detecting a position of an object in space which is both reliable and may be manufactured at low cost.

Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a

position of an object in space, preferably at a low technical effort and with low requirements in terms of technical resources and cost.

Summary of the invention

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This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

10

As used herein, the expressions "have", "comprise" and "contain" as well as grammatical variations thereof are used in a non-exclusive way. Thus, the expression "A has B" as well as the expression "A comprises B" or "A contains B" may both refer to the fact that, besides B, A contains one or more further components and/or constituents, and to the case in which, besides B, no other components, constituents or elements are present in A.

15

In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed. As used herein, the term position refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

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The detector comprises:

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- at least one optical sensor, the optical sensor being adapted to detect a light beam traveling from the object towards the detector, the optical sensor having at least one matrix of pixels; and
- at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

40

As used herein, an optical sensor generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by a light beam. The

optical sensor may be adapted, as outlined in further detail below, to determine at least one longitudinal coordinate of the object and/or of at least one part of the object, such as at least one part of the object from which the at least one light beam travels towards the detector.

5 As further used herein, a pixel generally refers to a light-sensitive element of the optical sensor, such as a minimum uniform unit of the optical sensor adapted to generate a light signal. As an example, each pixel may have a light-sensitive area of $1\ \mu\text{m}^2$ to $5\,000\,000\ \mu\text{m}^2$, preferably $100\ \mu\text{m}^2$ to $4\,000\,000\ \mu\text{m}^2$, preferably $1\,000\ \mu\text{m}^2$ to $1\,000\,000\ \mu\text{m}^2$ and more preferably $2\,500\ \mu\text{m}^2$ to $50\,000\ \mu\text{m}^2$. Still, other embodiments are feasible. The expression matrix generally refers to an
10 arrangement of a plurality of the pixels in space, which may be a linear arrangement or an areal arrangement. Thus, generally, the matrix preferably may be selected from the group consisting of a one-dimensional matrix and a two-dimensional matrix. As an example, the matrix may comprise 100 to $100\,000\,000$ pixels, preferably 1 000 to $1\,000\,000$ pixels and, more preferably, 10 000 to $500\,000$ pixels. Most preferably, the matrix is a rectangular matrix having pixels
15 arranged in rows and columns.

As further used herein, the term evaluation device generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor. Thus, as an example, the at least
20 one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands.

The optical sensor may be adapted to generate at least one signal indicating an intensity of illumination for each of the pixels. Thus, as an example, the optical sensor may be adapted to
25 generate at least one electronic signal for each of the pixels, each signal indicating the intensity of illumination for the respective pixel. The signal may be an analogue and/or a digital signal. Further, the detector may comprise one or more signal processing devices, such as one or more filters and/or analogue-digital-converters for processing and/or preprocessing the at least one signal.

30 The evaluation device may be adapted to compare, for each of the pixels, the signal to at least one threshold in order to determine whether the pixel is an illuminated pixel or not. This at least one threshold may be an individual threshold for each of the pixels or may be a threshold which is a uniform threshold for the whole matrix. In case a plurality of optical sensors is provided, at
35 least one threshold may be provided for each of the optical sensors and/or for a group comprising at least two of the optical sensors, wherein, for two optical sensors, their respective thresholds may be identical or different. Thus, for each of the optical sensors, an individual threshold may be provided.

40 As will be outlined in further detail below, the threshold may be predetermined and/or fixed. Alternatively, the at least one threshold may be variable. Thus, the at least one threshold may be determined individually for each measurement or groups of measurements. Thus, at least one algorithm may be provided adapted to determine the threshold.

The evaluation device generally may be adapted to determine at least one pixel having the highest illumination out of the pixels by comparing the signals of the pixels. Thus, the detector generally may be adapted to determine one or more pixels and/or an area or region of the matrix having the highest intensity of the illumination by the light beam. As an example, in this way, a center of illumination by the light beam may be determined.

The highest illumination and/or the information about the at least one area or region of highest illumination may be used in various ways. Thus, as outlined above, the at least one above-mentioned threshold may be a variable threshold. As an example, the evaluation device may be adapted to choose the above-mentioned at least one threshold as a fraction of the signal of the at least one pixel having the highest illumination. Thus, the evaluation device may be adapted to choose the threshold by multiplying the signal of the at least one pixel having the highest illumination with a factor of $1/e^2$. As will be outlined in further detail below, this option is particularly preferred in case Gaussian propagation properties are assumed for the at least one light beam, since the threshold $1/e^2$ generally determines the borders of a light spot having a beam radius or beam waist w generated by a Gaussian light beam on the optical sensor.

The evaluation device may be adapted to determine the longitudinal coordinate of the object by using a predetermined relationship between the number N of the pixels which are illuminated by the light beam and the longitudinal coordinate of the object. Thus, generally, the diameter of the light beam, due to propagation properties generally known to the skilled person, changes with propagation, such as with a longitudinal coordinate of the propagation. The relationship between the number of illuminated pixels and the longitudinal coordinate of the object may be an empirically determined relationship and/or may be analytically determined. Thus, as an example, a calibration process may be used for determining the relationship between the number of illuminated pixels and the longitudinal coordinate. Additionally or alternatively, as mentioned above, the predetermined relationship may be based on the assumption of the light beam being a Gaussian light beam. The light beam may be a monochromatic light beam having a precisely one wavelength λ or may be a light beam having a plurality of wavelengths or a wavelength spectrum, wherein, as an example, a central wavelength of the spectrum and/or a wavelength of a characteristic peak of the spectrum may be chosen as the wavelength λ of the light beam. In case a plurality of optical sensors is used, specific predetermined relationships between the respective number of illuminated pixels of the optical sensors and the longitudinal coordinate may be used. Thus, as an example, for each optical sensor, a predetermined relationship between the number of illuminated pixels and the longitudinal coordinate may be provided. As will be outlined in further detail below, the optical sensors may be arranged in a stack. Further, the optical sensors may have different properties, such as different geometries such as different geometries of the matrix of pixels and/or of the pixels. Further, the optical sensors may have differing sensitivities, such as differing spectral sensitivities. By providing specific predetermined relationships between the respective numbers of illuminated pixels of the optical sensors, these differing properties of the optical sensors may be taken into account.

As an example of an analytically determined relationship, the predetermined relationship, which may be derived by assuming Gaussian properties of the light beam, may be:

$$N \sim \pi \cdot w_0^2 \cdot \left(1 + \left(\frac{z}{z_0}\right)^2\right), \quad (1)$$

5 wherein z is the longitudinal coordinate,
wherein w_0 is a minimum beam radius of the light beam when propagating in space,
wherein z_0 is a Rayleigh-length of the light beam with $z_0 = \pi \cdot w_0^2 / \lambda$, λ being the wavelength
of the light beam.

10 This relationship may generally be derived from the general equation of an intensity I of a
Gaussian light beam traveling along a z -axis of a coordinate system, with r being a coordinate
perpendicular to the z -axis and E being the electric field of the light beam:

$$I(r, z) = |E(r, z)|^2 = I_0 \cdot (w_0/w(z))^2 \cdot e^{-2r^2/w(z)^2} \quad (2)$$

15 The beam radius w of the transversal profile of the Gaussian light beam generally representing
a Gaussian curve is defined, for a specific z -value, as a specific distance from the z -axis at
which the amplitude E has dropped to a value of $1/e$ (approx. 36%) and at which the intensity I
has dropped to $1/e^2$. The minimum beam radius, which, in the Gaussian equation given above
(which may also occur at other z -values, such as when performing a z -coordinate
20 transformation), occurs at coordinate $z = 0$, is denoted by w_0 . Depending on the z -coordinate,
the beam radius generally follows the following equation when light beam propagates along the
 z -axis:

$$w(z) = w_0 \cdot \sqrt{1 + \left(\frac{z}{z_0}\right)^2} \quad (3)$$

25 With the number N of illuminated pixels being proportional to the illuminated area A of the
optical sensor:

$$N \sim A \quad (4)$$

or, in case a plurality of optical sensors $i = 1, \dots, n$ is used, with the number N_i of illuminated
30 pixels for each optical sensor being proportional to the illuminated area A_i of the respective
optical sensor

$$N_i \sim A_i \quad (4')$$

and the general area of a circle having a radius w :

$$A = \pi \cdot w^2, \quad (5)$$

the following relationship between the number of illuminated pixels and the z -coordinate may be derived:

$$N \sim \pi \cdot w_0^2 \cdot \left(1 + \left(\frac{z}{z_0}\right)^2\right) \quad (6)$$

5

or

$$N_i \sim \pi \cdot w_0^2 \cdot \left(1 + \left(\frac{z}{z_0}\right)^2\right), \quad (6')$$

respectively, with $z_0 = \pi \cdot w_0^2 / \lambda$, as mentioned above. Thus, with N or N_i , respectively, being
 10 the number of pixels within a circle being illuminated at an intensity $I \geq I_0/e^2$, as an example, N or N_i may be determined by simple counting of pixels and/or other methods, such as a histogram analysis. In other words, a well-defined relationship between the z -coordinate and the number of illuminated pixels N or N_i , respectively, may be used for determining the longitudinal coordinate z of the object and/or of at least one point of the object, such as at least
 15 one longitudinal coordinate of at least one beacon device being one of integrated into the object and/or attached to the object.

In the equations given above, such as in equation (1), it is assumed that the light beam has a focus at position $z = 0$. It shall be noted however, that a coordinate transformation of the z -
 20 coordinate is possible, such as by adding and/or subtracting a specific value. Thus, as an example, the position of the focus typically is dependent on the distance of the object from the detector and/or on other properties of the light beam. Thus, by determining the focus and/or the position of the focus, a position of the object, specifically a longitudinal coordinate of the object, may be determined, such as by using an empirical and/or an analytical relationship between a
 25 position of the focus and a longitudinal coordinate of the object and/or the beacon device. Further, imaging properties of the at least one optional transfer device, such as the at least one optional lens, may be taken into account. Thus, as an example, in case beam properties of the light beam being directed from the object towards the detector are known, such as in case emission properties of an illuminating device contained in a beacon device are known, by using
 30 appropriate Gaussian transfer matrices representing a propagation from the object to the transfer device, representing imaging of the transfer device and representing beam propagation from the transfer device to the at least one optical sensor, a correlation between a beam waist and a position of the object and/or the beacon device may easily be determined analytically. Additionally or alternatively, a correlation may empirically be determined by appropriate
 35 calibration measurements.

As outlined above, the matrix of pixels preferably may be a two-dimensional matrix. However, other embodiments are feasible, such as one-dimensional matrices. More preferably, as outlined above, the matrix of pixels is a rectangular matrix.

- 5 The detector may comprise precisely one optical sensor or may comprise a plurality of the optical sensors. In case a plurality of the optical sensors is provided, the optical sensors may arrange in various ways. Thus, as an example, the light beam may be split into two or more beams, each beam being directed towards one or more of the optical sensors. Additionally or alternatively, two or more of the optical sensors may be stacked along an axis, such as along an
10 optical axis of the detector, preferably with their sensitive areas face towards the same direction. Thus, as an example, the detector may comprise a plurality of the optical sensors, wherein the optical sensors are stacked along an optical axis of the detector.

- As an example, the detector may comprise n optical sensors, with $n > 1$. The evaluation device
15 generally may be adapted to determine the number N_i of the pixels which are illuminated by the light beam for each of the optical sensors, wherein $i \in \{1, n\}$ denotes the respective optical sensor. As used herein, "each" refers to the fact that a number of illuminated pixels is determined for each optical sensor part of a plurality of the optical sensors, notwithstanding the fact that optical sensors may exist which are not illuminated by the light beam and/or which are
20 used for other purposes, and, consequently, for which no number of illuminated pixels is determined.

- The evaluation device may be adapted to compare the number N_i of pixels which are illuminated by the light beam for each optical sensor with at least one neighboring optical
25 sensor, thereby resolving an ambiguity in the longitudinal coordinate of the object.

- The sensor signals of the optical sensors may be used in various ways. Thus, as an example, the redundancy of the sensor signals may be used for eliminating unknown information regarding the power of the light beam. Thus, the sensor signals may be normalized, such as by
30 setting the strongest sensor signal to value 1 or 100 and providing the remaining sensor signals as fractions of this strongest sensor signal. Thus, generally, the evaluation device may be adapted to normalize sensor signals of the optical sensors for a power of the light beam. Additionally or alternatively, however, as outlined above, a normalization may take place within each optical sensor, such as for setting appropriate thresholds for determining pixels which are
35 illuminated by the light beam and for determining pixels which are not illuminated by the light beam.

- The at least one optical sensor generally may be transparent or may be intransparent. Thus, as an example, the optical sensor may be transparent and adapted to transmit at least 10%,
40 preferably at least 20% and, more preferably, at least 50% of the power of the light beam. In case a plurality of the optical sensors is provided, such as in a stacked fashion, preferably, at least one of the optical sensors is transparent.

In case a plurality of the optical sensors is provided, which may be arranged in the stacked fashion and/or in another arrangement, the optical sensors may have identical spectral sensitivities or may provide different spectral sensitivities. Thus, as an example, at least two of the optical sensors may have a differing spectral sensitivity. As used herein, the term spectral sensitivity generally refers to the fact that the sensor signal of the optical sensor, for the same power of the light beam, may vary with the wavelength of the light beam. Thus, generally, at least two of the optical sensors may differ with regard to their spectral properties. This embodiment generally may be achieved by using different types of absorbing materials for the optical sensors, such as different types of dyes or other absorbing materials. Additionally or alternatively, differing spectral properties of the optical sensors may be generated by other means implemented into the optical sensors and/or into the detector, such as by using one or more wavelength-selective elements, such as one or more filters (such as color filters) in front of the optical sensors and/or by using one or more prisms and/or by using one or more dichroitic mirrors. Thus, in case a plurality of optical sensors is provided, at least one of the optical sensors may comprise a wavelength-selective element such as a color filter, having a specific transmission or reflection characteristic, thereby generating differing spectral properties of the optical sensors. Further, as will be outlined in further detail below, in case a plurality of optical sensors is used, these optical sensors may all be organic optical sensors, may all be inorganic optical sensors, may all be hybrid organic – inorganic optical sensors or may comprise an arbitrary combination of at least two optical sensors selected from the group consisting of organic optical sensors, inorganic optical sensors and hybrid organic – inorganic optical sensors.

In case a plurality of the optical sensors is used, wherein at least two of the optical sensors differ with regard to their respective spectral sensitivity, the evaluation device generally may be adapted to determine a color of the light beam by comparing sensor signals of the optical sensors having the differing spectral sensitivity. As used herein, the expression “determine a color” generally refers to the step of generating at least one item of spectral information about the light beam. The at least one item of spectral information may be selected from the group consisting of a wavelength, specifically a peak wavelength; color coordinates, such as CIE coordinates.

The determination of the color of the light beam may be performed in various ways which are generally known to the skilled person. Thus, the spectral sensitivities of the optical sensors may span a coordinate system in color space, and the signals provided by the optical sensors may provide a coordinate in this color space, as known to the skilled person for example from the way of determining CIE coordinates.

As an example, the detector may comprise two, three or more optical sensors in a stack. Thereof, at least two, preferably at least three, of the optical sensors may have differing spectral sensitivities. Further, the evaluation device may be adapted to generate at least one item of color information for the light beam by evaluating the signals of the optical sensors having differing spectral sensitivities.

As an example, at least three optical sensors being spectrally sensitive optical sensors may be contained in the stack. Thus, e.g., the spectrally sensitive optical sensors may comprise at least one red sensitive optical sensor, the red sensitive optical sensor having a maximum absorption wavelength λ_r in a spectral range $600 \text{ nm} < \lambda_r < 780 \text{ nm}$, wherein the spectrally sensitive optical sensors further comprise at least one green sensitive optical sensor, the green sensitive optical sensor having a maximum absorption wavelength λ_g in a spectral range $490 \text{ nm} < \lambda_g < 600 \text{ nm}$, wherein the spectrally sensitive optical sensors further may comprise at least one blue sensitive optical sensor, the blue sensitive optical sensor having a maximum absorption wavelength λ_b in a spectral range $380 \text{ nm} < \lambda_b < 490 \text{ nm}$. As an example, the red sensitive optical sensor, the green sensitive optical sensor and the blue sensitive optical sensor, in this order or in a different order, may be the first optical sensors of the optical sensor stack, facing towards the object.

The evaluation device may be adapted to generate at least two color coordinates, preferably at least three color coordinates, wherein each of the color coordinates is determined by dividing a signal of one of the spectrally sensitive optical sensors by a normalization value. As an example, the normalization value may contain a sum of the signals of all spectrally sensitive optical sensors. Additionally or alternatively, the normalization value may contain a detector signal of a white detector.

The at least one item of color information may contain the color coordinates. The at least one item of color information may, as an example, contain CIE coordinates.

In addition to the preferred at least two, more preferably at least three, spectrally sensitive optical sensors, the stack further may comprise at least one white detector, wherein the white detector may be adapted to absorb light in an absorption range of all spectrally sensitive detectors. Thus, as an example, the white detector may have an absorption spectrum absorbing light all over the visible spectral range.

The spectrally sensitive optical sensors each may comprise at least one dye, the dye being adapted to absorb light in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range, with a non-uniform absorption spectrum. As an example, the dyes each may have an absorption spectrum having at least one absorption peak. The absorption peak, as a solid, as a film or as sensitizer on a scaffold material (e.g. TiO_2), may have a width, such as a full width at half maximum (FWHM) which may be broad, such as by using dyes absorbing all over one or more of the visible, the infrared or the ultraviolet spectral range, or may be narrow, such as by using dyes having an FWHM of no more than 300 nm, preferably of no more than 200 nm, more preferably of no more than 80 nm, of no more than 60 nm or of no more than 40 nm. The absorption peaks of the dyes may be spaced apart by at least 60 nm, preferably at least 80 nm and, more preferably, by at least 100 nm. Other embodiments are feasible.

In case the detector comprises a plurality of optical sensors, as outlined above, the optical sensors preferably may be arranged in a stack. In case a plurality of optical sensors is provided,

the optical sensors may be identical, or at least two of the optical sensors may differ with regard to at least one property, such as with regard to a geometric property, a property of device setup or, as outlined above, spectral sensitivity.

5 As an example, as outlined above, the plurality of optical sensors may provide differing spectral sensitivities, such as an optical sensor with sensitivity in the red spectral range (R), an optical sensor with spectral sensitivity in the green spectral range (G) and an optical sensor with spectral sensitivity in the blue spectral range (B). As outlined above, the differing spectral sensitivities may be provided by various means, such as by providing absorbing materials
10 having differing spectral sensitivities and/or by providing one or more wavelength-selective elements. Thus, in a stack of optical sensors and/or in another arrangement of a plurality of optical sensors, various combinations of optical sensors having differing spectral sensitivities may be provided. As an example, an RGB stack may be provided, in the given order or in a different order.

15 Further, in case a plurality of optical sensors is provided, the plurality of optical sensors may differ with regard to device setup and/or with regard to the materials used in the optical sensors. Specifically, the optical sensors may differ with regard to their organic or inorganic in nature. Thus, the plurality of optical sensors and, more specifically, the stack, may comprise one or
20 more organic optical sensors, one or more inorganic optical sensors, one or more hybrid organic-inorganic optical sensors or an arbitrary combination of at least two of these optical sensors. Thus, as an example, the stack may consist of organic optical sensors only, may consist of inorganic optical sensors only or may consist of hybrid organic-inorganic optical sensors, only. Additionally or alternatively, the stack may comprise at least one organic optical
25 sensor and at least one inorganic optical sensor or may comprise at least one organic optical sensor and at least one hybrid organic-inorganic optical sensor or may comprise at least one organic optical sensor and at least one hybrid organic-inorganic optical sensor. Preferably, the stack may comprise at least one organic optical sensor and, further, preferably on a side of the stack furthest away from the object, at least one inorganic optical sensor. Thus, as an example,
30 as will be outlined in further detail below, the stack may comprise at least one organic optical sensor, such as at least one DSC or sDSC, and, further, at least one inorganic sensor such as a CCD or CMOS sensor chip more preferably on a side of the stack furthest away from the object.

Further, in case a plurality of optical sensors is provided, the optical sensors may differ with
35 regard to their transparency. Thus, as an example, all optical sensors may fully or partially be transparent. Alternatively, all optical sensors may fully or partially be intransparent (also referred to as opaque). Further, specifically in case the plurality of optical sensors is arranged as a stack, a combination of at least one at least partially transparent optical sensor and at least one at least partially intransparent optical sensor may be used. Thus, the stack may comprise one or
40 more transparent optical sensors and, additionally, preferably on a side of the stack furthest away from the object, at least one intransparent optical sensor. Thus, as outlined above, one or more at least partially transparent optical sensors may be provided by using one or more transparent organic optical sensors such as one or more transparent DSCs or sDSCs.

Additionally or alternatively, at least partially transparent optical sensors may be provided by using inorganic sensors, such as very thin inorganic optical sensors or inorganic optical sensors having a bandgap which is designed such that at least a part of the incident light beam is transmitted. Intransparent optical sensors may be provided by using intransparent electrodes and/or intransparent absorbing materials, such as organic and/or inorganic materials. As an example, an organic optical sensor may be provided having a thick metal electrode, such as a metal electrode having a thickness of more than 50 nm, preferably more than 100 nm. Additionally or alternatively, an inorganic optical sensor may be provided having an intransparent semiconductor material. As an example, typical CCD or CMOS camera chips provide intransparent properties. As an example, the stack may comprise one or more at least partially transparent DSCs or sDSCs and, on a side furthest away from the object, an intransparent CMOS or CCD chip. Other embodiments are feasible.

As outlined above, the detector may further comprise at least one transfer device. The transfer device preferably may be positioned in a light path in between the object and the detector. As used herein, a transfer device generally is an arbitrary optical element adapted to guide the light beam onto the optical sensor. The guiding may take place with unmodified properties of the light beam or may take place with imaging or modifying properties. Thus, generally, the transfer device might have imaging properties and/or beam-shaping properties, i.e. might change a beam waist and/or a widening angle of the light beam and/or a shape of the cross-section of the light beam when the light beam passes the transfer device. The transfer device, as an example, may comprise one or more elements selected from the group consisting of a lens and a mirror. The mirror may be selected from the group consisting of a planar mirror, a convex mirror and a concave mirror. Additionally or alternatively, one or more prisms may be comprised. Additionally or alternatively, one or more wavelength-selective elements may be comprised, such as one or more filters, specifically color filters, and/or one or more dichroitic mirrors. Again, additionally or alternatively, the transfer device may comprise one or more diaphragms, such as one or more pinhole diaphragms and/or iris diaphragms.

The transfer device can for example comprise one or a plurality of mirrors and/or beam splitters and/or beam deflecting elements in order to influence a direction of the electromagnetic radiation. Alternatively or additionally, the transfer device can comprise one or a plurality of imaging elements which can have the effect of a converging lens and/or a diverging lens. By way of example, the optional transfer device can have one or a plurality of lenses and/or one or a plurality of convex and/or concave mirrors. Once again alternatively or additionally, the transfer device can have at least one wavelength-selective element, for example at least one optical filter. Once again alternatively or additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The above-mentioned optional embodiments of the optional transfer device can, in principle, be realized individually or in any desired combination. The at least one transfer device, as an example, may be positioned in front of the detector, i.e. on a side of the detector facing towards the object.

In addition to the at least one longitudinal coordinate of the object, at least one transversal coordinate of the object may be determined. Thus, generally, the evaluation device may further be adapted to determine at least one transversal coordinate of the object by determining a position of the light beam on the matrix of pixels. As an example, the evaluation device may be adapted to determine a center of illumination of the at least one matrix by the light beam, wherein the at least one transversal coordinate of the object is determined by evaluating at least one coordinate of the center of illumination. Thus, the coordinate of the center of illumination may be a pixel coordinate of the center of illumination. As an example, the matrix may comprise rows and columns of pixels, wherein the row number of the light beam and/or the center of the light beam within the matrix may provide an x-coordinate, and wherein the column number of the light beam and/or the center of the light beam within the matrix may provide a y-coordinate.

By providing one or more transversal coordinates, such as x- and/or y-coordinates, the evaluation device is adapted to provide at least one three-dimensional position of the object.

Further options of the present invention refer to potential embodiments of the at least one optical sensor. Generally, an arbitrary optical sensor device having a matrix of pixels used, such as an optical sensor device selected from the group consisting of: an inorganic semiconductor sensor device such as a CCD chip and/or a CMOS chip; an organic semiconductor sensor device. In the latter case, as an example, the optical sensor may, for example, comprise at least one organic photovoltaic device having a matrix of pixels. As used herein, an organic photovoltaic device generally refers to a device having at least one organic photosensitive element and/or at least one organic layer. Therein, generally, any type of organic photovoltaic device may be used, such as organic solar cells and/or an arbitrary device having at least one organic photosensitive layer. As an example, an organic solar cell and/or a dye-sensitized organic solar cell may be comprised. Further, a hybrid device may be used, such as inorganic – organic photovoltaic devices. The detector may comprise one or more organic photovoltaic devices and/or may comprise one or more inorganic photovoltaic devices and/or may comprise a combination consisting of one or more organic photovoltaic devices and one or more inorganic photovoltaic devices.

The optical sensor specifically may comprise at least one dye-sensitized solar cell having at least one patterned electrode. For potential embodiments of dye-sensitized solar cells, reference may be made to WO 2012/110924 A1 and to US provisional applications number 61/739,173 and 61/749,964. The device setups of the dye-sensitized solar cells disclosed therein, specifically of the optical sensors disclosed therein, may generally also be applied to the present invention, with the proviso that at least one of the electrodes of these dye-sensitized solar cells is patterned in order to provide a plurality of pixels. Thus, as an example, one or both of the first and second electrodes of the dye-sensitized solar cells disclosed in the named documents may be patterned. As an example, the first electrode may be patterned to provide a plurality of parallel horizontal stripes, and the second electrode may be patterned to provide a plurality of parallel vertical stripes, or vice versa. Thus, at each crossing point of a stripe of the first electrode and a stripe of the second electrode, a pixel is provided which, by electrically

contacting the respective stripes of the electrodes and measuring an electrical current and/or voltage, may be read out.

The optical sensor specifically may comprise at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode, with the at least one n-semiconducting metal oxide, the at least one dye and the at least one p-semiconducting organic material being embedded in between the first electrode and the second electrode.

As mentioned above and as will be outlined in further detail below, generally, for potential embodiments of materials or layer combinations which may be used for the first electrode, the at least one n-semiconducting metal oxide, the at least one dye, the at least one p-semiconducting organic material and the at least one second electrode, reference may be made to the above-mentioned WO 2012/110924 A1 as well as to US provisional applications 61/739,173, filed on December 19, 2012, and 61/749,964, filed on January 8, 2013. Still, other embodiments are feasible. Thus, as an example, the n-semiconducting metal oxide may comprise a nanoporous layer of a metal oxide and a dense layer of a metal oxide. The metal oxide preferably may be or may comprise titanium dioxide. For potential dyes, reference may be made to the dyes mentioned in the above-mentioned documents, such as to the dye ID 504. Further, with regard to potential p-semiconducting organic materials, as an example, spiro-MeOTAD may be used, as disclosed in one or more of the above-mentioned documents. Similarly, for potential electrode materials, both transparent and intransparent, reference may be made to these documents. As outlined above, other embodiments are feasible.

Both the first electrode and the second electrode may be transparent. However, other electrode setups may be possible.

As outlined above, preferably, at least one of the first electrode and the second electrode may be a patterned electrode. Thus, as an example, one or both of the first electrode and the second electrode may comprise a plurality of electrode stripes, such as horizontal electrode stripes and/or vertical electrode stripes. As an example for pixelated optical sensors which may also be used in the context of the present invention, reference may be made to EP patent application No. 13171898.3, the full content of which is herewith included by reference. Still, other pixelated optical sensors may be used.

As an example, as outlined above, the first electrode may comprise a plurality of electrode stripes and the second electrode may comprise a plurality of electrode stripes, wherein the electrode stripes of the first electrode are oriented perpendicular to the electrode stripes of the second electrode. As outlined above, at each crossing point of electrode stripes of the first electrode and an electrode stripe of the second electrode, a pixel is formed which may be read out independently.

The detector may further comprise appropriate reading electronics, such as a reading electronics adapted to contact one of the electrode stripes of the first electrode and to contact one of the electrode stripes of the second electrode and to measure a current through the stripes and/or to measure a voltage at the respective stripes. In order to read out the pixels, a sequential reading scheme may be chosen and/or a multiplexing scheme. Thus, as an example, all pixels within one row may be read out simultaneously, before switching to the next row of the matrix, thereby sequentially reading out all pixel rows. Alternatively, a row multiplexing be chosen, reading out all pixels of one column spontaneously, before switching to the next column. However, other readout schemes are possible, such as readout schemes using a plurality of transistors. Generally, passive matrix and/or active matrix readout schemes may be used. As an example of read-out schemes which may be used, reference may be made to US 2007/0080925 A1. Other readout schemes are feasible.

For potential electrode materials, reference may be made to WO 2012/110924 A1 and to US provisional applications number 61/739,173 and 61/749,964. Other embodiments are possible. Specifically, at least one of the first electrode and the second electrode may comprise an electrically conductive polymer. Specifically, a transparent electrically conductive polymer may be used. For potential embodiments of electrically conductive polymers, reference may be made to the named documents.

It shall be noted, however, that any other type of optical sensors, preferably transparent optical sensors, having a plurality of pixels, may be used for the present invention.

In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments described above and/or as disclosed in one or more of the embodiments disclosed in further detail below.

The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector. As used herein and as will be disclosed in further detail below, a beacon device generally refers to an arbitrary device adapted to direct at least one light beam towards the detector. The beacon device may fully or partially be embodied as an active beacon device, comprising at least one illumination source for generating the light beam. Additionally or alternatively, the beacon device may fully or partially be embodied as a passive beacon device comprising at least one reflective element adapted to reflect a primary light beam generated independently from the beacon device towards the detector.

The beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Thus, the beacon device may be attached to the object by an arbitrary attachment means, such as one or more connecting elements. Additionally or alternatively, the object may be adapted to hold the beacon device, such as by one or more appropriate holding means. Additionally or alternatively, again, the beacon device may fully or

partially be integrated into the object and, thus, may form part of the object or even may form the object.

Generally, with regard to potential embodiments of the beacon device, reference may be made to one or more of US provisional applications 61/739,173, filed on December 19, 2012, and 61/749,964, filed on January 8, 2013. Still, other embodiments are feasible.

As outlined above, the beacon device may fully or partially be embodied as an active beacon device and may comprise at least one illumination source. Thus, as an example, the beacon device may comprise a generally arbitrary illumination source, such as an illumination source selected from the group consisting of a light-emitting diode (LED), a light bulb, an incandescent lamp and a fluorescent lamp. Other embodiments are feasible.

Additionally or alternatively, as outlined above, the beacon device may fully or partially be embodied as a passive beacon device and may comprise at least one reflective device adapted to reflect a primary light beam generated by an illumination source independent from the object. Thus, in addition or alternatively to generating the light beam, the beacon device may be adapted to reflect a primary light beam towards the detector.

The detector system may comprise one, two, three or more beacon devices. Thus, generally, in case the object is a rigid object which, at least on a microscope scale, thus not change its shape, preferably, at least two beacon devices may be used. In case the object is fully or partially flexible or is adapted to fully or partially change its shape, preferably, three or more beacon devices may be used. Generally, the number of beacon devices may be adapted to the degree of flexibility of the object. Preferably, the detector system comprises at least three beacon devices.

The object generally may be a living or non-living object. The detector system even may comprise the at least one object, the object thereby forming part of the detector system.

Preferably, however, the object may move independently from the detector, in at least one spatial dimension.

The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the present invention, such as to one or more of the embodiments disclosed above and/or according to one or more of the
5 embodiments disclosed in further detail below. The beacon devices are adapted to be at least one of directly or indirectly attached to the user and held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system. The human-machine interface further is designed to assign to the position at least one item of information.

10 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the present invention. The entertainment device further is designed to enable at least one item of information to be input by a player by means of the
15 human-machine interface. The entertainment device further is designed to vary the entertainment function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system
20 according to the present invention, such as to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The tracking system further comprises at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

25 In a further aspect of the present invention, a method for determining a position of at least one object is disclosed. The method comprises the following steps, which may be performed in the given order or in a different order. Further, two or more or even all of the method steps may be performed simultaneously and/or overlapping in time. Further, one, two or more or even all of the method steps may be performed repeatedly. The method may further comprise additional
30 method steps. The method comprises the following method steps:

- at least one detection step, wherein at least one light beam traveling from the object to a detector is detected by at least one optical sensor of the detector, the at least one optical sensor having a matrix of pixels; and
- at least one evaluation step, wherein a number N of pixels of the optical sensor is
35 determined which are illuminated by the light beam, wherein at least one longitudinal coordinate of the object is determined by using the number N of pixels which are illuminated by the light beam.

40 The method preferably may be performed by using at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Thus, for preferred optional embodiments of the method, reference may be made to the disclosure of the detector. The method may further be performed by using the detector system, the human-

machine interface, the entertainment device or the tracking system according to the present invention. Other embodiments are feasible.

The method specifically may be performed in such a way that the optical sensor generates at least one signal indicating an intensity of illumination for each of the pixels. Thus, for each of the pixels, an analog and/or digital intensity signal may be generated. In case a digital intensity signal is generated directly or e.g. after analog-digital conversion, the digital signal for each pixel may be a one-bit signal or, preferably, may be a signal having an information depth of more than one bit, such as 4 bit, 8 bit, 16 bit or a different number of bits.

In the evaluation step, for each of the pixels, the signal of the respective pixel may be compared to at least one threshold in order to determine whether the pixel is an illuminated pixel or not.

As outlined above, the threshold may be a predetermined threshold or may be a variable threshold which may be determined according to at least one predetermined algorithm.

In the evaluation step, at least one pixel having the highest illumination out of the pixels is determined by comparing the signals of the pixels. Thus, one or more of the pixels may be determined which have the highest illumination out of the pixels of the matrix.

The knowledge about the one or more pixels having the highest illumination may be used in various ways. Thus, as an example, this information may be used for determining the above-mentioned threshold. As an example, the threshold may be chosen as a fraction of the signal of the at least one pixel having the highest illumination. As outlined above, as an example, the threshold may be chosen by multiplying the signal of the at least one pixel having the highest illumination with a factor of $1/e^2$.

The longitudinal coordinate of the object may be determined by using a predetermined relationship between the number N of pixels which are illuminated by the light beam and the longitudinal coordinate. As outlined above, the predetermined relationship may be an empirical relationship and/or an analytical relationship. As an example, the predetermined relationship may be based on the assumption of the light beam being a Gaussian light beam. Thus, as explained above with respect to the detector, the predetermined relationship may be:

$$N \sim \pi \cdot w_0^2 \cdot \left(1 + \left(\frac{z}{z_0}\right)^2\right),$$

wherein z is the longitudinal coordinate,

wherein w_0 is a minimum beam radius of the light beam when propagating in space,

wherein z_0 is a Rayleigh-length of the light beam with $z_0 = \pi \cdot w_0^2 / \lambda$, λ being the wavelength of the light beam.

Again, preferably, the matrix of pixels may be a two-dimensional matrix. More preferably, the matrix of pixels may be a rectangular matrix.

5 As for the detector, the detector may comprise one optical sensor or, preferably, a plurality of the optical sensors. The optical sensors may be stacked along an optical axis of the detector.

The detector may comprise n optical sensors. Therein, the number N_i of pixels which are illuminated by the light beam may be determined for each of the optical sensors, wherein $i \in \{1, n\}$ denotes the respective optical sensor.

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The number N_i of pixels which are illuminated by the light beam for each optical sensor may be compared with at least one neighboring optical sensor, thereby resolving an ambiguity in the longitudinal coordinate of the object. Further, additionally or alternatively, the sensor signals of the optical sensors may be normalized for a power of the light beam.

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In a further aspect of the present invention, a use of the detector according to the present invention, such as to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below, is disclosed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an
20 entertainment application; a security application; a safety application; a human-machine interface application; a tracking application; a photography application, such as an application for digital photography for arts, documentation or technical purposes; a use in combination with at least one time-of-flight detector.

25 Thus, generally, the detector according to the present invention may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such
30 as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane
35 application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application; a use in combination with at least one time-of-flight detector. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for
40 cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for household applications and/or for robots used in manufacturing technology.

Thus, as for the optical detectors and devices disclosed in WO 2012/110924 A1 or in US provisional applications 61/739,173, filed on December 19, 2012, 61/749,964, filed on January 8, 2013, and 61/867,169, filed on August 19, 2013, and international patent application PCT/IB2013/061095, filed on December 18, 2013, the detector, the detector system, the
5 human-machine interface, the entertainment device, the tracking system or the camera according to the present invention (in the following simply referred to as “the devices according to the present invention” may be used for a plurality of application purposes, such as one or more of the purposes disclosed in further detail in the following.

Thus, firstly, the devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the
15 devices according to the present invention may be used as cameras and/or sensors, such as in combination with mobile software for scanning environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input
20 devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus, specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to
25 the present invention may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to
30 the present invention may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices
35 like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences

40 Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present

invention may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at.

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Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

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Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analog electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or

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museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The

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devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking

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of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

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Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention

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may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in combination with image intensifiers or image enhancement devices and/or photomultipliers.

- 5 Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally
- 10 require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to
- 15 the general need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.
- 20 Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, rear cross traffic alert, and other automotive and traffic applications.
- 25 Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in
- 30 an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury.
- 35 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with
- 40 infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor

signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established stereovision systems such as LIDAR. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on a window screen, on a front hood, on bumpers, on lights, on mirrors or other places the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions.

Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used for detecting free parking spaces in parking lots.

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endoscopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. These applications are specifically valuable e.g. in medical treatments and long-distance diagnosis and tele-medicine.

Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball,

volleyball, rugby, sumo, judo, fencing, boxing etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used in to monitor the filling level of tanks, silos etc.

Further, the devices according to the present invention may be used in the polls, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at

least one detector according to the present invention, for monitoring the speed and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft at landing or starting. Wherein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving in dangerous but well defined routes, such as mining vehicles.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more the devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racquet, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more the devices according to the present invention
5 may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. the devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction,
10 cartography, real estate management, land surveying or the like.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy
15 of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges.

As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or
20 packing containers or vehicles. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention may be used for identifying a size of material, object or tools, such as for optimal
25 material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to the present invention may be used for analyzing 3D-quality marks. Further, the
30 devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-
35 counterfeiting purposes.
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Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the

detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As used herein, a detector for determining a position of at least one object generally is a device adapted for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device adapted for detecting light beams, such as the light beams traveling from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

As used herein, an optical sensor generally is a device which is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the optical sensor by the light beam. Preferably, as outlined above and as will be outlined in further detail below, the detector according to the present invention may comprise a plurality of optical sensors, preferably as a sensor stack.

The at least one optical sensor may comprise one or more photo detectors, preferably one or more organic photodetectors and, most preferably, one or more dye-sensitized organic solar cells (DSCs, also referred to as dye solar cells), such as one or more solid dye-sensitized organic solar cells (sDSCs). Thus, preferably, the detector may comprise one or more DSCs (such as one or more sDSCs) acting as the at least one optical sensor, preferably a stack of a plurality of DSCs (preferably a stack of a plurality of sDSCs) acting as the at least one optical sensor. Additionally or alternatively, the detector may comprise other types of optical sensors, as outlined in further detail above.

As outlined above, the detector according to the present invention specifically may comprise a plurality of the optical sensors, which are also referred to as pixelated sensors or pixelated optical sensors. Specifically, the detector may comprise a stack of the optical sensors, such as a stack of transparent optical sensors.

Thus, the detector may comprise at least one stack of the optical sensors, and the detector may be adapted to acquire a three-dimensional image of a scene within a field of view of the detector. The optical sensors of the stack may have identical spectral properties, such as identical or uniform absorption spectra. Alternatively, the optical sensors of the stack may have differing spectral properties. Thus, the stack may comprise at least one first optical sensor having a first spectral sensitivity and at least one second optical sensor having a second spectral sensitivity, wherein the first spectral sensitivity and the second spectral sensitivity are different. The stack specifically may comprise optical sensors having differing spectral properties in an alternating sequence. The detector may be adapted to acquire a multicolor three-dimensional image, preferably a full-color three-dimensional image, by evaluating sensor signals of the optical sensors having differing spectral properties.

Thus, generally, the pixelated sensors, specifically the transparent pixelated sensors, may be used to record images at different distances to the detector, such as at different distances to the at least one optional transfer device and, more specifically, to at least one lens of the detector. If more than one pixelated sensor is used, several images at different distances to the detector may be recorded simultaneously. Preferably the distances to the lens are as such, that different parts of the images are in focus. Thus, the images can be used in image-processing techniques known as focus stacking, z-stacking, focal plane merging. One application of these techniques is obtaining images with greater depth of field, which is especially helpful in imaging techniques with typically very shallow depth of field such as macro photography or optical microscopy. Another application is to obtain distance information using algorithms, convolution based algorithms such as depth from focus or depth from defocus. Another application is to optimize the images to obtain a greater artistic or scientific merit.

The detector having the plurality of pixelated sensors also may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein,

generally refers to the spatial light propagation of light inside the detector such as inside camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of pixelated sensors may record images at different
5 distances from the lens. Using, e.g., convolution-based algorithms such as “depth from focus” or “depth from defocus”, the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further
10 information may be extracted.

Once the light propagation inside the detector, such as behind a lens of the detector, is modeled and/or recorded, this knowledge of light propagation provides a large number of advantages. Thus, the light-field may be recorded in terms of beam parameters for one or more light beams
15 of a scene captured by the detector. As an example, for each light beam recorded, two or more beam parameters may be recorded, such as one or more Gaussian beam parameters, e.g. a beam waist, a minimum beam waist as a focal point, a Rayleigh length, or other beam parameters. Several representations of light beams may be used and beam parameters may be chosen accordingly.

This knowledge of light propagation, as an example, allows for slightly modifying the observer position after recording an image stack using image processing techniques. In a single image, an object may be hidden behind another object and is not visible. However, if the light scattered by the hidden object reaches the lens and through the lens one or more of the sensors, the
25 object may be made visible, by changing the distance to the lens and/or the image plane relative to the optical axis, or even using non-planar image planes. The change of the observer position may be compared to looking at a hologram, in which changing the observer position slightly changes the image.

The knowledge of light propagation inside the detector, such as by modeling the light propagation behind the lens, may further allow for storing the image information in a more compact way as compared to conventional technology of storing each image recorded by each individual optical sensor. The memory demand for storing all images of each optical sensor typically scales with the number of sensors times the number of pixels. The memory demand of
35 the light propagation scales with the number of modeled light beams times the number of parameters per light beam. Typical model functions for light beams may be Gaussians, Lorentzians, Bessel functions, especially spherical Bessel functions, other functions typically used for describing diffraction effects in physics, or typical spread functions used in depth from defocus techniques such as point spread functions, line spread functions or edge spread
40 functions.

The use of several pixelated sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and

challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

As outlined above, the detector according to the present invention may further allow for color detection. For color detection in stacks of several pixelated sensors, the single stacks may have pixels that have different absorption properties, equal or similar to the so-called Bayer pattern, and color information may be obtained by interpolation techniques. A further method is to use sensors of alternating color, wherein different sensors in the stack may record different colors. In a Bayer pattern, color may be interpolated between same-color pixels. In a stack of sensors, the image information such as color and brightness, etc., can also be obtained by interpolation techniques.

The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Thus, as an example, the evaluation device may comprise one or more measurement devices for measuring electrical currents through and/or electrical voltages of the pixels. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.

The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained

by the optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses.

The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analog interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or

appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally refers to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

As outlined above, preferably, specifically in case a plurality of optical sensors is provided, at least one of the optical sensors is a transparent optical sensor. Thus, in case a stack of the optical sensors is provided, preferably all optical sensors of the plurality and/or the stack or all optical sensors of the plurality and/or the stack but one optical sensor are transparent. As an example, in case a stack of optical sensors is provided, wherein the optical sensors are arranged along the optical axis of the detector, preferably all optical sensors but the last optical sensor furthest away from the object may be transparent optical sensors. The last optical sensor, i.e. the optical sensor on the side of the stack facing away from the object, may be a transparent optical sensor or an intransparent optical sensor. Exemplary embodiments will be given below.

As outlined above, the optical sensor preferably comprises an organic photodetector such as an organic solar cell and/or an sDSC. In order to provide a transparent optical sensor, the optical sensor may have two transparent electrodes. Thus, at least one first electrode and/or the at least one second electrode of the optical sensor preferably may fully or partially be transparent. In order to provide a transparent electrode, a transparent conductive oxide may be used (TCO), such as indium-doped tin oxide (ITO) and/or fluorine-doped tin oxide (FTO). Additionally or alternatively, metal layers may be used, such as thin layers of one or more metal selected from the group consisting of Al, Ag, Au and Pt, such as metal layers having a thickness of below 50 nm, preferably below 40 nm. In order to support connectivity, additionally or alternatively, conductive organic materials may be used, such as conductive polymers. Thus, as an example, one or more of the electrodes of the at least one optical sensor may comprise one or more transparent conductive polymers. As an example, one or more conductive polymer films having a surface conductivity of at least 0.00001 S/cm, at least 0.001 S/cm or at least 0.01 S/cm may be used, preferably of at least 0.1 S/cm or, more preferably, of at least 1 S/cm or even at least 10 S/cm or at least 100 S/cm. As an example, the at least one conductive polymer may be selected from the group consisting of: a poly-3,4-ethylenedioxythiophene (PEDOT), preferably PEDOT being electrically doped with at least one counter ion, more preferably PEDOT doped with sodium polystyrene sulfonate (PEDOT:PSS); a polyaniline (PANI); a polythiophene. Preferably, the conductive polymer provides an electric resistivity of 0.1 – 20 k Ω between the partial electrodes, preferably an electric resistivity of 0.5 – 5.0 k Ω and, more preferably, an electric resistivity of 1.0 – 3.0 k Ω . Generally, as used herein, a conductive material may be a material which has a specific electrical resistance of less than 10⁴, less than 10³, less than 10², or of less than 10 Ω m. Preferably, the conductive material has a specific electrical resistance of less than 10⁻¹, less than 10⁻², less than 10⁻³, less than 10⁻⁵, or less than 10⁻⁶ Ω m. Most preferably, the specific electrical resistance of the conductive material is less than 5 x 10⁻⁷ Ω m or is less than 1 x 10⁻⁷ Ω m, particularly in the range of the specific electrical resistance of aluminum.

Generally, the optical sensor may comprise an arbitrary optical sensor having a plurality of pixels arranged in a matrix. The optical sensor, as outlined above and for reasons of example only, may comprise at least one semiconductor detector, in particular an organic semiconductor detector comprising at least one organic material, preferably an organic solar cell and

particularly preferably a dye solar cell or dye-sensitized solar cell, in particular a solid dye solar cell or a solid dye-sensitized solar cell. Preferably, the optical sensor is or comprises a DSC or sDSC. Thus, preferably, the optical sensor comprises at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode. In a preferred embodiment, the optical sensor comprises at least one DSC or, more preferably, at least one sDSC. As outlined above, preferably, the at least one optical sensor is a transparent optical sensor or comprises at least one transparent optical sensor. Thus, preferably, both the first electrode and the second electrode are transparent or, in case a plurality of optical sensors is provided, at least one of the optical sensors is designed such that both the first electrode and the second electrode are transparent. As outlined above, in case a stack of optical sensors is provided, preferably all optical sensors of the stack are transparent but the last optical sensor of the stack furthest away from the object. The last optical sensor may be transparent or intransparent. In the latter case, the last optical sensor may be designed such that its electrode facing towards the object is transparent, whereas its electrode facing away from the object may be intransparent.

As outlined above, the detector preferably has a plurality of optical sensors. More preferably, the plurality of optical sensors is stacked, such as along the optical axis of the detector. Thus, the optical sensors may form an optical sensor stack. The optical sensor stack preferably may be oriented such that the sensor regions of the optical sensors are oriented perpendicular to the optical axis. Thus, as an example, sensor areas or sensor surfaces of the single optical sensors may be oriented in parallel, wherein slight angular tolerances might be tolerable, such as angular tolerances of no more than 10° , preferably of no more than 5° .

In case stacked optical sensors are provided, the at least one transversal optical sensor preferably fully or partially may be located on a side of the stacked optical sensors facing towards the object. The at least one optional transversal optical sensor may as well be part of the sensor stack. Generally, any other arrangement is feasible. The optical sensors preferably are arranged such that the at least one light beam traveling from the object towards the detector illuminates all optical sensors, preferably sequentially. For the purpose of normalizing the sensor signals of the optical sensors, use may be made of the fact that, in case the single longitudinal sensor signals are generated by one and the same light beam, differences in the single longitudinal sensor signals are generally only due to differences in the cross-sections of the light beam at the location of the respective sensor regions of the single optical sensors. Thus, by comparing the single longitudinal sensor signals, information on a beam cross-section may be generated even if the overall power of the light beam is unknown.

Further, the above-mentioned stacking of the optical sensors and the generation of a plurality of longitudinal sensor signals by these stacked optical sensors may be used by the evaluation device in order to resolve an ambiguity in a known relationship between a beam cross-section of the light beam and the longitudinal coordinate of the beacon device. Thus, even if the beam properties of the light beam propagating from the beacon device to the detector are known fully

or partially, it is known that, in many beams, the beam cross-section narrows before reaching a focal point and, afterwards, widens again. Such is the case e.g. for Gaussian light beams. Thus, before and after a focal point in which the light beam has the narrowest beam cross-section, positions along the axis of propagation of the light beam occur in which the light beam has the same cross-section. Thus, as an example, at a distance z_0 before and after the focal point, the cross-section of the light beam is identical. Thus, in case only one optical sensor is used, a specific cross-section of the light beam might be determined, in case the overall power or intensity of the light beam is known. By using this information, the distance z_0 of the respective optical sensor from the focal point might be determined. However, in order to determine whether the respective optical sensor is located before or behind the focal point, additional information may be required, such as a history of movement of the object and/or the detector and/or information on whether the detector is located before or behind the focal point. In typical situations, this additional information may not be available. Therefore, by using a plurality of optical sensors, additional information may be gained in order to resolve the above-mentioned ambiguity. Thus, in case the evaluation device, by evaluating the longitudinal sensor signals, recognizes that the beam cross-section of the light beam on a first optical sensor is larger than the beam cross-section of the light beam on a second optical sensor, wherein the second optical sensor is located behind the first optical sensor, the evaluation device may determine that the light beam is still narrowing and that the location of the first optical sensor is situated before the focal point of the light beam. Contrarily, in case the beam cross-section of the light beam on the first optical sensor is smaller than the beam cross-section of the light beam on the second optical sensor, the evaluation device may determine that the light beam is widening and that the location of the second optical sensor is situated behind the focal point. Thus, generally, the evaluation device may be adapted to recognize whether the light beam widens or narrows, by comparing the longitudinal sensor signals of different longitudinal sensors.

The optical sensor stack preferably may comprise at least three optical sensors, more preferably at least four optical sensors, even more preferably at least five optical sensors or even at least six optical sensors. By tracking the longitudinal sensor signals of the optical sensors, even a beam profile of the light beam might be evaluated.

As used herein and as used in the following, the diameter of the light beam or, equivalently, a beam waist or twice the beam waist of the light beam might be used to characterize the beam cross-section of the light beam at a specific location. As outlined above, a known relationship might be used between the longitudinal position of the object and/or the respective beacon device, i.e. the beacon device emitting and/or reflecting the light beam, and the beam cross-section in order to determine the longitudinal coordinate of the beacon device by evaluating the at least one longitudinal sensor signal. As an example, as outlined above, a Gaussian relationship might be used, assuming that the light beam propagates at least approximately in a Gaussian manner. For this purpose, the light beam might be shaped appropriately, such as by using an illumination source generating a light beam having known propagation properties, such as a known Gaussian profile. For this purpose, the illumination source itself may generate the light beam having the known properties, which, for example, is the case for many types of

lasers, as the skilled person knows. Additionally or alternatively, the illumination source and/or the detector may have one or more transfer devices, such as one or more beam-shaping elements, such as one or more lenses and/or one or more diaphragms, in order to provide a light beam having known properties, as the skilled person will recognize. Thus, as an example, one or more transfer devices may be provided, such as one or more transfer devices having known beam-shaping properties. Additionally or alternatively, the illumination source and/or the detector, such as the at least one optional transfer device, may have one or more wavelength-selective elements, such as one or more filters, such as one or more filter elements for filtering out wavelengths outside an excitation maximum of the at least one transversal optical sensor and/or the at least one optical sensor.

Thus, generally, the evaluation device may be adapted to compare the beam cross-section and/or the diameter of the light beam with known beam properties of the light beam in order to determine the at least one item of information on the longitudinal position of the object, preferably from a known dependency of a beam diameter of the light beam on at least one propagation coordinate in a direction of propagation of the light beam and/or from a known Gaussian profile of the light beam.

As outlined above, the at least one optical sensor or pixelated sensor of the detector, as an example, may be or may comprise at least one organic optical sensor. As an example, the at least one optical sensor may be or may comprise at least one organic solar cell, such as at least one dye-sensitized solar cell (DSC), preferably at least one solid DSC or sDSC. Specifically, the at least one optical sensor may be or may comprise at least one optical sensor capable of showing an effect of the sensor signal being dependent on a photon density or flux of photons. In FiP sensors, given the same total power p of illumination, the sensor signal i is generally dependent on a flux F of photons, i.e. the number of photons per unit area. In other words, the at least one optical sensor may comprise at least one optical sensor which is defined as a FiP sensor, i.e. as an optical sensor capable of providing a sensor signal, the sensor having at least one sensor region, such as a plurality of sensor regions like e.g. pixels, wherein the sensor signal, given the same total power of illumination of the sensor region by the light beam, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area. This effect including potential embodiments of optical sensors exhibiting this effect (such as sDSCs) is disclosed in further detail in WO 2012/110924 A1, in US provisional applications 61/739,173, filed on December 19, 2012, 61/749,964, filed on January 8, 2013, and 61/867,169, filed on August 19, 2013, and international patent application PCT/IB2013/061095, filed on December 18, 2013. The embodiments of optical sensors exhibiting the FiP effect as disclosed in these prior art documents, which all are included herewith by reference, may also be used as optical sensors in the detector according to the present invention, besides the fact that the optical sensors or at least one of the optical sensors are pixelated. Thus, the optical sensors as used in one or more of the above-mentioned prior art documents, in a pixelated fashion, may also be used in the context of the present invention. A pixelation may simply be achieved by an appropriate patterning of the first and/or second

electrodes of these optical sensors. Thus, each of the pixels of the pixelated optical sensors exhibiting the above-mentioned FiP-effect may, by itself, form a FiP sensor.

Thus, the detector according to the present invention specifically may fully or partially be embodied as a pixelated FiP camera, i.e. as a camera in which the at least one optical sensor or, in case a plurality of optical sensors is provided, at least one of the optical sensors, are embodied as pixelated FiP sensors. In pixelated FiP-cameras, pictures may be recorded in a similar way as disclosed above in the setup of the light-field camera. Thus, the detector may comprise a stack of optical sensors, each optical sensor being embodied as a pixelated FiP sensor. Pictures may be recorded at different distances from the lens. A depth can be calculated from these pictures using approaches such as depth-from-focus and/or depth-from-defocus.

The FiP measurement typically necessitates two or more FiP sensors such as organic solar cells exhibiting the FiP effect. The photon density on the different cells may vary as such, that a current ratio of at least 1/100 is obtained between a cell close to focus and a cell out of focus. If the ratio is closer to 1, the measurement may be imprecise.

The at least one evaluation device may specifically be embodied to compare signals generated by pixels of different optical sensors, the pixels being located on a line parallel to an optical axis of the detector. A light cone of the light beam might cover a single pixel in the focus region. In the out-of-focus region, only a small part of the light cone will cover the pixel. Thus, in a stack of pixelated FiP sensors, the signal of the pixel of the sensor being out of focus will generally be much smaller than the signal of the pixel of the sensor being in focus. Consequently, the signal ratio will improve. For a calculation of the distance between the object and the detector, more than two optical sensors may be used in order to further increase the precision.

Thus, generally, the at least one optical sensor may comprise at least one stack of optical sensors, each optical sensor having at least one sensor region and being capable of providing at least one sensor signal, wherein the sensor signal, given the same total power of illumination of the sensor region by the light beam, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area, wherein the evaluation device may be adapted to compare at least one sensor signal generated by at least one pixel of a first one of the optical sensors with at least one sensor signal generated by at least one pixel of a second one of the optical sensors, specifically for determining a distance between the object and the detector and/or a z-coordinate of the object. The evaluation device may further be adapted for evaluating the sensor signals of the pixels. Thus, one or more evaluation algorithms may be used. Additionally or alternatively, other means of evaluation may be used, such as by using one or more lookup tables, such as one or more lookup tables comprising FiP sensor signal values or ratios thereof and corresponding z-coordinates of the object and/or corresponding distances between the object and the detector. An analysis of several FiP-signals, taking into account the distance to the lens and/or a distance between the optical

sensors may also result in information regarding the light beam, such as the spread of the light beam and, thus, the conventional FiP-distance.

As outlined above, the present invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine. The human-machine interface as proposed may make use of the fact that the above-mentioned detector in one or more of the embodiments mentioned above or as mentioned in further detail below may be used by one or more users for providing information and/or commands to a machine. Thus, preferably, the human-machine interface may be used for inputting control commands.

Generally, as used herein, the at least one position of the user may imply one or more items of information on a position of the user as a whole and/or one of or more body parts of the user. Thus, preferably, the position of the user may imply one or more items of information on a position of the user as provided by the evaluation device of the detector. The user, a body part of the user or a plurality of body parts of the user may be regarded as one or more objects the position of which may be detected by the at least one detector device. Therein, precisely one detector may be provided, or a combination of a plurality of detectors may be provided. As an example, a plurality of detectors may be provided for determining positions of a plurality of body parts of the user and/or for determining a position of at least one body part of the user.

The detector according to the present invention, comprising the at least one optical sensor and the at least one evaluation device, may further be combined with one or more other types of sensors or detectors. Thus, the detector comprising the at least one optical sensor having the matrix of pixels (in the following also simply referred to as the pixelated optical sensor and/or the pixelated sensor) and the at least one evaluation device may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

Specifically, the detector according to the present invention may further comprise at least one time-of-flight (ToF) detector adapted for detecting at least one distance between the at least one object and the detector by performing at least one time-of-flight measurement. As used herein, a time-of-flight measurement generally refers to a measurement based on a time a signal needs for propagating between two objects or from one object to a second object and back. In the present case, the signal specifically may be one or more of an acoustic signal or an electromagnetic signal such as a light signal. A time-of-flight detector consequently refers to a detector adapted for performing a time-of-flight measurement. Time-of-flight measurements are well-known in various fields of technology such as in commercially available distance

measurement devices or in commercially available flow meters, such as ultrasonic flow meters. Time-of-flight detectors even may be embodied as time-of-flight cameras. These types of cameras are commercially available as range-imaging camera systems, capable of resolving distances between objects based on the known speed of light.

5 Presently available ToF detectors generally are based on the use of a pulsed signal, optionally in combination with one or more light sensors such as CMOS-sensors. A sensor signal produced by the light sensor may be integrated. The integration may start at two different points in time. The distance may be calculated from the relative signal intensity between the two
10 integration results.

Further, as outlined above, ToF cameras are known and may generally be used, also in the context of the present invention. These ToF cameras may contain pixelated light sensors. However, since each pixel generally has to allow for performing two integrations, the pixel
15 construction generally is more complex and the resolutions of commercially available ToF cameras is rather low (typically 200x200 pixels). Distances below ~40 cm and above several meters typically are difficult or impossible to detect. Furthermore, the periodicity of the pulses leads to ambiguous distances, as only the relative shift of the pulses within one period is measured.

20 ToF detectors, as standalone devices, typically suffer from a variety of shortcomings and technical challenges. Thus, in general, ToF detectors and, more specifically, ToF cameras suffer from rain and other transparent objects in the light path, since the pulses might be reflected too early, objects behind the raindrop are hidden, or in partial reflections the
25 integration will lead to erroneous results. Further, in order to avoid errors in the measurements and in order to allow for a clear distinction of the pulses, low light conditions are preferred for ToF-measurements. Bright light such as bright sunlight can make a ToF-measurement impossible. Further, the energy consumption of typical ToF cameras is rather high, since pulses must be bright enough to be back-reflected and still be detectable by the camera. The
30 brightness of the pulses, however, may be harmful for eyes or other sensors or may cause measurement errors when two or more ToF measurements interfere with each other. In summary, current ToF detectors and, specifically, current ToF-cameras suffer from several disadvantages such as low resolution, ambiguities in the distance measurement, limited range of use, limited light conditions, sensitivity towards transparent objects in the light path, sensitivity
35 towards weather conditions and high energy consumption. These technical challenges generally lower the aptitude of present ToF cameras for daily applications such as for safety applications in cars, cameras for daily use or human-machine-interfaces, specifically for use in gaming applications.

40 In combination with the detector according to the present invention, having the at least one optical sensor comprising the matrix of pixels and as well as the above-mentioned principle of evaluating the number of illuminated pixels, the advantages and capabilities of both systems may be combined in a fruitful way. Thus, the detector may provide advantages at bright light

conditions, while the ToF detector generally provides better results at low-light conditions. A combined device, i.e. a detector according to the present invention further including at least one ToF detector, therefore provides increased tolerance with regard to light conditions as compared to both single systems. This is especially important for safety applications, such as in cars or other vehicles.

Specifically, the detector may be designed to use at least one ToF measurement for correcting at least one measurement performed by using the detector according to the present invention and vice versa. Further, the ambiguity of a ToF measurement may be resolved by using the detector. A measurement using the pixelated detector specifically may be performed whenever an analysis of ToF measurements results in a likelihood of ambiguity. Additionally or alternatively, measurements using the pixelated detector may be performed continuously in order to extend the working range of the ToF detector into regions which are usually excluded due to the ambiguity of ToF measurements. Additionally or alternatively, the pixelated detector may cover a broader or an additional range to allow for a broader distance measurement region. The pixelated detector, specifically the pixelated camera, may further be used for determining one or more important regions for measurements to reduce energy consumption or to protect eyes. Thus the pixelated detector may be adapted for detecting one or more regions of interest. Additionally or alternatively, the pixelated detector may be used for determining a rough depth map of one or more objects within a scene captured by the detector, wherein the rough depth map may be refined in important regions by one or more ToF measurements. Further, the pixelated detector may be used to adjust the ToF detector, such as the ToF camera, to the required distance region. Thereby, a pulse length and/or a frequency of the ToF measurements may be pre-set, such as for removing or reducing the likelihood of ambiguities in the ToF measurements. Thus, generally, the pixelated detector may be used for providing an autofocus for the ToF detector, such as for the ToF camera.

As outlined above, a rough depth map may be recorded by the pixelated detector, such as the pixelated camera. Further, the rough depth map, containing depth information or z-information regarding one or more objects within a scene captured by the detector, may be refined by using one or more ToF measurements. The ToF measurements specifically may be performed only in important regions. Additionally or alternatively, the rough depth map may be used to adjust the ToF detector, specifically the ToF camera.

Further, the use of the pixelated detector in combination with the at least one ToF detector may solve the above-mentioned problem of the sensitivity of ToF detectors towards the nature of the object to be detected or towards obstacles or media within the light path between the detector and the object to be detected, such as the sensitivity towards rain or weather conditions. A combined pixelated/ToF measurement may be used to extract the important information from ToF signals, or measure complex objects with several transparent or semi-transparent layers. Thus, objects made of glass, crystals, liquid structures, phase transitions, liquid motions, etc. may be observed. Further, the combination of a pixelated detector and at least one ToF detector will still work in rainy weather, and the overall detector will generally be less dependent

on weather conditions. As an example, measurement results provided by the pixelated detector may be used to remove the errors provoked by rain from ToF measurement results, which specifically renders this combination useful for safety applications such as in cars or other vehicles.

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The implementation of at least one ToF detector into the detector according to the present invention may be realized in various ways. Thus, the at least one pixelated detector and the at least one ToF detector may be arranged in a sequence, within the same light path. As an example, at least one transparent pixelated detector may be placed in front of at least one ToF detector. Additionally or alternatively, separate light paths or split light paths for the pixelated detector and the ToF detector may be used. Therein, as an example, light paths may be separated by one or more beam-splitting elements, such as one or more of the beam splitting elements listed above or listed in further detail below. As an example, a separation of beam paths by wavelength-selective elements may be performed. Thus, e.g., the ToF detector may make use of infrared light, whereas the pixelated detector may make use of light of a different wavelength. In this example, the infrared light for the ToF detector may be separated off by using a wavelength-selective beam splitting element such as a hot mirror. Additionally or alternatively, light beams used for the measurement using the pixelated detector and light beams used for the ToF measurement may be separated by one or more beam-splitting elements, such as one or more semitransparent mirrors, beam-splitter cubes, polarization beam splitters or combinations thereof. Further, the at least one pixelated detector and the at least one ToF detector may be placed next to each other in the same device, using distinct optical pathways. Various other setups are feasible.

25 The at least one optional ToF detector may be combined with basically any of the embodiments of the detector according to the present invention. Specifically, the at least one ToF detector which may be a single ToF detector or a ToF camera, may be combined with a single optical sensor or with a plurality of optical sensors such as a sensor stack. Further, the detector may also comprise one or more imaging devices such as one or more inorganic imaging devices like
30 CCD chips and/or CMOS chips, preferably one or more full-color CCD chips or full-color CMOS chips. Additionally or alternatively, the detector may further comprise one or more thermographic cameras.

As outlined above, the human-machine interface may comprise a plurality of beacon devices
35 which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the
40 beacon device on a body part.

The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined

above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam.

Additionally or alternatively, the beacon device may fully or partially be designed as a passive

5 beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one
10 beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will
15 be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device.

Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least
20 one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or
25 more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an
30 example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

35 The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of
40 information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit; a stick that may be held by hand; a bat; a club; a racket; a cane; a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device.

The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the

information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which
5 might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is adapted to gather information on
10 a series of past positions of the at least one object and/or at least one part of the object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing
15 device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as
20 at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group
25 of data or data pair comprising at least one position information and at least one time information.

The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device
30 and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual
40 position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The optional transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The optional transfer device can also, as explained in even greater detail below, be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the

illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation
5 can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or
10 fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be
15 regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source independently from each other and generally may emit light in at least one of: the ultraviolet spectral range,
20 preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Most preferably, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects
30 can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

The evaluation device can comprise in particular at least one data processing device, in particular an electronic data processing device, which can be designed to generate the at least one item of information on the position of the object. Thus, the evaluation device may be designed to use the number of illuminated pixels of the optical sensor or of each optical sensor
40 as input variables and to generate the at least one item of information on the position of the object by processing these input variables. The processing can be done in parallel, subsequently or even in a combined manner. The evaluation device may use an arbitrary process for generating these items of information, such as by calculation and/or using at least

one stored and/or known relationship. The relationship can be a predetermined analytical relationship or can be determined or determinable empirically, analytically or else semi-empirically. Particularly preferably, the relationship comprises at least one calibration curve, at least one set of calibration curves, at least one function or a combination of the possibilities mentioned. One or a plurality of calibration curves can be stored for example in the form of a set of values and the associated function values thereof, for example in a data storage device and/or a table. Alternatively or additionally, however, the at least one calibration curve can also be stored for example in parameterized form and/or as a functional equation.

By way of example, the evaluation device can be designed in terms of programming for the purpose of determining the items of information. The evaluation device can comprise in particular at least one computer, for example at least one microcomputer. Furthermore, the evaluation device can comprise one or a plurality of volatile or nonvolatile data memories. As an alternative or in addition to a data processing device, in particular at least one computer, the evaluation device can comprise one or a plurality of further electronic components which are designed for determining the items of information, for example an electronic table and in particular at least one look-up table and/or at least one application-specific integrated circuit (ASIC).

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector comprising:

- at least one optical sensor, the optical sensor being adapted to detect a light beam traveling from the object towards the detector, the optical sensor having at least one matrix of pixels; and
- at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

Embodiment 2: The detector according to the preceding embodiment, wherein the optical sensor is adapted to generate at least one signal indicating an intensity of illumination for each of the pixels.

Embodiment 3: The detector according to the preceding embodiment, wherein the evaluation device is adapted to compare, for each of the pixels, the signal to at least one threshold in order to determine whether the pixel is an illuminated pixel or not.

Embodiment 4: The detector according to the preceding embodiment, wherein the evaluation device is adapted to determine at least one pixel having the highest illumination out of the pixels by comparing the signals of the pixels.

Embodiment 5: The detector according to the preceding embodiment, wherein the evaluation device is further adapted to choose the threshold as a fraction of the signal of the at least one pixel having the highest illumination.

- 5 Embodiment 6: The detector according to the preceding embodiment, wherein the evaluation device is adapted to choose the threshold by multiplying the signal of the at least one pixel having the highest illumination with a factor of $1/e^2$.

- 10 Embodiment 7: The detector according to any one of the preceding embodiments, wherein the evaluation device is adapted to determine the longitudinal coordinate of the object by using a predetermined relationship between the number N of pixels which are illuminated by the light beam and the longitudinal coordinate.

- 15 Embodiment 8: The detector according to the preceding embodiment, wherein the predetermined relationship is based on the assumption of the light beam being a Gaussian light beam.

- Embodiment 9: The detector according to any one of the two preceding embodiments, wherein the predetermined relationship is

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$$N \sim \pi \cdot w_0^2 \cdot \left(1 + \left(\frac{z}{z_0}\right)^2\right),$$

- wherein z is the longitudinal coordinate,
 wherein w_0 is a minimum beam radius of the light beam when propagating in space,
 25 wherein z_0 is a Rayleigh-length of the light beam with $z_0 = \pi \cdot w_0^2 / \lambda$, λ being the wavelength of the light beam.

- Embodiment 10: The detector according to any one of the preceding embodiments, wherein the matrix of pixels is a two-dimensional matrix.

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- Embodiment 11: The detector according to any one of the preceding embodiments, wherein the matrix of pixels is a rectangular matrix.

- Embodiment 12: The detector according to any one of the preceding embodiments, wherein the detector comprises a plurality of the optical sensors.

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- Embodiment 13: The detector according to the preceding embodiment, wherein the optical sensors are stacked along an optical axis of the detector.

- 40 Embodiment 14: The detector according to any one of the two preceding embodiments, wherein the detector comprises n optical sensors, wherein the evaluation device is adapted to determine

the number N_i of pixels which are illuminated by the light beam for each of the optical sensors, wherein $i \in \{1, n\}$ denotes the respective optical sensor.

Embodiment 15: The detector according to any one of the three preceding embodiments, wherein the evaluation device is adapted to compare the number N_i of pixels which are illuminated by the light beam for each optical sensor with at least one neighboring optical sensor, thereby resolving an ambiguity in the longitudinal coordinate of the object.

Embodiment 16: The detector according to any one of the four preceding embodiments, wherein the evaluation device is adapted to normalize sensor signals of the optical sensors for a power of the light beam.

Embodiment 17: The detector according to any one of the five preceding embodiments, wherein at least one of the optical sensors is transparent.

Embodiment 18: The detector according to any one of the six preceding embodiments, wherein at least two of the optical sensors have a differing spectral sensitivity, wherein the evaluation device is adapted to determine a color of the light beam by comparing sensor signals of the optical sensors having the differing spectral sensitivity.

Embodiment 19: The detector according to any one of the preceding embodiments, wherein the evaluation device is further adapted to determine at least one transversal coordinate of the object by determining a position of the light beam on the matrix of pixels.

Embodiment 20: The detector according to the preceding embodiment, wherein the evaluation device is adapted to determine a center of illumination of the matrix by the light beam, wherein the at least one transversal coordinate of the object is determined by evaluating at least one coordinate of the center of illumination.

Embodiment 21: The detector according to the preceding embodiment, wherein the coordinate of the center of illumination is a pixel coordinate of the center of illumination.

Embodiment 22: The detector according to any one of the three preceding embodiments, wherein the evaluation device is adapted to provide at least one three-dimensional position of the object.

Embodiment 23: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one transfer device, the transfer device being adapted to guide the light beam onto the optical sensor.

Embodiment 24: The detector according to the preceding embodiment, wherein the transfer device has imaging properties.

Embodiment 25: The detector according to any one of the two preceding embodiments, wherein the transfer device comprises at least one element selected from the group consisting of a lens, a mirror, a prism, a wavelength-selective element; a diaphragm.

- 5 Embodiment 26: The detector according to any one of the preceding embodiments, wherein the optical sensor comprises at least one organic photovoltaic device.

10 Embodiment 27: The detector according to any one of the preceding embodiments, wherein the optical sensor comprises at least one dye-sensitized solar cell having at least one patterned electrode.

15 Embodiment 28: The detector according to any one of the preceding embodiments, wherein the optical sensor comprises at least one first electrode, at least one second electrode and at least one light-sensitive layer embedded in between the first electrode and the second electrode.

20 Embodiment 29: The detector according to the preceding embodiment, wherein the first electrode comprises a plurality of first electrode stripes and wherein the second electrode comprises a plurality of second electrode stripes, wherein the first electrode stripes are oriented perpendicular to the second electrode stripes.

25 Embodiment 30: The detector according to any one of the preceding embodiments, wherein the optical sensor comprises at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode, with the at least one n-semiconducting metal oxide, the at least one dye and the at least one p-semiconducting organic material being embedded in between the first electrode and the second electrode.

30 Embodiment 31: The detector according to the preceding embodiment, wherein both the first electrode and the second electrode are transparent.

Embodiment 32: The detector according to any one of the two preceding embodiments, wherein at least one of the first electrode and the second electrode is the patterned electrode.

35 Embodiment 33: The detector according to any one of the three preceding embodiments, wherein the first electrode comprises a plurality of first electrode stripes and wherein the second electrode comprises a plurality of second electrode stripes, wherein the first electrode stripes are oriented perpendicular to the second electrode stripes.

40 Embodiment 34: The detector according to any one of the four preceding embodiments, wherein at least one of the first electrode and the second electrode comprise an electrically conductive polymer.

Embodiment 35: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one stack of the optical sensors, wherein the detector is adapted to acquire a three-dimensional image of a scene within a field of view of the detector.

- 5 Embodiment 36: The detector according to the preceding embodiment, wherein the optical sensors of the stack have differing spectral properties.

- Embodiment 37: The detector according to the preceding embodiment, wherein the stack comprises at least one first optical sensor having a first spectral sensitivity and at least one
10 second optical sensor having a second spectral sensitivity, wherein the first spectral sensitivity and the second spectral sensitivity are different.

- Embodiment 38: The detector according to any one of the two preceding embodiments, wherein the stack comprises optical sensors having differing spectral properties in an alternating
15 sequence.

- Embodiment 39: The detector according to any one of the three preceding embodiments, wherein the detector is adapted to acquire a multicolor three-dimensional image, preferably a full-color three-dimensional image, by evaluating sensor signals of the optical sensors having
20 differing spectral properties.

- Embodiment 40: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one time-of-flight detector adapted for detecting at least one distance between the at least one object and the detector by performing at least one time-of-flight measurement.
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- Embodiment 41: The detector according to any one of the preceding embodiments, wherein the at least one optical sensor comprises at least one optical sensor having at least one sensor region and being capable of providing at least one sensor signal, wherein the sensor signal,
30 given the same total power of illumination of the sensor region by the light beam, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area.

- Embodiment 42: The detector according to any one of the preceding embodiments, wherein the
35 at least one optical sensor comprises at least one stack of optical sensors, each optical sensor having at least one sensor region and being capable of providing at least one sensor signal, wherein the sensor signal, given the same total power of illumination of the sensor region by the light beam, is dependent on a geometry of the illumination, in particular on a beam cross section of the illumination on the sensor area, wherein the evaluation device is adapted to compare at
40 least one sensor signal generated by at least one pixel of a first one of the optical sensors with at least one sensor signal generated by at least one pixel of a second one of the optical sensors.

Embodiment 43: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the
5 object, holdable by the object and integratable into the object.

Embodiment 44: The detector system according to the preceding embodiment, wherein the beacon device comprises at least one illumination source.

10 Embodiment 45: The detector system according to any one of the two preceding embodiments, wherein the beacon device comprises at least one reflective device adapted to reflect a primary light beam generated by an illumination source independent from the object.

Embodiment 46: The detector system according to any one of the three preceding
15 embodiments, wherein the detector system comprises at least two beacon devices, preferably at least three beacon devices.

Embodiment 47: The detector system according to any one of the four preceding embodiments, wherein the detector system further comprises the at least one object.
20

Embodiment 48: The detector system according to the preceding embodiment, wherein the object is a rigid object.

Embodiment 49: The detector system according to any one of the two preceding embodiments,
25 wherein the object is selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

Embodiment 50: A human-machine interface for exchanging at least one item of information
30 between a user and a machine, wherein the human-machine interface comprises at least one detector system according to any one of the preceding embodiments referring to a detector system, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system,
35 wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 51: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to
40 the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 52: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 53: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 54: A method for determining a position of at least one object, the method comprising the following steps:

- at least one detection step, wherein at least one light beam traveling from the object to a detector is detected by at least one optical sensor of the detector, the at least one optical sensor having at least one matrix of pixels; and
- at least one evaluation step, wherein a number N of pixels of the optical sensor is determined which are illuminated by the light beam, wherein at least one longitudinal coordinate of the object is determined by using the number N of pixels which are illuminated by the light beam.

Embodiment 55: The method according to the preceding embodiment, wherein the optical sensor generates at least one signal indicating an intensity of illumination for each of the pixels.

Embodiment 56: The method according to the preceding embodiment, wherein, in the evaluation step, for each of the pixels, the signal is compared to at least one threshold in order to determine whether the pixel is an illuminated pixel or not.

Embodiment 57: The method according to the preceding embodiment, wherein, in the evaluation step, at least one pixel having the highest illumination out of the pixels is determined by comparing the signals of the pixels.

Embodiment 58: The method according to the preceding embodiment, wherein the threshold is chosen as a fraction of the signal of the at least one pixel having the highest illumination.

Embodiment 59: The method according to the preceding embodiment, wherein the threshold is chosen by multiplying the signal of the at least one pixel having the highest illumination with a factor of $1/e^2$.

Embodiment 60: The method according to any one of the preceding method embodiments, wherein the longitudinal coordinate of the object is determined by using a predetermined relationship between the number N of pixels which are illuminated by the light beam and the longitudinal coordinate.

Embodiment 61: The method according to the preceding embodiment, wherein the predetermined relationship is based on the assumption of the light beam being a Gaussian light beam.

- 5 Embodiment 62: The method according to any one of the two preceding embodiments, wherein the predetermined relationship is

$$N \sim \pi \cdot w_0^2 \cdot \left(1 + \left(\frac{z}{z_0}\right)^2\right),$$

- 10 wherein z is the longitudinal coordinate,
wherein w_0 is a minimum beam radius of the light beam when propagating in space,
wherein z_0 is a Rayleigh-length of the light beam with $z_0 = \pi \cdot w_0^2 / \lambda$, λ being the wavelength of the light beam.

- 15 Embodiment 63: The method according to any one of the preceding method embodiments, wherein the matrix of pixels is a two-dimensional matrix.

Embodiment 64: The method according to any one of the preceding method embodiments, wherein the matrix of pixels is a rectangular matrix.

- 20 Embodiment 65: The method according to any one of the preceding method embodiments, wherein the detector comprises a plurality of the optical sensors.

- 25 Embodiment 66: The method according to the preceding embodiment, wherein the optical sensors are stacked along an optical axis of the detector.

- Embodiment 67: The method according to any one of the two preceding embodiments, wherein the detector comprises n optical sensors, wherein the number N_i of pixels which are illuminated by the light beam is determined for each of the optical sensors, wherein $i \in \{1, n\}$ denotes the
30 respective optical sensor.

- Embodiment 68: The method according to any one of the three preceding embodiments, wherein the number N_i of pixels which are illuminated by the light beam for each optical sensor is compared with at least one neighboring optical sensor, thereby resolving an ambiguity in the
35 longitudinal coordinate of the object.

Embodiment 69: The method according to any one of the four preceding embodiments, wherein the sensor signals of the optical sensors are normalized for a power of the light beam.

- 40 Embodiment 70: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a safety

application; a human-machine interface application; a tracking application; a photography application; a use in combination with at least one time-of-flight detector.

Brief description of the figures

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Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented alone or with several in combination. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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Figure 1 shows an exemplary embodiment of a detector, a detector system, a human-machine interface, an entertainment device and a tracking system according to the present invention;

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Figure 2A shows an exemplary embodiment of a detector according to the present invention;

Figure 2B shows an exemplary embodiment of determining a number N of pixels of an optical sensor of the detector according to Figure 2A;

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Figures 3A and 3B show typical propagation properties of a Gaussian beam;

Figure 3C shows spectral sensitivities of three optical sensor devices;

Figures 4A to 4C show various views of an optical sensor which may be used in the detector according to the present invention;

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Figure 5 shows another embodiment of a detector, a camera and of determining a position of an object;

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Figure 6 shows an embodiment of the detector to be used as a light-field camera; and

Figure 7 shows an exemplary arrangement of an implementation of a time-of-flight detector into the detector.

40

Exemplary embodiments

Fig. 1 shows, in a highly schematic illustration, an exemplary embodiment of a detector 110, having a plurality of optical sensors 112. The detector 110 specifically may be embodied as a camera 111 or may be part of a camera 111. The camera 111 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 1 further shows an embodiment of a detector system 114, which, besides the at least one detector 110, comprises one or more beacon devices 116, which, in this exemplary embodiment, are attached and/or integrated into an object 118, the position of which shall be detected by using the detector 110. Fig. 1 further shows an exemplary embodiment of a human-machine interface 120, which comprises the at least one detector system 114, and, further, an entertainment device 122, which comprises the human-machine interface 120. The figure further shows an embodiment of a tracking system 124 for tracking a position of the object 118, which comprises the detector system 114. The components of the devices and systems shall be explained in further detail in the following.

The detector 110, besides the one or more optical sensors 112, comprises at least one evaluation device 126. The evaluation device 126 may be connected to the optical sensors 112 by one or more connectors 128 and/or one or more interfaces. Further, the connector 128 may comprise one or more drivers and/or one or more measurement devices for generating sensor signals, as will be explained with regard to Figs. 2A and 2B below. Further, instead of using the at least one optional connector 128, the evaluation device 126 may fully or partially be integrated into the optical sensors 112 and/or into a housing 130 of the detector 110. Additionally or alternatively, the evaluation device 126 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 118, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element 132, the position of which may be manipulated by a user 134. As an example, the object 118 may be or may comprise a bat, a record, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 118 are possible. Further, the user 134 himself or herself may be considered as the object 118, the position of which shall be detected.

As outlined above, the detector 110 comprises the plurality of optical sensors 112. The optical sensors 112 may be located inside the housing 130 of the detector 110. Further, at least one transfer device 136 may be comprised, such as one or more optical systems, preferably comprising one or more lenses 138. An opening 140 inside the housing 130, which, preferably, is located concentrically with regard to an optical axis 142 of the detector 110, preferably defines a direction of view 144 of the detector 110. A coordinate system 146 may be defined, in which a direction parallel or antiparallel to the optical axis 142 is defined as a longitudinal direction, whereas directions perpendicular to the optical axis 142 may be defined as transversal directions. In the coordinate system 146, symbolically depicted in Fig. 1, a

longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 146 are feasible.

The detector 110 may comprise one or more of the optical sensors 112. Preferably, as depicted in Fig. 1, a plurality of optical sensors 112 is comprised, which, more preferably, are stacked along the optical axis 142, in order to form a sensor stack 148. In the embodiment shown in Fig. 1, five optical sensors 112 are depicted. It shall be noted, however, that embodiments having a different number of optical sensors 112 are feasible.

The optical sensors 112 or, at least, the optical sensors 112 besides the optical sensor 112 facing away from the object 118, preferably are transparent to light beams 150 traveling from the object 118 and/or one or more of the beacon devices 116 towards the detector 110, such that the at least one light beam 150 sequentially passes the optical sensors 112.

The detector 110 is adapted for determining a position of the at least one object 118. For this purpose, as will be explained with respect to Fig. 2A and the exemplary embodiment of one of the optical sensors 112 depicted therein, each of the optical sensors 112 comprises a matrix 152 of pixels 154. In this exemplary embodiment, the matrix 152 is a rectangular matrix, in which the pixels 154 are arranged in rows in an x-dimension and columns in a y-dimension, as symbolically depicted by the coordinate system 146 depicted in Fig. 2A. The plane of the matrix 152 may be perpendicular to the optical axis 142 of the detector 110 and, thus, may be perpendicular to the longitudinal coordinate z. However, other embodiments are feasible, such as embodiments having non-planar optical sensors 112 and/or embodiments having non-rectangular matrices of pixels 154.

The detector 110 is adapted to determine a position of the object 118, and the optical sensor 112 is adapted to detect the light beam 150 traveling from the object 118 towards the detector 110, specifically from one or more of the beacon devices 116. The light beam 150, directly and/or after being modified by the transfer device 136, such as being focused by the lens 138, creates a light spot 156 on a sensor surface 158 of the optical sensor 112 or of each of the optical sensors 112. Each of the pixels 154 may be adapted to generate an individual signal, also referred to as a sensor signal or a pixel signal, which represents an intensity of illumination of the respective pixel. Thus, as an example, in Fig. 2A, a multiplexing measuring scheme is depicted which may be used for generating sensor signals for each of the pixels 154. Thus, each of the columns of the matrix 152 may be connected to a respective current measurement device 160. A switch 162 may be provided for contacting each of the rows of the matrix 152. Thus, a multiplexing measurement scheme may be implemented in which the rows of the matrix 152 are contacted sequentially. Thus, in a first step, the uppermost row of the matrix 152 may be contacted by switch 162, thereby allowing for measuring electrical currents through each of the pixels of the uppermost row of the matrix 152. The currents may be provided in an analogue format and/or may be transformed into a digital format, such as by providing one or more analogue-digital-converters. Thus, measurement values for each of the pixels of the uppermost pixel row of matrix 152 may be generated, such as by providing 4-bit grayscale values, 8-bit

grayscale values or other information formats. The respective information values representing the sensor signals of the pixels 154 may be provided to the evaluation device 126, which may comprise one or more volatile and/or non-volatile data memories 164. Subsequently, by switching switch 162 to contact the second row of the matrix 152, sensor signals for each bit of the second row are generated, followed by sensor values of the subsequent rows. After finishing one measurement of the complete matrix 152, the routine may start anew, such as by contacting the first row of the matrix 152 again. Thus, by using this multiplexing scheme or other multiplexing schemes, sensor signals for each of the pixels 154 may be generated. Since the multiplexing may be performed at a high repetition rate, it may be assumed that neither the intensity of the light beam 150 nor the position of the light spot 156 changes significantly during one multiplexing cycle. However, it shall be noted that, specifically for fast moving objects 118, other schemes for generating sensor values may be used, such as measurement schemes creating sensor values for each pixel 154 of the matrix 152 simultaneously.

As outlined above, preferably, the matrix 152 preferably contains at least 10 pixel rows and at least 10 pixel columns. Thus, as an example, at least 20 pixel rows and at least 20 pixel columns may be present, preferably at least 50 pixel rows and 50 pixel columns and, more preferably, at least 100 pixel rows and 100 pixel columns. Thus, specifically, standard formats may be used, such as VGA and/or SVGA.

The sensor signals provided by the pixels 154 may be used to determine the position of the object 118. Thus, firstly, as outlined in Fig. 2A, the sensor signals of the pixels 154 may be compared, in order to determine the one or more pixels having the highest intensity of illumination by the light beam 150. This center of illumination, such as the center of the light spot 156, may be used for determining coordinates $x_{\max}$ and $y_{\max}$, representing transversal coordinates of the light spot 156. By using known imaging equations, such as the well-known lens equation, a transversal coordinate of the object 118 and/or the respective beacon device 116 emitting the light beam 150 in the coordinate system 146 may be determined from the coordinates $x_{\max}$, $y_{\max}$. Thus, by determining a transversal position of the light spot 156 on the sensor surface 158 of the optical sensor 112, a transversal position of the object 118 and/or a part of the object 118 may be determined.

Further, as outlined above and as will be explained in further detail below, the detector 110 is adapted to determine a longitudinal coordinate of the object 118 and/or of the at least one beacon device 116. For this purpose, a diameter and/or equivalent diameter of the light spot 156 may be evaluated, as will be explained in the following.

Thus, the evaluation device 126 is adapted to determine a number N of pixels 152 which are illuminated by the light beam 150. For this purpose, a threshold method may be used, in which the sensor signals of each of the pixels 154 are compared to one or more thresholds determining whether the respective pixel 154 is illuminated or not. The one or more thresholds may determine a borderline 166 of the light spot 156, as depicted in Fig. 2A. As an example, assuming a Gaussian illumination with a typical Gaussian intensity profile, the borderline 166

may be chosen as a line at which the intensity of the light spot 156 has dropped from a central intensity I_0 (which is the intensity at pixel coordinates $x_{\max}$, $y_{\max}$) to $1/e^2 \cdot I_0$.

The threshold method may easily be implemented, as an example, by using a histogram analysis of the sensor values of one image (such as of one scan of a multiplexing scheme and/or of one image of pixels simultaneously taken), as symbolically depicted in Fig. 2B. It shall be noted that the histogram analysis in Fig. 2B does not fully correspond to the image as depicted in Fig. 2A. In Fig. 2B, on a horizontal axis, the sensor signals of the pixels 154 acquired in one image, denoted by " I " (non withstanding the fact that other sensor signals than currents may be used, such as bit values or grayscale values), are given. On the vertical axis, denoted by "#", the counts for each of the sensor signals are given, i.e. the number of pixels 154 providing the respective sensor signal I . Thus, as an example, grayscale values may be given on the horizontal axis, and the number of pixels showing the respective grayscale values in one image may be given on the vertical axis. The highest sensor signal noted within this image is marked as I_0 . By providing an appropriate threshold $1/e^2 \cdot I_0$ (and/or the closest integer value to this threshold, such as the next integer value above $1/e^2 \cdot I_0$ and/or the next integer value below $1/e^2 \cdot I_0$ – within the present invention, these options shall be enclosed), as symbolically depicted in Fig. 2B, the pixel counts in this histogram analysis may be divided into counts for non-illuminated pixels 154 (denoted by reference number 168 in Fig. 2B and marked by white bars), i.e. the sensor signals of pixels 154 outside the borderline 166 in Fig. 2A, and counts for illuminated pixels (denoted by reference number 170 in Fig. 2B and marked by filled bars), i.e. pixels 154 within the borderline 166 in Fig. 2A. Thus, by using this threshold method and/or other threshold methods, illuminated pixels and non-illuminated pixels may be distinguished, such as by using an appropriate histogram analysis.

This distinguishing of illuminated and non-illuminated pixels allows for counting the number N of the pixels 154 which are illuminated by the light beam 150. Thus, an integration over the illuminated pixels 170 in Fig. 2B and their respective counts leads to the number N of illuminated pixels. Other methods for determining the number N of illuminated pixels may be used additionally or alternatively.

As given in equation (4) or, for a plurality of optical sensors 112, in equation (4') above, the number N of illuminated pixels is proportional to the area of the light spot 156. Thus, since the diameter of any type of light beam 150 varies with propagation, by evaluating the number N of illuminated pixels, a longitudinal coordinate of the object 118 or, specifically, of one or more beacon devices 116 emitting the respective light beam 150 may be determined. As an example, by assuming Gaussian properties of the light beam 150, equations (6) and/or (6') given above may be used. As an example, the light beam 150 itself may have Gaussian properties.

Additionally or alternatively, the at least one transfer device 136 with the at least one optional lens 138 may be used for beam-shaping wherein, still, the spatial information on the longitudinal position of the object 118 or, specifically, the respective beacon device 116, is contained in the propagation properties of the shaped light beam 150.

In case the detector 110 has a narrow viewing angle, the distance between the object 118 and the detector 110 may be considered a distance in the z-dimension, only. However, since, by using the matrix 152 and e.g. the algorithm given above, transversal coordinates x and/or y may be determined in addition, the full traveling distance of the light beam 150 may easily be
5 calculated, taking into account an offset of the respective beacon device 116 from the optical axis 152. Specifically, for objects which are located off-axis, reference may be made to the explanations regarding figure 5 below.

As outlined above, preferably, a plurality of the optical sensors 112 is provided, such as by
10 providing the sensor stack 148. The redundancy of the optical sensors 112 may be used in various ways.

Thus, as outlined above, by determining the number N of illuminated pixels for one of the optical sensors 112, a beam waste may be determined. However, as may easily be derived from one
15 or more of equations (3), (6) or (6') given above, the longitudinal coordinate z derived thereby is ambiguous with respect to the focal point. Thus, by simply determining one beam waste and/or one number N of illuminated pixels, uncertainty may arise whether the respective image was taken at a specific distance z before or after a focal point of the Gaussian light beam 150. This ambiguousness may be resolved in various ways. Thus, firstly, a movement of the detector 110
20 and/or the object 118 may be tracked, such as by using a series of images and/or a track controller 172 of the tracking system 124. Thus, a history of movements of the object 118 may be tracked, providing additional spatial information of the object 118 may allow for determining whether the respective optical sensor 112 is positioned before or after a focal point of the light beam 150. Additionally or alternatively, however, as will be explained with respect to Figs. 3A
25 and 3B, a redundancy of information provided by the optical sensor stack 148 may be used for resolving this ambiguousness of the longitudinal coordinate. Thus, in Fig. 3A, a side view of a simplified beam path of the light beam 150, traveling from one or more of the beacon devices 116 towards the detector 110, is depicted. As can be seen, due to Gaussian beam propagation properties, the light beam 150 within the sensor stack 148 narrows, up to a focal point 174,
30 which, in this exemplary embodiment, occurs close to the middle one of the optical sensors 112. Other embodiments of the beam path are feasible. In Fig. 3B, views of the sensor surfaces 158 of the optical sensors and the respective light spots 156 for each of the optical sensors 112 of the setup in Fig. 3A are given. The optical sensors 112 are numbered by numbers 1 through 5, as in Fig. 3A. As can be seen, the light spot 156 in the middle optical sensor 112, close to the
35 focal point 174, is smallest, whereas the diameter of the light spots 156 to the right and to the left of this middle sensor (sensor number 3) widens. As can be seen by comparing the diameter of the light spots 156 of the optical sensors 1 and 5 or 2 and 4, the diameter is ambiguous. However, by comparing a specific diameter with diameters of light spots of neighboring optical sensors, it may be determined whether the light beam widens or narrows, i.e. whether the
40 respective optical sensor 112 is positioned before or after the focal point 174. Thus, the above-mentioned ambiguousness may be resolved, and a z-coordinate may be determined, such as in the coordinate system 146 and/or in another coordinate system.

The optical sensors 112 of the sensor stack 148, as outlined above, preferably are transparent to the light beam 150. For the last optical sensor 112 of the sensor stack 148, facing away from the object 118, such as the optical sensor 112 named "5" in Fig. 3A, a transparency not necessarily has to be present. Thus, this last optical sensor 112 may also be intransparent.

As further outlined above, providing a plurality of the optical sensors 112, such as in a stacked fashion, may, additionally or alternatively, also be used for other purposes. Thus, the optical sensors 112 may provide different spectral sensitivities, in order to provide at least one information on a color of the light beam 150. Thus, in Fig. 3C, extinction coefficients of three of the optical sensors 112 are given as a function of the wavelength λ . These extinction coefficients or any other measure indicating a spectrum of absorption of the respective optical sensors 112, may be adjusted by providing appropriate absorptive materials, such as appropriate dyes, within the optical sensors 112. As an example, in case the optical sensors 112 comprise dye-sensitized solar cells (DSCs, specifically sDSCs), an appropriate dye may be chosen. As an example, in Fig. 3C, different spectral sensitivities (such as normalized sensitivities) ε are given for optical sensors 1, 2 and 3, as an example, as a function of the wavelength λ . Assuming that the total power of the light beam remains identical for all light spots 156 on the sensor surfaces 158, or, with known attenuation of the light beam 150 after passing a specific optical sensor 112, a ratio of the sensor signals of the respective optical sensors 112 having different absorption properties may be used for determining a color of the light beam 150. As an example, for each of the optical sensors 112, a total sensor signal may be determined by adding the sensor signals of each of the pixels 154. Alternatively, a respective representative sensor signal for each of the optical sensors 112 may be determined, such as a peak value or maximum value of the sensor signals. Again, alternatively, the sensor signals of the pixels 154 within the light spots 156 may be integrated, thereby generating a representative sensor signal for each of the optical sensors 112. In the exemplary embodiment depicted in Fig. 3C, an information on a green component of the light beam 150, e.g., may be determined by dividing the sensor signal of the third optical sensor 112 (sensor number 3) by a sum of the sensor signals of optical sensors 1, 2 and 3. Similarly, a yellow component of the light beam 150 may be determined by dividing the sensor signal of the first optical sensor by a sum of the sensor signals of optical sensors 1, 2 and 3. Again, similarly, a red component of the light beam 150 may be determined by dividing the sensor signal of the second optical sensor 112 by a sum of the sensor signals of optical sensors 1, 2 and 3. Other embodiments and/or algorithms for determining colors are feasible. Thus, as an example, the absorption spectra of three of the optical sensors 112 may be similar to the absorption materials used as a basis of the above-mentioned CIE coordinate system, thereby directly allowing for determining CIE coordinates of the light beam 150. It shall be noted that the determination of the color of the light beam 150 is independent from the above-described determination of the longitudinal coordinate of the object 118, since the above-mentioned algorithm is simply based on a counting of illuminated and non-illuminated pixels, independent from the color of the light beam 150. Thus, e.g. in the threshold method and histogram analysis described with regard to Figs. 2A and 2B above, an internal normalization of the intensity of the light beam and/or the color of the light beam may take place, since, as outlined above, the threshold may be chosen as a function and/or fraction of a

maximum intensity and/or of a maximum sensor signal. Thus, the determination of the longitudinal coordinate by using the above-mentioned pixel count is independent from the fact that the respective optical sensors 112 within the sensor stack 148 may have different spectral absorption properties.

As outlined above, the determination of a position of the object 118 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 120, in order to provide at least one item of information to a machine 176. In the embodiment schematically depicted in Fig. 1, the machine 176 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 126 even may fully or partially be integrated into the machine 176, such as into the computer. The same holds true for the track controller 172, which may also fully or partially form part of the computer of the machine 176.

Similarly, as outlined above, the human-machine interface 120 may form part of an entertainment device 122. The machine 176, specifically the computer, may also form part of the entertainment device 122. Thus, by means of the user 134 functioning as the object 118 and/or by means of the user 134 handling a control device 132 functioning as the object 118, the user 134 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment function, such as controlling the course of a computer game.

As outlined above, the one optical sensor 112 and/or one or more of the optical sensors 112 preferably may fully or partially be transparent with regard to the light beam 150. In Figs. 4A to 4C, an exemplary setup of a transparent optical sensor 112 is depicted in various views.

Therein, Fig. 4A shows a top view, Fig. 4B shows a cross-sectional view along line A-A in Fig. 4A, and Fig. 4C shows a cross-sectional view along line B-B in Fig. 4A.

The optical sensor 112 may comprise a transparent substrate 178, such as a glass substrate and/or a plastic substrate. For potential details of the substrate 178, reference may be made to documents WO 2012/110924 A1 and US provisional applications no. 61/739,173 and/or 61/749,964. However, other embodiments are feasible. The illumination by the light beam 150 may take place through the substrate 178 and/or from an opposite side. Thus, the bottom side of the substrate 178 in Fig. 4B may form the sensor surface 158. Alternatively, an illumination from the opposing surface may take place.

On top of the substrate 178, a first electrode 180 is deposited, which, in this embodiment, may comprise a plurality of first electrode stripes 182. Preferably, the first electrode 180 is fully or partially transparent. Thus, as an example, the first electrode 180 may fully or partially be made of a transparent conductive oxide, such as fluorine-doped tin oxide (FTO) and/or indium-doped tin oxide (ITO). For further details of the first electrode 180, reference may be made to WO 2012/110924 A1 and/or one or more of US provisional applications no. 61/739,173 and/or 61/749,964. However, other embodiments are feasible. A patterning of the first electrode stripes 182 may take place by appropriate patterning techniques which are generally known to the

skilled person in the field of display technology, e.g. etching and/or lithographic techniques. Thus, as an example, a large-area coating by the material of the first electrode 180 on the substrate 178 may be provided, wherein the areas of the first electrode stripes 182 may be covered by photoresist and wherein the uncovered regions may be etched by an appropriate
5 etching means, as known to the skilled person in the technical field of display manufacturing, such as LCD manufacturing.

On top of the first electrode 180, one or more light-sensitive layers 184, such as a light-sensitive layer setup comprising one, two, three or more layers, are deposited. As an example, the light-
10 sensitive layers 184 may comprise a layer setup of a dye-sensitized solar cell (DSC), more specifically of a solid dye-sensitized solar cell (sDSC), such as disclosed in WO 2012/110924 A1 and/or one or more of the US provisional applications 61/739,173 and/or 61/749,964. Thus, the light-sensitive layers 184 may comprise one or more layers of an n-semiconducting metal oxide, preferably a nanoporous metal oxide, such as TiO_2 , which may directly or indirectly be
15 deposited on top of the first electrode 180. Further, the n-semiconducting metal oxide may fully or partially be sensitized with one or more dyes, such as one or more organic dyes, preferably one or more of the dyes disclosed in WO 2012/110924 A1 and/or one or more of US provisional applications no. 61/739,173 and/or 61/749,964. Other embodiments are feasible.

On top of the dye-sensitized n-semiconducting metal oxide, one or more layers of a p-semiconducting and/or conducting material may be deposited. Thus, preferably, one or more solid p-semiconducting organic materials may be used which may directly or indirectly be deposited on top of the n-semiconducting metal oxide. As an example, reference may be made to one or more of the p-semiconducting materials as disclosed in WO 2012/110924 A1 and/or
20 as disclosed in one or more of US provisional applications no. 61/739,173 and/or 61/749,964. As a preferred example, Spiro-MeOTAD may be used.

It shall be noted that the named light-sensitive layers 184, which preferably may comprise one or more organic light-sensitive layers 184, may also be provided in a different layer setup. Thus,
30 basically, any type of light-sensitive material, such as an organic, inorganic or hybrid layer setup, may be used, which is adapted to provide an electric signal in accordance with an illumination of the layer setup.

As can be seen specifically in the top view of Fig. 4A, the one or more light-sensitive layers 184 preferably are patterned such that one or more contact areas 186 for contacting the first electrode stripes 182 remain uncovered by the light-sensitive layer 184. This patterning may be performed in various ways. Thus, a large-area coating of the light-sensitive layers 184 may be applied, and the contact areas 186 may, subsequently, be uncovered, such as by laser ablation and/or mechanical ablation. Additionally or alternatively, however, the one or more light-
40 sensitive layers 184 may fully or partially be applied to the setup in a patterned way, such as by using appropriate printing techniques. Combinations of the named techniques are feasible.

On top of the at least one light-sensitive layer 184, at least one second electrode 188 is deposited. Again, this at least one second electrode 188 preferably may comprise a plurality of electrode stripes, which, in this embodiment, are denoted by reference number 190 (second electrode stripes). As can be seen specifically in the top view of Fig. 4A, the second electrode stripes 190 preferably are oriented essentially perpendicular to the first electrode stripes 182, such as at an angle of $90^\circ \pm 20^\circ$, preferably $90^\circ \pm 10^\circ$ and more preferably $90^\circ \pm 5^\circ$. It shall be noted, however, that other electrode geometries for the first electrode 180 and the second electrode 188 are feasible.

As can be seen in the top view of Fig. 4A, each of the second electrode stripes 190 comprises at least one contact area 192 which allows for an electrical contacting of the second electrode stripes 190.

As further may be derived from the top view in Fig. 4A, the layer setup of the optical sensor 112 provides a plurality of areas in which the second electrode stripes 190 cross the first electrode stripes 182. Each of these areas, by itself, forms an individual optical sensor, which is also referred to as a pixel 154 and which may be contacted electrically by electrically contacting the appropriate contact areas 186, 192 of the respective electrode stripes 182, 190. Thus, by measuring an electrical current through these individual optical sensors, each of the pixels 154 may provide an individual optical signal, as explained above. In this embodiment, the pixels 154 may be arranged in a rectangular setup, forming a rectangular matrix 152. It shall be noted, however, that other setups are feasible, such as non-rectangular matrix setups. Thus, as an example, honeycomb structures or other geometric setups may be realized.

In addition to the layer setup shown in Figs. 4A to 4C, the optical sensor 112 may comprise one or more encapsulation elements, such as one or more encapsulation layers and/or one or more cover elements, such as glass lids and/or plastic lids. The latter, for example, may be glued on top of the layer setup shown e.g. in Figs. 4B and 4C, preferably by leaving open the contact areas 186, 192.

The second electrode stripes 190 preferably may comprise one or more metal layers, such as one or more layers of a metal selected from the group consisting of: Al, Ag, Au, Pt, Cu. Additionally or alternatively, combinations of two or more metals may be used, such as metal alloys. As an example, one or more metal alloys selected from the group of NiCr, AlNiCr, MoNb and AlNd may be used. Still, other embodiments are feasible. Preferably, as for the first electrode stripes 182, the second electrode stripes 190 may fully or partially be transparent. This transparency may be realized in various ways. Thus, as an example, thin metal layers may be used, such as metal layers having a thickness of below 50 nm, such as a thickness of ≤ 30 nm or ≤ 20 nm. At these layer thicknesses, the typical metals still are transparent. Additionally or alternatively, however, non-metallic conductive materials may be used, such as conductive polymers. As an example, PEDOT:PSS and/or PANI may be used. For further potential details of the setup of the second electrode 188, reference may be made to WO 2012/110924 A1, US 61/739,173 and/or 61/749,964, as mentioned above.

The second electrode stripes 190 may be applied to the layer setup by using typical application techniques. Thus, as an example, one or more metal layers may be deposited by using physical vapor deposition (such as evaporation and/or sputtering). Conductive non-metallic materials, such as conductive polymers, may e.g. be applied by using typical coating techniques, such as spin-coating and/or printing. Other techniques are feasible. The patterning of the second electrode stripes 190 may be performed in various ways. Thus, when using evaporation techniques and/or vacuum deposition techniques, a mask technique may be used, such as evaporation through shadow masks. Additionally or alternatively, printing may be performed in a patterned way. Thus, as an example, screen-printing and/or inkjet-printing may be used for patterning conductive polymers. Again, additionally or alternatively, one or more separating patterns may be provided on the layer setup and/or on the substrate 178, such as photoresist patterns, which sub-divide the second electrode 188 into the second electrode stripes 190.

It shall further be noted that the layer setup of the first electrode 180, the one or more light-sensitive layers 184 and the second electrode 188 may as well be inverted. Thus, as an example, the layer setup of the DSC, specifically the sDSC, may be inverted, as compared to the layer setup described above. Further, additionally or alternatively, the setup of the electrodes 180, 188 may be inverted, thus providing the second electrode 188 on the substrate 178, providing the one or more light-sensitive layers 184 directly or indirectly on top of this second electrode, and providing the first electrode 180 on top of this at least one light-sensitive layer 184. Various variations of the setup are feasible. Further, it shall be noted that one or more of electrodes 180, 188 may as well be intransparent. Thus, as explained above, a detector 110 having only one optical sensor 112 is feasible. In this case, the optical sensor 112 not necessarily has to be transparent. Thus, as an example, the second electrode 188 may be intransparent, such as by using thick metal layers, in case light is transmitted into the optical sensor 112 via sensor surface 158. In case light is transmitted into the optical sensor 112 from the other side, the first electrode 180 may be an intransparent electrode. Further, in case a sensor stack 148 is used, as e.g. in the setup of Fig. 1, the last optical sensor 112 of the sensor stack 148, facing away from the object 118, not necessarily has to be transparent. Thus, an intransparent optical sensor 112 may be used.

In figure 5, in addition to the explanations given above with regard to figures 2A to 3B, a further embodiment of a detector 110 and of a camera 111 are shown in a partial perspective view, which, in the following, will be used for further explaining an embodiment for determining the position of at least one object 118 (not shown in this figure) emitting a light beam 150. The detector 110 and/or the camera 111 may be part of a detector system 114, a human-machine interface 120, an entertainment device 122 or a tracking system 124 and may comprise additional components which are not depicted in figure 5. Thus, as an example, the evaluation device 126 is not depicted. With regard to potential embodiments of the evaluation device 126 and/or with regard to further details, reference may be made to the embodiments shown above.

As can be seen, in this preferred embodiment, the detector 110, again, comprises a plurality of optical sensors 112, which, again, are arranged in a sensor stack 148. For potential

embodiments of the optical sensors 112 and the sensor stack 148, reference may be made to the embodiments disclosed above.

The sensor stack 148 comprises at least two optical sensors 112, wherein, in this embodiment, only two optical sensors 112 are shown, one optical sensor 112 facing towards the object 118, of the last optical sensor 112 in figure 5) and one optical sensor 112 facing away from the object 118 (right optical sensor 112). Preferably, at least one of the optical sensors 112 is at least partially transparent with regard to the light beam 150, such that at least part of the light beam 150, in an unattenuated fashion or in an attenuated fashion, with unchanged spectral properties or modified spectral properties, may pass through the optical sensor 112. Thus, in figure 5, the left optical sensor 112 may fully or partially be transparent, whereas the right optical sensor 112 may be intransparent or transparent.

The light beam 150 may propagate in a direction of propagation 194, along an axis of propagation 196, which may be parallel or nonparallel to the z-axis which, preferably, is oriented orthogonally to the sensor surfaces 158 of the optical sensors 112.

As outlined above with regard to figures 2A and 2B, and as outlined above with regard to figures 3A and 3B, light spots 156 created by the light beam 150 on the optical sensors 112 may be evaluated. Thus, as outlined above, for one or more of the light spots 156, a center 198 may be determined, at least within the boundaries of resolution given by the pixels 154. As an example, in figure 5, the optical sensors 112 contain matrices 152 of pixels 154, wherein each pixel is characterized by its row (symbolically denoted by row identifier A to I in figure 5) and its column (symbolically denoted by column identifier 1 to 7 in figure 5). Other embodiments of identifying pixel coordinates are feasible, such as by using numbers both as row identifiers and as column identifiers. Thus, in the exemplary embodiment shown in figure 5, the center 198 of the light spot 156 for the left optical sensor 112 may be identified to be located in between rows D and E and in between columns 4 and 5, whereas the center 198 of the light spot 156 on the right optical sensor 112 may be identified to be located in row D and column 6. Thus, by connecting the centers 198, the axis of propagation 196 of the light beam 150 may easily be determined. Consequently, the direction of the object 118 with regard to the detector 110 may be determined. Thus, since the center 198 of the light spot 156 on the right optical sensor 112 is shifted towards the right (i.e. towards higher column numbers), it may be determined that the object 118 is located off centered from the z-Axis towards the right.

Further, as outlined above, by evaluating beam waist w_0 , a longitudinal coordinate of the object 118 may be determined. Thus, as an example, the beam waist may be dependent on the longitudinal coordinate according to one or more of the above-mentioned relationships, specifically according to a Gaussian relationship. In case the direction of propagation 194 is non-parallel to the optical axis or z-coordinate, as depicted in figure 5, the longitudinal coordinate of the object 118 may be a coordinate along the axis of propagation 196. Since, e.g. by comparing the coordinates of the centers 198, the axis of propagation 196 may easily be determined, and angular relationship between the z-axis and the axis of propagation 196 is known, and, thus, a coordinate transformation is easily possible. Thus, generally, by evaluating

the pixel counts of one or more optical sensors 112, the position of the object 118 may be determined. Further, since each of the optical sensors 112 may be used for generating an image of the object 118 and since the longitudinal coordinate of the object 118 and/or of one or more points of the object 118 are known, a three-dimensional image of the object 118 may be generated.

In Figure 6, a schematic setup of a detector 110 according to the present invention to be used as a light-field camera is shown. Basically, the setup shown in Figure 6 may correspond to the embodiment shown in Figures 1 or any other of the embodiments shown herein. The detector 110 comprises a sensor stack 148 of optical sensors 112, also referred to as pixelated sensors, which specifically may be transparent. As an example, pixelated organic optical sensors may be used, such as organic solar cells, specifically sDSCs. In addition, the detector 110 may comprise at least one transfer device 136 such as at least one lens 138 or lens system, adapted for imaging objects 118. Additionally, in this embodiment or other embodiments, the detector 110 may comprise at least one imaging device 196, such as a CCD and/or a CMOS imaging device.

As outlined above, the detector 110 in the embodiment shown herein is suited to act as a light-field camera. Thus, light-beams 150 propagating from various objects 118 or beacon devices 116, symbolically denoted by A, B and C in Figure 6, are focused by the transfer device 136 into corresponding images, denoted by A', B' and C' in Figure 6. By using the stack 148 of optical sensors 112, a three-dimensional image may be captured. Thus, specifically in case the optical sensors 112 are FiP-sensors, i.e. sensors for which the sensor signals are dependent on the photon density, the focal points for each of the light beams 150 may be determined by evaluating sensor signals of neighboring optical sensors 112. Thus, by evaluating the sensor signals of the stack 148, beam parameters of the various light beams 150 may be determined, such as a focal position, spreading parameters or other parameters. Thus, as an example, each light beam 150 and/or one or more light beams 150 of interest may be determined in terms of their beam parameters and may be represented by a parameter representation and/or vector representation. Thus, since the optical qualities and properties of the transfer device 136 are generally known, as soon as the beam parameters of the light beams 150 are determined by using the stack 148, a scene captured by the optical detector 110, containing one or more objects 118, may be represented by a simplified set of beam parameters. For further details of the light-field camera shown in Figure 6, reference may be made to the description of the various possibilities given above.

Further, as outlined above, the optical sensors 112 of the stack 148 of optical sensors may have identical or different wavelength sensitivities. Thus, the stack 148 may, besides the optional imaging device 196, comprise two types of optical sensors 112, such as in an alternating fashion. Therein, a first type and a second type of optical sensors 112 may be provided in the stack 148. The optical sensors 112 of the first type and the second type specifically may be arranged in an alternating fashion along the optical axis 142. The optical sensors 112 of the first type may have a first spectral sensitivity, such as a first absorption spectrum, such as a first

absorption spectrum defined by a first dye, and the optical sensors 112 of the second type may have a second spectral sensitivity different from the first spectral sensitivity, such as a second absorption spectrum, such as a second absorption spectrum defined by a second dye. By evaluating sensor signals of these two or more types of optical sensors 112, color information may be obtained. Thus, in addition to the beam parameters which may be derived, the two or more types of optical sensors 112 may allow for deriving additional color information, such as for deriving a full-color three-dimensional image. Thus, as an example, color information may be derived by comparing the sensor signals of the optical sensors 112 of different color with values stored in a look-up table. Thus, the setup of Figure 6 may be embodied as a monochrome, a full-color or multicolor light-field camera.

As outlined above, the detector 110 may further comprise one or more time-of-flight detectors. This possibility is shown in Figure 7. The detector 110, firstly, comprises at least one component comprising the one or more pixelated optical sensors 112, such as a sensor stack 148. In the embodiment shown in Figure 7, the at least one unit comprising the optical sensors 112 is denoted as a camera 111. It shall be noted, however, that other embodiments are feasible. For details of potential setups of the camera 111, reference may be made to the setups shown above, such as the embodiment shown in Figure 1, or other embodiments of the detector 110. Basically any setup of the detector 110 as disclosed above may also be used in the context of the embodiment shown in Figure 7.

Further, the detector 110 comprises at least one time-of-flight (ToF) detector 198. As shown in Figure 7, the ToF detector 198 may be connected to the evaluation device 126 of the detector 110 or may be provided with a separate evaluation device. As outlined above, the ToF detector 198 may be adapted, by emitting and receiving pulses 200, as symbolically depicted in Figure 7, to determine a distance between the detector 110 and the object 118 or, in other words, a z-coordinate along the optical axis 142.

The at least one optional ToF detector 198 may be combined with the at least one detector having the pixelated optical sensors 112 such as the camera 111 in various ways. Thus, as an example and as shown in Figure 7, the at least one camera 111 may be located in a first partial beam path 202, and the ToF detector 198 may be located in a second partial beam path 204. The partial beam paths 202, 204 may be separated and/or combined by at least one beam-splitting element 206. As an example, the beam-splitting element 206 may be a wavelength-indifferent beam-splitting element 206, such as a semi-transparent mirror. Additionally or alternatively, a wavelength-dependency may be provided, thereby allowing for separating different wavelengths. As an alternative, or in addition to the setup shown in Figure 7, other setups of the ToF detector 198 may be used. Thus, the camera 111 and the ToF detector 198 may be arranged in line, such as by arranging the ToF detector 198 behind the camera 111. In this case, preferably, no intransparent optical sensor is provided in the camera 111, and all optical sensors 112 are at least partially transparent. Again, as an alternative or in addition, the ToF detector 198 may also be arranged independently from the camera 111, and different light paths may be used, without combining the light paths. Various setups are feasible.

As outlined above, the ToF detector 198 and the camera 111 may be combined in a beneficial way, for various purposes, such as for resolving ambiguities, for increasing the range of weather conditions in which the optical detector 110 may be used, or for extending a distance range between the object 118 and the optical detector 110. For further details, reference may be made to the description above.

List of reference numbers

110	detector
111	camera
112	optical sensor
114	detector system
116	beacon device
118	object
120	human-machine interface
122	entertainment device
124	tracking system
126	evaluation device
128	connector
130	housing
132	control device
134	user
136	transfer device
138	lens
140	opening
142	optical axis
144	direction of view
146	coordinate system
148	sensor stack
150	light beam
152	matrix
154	pixel
156	light spot
158	sensor surface
160	current measurement device
162	switch
164	data memory
166	borderline
168	non-illuminated pixels
170	illuminated pixels
172	track controller
174	focal point
176	machine
178	substrate
180	first electrode
182	first electrode stripes
184	light-sensitive layer
186	contact area
188	second electrode

190 second electrode stripes
192 contact area
194 direction of propagation
192 axis of propagation
194 center
196 imaging device
198 time-of-flight detector
200 pulses
202 first partial beam path
204 second partial beam path
206 beam-splitting element

Patent claims

1. A detector (110) for determining a position of at least one object (118), the detector (110) comprising:
 - 5 - at least one optical sensor (112), the optical sensor (112) being adapted to detect a light beam (150) traveling from the object (118) towards the detector (110), the optical sensor (112) having at least one matrix (152) of pixels (154); and
 - 10 - at least one evaluation device (126), the evaluation device (126) being adapted to determine a number N of pixels (154) of the optical sensor (112) which are illuminated by the light beam (150), the evaluation device (126) further being adapted to determine at least one longitudinal coordinate of the object (118) by using the number N of pixels (154) which are illuminated by the light beam (150).
- 15 2. The detector (110) according to the preceding claim, wherein the optical sensor (112) is adapted to generate at least one signal indicating an intensity of illumination for each of the pixels (154).
- 20 3. The detector (110) according to the preceding claim, wherein the evaluation device (126) is adapted to compare, for each of the pixels (154), the signal to at least one threshold in order to determine whether the pixel is an illuminated pixel or not.
- 25 4. The detector (110) according to the preceding claim, wherein the evaluation device (126) is adapted to determine at least one pixel having the highest illumination out of the pixels (154) by comparing the signals of the pixels (154).
5. The detector (110) according to the preceding claim, wherein the evaluation device (126) is further adapted to choose the threshold as a fraction of the signal of the at least one pixel having the highest illumination.
- 30 6. The detector (110) according to the preceding claim, wherein the evaluation device (126) is adapted to choose the threshold by multiplying the signal of the at least one pixel having the highest illumination with a factor of $1/e^2$.
- 35 7. The detector (110) according to any one of the preceding claims, wherein the evaluation device (126) is adapted to determine the longitudinal coordinate of the object (118) by using a predetermined relationship between the number N of pixels (154) which are illuminated by the light beam (150) and the longitudinal coordinate.
- 40 8. The detector (110) according to the preceding claim, wherein the predetermined relationship is based on the assumption of the light beam (150) being a Gaussian light beam (150).

9. The detector (110) according to any one of the two preceding claims, wherein the predetermined relationship is

$$N \sim \pi \cdot w_0^2 \cdot \left(1 + \left(\frac{z}{z_0}\right)^2\right),$$

5

wherein z is the longitudinal coordinate,
 wherein w_0 is a minimum beam radius of the light beam (150) when propagating in space,
 wherein z_0 is a Rayleigh-length of the light beam (150) with $z_0 = \pi \cdot w_0^2 / \lambda$, λ being the
 wavelength of the light beam (150).

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10. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises a plurality of the optical sensors (112).

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11. The detector (110) according to the preceding claim, wherein the detector (110) comprises n optical sensors (112), wherein the evaluation device (126) is adapted to determine the number N_i of pixels (154) which are illuminated by the light beam (150) for each of the optical sensors (112), wherein $i \in \{1, n\}$ denotes the respective optical sensor (112).

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12. The detector (110) according to any one of the two preceding claims, wherein the evaluation device (126) is adapted to compare the number N_i of pixels (154) which are illuminated by the light beam (150) for each optical sensor (112) with at least one neighboring optical sensor (112), thereby resolving an ambiguity in the longitudinal coordinate of the object (118).

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13. The detector (110) according to any one of the three preceding claims, wherein at least one of the optical sensors (112) is transparent.

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14. The detector (110) according to any one of the four preceding claims, wherein at least two of the optical sensors (112) have a differing spectral sensitivity, wherein the evaluation device (126) is adapted to determine a color of the light beam (150) by comparing sensor signals of the optical sensors (112) having the differing spectral sensitivity.

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15. The detector (110) according to any one of the preceding claims, wherein the evaluation device (126) is further adapted to determine at least one transversal coordinate of the object (118) by determining a position of the light beam (150) on the matrix (152) of pixels (154).

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16. The detector (110) according to any one of the preceding claims, wherein the detector (110) further comprises at least one transfer device (136), the transfer device (136) being adapted to guide the light beam (150) onto the optical sensor (112).

17. The detector (110) according to any one of the preceding claims, wherein the optical sensor (112) comprises at least one first electrode (180), at least one second electrode (188) and at least one light-sensitive layer (184) embedded in between the first electrode (180) and the second electrode (188), wherein the first electrode (180) comprises a plurality of first electrode stripes (182) and wherein the second electrode (188) comprises a plurality of second electrode stripes (190), wherein the first electrode stripes (182) are oriented perpendicular to the second electrode stripes (190).
18. A detector system (114) for determining a position of at least one object (118), the detector system (114) comprising at least one detector (110) according to any one of the preceding claims, the detector system (114) further comprising at least one beacon device (116) adapted to direct at least one light beam (150) towards the detector (110), wherein the beacon device (116) is at least one of attachable to the object (118), holdable by the object (118) and integratable into the object (118).
19. A human-machine interface (120) for exchanging at least one item of information between a user (134) and a machine (176), wherein the human-machine interface (120) comprises at least one detector system (114) according to the preceding claim, wherein the at least one beacon device (116) is adapted to be at least one of directly or indirectly attached to the user (134) and held by the user (134), wherein the human-machine interface (120) is designed to determine at least one position of the user (134) by means of the detector system (114), wherein the human-machine interface (120) is designed to assign to the position at least one item of information.
20. An entertainment device (122) for carrying out at least one entertainment function, wherein the entertainment device (122) comprises at least one human-machine interface (120) according to the preceding claim, wherein the entertainment device (122) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (120), wherein the entertainment device (122) is designed to vary the entertainment function in accordance with the information.
21. A tracking system (124) for tracking a position of at least one movable object (118), the tracking system (124) comprising at least one detector system (114) according to any one of the preceding claims referring to a detector system (114), the tracking system (124) further comprising at least one track controller (172), wherein the track controller (172) is adapted to track a series of positions of the object (118) at specific points in time.
22. A camera (111) for imaging at least one object (118), the camera (111) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).

23. A method for determining a position of at least one object (118), the method comprising the following steps:
- at least one detection step, wherein at least one light beam (150) traveling from the object (118) to a detector (110) is detected by at least one optical sensor (112) of the detector (110), the at least one optical sensor (112) having at least one matrix (152) of pixels (154); and
 - at least one evaluation step, wherein a number N of pixels (154) of the optical sensor (112) is determined which are illuminated by the light beam (150), wherein at least one longitudinal coordinate of the object (118) is determined by using the number N of pixels (154) which are illuminated by the light beam (150).
24. A use of the detector (110) according to any one of the preceding claims relating to a detector (110), for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a safety application; a human-machine interface (120) application; a tracking application; a photography application; a use in combination with at least one time-of-flight detector (198).

FIG.1

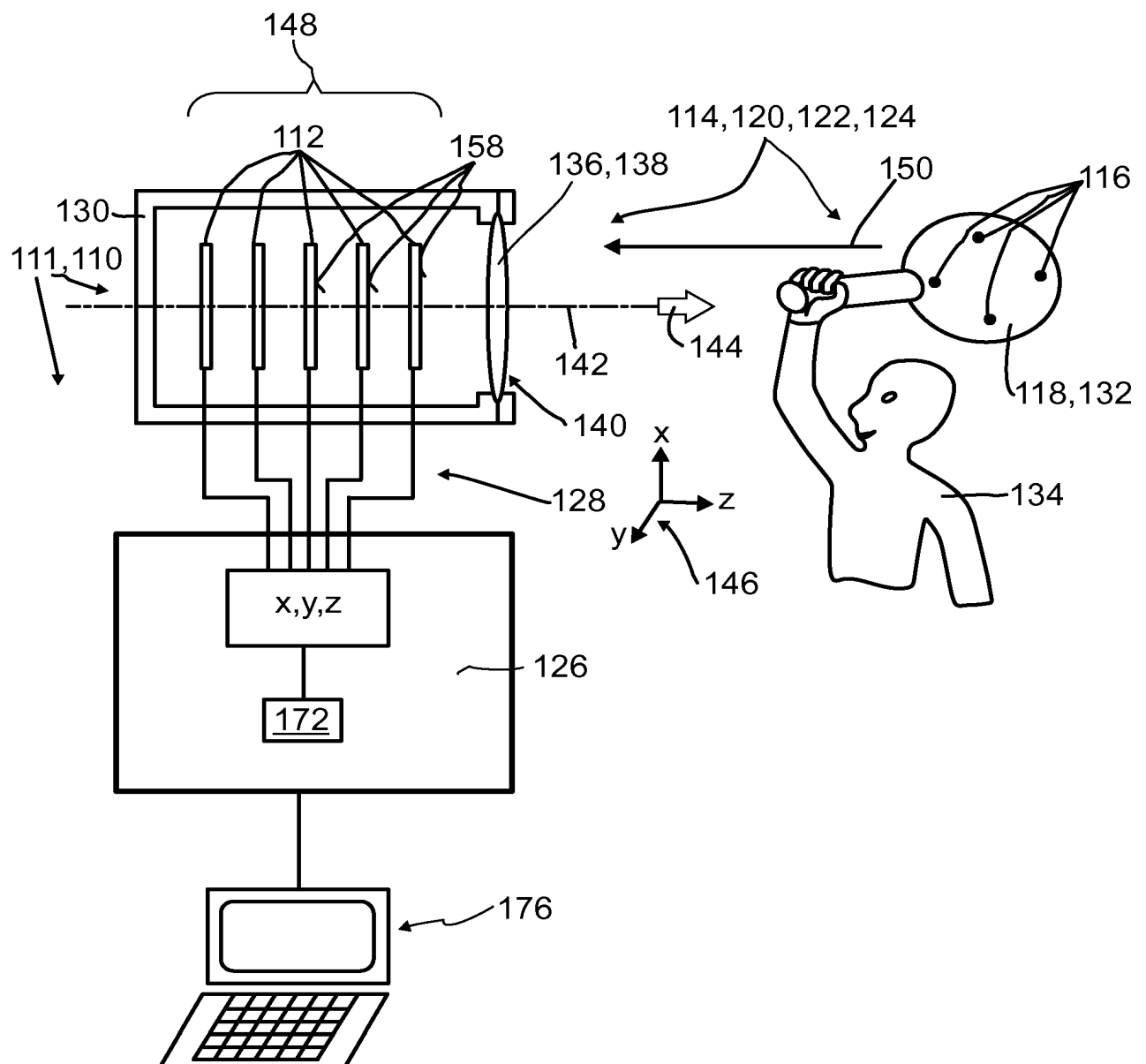


FIG.2A

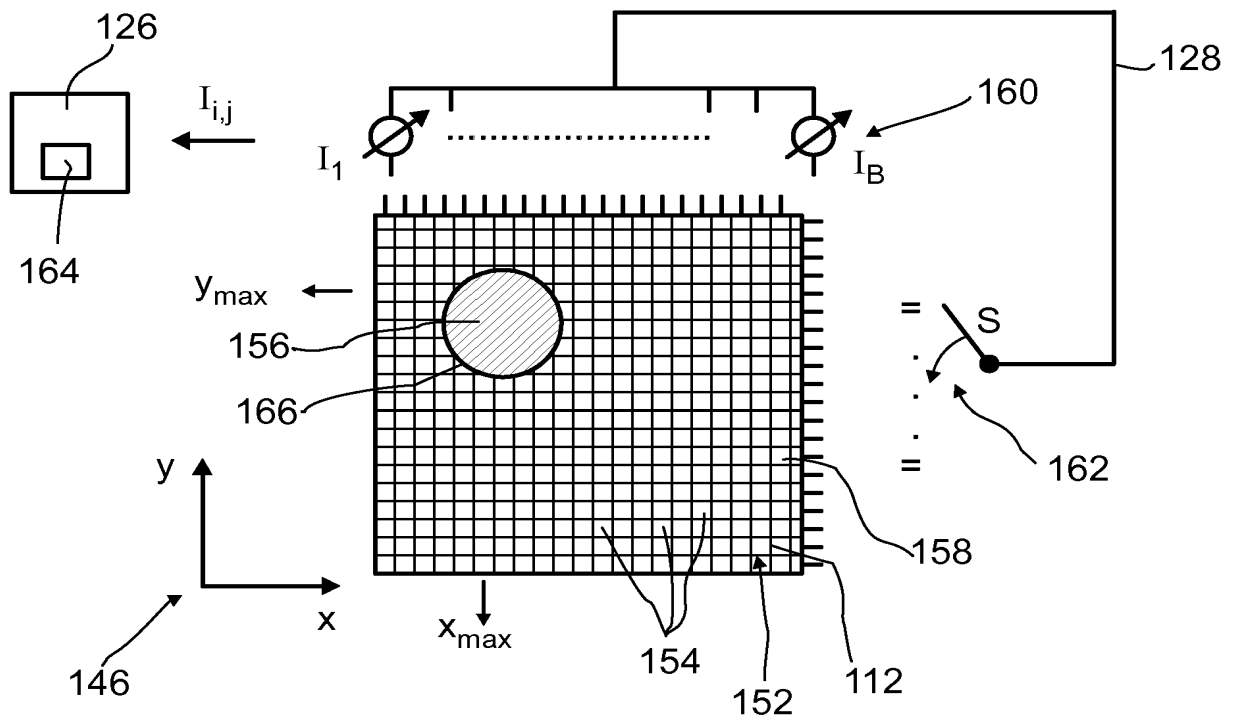


FIG.2B

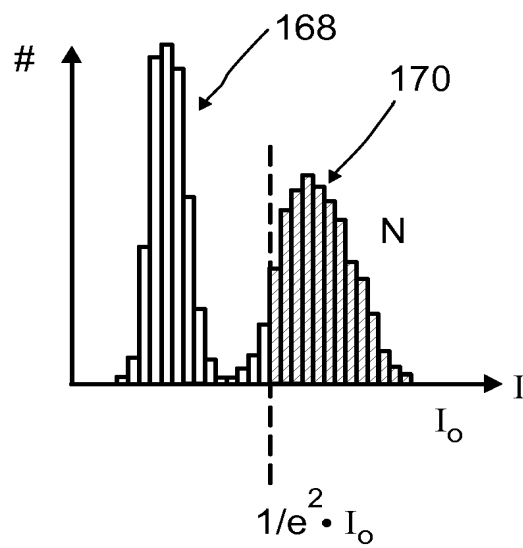


FIG.3A

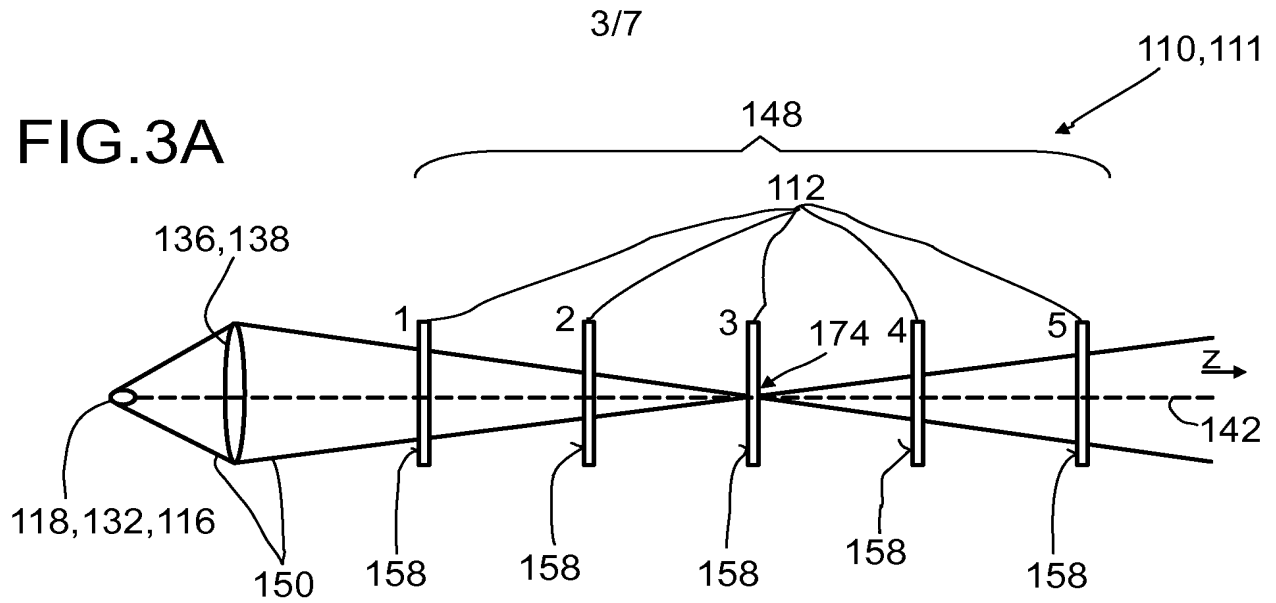


FIG.3B

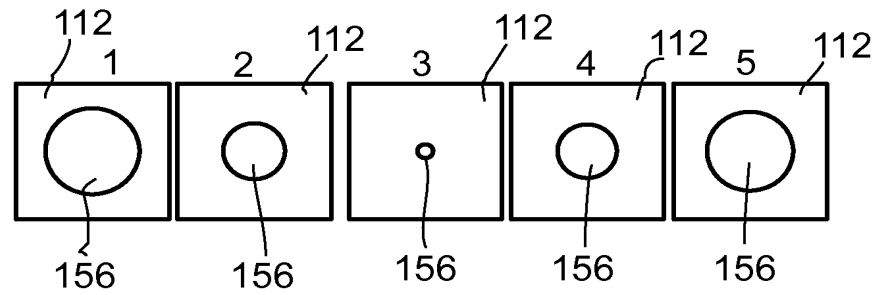


FIG.3C

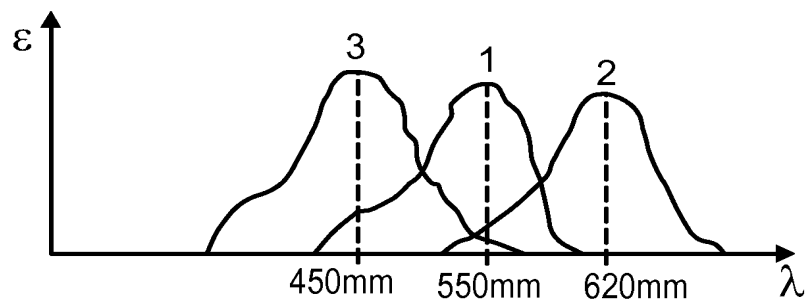


FIG.4B

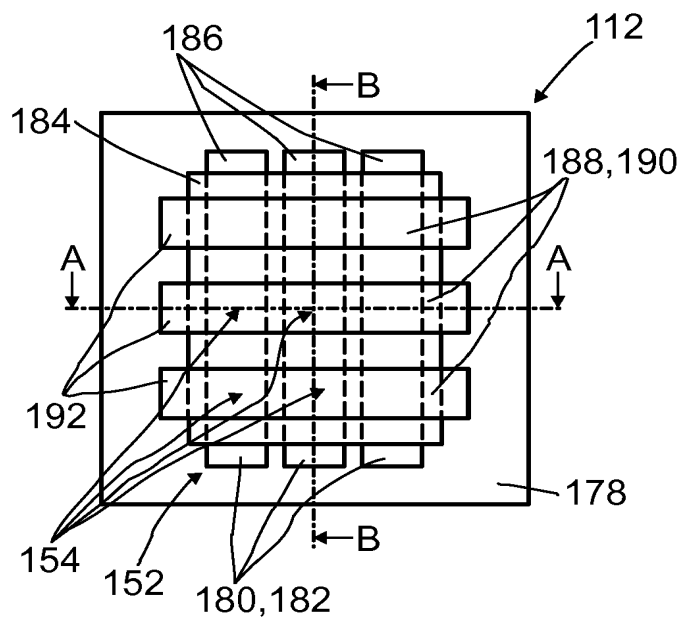
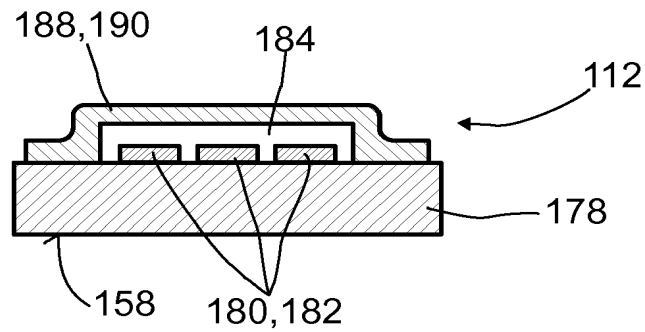


FIG.4A

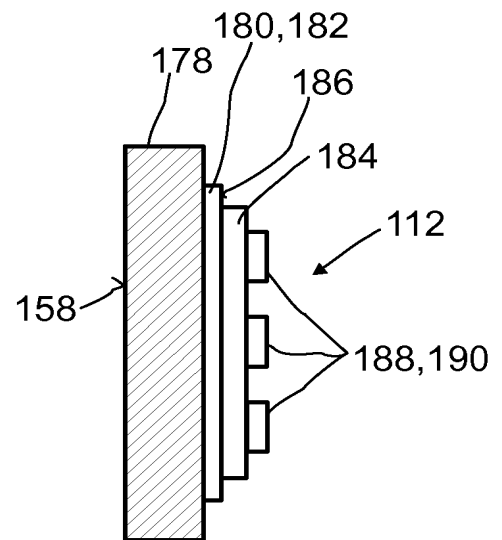


FIG.4C

FIG.5

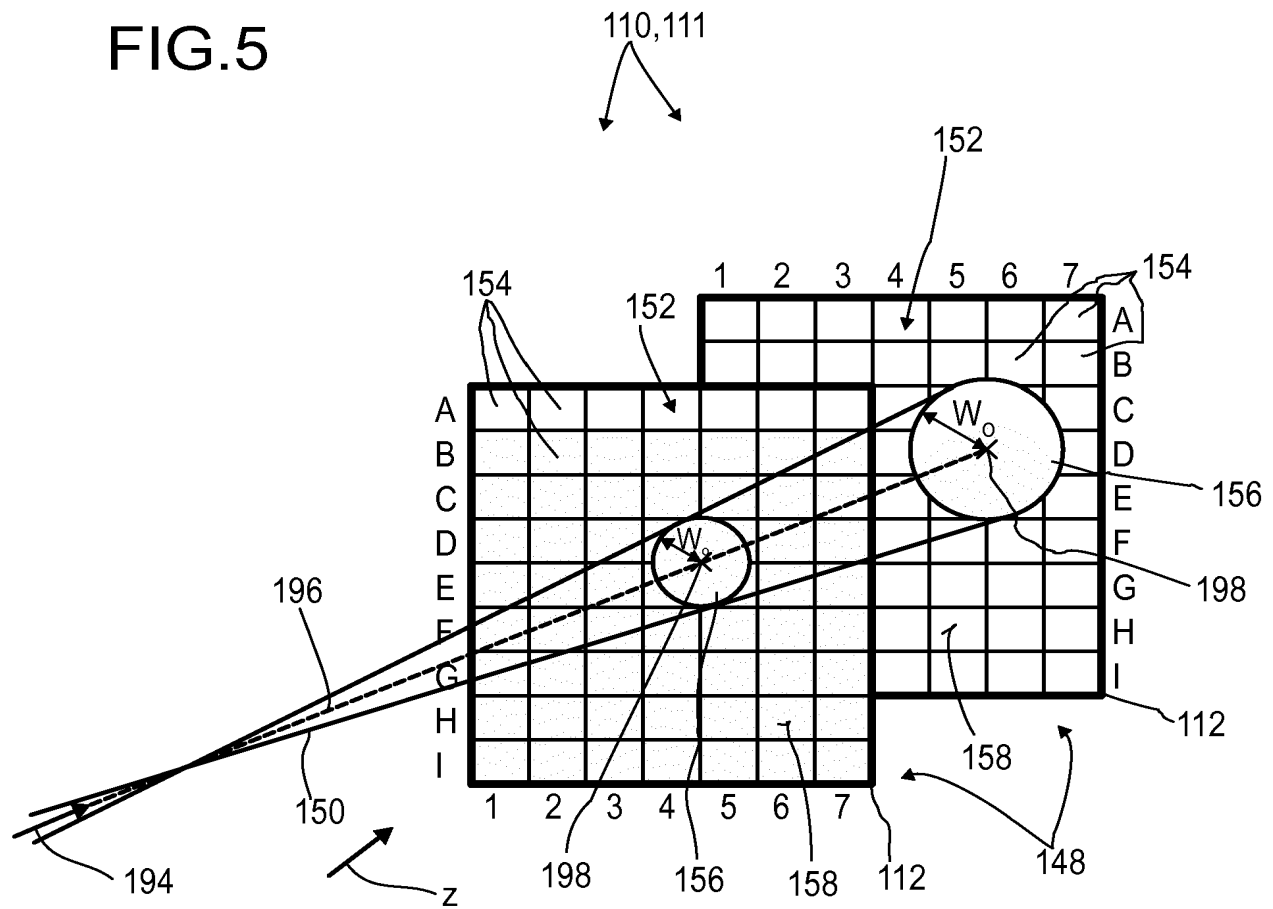


FIG.6

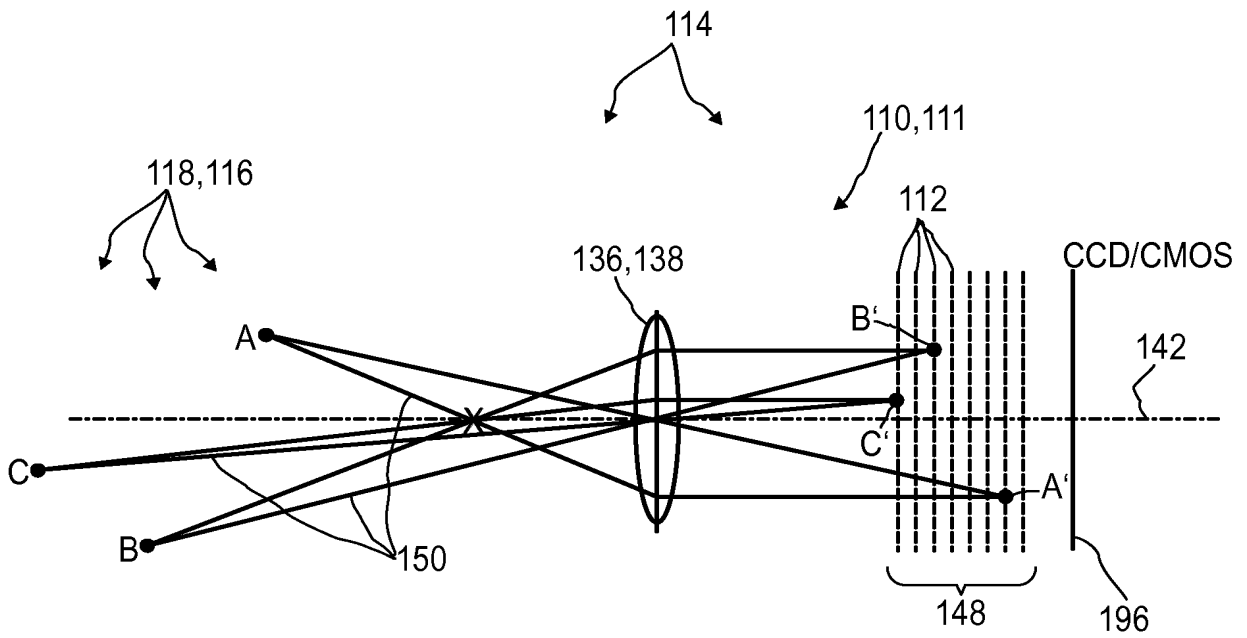
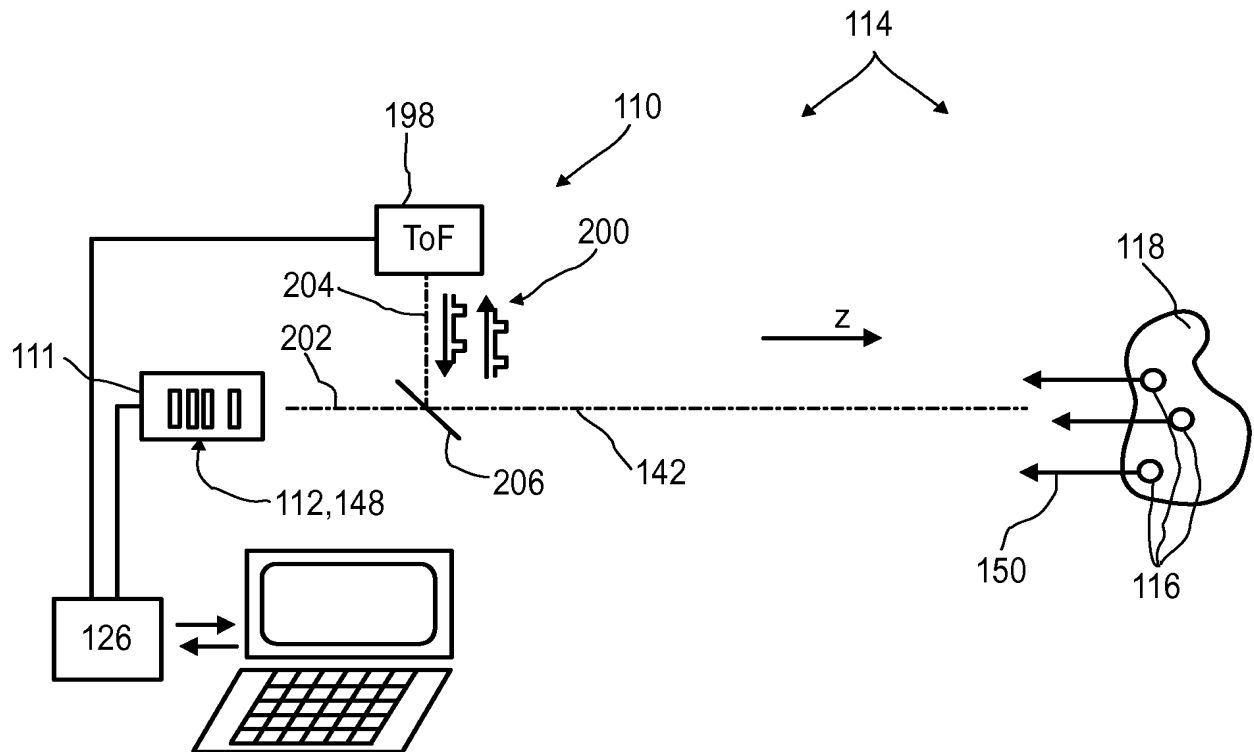


FIG.7



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2014/061695

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S5/16 G01S11/12 G01S17/46 G01C3/32
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S G01C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	HIURA S ET AL: "Depth measurement by the multi-focus camera", COMPUTER VISION AND PATTERN RECOGNITION, 1998. PROCEEDINGS. 1998 IEEE COMPUTER SOCIETY CONFERENCE ON SANTA BARBARA, CA, USA 23-25 JUNE 1998, LOS ALAMITOS, CA, USA, IEEE COMPUT. SOC, US, 23 June 1998 (1998-06-23), pages 953-959, XP010291663, DOI: 10.1109/CVPR.1998.698719 ISBN: 978-0-8186-8497-5 paragraphs [0002], [0003] -----	1-12, 14-21, 23,24
X	WO 2006/134370 A1 (MBDA UK LTD [GB]; JENNINGS MARTYN ROBERT [GB]; MILLER LEE DOUGLAS [GB]) 21 December 2006 (2006-12-21) the whole document ----- -/-	1-10,15, 16,18-24



Further documents are listed in the continuation of Box C.



See patent family annex.

* Special categories of cited documents :

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"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search

27 August 2014

Date of mailing of the international search report

03/09/2014

Name and mailing address of the ISA/

European Patent Office, P.B. 5818 Patentlaan 2
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Fax: (+31-70) 340-3016

Authorized officer

Damp, Stephan

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2014/061695

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	EP 0 185 450 A2 (TOSHIBA KK [JP]) 25 June 1986 (1986-06-25) paragraphs [0020], [0021] -----	1-24
X	CA 2 196 563 A1 (UNIV COLUMBIA [JP]) 19 December 1996 (1996-12-19) pages 2-14 -----	1-12, 14-16, 18-24
A	KUTHIRUMMAL S ET AL: "Flexible Depth of Field Photography", IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE COMPUTER SOCIETY, USA, vol. 33, no. 1, 1 January 2011 (2011-01-01), pages 58-71, XP011373495, ISSN: 0162-8828, DOI: 10.1109/TPAMI.2010.66 the whole document -----	1-24
A	US 2012/242867 A1 (SHUSTER GARY S [US]) 27 September 2012 (2012-09-27) paragraphs [0045] - [0049] -----	1-24
A	US 2010/283868 A1 (CLARK LLOYD DOUGLAS [US] ET AL) 11 November 2010 (2010-11-11) the whole document -----	1-24

INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2014/061695

Patent document cited in search report		Publication date	Patent family member(s)	Publication date
WO 2006134370	A1	21-12-2006	NONE	
EP 0185450	A2	25-06-1986	DE 3576434 D1 EP 0185450 A2 JP S61135280 A US 4675535 A	12-04-1990 25-06-1986 23-06-1986 23-06-1987
CA 2196563	A1	19-12-1996	NONE	
US 2012242867	A1	27-09-2012	US 2012242867 A1 US 2014078376 A1	27-09-2012 20-03-2014
US 2010283868	A1	11-11-2010	NONE	

PCTNOTIFICATION CONCERNING
TRANSMITTAL OF COPY OF INTERNATIONAL
PRELIMINARY REPORT ON PATENTABILITY
(CHAPTER I OF THE PATENT COOPERATION TREATY)

(PCT Rule 44bis.1(c))

From the INTERNATIONAL BUREAU

To:

ALTMANN STÖSSEL DICK PATENTANWÄLTE
PARTG MBB
Theodor-Heuss-Anlage 2
68165 Mannheim
ALLEMAGNE

Date of mailing (<i>day/month/year</i>) 30 September 2021 (30.09.2021)		
Applicant's or agent's file reference B182037PC		IMPORTANT NOTICE
International application No. PCT/EP2020/056759	International filing date (<i>day/month/year</i>) 13 March 2020 (13.03.2020)	Priority date (<i>day/month/year</i>) 15 March 2019 (15.03.2019)
Applicant TRINAMIX GMBH		

The International Bureau transmits herewith a copy of the international preliminary report on patentability (Chapter I of the Patent Cooperation Treaty)

The International Bureau of WIPO 34, chemin des Colombettes 1211 Geneva 20, Switzerland	Authorized officer Nora Lindner e-mail pct.team5@wipo.int Telephone No. +41 22 338 74 05
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PATENT COOPERATION TREATY

PCT

INTERNATIONAL PRELIMINARY REPORT ON PATENTABILITY

(Chapter I of the Patent Cooperation Treaty)

(PCT Rule 44*bis*)

Applicant's or agent's file reference B182037PC	FOR FURTHER ACTION	See item 4 below
International application No. PCT/EP2020/056759	International filing date (<i>day/month/year</i>) 13 March 2020 (13.03.2020)	Priority date (<i>day/month/year</i>) 15 March 2019 (15.03.2019)
International Patent Classification (8th edition unless older edition indicated) See relevant information in Form PCT/ISA/237		
Applicant TRINAMIX GMBH		

1.	This international preliminary report on patentability (Chapter I) is issued by the International Bureau on behalf of the International Searching Authority under Rule 44 <i>bis</i> . I(a).																								
2.	This REPORT consists of a total of 8 sheets, including this cover sheet.																								
In the attached sheets, any reference to the written opinion of the International Searching Authority should be read as a reference to the international preliminary report on patentability (Chapter I) instead.																									
3.	This report contains indications relating to the following items: <table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 10%; text-align: center;">☒</td> <td style="width: 30%;">Box No. I</td> <td style="width: 80%;">Basis of the report</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. II</td> <td>Priority</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. III</td> <td>Non-establishment of opinion with regard to novelty, inventive step and industrial applicability</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. IV</td> <td>Lack of unity of invention</td> </tr> <tr> <td style="text-align: center;">☒</td> <td>Box No. V</td> <td>Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. VI</td> <td>Certain documents cited</td> </tr> <tr> <td style="text-align: center;">☐</td> <td>Box No. VII</td> <td>Certain defects in the international application</td> </tr> <tr> <td style="text-align: center;">☒</td> <td>Box No. VIII</td> <td>Certain observations on the international application</td> </tr> </table>	☒	Box No. I	Basis of the report	☐	Box No. II	Priority	☐	Box No. III	Non-establishment of opinion with regard to novelty, inventive step and industrial applicability	☐	Box No. IV	Lack of unity of invention	☒	Box No. V	Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement	☐	Box No. VI	Certain documents cited	☐	Box No. VII	Certain defects in the international application	☒	Box No. VIII	Certain observations on the international application
☒	Box No. I	Basis of the report																							
☐	Box No. II	Priority																							
☐	Box No. III	Non-establishment of opinion with regard to novelty, inventive step and industrial applicability																							
☐	Box No. IV	Lack of unity of invention																							
☒	Box No. V	Reasoned statement under Article 35(2) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement																							
☐	Box No. VI	Certain documents cited																							
☐	Box No. VII	Certain defects in the international application																							
☒	Box No. VIII	Certain observations on the international application																							
4.	The International Bureau will communicate this report to designated Offices in accordance with Rules 44 <i>bis</i> .3(c) and 93 <i>bis</i> .1 but not, except where the applicant makes an express request under Article 23(2), before the expiration of 30 months from the priority date (Rule 44 <i>bis</i> .2).																								

The International Bureau of WIPO 34, chemin des Colombettes 1211 Geneva 20, Switzerland	Date of issuance of this report 16 September 2021 (16.09.2021) Authorized officer Nora Lindner e-mail pct.team5@wipo.int
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PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

PCT

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see form PCT/ISA/220

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY (PCT Rule 43bis.1)

Date of mailing
(day/month/year) see form PCT/ISA/210 (second sheet)

Applicant's or agent's file reference
see form PCT/ISA/220

FOR FURTHER ACTION
See paragraph 2 below

International application No.
PCT/EP2020/056759

International filing date (day/month/year)
13.03.2020

Priority date (day/month/year)
15.03.2019

International Patent Classification (IPC) or both national classification and IPC
INV. G01S7/48 G01N21/47 G01S17/46 G06K9/00 G06T7/00 G06T7/521

Applicant
TRINAMIX GMBH

1. This opinion contains indications relating to the following items:

- ☒ Box No. I Basis of the opinion
- ☐ Box No. II Priority
- ☐ Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- ☐ Box No. IV Lack of unity of invention
- ☒ Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- ☐ Box No. VI Certain documents cited
- ☐ Box No. VII Certain defects in the international application
- ☒ Box No. VIII Certain observations on the international application

2. FURTHER ACTION

If a demand for international preliminary examination is made, this opinion will usually be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1bis(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA:



European Patent Office
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Fax: +49 89 2399 - 4465

Date of completion of
this opinion

see form
PCT/ISA/210

Authorized Officer

Grübl, Alexander

Telephone No. +49 89 2399-0



Box No. I Basis of the opinion

1. With regard to the **language**, this opinion has been established on the basis of:
 - ☒ the international application in the language in which it was filed.
 - ☐ a translation of the international application into , which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1 (b)).
2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(a))
3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:
 - a. ☐ forming part of the international application as filed:
 - ☐ in the form of an Annex C/ST.25 text file.
 - ☐ on paper or in the form of an image file.
 - b. ☐ furnished together with the international application under PCT Rule 13*ter*.1(a) for the purposes of international search only in the form of an Annex C/ST.25 text file.
 - c. ☐ furnished subsequent to the international filing date for the purposes of international search only:
 - ☐ in the form of an Annex C/ST.25 text file (Rule 13*ter*.1(a)).
 - ☐ on paper or in the form of an image file (Rule 13*ter*.1(b) and Administrative Instructions, Section 713).
4. ☐ In addition, in the case that more than one version or copy of a sequence listing has been filed or furnished, the required statements that the information in the subsequent or additional copies is identical to that forming part of the application as filed or does not go beyond the application as filed, as appropriate, were furnished.
5. Additional comments:

**WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY**

International application No.
PCT/EP2020/056759

Box No. V Reasoned statement under Rule 43bis.1(a)(i) with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

1. Statement

Novelty (N)	Yes: Claims	<u>4, 9-14, 16-20, 23</u>
	No: Claims	<u>1-3, 5-8, 15, 21, 22, 24, 25</u>
Inventive step (IS)	Yes: Claims	
	No: Claims	<u>1-25</u>
Industrial applicability (IA)	Yes: Claims	<u>1-25</u>
	No: Claims	

2. Citations and explanations

see separate sheet

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

see separate sheet

Re Item V

Reasoned statement with regard to novelty, inventive step or industrial applicability; citations and explanations supporting such statement

Reference is made to the following documents:

- D1 DE 198 46 619 A1 (BYK GARDNER GMBH [DE]) 27 April 2000 (2000-04-27)
- D2 CN 108 363 482 A (JIANGSU SIDIANLING ROBOT CO LTD) 3 August 2018 (2018-08-03)
- D3 US 2018/033146 A1 (BLEYER MICHAEL [US] ET AL) 1 February 2018 (2018-02-01)
- D4 WO 2019/042956 A1 (TRINAMIX GMBH [DE]) 7 March 2019 (2019-03-07)

- 1 The present application does not meet the criteria of Article 33(1) PCT, because the subject-matter of independent claims 1, 24 and 25 is not new in the sense of Article 33(2) PCT.

Claim 1

A detector for identifying at least one material property comprising

- at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object (D1: claim 1; D2: claim 1 - camera; D3: claim 1, claim 11 - imaging system)

- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image (D1: claim 1; D2: claim 1 - processor, depth information; D3: claim 1, claim 11 - intensity profile determination),

wherein the evaluation device is configured for determining at least one distance feature by applying at least one distance dependent image filter to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof, being one of the

depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, (D1: claim 29; D2: claim 1 - triangulation; D3: claims 1, 4 - depth map, triangulation)

wherein the evaluation device is configured for determining at least one material feature by applying at least one material dependent image filter to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Mother which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof (D1: claim 1, col. 14, l. 38-66; D2: claim 1 - median filter; D3: claim 6, [0036]),

wherein the evaluation device is configured for determining a longitudinal coordinate and the material property by evaluating the distance feature and the material feature (D1: claims 1 and 29; D2: claim 1 - median filtering, processing depth image; D3: claim 6, [0037]).

The method according to **claim 24** is disclosed in the same manner.

Claim 25

A use of the detector according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography

- application; a machine vision application; a robotics application; a quality control application; a manufacturing application (D1: col 2, l. 31-38; D2: claim 1; D3: [0013]).
- 2 The present application does not meet the criteria of Article 33(1) PCT, because the subject-matter of independent **claim 23** does not involve an inventive step in the sense of Article 33(3) PCT.
- Inertial measurement units are generally known, being adapted to receive data of a sensor as defined in claim 1, a material property detector, does not allow to establish a technical effect resulting from this feature.
- 3 The dependent claims do not contain particular features which appear suitable for rendering the independent claims clear, new and defining inventive subject matter:
- Claims 2, 3, 5-7 relate to further vague definitions of the filters, not adding any more substance.
- Claim 4 relates to material properties usually determined: e.g. D1: column 14, lines 38-45; D3: claim 1.
- Claim 8: see D3, claim 6
- Claims 9-11: Recursive, iterative or simultaneous application of filters and applying a further filter are considered usual alternatives.
- Claims 12, 13: Using combined signals from sensor elements or a convolution based algorithm are considered common measures.
- Claim 14: A CMOS sensor is a common matrix sensor.
- Claim 15: D1, claim 7
- Claim 16: A diffractive optical element is a usual means for producing such patterns.
- The features of claims 17-20 are known from D4, claims 19-22, which the skilled person would turn to in order to find implementations.
- Claim 21: D2, claim 1
- Claim 22: D3, claim 1

Re Item VIII

Certain observations on the international application

The claims are not clear (Article 6 PCT) and the disclosure of the invention defined therein is not sufficient for enabling the skilled person to carry it out (Article 5 PCT).

The independent claims 1 and 24 relate to determining a longitudinal coordinate and a material property. It is neither clear how it is arrived at the desired distance feature and material feature nor which "further" filters are covered by the claim. The vague indication of correlation with one of the listed filters is an infinite definition of filters which does not allow to put into practice the invention or determine what is covered by the claims. The list of filters contains filters which are not commonly known and lack a definition in the claims.

The sign φ_{1m} in claim 10 is not defined.

Claim 23: As the detector as defined in claim 1 determines a material property, it is not clear how it can be used in an inertial measurement unit. A disclosure in this respect is also missing in the description.

Electronic Acknowledgement Receipt

EFS ID:	43948286
Application Number:	17439492
International Application Number:	
Confirmation Number:	9002
Title of Invention:	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY
First Named Inventor/Applicant Name:	Friedrich Schick
Customer Number:	76656
Filer:	CHRISTOPHER M GOFF/Christine Gilliland
Filer Authorized By:	CHRISTOPHER M GOFF
Attorney Docket Number:	27843-1809 182037US01
Receipt Date:	05-OCT-2021
Filing Date:	
Time Stamp:	11:47:35
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Information Disclosure Statement (IDS) Form (SB08)	2021-10-05_Supp-IDS_27843-1809-50034715.pdf	1097333	no	5
			6e07a9f65af7ae63937f3d570cfe251278ed bfb0		

Warnings:

Information:					
2	Foreign Reference	WO2014097181A1-50034834.pdf	9274093	no	146
			7dc9091ac7f5941011acd1c633b1afae0d00729f		
Warnings:					
Information:					
3	Foreign Reference	WO2018091640A2.pdf	15572435	no	258
			01c58c0a86a4aa5bc35733969e9d2629ab1d6956		
Warnings:					
Information:					
4	Foreign Reference	WO2018091649A1.pdf	7187528	no	109
			df8c2044721145c61c8542afef8cf620db14ff08f		
Warnings:					
Information:					
5	Foreign Reference	WO2018091638A1.pdf	6885816	no	106
			b8449ae88effd58a93df0c1b209b94f2ad9e3d16		
Warnings:					
Information:					
6	Foreign Reference	WO2012110924A1.pdf	5201933	no	120
			91978800a1aa8d9fa253c032a143407a91e9dcb7		
Warnings:					
Information:					
7	Foreign Reference	WO2014198629A1.pdf	5128912	no	88
			4ff338f0c5de1c9f7946b530c510fc961a8bede0		
Warnings:					
Information:					
8	Other Reference-Patent/App/Search documents	IPRP.pdf	329426	no	9
			e53f99e974624af810dc34476763d30cf03172b4		
Warnings:					
Information:					

9	Non Patent Literature	Bunke1997.pdf	7925563	no	32
			f045fde3347b2486bca1ba1bca608a17b5c398f9		

Warnings:

Information:

10	Non Patent Literature	Eichler1991.pdf	10143752	no	96
			0e33c1eaac9a1bb02a5e7bd2f73893ab840824b6		

Warnings:

Information:

Total Files Size (in bytes):			68746791
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/439,492	09/15/2021		2580	27843-1809 182037US01	25	1

CONFIRMATION NO. 9002

FILING RECEIPT



0000000131694238

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Date Mailed: 02/09/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

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Applicant(s)

tinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

Power of Attorney: The patent practitioners associated with Customer Number 76656

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/EP2020/056759 03/13/2020

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)
EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 02/07/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/439,492**

Projected Publication Date: 05/19/2022

Non-Publication Request: No

Early Publication Request: No
Title

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

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U.S. APPLICATION NO.	FIRST NAMED INVENTOR	ATTY. DOCKET NO.
17/439,492	Friedrich Schick	27843-1809 182037US01

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

INTERNATIONAL APPLICATION NO.	
PCT/EP2020/056759	
I.A. FILING DATE	PRIORITY DATE
03/13/2020	03/15/2019

CONFIRMATION NO. 9002
371 ACCEPTANCE LETTER



Date Mailed: 02/09/2022

NOTICE OF ACCEPTANCE OF APPLICATION UNDER 35 U.S.C 371 AND 37 CFR 1.495

The applicant is hereby advised that the United States Patent and Trademark Office, in its capacity as a Designated / Elected Office (37 CFR 1.495), has ACCEPTED the above identified international application for national patentability examination in the United States Patent and Trademark Office.

The United States Application Number assigned to the application is shown above. A Filing Receipt will be issued for the present application in due course. **THE DATE APPEARING ON THE FILING RECEIPT AS THE "FILING DATE or 371(c) DATE" IS THE DATE ON WHICH THE LAST OF THE 35 U.S.C. 371 (c)(1) and (c)(2) REQUIREMENTS HAS BEEN RECEIVED IN THE OFFICE. THIS DATE IS SHOWN BELOW.** The filing date of the above identified application is the international filing date of the international application (Article 11(3) and 35 U.S.C. 363)

09/15/2021
DATE OF RECEIPT OF 35 U.S.C.
371(c)(1) and (c)(2) REQUIREMENTS

The following items have been received:

- Copy of the International Application filed on 09/15/2021
- Copy of the International Search Report filed on 09/15/2021
- Preliminary Amendments filed on 09/15/2021
- Information Disclosure Statements filed on 09/15/2021
- Inventor's Oath or Declaration filed on 09/15/2021
- Request for Immediate Examination filed on 09/15/2021
- U.S. Basic National Fees filed on 09/15/2021
- Authorize Access to Search Results filed on 09/15/2021
- Priority Documents filed on 09/15/2021
- Power of Attorney filed on 09/15/2021
- Specification filed on 09/15/2021
- Claims filed on 09/15/2021
- Abstracts filed on 09/15/2021
- Drawings filed on 09/15/2021
- Authorization to Permit Access filed on 09/15/2021
- Application Data Sheet (37 CFR 1.76) filed on 09/15/2021

Applicant is reminded that any communications to the United States Patent and Trademark Office must be mailed to the address given in the heading and include the U.S. application no. shown above (37 CFR 1.5)

PAULETTE R KIDWELL

Telephone: (571) 272-0398

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/439,492

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	540
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(j))	25 minus 20 =	*			OR	x 100 =	500
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x 480 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						420
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	2580

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

MULTIPLE DEPENDENT CLAIM FEE CALCULATION SHEET Substitute for Form PTO-1360 (For use with Form PTO/SB/06)							Application Number		Filing Date				
							17439492						
							Applicant(s) Friedrich Schick						
							* May be used for additional claims or amendments						
CLAIMS	AS FILED		AFTER FIRST AMENDMENT		AFTER SECOND AMENDMENT			*		*		*	
	Indep	Depend	Indep	Depend	Indep	Depend		Indep	Depend	Indep	Depend	Indep	Depend
1	1		1					51					
2		1		1				52					
3		(1)		1				53					
4		(1)		1				54					
5		(1)		1				55					
6		(1)		1				56					
7		(1)		1				57					
8		(1)		1				58					
9		(1)		1				59					
10		(1)		1				60					
11		(1)		1				61					
12		(1)		1				62					
13		(1)		1				63					
14		(1)		1				64					
15		(1)		1				65					
16		(1)		1				66					
17		(1)		1				67					
18		(1)		1				68					
19		(1)		1				69					
20		(1)		1				70					
21		(1)		1				71					
22		(1)		1				72					
23		(1)		1				73					
24		(1)		1				74					
25		(1)		1				75					
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44								94					
45								95					
46								96					
47								97					
48								98					
49								99					
50								100					
Total Indep	1		1		0								
Total Depend	24	↙	24	↙	0	↙							
Total Claims	25		25		0								

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of: Friedrich Schick et al.

Serial No.: 17/439,492

Filed: September 15, 2021

Confirmation No.: 9002

For: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

February 9, 2022

To The Commissioner for Patents,

Sir:

REQUEST FOR CORRECTED FILING RECEIPT

Applicant respectfully requests that the Filing Receipt in the above-referenced application be corrected to reflect the correct spelling of the applicant as shown on the Application Data Sheet.

A marked-up copy of the Filing Receipt reflecting the requested change, and the Application Data Sheet (ADS) as filed on September 15, 2021 showing the correct spelling, are attached.

Applicants further request that a new, corrected filing receipt be issued.

Since this was a Patent Office error it is believed that no fee is due. The Commissioner is hereby authorized to charge any deficiency or credit any overpayment of fees for this Request to Deposit Account No. 012384.

Respectfully submitted,

/Christopher M. Goff/

Christopher M. Goff, Reg. No. 41,785
Armstrong Teasdale LLP
One Metropolitan Square, Suite 2600
St. Louis, MO 63102
314) 621-5070



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/439,492	09/15/2021		2580	27843-1809 182037US01	25	1

CONFIRMATION NO. 9002

FILING RECEIPT



0000000131694238

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Date Mailed: 02/09/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Friedrich Schick, Ludwigshafen, GERMANY;
Peter Schillen, Ludwigshafen, GERMANY;
Patrick Schindler, Ludwigshafen, GERMANY;
Andre Schmidt, Ludwigshafen, GERMANY;
Michael Eberspach, Ludwigshafen, GERMANY;
Christian Lennartz, Ludwigshafen, GERMANY;
Robert Send, Karlsruhe, GERMANY;
Lars Diesselberg, Karlsruhe, GERMANY;
Heiko Hengen, Steinweiler, GERMANY;
Ingmar Bruder, Ludwigshafen, GERMANY;
Jakob Unger, Ludwigshafen, GERMANY;
Christian Bonsignore, Ludwigshafen, GERMANY;

Applicant(s)

~~trinamix GmbH~~, Ludwigshafen am Rhein, GERMANY; trinamix GmbH

Power of Attorney: The patent practitioners associated with Customer Number 76656

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/EP2020/056759 03/13/2020

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)
EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 02/07/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/439,492**

Projected Publication Date: 05/19/2022

Non-Publication Request: No

Early Publication Request: No
Title

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER

Title 35, United States Code, Section 184

Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor

community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Friedrich		Schick		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen	Country of Residence ⁱ	DE		
Mailing Address of Inventor:					
Address 1		Industriestrasse 35			
Address 2					
City	Ludwigshafen	State/Province			
Postal Code	67063	Country ⁱ	DE		
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Peter		Schillen		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen	Country of Residence ⁱ	DE		
Mailing Address of Inventor:					
Address 1		Industriestrasse 35			
Address 2					
City	Ludwigshafen	State/Province			
Postal Code	67063	Country ⁱ	DE		
Inventor 3					Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Patrick		Schindler	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Ludwigshafen	Country of Residence ⁱ	DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country ⁱ	DE	
Inventor 4				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Andre		Schmidt	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Ludwigshafen	Country of Residence ⁱ	DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country ⁱ	DE	
Inventor 5				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Michael		Eberspach	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Ludwigshafen	Country of Residence ⁱ	DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen	State/Province		
Postal Code	67063	Country ⁱ	DE	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Inventor 6					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Christian		Lennartz		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen	Country of Residence ⁱ		DE	

Mailing Address of Inventor:

Address 1	Industriestrasse 35				
Address 2					
City	Ludwigshafen	State/Province			
Postal Code	67063	Country ⁱ	DE		

Inventor 7					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Robert		Send		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Karlsruhe	Country of Residence ⁱ		DE	

Mailing Address of Inventor:

Address 1	Luisenstr. 25				
Address 2					
City	Karlsruhe	State/Province			
Postal Code	76137	Country ⁱ	DE		

Inventor 8					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Lars		Diesselberg		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Karlsruhe	Country of Residence ⁱ		DE	

Mailing Address of Inventor:

Address 1	Geibelstr. 22				
Address 2					

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

City	Karlsruhe	State/Province	
Postal Code	76185	Country ⁱ	DE
Inventor 9	Remove		
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Heiko		Hengen
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service			
City	Steinweiler	Country of Residence ⁱ	DE

Mailing Address of Inventor:			
Address 1	Siedlerweg 8		
Address 2			
City	Steinweiler	State/Province	
Postal Code	76872	Country ⁱ	DE
Inventor 10	Remove		
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Ingmar		Bruder
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service			
City	Ludwigshafen	Country of Residence ⁱ	DE

Mailing Address of Inventor:			
Address 1	Industriestrasse 35		
Address 2			
City	Ludwigshafen	State/Province	
Postal Code	67063	Country ⁱ	DE
Inventor 11	Remove		
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Jakob		Unger
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service			
City	Ludwigshafen	Country of Residence ⁱ	DE

Mailing Address of Inventor:			
-------------------------------------	--	--	--

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Address 1		Industriestrasse 35	
Address 2			
City	Ludwigshafen	State/Province	
Postal Code	67063	Country ⁱ	DE
Inventor 12			Remove
Legal Name			
Prefix	Given Name	Middle Name	Family Name
	Christian		Bonsignore
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service			
City	Ludwigshafen	Country of Residence ⁱ	DE

Mailing Address of Inventor:			
Address 1		Industriestrasse 35	
Address 2			
City	Ludwigshafen	State/Province	
Postal Code	67063	Country ⁱ	DE
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add			

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence information of this application.			
Customer Number	76656		
Email Address	uspatents@armstrongteasdale.com	Add Email	Remove Email

Application Information:

Title of the Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		
Attorney Docket Number	27843-1809 182037US01	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	2	Suggested Figure for Publication (if any)	1

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

Request Not to Publish. I hereby request that the attached application not be published under

☐ 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	76656		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	a 371 of international	PCT/EP2020/056759	2020-03-13

Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the **Add** button.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	 Access Code ⁱ (if applicable)
19163250.4	EP	2019-03-15	

Additional Foreign Priority Data may be generated within this form by selecting the **Add** button.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant 1

If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.

[Clear](#)

☒ Assignee
 ☐ Legal Representative under 35 U.S.C. 117
 ☐ Joint Inventor

☐ Person to whom the inventor is obligated to assign.
 ☐ Person who shows sufficient proprietary interest

If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:

Name of the Deceased or Legally Incapacitated Inventor:

If the Applicant is an Organization check here. ☒

Organization Name

Mailing Address Information For Applicant:

Address 1		Industriestrasse 35	
Address 2			
City		Ludwigshafen am Rhein	State/Province
Country	DE	Postal Code	67063
Phone Number		Fax Number	
Email Address			

Additional Applicant Data may be generated within this form by selecting the Add button.

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

Assignee 1

Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.

If the Assignee or Non-Applicant Assignee is an Organization check here. ☐

Prefix	Given Name	Middle Name	Family Name	Suffix

Mailing Address Information For Assignee including Non-Applicant Assignee:

Address 1			
Address 2			
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Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button.

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). **However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).**

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Christopher M. Goff/		Date (YYYY-MM-DD)	2021-09-15
First Name	Christopher	Last Name	M. Goff	Registration Number 41785

Additional Signature may be generated within this form by selecting the Add button.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-1809 182037US01
		Application Number	
Title of Invention	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Electronic Acknowledgement Receipt

EFS ID:	44957972
Application Number:	17439492
International Application Number:	
Confirmation Number:	9002
Title of Invention:	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY
First Named Inventor/Applicant Name:	Friedrich Schick
Customer Number:	76656
Filer:	CHRISTOPHER M GOFF/Darlene Pearman
Filer Authorized By:	CHRISTOPHER M GOFF
Attorney Docket Number:	27843-1809 182037US01
Receipt Date:	09-FEB-2022
Filing Date:	15-SEP-2021
Time Stamp:	16:54:03
Application Type:	U.S. National Stage under 35 USC 371

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Request for Corrected Filing Receipt	RequestForCorrectedFR-52214 453.pdf	924163	no	17
			1b38a1339348409ed8d292dd64366be497 092f43		

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Total Files Size (in bytes):	924163
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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/439,492	09/15/2021		2580	27843-1809 182037US01	25	1

CONFIRMATION NO. 9002
CORRECTED FILING RECEIPT

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105



0000000131778833

Date Mailed: 02/11/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

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Applicant(s)

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Power of Attorney: The patent practitioners associated with Customer Number 76656

Domestic Priority data as claimed by applicant

This application is a 371 of PCT/EP2020/056759 03/13/2020

Foreign Applications (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)
EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

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If Required, Foreign Filing License Granted: 02/07/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/439,492**

Projected Publication Date: 05/19/2022

Non-Publication Request: No

Early Publication Request: No
Title

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/439,492	09/15/2021	Friedrich Schick	27843-1809 182037US01

CONFIRMATION NO. 9002

PUBLICATION NOTICE



OC000000133957228

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Title:DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

Publication No.US-2022-0157044-A1

Publication Date:05/19/2022

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The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

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Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

<i>Notice of References Cited</i>	Application/Control No. 17/439,492		Applicant(s)/Patent Under Reexamination Schick et al.	
	Examiner FRANTZ B JEAN		Art Unit 2454	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20180033146-A1	02-2018	Bleyer; Michael	G01B11/2513	1/1
*	B	US-20180276843-A1	09-2018	SEND; Robert	G02B3/14	1/1
*	C	US-20170363465-A1	12-2017	SEND; Robert	G01S17/46	1/1
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N	DE1984646619	04-2000	DE	BYK Gardner GMBH	G01N 21/55
	O	CN108363482	08-2018	CN	JIANGSU SIDIANLING ROBOT CO LT	H04N 21/422
	P	WO2019042956 A!	03-2019	WO	TRINAMIX GMBH	G01S 7/48
	Q	WO-2014097181-A1	06-2014	WO	BRUDER I	G01B11/14
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 17/439,492	Applicant(s)/Patent Under Reexamination Schick et al.
	Examiner FRANTZ B JEAN	Art Unit 2454

CPC - Searched*		
Symbol	Date	Examiner
G01S17/46 ; G06T7/521 ; G06T7/73 ; G06V10/60 ; G06V10/145 ; G06V10/764 ; H04N23/74 ; H04N23/56 ; G01B11/22 ; G01C21/16 ; G01S17/66 ; G06T5/002 ; G06T5/20 ; ; G06T7/20	09/09/2023	FJ

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Derwent, East, EPO, JPO, OCR, IBMTDBD, PGPUB, Google	09/08/2023	FJ

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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<i>Index of Claims</i> 	Application/Control No. 17/439,492	Applicant(s)/Patent Under Reexamination Schick et al.
	Examiner FRANTZ B JEAN	Art Unit 2454

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☐ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	09/09/2023								
	1	✓								
	2	✓								
	3	✓								
	4	✓								
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	22	✓								
	23	✓								
	24	✓								
	25	✓								

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	2516	(Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore)).inv.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:39 AM
L2	440018	(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:49 AM
L3	757	(G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:01 AM
L4	183885	(detect\$3 OR sen\$3) near10 (identif\$ OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP,	OR	ON	ON	2023/09/09 02:34 AM

		classify\$3) WITH (material OR property)	AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L5	249451	longitudinal WITH coordinate AND (material OR property OR evaluat\$3 OR distance)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L6	2430	L4 AND I5	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L7	11	L6 AND I3	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO,	OR	ON	ON	2023/09/09 02:38 AM

L8	1872340	(reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)	RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 09:57 AM
L9	1341757	filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss)	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:00 AM
L10	113287	L8 AND I9	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:01 AM
L11	67	L10 AND I6	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU,	OR	ON	ON	2023/09/09 10:02 AM

L12	0	L11 AND I1	CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM
L13	35	L11 AND I2	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM
L14	11	L11 AND I3	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM

			WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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PE2E SEARCH - Search History (Interference)

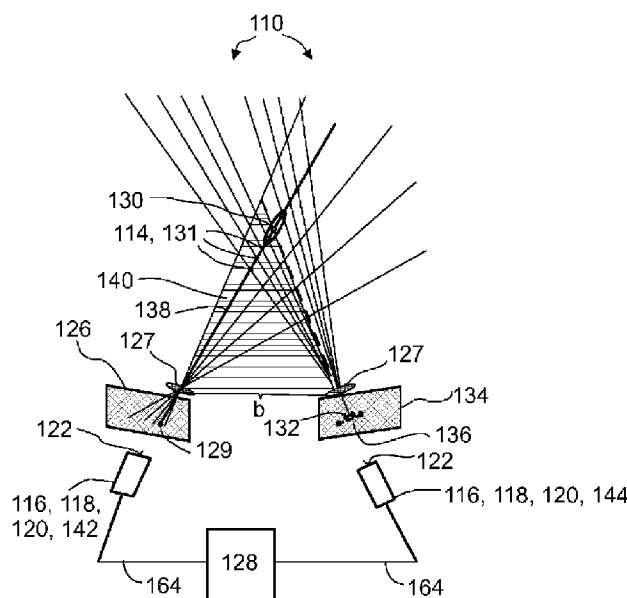
There are no Interference searches to show.



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(54) Title: DETECTOR FOR DETERMINING A POSITION OF AT LEAST ONE OBJECT

FIG.2



(57) Abstract: A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises -at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein each optical sensor (120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (122) by a reflection light beam (182) propagating from the object (112) to the detector (110), wherein the sensor element (116) is adapted to determine at least one reflection image (126); -at least one evaluation device (128), wherein the evaluation device (128) is adapted to select at least one reflection feature of the reflection image (126), wherein the evaluation device (128) is configured for determining at least one longitudinal region (130) of the selected reflection feature of the reflection image (134) by evaluating a combined signal Q from the sensor signals, wherein the evaluation device (128) is adapted to determine at least one

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displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130), wherein the evaluation device (128) is adapted to match the selected reflection feature with at least one reference feature within the displacement region (132).

Detector for determining a position of at least one object

Description

5 Field of the invention

The invention relates to a detector and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

Prior art

Optical 3D sensing methods generally may determine unreliable results in case of environment causing multiple reflections, with biasing light sources, or reflective measurement objects. Furthermore, 3D sensing methods, such as triangulation methods using structured light or stereo cameras, with imaging capabilities often suffer from a high demand of computational power to solve correspondence problems. Necessary computational power may result in high costs for processors or Field Programmable Gate Arrays (FPGAs), for heat removal in view of ventilation requirements or water proof housing difficulties, for electrical power consumption, in particular in mobile devices, and, in addition, in measurement uncertainties. The high demand in computational power may be prohibitive to realize real-time applications, high framerates or even standards video speed frame rates of 25 frames per second.

A large number of optical devices are known from the prior art using triangulation imaging methods. For example, structured light methods or stereo methods are known. For example, passive stereo methods using two cameras in a fixed relative orientation or active stereo technologies, where an additional light projector is used. Another example is the structures light approach, where one light projector and one camera in a fixed relative orientation are used. In order to determine a depth image via triangulation, the correspondence problem has to be solved first. Therefore, in passive stereo camera techniques enough corresponding feature points have to be identified in both camera views. In structured light approaches correspondences between pre-stored and projected pseudo random light patterns have to be determined. For a robust solution of these correspondence problems computational imaging algorithms such as algorithms scaling approximately quadratically with the number of points in the projected point pattern have to be employed. In structured light methods, for example, using a stereo system comprising two detectors having a fixed relative distance, a light source projects a pat-

tern such as points, pseudo-random, random, non-periodic or irregular point patterns, or the like. Each of the detectors generates an image of a reflection pattern and an image analysis task is to identify corresponding features in the two images. Due to fixed relative position a corresponding feature point selected in one of the two images lies along an epipolar line in the other image. However, solving the so-called correspondence problem may be difficult. In stereo and triangulation systems, the distance of all feature points along the epipolar line have to have reasonable correspondence among each other. A correspondence decision cannot be made one after another. If one correspondence is wrong, this has implications for other feature points, such as invisibility. This usually yields quadratic scaling evaluation algorithms.

For example, US 2008/0240502 A1 and US 2010/0118123 A1 describe an apparatus for mapping an object including an illumination assembly, which includes a single transparency containing a fixed pattern of spots. A light source transilluminates the single transparency with optical radiation so as to project the pattern onto the object. An image capture assembly captures an image of the pattern that is projected onto the object using the single transparency. A processor processes the image captured by the image capture assembly as to reconstruct a three-dimensional map of the object.

US 2008/118143 A1 describes a method and apparatus for obtaining an image to determine a three dimensional shape of a stationary or moving object using a bi dimensional coded light pattern having a plurality of distinct identifiable feature types. The coded light pattern is projected on the object such that each of the identifiable feature types appears at most once on predefined sections of distinguishable epipolar lines. An image of the object is captured and the reflected feature types are extracted along with their location on known epipolar lines in the captured image. Displacements of the reflected feature types along their epipolar lines from reference coordinates thereupon determine corresponding three dimensional coordinates in space and thus a 3D mapping or model of the shape of the object at any point in time.

US 2009/066929 A1 describes establishing of the correspondence of the points for creation of a range image with stereo cameras. The scene is illuminated twice; thereof at least once with a random or pseudo random pattern. For both cameras, an image is taken for each of the illuminations and the quotient of brightnesses is calculated pixelwise. The correspondence is established on the basis of a comparison of the quotient of pixels on epipolar lines of different cameras. The illumination pattern is preferably highly modulated along the epipolar line; transversally or diagonally to it, it is not or only slightly modulated. For illumination, a projection unit is used which in a preferred arrangement comprises two superposed grating patterns which have a distance to each other, with at least one varying in a pseudo random manner, and with two closely neighboring light sources which shine through the gratings and thereby generate different pseudo random patterns, especially moiré patterns.

Despite the advantages implied by the above-mentioned devices and detectors, several technical challenges remain. Thus, the evaluation algorithms used require high computation power which is a severe cost driver. Furthermore, due to energy consumption and heat production of

the required computational resources, the computational demand limits use of such 3D sensor methods in outdoor and mobile applications.

Problem addressed by the invention

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It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

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Summary of the invention

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This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

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As used in the following, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

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Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, notwithstanding the fact that the respective feature or element may be present once or more than once.

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Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way.

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The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without

any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for determining a position of at least one object is disclosed. As used herein, the term "object" refers to a point or region emitting at least one light beam. The light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is adapted to determine at least one reflection image;
- at least one evaluation device, wherein the evaluation device is adapted to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals, wherein the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as

a combination of several devices. As further used herein, the term "matrix" generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

10 The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane.

15 As used herein, an "optical sensor" generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a "light-sensitive area" generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term "the optical sensors each having at least one light sensitive area" refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. Thus, the term "optical sensor" furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one CCD and/or CMOS device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible. Thus, as an example, an optical device comprising two, three, four or more than four light-sensitive areas may be used which is regarded as two, three, four or more than four optical sensors in the context of the present invention. As outlined above, the sensor element comprises a matrix of optical sensors. Thus, as an example, the optical sensors may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

- Specifically, the optical sensors may be or may comprise inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Au photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

- The matrix may be composed of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

- Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term "essentially perpendicular", reference may be made to the definition given above. Thus, as an example, tolerances of less than 20°, specifically less than 10° or even less than 5°, may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Preferably, the sensor element may be oriented essentially perpendicular to an optical axis of the detector. Again, with respect to the term "essentially perpendicular", reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The reflection light beam may propagate from the object towards the detector. As will be outlined in further detail below, the reflection light beam may originate from the object or may originate from an illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. The illumination source may be or may comprise at least one multiple beam light source. For example, the illumination source may comprise at least one laser source and one or more diffractive optical elements (DOEs). Additionally or alternatively, the detector may use radiation already present in the scene such as from at least one ambient light source.

The reflection light beam specifically may fully illuminate the at least one optical sensor from which the center signal is generated, such that the at least one optical sensor from which the center signal arises is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the at least one optical sensor from which the sensor signal arises. Contrarily, preferably, the reflection light beam specifically may create a light spot on the entire matrix which is smaller than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further de-

tail below. As further used herein, a “light spot” generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range

from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may
5 be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a
10 CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as com-
15 mercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The detector may be adapted to determine at least one distance information of the object by using triangulation and/or structured light techniques. In known 3D sensing devices, such as
20 devices using triangulation or structured light techniques, due to correspondence problems regular, constant or periodic pattern are not suitable since each measured point has to be assigned to one reference point of a reference pattern.

As used herein, the term "reflection image" refers to an image determined by the optical sensor comprising at least one reflection feature. As used herein, the term "reflection feature" refers to
25 a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. The reflection image may comprise the at least one reflection pattern comprising the at least one reflection feature. As used herein, the term "illumination feature" refers to at least one arbitrary shaped feature generated by at least one ambient light source or at least one illumination source adapted to illuminate the object. As used herein, the
30 term "determining at least one reflection image" refers to one or more of imaging, recording and generating of the reflection image. As used herein, the term "select at least one reflection feature" refers to one or more of identifying, determining and choosing at least one reflection feature of the reflection image. The detector may be adapted to determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection image from the com-
35 bined signal. Thus, the detector may be adapted to pre-classify the at least one reflection feature of the reflection image. This allows using illumination patterns comprising regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal patterns may be preferred.

40 The evaluation device may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image

processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals. The evaluation device may be adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q, wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \varepsilon$. As used herein, the term "longitudinal region" refers to at least one uncertainty interval defined by the longitudinal coordinate z and the measurement uncertainty $\pm \varepsilon$ of the determination of the longitudinal coordinate from the combined signal Q. Error ε may depend on measurement uncertainty of the optical sensor. The measurement uncertainty of the optical sensors may be pre-determined and/or estimated and/or may be deposited in at least one data storage unit of the evaluation device. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$. As used herein, the term "combined signal Q" refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region. For example, the evaluation device may be configured for deriving the combined

$$\text{signal Q by } Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. As used herein, the term "beam profile" relates to a spatial distribution, in particular in at least one plane per-

pendicular to the propagation of the light beam, of an intensity of the light beam. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_0)$ and beam shape $S(x,y;z_0)$, $E(x,y;z_0)=L \cdot S$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_0 independent from the object size. Thus, the combined signal allows determination of the distance z_0 independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q . The light-sensitive areas may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term "essentially edge information" generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts

of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

- 5 Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area
10 may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam
15 profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 25 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
30 cuts may result in improved distance information.

In one embodiment, the light beam propagating from the object to the detector may illuminate the sensor element with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The
35 central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

For example, the light beam propagating from the object to the detector may illuminate the sensor element with at least one point pattern. The point pattern may be generated by the object, for example in response to an illumination by the at least one light source with an illumination pattern comprising the at least one point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The light beam propagating from the object to the detector may illuminate the sensor element with a reflection pattern comprising both point patterns and line patterns. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

The evaluation device may be configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

For example, the evaluation device may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;

- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the at least one longitudinal coordinate z of the selected feature by evaluating the combined signal.

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As explained, e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot may be characterized by two measurement variables: a measurement signal measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their respective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming the combined signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the quotient signal for deriving the longitudinal coordinate.

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The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus, may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.

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The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, as will be outlined in further detail below, the detector according to the present

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invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

Consequently, in accordance with the present invention, the term "center signal" generally refers to the at least one sensor signal comprising essentially center information of the beam profile. For example, the center signal may be the signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. As used herein, the term "highest sensor signal" refers to one or both of a local maximum or a maximum in a region of interest. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an in-

tegral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

- 5 Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix.
- 10 When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of
- 15 edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.
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Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment

25 will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

- 30 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
- 35 cuts may result in an improved distance information.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which

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contains at least one item of information on a comparison between the center signal and the sum signal.

The evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the object by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

Raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations, however, are feasible.

As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be per-

formed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a

quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the combined signal Q . This determining may be a one-step process, such as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined sig-

nal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

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The evaluation device may be configured for using at least one predetermined relationship between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

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The combined signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider. The divider may fully or partially be integrated into the sensor element answers or may fully or partially be embodied independent from the sensor element.

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The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term "transfer device", also denoted as "transfer system", may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term "focal length" of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a "focus" which may also be denoted as "focal point". Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer

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device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term "optical axis of the transfer device" generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information

of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

As outlined above, the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region. As used herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. As outlined above, the evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. Relative position of the reference image and reflection image may be known. For example, relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

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As used herein, the term "displacement region" refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the refer-

ence feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term "reference feature" refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \varepsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance measurement using the combined signal Q may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device may determine a displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm \varepsilon$ from the combined signal Q to determine the displacement region along the epipolar line corresponding to $z \pm \varepsilon$. The evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region. As used herein, the term "matching" refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be adapted to match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement

region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be configured to perform the following steps:

- 5 - Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
- 10 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

- 15 Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ϵ -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or
- 20 reference feature in shorter distance and/or ϵ -weighted distance to the reference feature and/or reflection feature.

- Preferably, the detector may be adapted to pre-classify the selected reflection feature using the combined signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference image may have a relative distance to each other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line. For example, the illumination pattern may comprise at least one hexagonal pattern. Preferably, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Preferably, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point.
- 30 The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other.

- 40 The evaluation device may be adapted to determine a displacement of the matched reference feature and the selected reflection feature. As used herein, the term "displacement" refers to difference between a position in the reference image to a position in the reflection image. The evaluation device may be adapted to determine a longitudinal information of the matched fea-

ture using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term "longitudinal information" refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a distance value. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship. The evaluation device may be adapted to store parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector. The evaluation device may be adapted to generate the parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector such as by performing a Tsai camera calibration. The evaluation device may be adapted to compute and/or estimate parameters such as the focal length of the transfer device, the radial lens distortion coefficient, the coordinates of the center of radial lens distortion, scale factors to account for any uncertainty due to imperfections in hardware timing for scanning and digitization, rotation angles for the transformation between the world and camera coordinates, translation components for the transformation between the world and camera coordinates, aperture angles, image sensor format, principal point, skew coefficients, camera center, camera heading, baseline, rotation or translation parameters between camera and/or illumination source, apertures, focal distance, or the like.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce the possible number of solutions along the epipolar line significantly. The number of possible solutions may even be reduced to one. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature

may enhance accuracy and/or speed and/or may lower computational demand in comparison to conventional triangulation systems. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be reduced to increase the light intensity in each illumination point such as to compete with ambient light while complying to eye safety regulations. A reduced number of illumination features in a conventional triangulation system might increase the difficulty to match reflection features and reference features. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be increased, such as to increase the resolution of the distance measurement, such as to increase the resolution of the obtained depth map without increasing the processing power of the evaluation device such as in a mobile application.

The detector may further comprise an illumination source. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source may be adapted to generate at least one illumination pattern for illumination of the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "illumination pattern" refers to a pattern which illuminates the object. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern

such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that a hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination device may be attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-

frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

Specifically, the illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.

The illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The sensor element may be adapted to determine at least one reflection pattern. As used herein, the term "reflection pattern" refers to a response pattern generated by reflection or scattering of light at the surface of the object, in particular generated by the object in response to illumination by the illumination pattern. As outlined above, the illumination pattern comprises at least one feature adapted to illuminate the object. As outlined above, the illumination feature may be generated by ambient light or by the at least one illumination source. The reflection pattern may comprise at least one feature corresponding to at least one feature of the illumination pattern.

The reflection pattern may comprise, in comparison to the illumination pattern, at least one distorted pattern, wherein the distortion depends on the distance of the object, such as surface properties of the object. The evaluation device may be adapted to select at least one feature of the reflection pattern and to determine the longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals, as described above.

For example, the reference image may be an image of the illumination pattern at an image plane at a position of the illumination source. The evaluation device may be adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The illumination source and the sensor element may be separated by a fixed distance.

For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to

- determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be adapted to select at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The evaluation device may be adapted to select the at least one reflection feature in the reflection pattern and to determine the longitudinal region of the selected feature by evaluating the combined signal Q from the sensor signals. The evaluation device may be adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region.
- 15 As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

- In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein,
5 the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

- 10 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment
15 function in accordance with the information.

- In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system
20 according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

- 25 In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

- 30 In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further
35 comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further
40 used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different

from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for
5 scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the
10 scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to
15 emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order
20 to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.
25

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic
30 and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other
35 properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector,
40 such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least

one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in

order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as

such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a drill, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a

wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one in-

candescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is pro-

duced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used. The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method comprises the following method steps:

- generating at least two sensor signals in response to an illumination of at least one sensor element having a matrix of optical sensors by at least one reflection light beam propagating from the object to the detector, the optical sensors each having a light-sensitive area;
- determining at least one reflection image by using the sensor element;
- selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal region of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
- determining at least one displacement region in at least one reference image corresponding to the longitudinal region;
- matching the selected reflection feature with at least one reference feature within the displacement region.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

The method may further comprise determining a displacement of the matched reference feature and the selected reflection feature. A longitudinal information of the matched feature may be determined using a predetermined relationship between a longitudinal coordinate and the displacement.

The deriving of the combined signal Q may comprise one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

5 In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

15 The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

25 As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

30 Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for house-

hold applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

5 The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with
10 mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus,
15 specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or
20 other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance,
25 scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head-mounted displays.

35 Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention
40 may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries,

obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in

combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

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Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

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Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems, or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present

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invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.

5 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. 10 Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, 20 such as in the windows like rear window, side window or front window, on the bumpers or on the lights. 25

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions. 30

35 Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used for detecting free parking spaces in parking lots. 40

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endo-

scopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric surgery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment. Further, the devices according to the present invention may be used in wearable robots such as in exoskeletons or prosthesis or the like. Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe. In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

Further, the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

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Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

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Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

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The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

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The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

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Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detec-

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tion etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore, multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection, such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections, inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft when landing or starting. Therein, fixed, known active targets may be used for precise

guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or accelerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racket, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and/or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both

from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the present invention may be used in drones or multicopters to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concertized movements of several drones or recharging or refueling in the air or the like.

Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as dog or cat litter robots, charging robot for electrical vehicles, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, robotic shopping carts, luggage carrying robots, line following robots, laundry robots, ironing robots, window washing robots, toy robots, patient monitoring robots, baby monitoring robots, elderly monitoring robots, children monitoring robots, transport robots, telepresence robots, professional service robots, programmable toy robots, pathfinder robots, social robots providing company to less mobile people, following robots, smart card following robots, psychotherapy robots, or robots translating speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the

present invention may be used in humanoid robots, especially in the context of using humanoid hands to pick up or hold or place objects. Further, the devices according to the present invention may be used in combination with audio interfaces especially in combination with household robots which may serve as a digital assistant with interfaces to online or offline computer applications. Further, the devices according to the present invention may be used in robots that can control switches and buttons in industrial and household purposes. Further, the devices according to the present invention may be used in smart home robots such as Mayfield's Kuri. Further the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate insecure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

Further, devices according to the present invention may be used for the inspection of adhesive beads, sealing beads, or the like, such as to recognize disruptions, slubs, contractions, asymmetries, local defects, or the like. Further, devices according to the present invention may be used to count objects such as dry fruits on a conveyer belt, such as in difficult situations, such as when fruit of similar color and shape may be in direct contact with each other. Further, devices according to the present invention may be used in quality control of die cast or injection molded parts such as to ensure flawless casting or molding, recognize surface damages, worn out toolings or the like. Further, devices according to the present invention may be used for laserscribing such as for quality control and positioning of the laser. Further, devices according to the present invention may be used for sorting systems, such as to detect position, rotation, and shape of an object, compare it to a database of objects, and classify the object. Further, devices according to the present invention may be used for stamping part inspection, packaging inspection, such as food and pharma packaging inspection, filament inspection, or the like.

Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that are can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.

Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition devices, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the

presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.

- 5 Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.
- 10 The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may
- 15 be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or greenhouse or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.
- 20 Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific
- 25 groups of animals such as bees.

Further, devices according to the present invention may be used to guide users during a shaving, hair cutting, or cosmetics procedure, or the like. Further, devices according to the present invention may be used to record or monitor what is played on an instrument, such as a violin.

- 30 Further, devices according to the present invention may be used in smart household appliances such as a smart refrigerator, such as to monitor the contents of the refrigerator and transmit notifications depending on the contents. Further, devices according to the present invention may be used for monitoring or tracking populations of humans, animals, or plants, such as deer or tree populations in forests. Further, devices according to the present invention may be used
- 35 in harvesting machines, such as for harvesting crops, flowers or fruits, such as grapes, corn, hops, apples, grains, rice, strawberries, asparagus, tulips, roses, soy beans, or the like. Further, devices according to the present invention may be used to monitor the growth of plants, animals, algae, fish, or the like, such as in breeding, food production, agriculture or research applications, to control irrigation, fertilization, humidity, temperature, use of herbicides, insecticides,
- 40 fungicides, rodenticides, or the like. Further, devices according to the present invention may be used in feeding machines for animals or pets, such as for cows, pigs, cats, dogs, birds, fish, or the like. Further, devices according to the present invention may be used in animal product production processes, such as for collecting milk, eggs, fur, meat, or the like, such as in automated

milking or butchering processes. Further, devices according to the present invention may be used for automated seeding machines, or sowing machines, or planting machines such as for planting corn, garlic, trees, salad or the like. Further, devices according to the present invention may be used to assess or monitor weather phenomena, such as clouds, fog, or the like, or to
5 warn from danger of avalanches, tsunamis, gales, earthquakes, thunder storms, or the like. Further, devices according to the present invention may be used to measure motions, shocks, concussions, or the like such as to monitor earthquake risk. Further, devices according to the present invention may be used in traffic technology to monitor dangerous crossings, to control traffic lights depending on traffic, to monitor public spaces, to monitor roads, gyms, stadiums, ski
10 resorts, public events, or the like. Further, devices according to the present invention may be used in medical applications such as to monitor or analyze tissues, medical or biological assays, changes in tissues such as in moles or melanoma or the like, to count bacteria, blood cells, cells, algae, or the like, for retina scans, breath or pulse measurements, gastroscopy, patient surveillance, or the like. Further, devices according to the present invention may be used to
15 monitor the shape, size, or circumference of drops, streams, jets, or the like or to analyze, assess, or monitor profiles or gas or liquid currents such as in a wind channel, or the like. Further, devices according to the present invention may be used to warn drivers such as car or train drivers when they are getting sick or tired or the like. Further, devices according to the present invention may be used in material testing to recognize strains or tensions or fissures, or the like.
20 Further, devices according to the present invention may be used in sailing to monitor and optimize sail positions such as automatically. Further, devices according to the present invention may be used for fuel level gauges.

Further, the devices according to the present invention may be combined with sensors to detect
25 chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or viruses or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or
30 for pest control in agriculture.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices
35 according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting
40 vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further, the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location.

- 5 Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location specific information, such as the current position on a map, or information on goods sold, or the like.
- 10 Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.
- 15 Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices according to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.
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- As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further,
- 25 the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, devices according to the present invention may be recalibrated using bar codes, QR-codes or prerecorded symbols of known size by using the at least one image matrix and comparing the prerecorded size with a measured property of the recorded image of
- 30 the bar code, QR code or prerecorded symbol, such as by comparing the width or height of the symbol with a prerecorded value. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention
- 35 may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to
- 40 the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the

like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

5 Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least
10 one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally
15 refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital
20 video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at
25 least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

30 As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively,
35 however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field
40 of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device adapted for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based algorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared. Further, the devices according to the present invention may be used in the context of gesture recognition. In this context, gesture recognition in combination with devices according to the present invention may, in particular, be used as a human-machine interface for transmitting information via motion of a body, of body parts or of objects to a machine. Herein, the information may, preferably, be transmitted via a motion of hands or hand parts, such as fingers, in particular, by pointing at objects, applying sign language, such as for deaf people, making signs for numbers, approval, disapproval, or the like, by waving the hand, such as when asking someone to approach, to leave, or to greet a person, to press an object, to take an object, or, in the field of sports or music, in a hand or finger exercise, such as a warm-up exercise. Further, the information may be transmitted by motion of arms or legs, such as rotating, kicking, grabbing, twisting, rotating, scrolling, browsing, pushing, bending, punching, shaking, arms, legs, both arms, or both legs, or a combination of arms and legs, such as for a purpose of sports or music, such as for entertainment, exercise, or training function of a machine. Further, the information may be transmitted by motion of the

whole body or major parts thereof, such as jumping, rotating, or making complex signs, such as sign language used at airports or by traffic police in order to transmit information, such as “turn right”, “turn left”, “proceed”, “slow down”, “stop”, or “stop engines”, or by pretending to swim, to dive, to run, to shoot, or the like, or by making complex motions or body positions such as in
5 yoga, pilates, judo, karate, dancing, or ballet. Further, the information may be transmitted by using a real or mock-up device for controlling a virtual device corresponding to the mock-up device, such as using a mock-up guitar for controlling a virtual guitar function in a computer program, using a real guitar for controlling a virtual guitar function in a computer program, using
10 a real or a mock-up book for reading an e-book or moving pages or browsing through in a virtual document, using a real or mock-up pen for drawing in a computer program, or the like. Further, the transmission of the information may be coupled to a feedback to the user, such as a sound, a vibration, or a motion.

15 In the context of music and/or instruments, devices according to the present invention in combination with gesture recognition may be used for exercising purposes, control of instruments, recording of instruments, playing or recording of music via use of a mock-up instrument or by only pretending to have a instrument present such as playing air guitar, such as to avoid noise or make recordings, or, for conducting of a virtual orchestra, ensemble, band, big band, choir, or the like, for practicing, exercising, recording or entertainment purposes or the like.

20 Further, in the context of safety and surveillance, devices according to the present invention in combination with gesture recognition may be used to recognize motion profiles of persons, such as recognizing a person by the way of walking or moving the body, or to use hand signs or movements or signs or movements of body parts or the whole body as access or identification
25 control such as a personal identification sign or a personal identification movement.

Further, in the context of smart home applications or internet of things, devices according to the present invention in combination with gesture recognition may be used for central or non-central
30 control of household devices which may be part of an interconnecting network of home appliances and/or household devices, such as refrigerators, central heating, air condition, microwave ovens, ice cube makers, or water boilers, or entertainment devices, such as television sets, smart phones, game consoles, video recorders, DVD players, personal computers, laptops, tablets, or combinations thereof, or a combination of household devices and entertainment devices.

35 Further, in the context of virtual reality or of augmented reality, devices according to the present invention in combination with gesture recognition may be used to control movements or function of the virtual reality application or of the augmented reality application, such as playing or controlling a game using signs, gestures, body movements or body part movements or the like,
40 moving through a virtual world, manipulating virtual objects, practicing, exercising or playing sports, arts, crafts, music or games using virtual objects such as a ball, chess figures, go stones, instruments, tools, brushes.

Further, in the context of medicine, devices according to the present invention in combination with gesture recognition may be used to support rehabilitation training, remote diagnostics, or to monitor or survey surgery or treatment, to overlay and display medical images with positions of medical devices, or to overlay display prerecorded medical images such as from magnetic resonance tomography or x-ray or the like with images from endoscopes or ultra sound or the like that are recorded during a surgery or treatment.

Further, in the context of manufacturing and process automation, devices according to the present invention in combination with gesture recognition may be used to control, teach, or program robots, drones, unmanned autonomous vehicles, service robots, movable objects, or the like, such as for programming, controlling, manufacturing, manipulating, repairing, or teaching purposes, or for remote manipulating of objects or areas, such as for safety reasons, or for maintenance purposes.

Further, in the context of business intelligence metrics, devices according to the present invention in combination with gesture recognition may be used for people counting, surveying customer movements, areas where customers spend time, objects, customers test, take, probe, or the like.

Further, devices according to the present invention may be used in the context of do-it-yourself or professional tools, especially electric or motor driven tools or power tools, such as drilling machines, saws, chisels, hammers, wrenches, staple guns, disc cutters, metals shears and nibblers, angle grinders, die grinders, drills, hammer drills, heat guns, wrenches, sanders, engravers, nailers, jig saws, biscuit joiners, wood routers, planers, polishers, tile cutters, washers, rollers, wall chasers, lathes, impact drivers, jointers, paint rollers, spray guns, morticers, or welders, in particular, to support precision in manufacturing, keeping a minimum or maximum distance, or for safety measures.

Further, the devices according to the present invention may be used to aid visually impaired persons. Further, devices according to the present invention may be used in touch screen such as to avoid direct contact such as for hygienic reasons, which may be used in retail environments, in medical applications, in production environments, or the like. Further, devices according to the present invention may be used in agricultural production environments such as in stable cleaning robots, egg collecting machines, milking machines, harvesting machines, farm machinery, harvesters, forwarders, combine harvesters, tractors, cultivators, ploughs, destoners, harrows, strip tills, broadcast seeders, planters such as potato planters, manure spreaders, sprayers, sprinkler systems, swathers, balers, loaders, forklifts, mowers, or the like.

Further, devices according to the present invention may be used for selection and/or adaption of clothing, shoes, glasses, hats, prosthesis, dental braces, for persons or animals with limited communication skills or possibilities, such as children or impaired persons, or the like. Further, devices according to the present invention may be used in the context of warehouses, logistics, distribution, shipping, loading, unloading, smart manufacturing, industry 4.0, or the like. Further,

in a manufacturing context, devices according to the present invention may be used in the context of processing, dispensing, bending, material handling, or the like.

5 The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as
10 one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

15 The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.
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The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the
25 optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted
30 information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks
35 such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

40 The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue

interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces
5 ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or
10 more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication
15 device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining
20 and or processing 3D information. Further, devices according to the present invention may be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display
30 device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design
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may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

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As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

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The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

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The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

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The human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

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The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance

with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller
5 might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally,
10 the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may
15 fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as
20 at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time infor-
25 mation.

The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the ob-
30 ject itself or a part of the object, such as at least one control element comprising the beacon devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of:
35 a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
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As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The proposed devices and methods provide a large number of advantages over known detectors of this kind. Thus, the detector generally may avoid the shortcomings of the known prior art systems disclosed above. Specifically, the detector may avoid the use of FiP sensors, thereby allowing for e.g. using simple and cheap and commercially available semiconductor sensors such as silicon photodiodes. These photodiodes generally do not show a luminance dependency, and the method disclosed above is generally independent from the brightness of the scenery and/or the brightness of the light spot on the light beam. Consequently, a range of measurement in terms of luminance or total power of the light beam entering the detector is generally larger in the present invention as compared to many of the devices disclosed above. Further, the measurement by using the detector according to the present invention is generally independent from the target spot size, i.e. from either the size of the object, the size of a light spot projected onto the object or the size of a beacon device being one or more of attached to the object, integrated into the object or held by the object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional option of measuring one or more transversal coordinates, thereby integrating the functionality of a PSD.

The result of the measurement, i.e. the position determined by the detector such as the at least one longitudinal coordinate, may be rendered widely independent from the brightness of the scenery and/or the brightness of the object, the brightness of the at least one beacon device, the brightness of the at least one illumination source or the total power of the light beam propagating from the object to the detector. Further, due to this independence and wide range of measurement in terms of brightness, reflective objects or non-reflective objects may be used.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention, such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the quotient signal, as mentioned above, contrarily, provides a wide range of

continuously and monotonously decreasing or increasing functions which may be used to determine the longitudinal coordinate from the quotient signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the quotient signal from the total power of the light beam, at least as long as no saturation of one or both of the optical sensors is reached, this also provides a very wide range of measurement in terms of brightness, i.e. in terms of total power of the light beam.

The light beam, within the detector, generally may propagate along an optical axis of the detector. The light beam, however, may also propagate in other ways than along the optical axis. As an example, an illumination light beam may be generated which propagates along the optical axis or which propagates parallel to the optical axis or at an angle to the optical axis which is different from 0° , such as an angle of 1° to 20° . Other embodiments are feasible.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector comprising

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is adapted to determine at least one reflection image;
- at least one evaluation device, wherein the evaluation device is adapted to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals, wherein the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region.

Embodiment 2: The detector according to the preceding embodiment, wherein the evaluation device is adapted to determine a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region.

Embodiment 5: The detector according to any one of the preceding embodiments, wherein the evaluation device is adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q , wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm\epsilon$.

Embodiment 6: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image.

Embodiment 7: The detector according to the preceding embodiment, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 8: The detector according to the preceding embodiment, wherein the evaluation device is configured to perform the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is adapted to match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z .

Embodiment 10: The detector according to the preceding embodiments, wherein the evaluation algorithm is a linear scaling algorithm.

Embodiment 11: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o.

Embodiment 12: The detector according to the preceding embodiment, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the reflection light beam.

Embodiment 13: The detector according to any of the two preceding embodiments, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

Embodiment 14: The detector according to any one of the three preceding embodiments, wherein the light sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile.

Embodiment 16: The detector according to any one of the two preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.

Embodiment 17: The detector according to any one of the two preceding embodiments, wherein the beam profile is segmented by a secant or chord passing the center of the beam profile by a given distance, whereas the first segment comprises the center information, whereas the second segment comprises essentially edge information.

Embodiment 18: The detector according to the preceding embodiment, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

Embodiment 19: The detector according to any one of the two preceding embodiments, wherein the evaluation device is configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the at least one evaluation device is configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the longitudinal region by evaluating the combined signal.

Embodiment 21: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor

signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.

5 Embodiment 23: The detector according to any one of the three preceding embodiments, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.

10 Embodiment 24: The detector according to any one of the preceding embodiment, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

15 Embodiment 25: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one optical element, wherein the optical element is at least one element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element.

20 Embodiment 26: The detector according to any one of the six preceding embodiments, wherein the evaluation device is further configured for determining at least one transversal coordinate of the object by evaluating a transversal position of the at least one optical sensor having the highest sensor signal.

25 Embodiment 27: The detector according to any one of the preceding embodiments, wherein the sensor element comprises at least one CMOS sensor.

Embodiment 28: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object.

30 Embodiment 29: The detector according to preceding embodiment, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

40 Embodiment 30: The detector according to any one of the two preceding embodiments, wherein the sensor element is adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the

longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals.

5 Embodiment 31: The detector according to the preceding embodiment, wherein the reference image is an image of the illumination pattern at an image plane at a position of the illumination source, wherein the evaluation device is adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern, wherein the evaluation device is adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region.

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Embodiment 32: The detector according to any one of the four preceding embodiments, wherein the illumination source and the sensor element are separated by a fixed distance.

15 Embodiment 33: The detector according to any one of the five preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors.

20 Embodiment 34: The detector according to the preceding embodiment, where at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein a relative distance between the first sensor element and the second element is fixed.

25 Embodiment 35: The detector according to any one of the two preceding embodiments, wherein the at least one first sensor element is adapted to determine at least one first reflection pattern and the at least one second sensor element is adapted to determine at least one second reflection pattern, wherein the evaluation device is adapted to select at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

30 Embodiment 36: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

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40 Embodiment 37: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 38: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein
5 the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 39: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding
10 embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 40: A scanning system for determining a depth profile of a scenery, the scanning
15 system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 41: A camera for imaging at least one object, the camera comprising at least one
20 detector according to any one of the preceding embodiments referring to a detector.

Embodiment 42: A method for determining a position of at least one object by using at least one detector according to any one of the preceding embodiments referring to a detector, the method comprising the following steps:

- 25 - generating at least two sensor signals in response to an illumination of at least one sensor element having a matrix of optical sensors by at least one reflection light beam propagating from the object to the detector, the optical sensors each having a light-sensitive area;
- determining at least one reflection image by using the sensor element;
- 30 - selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal region of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
- determining at least one displacement region in at least one reference image corresponding to the longitudinal region;
- 35 - matching the selected reflection feature with at least one reference feature within the displacement region.

Embodiment 43: The method according to the preceding embodiment, wherein a displacement
40 of the matched reference feature and the selected reflection feature is determined, wherein a longitudinal information of the matched feature is determined using a predetermined relationship between a longitudinal coordinate and the displacement.

Embodiment 44: The method according to any one of the preceding embodiments referring to a method, wherein the deriving of the combined signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

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Embodiment 45: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; an outdoor application; a mobile application; a communication application; a tracking application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

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Brief description of the figures

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Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary

embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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Figure 1 shows a first embodiment of a detector according to the present invention;

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Figure 2 shows a further embodiment of the detector according to the present invention;

Figure 3 shows an embodiment of a detector system, camera, entertainment device, tracking system;

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Figure 4 shows an embodiment of determining a longitudinal coordinate z using a line pattern;

Figures 5A to C show three embodiments of a hexagonal illumination pattern; and

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Figure 6 shows an embodiment of a scanning system.

Detailed description of the embodiments:

Fig. 1 shows in a highly schematic fashion a first embodiment of a detector 110 for determining a position of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.

The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is

given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

In the embodiment shown in Figure 1, the detector 110 further comprises an illumination source 124. As an example, the illumination source 124 may be configured for generating an illuminating light beam for illuminating the object 112. The detector 110 may be configured such that the illuminating light beam propagates from the detector 110 towards the object 112 along an optical axis of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source 124 may be adapted to generate at least one illumination pattern for illumination of the object 112. Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; at least one pattern comprising at least one pre-known feature. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points. The illumination source 124 may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source 124 may comprise at least one light source adapted to generate the illumination pattern directly. The illumination source 124 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source 124 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source 124.

In the embodiment shown in Figure 1, the illumination source 124 may illuminate the at least one object 112 with the illumination pattern. The illumination pattern may comprise a plurality of points as image features. These points are illustrated as light beams 115 emerging from the illumination source 124.

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Each optical sensor 120 is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam propagating from the object 112 to the detector 110.

10 Furthermore, the sensor element 116 is adapted to determine at least one reflection image 126. The matrix 118 may comprise the reflection image 126. The reflection image 126 may comprise points as reflection features. These points result from reflection light beams 125 originating from the at least one object 112. Crossings 131 of the illumination light beams and the reflection light beams 125 correspond to possible positions 114 of the at least one object 112.

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The detector 127 may comprise at least one transfer device 127 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam de-
20 flection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 127 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

25 The detector 110 comprises at least one evaluation device 128. The evaluation device 126 is adapted to select at least one reflection feature of the reflection image 126, for example selected reflection feature 129. The evaluation device 128 may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image
30 analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency de-
35 composition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a blob analysis; applying an edge filter; applying a low-pass filter; applying a Fourier transformation; apply-
40 ing a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be deter-

mined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

If the selected reflection feature 129 is matched to a corresponding reference feature, the corresponding light beams are matched and the distance of the object 112 can be calculated. For a selected reflection feature 129 and a corresponding reflection light beam, the number of corresponding illumination light beams, and thus the number of corresponding reference features 112, is reduced after determining a combined signal Q.

- 5 The evaluation device 128 is configured for determining at least one longitudinal region 130 of the selected reflection feature 129 of the reflection image by evaluating the combined signal Q from the sensor signals. The evaluation device 128 may be adapted to determine at least one longitudinal coordinate z of the selected reflection feature 129 by evaluating the combined signal Q, wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \varepsilon$. Error ε may depend on measurement uncertainty of the optical sensor. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$. The evaluation device 128 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device 128 may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region. For example, the evaluation device 128 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance L(z_o) and beam shape S(x,y;z_o), E(x,y;z_o)=L•S. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. The light-sensitive areas 122 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device 128 may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably, the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1. Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 128 may be adapted for determining an area integral of the beam profile. The evaluation device 128 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 128 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be

replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 5 Additionally or alternatively, the evaluation device 128 may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
10 cuts may result in improved distance information.

For example, the evaluation device 128 may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor 120 having the highest sensor signal and forming at least one center signal;
- 15 b) evaluating the sensor signals of the optical sensors 120 of the matrix 118 and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the at least one longitudinal coordinate z of the selected feature by evaluating
20 the combined signal.

- For example, the center signal may be the signal of the at least one optical sensor 118 having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors 120 of the entire matrix 118 or of a region of interest within the matrix 118, wherein the region of
25 interest may be predetermined or determinable within an image generated by the optical sensors of the matrix 118. The center signal may arise from a single optical sensor 120 or from a group of optical sensors 120, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors 120 may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors 120 from which the center signal arises
30 may be a group of neighboring optical sensors, such as optical sensors 120 having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors 120 from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The
35 evaluation device 128 may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal.

- The determination of the center signal may be performed electronically, since a comparison of
40 sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. The detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be performed fully or partially electronically or fully or partially by using one or more software

algorithms. Specifically, the evaluation device 128 may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at
5 least one sensor element and/or may fully or partially be embodied independently from the sensor element.

Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance
10 from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors 120; a sum of sensor signals from a group of optical sensors 120 containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors 120; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors 120 containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.
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For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix 118 or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable
25 within an image generated by the optical sensors 120 of the matrix 118. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors 120 from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device 128 may be adapted to determine the sum signal by integrating signals of the entire matrix 118, or of the region of interest within the matrix 118. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by
30 geometric considerations.
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The sum signal may be derived from all sensor signals of the matrix 118, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors 120 contributing to the center signal excluded. In every case, a reliable
40 sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix 118; a sum of all sensor signals of the matrix 118; an integral of all sensor signals of the matrix 118; an average over all

sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; a sum of all sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; an integral of all sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; a sum of sensor signals of optical sensors 120 within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor 120 having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor 120 having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors 120 being located within a predetermined range from the optical sensor 120 having the highest sensor signal. Other options, however, exist. The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device 128 may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal. The evaluation device 128 may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device 128 may be configured to determine the at least one coordinate z of the by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

The comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a

quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device 128 may be configured for forming the one or more quotient signals. The evaluation device 128 may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient
5 signal.

The evaluation device 128 specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and
10 due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is
15 due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The
20 quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum
25 signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by
30 analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

The evaluation device 128 may be configured for using at least one predetermined relationship
40 between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 128 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The sensor element 116 may be adapted to determine at least one reflection pattern. The reflection pattern may comprise at least one feature corresponding to at least one feature of the illumination pattern. The reflection pattern may comprise, in comparison to the illumination pattern, at least one distorted pattern, wherein the distortion depends on the distance of the object, such as surface properties of the object. The evaluation device 128 may be adapted to select at least one feature of the reflection pattern and to determine the longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals, as described above.

The evaluation device 128 is adapted to determine at least one displacement region 132 in at least one reference image 134 corresponding to the longitudinal region 130. The longitudinal region corresponds to the displacement region 132. The reference image 134 and the reflection image 126 may be images of the object 112 determined at different spatial positions having a fixed distance b, also called baseline. The evaluation device 128 may be adapted to determine an epipolar line 136 in the reference image 134. Relative position of the reference image 134 and reflection image 126 may be known. For example, relative position of the reference image 134 and reflection image 126 may be stored within at least one storage unit of the evaluation device 128. The evaluation device 128 may be adapted to determine a straight line 138 extending from the selected reflection feature 129 of the reflection image 126. The straight line 138 may comprise possible object positions 114. The straight line 138 and the baseline b span an epipolar plane 140. As the reference image 134 is determined at a different relative position from the reflection image, the corresponding reference features may be imaged on the epipolar line 136 in the reference image 134.

For example, in the embodiment shown in Figure 1, the reference image 134 may be an image of the illumination pattern at an image plane at a position of the illumination source 124. The evaluation device 128 may be adapted to determine the displacement region 132 in the reference image 134 corresponding to the longitudinal region 130 of the selected feature 129 of the reflection pattern. The evaluation device 128 may be adapted to match the selected feature 129 of the reflection pattern with at least one feature of the reference pattern within the displacement region 132. The illumination source 124 and the sensor element 116 may be separated by a fixed distance.

The displacement region 132 may extend along the epipolar line 136. The evaluation device 128 may be adapted to determine the reference feature along the epipolar line 136 corresponding to the longitudinal coordinate z and to determine an extent of the displacement region 132 along the epipolar line 136 corresponding to the error interval $\pm\epsilon$. The evaluation device 128 is adapted to match the selected reflection feature 129 with at least one reference feature within the displacement region 132. The evaluation device 128 may be adapted to match the selected feature of the reflection image with the reference feature within the displacement region 132 by using at least one evaluation algorithm considering the determined longitudinal coordinate z. The evaluation algorithm may be a linear scaling algorithm.

The evaluation device 128 may be adapted to determine a displacement of the matched reference feature and the selected reflection feature 129. The evaluation device 128 may be adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The evaluation device 128 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature 129 in the reflection image 126 and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, a longitudinal coordinate of the corresponding object feature 114 may be determined by triangulation. Thus, the evaluation device 128 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. For example, Figure 1 shows possible projection lines resulting in reflection features in the reflection image 126 and corresponding to reference features in the reference image 134. In addition, intersection points of these lines are shown corresponding to object positions. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device 128 may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

In Figure 2, an embodiment of the detector 110 is shown, wherein the detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element 142 and at least one second sensor element 144 may be positioned at different spatial positions. A relative distance b between the first sensor element 142 and the second element 144 may be fixed. The at least one first sensor element 142 may be adapted to determine at least one first reflection pattern and the at least one second sensor element 144 may be adapted to determine at least one second reflection pattern. The evaluation device 128 may be adapted to select at least one image determined by the first sensor element 142 or the second sensor element 144 as reflection image and to select at least one image determined by the other one of the first sensor element 142 or the second sensor element 144 as reference image. In Figure 2, the image of the first sensor element 142 is selected as reflection image 126, whereas the image of the second sensor element 144 is selected as reference image 134. The evaluation device 128 may be adapted to select the at least one reflection feature in the reflection pattern and to determine the longitudinal region 130 of the selected feature by evaluating the combined signal Q from the sensor signals, as outlined with respect to Figure 1. The evaluation device 128 may be adapted to determine the displacement region 132 in the reference image 134 corresponding to the longitudinal region 120 of the selected feature 129 of the reflection pattern. The evaluation device 128 may be adapted to match the selected feature 129

of the reflection pattern with at least one feature of the reference pattern within the displacement region 132, as outlined with respect to Figure 1.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region 132 corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce possible number of solutions along the epipolar line 136 significantly. The number of solutions may even be reduced to one. Subsequent matching of the selected reflection feature and reference feature may enhance accuracy. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature 129 and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices, where battery life or ventilation or heat management is an issue. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature 129 and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may enhance accuracy in comparison the triangulation systems. Furthermore, it may be possible to use denser pattern in order to enhance further resolution or to use a less dense pattern in order to increase power per point.

Figure 3 shows, in a highly schematic illustration, an exemplary embodiment of the detector 110, for example according to the embodiment shown in Figure 2. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 3 further shows an embodiment of a detector system 148, which, besides the at least one detector 110, comprises one or more beacon devices 150, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 3 further shows an exemplary embodiment of a human-machine interface 152, which comprises the at least one detector system 148 and, further, an entertainment device 154, which comprises the human-machine interface 152. The figure further shows an embodiment of a tracking system 156 for tracking a position of the object 112, which comprises the detector system 148. The components of the devices and systems shall be explained in further detail below.

Fig. 3 further shows an exemplary embodiment of a scanning system 158 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 158 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well as, optionally, at least one further illumination source, not depicted here. The illumination source

124, generally, is configured to emit at least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 150 and/or on a surface of the object 112. The scanning system 158 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 158, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 3 is shown in Fig. 2. Thus, the detector 110, besides the optical sensors 120 comprises at least one evaluation device 128, having e.g. at least one divider 160 and/or the at least one position evaluation device 162, as symbolically depicted in Fig. 3. The components of the evaluation device 128 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 128 may be interconnected by one or more connectors 164 and/or by one or more interfaces, as symbolically depicted in Fig. 3. Further, the one or more connectors 164 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 164, the evaluation device 128 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 166 of the detector 110. Additionally or alternatively, the evaluation device 128 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 168, the position of which may be manipulated by a user 170. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 170 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 166. Further, the detector 110 may comprise the at least one transfer device 127 such as one or more optical systems, preferably comprising one or more lenses. An opening 172 inside the housing 166, which, preferably, is located concentrically with regard to an optical axis 174 of the detector 110, preferably defines a direction of view 176 of the detector 110. A coordinate system 178 may be defined, in which a direction parallel or anti-parallel to the optical axis 174 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 174 may be defined as transversal directions. In the coordinate system 178, symbolically depicted in Fig. 3, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

One or more light beams 180, such as light beams of the illumination pattern, are propagating from the object 112 and/or from one or more of the beacon devices 150, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. In case no illumination source 124 is used, the beacon devices 150 and/or at least one of these
5 beacon devices 150 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. Alternatively, ambient light sources may be used.

The light beam 180, for example generated by the illumination source 124 such as the at least one illumination pattern, may be directed towards the object by using one or more optical elements. The object 112 may in response to the illumination generate the at least one reflection
10 light beam 182. For details of the evaluation, reference may be made to Fig. 1 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 152, in order to
15 provide at least one item of information to a machine 184. In the embodiments schematically depicted in Fig. 3, the machine 184 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 128 may even be fully or partially integrated into the machine 184, such as into the computer.

As outlined above, Fig. 3 also depicts an example of a tracking system 156, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 156 comprises the detector 110 and at least one track controller 186. The track controller 186 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 186 may be an independent device and/or may be fully or partially integrated
20 into the machine 184, specifically the computer, as indicated in Fig. 3 and/or into the evaluation device 128.

Similarly, as outlined above, the human-machine interface 152 may form part of an entertainment device 154. The machine 184, specifically the computer, may also form part of the entertainment device 154. Thus, by means of the user 170 functioning as the object 112 and/or by
30 means of the user 170 handling a control device functioning as the object 112, the user 170 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Referring to Figure 4, the detector 110 such as the detector as described with respect to Figures 1 to 3 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 120. For example, the detector 110 may comprise a plurality of optical sensors arranged in the matrix 118. The detector 110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector 110 may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. As outlined above, this principle is
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called Distance by Photon Ratio (DPR). In one embodiment, the light beam 180 may illuminate the sensor element 116 with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 128 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 4, the reflection light beam 182 propagating from the object 112 to the detector 110 may illuminate the matrix 118 with at least one line pattern 188. The line pattern 188 may be generated by the object 112 for example in response to an illumination by the at least one illumination source 124 with an illumination pattern comprising at least one illumination line pattern. A1 may correspond to an area with a full line width of the line pattern 188 in the matrix 118. The line pattern 188 in the matrix 118 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a line width in the matrix 118 is increased. In particular, the line width of the line pattern 188 in the matrix 118 may change from one column to another column. A2 may be a central area of the line pattern 188 in the matrix 118. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern 188 may be segmented in the matrix 118. Each of the columns may comprise center information of intensity in the central area of the line pattern 188 and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern 188.

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Figures 5A to C show three embodiments of a hexagonal illumination pattern. The illumination source 124 may be adapted to generate at least one illumination pattern for illuminating the object 112. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. Specifically, the illumination source 124 may comprise at least one laser

and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination pattern may comprise at least one feature such as a point or symbol. The illumination pattern may comprise a plurality of features. The illumination pattern may com-
5 prise an arrangement of periodic or non-periodic features. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one qua-
10 siperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to
15 project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature
20 may depend on the circle of confusion in the image.

The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on an epipolar line. As shown in Figure 5A, the illumination pattern may comprise at least one hexagonal pattern, wherein the individual points are positioned on
25 epipolar lines 190. As shown in Figure 5B, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Such a positioning of the illumination features allows enhancing distance between the individual points on each epipolar line. For example as shown in Figure 5C, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are dis-
30 placed by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other

35
Figure 6 shows an embodiment of a scanning system 158. The scanning system 158 may be adapted as a line scanning device. In particular, the scanning system 158 may comprise at least one sensor line or row of optical sensors 120. Furthermore, the scanning system 158 may comprise the at least one transfer device 127 and the at least one illumination source 124. Triangu-
40 lation systems require a sufficient baseline, however due to the baseline in the near field no detection may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-

zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. Further, these near field and far field problems can be overcome by using the scanning system 158 of Figure 6. The scanning system 158 may be adapted to detect a plurality of light beams 182 propagating from the object 112 to the scanning system 158 on the CMOS line. The light beams 182 may be generated at different position on the object 112 or by movement of the object 112. The scanning system 158 may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the combined signal Q as described above.

List of reference numbers

110	Detector
112	Object
114	object positions
116	sensor element
118	Matrix
120	optical sensor
122	light sensitive area
124	illumination source
126	reflection image
127	transfer device
128	evaluation device
129	selected reflection feature
130	longitudinal region
131	Crossing
132	displacement region
134	reference image
136	epipolar line
138	straight line
140	epipolar plane
142	first sensor element
144	second sensor element
146	Camera
148	detector system
150	beacon devices
152	human-machine-interface
154	entertainment device
156	tracking system
158	scanning system
160	Divider
162	position evaluation device
164	Connector
166	Housing
168	control device
170	User
172	Opening
174	optical axis
176	direction of view
178	coordinate system
182	reflection light beam
184	Machine
186	track controller

188 Line pattern
190 Epipolar line

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WO 2014/198629 A1
R.A. Street (Ed.): Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349 X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer,
10 Berlin Heidelberg, 1997, chapter 2

Patent claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) comprising
 - 5 - at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein each optical sensor (120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (122) by a reflection light beam (182) propagating from the object (112) to the detector (110), wherein the sensor element (116) is adapted to determine at least one reflection image (126);
 - 10 - at least one evaluation device (128), wherein the evaluation device (128) is adapted to select at least one reflection feature of the reflection image (126), wherein the evaluation device (128) is configured for determining at least one longitudinal region (130) of the selected reflection feature of the reflection image (134) by evaluating a combined signal Q from the sensor signals, wherein the evaluation device (128) is adapted to determine at least one displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130), wherein the evaluation device (128) is adapted to match the selected reflection feature with at least one reference feature within the displacement region (132).
 - 15
 - 20
2. The detector (110) according to the preceding claim, wherein the evaluation device (128) is adapted to determine a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device (128) is adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement.
- 25
3. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.
- 30
4. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region.
- 35
5. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q, wherein the longitudinal region (130) is given by the longitudinal coordinate z and an error interval $\pm\epsilon$.
- 40
6. The detector (110) according any one of the preceding claims, wherein the reference image (134) and the reflection image (126) are images of the object determined at different spatial

positions having a fixed distance, wherein the evaluation device (128) is adapted to determine an epipolar line (136) in the reference image (134).

7. The detector (110) according to the two preceding claims, wherein the displacement region (132) extends along the epipolar line (136), wherein the evaluation device (128) is adapted to determine the reference feature along the epipolar line (136) corresponding to the longitudinal coordinate z and to determine an extent of the displacement region (132) along the epipolar line (136) corresponding to the error interval $\pm \epsilon$.
8. The detector (110) according to the preceding claim, wherein the evaluation device (128) is configured to perform the following steps:
 - Determining the displacement region for the image position of each reflection feature;
 - Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.
9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is adapted to match the selected feature of the reflection image (126) with the reference feature within the displacement region (132) by using at least one evaluation algorithm considering the determined longitudinal coordinate z , wherein the evaluation algorithm is a linear scaling algorithm.
10. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are different areas of at least one beam profile of the reflection light beam (182) at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the reflection light beam.

11. The detector (110) according to the preceding claim, wherein the light-sensitive areas (122) are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.
12. The detector (110) according to the preceding claim, wherein the evaluation device (128) is configured to determine the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
13. The detector (110) according to any one of the preceding claims, wherein the at least one evaluation device (128) is configured for evaluating the sensor signals, by
- a) determining at least one optical sensor (120) having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors (120) of the matrix (118) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining the longitudinal region by evaluating the combined signal.
14. The detector (110) according to the preceding claim, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.
15. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is adapted to generate at least one illumination pattern for illumination of the object (112), wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; at least one pattern comprising at least one pre-known feature.
16. The detector (110) according to the preceding claim, wherein the sensor element (116) is adapted to determine at least one reflection pattern, wherein the evaluation device (128) is adapted to select at least one feature of the reflection pattern and to determine the longitudinal region (130) of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals.

17. The detector (110) according to the preceding claim, wherein the reference image (134) is an image of the illumination pattern at an image plane at a position of the illumination source (124), wherein the evaluation device (128) is adapted to determine the displacement region (132) in the reference image (134) corresponding to the longitudinal region (130) of the selected feature of the reflection pattern, wherein the evaluation device (128) is adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region (132).
18. The detector (110) according to any one of the three preceding claims, wherein the detector (110) comprises at least two sensor elements (116) each having a matrix (118) of optical sensors (120), where at least one first sensor element (142) and at least one second sensor element (144) are positioned at different spatial positions, wherein a relative distance between the first sensor element (142) and the second element (144) is fixed, wherein the at least one first sensor element (142) is adapted to determine at least one first reflection pattern and the at least one second sensor element (144) is adapted to determine at least one second reflection pattern, wherein the evaluation device (128) is adapted to select at least one image determined by the first sensor element (142) or the second sensor element (144) as reflection image (126) and to select at least one image determined by the other one of the first sensor element (142) or the second sensor element (144) as reference image (134).
19. A detector system (148) for determining a position of at least one object (112), the detector system (148) comprising at least one detector (110) according to any one of the preceding claims, the detector system (148) further comprising at least one beacon device (150) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (150) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
20. A human-machine interface (152) for exchanging at least one item of information between a user (170) and a machine (184), wherein the human-machine interface (152) comprises at least one detector system (148) according to the preceding claim, wherein the at least one beacon device (150) is adapted to be at least one of directly or indirectly attached to the user (170) and held by the user (170), wherein the human-machine interface (152) is designed to determine at least one position of the user (170) by means of the detector system (148), wherein the human-machine interface (152) is designed to assign to the position at least one item of information.
21. An entertainment device (154) for carrying out at least one entertainment function, wherein the entertainment device (154) comprises at least one human-machine interface (152) according to the preceding claim, wherein the entertainment device (154) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (152), wherein the entertainment device (154) is designed to vary the entertainment function in accordance with the information.

22. A tracking system (156) for tracking a position of at least one movable object (112), the tracking system (156) comprising at least one detector system (148) according to any one of the preceding claims referring to a detector system (148), the tracking system (156) further comprising at least one track controller (186), wherein the track controller (186) is adapted to track a series of positions of the object (112) at specific points in time.
23. A scanning system (158) for determining a depth profile of a scenery, the scanning system comprising at least one detector (110) according to any of the preceding claims referring to a detector, the scanning system (158) further comprising at least one illumination source (124) adapted to scan the scenery with at least one light beam.
24. A camera (146) for imaging at least one object (112), the camera (146) comprising at least one detector (110) according to any one of the preceding claims referring to a detector.
25. A method for determining a position of at least one object (112) by using at least one detector (110) according to any one of the preceding claims referring to a detector, the method comprising the following steps:
- generating at least two sensor signals in response to an illumination of at least one sensor element (116) having a matrix (118) of optical sensors (120) by at least one reflection light beam (182) propagating from the object (112) to the detector (110), the optical sensors (120) each having a light-sensitive area (122);
 - determining at least one reflection image (126) by using the sensor element (116);
 - selecting at least one reflection feature of the reflection image (126) and evaluating the sensor signals by using at least one evaluation device (128), thereby, determining at least one longitudinal region (130) of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
 - determining at least one displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130);
 - matching the selected reflection feature with at least one reference feature within the displacement region (132).
26. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application;

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FIG.1

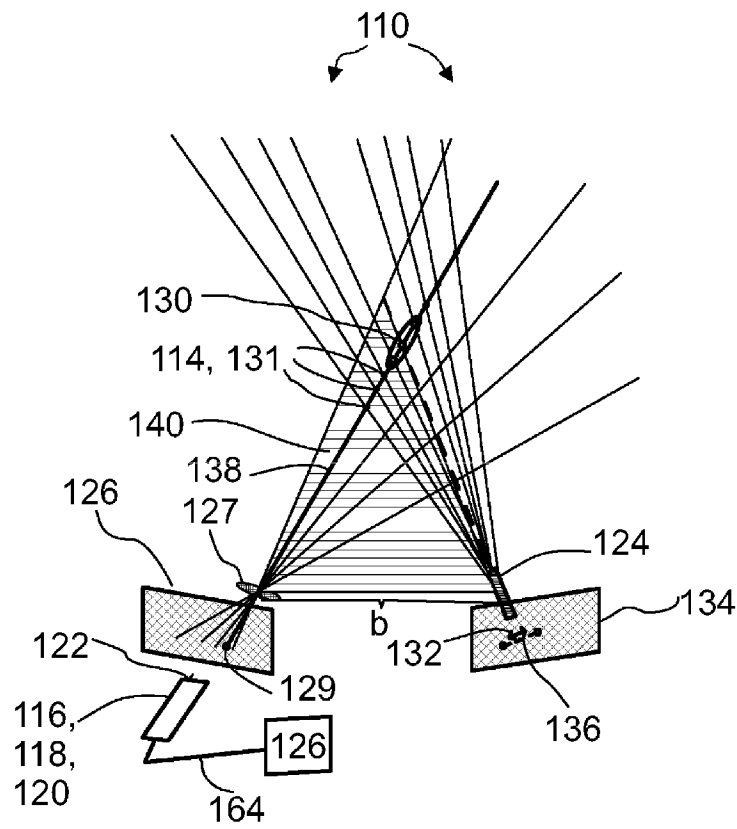


FIG.2

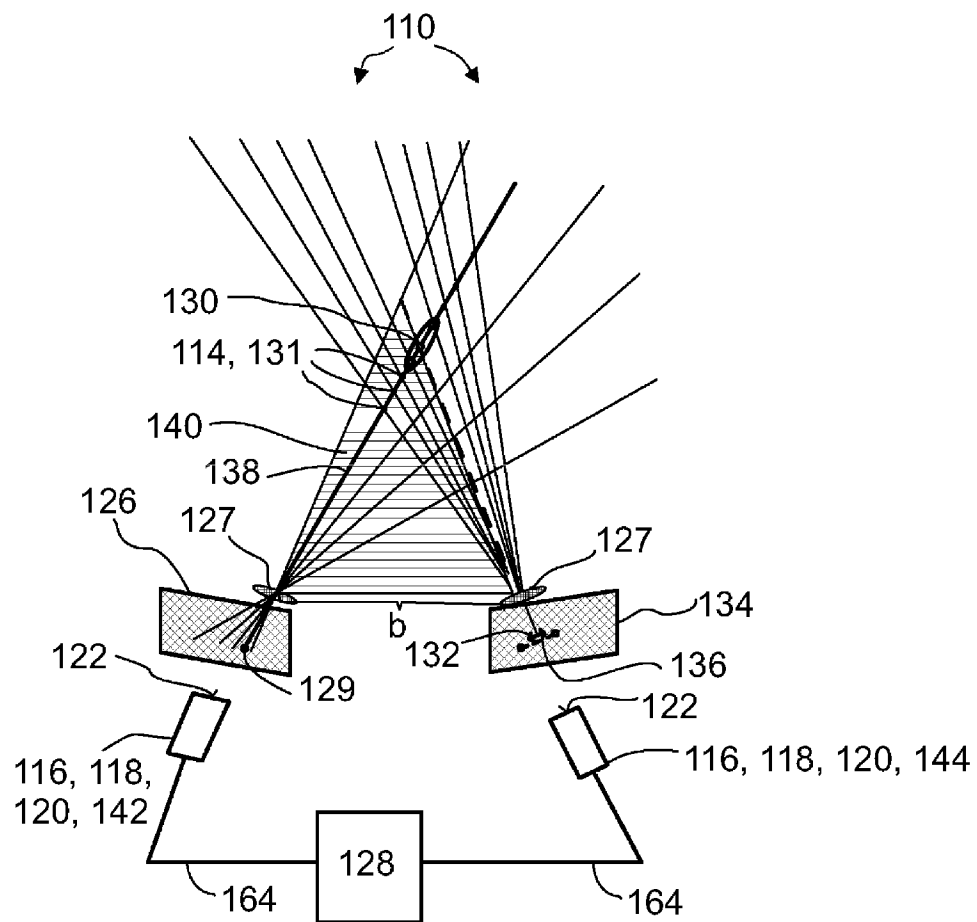
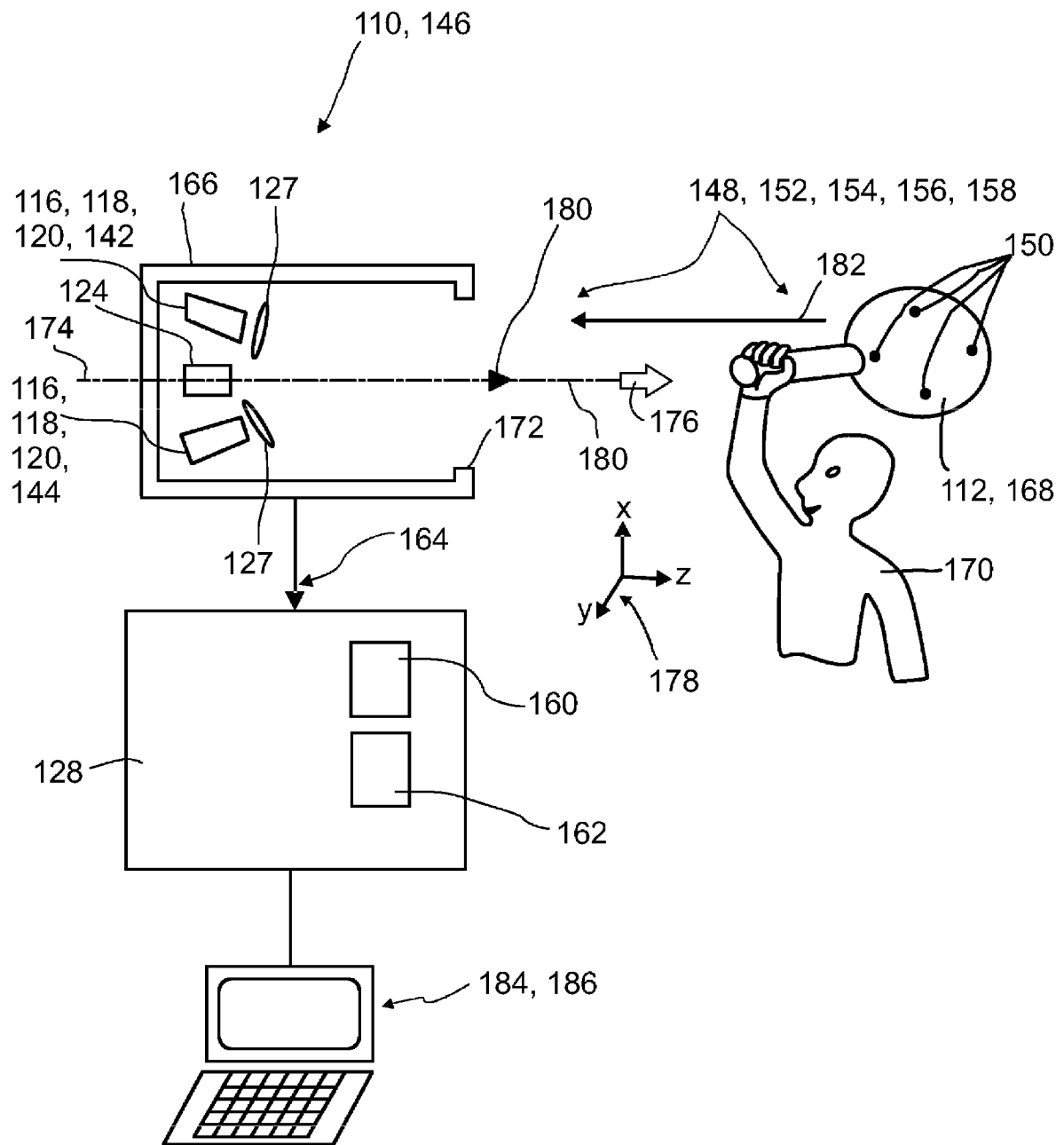


FIG.3



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FIG.4

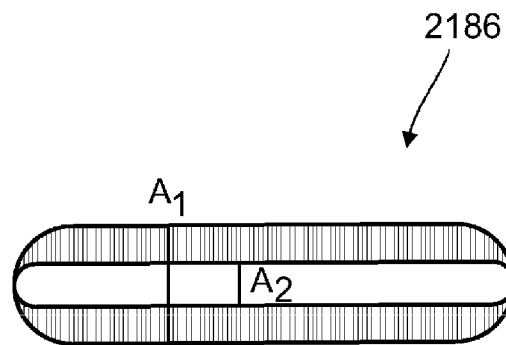


FIG.5A

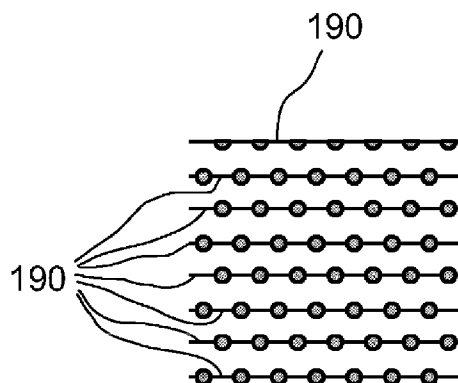


FIG.5B

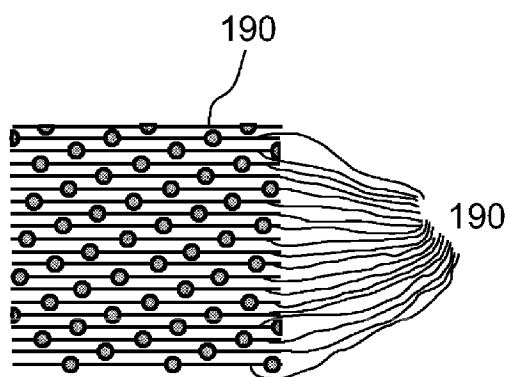


FIG.5C

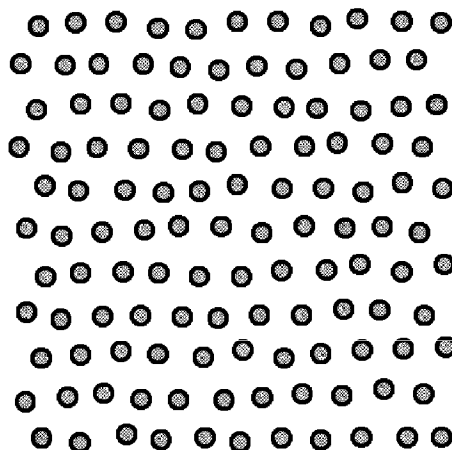
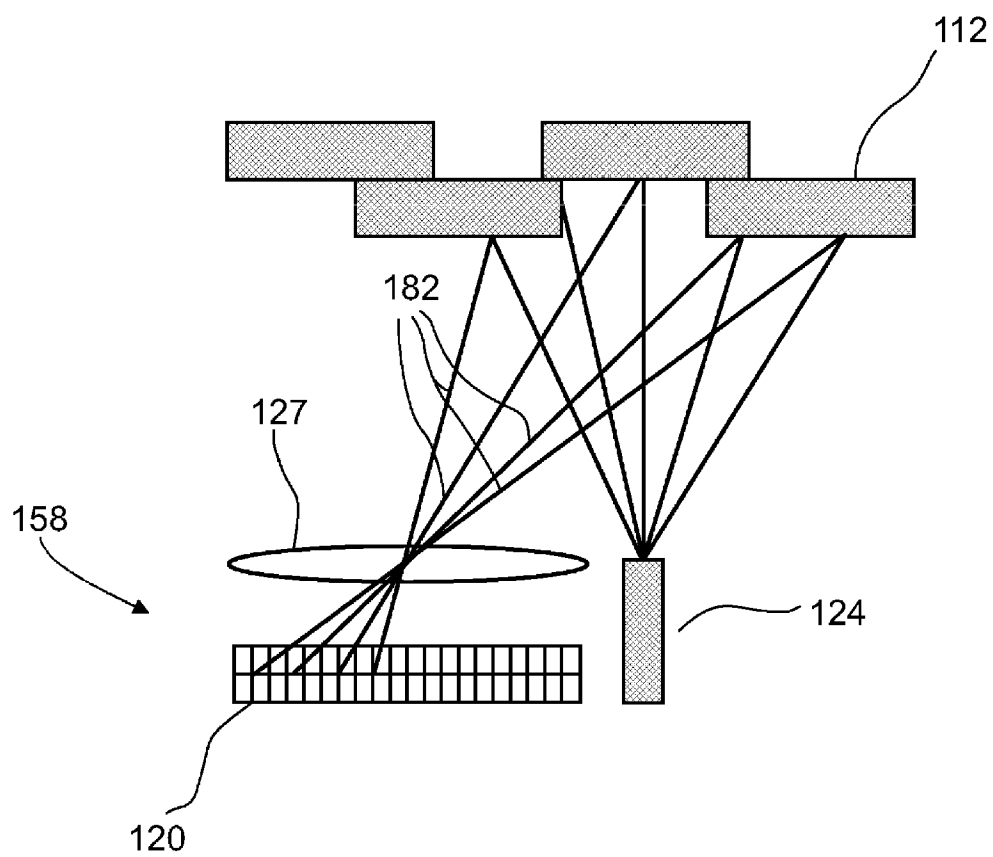


FIG.6



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2018/073067

A. CLASSIFICATION OF SUBJECT MATTER

INV. G01S7/48 G01S17/46
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

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C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2008/118143 A1 (GORDON EYAL [IL] ET AL) 22 May 2008 (2008-05-22) abstract; figures 1-25 paragraph [0309] - paragraph [0374] -----	1-26
X	US 2009/066929 A1 (TROPF HERMANN [DE]) 12 March 2009 (2009-03-12) abstract paragraph [0018] - paragraph [0110] -----	1,13, 19-26



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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

PCT/EP2018/073067

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2008118143 A1	22-05-2008	US 2008118143 A1	22-05-2008
		US 2012063672 A1	15-03-2012

US 2009066929 A1	12-03-2009	EP 1971820 A1	24-09-2008
		US 2009066929 A1	12-03-2009
		WO 2007079805 A1	19-07-2007



A detector (110) for identifying at least one mate



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"116" (and any subsequent words) was ignored because we limit queries to 32 words.



Google Patents

<https://patents.google.com/patent/>

Detector for identifying at least one material property

The detector (110) comprises at least one sensor element (116) comprising a matrix (118) ... the optical sensors (120) each having a light-sensitive area (122).

<https://patents.google.com/patent/>

Detector for optically detecting at least one object

A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises: at least one transversal optical ...



Justia

<https://patents.justia.com/assignee/trinamix-gmbh>

Patents Assigned to Trinamix GmbH

Abstract: A detector (110) for determining a position of at least one object (112) is proposed.

The detector (110) comprises: —at least two optical sensors (118 ...

<https://patents.justia.com/inventor/christian-lennartz>

Christian Lennartz Inventions, Patents and Patent Applications

The detector (110) comprises: —at least two optical sensors (118, 120, 176), each optical sensor (118, 120, 176) having a light-sensitive area (122, 124), ...



World Intellectual Property Organization

<https://patentscope.wipo.int/search/detail/docId/W...>

Detector for an optical detection of at least one object

Aug 4, 2016 — The detector (110) comprises: at least one longitudinal optical sensor (114), wherein the longitudinal optical sensor (114) has at least one ...

Missing: property m (116) matrix (118) sensors (120); (120) sensitive (122);

Related questions

What are the detectors used in optical sensors?

What are the two types of optical sensors?

What is an example of an optical sensor?

What are optical air sensors and how do they work?

Feedback



Google

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US9557856B2 - Optical detector - Google Patents

The detector comprises at least one optical sensor being adapted to detect a light beam traveling from the object towards the detector, the optical sensor ...



NASA (.gov)

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(12) United States Patent

by V Chirayath · 2018 · Cited by 3 --- Provided is a system and method for active multispectral imaging having a transmitter that uses narrowband optical radiation to dynamically ...
15 pages

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<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6473273/> }

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EP004224206A3* - EP 4 224 206 A3

Aug 30, 2023 --- at least one sensor element (114) having a matrix (116) of optical sensors (118), the optical sensors (118) each having a light-sensitive ...

Missing: material property m

 Images for A detector (110) for identifying at least one ... }



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Application Number	17439492
Filing Date	2021-09-15
First Named Inventor	Schick, Friedrich
Art Unit	2454
Examiner Name	Frantz B Jean
Attorney Docket Number	27843-1809 182037US01

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	1	20160206216	A1	2016-07-21	KIRENKO	
	2	20160155006	A1	2016-06-02	MAKKAPATI et al.	

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	1	2014097181	WO	A1	2014-06-26	BASF SE et al.		
	2	2018091640	WO	A2	2018-05-24	TRINAMIX GMBH		

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Art Unit	2454
Examiner Name	Frantz B Jean
Attorney Docket Number	27843-1809 182037US01

3	2018091649	WO	A1	2018-05-24	TRINAMIX GMBH		
4	2018091638	WO	A1	2018-05-24	TRINAMIX GMBH		
5	2012110924	WO	A1	2012-08-23	BASF SE et al.		
6	2014198629	WO	A1	2014-12-18	BASF SE		

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Preliminary Report on Patentability for corresponding PCT/EP2020/056759 mailed 30 Sept. 2021, 9 Pages	
	2	X. Jiang, H. Bunke: "Dreidimensionales Computersehen" Springer, Berlin Heidelberg, 1997, Chapter 2. 32 pages. Cited in specification, no English translation available.	
	3	EICHLER et al., "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 171 to 266, Springer Verlag, ISBN 0939-0979. Cited in specification, no English translation available.	

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Examiner Signature	/FRANTZ B JEAN/	Date Considered	09/08/2023
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Application Number		17439492
Filing Date		2021-09-15
First Named Inventor	Schick, Friedrich	
Art Unit	2454	
Examiner Name	Frantz B Jean	
Attorney Docket Number	27843-1809 182037US01	

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Application Number	17439492
Filing Date	2021-09-15
First Named Inventor	Schick, Friedrich
Art Unit	2454
Examiner Name	Frantz B Jean
Attorney Docket Number	27843-1809 182037US01

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Name/Print	Christopher M. Goff	Registration Number	41785

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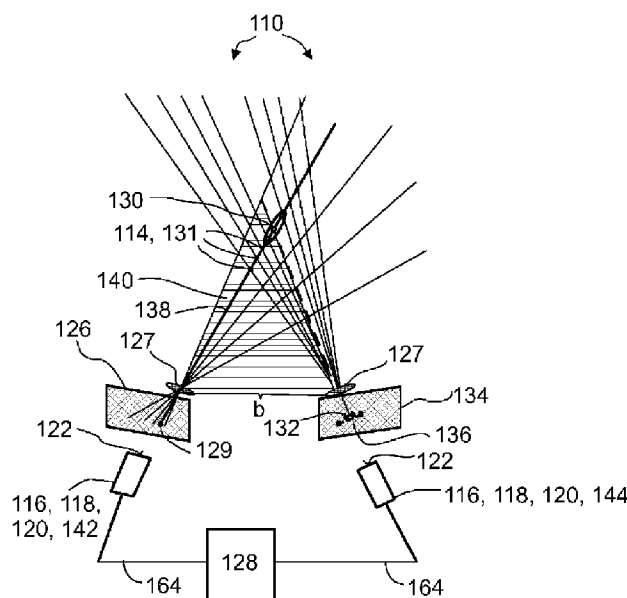
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(54) Title: DETECTOR FOR DETERMINING A POSITION OF AT LEAST ONE OBJECT

FIG.2



(57) Abstract: A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises -at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein each optical sensor (120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (122) by a reflection light beam (182) propagating from the object (112) to the detector (110), wherein the sensor element (116) is adapted to determine at least one reflection image (126); -at least one evaluation device (128), wherein the evaluation device (128) is adapted to select at least one reflection feature of the reflection image (126), wherein the evaluation device (128) is configured for determining at least one longitudinal region (130) of the selected reflection feature of the reflection image (134) by evaluating a combined signal Q from the sensor signals, wherein the evaluation device (128) is adapted to determine at least one

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displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130), wherein the evaluation device (128) is adapted to match the selected reflection feature with at least one reference feature within the displacement region (132).

Detector for determining a position of at least one object

Description

5 Field of the invention

The invention relates to a detector and a method for determining a position of at least one object. The invention further relates to a human-machine interface for exchanging at least one item of information between a user and a machine, an entertainment device, a tracking system, a camera, a scanning system and various uses of the detector device. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, photography such as digital photography or video photography for arts, documentation or technical purposes, medical technology or in the sciences. Further, the invention specifically may be used for scanning one or more objects and/or for scanning a scenery, such as for generating a depth profile of an object or of a scenery, e.g. in the field of architecture, metrology, archaeology, arts, medicine, engineering or manufacturing. However, other applications are also possible.

Prior art

Optical 3D sensing methods generally may determine unreliable results in case of environment causing multiple reflections, with biasing light sources, or reflective measurement objects. Furthermore, 3D sensing methods, such as triangulation methods using structured light or stereo cameras, with imaging capabilities often suffer from a high demand of computational power to solve correspondence problems. Necessary computational power may result in high costs for processors or Field Programmable Gate Arrays (FPGAs), for heat removal in view of ventilation requirements or water proof housing difficulties, for electrical power consumption, in particular in mobile devices, and, in addition, in measurement uncertainties. The high demand in computational power may be prohibitive to realize real-time applications, high framerates or even standards video speed frame rates of 25 frames per second.

A large number of optical devices are known from the prior art using triangulation imaging methods. For example, structured light methods or stereo methods are known. For example, passive stereo methods using two cameras in a fixed relative orientation or active stereo technologies, where an additional light projector is used. Another example is the structures light approach, where one light projector and one camera in a fixed relative orientation are used. In order to determine a depth image via triangulation, the correspondence problem has to be solved first. Therefore, in passive stereo camera techniques enough corresponding feature points have to be identified in both camera views. In structured light approaches correspondences between pre-stored and projected pseudo random light patterns have to be determined. For a robust solution of these correspondence problems computational imaging algorithms such as algorithms scaling approximately quadratically with the number of points in the projected point pattern have to be employed. In structured light methods, for example, using a stereo system comprising two detectors having a fixed relative distance, a light source projects a pat-

tern such as points, pseudo-random, random, non-periodic or irregular point patterns, or the like. Each of the detectors generates an image of a reflection pattern and an image analysis task is to identify corresponding features in the two images. Due to fixed relative position a corresponding feature point selected in one of the two images lies along an epipolar line in the other image. However, solving the so-called correspondence problem may be difficult. In stereo and triangulation systems, the distance of all feature points along the epipolar line have to have reasonable correspondence among each other. A correspondence decision cannot be made one after another. If one correspondence is wrong, this has implications for other feature points, such as invisibility. This usually yields quadratic scaling evaluation algorithms.

For example, US 2008/0240502 A1 and US 2010/0118123 A1 describe an apparatus for mapping an object including an illumination assembly, which includes a single transparency containing a fixed pattern of spots. A light source transilluminates the single transparency with optical radiation so as to project the pattern onto the object. An image capture assembly captures an image of the pattern that is projected onto the object using the single transparency. A processor processes the image captured by the image capture assembly as to reconstruct a three-dimensional map of the object.

US 2008/118143 A1 describes a method and apparatus for obtaining an image to determine a three dimensional shape of a stationary or moving object using a bi dimensional coded light pattern having a plurality of distinct identifiable feature types. The coded light pattern is projected on the object such that each of the identifiable feature types appears at most once on predefined sections of distinguishable epipolar lines. An image of the object is captured and the reflected feature types are extracted along with their location on known epipolar lines in the captured image. Displacements of the reflected feature types along their epipolar lines from reference coordinates thereupon determine corresponding three dimensional coordinates in space and thus a 3D mapping or model of the shape of the object at any point in time.

US 2009/066929 A1 describes establishing of the correspondence of the points for creation of a range image with stereo cameras. The scene is illuminated twice; thereof at least once with a random or pseudo random pattern. For both cameras, an image is taken for each of the illuminations and the quotient of brightnesses is calculated pixelwise. The correspondence is established on the basis of a comparison of the quotient of pixels on epipolar lines of different cameras. The illumination pattern is preferably highly modulated along the epipolar line; transversally or diagonally to it, it is not or only slightly modulated. For illumination, a projection unit is used which in a preferred arrangement comprises two superposed grating patterns which have a distance to each other, with at least one varying in a pseudo random manner, and with two closely neighboring light sources which shine through the gratings and thereby generate different pseudo random patterns, especially moiré patterns.

Despite the advantages implied by the above-mentioned devices and detectors, several technical challenges remain. Thus, the evaluation algorithms used require high computation power which is a severe cost driver. Furthermore, due to energy consumption and heat production of

the required computational resources, the computational demand limits use of such 3D sensor methods in outdoor and mobile applications.

Problem addressed by the invention

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It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Specifically, it is an object of the present invention to provide devices and methods which reliably may determine a position of an object in space, preferably with a low technical effort and with low requirements in terms of technical resources and cost.

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Summary of the invention

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This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

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As used in the following, the terms "have", "comprise" or "include" or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions "A has B", "A comprises B" and "A includes B" may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

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Further, it shall be noted that the terms "at least one", "one or more" or similar expressions indicating that a feature or element may be present once or more than once typically will be used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions "at least one" or "one or more" will not be repeated, notwithstanding the fact that the respective feature or element may be present once or more than once.

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Further, as used in the following, the terms "preferably", "more preferably", "particularly", "more particularly", "specifically", "more specifically" or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without

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any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a detector for determining a position of at least one object is disclosed. As used herein, the term "object" refers to a point or region emitting at least one light beam. The light beam may originate from the object, such as by the object and/or at least one illumination source integrated or attached to the object emitting the light beam, or may originate from a different illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object. As used herein, the term "position" refers to at least one item of information regarding a location and/or orientation of the object and/or at least one part of the object in space. Thus, the at least one item of information may imply at least one distance between at least one point of the object and the at least one detector. As will be outlined in further detail below, the distance may be a longitudinal coordinate or may contribute to determining a longitudinal coordinate of the point of the object. Additionally or alternatively, one or more other items of information regarding the location and/or orientation of the object and/or at least one part of the object may be determined. As an example, additionally, at least one transversal coordinate of the object and/or at least one part of the object may be determined. Thus, the position of the object may imply at least one longitudinal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one transversal coordinate of the object and/or at least one part of the object. Additionally or alternatively, the position of the object may imply at least one orientation information of the object, indicating an orientation of the object in space.

The detector comprises:

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is adapted to determine at least one reflection image;
- at least one evaluation device, wherein the evaluation device is adapted to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals, wherein the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as

a combination of several devices. As further used herein, the term "matrix" generally refers to an arrangement of a plurality of elements in a predetermined geometrical order. The matrix, as will be outlined in further detail below, specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. It shall be outlined, however, that other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix may be a single row of pixels. Other arrangements are feasible.

10 The optical sensors of the matrix specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas of all optical sensors of the matrix specifically may be located in a common plane, the common plane preferably facing the object, such that a light beam propagating from the object to the detector may generate a light spot on the common plane.

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As used herein, an "optical sensor" generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a "light-sensitive area" generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination the at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. As used herein, the term "the optical sensors each having at least one light sensitive area" refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. Thus, the term "optical sensor" furthermore refers to a light-sensitive device configured to generate one output signal, whereas, herein, a light-sensitive device configured to generate two or more output signals, for example at least one CCD and/or CMOS device, is referred to as two or more optical sensors. As will further be outlined in detail below, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the detector specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible. Thus, as an example, an optical device comprising two, three, four or more than four light-sensitive areas may be used which is regarded as two, three, four or more than four optical sensors in the context of the present invention. As outlined above, the sensor element comprises a matrix of optical sensors. Thus, as an example, the optical sensors may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

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As outlined above, the optical sensors specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors may be sensitive in the infrared spectral range. All of the optical sensors of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical optical sensors of the matrix specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors may be identical in size and/or with regard to their electronic or optoelectronic properties.

Specifically, the optical sensors may be or may comprise inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available from Hamamatsu Photonics Deutschland GmbH, D-82211 Herrsching am Ammersee, Germany. Thus, as an example, the optical sensors may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensors may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge: Au photodiode, a Ge: Hg photodiode, a Ge: Cu photodiode, a Ge: Zn photodiode, a Si: Ga photodiode, a Si: As photodiode. Additionally or alternatively, the optical sensors may comprise at least one bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The matrix may be composed of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

Thus, generally, the optical sensors of the detector may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector may comprise an array of optical sensors, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular, wherein, with respect to the term "essentially perpendicular", reference may be made to the definition given above. Thus, as an example, tolerances of less than 20°, specifically less than 10° or even less than 5°, may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100 optical sensors, more preferably at least 500 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Preferably, the sensor element may be oriented essentially perpendicular to an optical axis of the detector. Again, with respect to the term "essentially perpendicular", reference may be made to the definition and the tolerances given above. The optical axis may be a straight optical axis or may be bent or even split, such as by using one or more deflection elements and/or by using one or more beam splitters, wherein the essentially perpendicular orientation, in the latter cases, may refer to the local optical axis in the respective branch or beam path of the optical setup.

The reflection light beam may propagate from the object towards the detector. As will be outlined in further detail below, the reflection light beam may originate from the object or may originate from an illumination source, such as from an illumination source directly or indirectly illuminating the object, wherein the light beam is reflected or scattered by the object and, thereby, is at least partially directed towards the detector. The detector may be used in active and/or passive illumination scenarios. For example, the at least one illumination source may be adapted to illuminate the object, for example, by directing a light beam towards the object, which reflects the light beam. The illumination source may be or may comprise at least one multiple beam light source. For example, the illumination source may comprise at least one laser source and one or more diffractive optical elements (DOEs). Additionally or alternatively, the detector may use radiation already present in the scene such as from at least one ambient light source.

The reflection light beam specifically may fully illuminate the at least one optical sensor from which the center signal is generated, such that the at least one optical sensor from which the center signal arises is fully located within the light beam, with a width of the light beam being larger than the light-sensitive area of the at least one optical sensor from which the sensor signal arises. Contrarily, preferably, the reflection light beam specifically may create a light spot on the entire matrix which is smaller than the matrix, such that the light spot is fully located within the matrix. This situation may easily be adjusted by a person skilled in the art of optics by choosing one or more appropriate lenses or elements having a focusing or defocusing effect on the light beam, such as by using an appropriate transfer device as will be outlined in further de-

tail below. As further used herein, a “light spot” generally refers to a visible or detectable round or non-round illumination of an article, an area or object by a light beam.

As further used herein, a “sensor signal” generally refers to a signal generated by an optical sensor in response to the illumination by the light beam. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the detector, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like.

The light-sensitive areas specifically may be oriented towards the object. As used herein, the term “is oriented towards the object” generally refers to the situation that the respective surfaces of the light-sensitive areas are fully or partially visible from the object. Specifically, at least one interconnecting line between at least one point of the object and at least one point of the respective light-sensitive area may form an angle with a surface element of the light-sensitive area which is different from 0° , such as an angle in the range of 20° to 90° , preferably 80° to 90° such as 90° . Thus, when the object is located on the optical axis or close to the optical axis, the light beam propagating from the object towards the detector may be essentially parallel to the optical axis. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however. The transfer device may be configured for one or more of adjusting, defining and determining the beam profile, in particular a shape of the beam profile.

The optical sensors may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensors may be sensitive in the visible spectral range

from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensors may be sensitive in the near infrared region. Specifically, the optical sensors may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensors, specifically, may
5 be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensors each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensors may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a
10 CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. As will be outlined in further detail below, the photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, as will be outlined in further detail below, one or more photodiodes may be used, such as com-
15 mercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The detector may be adapted to determine at least one distance information of the object by using triangulation and/or structured light techniques. In known 3D sensing devices, such as
20 devices using triangulation or structured light techniques, due to correspondence problems regular, constant or periodic pattern are not suitable since each measured point has to be assigned to one reference point of a reference pattern.

As used herein, the term "reflection image" refers to an image determined by the optical sensor comprising at least one reflection feature. As used herein, the term "reflection feature" refers to
25 a feature in an image plane generated by the object in response to illumination, for example with at least one illumination feature. The reflection image may comprise the at least one reflection pattern comprising the at least one reflection feature. As used herein, the term "illumination feature" refers to at least one arbitrary shaped feature generated by at least one ambient light source or at least one illumination source adapted to illuminate the object. As used herein, the
30 term "determining at least one reflection image" refers to one or more of imaging, recording and generating of the reflection image. As used herein, the term "select at least one reflection feature" refers to one or more of identifying, determining and choosing at least one reflection feature of the reflection image. The detector may be adapted to determine the longitudinal coordinate of an object point for at least one reflection feature of the reflection image from the com-
35 bined signal. Thus, the detector may be adapted to pre-classify the at least one reflection feature of the reflection image. This allows using illumination patterns comprising regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal patterns may be preferred.

40 The evaluation device may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image

processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

The evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals. The evaluation device may be adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q, wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \varepsilon$. As used herein, the term "longitudinal region" refers to at least one uncertainty interval defined by the longitudinal coordinate z and the measurement uncertainty $\pm \varepsilon$ of the determination of the longitudinal coordinate from the combined signal Q. Error ε may depend on measurement uncertainty of the optical sensor. The measurement uncertainty of the optical sensors may be pre-determined and/or estimated and/or may be deposited in at least one data storage unit of the evaluation device. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$. As used herein, the term "combined signal Q" refers to a signal which is generated by combining the sensor signals, in particular by one or more of dividing the sensor signals, dividing multiples of the sensor signals or dividing linear combinations of the sensor signals. The evaluation device may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region. For example, the evaluation device may be configured for deriving the combined

$$\text{signal Q by } Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy},$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o . Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. As used herein, the term "beam profile" relates to a spatial distribution, in particular in at least one plane per-

pendicular to the propagation of the light beam, of an intensity of the light beam. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance $L(z_0)$ and beam shape $S(x,y;z_0)$, $E(x,y;z_0)=L \cdot S$. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_0 independent from the object size. Thus, the combined signal allows determination of the distance z_0 independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the sensor position used for determining the combined signal Q . The light-sensitive areas may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term "essentially center information" generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term "essentially edge information" generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts

of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1.

- 5 Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area
10 may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device may be adapted for determining an area integral of the beam profile. The evaluation device may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam
15 profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 25 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
30 cuts may result in improved distance information.

In one embodiment, the light beam propagating from the object to the detector may illuminate the sensor element with at least one pattern comprising at least one feature point. As used herein, the term "feature point" refers to at least one at least partially extended feature of the pattern. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The
35 central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

For example, the light beam propagating from the object to the detector may illuminate the optical sensors with at least one line pattern. The line pattern may be generated by the object, for example in response to an illumination by the at least one illumination source with an illumination pattern comprising the at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensors may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensors. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

For example, the light beam propagating from the object to the detector may illuminate the sensor element with at least one point pattern. The point pattern may be generated by the object, for example in response to an illumination by the at least one light source with an illumination pattern comprising the at least one point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The light beam propagating from the object to the detector may illuminate the sensor element with a reflection pattern comprising both point patterns and line patterns. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

The evaluation device may be configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information. Thus, essentially, photon ratios may be used as the physical basis of the method.

For example, the evaluation device may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;

- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the at least one longitudinal coordinate z of the selected feature by evaluating the combined signal.

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As explained, e.g. in WO 2012/110924 A1 or WO 2014/097181 A1, typically, a predetermined or determinable relationship exists between a size of a light spot, such as a diameter of the light spot, a beam waist or an equivalent diameter, and the longitudinal coordinate of the object from which the light beam propagates towards the detector. Without wishing to be bound by this theory, the light spot may be characterized by two measurement variables: a measurement signal measured in a small measurement patch in the center or close to the center of the light spot, also referred to as the center signal, and an integral or sum signal integrated over the light spot, with or without the center signal. For a light beam having a certain total power which does not change when the beam is widened or focused, the sum signal should be independent from the spot size of the light spot, and, thus, should, at least when linear optical sensors within their respective measurement range are used, be independent from the distance between the object and the detector. The center signal, however, is dependent on the spot size. Thus, the center signal typically increases when the light beam is focused, and decreases when the light beam is defocused. By comparing the center signal and the sum signal, thus, an item of information on the size of the light spot generated by the light beam and, thus, on the longitudinal coordinate of the object may be generated. The comparing of the center signal and the sum signal, as an example, may be done by forming the combined signal Q out of the center signal and the sum signal and by using a predetermined or determinable relationship between the longitudinal coordinate and the quotient signal for deriving the longitudinal coordinate.

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The use of a matrix of optical sensors provides a plurality of advantages and benefits. Thus, the center of the light spot generated by the light beam on the sensor element, such as on the common plane of the light-sensitive areas of the optical sensors of the matrix of the sensor element, may vary with a transversal position of the object. By using a matrix of optical sensors, the detector according to the present invention may adapt to these changes in conditions and, thus, may determine the center of the light spot simply by comparing the sensor signals. Consequently, the detector according to the present invention may, by itself, choose the center signal and determine the sum signal and, from these two signals, derive a combined signal which contains information on the longitudinal coordinate of the object. By evaluating the combined signal, the longitudinal coordinate of the object may, thus, be determined. The use of the matrix of optical sensors, thus, provides a significant flexibility in terms of the position of the object, specifically in terms of a transversal position of the object.

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The transversal position of the light spot on the matrix of optical sensors, such as the transversal position of the at least one optical sensor generating the sensor signal, may even be used as an additional item of information, from which at least one item of information on a transversal position of the object may be derived, as e.g. disclosed in WO 2014/198629 A1. Additionally or alternatively, as will be outlined in further detail below, the detector according to the present

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invention may contain at least one additional transversal detector for, in addition to the at least one longitudinal coordinate, detecting at least one transversal coordinate of the object.

Consequently, in accordance with the present invention, the term "center signal" generally refers to the at least one sensor signal comprising essentially center information of the beam profile. For example, the center signal may be the signal of the at least one optical sensor having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. As used herein, the term "highest sensor signal" refers to one or both of a local maximum or a maximum in a region of interest. The center signal may arise from a single optical sensor or, as will be outlined in further detail below, from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors from which the center signal arises may be a group of neighboring optical sensors, such as optical sensors having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The evaluation device may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the optical sensor in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from optical sensors in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an in-

tegral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

- 5 Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix.
- 10 When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of
- 15 edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.
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Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment

25 will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

- 30 Additionally or alternatively, the evaluation device may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized for example by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
- 35 cuts may result in an improved distance information.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which

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contains at least one item of information on a comparison between the center signal and the sum signal.

The evaluation device may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device is configured to determine the at least one coordinate z of the object by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

Raw sensor signals of the optical sensors may be used for evaluation or secondary sensor signals derived thereof. As used herein, the term "secondary sensor signal" generally refers to a signal, such as an electronic signal, more preferably an analogue and/or a digital signal, which is obtained by processing one or more raw signals, such as by filtering, averaging, demodulating or the like. Thus, image processing algorithms may be used for generating secondary sensor signals from the totality of sensor signals of the matrix or from a region of interest within the matrix. Specifically, the detector, such as the evaluation device, may be configured for transforming the sensor signals of the optical sensor, thereby generating secondary optical sensor signals, wherein the evaluation device is configured for performing steps a)-d) by using the secondary optical sensor signals. The transformation of the sensor signals specifically may comprise at least one transformation selected from the group consisting of: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors. As an example, a vehicle, a person or another type of predetermined object may be determined by automatic image recognition within an image, i.e. within the totality of sensor signals generated by the optical sensors, and the region of interest may be chosen such that the object is located within the region of interest. In this case, the evaluation, such as the determination of the longitudinal coordinate, may be performed for the region of interest, only. Other implementations, however, are feasible.

As outlined above, the detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be per-

formed fully or partially electronically or fully or partially by using one or more software algorithms. Specifically, the evaluation device may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at least one sensor element and/or may fully or partially be embodied independently from the sensor element.

As outlined above, the sum signal may be derived from all sensor signals of the matrix, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors contributing to the center signal excluded. In every case, a reliable sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal. Other options, however, exist.

The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

As outlined above, the comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a

quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device may be configured for forming the one or more quotient signals. The evaluation device may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient signal.

The evaluation device specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams, by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.

Thus, generally, the evaluation device may be configured for determining the longitudinal coordinate by evaluating the combined signal Q . This determining may be a one-step process, such as by directly combining the center signal and the sum signal and deriving the longitudinal coordinate thereof, or may be a multiple step process, such as by firstly deriving the combined sig-

nal from the center signal and the sum signal and, secondly, by deriving the longitudinal coordinate from the combined signal. Both options, i.e. the option of steps c) and d) being separate and independent steps and the option of steps c) and d) being fully or partially combined, shall be comprised by the present invention.

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The evaluation device may be configured for using at least one predetermined relationship between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

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The combined signal Q may be determined by using various means. As an example, a software means for deriving the quotient signal, a hardware means for deriving the quotient signal, or both, may be used and may be implemented in the evaluation device. Thus, the evaluation device, as an example, may comprise at least one divider, wherein the divider is configured for deriving the quotient signal. The divider may fully or partially be embodied as one or both of a software divider or a hardware divider. The divider may fully or partially be integrated into the sensor element answers or may fully or partially be embodied independent from the sensor element.

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The detector may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term "transfer device", also denoted as "transfer system", may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensors. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. As used herein, the term "focal length" of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a "focus" which may also be denoted as "focal point". Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer

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device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. In particular, the detector and the transfer device have a common optical axis. As used herein, the term "optical axis of the transfer device" generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The optical axis of the detector may be a line of symmetry of the optical setup of the detector. The detector comprises at least one transfer device, preferably at least one transfer system having at least one lens. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, as will also be outlined in further detail below, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the detector may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. The optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate l is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate l . Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

As further used herein, the term "evaluation device" generally refers to an arbitrary device adapted to perform the named operations, preferably by using at least one data processing device and, more preferably, by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one evaluation device may comprise at least one data processing device having a software code stored thereon comprising a number of computer commands. The evaluation device may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations.

The above-mentioned operations, including determining the at least one longitudinal coordinate of the object, are performed by the at least one evaluation device. Thus, as an example, one or more of the above-mentioned relationships may be implemented in software and/or hardware, such as by implementing one or more lookup tables. Thus, as an example, the evaluation device may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation, in order to determine the at least one longitudinal coordinate of the object. Additionally or alternatively, however, the evaluation device may also fully or partially be embodied by hardware.

As outlined above, by evaluating the center signal and the sum signal, the detector may be enabled to determine the at least one longitudinal coordinate of the object, including the option of determining the longitudinal coordinate of the whole object or of one or more parts thereof. In addition, however, other coordinates of the object, including one or more transversal coordinates and/or rotational coordinates, may be determined by the detector, specifically by the evaluation device. Thus, as an example, one or more transversal sensors may be used for determining at least one transversal coordinate of the object. As outlined above, the position of the at least one optical sensor from which the center signal arises may provide information on the at least one transversal coordinate of the object, wherein, as an example, a simple lens equation may be used for optical transformation and for deriving the transversal coordinate. Additionally or alternatively, one or more additional transversal sensors may be used and may be comprised by the detector. Various transversal sensors are generally known in the art, such as the transversal sensors disclosed in WO 2014/097181 A1 and/or other position-sensitive devices (PSDs), such as quadrant diodes, CCD or CMOS chips or the like. Additionally or alternatively, as an example, the detector according to the present invention may comprise one or more PSDs disclosed in R.A. Street (Ed.): *Technology and Applications of Amorphous Silicon*, Springer-Verlag Heidelberg, 2010, pp. 346-349. Other embodiments are feasible. These devices may generally also be implemented into the detector according to the present invention. As an example, a part of the light beam may be split off within the detector, by at least one beam splitting element. The split-off portion, as an example, may be guided towards a transversal sensor, such as a CCD or CMOS chip or a camera sensor, and a transversal position of a light spot generated by the split-off portion on the transversal sensor may be determined, thereby determining at least one transversal coordinate of the object. Consequently, the detector according to the present invention may either be a one-dimensional detector, such as a simple distance measurement device, or may be embodied as a two-dimensional detector or even as a three-dimensional detector. Further, as outlined above or as outlined in further detail below, by scanning a scenery or an environment in a one-dimensional fashion, a three-dimensional image may also be created. Consequently, the detector according to the present invention specifically may be one of a one-dimensional detector, a two-dimensional detector or a three-dimensional detector. The evaluation device may further be configured to determine at least one transversal coordinate x , y of the object. The evaluation device may be adapted to combine the information

of the longitudinal coordinate and the transversal coordinate and to determine a position of the object in space.

As outlined above, the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region. As used herein, the term "reference image" refers to an image different from the reflection image which is determined at a different spatial position compared to the reflection image. The reference image may be determined by one or more of recording at least one reference feature, imaging the at least one reference feature, calculating of the reference image. The reference image and the reflection image may be images of the object determined at different spatial positions having a fixed distance. The distance may be a relative distance, also called baseline. The evaluation device may be adapted to determine the at least one reference feature in the at least one reference image corresponding to the at least one reflection feature. As outlined above, the evaluation device may be adapted to perform an image analysis and to identify features of the reflection image. The evaluation device may be adapted to identify at least one reference feature in the reference image having an essentially identical longitudinal coordinate as the selected reflection feature. The term "essentially identical" refers to identical within 10%, preferably 5%, most preferably 1%. The reference feature corresponding to the reflection feature may be determined using epipolar geometry. For description of epipolar geometry reference is made, for example, to chapter 2 in X. Jiang, H. Bunke: „Dreidimensionales Computersehen“ Springer, Berlin Heidelberg, 1997. Epipolar geometry may assume that the reference image and the reflection image may be images of the object determined at different spatial positions and/or spatial orientations having a fixed distance. The evaluation device may be adapted to determine an epipolar line in the reference image. Relative position of the reference image and reflection image may be known. For example, relative position of the reference image and reflection image may be stored within at least one storage unit of the evaluation device. The evaluation device may be adapted to determine a straight line extending from the selected reflection feature of the reflection image. The straight line may comprise possible object features corresponding to the selected feature. The straight line and the baseline span an epipolar plane. As the reference image is determined at a different relative position from the reflection image, the corresponding possible object features may be imaged on a straight line, called epipolar line, in the reference image. Thus, a feature of the reference image corresponding to the selected feature of the reflection image lies on the epipolar line. Due to distortions of the image or changes in the system parameters such as due to ageing, temperature changes, mechanical stress or the like, epipolar lines may intersect or be very close to each other and/or the correspondence between reference feature and reflection feature may be unclear. Further, each known position or object in the real world may be projected onto the reference image and vice versa. The projection may be known due to a calibration of the detector, whereas the calibration is comparable to a teaching of the epipolar geometry of the specific camera.

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As used herein, the term "displacement region" refers to a region in the reference image in which the reference feature corresponding to the selected reflection feature may be imaged. Specifically, the displacement region may be a region in the reference image in which the refer-

ence feature corresponding to the selected reflection feature is expected to be located in the reference image. Depending on the distance to the object, an image position of the reference feature corresponding to the reflection feature may be displaced within the reference image compared to an image position of the reflection feature in the reflection image. The displacement region may comprise only one reference feature. The displacement region may also comprise more than one reference feature. The displacement region may comprise an epipolar line or a section of an epipolar line. The displacement region may comprise more than one epipolar line or more sections of more than one epipolar line. As used herein, the term "reference feature" refers to at least one feature of the reference image. The displacement region may extend along the epipolar line, orthogonal to an epipolar line, or both. The evaluation device may be adapted to determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm \varepsilon$ or orthogonal to an epipolar line. The measurement uncertainty of the distance measurement using the combined signal Q may result in a displacement region which is non-circular since the measurement uncertainty may be different for different directions. Specifically, the measurement uncertainty along the epipolar line or epipolar lines may be greater than the measurement uncertainty in an orthogonal direction with respect to the epipolar line or epipolar lines. The displacement region may comprise an extend in an orthogonal direction with respect to the epipolar line or epipolar lines. The evaluation device may determine a displacement region around the image position of the reflection feature. The evaluation device may be adapted to determine the longitudinal coordinate z for the reflection feature and an error interval $\pm \varepsilon$ from the combined signal Q to determine the displacement region along the epipolar line corresponding to $z \pm \varepsilon$. The evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region. As used herein, the term "matching" refers to determining and/or evaluating corresponding reference and reflection features. The evaluation device may be adapted to match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z . The evaluation algorithm may be a linear scaling algorithm. The evaluation device may be adapted to determine the epipolar line closest to and/or within the displacement region. The evaluation device may be adapted to determine the epipolar line closest to the image position of the reflection feature. The extent of the displacement region along the epipolar line may be larger than the extent of the displacement region orthogonal to the epipolar line. The evaluation device may be adapted to determine an epipolar line before determining a corresponding reference feature. The evaluation device may determine a displacement region around the image position of each reflection feature. The evaluation device may be adapted to assign an epipolar line to each displacement region of each image position of the reflection features, such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line. The evaluation device may be adapted to determine the reference feature corresponding to the image position of the reflection feature by determining the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement

region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Additionally or alternatively, the evaluation device may be configured to perform the following steps:

- 5 - Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
- 10 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

- 15 Additionally or alternatively, the evaluation device may be adapted to decide between more than one epipolar line and/or reference feature to be assigned to a reflection feature such as by comparing distances of reflection features and/or epipolar lines within the reference image and/or by comparing error weighted distances, such as ϵ -weighted distances of reflection features and/or epipolar lines within the reference image and assigning the epipolar line and/or
- 20 reference feature in shorter distance and/or ϵ -weighted distance to the reference feature and/or reflection feature.

- Preferably, the detector may be adapted to pre-classify the selected reflection feature using the combined signal Q such that an unambiguous assignment to one reference feature is possible. In particular, illumination features of the illumination pattern may be arranged such that corresponding reference features of the reference image may have a relative distance to each other as large as possible on the epipolar line. The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on the epipolar line. For example, the illumination pattern may comprise at least one hexagonal pattern. Preferably, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Preferably, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are displaced by a random distance from the regular position, for example orthogonal to the epipolar line of the point.
- 30 The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other.

- 40 The evaluation device may be adapted to determine a displacement of the matched reference feature and the selected reflection feature. As used herein, the term "displacement" refers to difference between a position in the reference image to a position in the reflection image. The evaluation device may be adapted to determine a longitudinal information of the matched fea-

ture using a predetermined relationship between a longitudinal coordinate and the displacement. As used herein, the term "longitudinal information" refers to information relating to the longitudinal coordinate. For example, the longitudinal information may be a distance value. The evaluation device may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature in the reflection image and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, longitudinal coordinate of the corresponding object feature may be determined by triangulation. Thus, the evaluation device may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship. The evaluation device may be adapted to store parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector. The evaluation device may be adapted to generate the parameters for an intrinsic and/or extrinsic calibration of the camera and/or the detector such as by performing a Tsai camera calibration. The evaluation device may be adapted to compute and/or estimate parameters such as the focal length of the transfer device, the radial lens distortion coefficient, the coordinates of the center of radial lens distortion, scale factors to account for any uncertainty due to imperfections in hardware timing for scanning and digitization, rotation angles for the transformation between the world and camera coordinates, translation components for the transformation between the world and camera coordinates, aperture angles, image sensor format, principal point, skew coefficients, camera center, camera heading, baseline, rotation or translation parameters between camera and/or illumination source, apertures, focal distance, or the like.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce the possible number of solutions along the epipolar line significantly. The number of possible solutions may even be reduced to one. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature

may enhance accuracy and/or speed and/or may lower computational demand in comparison to conventional triangulation systems. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be reduced to increase the light intensity in each illumination point such as to compete with ambient light while complying to eye safety regulations. A reduced number of illumination features in a conventional triangulation system might increase the difficulty to match reflection features and reference features. Further, the number of illumination features such as the number of illumination points in the illumination pattern may be increased, such as to increase the resolution of the distance measurement, such as to increase the resolution of the obtained depth map without increasing the processing power of the evaluation device such as in a mobile application.

The detector may further comprise an illumination source. As an example, the illumination source may be configured for generating an illuminating light beam for illuminating the object. The detector may be configured such that the illuminating light beam propagates from the detector towards the object along an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source may be adapted to generate at least one illumination pattern for illumination of the object. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. The detector may be configured such that the illumination pattern propagates from the detector, in particular from at least one opening of the housing, towards the object along and/or parallel to an optical axis of the detector. For this purpose, the detector may comprise at least one reflective element, preferably at least one prism, for deflecting the illumination pattern such that it propagates along or parallel to the optical axis. Specifically, the illumination source may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. As used herein, the term "pattern" refers to an arbitrary known or pre-determined arrangement comprising at least one arbitrarily shaped feature. The pattern may comprise at least one feature such as a point or symbol. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. As used herein, the term "illumination pattern" refers to a pattern which illuminates the object. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern

such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that a hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image.

The illumination source may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source may comprise at least one light source adapted to generate the illumination pattern directly. For example, the illumination source may comprise at least one laser source. For example, the illumination source may comprise at least one line laser. The line laser may be adapted to send a laser line to the object, for example a horizontal or vertical laser line. The illumination source may comprise a plurality of line lasers. For example, the illumination source may comprise at least two line lasers which may be arranged such that the illumination pattern comprises at least two parallel or crossing lines. The illumination source may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source. The illumination source may be one of attached to or integrated into a mobile device such as a smartphone. The illumination source may be used for further functions that may be used in determining an image such as for an autofocus function. The illumination device may be attached to a mobile device such as by using a connector such as a USB- or phone-connector such as the headphone jack.

The illumination source, specifically, may be configured for emitting light in the infrared spectral range. It shall be noted, however, that other spectral ranges are feasible, additionally or alternatively. Further, the illumination source specifically may be configured for emitting modulated or non-modulated light. In case a plurality of illumination sources is used, the different illumination sources may have different modulation frequencies which, as outlined in further detail below, later on may be used for distinguishing the light beams. The detector may be configured for evaluating a single light beam or a plurality of light beams. In case a plurality of light beams propagates from the object to the detector, means for distinguishing the light beams may be provided. Thus, the light beams may have different spectral properties, and the detector may comprise one or more wavelength selective elements for distinguishing the different light beams. Each of the light beams may then be evaluated independently. The wavelength selective elements, as an example, may be or may comprise one or more filters, one or more prisms, one or more gratings, one or more dichroitic mirrors or arbitrary combinations thereof. Further, additionally or alternatively, for distinguishing two or more light beams, the light beams may be modulated in a specific fashion. Thus, as an example, the light beams may be frequency modulated, and the sensor signals may be demodulated in order to distinguish partially the sensor signals originating from the different light beams, in accordance with their demodulation frequencies. These techniques generally are known to the skilled person in the field of high-

frequency electronics. Generally, the evaluation device may be configured for distinguishing different light beams having different modulations.

Specifically, the illumination source and the optical sensors may be arranged in a common plane or in different planes. The illumination source and the optical sensors may have different spatial orientation. In particular, the illumination source and the sensor element may be arranged in a twisted arrangement.

The illumination source may be adapted to generate and/or to project a cloud of points such that a plurality of illuminated regions is generated on the matrix of optical sensors, for example the CMOS detector. Additionally, disturbances may be present on the matrix of optical sensors such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The evaluation device may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate of the object. For example, the evaluation device may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The sensor element may be adapted to determine at least one reflection pattern. As used herein, the term "reflection pattern" refers to a response pattern generated by reflection or scattering of light at the surface of the object, in particular generated by the object in response to illumination by the illumination pattern. As outlined above, the illumination pattern comprises at least one feature adapted to illuminate the object. As outlined above, the illumination feature may be generated by ambient light or by the at least one illumination source. The reflection pattern may comprise at least one feature corresponding to at least one feature of the illumination pattern.

The reflection pattern may comprise, in comparison to the illumination pattern, at least one distorted pattern, wherein the distortion depends on the distance of the object, such as surface properties of the object. The evaluation device may be adapted to select at least one feature of the reflection pattern and to determine the longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals, as described above.

For example, the reference image may be an image of the illumination pattern at an image plane at a position of the illumination source. The evaluation device may be adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region. The illumination source and the sensor element may be separated by a fixed distance.

For example, the detector may comprise at least two sensor elements each having a matrix of optical sensors. At least one first sensor element and at least one second sensor element may be positioned at different spatial positions. A relative distance between the first sensor element and the second element may be fixed. The at least one first sensor element may be adapted to

- determine at least one first reflection pattern, in particular at least one first reflection feature, and the at least one second sensor element may be adapted to determine at least one second reflection pattern, in particular at least one second reflection feature. The evaluation device may be adapted to select at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image. The evaluation device may be adapted to select the at least one reflection feature in the reflection pattern and to determine the longitudinal region of the selected feature by evaluating the combined signal Q from the sensor signals. The evaluation device may be adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern. The evaluation device may be adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region.
- 15 As outlined above, the detector may further comprise one or more additional elements such as one or more additional optical elements. Further, the detector may fully or partially be integrated into at least one housing.

- In a further aspect of the present invention, a detector system for determining a position of at least one object is disclosed. The detector system comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above or according to one or more of the embodiments disclosed in further detail below. The detector system further comprises at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object. Further details regarding the beacon device will be given below, including potential embodiments thereof. Thus, the at least one beacon device may be or may comprise at least one active beacon device, comprising one or more illumination sources such as one or more light sources like lasers, LEDs, light bulbs or the like. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Alternatively, as outlined above, the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. The light emitted by the one or more beacon devices may be non-modulated or may be modulated, as outlined above, in order to distinguish two or more light beams. Additionally or alternatively, the at least one beacon device may be adapted to reflect one or more light beams towards the detector, such as by comprising one or more reflective elements. Further, the at least one beacon device may be or may comprise one or more scattering elements adapted for scattering a light beam. Therein, elastic or inelastic scattering may be used. In case the at least one beacon device is adapted to reflect and/or scatter a primary light beam towards the detector, the beacon device may be adapted to leave the spectral properties of the light beam unaffected or, alternatively, may be adapted to change the spectral properties of the light beam, such as by modifying a wavelength of the light beam.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is disclosed. The human-machine interface comprises at least one detector system according to the embodiments disclosed above and/or according to one or more of the embodiments disclosed in further detail below. Therein,
5 the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user or held by the user. The human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

10 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. The entertainment device comprises at least one human-machine interface according to the embodiment disclosed above and/or according to one or more of the embodiments disclosed in further detail below. The entertainment device is configured to enable at least one item of information to be input by a player by means of the human-machine interface. The entertainment device is further configured to vary the entertainment
15 function in accordance with the information.

In a further aspect of the present invention, a tracking system for tracking a position of at least one movable object is disclosed. The tracking system comprises at least one detector system
20 according to one or more of the embodiments referring to a detector system as disclosed above and/or as disclosed in further detail below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time.

25 In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to any one of the embodiments referring to a detector as disclosed above or as disclosed in further detail below.

30 In a further aspect of the present invention, a scanning system for determining a depth profile of a scenery, which may also imply determining at least one position of at least one object, is provided. The scanning system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The scanning system further
35 comprises at least one illumination source adapted to scan the scenery with at least one light beam, which may also be referred to as an illumination light beam or scanning light beam. As used herein, the term "scenery" generally refers to a two-dimensional or three-dimensional range which is visible by the detector, such that at least one geometric or spatial property of the two-dimensional or three-dimensional range may be evaluated with the detector. As further
40 used herein, the term "scan" generally refers to a consecutive measurement in different regions. Thus, the scanning specifically may imply at least one first measurement with the illumination light beam being oriented or directed in a first fashion, and at least one second measurement with the illumination light beam being oriented or directed in a second fashion which is different

from the first fashion. The scanning may be a continuous scanning or a stepwise scanning. Thus, in a continuous or stepwise fashion, the illumination light beam may be directed into different regions of the scenery, and the detector may be detected to generate at least one item of information, such as at least one longitudinal coordinate, for each region. As an example, for
5 scanning an object, one or more illumination light beams may, continuously or in a stepwise fashion, create light spots on the surface of the object, wherein longitudinal coordinates are generated for the light spots. Alternatively, however, a light pattern may be used for scanning. The scanning may be a point scanning or a line scanning or even a scanning with more complex light patterns. The illumination source of the scanning system may be distinct from the optional illumination source of the detector. Alternatively, however, the illumination source of the
10 scanning system may also be fully or partially identical with or integrated into the at least one optional illumination source of the detector.

Thus, the scanning system may comprise at least one illumination source which is adapted to
15 emit the at least one light beam being configured for the illumination of the at least one dot located at the at least one surface of the at least one object. As used herein, the term "dot" refers to an area, specifically a small area, on a part of the surface of the object which may be selected, for example by a user of the scanning system, to be illuminated by the illumination source. Preferably, the dot may exhibit a size which may, on one hand, be as small as possible in order
20 to allow the scanning system to determine a value for the distance between the illumination source comprised by the scanning system and the part of the surface of the object on which the dot may be located as exactly as possible and which, on the other hand, may be as large as possible in order to allow the user of the scanning system or the scanning system itself, in particular by an automatic procedure, to detect a presence of the dot on the related part of the surface of the object.
25

For this purpose, the illumination source may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic
30 and/or inorganic light-emitting diode. As an example, the light emitted by the illumination source may have a wavelength of 300-500 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm may be used. On account of their generally defined beam profiles and other
35 properties of handleability, the use of at least one laser source as the illumination source is particularly preferred. Herein, the use of a single laser source may be preferred, in particular in a case in which it may be important to provide a compact scanning system that might be easily storable and transportable by the user. The illumination source may, thus, preferably be a constituent part of the detector and may, therefore, in particular be integrated into the detector,
40 such as into the housing of the detector. In a preferred embodiment, particularly the housing of the scanning system may comprise at least one display configured for providing distance-related information to the user, such as in an easy-to-read manner. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least

one button which may be configured for operating at least one function related to the scanning system, such as for setting one or more operation modes. In a further preferred embodiment, particularly the housing of the scanning system may, in addition, comprise at least one fastening unit which may be configured for fastening the scanning system to a further surface, such as a rubber foot, a base plate or a wall holder, such as a base plate or holder comprising a magnetic material, in particular for increasing the accuracy of the distance measurement and/or the handleability of the scanning system by the user.

Particularly, the illumination source of the scanning system may, thus, emit a single laser beam which may be configured for the illumination of a single dot located at the surface of the object. By using at least one of the detectors according to the present invention at least one item of information about the distance between the at least one dot and the scanning system may, thus, be generated. Hereby, preferably, the distance between the illumination system as comprised by the scanning system and the single dot as generated by the illumination source may be determined, such as by employing the evaluation device as comprised by the at least one detector. However, the scanning system may, further, comprise an additional evaluation system which may, particularly, be adapted for this purpose. Alternatively or in addition, a size of the scanning system, in particular of the housing of the scanning system, may be taken into account and, thus, the distance between a specific point on the housing of the scanning system, such as a front edge or a back edge of the housing, and the single dot may, alternatively, be determined. The illumination source may be adapted to generate and/or to project a cloud of points, for example the illumination source may comprise one or more of at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources.

Alternatively, the illumination source of the scanning system may emit two individual laser beams which may be configured for providing a respective angle, such as a right angle, between the directions of an emission of the beams, whereby two respective dots located at the surface of the same object or at two different surfaces at two separate objects may be illuminated. However, other values for the respective angle between the two individual laser beams may also be feasible. This feature may, in particular, be employed for indirect measuring functions, such as for deriving an indirect distance which may not be directly accessible, such as due to a presence of one or more obstacles between the scanning system and the dot or which may otherwise be hard to reach. By way of example, it may, thus, be feasible to determine a value for a height of an object by measuring two individual distances and deriving the height by using the Pythagoras formula. In particular for being able to keep a predefined level with respect to the object, the scanning system may, further, comprise at least one leveling unit, in particular an integrated bubble vial, which may be used for keeping the predefined level by the user.

As a further alternative, the illumination source of the scanning system may emit a plurality of individual laser beams, such as an array of laser beams which may exhibit a respective pitch, in particular a regular pitch, with respect to each other and which may be arranged in a manner in

order to generate an array of dots located on the at least one surface of the at least one object. For this purpose, specially adapted optical elements, such as beam-splitting devices and mirrors, may be provided which may allow a generation of the described array of the laser beams. In particular, the illumination source may be directed to scan an area or a volume by using one or more movable mirrors to redirect the light beam in a periodic or non-periodic fashion.

Thus, the scanning system may provide a static arrangement of the one or more dots placed on the one or more surfaces of the one or more objects. Alternatively, the illumination source of the scanning system, in particular the one or more laser beams, such as the above described array of the laser beams, may be configured for providing one or more light beams which may exhibit a varying intensity over time and/or which may be subject to an alternating direction of emission in a passage of time, in particular by moving one or more mirrors, such as the micro-mirrors comprised within the mentioned array of micro-mirrors. As a result, the illumination source may be configured for scanning a part of the at least one surface of the at least one object as an image by using one or more light beams with alternating features as generated by the at least one illumination source of the scanning system. In particular, the scanning system may, thus, use at least one row scan and/or line scan, such as to scan the one or more surfaces of the one or more objects sequentially or simultaneously. Thus, the scanning system may be adapted to measure angles by measuring three or more dots, or the scanning system may be adapted to measure corners or narrow regions such as a gable of a roof, which may be hardly accessible using a conventional measuring stick. As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step. As non-limiting examples, the scanning system may be used in industrial safety curtain applications. As non-limiting examples, the scanning system may be used to perform sweeping, vacuuming, mopping, or waxing functions, or yard or garden care functions such as mowing or raking. As non-limiting examples, the scanning system may employ an LED illumination source with collimated optics and may be adapted to shift the frequency of the illumination source to a different frequency to obtain more accurate results and/or employ a filter to attenuate certain frequencies while transmitting others. As non-limiting examples, the scanning system and/or the illumination source may be rotated as a whole or rotating only a particular optics package such as a mirror, beam splitter or the like, using a dedicated motor as such that in operation, the scanning system may have a full 360 degree view or even be moved and or rotated out of plane to further increase the scanned area. Further, the illumination source may be actively aimed in a predetermined direction. Further, to allow the rotation of wired electrical systems, slip rings, optical data transmission, or inductive couplings may be employed.

As a non-limiting example, the scanning system may be attached to a tripod and point towards an object or region with several corners and surfaces. One or more flexibly movable laser sources are attached to the scanning system. The one or more laser sources are moved as

such that they illuminate points of interest. The position of the illuminated points with respect to the scanning system is measured when pressing a designated button on the scanning system and the position information is transmitted via a wireless interface to a mobile phone. The position information is stored in a mobile phone application. The laser sources are moved to illuminate further points of interest the position of which are measured and transmitted to the mobile phone application. The mobile phone application may transform the set of points into a 3d model by connecting adjacent points with planar surfaces. The 3d model may be stored and processed further. The distances and or angles between the measured points or surfaces may be displayed directly on a display attached to a scanning system or on the mobile phone to which the position information is transmitted.

As a non-limiting example, a scanning system may comprise two or more flexible movable laser sources to project points and further one movable laser source projecting a line. The line may be used to arrange the two or more laser spots along a line and the display of the scanning system may display the distance between the two or more laser spots that may be arranged along the line, such as at equal distance. In the case of two laser spots, a single laser source may be used whereas the distance of the projected points is modified using one or more beam-splitters or prisms, where a beam-splitter or prism can be moved as such that the projected laser spots move apart or closer together. Further, the scanning system may be adapted to project further patterns such as a right angle, a circle, a square, a triangle, or the like, along which a measurement can be done by projecting laser spots and measuring their position.

As a non-limiting example, the scanning system may be adapted as a line scanning device. In particular, the scanning system may comprise at least one sensor line or row. Triangulation systems require a sufficient baseline such that in the near field no detection may be possible. Near field detection may be possible if the laser spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. These near field and far field problems can be overcome by using the detector according to the present invention. In particular, the detector may comprise a CMOS line of optical sensors. The scanning system may be adapted to detect a plurality of light beams propagating from the object to the detector on the CMOS line. The light beams may be generated at different positions on the object or by movement of the illumination source. The scanning system may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the quotient signal Q as described above and in more detail below.

As a non-limiting example, the scanning system may be adapted to support the work with tools, such as wood or metal processing tools, such as a saw, a drill, or the like. Thus, the scanning system may be adapted to measure the distance in two opposite directions and display the two measured distances or the sum of the distances in a display. Further, the scanning system may be adapted to measure the distance to the edge of a surface as such that when the scanning system is placed on the surface, a laser point is moved automatically away from the scanning system along the surface, until the distance measurement shows a sudden change due to a corner or the edge of a surface. This makes it possible to measure the distance of the end of a

wood plank while the scanning system is placed on the plank but remote from its end. Further, the scanning system may measure the distance of the end of a plank in one direction and project a line or circle or point in a designated distance in the opposite direction. The scanning system may be adapted to project the line or circle or point in a distance depending on the distance measured in the opposite direction such as depending on a predetermined sum distance. This allows working with a tool such as a saw or driller at the projected position while placing the scanning system in a safe distance from the tool and simultaneously perform a process using the tool in a predetermined distance to the edge of the plank. Further, the scanning system may be adapted to project points or lines or the like in two opposite directions in a predetermined distance. When the sum of the distances is changed, only one of the projected distances changes.

As a non-limiting example, the scanning system may be adapted to be placed onto a surface, such as a surface on which a task is performed, such as cutting, sawing, drilling, or the like, and to project a line onto the surface in a predetermined distance that can be adjusted such as with buttons on the scanning system.

As non-limiting examples, the scanning system may be used in safety laser scanners, e.g. in production environments, and/or in 3D-scanning devices as used for determining the shape of an object, such as in connection to 3D-printing, body scanning, quality control, in construction applications, e.g. as range meters, in logistics applications, e.g. for determining the size or volume of a parcel, in household applications, e.g. in robotic vacuum cleaners or lawn mowers, or in other kinds of applications which may include a scanning step.

The transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the optical sensor. The transfer device can also be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one linear combination of Gaussian beams, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient illumination source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one in-

candescent lamp and/or at least one semiconductor illumination source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the at least one beacon device or into one or more of the beacon devices and/or into the object or connected or spatially coupled to the object.

The light emerging from the one or more optional beacon devices can accordingly, alternatively or additionally from the option that said light originates in the respective beacon device itself, emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the beacon device can be emitted by the beacon device itself and/or be reflected by the beacon device and/or be scattered by the beacon device before it is fed to the detector. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary illumination source, for example by the object or a partial region of the object being excited to generate luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The beacon devices and/or the at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers, more preferably in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. For thermal imaging applications the target may emit light in the far infrared spectral range, preferably in the range of 3.0 micrometers to 20 micrometers. For example, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. For example, the at least one illumination source is adapted to emit light in the infrared spectral range. Other options, however, are feasible.

The feeding of the light beam to the optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is pro-

duced on the optional sensor area of the optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the transfer device may be designed in such a way that the light spot, for example in the case of an object arranged within a visual range of the detector, is arranged completely on a sensor region and/or on a sensor area of the optical sensor. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

In a further aspect, the present invention discloses a method for determining a position of at least one object by using a detector, such as a detector according to the present invention, such as according to one or more of the embodiments referring to a detector as disclosed above or as disclosed in further detail below. Still, other types of detectors may be used. The method comprises the following method steps, wherein the method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The method comprises the following method steps:

- generating at least two sensor signals in response to an illumination of at least one sensor element having a matrix of optical sensors by at least one reflection light beam propagating from the object to the detector, the optical sensors each having a light-sensitive area;
- determining at least one reflection image by using the sensor element;
- selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal region of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
- determining at least one displacement region in at least one reference image corresponding to the longitudinal region;
- matching the selected reflection feature with at least one reference feature within the displacement region.

For details, options and definitions, reference may be made to the detector as discussed above. Thus, specifically, as outlined above, the method may comprise using the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below.

The method may further comprise determining a displacement of the matched reference feature and the selected reflection feature. A longitudinal information of the matched feature may be determined using a predetermined relationship between a longitudinal coordinate and the displacement.

The deriving of the combined signal Q may comprise one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

5 In a further aspect of the present invention, use of the detector according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space; a homing or tracking beacon detector for vehicles; an outdoor application; a mobile application; a communication application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

15 The object generally may be a living or non-living object. The detector or the detector system even may comprise the at least one object, the object thereby forming part of the detector system. Preferably, however, the object may move independently from the detector, in at least one spatial dimension. The object generally may be an arbitrary object. In one embodiment, the object may be a rigid object. Other embodiments are feasible, such as embodiments in which the object is a non-rigid object or an object which may change its shape.

25 As will be outlined in further detail below, the present invention may specifically be used for tracking positions and/or motions of a person, such as for the purpose of controlling machines, gaming or simulation of sports. In this or other embodiments, specifically, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat; an article of clothing; a hat; a shoe.

Thus, generally, the devices according to the present invention, such as the detector, may be applied in various fields of uses. Specifically, the detector may be applied for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; a mapping application for generating maps of at least one space, such as at least one space selected from the group of a room, a building and a street; a mobile application; a webcam; an audio device; a dolby surround audio system; a computer peripheral device; a gaming application; a camera or video application; a security application; a surveillance application; an automotive application; a transport application; a medical application; a sports' application; a machine vision application; a vehicle application; an airplane application; a ship application; a spacecraft application; a building application; a construction application; a cartography application; a manufacturing application. Additionally or alternatively, applications in local and/or global positioning systems may be named, especially landmark-based positioning and/or navigation, specifically for use in cars or other vehicles (such as trains, motorcycles, bicycles, trucks for cargo transportation), robots or for use by pedestrians. Further, indoor positioning systems may be named as potential applications, such as for house-

hold applications and/or for robots used in manufacturing, logistics, surveillance, or maintenance technology.

5 The devices according to the present invention may be used in mobile phones, tablet computers, laptops, smart panels or other stationary or mobile or wearable computer or communication applications. Thus, the devices according to the present invention may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, the devices according to the present invention may be used as cameras and/or sensors, such as in combination with
10 mobile software for scanning and/or detecting environment, objects and living beings. The devices according to the present invention may even be combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. The devices according to the present invention may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with voice and/or gesture recognition. Thus,
15 specifically, the devices according to the present invention acting as human-machine interfaces, also referred to as input devices, may be used in mobile applications, such as for controlling other electronic devices or components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one device according to the present invention may be used for controlling a television set, a game console, a music player or music device or
20 other entertainment devices.

Further, the devices according to the present invention may be used in webcams or other peripheral devices for computing applications. Thus, as an example, the devices according to the present invention may be used in combination with software for imaging, recording, surveillance,
25 scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, the devices according to the present invention are particularly useful for giving commands by facial expressions and/or body expressions. The devices according to the present invention can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, microphone etc. Further, the devices according to the present invention may be used in applications for gaming, such as by using a webcam. Further, the devices according to the present invention may be used in virtual training applications and/or video conferences. Further, devices according to the present invention may be used to recognize or track hands, arms, or objects used in a virtual or augmented reality application, especially when wearing head-mounted displays.

35 Further, the devices according to the present invention may be used in mobile audio devices, television devices and gaming devices, as partially explained above. Specifically, the devices according to the present invention may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, the devices according to the present invention
40 may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications and/or for recognizing whether a display is being looked at and/or from which perspective a display is being looked at. Further, devices according to the present invention may be used to explore a room, boundaries,

obstacles, in connection with a virtual or augmented reality application, especially when wearing a head-mounted display.

Further, the devices according to the present invention may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of the devices according to the present invention in mobile applications such as mobile phones, as disclosed above.

Further, the devices according to the present invention may be used for security or surveillance applications. Thus, as an example, at least one device according to the present invention can be combined with one or more digital and/or analogue electronics that will give a signal if an object is within or outside a predetermined area (e.g. for surveillance applications in banks or museums). Specifically, the devices according to the present invention may be used for optical encryption. Detection by using at least one device according to the present invention can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. The devices according to the present invention may further be combined with an active infrared light source to allow detection in low light surroundings. The devices according to the present invention are generally advantageous as compared to active detector systems, specifically since the devices according to the present invention avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector devices. Thus, generally, the devices according to the present invention may be used for an unrecognized and undetectable tracking of moving objects. Additionally, the devices according to the present invention generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using the devices according to the present invention, the devices according to the present invention generally may be used for facial, body and person recognition and identification. Therein, the devices according to the present invention may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, the devices according to the present invention may be used in security devices and other personalized applications.

Further, the devices according to the present invention may be used as 3D barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, the devices according to the present invention generally can be used for surveillance and monitoring of spaces and areas. Thus, the devices according to the present invention may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, the devices according to the present invention may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in

combination with image intensifiers or image enhancement devices and/or photomultipliers. Further, the devices according to the present invention may be used in public spaces or crowded spaces to detect potentially hazardous activities such as commitment of crimes such as theft in a parking lot or unattended objects such as unattended baggage in an airport.

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Further, the devices according to the present invention may advantageously be applied in camera applications such as video and camcorder applications. Thus, the devices according to the present invention may be used for motion capture and 3D-movie recording. Therein, the devices according to the present invention generally provide a large number of advantages over conventional optical devices. Thus, the devices according to the present invention generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing the devices according to the present invention having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are voluminous, such as due to the general need for voluminous beam-splitters. Further, the devices according to the present invention generally may be used for focus/autofocus devices, such as autofocus cameras. Further, the devices according to the present invention may also be used in optical microscopy, especially in confocal microscopy.

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Further, the devices according to the present invention generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, the devices according to the present invention may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, traffic sign detection, traffic sign recognition, lane recognition, rear cross traffic alert, light source recognition for adapting the head light intensity and range depending on approaching traffic or vehicles driving ahead, adaptive frontlighting systems, automatic control of high beam head lights, adaptive cut-off lights in front light systems, glare-free high beam front lighting systems, marking animals, obstacles, or the like by headlight illumination, rear cross traffic alert, and other driver assistance systems such as advanced driver assistance systems, or other automotive and traffic applications. Further, devices according to the present invention may be used in driver assistance systems anticipating maneuvers of the driver beforehand for collision avoidance or the like. Further, the devices according to the present invention can also be used for velocity and/or acceleration measurements, such as by analyzing a first and second time-derivative of position information gained by using the detector according to the present invention. This feature generally may be applicable in automotive technology, transportation technology or general traffic technology. Applications in other fields of technology are feasible. A specific application in an indoor positioning system may be the detection of positioning of passengers in transportation, more specifically to electronically control the use of safety systems such as airbags. The use of an airbag may be prevented in case the passenger is located as such, that the use of an airbag will cause a severe injury. Further, in vehicles such as cars, trains, planes or the like, especially in autonomous vehicles, devices according to the present

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invention may be used to determine whether a driver pays attention to the traffic or is distracted, or asleep, or tired, or incapable of driving such as due to the consumption of alcohol or the like.

5 In these or other applications, generally, the devices according to the present invention may be used as standalone devices or in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, the devices according to the present invention may be used for autonomous driving and safety issues. Further, in these applications, the devices according to the present invention may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. In these applications, the generally passive nature of the devices according to the present invention is advantageous. Thus, since the devices according to the present invention generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. The devices according to the present invention specifically may be used in combination with recognition software, such as standard image recognition software. 10 Thus, signals and data as provided by the devices according to the present invention typically are readily processable and, therefore, generally require lower calculation power than established 3D measurement systems. Given the low space demand, the devices according to the present invention such as cameras may be placed at virtually any place in a vehicle, such as on or behind a window screen, on a front hood, on bumpers, on lights, on mirrors or other places and the like. Various detectors according to the present invention such as one or more detectors based on the effect disclosed within the present invention can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts. Thus, various devices according to the present invention may be combined with one or more other devices according to the present invention and/or conventional sensors, 20 such as in the windows like rear window, side window or front window, on the bumpers or on the lights. 25

A combination of at least one device according to the present invention such as at least one detector according to the present invention with one or more rain detection sensors is also possible. This is due to the fact that the devices according to the present invention generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one device according to the present invention with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions. 30

35 Further, the devices according to the present invention generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, the devices according to the present invention may be used for detecting free parking spaces in parking lots. 40

Further, the devices according to the present invention may be used in the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endo-

scopes, may be named, since, as outlined above, the devices according to the present invention may require a low volume only and may be integrated into other devices. Specifically, the devices according to the present invention having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, the devices according to the present invention may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. This may allow an instant overlay of the position of a medical device, such as an endoscope or a scalpel, with results from medical imaging, such as obtained from magnetic resonance imaging, x-ray imaging, or ultrasound imaging. These applications are specifically valuable e.g. in medical treatments where precise location information is important such as in brain surgery and long-distance diagnosis and tele-medicine. Further, the devices according to the present invention may be used in 3D-body scanning. Body scanning may be applied in a medical context, such as in dental surgery, plastic surgery, bariatric surgery, or cosmetic plastic surgery, or it may be applied in the context of medical diagnosis such as in the diagnosis of myofascial pain syndrome, cancer, body dysmorphic disorder, or further diseases. Body scanning may further be applied in the field of sports to assess ergonomic use or fit of sports equipment. Further, the devices according to the present invention may be used in wearable robots such as in exoskeletons or prosthesis or the like. Body scanning may further be used in the context of clothing, such as to determine a suitable size and fitting of clothes. This technology may be used in the context of tailor-made clothes or in the context of ordering clothes or shoes from the internet or at a self-service shopping device such as a micro kiosk device or customer concierge device. Body scanning in the context of clothing is especially important for scanning fully dressed customers.

Further, the devices according to the present invention may be used in the context of people counting systems, such as to count the number of people in an elevator, a train, a bus, a car, or a plane, or to count the number of people passing a hallway, a door, an aisle, a retail store, a stadium, an entertainment venue, a museum, a library, a public location, a cinema, a theater, or the like. Further, the 3D-function in the people counting system may be used to obtain or estimate further information about the people that are counted such as height, weight, age, physical fitness, or the like. This information may be used for business intelligence metrics, and/or for further optimizing the locality where people may be counted to make it more attractive or safe. In a retail environment, the devices according to the present invention in the context of people counting may be used to recognize returning customers or cross shoppers, to assess shopping behavior, to assess the percentage of visitors that make purchases, to optimize staff shifts, or to monitor the costs of a shopping mall per visitor. Further, people counting systems may be used for anthropometric surveys. Further, the devices according to the present invention may be used in public transportation systems for automatically charging passengers depending on the length of transport. Further, the devices according to the present invention may be used in playgrounds for children, to recognize injured children or children engaged in dangerous activities, to allow additional interaction with playground toys, to ensure safe use of playground toys or the like.

Further, the devices according to the present invention may be used in construction tools, such as a range meter that determines the distance to an object or to a wall, to assess whether a surface is planar, to align objects or place objects in an ordered manner, or in inspection cameras for use in construction environments or the like.

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Further, the devices according to the present invention may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, the devices according to the present invention may be applied in the fields of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing, golf, car racing, laser tag, battlefield simulation etc. The devices according to the present invention can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

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Further, the devices according to the present invention may be used in the field of auto racing or car driver training or car safety training or the like to determine the position of a car or the track of a car, or the deviation from a previous track or an ideal track or the like.

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The devices according to the present invention may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

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The devices according to the present invention further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the devices according to the present invention may also be applied for distance diagnostics.

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Further, the devices according to the present invention may be applied in the field of machine vision. Thus, one or more of the devices according to the present invention may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, the devices according to the present invention may allow for autonomous movement and/or autonomous detection of failures in parts. The devices according to the present invention may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. In robotics, the safe and direct interaction of humans and robots is often an issue, as robots may severely injure humans when they are not recognized. Devices according to the present invention may help robots to position objects and humans better and faster and allow a safe interaction. Given the passive nature of the devices according to the present invention, the devices according to the present invention may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detec-

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tion etc. One particular advantage of the devices according to the present invention is the low likelihood of signal interference. Therefore, multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, the devices according to the present invention generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. The devices according to the present invention can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, the devices according to the present invention may be used for assessment of surface quality, such as for surveying the surface evenness of a product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible. In a manufacturing environment, the devices according to the present invention are especially useful for processing natural products such as food or wood, with a complex 3-dimensional structure to avoid large amounts of waste material. Further, devices according to the present invention may be used to monitor the filling level of tanks, silos etc. Further, devices according to the present invention may be used to inspect complex products for missing parts, incomplete parts, loose parts, low quality parts, or the like, such as in automatic optical inspection, such as of printed circuit boards, inspection of assemblies or sub-assemblies, verification of engineered components, engine part inspections, wood quality inspection, label inspections, inspection of medical devices, inspection of product orientations, packaging inspections, food pack inspections, or the like.

Further, the devices according to the present invention may be used in vehicles, trains, airplanes, ships, spacecraft and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircraft, vehicles and the like may be named. The use of at least one device according to the present invention, such as at least one detector according to the present invention, for monitoring the speed and/or the direction of moving objects is feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one device according to the present invention, such as the at least one detector according to the present invention, specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one device according to the present invention can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. The devices according to the present invention generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. The devices according to the present invention are particularly useful but not limited to e.g. speed control and air traffic control devices. Further, the devices according to the present invention may be used in automated tolling systems for road charges.

The devices according to the present invention generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircraft when landing or starting. Therein, fixed, known active targets may be used for precise

guidance. The same can be used for vehicles driving on dangerous but well defined routes, such as mining vehicles. Further, the devices according to the present invention may be used to detect rapidly approaching objects, such as cars, trains, flying objects, animals, or the like. Further, the devices according to the present invention can be used for detecting velocities or accelerations of objects, or to predict the movement of an object by tracking one or more of its position, speed, and/or acceleration depending on time.

Further, as outlined above, the devices according to the present invention may be used in the field of gaming. Thus, the devices according to the present invention can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of the devices according to the present invention for giving commands are feasible, such as by using one or more of the devices according to the present invention for gesture or facial recognition. The devices according to the present invention may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more devices according to the present invention with one or more IR or VIS light sources is possible. A combination of a detector according to the present invention with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racket, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, the devices according to the present invention may be used to detect and/or track objects that emit light by themselves, such as due to high temperature or further light emission processes. The light emitting part may be an exhaust stream or the like. Further, the devices according to the present invention may be used to track reflecting objects and analyze the rotation or orientation of these objects.

Further, the devices according to the present invention generally may be used in the field of building, construction and cartography. Thus, generally, one or more devices according to the present invention may be used in order to measure and/or monitor environmental areas, e.g. countryside or buildings. Therein, one or more devices according to the present invention may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. The devices according to the present invention can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both

from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like. As an example, the devices according to the present invention may be used in drones or multicopters to monitor buildings, production sites, chimneys, agricultural production environments such as fields, production plants, or landscapes, to support rescue operations, to support work in dangerous environments, to support fire brigades in a burning location indoors or outdoors, or to find or monitor one or more persons or animals, or the like, or for entertainment purposes, such as a drone following and recording one or more persons doing sports such as skiing or cycling or the like, which could be realized by following a helmet, a mark, a beacon device, or the like. Devices according to the present invention could be used recognize obstacles, follow a predefined route, follow an edge, a pipe, a building, or the like, or to record a global or local map of the environment. Further, devices according to the present invention could be used for indoor or outdoor localization and positioning of drones, for stabilizing the height of a drone indoors where barometric pressure sensors are not accurate enough, or for the interaction of multiple drones such as concertized movements of several drones or recharging or refueling in the air or the like.

Further, the devices according to the present invention may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support. Further, the devices according to the present invention may be used in heating or cooling systems such as an air-conditioning system, to locate which part of the room should be brought to a certain temperature or humidity, especially depending on the location of one or more persons. Further, the devices according to the present invention may be used in domestic robots, such as service or autonomous robots which may be used for household chores. The devices according to the present invention may be used for a number of different purposes, such as to avoid collisions or to map the environment, but also to identify a user, to personalize the robot's performance for a given user, for security purposes, or for gesture or facial recognition. As an example, the devices according to the present invention may be used in robotic vacuum cleaners, floor-washing robots, dry-sweeping robots, ironing robots for ironing clothes, animal litter robots, such as dog or cat litter robots, charging robot for electrical vehicles, security robots that detect intruders, robotic lawn mowers, automated pool cleaners, rain gutter cleaning robots, robotic shopping carts, luggage carrying robots, line following robots, laundry robots, ironing robots, window washing robots, toy robots, patient monitoring robots, baby monitoring robots, elderly monitoring robots, children monitoring robots, transport robots, telepresence robots, professional service robots, programmable toy robots, pathfinder robots, social robots providing company to less mobile people, following robots, smart card following robots, psychotherapy robots, or robots translating speech to sign language or sign language to speech. In the context of less mobile people, such as elderly persons, household robots with the devices according to the present invention may be used for picking up objects, transporting objects, and interacting with the objects and the user in a safe way. Further, the devices according to the

present invention may be used in humanoid robots, especially in the context of using humanoid hands to pick up or hold or place objects. Further, the devices according to the present invention may be used in combination with audio interfaces especially in combination with household robots which may serve as a digital assistant with interfaces to online or offline computer applications. Further, the devices according to the present invention may be used in robots that can control switches and buttons in industrial and household purposes. Further, the devices according to the present invention may be used in smart home robots such as Mayfield's Kuri. Further the devices according to the present invention may be used in robots operating with hazardous materials or objects or in dangerous environments. As a non-limiting example, the devices according to the present invention may be used in robots or unmanned remote-controlled vehicles to operate with hazardous materials such as chemicals or radioactive materials especially after disasters, or with other hazardous or potentially hazardous objects such as mines, unexploded arms, or the like, or to operate in or to investigate insecure environments such as near burning objects or post disaster areas or for manned or unmanned rescue operations in the air, in the sea, underground, or the like.

Further, devices according to the present invention may be used for the inspection of adhesive beads, sealing beads, or the like, such as to recognize disruptions, slubs, contractions, asymmetries, local defects, or the like. Further, devices according to the present invention may be used to count objects such as dry fruits on a conveyer belt, such as in difficult situations, such as when fruit of similar color and shape may be in direct contact with each other. Further, devices according to the present invention may be used in quality control of die cast or injection molded parts such as to ensure flawless casting or molding, recognize surface damages, worn out toolings or the like. Further, devices according to the present invention may be used for laserscribing such as for quality control and positioning of the laser. Further, devices according to the present invention may be used for sorting systems, such as to detect position, rotation, and shape of an object, compare it to a database of objects, and classify the object. Further, devices according to the present invention may be used for stamping part inspection, packaging inspection, such as food and pharma packaging inspection, filament inspection, or the like.

Further, devices according to the present invention may be used for navigation purposes, where Global Positioning Systems (GPS) are not sufficiently reliable. GPS signals commonly use radio waves that are can be blocked or difficult to receive indoors or outdoors in valleys or in forests below the treeline. Further, especially in unmanned autonomous vehicles, the weight of the system may be critical. Especially unmanned autonomous vehicles need high-speed position data for reliable feedback and stability of their control systems. Using devices according to the present invention may allow short time response and positioning without adding weight due to a heavy device.

Further, the devices according to the present invention may be used in household, mobile or entertainment devices, such as a refrigerator, a microwave, a washing machine, a window blind or shutter, a household alarm, an air condition devices, a heating device, a television, an audio device, a smart watch, a mobile phone, a phone, a dishwasher, a stove or the like, to detect the

presence of a person, to monitor the contents or function of the device, or to interact with the person and/or share information about the person with further household, mobile or entertainment devices.

- 5 Further, the devices according to the present invention may be used to support elderly or disabled persons or persons with limited or no vision, such as in household chores or at work such as in devices for holding, carrying, or picking objects, or in a safety system with optical or acoustical signals signaling obstacles in the environment.
- 10 The devices according to the present invention may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, the devices according to the present invention may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices. Further, the devices according to the present invention may
- 15 be used to monitor the growth of plants in a field or greenhouse, in particular to adjust the amount of water or fertilizer or crop protection products for a given region in the field or greenhouse or even for a given plant. Further, in agricultural biotechnology, the devices according to the present invention may be used to monitor the size and shape of plants.
- 20 Further, devices according to the present invention may be used to automatically remove weeds such as with mechanical means, such as to avoid the use of herbicides. Further, devices according to the present invention may be used in the field of agriculture, in particular to detect and/or locate specific insects such as to decide whether or not to apply a crop protection or fertilization substance, such as to reduce the amount of applied substance or to protect specific
- 25 groups of animals such as bees.

Further, devices according to the present invention may be used to guide users during a shaving, hair cutting, or cosmetics procedure, or the like. Further, devices according to the present invention may be used to record or monitor what is played on an instrument, such as a violin.

- 30 Further, devices according to the present invention may be used in smart household appliances such as a smart refrigerator, such as to monitor the contents of the refrigerator and transmit notifications depending on the contents. Further, devices according to the present invention may be used for monitoring or tracking populations of humans, animals, or plants, such as deer or tree populations in forests. Further, devices according to the present invention may be used
- 35 in harvesting machines, such as for harvesting crops, flowers or fruits, such as grapes, corn, hops, apples, grains, rice, strawberries, asparagus, tulips, roses, soy beans, or the like. Further, devices according to the present invention may be used to monitor the growth of plants, animals, algae, fish, or the like, such as in breeding, food production, agriculture or research applications, to control irrigation, fertilization, humidity, temperature, use of herbicides, insecticides, fungicides, rodenticides, or the like. Further, devices according to the present invention may be
- 40 used in feeding machines for animals or pets, such as for cows, pigs, cats, dogs, birds, fish, or the like. Further, devices according to the present invention may be used in animal product production processes, such as for collecting milk, eggs, fur, meat, or the like, such as in automated

milking or butchering processes. Further, devices according to the present invention may be used for automated seeding machines, or sowing machines, or planting machines such as for planting corn, garlic, trees, salad or the like. Further, devices according to the present invention may be used to assess or monitor weather phenomena, such as clouds, fog, or the like, or to warn from danger of avalanches, tsunamis, gales, earthquakes, thunder storms, or the like. Further, devices according to the present invention may be used to measure motions, shocks, concussions, or the like such as to monitor earthquake risk. Further, devices according to the present invention may be used in traffic technology to monitor dangerous crossings, to control traffic lights depending on traffic, to monitor public spaces, to monitor roads, gyms, stadiums, ski resorts, public events, or the like. Further, devices according to the present invention may be used in medical applications such as to monitor or analyze tissues, medical or biological assays, changes in tissues such as in moles or melanoma or the like, to count bacteria, blood cells, cells, algae, or the like, for retina scans, breath or pulse measurements, gastroscopy, patient surveillance, or the like. Further, devices according to the present invention may be used to monitor the shape, size, or circumference of drops, streams, jets, or the like or to analyze, assess, or monitor profiles or gas or liquid currents such as in a wind channel, or the like. Further, devices according to the present invention may be used to warn drivers such as car or train drivers when they are getting sick or tired or the like. Further, devices according to the present invention may be used in material testing to recognize strains or tensions or fissures, or the like. Further, devices according to the present invention may be used in sailing to monitor and optimize sail positions such as automatically. Further, devices according to the present invention may be used for fuel level gauges.

Further, the devices according to the present invention may be combined with sensors to detect chemicals or pollutants, electronic nose chips, microbe sensor chips to detect bacteria or viruses or the like, Geiger counters, tactile sensors, heat sensors, or the like. This may for example be used in constructing smart robots which are configured for handling dangerous or difficult tasks, such as in treating highly infectious patients, handling or removing highly dangerous substances, cleaning highly polluted areas, such as highly radioactive areas or chemical spills, or for pest control in agriculture.

One or more devices according to the present invention can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of the devices according to the present invention, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, the devices according to the present invention may be used in inspections and maintenance, such as pipeline inspection gauges. Further, in a production environment, the devices according to the present invention may be used to work with objects of a badly defined shape such as naturally grown objects, such as sorting vegetables or other natural products by shape or size or cutting products such as meat or objects that are manufactured with a precision that is lower than the precision needed for a processing step.

Further, the devices according to the present invention may be used in local navigation systems to allow autonomously or partially autonomously moving vehicles or multicopters or the like through an indoor or outdoor space. A non-limiting example may comprise vehicles moving through an automated storage for picking up objects and placing them at a different location.

- 5 Indoor navigation may further be used in shopping malls, retail stores, museums, airports, or train stations, to track the location of mobile goods, mobile devices, baggage, customers or employees, or to supply users with a location specific information, such as the current position on a map, or information on goods sold, or the like.
- 10 Further, the devices according to the present invention may be used to ensure safe driving of motorcycles such as driving assistance for motorcycles by monitoring speed, inclination, upcoming obstacles, unevenness of the road, or curves or the like. Further, the devices according to the present invention may be used in trains or trams to avoid collisions.
- 15 Further, the devices according to the present invention may be used in handheld devices, such as for scanning packaging or parcels to optimize a logistics process. Further, the devices according to the present invention may be used in further handheld devices such as personal shopping devices, RFID-readers, handheld devices for use in hospitals or health environments such as for medical use or to obtain, exchange or record patient or patient health related information, smart badges for retail or health environments, or the like.
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- As outlined above, the devices according to the present invention may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further,
- 25 the devices according to the present invention may be used in logistics applications. Thus, the devices according to the present invention may be used for optimized loading or packing containers or vehicles. Further, devices according to the present invention may be recalibrated using bar codes, QR-codes or prerecorded symbols of known size by using the at least one image matrix and comparing the prerecorded size with a measured property of the recorded image of
- 30 the bar code, QR code or prerecorded symbol, such as by comparing the width or height of the symbol with a prerecorded value. Further, the devices according to the present invention may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, the devices according to the present invention
- 35 may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, the devices according to the present invention may be used for process control in production, e.g. for observing filling level of tanks. Further, the devices according to the present invention may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, the devices according to
- 40 the present invention may be used for analyzing 3D-quality marks. Further, the devices according to the present invention may be used in manufacturing tailor-made goods such as tooth inlays, dental braces, prosthesis, clothes or the like. The devices according to the present invention may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the

like. Further, the devices according to the present invention may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

5 Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term photography generally refers to the technology of acquiring image information of at least
10 one object. As further used herein, a camera generally is a device adapted for performing photography. As further used herein, the term digital photography generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term 3D photography generally
15 refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital
20 video sequences.

Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at
25 least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

30 As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively,
35 however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

As outlined above and as will be outlined in further detail below, the present invention preferably may be applied in the field of human-machine interfaces, in the field of sports and/or in the field
40 of computer games. Thus, preferably, the object may be selected from the group consisting of: an article of sports equipment, preferably an article selected from the group consisting of a racket, a club, a bat, an article of clothing, a hat, a shoe. Other embodiments are feasible.

As used herein, the object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

With regard to the coordinate system for determining the position of the object, which may be a coordinate system of the detector, the detector may constitute a coordinate system in which an optical axis of the detector forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The detector may be a device configured for providing at least one item of information on the position of the at least one object and/or a part thereof. Thus, the position may refer to an item of information fully describing the position of the object or a part thereof, preferably in the coordinate system of the detector, or may refer to a partial information, which only partially describes the position. The detector generally may be a device adapted for detecting light beams, such as the light beams propagating from the beacon devices towards the detector.

The evaluation device and the detector may fully or partially be integrated into a single device. Thus, generally, the evaluation device also may form part of the detector. Alternatively, the evaluation device and the detector may fully or partially be embodied as separate devices. The detector may comprise further components.

The detector may be a stationary device or a mobile device. Further, the detector may be a stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The detector specifically may be used to record a light-field behind a lens or lens system of the detector, comparable to a plenoptic or light-field camera. Thus, specifically, the detector may be embodied as a light-field camera adapted for acquiring images in multiple focal planes, such as simultaneously. The term light-field, as used herein, generally refers to the spatial light propagation of light inside the detector such as inside camera. The detector according to the present invention, specifically having a stack of optical sensors, may have the capability of directly recording a light-field within the detector or camera, such as behind a lens. The plurality of sensors may record images at different distances from the lens. Using, e.g., convolution-based algorithms such as "depth from focus" or "depth from defocus", the propagation direction, focus points, and spread of the light behind the lens can be modeled. From the modeled propagation of light behind the lens, images at various distances to the lens can be extracted, the depth of field can be optimized, pictures that are in focus at various distances can be extracted, or distances of objects can be calculated. Further information may be extracted.

The use of several optical sensors further allows for correcting lens errors in an image processing step after recording the images. Optical instruments often become expensive and challenging in construction, when lens errors need to be corrected. These are especially problematic in microscopes and telescopes. In microscopes, a typical lens error is that rays of varying distance to the optical axis are distorted differently (spherical aberration). In telescopes, varying the focus may occur from differing temperatures in the atmosphere. Static errors such as spherical aberration or further errors from production may be corrected by determining the errors in a calibration step and then using a fixed image processing such as fixed set of pixels and sensor, or more involved processing techniques using light propagation information. In cases in which lens errors are strongly time-dependent, i.e. dependent on weather conditions in telescopes, the lens errors may be corrected by using the light propagation behind the lens, calculating extended depth of field images, using depth from focus techniques, and others.

The detector according to the present invention may further allow for color detection. For color detection, a plurality of optical sensors having different spectral properties may be used, and sensor signals of these optical sensors may be compared. Further, the devices according to the present invention may be used in the context of gesture recognition. In this context, gesture recognition in combination with devices according to the present invention may, in particular, be used as a human-machine interface for transmitting information via motion of a body, of body parts or of objects to a machine. Herein, the information may, preferably, be transmitted via a motion of hands or hand parts, such as fingers, in particular, by pointing at objects, applying sign language, such as for deaf people, making signs for numbers, approval, disapproval, or the like, by waving the hand, such as when asking someone to approach, to leave, or to greet a person, to press an object, to take an object, or, in the field of sports or music, in a hand or finger exercise, such as a warm-up exercise. Further, the information may be transmitted by motion of arms or legs, such as rotating, kicking, grabbing, twisting, rotating, scrolling, browsing, pushing, bending, punching, shaking, arms, legs, both arms, or both legs, or a combination of arms and legs, such as for a purpose of sports or music, such as for entertainment, exercise, or training function of a machine. Further, the information may be transmitted by motion of the

whole body or major parts thereof, such as jumping, rotating, or making complex signs, such as sign language used at airports or by traffic police in order to transmit information, such as “turn right”, “turn left”, “proceed”, “slow down”, “stop”, or “stop engines”, or by pretending to swim, to dive, to run, to shoot, or the like, or by making complex motions or body positions such as in
5 yoga, pilates, judo, karate, dancing, or ballet. Further, the information may be transmitted by using a real or mock-up device for controlling a virtual device corresponding to the mock-up device, such as using a mock-up guitar for controlling a virtual guitar function in a computer program, using a real guitar for controlling a virtual guitar function in a computer program, using
10 a real or a mock-up book for reading an e-book or moving pages or browsing through in a virtual document, using a real or mock-up pen for drawing in a computer program, or the like. Further, the transmission of the information may be coupled to a feedback to the user, such as a sound, a vibration, or a motion.

15 In the context of music and/or instruments, devices according to the present invention in combination with gesture recognition may be used for exercising purposes, control of instruments, recording of instruments, playing or recording of music via use of a mock-up instrument or by only pretending to have a instrument present such as playing air guitar, such as to avoid noise or make recordings, or, for conducting of a virtual orchestra, ensemble, band, big band, choir, or the like, for practicing, exercising, recording or entertainment purposes or the like.

20 Further, in the context of safety and surveillance, devices according to the present invention in combination with gesture recognition may be used to recognize motion profiles of persons, such as recognizing a person by the way of walking or moving the body, or to use hand signs or movements or signs or movements of body parts or the whole body as access or identification
25 control such as a personal identification sign or a personal identification movement.

Further, in the context of smart home applications or internet of things, devices according to the present invention in combination with gesture recognition may be used for central or non-central
30 control of household devices which may be part of an interconnecting network of home appliances and/or household devices, such as refrigerators, central heating, air condition, microwave ovens, ice cube makers, or water boilers, or entertainment devices, such as television sets, smart phones, game consoles, video recorders, DVD players, personal computers, laptops, tablets, or combinations thereof, or a combination of household devices and entertainment devices.

35 Further, in the context of virtual reality or of augmented reality, devices according to the present invention in combination with gesture recognition may be used to control movements or function of the virtual reality application or of the augmented reality application, such as playing or controlling a game using signs, gestures, body movements or body part movements or the like,
40 moving through a virtual world, manipulating virtual objects, practicing, exercising or playing sports, arts, crafts, music or games using virtual objects such as a ball, chess figures, go stones, instruments, tools, brushes.

Further, in the context of medicine, devices according to the present invention in combination with gesture recognition may be used to support rehabilitation training, remote diagnostics, or to monitor or survey surgery or treatment, to overlay and display medical images with positions of medical devices, or to overlay display prerecorded medical images such as from magnetic resonance tomography or x-ray or the like with images from endoscopes or ultra sound or the like that are recorded during a surgery or treatment.

Further, in the context of manufacturing and process automation, devices according to the present invention in combination with gesture recognition may be used to control, teach, or program robots, drones, unmanned autonomous vehicles, service robots, movable objects, or the like, such as for programming, controlling, manufacturing, manipulating, repairing, or teaching purposes, or for remote manipulating of objects or areas, such as for safety reasons, or for maintenance purposes.

Further, in the context of business intelligence metrics, devices according to the present invention in combination with gesture recognition may be used for people counting, surveying customer movements, areas where customers spend time, objects, customers test, take, probe, or the like.

Further, devices according to the present invention may be used in the context of do-it-yourself or professional tools, especially electric or motor driven tools or power tools, such as drilling machines, saws, chisels, hammers, wrenches, staple guns, disc cutters, metals shears and nibblers, angle grinders, die grinders, drills, hammer drills, heat guns, wrenches, sanders, engravers, nailers, jig saws, biscuit joiners, wood routers, planers, polishers, tile cutters, washers, rollers, wall chasers, lathes, impact drivers, jointers, paint rollers, spray guns, morticers, or welders, in particular, to support precision in manufacturing, keeping a minimum or maximum distance, or for safety measures.

Further, the devices according to the present invention may be used to aid visually impaired persons. Further, devices according to the present invention may be used in touch screen such as to avoid direct contact such as for hygienic reasons, which may be used in retail environments, in medical applications, in production environments, or the like. Further, devices according to the present invention may be used in agricultural production environments such as in stable cleaning robots, egg collecting machines, milking machines, harvesting machines, farm machinery, harvesters, forwarders, combine harvesters, tractors, cultivators, ploughs, destoners, harrows, strip tills, broadcast seeders, planters such as potato planters, manure spreaders, sprayers, sprinkler systems, swathers, balers, loaders, forklifts, mowers, or the like.

Further, devices according to the present invention may be used for selection and/or adaption of clothing, shoes, glasses, hats, prosthesis, dental braces, for persons or animals with limited communication skills or possibilities, such as children or impaired persons, or the like. Further, devices according to the present invention may be used in the context of warehouses, logistics, distribution, shipping, loading, unloading, smart manufacturing, industry 4.0, or the like. Further,

in a manufacturing context, devices according to the present invention may be used in the context of processing, dispensing, bending, material handling, or the like.

5 The evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as
10 one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the evaluation device may comprise one or more data storage devices. Further, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

15 The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program adapted for performing or supporting one or more or even all of the method steps of the method according to the present invention. As an example, one or more algorithms may be implemented which, by using the sensor signals as input variables, may determine the position of the object.
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The evaluation device can be connected to or may comprise at least one further data processing device that may be used for one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the
25 optical sensor and/or by the evaluation device. The data processing device, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted
30 information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAP™), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks
35 such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

40 The evaluation device and/or the data processing device may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue

interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The evaluation device and/or the data processing device may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces
5 ports. The evaluation device and the data processing device may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The evaluation device and/or the data processing device may be connected by or have one or
10 more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

Possible embodiments of a single device incorporating one or more of the detectors according to the present invention, the evaluation device or the data processing device, such as incorporating one or more of the optical sensor, optical systems, evaluation device, communication
15 device, data processing device, interfaces, system on a chip, display devices, or further electronic devices, are: mobile phones, personal computers, tablet PCs, televisions, game consoles or further entertainment devices. In a further embodiment, the 3D-camera functionality which will be outlined in further detail below may be integrated in devices that are available with conventional 2D-digital cameras, without a noticeable difference in the housing or appearance of the device, where the noticeable difference for the user may only be the functionality of obtaining and or processing 3D information. Further, devices according to the present invention may
20 be used in 360° digital cameras or surround view cameras.

Specifically, an embodiment incorporating the detector and/or a part thereof such as the evaluation device and/or the data processing device may be: a mobile phone incorporating a display
30 device, a data processing device, the optical sensor, optionally the sensor optics, and the evaluation device, for the functionality of a 3D camera. The detector according to the present invention specifically may be suitable for integration in entertainment devices and/or communication devices such as a mobile phone.

A further embodiment of the present invention may be an incorporation of the detector or a part thereof such as the evaluation device and/or the data processing device in a device for use in automotive, for use in autonomous driving or for use in car safety systems such as Daimler's Intelligent Drive system, wherein, as an example, a device incorporating one or more of the optical sensors, optionally one or more optical systems, the evaluation device, optionally a communication device, optionally a data processing device, optionally one or more interfaces, optionally a system on a chip, optionally one or more display devices, or optionally further electronic devices may be part of a vehicle, a car, a truck, a train, a bicycle, an airplane, a ship, a motorcycle. In automotive applications, the integration of the device into the automotive design
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may necessitate the integration of the optical sensor, optionally optics, or device at minimal visibility from the exterior or interior. The detector or a part thereof such as the evaluation device and/or the data processing device may be especially suitable for such integration into automotive design.

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As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1 mm, preferably in the range of 780 nm to 3.0 micrometers. The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

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The term light beam generally may refer to an amount of light emitted and/or reflected into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beams may be or may comprise one or more Gaussian light beams such as a linear combination of Gaussian light beams, which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

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The detector according to the present invention may further be combined with one or more other types of sensors or detectors. Thus, the detector may further comprise at least one additional detector. The at least one additional detector may be adapted for detecting at least one parameter, such as at least one of: a parameter of a surrounding environment, such as a temperature and/or a brightness of a surrounding environment; a parameter regarding a position and/or orientation of the detector; a parameter specifying a state of the object to be detected, such as a position of the object, e.g. an absolute position of the object and/or an orientation of the object in space. Thus, generally, the principles of the present invention may be combined with other measurement principles in order to gain additional information and/or in order to verify measurement results or reduce measurement errors or noise.

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The human-machine interface may comprise a plurality of beacon devices which are adapted to be at least one of directly or indirectly attached to the user and held by the user. Thus, the beacon devices each may independently be attached to the user by any suitable means, such as by an appropriate fixing device. Additionally or alternatively, the user may hold and/or carry the at least one beacon device or one or more of the beacon devices in his or her hands and/or by wearing the at least one beacon device and/or a garment containing the beacon device on a body part.

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The beacon device generally may be an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as outlined above or as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user in a direct or indirect way and/or may be carried or held by the user. The attachment may take place by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

Additionally or alternatively, the beacon devices may be at least one of attached to an object and integrated into an object held by the user, which, in the sense of the present invention, shall be included into the meaning of the option of the user holding the beacon devices. Thus, as will be outlined in further detail below, the beacon devices may be attached to or integrated into a control element which may be part of the human-machine interface and which may be held or carried by the user, and of which the orientation may be recognized by the detector device. Thus, generally, the present invention also refers to a detector system comprising at least one detector device according to the present invention and which, further, may comprise at least one object, wherein the beacon devices are one of attached to the object, held by the object and integrated into the object. As an example, the object preferably may form a control element, the orientation of which may be recognized by a user. Thus, the detector system may be part of the human-machine interface as outlined above or as outlined in further detail below. As an example, the user may handle the control element in a specific way in order to transmit one or more items of information to a machine, such as in order to transmit one or more commands to the machine.

Alternatively, the detector system may be used in other ways. Thus, as an example, the object of the detector system may be different from a user or a body part of the user and, as an example, may be an object which moves independently from the user. As an example, the detector system may be used for controlling apparatuses and/or industrial processes, such as manufacturing processes and/or robotics processes. Thus, as an example, the object may be a machine and/or a machine part, such as a robot arm, the orientation of which may be detected by using the detector system.

The human-machine interface may be adapted in such a way that the detector device generates at least one item of information on the position of the user or of at least one body part of the user. Specifically in case a manner of attachment of the at least one beacon device to the user is known, by evaluating the position of the at least one beacon device, at least one item of information on a position and/or an orientation of the user or of a body part of the user may be gained.

The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The object, which may form part of the detector system, may generally have an arbitrary shape. Preferably, the object being part of the detector system, as outlined above, may be a control element which may be handled by a user, such as manually. As an example, the control element may be or may comprise at least one element selected from the group consisting of: a glove, a jacket, a hat, shoes, trousers and a suit, a stick that may be held by hand, a bat, a club, a racket, a cane, a toy, such as a toy gun. Thus, as an example, the detector system may be part of the human-machine interface and/or of the entertainment device.

As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device. The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one orientation of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or orientation and/or action required for gaming. As an example, one or more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

The entertainment device as a part or as a whole, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance

with the at least one item of information. Thus, the entertainment device might include one or more controllers which might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller
5 might include one or more data processing devices, such as one or more computers and/or microcontrollers.

As further used herein, a tracking system is a device which is adapted to gather information on a series of past positions of the at least one object and/or at least one part of the object. Additionally,
10 the tracking system may be adapted to provide information on at least one predicted future position and/or orientation of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at least one computer or microcontroller. Again, the at least one track controller may
15 fully or partially comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as
20 at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The track controller is adapted to track a series of positions of the object at specific points in time, such as by recording groups of data or data pairs, each group of data or data pair comprising at least one position information and at least one time information.
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The tracking system may further comprise the at least one detector system according to the present invention. Thus, besides the at least one detector and the at least one evaluation device and the optional at least one beacon device, the tracking system may further comprise the object itself or a part of the object, such as at least one control element comprising the beacon
30 devices or at least one beacon device, wherein the control element is directly or indirectly attachable to or integratable into the object to be tracked.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of:
35 a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
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As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

The proposed devices and methods provide a large number of advantages over known detectors of this kind. Thus, the detector generally may avoid the shortcomings of the known prior art systems disclosed above. Specifically, the detector may avoid the use of FiP sensors, thereby allowing for e.g. using simple and cheap and commercially available semiconductor sensors such as silicon photodiodes. These photodiodes generally do not show a luminance dependency, and the method disclosed above is generally independent from the brightness of the scenery and/or the brightness of the light spot on the light beam. Consequently, a range of measurement in terms of luminance or total power of the light beam entering the detector is generally larger in the present invention as compared to many of the devices disclosed above. Further, the measurement by using the detector according to the present invention is generally independent from the target spot size, i.e. from either the size of the object, the size of a light spot projected onto the object or the size of a beacon device being one or more of attached to the object, integrated into the object or held by the object.

The detector according to the present invention may be realized as a simple device combining the functionality of distance measurement or measurement of z-coordinates, with the additional option of measuring one or more transversal coordinates, thereby integrating the functionality of a PSD.

The result of the measurement, i.e. the position determined by the detector such as the at least one longitudinal coordinate, may be rendered widely independent from the brightness of the scenery and/or the brightness of the object, the brightness of the at least one beacon device, the brightness of the at least one illumination source or the total power of the light beam propagating from the object to the detector. Further, due to this independence and wide range of measurement in terms of brightness, reflective objects or non-reflective objects may be used.

When mentioning a range of measurement, the range of measurement may both refer to a range of brightness which may be used with the detector according to the present invention, such as a range of total powers of the light beam, or may refer to a range of distances between the detector and the object which may be measured. Conventional detectors, such as according to one or more of the documents listed above, are typically limited in both ranges of measurement. The use of the quotient signal, as mentioned above, contrarily, provides a wide range of

continuously and monotonously decreasing or increasing functions which may be used to determine the longitudinal coordinate from the quotient signal. Consequently, a very wide range of measurement in terms of distance between the object and the detector is given. Similarly, due to the general independence of the quotient signal from the total power of the light beam, at least as long as no saturation of one or both of the optical sensors is reached, this also provides a very wide range of measurement in terms of brightness, i.e. in terms of total power of the light beam.

The light beam, within the detector, generally may propagate along an optical axis of the detector. The light beam, however, may also propagate in other ways than along the optical axis. As an example, an illumination light beam may be generated which propagates along the optical axis or which propagates parallel to the optical axis or at an angle to the optical axis which is different from 0° , such as an angle of 1° to 20° . Other embodiments are feasible.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1: A detector for determining a position of at least one object, the detector comprising

- at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each optical sensor is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area by a reflection light beam propagating from the object to the detector, wherein the sensor element is adapted to determine at least one reflection image;
- at least one evaluation device, wherein the evaluation device is adapted to select at least one reflection feature of the reflection image, wherein the evaluation device is configured for determining at least one longitudinal region of the selected reflection feature of the reflection image by evaluating a combined signal Q from the sensor signals, wherein the evaluation device is adapted to determine at least one displacement region in at least one reference image corresponding to the longitudinal region, wherein the evaluation device is adapted to match the selected reflection feature with at least one reference feature within the displacement region.

Embodiment 2: The detector according to the preceding embodiment, wherein the evaluation device is adapted to determine a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device is adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement.

Embodiment 3: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

Embodiment 4: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region.

Embodiment 5: The detector according to any one of the preceding embodiments, wherein the evaluation device is adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q , wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm\epsilon$.

Embodiment 6: The detector according to the preceding embodiment, wherein the reference image and the reflection image are images of the object determined at different spatial positions having a fixed distance, wherein the evaluation device is adapted to determine an epipolar line in the reference image.

Embodiment 7: The detector according to the preceding embodiment, wherein the displacement region extends along the epipolar line, wherein the evaluation device is adapted to determine the reference feature along the epipolar line corresponding to the longitudinal coordinate z and to determine an extent of the displacement region along the epipolar line corresponding to the error interval $\pm\epsilon$.

Embodiment 8: The detector according to the preceding embodiment, wherein the evaluation device is configured to perform the following steps:

- Determining the displacement region for the image position of each reflection feature;
- Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
- Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.

Embodiment 9: The detector according to any one of the preceding embodiments, wherein the evaluation device is adapted to match the selected feature of the reflection image with the reference feature within the displacement region by using at least one evaluation algorithm considering the determined longitudinal coordinate z .

Embodiment 10: The detector according to the preceding embodiments, wherein the evaluation algorithm is a linear scaling algorithm.

Embodiment 11: The detector according to any one of the preceding embodiments, wherein the evaluation device is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o.

Embodiment 12: The detector according to the preceding embodiment, wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the reflection light beam.

Embodiment 13: The detector according to any of the two preceding embodiments, wherein the beam profile is selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

Embodiment 14: The detector according to any one of the three preceding embodiments, wherein the light sensitive areas are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.

Embodiment 15: The detector according to the preceding embodiment, wherein the evaluation device is configured to determine the first area of the beam profile and the second area of the beam profile.

Embodiment 16: The detector according to any one of the two preceding embodiments, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile.

Embodiment 17: The detector according to any one of the two preceding embodiments, wherein the beam profile is segmented by a secant or chord passing the center of the beam profile by a given distance, whereas the first segment comprises the center information, whereas the second segment comprises essentially edge information.

Embodiment 18: The detector according to the preceding embodiment, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.

Embodiment 19: The detector according to any one of the two preceding embodiments, wherein the evaluation device is configured to derive the combined signal Q by one or more of dividing the edge information and the center information, dividing multiples of the edge information and the center information, dividing linear combinations of the edge information and the center information.

Embodiment 20: The detector according to any one of the preceding embodiments, wherein the at least one evaluation device is configured for evaluating the sensor signals, by

- a) determining at least one optical sensor having the highest sensor signal and forming at least one center signal;
- b) evaluating the sensor signals of the optical sensors of the matrix and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the longitudinal region by evaluating the combined signal.

Embodiment 21: The detector according to the preceding embodiment, wherein the center signal is selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Embodiment 22: The detector according to any one of the two preceding embodiments, wherein the sum signal is selected from the group consisting of: an average over all sensor signals of the matrix; a sum of all sensor signals of the matrix; an integral of all sensor signals of the matrix; an average over all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; an integral of all sensor signals of the matrix except for sensor signals from those optical sensors contributing to the center signal; a sum of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor

signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor having the highest sensor signal.

5 Embodiment 23: The detector according to any one of the three preceding embodiments, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.

10 Embodiment 24: The detector according to any one of the preceding embodiment, wherein the evaluation device comprises at least one divider, wherein the divider is configured for deriving the quotient signal.

15 Embodiment 25: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one optical element, wherein the optical element is at least one element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element.

20 Embodiment 26: The detector according to any one of the six preceding embodiments, wherein the evaluation device is further configured for determining at least one transversal coordinate of the object by evaluating a transversal position of the at least one optical sensor having the highest sensor signal.

25 Embodiment 27: The detector according to any one of the preceding embodiments, wherein the sensor element comprises at least one CMOS sensor.

30 Embodiment 28: The detector according to any one of the preceding embodiments, wherein the detector comprises at least one illumination source, wherein the illumination source is adapted to generate at least one illumination pattern for illumination of the object.

35 Embodiment 29: The detector according to preceding embodiment, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern or a random point pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern, in particular a rotated hexagonal pattern and/or a displaced hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines.

40 Embodiment 30: The detector according to any one of the two preceding embodiments, wherein the sensor element is adapted to determine at least one reflection pattern, wherein the evaluation device is adapted to select at least one feature of the reflection pattern and to determine the

longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals.

5 Embodiment 31: The detector according to the preceding embodiment, wherein the reference image is an image of the illumination pattern at an image plane at a position of the illumination source, wherein the evaluation device is adapted to determine the displacement region in the reference image corresponding to the longitudinal region of the selected feature of the reflection pattern, wherein the evaluation device is adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region.

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Embodiment 32: The detector according to any one of the four preceding embodiments, wherein the illumination source and the sensor element are separated by a fixed distance.

15 Embodiment 33: The detector according to any one of the five preceding embodiments, wherein the detector comprises at least two sensor elements each having a matrix of optical sensors.

20 Embodiment 34: The detector according to the preceding embodiment, where at least one first sensor element and at least one second sensor element are positioned at different spatial positions, wherein a relative distance between the first sensor element and the second element is fixed.

25 Embodiment 35: The detector according to any one of the two preceding embodiments, wherein the at least one first sensor element is adapted to determine at least one first reflection pattern and the at least one second sensor element is adapted to determine at least one second reflection pattern, wherein the evaluation device is adapted to select at least one image determined by the first sensor element or the second sensor element as reflection image and to select at least one image determined by the other one of the first sensor element or the second sensor element as reference image.

30 Embodiment 36: A detector system for determining a position of at least one object, the detector system comprising at least one detector according to any one of the preceding embodiments, the detector system further comprising at least one beacon device adapted to direct at least one light beam towards the detector, wherein the beacon device is at least one of attachable to the object, holdable by the object and integratable into the object.

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40 Embodiment 37: A human-machine interface for exchanging at least one item of information between a user and a machine, wherein the human-machine interface comprises at least one detector system according to the preceding embodiment, wherein the at least one beacon device is adapted to be at least one of directly or indirectly attached to the user and held by the user, wherein the human-machine interface is designed to determine at least one position of the user by means of the detector system, wherein the human-machine interface is designed to assign to the position at least one item of information.

Embodiment 38: An entertainment device for carrying out at least one entertainment function, wherein the entertainment device comprises at least one human-machine interface according to the preceding embodiment, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein
5 the entertainment device is designed to vary the entertainment function in accordance with the information.

Embodiment 39: A tracking system for tracking a position of at least one movable object, the tracking system comprising at least one detector system according to any one of the preceding
10 embodiments referring to a detector system, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object at specific points in time.

Embodiment 40: A scanning system for determining a depth profile of a scenery, the scanning
15 system comprising at least one detector according to any of the preceding embodiments referring to a detector, the scanning system further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

Embodiment 41: A camera for imaging at least one object, the camera comprising at least one
20 detector according to any one of the preceding embodiments referring to a detector.

Embodiment 42: A method for determining a position of at least one object by using at least one detector according to any one of the preceding embodiments referring to a detector, the method comprising the following steps:

- 25 - generating at least two sensor signals in response to an illumination of at least one sensor element having a matrix of optical sensors by at least one reflection light beam propagating from the object to the detector, the optical sensors each having a light-sensitive area;
- determining at least one reflection image by using the sensor element;
- 30 - selecting at least one reflection feature of the reflection image and evaluating the sensor signals by using at least one evaluation device, thereby, determining at least one longitudinal region of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
- determining at least one displacement region in at least one reference image corresponding to the longitudinal region;
- 35 - matching the selected reflection feature with at least one reference feature within the displacement region.

Embodiment 43: The method according to the preceding embodiment, wherein a displacement
40 of the matched reference feature and the selected reflection feature is determined, wherein a longitudinal information of the matched feature is determined using a predetermined relationship between a longitudinal coordinate and the displacement.

Embodiment 44: The method according to any one of the preceding embodiments referring to a method, wherein the deriving of the combined signal Q comprises one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.

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Embodiment 45: A use of the detector according to any one of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; an outdoor application; a mobile application; a communication application; a tracking application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application.

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Brief description of the figures

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Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary

embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

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Specifically, in the figures:

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Figure 1 shows a first embodiment of a detector according to the present invention;

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Figure 2 shows a further embodiment of the detector according to the present invention;

Figure 3 shows an embodiment of a detector system, camera, entertainment device, tracking system;

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Figure 4 shows an embodiment of determining a longitudinal coordinate z using a line pattern;

Figures 5A to C show three embodiments of a hexagonal illumination pattern; and

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Figure 6 shows an embodiment of a scanning system.

Detailed description of the embodiments:

Fig. 1 shows in a highly schematic fashion a first embodiment of a detector 110 for determining a position of at least one object 112. The detector 110 comprises at least one sensor element 116 having a matrix 118 of optical sensors 120. The optical sensors 120 each have a light-sensitive area 122.

The sensor element 116 may be formed as a unitary, single device or as a combination of several devices. The matrix 118 specifically may be or may comprise a rectangular matrix having one or more rows and one or more columns. The rows and columns specifically may be arranged in a rectangular fashion. However, other arrangements are feasible, such as nonrectangular arrangements. As an example, circular arrangements are also feasible, wherein the elements are arranged in concentric circles or ellipses about a center point. For example, the matrix 118 may be a single row of pixels. Other arrangements are feasible.

The optical sensors 120 of the matrix 118 specifically may be equal in one or more of size, sensitivity and other optical, electrical and mechanical properties. The light-sensitive areas 122 of all optical sensors 120 of the matrix 118 specifically may be located in a common plane, the common plane preferably facing the object 112, such that a light beam propagating from the object to the detector 110 may generate a light spot on the common plane. The light-sensitive area 122 may specifically be located on a surface of the respective optical sensor 120. Other embodiments, however, are feasible. The optical sensors 120 may comprise for example, at least one CCD and/or CMOS device. As an example, the optical sensors 120 may be part of or constitute a pixelated optical device. As an example, the optical sensors may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area 122.

The optical sensors 120 specifically may be or may comprise photodetectors, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensors 120 may be sensitive in the infrared spectral range. All of the optical sensors 120 of the matrix 118 or at least a group of the optical sensors 120 of the matrix 118 specifically may be identical. Groups of identical optical sensors 120 of the matrix 118 specifically may be provided for different spectral ranges, or all optical sensors may be identical in terms of spectral sensitivity. Further, the optical sensors 120 may be identical in size and/or with regard to their electronic or optoelectronic properties. The matrix 118 may be composed of independent optical sensors 120. Thus, a matrix 118 of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip.

The optical sensors 120 may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the detector 110 may comprise an array of optical sensors 120, such as a rectangular array, having m rows and n columns, with m, n, independently, being positive integers. Preferably, more than one column and more than one row is

given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, 4:3, 16:9 or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix 118 specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. In order to provide a wide range of view, the matrix 118 specifically may have at least 10 rows, preferably at least 50 rows, more preferably at least 100 rows. Similarly, the matrix may have at least 10 columns, preferably at least 50 columns, more preferably at least 100 columns. The matrix 118 may comprise at least 50 optical sensors 120, preferably at least 100 optical sensors 120, more preferably at least 500 optical sensors 120. The matrix 118 may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible.

In the embodiment shown in Figure 1, the detector 110 further comprises an illumination source 124. As an example, the illumination source 124 may be configured for generating an illuminating light beam for illuminating the object 112. The detector 110 may be configured such that the illuminating light beam propagates from the detector 110 towards the object 112 along an optical axis of the detector 110. For this purpose, the detector 110 may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis.

The illumination source 124 may be adapted to generate at least one illumination pattern for illumination of the object 112. Specifically, the illumination source 124 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The pattern may comprise a plurality of features. The pattern may comprise an arrangement of periodic or non-periodic features. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; at least one pattern comprising at least one pre-known feature. For example, the illumination source 124 may be adapted to generate and/or to project a cloud of points. The illumination source 124 may comprise one or more of at least one light projector; at least one digital light processing (DLP) projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. The illumination source 124 may comprise at least one light source adapted to generate the illumination pattern directly. The illumination source 124 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination source 124 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination source 124.

In the embodiment shown in Figure 1, the illumination source 124 may illuminate the at least one object 112 with the illumination pattern. The illumination pattern may comprise a plurality of points as image features. These points are illustrated as light beams 115 emerging from the illumination source 124.

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Each optical sensor 120 is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area 122 by a reflection light beam propagating from the object 112 to the detector 110.

10 Furthermore, the sensor element 116 is adapted to determine at least one reflection image 126. The matrix 118 may comprise the reflection image 126. The reflection image 126 may comprise points as reflection features. These points result from reflection light beams 125 originating from the at least one object 112. Crossings 131 of the illumination light beams and the reflection light beams 125 correspond to possible positions 114 of the at least one object 112.

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The detector 127 may comprise at least one transfer device 127 comprising one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam de-
20 flection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. In particular, the transfer device 127 may comprise at least one collimating lens adapted to focus at least one object point in an image plane.

25 The detector 110 comprises at least one evaluation device 128. The evaluation device 126 is adapted to select at least one reflection feature of the reflection image 126, for example selected reflection feature 129. The evaluation device 128 may be adapted to perform at least one image analysis and/or image processing in order to identify the reflection feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image
30 analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency de-
35 composition; a singular value decomposition; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a blob analysis; applying an edge filter; applying a low-pass filter; applying a Fourier transformation; apply-
40 ing a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be deter-

mined manually by a user or may be determined automatically, such as by recognizing an object within an image generated by the optical sensors.

If the selected reflection feature 129 is matched to a corresponding reference feature, the corresponding light beams are matched and the distance of the object 112 can be calculated. For a selected reflection feature 129 and a corresponding reflection light beam, the number of corresponding illumination light beams, and thus the number of corresponding reference features 112, is reduced after determining a combined signal Q.

- 5 The evaluation device 128 is configured for determining at least one longitudinal region 130 of the selected reflection feature 129 of the reflection image by evaluating the combined signal Q from the sensor signals. The evaluation device 128 may be adapted to determine at least one longitudinal coordinate z of the selected reflection feature 129 by evaluating the combined signal Q, wherein the longitudinal region is given by the longitudinal coordinate z and an error interval $\pm \varepsilon$. Error ε may depend on measurement uncertainty of the optical sensor. For example, the error interval may be $\pm 10\%$, preferably $\pm 5\%$, more preferably $\pm 1\%$. The evaluation device 128 may be configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals. The evaluation device 128 may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region. For example, the evaluation device 128 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- wherein x and y are transversal coordinates, A1 and A2 are different areas of at least one beam profile of the reflection light beam at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. Area A1 and area A2 may differ. In particular, A1 and A2 are not congruent. Thus, A1 and A2 may differ in one or more of the shape or content. The beam profile may be a transverse intensity profile of the light beam. The beam profile may be a cross section of the light beam. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles. Generally the beam profile is dependent on luminance L(z_o) and beam shape S(x,y;z_o), E(x,y;z_o)=L•S. Thus, by deriving the combined signal it may allow determining the longitudinal coordinate independent from luminance. In addition, using the combined signal allows determination of the distance z_o independent from the object size. Thus, the combined signal allows determination of the distance z_o independent from the material properties and/or reflective properties and/or scattering properties of the object and independent from alterations of the light source such as by manufacturing precision, heat, water, dirt, damages on the lens, or the like.

Each of the sensor signals may comprise at least one information of at least one area of the beam profile of the light beam. The light-sensitive areas 122 may be arranged such that a first sensor signal comprises information of a first area of the beam profile and a second sensor signal comprises information of a second area of the beam profile. The first area of the beam profile and second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area.

The evaluation device 128 may be configured to determine and/or to select the first area of the beam profile and the second area of the beam profile. The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile may comprise essentially center information of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. Preferably, the center information has a proportion of edge information of less than 10 %, more preferably of less than 5%, most preferably the center information comprises no edge content. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10 %, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region. The first area of the beam profile may be area A2 and the second area of the beam profile may be area A1. Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The evaluation device 128 may be adapted for determining an area integral of the beam profile. The evaluation device 128 may be adapted to determine the edge information by integrating and/or summing of the first area. The evaluation device 128 may be adapted to determine the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the evaluation device may be adapted to determine an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be

replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

- 5 Additionally or alternatively, the evaluation device 128 may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the combined signal Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or
10 cuts may result in improved distance information.

For example, the evaluation device 128 may be configured for evaluating the sensor signals, by

- a) determining at least one optical sensor 120 having the highest sensor signal and forming at least one center signal;
- 15 b) evaluating the sensor signals of the optical sensors 120 of the matrix 118 and forming at least one sum signal;
- c) determining at least one combined signal by combining the center signal and the sum signal; and
- d) determining the at least one longitudinal coordinate z of the selected feature by evaluating
20 the combined signal.

- For example, the center signal may be the signal of the at least one optical sensor 118 having the highest sensor signal out of the plurality of sensor signals generated by the optical sensors 120 of the entire matrix 118 or of a region of interest within the matrix 118, wherein the region of
25 interest may be predetermined or determinable within an image generated by the optical sensors of the matrix 118. The center signal may arise from a single optical sensor 120 or from a group of optical sensors 120, wherein, in the latter case, as an example, the sensor signals of the group of optical sensors 120 may be added up, integrated or averaged, in order to determine the center signal. The group of optical sensors 120 from which the center signal arises
30 may be a group of neighboring optical sensors, such as optical sensors 120 having less than a predetermined distance from the actual optical sensor having the highest sensor signal, or may be a group of optical sensors generating sensor signals being within a predetermined range from the highest sensor signal. The group of optical sensors 120 from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The
35 evaluation device 128 may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of optical sensors around the optical sensor having the highest sensor signal.

- The determination of the center signal may be performed electronically, since a comparison of
40 sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. The detection of the center of the light spot, i.e. the detection of the center signal and/or of the at least one optical sensor from which the center signal arises, may be performed fully or partially electronically or fully or partially by using one or more software

algorithms. Specifically, the evaluation device 128 may comprise at least one center detector for detecting the at least one highest sensor signal and/or for forming the center signal. The center detector specifically may fully or partially be embodied in software and/or may fully or partially be embodied in hardware. The center detector may fully or partially be integrated into the at
5 least one sensor element and/or may fully or partially be embodied independently from the sensor element.

Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance
10 from the highest sensor signal; an average of sensor signals from a group of optical sensors containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors 120; a sum of sensor signals from a group of optical sensors 120 containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors 120; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors 120 containing the optical sensor 120 having the highest sensor signal and a predetermined group of neighboring optical sensors; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.
20

For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix 118 or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable
25 within an image generated by the optical sensors 120 of the matrix 118. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors 120 from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The evaluation device 128 may be adapted to determine the sum signal by integrating signals of the entire matrix 118, or of the region of interest within the matrix 118. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by
30 geometric considerations.
35

The sum signal may be derived from all sensor signals of the matrix 118, from the sensor signals within a region of interest or from one of these possibilities with the sensor signals arising from the optical sensors 120 contributing to the center signal excluded. In every case, a reliable
40 sum signal may be generated which may be compared with the center signal reliably, in order to determine the longitudinal coordinate. Generally, the sum signal may be selected from the group consisting of: an average over all sensor signals of the matrix 118; a sum of all sensor signals of the matrix 118; an integral of all sensor signals of the matrix 118; an average over all

sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; a sum of all sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; an integral of all sensor signals of the matrix 118 except for sensor signals from those optical sensors 120 contributing to the center signal; a sum of sensor signals of optical sensors 120 within a predetermined range from the optical sensor having the highest sensor signal; an integral of sensor signals of optical sensors within a predetermined range from the optical sensor 120 having the highest sensor signal; a sum of sensor signals above a certain threshold of optical sensors being located within a predetermined range from the optical sensor 120 having the highest sensor signal; an integral of sensor signals above a certain threshold of optical sensors 120 being located within a predetermined range from the optical sensor 120 having the highest sensor signal. Other options, however, exist. The summing may be performed fully or partially in software and/or may be performed fully or partially in hardware. A summing is generally possible by purely electronic means which, typically, may easily be implemented into the detector. Thus, in the art of electronics, summing devices are generally known for summing two or more electrical signals, both analogue signals and digital signals. Thus, the evaluation device 128 may comprise at least one summing device for forming the sum signal. The summing device may fully or partially be integrated into the sensor element or may fully or partially be embodied independently from the sensor element. The summing device may fully or partially be embodied in one or both of hardware or software.

The combined signal may be a signal which is generated by combining the center signal and the sum signal. Specifically, the combination may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the combined signal may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal. The evaluation device 128 may be configured to determine the at least one longitudinal coordinate z of the object by using at least one known, determinable or predetermined relationship between the sensor signals. In particular, the evaluation device 128 may be configured to determine the at least one coordinate z of the by using at least one known, determinable or predetermined relationship between a quotient signal derived from the sensor signals and the longitudinal coordinate.

The comparison between the center signal and the sum signal specifically may be performed by forming one or more quotient signals. Thus, generally, the combined signal may be a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa; forming a quotient of the center signal and a linear combination of the sum signal and the center signal or vice versa; forming a quotient of the sum signal and a linear combination of the sum signal and the center signal or vice versa; forming a

quotient of an exponentiation of the center signal and an exponentiation of the sum signal or vice versa. Other options, however, exist. The evaluation device 128 may be configured for forming the one or more quotient signals. The evaluation device 128 may further be configured for determining the at least one longitudinal coordinate by evaluating the at least one quotient
5 signal.

The evaluation device 128 specifically may be configured for using at least one predetermined relationship between the combined signal Q and the longitudinal coordinate, in order to determine the at least one longitudinal coordinate. Thus, due to the reasons disclosed above and
10 due to the dependency of the properties of the light spot on the longitudinal coordinate, the combined signal Q typically is a monotonous function of the longitudinal coordinate of the object and/or of the size of the light spot such as the diameter or equivalent diameter of the light spot. Thus, as an example, specifically in case linear optical sensors are used, a simple quotient of the sensor signal s_{center} and the sum signal s_{sum} $Q = s_{\text{center}}/s_{\text{sum}}$ may be a monotonously decreasing function of the distance. Without wishing to be bound by this theory, it is believed that this is
15 due to the fact that, in the preferred setup described above, both the center signal s_{center} and the sum signal s_{sum} decrease as a square function with increasing distance to the light source, since the amount of light reaching the detector decreases. Therein, however, the center signal s_{center} decreases more rapidly than the sum signal s_{sum} , since, in the optical setup as used in the experiments, the light spot in the image plane grows and, thus, is spread over a larger area. The
20 quotient of the center signal and the sum signal, thus, continuously decreases with increasing diameter of the light beam or diameter of the light spot on the light-sensitive areas of the optical sensors of the matrix. The quotient, further, is typically independent from the total power of the light beam, since the total power of the light beam forms a factor both in the center signal and in the sum sensor signal. Consequently, the combined signal Q may form a secondary signal which provides a unique and unambiguous relationship between the center signal and the sum signal and the size or diameter of the light beam. Since, on the other hand, the size or diameter of the light beam is dependent on a distance between the object, from which the light beam propagates towards the detector, and the detector itself, i.e. dependent on the longitudinal coordinate of the object, a unique and unambiguous relationship between the center signal and the sum signal on the one hand and the longitudinal coordinate on the other hand may exist. For the latter, reference e.g. may be made to one or more of the above-mentioned prior art documents, such as WO 2014/097181 A1. The predetermined relationship may be determined by analytical considerations, such as by assuming a linear combination of Gaussian light beams,
25 by empirical measurements, such as measurements measuring the combined signal and/or the center signal and the sum signal or secondary signals derived thereof as a function of the longitudinal coordinate of the object, or both.
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The evaluation device 128 may be configured for using at least one predetermined relationship
40 between the combined signal and the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The evaluation device 128 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The sensor element 116 may be adapted to determine at least one reflection pattern. The reflection pattern may comprise at least one feature corresponding to at least one feature of the illumination pattern. The reflection pattern may comprise, in comparison to the illumination pattern, at least one distorted pattern, wherein the distortion depends on the distance of the object, such as surface properties of the object. The evaluation device 128 may be adapted to select at least one feature of the reflection pattern and to determine the longitudinal region of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals, as described above.

The evaluation device 128 is adapted to determine at least one displacement region 132 in at least one reference image 134 corresponding to the longitudinal region 130. The longitudinal region corresponds to the displacement region 132. The reference image 134 and the reflection image 126 may be images of the object 112 determined at different spatial positions having a fixed distance b, also called baseline. The evaluation device 128 may be adapted to determine an epipolar line 136 in the reference image 134. Relative position of the reference image 134 and reflection image 126 may be known. For example, relative position of the reference image 134 and reflection image 126 may be stored within at least one storage unit of the evaluation device 128. The evaluation device 128 may be adapted to determine a straight line 138 extending from the selected reflection feature 129 of the reflection image 126. The straight line 138 may comprise possible object positions 114. The straight line 138 and the baseline b span an epipolar plane 140. As the reference image 134 is determined at a different relative position from the reflection image, the corresponding reference features may be imaged on the epipolar line 136 in the reference image 134.

For example, in the embodiment shown in Figure 1, the reference image 134 may be an image of the illumination pattern at an image plane at a position of the illumination source 124. The evaluation device 128 may be adapted to determine the displacement region 132 in the reference image 134 corresponding to the longitudinal region 130 of the selected feature 129 of the reflection pattern. The evaluation device 128 may be adapted to match the selected feature 129 of the reflection pattern with at least one feature of the reference pattern within the displacement region 132. The illumination source 124 and the sensor element 116 may be separated by a fixed distance.

The displacement region 132 may extend along the epipolar line 136. The evaluation device 128 may be adapted to determine the reference feature along the epipolar line 136 corresponding to the longitudinal coordinate z and to determine an extent of the displacement region 132 along the epipolar line 136 corresponding to the error interval $\pm\epsilon$. The evaluation device 128 is adapted to match the selected reflection feature 129 with at least one reference feature within the displacement region 132. The evaluation device 128 may be adapted to match the selected feature of the reflection image with the reference feature within the displacement region 132 by using at least one evaluation algorithm considering the determined longitudinal coordinate z. The evaluation algorithm may be a linear scaling algorithm.

The evaluation device 128 may be adapted to determine a displacement of the matched reference feature and the selected reflection feature 129. The evaluation device 128 may be adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement. For example, the longitudinal information may be a distance value. The evaluation device 128 may be adapted to determine the pre-determined relationship by using triangulation methods. In case position of the selected reflection feature 129 in the reflection image 126 and position of the matched reference feature and/or relative displacement of the selected reflection feature and the matched reference feature are known, a longitudinal coordinate of the corresponding object feature 114 may be determined by triangulation. Thus, the evaluation device 128 may be adapted to select, for example subsequent and/or column by column, a reflection feature and to determine for each potential position of the reference feature the corresponding distance value using triangulation. For example, Figure 1 shows possible projection lines resulting in reflection features in the reflection image 126 and corresponding to reference features in the reference image 134. In addition, intersection points of these lines are shown corresponding to object positions. Displacement and corresponding distance value may be stored in at least one storage device of the evaluation device. The evaluation device 128 may, as an example, comprise at least one data processing device, such as at least one processor, at least one DSP, at least one FPGA and/or at least one ASIC. Further, for storing the at least one predetermined or determinable relationship between the longitudinal coordinate z and the displacement, the at least one data storage device may be provided, such as for providing one or more look-up tables for storing the predetermined relationship.

In Figure 2, an embodiment of the detector 110 is shown, wherein the detector 110 may comprise at least two sensor elements 116 each having a matrix 118 of optical sensors 120. At least one first sensor element 142 and at least one second sensor element 144 may be positioned at different spatial positions. A relative distance b between the first sensor element 142 and the second element 144 may be fixed. The at least one first sensor element 142 may be adapted to determine at least one first reflection pattern and the at least one second sensor element 144 may be adapted to determine at least one second reflection pattern. The evaluation device 128 may be adapted to select at least one image determined by the first sensor element 142 or the second sensor element 144 as reflection image and to select at least one image determined by the other one of the first sensor element 142 or the second sensor element 144 as reference image. In Figure 2, the image of the first sensor element 142 is selected as reflection image 126, whereas the image of the second sensor element 144 is selected as reference image 134. The evaluation device 128 may be adapted to select the at least one reflection feature in the reflection pattern and to determine the longitudinal region 130 of the selected feature by evaluating the combined signal Q from the sensor signals, as outlined with respect to Figure 1. The evaluation device 128 may be adapted to determine the displacement region 132 in the reference image 134 corresponding to the longitudinal region 120 of the selected feature 129 of the reflection pattern. The evaluation device 128 may be adapted to match the selected feature 129

of the reflection pattern with at least one feature of the reference pattern within the displacement region 132, as outlined with respect to Figure 1.

Using the combined sensor signal allows to estimate distances, such as the longitudinal coordinate z within the error interval. By determining the displacement region 132 corresponding to the estimated longitudinal coordinate and the corresponding error interval allows to reduce possible number of solutions along the epipolar line 136 significantly. The number of solutions may even be reduced to one. Subsequent matching of the selected reflection feature and reference feature may enhance accuracy. Determining of the longitudinal coordinate z and the error interval may be performed during a pre-evaluation before matching the selected reflection feature 129 and reference feature. This may allow reducing the computational demand such that it is possible to significantly reduce costs and to allow a use in mobile device or outdoor devices, where battery life or ventilation or heat management is an issue. Furthermore, generally in triangulation systems the baseline has to be large in order to detect large distances. Pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature 129 and reference feature may allow using short baselines such that it may be possible to provide a compact device. Furthermore, pre-evaluation of the longitudinal coordinate z and error interval using the combined sensor signal and subsequent matching of the selected reflection feature and reference feature may enhance accuracy in comparison the triangulation systems. Furthermore, it may be possible to use denser pattern in order to enhance further resolution or to use a less dense pattern in order to increase power per point.

Figure 3 shows, in a highly schematic illustration, an exemplary embodiment of the detector 110, for example according to the embodiment shown in Figure 2. The detector 110 specifically may be embodied as a camera 146 and/or may be part of a camera 146. The camera 146 may be made for imaging, specifically for 3D imaging, and may be made for acquiring standstill images and/or image sequences such as digital video clips. Other embodiments are feasible. Fig. 3 further shows an embodiment of a detector system 148, which, besides the at least one detector 110, comprises one or more beacon devices 150, which, in this example, may be attached and/or integrated into an object 112, the position of which shall be detected by using the detector 110. Fig. 3 further shows an exemplary embodiment of a human-machine interface 152, which comprises the at least one detector system 148 and, further, an entertainment device 154, which comprises the human-machine interface 152. The figure further shows an embodiment of a tracking system 156 for tracking a position of the object 112, which comprises the detector system 148. The components of the devices and systems shall be explained in further detail below.

Fig. 3 further shows an exemplary embodiment of a scanning system 158 for scanning a scenery comprising the object 112, such as for scanning the object 112 and/or for determining at least one position of the at least one object 112. The scanning system 158 comprises the at least one detector 110, and, further, optionally, the at least one illumination source 124 as well as, optionally, at least one further illumination source, not depicted here. The illumination source

124, generally, is configured to emit at least one illumination light beam, such as for illumination of at least one dot, e.g. a dot located on one or more of the positions of the beacon devices 150 and/or on a surface of the object 112. The scanning system 158 may be designed to generate a profile of the scenery including the object 112 and/or a profile of the object 112, and/or may be designed to generate at least one item of information about the distance between the at least one dot and the scanning system 158, specifically the detector 110, by using the at least one detector 110.

As outlined above, an exemplary embodiment of the detector 110 which may be used in the setup of Fig. 3 is shown in Fig. 2. Thus, the detector 110, besides the optical sensors 120 comprises at least one evaluation device 128, having e.g. at least one divider 160 and/or the at least one position evaluation device 162, as symbolically depicted in Fig. 3. The components of the evaluation device 128 may fully or partially be integrated into a distinct device and/or may fully or partially be integrated into other components of the detector 110. Besides the possibility of fully or partially combining two or more components, one or more of the optical sensors 120 and one or more of the components of the evaluation device 128 may be interconnected by one or more connectors 164 and/or by one or more interfaces, as symbolically depicted in Fig. 3. Further, the one or more connectors 164 may comprise one or more drivers and/or one or more devices for modifying or preprocessing sensor signals. Further, instead of using the at least one optional connector 164, the evaluation device 128 may fully or partially be integrated into one or both of the optical sensors 120 and/or into a housing 166 of the detector 110. Additionally or alternatively, the evaluation device 128 may fully or partially be designed as a separate device.

In this exemplary embodiment, the object 112, the position of which may be detected, may be designed as an article of sports equipment and/or may form a control element or a control device 168, the position of which may be manipulated by a user 170. As an example, the object 112 may be or may comprise a bat, a racket, a club or any other article of sports equipment and/or fake sports equipment. Other types of objects 112 are possible. Further, the user 170 himself or herself may be considered as the object 112, the position of which shall be detected.

As outlined above, the detector 110 comprises optical sensors 120. The optical sensors 120 may be located inside the housing 166. Further, the detector 110 may comprise the at least one transfer device 127 such as one or more optical systems, preferably comprising one or more lenses. An opening 172 inside the housing 166, which, preferably, is located concentrically with regard to an optical axis 174 of the detector 110, preferably defines a direction of view 176 of the detector 110. A coordinate system 178 may be defined, in which a direction parallel or anti-parallel to the optical axis 174 may be defined as a longitudinal direction, whereas directions perpendicular to the optical axis 174 may be defined as transversal directions. In the coordinate system 178, symbolically depicted in Fig. 3, a longitudinal direction is denoted by z, and transversal directions are denoted by x and y, respectively. Other types of coordinate systems are feasible, such as non-Cartesian coordinate systems.

One or more light beams 180, such as light beams of the illumination pattern, are propagating from the object 112 and/or from one or more of the beacon devices 150, towards the detector 110. The detector 110 is configured for determining a position of the at least one object 112. In case no illumination source 124 is used, the beacon devices 150 and/or at least one of these
5 beacon devices 150 may be or may comprise active beacon devices with an integrated illumination source such as a light-emitting diode. Alternatively, ambient light sources may be used.

The light beam 180, for example generated by the illumination source 124 such as the at least one illumination pattern, may be directed towards the object by using one or more optical elements. The object 112 may in response to the illumination generate the at least one reflection
10 light beam 182. For details of the evaluation, reference may be made to Fig. 1 above.

As outlined above, the determination of the position of the object 112 and/or a part thereof by using the detector 110 may be used for providing a human-machine interface 152, in order to
15 provide at least one item of information to a machine 184. In the embodiments schematically depicted in Fig. 3, the machine 184 may be a computer and/or may comprise a computer. Other embodiments are feasible. The evaluation device 128 may even be fully or partially integrated into the machine 184, such as into the computer.

As outlined above, Fig. 3 also depicts an example of a tracking system 156, configured for tracking the position of the at least one object 112 and/or of parts thereof. The tracking system 156 comprises the detector 110 and at least one track controller 186. The track controller 186 may be adapted to track a series of positions of the object 112 at specific points in time. The track controller 186 may be an independent device and/or may be fully or partially integrated
20 into the machine 184, specifically the computer, as indicated in Fig. 3 and/or into the evaluation device 128.

Similarly, as outlined above, the human-machine interface 152 may form part of an entertainment device 154. The machine 184, specifically the computer, may also form part of the entertainment device 154. Thus, by means of the user 170 functioning as the object 112 and/or by
30 means of the user 170 handling a control device functioning as the object 112, the user 170 may input at least one item of information, such as at least one control command, into the computer, thereby varying the entertainment functions, such as controlling the course of a computer.

Referring to Figure 4, the detector 110 such as the detector as described with respect to Figures 1 to 3 may be adapted to determine depth information, in particular absolute depth information, from a radiance ratio of at least two asymmetric regions of a light beam profile on the at least two optical sensors 120. For example, the detector 110 may comprise a plurality of optical sensors arranged in the matrix 118. The detector 110 may be adapted to determine depth information from a radiance ratio of at least two asymmetric regions within an enclosed, in particular, defocused beam profile captured by a single matrix of optical sensors such as a CMOS detector. In particular, the detector 110 may be adapted to determine the depth information using the radiance ratio independent of a certain object size range. As outlined above, this principle is
40

called Distance by Photon Ratio (DPR). In one embodiment, the light beam 180 may illuminate the sensor element 116 with at least one pattern comprising at least one feature point. The feature point may be selected from the group consisting of: at least one point, at least one line, at least one edge. The pattern may be generated by the object, for example, in response to an illumination by the at least one light source with an illumination pattern comprising the at least one pattern. The evaluation device 128 may be configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

- wherein x and y are transversal coordinates, A1 and A2 are areas of the beam profile at the sensor position, and E(x,y,z_o) denotes the beam profile given at the object distance z_o. A1 may correspond to a full or complete area of a feature point on the optical sensors. A2 may be a central area of the feature point on the optical sensors. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In the embodiment shown in Figure 4, the reflection light beam 182 propagating from the object 112 to the detector 110 may illuminate the matrix 118 with at least one line pattern 188. The line pattern 188 may be generated by the object 112 for example in response to an illumination by the at least one illumination source 124 with an illumination pattern comprising at least one illumination line pattern. A1 may correspond to an area with a full line width of the line pattern 188 in the matrix 118. The line pattern 188 in the matrix 118 may be widened and/or displaced compared to the line pattern of the illumination pattern such that a line width in the matrix 118 is increased. In particular, the line width of the line pattern 188 in the matrix 118 may change from one column to another column. A2 may be a central area of the line pattern 188 in the matrix 118. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern 188 may be segmented in the matrix 118. Each of the columns may comprise center information of intensity in the central area of the line pattern 188 and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern 188.

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Figures 5A to C show three embodiments of a hexagonal illumination pattern. The illumination source 124 may be adapted to generate at least one illumination pattern for illuminating the object 112. Additionally or alternatively, the illumination pattern may be generated by at least one ambient light source. Specifically, the illumination source 124 may comprise at least one laser

and/or laser source. Various types of lasers may be employed, such as semiconductor lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination pattern may comprise at least one feature such as a point or symbol. The illumination pattern may comprise a plurality of features. The illumination pattern may com-
5 prise an arrangement of periodic or non-periodic features. The illumination pattern may be generated by ambient light, such as by at least one ambient light source, or by the at least one illumination source. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern, a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one qua-
10 siperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination source may be adapted to generate and/or to
15 project a cloud of points. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern, or a pattern comprising further convex tilings. The illumination pattern may comprise as much as possible features per area such that hexagonal pattern may be preferred. A distance between two features of the illumination pattern and/or an area of the at least one illumination feature
20 may depend on the circle of confusion in the image.

The illumination features of the illumination pattern may be arranged such that only few reference features are positioned on an epipolar line. As shown in Figure 5A, the illumination pattern may comprise at least one hexagonal pattern, wherein the individual points are positioned on
25 epipolar lines 190. As shown in Figure 5B, the illumination pattern may comprise at least one hexagonal pattern, wherein the pattern is rotated relative to the baseline. Such a positioning of the illumination features allows enhancing distance between the individual points on each epipolar line. For example as shown in Figure 5C, the illumination pattern may comprise at least one displaced hexagonal pattern, wherein individual points of the hexagonal pattern are dis-
30 placed by a random distance from the regular position, for example orthogonal to the epipolar line of the point. The displacement of the individual points may be smaller than half of the distance between two parallel epipolar lines, preferably smaller than one fourth of the distance between two parallel epipolar lines. The displacement of the individual points may be as such that two points are not displaced above each other

35 Figure 6 shows an embodiment of a scanning system 158. The scanning system 158 may be adapted as a line scanning device. In particular, the scanning system 158 may comprise at least one sensor line or row of optical sensors 120. Furthermore, the scanning system 158 may comprise the at least one transfer device 127 and the at least one illumination source 124. Triangu-
40 lation systems require a sufficient baseline, however due to the baseline in the near field no detection may be possible. Near field detection may be possible if the light spot is tilted in direction of the transfer device. However, the tilting leads to that the light spot will move out of the field of view which limits detection in far field regions. Thus, in triangulation systems, the non-

zero baseline will always lead to a substantial reduction in the measurement range, in the near field, and/or in the far field. Reducing the baseline as possible with the detector according to the present invention will thus always increase the measurement range. Further, these near field and far field problems can be overcome by using the scanning system 158 of Figure 6. The scanning system 158 may be adapted to detect a plurality of light beams 182 propagating from the object 112 to the scanning system 158 on the CMOS line. The light beams 182 may be generated at different position on the object 112 or by movement of the object 112. The scanning system 158 may be adapted to determine at least one longitudinal coordinate for each of the light points by determining the combined signal Q as described above.

List of reference numbers

110	Detector
112	Object
114	object positions
116	sensor element
118	Matrix
120	optical sensor
122	light sensitive area
124	illumination source
126	reflection image
127	transfer device
128	evaluation device
129	selected reflection feature
130	longitudinal region
131	Crossing
132	displacement region
134	reference image
136	epipolar line
138	straight line
140	epipolar plane
142	first sensor element
144	second sensor element
146	Camera
148	detector system
150	beacon devices
152	human-machine-interface
154	entertainment device
156	tracking system
158	scanning system
160	Divider
162	position evaluation device
164	Connector
166	Housing
168	control device
170	User
172	Opening
174	optical axis
176	direction of view
178	coordinate system
182	reflection light beam
184	Machine
186	track controller

188 Line pattern
190 Epipolar line

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10 Berlin Heidelberg, 1997, chapter 2

Patent claims

1. A detector (110) for determining a position of at least one object (112), the detector (110) comprising
 - 5 - at least one sensor element (116) having a matrix (118) of optical sensors (120), the optical sensors (120) each having a light-sensitive area (122), wherein each optical sensor (120) is designed to generate at least one sensor signal in response to an illumination of its respective light-sensitive area (122) by a reflection light beam (182) propagating from the object (112) to the detector (110), wherein the sensor element (116) is adapted to determine at least one reflection image (126);
 - 10 - at least one evaluation device (128), wherein the evaluation device (128) is adapted to select at least one reflection feature of the reflection image (126), wherein the evaluation device (128) is configured for determining at least one longitudinal region (130) of the selected reflection feature of the reflection image (134) by evaluating a combined signal Q from the sensor signals, wherein the evaluation device (128) is adapted to determine at least one displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130), wherein the evaluation device (128) is adapted to match the selected reflection feature with at least one reference feature within the displacement region (132).
 - 15
 - 20
2. The detector (110) according to the preceding claim, wherein the evaluation device (128) is adapted to determine a displacement of the matched reference feature and the selected reflection feature, wherein the evaluation device (128) is adapted to determine a longitudinal information of the matched feature using a predetermined relationship between a longitudinal coordinate and the displacement.
- 25
3. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals.
- 30
4. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for using at least one predetermined relationship between the combined signal Q and the longitudinal region for determining the longitudinal region.
- 35
5. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is adapted to determine at least one longitudinal coordinate z of the selected reflection feature by evaluating the combined signal Q, wherein the longitudinal region (130) is given by the longitudinal coordinate z and an error interval $\pm\epsilon$.
- 40
6. The detector (110) according any one of the preceding claims, wherein the reference image (134) and the reflection image (126) are images of the object determined at different spatial

positions having a fixed distance, wherein the evaluation device (128) is adapted to determine an epipolar line (136) in the reference image (134).

7. The detector (110) according to the two preceding claims, wherein the displacement region (132) extends along the epipolar line (136), wherein the evaluation device (128) is adapted to determine the reference feature along the epipolar line (136) corresponding to the longitudinal coordinate z and to determine an extent of the displacement region (132) along the epipolar line (136) corresponding to the error interval $\pm\epsilon$.
8. The detector (110) according to the preceding claim, wherein the evaluation device (128) is configured to perform the following steps:
 - Determining the displacement region for the image position of each reflection feature;
 - Assigning an epipolar line to the displacement region of each reflection feature such as by assigning the epipolar line closest to a displacement region and/or within a displacement region and/or closest to a displacement region along a direction orthogonal to the epipolar line;
 - Assigning and/or determining at least one reference feature to each reflection feature such as by assigning the reference feature closest to the assigned displacement region and/or within the assigned displacement region and/or closest to the assigned displacement region along the assigned epipolar line and/or within the assigned displacement region along the assigned epipolar line.
9. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is adapted to match the selected feature of the reflection image (126) with the reference feature within the displacement region (132) by using at least one evaluation algorithm considering the determined longitudinal coordinate z , wherein the evaluation algorithm is a linear scaling algorithm.
10. The detector (110) according to any one of the preceding claims, wherein the evaluation device (128) is configured for deriving the combined signal Q by

$$Q(z_o) = \frac{\iint_{A_1} E(x, y; z_o) dx dy}{\iint_{A_2} E(x, y; z_o) dx dy}$$

wherein x and y are transversal coordinates, A_1 and A_2 are different areas of at least one beam profile of the reflection light beam (182) at the sensor position, and $E(x, y, z_o)$ denotes the beam profile given at the object distance z_o , wherein each of the sensor signals comprises at least one information of at least one area of the beam profile of the reflection light beam.

11. The detector (110) according to the preceding claim, wherein the light-sensitive areas (122) are arranged such that a first sensor signal comprises information of a first area of the beam profiles and a second sensor signal comprises information of a second area of the beam profile, wherein the first area of the beam profile and the second area of the beam profile are one or both of adjacent or overlapping regions.
12. The detector (110) according to the preceding claim, wherein the evaluation device (128) is configured to determine the first area of the beam profile and the second area of the beam profile, wherein the first area of the beam profile comprises essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, wherein the edge information comprises information relating to a number of photons in the first area of the beam profile and the center information comprises information relating to a number of photons in the second area of the beam profile.
13. The detector (110) according to any one of the preceding claims, wherein the at least one evaluation device (128) is configured for evaluating the sensor signals, by
- a) determining at least one optical sensor (120) having the highest sensor signal and forming at least one center signal;
 - b) evaluating the sensor signals of the optical sensors (120) of the matrix (118) and forming at least one sum signal;
 - c) determining at least one combined signal by combining the center signal and the sum signal; and
 - d) determining the longitudinal region by evaluating the combined signal.
14. The detector (110) according to the preceding claim, wherein the combined signal is a quotient signal Q , derived by one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa.
15. The detector (110) according to any one of the preceding claims, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is adapted to generate at least one illumination pattern for illumination of the object (112), wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; at least one pattern comprising at least one pre-known feature.
16. The detector (110) according to the preceding claim, wherein the sensor element (116) is adapted to determine at least one reflection pattern, wherein the evaluation device (128) is adapted to select at least one feature of the reflection pattern and to determine the longitudinal region (130) of the selected feature of the reflection pattern by evaluating the combined signal Q from the sensor signals.

17. The detector (110) according to the preceding claim, wherein the reference image (134) is an image of the illumination pattern at an image plane at a position of the illumination source (124), wherein the evaluation device (128) is adapted to determine the displacement region (132) in the reference image (134) corresponding to the longitudinal region (130) of the selected feature of the reflection pattern, wherein the evaluation device (128) is adapted to match the selected feature of the reflection pattern with at least one feature of the reference pattern within the displacement region (132).
18. The detector (110) according to any one of the three preceding claims, wherein the detector (110) comprises at least two sensor elements (116) each having a matrix (118) of optical sensors (120), where at least one first sensor element (142) and at least one second sensor element (144) are positioned at different spatial positions, wherein a relative distance between the first sensor element (142) and the second element (144) is fixed, wherein the at least one first sensor element (142) is adapted to determine at least one first reflection pattern and the at least one second sensor element (144) is adapted to determine at least one second reflection pattern, wherein the evaluation device (128) is adapted to select at least one image determined by the first sensor element (142) or the second sensor element (144) as reflection image (126) and to select at least one image determined by the other one of the first sensor element (142) or the second sensor element (144) as reference image (134).
19. A detector system (148) for determining a position of at least one object (112), the detector system (148) comprising at least one detector (110) according to any one of the preceding claims, the detector system (148) further comprising at least one beacon device (150) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (150) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112).
20. A human-machine interface (152) for exchanging at least one item of information between a user (170) and a machine (184), wherein the human-machine interface (152) comprises at least one detector system (148) according to the preceding claim, wherein the at least one beacon device (150) is adapted to be at least one of directly or indirectly attached to the user (170) and held by the user (170), wherein the human-machine interface (152) is designed to determine at least one position of the user (170) by means of the detector system (148), wherein the human-machine interface (152) is designed to assign to the position at least one item of information.
21. An entertainment device (154) for carrying out at least one entertainment function, wherein the entertainment device (154) comprises at least one human-machine interface (152) according to the preceding claim, wherein the entertainment device (154) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (152), wherein the entertainment device (154) is designed to vary the entertainment function in accordance with the information.

22. A tracking system (156) for tracking a position of at least one movable object (112), the tracking system (156) comprising at least one detector system (148) according to any one of the preceding claims referring to a detector system (148), the tracking system (156) further comprising at least one track controller (186), wherein the track controller (186) is adapted to track a series of positions of the object (112) at specific points in time.
23. A scanning system (158) for determining a depth profile of a scenery, the scanning system comprising at least one detector (110) according to any of the preceding claims referring to a detector, the scanning system (158) further comprising at least one illumination source (124) adapted to scan the scenery with at least one light beam.
24. A camera (146) for imaging at least one object (112), the camera (146) comprising at least one detector (110) according to any one of the preceding claims referring to a detector.
25. A method for determining a position of at least one object (112) by using at least one detector (110) according to any one of the preceding claims referring to a detector, the method comprising the following steps:
- generating at least two sensor signals in response to an illumination of at least one sensor element (116) having a matrix (118) of optical sensors (120) by at least one reflection light beam (182) propagating from the object (112) to the detector (110), the optical sensors (120) each having a light-sensitive area (122);
 - determining at least one reflection image (126) by using the sensor element (116);
 - selecting at least one reflection feature of the reflection image (126) and evaluating the sensor signals by using at least one evaluation device (128), thereby, determining at least one longitudinal region (130) of the selected reflection feature, wherein the evaluating comprises evaluating a combined signal Q from the sensor signals;
 - determining at least one displacement region (132) in at least one reference image (134) corresponding to the longitudinal region (130);
 - matching the selected reflection feature with at least one reference feature within the displacement region (132).
26. A use of the detector (110) according to any one of the preceding claims relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application;

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FIG.1

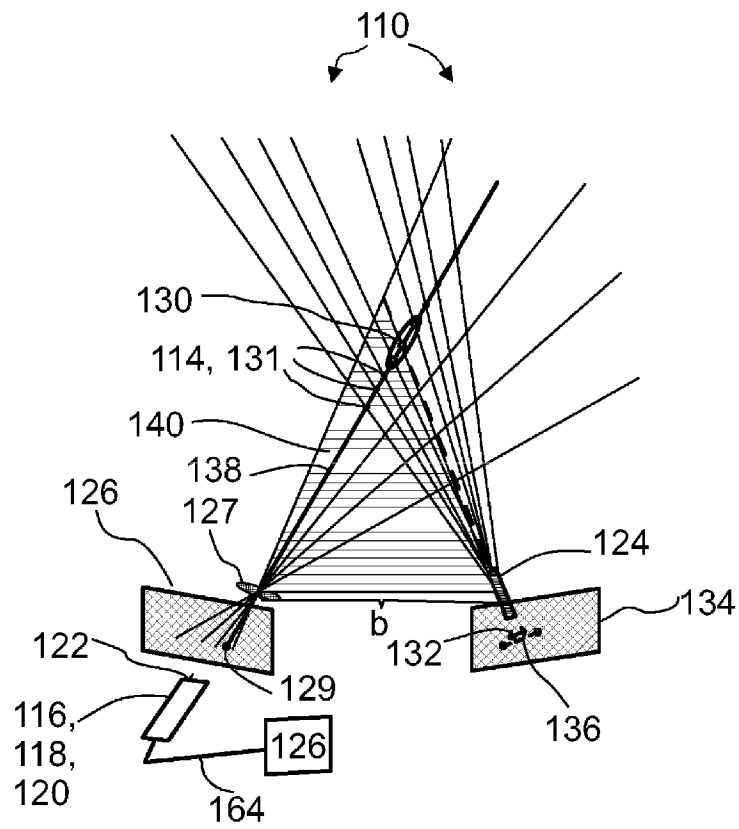


FIG.2

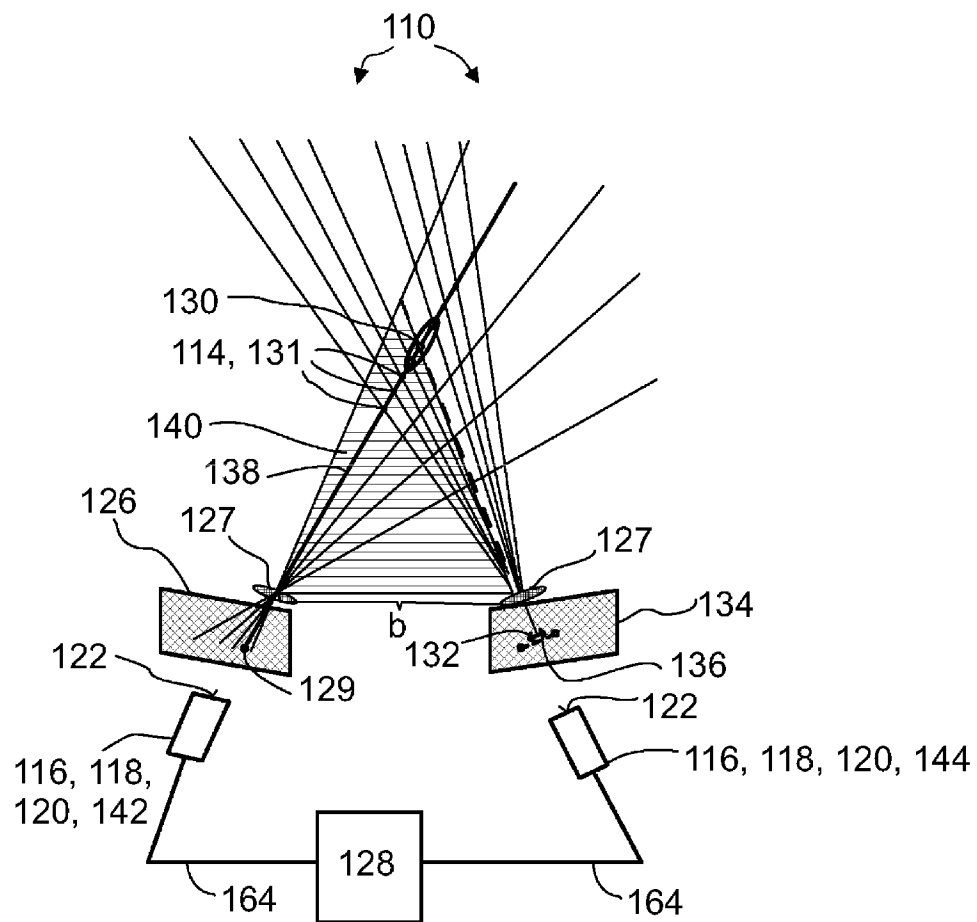
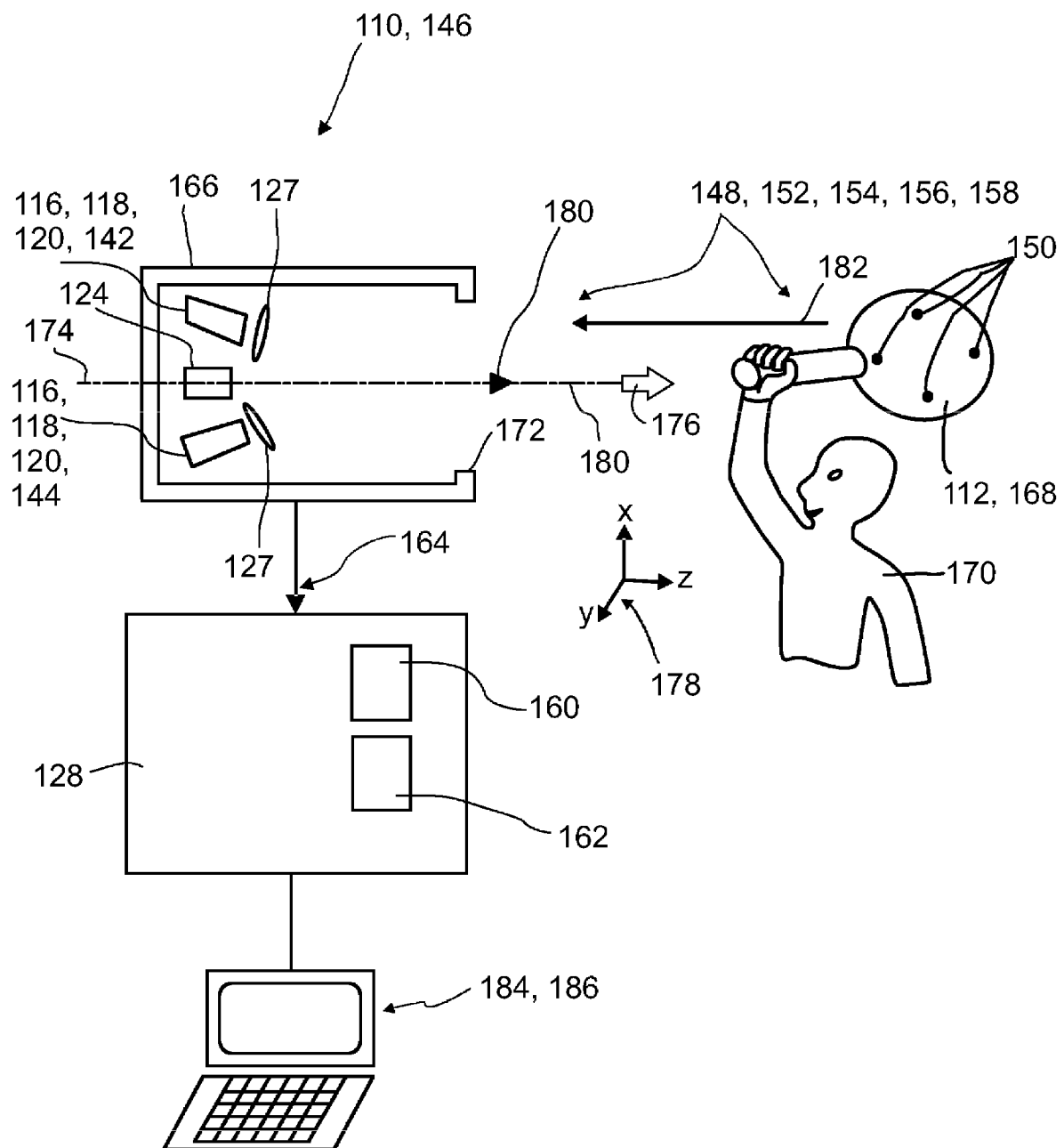


FIG.3



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FIG.4

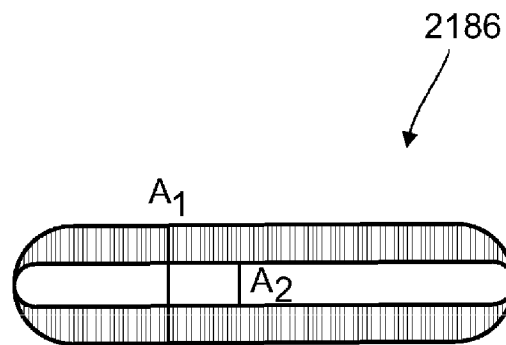


FIG.5A

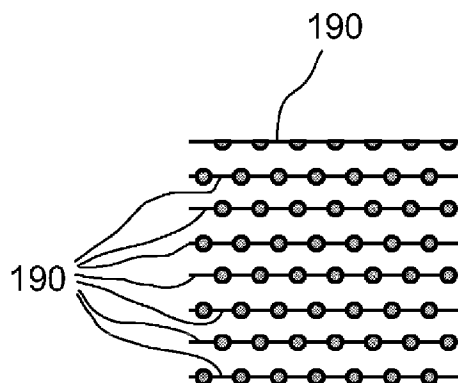


FIG.5B

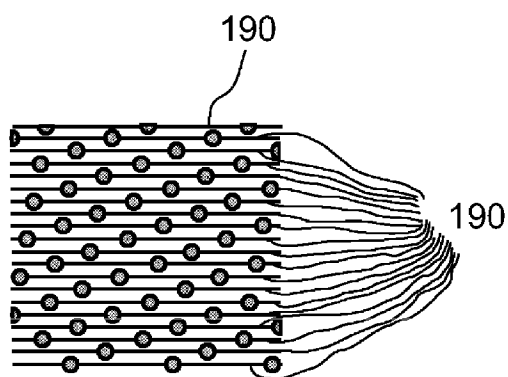


FIG.5C

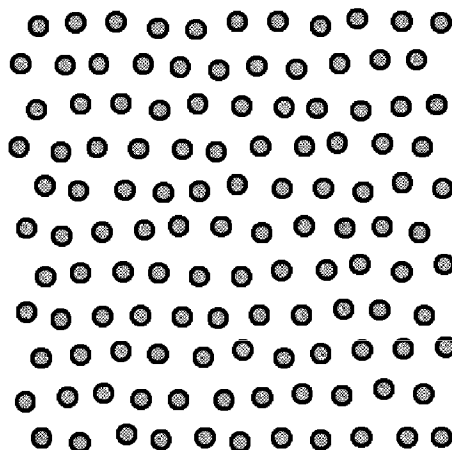
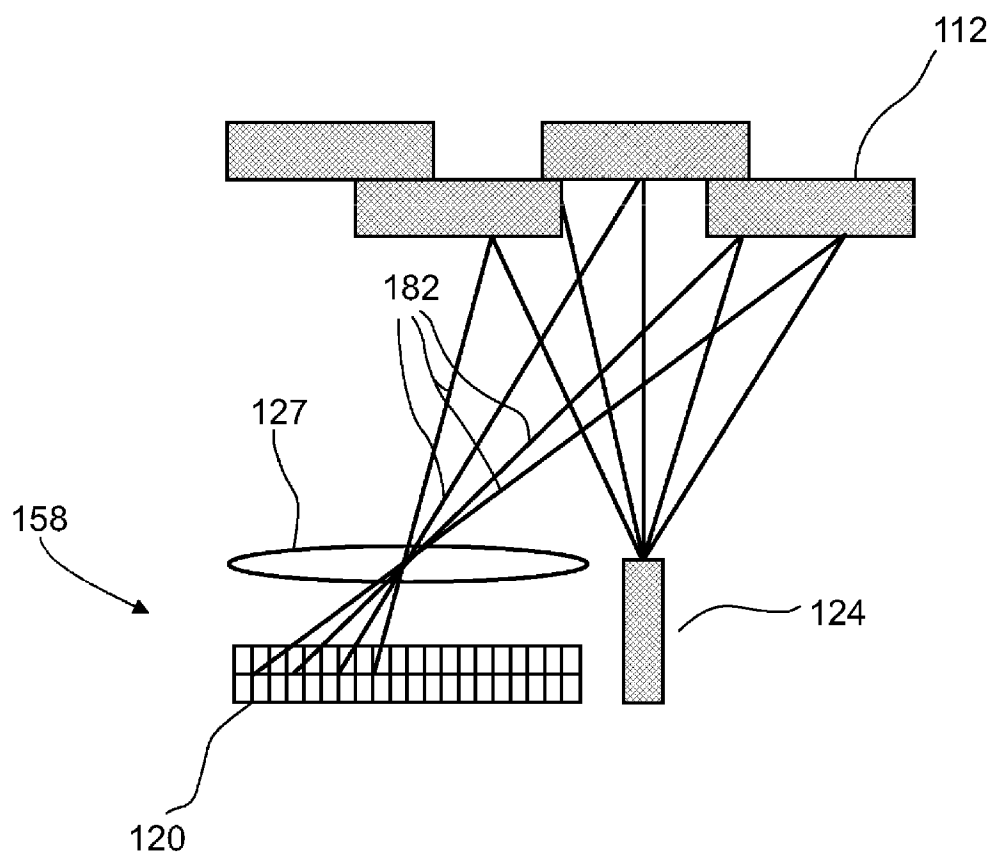


FIG.6



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2018/073067

A. CLASSIFICATION OF SUBJECT MATTER

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ADD.

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B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01S

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal, INSPEC, WPI Data

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2008/118143 A1 (GORDON EYAL [IL] ET AL) 22 May 2008 (2008-05-22) abstract; figures 1-25 paragraph [0309] - paragraph [0374] -----	1-26
X	US 2009/066929 A1 (TROPF HERMANN [DE]) 12 March 2009 (2009-03-12) abstract paragraph [0018] - paragraph [0110] -----	1,13, 19-26



Further documents are listed in the continuation of Box C.



See patent family annex.

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Date of the actual completion of the international search

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

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Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2008118143 A1	22-05-2008	US 2008118143 A1	22-05-2008
		US 2012063672 A1	15-03-2012

US 2009066929 A1	12-03-2009	EP 1971820 A1	24-09-2008
		US 2009066929 A1	12-03-2009
		WO 2007079805 A1	19-07-2007

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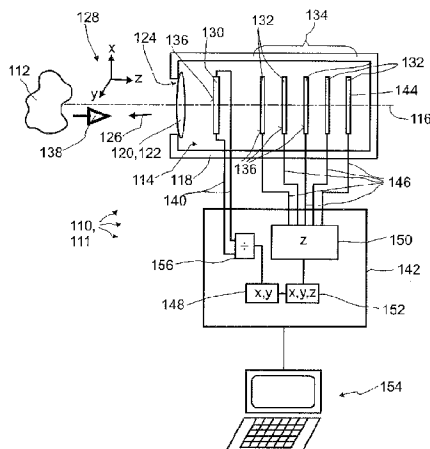
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[Continued on next page]

(54) Title: DETECTOR FOR OPTICALLY DETECTING AT LEAST ONE OBJECT

FIG.1A



(57) Abstract: A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises: - at least one transversal optical sensor (130), the transversal optical sensor (130) being adapted to determine a transversal position of at least one light beam (138) traveling from the object (112) to the detector (110), the transversal position being a position in at least one dimension perpendicular to an optical axis (116) of the detector (110), the transversal optical sensor (130) being adapted to generate at least one transversal sensor signal; - at least one longitudinal optical sensor (132), wherein the longitudinal optical sensor (132) has at least one sensor region (136), wherein the longitudinal optical sensor (132) is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region (136) by the light beam (138), wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam (138) in the sensor region (136); - at least one evaluation device (142), wherein the evaluation device (142) is designed to generate at least one item of information on a transversal position of the object (112) by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object (112) by evaluating the longitudinal sensor signal.



Published:

— with international search report (Art. 21(3))

— before the expiration of the time limit for amending the claims and to be republished in the event of receipt of amendments (Rule 48.2(h))

Detector for optically detecting at least one objectDescription5 Field of the invention

The invention relates to a detector for determining a position of at least one object. Furthermore, the invention relates to a human-machine interface, an entertainment device, a tracking system and a camera. Furthermore, the invention relates to a method for optically detecting a position
10 of at least one object and to various uses of the detector. Such devices, methods and uses can be employed for example in various areas of daily life, gaming, traffic technology, production technology, security technology, medical technology or in the sciences. Additionally or alternatively, the application may be applied in the field of mapping of spaces, such as for generating maps of one or more rooms, one or more buildings or one or more streets. However,
15 other applications are also possible in principle.

Prior art

A large number of optical sensors and photovoltaic devices are known from the prior art. While
20 photovoltaic devices are generally used to convert electromagnetic radiation, for example, ultraviolet, visible or infrared light, into electrical signals or electrical energy, optical detectors are generally used for picking up image information and/or for detecting at least one optical parameter, for example, a brightness.

25 A large number of optical sensors which can be based generally on the use of inorganic and/or organic sensor materials are known from the prior art. Examples of such sensors are disclosed in US 2007/0176165 A1, US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 or else in numerous other prior art documents. To an increasing extent, in particular for cost reasons and for reasons of large-area processing, sensors comprising at least one organic sensor material
30 are being used, as described for example in US 2007/0176165 A1. In particular, so-called dye solar cells are increasingly of importance here, which are described generally, for example in WO 2009/013282 A1.

35 As a further example, WO 2013/144177 A1 discloses quinolinium dyes having a fluorinated counter anion, an electrode layer which comprises a porous film made of oxide semiconductor fine particles sensitized with these kinds of quinolinium dyes having a fluorinated counter anion, a photoelectric conversion device which comprises such a kind of electrode layer, and a dye sensitized solar cell which comprises such a photoelectric conversion device.

40 A large number of detectors for detecting at least one object are known on the basis of such optical sensors. Such detectors can be embodied in diverse ways, depending on the respective purpose of use. Examples of such detectors are imaging devices, for example, cameras and/or

microscopes. High-resolution confocal microscopes are known, for example, which can be used in particular in the field of medical technology and biology in order to examine biological samples with high optical resolution. Further examples of detectors for optically detecting at least one object are distance measuring devices based, for example, on propagation time methods of corresponding optical signals, for example laser pulses. Further examples of detectors for optically detecting objects are triangulation systems, by means of which distance measurements can likewise be carried out.

Proceeding from such known detectors and methods for optically detecting objects, it can be ascertained that in many cases a considerable technical outlay has to be implemented in order to carry out this object detection with sufficient precision.

By way of example, in microscopy a considerable outlay in respect of apparatus is required in order to obtain correct focusing of a light beam and/or in order to obtain depth information about the sample to be imaged.

Another example for recording depth detection are active sensors which usually function by emitting a signal and measuring reflections of the signal caused by the at least one object. The active sensors have found a wide dissemination, particularly in applications related to computer vision. M.R. Andersen, T. Jensen, P. Lisouski, A.K. Mortensen, M.K. Hansen, T. Gregersen, and P. Ahrendt, *Kinect Depth Sensor Evaluation for Computer Vision Applications*, Technical report ECE-TR-6, 2012, Dept. of Engineering, Aarhus University, Denmark, describe a depth sensor which projects an infrared (IR) point pattern onto at least one object and measures its reflection pattern by use of an IR camera. According to their study, the authors could observe that this depth sensor required a settling time of approximately 30 seconds from the time the device was powered until a reliable depth signal was acquired, in particular owing to a large amount of computing power being required for performing this kind of procedure. Equivalently, a similar settling time could be observed when the depth sensor was rotated and quickly pointed towards a different scene.

A further problem arises when more than one active sensor is employed for acquiring depth information, particularly for extending a measurement range. As soon as signals which originate from one or more different or identical active sensors overlap, the origin of the signals becomes unidentifiable, so that in the region where the signals overlap no depth information can be recorded. A similar problem arises when two or more active sensors point towards each other. Particularly, since the measurement volume of an optical sensor can be described as an approximate cone, a large measurement volume can only be covered by a number of active sensors which, however, results in overlapping volumes and, thus, uncovered regions.

Distance measurements, by contrast, are based in many cases on technically inadequate assumptions such as, for example, the assumption of a specific size of an object in an image evaluation. Other methods are based in turn on complex pulse sequences, such as, for

example, distance measurements by means of laser pulses. Yet other methods are based on the use of a plurality of detectors such as, for example, triangulation methods.

5 In WO 2005/106965 A1, a setup of an organic solar cell is disclosed. A photo current is generated in response to incident light. Further, a method for producing the organic solar cell is disclosed. Therein, reference is made to the fact that defects or traps may diminish the efficiency of the organic solar cell.

10 Various position detectors are known in the art. Thus, in JP 8-159714 A, a distance measurement device is disclosed. Therein, by using a detector and a shadow-forming element, a distance between an object and the detector is determined based on the fact that shadow formation of objects depends on the distance. In US 2008/0259310 A1, an optical position detector is disclosed. The position of a transmission system is determined by using a variety of known distances and measured angles. In US 2005/0184301 A1, a distance measurement
15 device is disclosed. The measurement device makes use of a plurality of light-emitting diodes having different wavelengths. In CN 101650173 A, a position detector is disclosed which is based on the use of geometric principles. Further, in JP 10-221064 A, a complex optical setup is disclosed which is similar to optical setups used in holography.

20 In US 4,767,211, a device and a method for optical measurement and imaging are disclosed. Therein, a ratio of reflected light traveling along an optical axis and reflected light traveling off-axis is determined by using different photo-detectors and a divider. By using this principal, depressions in a sample may be detected.

25 In US 4,647,193, a range of a target object is determined by using a detector having multiple components. The detector is placed away from a focal plane of a lens. The size of a light spot of light from the object varies with the range of the object and, thus, is dependent on the range of the object. By using different photo-detectors, the size of the light spot and, thus, the range of the object may be determined by comparing signals generated by the photo-detectors.

30 In US 6,995,445 and US 2007/0176165 A1, a position sensitive organic detector is disclosed. Therein, a resistive bottom electrode, is used which is electrically contacted by using at least two electrical contacts. By forming a current ratio of the currents from the electric contacts, a position of a light spot on the organic detector may be detected.

35 In US 2007/0080925 A1, a low power consumption display device is disclosed. Therein, photoactive layers are utilized which both respond to electrical energy to allow a display device to display information and which generate electrical energy in response to incident radiation. Display pixels of a single display device may be divided into displaying and generating pixels.
40 The displaying pixels may display information and the generating pixels may generate electrical energy. The generated electrical energy may be used to provide power to drive an image.

US provisional applications 61/739,173, filed on December 19, 2012, and 61/749,964, filed on January 8, 2013, the full content of which is herewith included by reference, disclose a method and a detector for determining a position of at least one object, by using at least one transversal optical sensor and at least one optical sensor. Specifically, the use of sensor stacks is disclosed, in order to determine a longitudinal position of the object with a high degree of accuracy and without ambiguity.

European patent application number EP 13171898.3, filed on June 13, 2013, the full content of which is herewith included by reference, discloses an optical detector comprising an optical sensor having a substrate and at least one photosensitive layer setup disposed thereon. The photosensitive layer setup has at least one first electrode at least one second electrode and at least one photovoltaic material sandwiched in between the first electrode and the second electrode. The photovoltaic material comprises at least one organic material. The first electrode comprises a plurality of first electrode stripes, and the second electrode comprises a plurality of second electrode stripes, wherein the first electrode stripes and the second electrode stripes intersect such that a matrix of pixels is formed at intersections of the first electrode stripes and the second electrode stripes. The optical detector further comprises at least one readout device, the readout device comprising a plurality of electrical measurement devices being connected to the second electrode stripes and a switching device for subsequently connecting the first electrode stripes to the electrical measurement devices.

European patent application number EP 13171900.7, also filed on June 13, 2013, the full content of which is herewith included by reference, discloses a detector device for determining an orientation of at least one object, comprising at least two beacon devices being adapted to be at least one of attached to the object, held by the object and integrated into the object, the beacon devices each being adapted to direct light beams towards a detector, and the beacon devices having predetermined coordinates in a coordinate system of the object. The detector device further comprises at least one detector adapted to detect the light beams traveling from the beacon devices towards the detector and at least one evaluation device, the evaluation device being adapted to determine longitudinal coordinates of each of the beacon devices in a coordinate system of the detector. The evaluation device is further adapted to determine an orientation of the object in the coordinate system of the detector by using the longitudinal coordinates of the beacon devices.

European patent application number EP 13171901.5, filed on June 13, 2013, the full content of which is herewith included by reference, discloses a detector for determining a position of at least one object. The detector comprises at least one optical sensor being adapted to detect a light beam traveling from the object towards the detector, the optical sensor having at least one matrix of pixels. The detector further comprises at least one evaluation device, the evaluation device being adapted to determine a number N of pixels of the optical sensor which are illuminated by the light beam. The evaluation device is further adapted to determine at least one

longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam.

5 In WO 2012/110924 A1, on which the present invention is based and the content of which is herewith included by reference, a detector for optically detecting at least one object is proposed. The detector comprises at least one optical sensor. The optical sensor has at least one sensor region. The optical sensor is designed to generate at least one sensor signal in a manner dependent on an illumination of the sensor region. The sensor signal, given the same total power of the illumination, is dependent on a geometry of the illumination, in particular on a
10 beam cross section of the illumination on the sensor area. The detector furthermore has at least one evaluation device. The evaluation device is designed to generate at least one item of geometrical information from the sensor signal, in particular at least one item of geometrical information about the illumination and/or the object.

15 Despite the advantages implied by the above-mentioned devices and detectors specifically by the detector disclosed in WO 2012/110924 A1, there still is a need for a simple, cost-efficient and, still, reliable spatial detector. Thus, an improved spatial resolution of an object in conjunction with the possibility of tracking the object in space is desirable.

20 Problem addressed by the invention

Therefore, a problem addressed by the present invention is that of specifying devices and methods for optically detecting at least one object which at least substantially avoid the disadvantages of known devices and methods of this type. In particular, an improved detector
25 for determining the position of an object in space and, preferably, for reliably tracking the object in space is desirable.

Summary of the invention

30 This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

35 As used herein, the expressions "have", "comprise" and "contain" as well as grammatical variations thereof are used in a non-exclusive way. Thus, the expression "A has B" as well as the expression "A comprises B" or "A contains B" may both refer to the fact that, besides B, A contains one or more further components and/or constituents, and to the case in which, besides B, no other components, constituents or elements are present in A.

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In a first aspect of the present invention, a detector for determining a position of at least one object is disclosed.

The object generally may be an arbitrary object, chosen from a living object and a non-living object. Thus, as an example, the at least one object may comprise one or more articles and/or one or more parts of an article. Additionally or alternatively, the object may be or may comprise
5 one or more living beings and/or one or more parts thereof, such as one or more body parts of a human being, e.g. a user, and/or an animal.

As used herein, a position generally refers to an arbitrary item of information on a location and/or orientation of the object in space. For this purpose, as an example, one or more
10 coordinate systems may be used, and the position of the object may be determined by using one, two, three or more coordinates. As an example, one or more Cartesian coordinate systems and/or other types of coordinate systems may be used. In one example, the coordinate system may be a coordinate system of the detector in which the detector has a predetermined position and/or orientation. As will be outlined in further detail below, the detector may have an optical
15 axis, which may constitute a main direction of view of the detector. The optical axis may form an axis of the coordinate system, such as a z-axis. Further, one or more additional axes may be provided, preferably perpendicular to the z-axis.

Thus, as an example, the detector may constitute a coordinate system in which the optical axis
20 forms the z-axis and in which, additionally, an x-axis and a y-axis may be provided which are perpendicular to the z-axis and which are perpendicular to each other. As an example, the detector and/or a part of the detector may rest at a specific point in this coordinate system, such as at the origin of this coordinate system. In this coordinate system, a direction parallel or antiparallel to the z-axis may be regarded as a longitudinal direction, and a coordinate along the
25 z-axis may be considered a longitudinal coordinate. An arbitrary direction perpendicular to the longitudinal direction may be considered a transversal direction, and an x- and/or y-coordinate may be considered a transversal coordinate.

Alternatively, other types of coordinate systems may be used. Thus, as an example, a polar
30 coordinate system may be used in which the optical axis forms a z-axis and in which a distance from the z-axis and a polar angle may be used as additional coordinates. Again, a direction parallel or antiparallel to the z-axis may be considered a longitudinal direction, and a coordinate along the z-axis may be considered a longitudinal coordinate. Any direction perpendicular to the z-axis may be considered a transversal direction, and the polar coordinate and/or the polar
35 angle may be considered a transversal coordinate.

As used herein, a detector for determining a position of at least one object generally is a device adapted for providing at least one item of information on the position of the at least one object. The detector may be a stationary device or a mobile device. Further, the detector may be a
40 stand-alone device or may form part of another device, such as a computer, a vehicle or any other device. Further, the detector may be a hand-held device. Other embodiments of the detector are feasible.

The detector may be adapted to provide the at least one item of information on the position of the at least one object in any feasible way. Thus, the information may e.g. be provided electronically, visually, acoustically or in any arbitrary combination thereof. The information may
5 further be stored in a data storage of the detector or a separate device and/or may be provided via at least one interface, such as a wireless interface and/or a wire-bound interface.

The detector comprises:

- at least one transversal optical sensor, the transversal optical sensor being adapted to
10 determine a transversal position of at least one light beam traveling from the object to the detector, the transversal position being a position in at least one dimension perpendicular an optical axis of the detector, the transversal optical sensor being adapted to generate at least one transversal sensor signal;
- at least one longitudinal optical sensor, wherein the longitudinal optical sensor has at
15 least one sensor region, wherein the longitudinal optical sensor is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region;
- at least one evaluation device, wherein the evaluation device is designed to generate at
20 least one item of information on a transversal position of the object by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

As will be outlined in further detail below, the components listed above may be separate
25 components. Alternatively, two or more of the components listed above may be integrated into one component. Thus, the at least one transversal optical sensor and the at least one longitudinal optical sensor may at least partially be integrated into one optical sensor. Alternatively, at least one longitudinal optical sensor may be provided which is separate from at least one transversal optical sensor. Further, the at least one evaluation device may be formed
30 as a separate evaluation device independent from the at least one transversal optical sensor and the at least one longitudinal optical sensor, but may preferably be connected to the at least one transversal optical sensor and the at least one longitudinal optical sensor in order to receive the transversal sensor signal and the longitudinal sensor signal. Alternatively, the at least one evaluation device may fully or partially be integrated into the at least one transversal optical
35 sensor and/or the at least one longitudinal optical sensor.

As used herein, the term transversal optical sensor generally refers to a device which is adapted to determine a transversal position of at least one light beam traveling from the object to the detector. With regard to the term transversal position, reference may be made to the
40 definition given above. Thus, preferably, the transversal position may be or may comprise at least one coordinate in at least one dimension perpendicular to an optical axis of the detector. As an example, the transversal position may be a position of a light spot generated by the light

beam in a plane perpendicular to the optical axis, such as on a light-sensitive sensor surface of the transversal optical sensor. As an example, the position in the plane may be given in Cartesian coordinates and/or polar coordinates. Other embodiments are feasible.

- 5 For potential embodiments of the transversal optical sensor, reference may be made to the position sensitive organic detector as disclosed in US 6,995,445 and US 2007/0176165 A1. However, other embodiments are feasible and will be outlined in further detail below.

10 The at least one transversal sensor signal generally may be an arbitrary signal indicative of the transversal position. As an example, the transversal sensor signal may be or may comprise a digital and/or an analog signal. As an example, the transversal sensor signal may be or may comprise a voltage signal and/or a current signal. Additionally or alternatively, the transversal sensor signal may be or may comprise digital data. The transversal sensor signal may comprise a single signal value and/or a series of signal values. The transversal sensor signal may further
15 comprise an arbitrary signal which is derived by combining two or more individual signals, such as by averaging two or more signals and/or by forming a quotient of two or more signals, as will be outlined in further detail below.

As used herein, a longitudinal optical sensor generally is a device which is designed to generate
20 at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region. For potential embodiments of the longitudinal optical sensor, reference may be made to the optical sensor as disclosed in WO 2012/110924 A1. Preferably, however, as will be outlined in
25 further detail below, the detector according to the present invention comprises a plurality of optical sensors, such as a plurality of optical sensors as disclosed in WO 2012/110924 A1, preferably as a sensor stack.

Thus, as an example, the detector according to the present invention may comprise a stack of
30 optical sensors as disclosed in WO 2012/110924 A1, in combination with one or more transversal optical sensors. As an example, one or more transversal optical sensors may be disposed on a side of the stack of longitudinal optical sensors facing towards the object. Alternatively or additionally, one or more transversal optical sensors may be disposed on a side of the stack of longitudinal optical sensors facing away from the object. Again, additionally or
35 alternatively, one or more transversal article sensors may be interposed in between the longitudinal optical sensors of the stack.

As will further be outlined below, preferably, both the at least one transversal optical sensor longitudinal optical sensor and the at least one longitudinal optical sensor may comprise one or
40 more photo detectors, preferably one or more organic photodetectors and, most preferably, one or more dye-sensitized organic solar cells (DSCs, also referred to as dye solar cells), such as one or more solid dye-sensitized organic solar cells (s-DSCs). Thus, preferably, the detector

may comprise one or more DSCs (such as one or more sDSCs) acting as the at least one transversal optical sensor and one or more DSCs (such as one or more sDSCs) acting as the at least one longitudinal optical sensor, preferably a stack of a plurality of DSCs (preferably a stack of a plurality of sDSCs) acting as the at least one longitudinal optical sensor.

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As used herein, the term evaluation device generally refers to an arbitrary device designed to generate the at least one item of information on the transversal position of the object and the longitudinal position of the object. As an example, the evaluation device may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the transversal sensor signal and/or the longitudinal sensor signal, such as one or more AD-converters and/or one or more filters. Further, the evaluation device may comprise one or more data storage devices. Further, as outlined above, the evaluation device may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

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The at least one evaluation device may be adapted to perform at least one computer program, such as at least one computer program performing or supporting the step of generating the at least one item of information on the transversal position and/or the step of generating the at least one item of information on the longitudinal position. As an example, one or more algorithms may be implemented which, by using the transversal sensor signal and/or the longitudinal sensor signal as input variables, may perform a predetermined transformation into the transversal position and/or the longitudinal position of the object.

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As outlined above, preferably, the transversal optical sensor is a photo detector having at least one first electrode, at least one second electrode and at least one photovoltaic material, wherein the photovoltaic material is embedded in between the first electrode and the second electrode. As used herein, a photovoltaic material generally is a material or combination of materials adapted to generate electric charges in response to an illumination of the photovoltaic material with light.

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As used herein, the term light generally refers to electromagnetic radiation in one or more of the visible spectral range, the ultraviolet spectral range and the infrared spectral range. Therein, the term visible spectral range generally refers to a spectral range of 380 nm to 780 nm. The term infrared (IR) spectral range generally refers to electromagnetic radiation in the range of 780 nm to 1000 μm , preferably in the range of 780 nm to 3.0 μm . The term ultraviolet spectral range generally refers to electromagnetic radiation in the range of 1 nm to 380 nm, preferably in the range of 100 nm to 380 nm. Preferably, light as used within the present invention is visible light, i.e. light in the visible spectral range.

The term light beam generally refers to an amount of light emitted into a specific direction. Thus, the light beam may be a bundle of the light rays having a predetermined extension in a direction perpendicular to a direction of propagation of the light beam. Preferably, the light beam may be or may comprise one or more Gaussian light beams which may be characterized by one or more Gaussian beam parameters, such as one or more of a beam waist, a Rayleigh-length or any other beam parameter or combination of beam parameters suited to characterize a development of a beam diameter and/or a beam propagation in space.

Preferably, the second electrode of the transversal optical sensor may be a split electrode having at least two partial electrodes, wherein the transversal optical sensor has a sensor area, wherein the at least one transversal sensor signal indicates a position of the light beam in the sensor area. Thus, as outlined above, the transversal optical sensor may be or may comprise one or more photo detectors, preferably one or more organic photo detectors, more preferably one or more DSCs or sDSCs. The sensor area may be a surface of the photo detector facing towards the object. The sensor area preferably may be oriented perpendicular to the optical axis. Thus, the transversal sensor signal may indicate a position of a light spot generated by the light beam in a plane of the sensor area of the transversal optical sensor.

Generally, as used herein, the term partial electrode refers to an electrode out of a plurality of electrodes, adapted for measuring at least one current and/or voltage signal, preferably independent from other partial electrodes. Thus, in case a plurality of partial electrodes is provided, the second electrode is adapted to provide a plurality of electric potentials and/or electric currents and/or voltages via the at least two partial electrodes, which may be measured and/or used independently.

When using at least one transversal optical sensor having at least one split electrode having two or more partial electrodes as a second electrode, currents through the partial electrodes may be dependent on a position of the light beam in the sensor area. This may generally be due to the fact that Ohmic losses or resistive losses may occur on the way from a location of generation of electrical charges due to the impinging light to the partial electrodes. Thus, besides the partial electrodes, the second electrode may comprise one or more additional electrode materials connected to the partial electrodes, wherein the one or more additional electrode materials provide an electrical resistance. Thus, due to the Ohmic losses on the way from the location of generation of the electric charges to the partial electrodes through with the one or more additional electrode materials, the currents through the partial electrodes depend on the location of the generation of the electric charges and, thus, to the position of the light beam in the sensor area. For details of this principle of determining the position of the light beam in the sensor area, reference may be made to the preferred embodiments below and/or to the physical principles and device options as disclosed e.g. in US 6,995,445 and/or US 2007/0176165 A1.

The transversal optical sensor may further be adapted to generate the transversal sensor signal in accordance with the electrical currents through the partial electrodes. Thus, a ratio of electric currents through two horizontal partial electrodes may be formed, thereby generating an x-coordinate, and/or a ratio of electric currents through two vertical partial electrodes may be formed, thereby generating a y-coordinate. The detector, preferably the transversal optical sensor and/or the evaluation device, may be adapted to derive the information on the transversal position of the object from at least one ratio of the currents through the partial electrodes. Other ways of generating position coordinates by comparing currents through the partial electrodes are feasible.

The partial electrodes generally may be defined in various ways, in order to determine a position of the light beam in the sensor area. Thus, two or more horizontal partial electrodes may be provided in order to determine a horizontal coordinate or x-coordinate, and two or more vertical partial electrodes may be provided in order to determine a vertical coordinate or y-coordinate. Thus, the partial electrodes may be provided at a rim of the sensor area, wherein an interior space of the sensor area remains free and may be covered by one or more additional electrode materials. As will be outlined in further detail below, the additional electrode material preferably may be a transparent additional electrode material, such as a transparent metal and/or a transparent conductive oxide and/or, most preferably, a transparent conductive polymer.

Further preferred embodiments may refer to the photovoltaic material. Thus, the photovoltaic material of the transversal optical sensor may comprise at least one organic photovoltaic material. Thus, generally, the transversal optical sensor may be an organic photo detector. Preferably, the organic photo detector may be a dye-sensitized solar cell. The dye-sensitized solar cell preferably may be a solid dye-sensitized solar cell, comprising a layer setup embedded in between the first electrode and the second electrode, the layer setup comprising at least one n-semiconducting metal oxide, at least one dye, and at least one solid p-semiconducting organic material. Further details and optional embodiments of the dye-sensitized solar cell (DSC) will be disclosed below.

The at least one first electrode of the transversal optical sensor preferably is transparent. As used in the present invention, the term transparent generally refers to the fact that the intensity of light after transmission through the transparent object equals to or exceeds 10%, preferably 40% and, more preferably, 60% of the intensity of light before transmission through the transparent object. More preferably, the at least one first electrode of the transversal optical sensor may fully or partially be made of at least one transparent conductive oxide (TCO). As an example, indium-doped tin oxide (ITO) and/or fluorine-doped tin oxide (FTO) may be named. Further examples will be given below.

Further, the at least one second electrode of the transversal optical sensor preferably may fully or partially be transparent. Thus, specifically, the at least one second electrode may comprise

two or more partial electrodes and at least one additional electrode material contacting the two or more partial electrodes. The two or more partial electrodes may be intransparent. As an example, the two or more partial electrodes may fully or partially be made of a metal. Thus, the two or more partial electrodes preferably are located at a rim of the sensor area. The two or more partial electrodes, however, may electrically be connected by the at least one additional electrode material which, preferably, is transparent. Thus, the second electrode may comprise an intransparent rim having the two or more partial electrodes and a transparent inner area having the at least one transparent additional electrode material. More preferably, the at least one second electrode of the transversal optical sensor, such as the above-mentioned at least one additional electrode material, may fully or partially be made of at least one conductive polymer, preferably a transparent conductive polymer. As an example, conductive polymers having an electrical conductivity of at least 0.01 S/cm may be used, preferably of at least 0.1 S/cm or, more preferably, of at least 1 S/cm or even at least 10 S/cm or at least 100 S/cm. As an example, the at least one conductive polymer may be selected from the group consisting of: a poly-3,4-ethylenedioxythiophene (PEDOT), preferably PEDOT being electrically doped with at least one counter ion, more preferably PEDOT doped with sodium polystyrene sulfonate (PEDOT:PSS); a polyaniline (PANI); a polythiophene.

As outlined above, the conductive polymer may provide an electrical connection between the at least two partial electrodes. The conductive polymer may provide an Ohmic resistivity, allowing for determining the position of charge generation. Preferably, the conductive polymer provides an electric resistivity of 0.1 – 20 k Ω between the partial electrodes, preferably an electric resistivity of 0.5 – 5.0 k Ω and, more preferably, an electric resistivity of 1.0 – 3.0 k Ω .

Generally, as used herein, a conductive material may be a material which have a specific electrical resistance of less than 10^4 , less than 10^3 , less than 10^2 , or of less than 10 Ω m. Preferably, the conductive material has a specific electrical resistance of less than 10^{-1} , less than 10^{-2} , less than 10^{-3} , less than 10^{-5} , or less than 10^{-6} Ω m. Most preferably, the specific electrical resistance of the conductive material is less than 5×10^{-7} Ω m or is less than 1×10^{-7} Ω m, particularly in the range of the specific electrical resistance of aluminum.

As outlined above, preferably, at least one of the transversal optical sensor and the longitudinal optical sensor is a transparent optical sensor. Thus, the at least one transversal optical sensor may be a transparent transversal optical sensor and/or may comprise at least one transparent transversal optical sensor. Additionally or alternatively, the at least one longitudinal optical sensor may be a transparent longitudinal optical sensor and/or may comprise at least one transparent longitudinal optical sensor. In case a plurality of longitudinal optical sensors is provided, such as a stack of longitudinal optical sensors, preferably all longitudinal optical sensors of the plurality and/or the stack or all longitudinal optical sensors of the plurality and/or the stack but one longitudinal optical sensor are transparent. As an example, in case a stack of longitudinal optical sensors is provided, wherein the longitudinal optical sensors are arranged along the optical axis of the detector, preferably all longitudinal optical sensors but the last

longitudinal optical sensor facing away from the object may be transparent longitudinal optical sensors. The last longitudinal optical sensor, i.e. the longitudinal optical sensor on the side of the stack facing away from the object, may be a transparent longitudinal optical sensor or an intransparent longitudinal optical sensor. Exemplary embodiments will be given below.

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In case one of the transversal optical sensor and the longitudinal optical sensor is a transparent optical sensor or comprises at least one transparent optical sensor, the light beam may pass through the transparent optical sensor before impinging on the other one of the transversal optical sensor and the longitudinal optical sensor. Thus, the light beam from the object may subsequently reach the transversal optical sensor and the longitudinal optical sensor or vice versa.

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Further embodiments refer to the relationship between the transversal optical sensor and the longitudinal optical sensor. Thus, in principle, the transversal optical sensor and the longitudinal optical sensor at least partially may be identical, as outlined above. Preferably, however, the transversal optical sensor and the longitudinal optical sensor at least partially may be independent optical sensors, such as independent photo detectors and, more preferably, independent DSCs or sDSCs.

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As outlined above, the transversal optical sensor and the longitudinal optical sensor preferably may be stacked along the optical axis. Thus, a light beam travelling along the optical axis may both impinge on the transversal optical sensor and on the longitudinal optical sensor, preferably subsequently. Thus, the light beam may subsequently pass through the transversal optical sensor and the longitudinal optical sensor or vice versa.

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Further embodiments of the present invention referred to the nature of the light beam which propagates from the object to the detector. The light beam might be admitted by the object itself, i.e. might originate from the object. Additionally or alternatively, another origin of the light beam is feasible. Thus, as will be outlined in further detail below, one or more illumination sources might be provided which illuminate the object, such as by using one or more primary rays or beams, such as one or more primary rays or beams having a predetermined characteristic. In the latter case, the light beam propagating from the object to the detector might be a light beam which is reflected by the object and/or a reflection device connected to the object.

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As outlined above, the at least one longitudinal sensor signal, given the same total power of the illumination by the light beam, is dependent on a beam cross-section of the light beam in the sensor region of the at least one longitudinal optical sensor. As used herein, the term beam cross-section generally refers to a lateral extension of the light beam or a light spot generated by the light beam at a specific location. In case a circular light spot is generated, a radius, a diameter or a Gaussian beam waist or twice the Gaussian beam waist may function as a measure of the beam cross-section. In case non-circular light-spots are generated, the cross-

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section may be determined in any other feasible way, such as by determining the cross-section of a circle having the same area as the non-circular light spot, which is also referred to as the equivalent beam cross-section.

5 Thus, given the same total power of the illumination of the sensor region by the light beam, a light beam having a first beam diameter or beam cross-section may generate a first longitudinal sensor signal, whereas a light beam having a second beam diameter or beam-cross section being different from the first beam diameter or beam cross-section generates a second longitudinal sensor signal being different from the first longitudinal sensor signal. Thus, by
10 comparing the longitudinal sensor signals, an information or at least one item of information on the beam cross-section, specifically on the beam diameter, may be generated. For details of this effect, reference may be made to WO 2012/110924 A1. Specifically in case one or more beam properties of the light beam propagating from the object to the detector are known, the at least one item of information on the longitudinal position of the object may thus be derived from
15 a known relationship between the at least one longitudinal sensor signal and a longitudinal position of the object. The known relationship may be stored in the evaluation device as an algorithm and/or as one or more calibration curves. As an example, specifically for Gaussian beams, a relationship between a beam diameter or beam waist and a position of the object may easily be derived by using the Gaussian relationship between the beam waist and a longitudinal
20 coordinate.

The above-mentioned effect, which is also referred to as the FiP-effect (alluding to the effect that the beam cross section ϕ influences the electric power P generated by the longitudinal optical sensor) may depend on or may be emphasized by an appropriate modulation of the light
25 beam, as disclosed in WO 2012/110924 A1. Thus, preferably, the detector may furthermore have at least one modulation device for modulating the illumination. The detector may be designed to detect at least two longitudinal sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies. In this case, the evaluation device may be designed to generate the at least one item of information on
30 the longitudinal position of the object by evaluating the at least two longitudinal sensor signals.

Generally, the longitudinal optical sensor may be designed in such a way that the at least one longitudinal sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination. Further details and exemplary
35 embodiments will be given below. This property of frequency dependency is specifically provided in DSCs and, more preferably, in sDSCs. However, other types of optical sensors, preferably photo detectors and, more preferably, organic photo detectors may exhibit this effect.

Preferably, the transversal optical sensor and the longitudinal optical sensor both are thin film
40 devices, having a layer setup of layer including electrode and photovoltaic material, the layer setup having a thickness of preferably no more than 1 mm, more preferably of at most 500 μm or even less. Thus, the sensor region of the transversal optical sensor and/or the sensor region

of the longitudinal optical sensor preferably each may be or may comprise a sensor area, which may be formed by a surface of the respective device, wherein the surface may face towards the object or may face away from the object. Hereby, it may further be feasible to arrange the at least one transversal optical sensor and the at least one longitudinal optical sensor in a way that some surfaces comprising the sensor regions may face towards the object where other surfaces may face away from the object. Such a kind of arrangement of the respective devices, which might be helpful to optimize the path of the light beam through the stack and/or to reduce reflections within the light path, may, for any reason or purpose, be implemented in an alternating manner, such as with one, two, three or more devices where the sensor regions may face towards the object alternating with one, two, three or more other devices where the sensor regions may face away from the object.

Preferably, the sensor region of the transversal optical sensor and/or the sensor region of the longitudinal optical sensor may be formed by one continuous sensor region, such as one continuous sensor area or sensor surface per device. Thus, preferably, the sensor region of the longitudinal optical sensor or, in case a plurality of longitudinal optical sensors is provided (such as a stack of longitudinal optical sensors), each sensor region of the longitudinal optical sensor, may be formed by exactly one continuous sensor region. The longitudinal sensor signal preferably is a uniform sensor signal for the entire sensor region of the longitudinal optical sensor or, in case a plurality of longitudinal optical sensors is provided, is a uniform sensor signal for each sensor region of each longitudinal optical sensor.

The at least one transversal optical sensor and/or the at least one longitudinal optical sensor each, independently, may have a sensor region providing a sensitive area, also referred to as a sensor area, of at least 1 mm², preferably of at least 5 mm², such as a sensor area of 5 mm² to 1000 cm², preferably a sensor area of 7 mm² to 100 cm², more preferably a sensor area of 1 cm². The sensor area preferably has a rectangular geometry, such as a square geometry. However, other geometries and/or sensor areas are feasible.

The longitudinal sensor signal preferably may be selected from the group consisting of a current (such as a photocurrent) and a voltage (such as a photo voltage). Similarly, the transversal sensor signal preferably may be selected from the group consisting of a current (such as a photocurrent) and a voltage (such as a photo voltage) or any signal derived thereof, such as a quotient of currents and/or voltages. Further, longitudinal sensor signal and/or the transversal sensor signal may be preprocessed, in order to derive refined sensor signals from raw sensor signals, such as by averaging and/or filtering.

Generally, the longitudinal optical sensor may comprise at least one semiconductor detector, in particular an organic semiconductor detector comprising at least one organic material, preferably an organic solar cell and particularly preferably a dye solar cell or dye-sensitized solar cell, in particular a solid dye solar cell or a solid dye-sensitized solar cell. Preferably, the longitudinal optical sensor is or comprises a DSC or sDSC. Thus, preferably, the longitudinal

optical sensor comprises at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode. In a preferred embodiment, the longitudinal optical sensor comprises at least one DSC or, more preferably, at least one sDSC. As outlined above, preferably, the at least one longitudinal optical sensor is a transparent longitudinal optical sensor or comprises at least one transparent longitudinal optical sensor. Thus, preferably, both the first electrode and the second electrode are transparent or, in case a plurality of longitudinal optical sensors is provided, at least one of the longitudinal optical sensors is designed such that both the first electrode and the second electrode are transparent.

As outlined above, in case a stack of longitudinal optical sensors is provided, preferably some or even all longitudinal optical sensors of the stack are transparent but the last longitudinal optical sensor of the stack. The last longitudinal optical sensor of the stack, i.e. the longitudinal optical sensor of the stack furthest away from the object, may be transparent or intransparent.

The stack may, besides the at least one transversal optical sensor and the at least one longitudinal optical sensor, comprise one or more further optical sensors which may function as one or more of a transversal optical sensor, a longitudinal optical sensor (also referred to as an imaging device) and an imaging sensor.

Thus, it may be possible to locate an imaging device in the optical path of the light beam in a manner that the light beam travels through the stack of the transparent longitudinal optical sensors until it impinges on the imaging device.

Thus, generally, the detector may further comprise at least one imaging device, i.e. a device capable of acquiring at least one image. The imaging device can be embodied in various ways. Thus, the imaging device can be for example part of the detector in a detector housing. Alternatively or additionally, however, the imaging device can also be arranged outside the detector housing, for example as a separate imaging device. Alternatively or additionally, the imaging device can also be connected to the detector or even be part of the detector. In a preferred arrangement, the stack of the transparent longitudinal optical sensors and the imaging device are aligned along a common optical axis along which the light beam travels. However, other arrangements are possible.

In addition, the detector may comprise at least one transfer device, such as an optical lens, which will be described later in more detail, and which may further be arranged along the common optical axis. By way of example, the light beam which emerges from the object may in this case travel first through the at least one transfer device and thereafter through the stack of the transparent longitudinal optical sensors until it finally impinges on the imaging device.

As used herein, an imaging device is generally understood as a device which can generate a one-dimensional, a two-dimensional, or a three-dimensional image of the object or of a part thereof. In particular, the detector, with or without the at least one optional imaging device, can

be completely or partly used as a camera, such as an IR camera, or an RGB camera, i.e. a camera which is designed to deliver three basic colors which are designated as red, green, and blue, on three separate connections. Thus, as an example, the at least one imaging device may be or may comprise at least one imaging device selected from the group consisting of: a
5 pixelated organic camera element, preferably a pixelated organic camera chip; a pixelated inorganic camera element, preferably a pixelated inorganic camera chip, more preferably a CCD- or CMOS-chip; a monochrome camera element, preferably a monochrome camera chip; a multicolor camera element, preferably a multicolor camera chip; a full-color camera element, preferably a full-color camera chip. The imaging device may be or may comprise at least one
10 device selected from the group consisting of a monochrome imaging device, a multi-chrome imaging device and at least one full color imaging device. A multi-chrome imaging device and/or a full color imaging device may be generated by using filter techniques and/or by using intrinsic color sensitivity or other techniques, as the skilled person will recognize. Other embodiments of the imaging device are also possible.

15 The imaging device may be designed to image a plurality of partial regions of the object successively and/or simultaneously. By way of example, a partial region of the object can be a one-dimensional, a two-dimensional, or a three-dimensional region of the object which is delimited for example by a resolution limit of the imaging device and from which electromagnetic
20 radiation emerges. In this context, imaging should be understood to mean that the electromagnetic radiation which emerges from the respective partial region of the object is fed into the imaging device, for example by means of the at least one optional transfer device of the detector. The electromagnetic rays can be generated by the object itself, for example in the form of a luminescent radiation. Alternatively or additionally, the at least one detector may
25 comprise at least one illumination source for illuminating the object.

In particular, the imaging device can be designed to image sequentially, for example by means of a scanning method, in particular using at least one row scan and/or line scan, the plurality of partial regions sequentially. However, other embodiments are also possible, for example
30 embodiments in which a plurality of partial regions is simultaneously imaged. The imaging device is designed to generate, during this imaging of the partial regions of the object, signals, preferably electronic signals, associated with the partial regions. The signal may be an analogue and/or a digital signal. By way of example, an electronic signal can be associated with each partial region. The electronic signals can accordingly be generated simultaneously or else
35 in a temporally staggered manner. By way of example, during a row scan or line scan, it is possible to generate a sequence of electronic signals which correspond to the partial regions of the object, which are strung together in a line, for example. Further, the imaging device may comprise one or more signal processing devices, such as one or more filters and/or analogue-digital-converters for processing and/or preprocessing the electronic signals.

40 As outlined above, the at least one longitudinal optical sensor may be transparent or intransparent, or may comprise at least one transparent longitudinal optical sensor. A

combination of at least one transparent and at least one intransparent longitudinal optical sensor is possible.

5 In a further preferred embodiment, the last longitudinal optical sensor may be intransparent. For this purpose, at least those parts of the last longitudinal optical sensor which are subject to be illuminated by the light beam which travels from the object and which may impinge on the last longitudinal optical sensor may comprise an optical sensor material, preferably an inorganic optical sensor material, and/or an organic optical sensor material, and/or a hybrid organic – inorganic optical sensor material, which exhibits intransparent optical properties. Intransparency
10 may also be achieved by using at least one intransparent electrode. In this embodiment, the last longitudinal optical sensor may be designed such that its electrode facing towards the object is transparent, whereas its electrode facing away from the object may be intransparent, or vice-versa. Additionally or alternatively, a respective material which exhibits intransparent optical properties may be selected for at least one of the n-semiconducting metal oxides, for at least
15 one of the dyes, and/or for at least one of the p-semiconducting organic materials which may comprise the last longitudinal optical sensor.

As outlined above, the detector may comprise at least one imaging device. The imaging device may fully or partially be embodied as an independent imaging device, independent from the at
20 least one transversal optical sensor and the at least one longitudinal optical sensor. Additionally or alternatively, the at least one optional imaging device may fully or partially be integrated into one or both of the at least one transversal optical sensor and the at least one longitudinal optical sensor. Thus, as an example, the imaging device may be used for determining a transversal position of a light spot and, thus, may be used as a transversal optical sensor or as a part
25 thereof.

As outlined above, the detector may comprise a stack of at least two optical sensors, the at least two optical sensors comprising the at least one transversal optical sensor and the at least one longitudinal optical sensor and, optionally, the at least one imaging device. Thus, as an
30 example, the stack may comprise the at least one transversal optical sensor, the at least one longitudinal optical sensor (preferably at least one transparent longitudinal optical sensor) and, optionally, in a position furthest away from the object, at least one imaging device, preferably at least one intransparent imaging device such as a CCD or CMOS chip.

35 The stack of at least two optical sensors optionally may partially or fully be immersed in an oil, in a liquid and/or in a solid material in order to avoid and/or decrease reflections at interfaces. Hereby, the oil, the liquid, and/or the solid material may preferably be transparent, preferentially to a high degree, at least over a part of the ultraviolet, visible, and/or infrared spectral range. In a preferred embodiment, the solid material may be generated by inserting at least one curable
40 substance into a region between at least two optical sensors and treating the curable substance with a treatment, such as by incident light, particularly with light within the ultraviolet range, and/or by an application of a temperature above or below room temperature, by which treatment

the curable substance might be cured, preferentially by hardening the curable substance into the solid material. Alternatively, at least two different curable substances may be inserted into a region between at least two optical sensors, whereby the two different curable substances are selected in a manner that they begin to set into the solid material with or without the treatment as indicated above. However, further treatments and/or other procedures of providing the transparent solid material may be possible. Thus, at least one of the optical sensors of the stack may fully or partially be immersed in the oil and/or the liquid and/or covered with the solid material.

Alternatively or additionally, the region between the at least two optical sensors may be partially or fully filled with a substance, such as the oil, the liquid and/or the solid material. Hereby, the substance may preferably exhibit a refractive index with a value which may differ from that of the optical sensors adjoining to the substance on one or both sides of the region. However, inserting the additional substance in the regions may require the optical sensors within the stack to observe a minimum spacing between them.

In case a stack of at least two optical sensors is used, the last optical sensor of the stack may be transparent or intransparent. Thus, an intransparent inorganic optical sensor may be used in a position furthest away from the object. As an example, the last optical sensor of the stack may be or may comprise the at least one optional imaging device, such as at least one CCD or CMOS chip, preferably a full-color CCD or CMOS chip.

Thus, the intransparent last optical sensor may be used as an imaging device, wherein the imaging device is impinged by the light beam after the light beam has traveled before through the stack of the transparent optical sensors until it impinges the imaging device. In particular, the imaging device can be completely or partly used as a camera, such as an IR camera, or an RGB camera, as described above. Hereby, the intransparent last optical sensor can be embodied in various ways as imaging device. Thus, the intransparent last optical sensor can be for example part of the detector in a detector housing. Alternatively or additionally, however, the intransparent last optical sensor can also be arranged outside the detector housing, for example as a separate imaging device.

The stack comprising the at least one transversal optical sensor, the at least one longitudinal optical sensor and the optional at least one imaging device may be designed such that the elements of the stack are arranged along an optical axis of the detector. The last element of the stack may be an intransparent optical sensor, preferably selected from the group consisting of an intransparent transversal optical sensor, an intransparent longitudinal optical sensor and an intransparent imaging device such as an intransparent CCD or CMOS chip.

In a preferred arrangement, the stack comprising the at least one transversal optical sensor, the at least one longitudinal optical sensor and, optionally, the at least one imaging device may be arranged along a common optical axis of the detector, along which the light beam may travel. In

case the stack contains a plurality of optical sensors, the optical sensors comprising the at least one transversal optical sensor, the at least one longitudinal optical sensor and, optionally, the at least one imaging device, wherein at least one of the optical sensors is a transparent optical sensor and wherein at least one of the optical sensors is an intransparent optical sensor, the transparent optical sensor and the intransparent optical sensor, the latter preferably being positioned furthest away from the object, may be arranged along the optical axis of the detector. However, other arrangements are possible.

In a further preferred embodiment, the intransparent last optical sensor is having at least one matrix of pixels, wherein a 'matrix' generally refers to an arrangement of a plurality of the pixels in space, which may be a linear arrangement or an areal arrangement. Generally, the matrix may, thus, preferably be selected from the group consisting of a one-dimensional matrix and a two-dimensional matrix. As an example, the matrix may comprise 100 to 100 000 000 pixels, preferably 1 000 to 1 000 000 pixels and, more preferably, 10 000 to 500 000 pixels. Most preferably, the matrix is a rectangular matrix having pixels arranged in rows and columns.

As further used herein, a pixel generally refers to a light-sensitive element of an optical sensor, such as a minimum uniform unit of the optical sensor adapted to generate a light signal. As an example, each pixel may have a light-sensitive area of $1\ \mu\text{m}^2$ to $5\,000\,000\ \mu\text{m}^2$, preferably $100\ \mu\text{m}^2$ to $4\,000\,000\ \mu\text{m}^2$, preferably $1\,000\ \mu\text{m}^2$ to $1\,000\,000\ \mu\text{m}^2$ and more preferably $2\,500\ \mu\text{m}^2$ to $50\,000\ \mu\text{m}^2$. Still, other embodiments are feasible. The intransparent last optical sensor may be adapted to generate at least one signal which indicates an intensity of illumination for each of the pixels. Thus, as an example, the intransparent last optical sensor may be adapted to generate at least one electronic signal for each of the pixels, whereby each signal indicates the intensity of illumination for the respective pixel. The signal may be an analogue and/or a digital signal. Further, the detector may comprise one or more signal processing devices, such as one or more filters and/or analogue-digital-converters for processing and/or preprocessing the at least one signal.

The intransparent last optical sensor having a matrix of pixels can be selected from the group consisting of: an inorganic semiconductor sensor device such as a CCD chip and/or a CMOS chip; an organic semiconductor sensor device. In the latter case, as an example, the optical sensor may, for example, comprise at least one organic photovoltaic device having a matrix of pixels. As used herein, an organic photovoltaic device generally refers to a device having at least one organic photosensitive element and/or at least one organic layer. Therein, generally, any type of organic photovoltaic device may be used, such as organic solar cells and/or an arbitrary device having at least one organic photosensitive layer. As an example, an organic solar cell and/or a dye-sensitized organic solar cell may be comprised. Further, a hybrid device may be used, such as inorganic – organic photovoltaic devices.

Further preferred embodiments refer to the evaluation device. Thus, the evaluation device may be designed to generate the at least one item of information on the longitudinal position of the

object from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object with respect to the detector, preferably taking account of a known power of the illumination and optionally taking account of a modulation frequency with which the illumination is modulated.

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In a further preferred embodiment, the detector furthermore may comprise at least one transfer device, wherein the transfer device is designed to feed light emerging from the object to the transversal optical sensor and the longitudinal optical sensor, preferably subsequently. Details and preferred embodiments will be given below.

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As outlined above, the light beam propagating from the object of the detector may originate from the object or may originate from any other source. Thus, the object itself may emit the light beam. Additionally or alternatively, the object might be illuminated by using an illumination source generating primary light, wherein the object elastically or inelastically reflects the primary light, thereby generating the light beam propagating to the detector. The illumination source itself may be part of the detector. Thus, the detector may comprise at least one illumination source. The illumination source generally may be selected from: an illumination source, which is at least partly connected to the object and/or is at least partly identical to the object; an illumination source which is designed to at least partly illuminate the object with a primary radiation, preferably primary light, wherein the light beam preferably is generated by a reflection of the primary radiation on the object and/or by light emission by the object itself, stimulated by the primary radiation.

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As outlined above, the detector preferably has a plurality of longitudinal optical sensors. More preferably, the plurality of longitudinal optical sensors is stacked, such as along the optical axis of the detector. Thus, the longitudinal optical sensors may form a longitudinal optical sensor stack. The longitudinal optical sensor stack preferably may be oriented such that the sensor regions of the longitudinal optical sensors are oriented perpendicular to the optical axis. Thus, as an example, sensor areas or sensor surfaces of the single longitudinal optical sensors may be oriented in parallel, wherein slight angular tolerances might be tolerable, such as angular tolerances of no more than 10°, preferably of no more than 5°.

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In case stacked longitudinal optical sensors are provided, the at least one transversal optical sensor preferably fully or partially is located on a side of the stacked longitudinal optical sensors facing the object. However, other embodiments are feasible. Thus, embodiments in which the at least one transversal optical sensor is fully or partially located on a side of the transversal optical sensor stack facing away from the object. Again, additionally or alternatively, embodiments are feasible in which the at least one transversal optical sensor is located fully or partially in between the longitudinal optical sensor stack.

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The longitudinal optical sensors preferably are arranged such that a light beam from the object illuminates all longitudinal optical sensors, preferably sequentially. Specifically in this case,

preferably, at least one longitudinal sensor signal is generated by each longitudinal optical sensor. This embodiment is specifically preferred since the stacked setup of the longitudinal optical sensors allows for an easy and efficient normalization of the signals, even if an overall power or intensity of the light beam is unknown. Thus, the single longitudinal sensor signals may be known to be generated by one and the same light beam. Thus, the evaluation device may be adapted to normalize the longitudinal sensor signals and to generate the information on the longitudinal position of the object independent from an intensity of the light beam. For this purpose, use may be made of the fact that, in case the single longitudinal sensor signals are generated by one and the same light beam, differences in the single longitudinal sensor signals are only due to differences in the cross-sections of the light beam at the location of the respective sensor regions of the single longitudinal optical sensors. Thus, by comparing the single longitudinal sensor signals, information on a beam cross-section may be generated even if the overall power of the light beam is unknown. From the beam cross-section, information regarding the longitudinal position of the object may be gained, specifically by making use of a known relationship between the cross-section of the light beam and the longitudinal position of the object.

Further, the above-mentioned stacking of the longitudinal optical sensors and the generation of a plurality of longitudinal sensor signals by these stacked longitudinal optical sensors may be used by the evaluation device in order to resolve an ambiguity in a known relationship between a beam cross-section of the light beam and the longitudinal position of the object. Thus, even if the beam properties of the light beam propagating from the object to the detector are known fully or partially, it is known that, in many beams, the beam cross-section narrows before reaching a focal point and, afterwards, widens again. Thus, before and often as a focal point in which the light beam has the narrowest beam cross-section, positions along the axis of propagation of the light beam occur in which the light beam has the same cross-section. Thus, as an example, at a distance z_0 before and after the focal point, the cross-section of the light beam is identical. Thus, in case only one longitudinal optical sensor is used, a specific cross-section of the light beam might be determined, in case the overall power or intensity of the light beam is known. By using this information, the distance z_0 of the respective longitudinal optical sensor from the focal point might be determined. However, in order to determine whether the respective longitudinal optical sensor is located before or behind the focal point, additional information is required, such as a history of movement of the object and/or the detector and/or information on whether the detector is located before or behind the focal point. In typical situations, this additional information may not be provided. Therefore, by using a plurality of longitudinal optical sensors, additional information may be gained in order to resolve the above-mentioned ambiguity. Thus, in case the evaluation device, by evaluating the longitudinal sensor signals, recognizes that the beam cross-section of the light beam on a first longitudinal optical sensor is larger than the beam cross-section of the light beam on a second longitudinal optical sensor, wherein the second longitudinal optical sensor is located behind the first longitudinal optical sensor, the evaluation device may determine that the light beam is still narrowing and that the location of the first longitudinal optical sensor is situated before the focal point of the

light beam. Contrarily, in case the beam cross-section of the light beam on the first longitudinal optical sensor is smaller than the beam cross-section of the light beam on the second longitudinal optical sensor, the evaluation device may determine that the light beam is widening and that the location of the second longitudinal optical sensor is situated behind the focal point.

- 5 Thus, generally, the evaluation device may be adapted to recognize whether the light beam widens or narrows, by comparing the longitudinal sensor signals of different longitudinal sensors.

- 10 In addition to the at least one longitudinal coordinate of the object, at least one transversal coordinate of the object may be determined. Thus, generally, the evaluation device may further be adapted to determine at least one transversal coordinate of the object by determining a position of the light beam on the at least one transversal optical sensor, which may be a pixelated, a segmented or a large-area transversal optical sensor, as will be outlined in further detail below.

- 15 Thus, in case a pixelated transversal optical sensor is used and/or in case the at least one transversal optical sensor comprises at least one pixelated optical sensor having a matrix of pixels, the evaluation device may be adapted to determine a center of illumination of the at least one matrix by the light beam, wherein the at least one transversal coordinate of the object is determined by evaluating at least one coordinate of the center of illumination. Thus, the coordinate of the center of illumination may be a pixel coordinate of the center of illumination. As an example, the matrix may comprise rows and columns of pixels, wherein the row number of the light beam and/or the center of the light beam within the matrix may provide an x-coordinate, and wherein the column number of the light beam and/or the center of the light beam within the matrix may provide a y-coordinate.

- The detector, as outlined above, may comprise at least one stack of optical sensors, the optical sensors comprising the at least one transversal optical sensor and the at least one longitudinal optical sensor, and, optionally, the at least one imaging device. The stack of optical sensors may comprise at least one longitudinal optical sensor stack, being a stack of longitudinal optical sensors, having at least two longitudinal optical sensors in a stacked fashion. The longitudinal optical sensor stack preferably may comprise at least three longitudinal optical sensors, more preferably at least four longitudinal optical sensors, even more preferably at least five longitudinal optical sensors or even at least six longitudinal optical sensors. By tracking the longitudinal sensor signals of the longitudinal optical sensors, even a beam profile of the light beam might be evaluated.

- In case a plurality of the longitudinal optical sensors is used, wherein the plurality of the optical sensors may be arranged in the stacked fashion and/or in another arrangement, the longitudinal optical sensors may have identical spectral sensitivities or may provide different spectral sensitivities. Thus, as an example, at least two of the longitudinal optical sensors may have a differing spectral sensitivity. As used herein, the term spectral sensitivity generally refers to the

fact that the sensor signal of the optical sensor, for the same power of the light beam, may vary with the wavelength of the light beam. Thus, generally, at least two of the optical sensors may differ with regard to their spectral properties. This embodiment generally may be achieved by using different types of absorbing materials for the optical sensors, such as different types of dyes or other absorbing materials.

Preferably, the at least one transversal optical sensor uses at least one transparent substrate. Similarly; preferably, the at least one longitudinal optical sensor uses at least one transparent substrate. In case a plurality of longitudinal optical sensors is used, such as a stack of longitudinal optical sensors, preferably, at least one of these longitudinal optical sensors uses a transparent substrate. Herein, the substrates employed for the plurality of optical sensors may exhibit identical properties or may differ from each other, in particular, with regard to a geometrical quantity and/or a material quantity related to the substrates, such as the thickness, the shape, and/or the refractive index of each substrate. Thus, identical planar glass plates may be used for the plurality of optical sensors within the stack. On the other hand, different substrates may be employed for some optical sensors or for each optical sensor within a plurality of optical sensors, in particular, for a purpose of optimizing the light path within the stack, especially for guiding the light path along regions on the optical axis which might particularly be suited for exploiting the FIP-effect as described elsewhere in this application. Within this regard, the thickness of some substrates or of each substrate, which may be defined by the light path as traversed by the light beam travelling through the respective substrate, may thus be varied, in particular, for reducing or else increasing or even maximizing reflections of the light beam.

In addition or alternatively, the substrates employed for the plurality of optical sensors may differ by exhibiting a different shape which may be selected from the group comprising a planar, a planar-convex, a planar-concave, a biconvex, a biconcave or any other form which may be employed for optical purposes, such as lenses or prisms. Herein, the substrates may be rigid or else flexible. Suitable substrates are, as well as metal foils, in particular plastic sheets or films and especially glass sheets or glass films. Shape-changing materials, such as shape-changing polymers, constitute an example of materials which may preferentially be employed as flexible substrates. Furthermore, the substrate may be covered or coated, in particular, for the purpose of reducing and/or modifying reflections of the incident light beam. As an example, the substrate may be shaped in a manner that it might exhibit a mirror effect, such as that of a dichroic mirror, which might particularly be useful in a setup where a splitting of the optical axis behind the substrate may be required for any purpose.

Additionally or alternatively, differing spectral properties of the optical sensors may be generated by other means implemented into the optical sensors and/or into the detector, such as by using one or more wavelength-selective elements, such as one or more filters (such as color filters) in front of the optical sensors and/or by using one or more prisms and/or by using one or more dichroitic mirrors and/or by using one or more color conversion elements. Thus, in

case a plurality of longitudinal optical sensors is provided, at least one of the longitudinal optical sensors may comprise a wavelength-selective element such as a color filter, having a specific transmission or reflection characteristic, thereby generating differing spectral properties of the optical sensors. Further, the longitudinal optical sensors may all be organic optical sensors, may
5 all be inorganic optical sensors, may all be hybrid organic – inorganic optical sensors or may comprise an arbitrary combination of at least two optical sensors selected from the group consisting of organic optical sensors, inorganic optical sensors and hybrid organic – inorganic optical sensors.

10 In case the plurality of the longitudinal optical sensors is used, wherein at least two of the longitudinal optical sensors differ with regard to their respective spectral sensitivity, the evaluation device generally may be adapted to determine a color of the light beam by comparing sensor signals of the longitudinal optical sensors having the differing spectral
15 sensitivity. As used herein, the expression “determine a color” generally refers to the step of generating at least one item of spectral information about the light beam. The at least one item of spectral information may be selected from the group consisting of a wavelength, specifically a peak wavelength; color coordinates, such as CIE coordinates. As further used herein, a “color” of the light beam generally refers to a spectral composition of the light beam. Specifically, the color of the light beam may be given in any arbitrary color coordinate system and/or in spectral
20 units such as by giving a wavelength of a dominant peak of a spectrum of the light. Other embodiments are feasible. In case the light beam is a narrow-band light beam such as a laser light beam and/or a light beam generated by a semiconductor device such as a light-emitting diode, the peak wavelength of the light beam may be given to characterize the color of the light beam. The determination of the color of the light beam may be performed in various ways which
25 are generally known to the skilled person. Thus, the spectral sensitivities of the longitudinal optical sensors may span a coordinate system in color space, and the signals provided by the optical sensors may provide a coordinate in this color space, as known to the skilled person for example from the way of determining CIE coordinates. As an example, the detector may comprise two, three or more longitudinal optical sensors in a stack. Thereof, at least two,
30 preferably at least three, of the optical sensors may have differing spectral sensitivities, whereby three different longitudinal optical sensors with maximum absorption wavelengths in a spectral range between 600 nm and 780 nm (red), between 490 nm and 600 nm (green), and between 380 nm and 490 nm (blue) are generally preferred. Further, the evaluation device may be adapted to generate at least one item of color information for the light beam by evaluating the
35 signals of the longitudinal optical sensors having differing spectral sensitivities.

The evaluation device may be adapted to generate at least two color coordinates, preferably at least three color coordinates, wherein each of the color coordinates is determined by dividing a signal of one of the spectrally sensitive optical sensors by a normalization value. As an
40 example, the normalization value may contain a sum of the signals of all spectrally sensitive optical sensors. Additionally or alternatively, the normalization value may contain a detector signal of a white detector. The at least one item of color information may contain the color

coordinates. The at least one item of color information may, as an example, contain CIE coordinates.

Furthermore, in addition to the preferred at least two, more preferably at least three, spectrally sensitive longitudinal optical sensors, wherein at least two of the optical sensors differ with regard to their respective spectral sensitivity, particularly wherein the spectral sensitivities of the at least two longitudinal optical sensors may span a coordinate system in color space, the stack may comprise a last longitudinal optical sensor, wherein the last longitudinal optical sensor may be intransparent. An intransparency of the last longitudinal optical sensor may be achieved by selecting an optically intransparent material from the group consisting of: organic optical sensor materials, inorganic optical sensor materials and hybrid organic – inorganic optical sensor materials.

In a preferred embodiment, the intransparent last longitudinal optical sensor may be configured to exhibit an absorption spectrum which either does not vary at all or varies only to minor extent over the spectral range of the at least two different optical sensors. Whereas each of the at least two different optical sensors displays a specific spectral sensitivity which may, as described above, vary considerably over their spectral range allowing them to be sensitive to a specific color, the intransparent last longitudinal optical sensor may, thus, absorb substantially all colors over the spectral ranges of the at least two of the longitudinal optical sensors have a differing spectral sensitivity. By this property, the last longitudinal optical sensor may be described as a white detector, wherein the white detector may be adapted to absorb light in an absorption range covering the visible spectral range. This arrangement has the advantage that, irrespective of the specific color, each beam which propagates through the at least two different optical sensors until it impinges the intransparent last longitudinal optical sensor, will be recorded by at least two different optical sensors, i.e. by at least one of the at least two different optical sensors which is sensitive to the color as the first longitudinal optical sensor and by the intransparent last longitudinal optical sensor as the second longitudinal optical sensor. As described above, by recording at least one object by both the first longitudinal optical sensor and the second longitudinal optical sensor it will be possible to resolve the ambiguity in the known relationship between the beam cross-section of the light beam and the longitudinal position of the at least one object, wherein the resolving of the ambiguity may work separately for each color which may be recorded. In addition, it may be advantageous to set a first signal recorded by the first longitudinal optical sensor in relationship to a second signal recorded by the last or second longitudinal optical sensor, such as by calculating a quotient or another related relationship. A result obtained by creating such a relationship may particularly facilitate the recognition of a certain color by this arrangement.

As outlined above, the intransparent last longitudinal optical sensor may be a large-area sensor having a single sensitive area or may comprise at least one matrix of pixels, i.e. may be a pixelated optical sensor or imaging sensor. In a further preferred embodiment, the pixels themselves may be equipped with different spectral sensitivity which may allow them to become

sensitive to a specific color. Hereby, the specific colors may be distributed over the area of the last longitudinal optical sensor in any fashion, for example in a random fashion. However, an arrangement is preferred wherein pixels with sensitivity to a specific color are placed in an alternating fashion over the area of the intransparent last longitudinal optical sensor. By way of example, two, three, or four pixels with different sensitivity, e.g. three pixels with a specific sensitivity to red, green, and blue, alternate each other in a one-dimensional or, preferably, in a two-dimensional manner, over the area of the last longitudinal optical sensor. This arrangement may further be interrupted by white pixels, i.e. pixels which exhibit a spectral sensitivity over a large spectral range, e.g. over the visible spectrum, or over a spectrum which may include infrared or ultraviolet shares.

In a further preferred embodiment, the detector may be adapted, such as by using at least one optically sensitive element. As further used herein, the "optically sensitive element" may be considered as an arbitrary optical element which may be sensitive to a specific value of an optical property or to a specific range of values of the optical property of the light beam which may impinge on the optically sensitive element in a manner that the specific value of the optical property or to the specific range of values of the optical property is preferred with regard to other values of the light beam. The optical property as employed here may be selected from the group referring to the light beam and consisting of: a wavelength, a phase, and a polarization. Thus, the optically sensitive element may be designated as a wavelength-sensitive element, a phase-sensitive element, and/or a polarization-sensitive element, respectively. Examples for wavelength-sensitive elements may comprise one or more of a prism, a grating, a dichroitic mirror, a color wheel, or a color drum.

By using the optically sensitive element, the light beam may be influenced by an optical effect related to the at least one optical property immediately before an interaction, during the interaction and/or immediately after the interaction of the light beam with the optically sensitive element. Consequently, an expression like "passing through the optically sensitive element" may refer to a time period during which the light beam may interact with the respective element. In general, the optically sensitive element may induce the optical effect on the light beam impinging on the optically sensitive element, such as by altering a transmission or a reflectance of the impinging light beam.

Within this particular embodiment, for sequentially detecting detector signals for the different optical properties, such as different colors, different phases, and/or different polarizations, the at least one optically sensitive element may be adapted to sequentially influence the light beam. As an example for a sequential process, a rotating filter wheel may be used having filter segments of different transmission properties, for periodically influencing the light beam. Thus, each cycle of rotation of the filter wheel may be split into time segments, wherein each segment may correspond to a different optical property, such as to a different color, a different phase and/or a different polarization. With regard to a filter wheel wherein each segment corresponds to a different polarization, the filter wheel may preferably exhibit an elliptical polarization, in

particular a circular polarization, which may be changing along the circumference of the filter wheel, for example in a discrete or in a continuous manner. However, other embodiments, such as filter wheels which may employ different orientations of a linear polarization within different segments, may be feasible.

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In general, at least one optical sensor may be placed behind the filter wheel, in order to generate at least one combined detector signal. By evaluating the at least one combined detector signal in a time-resolved fashion, such as by using a phase-sensitive detection, the combined detector signal may be split into partial detector signals corresponding to the different
10 time segments and, thus, corresponding to the different colors of the light beam. Thereby, detector signals for each color, each phase and/or each polarization may be generated, which may correspond to the light beams impinging on the detector. Collecting the data for different optical properties, i.e. different colors or phases or polarizations, from the broadly absorbing stack of optical sensors, may result in acquiring the overall distribution. Consequently, the use
15 of a filter wheel may, therefore, allow determining simultaneously the respective optical property, such as color, phase, or polarization, and intensity and depth by using the detector according to the present invention, thereby avoiding the previous necessity to employ solar cells exhibiting different absorption spectra.

20 Generally, as outlined above, the evaluation device may be adapted to generate the at least one item of information on the longitudinal position of the object by determining a diameter of the light beam from the at least one longitudinal sensor signal. As used herein and as used in the following, the diameter of the light beam or, equivalently, a beam waist of the light beam might be used to characterize the beam cross-section of the light beam at a specific location. As
25 outlined above, a known relationship might be used between the longitudinal position of the object and the beam cross-section in order to determine the longitudinal position of the object by evaluating the at least one longitudinal sensor signal. As an example, as outlined above, a Gaussian relationship might be used, assuming that the light beam propagates at least approximately in a Gaussian Manner. For this purpose, the light beam might be shaped
30 appropriately, such as by using an illumination source generating a light beam having known propagation properties, such as a known Gaussian profile. For this purpose, the illumination source itself may generate the light beam having the known properties, which, for example, is the case for many types of lasers, as the skilled person knows. Additionally or alternatively, the illumination source and/or the detector may have one or more beam-shaping elements, such as
35 one or more lenses and/or one or more diaphragms, in order to provide a light beam having known properties, as the skilled person will recognize. Thus, as an example, one or more transfer elements may be provided, such as one or more transfer elements having known beam-shaping properties. Additionally or alternatively, the illumination source and/or the detector, such as the at least one optional transfer element, may have one or more wavelength-
40 selective elements, such as one or more filters, such as one or more filter elements for filtering out wavelengths outside an excitation maximum of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor.

Thus, generally, the evaluation device may be adapted to compare the beam cross-section and/or the diameter of the light beam with known beam properties of the light beam in order to determine the at least one item of information on the longitudinal position of the object,
5 preferably from a known dependency of a beam diameter of the light beam on at least one propagation coordinate in a direction of propagation of the light beam and/or from a known Gaussian profile of the light beam.

In a specific embodiment, the evaluation device is adapted to determine a number N of pixels of
10 at least one pixelated optical sensor of the detector, such as of the last longitudinal optical sensor, which are illuminated by the light beam, the evaluation device further being adapted to determine at least one longitudinal coordinate of the object by using the number N of pixels which are illuminated by the light beam. Therefore, the evaluation device may be adapted to compare, for each of the pixels, the signal to at least one threshold in order to determine
15 whether the pixel is an illuminated pixel or not. This at least one threshold may be an individual threshold for each of the pixels or may be a threshold which is a uniform threshold for the whole matrix. In case a plurality of optical sensors is provided, at least one threshold may be provided for each of the optical sensors and/or for a group comprising at least two of the optical sensors, wherein, for two optical sensors, their respective thresholds may be identical or different. Thus,
20 for each of the optical sensors, an individual threshold may be provided. The threshold may be predetermined and/or fixed. Alternatively, the at least one threshold may be variable. Thus, the at least one threshold may be determined individually for each measurement or groups of measurements. Thus, at least one algorithm may be provided adapted to determine the threshold.

25 The evaluation device may generally be adapted to determine at least one pixel having the highest illumination out of the pixels by comparing the signals of the pixels. Thus, the detector generally may be adapted to determine one or more pixels and/or an area or region of the matrix having the highest intensity of the illumination by the light beam. As an example, in this
30 way, a center of illumination by the light beam may be determined. The highest illumination and/or the information about the at least one area or region of highest illumination may be used in various ways. Thus, as outlined above, the at least one above-mentioned threshold may be a variable threshold. As an example, the evaluation device may be adapted to choose the above-mentioned at least one threshold as a fraction of the signal of the at least one pixel having the
35 highest illumination. Thus, the evaluation device may be adapted to choose the threshold by multiplying the signal of the at least one pixel having the highest illumination with a factor of $1/e^2$. This option is particularly preferred in case Gaussian propagation properties are assumed for the at least one light beam, since the threshold $1/e^2$ generally determines the borders of a light spot having a beam radius or beam waist generated by a Gaussian light beam on the
40 optical sensor.

A further aspect of the present invention makes use of at least two detectors according to the present invention, wherein each of such detectors may be selected as of at least one detector according to one or more of the embodiments disclosed above or disclosed in further detail below. Thus, for optional embodiments of the method, reference might be made to the
5 respective embodiments of the detector.

In a preferred embodiment, the at least one object might be illuminated by using at least one illumination source which generates primary light, wherein the at least one object elastically or inelastically reflects the primary light, thereby generating a plurality of light beams which
10 propagate to one of the at least two detectors. The at least one illumination source may form or may not form a constituent part of each of the at least two detectors which. Thus, the at least one illumination source may be formed independently of the at least two detectors and may, therefore, particularly be located in at least one position which is separated from the at least two detectors. By way of example, the at least one illumination source itself may be or may
15 comprise an ambient light source and/or may be or may comprise an artificial illumination source. This embodiment is preferably suited for an application in which at least two detectors, preferentially two identical detectors, are employed for acquiring depth information, in particular, for the purpose to providing a measurement volume which extends the inherent measurement volume of a single detector.

The inherent measurement volume of a single detector may, in many cases, be described as an approximate half-cone, wherein a first object which is located within the inherent measurement volume may be detected by the single detector whereas a second object which is located outside the inherent measurement volume can principally not be detected by the single
20 detector. The conical surface of the approximate half-cone can be considered to be formed by a virtual reverse light beam which would be emitted by the at least one optical sensor. The virtual reverse light beam would, thus, emerge from the surface of the at least one optical sensor which, however, does not constitute a point source but rather an extended area. From simple geometrical considerations it can be deduced that no virtual reverse light beam emitted by the
25 at least one optical sensor in this manner will be able to reach all positions in all directions in the volume which surrounds the at least one optical sensor. However, the locations which may be virtually hit by the virtual reverse light beam form an approximate half-cone which can be described as the inherent measurement volume of the single detector.

Therefore, in order to be able to cover a large measurement volume which exceeds the inherent measurement volume of a single detector, at least two detectors may be used, wherein the at least two detectors may be identical or may differ from each other with respect to at least one certain technical property as described elsewhere. In general, the large measurement volume comprises an overlapping volume, which denotes a region in space, which may be coherent or
30 not, wherein double or even multiple detection may occur, i.e. a specific object may be detected by two or more detectors independently, at the same time or at different times. Even in case when two or more detectors, particularly two or more identical detectors, are employed, the
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double or multiple detection of a specific object does not destroy a reliable acquisition of the depth information about the specific object in the overlapping volume. Since the at least one illumination source may be formed independently of the at least two detectors, typically, no relationship exists between a specific illumination source and a specific detector. Therefore, a reliable acquisition of the depth information will also be possible when at least two of the detectors point towards each other. As a result of this disconnection between the specific illumination source and the specific detector, the recording of the depth information about the specific object will not be impaired when the specific object may be located in the overlapping volume. On the contrary, the depth information in the overlapping volume which concerns a specific object may independently been acquired by more than one detector at the same time and can, thus, be used to improve the accuracy of the depth measurement for the specific object. By way of example, this improvement may be achieved by comparing respective depth values which are recorded simultaneously or successively for the same object by at least two separate single detectors.

In a further aspect of the present invention, a human-machine interface for exchanging at least one item of information between a user and a machine is proposed. The human-machine interface as proposed may make use of the fact that the above-mentioned detector in one or more of the embodiments mentioned above or as mentioned in further detail below may be used by one or more users for providing information and/or commands to a machine. Thus, preferably, the human-machine interface may be used for inputting control commands.

The human-machine interface comprises at least one detector according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments as disclosed in further detail below, wherein the human-machine interface is designed to generate at least one item of geometrical information of the user by means of the detector wherein the human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command.

Generally, as used herein, the at least one item of geometrical information of the user may imply one or more items of information on a transversal position and/or on a longitudinal position of the user and/or one or more body parts of the user. Thus, preferably, the geometrical information of the user may imply one or more items of information on a transversal position and/or a longitudinal position as provided by the evaluation device of the detector. The user, a body part of the user or a plurality of body parts of the user may be regarded as one or more objects which may be detected by the at least one detector. Therein, precisely one detector may be provided, or a combination of a plurality of detectors may be provided. As an example, a plurality of detectors may be provided for determining positions of a plurality of body parts of the user and/or for determining an orientation of at least one body part of the user. The human-machine interface may comprise one or more detectors, wherein, in case a plurality of detectors is provided, the detectors may be identical or may differ. Herein, in case a plurality of detectors

is used, the plurality of detectors, particularly the plurality of identical detectors, still allows for a reliable acquisition of depth information about the at least one object in an overlapping volume which may be recorded, as described above, by the plurality of detectors.

- 5 Thus, preferably, the at least one item of geometrical information of the user is selected from the group consisting of: a position of a body of the user; a position of at least one body part of the user; an orientation of a body of the user; an orientation of at least one body part of the user.
- 10 The human-machine interface may further comprise at least one beacon device connectable to the user. As used herein, a beacon device generally is an arbitrary device which may be detected by the at least one detector and/or which facilitates detection by the at least one detector. Thus, as will be outlined in further detail below, the beacon device may be an active beacon device adapted for generating the at least one light beam to be detected by the
- 15 detector, such as by having one or more illumination sources for generating the at least one light beam. Additionally or alternatively, the beacon device may fully or partially be designed as a passive beacon device, such as by providing one or more reflective elements adapted to reflect a light beam generated by a separate illumination source. The at least one beacon device may permanently or temporarily be attached to the user. The attachment may take place
- 20 by using one or more attachment means and/or by the user himself or herself, such as by the user holding the at least one beacon device by hand and/or by the user wearing the beacon device.

- The human-machine interface may be adapted such that the detector may generate an
- 25 information on the position of the at least one beacon device. Specifically in case a manner of attachment of the at least one beacon device to the user is known, from the at least one item of information on the position of the at least one beacon device at least one item of information regarding a position and/or an orientation of the user or one or more body parts of the user may be gained.

- 30 The beacon device preferably is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user. As outlined above, the beacon device may fully or partially be designed as an active beacon device. Thus, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be
- 35 transmitted to the detector, preferably at least one light beam having known beam properties. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

- 40 The beacon device preferably may comprise at least one of: a garment to be worn by the user, preferably a garment selected from the group consisting of a glove, a jacket, a hat, shoes,

trousers and a suit; a stick that may be held by hand; a bat; a club; a racket; a cane; a toy, such as a toy gun.

5 In a further aspect of the present invention, an entertainment device for carrying out at least one entertainment function is disclosed. As used herein, an entertainment device is a device which may serve the purpose of leisure and/or entertainment of one or more users, in the following also referred to as one or more players. As an example, the entertainment device may serve the purpose of gaming, preferably computer gaming. Additionally or alternatively, the entertainment device may also be used for other purposes, such as for exercising, sports, physical therapy or
10 motion tracking in general. Thus, the entertainment device may be implemented into a computer, a computer network or a computer system or may comprise a computer, a computer network or a computer system which runs one or more gaming software programs.

15 The entertainment device comprises at least one human-machine interface according to the present invention, such as according to one or more of the embodiments disclosed above and/or according to one or more of the embodiments disclosed below. The entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface. The at least one item of information may be transmitted to and/or may be used by a controller and/or a computer of the entertainment device.

20 The at least one item of information preferably may comprise at least one command adapted for influencing the course of a game. Thus, as an example, the at least one item of information may include at least one item of information on at least one of a movement, the orientation and a position of the player and/or of one or more body parts of the player, thereby allowing for the player to simulate a specific position and/or action required for gaming. As an example, one or
25 more of the following movements may be simulated and communicated to a controller and/or a computer of the entertainment device: dancing; running; jumping; swinging of a racket; swinging of a bat; swinging of a club; pointing of an object towards another object, such as pointing of a toy gun towards a target.

30 The entertainment device, preferably a controller and/or a computer of the entertainment device, is designed to vary the entertainment function in accordance with the information. Thus, as outlined above, a course of a game might be influenced in accordance with the at least one item of information. Thus, the entertainment device might include one or more controllers which
35 might be separate from the evaluation device of the at least one detector and/or which might be fully or partially identical to the at least one evaluation device or which might even include the at least one evaluation device. Preferably, the at least one controller might include one or more data processing devices, such as one or more computers and/or microcontrollers.

40 In a further embodiment of the present invention, the entertainment device may be part of an equipment, the equipment being a mobile piece or, in particular, an immobile piece, wherein the equipment may at least partially incorporate the entertainment device. The equipment may

include a single, separate piece being located at a position, either a fixed position or a position at least intermittently subject to a variation, but the equipment may also comprise at least two pieces, preferably two to ten pieces, such as three, four, five, or six pieces, wherein the at least two pieces may be distributed over at least two positions differing from each other within an area, such as a room or a part thereof. Hereby, the entertainment device may be part of the equipment, wherein preferably some or each piece of the equipment may exhibit a part of the entertainment device, e.g. in such a manner that some or each piece of the equipment may comprise at least one detector according to the present invention or a part thereof, such as a sensor. As used herein, "immobile equipment" may include an immobile electronic article, in particular, designated as consumer electronics, wherein "consumer electronics" comprises electronic articles preferentially intended for everyday use, mainly in entertainment, communications and office matters, such as radio receivers, monitors, television sets, audio players, video players, personal computers and/or telephones. A particular example constituting the immobile equipment may be a surround system, which may for example be formed by a number of two, three, four, five, six or more separate pieces of the equipment, such as individual monitors or audio players, including loudspeakers, which may be distributed over an area, preferentially in a specific fashion, such as forming an arc-shaped assembly enclosing a room or a part thereof.

Additionally or alternatively, the entertainment device or a part thereof, such as one, some or each piece of the equipment, may further be equipped with one or more of the following devices: a photographic device, such as a camera, particularly a 2D camera, a picture analysis software, particularly a 2D picture analysis software, and a reference object, particularly a geometrically symmetric reference object, such as a book or a specially formed toy. Hereby, the reference object may also be a part of the immobile equipment and fulfill a further function of the entertainment device as described above and/or below, wherein the reference object may further comprise a detector, a 2D camera or another photographic device. Preferably, a constructive interaction of the photographic device, the picture analysis software and the particularly symmetric reference object, may facilitate an alignment of a 2D picture of an object in question as recorded by the photographic device with the 3D position of the same object as determined by the at least one detector.

In a further embodiment of the present invention, the object which may constitute a target of the at least one detector comprised within the at least one human-machine interface of the entertainment device, may be part of a controller as comprised within mobile equipment, wherein the mobile equipment may be configured to control another mobile equipment or immobile equipment. As used herein, "mobile equipment" may, thus, include mobile electronic articles, in particular, designated as consumer electronics, such as mobile phones, radio receivers, video recorders, audio players, digital cameras, camcorders, mobile computers, video game consoles and/or other devices adapted for remote control. This embodiment may particularly allow controlling immobile equipment with any kind of mobile equipment, preferably with a lesser number of pieces of equipment. As a non-limiting example, it may, thus, be

possible to simultaneously control for example both a game console and a television set by using a mobile phone.

5 Additionally or alternatively, the object which may constitute the target of the detector may further be equipped with an additional sensor (apart from the sensors as comprised within the detector) particularly configured for determining a physical and/or chemical quantity related to the object, such as an inertia sensor for measuring the inertial motion of the object, or an acceleration sensor for determining the acceleration of the object. However, besides these preferred examples, other kinds of sensors adapted for acquiring further parameter related to
10 the object, such as a vibrational sensor for determining vibrations of the object, a temperature sensor for recording the temperature of the object, or a humidity sensor for recording the humidity of the object may be employed. An application of the additional sensor within the object may allow improving the quality and/or the scope of the detection of the position of the object. As a non-limiting example, the additional inertia sensor and/or acceleration sensor may
15 particularly be configured to record additional movements of the object, such as a rotation of the object, which may particularly be employed for increasing the accuracy of the object detection. Moreover, the additional inertia sensor and/or acceleration sensor may preferentially still be addressed in a case where the object being equipped with at least one of these sensors may leave a visual range of the detector comprised within the human-machine interface of the
20 entertainment device. In this case, it might, nevertheless, be possible to follow the object after the object may have left the visual range of the detector by still being able to recording signals emitted from at least one of these sensors and using these signals for determining the location of the object by taking into account its actual inertia and acceleration values and calculating the position therefrom.

25 Additionally or alternatively, the object which may constitute the target of the detector may further be equipped with a further feature allowing both to simulate and/or to stimulate motions, for example by simulating motions of the object, wherein the object may be virtual or real and wherein the object may be controlled by a controller, and/or by stimulating motions of the object
30 by applying the controller accordingly. This feature may particularly be employed for providing a more realistic entertainment experience for a user. As an illustrating example, a steering wheel as employed in an entertainment device may vibrate, wherein the amplitude of the vibrations may depend on a nature of a ground on which a virtual car may be driving on. As a further embodiment, the motion of the object may be stimulated by employing a gyroscope, which may
35 be used for example for a stabilization of flying vehicles as described under the address en.wikipedia.org/wiki/gyroscope.

In a further embodiment of the present invention, the object which may constitute the target of the at least one detector may be equipped with at least one modulation device for modulating
40 the illumination, in particular for a periodic modulation. Preferably, the object may comprise at least one illumination source, which may be part of the object or, alternatively or additionally, be held by the object or attached thereto and which may act as a beacon in a manner as described

elsewhere in this application. The illumination may be adapted to generate at least one light beam to be transmitted to the detector, whereby the illumination source comprises a modulation device for modulating the illumination and/or the modulation device may be a separate device configured for controlling the emission of the illumination source. According to this embodiment, 5 apart from the fundamental modulation of the illumination source, the modulation device may generate additional modulation frequencies, also be denoted as "overtones", which might be used for transmitting any item of additional information or data from the object to the detector. This embodiment, which might also be designated as a "modulating retro-reflector", may open the way to employ the object equipped with the modulation device configured for generating 10 both the fundamental and the additional modulation frequencies as remote control. Further, an already existing remote control may be replaced by the object equipped with the described modulation device. Against this background, this kind of object and a piece configured for remote control may interchangeably be used within an arrangement incorporating a detector according to the present invention.

15 In a further embodiment of the present invention, the entertainment device may further be equipped with additional items such as items commonly used within such an environment. Glasses or other devices being configured for creating a 3D vision within the mind of a player may constitute a particular example.

20 In a further embodiment of the present invention, the entertainment device may further be equipped with an augmented reality application. As further used herein, "augmented reality" may describe a live perception of reality comprising elements which may be modified by computer-generated data primarily related to physical phenomena such as sound, images, or 25 others. An example may be vision glasses which are particularly adapted for augmented reality applications. Another example may include an arrangement comprising at least two detectors, preferably a multitude thereof, wherein the detectors may particularly be arranged to cover an area within a room or a preferably large part thereof, a photographic device, such as a camera, particularly a 2D camera, and an augmented reality application, wherein the arrangement might 30 be employed to transform a real area into a playing field, which may also be designated as an entertainment field.

In a further aspect of the present invention, a tracking system for tracking the position of at least one movable object is provided. As used herein, a tracking system is a device which is adapted 35 to gather information on a series of past positions of the at least one object or at least one part of an object. Additionally, the tracking system may be adapted to provide information on at least one predicted future position of the at least one object or the at least one part of the object. The tracking system may have at least one track controller, which may fully or partially be embodied as an electronic device, preferably as at least one data processing device, more preferably as at 40 least one computer or microcontroller. Again, the at least one track controller may comprise the at least one evaluation device and/or may be part of the at least one evaluation device and/or may fully or partially be identical to the at least one evaluation device.

The tracking system comprises at least one detector according to the present invention, such as at least one detector as disclosed in one or more of the embodiments listed above and/or as disclosed in one or more of the embodiments below. The tracking system further comprises at least one track controller. The tracking system may comprise one, two or more detectors, particularly two or more identical detectors, which allow for a reliable acquisition of depth information about the at least one object in an overlapping volume between the two or more detectors. The track controller is adapted to track a series of positions of the object, each position comprising at least one item of information on a transversal position of the object at a specific point in time and at least one item of information on a longitudinal position of the object at a specific point in time.

The tracking system may further comprise at least one beacon device connectable to the object. For a potential definition of the beacon device, reference may be made to the disclosure above. The tracking system preferably is adapted such that the detector may generate an information on the position of the object of the at least one beacon device. For potential embodiments of the beacon device, reference may be made to the disclosure above. Thus, again, the beacon device may fully or partially be embodied as an active beacon device and/or as a passive beacon device. As an example, the beacon device may comprise at least one illumination source adapted to generate at least one light beam to be transmitted to the detector. Additionally or alternatively, the beacon device may comprise at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

The tracking system may be adapted to initiate one or more actions of the tracking system itself and/or of one or more separate devices. For the latter purpose, the tracking system, preferably the track controller, may have one or more wireless and/or wire-bound interfaces and/or other types of control connections for initiating at least one action. Preferably, the at least one track controller may be adapted to initiate at least one action in accordance with at least one actual position of the object. As an example, the action may be selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.

As an example of application of a tracking system, the tracking system may be used for continuously pointing at least one first object to at least one second object even though the first object and/or the second object might move. Potential examples, again, may be found in industrial applications, such as in robotics and/or for continuously working on an article even though the article is moving, such as during manufacturing in a manufacturing line or assembly line. Additionally or alternatively, the tracking system might be used for illumination purposes, such as for continuously illuminating the object by continuously pointing an illumination source to the object even though the object might be moving. Further applications might be found in

communication systems, such as in order to continuously transmit information to a moving object by pointing a transmitter towards the moving object.

5 In a further aspect of the present invention, a camera for imaging at least one object is disclosed. The camera comprises at least one detector according to the present invention, such as disclosed in one or more of the embodiments given above or given in further detail below.

10 Thus, specifically, the present application may be applied in the field of photography. Thus, the detector may be part of a photographic device, specifically of a digital camera. Specifically, the detector may be used for 3D photography, specifically for digital 3D photography. Thus, the detector may form a digital 3D camera or may be part of a digital 3D camera. As used herein, the term "photography" generally refers to the technology of acquiring image information of at least one object. As further used herein, a "camera" generally is a device adapted for performing photography. As further used herein, the term "digital photography" generally refers to the technology of acquiring image information of at least one object by using a plurality of light-sensitive elements adapted to generate electrical signals indicating an intensity and/or color of illumination, preferably digital electrical signals. As further used herein, the term "3D photography" generally refers to the technology of acquiring image information of at least one object in three spatial dimensions. Accordingly, a 3D camera is a device adapted for performing 20 3D photography. The camera generally may be adapted for acquiring a single image, such as a single 3D image, or may be adapted for acquiring a plurality of images, such as a sequence of images. Thus, the camera may also be a video camera adapted for video applications, such as for acquiring digital video sequences.

25 Thus, generally, the present invention further refers to a camera, specifically a digital camera, more specifically a 3D camera or digital 3D camera, for imaging at least one object. As outlined above, the term imaging, as used herein, generally refers to acquiring image information of at least one object. The camera comprises at least one detector according to the present invention. The camera, as outlined above, may be adapted for acquiring a single image or for 30 acquiring a plurality of images, such as image sequence, preferably for acquiring digital video sequences. Thus, as an example, the camera may be or may comprise a video camera. In the latter case, the camera preferably comprises a data memory for storing the image sequence.

35 As used within the present invention, the expression "position" generally refers to at least one item of information regarding one or more of an absolute position and an orientation of one or more points of the object. Thus, specifically, the position may be determined in a coordinate system of the detector, such as in a Cartesian coordinate system. Additionally or alternatively, however, other types of coordinate systems may be used, such as polar coordinate systems and/or spherical coordinate systems.

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In a further aspect of the present invention, a method for determining a position of at least one object is disclosed. The method preferably may make use of at least one detector according to

the present invention, such as of at least one detector according to one or more of the embodiments disclosed above or disclosed in further detail below. Thus, for optional embodiments of the method, reference might be made to the embodiments of the detector.

- 5 The method comprises the following steps, which may be performed in the given order or in a different order. Further, additional method steps might be provided which are not listed. Further, two or more or even all of the method steps might be performed at least partially simultaneously. Further, two or more or even all of the method steps might be performed twice or even more than twice, repeatedly.

10

In a first method step, which might also be referred to as a step of determining at least one transversal position, at least one transversal optical sensor is used. The transversal optical sensor determines a transversal position of at least one light beam traveling from the object to the detector, wherein the transversal position is a position in at least one dimension
15 perpendicular to an optical axis of the detector. The transversal optical sensor generates at least one transversal sensor signal.

20

In a further method step, which might also be referred to as a step of determining at least one longitudinal position, at least one longitudinal optical sensor is used. The longitudinal optical sensor has at least one sensor region. The longitudinal optical sensor generates at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam. The longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region.

25

In a further method step, which might also be referred to as an evaluation step, at least one evaluation device is used. The evaluation device generates at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and wherein the evaluation device further generates at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

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In a further aspect of the present invention, a use of a detector according to the present invention is disclosed. Therein, a use of the detector for a purpose of use is proposed, selected from the group consisting of: a distance measurement, in particular in traffic technology; a position measurement, in particular in traffic technology; an entertainment application; a security
35 application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space.

40

In the following, some additional remarks regarding potential embodiments of the detector, the human-machine interface, the tracking system and the method according to the present invention are given. As outlined above, preferably, for potential details of the setups of the at least one transversal optical detector and the at least one longitudinal optical detector,

reference may be made to WO 2012/110924 A1, specifically with regard to potential electrode materials, organic materials, inorganic materials, layer setups and further details.

5 The object can generally be a living or else inanimate object. Examples of objects which can be detected completely or partly by means of the detector are described in even greater detail below.

Further, with regard to potential embodiments of the optional transfer device, reference may be made to WO 2012/110924 A1. Thus, this optional transfer device can comprise for example at
10 least one beam path. The transfer device can for example comprise one or a plurality of mirrors and/or beam splitters and/or beam deflecting elements in order to influence a direction of the electromagnetic radiation. Alternatively or additionally, the transfer device can comprise one or a plurality of imaging elements which can have the effect of a converging lens and/or a diverging lens. By way of example, the optional transfer device can have one or a plurality of
15 lenses and/or one or a plurality of convex and/or concave mirrors. Once again alternatively or additionally, the transfer device can have at least one wavelength-selective element, for example at least one optical filter. Once again alternatively or additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The
20 abovementioned optional embodiments of the optional transfer device can, in principle, be realized individually or in any desired combination.

Further, generally, it shall be noted that, in the context of the present invention, an optical
25 sensor may refer to an arbitrary element which is designed to convert at least one optical signal into a different signal form, preferably into at least one electrical signal, for example a voltage signal and/or a current signal. In particular the optical sensor can comprise at least one optical-electrical converter element, preferably at least one photodiode and/or at least one solar cell. As is explained in even greater detail below, in the context of the present invention, preference is attached particularly to a use of at least one organic optical sensor, that is to say an optical
30 sensor which comprises at least one organic material, for example at least one organic semiconductor material.

In the context of the present invention, a sensor region should be understood to mean a two-dimensional or three-dimensional region which preferably, but not necessarily, is continuous
35 and can form a continuous region, wherein the sensor region is designed to vary at least one measurable property, in a manner dependent on the illumination. By way of example, said at least one property can comprise an electrical property, for example, by the sensor region being designed to generate, solely or in interaction with other elements of the optical sensor, a photo voltage and/or a photocurrent and/or some other type of signal. In particular, the sensor region
40 can be embodied in such a way that it generates a uniform, preferably a single, signal in a manner dependent on the illumination of the sensor region. The sensor region can thus be the smallest unit of the optical sensor for which a uniform signal, for example, an electrical signal, is

generated, which preferably can no longer be subdivided to partial signals, for example for partial regions of the sensor region. The transversal optical sensor and/or the longitudinal optical sensor each can have one or else a plurality of such sensor regions, the latter case for example by a plurality of such sensor regions being arranged in a two-dimensional and/or three-dimensional matrix arrangement.

The at least one sensor region can comprise for example at least one sensor area, that is to say a sensor region whose lateral extent considerably exceeds the thickness of the sensor region, for example by at least a factor of 10, preferably by at least a factor of 100 and particularly preferably by at least a factor of 1000. Examples of such sensor areas can be found in organic or inorganic photovoltaic elements, for example, in accordance with the prior art described above, or else in accordance with the exemplary embodiments described in even greater detail below. The detector can have one or a plurality of such optical sensors and/or sensor regions. By way of example, a plurality of optical sensors can be arranged linearly in a spaced-apart manner or in a two-dimensional arrangement or else in a three-dimensional arrangement, for example by a stack of photovoltaic elements being used, preferably organic photovoltaic elements, preferably a stack in which the sensor areas of the photovoltaic elements are arranged parallel to one another. Other embodiments are also possible.

The optional transfer device can, as explained above, be designed to feed light propagating from the object to the detector to the transversal optical sensor and/or the longitudinal optical sensor, preferably successively. As explained above, this feeding can optionally be effected by means of imaging or else by means of non-imaging properties of the transfer device. In particular the transfer device can also be designed to collect the electromagnetic radiation before the latter is fed to the transversal and/or longitudinal optical sensor. The optional transfer device can also, as explained in even greater detail below, be wholly or partly a constituent part of at least one optional illumination source, for example by the illumination source being designed to provide a light beam having defined optical properties, for example having a defined or precisely known beam profile, for example at least one Gaussian beam, in particular at least one laser beam having a known beam profile.

For potential embodiments of the optional illumination source, reference may be made to WO 2012/110924 A1. Still, other embodiments are feasible. Light emerging from the object can originate in the object itself, but can also optionally have a different origin and propagate from this origin to the object and subsequently toward the transversal and/or longitudinal optical sensor. The latter case can be effected for example by at least one illumination source being used. This illumination source can for example be or comprise an ambient light source and/or may be or may comprise an artificial illumination source. By way of example, the detector itself can comprise at least one illumination source, for example at least one laser and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. On account of their generally defined beam profiles and other properties of handleability, the use of one or a

plurality of lasers as illumination source or as part thereof, is particularly preferred. The illumination source itself can be a constituent part of the detector or else be formed independently of the detector. The illumination source can be integrated in particular into the detector, for example a housing of the detector. Alternatively or additionally, at least one illumination source can also be integrated into the object or connected or spatially coupled to the object.

The light emerging from the object can accordingly, alternatively or additionally from the option that said light originates in the object itself emerge from the illumination source and/or be excited by the illumination source. By way of example, the electromagnetic light emerging from the object can be emitted by the object itself and/or be reflected by the object and/or be scattered by the object before it is fed to the optical sensor. In this case, emission and/or scattering of the electromagnetic radiation can be effected without spectral influencing of the electromagnetic radiation or with such influencing. Thus, by way of example, a wavelength shift can also occur during scattering, for example according to Stokes or Raman. Furthermore, emission of light can be excited, for example, by a primary light source, for example by the object or a partial region of the object being excited to effect luminescence, in particular phosphorescence and/or fluorescence. Other emission processes are also possible, in principle. If a reflection occurs, then the object can have for example at least one reflective region, in particular at least one reflective surface. Said reflective surface can be a part of the object itself, but can also be for example a reflector which is connected or spatially coupled to the object, for example a reflector plaque connected to the object. If at least one reflector is used, then it can in turn also be regarded as part of the detector which is connected to the object, for example, independently of other constituent parts of the detector.

The at least one illumination source of the detector can generally be adapted to the emission and/or reflective properties of the object, for example in terms of its wavelength. Various embodiments are possible.

The at least one optional illumination source generally may emit light in at least one of: the ultraviolet spectral range, preferably in the range of 200 nm to 380 nm; the visible spectral range (380 nm to 780 nm); the infrared spectral range, preferably in the range of 780 nm to 3.0 micrometers. Most preferably, the at least one illumination source is adapted to emit light in the visible spectral range, preferably in the range of 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm.

The feeding of the light to the transversal and/or longitudinal optical sensor can be effected in particular in such a way that a light spot, for example having a round, oval or differently configured cross section, is produced on the optional sensor area of the transversal and/or longitudinal optical sensor. By way of example, the detector can have a visual range, in particular a solid angle range and/or spatial range, within which objects can be detected. Preferably, the optional transfer device is designed in such a way that the light spot, for example

in the case of an object arranged within a visual range of the detector, is arranged completely on the sensor region, in particular the sensor area. By way of example, a sensor area can be chosen to have a corresponding size in order to ensure this condition.

5 The at least one longitudinal optical sensor, as outlined above, can be designed for example in such a way that the longitudinal sensor signal, given the same power of the illumination, that is to say for example given the same integral over the intensity of the illumination on the sensor area, is dependent on the geometry of the illumination, that is to say for example on the diameter and/or the equivalent diameter for the sensor spot. By way of example, the longitudinal
10 optical sensor can be designed in such a way that upon a doubling of the beam cross section given the same total power, a signal variation occurs by at least a factor of 3, preferably by at least a factor of 4, in particular a factor of 5 or even a factor of 10. This condition can hold true for example for a specific focusing range, for example for at least one specific beam cross section. Thus, by way of example, the longitudinal sensor signal can have, between at least one
15 optimum focusing at which the signal can have for example at least one global or local maximum and a focusing outside said at least one optimum focusing, a signal difference by at least a factor of 3, preferably by at least a factor of 4, in particular a factor of 5 or even a factor of 10. In particular, the longitudinal sensor signal can have as a function of the geometry of the illumination, for example of the diameter or equivalent diameter of a light spot, at least one
20 pronounced maximum, for example with a boost by at least a factor of 3, particularly preferably by at least a factor of 4 and particularly preferably by at least a factor of 10. Consequently, the longitudinal optical sensor may be based on the above-mentioned FiP-effect, which is disclosed in great detail in WO 2012/110924 A1. Thus, specifically in sDSCs, the focusing of the light beam may play a decisive role, i.e. the cross-section or cross-sectional area on which a certain
25 number of photons (nph) is incident. The more tightly the light beam is focused, i.e. the smaller its cross-section, the higher the photo current may be. The term 'FiP' expresses the relationship between the cross-section ϕ (Fi) of the incident beam and the solar cell's power (P).

30 The at least one longitudinal optical sensor is combined with at least one transversal optical sensor in order to preferably provide appropriate position information of the object.

Such effects of the dependence of the at least one longitudinal sensor signal on a beam geometry, preferably a beam cross-section of the at least one light beam, were observed in the context of the investigations leading to the present invention in particular in the case of organic
35 photovoltaic components, that is to say photovoltaic components, for example, solar cells, which comprise at least one organic material, for example at least one organic p-semiconducting material and/or at least one organic dye. By way of example, such effects, as is explained in even greater detail below by way of example, were observed in the case of dye solar cells, that is to say components which have at least one first electrode, at least one n-semiconducting
40 metal oxide, at least one dye, at least one p-semiconducting organic material, preferably a solid organic p-type semiconductor, and at least one second electrode. Such dye solar cells, preferably solid dye solar cells (solid dye sensitized solar cells, sDSC), are known in principle in

numerous variations from the literature. The described effect of the dependence of the sensor signal on a geometry of the illumination on the sensor area and a use of this effect have not, however, been described heretofore.

- 5 In particular, the at least one longitudinal optical sensor can be designed in such a way that the sensor signal, given the same total power of the illumination, is substantially independent of a size of the sensor region, in particular of a size of the sensor area, in particular as long as the light spot of the illumination lies completely within the sensor region, in particular the sensor area. Consequently, the longitudinal sensor signal can be dependent exclusively on a focusing
10 of the electromagnetic rays on the sensor area. In particular the sensor signal can be embodied in such a way that a photocurrent and/or a photo voltage per sensor area have/has the same values given the same illumination, for example the same values given the same size of the light spot.
- 15 The evaluation device can comprise in particular at least one data processing device, in particular an electronic data processing device, which can be designed to generate the at least one item of information on the transversal position of the object by evaluating the at least one transversal sensor signal and to generate the at least one item of information on the longitudinal position of the object by evaluating the at least one longitudinal sensor signal. Thus, the
20 evaluation device is designed to use the at least one transversal sensor signal and the at least one longitudinal sensor signal as input variables and to generate the items of information on the transversal position and the longitudinal position of the object by processing these input variables. The processing can be done in parallel, subsequently or even in a combined manner. The evaluation device may use an arbitrary process for generating these items of information,
25 such as by calculation and/or using at least one stored and/or known relationship. Besides the at least one transversal sensor signal and at least one longitudinal sensor signal, one or a plurality of further parameters and/or items of information can influence said relationship, for example at least one item of information about a modulation frequency. The relationship can be determined or determinable empirically, analytically or else semi-empirically. Particularly
30 preferably, the relationship comprises at least one calibration curve, at least one set of calibration curves, at least one function or a combination of the possibilities mentioned. One or a plurality of calibration curves can be stored for example in the form of a set of values and the associated function values thereof, for example in a data storage device and/or a table. Alternatively or additionally, however, the at least one calibration curve can also be stored for
35 example in parameterized form and/or as a functional equation. Separate relationships for processing the at least one transversal sensor signal into the at least one item of information on the transversal position and for processing the at least one longitudinal sensor signal into the at least one item of information on the longitudinal position may be used. Alternatively, at least one combined relationship for processing the sensor signals is feasible. Various possibilities are
40 conceivable and can also be combined.

By way of example, the evaluation device can be designed in terms of programming for the purpose of determining the items of information. The evaluation device can comprise in particular at least one computer, for example at least one microcomputer. Furthermore, the evaluation device can comprise one or a plurality of volatile or nonvolatile data memories. As an
5 alternative or in addition to a data processing device, in particular at least one computer, the evaluation device can comprise one or a plurality of further electronic components which are designed for determining the items of information, for example an electronic table and in particular at least one look-up table and/or at least one application-specific integrated circuit (ASIC).

10 The combination of the at least one item of information on the transversal position and the at least one item of information on the longitudinal position allows for a multiplicity of possible uses of the detector, which will be described by way of example hereinafter. As outlined in great detail in WO 2012/110924 A1, a cross-section of the light beam, resulting in a specific diameter
15 or equivalent diameter of a light spot on the sensor region of the at least one longitudinal optical sensor, can be dependent on a distance between the object and the detector and/or on the optional transfer device of the detector, for example at least one detector lens. By way of example, a variation of the distance between the object and a lens of the optional transfer device can lead to a defocusing of the illumination on the sensor region, accompanied by a
20 change in the geometry of the illumination, for example a widening of a light spot, which can result in a correspondingly altered longitudinal sensor signal or a multiplicity of altered longitudinal sensor signals, in case a plurality of longitudinal optical sensors is used. Even without a transfer device, by way of example, from a known beam profile from the sensor signal and/or a variation thereof, for example, by means of a known beam profile and/or a known
25 propagation of the light beam, it is possible to deduce a defocusing and/or the geometrical information. By way of example, given a known total power of the illumination, it is thus possible to deduce from the longitudinal sensor signal of the longitudinal optical sensor a geometry of the illumination and therefrom in turn the geometrical information, in particular at least one item of location information of the object.

30 Similarly, the at least one transversal optical sensor allows for an easy detection of a transversal position of the object. For this purpose, use may be made of the fact that a change in a transversal position of the object generally will lead to a change in a transversal position of the light beam in the sensor region of the at least one transversal optical sensor. Thus, for
35 example by detecting a transversal position of a light spot generated by the light beam impinging on a sensor region, such as a sensor area, of the transversal optical sensor, the transversal position or at least one item of information on the transversal position of the object may be generated. Thus, the position of the light spot may be determined by comparing currents and/or voltage signals of partial electrodes of the transversal optical sensor, such as by
40 forming at least one ratio of at least two currents through at least two different partial electrodes. For this measurement principle, as an example, reference may be made to US 6,995,445 and/or US 2007/0176165 A1. The above-mentioned at least one relationship between the at

least one transversal sensor signal and the at least one item of information on the transversal position of the object may include a known relationship between a transversal position of the light spot on a sensor region of the transversal optical sensor and a transversal position of the object. For this purpose, known imaging properties of the detector, specifically known imaging properties of at least one transfer device of the detector, may be used. Thus, as an example, the transfer device may include at least one lens, and the known imaging properties may make use of a known lens equation of the lens, thereby allowing for a transformation of at least one transversal coordinate of a light spot into at least one transversal coordinate of the object, as the skilled person will recognize. Therein, the known relationship may also make use of additional information, such as the at least one longitudinal sensor signal and/or the at least one item of information on the longitudinal position of the object. Thus, by using the at least one longitudinal sensor signal, the evaluation device may, as an example, first determine at least one item of information on a lateral position of the object, such as at least one distance between the object and the detector, specifically the transfer device and, more preferably, at least one lens of the transfer device. This item of information on the longitudinal position of the object may then be used in the lens equation in order to transform the at least one transversal sensor signal into the at least one item of information on the transversal position of the object, such as by transforming the at least one transversal coordinate of the light spot in the sensor region of the at least one transversal optical sensor into at least one transversal coordinate of the object. Other algorithms are feasible.

As outlined above, the total intensity of total power of the light beam is often unknown, since this total power e.g. may depend on the properties of the object, such as reflecting properties, and/or may depend on a total power of an illumination source and/or may depend on a large number of environmental conditions. Since the above-mentioned known relationship between the at least one longitudinal optical sensor signal and a beam cross-section of the light beam in the at least one sensor region of the at least one longitudinal optical sensor and, thus, a known relationship between the at least one longitudinal optical sensor signal and the at least one item of information on the longitudinal position of the object may depend on the total power of total intensity of the light beam, various ways of overcoming this uncertainty are feasible. Thus, as outlined in great detail in WO 2012/110924 A1, a plurality of longitudinal sensor signals may be detected by the same longitudinal optical sensor, such as by using different modulation frequencies of an illumination of the object. Thus, at least two longitudinal sensor signals may be acquired at different frequencies of a modulation of the illumination, wherein, from the at least two sensor signals, for example by comparison with corresponding calibration curves, it is possible to deduce the total power and/or the geometry of the illumination, and/or therefrom, directly or indirectly, to deduce the at least one item of information on the longitudinal position of the object.

Additionally or alternatively, however, as outlined above, the detector may comprise a plurality of longitudinal optical sensors, each longitudinal optical sensor being adapted to generate at least one longitudinal sensor signal. The longitudinal sensor signals generated by the

longitudinal optical sensors may be compared, in order to gain information on the total power and/or intensity of the light beam and/or in order to normalize the longitudinal sensor signals and/or the at least one item of information on the longitudinal position of the object for the total power and/or total intensity of the light beam. Thus, as an example, a maximum value of the longitudinal optical sensor signals may be detected, and all longitudinal sensor signals may be divided by this maximum value, thereby generating normalized longitudinal optical sensor signals, which, then, may be transformed by using the above-mentioned known relationship, into the at least one item of longitudinal information on the object. Other ways of normalization are feasible, such as a normalization using a mean value of the longitudinal sensor signals and dividing all longitudinal sensor signals by the mean value. Other options are possible. Each of these options is suited to render the transformation independent from the total power and/or intensity of the light beam. In addition, information on the total power and/or intensity of the light beam might be generated.

The detector described can advantageously be developed in various ways. Thus, the detector can furthermore have at least one modulation device for modulating the illumination, in particular for periodic modulation, in particular a periodic beam interrupting device. A modulation of the illumination should be understood to mean a process in which a total power of the illumination is varied, preferably periodically, in particular with one or a plurality of modulation frequencies. In particular, a periodic modulation can be effected between a maximum value and a minimum value of the total power of the illumination. The minimum value can be 0, but can also be > 0 , such that, by way of example, complete modulation does not have to be effected. The modulation can be effected for example in a beam path between the object and the optical sensor, for example by the at least one modulation device being arranged in said beam path.

Alternatively or additionally, however, the modulation can also be effected in a beam path between an optional illumination source - described in even greater detail below - for illuminating the object and the object, for example by the at least one modulation device being arranged in said beam path. A combination of these possibilities is also conceivable. The at least one modulation device can comprise for example a beam chopper or some other type of periodic beam interrupting device, for example comprising at least one interrupter blade or interrupter wheel, which preferably rotates at constant speed and which can thus periodically interrupt the illumination. Alternatively or additionally, however, it is also possible to use one or a plurality of different types of modulation devices, for example modulation devices based on an electro-optical effect and/or an acousto-optical effect. Once again alternatively or additionally, the at least one optional illumination source itself can also be designed to generate a modulated illumination, for example by said illumination source itself having a modulated intensity and/or total power, for example a periodically modulated total power, and/or by said illumination source being embodied as a pulsed illumination source, for example as a pulsed laser. Thus, by way of example, the at least one modulation device can also be wholly or partly integrated into the illumination source. Various possibilities are conceivable.

The detector can be designed in particular to detect at least two sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies. The evaluation device can be designed to generate the geometrical information from the at least two sensor signals. As described above, in this way, by way of
5 example, it is possible to resolve ambiguities and/or it is possible to take account of the fact that, for example, a total power of the illumination is generally unknown.

Further possible embodiments of the detector relate to the embodiment of the at least one optional transfer device. As explained above, said at least one transfer device can have imaging
10 properties or else can be embodied as a pure non-imaging transfer device, which has no influence on a focusing of the illumination. It is particularly preferred, however, if the transfer device has at least one imaging element, for example at least one lens and/or at least one curved mirror, since, in the case of such imaging elements, for example, a geometry of the illumination on the sensor region can be dependent on a relative positioning, for example a
15 distance, between the transfer device and the object. Generally, it is particularly preferred if the transfer device is designed in such a way that the electromagnetic radiation which emerges from the object is transferred completely to the sensor region, for example is focused completely onto the sensor region, in particular the sensor area, in particular if the object is arranged in a visual range of the detector.

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As explained above, the optical sensor can furthermore be designed in such a way that the sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination. The detector can be embodied, in particular, as explained above, in such a way that sensor signals at different modulation frequencies are
25 picked up, for example in order to generate one or a plurality of further items of information about the object. As described above, by way of example, a sensor signal at at least two different modulation frequencies can, in each case, be picked up, wherein, by way of example, in this way, a lack of information about a total power of the illumination can be supplemented. By way of example, by comparing the at least two sensor signals picked up at different
30 modulation frequencies with one or a plurality of calibration curves, which can be stored for example in a data storage device of the detector, even in the case of an unknown total power of the illumination, it is possible to deduce a geometry of the illumination, for example a diameter or an equivalent diameter of a light spot on the sensor area. For this purpose, by way of example, it is possible to use the at least one evaluation device described above, for example at
35 least one data processing data, which can be designed to control such picking-up of sensor signals at different frequencies and which can be designed to compare said sensor signals with the at least one calibration curve in order to generate therefrom the geometrical information, for example information about a geometry of the illumination, for example information about a diameter or equivalent diameter of a light spot of the illumination on a sensor area of the optical
40 sensor. Furthermore, as is explained in even greater detail below, the evaluation device can alternatively or additionally be designed to generate at least one item of geometrical information about the object, for example at least one item of location information. This generation of the at

least one item of geometrical information, as explained above, can be effected for example taking account of at least one known relationship between a positioning of the object relative to the detector and/or the transfer device or a part thereof and a size of a light spot, for example empirically, semi-empirically or analytically using corresponding imaging equations.

5

In contrast to known detectors, in which a spatial resolution and/or imaging of objects is also generally tied to the fact that the smallest possible sensor areas are used, for example the smallest possible pixels in the case of CCD chips, the sensor region of the proposed detector can be embodied in a very large fashion, in principle, since for example the geometrical
10 information, in particular the at least one item of location information, about the object can be generated from a known relationship for example between the geometry of the illumination and the sensor signal. Accordingly, the sensor region can have for example a sensor area, for example an optical sensor area, which is at least 0.001 mm^2 , in particular at least 0.01 mm^2 , preferably at least 0.1 mm^2 , more preferably at least 1 mm^2 , more preferably at least 5 mm^2 ,
15 more preferably at least 10 mm^2 , in particular at least 100 mm^2 or at least 1000 mm^2 or even at least $10\,000 \text{ mm}^2$. In particular, sensor areas of 100 cm^2 or more can be used. The sensor area can generally be adapted to the application. In particular, the sensor area should be chosen in such a way that, at least if the object is situated within a visual range of the detector, preferably within a predefined viewing angle and/or a predefined distance from the detector, the light spot
20 is always arranged within the sensor area. In this way, it can be ensured that the light spot is not trimmed by the limits of the sensor region, as a result of which signal corruption could occur.

As described above, the sensor region can be in particular a continuous sensor region, in particular a continuous sensor area, which can preferably generate a uniform, in particular a
25 single, sensor signal. Consequently, the sensor signal can be in particular a uniform sensor signal for the entire sensor region, that is to say a sensor signal to which each partial region of the sensor region contributes, for example additively. The sensor signal can generally, as explained above, in particular be selected from the group consisting of a photocurrent and a photo voltage.

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The optical sensor can comprise in particular at least one semiconductor detector and/or be at least one semiconductor detector. In particular, the optical sensor can comprise at least one organic semiconductor detector or be at least one organic semiconductor detector, that is to say a semiconductor detector comprising at least one organic semiconducting material and/or at
35 least one organic sensor material, for example at least one organic dye. Preferably, the organic semiconductor detector can comprise at least one organic solar cell and particularly preferably a dye solar cell, in particular a solid dye solar cell. Exemplary embodiments of such preferred solid dye solar cells are explained in even greater detail below.

40 In particular, the optical sensor can comprise at least one first electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material, preferably at least one solid p-semiconducting organic material, and at least one second

electrode. Generally, however, it is pointed out that the described effect in which the sensor signal, given a constant total power, is dependent on a geometry of the illumination of the sensor region is with high probability not restricted to organic solar cells and in particular not to dye solar cells. Without intending to restrict the scope of protection of the invention by this theory, and without the invention being bound to the correctness of this theory, it is supposed that generally photovoltaic elements are suitable as optical sensors in which at least one semiconducting material having trap states is used. Consequently, the optical sensor can comprise at least one n-semiconducting material and/or at least one p-semiconducting material which can have for example a conduction band and a valence band, wherein, in the case of organic materials, conduction band and valence band should correspondingly be replaced by LUMO (lowest unoccupied molecular orbital) and HOMO (highest occupied molecular orbital). Trap states should be understood to mean energetically possible states which are disposed between the conduction band (or LUMO) and the valence band (or HOMO) and which can be occupied by charge carriers. By way of example, it is possible to provide trap states for hole conduction which are disposed at at least one distance ΔE_h above the valence band (or HOMO) and/or trap states for electron conduction which are disposed at at least one distance ΔE_e below the conduction band (or LUMO). Such traps can be achieved for example by impurities and/or defects, which can optionally also be introduced in a targeted manner, or can be present intrinsically. By way of example, in the case of a low intensity, that is to say for example in the case of a light spot having a large diameter, only a low current can flow, since firstly the trap states are occupied before holes in the conduction band or electrons in the valence band contribute to a photocurrent. It is only starting from a higher intensity, that is to say for example starting from a more intense focusing of the light spot in the sensor region, that a considerable photocurrent can then flow. The described frequency dependence can be explained for example by the fact that charge carriers leave the traps again after a residence duration τ , such that the described effect occurs only in the case of modulated illumination with a high modulation frequency.

By way of example, the detector can be designed to bring about a modulation of the illumination of the object and/or at least one sensor region of the detector, such as at least one sensor region of the at least one longitudinal optical sensor, with a frequency of 0.05 Hz to 1 MHz, such as 0.1 Hz to 10 kHz. As outlined above, for this purpose, the detector may comprise at least one modulation device, which may be integrated into the at least one optional illumination source and/or may be independent from the illumination source. Thus, at least one illumination source might, by itself, be adapted to generate the above-mentioned modulation of the illumination, and/or at least one independent modulation device may be present, such as at least one chopper and/or at least one device having a modulated transmissibility, such as at least one electro-optical device and/or at least one acousto-optical device.

The above-mentioned trap states can be present for example with a density of 10^{-5} to 10^{-1} , relative to the n-semiconducting material and/or the p-semiconducting material and/or the dye.

The energy differences ΔE with respect to the conduction band and with respect to the valence band can be in particular 0.05-0.3 eV.

5 The detector has, as described above, at least one evaluation device. In particular, the at least one evaluation device can also be designed to completely or partly control or drive the detector, for example by the evaluation device being designed to control one or a plurality of modulation devices of the detector and/or to control at least one illumination source of the detector. The evaluation device can be designed, in particular, to carry out at least one measurement cycle in which one or a plurality of sensor signals, such as a plurality of longitudinal sensor signals, are
10 picked up, for example a plurality of sensor signals of successively at different modulation frequencies of the illumination.

However, unlike in conventional semiconductor devices in which, e.g. the absorption of incident light may occur in a p-type conductor, the absorption of incident light in the present detector
15 may be spatially separated from the movement of the charge carriers in dye sensitized solar cells (DSCs), where incident light may cause the light-absorbing organic dye to switch into an excited state such as Frenkel excitons, i.e. excited, strongly bound electron-hole pairs. As long as the energy levels of both the p-type conductor and the n-type conductor are well-matched to the energy states of the excited light-absorbing organic dye, the excitons may be separated
20 and, thus, the electrons and the holes may travel through the n-type and the p-type conductor, respectively, to the appropriate contact electrodes. Hereby, the moving charge carriers may be majority charge carriers, i.e. in the n-type conductor electrons, in the p-type conductor holes are traveling. Since the light-absorbing organic dye itself is a non-conducting substance, an effective charge transport may depend on an extent to which molecules of the light-absorbing
25 organic dye are in close contact with both the p-type conductor and the n-type conductor. For potential details of DSCs, reference may be made e.g. to U. Bach, M. Grätzel, D. Lupo, P. Comte, J. E. Moser, F. Weissörtel, J. Salbeck, and H. Spreitzer. "Solid-state dye-sensitized mesoporous TiO₂ solar cells with high proton-to-electron conversion efficiencies". *Nature*, Vol. 395, no. 6702, pp. 583-585, 1998.

30 As described above, when light falls on the cell, it may be absorbed by the light-absorbing organic dye and an exciton may be created. If the energy of the absorbed photons is greater than an energy gap between the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) of the light-absorbing organic dye, an electron of the
35 HOMO may be raised to the LUMO of the photoexcited dye and charge separation may take place at the boundary between the dye and the semiconductor which nanoporous titanium dioxide. From there, the electron may travel within femtoseconds to picoseconds further into the conduction band of nanoporous titanium dioxide. Preferably, the energy level of the excited state is matched to the lower limit of the conduction band of the titanium dioxide in order to
40 minimize energy losses during electron transition and the LUMO level should extend sufficiently beyond the lower limit of the conduction band of the titanium dioxide. The oxidation potential of the hole conductor should extend over the HOMO level of the dye allowing it to transport away

the holes of the excited dye. If a load is connected in an external circuit, a current may flow across the titanium dioxide and the anode. The reduced dye may be regenerated through electron donation by the organic p-type conductor to the dye, which may prevent recombination of the electrons from the conduction band of the titanium dioxide with the oxidised dye. The p-type conductor, in turn, may be regenerated through the counter-electrode, which may ensure a constant conversion of energy from incident light into electrical energy without requiring a permanent chemical change.

The evaluation device is designed, as described above, to generate at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal. Said position of the object can be static or may even comprise at least one movement of the object, for example a relative movement between the detector or parts thereof and the object or parts thereof. In this case, a relative movement can generally comprise at least one linear movement and/or at least one rotational movement. Items of movement information can for example also be obtained by comparison of at least two items of information picked up at different times, such that for example at least one item of location information can also comprise at least one item of velocity information and/or at least one item of acceleration information, for example at least one item of information about at least one relative velocity between the object or parts thereof and the detector or parts thereof. In particular, the at least one item of location information can generally be selected from: an item of information about a distance between the object or parts thereof and the detector or parts thereof, in particular an optical path length; an item of information about a distance or an optical distance between the object or parts thereof and the optional transfer device or parts thereof; an item of information about a positioning of the object or parts thereof relative to the detector or parts thereof; an item of information about an orientation of the object and/or parts thereof relative to the detector or parts thereof; an item of information about a relative movement between the object or parts thereof and the detector or parts thereof; an item of information about a two-dimensional or three-dimensional spatial configuration of the object or of parts thereof, in particular a geometry or form of the object. Generally, the at least one item of location information can therefore be selected for example from the group consisting of: an item of information about at least one location of the object or at least one part thereof; information about at least one orientation of the object or a part thereof; an item of information about a geometry or form of the object or of a part thereof, an item of information about a velocity of the object or of a part thereof, an item of information about an acceleration of the object or of a part thereof, an item of information about a presence or absence of the object or of a part thereof in a visual range of the detector.

The at least one item of location information can be specified for example in at least one coordinate system, for example a coordinate system in which the detector or parts thereof rest. Alternatively or additionally, the location information can also simply comprise for example a

distance between the detector or parts thereof and the object or parts thereof. Combinations of the possibilities mentioned are also conceivable.

As outlined above, the detector may comprise at least one illumination source. The illumination source can be embodied in various ways. Thus, the illumination source can be for example part of the detector in a detector housing. Alternatively or additionally, however, the at least one illumination source can also be arranged outside a detector housing, for example as a separate light source. The illumination source can be arranged separately from the object and illuminate the object from a distance. Alternatively or additionally, the illumination source can also be connected to the object or even be part of the object, such that, by way of example, the electromagnetic radiation emerging from the object can also be generated directly by the illumination source. By way of example, at least one illumination source can be arranged on and/or in the object and directly generate the electromagnetic radiation by means of which the sensor region is illuminated. By way of example, at least one infrared emitter and/or at least one emitter for visible light and/or at least one emitter for ultraviolet light can be arranged on the object. By way of example, at least one light emitting diode and/or at least one laser diode can be arranged on and/or in the object. The illumination source can comprise in particular one or a plurality of the following illumination sources: a laser, in particular a laser diode, although in principle, alternatively or additionally, other types of lasers can also be used; a light emitting diode; an incandescent lamp; an organic light source, in particular an organic light emitting diode. Alternatively or additionally, other illumination sources can also be used. It is particularly preferred if the illumination source is designed to generate one or more light beams having a Gaussian beam profile, as is at least approximately the case for example in many lasers. However, other embodiments are also possible, in principle.

As outlined above, a further aspect of the present invention proposes a human-machine interface for exchanging at least one item of information between a user and a machine. A human-machine interface should generally be understood to mean a device by means of which such information can be exchanged. The machine can comprise in particular a data processing device. The at least one item of information can generally comprise for example data and/or control commands. Thus, the human-machine interface can be designed in particular for the inputting of control commands by the user.

The human-machine interface has at least one detector in accordance with one or a plurality of the embodiments described above. The human-machine interface is designed to generate at least one item of geometrical information of the user by means of the detector wherein the human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command. By way of example, said at least one item of geometrical information can be or comprise an item of location information and/or position information and/or orientation information about a body and/or at least one body part of the user, for example an item of location information about a hand posture and/or a posture of some other body part of the user.

In this case, the term user should be interpreted broadly and can for example also encompass one or a plurality of articles directly influenced by the user. Thus, the user can for example also wear one or a plurality of gloves and/or other garments, wherein the geometrical information is at least one item of geometrical information of this at least one garment. By way of example, such garments can be embodied as reflective to a primary radiation emerging from at least one illumination source, for example by the use of one or a plurality of reflectors. Once again alternatively or additionally, the user can for example spatially move one or a plurality of articles whose geometrical information can be detected, which is likewise also intended to be subsumable under generation of at least one item of geometrical information of the user. By way of example, the user can move at least one reflective rod and/or some other type of article, for example by means of said user's hand.

The at least one item of geometrical information can be static, that is to say can for example once again comprise a snapshot, but can also for example once again comprise a series of sequential items of geometrical information and/or at least one movement. By way of example, at least two items of geometrical information picked up at different times can be compared, such that, by way of example, the at least one item of geometrical information can also comprise at least one item of information about a velocity and/or an acceleration of a movement. Accordingly, the at least one item of geometrical information can for example comprise at least one item of information about at least one body posture and/or about at least one movement of the user.

The human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command. As explained above, the term information should in this case be interpreted broadly and can comprise for example data and/or control commands. By way of example, the human-machine interface can be designed to assign the at least one item of information to the at least one item of geometrical information, for example by means of a corresponding assignment algorithm and/or a stored assignment specification. By way of example, a unique assignment between a set of items of geometrical information and corresponding items of information can be stored. In this way, for example by means of a corresponding body posture and/or movement of the user, an inputting of at least one item of information can be effected.

Such human-machine interfaces can generally be used in the machine control or else for example in virtual reality. By way of example, industrial controllers, manufacturing controllers, machine controllers in general, robot controllers, vehicle controllers or similar controllers can be made possible by means of the human-machine interface having the one or the plurality of detectors. However, the use of such a human-machine interface in consumer electronics is particularly preferred.

Accordingly, as outlined above, a further aspect of the present invention proposes an entertainment device for carrying out at least one entertainment function, in particular a game. The entertainment function can comprise in particular at least one game function. By way of example, one or a plurality of games can be stored which can be influencable by a user, who in this context is also called a player hereinafter. By way of example, the entertainment device can comprise at least one display device, for example at least one screen and/or at least one projector and/or at least one set of display spectacles.

The entertainment device furthermore comprises at least one human-machine interface in accordance with one or more of the embodiments described above. The entertainment device is designed to enable at least one item of information of a player to be input by means of the human-machine interface. By way of example, the player, as described above, can adopt or alter one or a plurality of body postures for this purpose. This includes the possibility of the player for example using corresponding articles for this purpose, for example garments such as e.g. gloves, for example garments which are equipped with one or a plurality of reflectors for reflecting the electromagnetic radiation of the detector. The at least one item of information can comprise for example, as explained above, one or a plurality of control commands. By way of example, in this way, changes in direction can be performed, inputs can be confirmed, a selection can be made from a menu, specific game options can be initiated, movements can be influenced in a virtual space or similar instances of influencing or altering the entertainment function can be performed.

The above-described detector, the method, the human-machine interface and the entertainment device and also the proposed uses have considerable advantages over the prior art. Thus, generally, a simple and, still, efficient detector for determining a position of at least one object in space may be provided. Therein, as an example, three-dimensional coordinates of an object or part of an object may be determined in a fast and efficient way. Specifically the combination of the at least one transversal optical sensor and the at least one longitudinal optical sensor, which each may be designed in a cost-efficient way, may lead to a compact, cost-efficient and, still, highly precise device. Both the at least one transversal optical sensor and the at least one longitudinal optical sensor preferably may fully or partially be designed as organic photovoltaic devices, such as by using dye-sensitized solar cells for each of these optical sensors, preferably sDSCs.

As compared to devices known in the art, which of them are based on complex triangulation methods, the detector as proposed provides a high degree of simplicity, specifically with regard to an optical setup of the detector. Thus, in principle, a simple combination of one, two or more sDSCs, preferably in combination with a suited transfer device, specifically a suited lens, and in conjunction with an appropriate evaluation device, is sufficient for high precision position detection.

This high degree of simplicity, in combination with the possibility of high precision measurements, is specifically suited for machine control, such as in human-machine interfaces and, more preferably, in gaming. Thus, cost-efficient entertainment devices may be provided which may be used for a large number of gaming purposes.

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Thus, as for the optical detectors and devices disclosed in WO 2012/110924 A1 or in US provisional applications 61/739,173, filed on December 19, 2012, and 61/749,964, filed on January 8, 2013, the optical detector, the detector system, the human-machine interface, the entertainment device, the tracking system or the camera according to the present invention (in
10 the following simply referred to as "the devices according to the present invention" or "Fip-devices") may be used for a plurality of application purposes, such as one or more of the purposes disclosed in further detail in the following.

Thus, firstly, Fip-devices may be used in mobile phones, tablet computers, laptops, smart
15 panels or other stationary or mobile computer or communication applications. Thus, Fip-devices may be combined with at least one active light source, such as a light source emitting light in the visible range or infrared spectral range, in order to enhance performance. Thus, as an example, Fip-devices may be used as cameras and/or sensors, such as in combination with mobile software for scanning environment, objects and living beings. Fip-devices may even be
20 combined with 2D cameras, such as conventional cameras, in order to increase imaging effects. Fip-devices may further be used for surveillance and/or for recording purposes or as input devices to control mobile devices, especially in combination with gesture recognition. Thus, specifically, Fip-devices acting as human-machine interfaces, also referred to as FiP input devices, may be used in mobile applications, such as for controlling other electronic devices or
25 components via the mobile device, such as the mobile phone. As an example, the mobile application including at least one Fip-device may be used for controlling a television set, a game console, a music player or music device or other entertainment devices.

Further, Fip-devices may be used in webcams or other peripheral devices for computing
30 applications. Thus, as an example, Fip-devices may be used in combination with software for imaging, recording, surveillance, scanning or motion detection. As outlined in the context of the human-machine interface and/or the entertainment device, Fip-devices are particularly useful for giving commands by facial expressions and/or body expressions. Fip-devices can be combined with other input generating devices like e.g. mouse, keyboard, touchpad, etc. Further,
35 Fip-devices may be used in applications for gaming, such as by using a webcam. Further, Fip-devices may be used in virtual training applications and/or video conferences.

Further, Fip-devices may be used in mobile audio devices, television devices and gaming
40 devices, as partially explained above. Specifically, Fip-devices may be used as controls or control devices for electronic devices, entertainment devices or the like. Further, Fip-devices may be used for eye detection or eye tracking, such as in 2D- and 3D-display techniques, especially with transparent displays for augmented reality applications.

Further, Fip-devices may be used in or as digital cameras such as DSC cameras and/or in or as reflex cameras such as SLR cameras. For these applications, reference may be made to the use of Fip-devices in mobile applications such as mobile phones, as disclosed above.

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Further, Fip-devices may be used for security and surveillance applications. Specifically, Fip-devices may be used for optical encryption. Fip-based detection can be combined with other detection devices to complement wavelengths, such as with IR, x-ray, UV-VIS, radar or ultrasound detectors. Fip-devices may further be combined with an active infrared light source to allow detection in low light surroundings. Fip-devices such as FIP-based sensors are generally advantageous as compared to active detector systems, specifically since Fip-devices avoid actively sending signals which may be detected by third parties, as is the case e.g. in radar applications, ultrasound applications, LIDAR or similar active detector device is. Thus, generally, Fip-devices may be used for an unrecognized and undetectable tracking of moving objects. Additionally, Fip-devices generally are less prone to manipulations and irritations as compared to conventional devices.

Further, given the ease and accuracy of 3D detection by using Fip-devices, Fip-devices generally may be used for facial, body and person recognition and identification. Therein, Fip-devices may be combined with other detection means for identification or personalization purposes such as passwords, finger prints, iris detection, voice recognition or other means. Thus, generally, Fip-devices may be used in security devices and other personalized applications.

Further, Fip-devices may be used as 3D-Barcode readers for product identification.

In addition to the security and surveillance applications mentioned above, Fip-devices generally can be used for surveillance and monitoring of spaces and areas. Thus, Fip-devices may be used for surveying and monitoring spaces and areas and, as an example, for triggering or executing alarms in case prohibited areas are violated. Thus, generally, Fip-devices may be used for surveillance purposes in building surveillance or museums, optionally in combination with other types of sensors, such as in combination with motion or heat sensors, in combination with image intensifiers or image enhancement devices and/or photomultipliers.

Further, Fip-devices may advantageously be applied in camera applications such as video and camcorder applications. Thus, Fip-devices may be used for motion capture and 3D-movie recording. Therein, Fip-devices generally provide a large number of advantages over conventional optical devices. Thus, Fip-devices generally require a lower complexity with regard to optical components. Thus, as an example, the number of lenses may be reduced as compared to conventional optical devices, such as by providing Fip-devices having one lens only. Due to the reduced complexity, very compact devices are possible, such as for mobile use. Conventional optical systems having two or more lenses with high quality generally are

voluminous, such as due to the general need for voluminous beam-splitters. Further, Fip-devices generally may be used for focus/autofocus devices, such as autofocus cameras. Further, Fip-devices may also be used in optical microscopy, especially in confocal microscopy.

- 5 Further, Fip-devices generally are applicable in the technical field of automotive technology and transport technology. Thus, as an example, Fip-devices may be used as distance and surveillance sensors, such as for adaptive cruise control, emergency brake assist, lane departure warning, surround view, blind spot detection, rear cross traffic alert, and other automotive and traffic applications. Therein, Fip-devices may be used as standalone devices or
- 10 in combination with other sensor devices, such as in combination with radar and/or ultrasonic devices. Specifically, Fip-devices may be used for autonomous driving and safety issues. Further, in these applications, Fip-devices may be used in combination with infrared sensors, radar sensors, which are sonic sensors, two-dimensional cameras or other types of sensors. Advantage, in these applications, the generally passive nature of typical Fip-devices is
- 15 advantageous. Thus, since Fip-devices generally do not require emitting signals, the risk of interference of active sensor signals with other signal sources may be avoided. Fip-devices specifically may be used in combination with recognition software, such as standard image recognition software. Thus, signals and data as provide by Fip-devices typically are readily processable and, therefore, generally require lower calculation power than established
- 20 stereovision systems such as LIDAR. Given the low space demand, Fip-devices such as cameras using the FiP-effect may be placed at virtually any place in a vehicle, such as on a window screen, on a front hood, on bumpers, on lights, on mirrors or other places the like. Various detectors based on the FiP-effect can be combined, such as in order to allow autonomously driving vehicles or in order to increase the performance of active safety concepts.
- 25 Thus, various FiP-based sensors may be combined with other Fip-based sensors and/or conventional sensors, such as in the windows like rear window, side window or front window, on the bumpers or on the lights.

- A combination of a FiP-sensor with one or more rain detection sensors is also possible. This is
- 30 due to the fact that Fip-devices generally are advantageous over conventional sensor techniques such as radar, specifically during heavy rain. A combination of at least one Fip-device with at least one conventional sensing technique such as radar may allow for a software to pick the right combination of signals according to the weather conditions.

- 35 Further, Fip-devices generally may be used as break assist and/or parking assist and/or for speed measurements. Speed measurements can be integrated in the vehicle or may be used outside the vehicle, such as in order to measure the speed of other cars in traffic control. Further, Fip-devices may be used for detecting free parking spaces in parking lots.

- 40 Further, Fip-devices may be used is the fields of medical systems and sports. Thus, in the field of medical technology, surgery robotics, e.g. for use in endoscopes, may be named, since, as outlined above, Fip-devices may require a low volume only and may be integrated into other

devices. Specifically, Fip-devices having one lens, at most, may be used for capturing 3D information in medical devices such as in endoscopes. Further, Fip-devices may be combined with an appropriate monitoring software, in order to enable tracking and analysis of movements. These applications are specifically valuable e.g. in medical treatments and long-distance diagnosis and tele-medicine.

Further, Fip-devices may be applied in the field of sports and exercising, such as for training, remote instructions or competition purposes. Specifically, Fip-devices may be applied in the field of dancing, aerobic, football, soccer, basketball, baseball, cricket, hockey, track and field, swimming, polo, handball, volleyball, rugby, sumo, judo, fencing, boxing etc. Fip-devices can be used to detect the position of a ball, a bat, a sword, motions, etc., both in sports and in games, such as to monitor the game, support the referee or for judgment, specifically automatic judgment, of specific situations in sports, such as for judging whether a point or a goal actually was made.

Fip devices may further be used to support a practice of musical instruments, in particular remote lessons, for example lessons of string instruments, such as fiddles, violins, violas, celli, basses, harps, guitars, banjos, or ukuleles, keyboard instruments, such as pianos, organs, keyboards, harpsichords, harmoniums, or accordions, and/or percussion instruments, such as drums, timpani, marimbas, xylophones, vibraphones, bongos, congas, timbales, djembes or tablas.

Fip-devices further may be used in rehabilitation and physiotherapy, in order to encourage training and/or in order to survey and correct movements. Therein, the Fip-devices may also be applied for distance diagnostics.

Further, Fip-devices may be applied in the field of machine vision. Thus, one or more Fip-devices may be used e.g. as a passive controlling unit for autonomous driving and or working of robots. In combination with moving robots, Fip-devices may allow for autonomous movement and/or autonomous detection of failures in parts. Fip-devices may also be used for manufacturing and safety surveillance, such as in order to avoid accidents including but not limited to collisions between robots, production parts and living beings. Given the passive nature of Fip-devices, Fip-devices may be advantageous over active devices and/or may be used complementary to existing solutions like radar, ultrasound, 2D cameras, IR detection etc. One particular advantage of Fip-devices is the low likelihood of signal interference. Therefore multiple sensors can work at the same time in the same environment, without the risk of signal interference. Thus, Fip-devices generally may be useful in highly automated production environments like e.g. but not limited to automotive, mining, steel, etc. Fip-devices can also be used for quality control in production, e.g. in combination with other sensors like 2-D imaging, radar, ultrasound, IR etc., such as for quality control or other purposes. Further, Fip-devices may be used for assessment of surface quality, such as for surveying the surface evenness of a

product or the adherence to specified dimensions, from the range of micrometers to the range of meters. Other quality control applications are feasible.

Further, Fip-devices may be used in the polls, airplanes, ships, spacecrafts and other traffic applications. Thus, besides the applications mentioned above in the context of traffic applications, passive tracking systems for aircrafts, vehicles and the like may be named. Detection devices based on the FiP-effect for monitoring the speed and/or the direction of moving objects are feasible. Specifically, the tracking of fast moving objects on land, sea and in the air including space may be named. The at least one Fip-detector specifically may be mounted on a still-standing and/or on a moving device. An output signal of the at least one Fip-device can be combined e.g. with a guiding mechanism for autonomous or guided movement of another object. Thus, applications for avoiding collisions or for enabling collisions between the tracked and the steered object are feasible. Fip-devices generally are useful and advantageous due to the low calculation power required, the instant response and due to the passive nature of the detection system which generally is more difficult to detect and to disturb as compared to active systems, like e.g. radar. Fip-devices are particularly useful but not limited to e.g. speed control and air traffic control devices.

Fip-devices generally may be used in passive applications. Passive applications include guidance for ships in harbors or in dangerous areas, and for aircrafts at landing or starting. Wherein, fixed, known active targets may be used for precise guidance. The same can be used for vehicles driving in dangerous but well defined routes, such as mining vehicles.

Further, as outlined above, Fip-devices may be used in the field of gaming. Thus, Fip-devices can be passive for use with multiple objects of the same or of different size, color, shape, etc., such as for movement detection in combination with software that incorporates the movement into its content. In particular, applications are feasible in implementing movements into graphical output. Further, applications of Fip-devices for giving commands are feasible, such as by using one or more Fip-devices for gesture or facial recognition. Fip-devices may be combined with an active system in order to work under e.g. low light conditions or in other situations in which enhancement of the surrounding conditions is required. Additionally or alternatively, a combination of one or more Fip-devices with one or more IR or VIS light sources is possible, such as with a detection device based on the FiP effect. A combination of a FiP-based detector with special devices is also possible, which can be distinguished easily by the system and its software, e.g. and not limited to, a special color, shape, relative position to other devices, speed of movement, light, frequency used to modulate light sources on the device, surface properties, material used, reflection properties, transparency degree, absorption characteristics, etc. The device can, amongst other possibilities, resemble a stick, a racquet, a club, a gun, a knife, a wheel, a ring, a steering wheel, a bottle, a ball, a glass, a vase, a spoon, a fork, a cube, a dice, a figure, a puppet, a teddy, a beaker, a pedal, a switch, a glove, jewelry, a musical instrument or an auxiliary device for playing a musical instrument, such as a plectrum, a drumstick or the like. Other options are feasible.

Further, Fip-devices generally may be used in the field of building, construction and cartography. Thus, generally, FiP-based devices may be used in order to measure and/or monitor environmental areas, e.g. country side or buildings. Therein, one or more Fip-devices may be combined with other methods and devices or can be used solely in order to monitor progress and accuracy of building projects, changing objects, houses, etc. FiP-devices can be used for generating three-dimensional models of scanned environments, in order to construct maps of rooms, streets, houses, communities or landscapes, both from ground or from air. Potential fields of application may be construction, cartography, real estate management, land surveying or the like.

FiP-devices may be used within an interconnecting network of home appliances such as CHAIN (Cedec Home Appliances Interoperating Network) to interconnect, automate, and control basic appliance-related services in a home, e.g. energy or load management, remote diagnostics, pet related appliances, child related appliances, child surveillance, appliances related surveillance, support or service to elderly or ill persons, home security and/or surveillance, remote control of appliance operation, and automatic maintenance support.

FiP devices may further be used in agriculture, for example to detect and sort out vermin, weeds, and/or infected crop plants, fully or in parts, wherein crop plants may be infected by fungus or insects. Further, for harvesting crops, FiP detectors may be used to detect animals, such as deer, which may otherwise be harmed by harvesting devices.

FiP-based devices can further be used for scanning of objects, such as in combination with CAD or similar software, such as for additive manufacturing and/or 3D printing. Therein, use may be made of the high dimensional accuracy of Fip-devices, e.g. in x-, y- or z- direction or in any arbitrary combination of these directions, such as simultaneously. Further, Fip-devices may be used in inspections and maintenance, such as pipeline inspection gauges.

As outlined above, Fip-devices may further be used in manufacturing, quality control or identification applications, such as in product identification or size identification (such as for finding an optimal place or package, for reducing waste etc.). Further, Fip-devices may be used in logistics applications. Thus, Fip-devices may be used for optimized loading or packing containers or vehicles. Further, Fip-devices may be used for monitoring or controlling of surface damages in the field of manufacturing, for monitoring or controlling rental objects such as rental vehicles, and/or for insurance applications, such as for assessment of damages. Further, Fip-devices may be used for identifying a size of material, object or tools, such as for optimal material handling, especially in combination with robots. Further, Fip-devices may be used for process control in production, e.g. for observing filling level of tanks. Further, Fip-devices may be used for maintenance of production assets like, but not limited to, tanks, pipes, reactors, tools etc. Further, Fip-devices may be used for analyzing 3D-quality marks. Further, Fip-devices may be used in manufacturing tailor-made goods such as tooth inlays, dental braces,

prosthesis, clothes or the like. Fip-devices may also be combined with one or more 3D-printers for rapid prototyping, 3D-copying or the like. Further, Fip-devices may be used for detecting the shape of one or more articles, such as for anti-product piracy and for anti-counterfeiting purposes.

5

As outlined above, preferably, the at least one transversal optical sensor and/or the at least one longitudinal optical sensor both may comprise at least one organic semiconductor detector, particularly preferably at least one dye solar cell, DSC or sDSC. In particular, the transversal optical sensor and/or the longitudinal optical sensor each may comprise at least one first
10 electrode, at least one n-semiconducting metal oxide, at least one dye, at least one p-semiconducting organic material and at least one second electrode, preferably in the stated order. The stated elements can be present as layers in a layer construction, for example. The layer construction can be applied for example to a substrate, preferably a transparent substrate, for example a glass substrate.

15

Preferred embodiments of the above-mentioned elements of the preferred optical sensor are described below by way of example, wherein these embodiments can be used in any desired combination. However, numerous other configurations are also possible, in principle, wherein reference can be made for example to WO 2012/110924 A1, US 2007/0176165 A1,
20 US 6,995,445 B2, DE 2501124 A1, DE 3225372 A1 and WO 2009/013282 A1 cited above.

As outlined above, the at least one transversal optical sensor may be designed as a dye-sensitized solar cell (DSC), preferably a solid dye-sensitized solar cell (sDSC). Similarly, the at least one longitudinal optical sensor may be designed as at least one dye-sensitized solar cell
25 (DSC) or may comprise at least one dye-sensitized solar cell (DSC), preferably a solid dye-sensitized solar cell (sDSC). More preferably, the at least one longitudinal optical sensor comprises a stack of DSCs, preferably a stack of sDSCs. Preferred components of the DSCs or sDSCs will be disclosed in the following. However, it shall be understood that other embodiments are feasible.

30

First electrode and n-semiconductive metal oxide

Generally, for preferred embodiments of the first electrode and the n-semiconductive metal oxide, which may be used in a layer setup of the transversal optical sensor and/or the
35 longitudinal optical sensor, reference may be made to WO 2012/110924 A1. The n-semiconductive metal oxide used in the dye solar cell of the transversal optical sensor and/or the longitudinal optical sensor may be a single metal oxide or a mixture of different oxides. It is also possible to use mixed oxides. The n-semiconductive metal oxide may especially be porous and/or be used in the form of a nanoparticulate oxide, nanoparticles in this context being
40 understood to mean particles which have an average particle size of less than 0.1 micrometer. A nanoparticulate oxide is typically applied to a conductive substrate (i.e. a carrier with a

conductive layer as the first electrode) by a sintering process as a thin porous film with large surface area.

Preferably, the at least one transversal optical sensor uses at least one transparent substrate.

- 5 Similarly, preferably, the at least one longitudinal optical sensor uses at least one transparent substrate. In case a plurality of longitudinal optical sensors is used, such as a stack of longitudinal optical sensors, preferably, at least one of these longitudinal optical sensors uses a transparent substrate. Thus, as an example, all longitudinal optical sensors but the last longitudinal optical sensor facing away from the object, each may use a transparent substrate.
- 10 The last longitudinal optical sensor may either use a transparent or an intransparent substrate.

Similarly, the at least one transversal optical sensor uses at least one transparent first electrode. Further, the at least one longitudinal optical sensor may use at least one transparent first electrode. In case a plurality of longitudinal optical sensors is used, such as a stack of longitudinal optical sensors, preferably, at least one of these longitudinal optical sensors uses a transparent first electrode. Thus, as an example, all longitudinal optical sensors but the last longitudinal optical sensor facing away from the object each may use a transparent first electrode. The last longitudinal optical sensor may either use a transparent or an intransparent first electrode.

- 20 The substrate may be rigid or else flexible. Suitable substrates (also referred to hereinafter as carriers) are, as well as metal foils, in particular plastic sheets or films and especially glass sheets or glass films. Particularly suitable electrode materials, especially for the first electrode according to the above-described, preferred structure, are conductive materials, for example transparent conductive oxides (TCOs), for example fluorine- and/or indium-doped tin oxide (FTO or ITO) and/or aluminum-doped zinc oxide (AZO), carbon nanotubes or metal films.
- 25 Alternatively or additionally, it would, however, also be possible to use thin metal films which still have a sufficient transparency. In case an intransparent first electrode is desired and used, thick metal films may be used.

- 30 The substrate can be covered or coated with these conductive materials. Since generally, only a single substrate is required in the structure proposed, the formation of flexible cells is also possible. This enables a multitude of end uses which would be achievable only with difficulty, if at all, with rigid substrates, for example use in bank cards, garments, etc.

- 35 The first electrode, especially the TCO layer, may additionally be covered or coated with a solid metal oxide buffer layer (for example of thickness 10 to 200 nm), in order to prevent direct contact of the p-type semiconductor with the TCO layer (see Peng *et al.*, Coord. Chem. Rev. 248, 1479 (2004)). The inventive use of solid p-semiconducting electrolytes, in the case of which contact of the electrolyte with the first electrode is greatly reduced compared to liquid or gel-form electrolytes, however, makes this buffer layer unnecessary in many cases, such that it is possible in many cases to dispense with this layer, which also has a current-limiting effect
- 40

and can also worsen the contact of the n-semiconducting metal oxide with the first electrode. This enhances the efficiency of the components. On the other hand, such a buffer layer can in turn be utilized in a controlled manner in order to match the current component of the dye solar cell to the current component of the organic solar cell. In addition, in the case of cells in which the buffer layer has been dispensed with, especially in solid cells, problems frequently occur with unwanted recombinations of charge carriers. In this respect, buffer layers are advantageous in many cases specifically in solid cells.

As is well known, thin layers or films of metal oxides are generally inexpensive solid semiconductor materials (n-type semiconductors), but the absorption thereof, due to large band gaps, is typically not within the visible region of the electromagnetic spectrum, but rather usually in the ultraviolet spectral region. For use in solar cells, the metal oxides therefore generally, as is the case in the dye solar cells, have to be combined with a dye as a photosensitizer, which absorbs in the wavelength range of sunlight, i.e. at 300 to 2000 nm, and, in the electronically excited state, injects electrons into the conduction band of the semiconductor. With the aid of a solid p-type semiconductor used additionally in the cell as an electrolyte, which is in turn reduced at the counter electrode, electrons can be recycled to the sensitizer, such that it is regenerated.

Of particular interest for use in organic solar cells are semiconductors zinc oxide, tin dioxide, titanium dioxide or mixtures of these metal oxides. The metal oxides can be used in the form of nanocrystalline porous layers. These layers have a large surface area which is coated with the dye as a sensitizer, such that a high absorption of sunlight is achieved. Metal oxide layers which are structured, for example nanorods, give advantages such as higher electron mobilities or improved pore filling by the dye.

The metal oxide semiconductors can be used alone or in the form of mixtures. It is also possible to coat a metal oxide with one or more other metal oxides. In addition, the metal oxides may also be applied as a coating to another semiconductor, for example GaP, ZnP or ZnS.

Particularly preferred semiconductors are zinc oxide and titanium dioxide in the anatase polymorph, which is preferably used in nanocrystalline form.

In addition, the sensitizers can advantageously be combined with all n-type semiconductors which typically find use in these solar cells. Preferred examples include metal oxides used in ceramics, such as titanium dioxide, zinc oxide, tin(IV) oxide, tungsten(VI) oxide, tantalum(V) oxide, niobium(V) oxide, cesium oxide, strontium titanate, zinc stannate, complex oxides of the perovskite type, for example barium titanate, and binary and ternary iron oxides, which may also be present in nanocrystalline or amorphous form.

Due to the strong absorption that customary organic dyes and phthalocyanines and porphyrins have, even thin layers or films of the n-semiconducting metal oxide are sufficient to absorb the

required amount of dye. Thin metal oxide films in turn have the advantage that the probability of unwanted recombination processes falls and that the internal resistance of the dye subcell is reduced. For the n-semiconducting metal oxide, it is possible with preference to use layer thicknesses of 100 nm up to 20 micrometers, more preferably in the range between 500 nm and approx. 3 micrometers.

Dye

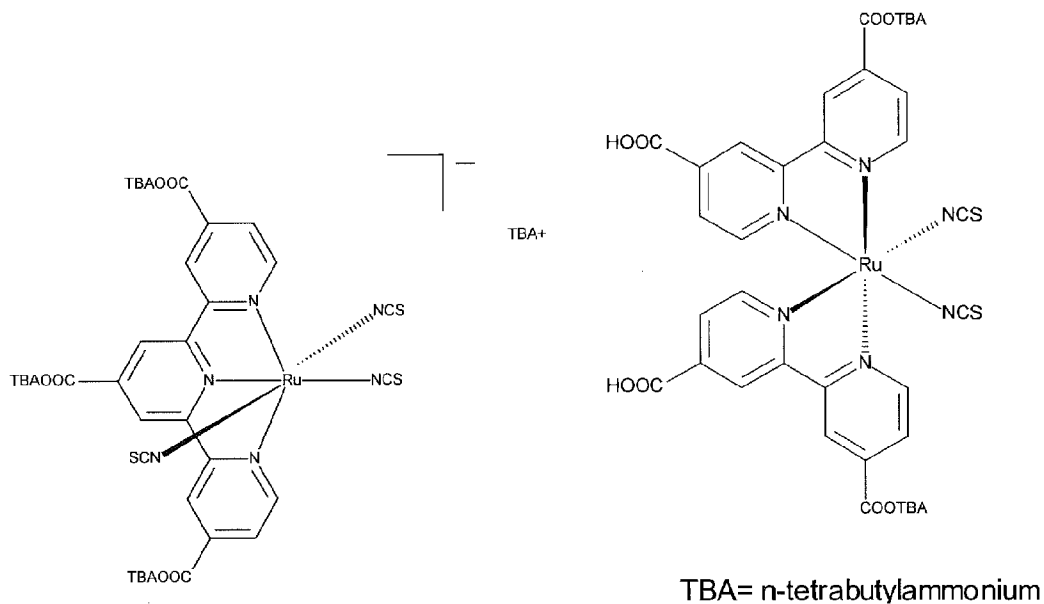
In the context of the present invention, as usual in particular for DSCs, the terms "dye", "sensitizer dye" and "sensitizer" are used essentially synonymously without any restriction of possible configurations. Numerous dyes which are usable in the context of the present invention are known from the prior art, and so, for possible material examples, reference may also be made to the above description of the prior art regarding dye solar cells. As a preferred example, one or more of the dyes disclosed in WO 2012/110924 A1 may be used.

Additionally or alternatively, one or more quinolinium dyes with fluorinated counter anion may be used in the detector according to the present invention, such as one or more of the dyes as disclosed in WO 2013/144177 A1. Specifically, one or more of the dyes disclosed in the following may be used in the at least one longitudinal optical sensor and/or in the at least one transversal optical sensor. Details of these dyes and details of the disclosure of these unpublished applications will be given below. Specifically, dye D-5, which will be described below in more detail, may be used. However, one or more other dyes may be used in addition or alternatively.

All dyes listed and claimed may in principle also be present as pigments. Dye-sensitized solar cells based on titanium dioxide as a semiconductor material are described, for example, in US 4 927 721 A, Nature 353, p. 737-740 (1991) and US 5 350 644 A, and also Nature 395, p. 583-585 (1998) and EP 1 176 646 A1. The dyes described in these documents can in principle also be used advantageously in the context of the present invention. These dye solar cells preferably comprise monomolecular films of transition metal complexes, especially ruthenium complexes, which are bonded to the titanium dioxide layer via acid groups as sensitizers.

Dyes for use in dye-sensitized solar cells which may comprise ruthenium complexes have been rather of academic interest so far, particularly due to the high costs of ruthenium. However, a dye-sensitized solar cell which may be used in the detector according to the present invention would only require such a small amount of ruthenium that the cost argument may easily be overruled by its attractive features for use within the present method for determining a position of at least one object, particularly in a case where at least one light beam which travels from the object might involve a spectral range which may at least partially include a part of the infrared (IR) region, i.e. a part of the electromagnetic spectrum ranging from approximately 750 nm to 1000 μm , preferentially a part thereof generally denoted as near infrared (NIR) region, which is usually considered to range from approximately 750 nm to 1.5 μm . Examples of known

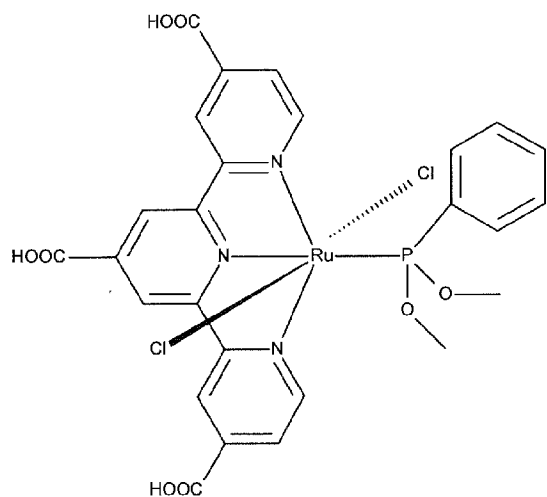
ruthenium complexes which might be suitable for an application within the detector according to the present invention are:



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Another example may be found in T. Kinoshita, J.T. Dy, S. Uchida, T. Kubo, and H. Segawa, *Wideband dye-sensitized solar cells employing a phosphine-coordinated ruthenium sensitizer*, Nature Photonics, 7, 535-539 (2013), wherein a phosphine-coordinated ruthenium complex is described, which exhibits a strong absorption in the NIR, particularly within the range from 750 nm to 950 nm, which may thus, yield a dye-sensitized solar cell with a promising efficiency:

10



15 Owing to weak absorption properties of most known dyes within the IR region, including the NIR region, dyes comprising ruthenium complexes may, thus, be able to extend the scope of the

detector according to the present invention into the IR region, in particular into the NIR region, e.g. as being used as active depth sensors, particularly in applications related to computer vision, wherein IR light may play an important role, as described elsewhere in this application.

5 Many sensitizers which have been proposed include metal-free organic dyes, which are likewise also usable in the context of the present invention. High efficiencies of more than 4%, especially in solid dye solar cells, can be achieved, for example, with indoline dyes (see, for example, Schmidt-Mende *et al.*, Adv. Mater. 2005, 17, 813). US-A-6 359 211 describes the use, also implementable in the context of the present invention, of cyanine, oxazine, thiazine and acridine
10 dyes which have carboxyl groups bonded via an alkylene radical for fixing to the titanium dioxide semiconductor.

Organic dyes now achieve efficiencies of almost 12.1% in liquid cells (see, for example, P. Wang *et al.*, ACS. Nano 2010). Pyridinium-containing dyes have also been reported, can be
15 used in the context of the present invention and exhibit promising efficiencies.

Particularly preferred sensitizer dyes in the dye solar cell proposed are the perylene derivatives, terrylene derivatives and quaterrylene derivatives described in DE 10 2005 053 995 A1 or WO 2007/054470 A1. The use of these dyes, which is also possible in the context of the
20 present invention, leads to photovoltaic elements with high efficiencies and simultaneously high stabilities.

The rylenes exhibit strong absorption in the wavelength range of sunlight and can, depending on the length of the conjugated system, cover a range from about 400 nm (perylene derivatives I from DE 10 2005 053 995 A1) up to about 900 nm (quaterrylene derivatives I from
25 DE 10 2005 053 995 A1). Rylene derivatives I based on terrylene absorb, according to the composition thereof, in the solid state adsorbed onto titanium dioxide, within a range from about 400 to 800 nm. In order to achieve very substantial utilization of the incident sunlight from the visible into the near infrared region, it is advantageous to use mixtures of different rylene
30 derivatives I. Occasionally, it may also be advisable also to use different rylene homologs.

The rylene derivatives I can be fixed easily and in a permanent manner to the n-semiconducting metal oxide film. The bonding is effected via the anhydride function (x1) or the carboxyl groups –COOH or –COO– formed in situ, or via the acid groups A present in the imide or condensate
35 radicals ((x2) or (x3)). The rylene derivatives I described in DE 10 2005 053 995 A1 have good suitability for use in dye-sensitized solar cells in the context of the present invention.

It is particularly preferred when the dyes, at one end of the molecule, have an anchor group which enables the fixing thereof to the n-type semiconductor film. At the other end of the
40 molecule, the dyes preferably comprise electron donors Y which facilitate the regeneration of the dye after the electron release to the n-type semiconductor, and also prevent recombination with electrons already released to the semiconductor.

For further details regarding the possible selection of a suitable dye, it is possible, for example, again to refer to DE 10 2005 053 995 A1. By way of example, it is possible especially to use ruthenium complexes, porphyrins, other organic sensitizers, and preferably rylenes.

5

The dyes can be fixed onto or into the n-semiconducting metal oxide films in a simple manner. For example, the n-semiconducting metal oxide films can be contacted in the freshly sintered (still warm) state over a sufficient period (for example about 0.5 to 24 h) with a solution or suspension of the dye in a suitable organic solvent. This can be accomplished, for example, by
10 immersing the metal oxide-coated substrate into the solution of the dye.

If combinations of different dyes are to be used, they may, for example, be applied successively from one or more solutions or suspensions which comprise one or more of the dyes. It is also possible to use two dyes which are separated by a layer of, for example, CuSCN (on this
15 subject see, for example, Tennakone, K.J., Phys. Chem. B. 2003, 107, 13758). The most convenient method can be determined comparatively easily in the individual case.

In the selection of the dye and of the size of the oxide particles of the n-semiconducting metal oxide, the organic solar cell should be configured such that a maximum amount of light is
20 absorbed. The oxide layers should be structured such that the solid p-type semiconductor can efficiently fill the pores. For instance, smaller particles have greater surface areas and are therefore capable of adsorbing a greater amount of dyes. On the other hand, larger particles generally have larger pores which enable better penetration through the p-conductor.

25 p-semiconducting organic material

As described above, the at least one DSC or sDSC of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor can comprise in particular at least one p-semiconducting organic material, preferably at least one solid p-semiconducting material, which
30 is also designated hereinafter as p-type semiconductor or p-type conductor. Hereinafter, a description is given of a series of preferred examples of such organic p-type semiconductors which can be used individually or else in any desired combination, for example in a combination of a plurality of layers with a respective p-type semiconductor, and/or in a combination of a plurality of p-type semiconductors in one layer.

35

In order to prevent recombination of the electrons in the n-semiconducting metal oxide with the solid p-conductor, it is possible to use, between the n-semiconducting metal oxide and the p-type semiconductor, at least one passivating layer which has a passivating material. This layer should be very thin and should as far as possible cover only the as yet uncovered sites of the n-semiconducting metal oxide. The passivation material may, under some circumstances, also be
40 applied to the metal oxide before the dye. Preferred passivation materials are especially one or more of the following substances: Al_2O_3 ; silanes, for example CH_3SiCl_3 ; Al^{3+} ; 4-tert-butylpyridine

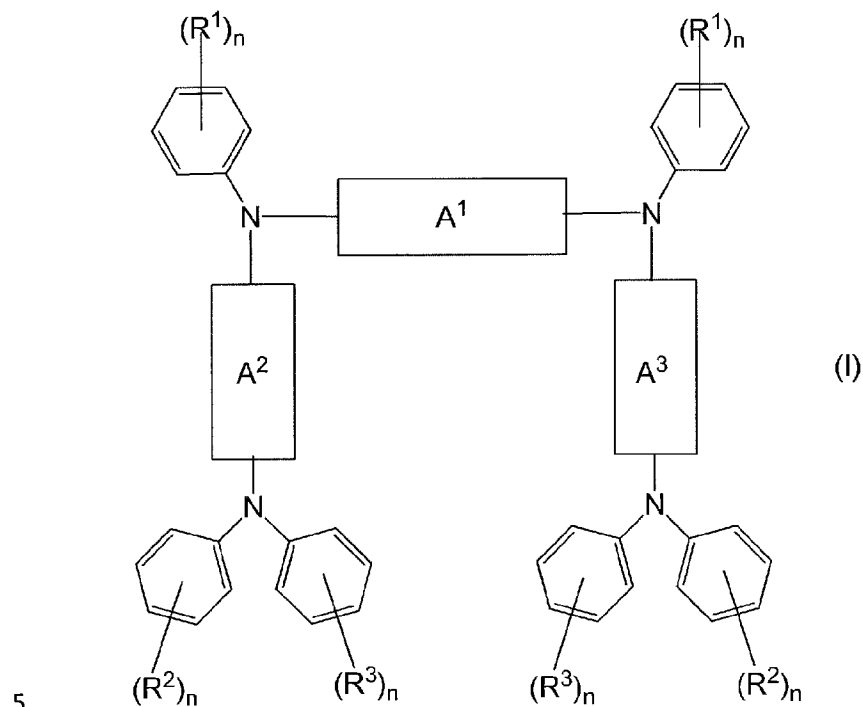
(TBP); MgO; GBA (4-guanidinobutyric acid) and similar derivatives; alkyl acids; hexadecylmalonic acid (HDMA).

As described above, in the context of the organic solar cell, preferably one or more solid organic
5 p-type semiconductors are used – alone or else in combination with one or more further p-type
semiconductors which are organic or inorganic in nature. In the context of the present invention,
a p-type semiconductor is generally understood to mean a material, especially an organic
material, which is capable of conducting holes, that is to say positive charge carriers. More
particularly, it may be an organic material with an extensive π -electron system which can be
10 oxidized stably at least once, for example to form what is called a free-radical cation. For
example, the p-type semiconductor may comprise at least one organic matrix material which
has the properties mentioned. Furthermore, the p-type semiconductor can optionally comprise
one or a plurality of dopants which intensify the p-semiconducting properties. A significant
parameter influencing the selection of the p-type semiconductor is the hole mobility, since this
15 partly determines the hole diffusion length (cf. Kumara, G., Langmuir, 2002, 18, 10493-10495).
A comparison of charge carrier mobilities in different spiro compounds can be found, for
example, in T. Saragi, Adv. Funct. Mater. 2006, 16, 966-974.

Preferably, in the context of the present invention, organic semiconductors are used (i.e. low
20 molecular weight, oligomeric or polymeric semiconductors or mixtures of such semiconductors).
Particular preference is given to p-type semiconductors which can be processed from a liquid
phase. Examples here are p-type semiconductors based on polymers such as polythiophene
and polyarylamines, or on amorphous, reversibly oxidizable, nonpolymeric organic compounds,
such as the spirobifluorenes mentioned at the outset (cf., for example, US 2006/0049397 and
25 the spiro compounds disclosed therein as p-type semiconductors, which are also usable in the
context of the present invention). Preference is given to using low molecular weight organic
semiconductors, such as the low molecular weight p-type semiconducting materials as
disclosed in WO 2012/110924 A1, preferably spiro-MeOTAD, and/or one or more of the p-type
semiconducting materials disclosed in Leijtens et al., ACS Nano, VOL. 6, NO. 2, 1455-1462
30 (2012). Additionally or alternatively, one or more of the p-type semiconducting materials as
disclosed in WO 2010/094636 A1 may be used, the full content of which is herewith included by
reference. In addition, reference may also be made to the remarks regarding the p-
semiconducting materials and dopants from the above description of the prior art.

35 The p-type semiconductor is preferably producible or produced by applying at least one p-
conducting organic material to at least one carrier element, wherein the application is effected
for example by deposition from a liquid phase comprising the at least one p-conducting organic
material. The deposition can in this case once again be effected, in principle, by any desired
deposition process, for example by spin-coating, knife-coating, printing or combinations of the
40 stated and/or other deposition methods.

The organic p-type semiconductor may especially comprise at least one spiro compound and/or especially be selected from: a spiro compound, especially spiro-MeOTAD; a compound with the structural formula:



in which

10 A^1 , A^2 , A^3 are each independently optionally substituted aryl groups or heteroaryl groups,

R^1 , R^2 , R^3 are each independently selected from the group consisting of the substituents $-R$, $-OR$, $-NR_2$, $-A^4-OR$ and $-A^4-NR_2$,

where R is selected from the group consisting of alkyl, aryl and heteroaryl,

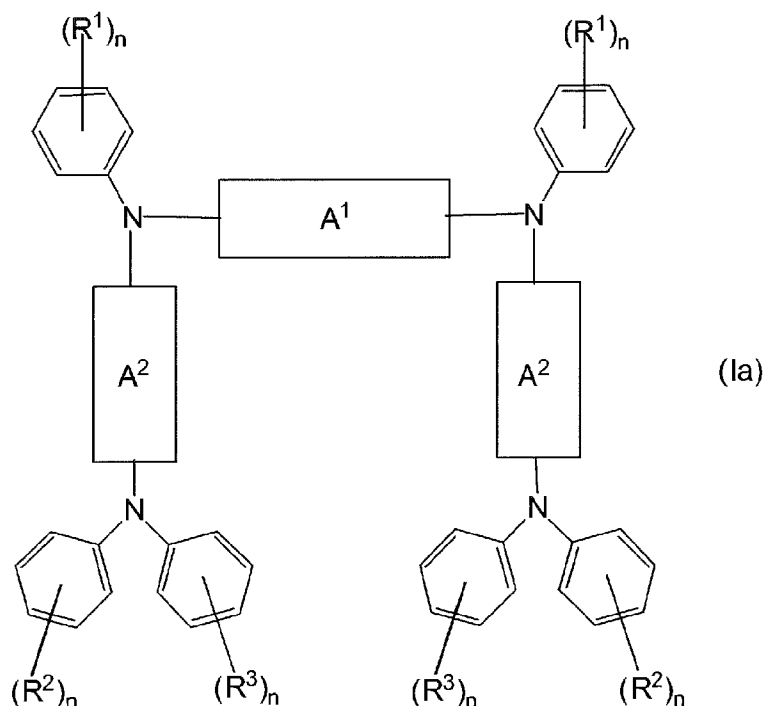
and

where A^4 is an aryl group or heteroaryl group, and

20 where n at each instance in formula I is independently a value of 0, 1, 2 or 3,

with the proviso that the sum of the individual n values is at least 2 and at least two of the R^1 , R^2 and R^3 radicals are $-OR$ and/or $-NR_2$.

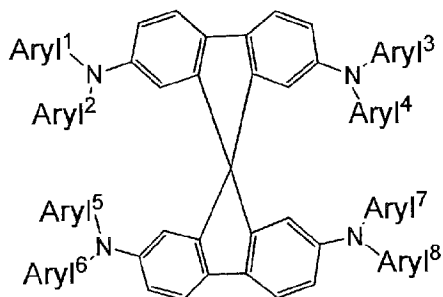
25 Preferably, A^2 and A^3 are the same; accordingly, the compound of the formula (I) preferably has the following structure (Ia)



5 More particularly, as explained above, the p-type semiconductor may thus have at least one low molecular weight organic p-type semiconductor. A low molecular weight material is generally understood to mean a material which is present in monomeric, nonpolymerized or nonoligomerized form. The term "low molecular weight" as used in the present context preferably means that the p-type semiconductor has molecular weights in the range from 100 to
 10 25 000 g/mol. Preferably, the low molecular weight substances have molecular weights of 500 to 2000 g/mol.

In general, in the context of the present invention, p-semiconducting properties are understood to mean the property of materials, especially of organic molecules, to form holes and to
 15 transport these holes and/or to pass them on to adjacent molecules. More particularly, stable oxidation of these molecules should be possible. In addition, the low molecular weight organic p-type semiconductors mentioned may especially have an extensive π -electron system. More particularly, the at least one low molecular weight p-type semiconductor may be processable from a solution. The low molecular weight p-type semiconductor may especially comprise at
 20 least one triphenylamine. It is particularly preferred when the low molecular weight organic p-type semiconductor comprises at least one spiro compound. A spiro compound is understood to mean polycyclic organic compounds whose rings are joined only at one atom, which is also referred to as the spiro atom. More particularly, the spiro atom may be sp^3 -hybridized, such that the constituents of the spiro compound connected to one another via the spiro atom are, for
 25 example, arranged in different planes with respect to one another.

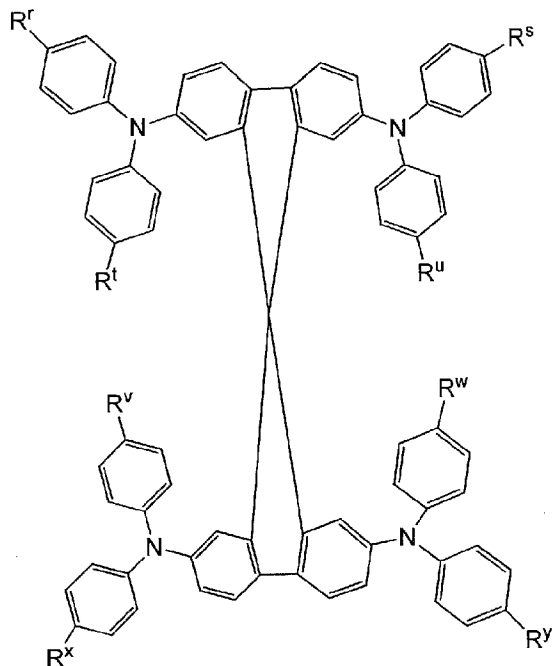
More preferably, the spiro compound has a structure of the following formula:



- 5 where the aryl¹, aryl², aryl³, aryl⁴, aryl⁵, aryl⁶, aryl⁷ and aryl⁸ radicals are each independently selected from substituted aryl radicals and heteroaryl radicals, especially from substituted phenyl radicals, where the aryl radicals and heteroaryl radicals, preferably the phenyl radicals, are each independently substituted, preferably in each case by one or more substituents selected from the group consisting of -O-alkyl, -OH, -F, -Cl, -Br and -I, where alkyl is preferably
- 10 methyl, ethyl, propyl or isopropyl. More preferably, the phenyl radicals are each independently substituted, in each case by one or more substituents selected from the group consisting of -O-Me, -OH, -F, -Cl, -Br and -I.

Further preferably, the spiro compound is a compound of the following formula:

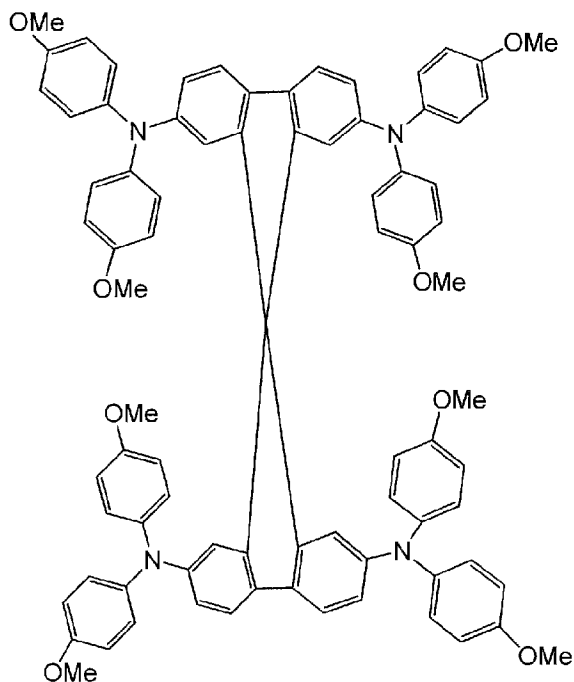
15



- where R^r, R^s, R^t, R^u, R^v, R^w, R^x and R^y are each independently selected from the group consisting of -O-alkyl, -OH, -F, -Cl, -Br and -I, where alkyl is preferably methyl, ethyl, propyl or
- 20 isopropyl. More preferably, R^r, R^s, R^t, R^u, R^v, R^w, R^x and R^y are each independently selected from the group consisting of -O-Me, -OH, -F, -Cl, -Br and -I.

More particularly, the p-type semiconductor may comprise spiro-MeOTAD or consist of spiro-MeOTAD, i.e. a compound of the formula below, commercially available, for example, from Merck KGaA, Darmstadt, Germany:

5



Alternatively or additionally, it is also possible to use other p-semiconducting compounds, especially low molecular weight and/or oligomeric and/or polymeric p-semiconducting compounds.

10

In an alternative embodiment, the low molecular weight organic p-type semiconductor comprises one or more compounds of the above-mentioned general formula I, for which reference may be made, for example, to PCT application number PCT/EP2010/051826, which will be published after the priority date of the present application. The p-type semiconductor may comprise the at least one compound of the above-mentioned general formula I additionally or alternatively to the spiro compound described above.

15

The term "alkyl" or "alkyl group" or "alkyl radical" as used in the context of the present invention is understood to mean substituted or unsubstituted C₁-C₂₀-alkyl radicals in general. Preference is given to C₁- to C₁₀-alkyl radicals, particular preference to C₁- to C₈-alkyl radicals. The alkyl radicals may be either straight-chain or branched. In addition, the alkyl radicals may be substituted by one or more substituents selected from the group consisting of C₁-C₂₀-alkoxy, halogen, preferably F, and C₆-C₃₀-aryl which may in turn be substituted or unsubstituted.

20

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Examples of suitable alkyl groups are methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl and octyl, and also isopropyl, isobutyl, isopentyl, sec-butyl, tert-butyl, neopentyl, 3,3-dimethylbutyl, 2-

ethylhexyl, and also derivatives of the alkyl groups mentioned substituted by C₆-C₃₀-aryl, C₁-C₂₀-alkoxy and/or halogen, especially F, for example CF₃.

The term "aryl" or "aryl group" or "aryl radical" as used in the context of the present invention is understood to mean optionally substituted C₆-C₃₀-aryl radicals which are derived from monocyclic, bicyclic, tricyclic or else multicyclic aromatic rings, where the aromatic rings do not comprise any ring heteroatoms. The aryl radical preferably comprises 5- and/or 6-membered aromatic rings. When the aryls are not monocyclic systems, in the case of the term "aryl" for the second ring, the saturated form (perhydro form) or the partly unsaturated form (for example the dihydro form or tetrahydro form), provided the particular forms are known and stable, is also possible. The term "aryl" in the context of the present invention thus comprises, for example, also bicyclic or tricyclic radicals in which either both or all three radicals are aromatic, and also bicyclic or tricyclic radicals in which only one ring is aromatic, and also tricyclic radicals in which two rings are aromatic. Examples of aryl are: phenyl, naphthyl, indanyl, 1,2-dihydronaphthenyl, 1,4-dihydronaphthenyl, fluorenyl, indenyl, anthracenyl, phenanthrenyl or 1,2,3,4-tetrahydronaphthyl. Particular preference is given to C₆-C₁₀-aryl radicals, for example phenyl or naphthyl, very particular preference to C₆-aryl radicals, for example phenyl. In addition, the term "aryl" also comprises ring systems comprising at least two monocyclic, bicyclic or multicyclic aromatic rings joined to one another via single or double bonds. One example is that of biphenyl groups.

The term "heteroaryl" or "heteroaryl group" or "heteroaryl radical" as used in the context of the present invention is understood to mean optionally substituted 5- or 6-membered aromatic rings and multicyclic rings, for example bicyclic and tricyclic compounds having at least one heteroatom in at least one ring. The heteroaryls in the context of the invention preferably comprise 5 to 30 ring atoms. They may be monocyclic, bicyclic or tricyclic, and some can be derived from the afore-mentioned aryl by replacing at least one carbon atom in the aryl base skeleton with a heteroatom. Preferred heteroatoms are N, O and S. The heteroaryl radicals more preferably have 5 to 13 ring atoms. The base skeleton of the heteroaryl radicals is especially preferably selected from systems such as pyridine and five-membered heteroaromatics such as thiophene, pyrrole, imidazole or furan. These base skeletons may optionally be fused to one or two six-membered aromatic radicals. In addition, the term "heteroaryl" also comprises ring systems comprising at least two monocyclic, bicyclic or multicyclic aromatic rings joined to one another via single or double bonds, where at least one ring comprises a heteroatom. When the heteroaryls are not monocyclic systems, in the case of the term "heteroaryl" for at least one ring, the saturated form (perhydro form) or the partly unsaturated form (for example the dihydro form or tetrahydro form), provided the particular forms are known and stable, is also possible. The term "heteroaryl" in the context of the present invention thus comprises, for example, also bicyclic or tricyclic radicals in which either both or all three radicals are aromatic, and also bicyclic or tricyclic radicals in which only one ring is aromatic, and also tricyclic radicals in which two rings are aromatic, where at least one of the rings, i.e. at least one aromatic or one nonaromatic ring has a heteroatom. Suitable fused heteroaromatics are, for example,

carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl. The base skeleton may be substituted at one, more than one or all substitutable positions, suitable substituents being the same as have already been specified under the definition of C₆-C₃₀-aryl. However, the hetaryl radicals are preferably unsubstituted. Suitable hetaryl radicals are, for example, pyridin-2-yl, pyridin-3-yl, pyridin-4-yl, thiophen-2-yl, thiophen-3-yl, pyrrol-2-yl, pyrrol-3-yl, furan-2-yl, furan-3-yl and imidazol-2-yl and the corresponding benzofused radicals, especially carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl.

In the context of the invention the term "optionally substituted" refers to radicals in which at least one hydrogen radical of an alkyl group, aryl group or heteroaryl group has been replaced by a substituent. With regard to the type of this substituent, preference is given to alkyl radicals, for example methyl, ethyl, propyl, butyl, pentyl, hexyl, heptyl and octyl, and also isopropyl, isobutyl, isopentyl, sec-butyl, tert-butyl, neopentyl, 3,3-dimethylbutyl and 2-ethylhexyl, aryl radicals, for example C₆-C₁₀-aryl radicals, especially phenyl or naphthyl, most preferably C₆-aryl radicals, for example phenyl, and hetaryl radicals, for example pyridin-2-yl, pyridin-3-yl, pyridin-4-yl, thiophen-2-yl, thiophen-3-yl, pyrrol-2-yl, pyrrol-3-yl, furan-2-yl, furan-3-yl and imidazol-2-yl, and also the corresponding benzofused radicals, especially carbazolyl, benzimidazolyl, benzofuryl, dibenzofuryl or dibenzothiophenyl. Further examples include the following substituents: alkenyl, alkynyl, halogen, hydroxyl.

The degree of substitution here may vary from monosubstitution up to the maximum number of possible substituents.

Preferred compounds of the formula I for use in accordance with the invention are notable in that at least two of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. The at least two radicals here may be only -OR radicals, only -NR₂ radicals, or at least one -OR and at least one -NR₂ radical.

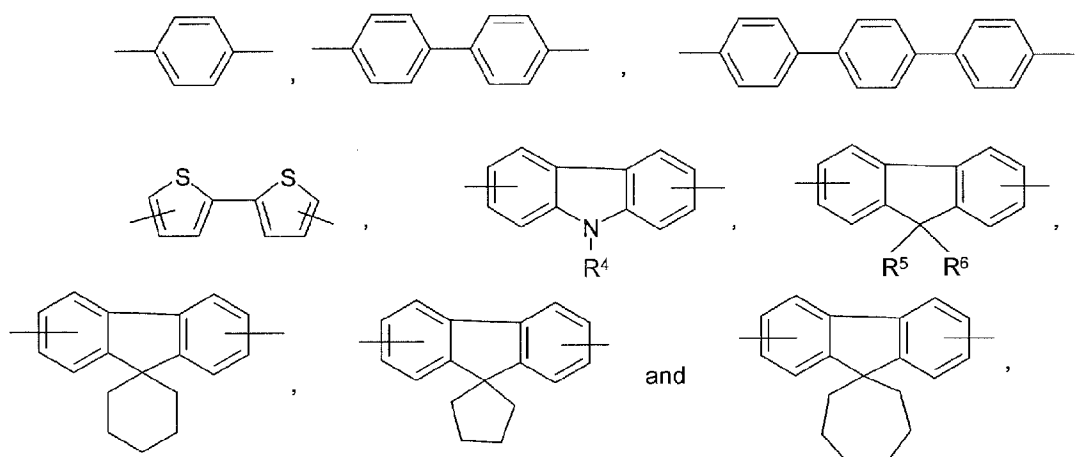
Particularly preferred compounds of the formula I for use in accordance with the invention are notable in that at least four of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. The at least four radicals here may be only -OR radicals, only -NR₂ radicals or a mixture of -OR and -NR₂ radicals.

Very particularly preferred compounds of the formula I for use in accordance with the invention are notable in that all of the R¹, R² and R³ radicals are para-OR and/or -NR₂ substituents. They may be only -OR radicals, only -NR₂ radicals or a mixture of -OR and -NR₂ radicals.

In all cases, the two R in the -NR₂ radicals may be different from one another, but they are preferably the same.

Preferably, A¹, A² and A³ are each independently selected from the group consisting of

76



in which

5

m is an integer from 1 to 18,

R⁴ is alkyl, aryl or heteroaryl, where R⁴ is preferably an aryl radical, more preferably a phenyl radical,

10

R⁵, R⁶ are each independently H, alkyl, aryl or heteroaryl,

where the aromatic and heteroaromatic rings of the structures shown may optionally have further substitution. The degree of substitution of the aromatic and heteroaromatic rings here may vary from monosubstitution up to the maximum number of possible substituents.

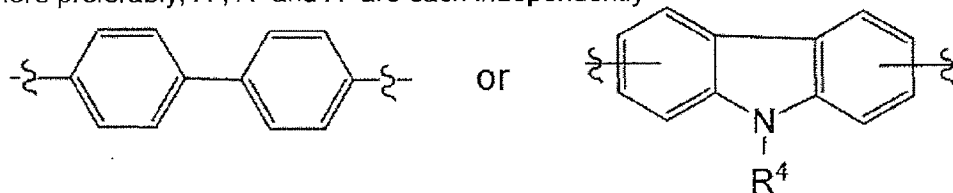
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Preferred substituents in the case of further substitution of the aromatic and heteroaromatic rings include the substituents already mentioned above for the one, two or three optionally substituted aromatic or heteroaromatic groups.

20

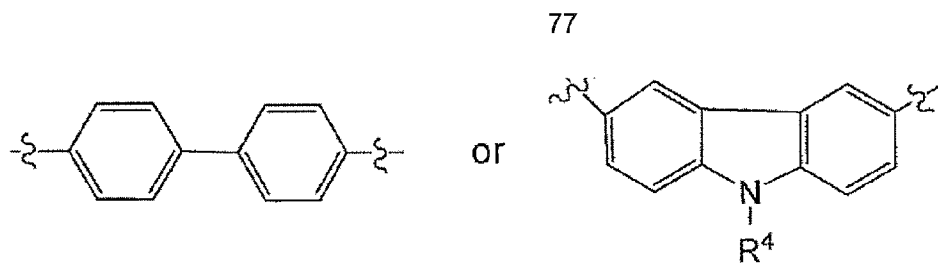
Preferably, the aromatic and heteroaromatic rings of the structures shown do not have further substitution.

More preferably, A¹, A² and A³ are each independently



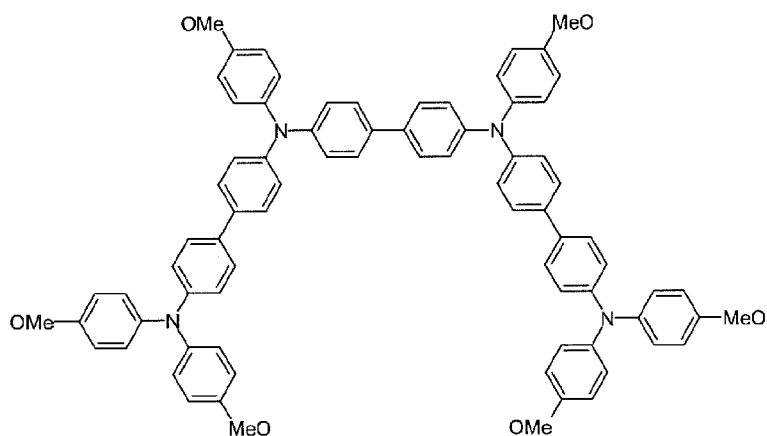
25

more preferably



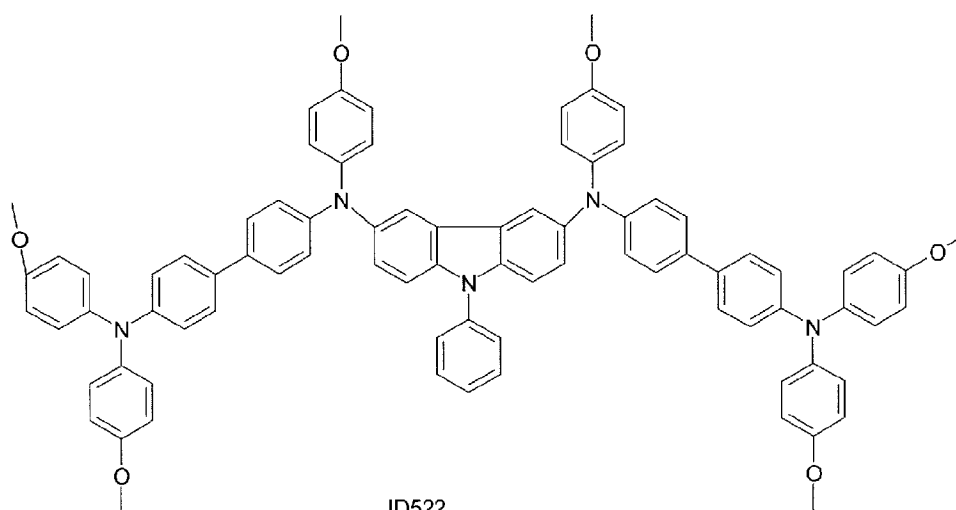
More preferably, the at least one compound of the formula (I) has one of the following structures:

5



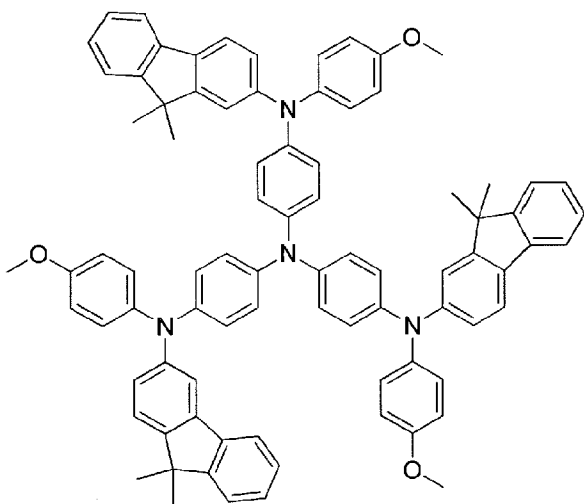
ID367

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ID522

In an alternative embodiment, the organic p-type semiconductor comprises a compound of the type ID322 having the following structure:



The compounds for use in accordance with the invention can be prepared by customary methods of organic synthesis known to those skilled in the art. References to relevant (patent) literature can additionally be found in the synthesis examples adduced below.

Second electrode

a) General remarks

10

The second electrode may be a bottom electrode facing the substrate or else a top electrode facing away from the substrate. As outlined above, the second electrode may be fully or partially transparent or, else, may be intransparent. As used herein, the term partially transparent refers to the fact that the second electrode may comprise transparent regions and intransparent regions.

15

In case the second electrode is fully or partially transparent, the second electrode may comprise at least one transparent conductive electrode material, which may be selected from the group consisting of: an inorganic transparent conductive material; an organic transparent conductive material. As an example of an inorganic conductive transparent material, a metal oxide may be used, such as ITO and/or FTO. As an example of an organic transparent conductive material, one or more electrically conductive polymer materials may be used. As used herein, the term "transparent" refers to the actual layer or layer setup of the second electrode. Thus, the transparency may be generated by using thin layers, such as layers having a thickness of less than 100 nm, more preferably less than 50 nm.

20

One or more materials of the following group of materials may be used: at least one metallic material, preferably a metallic material selected from the group consisting of aluminum, silver, platinum, gold; at least one nonmetallic inorganic material, preferably LiF; at least one organic conductive material, preferably at least one electrically conductive polymer and, more preferably, at least one transparent electrically conductive polymer.

25

30

The second electrode may comprise one or more metals in a pure form and/or may comprise one or more metal alloys. The second electrode may further comprise a single layer and/or may comprise a layer setup of two or more layers, wherein, preferably at least one layer is a metal layer comprising one or more metals or metal alloys. As an example, the second electrode may comprise at least one metal selected from the group listed in the preceding paragraph in a pure form and/or as a component of an alloy. As an example, the second electrode may comprise at least one alloy selected from the group consisting of: a molybdenum alloy; a niobium alloy; a neodymium alloy; an aluminum alloy. Most preferably, the second electrode may comprise at least one alloy selected from the group consisting of: MoNb; AlNd; MoNb. As an example, a layer setup comprising two or more layers of two or more of the named alloys may be used, such as a layer setup comprising the following layers: MoNb / AlNd / MoNb. As an example, the following layer thicknesses may be used: MoNb 30nm / AlNd 100 nm / MoNb 30nm. Additionally or alternatively, however, other setups and/or other layer thicknesses may be used.

The second electrode may comprise at least one metal electrode, wherein one or more metals in pure form or as a mixture/alloy, such as especially aluminum or silver may be used.

Additionally or alternatively, nonmetallic materials may be used, such as inorganic materials and/or organic materials, both alone and in combination with metal electrodes. As an example, the use of inorganic/organic mixed electrodes or multilayer electrodes is possible, for example the use of LiF/Al electrodes. Additionally or alternatively, conductive polymers may be used. Thus, the second electrode of the at least one transversal optical sensor and/or the second electrode of the at least one longitudinal optical sensor preferably may comprise one or more conductive polymers.

As an example, one or more electrically conductive polymers may be used, selected from the group consisting of: polyaniline (PANI) and/or its chemical relatives; a polythiophene and/or its chemical relatives, such as poly(3-hexylthiophene) (P3HT) and/or PEDOT:PSS (poly(3,4-ethylenedioxythiophene) poly(styrenesulfonate)). Additionally or alternatively, one or more of the conductive polymers as disclosed in EP2507286 A2, EP2205657 A1 or EP2220141 A1

In addition or alternatively, inorganic conductive materials may be used, such as inorganic conductive carbon materials, such as carbon materials selected from the group consisting of: graphite, graphene, carbon nano-tubes, carbon nano-wires.

In addition, it is also possible to use electrode designs in which the quantum efficiency of the components is increased by virtue of the photons being forced, by means of appropriate reflections, to pass through the absorbing layers at least twice. Such layer structures are also referred to as "concentrators" and are likewise described, for example, in WO 02/101838 (especially pages 23-24).

The second electrode may be identical for the at least one transversal optical sensor and the at least one longitudinal optical sensor. Still, different setups of the second electrode for the transversal optical sensor and the longitudinal optical sensor may be used.

5 b) Second electrode of the transversal sensor device

Preferably, the second electrode for the at least one transversal sensor device is at least partially transparent. As an example, the second electrode of the transversal sensor device may comprise at least one transparent electrode layer which covers a sensor region, preferably a
10 sensor area, of the transversal optical sensor. As outlined above, the at least one transparent electrode layer preferably may comprise at least one layer of an electrically conductive polymer, preferably a transparent electrically conductive polymer.

Additionally, the second electrode of the transversal sensor device may comprise two or more
15 partial electrodes, which, preferably, may be made of one or more metals, such as one or more of the metals and/or metal alloys listed above. As an example, the two or more partial electrodes may form a frame which surrounds the sensor region, preferably the sensor area, of the transversal optical sensor. The frame may have a polygonal shape, such as a rectangular or, preferably, a square shape. Preferably, on each side of the polygon, preferably the rectangle
20 or square, one partial electrodes is provided, such as a partial electrode being formed as a bar fully or partially extending along the side.

The at least one electrically conductive polymer may have an electrical conductivity of at least one order of magnitude below electrical conductivity of the material of the partial electrodes,
25 preferably of at least two orders of magnitude below. The at least one electrically conductive polymer may electrically interconnect the partial electrodes. Thus, as outlined above, the partial electrodes may form a frame surrounding a sensor region, preferably a sensor area, of the transversal optical sensor. The at least one layer of the electrically conductive polymer may form a transparent, electrically conductive layer which fully or partially covers the sensor region
30 and which electrically contacts the partial electrodes. As an example, the partial electrodes may comprise metal strips or metal bars along the sides of a rectangle, wherein an inner region of the rectangle forms the sensor region, wherein the at least one layer of the electrically conductive polymer forms one or more transparent electrode layers fully or partially covering the inner region of the rectangle and electrically contacting the metal strips or bars.

35 In case two or more partial electrodes are used which, preferably, are electrically interconnected by at least one layer of an electrically conductive polymer, each of the partial electrodes may be contacted individually, such as by one or more electrical leads or contact pads. Thus, by electrically contacting the partial electrodes, an electrical current through each of the partial
40 electrodes may be measured individually, such as by using an individual current measurement device and/or by using a sequential measurement scheme for individually detecting the currents through the partial electrodes. The detector, for the purpose of measuring the currents through

the partial electrodes, may provide an appropriate measurement setup comprising one or more current measurement devices.

c) Second electrode of the longitudinal sensor device

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Generally, with regard to the at least one second electrode of the at least one longitudinal sensor device, the above-mentioned details regarding the transversal sensor device may apply mutatis mutandis. Again, the second electrode of the at least one longitudinal sensor device preferably is transparent. In case a plurality of longitudinal sensor devices is provided, such as
10 in a stack, preferably all second electrodes of the longitudinal sensor devices are transparent but the second electrode of the last longitudinal sensor device facing away from the object. The second electrode of the last longitudinal sensor device may be transparent or intransparent.

With regard to materials which may be used for the second electrode of the longitudinal sensor
15 device, reference may be made to the above-mentioned materials, which may be selected from metallic materials, nonmetallic inorganic materials and electrically conductive organic materials.

Again, the second electrode of the longitudinal optical sensor or, in case a plurality of longitudinal optical sensors is provided, of at least one of the longitudinal optical sensors may
20 optionally be subdivided into partial electrodes which may be contacted individually. However, since for the purpose of the at least one longitudinal optical sensor generally only one individual longitudinal sensor signal is required per longitudinal optical sensor, the second electrode of the at least one longitudinal optical sensor may as well be designed to provide a single sensor signal and, thus, may provide a single electrode contact only.

25

The second electrode of the longitudinal optical sensor, again, preferably may comprise one or more layers of an electrically conductive polymer, such as one or more of the polymers mentioned above. The at least one layer of the electrically conductive polymer, which preferably is transparent, may fully or partially cover the sensor region, preferably a sensor area, of the
30 longitudinal optical sensor. In addition, one or more contact pads may be provided which electrically contact the at least one electrically conductive polymer layer. This at least one contact pad for the second electrode of the longitudinal optical sensor preferably may be made of at least one metal, such as at least one of the methods mentioned above, and/or may fully or partially be made of at least one inorganic conductive material, such as one or more transparent
35 conductive oxides, such as one or more of the conductive oxides mentioned above with regard to the first electrode.

Encapsulation

40 The at least one transversal optical sensor and/or the at least one longitudinal optical sensor may further be encapsulated and/or packaged, in order to provide protection against

environmental influences, such as oxygen and/or humidity. Thereby, an increased long-term stability may be provided.

5 Therein, each of the optical sensors may be encapsulated individually. Thus, an individual encapsulation may be provided for each of the optical sensors, such as an encapsulation for the transversal optical sensor or each of the transversal optical sensors, and an individual encapsulation for the longitudinal optical sensor or for each of the longitudinal optical sensors. Additionally or alternatively, a plurality of optical sensors may be encapsulated as a group. Thus, an encapsulation may be provided which encapsulates more than one optical sensor,
10 such as a plurality of transversal optical sensors, a plurality of longitudinal optical sensors, or at least one transversal optical sensor and at least one longitudinal optical sensor.

For the purpose of encapsulation, various techniques may be used. Thus, the detector may comprise an airtight housing which protects the optical sensors. Additionally or alternatively,
15 specifically in case organic photodetectors and, more preferably, DSCs or sDSCs are used, an encapsulation by one or more lids interacting with a substrate of the optical sensors may be used. Thus, a lid made of a metal, a ceramic material or a glass material may be glued to a substrate of the optical sensors, wherein a layer setup is located in inner space of the lid. Two or more contact leads for contacting the at least one first electrode and the at least one second
20 electrode may be provided which may be contacted from the outside of the lid.

Various other encapsulation techniques may be used alternatively or in addition. Thus, encapsulation by one or more encapsulation layers may be provided. The at least one encapsulation layer may be deposited on top of a layer setup of the devices. Thus, one or more
25 organic and/or inorganic encapsulation materials such as one or more barrier materials may be used.

Synthesis examples:

30 Syntheses of various compounds which can be used in dye solar cells in the context of the present invention, in particular as p-type semiconductors, are listed by way of example in WO 2012/110924 A1, the content of which is herewith included by reference.

35 Overall, in the context of the present invention, the following embodiments are regarded as particularly preferred:

Embodiment 1: A detector for determining a position of at least one object, comprising:

- 40 - at least one transversal optical sensor, the transversal optical sensor being adapted to determine a transversal position of at least one light beam traveling from the object to the detector, the transversal position being a position in at least one dimension

perpendicular to an optical axis of the detector, the transversal optical sensor being adapted to generate at least one transversal sensor signal;

- at least one longitudinal optical sensor, wherein the longitudinal optical sensor has at least one sensor region, wherein the longitudinal optical sensor is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region;
- at least one evaluation device, wherein the evaluation device is designed to generate at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

Embodiment 2: The detector according to the preceding embodiment, wherein the transversal optical sensor is a photo detector having at least one first electrode, at least one second electrode and at least one photovoltaic material, wherein the photovoltaic material is embedded in between the first electrode and the second electrode, wherein the photovoltaic material is adapted to generate electric charges in response to an illumination of the photovoltaic material with light, wherein the second electrode is a split electrode having at least two partial electrodes, wherein the transversal optical sensor has a sensor region, wherein the at least one transversal sensor signal indicates a position of the light beam in the sensor region, preferably a sensor area.

Embodiment 3: The detector according to the preceding embodiment, wherein electrical currents through the partial electrodes are dependent on a position of the light beam in the sensor region.

Embodiment 4: The detector according to the preceding embodiment, wherein the transversal optical sensor is adapted to generate the transversal sensor signal in accordance with the electrical currents through the partial electrodes.

Embodiment 5: The detector according to any of the two preceding embodiments, wherein the detector, preferably the transversal optical sensor and/or the evaluation device, is adapted to derive the information on the transversal position of the object from at least one ratio of the currents through the partial electrodes.

Embodiment 6: The detector according to any of the four preceding embodiments, wherein at least four partial electrodes are provided.

Embodiment 7: The detector according to any of the five preceding embodiments, wherein the photovoltaic material comprises at least one organic photovoltaic material and wherein the transversal optical sensor is an organic photo detector.

Embodiment 8: The detector according to any of the six preceding embodiments, wherein the organic photo detector is a dye-sensitized solar cell.

5 Embodiment 9: The detector according to the preceding embodiment, wherein the dye-sensitized solar cell is a solid dye-sensitized solar cell, comprising a layer setup embedded in between the first electrode and the second electrode, the layer setup comprising at least one n-semiconducting metal oxide, at least one dye, and at least one solid p-semiconducting organic material.

10

Embodiment 10: The detector according to any of the eight preceding embodiments, wherein the first electrode at least partially is made of at least one transparent conductive oxide, wherein the second electrode at least partially is made of an electrically conductive polymer, preferably a transparent electrically conductive polymer.

15

Embodiment 11: The detector according to the preceding embodiment, wherein the conductive polymer is selected from the group consisting of: a poly-3,4-ethylenedioxythiophene (PEDOT), preferably PEDOT being electrically doped with at least one counter ion, more preferably PEDOT doped with sodium polystyrene sulfonate (PEDOT:PSS); a polyaniline (PANI); a

20

polythiophene.

Embodiment 12: The detector according to any of the two preceding embodiments, wherein the conductive polymer provides an electric resistivity of 0.1 – 20 k Ω between the partial electrodes, preferably an electric resistivity of 0.5 – 5.0 k Ω and, more preferably, an electric resistivity of 1.0

25

– 3.0 k Ω .

Embodiment 13: The detector according to any of the preceding embodiments, wherein at least one of the transversal optical sensor and the longitudinal optical sensor is a transparent optical sensor.

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Embodiment 14: The detector according to the preceding embodiment, wherein the light beam passes through the transparent optical sensor before impinging on the other one of the transversal optical sensor and the longitudinal optical sensor.

35 Embodiment 15: The detector according to any one of the preceding embodiments, wherein the detector further comprises at least one imaging device.

Embodiment 16: The detector according to the preceding claim, wherein the detector comprises a stack of optical sensors, the optical sensors comprising the at least one transversal optical

40

sensor and the at least one longitudinal optical sensor, the stack further comprising the imaging device.

Embodiment 17: The detector according to the preceding claim, wherein the imaging device is located in a position of the stack furthest away from the object.

5 Embodiment 18: The detector according to any of the three preceding embodiments, wherein the light beam passes through the at least one longitudinal optical sensor before illuminating the imaging device.

Embodiment 19: The detector according to any of the four preceding embodiments, wherein the imaging device comprises a camera.

10

Embodiment 20: The detector according to any of the five preceding embodiments, wherein the imaging device comprises at least one of: an inorganic camera; a monochrome camera; a multichrome camera; a full-color camera; a pixelated inorganic chip; a pixelated organic camera; a CCD chip, preferably a multi-color CCD chip or a full-color CCD chip; a CMOS chip; an IR camera; an RGB camera.

15

Embodiment 21: The detector according to any of the preceding embodiments, wherein the transversal optical sensor and the longitudinal optical sensor at least partially are identical.

20 Embodiment 22: The detector according to any of the preceding embodiments, wherein the transversal optical sensor and the longitudinal optical sensor at least partially are independent optical sensors.

Embodiment 23: The detector according to any of the preceding embodiments, wherein the transversal optical sensor and the longitudinal optical sensor are stacked along the optical axis such that a light beam travelling along the optical axis both impinges on the transversal optical sensor and on the longitudinal optical sensor.

25

Embodiment 24: The detector according to the preceding embodiment, wherein the light beam subsequently passes through the transversal optical sensor and the longitudinal optical sensor or vice versa.

30

Embodiment 25: The detector according to any of the preceding embodiments, wherein the detector furthermore has at least one modulation device for modulating the illumination.

35

Embodiment 26: The detector according to the preceding embodiment, wherein the detector is designed to detect at least two longitudinal sensor signals in the case of different modulations, in particular at least two sensor signals at respectively different modulation frequencies, wherein the evaluation device is designed to generate the at least one item of information on the longitudinal position of the object by evaluating the at least two longitudinal sensor signals.

40

Embodiment 27: The detector according to any of the preceding embodiments, wherein the longitudinal optical sensor is furthermore designed in such a way that the longitudinal sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination.

5

Embodiment 28: The detector according to any of the preceding embodiments, wherein the sensor region of the longitudinal optical sensor is exactly one continuous sensor region, wherein the longitudinal sensor signal is a uniform sensor signal for the entire sensor region.

10 Embodiment 29: The detector according to any of the preceding embodiments, wherein the sensor region of the transversal optical sensor and/or the sensor region of the longitudinal optical sensor is or comprises a sensor area, the sensor area being formed by a surface of the respective device, wherein the surface faces towards the object or faces away from the object.

15 Embodiment 30: The detector according to any of the preceding embodiments, wherein the longitudinal sensor signal is selected from the group consisting of a current and a voltage.

Embodiment 31: The detector according to any of the preceding embodiments, wherein the transversal sensor signal is selected from the group consisting of a current and a voltage or any
20 signal derived thereof.

Embodiment 32: The detector according to any of the preceding embodiments, wherein the longitudinal optical sensor comprises at least one semiconductor detector, in particular an organic semiconductor detector comprising at least one organic material, preferably an organic
25 solar cell and particularly preferably a dye solar cell or dye-sensitized solar cell, in particular a solid dye solar cell or a solid dye-sensitized solar cell.

Embodiment 33: The detector according to the preceding embodiment, wherein the longitudinal optical sensor comprises at least one first electrode, at least one n-semiconducting metal oxide,
30 at least one dye, at least one p-semiconducting organic material, preferably a solid p-semiconducting organic material, and at least one second electrode.

Embodiment 34: The detector according to the preceding embodiment, wherein both the first electrode and the second electrode are transparent.

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Embodiment 35: The detector according to any of the preceding embodiments, wherein the evaluation device is designed to generate the at least one item of information on the longitudinal position of the object from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object with respect to the detector, preferably taking
40 account of a known power of the illumination and optionally taking account of a modulation frequency with which the illumination is modulated.

Embodiment 36: The detector according to any of the preceding embodiments, furthermore comprising at least one transfer device, wherein the transfer device is designed to feed light emerging from the object to the transversal optical sensor and the longitudinal optical sensor.

- 5 Embodiment 37: The detector according to any of the preceding embodiments, furthermore comprising at least one illumination source.

Embodiment 38: The detector according to the preceding embodiment, wherein the illumination source is selected from: an illumination source, which is at least partly connected to the object
10 and/or is at least partly identical to the object; an illumination source which is designed to at least partly illuminate the object with a primary radiation, wherein the light beam preferably is generated by a reflection of the primary radiation on the object and/or by light emission by the object itself, stimulated by the primary radiation.

- 15 Embodiment 39: The detector according to any of the preceding embodiments, wherein the detector has a plurality of longitudinal optical sensors, wherein the longitudinal optical sensors are stacked.

Embodiment 40: The detector according to the preceding embodiment, wherein the longitudinal
20 optical sensors are stacked along the optical axis.

Embodiment 41: The detector according to any of the two preceding embodiments, wherein the longitudinal optical sensors form a longitudinal optical sensor stack, wherein the sensor regions of the longitudinal optical sensors are oriented perpendicular to the optical axis.

- 25 Embodiment 42: The detector according to any of the three preceding embodiments, wherein the transversal optical sensor is located on a side of the stacked longitudinal optical sensors facing the object.

30 Embodiment 43: The detector according to any of the four preceding embodiments, wherein the longitudinal optical sensors are arranged such that a light beam from the object illuminates all longitudinal optical sensors, preferably sequentially, wherein at least one longitudinal sensor signal is generated by each longitudinal optical sensor.

35 Embodiment 44: The detector according to any of the five preceding embodiments, wherein a last longitudinal optical sensor is arranged in a manner that the light beam first illuminates all other longitudinal optical sensors apart from the last longitudinal optical sensor, until the light beam finally illuminates the last longitudinal optical sensor.

- 40 Embodiment 45: The detector according to the preceding embodiment, wherein the last longitudinal optical sensor is intransparent with respect to the light beam.

Embodiment 46: The detector according to any of the seven preceding embodiments, wherein at least two of the longitudinal optical sensors have a differing spectral sensitivities.

5 Embodiment 47: The detector according to the preceding embodiment, wherein the differing spectral sensitivities are arranged over a spectral range allowing each of the at least two of the longitudinal optical sensors to be sensitive to a specific color.

10 Embodiment 48: The detector according to the preceding embodiment, wherein the longitudinal optical sensors comprise at least one first longitudinal optical sensor absorbing light in a first spectral range, wherein the longitudinal optical sensors further comprise at least one second longitudinal optical sensor absorbing light in a second spectral range being different from the first spectral range, wherein the longitudinal optical sensors further comprise at least one third longitudinal optical sensor absorbing light in a third spectral range which includes both the first spectral range and the second spectral range.

15 Embodiment 49: The detector according to any of the six preceding embodiments, wherein the evaluation device is adapted to normalize the longitudinal sensor signals and to generate the information on the longitudinal position of the object independent from an intensity of the light beam.

20 Embodiment 50: The detector according to any of the seven preceding embodiments, wherein the evaluation device is adapted to recognize whether the light beam widens or narrows, by comparing the longitudinal sensor signals of different longitudinal sensors.

25 Embodiment 51: The detector according to any of the preceding embodiments, wherein the stack of at least two optical sensors is partially or fully be immersed in an oil, in a liquid and/or in a solid material.

30 Embodiment 52: The detector according to the preceding embodiment, wherein the oil, the liquid, and/or the solid material are transparent at least over a part of the ultraviolet, visible, and/or infrared spectral range.

35 Embodiment 53: The detector according to any of the two preceding embodiments, wherein the solid material is generable by using at least one curable substance and treating the curable substance with a treatment, by which treatment the curable substance is cured into the solid material.

40 Embodiment 54: The detector according to any of the three preceding embodiments, wherein a region between the at least two optical sensors is partially or fully filled with a substance.

Embodiment 55: The detector according to the preceding embodiment, wherein the substance exhibits a refractive index differing from that of the optical sensors adjoining to the substance on one or both sides of the region.

- 5 Embodiment 56: The detector according to any of the preceding embodiments, wherein the at least one transversal optical sensor and/or the at least one longitudinal optical sensor uses at least two transparent substrates.

- 10 Embodiment 57: The detector according to the preceding embodiment, wherein the substrates exhibit identical properties.

- 15 Embodiment 58: The detector according to any of the two preceding embodiments, wherein the substrates differ from each other with regard to a geometrical quantity and/or to a material quantity related to the substrates.

Embodiment 59: The detector according to the preceding embodiment, wherein the substrates differ from each other by the thickness.

- 20 Embodiment 60: The detector according to any of the two preceding embodiments, wherein the substrates differ from each other by shape.

- 25 Embodiment 61: The detector according to the preceding embodiment, wherein the shape is selected from the group comprising a planar, a planar-convex, a planar-concave, a biconvex, a biconcave or any other form employed for optical purposes.

Embodiment 62: The detector according to any of the six preceding embodiments, wherein the substrates are rigid or flexible.

- 30 Embodiment 63: The detector according to any of the seven preceding embodiments, wherein the substrates are covered or coated.

Embodiment 63: The detector according to any of the eight preceding embodiments, wherein the substrate is shaped in a manner that it exhibits a mirror effect.

- 35 Embodiment 64: The detector according to the preceding embodiment, wherein the substrate is shaped in a manner that it exhibits the effect of a dichroic mirror.

- 40 Embodiment 65: The detector according to any of the preceding embodiments, wherein the evaluation device is adapted to generate the at least one item of information on the longitudinal position of the object by determining a diameter of the light beam from the at least one longitudinal sensor signal.

Embodiment 66: The detector according to the preceding embodiment, wherein the evaluation device is adapted to compare the diameter of the light beam with known beam properties of the light beam in order to determine the at least one item of information on the longitudinal position of the object, preferably from a known dependency of a beam diameter of the light beam on at least one propagation coordinate in a direction of propagation of the light beam and/or from a known Gaussian profile of the light beam.

Embodiment 67: The detector according to any of the preceding embodiments, wherein the detector comprises an optically sensitive element.

Embodiment 68: The detector according to the preceding embodiment, wherein the optically sensitive element is located between the transfer device and the optical sensor.

Embodiment 69: The detector according to any of the two preceding embodiments, wherein the optically sensitive element comprises a wavelength-sensitive element, a phase-sensitive element, and/or a polarization-sensitive element.

Embodiment 70: The detector according to the preceding embodiment, wherein the wavelength-sensitive element comprises one or more of a prism, a grating, a dichroitic mirror, a color wheel, or a color drum.

Embodiment 71: The detector according to the preceding embodiment, wherein the color wheel comprises a sequential color recapture wheel.

Embodiment 72: The detector according to any of the two preceding embodiments, wherein the color wheel or the color drum comprises two or more of at least one red, green, blue, white, cyan, yellow, or magenta segment.

Embodiment 73: The detector according to embodiment 54, wherein the polarization-sensitive element comprises a filter wheel using an elliptically polarizing filter.

Embodiment 74: The detector according to the preceding embodiment, wherein the polarization-sensitive element comprises a filter wheel using a circular polarizing filter.

Embodiment 75: The detector according to any of the two preceding embodiments, wherein the optically sensitive element comprises at least two wheels, at least one first wheel and at least one second wheel, wherein the first wheel constitutes a color wheel and wherein the second wheel constitutes an elliptically polarizing filter.

Embodiment 76: An arrangement comprising at least two detectors according to any of the preceding embodiments.

Embodiment 77: An arrangement according to the preceding embodiment, wherein the at least two detectors have identical optical properties.

Embodiment 78: The arrangement according to any of the two preceding embodiments, wherein
5 the arrangement further comprises at least one illumination source.

Embodiment 79: A human-machine interface for exchanging at least one item of information between a user and a machine, in particular for inputting control commands, wherein the human-machine interface comprises at least one detector according to any of the preceding
10 embodiments relating to a detector, wherein the human-machine interface is designed to generate at least one item of geometrical information of the user by means of the detector wherein the human-machine interface is designed to assign to the geometrical information at least one item of information, in particular at least one control command.

15 Embodiment 80: The human-machine interface according to the preceding embodiment, wherein the at least one item of geometrical information of the user is selected from the group consisting of: a position of a body of the user; a position of at least one body part of the user; an orientation of a body of the user; an orientation of at least one body part of the user.

20 Embodiment 81: The human-machine interface according to any of the two preceding embodiments, wherein the human-machine interface further comprises at least one beacon device connectable to the user, wherein the human-machine interface is adapted such that the detector may generate an information on the position of the at least one beacon device.

25 Embodiment 82: The human-machine interface according to the preceding embodiment, wherein the beacon device is one of a beacon device attachable to a body or a body part of the user and a beacon device which may be held by the user.

Embodiment 83: The human-machine interface according to the preceding embodiment,
30 wherein the beacon device comprises at least one illumination source adapted to generate at least one light beam to be transmitted to the detector.

Embodiment 84: The human-machine interface according to any of the two preceding
embodiments, wherein the beacon device comprises at least one reflector adapted to reflect
35 light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the detector.

Embodiment 85: The human-machine interface according to any of the three preceding
embodiments, wherein the beacon device comprises at least one of: a garment to be worn by
40 the user, preferably a garment selected from the group consisting of a glove, a jacket, a hat, shoes, trousers and a suit; a stick that may be held by hand; a bat; a club; a racket; a cane; a toy, such as a toy gun.

Embodiment 86: An entertainment device for carrying out at least one entertainment function, in particular a game, wherein the entertainment device comprises at least one human-machine interface according to any of the preceding embodiments referring to a human-machine
5 interface, wherein the entertainment device is designed to enable at least one item of information to be input by a player by means of the human-machine interface, wherein the entertainment device is designed to vary the entertainment function in accordance with the information.

10 Embodiment 87: A tracking system for tracking the position of at least one movable object, the tracking system comprising at least one detector according to any of the preceding embodiments referring to a detector, the tracking system further comprising at least one track controller, wherein the track controller is adapted to track a series of positions of the object,
15 each position comprising at least one item of information on a transversal position of the object at a specific point in time and at least one item of information on a longitudinal position of the object at a specific point in time.

Embodiment 88: The tracking system according to the preceding embodiment, wherein the tracking system further comprises at least one beacon device connectable to the object,
20 wherein the tracking system is adapted such that the detector may generate an information on the position of the object of the at least one beacon device.

Embodiment 89: The tracking system according to the preceding embodiment, wherein the beacon device comprises at least one illumination source adapted to generate at least one light
25 beam to be transmitted to the detector.

Embodiment 90: The tracking system according to any of the two preceding embodiments, wherein the beacon device comprises at least one reflector adapted to reflect light generated by an illumination source, thereby generating a reflected light beam to be transmitted to the
30 detector.

Embodiment 91: The tracking system according to any of the preceding embodiments referring to a tracking system, wherein the track controller is adapted to initiate at least one action in accordance with an actual position of the object.
35

Embodiment 92: The tracking system according to the preceding embodiment, wherein the action is selected from the group consisting of: a prediction of a future position of the object; pointing at least one device towards the object; pointing at least one device towards the detector; illuminating the object; illuminating the detector.
40

Embodiment 93: A camera for imaging at least one object, the camera comprising at least one detector according to any one of the preceding embodiments referring to a detector.

Embodiment 94: A method for determining a position of at least one object, in particular using a detector according to any of the preceding embodiments relating to a detector,

- 5 - wherein at least one transversal optical sensor of a detector is used, wherein the transversal optical sensor determines a transversal position of at least one light beam traveling from the object to the detector, the transversal position being a position in at least one dimension perpendicular to an optical axis of the detector, wherein the transversal optical sensor generates at least one transversal sensor signal;
- 10 - wherein at least one longitudinal optical sensor of the detector is used, wherein the longitudinal optical sensor has at least one sensor region, wherein the longitudinal optical sensor generates at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region by the light beam, wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam in the sensor region;
- 15 - wherein at least one evaluation device is used, wherein the evaluation device generates at least one item of information on a transversal position of the object by evaluating the transversal sensor signal and wherein the evaluation device further generates at least one item of information on a longitudinal position of the object by evaluating the longitudinal sensor signal.

20

Embodiment 95: The use of a detector according to any of the preceding embodiments relating to a detector, for a purpose of use, selected from the group consisting of: a distance measurement, in particular in traffic technology; a position measurement, in particular in traffic technology; an entertainment application; a security application; a human-machine interface application; a tracking application; a photography application; an imaging application or camera application; a mapping application for generating maps of at least one space.

25

Brief description of the figures

30

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented alone or with several in combination. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

35

Specifically, in the figures:

40

Figure 1A shows an exemplary embodiment of a detector according to the present invention;

- Figure 1B shows a further exemplary embodiment of a detector according to the present invention;
- 5 Figure 1C shows a further exemplary embodiment of a detector according to the present invention;
- Figure 2A and 2B show different views of an embodiment of a transversal detector which may be used in the detector of the present invention;
- 10 Figures 3A to 3D show principles of generating transversal sensor signals and deriving information on a transversal position of an object;
- Figures 4A to 4C show different views of embodiments of a longitudinal optical sensor which may be used in the detector according to the present invention;
- 15 Figures 5A to 5E show the principle of generating longitudinal sensor signals and deriving information on a longitudinal position of an object; and
- Figure 6 shows a schematic embodiment of a human-machine interface and of an entertainment device according to the present invention.
- 20

Exemplary embodiments

25 Detector

Figure 1A illustrates, in a highly schematic illustration, an exemplary embodiment of a detector 110 according to the invention, for determining a position of at least one object 112. The detector 110 preferably may form a camera 111 or may be part of a camera 111. Other
30 embodiments are feasible.

The detector 110 comprises a plurality of optical sensors 114, which, in the specific embodiment, are all stacked along an optical axis 116 of the detector 110. Specifically, the optical axis 116 may be an axis of symmetry and/or rotation of the setup of the optical sensors
35 114. The optical sensors 114 may be located inside a housing 118 of the detector 110. Further, at least one transfer device 120 may be comprised, such as one or more optical systems, preferably comprising one or more lenses 122. An opening 124 in the housing 118, which, preferably, is located concentrically with regard to the optical axis 116, preferably defines a direction of view 126 of the detector 110. A coordinate system 128 may be defined, in which a
40 direction parallel or antiparallel to the optical axis 116 is defined as a longitudinal direction, whereas directions perpendicular to the optical axis 116 may be defined as transversal directions. In the coordinate system 128, symbolically depicted in Fig. 1A, a longitudinal

direction is denoted by z and transversal directions are denoted by x and y, respectively. Other types of coordinate systems 128 are feasible.

5 The optical sensors 114 comprise at least one transversal optical sensor 130 and, in this embodiment, a plurality of longitudinal optical sensors 132. The longitudinal optical sensors 132 form a longitudinal optical sensor stack 134. In the embodiment shown in Fig. 1A, five longitudinal sensors 132 are depicted. It shall be noted, however, that embodiments having a different number of longitudinal optical sensors 132 are feasible.

10 The transversal optical sensor 132 comprises a sensor region 136, which, preferably, is transparent to a light beam 138 travelling from the object 112 to the detector 110. The transversal optical sensor 130 is adapted to determine a transversal position of the light beam 138 in one or more transversal directions, such as in direction x and/or in direction y. Therein, embodiments are feasible in which a transversal position in only one transversal direction is
15 determined, embodiments in which transversal positions in more than one transversal direction are determined by one and the same transversal optical sensor 130, and embodiments in which a transversal position in a first transversal direction is determined by a first transversal optical sensor and wherein at least one further transversal position in at least one further transversal direction is determined by at least one further transversal optical sensor.

20 The at least one transversal optical sensor 130 is adapted to generate at least one transversal sensor signal. This transversal sensor signal may be transmitted by one or more transversal signal leads 140 to at least one evaluation device 142 of the detector 110, which will be explained in further detail below.

25 The longitudinal optical sensors 132 each comprise at least one sensor region 136, too. Preferably, one, more or all of the longitudinal optical sensors 132 are transparent, but the last longitudinal optical sensor 144 of the longitudinal optical sensor stack 134, i.e. the longitudinal optical sensor 132 on the side of the stack 134 facing away from the object 112. This last
30 longitudinal sensor 144 may fully or partially be intransparent.

Each of the longitudinal optical sensors 132 is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the respective sensor region 136 by the light beam 138. The longitudinal sensor signals, given the same total power of the
35 illumination, is dependent on a beam cross-section of the light beam 138 in the respective sensor region 136, as will be outlined in further detail below. Via one or more longitudinal signal leads 146, the longitudinal sensor signals may be transmitted to the evaluation device 142. As will be outlined in further detail below, the evaluation device may be designed to generate at least one item of information on at least one transversal position of the object 112 by evaluating
40 the at least one transversal sensor signal and to generate at least one item of information on at least one longitudinal position of the object 112 by evaluating the longitudinal sensor signal. For this purpose, the evaluation device 142 may comprise one or more electronic devices and/or

one or more software components, in order to evaluate the sensor signals, which is symbolically denoted by transversal evaluation unit 148 (denoted by "xy") and longitudinal evaluation unit 150 (denoted by "z"). By combining results derived by these evolution units 148, 150, a position information 152, preferably a three-dimensional position information, may be generated (denoted by "x, y, z").

The evaluation device 142 may be part of a data processing device 154 and/or may comprise one or more data processing devices 154. The evaluation device 142 may be fully or partially integrated into the housing 118 and/or may fully or partially be embodied as a separate device which is electrically connected in a wireless or wire-bound fashion to the optical sensors 114. The evaluation device 142 may further comprise one or more additional components, such as one or more electronic hardware components and/or one or more software components, such as one or more measurement units (not depicted in Fig. 1A) and/or one or more transformation units 156. Symbolically, in Fig. 1A, one optional transformation unit 156 is depicted which may be adapted to transform at least two transversal sensor signals into a common signal or common information.

In the following, embodiments of the transversal optical sensor 130 and the at least one longitudinal optical sensor 132 are disclosed. It shall be noted, however, that other embodiments are feasible. Thus, in the embodiments disclosed hereinafter, the optical sensors 114 are all designed as solid dye-sensitized solar cells (sDSCs). It shall be noted, however, that other embodiments are feasible.

Fig. 1B illustrates, in a highly schematic illustration, a further exemplary embodiment of the detector 110 according to the present invention, for determining a position of the at least one object 112. In this particular embodiment, the detector 110 may comprise one or more illumination sources 192, which may include an ambient light source and/or an artificial light source, and/or may comprise one or more reflective elements which may, for example, be connected to the object 112 for reflecting one or more primary light beams 206, as indicated in Fig. 1B. In addition or alternatively, the light beam 138 which emerges from the object 112 can fully or partially be generated by the object 112 itself, for example in the form of a luminescent radiation.

The optical sensors 114 comprise at least one transversal optical sensor 130 and, in this embodiment, a plurality of longitudinal optical sensors 132. The longitudinal optical sensors 132 form a longitudinal optical sensor stack 134. In the embodiment shown in Fig. 1B, five longitudinal optical sensors 132 are depicted, wherein the last longitudinal optical sensor 144 comprises a single longitudinal optical sensor 132 out of the plurality of the longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 which faces away from the object 112. In this embodiment, it is particularly preferred to arrange the last longitudinal optical sensor 144 in the longitudinal optical sensor stack 134 in a manner that a light beam 138 first travels

through the plurality of the longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 until it impinges the last longitudinal optical sensor 144.

5 The last longitudinal optical sensor 144 may be configured in various ways. Thus, the last longitudinal optical sensor 144 can for example be part of the detector 110 within the detector housing 118. Alternatively, the last longitudinal optical sensor 144 may be arranged outside the detector housing 118. It shall be emphasized that embodiments having a different number of longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 are feasible.

10 Whereas the plurality of the longitudinal optical sensors 132 within the longitudinal optical sensor stack 134 with the exception of the last longitudinal optical sensor 144 preferably are at least partially transparent, particularly to enabling a high relative intensity at each of these longitudinal optical sensors 132, the last longitudinal optical sensor 144 may either be transparent or intransparent. In case the last longitudinal optical sensor 144 is transparent, it
15 may be feasible to further place an additional optical sensor behind the longitudinal optical sensor stack 134, such as a separate imaging device 157 in a manner that a light beam 138 first travels through the plurality of the longitudinal optical sensors 132 including the last longitudinal optical sensor 144 until it impinges on the imaging device 157.

20 The imaging device 157 may be configured in various ways. Thus, the imaging device 157 can for example be part of the detector 110 within the detector housing 118. Alternatively, the imaging device 157 may be separately located outside the detector housing 118. The imaging device 157 may be fully or partially transparent or intransparent. The imaging device 157 may be or may comprise an organic imaging device or an inorganic imaging device. Preferably, the
25 imaging device 157 may comprise at least one matrix of pixels, wherein the matrix of pixels is particularly selected from the group consisting of: an inorganic semiconductor sensor device such as a CCD chip and/or a CMOS chip; an organic semiconductor sensor device. The imaging device signal may be transmitted by one or more imaging device signal leads 159 to at least one evaluation device 142 of the detector 110, which will be explained in further detail
30 below.

In a further preferred embodiment, at least two of the longitudinal optical sensors 132 exhibit a differing spectral sensitivity. Preferably, a last longitudinal optical sensor 144, which preferably is intransparent, is configured to absorb all over the spectral ranges of the at least two of the
35 longitudinal optical sensors have the differing spectral sensitivity. By way of example, the differing spectral sensitivity of at least two of the longitudinal optical sensors 132 varies over spectral range in a manner that the at least two of the longitudinal optical sensors 132 are sensitive to a specific color whereas the intransparent last longitudinal optical sensor 144 is adapted to be sensitive to all colors of the visible spectra range. In a specific embodiment, the
40 longitudinal optical sensors 132 with the exception of the last longitudinal optical sensor 144 are each sensitive to a different specific color with little overlap. This feature particularly allows reliably acquiring depth information for a colored object 112 irrespective of the specific color of

the object 112, for which information at least two of the longitudinal optical sensors 132 are required, as described above. This feature is realized by the fact that for each color, absorption of the light beam 118 always occurs at least in two of the longitudinal optical sensors 132, i.e. in the one of the longitudinal optical sensor 132 which is sensitive to the specific color of the object 112, as well as in the last longitudinal optical sensor 144 being sensitive to all colors.

With respect to the other features as presented in an exemplary fashion in Fig. 1B, reference is made to the above description of Fig. 1A.

A further exemplary embodiment of the detector 110 according to the present invention is shown in Fig. 1C in a highly schematic manner. In this particular embodiment, the detector 110 may comprise one or more optically sensitive elements 161 which may particularly be placed along the optical axis 116 of the detector 110. As depicted herein, the optically sensitive element 161 may preferably be located between the transfer device 120, which here comprises one lens 122, and the optical sensor 114, which, in this embodiment, comprises at least one transversal optical sensor 130 and a plurality of longitudinal optical sensors 132 forming a longitudinal optical sensor stack 134. The optically sensitive element 161 may be configured in a number of ways, such as a wavelength-sensitive element, a phase-sensitive element, or a polarization-sensitive element.

As a non-limiting example, a wavelength-sensitive element comprising a color wheel 163 may be employed here as the optically sensitive element 161. Alternative wavelength-sensitive elements may comprise one or more of a prism, a grating, and/or a dichroitic mirror. The color-wheel 163 may especially comprise a circle around which at least two, preferably two to eight, separate segments are arranged which may differ from each other by their optical effect, particularly a grade of transmission, to the impinging light beam 138. Herein, the separate segments may each act as a filter which may allow determining a contribution of the colors red (R), green (G), and blue (B), also designated as "RGB color wheel". Alternatively, the color wheel 163 commonly designated as a "RGB-W color wheel" and additionally comprising a white (W) segment may be employed. As a further alternative, the color wheel 163 may comprise clear or transparent segments which may be used to increase luminous efficiency. Furthermore, the color wheel 163 may, alternatively or additionally, comprise one or more of the colors cyan, yellow, and magenta. A further alternative may comprise a so-called "color drum", wherein the segments are arranged on the inner surface of a drum where the incident light beam may be directed through in order to impinge an optical sensor.

Moreover, a sequential color recapture wheel (SCR wheel) may be generated from RGB dichroic coatings arranged in an Archimedes spiral pattern. Herein, the Archimedes spiral may exhibit the property that the boundaries between two segments may move at a constant speed in radial direction. Furthermore, it may be possible to include also here white or clear segments. Further examples for rotating color wheels, such as known from video beamer devices, and for color drums may be found under the address www.hcinema.de/farbrad.htm.

With respect to the other features as presented in an exemplary fashion in Fig. 1C, reference is made to the above descriptions concerning Figs. 1A and/or 1B.

5 In Figs. 2A and 2B, different views of a potential embodiment of a transversal optical sensor 130 are depicted. Therein, Fig. 2A shows a top-view on a layer setup of the transversal optical sensor 130, whereas Fig. 2B shows a partial cross-sectional view of the layer setup in a schematic setup. For alternative embodiments of the layer setup, reference may be made to the disclosure above.

10

The transversal optical sensor 130 comprises a transparent substrate 158, such as a substrate made of glass and/or a transparent plastic material. The setup further comprises a first electrode 160, an optical blocking layer 162, at least one n-semiconducting metal oxide 164, sensitized with at least one dye 166, at least one p-semiconducting organic material 168 and at least one second electrode 170. These elements are depicted in Fig. 2B. The setup may further comprise at least one encapsulation 172 which is not depicted in Fig. 2B and which is symbolically depicted in the top-view of Fig. 2A, which may cover a sensor region 136 of the transversal optical sensor 130.

15

20 As an exemplary embodiment, the substrate 158 may be made of glass, the first electrode 160 may fully or partially be made of fluorine-doped tin oxide (FTO), the blocking layer 162 may be made of dense titanium dioxide (TiO₂), the n-semiconducting metal oxide 164 may be made of nonporous titanium dioxide, the p-semiconducting organic material 168 may be made of spiro-MiOTAD, and the second electrode 170 may comprise PEDOT:PSS. Further, dye ID504, as e.g. disclosed in WO 2012/110924 A1, may be used. Other embodiments are feasible.

25

As depicted in Figs. 2A and 2B, the first electrode 160 may be a large-area electrode, which may be contacted by a single electrode contact 174. As depicted in the top-view in Fig. 2A, the electrode contacts 174 of the first electrode 160 may be located in corners of the transversal optical sensor 130. By providing more than one electrode contact 174, a redundancy may be generated, and resistive losses over the first electrode 160 might be eliminated, thereby generating a common signal for the first electrode 160.

30

Contrarily, the second electrode 170 comprises at least two partial electrodes 176. As can be seen in the top-view in Fig. 2A, the second electrode 170 may comprise at least two partial electrodes 178 for an x-direction, and at least two partial electrodes 180 for a y-direction via contact leads 182, these partial electrodes 176 may be contacted electrically through the encapsulation 172.

35

40 The partial electrodes 176, in this specific embodiment, form a frame which surrounds the sensor region 136. As an example, a rectangular or, more preferably, a square frame may be formed. By using appropriate current measurement devices, electrode currents through the

partial electrodes 176 may be determined individually, such as by current measurement devices implemented into the evaluation device 142. By comparing e.g. electrode currents through the two single x-partial electrodes 178, and by comparing the electrode currents through the individual y-partial electrodes 180, x- and y-coordinates of a light spot 184 generated by the light beam 138 in the sensor region 136 may be determined, as for the outlined with respect to Figs. 3A to 3D below.

In Figs. 3A to 3D, two different situations of a positioning of the object 112 are depicted. Thus, Fig. 3A and Fig. 3B show a situation in which the object 112 is located on the optical axis 116 of the detector 110. Therein, Fig. 3A shows a side-view and Fig. 3B shows a top-view onto the sensor region 136 of the transversal optical sensor 130. The longitudinal optical sensors 132 are not depicted in this setup.

In Fig. 3C and 3D, the setup of Figs. 3A and 3B is depicted in analogous views with the object 112 shifted in a transversal direction, to an off-axis position.

It shall be noted that, in Figs. 3A and 3C, the object 112 is depicted as the source of one or more light beams 138. As will be outlined in further detail below, specifically with respect the embodiment in Fig. 6, the detector 110 may as well comprise one or more illumination sources which may be connected to the object 112 and, thus, which may emit the light beams 138, and/or which might be adapted to illuminate the object 112 and, by the object 112 reflecting primary light beams, generate the light beams 138 by reflection and/or diffusion.

According to well-known imaging equations, the object 112 is imaged onto the sensor region 136 of the transversal optical sensor 130, thereby generating an image 186 of the object 112 on the sensor region 136, which, in the following, will be considered a light spot 184 and/or a plurality of light spots 184.

As can be seen in the partial images 3B and 3D, the light spot 184 on the sensor region 136 will lead, by generating charges in the layer setup of the sDSC, electrode currents, which, in each case, are denoted by i_1 to i_4 . Therein, electrode currents i_1, i_2 denote electrode currents through partial electrodes 180 in y-direction and electrode currents i_3, i_4 denote electrode currents through partial electrodes 178 in x-direction. These electrode currents may be measured by one or more appropriate electrode measurement devices simultaneously or sequentially. By evaluating these electrode currents, x- and y-coordinates may be determined. Thus, the following equations may be used:

$$x_0 = f\left(\frac{i_3 - i_4}{i_3 + i_4}\right) \text{ and } y_0 = f\left(\frac{i_1 - i_2}{i_1 + i_2}\right).$$

Therein, f might be an arbitrary known function, such as a simple multiplication of the quotient of the currents with a known stretch factor and/or an addition of an offset. Thus, generally, the

electrode currents i_1 to i_4 might form transversal sensor signals generated by the transversal optical sensor 130, whereas the evaluation device 142 might be adapted to generate information on a transversal position, such as at least one x-coordinate and/or at least one y-coordinate, by transforming the transversal sensor signals by using a predetermined or determinable transformation algorithm and/or a known relationship.

In Figs. 4A to 4C, various views of longitudinal optical sensors 132 are shown. Therein, Fig. 4A shows a cross-sectional view of a potential layer setup, and Figs. 4B and 4C show top-view of two embodiments of potential longitudinal optical sensors 132. Therein, Fig. 4C shows a potential embodiment of the last longitudinal optical sensor 144, wherein Fig. 4B shows potential embodiments of the remaining longitudinal optical sensors 132 of the longitudinal optical sensor stack 134. Thus, the embodiment in Fig. 4B may form a transparent longitudinal optical sensor 132, whereas the embodiment in Fig. 4C may be an intransparent longitudinal optical sensor 132. Other embodiments are feasible. Thus, the last longitudinal optical sensor 144, alternatively, might also be embodied as a transparent longitudinal optical sensor 132.

As can be seen in the schematic cross-sectional view in Fig. 4A, the longitudinal optical sensor 132 again, might be embodied as an organic photo-detector, preferably as an sDSC. Thus, similarly to the setup of Fig. 2B, a layer setup using a substrate 158, a first electrode 160, a blocking layer 162, an n-semiconducting metal oxide 164 being sensitized with a dye 166, a p-semiconducting organic material 168 and a second electrode 170 may be used. Additionally, an encapsulation 172 may be provided. For potential materials of the layers, reference may be made to Fig. 2B above. Additionally or alternatively, other types of materials may be used.

It shall be noted that, in Fig. 2B, an illumination from the top is symbolically depicted, i.e. an illumination by the light beam 138 from the side of the second electrode 170. Alternatively, an illumination from the bottom, i.e. from the side of the substrate 158 and through the substrate 158, may be used. The same holds true for the setup of Fig. 4A.

However, as depicted in Fig. 4A, in a preferred orientation of the longitudinal optical sensor 132, an illumination by the light beam 138 preferably takes place from the bottom, i.e. through the transparent substrate 158. This is due to the fact that the first electrode 160 may easily be embodied as a transparent electrode, such as by using a transparent conductive oxide, such as FTO. The second electrode 170, as will be outlined in further detail below may either be transparent or, specifically, for the last longitudinal optical sensor 144, intransparent.

In Fig. 4B and Fig. 4C, different setups of the second electrode 170 are depicted. Therein, in Fig. 4B, corresponding to the cross-sectional view of Fig. 4A, the first electrode 160 may be contacted by one or more electrode contacts 174, which, as an example, may comprise one or more metal pads, similar to the setup in Fig. 2B. These electrode contacts 174 may be located in the corners of the substrate 158. Other embodiments are feasible.

The second electrode 170, however, in the setup of Fig. 4B may comprise one or more layers of a transparent electrically conductive polymer 188. As an example, similar to the setup of Figs. 2A and 2B, PEDOT:PSS may be used. Further, one or more top contacts 190 may be provided, which may be made of a metallic material, such as aluminum and/or silver. By using one or
5 more contact leads 182, leading through the encapsulation 172, this top contact 190 may be electrically contacted.

In the exemplary embodiment shown in Fig. 4B, the top contact 190 forms a closed opened frame surrounding the sensor region 136. Thus, as opposed to the partial electrodes 176 in
10 Figs. 2A and 2B, only one top contact 190 is required. However, the longitudinal optical sensor 132 and the transversal optical sensor 130 may be combined in one single device, such as by providing partial electrodes in the setup of Figs. 4A to 4C. Thus, in addition to the FiP effect which will be outlined in further detail below, transversal sensor signals may be generated with the longitudinal optical sensor 132. Thereby, a combined transversal and longitudinal optical
15 sensor may be provided.

The use of the transparent electrically conductive polymer 188 allows for an embodiment of the longitudinal optical sensor 132 in which both the first electrode 160 and the second electrode 170 are at least partially transparent. The same, preferably, holds true for the transversal optical
20 sensor 130. In Fig. 4C, however, a setup of the longitudinal optical sensor 132 is disclosed which uses an intransparent second electrode 170. Thus, as an example, the second electrode 170 may be embodied by using one or more metal layers, such as aluminum and/or silver, instead of or in addition to the at least one electrically conductive polymer 188. Thus, as an example, the electrically conductive polymer 188 may be replaced or may be reinforced by one
25 or more metal layers which, preferably, may cover the full sensor region 136.

In Figs. 5A to 5E, the above-mentioned FiP effect shall be explained. Therein, Fig. 5A shows a side-view of a part of a detector 110, in a plain parallel to the optical axis 116, similar to the setup in Figs. 1, 3A and 3C. Of the detector 110, only the longitudinal optical sensors 132 and
30 the transfer device 120 are depicted. Not shown is at least one transversal optical sensor 130. This transversal optical sensor 130 may be embodied as a separate optical sensor 114 and/or may be combined with one or more of the longitudinal optical sensors 132.

Again, the measurement starts with an emission and/or reflection of one or more light beams
35 138 by at least one object 112. The object 112 may comprise an illumination source 192, which may be considered as a part of the detector 110. Additionally or alternatively, a separate illumination source 192 may be used.

Due to a characteristic of the light beam 138 itself and/or due to beam shaping characteristics of
40 the transfer device 120, preferably of the at least one lens 122, the beam properties of the light beam 138 in the region of the longitudinal optical sensors 132 at least partially are known. Thus,

as depicted in Fig. 5A, one or more focal points 194 might occur. In the focal point 194, a beam waist or a cross-section of the light beam 138 may assume a minimum value.

In Fig. 5B, in a top-view onto the sensor regions 136 of the longitudinal optical sensors 132 in Fig. 5A, a development of the light spots 184 generated by the light beam 138 impinging on the sensor regions 136 is depicted. As can be seen, close to the focal point 194, the cross-section of the light spot 184 assumes a minimum value.

In Fig. 5C, a photo current I of the longitudinal optical sensors 132 is given for the five cross-sections of the light spot 184 in Fig. 5B, in case longitudinal optical sensors 132 exhibiting the above-mentioned FiP effect are used. Thus, as an exemplary embodiment, five different photo currents I for the spot cross-sections as shown in Figs. 5B are shown for typical DSC devices preferably sDSC devices. The photo current I is depicted as a function of the area A of the light spot 184, which is a measure of the cross-section of the light spots 184.

As can be seen in Fig. 5C, the photo current I , even if all longitudinal optical sensors 132 are illuminated with the same total power of the illumination, the photo current I is dependent on the cross-section of the light beam 138, such as by providing a strong dependency on the cross-sectional area A and/or the beam waist of the light spot 184. Thus, the photo current is a function both of the power of the light beam 138 and of the cross-section of the light beam 138:

$$I=f(n,a).$$

Therein, I denotes the photo current provided by each longitudinal optical sensor 132, such as a photo current measured in arbitrary units, as a voltage over at least one measurement resistor and/or in amps. n denotes the overall number of photons impinging on the sensor region 136 and/or the overall power of the light beam in the sensor region 136. A denotes the beam cross-section of the light beam 138, provided in arbitrary units, as a beam waist, as a beam diameter of beam radius or as an area of the light spot 134. As an example, the beam cross-section may be calculated by the $1/e^2$ diameter of the light spot 184, i.e. a cross-sectional distance from a first point on a first side of a maximum intensity having an intensity of $1/e^2$ as compared to the maximum intensity of the light spot 184, to a point on the other side of the maximum having the same intensity. Other options of quantifying the beam cross-section are feasible.

The setup in Fig. 5C shows the photo current of a longitudinal optical sensor 132 according to the present invention which may be used in the detector 110 according to the present invention, showing the above-mentioned FiP effect. Contrarily, Fig. 5D in a diagram corresponding to the diagram of Fig. 5C, photo currents of traditional optical sensors are shown, for the same setup as depicted in Fig. 5A. As an example, silicon photo detectors may be used for this measurement. As can be seen, in these traditional measurements, the photo current or photo signal of the detectors is independent from the beam cross-section A .

Thus, by evaluating the photo currents and/or other types of longitudinal sensor signals of the longitudinal optical sensors 132 of the detector 110, the light beam 138 may be characterized. Since the optical characteristics of the light beam 138 depend on the distance of the object 112 from the detector 110, by evaluating these longitudinal sensor signals, a position of the object
 5 112 along the optical axis 116, i.e. a z-position, may be determined. For this purpose, the photo currents of the longitudinal optical sensors 132 may be transformed, such as by using one or more known relationships between the photo current I and the position of the object 112, into at least one item of information on a longitudinal position of the object 112, i.e. a z-position. Thus, as an example, the position of the focal point 194 may be determined by evaluating the sensor
 10 signals, and a correlation between the focal point 194 and a position of the object 112 in the z-direction may be used for generating the above-mentioned information. Additionally or alternatively, a widening and/or narrowing of the light beam 138 may be evaluated by comparing the sensor signals of the longitudinal sensors 132. As an example, known beam properties may be assumed, such as a beam propagation of the light beam 138 according to
 15 Gaussian laws, using one or more Gaussian beam parameters.

Further, the use of a plurality of longitudinal optical sensors 132 provides additional advantages as opposed to the use of a single longitudinal optical sensor 132. Thus, as outlined above, the overall power of the light beam 138 generally might be unknown. By normalizing the longitudinal
 20 sensor signals, such as to a maximum value, the longitudinal sensor signals might be rendered independent from the overall power of the light beam 138, and a relationship

$$I_n = g(A)$$

25 may be used by using normalized photo currents and/or normalized longitudinal sensor signals, which is independent from the overall power of the light beam 138.

Additionally, by using the plurality of longitudinal optical sensors 132, an ambiguity of the longitudinal sensor signals may be resolved. Thus, as can be seen by comparing the first and
 30 the last image in Fig. 5B and/or by comparing the second and the fourth image in Fig. 5B, and/or by comparing the corresponding photo currents in Fig. 5C, longitudinal optical sensors 132 being positioned at a specific distance before or behind the focal point 194 may lead to the same longitudinal sensor signals. A similar ambiguity might arise in case the light beam 138 weakens during propagations along the optical axis 116, which might generally be corrected
 35 empirically and/or by calculation. In order to resolve this ambiguity in the z-position, the plurality of longitudinal sensor signals clearly shows the position of the focal point and of the maximum. Thus, by e.g. comparing with one or more neighboring longitudinal sensor signals, it may be determined whether a specific longitudinal optical sensor 132 is located before or beyond a focal point on the longitudinal axis.

40

In Figure 5E, a longitudinal sensor signal for a typical example of an sDSC is depicted, in order to demonstrate the possibility of the longitudinal sensor signal and the above-mentioned FiP

effect being dependent on a modulation frequency. In this figure, a short-circuit current I_{sc} is given as the longitudinal sensor signal on the vertical axis, in arbitrary units, for a variety of modulation frequencies f . On the horizontal axis, a longitudinal coordinate z is depicted. The longitudinal coordinate z , given in micrometers, is chosen such that a position of a focus of the light beam on the z -axis is denoted by position 0, such that all longitudinal coordinates z on the horizontal axis are given as a distance to the focal point of the light beam. Consequently, since the beam cross-section of the light beam depends on the distance from the focal point, the longitudinal coordinate in Figure 5E denotes the beam cross-section in arbitrary units. As an example, a Gaussian light beam may be assumed, with known or determinable beam parameters, in order to transform the longitudinal coordinate into a specific beam waist or beam cross-section.

In this experiment, longitudinal sensor signals are provided for a variety of modulation frequencies of the light beam, for 0 Hz (no modulation), 7 Hz, 377 Hz and 777 Hz. As can be seen in the figure, for modulation frequency 0Hz, no FiP effect or only a very small FiP effect, which may not easily be distinguished from the noise of the longitudinal sensor signal, may be detected. For higher modulation frequencies, a pronounced dependency of the longitudinal sensor signal on the cross section of the light beam may be observed. Typically, modulation frequencies in the range of 0.1 Hz to 10 kHz may be used for the detector according to the present invention, such as modulation frequencies of 0.3 Hz.

Human-machine interface, entertainment device and tracking system:

In Fig. 6, an exemplary embodiment of a human-machine interface 196 according to the invention, which can simultaneously also be embodied as an exemplary embodiment of an entertainment device 198 according to the invention or which can be a constituent part of such an entertainment device 198, is depicted. Further, the human-machine interface 196 and/or the entertainment device 198 may also form an exemplary embodiment of a tracking system 199 adapted for tracking a user 200 and/or one or more body parts of the user 200. Thus, a motion of one or more of the body parts of the user 200 may be tracked.

By way of example, at least one detector 110, according to the present invention can once again be provided, for example, in accordance with one or more of the embodiments described above, with one or a plurality of optical sensors 114, which may comprise one or more transversal optical sensors 130 and one or more longitudinal optical sensors 132. Further elements of the detector 110 can be provided, which are not illustrated in Fig. 6, such as, for example, elements of an optional transfer device 120. For a potential embodiment, reference may be made to Fig. 1A and/or B. Furthermore, one or a plurality of illumination sources 192 may be provided. Generally, with regard to these possible embodiments of the detector 110, reference can be made for example to the description above.

The human-machine interface 196 can be designed to enable an exchange of at least one item of information between a user 200 and a machine 202, which is merely indicated in Fig. 6. For example, in exchange of control commands, and/or information may be performed by using the human-machine interface 196. The machine 202 can comprise, in principle, any desired device
5 having at least one function which can be controlled and/or influenced in some way. At least one evaluation device 142 of the at least one detector 110 and/or a part thereof can, as indicated in Fig. 6, be wholly or partially integrated into said machine 201, but can, in principle, also be formed fully or partially separately from the machine 202.

10 The human-machine interface 196 can be designed for example to generate, by means of the detector 110, at least one item of geometrical information of the user 200 by means of the detector 110 and can assign the geometrical information at least to one item of information, in particular at least one control command. For this purpose, by way of examples, by means of the detector 110, a movement and/or a change in posture of the user 200 can be identified. For
15 example, as indicated in Fig. 6, a hand movement and/or a specific hand posture of the user 200 may be detected. Additionally or alternatively, other types of geometrical information of the user 200 may be detected by the one or more detectors 110. For this purpose, one or more positions and/or one or more position information regarding the user 200 and/or one or more body parts of the user 200 may be identified by the at least one detector 110. It is then possible
20 to recognize, for example by comparison with a corresponding command list, that the user 200 would like to effect a specific input, for example would like to give the machine 202 a control command. As an alternative or an addition to direct diametrical information about the actual user 200, it is also possible, for example, to generate at least one item of geometrical
25 information about at least one beacon device 204 attached to the user 200, such as at least one item of geometrical information about a garment of the user 200 and/or a glove and/or an article moved by the user 200, such as a stick, a bat, a club, a racket, a cane, a toy, such as a toy gun. One or more beacon devices 204 may be used. The beacon device 204 may be embodied as an active beacon device and/or as a passive beacon device. Thus, the beacon device 204 may
30 comprise one or more illumination sources 192 and/or may comprise one or more reflective elements for reflecting one or more primary light beams 206, as indicated in Fig. 6.

The machine 202 can furthermore comprise one or a plurality of further human-machine interfaces, which need not necessarily be embodied according to the invention, for example, as indicated in Fig. 6, at least one display 208 and/or at least one keyboard 210. Additionally or
35 alternatively, other types of human-machine interfaces may be provided. The machine 202 can, in principle, be any desired type of machine or combination of machines, such as a personal computer.

40 The at least one evaluation device 142 and/or parts thereof may further function as a track controller 201 of the tracking system 199. Additionally or alternatively, one or more additional track controllers 201 may be provided, such as one or more additional data evaluation devices. The track controller 201 may be or may comprise one or more data memories, such as one or

more volatile and/or non-volatile memories. In this at least one data memory, a plurality of subsequent positions and/or orientation of one or more objects of parts of an object may be stored, in order to allow for storing a past trajectory. Additionally or alternatively, a future trajectory of the object and/or parts thereof may be predicted, such as by calculation, extrapolation or any other suitable algorithm. As an example, a past trajectory of an object or a part thereof may be extrapolated to future values, in order to predict at least one of a future position, a future orientation and a future trajectory of the object or a part thereof.

In the context of an entertainment device 198, said machine 202 can be designed for example to carry out at least one entertainment function, for example at least one game, in particular with at least one graphical display on the display 208 and, optionally, a corresponding audio output. The user 200 can input at least one item of information, for example via the human-machine interface 196 and/or one or more other interfaces, wherein the entertainment device 198 is designed to alter the entertainment function in accordance with the information. By way of example, specific movements or one or more virtual articles, for example of virtual persons in a game and/or movements of virtual vehicles in a game, may be controlled by means of corresponding movements of the user 200 and/or one or more body parts of the user 200, which, in turn, may be recognized by the detector 110. Other types of control of at least one entertainment function by the user 200, by means of the at least one detector 110, are also possible.

Exemplary embodiments of sDSCs for a 3-D position sensor:

The practical implementation of the FiP-effect of the sDSCs in the form of a 3-D sensor, and achieving good spatial resolution both in the x,y- and in the z-direction, typically may require the cells to have an active area of approximately 1 cm x 1 cm and meet certain requirements. Therefore, in the following, preferred requirements for the individual cells of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor are given. It shall be noted, however, that other embodiments are feasible.

Optical properties of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor:

As can be seen in Figures 5A to 5C, one particular current signal can imply two different spatial points (in front and behind the focus). In order to obtain unambiguous depth information on the z-axis, therefore, preferably at least two cells need to be arranged one behind the other. Unambiguous information is then derived from the ratio between the current signals of the two cells. For the sake of precise z-information, this sensor should have six cells stacked behind each other. This requires the cells to be transparent, i.e. the back electrode that normally consists of vapor-deposited silver across its entire area needs to be replaced by a transparent conducting material.

To ensure that sufficient illumination reaches the last cell and it supplies a useful current signal, the front five cells may have only low absorption at the excitation wavelength. The wavelength used for excitation should be around 700 nm.

5 Cross-resistance of the transversal optical sensors:

To achieve precise x,y resolution, there has to be a sufficient potential difference between each pair of opposite sides in this square cell. Fig. 2A shows such a transparent cell with which the x,y resolution is possible.

10

Even without a silver back electrode, sufficiently good electron transport from the p-type conductor into the oxidized dye has to be ensured across the entire surface area of the cells so that the dye is regenerated rapidly by a supply of electrons. Since the p-type conductor itself has very low conductance (10^{-5} S/cm), a conducting layer needs to be coated onto the p-type

15 conductor. Thanks to this additional layer, a defined cross-resistance R is to be achieved between the opposite sides of this square cell.

Transparency of the transversal optical sensors:

20 Due to their good conductance, normal solar cells have back electrodes (second electrodes) made of silver. The cells being developed here, however, have to be transparent, which is why the 1 cm² cell area typically requires a transparent back electrode. The material preferably used for this purpose is the conducting polymer poly(3,4-ethylenedioxythiophene)-poly(styrenesulfonate) (PEDOT:PSS) in aqueous dispersion. The conjugated polymer

25 PEDOT:PSS is highly transparent; it absorbs in the blue-green region (450-550 nm) only at considerable layer thicknesses, and only minimally in the red spectral range.

The additional PEDOT layer makes possible good electron transport in the p-type conductor. To improve the conductance of this layer and provide contacts, four silver electrodes 1 cm in length

30 are vapor-deposited around the square cell. The arrangement of the silver electrodes is shown in Fig. 3.3a. Fig. 3.3b shows a cell with a transparent PEDOT back electrode.

Extinction of the cells of the at least one transversal optical sensor and/or the at least one longitudinal optical sensor:

35

It is not only the back electrode that has to be transparent, but the entire cell. To ensure that a sufficient amount of light still reaches the last cell in the stack, the extinction of the front five cells should be as low as possible. This is determined first and foremost by the dye's absorption. The extinction of a solar cell, i.e. the absorption of light by the dye, has a decisive

40 effect on the cell's output current. Typically, the wavelength-dependent absorption spectrum has a maximum – the wavelength of maximum absorption is characteristic of the particular dye used. The more dye has been adsorbed in the np TiO₂ layer, the higher the cell's absorption.

The more dye molecules are adsorbed, the more electrons can reach the cb of the TiO₂ through optical excitation and the higher the current. A cell with higher extinction will, therefore, have a higher output current than one with low extinction.

- 5 The objective here is to obtain the maximum total current from the complete cell arrangement – which in the ideal case is divided equally among all the cells. Since the intensity of the light is attenuated by absorption in the cells, the ones located further back in the stack receive less and less light. In order, nonetheless, to obtain similar output currents from all six cells, it would make sense for the front cells to have lower extinction than the ones at the back. As a result, they will
- 10 stop less of the light reaching the following cells, which in turn will absorb a larger proportion of the already weakened light. Through optimal adjustment of the extinctions at the positions of the cells in the stack, in this way theoretically one could obtain the same current from all the cells. The extinction of a solar cell can be adjusted by staining with a dye and by controlling the thickness of the np TiO₂ layer.

15

Optimizing the extinction and the output current of the cells of the longitudinal optical sensor stack:

- The last cell in the stack preferably should absorb almost all the incident light. For this reason,
- 20 this cell should have maximum extinction. Starting with the current obtained under maximum extinction at the last cell, the extinctions of the front cells need to be so adjusted that all cells together supply a maximum total current, one that is distributed as uniformly as possible across all the cells.

- 25 Optimizing the stack's output currents is carried out as follows:

- The choice of dye
- Maximum extinction/maximum output current of the last cell
- Dye concentration for staining the last cell

30

- Staining time of the last cell
- Optimum thickness of the last cell's np TiO₂ layer
- Maximum output current of the complete stack
- Optimum thickness of the np TiO₂ layers of the front five cells

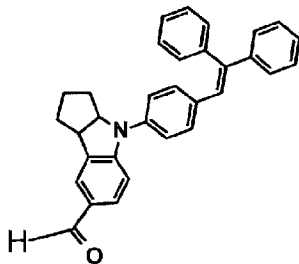
- 35 The extinction was measured with a Zeiss spectrometer MCS 501 UV-NIR using Zeiss lamp MCS 500. The results were evaluated with an Aspect Plus software program.

Choice of dye:

- 40 To begin with, a dye should be found that absorbs sufficiently at the excitation wavelength of approximately 700 nm. The ideal dye for solar cells typically has a broad absorption spectrum and should absorb completely the incident light below a wavelength of ca. 920 nm. In actual

fact, most dyes have an absorption maximum in the wavelength range between 450-600 nm; above 650 nm they usually absorb weakly or not at all.

5 The dye with which the first experiments were performed was ID504, as e.g. disclosed in WO 2012/110924 A1. This dye, however, turned out to exhibit only a low absorption in the range of 700 nm. Therefore, for the stack, the dye D-5 (also referred to as ID 1338) was used:



10 Preparation and properties of the Dye D-5 are disclosed in WO 2013/144177 A1.

Additionally or alternatively, however, other dyes may be used. The staining time, i.e. the duration of staining of the TiO₂-layer with the respective dye, turned out to have an influence on the absorption properties. The tests were performed with cells having np TiO₂ layers having a thickness of 1.3 microns. The absorption maximum of D-5 is around 550-560 nm which exhibits
15 an extinction $\epsilon \approx 59000$ at this maximum.

In this experimental series, the dye concentration was 0.3 mM and the staining time was increased so as to be 10 to 30 minutes. A pronounced increase in extinction at longer staining
20 times was observed, so finally staining times of 30 min. were used for D-5.

Still, even after optimizing the staining time, the absorption was determined to be rather very low. Therefore, generally, the absorption will have to be maximized by increasing the dye concentration, the staining time and the thickness of the np TiO₂ layer.

25

Dye concentration and staining time of the last cell in the longitudinal optical sensor stack:

Several experiments regarding the staining time and the dye concentration were performed. The standard concentration of the dye solution for a layer thickness of the TiO₂-layers of 1-2
30 microns was 0.5 mM. At these concentrations, the dye should already be present in excess. Here the dye concentration was increased to 0.7 mM. In order to prevent inhomogeneities across the cell's area, the dye solution was cleaned by removing undissolved dye particles and other impurities, using a 0.2 micron syringe filter, before placing the cells in it.

35 If the dye is present in excess, then after a 1 hour staining time a dye monolayer dye should have been adsorbed to the surface of the np TiO₂ layer, which leads to maximum absorption by

the dye being used. The maximum staining time tested here was 75 minutes, which was finally used for the cells.

Finally, cells having a layer thickness of the TiO₂-layer of 1.3 microns, a dye concentration of 0.7 mM and a staining time of 75 min was used. The cell's extinction turned out to be 0.4 at 700 nm.

The np TiO₂ layer thickness of the last cell of the longitudinal optical sensor stack:

Ultimately, the thickness of the nanoporous (np) layer and thus the TiO₂ surface area available for dye adsorption may be an important factor influencing the absorption behavior and therefore the cells' output current. So far, maximizing the extinction was done in cells with np TiO₂ layers whose thickness was 1.3 microns. Since more dye can be adsorbed in thicker np TiO₂ layers, the thickness of the TiO₂ layer was increased in steps to 3 microns, and the thickness at which the greatest output current occurred was determined.

The nanoporous TiO₂ layers were applied by spin coating. Spin coating is suitable for applying low-volatility substances dissolved in a high-volatility solvent (here: terpineol). As a starting product, TiO₂ paste made by Dyesol (DSL 18 NR-T) was used. This paste was mixed with terpineol, which decreases the viscosity of the paste. Depending on the composition ratio of the paste : terpineol mixture, and at a constant spin velocity of 4500 1/min, np TiO₂ layers of varying thicknesses are obtained. The higher the terpineol proportion, the lower the viscosity of the diluted paste and the thinner the cells will be.

The diluted TiO₂ paste was also cleaned using a 1.2 micron syringe filter to remove larger particles, before applying the paste the next day by spin coating on the cells coated with blocking layer.

When varying the np TiO₂ layer thickness, it should be noted that the concentration of the p-type conductor dissolved in chlorobenzene needs to be adjusted. Thicker np layers have a larger cavity volume that has to be filled with p-type conductor. For this reason, in the case of thicker np layers the amount of supernatant p-type conductor solution on top of the np layer is smaller. To ensure that the solid p-type conductor layer remaining on the np TiO₂ layer after spin coating has a constant thickness (the solvent evaporates during spin coating), higher p-type conductor concentrations are needed for thick np TiO₂ layers than for thin ones. The optimal p-type conductor concentrations are not known for all the TiO₂ layer thicknesses tested here. For this reason, the p-type conductor concentration is varied for the unknown layer thicknesses and the output currents compared for equal layer thicknesses but different p-type conductor concentrations.

The chosen starting value for the layer thickness variation was a 1.3 microns for the np TiO₂ layer. 1.3 microns corresponds to a TiO₂ paste : terpineol mass composition of 5 g : 5 g. A test

series with cells whose np TiO₂ layers are thicker than 1.3 microns will show at which layer thickness the largest output current is obtained from the last cell in the stack.

5 These cells were stained with the aforementioned optimized parameters for maximum extinction (D-5; c = 0.7 mM; staining time: 75 minutes). The extinction of these cells was found to be approximately 0.6 at 700 nm.

10 Since the last cell generally does not have to be transparent, the back electrode was vapor-deposited on the whole 1 cm² area directly onto the p-type conductor – without PEDOT.

The measurement results indicated that, as expected, the output currents of the cells with a whole-area back electrode (second electrode) are much higher. Highest output currents were obtained with a TiO₂ : terpineol mass ratio of 5 : 3. This corresponds to a TiO₂ layer thickness of 2-3 microns.

15 Therefore, in subsequent experiments, a TiO₂ paste : terpineol composition of 5 : 3 was used for the last cell in the stack. The back electrode was vapor-deposited across the entire 1 cm² cell area.

20 The np TiO₂ layer thickness of the front cells of the longitudinal optical sensor stack:

Starting from the maximum output current obtained with the last cell, the thicknesses of the front cells' np TiO₂ layers are to be so adjusted that every cell in the stack generates the maximum possible output current. This requires low extinction values in the front cells.

25 During the experiments, it turned out that, in practical terms, it is rather difficult to obtain reproducible low extinctions through the dye concentration and staining time parameters. In order to make cells with low, reproducible extinction, therefore, it makes sense to fabricate cells with thin np TiO₂ layers and to keep them in the dye solution for the time needed to ensure dye saturation of the np TiO₂ surface. The terpineol proportion in the TiO₂ layers was increased in a stepwise fashion. All the cells were stained under the same conditions. Since their extinction is intended to be decreased significantly, the dye concentration here was 0.5 mM, and the staining time was 60 minutes.

35 Surprisingly, in this series, the cell's output voltages turned out to begin with an increase in output current with decreasing np TiO₂ layer thickness. The optimum out of the tested TiO₂ paste dilutions turned out to be 5 : 6. At higher dilutions and therefore thinner np TiO₂ layers, the output current tends to decrease. The reason for the exception to this trend at a dilution of 5 : 9 is likely to be optimal adjustment of the p-type conductor concentration of 100 mg/ml for this layer thickness.

40

If, however, one considers the decrease in extinction relative to that of the output current, it makes sense to accept the lower output current in order to ensure that the following cells receive very much more light than would be the case with 5 : 6 dilution. Photographs of cells with TiO₂ : terpineol mixtures of 5 : 4.1, 5 : 6 and 5 : 10 were taken, which illustrated this effect.

- 5 Effects of inhomogeneity were observed. In order to achieve a homogeneous layer within a 1 cm² cell, the TiO₂ area was increased for later cells such that the region in which the TiO₂ banks up during spin coating lies outside the silver electrodes and thus outside the cell.

- 10 The construction of the cell stack with regard to the TiO₂ layer's thickness in the cells and their positioning in the stack, was carried out by testing various arrangements of cells with np TiO₂ layers of various thicknesses.

Preparation and the properties of a dye sensitized solar cell (DSC) prepared with the Dye D-5

- 15 FTO (tin oxide doped with fluorine) glass substrates (< 12 ohms/sq, A11DU80, supplied by AGC Fabritech Co., Ltd.) were used as the base material, which were successively treated with glass cleaner, Semico Clean (Furuuchi Chemical Corporation), fully deionized water and acetone, in each case for 5 min in an ultrasonic bath, then baked for 10 minutes in isopropanol and dried in a nitrogen flow.

- 20 A spray pyrolysis method was used to produce the solid TiO₂ buffer layer. Titanium oxide paste (PST-18NR, supplied by Catalysts & Chemicals Ind. Co., Ltd.) was applied onto the FTO glass substrate by screen printing method. After being dried for 5 minutes at 120°C, a working electrode layer having a thickness of 1.6 µm was obtained by applying heat treatment in air at 25 450°C for 30 minutes and 500°C for 30 minutes. Obtained working electrode is then treated with TiCl₄, as described by M. Grätzel et al., for example, in Grätzel M. et al., Adv. Mater. 2006, 18, 1202. After sintering the sample was cooled to 60 to 80°C. The sample was then treated with an additive as disclosed in WO 2012/001628 A1. 5 mM of the additive in ethanol was prepared and the intermediate was immersed for 17 hours, washed in a bath of pure ethanol, briefly dried in a 30 nitrogen stream and subsequently immersed in a 0.5 mM solution of dye D-5 in a mixture solvent of acetonitrile + t-butyl alcohol (1:1) for 2 hours so as to adsorb the dye. After removal from the solution, the specimen was subsequently washed in acetonitrile and dried in a nitrogen flow.

- 35 A p-type semiconductor solution was spin-coated on next. To this end a 0.165M 2,2',7,7'-tetrakis(N,N-di-p-methoxyphenyl-amine)-9,9'-spirobifluorene (spiro-MeOTAD) and 20mM LiN(SO₂CF₃)₂ (Wako Pure Chemical Industries, Ltd.) solution in chlorobenzene was employed. 20 µl/cm² of this solution was applied onto the specimen and allowed to act for 60 s. The supernatant solution was then spun off for 30 s at 2000 revolutions per minute. The substrate 40 was stored overnight under ambient conditions. Thus, the HTM was oxidized and for this reason the conductivity increased.

As the metal back electrode, Ag was evaporated by thermal metal evaporation in a vacuum at a rate of 0.5 nm/s in a pressure of 1×10^{-5} mbar, so that an approximately 100 nm thick Ag layer was obtained.

- 5 In order to determine the photo-electric power conversion efficiency η of the above photoelectric conversion device, the respective current/voltage characteristic such as short-circuit current density J_{sc} , open-circuit voltage V_{oc} and fill factor FF was obtained with a Source Meter Model 2400 (Keithley Instruments Inc.) under the illumination of an artificial sunlight (AM 1.5, 100 mW/cm² intensity) generated by a solar simulator (Peccell Technologies, Inc). As a result, the
- 10 DSC prepared with the Dye D-5 exhibited the following parameters:

J_{sc} [mA/cm ²]	V_{oc} [mV]	FF [%]	η [%]
10.5	721	59	4.5

Results of the optimized output currents for the longitudinal optical sensor stack:

- 15 The best result in terms of the cell stack's output currents were obtained when all five transparent cells of the longitudinal optical sensors step have an np TiO₂ layer thickness of 0.45 micrometers (i.e. a TiO₂ paste dilution of 5 : 10). These cells with a 0.45 micrometers np TiO₂ layer were stained for 60 minutes in a 0.5 mM dye solution. Only the last cell had a np TiO₂ layer just under 3 micrometers and was stained for 75 minutes (0.7 mM). Since the last cell
- 20 does not have to be transparent, the back electrode (second electrode) of the last cell was a vapor-deposited silver layer across the whole 1 cm² area so as to be able to pick up the maximum possible current. The following photo currents were observed with this stack, in order from the first cell to the last cell of the stack :

25 Current [μ A]: 37 9.7 7.6 4.0 1.6 1.9

- The first five cells were made identically. The last cell had a thicker np TiO₂ layer and a silver back electrode vapor-deposited across the whole cell area. One can see that the current of the second cell has already dropped to 1/4 of the first cell's. Even in these five highly transparent
- 30 cells, the last cell's current is only a fraction of the current in the first cell. The cells were excited with a red laser (690 nm, 1 mW) directed at the centre of the cell area.

- The current obtained with cells with a TiO₂ : terpineol dilution of 5 : 9, 5 : 8 or 5 : 7 (i.e. thicker cells) was at most 10 microamps larger than the current of a cell with 5 : 10 dilution of the TiO₂
- 35 paste. These cells, however, exhibited a significantly higher extinction, as a result of which the output current of the following cells decreases considerably.

- The cells with a TiO₂ dilution of 5 : 6, in which – compared with the TiO₂ dilutions of 5 : 9; 5 : 8 and 5 : 7 – a significantly higher current was obtained, nevertheless absorbed so much light that
- 40 no more light reaches the last cell of the stack. Even when placing just one of these cells in

position 5 with four preceding 450 nm thin cells, the output current of the last cell decreased considerably, such that the last cell supplies practically no more current.

It needs mentioning that each of these cells in the test stack was sealed with an additional glass plate for protection against ambient effects. This, however, created many additional interfaces at which the light beam of the 690 nm laser (1 mW) can be reflected and scattered, as a result of which the extinction of such a sealed cell is higher. In the later device the cell stack was kept in nitrogen, which is why the sealing becomes unnecessary and the cells lie directly on top of each other. This decreased the stack's extinction, since the losses resulting from scattering at the cover glasses no longer occur.

Cross-resistance of the transversal optical sensors:

A defined cross-resistance between the opposite sides of the square cell makes precise x,y resolution possible. The principle of x,y resolution is illustrated in Figures 3A to 3D. The cross-resistance across the area of the cell is determined by the PEDOT layer present between the p-type conductor and the silver electrodes bordering the cell. In the undoped state, PEDOT is a semiconductor. Conductivity is made possible by the system of conjugated double bonds extending across the entire molecule in combination with the doping with a negatively charged counter-ion. The PEDOTs used which were used for the present experiments were all doped with the negatively charged polymer polystyrene sulfonate (PSS). PEDOT:PSS is available in a wide range of embodiments as regards conductance, solid content, ionisation potential (IP), viscosity and pH.

Factors affecting the cross-resistance:

The PEDOT was also applied to the cells by spin coating. During the spinning process, the solvents ethanol and isopropanol evaporate whereas the low-volatility PEDOT remains on the substrate in the form of a film. The resistance of this layer depends on the conductance of the PEDOT being used and on the thickness of the layer:

$$R = \rho \cdot \frac{l}{A}$$

where ρ is the resistivity, l the distance across which the resistance is measured and A the cross-sectional area through which the charge carriers flow (A is a function of the PEDOT layer's thickness).

According to known principles of spin-coating, the layer thickness d to be expected when coating a non-Newtonian fluid can be determined by:

$$d = \sqrt[3]{\frac{3 \cdot x_s^3 \cdot u_k \cdot e}{2 \cdot (1 - x_s) \cdot \omega^2}}$$

$$\sim \sqrt[3]{\frac{x_s^3 \cdot u_k}{(1 - x_s) \cdot \omega^{3/2}}}$$

where x_s is the PEDOT percentage in the mixed diluted solution, u_k is the kinematic viscosity, e the evaporation rate of the solvent(s) and ω the angular velocity during spin coating. The evaporation rate is proportional to $\omega^{1/2}$.

The thickness of the PEDOT layer can, therefore, be affected by various parameters: the angular velocity, the viscosity of the PEDOT solution and the percentage of PEDOT in the solution. The angular velocity can be varied directly. The viscosity and the percentage of PEDOT in the solution can only be affected indirectly, i.e. via the ratio at which the PEDOT is mixed with ethanol and isopropanol.

The following parameters, therefore, can be used in order to adjust the cross-resistance, and they will be optimized in due course:

- The choice of PEDOT
- The layer thickness of the PEDOT
- The PEDOT/solvent ratio
- The spin speed during the PEDOT's spin coating
- The number of PEDOT layers
- The time interval Δt between applying and spin coating the PEDOT

Optimizing the cross-resistance:

The PEDOT solution was mixed with ethanol and isopropanol in a standard volumetric ratio of 1: 1 : 1, and larger particles removed with a 0.45 micrometer syringe filter. The entire cell was covered with this diluted PEDOT solution (approximately 900 microliters were needed per substrate) and spin coated at a speed of 2000 1/s. At this speed, 30 s turned out to suffice to remove and evaporate the solvents ethanol and isopropanol.

The aforementioned parameters were then varied systematically, with the objective of obtaining a cross-resistance of ca. 2 k Ω between the opposite electrodes of the square cell.

Choice of PEDOT:

The greatest impact on the PEDOT layer's cross-resistance turned out to come from the conductance of the PEDOT solution used. In order to obtain a first impression of the order of magnitude of the resistances of such a PEDOT layer across 1 cm, three PEDOT products with very different conductances were tested:

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- Clevios™ PVP AI 4083 from Heraeus
- Clevios™ PH 1000 from Heraeus
- Orgacon™ N-1005 from Sigma Aldrich

10 The relevant parameters of dynamic viscosity η_d , ionisation potential IP and resistivity ρ are summarised in Table 1. The IP is an important selection criterion for the PEDOT. The PEDOT's IP generally should be less than 5 eV in order to ensure good functionality of the cell.

	AI 4083	PH 1000	N-1005
η_d [mPas]	5 - 12	15 - 50	7 - 12
IP [eV]	5.2	4.8	N/A
ρ [Ωcm]	500 - 5000	0.0012	75 - 120

15

Table 1: Relevant parameters for various PEDOTs.

For these first tests, 1.3 micrometer np TiO₂ layers were coated onto FTO-free glass substrates. In this first experimental series, only 300 microliters of each of the three prepared PEDOT solutions were coated directly on np TiO₂ layers – without the staining or p-type conductor coating steps. For each PEDOT solution, three substrates with 1, 2 and 3 PEDOT layers were made. The resistance was measured by applying a layer of conducting silver paint at 1 cm spacing at several positions on each substrate.

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25 It is to be expected that the resistance of the substrate fabricated in this way, will be smaller than the resistance resulting from applying the PEDOT on a smooth p-type conductor layer.

As expected, the experiments showed a decrease in cross-resistance with an increasing number of layers and therefore increasing total thickness of the PEDOT. The cross-resistance of AI 4083 is in the M Ω range even with three layers, therefore it was not used in further tests. PH 1000 with two applied layers was in the required range. The cross-resistance of N 1005 is also in the k Ω range, and could be decreased through optimization. Since, however, it can be assumed that when applying the PEDOT on the smooth p-type conductor's surface the resistance will be higher than when applying it directly on the np TiO₂ layer as in this test series, further optimizations will focus on PH 1000.

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Applying several PEDOT layers:

A further option for increasing the total thickness of the PEDOT layer is to apply several PEDOT layers consecutively. Tests were conducted with 1 and 2 applied PEDOT layers. PH 1000 was mixed with ethanol and isopropanol at a 1 : 1 : 1 volumetric ratio. The cells were covered completely with 900 microliters of the PEDOT solution and the excess solution was removed by spin coating at an RPM of 2000 1/min.

Unlike the first experimental series, these tests were conducted on 'complete' cells, i.e. stained cells coated with p-type conductor. The cross-resistance was measured between two circular vapour-deposited electrodes ca. 2 mm apart, in order to exclude errors resulting from different contact resistances of PEDOT/conducting silver paint and PEDOT/vapour-deposited silver. In addition, cell efficiency measurements can be automated with this electrode arrangement. These test cells are very much simpler and quicker to make than the square transparent ones, but they suffice for the requirements of these tests (for one thing, the cross-resistance of the PEDOT layer(s) is to be measured here across a defined section; for another, the cells are to be tested for functionality, i.e. whether there is good contact between the p-type conductor and the PEDOT and the IP of the PEDOT matches the energy level of the p-type conductor). The cross-resistance across 1 cm is calculated using equation 3.1 – assuming equal layer thickness and therefore equal area A – through multiplication by the factor 5 (the resistivity of the solution is constant).

Table 2 shows the results of the resistance measurement between the two circular vapor-deposited back electrodes and the calculated cross-resistances across 1 cm for 1 and 2 PEDOT layers. The last column shows the efficiency of the cells. The smallest and largest measurements obtained in several tests are shown in each case.

Mixture ratio	Number of layers	RPM [1/min]	$R_{l=2mm}$ [k Ω]	$R_{l=1cm}$ [k Ω]	η [%]
1 : 1 : 1	1	2000	6 - 19	30 - 90	1.5 - 2.5
	2		0.18 - 0.27	0.9 - 1.35	0.4 - 0.6

Table 2: Results of resistance measurements for one and two PEDOT layers.

The difference in cross-resistances between one and two applied PEDOT layers can be seen clearly. For one PEDOT layer it is significantly higher than the required 2 k Ω , for two layers it is very much lower. It is also evident, however, that the efficiency of the cells with two applied PEDOT layers is very much smaller, which means that the contact between the two PEDOT layers is poor. It should be noted that the efficiency being measured here refers to the circular cell, i.e. to a cell with a back electrode vapor-deposited across the whole surface. The efficiency of the square transparent cells, therefore, will be much lower still, which is why the idea of several PEDOT layers applied consecutively has to be discarded. Further experiments will attempt to minimize the cross-resistance of one PEDOT layer only.

It is noticeable that the resistance with two applied PEDOT layers is smaller than in the first experimental series, in which the PEDOT was applied directly onto the np TiO₂ layer. Presumably, the reason for this discrepancy is that in the first experiments the layers were applied immediately after each other, even though the first PEDOT layer had not yet dried out completely. In this experiment, the cell was placed on the hotplate at 60 °C between applying the two layers.

As expected, in this experimental series – in which the PEDOT was applied onto the p-type conductor – the cross-resistance with only one applied PEDOT layer was higher than when the PEDOT was applied onto the rough surface of the np TiO₂.

Increasing the PEDOT concentration of the solution:

As mentioned above, usually, the PEDOT solution is mixed with ethanol and isopropanol at a volume ratio of 1 : 1 : 1 in order to decrease the viscosity of the solution and obtain homogeneous layers through spin coating. When the PEDOT proportion in the mixture is increased, the solution's viscosity rises. Due to the higher viscosity, an increase in the thickness of the PEDOT layer remaining on the cell after spin coating is expected (for comparison: η_d , ethanol, 20°C = 1.19 mPas; η_d , isopropanol, 20°C = 2.43 mPas; η_d , PEDOT = 5-50 mPas).

In order to investigate the practical effect of the PEDOT solution's viscosity and the amount of substance it contains on layer thickness and therefore on the cross-resistance, the PEDOT proportion was increased a little to begin with, and then significantly. The volumetric mixture ratios ethanol : isopropanol : PEDOT tested here were as follows:

1 : 1 : 1
1 : 1 : 2
1 : 1 : 5
1 : 1 : 10
2 : 2 : 1

Since a preliminary experiment showed that small variations in PEDOT concentration do not result in significant differences as far as the resistance is concerned, the PEDOT proportion in the mixed solutions was increased considerably. This was the first experimental series in which the construction of the cells and the arrangement of the electrodes corresponded to those of the actual square cells.

The thickness of the cells' np TiO₂ layer was 1.3 micrometers. Each time, a PEDOT layer with a different proportion of PH 1000 was applied. The PEDOT solution was spun for 90 seconds at 2000 or 1500 1/min. Then the PEDOT layer was dried for ca. 1 minute with the hot air blower, before the silver electrodes (ca. 2 micrometers thick) were vapor-deposited.

Mixture	Number	RPM	R
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ratio	of layers	[1/min]	[k Ω]
1 : 1 : 1	1	2000	22 - 39
1 : 1 : 5			31 - 51
1 : 1 : 10			29 - 36
1 : 1 : 1	1	1500	21 - 27
1 : 1 : 5			21 - 32
1 : 1 : 10			26 - 37

Table 3: Cross-resistances for various mixture ratios and spin speeds of PEDOT layers.

As can be seen in Table 3, the cross-resistance does not decrease as expected with the increasing PEDOT proportion in the applied solution. Both at an angular velocity of 2000 and of 1500 1/min, the cross-resistance increases with the increasing PEDOT concentration in the solution. However, it is also noticeable that the resistance tends to decrease with decreasing RPM for the same PEDOT proportion, but is still 10-15 orders of magnitude too high.

Adjusting the time interval between application and spin coating of the PEDOT (Δt) and minimizing the RPM during spin coating:

The classic method for increasing layer thickness during spin coating is to decrease the angular velocity. In this way, layer thickness can easily be increased and the cross-resistance decreased. In the experimental series thus far, this was the only variation that led to a sensible result. The angular velocity during spin coating cannot, however, be decreased to an arbitrary value, since at excessively low RPM the solvent no longer evaporates sufficiently quickly which results in inhomogeneous PEDOT layers.

Tests have shown, however, that the time interval between applying the PEDOT solution to the substrate and starting the spin coating (and thus, removing excess solution from the substrate) has a significant effect on the cross-resistance. Therefore, subsequently, the cross-resistance was minimized through iterative optimization of the two parameters Δt and angular velocity during spin coating.

Therefore, over several tests series, the time interval Δt between applying the PEDOT solution to the cell and the start of spin coating was increased in steps from 30 seconds to 2 minutes, later in combination with RPM optimization from 1 to 3 minutes and finally from 3.5 to 5 minutes. This involves decreasing the RPM from 2000 1/min to 350 1/min. When decreasing the RPM to under 1000 1/min, 30 seconds turned out to no longer suffice for complete evaporation of the solvent. This time was, therefore, extended to 2 minutes in every case. Thereafter, the cells were dried with the hot air blower for approximately 1 minute before the electrodes were vapor-deposited.

RPM [1/min] 2000 1000 750 600 500 450 400 350

The results of the optimization are summarized in Tables 4 to 7.

- 5 In the first experimental series of the final optimization (Table 4), in which to begin with the time interval Δt between applying the PEDOT solution on the cell and spinning at constant angular velocity was increased, at an RPM of 1000 1/min there seems to be an optimum at $\Delta t = 60$ s (4.1 - 4.2 k Ω). For this time interval and a further decrease in RPM, a new minimum was obtained for 600 1/min (2.6 - 2.7 k Ω).

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	1000	30	8.5 - 8.7
			60	4.1 - 4.2
			90	4.7 - 4.8
			120	5.4 - 5.8
		750	60	9.0 - 9.2
		600	60	2.6 - 2.7
		500	60	3.7 - 4.0

Table 4: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 1.

5

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	600	60	15.8 - 17.6
			90	11.0 - 11.6
			120	11.2 - 11.4
		500	60	6.0 - 6.8
			90	5.0
			120	6.4 - 6.8
			180	3.1 - 4.1

Table 5: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 2.

10

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	500	180	0.4
		450	180	0.6
		400	180	1.1 - 1.3
		350	180	2.0 - 2.7
		350	210	1.8 - 2.5
		350	240	1.6
		350	270	1.5 - 1.8
		350	300	1.8 - 1.9

Table 6: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 3.

5

Mixture ratio	Number of layers	RPM [1/min]	Δt [s]	R [k Ω]
1 : 1 : 1	1	500	180	1.3 - 2.4
		450	180	1.0 - 2.0

Table 7: Optimization of the cross-resistance by optimizing the time interval Δt and the angular velocity during PEDOT solution spin coating – experimental series 4.

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Since, however, the results do not differ substantially between 600 and 500 1/min, in the next experimental series the time interval Δt was increased further stepwise for both RPM values. The results are shown in Table 5.

15 A further decrease in RPM and increase in Δt did not exhibit further improvement. In fact, the cross-resistance even increased again at an RPM < 450 1/min (see Table 6).

Since the values for 500 and 450 1/min lie very close together, a last comparative test was performed (see Table 7).

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It was shown that the cross-resistance at an RPM of 450 1/min is slightly smaller than at 500 1/min. Since, however, no significant is achieved thereby and since PEDOT layers coated at an excessively low RPM are no longer homogeneous, 500 1/min was chosen as the optimal RPM. The time interval Δt is then 180 s.

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Generally speaking, it is noticeable that the resistance values in the last set of experimental series, where there was a time interval between applying the PEDOT solution to the cell and starting the spinning, do not fluctuate as much within one series for constant parameters as

before. In the last experimental series (Table 7), the two cross-resistances were measured on each of four cells (left-right and top-bottom) and the results varied only by ca. 1 k Ω . The fact that in some cases the results from different series vary significantly for the same experimental parameters, is likely to be due to the production of the PEDOT solution since the individual experimental series in themselves provide coherent results.

The spin coater's open lid during the time interval Δt may be an important interference factor in these experiments. In one experimental series, it was not closed immediately after applying the PEDOT solution to the cell but only before spin coating. The cross-resistances measured in this experimental series are very much higher and there is a considerable variation in their values between the substrates – but not on any one substrate. It could not be determined exactly why the cross-resistance is so strongly influenced by the time interval Δt before the PEDOT solution's spin coating. Perhaps some of the PEDOT solution dries and adheres to the cell during this time, resulting in a thicker PEDOT layer.

Results of the optimized parameters:

The minimum cross-resistance obtained in this way lies between 1 and 3 k Ω . The parameters that bring about a minimum cross-resistance here were:

- PEDOT: Clevios[®] PH 1000 from Heraeus
- Number of layers: 1
- PEDOT : ethanol : isopropanol ratio = 1 : 1 : 1
- Time interval between applying and spin coating the PEDOT: $\Delta t = 180$ s
- RPM during PEDOT spin coating: $n = 500$ 1/min ($t = 120$ s)

The final cells used in the experiments:

The cells used during the optimization process so far were made on thick 2.5 mm TEC 8 glass carriers, with the FTO layer already applied during the fabrication process. They have a highly homogeneous FTO layer, onto which the application of homogeneous np TiO₂ layers is possible. This makes possible the fabrication of cells that appear homogeneous to the human eye.

For the technical realization of the sensor stack, however, the cells were made on thin 1 mm special glass carriers made of quartz glass, which were coated subsequently with FTO. Therein, loss carriers with beveled edges were used. The beveled edges served as bases for cells contacts. The silver contacts were vapor-deposited up to the edge of the bevels. This made it possible to contact the cells directly adjacent to each other in the stack individually with pins.

The subsequently applied FTO layer on these special carriers exhibited in part inhomogeneities arising from the fabrication process. The fabrication of homogeneous cells on these carriers

turned out to be very difficult, as was shown by generating current diagrams of the final cells. Even in the first cell of the stack which supplied a homogeneous current signal across its entire area, four locations were identified that supplied lower currents due to inhomogeneities. Current diagrams were obtained by exciting the cells with a laser at a wavelength 690 nm. The laser scanned the cells at 1 mm intervals. The cells were scanned in their final arrangement as a stack, i.e. the current diagram of the last cells was recorded with five 'thin' cells located in front of the last cell.

At the excitation wavelength of 690 nm, the developed cells had an extinction of 0.13. At the maximum (at ca. 550 nm) the extinction of these cells was ca. 0.4. Despite this low absorption by the cells and the fact that the back electrode consists of a poorly conducting – unlike silver – transparent layer, the efficiency of such a cell is still ~0.3% (AM 1.5*), that of the last cell ~2%.

Figures 2A, 4B and 4C show the final cells on the 1 mm special glass carriers. The first cell in the stack, which acts as the transversal optical sensor, requires the special electrode arrangement for x,y-resolution. For cells 2-5, which form the longitudinal optical sensor stack, only the total current is needed for z-resolution, which is why the contacting silver electrodes are combined here into one electrode surrounding the cell. Otherwise, however, the first five cells were made identically.

The last cell of the longitudinal optical sensor stack is intended to absorb the remaining light preferably completely, which is why it was chosen to have a significantly higher extinction than the front cells. In addition, it has a back electrode covering its entire area in order to supply maximum output current.

The cell shown in Fig. 2A, which forms the transversal optical sensor, in this experiment, was used only once for x,y resolution at position 1 in the stack. The cells in Fig. 4B were used four times in the entire stack of the optical sensors, i.e. for positions 2-5 of the entire stack. The last cell, which is depicted in Fig. 4C, was used at position 6 of the entire stack of optical sensors. Thus, generally, a stack of optical sensors was formed with the first optical sensor being the transversal optical sensor (Fig. 2A), followed by four transparent longitudinal optical sensors (Fig. 4B) and a last longitudinal optical sensor having the set-top of Fig. 4C.

When illuminating one of these single transparent final cells with a red laser (690 nm, 1 mW), this cell supplied a current of 30–40 microamps. The last longitudinal optical sensor provided a current of approximately 70 μ A. The cross-resistance between any two opposite electrodes in the first cell on these special glass carriers turned out to be 0.1 and 0.3 k Ω .

Since the fabrication of transparent cells on the special glass carriers was problematic due to the poor FTO coating, these cells had to be fabricated in large numbers. The cells were screened, and only selected cells were used for the final setup of the detector forming a prototype 3-D sensor. For this screening procedure, specifically for the transversal optical

sensor, the cells were excited by a laser beam (690 nm, 1 mW) at the center of the cell. If the cells are homogenous, the currents at all four contacts are equal ($I_1 = I_2 = I_3 = I_4$). By comparing the currents, specific cells were selected for use in the prototype.

- 5 The x,y resolution achieved with the detector of the set-top turned out to be approximately 1 mm at a distance of 3 m. The z-resolution of this detective setup turned out to be approximately 1 cm.

List of reference numbers

	110	detector
5	111	camera
	112	object
	114	optical sensors
	116	optical axis
	118	housing
10	120	transfer device
	122	lens
	124	opening
	126	direction of view
	128	coordinate system
15	130	transversal optical sensor
	132	longitudinal optical sensor
	134	longitudinal optical sensor stack
	136	sensor region
	138	light beam
20	140	transversal signal lead
	142	evaluation device
	144	last longitudinal optical sensor
	146	longitudinal signal leads
	148	transversal evaluation unit
25	150	longitudinal evaluation unit
	152	position information
	154	data processing device
	156	transformation unit
	157	imaging device
30	158	substrate
	159	imaging device signal lead
	160	first electrode
	161	optically sensitive element
	162	blocking layer
35	163	color wheel
	164	n-semiconducting metal oxide
	166	dye
	168	p-semiconducting organic material
	170	second electrode
40	172	encapsulation
	174	electrode contact
	176	partial electrode

	178	partial electrode, x
	180	partial electrode, y
	182	contact leads
	184	light spot
5	186	image
	188	electrically conductive polymer
	190	top contact
	192	illumination source
	194	focal point
10	196	human-machine interface
	198	entertainment device
	199	tracking system
	200	user
	201	track controller
15	202	machine
	204	beacon device
	206	primary light beam
	208	display
	210	keyboard
20		

Patent claims

1. A detector (110) for determining a position of at least one object (112), comprising:
 - 5 - at least one transversal optical sensor (130), the transversal optical sensor (130) being adapted to determine a transversal position of at least one light beam (138) traveling from the object (112) to the detector (110), the transversal position being a position in at least one dimension perpendicular an optical axis (116) of the detector (110), the transversal optical sensor (130) being adapted to generate at least one transversal sensor signal;
 - 10 - at least one longitudinal optical sensor (132), wherein the longitudinal optical sensor (132) has at least one sensor region (136), wherein the longitudinal optical sensor (132) is designed to generate at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region (136) by the light beam (138), wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam (138) in the sensor region (136);
 - 15 - at least one evaluation device (142), wherein the evaluation device (142) is designed to generate at least one item of information on a transversal position of the object (112) by evaluating the transversal sensor signal and to generate at least one item of information on a longitudinal position of the object (112) by evaluating the longitudinal sensor signal.
 - 20
2. The detector (110) according to the preceding claim, wherein the transversal optical sensor (130) is a photo detector having at least one first electrode (160), at least one second electrode (170) and at least one photovoltaic material (164, 166, 168), wherein the photovoltaic material (164, 166, 168) is embedded in between the first electrode (160) and the second electrode (170), wherein the photovoltaic material (164, 166, 168) is adapted to generate electric charges in response to an illumination of the photovoltaic material (164, 166, 168) with light, wherein the second electrode (170) is a split electrode having at least two partial electrodes (176), wherein the transversal optical sensor (130) has a sensor region (136), wherein the at least one transversal sensor signal indicates a position of the light beam (138) in the sensor region (136).
 - 25
 - 30
3. The detector (110) according to the preceding claim, wherein electrical currents through the partial electrodes (176) are dependent on a position of the light beam (138) in the sensor region (136), wherein the transversal optical sensor (130) is adapted to generate the transversal sensor signal in accordance with the electrical currents through the partial electrodes (176).
 - 35
4. The detector (110) according to the preceding claim, wherein the detector (110) is adapted to derive the information on the transversal position of the object (112) from at least one ratio of the currents through the partial electrodes (176).
 - 40

5. The detector (110) according to any of the three preceding claims, wherein the photo detector is a dye-sensitized solar cell.
- 5 6. The detector (110) according to any of the four preceding claims, wherein the first electrode (160) at least partially is made of at least one transparent conductive oxide, wherein the second electrode (170) at least partially is made of an electrically conductive polymer (188), preferably a transparent electrically conductive polymer (188).
- 10 7. The detector (110) according to any of the preceding claims, wherein at least one of the transversal optical sensor (130) and the longitudinal optical sensor (132) is a transparent optical sensor.
- 15 8. The detector (110) according to any of the preceding claims, wherein the transversal optical sensor (130) and the longitudinal optical sensor (132) are stacked along the optical axis (116) such that a light beam (138) travelling along the optical axis (116) both impinges on the transversal optical sensor (130) and on the longitudinal optical sensor (132).
- 20 9. The detector (110) according to any of the preceding claims, wherein the longitudinal optical sensor (132) comprises at least one dye-sensitized solar cell.
- 25 10. The detector (110) according to the preceding claim, wherein the longitudinal optical sensor (132) comprises at least one first electrode (160), at least one n-semiconducting metal oxide (164), at least one dye (166), at least one p-semiconducting organic material (168), preferably a solid p-semiconducting organic material (168), and at least one second electrode (170).
- 30 11. The detector (110) according to the preceding claim, wherein both the first electrode (160) and the second electrode (170) are transparent.
- 35 12. The detector (110) according to any of the preceding claims, wherein the evaluation device (142) is designed to generate the at least one item of information on the longitudinal position of the object (112) from at least one predefined relationship between the geometry of the illumination and a relative positioning of the object (112) with respect to the detector (110).
13. The detector (110) according to any of the preceding claims, furthermore comprising at least one illumination source (192).
- 40 14. The detector (110) according to any of the preceding claims, wherein the detector (110) has a plurality of longitudinal optical sensors (132), wherein the longitudinal optical sensors (132) are stacked.

15. The detector (110) according to the preceding claim, wherein the longitudinal optical sensors (132) are arranged such that a light beam (138) from the object (112) illuminates all longitudinal optical sensors (132), wherein at least one longitudinal sensor signal is generated by each longitudinal optical sensor (132), wherein the evaluation device (142) is adapted to normalize the longitudinal sensor signals and to generate the information on the longitudinal position of the object (112) independent from an intensity of the light beam (138).
16. The detector (110) according to any of the two preceding claims, wherein a last longitudinal optical sensor (144) is arranged in a manner that the light beam (138) illuminates all other longitudinal optical sensors (132) apart from the last longitudinal optical sensor (144), until the light beam (138) impinges on the last longitudinal optical sensor (144), wherein the last longitudinal optical sensor (144) is intransparent with respect to the light beam (138).
17. The detector (110) according to any of the three preceding claims, wherein the stack of at least two optical sensors is partially or fully be immersed in an oil, in a liquid and/or in a solid material.
18. The detector (110) according to any of the preceding claims, wherein the at least one transversal optical sensor and/or the at least one longitudinal optical sensor uses at least two different transparent substrates.
19. The detector (110) according to any of the preceding claims, wherein the detector (110) further comprises at least one imaging device (157).
20. The detector (110) according to any of the preceding claims, wherein the detector (110) further comprises at least one optically sensitive element (159).
21. The detector (110) according to any of the preceding claims, wherein the optically sensitive element (161) comprises a color wheel (163), a color drum, and/or a filter wheel using an elliptically polarizing filter.
22. The detector (110) according to any of the preceding claims, wherein the evaluation device (142) is adapted to generate the at least one item of information on the longitudinal position of the object (112) by determining a diameter of the light beam (138) from the at least one longitudinal sensor signal.
23. The detector (110) according to the preceding claim, wherein the evaluation device (142) is adapted to compare the diameter of the light beam (138) with known beam properties of

the light beam (138) in order to determine the at least one item of information on the longitudinal position of the object (112).

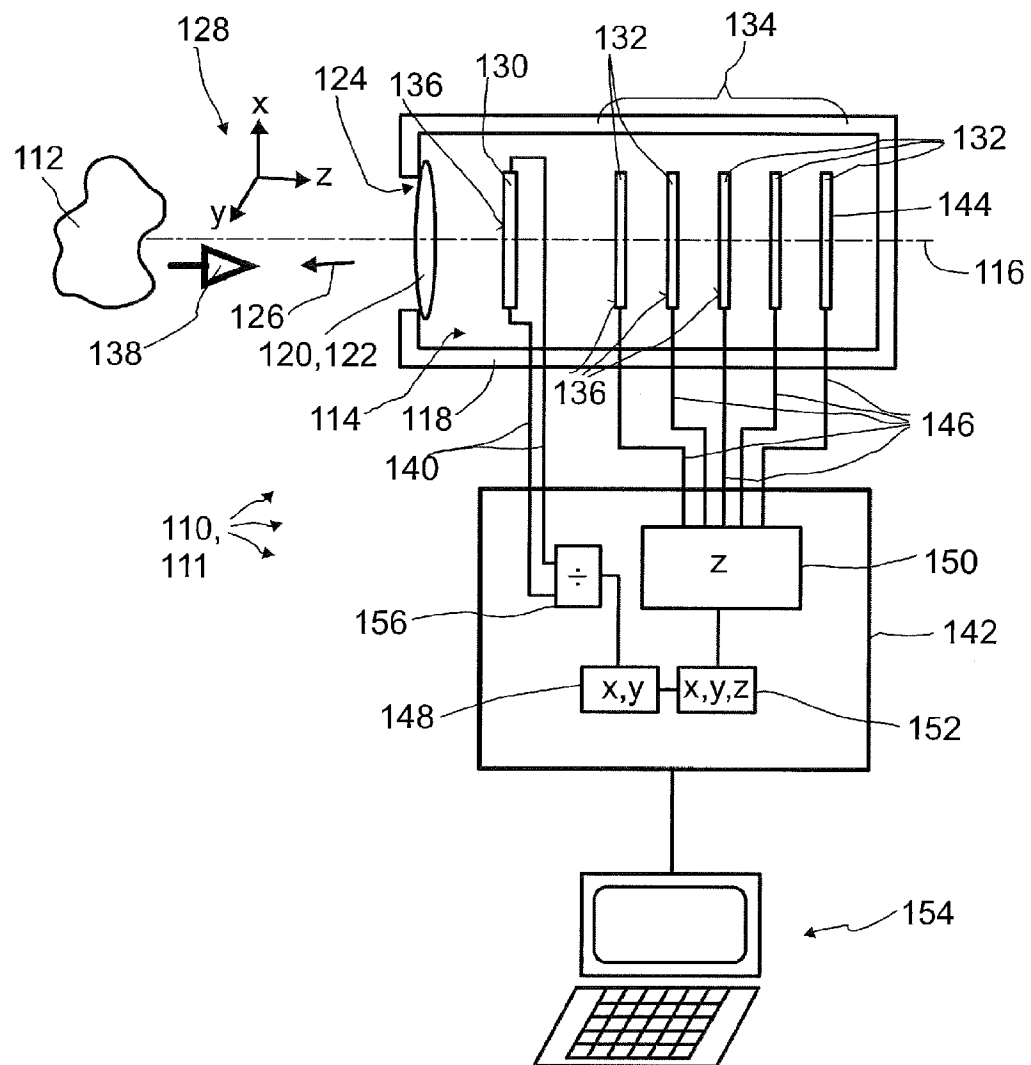
24. The detector (110) according to any of the preceding claims, wherein the longitudinal optical sensor (132) is furthermore designed in such a way that the longitudinal sensor signal, given the same total power of the illumination, is dependent on a modulation frequency of a modulation of the illumination.
25. A human-machine interface (196) for exchanging at least one item of information between a user (200) and a machine, wherein the human-machine interface (196) comprises at least one detector (110) according to any of the preceding claims relating to a detector (110), wherein the human-machine interface (196) is designed to generate at least one item of geometrical information of the user (200) by means of the detector (110) wherein the human-machine interface (196) is designed to assign to the geometrical information at least one item of information.
26. An entertainment device (198) for carrying out at least one entertainment function, wherein the entertainment device (198) comprises at least one human-machine interface (196) according to any of the preceding claims referring to a human-machine interface (196), wherein the entertainment device (198) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (196), wherein the entertainment device (198) is designed to vary the entertainment function in accordance with the information.
27. A tracking system (199) for tracking the position of at least one movable object (112), the tracking system (199) comprising at least one detector (110) according to any of the preceding claims referring to a detector (110), the tracking system (199) further comprising at least one track controller (201), wherein the track controller (201) is adapted to track a series of positions of the object (112), each position comprising at least one item of information on a transversal position of the object (112) at a specific point in time and at least one item of information on a longitudinal position of the object (112) at a specific point in time.
28. A camera (111) for imaging at least one object (112), the camera (111) comprising at least one detector (110) according to any one of the preceding claims referring to a detector (110).
29. A method for determining a position of at least one object (112),
- wherein at least one transversal optical sensor (130) of a detector (110) is used, wherein the transversal optical sensor (130) determines a transversal position of at least one light beam (138) traveling from the object (112) to the detector (110), the transversal position being a position in at least one dimension perpendicular to an optical axis (116) of the

- detector (110), wherein the transversal optical sensor (130) generates at least one transversal sensor signal;
- wherein at least one longitudinal optical sensor (132) of the detector (110) is used, wherein the longitudinal optical sensor (132) has at least one sensor region (136), wherein the longitudinal optical sensor (132) generates at least one longitudinal sensor signal in a manner dependent on an illumination of the sensor region (136) by the light beam (138), wherein the longitudinal sensor signal, given the same total power of the illumination, is dependent on a beam cross-section of the light beam (138) in the sensor region (136);
 - wherein at least one evaluation device (142) is used, wherein the evaluation device (142) generates at least one item of information on a transversal position of the object (112) by evaluating the transversal sensor signal and wherein the evaluation device (142) further generates at least one item of information on a longitudinal position of the object (112) by evaluating the longitudinal sensor signal.
30. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use, selected from the group consisting of: a distance measurement, in particular in traffic technology; a position measurement, in particular in traffic technology; a tracking application, in particular in traffic technology.
31. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as an entertainment application.
32. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as a camera (111), in particular in a security application.
33. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as a human-machine interface (196) application.
34. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use as a mapping application, in particular for generating maps of at least one space.
35. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in automated machine processes, selected from the group consisting of: a distance measurement; a position measurement; a tracking application.
36. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in high-precision metrology, in particular in analytics.

37. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in modeling of manufacturing parts.
 38. The use of a detector (110) according to any of the preceding claims relating to a detector (110), for a purpose of use in medical operations, in particular in endoscopic methods.
- 5

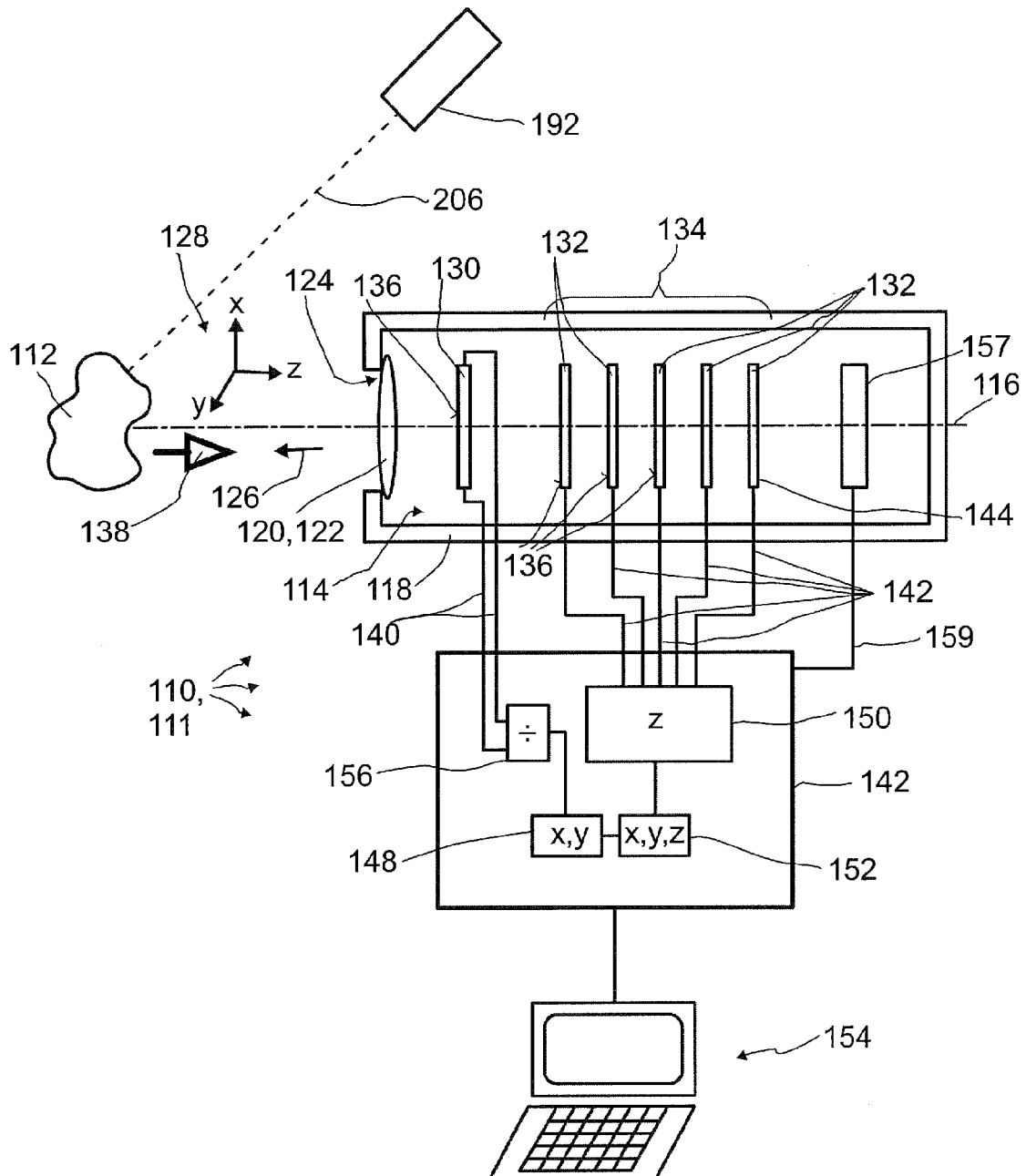
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FIG. 1A



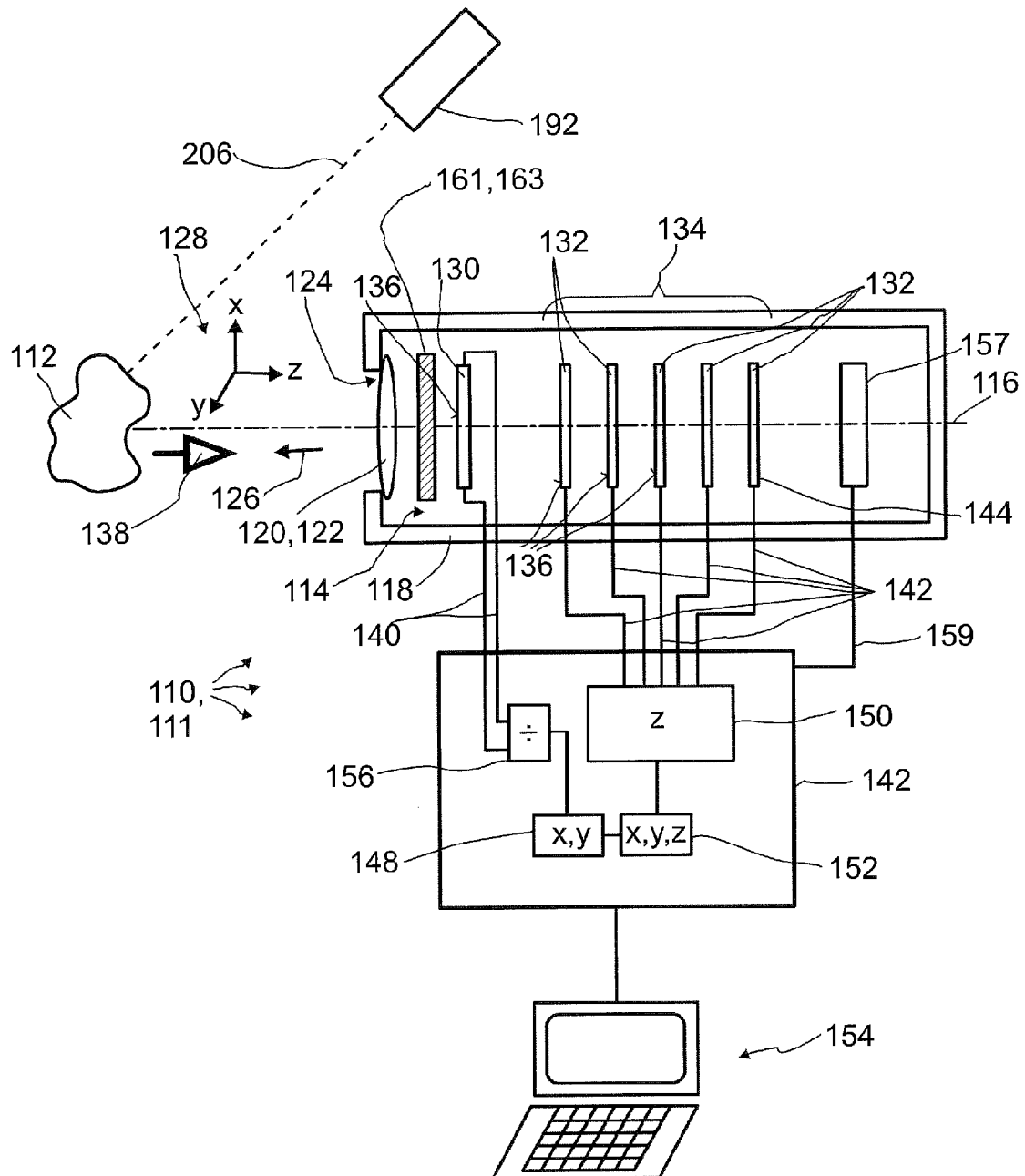
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FIG.1B



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FIG.1C



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FIG.2A

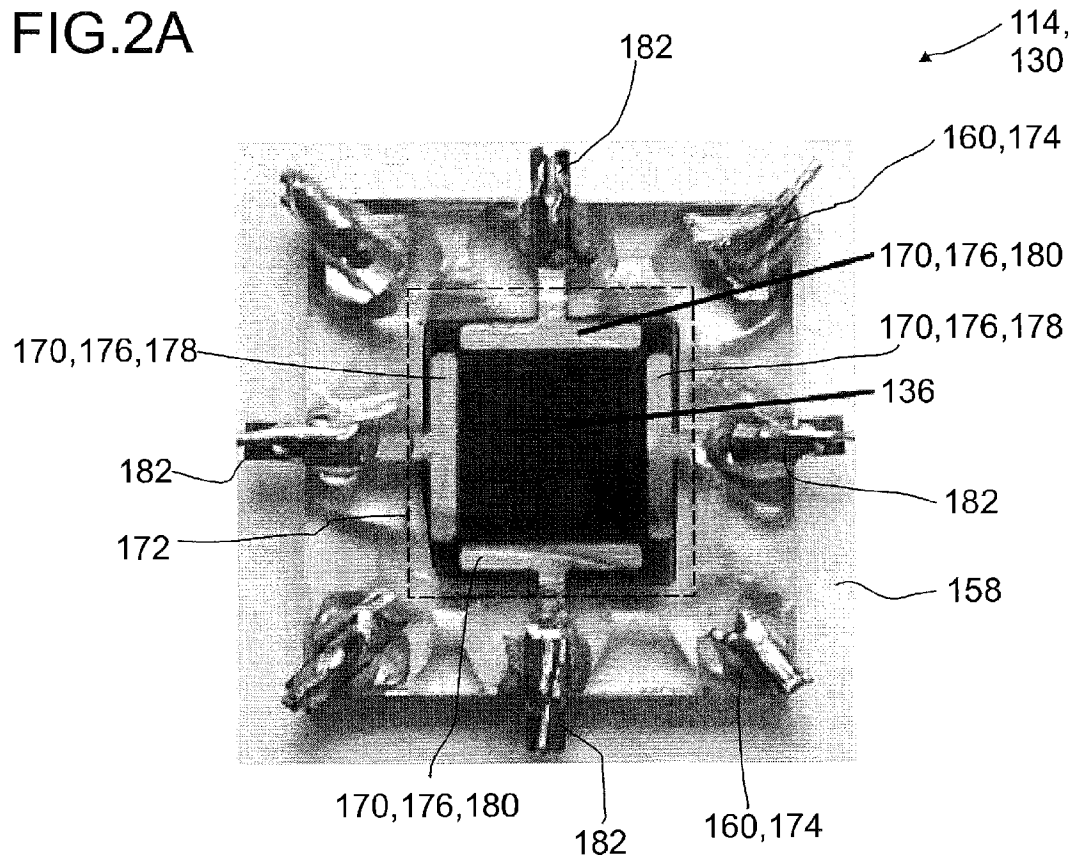
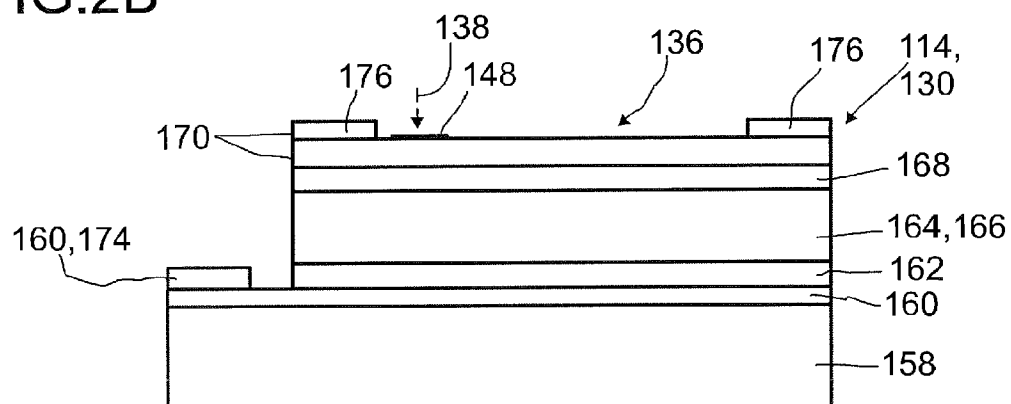


FIG.2B



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FIG.3A

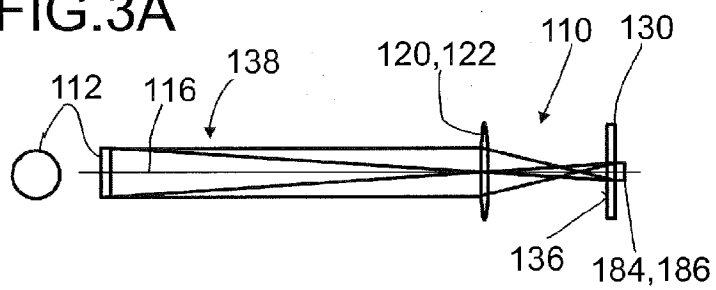


FIG.3B

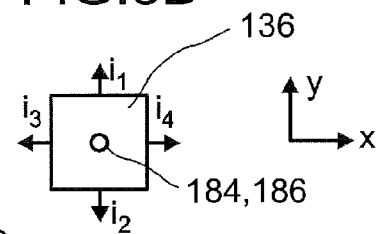


FIG.3C

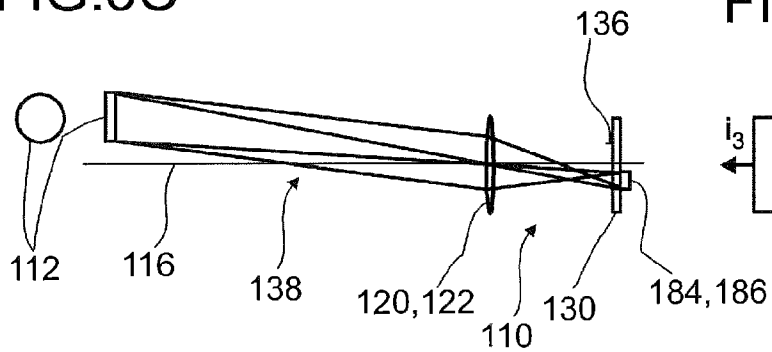
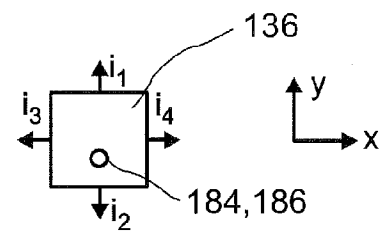


FIG.3D



6/8

FIG.4A

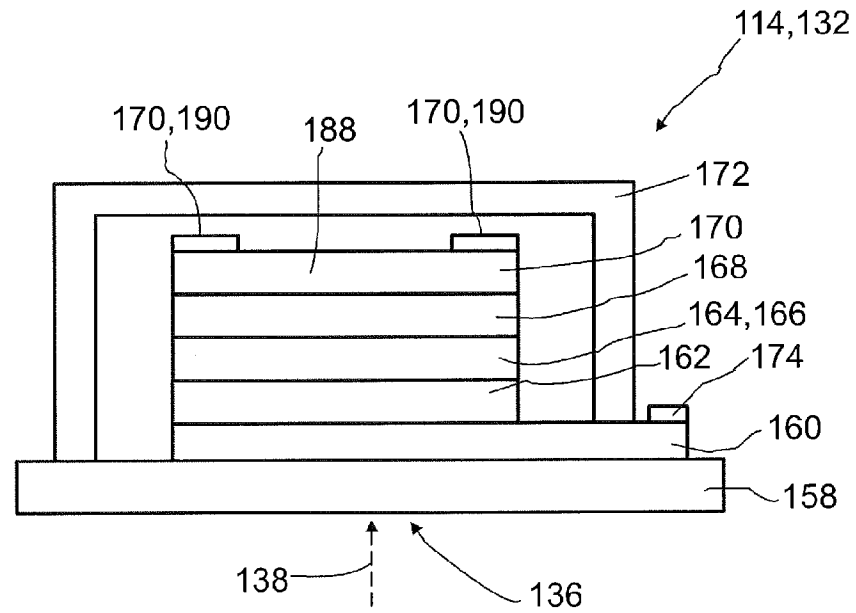


FIG.4B

FIG.4C

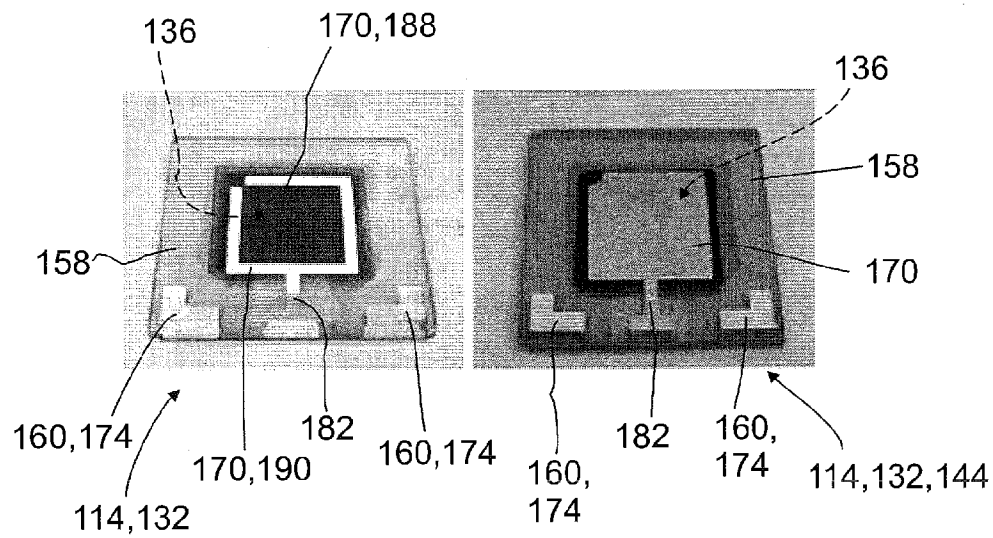


FIG.5A

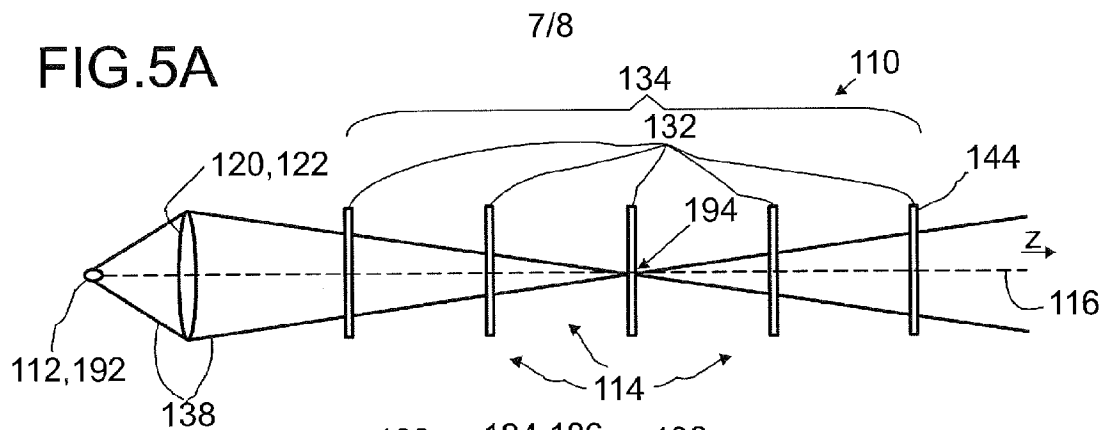


FIG.5B

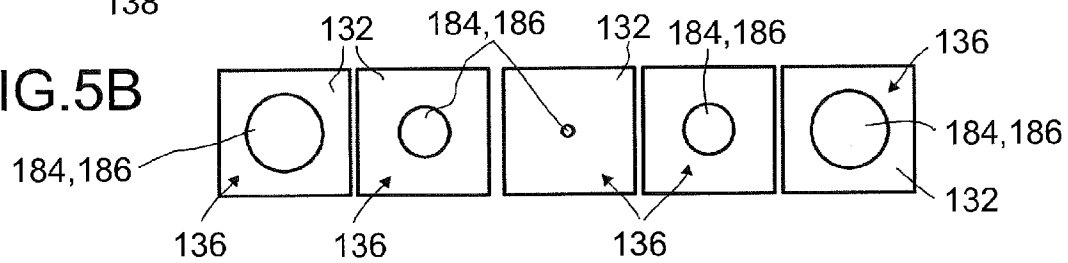


FIG.5C



FIG.5D

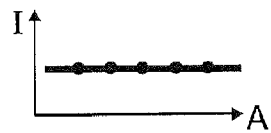


FIG.5E

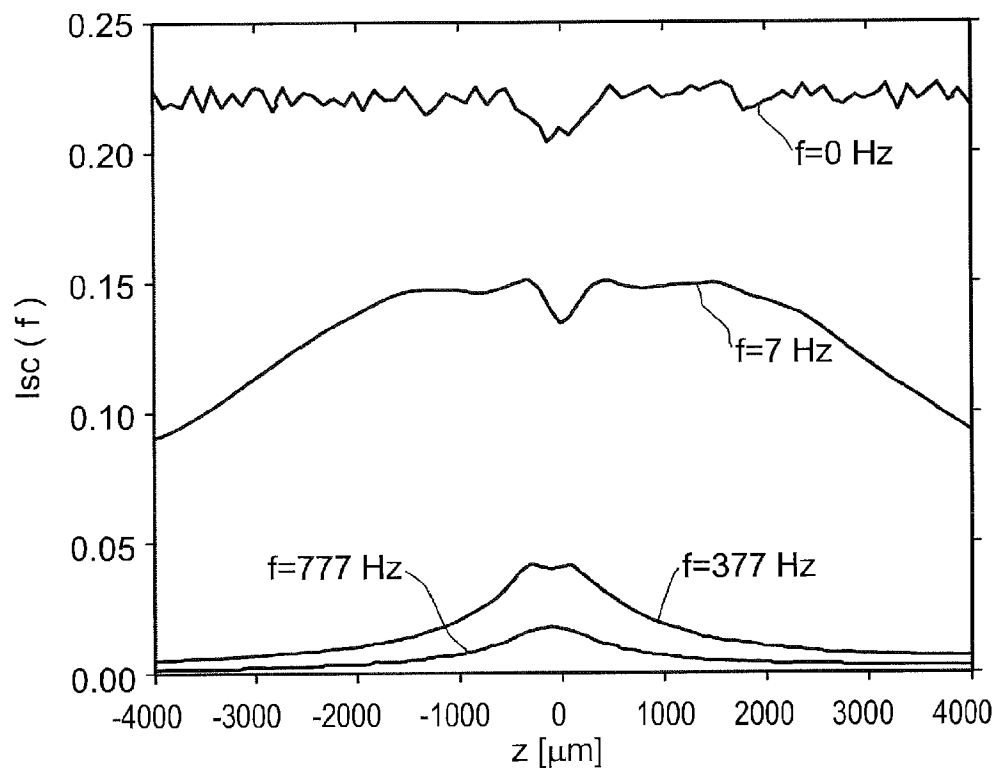
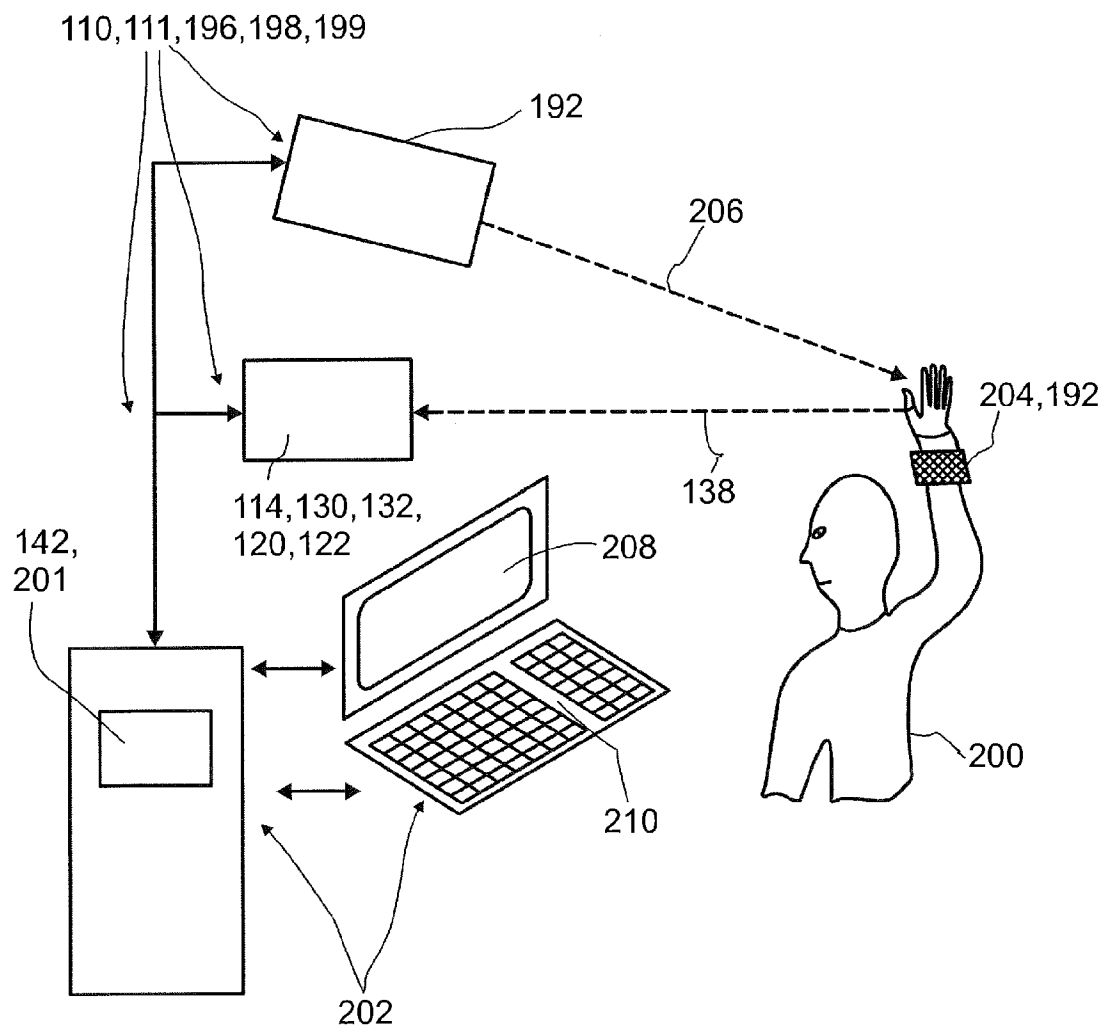


FIG. 6



INTERNATIONAL SEARCH REPORT

International application No.

PCT/IB2013/061095

A. CLASSIFICATION OF SUBJECT MATTER

G01C 3/06(2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G01C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

WPI;EPODOC;CNKI;IEEE;CNPAT:measur+, positon, coordinate?, depth, distance, transver+, cross, longitudinal, length+ w direction, sensor?, beam, optical, light, luminous, spot?, speckle?, facula?, focus+

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	US 2008259310A1 ((WADA, HIDEO)) 23 October 2008 (2008-10-23) description, paragraphs [0085]-[0086], [0105]-[0108]	1-38
A	CN 1896686A ((TOPCON CORPORATION)) 17 January 2007 (2007-01-17) the whole document	1-38
A	CN 102549381A ((ROBERT BOSCH GMBH)) 04 July 2012 (2012-07-04) the whole document	1-38
A	US 2009097010A1 ((YAMAGUCHI, AKIFUMI)) 16 April 2009 (2009-04-16) the whole document	1-38
A	US 2011194097A1 ((YAMAGUCHI, AKIFUMI ET AL.)) 11 August 2011 (2011-08-11) the whole document	1-38
A	JP 2005241340A ((SHARP K.K.)) 08 September 2005 (2005-09-08) the whole document	1-38

☐ Further documents are listed in the continuation of Box C.☒ See patent family annex.

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“E” earlier application or patent but published on or after the international filing date	“X” document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
“L” document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)	“Y” document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
“O” document referring to an oral disclosure, use, exhibition or other means	“&” document member of the same patent family
“P” document published prior to the international filing date but later than the priority date claimed	

Date of the actual completion of the international search

21 April 2014

Date of mailing of the international search report

16 May 2014

Name and mailing address of the ISA/

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No.

PCT/IB2013/061095

Patent document cited in search report		Publication date (day/month/year)	Patent family member(s)		Publication date (day/month/year)
US	2008259310A1	23 October 2008	CN	101290348A	22 October 2008
			JP	2008267898A	06 November 2008
CN	1896686A	17 January 2007	EP	1744122A2	17 January 2007
			JP	2007017406A	25 January 2007
			EP	1760430A2	07 March 2007
			US	2011096319A1	28 April 2011
			CN	1896684A	17 January 2007
			US	2007008515A1	11 January 2007
			US	2007010924A1	11 January 2007
			JP	2007248156A	27 September 2007
CN	102549381A	04 July 2012	WO	2011029645A1	17 March 2011
			DE	102009029372A1	24 March 2011
			EP	2475958A1	18 July 2012
			US	2012249998A1	04 October 2012
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			CN	101408622A	15 April 2009
US	2011194097A1	11 August 2011	JP	2011163901A	25 August 2011
			CN	102192723A	21 September 2011
			TW	201140111A	16 November 2011
JP	2005241340A	08 September 2005	CN	1677053A	05 October 2005
			US	2005184301A1	25 August 2005

INFORMATION DISCLOSURE STATEMENT BY APPLICANT (Not for submission under 37 CFR 1.99)	Application Number		17439492
	Filing Date		2021-09-15
	First Named Inventor	Schick, Friedrich	
	Art Unit	2454	
	Examiner Name	Frantz Jean	
	Attorney Docket Number	27843-1809 182037US01	

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	1	19846619	DE	A1	2000-04-27	BYK GARDNER GMBH	Machine Translation Appended	×
	2	108363482	CN	A	2018-08-03	JIANGSU SIDIANLING ROBOT CO LTD	Machine Translation Appended	☒
	3	2019042956	WO	A1	2019-03-07	TRINAMIX GMBH		☐

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17439492
Filing Date	2021-09-15
First Named Inventor	Schick, Friedrich
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Examiner Name	Frantz Jean
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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report and Written Opinion for corresponding PCT/EP2020/056759 mailed May 29, 2021, 11 Pages	
	2	R.A. STREET: Technology and Applications of Amorphous Silicon, Springer-Verlag Heidelberg, 2010, pp. 346-349	

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EXAMINER SIGNATURE

Examiner Signature	/FRANTZ B JEAN/	Date Considered	09/08/2023
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*EXAMINER: Initial if reference considered, whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**
(Not for submission under 37 CFR 1.99)

Application Number	17439492
Filing Date	2021-09-15
First Named Inventor	Schick, Friedrich
Art Unit	2454
Examiner Name	Frantz B Jean
Attorney Docket Number	27843-1809 182037US01

CERTIFICATION STATEMENT

Please see 37 CFR 1.97 and 1.98 to make the appropriate selection(s):

That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

See attached certification statement.

The fee set forth in 37 CFR 1.17 (p) has been submitted herewith.

☒ A certification statement is not submitted herewith.

SIGNATURE

A signature of the applicant or representative is required in accordance with CFR 1.33, 10.18. Please see CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2021-09-15
Name/Print	Christopher M. Goff	Registration Number	41785

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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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(12)发明专利申请

(10)申请公布号 CN 108363482 A

(43)申请公布日 2018.08.03

(21)申请号 201810025968.8

H04N 21/422(2011.01)

(22)申请日 2018.01.11

H04N 21/4223(2011.01)

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代理人 金方玮 董建林

(51)Int.Cl.

G06F 3/01(2006.01)

G06K 9/00(2006.01)

G06K 9/40(2006.01)

G06K 9/46(2006.01)

G06K 9/62(2006.01)

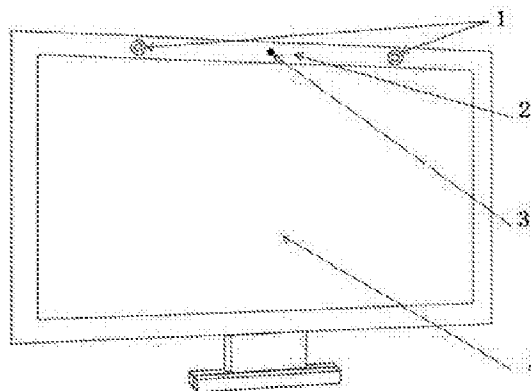
权利要求书2页 说明书5页 附图1页

(54)发明名称

一种基于双目结构光的三维手势控制智能电视的方法

(57)摘要

本发明公开了一种基于双目结构光的三维手势控制智能电视的方法,包括以下步骤:首先通过双目摄像头同步采集图像,根据获得的左右视图重建出具有深度信息的三维图像,并进行预处理,分割出三维手势后,再对手势动作进行识别,将识别出的手势动作转换为智能电视的操作指令,并执行。本发明采用双目视觉和结构光结合的方式,可以有效地提高环境抗干扰能力、降低光照条件的便会对指尖定位的影响;满足控制智能电视的实时性和有效性。



1. 一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,包括如下步骤:

步骤一,采集图像:通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片;

步骤二,重建图像三维:根据双目匹配算法,找出左视图、右视图之间各像素点的一一对应关系,根据他们之间的像素位置的横坐标差得到视差图,然后根据投影模型与三角测量原理,获得原始图像的深度信息;

步骤三,预处理深度图像:对深度图像进行中值滤波去除噪声点,再对深度图像中的内部空洞进行形态学处理;

步骤四,检测手势图像:对获得的图像深度数据进行分析,将人和背景分离开来,只选取人体的像素信息,将其余像素信息全部置零,获得人体的深度图像,最后获得手势的轮廓;

步骤五,识别三维手势:根据获得的手势轮廓,对手势轮廓提取中心距特征,中心距是指图形中必点到各边缘点的距离;设点 $O(x_0, y_0)$ 为手势的几何中心点,点集 $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ 为手势的边缘轮廓,不妨把手势的边缘轮廓看成有一个个边缘点连接而成,由此,几何中心点 $O(x_0, y_0)$ 和中心距 d 可由如下公式计算:

$$x_0 = \frac{1}{N} \sum_{i=1}^N x_i;$$

$$y_0 = \frac{1}{N} \sum_{i=1}^N y_i;$$

$$d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2};$$

再结合三维信息,利用基于贝叶斯网络的分类器进行识别;

步骤六,执行操作指令:根据在双目摄像头前做出的手势,智能电视执行相应的预先设定的操作指令,完成人机交互。

2. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,步骤一,采集图像:通过红外探测器探测操作环境的光照情况,若红外光过多则打开红外激光器;再通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片。

3. 根据权利要求2所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,若红外探测器探测操作环境的红外辐射强度小于 $0.2\text{W}/\text{m}^2 \cdot 850\text{nm}$,则打开红外激光器。

4. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,上述步骤三中的形态学处理即对图像边缘存在的毛刺和图像的内部空洞进行开、闭运算。

5. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法,其特征在于,上述步骤四中获得手势的轮廓的具体步骤为:通过像素的灰度值反映改点距离镜头的远近,统计出人体的深度图像的灰度分布直方图,灰度直方图较小的峰是手部区域的灰度分布,较大的峰是人体躯干的灰度分布,从而分割出人体的手部图像,再对手部图像使用Canny边缘算了进行边缘检测,从而获得手势的轮廓。

6. 根据权利要求1所述的一种基于双目结构光的三维手势控制智能电视的方法, 其特征在于, 上述步骤五中利用基于贝叶斯网络的分类器进行识别的具体步骤如下:

步骤a, 计算不同手势类别下所有特征的高斯分布信息, 即是计算所有的均值与方差;

步骤b, 对于测试的每一个输入样本, 求出它在手势1类别下, 各个特征的概率, 由于每个特征是相互独立的, 因而特征下的概率分布也满足独立性, 由此便可以得出该输入样本在手势1类别下的总概率; 以此类推, 求出该样本在所有手势类别下的不同概率;

步骤c, 比较输入样本在不同类别下得到概率值, 找出其最大概率值所属的类别, 便预测该类别就是输入样本所属的类别。

一种基于双目结构光的三维手势控制智能电视的方法

技术领域

[0001] 本发明涉及人机交互领域,特别是一种手势控制的方法。

背景技术

[0002] 随着互联网技术的发展,智能电视——一种具备开放式操作系统与芯片,拥有开放式应用平台,可实现双向人机交互功能,集影音、娱乐、数据等多种功能于一体,以满足用户多样化和个性化需求的电视产品成为了电视的潮流趋势。

[0003] 传统的按键式遥控器主要是由集成电路电板和用来产生不同讯息的按钮所组成。内装有一个中央处理器CPU,CPU在制造时就将电视机各种菜单密码信息输入其中,电视机的遥控发射器只要发出与之对应的密码就可以实现电视机的遥控了。

[0004] 智能电视的发展对电视的交互操作提出了更多的要求。首先便是需要简洁方便的文字输入、网页浏览和频道选取,传统的遥控器显然无法满足新的交互需求。如果在智能电视上外接鼠标和键盘,用户则需要配置多个控制终端,且无法在远距离(3~5米)和复杂环境下(鼠标、键盘需要相对平坦的桌面)完成交互,给用户操作带来了很多的障碍。这也是目前电视的人机交互落后于计算机和手持设备的原因。

[0005] 日前,基于视觉的电视人机交互成为一个主流技术,而手势识别是其中非常重要的研究领域。手势语言是一种依靠动作和视觉进行交际的特殊语言,它是包含信息量最多的人体语言。而手势识别的本质就是根据用户的手势,识别出所表达的意义。然而目前的手势检测、指尖定位等图像处理技术大部分是基于二维的,适合近距离、简单背景的,不符合智能电视的交互需求。而三维手势识别能够判断出三维空间内的手势变化,以更为自然、直观的交互方式得到广泛关注,但是日前来讲,一些三维手势识别设备使用了数据手套,虽然其识别率高,但也失去了手势交互的自然性,费用也很昂贵;一些利用飞行时间的设备,对硬件要求高、费用高昂、分辨率也较低;基于双目视觉的设备视角较小、在昏暗环境下以及特征不明显的情况下效果较差;而基于结构光的设备非常容易受到光照影响。可以说现有的手势识别设备都存在不同程度的缺陷。

发明内容

[0006] 为解决现有技术的不足,本发明的目的在于提供一种基于双目结构光的三维手势控制智能电视的方法,本发明通过双目摄像头和红外结构光结合的方式获得优质的双手的三维信息,然后进行实时手势分割、检测以及指尖定位,完成三维手势动作与电视指令的实时转换,满足电视人机交互操作实时性的要求。

[0007] 为了实现上述目标,本发明采用如下的技术方案:

[0008] 一种基于双目结构光的三维手势控制智能电视的方法,包括如下步骤:

[0009] 步骤一,采集图像:通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片;

[0010] 步骤二,重建图像三维:根据双目匹配算法,找出左视图、右视图之间各像素点的

一一对应关系,根据他们之间的像素位置的横坐标差得到视差图,然后根据投影模型与三角测量原理,获得原始图像的深度信息;

[0011] 步骤三,预处理深度图像;对深度图像进行中值滤波去除噪声点,并对深度图像中的内部空洞进行形态学处理;

[0012] 步骤四,检测手势图像;对获得的图像深度数据进行分析,将人和背景分离开来,只选取人体的像素信息,将其余像素信息全部置零,获得人体的深度图像,最后获得手势的轮廓;

[0013] 步骤五,识别三维手势;根据获得的手势轮廓,对手势轮廓提取中心距特征,中心距是指图形中必点到各边缘点的距离;设点 $O(x_0, y_0)$ 为手势的几何中心点,点集 $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ 为手势的边缘轮廓,不妨把手势的边缘轮廓看成有一个个边缘点连接而成,由此,几何中心点 $O(x_0, y_0)$ 和中心距 d 可由如下公式计算:

$$[0014] \quad x_0 = \frac{1}{N} \sum_{i=1}^N x_i;$$

$$[0015] \quad y_0 = \frac{1}{N} \sum_{i=1}^N y_i;$$

$$[0016] \quad d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2};$$

[0017] 再结合三维信息,利用基于贝叶斯网络的分类器进行识别;

[0018] 步骤六,执行操作指令;根据在双目摄像头前做出的手势,智能电视执行相应的预先设定的操作指令,完成人机交互。

[0019] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤一,采集图像;通过红外探测器探测操作环境的光照情况,若红外光过多则打开红外激光器;再通过设置于智能电视内的双目摄像头捕获摄像头前方图像,采集到的左视图、右视图分别储存进储存器,再由处理器分别处理储存器内的两张图片。

[0020] 前述的一种基于双目结构光的三维手势控制智能电视的方法,若红外探测器探测操作环境的红外辐射强度小于 $0.2W/m^2 \times 850nm$,则打开红外激光器。

[0021] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤三中的形态学处理即对图像边缘存在的毛刺和图像的内部空洞进行开、闭运算。

[0022] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤四中获得手势的轮廓的具体步骤为:通过像素的灰度值反映改点距离镜头的远近,统计出人体的深度图像的灰度分布直方图,灰度直方图较小的峰是手部区域的灰度分布,较大的峰是人体躯干的灰度分布,从而分割出人体的手部图像,再对手部图像使用Canny边缘算子进行边缘检测,从而获得手势的轮廓。

[0023] 前述的一种基于双目结构光的三维手势控制智能电视的方法,步骤五中利用基于贝叶斯网络的分类器进行识别的具体步骤如下:

[0024] 步骤a,计算不同手势类别下所有特征的高斯分布信息,即是计算所有的均值与方差;

[0025] 步骤b,对于测试的每一个输入样本,求出它在手势 l 类别下,各个特征的概率,由于每个特征是相互独立的,因而特征下的概率分布也满足独立性,由此便可以得出该输入

样本在手势1类别下的总概率;以此类推,求出该样本在所有手势类别下的不同概率;

[0026] 步骤c,比较输入样本在不同类别下得到概率值,找出其最大概率值所属的类别,便预测该类别就是输入样本所属的类别。

[0027] 本发明的有益之处在于:本发明提供一种基于双目结构光的三维手势控制智能电视的方法,本发明通过双目摄像头和红外结构光结合的方式获得优质的双手的三维信息,然后进行实时手势分割、检测以及指尖定位,完成三维手势动作与电视指令的实时转换,满足电视人机交互操作实时性的要求;

[0028] 本发明因为采用了双目视觉和结构光结合的方式,可以有效地提高环境抗干扰能力、降低光照条件的便会对指尖定位的影响,当在室内或环境昏暗时采用结构光测量深度信息,当太阳光照较强时则转为纯双目的方式。而且双目结构光的标定与激光器无关,当激光器损坏时替换较为方便;

[0029] 本发明通过对手势图像的前景分割减少了手势检测的遍历范围,可以满足电视人机交互操作实时性的要求。

附图说明

[0030] 图1是本发明的一种实施例的结构示意图;

[0031] 图中附图标记的含义:

[0032] 1红外摄像头,2红外探测器,3红外激光器,4电视。

具体实施方式

[0033] 以下结合附图和具体实施例对本发明作具体的介绍。

[0034] 如图1所示,本发明的所需的所有模块都集成在智能电视4当中。水平安装在智能电视4左右两边的是已经标定好的红外摄像头1,用于获取左右视角的图像。在两个红外摄像头1中间的是红外激光器3,用来发出红外条纹光,增加表面纹理,便于双目匹配,方便在昏暗条件下依旧能够准确识别手势。激光器旁边的是红外探测器2,用来监测操作环境的红外情况,以选择是否打开红外激光器3。而后续的图像三维重建、深度图像预处理、手势图像检测、三维手势识别以及人机交互则由智能电视4内的处理器来完成。

[0035] 一种基于双目结构光的三维手势控制智能电视4的方法,包括如下步骤:

[0036] 步骤一,采集图像;

[0037] 先通过红外探测器2探测操作环境的红外辐射情况,若红外辐射强度大于 $0.2\text{W}/\text{m}^2 \times 850\text{nm}$,激光器发出的编码光斑容易被淹没,则选用纯双目的方式;若红外辐射强度小于 $0.2\text{W}/\text{m}^2 \times 850\text{nm}$,编码光斑不容易被干扰,则打开红外激光器3,以提高双目匹配精度。

[0038] 再打开水平安装在智能电视4机内已经标定好的两个相同型号的红外摄像头1,双目摄像头以25帧每秒捕获摄像头前方图像,采集到的图像大小为 640×480 的RGB彩色空间的数字图像,并将双目摄像头拍摄的的左视图、右视图分别储存在智能电视4里两个独立的存储空间,再由处理器分别处理储存器内的两张图片。

[0039] 步骤二,重建图像三维;根据双目匹配算法,找出左视图、右视图之间各像素点的一一对应关系,根据他们之间的像素位置的横坐标差得到视差图,然后根据投影模型与三角测量原理,获得原始图像的深度信息;

[0040] 步骤三, 预处理深度图像; 由于获得的深度图像中手势边缘的深度值不稳定, 有许多0值噪声点, 直接处理会对算法的效果造成影响, 因此采用中值滤波去除噪声点, 降低目标边缘的模糊程度。再对深度图像中的内部空洞进行形态学处理; 形态学处理即对图像边缘存在的毛刺和图像的内部空洞进行开、闭运算。

[0041] 步骤四, 检测手势图像; 对获得的图像深度数据进行分析, 将人和背景分离开来, 只选取人体的像素信息, 将其余像素信息全部置零, 获得人体的深度图像, 最后获得手势的轮廓; 获得手势的轮廓的具体步骤为: 通过像素的灰度值反映改点距离镜头的远近, 统计出人体的深度图像的灰度分布直方图, 灰度直方图较小的峰是手部区域的灰度分布, 较大的峰是人体躯干的灰度分布, 从而分割出人体的手部图像, 再对手部图像使用Canny边缘算子进行边缘检测, 从而获得手势的轮廓。

[0042] 步骤五, 识别三维手势; 根据获得的手势轮廓, 对手势轮廓提取中心距特征(SCCD), 中心距是指图形中必点到各边缘点的距离; 设点 $O(x_0, y_0)$ 为手势的几何中心点, 点集 $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ 为手势的边缘轮廓, 不妨把手势的边缘轮廓看成有一个个边缘点连接而成, 由此, 几何中心点 $O(x_0, y_0)$ 和中心距 d 可由如下公式计算:

$$[0043] \quad x_0 = \frac{1}{N} \sum_{i=1}^N x_i;$$

$$[0044] \quad y_0 = \frac{1}{N} \sum_{i=1}^N y_i;$$

$$[0045] \quad d_i = \sqrt{(x_i - x_0)^2 + (y_i - y_0)^2};$$

[0046] 再结合三维信息, 利用基于贝叶斯网络的分类器进行识别;

[0047] 利用基于贝叶斯网络的分类器进行识别的具体步骤如下:

[0048] 步骤a, 计算不同手势类别下所有特征的高斯分布信息, 即是计算所有的均值与方差;

[0049] 步骤b, 对于测试的每一个输入样本, 求出它在手势 l 类别下, 各个特征的概率, 由于每个特征是相互独立的, 因而特征下的概率分布也满足独立性, 由此便可以得出该输入样本在手势 l 类别下的总概率; 以此类推, 求出该样本在所有手势类别下的不同概率;

[0050] 步骤c, 比较输入样本在不同类别下得到概率值, 找出其最大概率值所属的类别, 便预测该类别就是输入样本所属的类别。

[0051] 步骤六, 执行操作指令; 根据在双目摄像头前做出的手势, 智能电视4执行相应的预先设定的操作指令, 完成人机交互。

[0052] 本发明提供一种基于双目结构光的三维手势控制智能电视4的方法, 本发明通过双目摄像头和红外结构光结合的方式获得优质的双手的三维信息, 然后进行实时手势分割、检测以及指尖定位, 完成三维手势动作与电视4指令的实时转换, 满足电视4人机交互操作实时性的要求;

[0053] 本发明因为采用了双目视觉和结构光结合的方式, 可以有效地提高环境抗干扰能力、降低光照条件的便会对指尖定位的影响, 当在室内或环境昏暗时采用结构光测量深度信息, 当太阳光照较强时则转为纯双日的方式。而且双目结构光的标定与激光器无关, 当激光器损坏时替换较为方便;

[0054] 本发明通过对手势图像的前景分割减少了手势检测的遍历范围,可以满足电视4人机交互操作实时性的要求。

[0055] 以上显示和描述了本发明的基本原理、主要特征和优点。本行业的技术人员应该了解,上述实施例不以任何形式限制本发明,凡采用等同替换或等效变换的方式所获得的技术方案,均落在本发明的保护范围内。

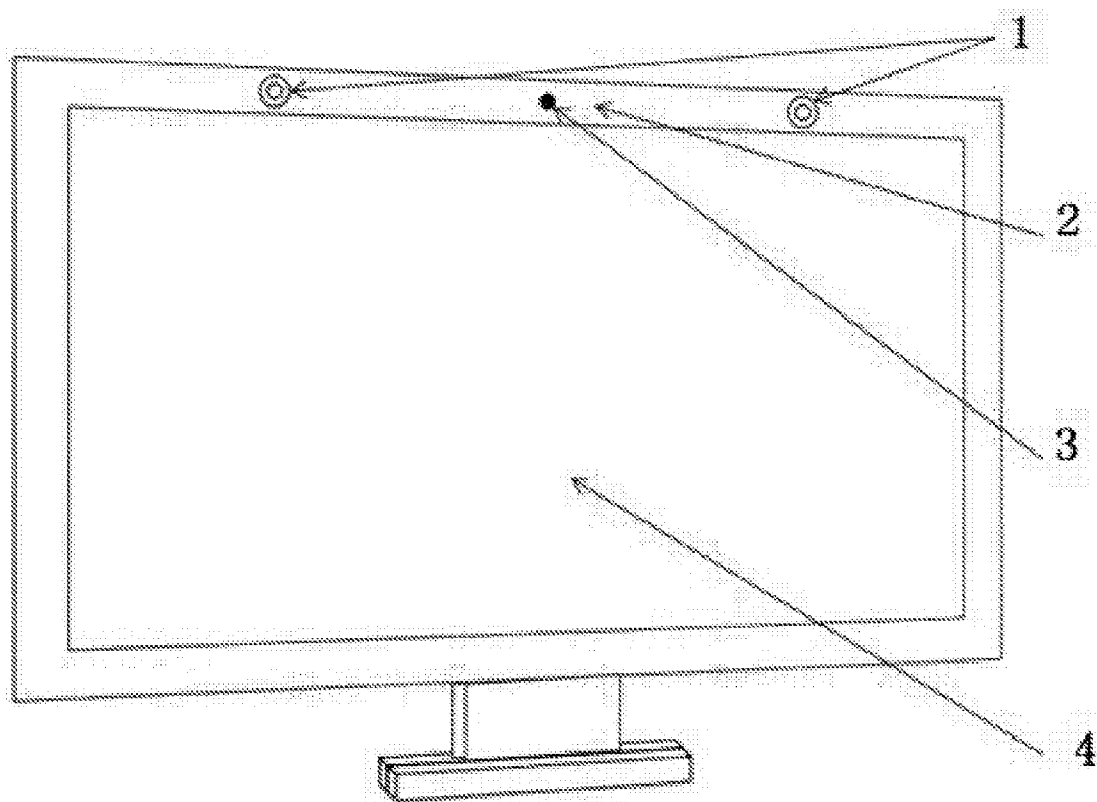
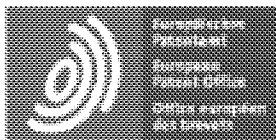


图1



Notice

This translation is machine-generated. It cannot be guaranteed that it is intelligible, accurate, complete, reliable or fit for specific purposes. Critical decisions, such as commercially relevant or financial decisions, should not be based on machine-translation output.

DESCRIPTION CN108363482A

¹⁰ Method for controlling smart TV with three-dimensional gesture based on binocular structured light

[0001]

¹⁴ Technical field

[0002]

¹⁸ The invention relates to the field of human-computer interaction, in particular to a method of gesture control.

[0003]

²² Background technique

[0004]

²⁶ With the development of Internet technology, smart TV-an open operating system and chip, with an open application platform, can achieve two-way human-computer interaction, and integrate multiple functions such as audio and video, entertainment, and data to meet TV products with diversified and individual needs of users have become the trend of TV.

[0005]

³³ The traditional push-button remote control is mainly composed of an integrated circuit board and buttons used to generate different messages.

³⁵ It is equipped with a central processing unit CPU, and the CPU inputs various menu password information of the TV when it is manufactured, and the remote control transmitter of the TV can realize the remote control of the TV as long as the corresponding password is issued.

[0006]

⁴¹ The development of smart TV puts forward more requirements on the interactive operation of the TV.

⁴² The first is the need for simple and convenient text input, web browsing and channel selection. The traditional remote control obviously cannot meet the new interactive needs.

⁴⁴ If an external mouse and keyboard are connected to the smart TV, the user needs to configure multiple control terminals, and it is not possible to complete the interaction in a long distance (3 to 5 meters) and in a complex environment (mouse and keyboard require a relatively flat desktop) to operate for the user Brings a lot of obstacles.

⁴⁸ This is also the reason why the current human-computer interaction of TV lags behind computers and handheld devices.

[0007]

⁵³ At present, vision-based television human-computer interaction has become a mainstream technology, and gesture recognition is a very important research field.

⁵⁵ Gesture language is a special language that relies on movement and vision to communicate. It is the human body language that contains the most information.

⁵⁷ The essence of gesture recognition is to recognize the meaning expressed according to the user's gestures.

However, most of the current image processing technologies such as gesture detection and fingertip positioning are based on two-dimensional, suitable for short-distance, simple background, and do not meet the interactive requirements of smart TVs. The three-dimensional gesture recognition can determine the gesture changes in the three-dimensional space, and has attracted widespread attention in a more natural and intuitive way of interaction. However, at present, some three-dimensional gesture recognition devices use data gloves. Although its recognition rate is high, it is also The naturalness of gesture interaction is lost, and the cost is also very expensive; some devices that use time of flight have high hardware requirements, high costs, and low resolution; devices based on binocular vision have small viewing angles, in dim environments, and features The effect is poor when it is not obvious; and equipment based on structured light is very susceptible to light. It can be said that the existing gesture recognition devices have various degrees of defects.

[0008]

⁷¹ Summary of the invention

[0009]

⁷⁵ In order to solve the shortcomings of the prior art, the purpose of the present invention is to provide a method for controlling a smart TV with a three-dimensional gesture based on binocular structured light. The present invention obtains high-quality three-dimensional information of both hands through a combination of binocular camera and infrared structured light. , And then perform real-time gesture segmentation, detection, and fingertip positioning to complete the real-time conversion of three-dimensional gesture actions and TV

commands to meet the real-time requirements of TV human-computer interactive operations.

[0010]

⁸⁴ In order to achieve the above objectives, the present invention adopts the following technical solutions:

[0011]

⁸⁸ A method for controlling a smart TV with a three-dimensional gesture based on binocular structured light includes the following steps:

[0012]

⁹³ Step one, capture images; capture images in front of the camera through the binocular camera set in the smart TV, and store the collected left view and right view into the memory respectively, and then the processor processes the two images in the memory respectively;

[0013]

⁹⁹ Step two, reconstruct the image in three dimensions; according to the binocular matching algorithm, find the one-to-one correspondence between the pixels of the left view and the right view, and obtain the disparity map according to the abscissa difference of the pixel positions between them, and then according to the projection model With the principle of triangulation, the depth information of the original image is obtained;

[0014]

¹⁰⁶ Step three, preprocess the depth image; perform median filtering on the depth image to remove noise points, and then perform morphological processing on the internal holes in the depth image;

[0015]

¹¹¹ Step 4: Detect the gesture image; analyze the obtained image depth data, separate the person from the background, select only the pixel information of the human body, and set all other pixel information to zero to obtain the depth image of the human body, and finally obtain the contour of the gesture ;

[0016]

¹¹⁷ Step 5: Recognize the three-dimensional gesture; extract the center distance feature of the gesture contour according to the obtained gesture contour. The center distance refers to the distance between the necessary point in the graph and each edge point; set the point $O(x_0, y_0)$ as the geometric center point of the gesture , The point set $C=\{(x_i, y_i)|i=1, 2, \dots, N\}$ is the edge contour of the gesture. It may be considered that the edge contour of the gesture is formed by connecting edge points. Therefore, the geometric center The point $O(x_0,$

y0) and the center distance d can be calculated by the following formula:

[0020]

¹²⁶ Combined with three-dimensional information, use the classifier based on Bayesian network for recognition;

[0021]

¹³⁰ Step 6, execute the operation instruction; according to the gesture made in front of the binocular camera, the smart TV executes the corresponding preset operation instruction to complete the human-computer interaction.

[0022]

¹³⁶ The aforementioned method for controlling a smart TV with three-dimensional gestures based on binocular structured light. Step one is to collect images; detect the lighting conditions of the operating environment through an infrared detector; if there is too much infrared light, turn on the infrared laser; The binocular camera in the TV captures the image in front of the camera, and the collected left view and right view are respectively stored in the memory, and then the two pictures in the memory are processed by the processor.

[0023]

¹⁴⁴ In the aforementioned method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, if the infrared detector detects that the infrared radiation intensity of the operating environment is less than $0.2\text{W}/\text{m}^2 \cdot 850\text{nm}$, the infrared laser is turned on.

[0024]

¹⁵⁰ In the aforementioned method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, the morphological processing in step 3 is to open and close the burr existing on the edge of the image and the internal cavity of the image.

[0025]

¹⁵⁶ In the aforementioned method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, the specific steps for obtaining the outline of the gesture in step 4 are: the gray value of the pixel reflects the distance of the changed point from the lens, and the depth image of the human body is calculated. The grayscale distribution histogram of the grayscale histogram, the smaller peak of the grayscale histogram is the gray distribution of the hand area, and the larger peak is the gray distribution of the human torso, so as to segment the human hand image, and then use the hand image. The Canny edge operator performs edge detection to obtain the contour of the gesture.

[0026]

¹⁶⁶ In the foregoing method for controlling a smart TV with a three-dimensional gesture based on binocular structured light, the specific steps of using a Bayesian network-based classifier for recognition in step 5 are as follows:

[0027]

¹⁷² Step a: Calculate the Gaussian distribution information of all features in different gesture categories, that is, calculate all the mean and variance;

[0028]

¹⁷⁷ Step b: For each input sample tested, find the probability of each feature in the gesture 1 category. Since each feature is independent of each other, the probability distribution under the feature also satisfies the independence, which can be obtained Find the total probability of the input sample in the gesture 1 category; and so on, find the different probabilities of the sample in all gesture categories;

[0029]

¹⁸⁴ Step c: Compare the probability values of the input samples under different categories, find the category to which the maximum probability value belongs, and predict that the category is the category to which the input sample belongs.

[0030]

¹⁹⁰ The advantage of the present invention is that the present invention provides a method for controlling a smart TV with a three-dimensional gesture based on binocular structured light. The present invention obtains high-quality three-dimensional information of both hands through a combination of binocular camera and infrared structured light, and then performs Real-time gesture segmentation, detection, and fingertip positioning, complete the real-time conversion of three-dimensional gesture actions and TV commands, and meet the real-time requirements of TV man-machine interactive operations;

[0031]

¹⁹⁹ Because the invention adopts the combination of binocular vision and structured light, it can effectively improve the environment's anti-interference ability and reduce the influence of light conditions on fingertip positioning. When indoors or when the environment is dim, the structured light is used to measure depth information , When the sun is strong, it turns to a pure binocular way.

²⁰³ Moreover, the calibration of binocular structured light has nothing to do with the laser, and it is more convenient to replace when the laser is damaged;

[0032]

²⁰⁸ The invention reduces the traversal range of the gesture detection by segmenting the foreground of the gesture image, and can meet the real-time requirements of the television human-computer interactive operation.

[0033]

²¹³ Description of the drawings

[0034]

²¹⁷ Figure 1 is a schematic structural diagram of an embodiment of the present invention;

[0035]

²²¹ The meaning of the reference signs in the figure:

[0036]

²²⁵ 1 Infrared camera, 2 infrared detectors, 3 infrared lasers, 4 TVs.

[0037]

²²⁹ Detailed ways

[0038]

²³³ Hereinafter, the present invention will be described in detail with reference to the drawings and specific embodiments.

[0039]

²³⁸ As shown in FIG. 1, all the modules required by the present invention are integrated in the smart TV 4.

²³⁹ Horizontally installed on the left and right sides of the smart TV 4 are calibrated infrared cameras 1 for obtaining images with left and right viewing angles.

²⁴¹ In the middle of the two infrared cameras 1 is an infrared laser 3, which is used to emit infrared fringe light, increase the surface texture, facilitate binocular matching, and facilitate accurate recognition of gestures in dim conditions.

²⁴⁴ Next to the laser is an infrared detector 2, which is used to monitor the infrared situation of the operating environment to choose whether to turn on the infrared laser 3.

²⁴⁶ The subsequent image 3D reconstruction, depth image preprocessing, gesture image detection, 3D gesture recognition, and human-computer interaction are completed by the processor in the smart TV 4.

[0040]

251 A method for controlling a smart TV 4 with a three-dimensional gesture based on binocular structured light includes the following steps:

[0041]

256 Step one, collect images;

[0042]

260 First detect the infrared radiation of the operating environment through the infrared detector 2. If the infrared radiation intensity is greater than $0.2W/m^2 \cdot 850nm$, the coded spot emitted by the laser is likely to be submerged, and the pure binocular method is used; if the infrared radiation intensity is less than $0.2W/m^2 \cdot 850nm$, the coded spot is not easy to be disturbed, then turn on the infrared laser 3 to improve the accuracy of binocular matching.

[0043]

268 Then turn on the two infrared cameras 1 of the same model that have been calibrated horizontally installed in the smart TV 4. The binocular camera captures the front image of the camera at 25 frames per second, and the collected image size is 640×480 in the RGB color space. Digital images, and the left view and right view taken by the binocular camera are respectively stored in two independent storage spaces in the smart TV 4, and then the two pictures in the storage are processed by the processor.

[0044]

276 Step two, reconstruct the image in three dimensions; according to the binocular matching algorithm, find the one-to-one correspondence between the pixels of the left view and the right view, and obtain the disparity map according to the abscissa difference of the pixel positions between them, and then according to the projection model With the principle of triangulation, the depth information of the original image is obtained;

[0045]

283 Step three, preprocess the depth image; because the depth value of the gesture edge in the obtained depth image is unstable, there are many zero-value noise points, direct processing will affect the effect of the algorithm, so the median filter is used to remove the noise points and reduce the target The degree of blurring of the edges.

287 Morphological processing is performed on the internal cavity in the depth image; the morphological processing is to open and close the burr existing on the edge of the image and the internal cavity of the image.

[0046]

²⁹² Step 4: Detect the gesture image; analyze the obtained image depth data, separate the person from the background, select only the pixel information of the human body, and set all other pixel information to zero to obtain the depth image of the human body, and finally obtain the contour of the gesture ; The specific steps to obtain the contour of the gesture are; the gray value of the pixel reflects the distance of the changed point from the lens, and the gray distribution histogram of the depth image of the human body is calculated. The smaller peak of the gray histogram is the hand area Gray-scale distribution, the larger peak is the gray-scale distribution of the human torso, thereby segmenting the human hand image, and then using the Canny edge operator to perform edge detection on the hand image to obtain the contour of the gesture.

[0047]

³⁰³ Step 5: Recognize the three-dimensional gesture; extract the center distance feature (SCCD) of the gesture contour according to the obtained gesture contour. The center distance refers to the distance from the edge point to the edge point in the figure; set point $O(x_0, y_0)$ as the gesture The geometric center point, point set $C = \{(x_i, y_i) | i = 1, 2, \dots, N\}$ is the edge contour of the gesture. It may be considered that the edge contour of the gesture is formed by connecting edge points. , The geometric center point $O(x_0, y_0)$ and the center distance d can be calculated by the following formula:

[0051]

³¹² Combined with three-dimensional information, use the classifier based on Bayesian network for recognition;

[0052]

³¹⁶ The specific steps of using a Bayesian network-based classifier for recognition are as follows:

[0053]

³²⁰ Step a: Calculate the Gaussian distribution information of all features in different gesture categories, that is, calculate all the mean and variance;

[0054]

³²⁵ Step b: For each input sample tested, find the probability of each feature in the gesture 1 category. Since each feature is independent of each other, the probability distribution under the feature also satisfies the independence, which can be obtained Find the total probability of the input sample in the gesture 1 category; and so on, find the different probabilities of the sample in all gesture categories;

[0055]

332 Step c: Compare the probability values of the input samples under different categories, find the category to which the maximum probability value belongs, and predict that the category is the category to which the input sample belongs.

[0056]

338 Step 6, execute the operation instruction; according to the gesture made in front of the binocular camera, the smart TV 4 executes the corresponding preset operation instruction to complete the human-computer interaction.

[0057]

344 The present invention provides a three-dimensional gesture control method for smart TV 4 based on binocular structured light. The present invention obtains high-quality three-dimensional information of both hands through the combination of binocular camera and infrared structured light, and then performs real-time gesture segmentation, detection and pointing. Pointer positioning, complete the real-time conversion of three-dimensional gesture actions and TV 4 instructions, and meet the real-time requirements of TV 4 man-machine interactive operation;

[0058]

353 Because the invention adopts the combination of binocular vision and structured light, it can effectively improve the environment's anti-interference ability and reduce the influence of light conditions on fingertip positioning. When indoors or when the environment is dim, the structured light is used to measure depth information , When the sun is strong, it turns to a pure binocular way.

357 Moreover, the calibration of binocular structured light has nothing to do with the laser, and it is more convenient to replace when the laser is damaged;

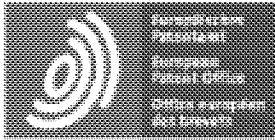
[0059]

362 The invention reduces the traversal range of the gesture detection by segmenting the foreground of the gesture image, and can meet the real-time requirements of the human-computer interaction operation of the TV 4.

[0060]

367 The basic principles, main features and advantages of the present invention have been shown and described above.

369 Those skilled in the industry should understand that the foregoing embodiments do not limit the present invention in any form, and all technical solutions obtained by equivalent substitutions or equivalent transformations fall within the protection scope of the present invention.



Notice

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CLAIMS CN108363482A

1.

¹³ A method for controlling a smart TV with a three-dimensional gesture based on binocular structured light is characterized in that it comprises the following steps:

¹⁵ Step one, capture images; capture images in front of the camera through the binocular camera set in the smart TV, and store the collected left view and right view into the memory respectively, and then the processor processes the two images in the memory respectively;

¹⁸ Step two, reconstruct the image in three dimensions; according to the binocular matching algorithm, find the one-to-one correspondence between the pixels of the left view and the right view, and obtain the disparity map according to the abscissa difference of the pixel positions between them, and then according to the projection model With the principle of triangulation, the depth information of the original image is obtained;

²² Step three, preprocess the depth image; perform median filtering on the depth image to remove noise points, and then perform morphological processing on the internal holes in the depth image;

²⁴ Step 4: Detect the gesture image; analyze the obtained image depth data, separate the person from the background, select only the pixel information of the human body, and set all other pixel information to zero to obtain the depth image of the human body, and finally obtain the contour of the gesture ;

²⁷ Step 5: Recognize the three-dimensional gesture; extract the center distance feature of the gesture contour according to the obtained gesture contour. The center distance refers to the distance between the necessary point in the graph and each edge point; set the point $O(x_0, y_0)$ as the geometric center point of the gesture , The point set $C=\{(x_i, y_i)|i=1, 2, \dots, N\}$ is the edge contour of the gesture. It may be considered that the edge contour of the gesture is formed by connecting edge points. Therefore, the geometric center The point $O(x_0, y_0)$ and the center distance d can be calculated by the following formula:

³³ Combined with three-dimensional information, use the classifier based on Bayesian network for recognition;

³⁴ Step 6, execute the operation instruction; according to the gesture made in front of the binocular camera, the smart TV executes the corresponding preset operation instruction to complete the human-computer interaction.

2.

⁴⁰ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, characterized in that, in step one, an image is acquired; Turn on the infrared laser; then capture the image in front of the camera through the binocular camera set in the smart TV, and store the collected left view and right view into the memory respectively, and then the processor processes the two images in the memory respectively.

3.

⁴⁸ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 2, wherein if the infrared radiation intensity of the operating environment detected by the infrared detector is less than $0.2W/m^2 \cdot 850nm$, the infrared laser is turned on .

4.

⁵⁴ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, wherein the morphological processing in the third step is to open the burr existing on the edge of the image and the internal cavity of the image. Close operation.

5.

⁶⁰ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, wherein the specific steps of obtaining the outline of the gesture in the above step 4 are: reflecting the change point distance through the gray value of the pixel The distance of the lens is to calculate the gray distribution histogram of the depth image of the human body. The smaller peak of the gray histogram is the gray distribution of the hand area, and the larger peak is the gray distribution of the human torso, so as to segment the human body. Then use Canny edge operator to perform edge detection on the hand image of the hand image to obtain the contour of the gesture.

6.

⁷⁰ The method for controlling a smart TV with a three-dimensional gesture based on binocular structured light according to claim 1, wherein the specific steps of using a Bayesian network-based classifier to perform recognition in the above step 5 are as follows:

⁷³ Step a: Calculate the Gaussian distribution information of all features in different gesture categories, that is, calculate all the mean and variance;

⁷⁵ Step b: For each input sample tested, find the probability of each feature in the gesture 1 category. Since each feature is independent of each other, the probability distribution under the feature also satisfies the independence, which can be obtained Find the total probability of the input sample in the gesture 1 category; and so on, find the different probabilities of the sample in all gesture categories;

⁷⁹ Step c: Compare the probability values of the input samples under different categories, find the category to

which the maximum probability value belongs, and predict that the category is the category to which the input sample belongs.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	2516	(Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore)).inv.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:39 AM
L2	440018	(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:49 AM
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		classify\$3) WITH (material OR property)	AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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			RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



UNITED STATES PATENT AND TRADEMARK OFFICE

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17/439,492	09/15/2021	Friedrich Schick	27843-1809 182037US01	9002

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Patent Docket Department
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EXAMINER

JEAN, FRANTZ B

ART UNIT	PAPER NUMBER
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2454

NOTIFICATION DATE	DELIVERY MODE
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09/25/2023

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

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DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This is a first office action in response to an application for letters patent filed on 15 September 2021.

Claims 1-25 are presented for examination.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 09/15/2021 and 10/15/2021 was filed before the mailing date of the first office action on the merits. The submission is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Claim Rejections - 35 USC § 112

The following is a quotation of 35 U.S.C. 112(b):

(b) **CONCLUSION.**—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1, 23-24 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claims 1 and 24 relate to determining a longitudinal coordinate and a material property. It is neither clear how it is arrived at the desired distance feature and material feature nor which "further" filters are covered by the claim. The vague indication of correlation with one of the listed filters is an infinite definition of filters which does not allow to put into practice the invention or determine what is covered by the claims. The list of filters contains filters which are not commonly known and lack a definition in the claims. The sign in claim 10 is not defined.

As per claim 23, the detector as defined in claim 1 determines a material property, it is not clear how it can be used in an inertial measurement unit. the disclosure in this respect is also missing in the description.

Claim Objections

Claims 1-25 are objected to because of the following informalities: the parenthesis and numeral numbers are unnecessary. Appropriate correction is required.

Claim Rejections - 35 USC § 102

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claim(s) 1-7, 9-16, and 24-25 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Schwartz et al. hereinafter Schwartz DE 19846619 A1.

As per claim 1, Schwartz teaches a detector for identifying at least one material property comprising - at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object (Schwartz: claim 1; CMOS sensor is a common matrix sensor);

- at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image (Schwartz: claim 1);

- intensity profile determination), wherein the evaluation device is configured for determining at least one distance feature by applying at least one distance dependent image filter to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof, being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, (Schwartz: claim 29); wherein the evaluation device is configured for determining at least one material feature by applying at least one material dependent image filter to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Mother which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof (Schwartz: claim 1, col. 14, lines 38-66), wherein the evaluation device is configured for determining a longitudinal coordinate and the material property by evaluating the distance feature and the material feature (Schwartz: claims 1 and 29).

The method according to claim 24 is disclosed in the same manner.

As per claim 25, a method of using the detector according to claim 1, relating to a detector, for a purpose of use, selected from the group consisting of: a position measurement in traffic technology; an entertainment application; a security application; a surveillance application; a safety application; a human-machine interface application; a logistics application; a tracking application; an outdoor application; a mobile application; a communication application; a

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photography application; a machine vision application; a robotics application; a quality control application; a manufacturing application (Schwartz: col 2, lines 31-38).

As per claim 2, Schwartz teaches the detector (110) according to claim 1, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance (Schwartz: claim 1, col. 14, lines 38-66).

As per claim 3, Schwartz inherently teaches the detector (110) according to claim 1, wherein $p < 0.075$ (Schwartz: claim 1, col. 14, lines 38-66).

As per claim 4, Schwartz teaches the detector (110) according to claim 1, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, and a speckle (Schwartz column 14, lines 38-45; material properties are usually determined).

As per claim 5, Schwartz inherently teaches the detector (110) according to claim 1, wherein the further distance dependent image filter D_{10} other correlates to one or more of the distance dependent image filters by $p > 0.60$. (Schwartz: claim 1, col. 14, lines 38-66).

As per claim 6, Schwartz inherently teaches the detector (110) according to claim 1, wherein the further material dependent image filter D_{20} other correlates to one or more of the material dependent image filters D_m by ~~$|p_{D_{20} \text{ other}, D_m}| \geq 0.60$~~ , preferably by ~~$|p_{D_{20} \text{ other}, D_m}| \geq 0.80$~~ . (Schwartz: claim 1, col. 14, lines 38-66).

As per claim 7, Schwartz inherently teaches the detector (110) according to claim 1, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between ... and z , m . (Schwartz: claim 1, col. 14, lines 38-66).

As per claim 9, Schwartz teaches the detector (110) according to claim 1, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image (Schwartz: claim 1, col.

14, lines 38-66; Recursive, iterative or simultaneous application of filters and applying a further filter are considered usual alternatives).

As per claim 10, Schwartz teaches the detector (110) according to claim 1, wherein the evaluation device (132) is configured for determining whether at least one of $(I_t$ or O_2 is a function of the feature p_z or T_{im} of the other image filter or whether at least one of $\&P$ or $(T)_2$ is a function of the at least one other image filter, wherein the evaluation device (132) is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image (Schwartz: claim 1, col. 14, lines 38-66; Recursive, iterative or simultaneous application of filters and applying a further filter are considered usual alternatives).

As per claim 11, Schwartz teaches the detector (110) according to claim 10 wherein the evaluation device (132) is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of $(I_z$ and y_{2m} to the reflection image (Schwartz: claim 1, col. 14, lines 38-66; Recursive, iterative or simultaneous application of filters and applying a further filter are considered usual alternatives).

As per claim 12, Schwartz teaches the detector (110) according to claim 1, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element (116), wherein the evaluation device (132) is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device (132) is configured for using at least one predetermined relationship between the combined signal Q and the distance feature y_{iz} for determining the distance feature (see Schwartz claim 1; it's using combined signals from sensor elements or a convolution based algorithm are considered common measures).

As per claim 13, Schwartz teaches the detector (110) according to claim 1, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature y_{iz} by optimizing at least one blurring function f_b , wherein the blurring function is optimized by varying the parameters of the at least one blurring function (see Schwartz claim 1; it's using combined signals from sensor elements or a convolution based algorithm are considered common measures).

As per claim 14, Schwartz teaches the detector (110) according to claim 1, wherein the sensor element (116) comprises at least one CMOS sensor (see Schwartz claim 1, a CMOS sensor is a common matrix sensor).

As per claim 15, Schwartz teaches the detector (110) according to claim 1, wherein the detector (110) comprises at least one illumination source (124), wherein the illumination source (124) is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising ~~such as~~ parallel or crossing lines (Schwartz claim 7).

As per claim 16, Schwartz inherently teaches the detector (110) according to claim 15, wherein the illumination source (124) comprises at least one laser source and at least one diffractive optical element (Schwartz claim 1, a diffractive optical element is a usual means for producing such patterns).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.
3. Resolving the level of ordinary skill in the pertinent art.
4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) 8 and 22 are rejected under 35 U.S.C. 103 as being unpatentable over Schwartz and Bleyer et al. hereinafter Bleyer US PUB 20180033146.

As per claim 8, Schwartz does not teach The detector (110) according to claim 1, wherein the material property m and/or the longitudinal coordinate z are determined by a function Bleyer discusses intensity profile corrected based on distance (see claim 6 of Bleyer). It would be obvious to a skilled artisan at the effective date and time of the invention as claimed to incorporate Bleyer feature into Schwartz's system to facilitate precise position and modification.

As per claim 22, Schwartz does not discuss a camera (136) for imaging at least one object (112), the camera (136) comprising at least one detector (110) according to claim 1. Bleyer discloses this feature (see Bleyer, claim 1 and background of the invention). See claim 8 reasoning.

Claim(s) 17-20 are rejected under 35 U.S.C. 103 as being unpatentable over Schwartz and Schindler et al. hereinafter Schindler WO 2019042956 A1.

As per claim 17, Schwartz does not disclose the detector system (138) that further comprising at least one beacon device (140) adapted to direct at least one light beam towards the detector (110), wherein the beacon device (140) is at least one of attachable to the object (112), holdable by the object (112) and integratable into the object (112). Schindler teaches this feature (see Schindler claim 19). A skilled artisan will use a beacon just for implementation purpose.

As per claim 18, Schwartz does not teach a human-machine interface (142) for exchanging at least one item of information between a user (160) and a machine (168), wherein the human-machine interface (142) comprises at least one detector system (138) according to claim 17, wherein the at least one beacon device (140) is adapted to be at least one of directly or indirectly attached to the user (160) and held by the user (160), wherein the human-machine interface (142) is designed to determine at least one position of the user (160) by means of the detector system (138), wherein the human-machine interface (142) is designed to assign to the position at least one item of information. Schindler teaches these features (see Schindler claim 20).

As per claim 19, Schwartz does not teach an entertainment device (144) for carrying out at least one entertainment function, wherein the entertainment device (144) comprises at least one human-machine interface (142) according to claim 18, wherein the entertainment device (144) is designed to enable at least one item of information to be input by a player by means of the human-machine interface (142), wherein the entertainment device (144) is designed to vary the entertainment function in accordance with the information. Schindler teaches these features (see Schindler claim 21).

As per claim 20, Schwartz does not teach a tracking system (146) for tracking a position of at least one movable object, the tracking system (146) comprising at least one detector system (138) according to claim 17 referring to a detector system, the tracking system (146) further comprising at least one track controller (170), wherein the track controller (170) is adapted to track a series of positions of the object (112) at specific points in time. Schindler teaches these features (see claim 22).

Claim(s) 21 is rejected under 35 U.S.C. 103 as being unpatentable over Schwartz and Jiangsu

CN108363482.

As per claim 21, Schwartz does not teach a scanning system (148) for determining a depth profile of a scenery, the scanning system (148) comprising at least one detector (110) according to claim 1, the scanning system (148) further comprising at least one illumination source adapted to scan the scenery with at least one light beam. Jiangsu teaches that feature (see Jiangsu claim 1).

Claim(s) 23 is rejected under 35 U.S.C. 103 as being unpatentable over Schwartz.

As per claim 23, Inertial measurement units are generally known, being adapted to receive data of a sensor as defined in claim 1, a material property detector, does not allow to establish a technical effect resulting from this feature.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to FRANTZ B JEAN whose telephone number is (571)272-3937. The examiner can normally be reached 8-5 M-F.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Glenton B Burgess can be reached on 5712723949. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/FRANTZ B JEAN/

Primary Examiner, Art Unit 2454

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 17/439,492	Filing Date 09/15/2021	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	11/13/2023	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	* 25	Minus	** 25	= 0	x \$100 =	0
	Independent (37 CFR 1.16(h))	* 1	Minus	*** 3	= 0	x \$480 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE	
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/SHERRY A DAVIS/	
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".							
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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IN the United States Patent & Trademark Office

First Inventor:	Friedrich Schick	Art Unit:	2454
Serial No.:	17/439,492	Confirmation No.:	9002
Date Filed:	September 15, 2021	Examiner:	Frantz B Jean
Title:	DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY		

November 13, 2023

AMENDMENT AND RESPONSE TO NON-FINAL OFFICE ACTION

TO THE COMMISSIONER FOR PATENTS:

In response to the Non-Final Office Action dated September 25, 2023, please consider the following remarks.

Amendments to the Claims begin on page 2.

Remarks begin on page 10.

Conclusion is on page 17.

AMENDMENTS TO THE CLAIMS

1. (Currently Amended) A detector $[(110)]$ for identifying at least one material property in comprising

at least one sensor element $[(116)]$ comprising a matrix $[(118)]$ of optical sensors $[(120)]$, the optical sensors $[(120)]$ each having a light-sensitive area $[(122)]$, wherein the sensor element $[(116)]$ is configured for recording at least one reflection image of a light beam originating from at least one object $[(112)]$, and

at least one evaluation device $[(132)]$ configured for determining the material property by evaluation of at least one beam profile of the reflection image,

wherein the evaluation device $[(132)]$ is configured for determining at least one distance feature φ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter, or a linear combination thereof; and a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

wherein the evaluation device $[(132)]$ is configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter; a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; and a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard

deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2\text{other}},\Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, and

wherein the evaluation device **[(132)]** is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ϕ_{1z} and the material feature ϕ_{2m} .

2. (Currently Amended) The detector **[(110)]** according claim 1, wherein the material dependent image filter is at least one filter that passes a hypothesis testing, wherein the hypothesis testing uses a Null-hypothesis that the filter does not distinguish between material classifiers and an alternative hypothesis that the filter distinguishes at least two material classifiers, wherein the filter passes the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance.

3. (Currently Amended) The detector **[(110)]** according to claim 1, wherein $p \leq 0.075$.

4. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the at least one material property is a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, and a speckle.

5. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the further distance dependent image filter $\Phi_{1\text{other}}$ correlates to one or more of the distance dependent image filters Φ_z by $|\rho_{\Phi_{1\text{other}},\Phi_z}| \geq 0.60$.

6. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the further material dependent image filter $\Phi_{2\text{other}}$ correlates to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$.

7. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the material property m and/or the longitudinal coordinate z are determined by using a predetermined relationship between ϕ_{1z} , ϕ_{2m} and z , m .

8. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the material property m and/or the longitudinal coordinate z are determined by a function $z(\phi_{1z}, \phi_{2m})$ and/or $m(\phi_{1z}, \phi_{2m})$.

9. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the evaluation device **[(132)]** is configured for applying the distance dependent image filter and the material dependent image filter simultaneous to the reflection image.

10. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the evaluation device **[(132)]** is configured for determining whether at least one of Φ_1 or Φ_2 is a function of the feature ϕ_{1z} or ϕ_{1m} or ϕ_{2m} of the other image filter or whether at least one of Φ_1 or Φ_2 is a function of the at least one other image filter, wherein the evaluation device **[(132)]** is configured for applying the distance dependent image filter and the material dependent image filter sequentially or recursively to the reflection image.

11. (Currently Amended) The detector **[(110)]** according to claim 10, wherein the evaluation device **[(132)]** is configured for determining at least one of z and/or m by applying at least one further filter depending on at least one of ϕ_{1z} and ϕ_{2m} to the reflection image.

12. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the depth-from-photon-ratio filter comprises evaluating a combined signal Q from at least two sensor signals of the sensor element **[(116)]**, wherein the evaluation device **[(132)]** is configured for deriving the combined signal Q by one or more of dividing the sensor signals, dividing multiples of the sensor signals, dividing linear combinations of the sensor signals, wherein the evaluation device **[(132)]** is configured for using at least

one predetermined relationship between the combined signal Q and the distance feature ϕ_{1z} for determining the distance feature ϕ_{1z} .

13. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the depth-from-defocus filter comprises using at least one convolution-based algorithm such as a depth-from-defocus algorithm, wherein the evaluation device is configured for determining the distance feature ϕ_{1z} by optimizing at least one blurring function f_a , wherein the blurring function is optimized by varying the parameters of the at least one blurring function.

14. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the sensor element **[(116)]** comprises at least one CMOS sensor.

15. (Currently Amended) The detector **[(110)]** according to claim 1, wherein the detector **[(110)]** comprises at least one illumination source **[(124)]**, wherein the illumination source **[(124)]** is configured for generating at least one illumination pattern for illumination of the object, wherein the illumination pattern comprises at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature; at least one regular pattern, at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern; at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines; and at least one line pattern comprising parallel or crossing lines.

16. (Currently Amended) The detector **(110)** according to claim 15, wherein the illumination source **[(124)]** comprises at least one laser source and at least one diffractive optical element.

17. (Currently Amended) A detector system **[(138)]**, the detector system **[(138)]** comprising at least one detector **[(110)]** according to claim 1, the detector system **[(138)]** further comprising at least one beacon device **[(140)]** adapted to direct

at least one light beam towards the detector **[[110]]**, wherein the beacon device **[[140]]** is at least one of attachable to the object **[[112]]**, holdable by the object **[[112]]** and integratable into the object **[[112]]**.

18. (Currently Amended) A human-machine interface **[[142]]** for exchanging at least one item of information between a user **[[160]]** and a machine **[[168]]**, wherein the human-machine interface **[[142]]** comprises at least one detector system **[[138]]** according to claim 17, wherein the at least one beacon device **[[140]]** is adapted to be at least one of directly or indirectly attached to the user **[[160]]** and held by the user **[[160]]**, wherein the human-machine interface **[[142]]** is designed to determine at least one position of the user **[[160]]** by means of the detector system **[[138]]**, wherein the human-machine interface **[[142]]** is designed to assign to the position at least one item of information.

19. (Currently Amended) An entertainment device **[[144]]** for carrying out at least one entertainment function, wherein the entertainment device **[[144]]** comprises at least one human-machine interface **[[142]]** according to claim 18, wherein the entertainment device **[[144]]** is designed to enable at least one item of information to be input by a player by means of the human-machine interface **[[142]]**, wherein the entertainment device **[[144]]** is designed to vary the entertainment function in accordance with the information.

20. (Currently Amended) A tracking system **[[146]]** for tracking a position of at least one movable object, the tracking system **[[146]]** comprising at least one detector system **[[138]]** according to claim 17 referring to a detector system, the tracking system **[[146]]** further comprising at least one track controller **[[170]]**, wherein the track controller **[[170]]** is adapted to track a series of positions of the object **[[112]]** at specific points in time.

21. (Currently Amended) A scanning system **[[148]]** for determining a depth profile of a scenery, the scanning system **[[148]]** comprising at least one detector **[[110]]** according to claim 1, the scanning system **[[148]]** further comprising at least one illumination source adapted to scan the scenery with at least one light beam.

22. (Currently Amended) A camera [(136)] for imaging at least one object [(112)], the camera [(136)] comprising at least one detector [(110)] according to claim 1.

23. (Currently Amended) An inertial measurement unit [(172)] for use in an electronic device, wherein the inertial measurement unit [(172)] is adapted to receive data determined by at least one detector [(110)] according to claim 1, wherein the inertial measurement unit [(172)] further is adapted to receive data determined by at least one further sensor selected from the group consisting of: a wheel speed sensor, a turn rate sensor, an inclination sensor, an orientation sensor, a motion sensor, a magneto hydro dynamic sensor, a force sensor, an angular sensor, an angular rate sensor, a magnetic field sensor, a magnetometer, an accelerometer, and a gyroscope, wherein the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property of the electronic device selected from the group consisting of: position in space, relative or absolute motion in space, rotation, acceleration, orientation, angle position, inclination, turn rate, and speed.

24. (Currently Amended) A method for determining at least one material property of at least one object [(112)] by using at least one detector [(110)] according to claim 1, the method comprising the following steps:

a) determining at least one reflection image of the object [(112)] by using at least one sensor element [(116)] having a matrix [(118)] of optical sensors [(120)], the optical sensors [(120)] each having a light-sensitive area [(122)]; and

b) determining the material property by evaluating at least one beam profile of the reflection image by using at least one evaluation device [(132)], wherein the evaluating comprises:

b1) determining at least one distance feature ϕ_{1z} by applying at least one distance dependent image filter Φ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof and a further distance

dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\Phi_{1other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof,

b2) determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter; a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; and a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, and

b3) determining a longitudinal coordinate z and the material property m by evaluating the distance feature φ_{1z} and the material feature φ_{2m} .

25. (Currently Amended) A method of using the detector **[(110)]** according to claim 1, the method comprising using the detector **[(110)]** for a purpose of use selected from the group consisting of: a position measurement in traffic technology; an entertainment application, a security application; a surveillance application, a safety application, a human-machine interface application; a logistics application; a tracking

application; an outdoor application; a mobile application; a communication application; a photography application; a machine vision application; a robotics application; a quality control application; and a manufacturing application.

REMARKS

I. Status of the Claims

Claims 1-25 are amended for clarity.

Claims 1-25 are pending and under examination. No new matter is added herein.

II. Rejection under 35 U.S.C. § 112

Claim 1, 23 and 24 stand rejected under 35 U.S.C. § 112(b) as allegedly being indefinite.

Applicant respectfully submits that claims 1 and 24 as recited are clear and definite.

The Office based the rejections in part on, “Claims 1 and 24 relate to determining a longitudinal coordinate and a material property. It is neither clear how it is arrived at the desired distance feature and material feature nor which "further" filters are covered by the claim. The vague indication of correlation with one of the listed filters is an infinite definition of filters which does not allow to put into practice the invention or determine what is covered by the claims. The list of filters contains filters which are not commonly known and lack a definition in the claims. The sign in claim 10 is not defined.” (see Page 3 of the Office Action dated 09/25/2023).

Without commenting on the appropriateness of the rejection regarding claim 10, claim 10 has been amended by correcting “ φ_{1m} ” to “ φ_{2m} ” to make it consistent with the other claims and the description. Accordingly, Applicant respectfully submits that amended claim 10 is clear and definite.

With respect to claim 23, Applicant respectfully submits that the claim requires that “the inertial measurement unit is adapted to receive data determined by at least one detector according to claim 1,” and also “the inertial measurement unit is adapted to determine by evaluating the data from the detector and the at least one further sensor at least one property.” Thus, in contrast to the assertions of the Office, it is

clear that the inertial measurement unit of claim 23 receives data and evaluates that data, in combination with further data, for its measurement. Thus, claim 23 is definite.

With respect to claims 1 and 24, Applicant respectfully submits that the further filters are described in each recitation by a functional correlation. These functional correlations also indicate how the longitudinal coordinates and material properties are determined. Applicant also respectfully submits that these functional correlations properly define the claim scope and should not be unduly restricted by the requirement of structure. Finally, Applicant also respectfully submits that the filters recited in the present claims are known to a person having ordinary skill in the art, in contrast to the Office's assertions that they are not commonly known. Thus, claims 1 and 24 are definite.

Moreover, reference is made herein to the description given on starting of page 15 to page 23 of the application as-filed, which provides a clear and detailed explanation regarding the distance dependent image filters. A detailed description of the determining of the material feature and the listed filters is given in the present application as-filed starting on page 24. A detailed description of Pearson correlation coefficients is found in the present application as-filed at page 15. These descriptions inform the present claims and support their definiteness.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejection and allowance of the claims.

III. Claim Objections

Claims 1-25 are objected to because of informalities.

Claims 1-25 are amended herein consistent with the comments of the Office. It is believed these amendments overcome the objections.

IV. Rejections under 35 U.S.C. § 102

Claim(s) 1-7, 9-16, and 24-25 are rejected under 35 U.S.C. 102(a)(1) as allegedly being unpatentable over DE Patent Application Publication No. 19846619 A1 (Schwartz).

Applicant respectfully disagrees and submits that claims 1-7, 9-16, and 24-25 are novel and patentable over Schwartz.

Schwartz fails to teach or suggest at least the following features of claim 1:

a. wherein the evaluation device (132) is configured for determining at least one distance feature ϕ_{1z} by applying at least one distance dependent image filter ϕ_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; or a further distance dependent image filter ϕ_1 other which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|\rho_{\phi_1 \text{ other}, \Phi_z}| \geq 0.40$ with Φ_z being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof;

b. wherein the evaluation device (132) is configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_2 other which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_2 \text{ other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter,

the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof.

The present application describes using **beam profile analysis** for determining a material property of an object. Usual image filters which can be used for beam profile analysis yield features that depend on both distance and material such that reliable identification and classification of the material is not possible. However, it was found that the listed image filters of claim 1 of the present application allow reliable identification and classification of the material.

As mentioned above, Schwartz fails to disclose feature a). In contrast to using a distance dependent filter such as a depth-from-photon-ratio filter, a depth-from-defocus filter or a further correlated distance dependent image filter, Schwartz describes determining **depth information** by using structured light techniques based on triangulation. However, these techniques cannot be considered as distance dependent image filter according to claim 1, i.e. depth-from-defocus or depth-from-photon or a correlated image filter. Thus, Schwartz is deficient at least for this reason.

With respect to feature b), Schwartz describes determining **reflectivity information**. However, it describes different techniques as recited in claim 1 of the present application. Schwartz describes a variety of different methods for determining a "structure surface appearance quality", however, none of the disclosed methods relates to determining a material feature by applying at least one material dependent image filter of the list given in claim 1. Accordingly, claims 1-7, 9-16, and 24-25 are novel and patentable over Schwartz.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

V. Rejections under 35 U.S.C. § 103

i. Claims 8 and 22 stand rejected under 35 U.S.C. 103 as allegedly being unpatentable over DE Patent Application Publication No. 19846619 A1 (Schwartz), and in view of US Patent Application Publication No. 20180033146 (Bleyer).

Applicants respectfully submits that claims 8 and 22 are patentable over Schwartz and Bleyer.

The Office based the rejections in part on, “As per claim 8, Schwartz does not teach the detector (110) according to claim 1, wherein the material property m and/or the longitudinal coordinate z are determined by a function... Bleyer discusses intensity profile corrected based on distance (see claim 6 of Bleyer). It would be obvious to a skilled artisan at the effective date and time of the invention as claimed to incorporate Bleyer feature into Schwartz’s system to facilitate precise position and modification” (see Page 8 of the Office Action dated 09/25/23).

The Office based the further rejections in part on, “As per claim 22, Schwartz does not discuss a camera (136) for imaging at least one object (112), the camera (136) comprising at least one detector (110) according to claim 1. Bleyer discloses this feature (see Bleyer, claim 1 and background of the invention). See claim 8 reasoning” (see Page 8 of the Office Action dated 09/25/23).

Applicant respectfully disagrees. Claims 8 and 22 depend from claim 1, and therefore claims 8 and 22 are patentable over Schwartz at least for the same reasons outlined above.

Bleyer does not cure the deficiencies of Schwartz. As outlined above, the distinguishing features a) and b) of the present application have the **synergistic technical effect** of allowing for reliably identifying at least one material property of an object. On the contrary, neither Schwartz nor Bleyer describe a technique based on beam profile analysis. Thus, Schwartz and Bleyer are deficient for at least this reason.

The objective technical problem to be solved would be increasing reliability for identifying at least one material property of an object. Starting from Schwartz, the skilled person, firstly, would have to modify the device and the disclosed method to use beam

profile analysis for identifying a material property. In a further step, the skilled person would have to notice that there are problems with image filters, and in an even further step, the skilled person would have to find the image filters solving the above-mentioned problem. However, the skilled person has no motivation for any of the named steps when starting with Schwartz and Bleyer.

Accordingly, it would not have been obvious to a person having ordinary skill in the art to modify Schwartz and Bleyer with elements not taught or suggested in Schwartz or Bleyer in order to arrive at the present claims. There is no teaching, suggestion, motivation, or other reason in Schwartz and Bleyer to make such modifications. Such modifications would require impermissible hindsight and could only have been arrived at by using the present application as a roadmap.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

ii. Claim 17-20 stand rejected under 35 U.S.C. 103 as allegedly being unpatentable over DE Patent Application Publication No. 19846619 A1 (Schwartz), and in view of WO Patent Application Publication No. 2019042956 A1 (Schindler).

Applicants respectfully submits that claims 17-20 are patentable over Schwartz and Schindler.

Claims 17-20 depend from claim 1, and are therefore patentable over Schwartz at least for the reasons set forth above.

Schindler does not cure the deficiencies of Schwartz. The Office relied on Schindler for allegedly teaching a detector system (138) that further comprising at least one beacon device (140), a human-machine interface (142) for exchanging at least one item of information between a user (160) and a machine (168), entertainment device (144) and a tracking system (146). See Page 9 of the Office Action dated 09/25/23.

However, without conceding to any assertions of the Office, Applicant respectfully submits that Schindler does not teach or suggest all the elements of claim 1 from which

claims 17 and 18-20 depend, and the Office has not cited Schindler as such. Thus, Schindler is deficient at least for this reason.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

iii. Claim 21 stands rejected under 35 U.S.C. 103 as allegedly being unpatentable over DE Patent Application Publication No. 19846619 A1 (Schwartz)), and in view of CN Patent Application Publication No. 108363482 (Jiangsu).

Applicants respectfully submits that claim 21 is patentable over Schwartz and Jiangsu.

Claim 21 depends from claim 1, and is therefore patentable over Schwartz at least for the reasons set forth above.

Jiangsu does not cure the deficiencies of Schwartz. The Office based the rejections in part on, “As per claim 21, Schwartz does not teach a scanning system (148) for determining a depth profile of a scenery, the scanning system (148) comprising at least one detector (110) according to claim 1, the scanning system (148) further comprising at least one illumination source adapted to scan the scenery with at least one light beam. Jiangsu teaches that feature (see Jiangsu clam 1)” (see Page 10 of the Office Action dated 09/25/23).

However, without conceding to any assertions of the Office, Applicant respectfully submits that Jiangsu does not teach or suggest all the elements of claim 1 from which 21 depends, and the Office has not cited Jiangsu as such. Thus, Jiangsu is deficient at least for this reason.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

iv. Claim 23 stands rejected under 35 U.S.C. 103 as allegedly being unpatentable over DE Patent Application Publication No. 19846619 A1 (Schwartz).

Applicants respectfully submits that claim 23 is patentable over Schwartz.

The Office based the rejections in part on, “As per claim 23, Inertial measurement units are generally known, being adapted to receive data of a sensor as defined in claim 1, a material property detector, does not allow to establish a technical effect resulting from this feature” (see Page 10 of the Office Action dated 09/25/23).

Claim 23 depends from claim 1, and is therefore patentable over Schwartz at least for the reasons set forth above.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

CONCLUSION

Applicant requests reconsideration and allowance of all pending claims. Applicant authorizes the Commissioner to charge any fees due in connection with this response to Deposit Account No. 01-2384 in the name of Armstrong Teasdale LLP. Please reference docket number 27843-1809.

Respectfully submitted,

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APPLICATION #
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Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

Application Information

APPLICATION TYPE Utility - U.S. National Stage under 35 USC 371

PATENT # -

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FIRST NAMED INVENTOR Friedrich Schick

INTL. APPLICATION # -

INTL. FILING DATE -

CORRESPONDENCE ADDRESS -

AUTHORIZED BY Christopher Goff

Documents

TOTAL DOCUMENTS: 3

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2023-11-13 Amendment and Response to Non-Final Office Action 27843-1809.pdf	17	-	281 KB
2023-11-13 Amendment and Response to Non-Final Office Action 27843-1809-A....pdf	(1-1) 1	Amendment/Request for Reconsideration-After Non-Final Rejection	108 KB
2023-11-13 Amendment and Response to Non-Final Office Action 27843-1809-CLM.pdf	(2-9) 8	Claims	204 KB

2023-11-13 Amendment and Response to Non-Final Office Action 27843-1809- REM.pdf	(10-17)	8	Applicant Arguments/Remarks Made in an Amendment	219 KB
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MESSAGE DIGEST(SHA-512)

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the

International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Search Notes 	Application/Control No. 17/439,492	Applicant(s)/Patent Under Reexamination Schick et al.
	Examiner FRANTZ B JEAN	Art Unit 2454

CPC - Searched*		
Symbol	Date	Examiner
G01S17/46 ; G06T7/521 ; G06T7/73 ; G06V10/60 ; G06V10/145 ; G06V10/764 ; H04N23/74 ; H04N23/56 ; G01B11/22 ; G01C21/16 ; G01S17/66 ; G06T5/002 ; G06T5/20 ; ; G06T7/20	09/09/2023	FJ

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
348	135	11/18/2023	FJ

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Derwent, East, EPO, JPO, OCR, IBMTDBD, PGPUB, Google	09/08/2023	FJ
updated search	11/18/2023	FJ

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
G06T	7/521,73	11/18/2023	FJ

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<i>Index of Claims</i> 	Application/Control No. 17/439,492	Applicant(s)/Patent Under Reexamination Schick et al.
	Examiner FRANTZ B JEAN	Art Unit 2454

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	09/09/2023	11/18/2023							
	1	✓	=							
	2	✓	=							
	3	✓	=							
	4	✓	=							
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	25	✓	=							

Issue Classification 	Application/Control No. 17/439,492	Applicant(s)/Patent Under Reexamination Schick et al.
	Examiner FRANTZ B JEAN	Art Unit 2454

CPC						
Symbol					Type	Version
G01S	/	17	/	46	F	2013-01-01
G06T	/	7	/	521	I	2017-01-01
G06T	/	7	/	73	I	2017-01-01
G06V	/	10	/	764	I	2022-01-01
G01B	/	11	/	22	I	2013-01-01
G01C	/	21	/	16	I	2013-01-01
G01S	/	17	/	66	I	2013-01-01
G06T	/	5	/	002	I	2013-01-01
G06T	/	5	/	20	I	2013-01-01
G06T	/	7	/	20	I	2013-01-01
H04N	/	23	/	56	I	2023-01-01
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G06V	/	10	/	60	I	2022-01-01
G06V	/	10	/	145	I	2022-01-01

CPC Combination Sets					
Symbol				Type	Set
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NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	25	
/FRANTZ B JEAN/ Primary Examiner, Art Unit 2454	18 November 2023	O.G. Print Claim(s)	O.G. Print Figure
(Primary Examiner)	(Date)	1	1

Issue Classification 	Application/Control No. 17/439,492	Applicant(s)/Patent Under Reexamination Schick et al.
	Examiner FRANTZ B JEAN	Art Unit 2454

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06T7/521	/	7	/ 521
G06T7/73	/	7	/ 73

NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
348	135

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE		Total Claims Allowed:	
(Assistant Examiner)	(Date)	25	
/FRANTZ B JEAN/ Primary Examiner, Art Unit 2454 (Primary Examiner)	18 November 2023 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 1

<i>Issue Classification</i> 	Application/Control No. 17/439,492	Applicant(s)/Patent Under Reexamination Schick et al.
	Examiner FRANTZ B JEAN	Art Unit 2454

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1		10		19										
	2		11		20										
	3		12		21										
	4		13		22										
	5		14		23										
	6		15		24										
	7		16		25										
	8		17												
	9		18												

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 25	
/FRANTZ B JEAN/ Primary Examiner, Art Unit 2454 (Primary Examiner) _____ (Date) 18 November 2023		O.G. Print Claim(s) 1	O.G. Print Figure 1

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About 5,090,000 results (1.05 seconds)

"of" (and any subsequent words) was ignored because we limit queries to 32 words.

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Lars Desselberg Inventions, Patents and Patent Applications

The detector includes at least one sensor element including a matrix of optical sensors, the optical sensors each having a light-sensitive area. The sensor ...

**Justia**<https://patents.justia.com/patent/> ⓘ

Detector for determining a position of at least one object

The detector comprises: at least one sensor element having a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein each ...

**Google Patents**<https://patents.google.com/patent/> ⓘ

Detector for an optical detection of at least one object

WQ 2014/198629 A1 discloses a detector for determining a position of at least one object comprising at least one longitudinal optical sensor, the optical sensor ...

**Google Patents**<https://patents.google.com/patent/> ⓘ

EP3542179B1 - Detector for optically detecting at least one object

In a further preferred embodiment the detector may comprise: at least two optical sensors, each optical sensor having a light-sensitive area, wherein each light ...

People also ask ⓘ

What are the detectors used in optical sensors?

What is the principle used by an optical device to detect the presence or absence of light?

What can a optical sensor be used for?

What are the characteristics of optical sensors?

[Feedback](#)**European Patent Office**<https://data.epo.org/rest/patents/document> ⓘ

EP004224206A2* - EP 4 224 206 A2

detector; at least two optical sensors, wherein each optical sensor has at least one light sensitive area, wherein each optical sensor is designed to ...

**Google**<https://google.com/patents/> ⓘ

US9389315B2 - Detector comprising a transversal optical sensor ...

A detector (110) for determining a position of at least one object (112) is proposed. The detector (110) comprises: at least one transversal optical ...



Google

<https://www.google.com/patents>

US9557856B2 - Optical detector - Google Patents

The optical detector according to claim 1, wherein the at least one optical sensor comprises at least one optical sensor having a layer setup comprising at ...



European Patent Office

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EP003676629B1* - EP 3 676 629 B1

two sensor elements each having a matrix of optical sensors. At least one ... sensors (120) each having a light-sensitive area. (122), wherein each optical ...



Digikey

<https://forum.digikey.com/.../Sensors/Transducers>

How do light convergent reflective Optical Sensors work?

Aug 4, 2023 — The light convergent reflective sensors are suitable for detecting objects without concern for the background conditions.



Google

<https://www.google.com/patents>

Integrated optical light sensitive active matrix liquid crystal ...

The device of claim 1 wherein each of said light sensitive elements include a transistor, ...
Image sensor having TFT gate electrode surrounding the photoelectric ...

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BIB DATA SHEET

CONFIRMATION NO. 9002

SERIAL NUMBER	FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.	
17/439,492	09/15/2021	348	2454	27843-1809 182037US01	
APPLICANTS trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;					
INVENTORS Friedrich Schick, Ludwigshafen, GERMANY; Peter Schillen, Ludwigshafen, GERMANY; Patrick Schindler, Ludwigshafen, GERMANY; Andre Schmidt, Ludwigshafen, GERMANY; Michael Eberspach, Ludwigshafen, GERMANY; Christian Lennartz, Ludwigshafen, GERMANY; Robert Send, Karlsruhe, GERMANY; Lars Diesselberg, Karlsruhe, GERMANY; Heiko Hengen, Steinweiler, GERMANY; Ingmar Bruder, Ludwigshafen, GERMANY; Jakob Unger, Ludwigshafen, GERMANY; Christian Bonsignore, Ludwigshafen, GERMANY;					
** CONTINUING DATA ***** This application is a 371 of PCT/EP2020/056759 03/13/2020					
** FOREIGN APPLICATIONS ***** EUROPEAN PATENT OFFICE (EPO) 19163250.4 03/15/2019					
** IF REQUIRED, FOREIGN FILING LICENSE GRANTED ** 02/07/2022					
Foreign Priority claimed ☒ Yes ☐ No 35 USC 119(a-d) conditions met ☒ Yes ☐ No Verified and Acknowledged <u>/FRANTZ B JEAN/</u> Examiner's Signature	☐ Met after Allowance FJ Initials	STATE OR COUNTRY GERMANY	SHEETS DRAWINGS 2	TOTAL CLAIMS 25	INDEPENDENT CLAIMS 1
ADDRESS Patent Docket Department Armstrong Teasdale LLP 7700 Forsyth Boulevard Suite 1800 St. Louis, MO 63105 UNITED STATES					
TITLE DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY					
FEES: Authority has been given in Paper No. _____ to charge/credit DEPOSIT ACCOUNT			☐ All Fees ☐ 1.16 Fees (Filing) ☐ 1.17 Fees (Processing Ext. of time)		

FILING FEE RECEIVED 0.00	No. _____ for following:	☐ 1.18 Fees (Issue)
		☐ Other _____
		☐ Credit

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	2516	(Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore)).inv.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:39 AM
L2	440018	(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:49 AM
L3	757	(G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:01 AM
L4	183885	(detect\$3 OR sen\$3) near10 (identif\$ OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP,	OR	ON	ON	2023/09/09 02:34 AM

		classify\$3) WITH (material OR property)	AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L5	249451	longitudinal WITH coordinate AND (material OR property OR evaluat\$3 OR distance)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L6	2430	L4 AND I5	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L7	11	L6 AND I3	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO,	OR	ON	ON	2023/09/09 02:38 AM

L8	1872340	(reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)	RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 09:57 AM
L9	1341757	filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss)	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:00 AM
L10	113287	L8 AND I9	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:01 AM
L11	67	L10 AND I6	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU,	OR	ON	ON	2023/09/09 10:02 AM

L12	0	L11 AND I1	CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM
L13	35	L11 AND I2	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM
L14	11	L11 AND I3	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM

L15	4	"20180033146".pn.	WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 06:13 PM
L16	2541	(Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore)).inv.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L17	454819	(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L18	791	(G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU,	OR	ON	ON	2023/11/18 08:50 PM

L19	188776	G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc. (detect\$3 OR sen\$3) near10 (identif\$ OR classify\$3) WITH (material OR property)	CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L20	259070	longitudinal WITH coordinate AND (material OR property OR evaluat\$3 OR distance)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L21	2529	L4 AND I5	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM

L22	12	L6 AND I3	VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L23	1891255	(reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L24	1363696	filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss)	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L25	114665	L8 AND I9	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB,	OR	ON	ON	2023/11/18 08:50 PM

L26	70	L10 AND I6	HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L27	27	L11 AND I1	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L28	36	L11 AND I2	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM

L29	12	L11 AND I3	IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM
L30	4	"20180033146".pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/11/18 08:50 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	2516	(Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore)).inv.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:39 AM
L2	440018	(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:49 AM
L3	757	(G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:01 AM
L4	183885	(detect\$3 OR sen\$3) near10 (identif\$ OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP,	OR	ON	ON	2023/09/09 02:34 AM

		classify\$3) WITH (material OR property)	AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L5	249451	longitudinal WITH coordinate AND (material OR property OR evaluat\$3 OR distance)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L6	2430	L4 AND I5	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L7	11	L6 AND I3	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO,	OR	ON	ON	2023/09/09 02:38 AM

			RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	2516	(Friedrichr near3 Schoick) OR (Peter near3 Schillen) OR ((Patrick near3 Schindler) OR (Andre near3 Schmidt) OR (Michael near3 Eberspach) OR (Christian near3 Lennartz) OR (Robert near3 Send) OR (Lars near3 Diesselberg) OR (Heiko near3 Hengen) OR (Ingmar near3 Bruder) OR (Jakob near3 Unger) OR (Christian near3 Bonsignore)).inv.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:39 AM
L2	440018	(348/135).ccls. OR (G01S17/46 OR G06T7/521 OR G06T7/73 OR G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56 OR G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 01:49 AM
L3	757	(G01S17/46 OR G06T7/521 OR G06T7/73).cpc. AND (G06V10/60 OR G06V10/145 OR G06V10/764 OR H04N23/74 OR H04N23/56).cpc. AND (G01B11/22 OR G01C21/16 OR G01S17/66 OR G06T5/002 OR G06T5/20 OR G06T7/20).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:01 AM
L4	183885	(detect\$3 OR sen\$3) near10 (identif\$ OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP,	OR	ON	ON	2023/09/09 02:34 AM

		classify\$3) WITH (material OR property)	AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
L5	249451	longitudinal WITH coordinate AND (material OR property OR evaluat\$3 OR distance)	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L6	2430	L4 AND I5	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 02:37 AM
L7	11	L6 AND I3	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO,	OR	ON	ON	2023/09/09 02:38 AM

L8	1872340	(reflect\$3 OR image OR object OR evaluat\$3 OR distance) near60 beam near5 (light OR profile)	RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 09:57 AM
L9	1341757	filter near6 (image OR median OR contrast OR energy OR homogeneity OR dissimilarity OR smoothnwss)	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:00 AM
L10	113287	L8 AND I9	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:01 AM
L11	67	L10 AND I6	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU,	OR	ON	ON	2023/09/09 10:02 AM

L12	0	L11 AND I1	CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM
L13	35	L11 AND I2	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2023/09/09 10:03 AM
L14	11	L11 AND I3	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN,	OR	ON	ON	2023/09/09 10:03 AM

			WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)				
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



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NOTICE OF ALLOWANCE AND FEE(S) DUE

76656 7590 12/01/2023
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

EXAMINER

JEAN, FRANTZ B

ART UNIT

PAPER NUMBER

2454

DATE MAILED: 12/01/2023

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/439,492	09/15/2021	Friedrich Schick	27843-1809 182037US01	9002

TITLE OF INVENTION: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/01/2024

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

76656 7590 12/01/2023
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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17/439,492 09/15/2021 Friedrich Schick 27843-1809 182037US01 9002

TITLE OF INVENTION: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/01/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
JEAN, FRANTZ B	2454	348-135000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/439,492	09/15/2021	Friedrich Schick	27843-1809 182037US01	9002

76656 7590 12/01/2023
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

EXAMINER

JEAN, FRANTZ B

ART UNIT	PAPER NUMBER
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2454

DATE MAILED: 12/01/2023

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 17/439,492	Applicant(s) Schick et al.	
	Examiner FRANTZ B JEAN	Art Unit 2454	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to the amendment filed on 13 November 2023.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-25. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some* c) ☐ None of the:

1. ☒ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____.	6. ☒ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____.	7. ☐ Other _____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	

/FRANTZ B JEAN/ Primary Examiner, Art Unit 2454	
--	--

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This office action is in response to applicant's arguments filed on 13 November 2023. Claims 1-25 are allowed over the prior art of record.

The following is an examiner's statement of reasons for allowance:

. The prior art of record fails to teach and suggest as a combination a detector for identifying at least one material property m comprising at least one sensor element comprising a matrix of optical sensors, the optical sensors each having a light-sensitive area, wherein the sensor element is configured for recording at least one reflection image of a light beam originating from at least one object; and at least one evaluation device configured for determining the material property by evaluation of at least one beam profile of the reflection image, wherein the evaluation device is configured for determining at least one distance feature Φ_{1z} by applying at least one distance dependent image filter D_1 to the reflection image, wherein the distance dependent image filter is at least one filter selected from the group consisting of: a depth-from-photon-ratio filter; a depth-from-defocus filter; or a linear combination thereof; and a further distance dependent image filter Φ_{1other} which correlates to the depth-from-photon-ratio filter and/or the depth-from-defocus filter or a linear combination thereof by $|p_{ot_{her}, o_z}| > 0.40$ with ... being one of the depth-from-photon-ratio filter or the depth-from-defocus filter or a linear combination thereof, wherein the evaluation device is configured for determining at least one material feature ... by applying at least one material dependent image filter O_2 to the reflection image, wherein the material dependent image filter is at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter; a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-

Art Unit: 2454

based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law's energy filter; a threshold area filter; or a linear combination thereof; and a further material dependent image filter $O_{z_0}th_{er}$ which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof by $|po2ot_{er,om}|^{20.40}$ with ... being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law's energy filter, or the threshold area filter, or a linear combination thereof, and wherein the evaluation device is configured for determining a longitudinal coordinate z and the material property m by evaluating the distance feature ... and the material feature

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Any inquiry concerning this communication or earlier communications from the examiner should be directed to FRANTZ B JEAN whose telephone number is (571)272-3937. The examiner can normally be reached 8-5 M-F.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Glenton B Burgess can be reached on 5712723949. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/FRANTZ B JEAN/
Primary Examiner, Art Unit 2454

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
17/439,492	02/29/2024 09:14:05 AM Z ET	27843-1809 182037US01

Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	9002	FILED BY	Barbara McGregor
PATENT CENTER #	64494470	FILING DATE	09/15/2021
CUSTOMER #	76656	FIRST NAMED INVENTOR	Friedrich Schick
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2024-02-29 Issue Fee Transmittal 27843-1809.pdf	1	Issue Fee Payment (PTO-85B)	158 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2024-02-29 Issue Fee Transmittal 27843-1809.pdf	1F1A361B2D97DF5111823475150E7CB843C0190E3CD6E0877 D9325229A5DC9221696A2B47DACB5EDBA24284E23F44F1C9 35D942A953207F5813BB7F3762A087C

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as

described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
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ELECTRONIC PAYMENT RECEIPT

APPLICATION # 17/439,492	RECEIPT DATE / TIME 02/29/2024 09:14:05 AM Z ET	ATTORNEY DOCKET # 27843-1809 182037US01
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Title of Invention

DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

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APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	9002	FILED BY	Barbara McGregor
PATENT CENTER #	64494470	AUTHORIZED BY	Christopher Goff
CUSTOMER #	76656	FILING DATE	09/15/2021
INTL. APPLICATION #	-	INTL. FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Friedrich Schick

Payment Information

PAYMENT METHOD DA / 012384	PAYMENT TRANSACTION ID E20242S014509611	PAYMENT AUTHORIZED BY Barbara McGregor
--------------------------------------	---	--

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1200.00	1	1200.00
TOTAL AMOUNT:				\$1,200.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

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New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

76656 7590 12/01/2023
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/439,492	09/15/2021	Friedrich Schick	27843-1809 182037US01	9002

TITLE OF INVENTION: DETECTOR FOR IDENTIFYING AT LEAST ONE MATERIAL PROPERTY

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/01/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
JEAN, FRANTZ B	2454	348-135000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Armstrong Teasdale LLP

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

TRINAMIX GMBH

LUDWIGSHAFEN AM RHEIN, GERMANY

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 012384

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Christopher M. Goff/

Date February 29, 2024

Typed or printed name Christopher M. Goff

Registration No. 41785



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/439,492	04/02/2024	11947013	27843-1809 182037US01	9002

76656 7590 03/13/2024
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 314 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Friedrich Schick, Ludwigshafen, GERMANY;
Peter Schillen, Ludwigshafen, GERMANY;
Patrick Schindler, Ludwigshafen, GERMANY;
Andre Schmidt, Ludwigshafen, GERMANY;
Michael Eberspach, Ludwigshafen, GERMANY;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

trinamiX GmbH, Ludwigshafen am Rhein, GERMANY;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit [SelectUSA.gov](https://selectusa.gov).



UNITED STATES PATENT AND TRADEMARK OFFICE

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EXAMINER

JEAN, FRANTZ B

ART UNIT	PAPER NUMBER
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2454

NOTIFICATION DATE	DELIVERY MODE
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Patent Docket Department
Armstrong Teasdale LLP
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